Local Mixed Holomorphic-Topological String Field Theory

The Formal-Darboux Stalk in Mixed Holomorphic-Topological Strings

Formal Local Hamiltonian BF Theory,
Reduced CE/PV Central Operations,
and the Trace Chart at N Dirac Branes on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$

the formal-Darboux stalk of the Mixed Holomorphic–Topological Deligne conjecture

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Abstract

The local mixed holomorphic–topological sector of the Bershadsky–Cecotti–Ooguri–Vafa B-twist [1] on $X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ is topological along $\mathbb{R}^2$ and holomorphic along $\mathbb{C}^2$ [25]. It is written here in the Costello–Li Batalin–Vilkovisky (BV) / factorization renormalization formalism [12][13][14]. At a formal holomorphic-symplectic Dirac brane vacuum the formal local closed fields are Hamiltonian BF fields on the formal symplectic polydisk, and the boundary fields form the Koszul model of the derived commuting variety of the Chan–Paton matrices. The mixed holomorphic–topological Deligne conjecture asserts that the physical bulk is the derived chiral E_2-centre of the primitive open boundary category. The theorem identifies the trace-sector formal-Darboux stalk at N Dirac branes.

The closed fields are $\mathcal{A} \in \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\mathfrak{h}$, with

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1, \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g,$$

and antifields $\mathcal{B}$ valued in the Grothendieck–Matlis residue dual

$$\mathfrak{h}^\vee_{\mathrm{cont}} = \mathrm{Ann}_{\mathbf{Res}}(\mathbb{C}\cdot 1) \subset H^2_{(z_1,z_2)}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2.$$

They form the shifted-cotangent BF Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$, with classical action

$$S = \int_X \big\langle \mathcal{B},\ (d + \bar\partial)\mathcal{A} + \tfrac{1}{2}[\mathcal{A}, \mathcal{A}]_{\mathfrak{h}} \big\rangle.$$

The closed local observables at the formal insertion point are $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$.

A stack of N Dirac branes on the worldline $\mathbb{R}_t \subset \mathbb{R}^2_{\mathrm{top}}$ has transverse Chan–Paton coordinates $\phi_1, \phi_2 \in \mathfrak{gl}_N$. The formal Darboux coordinate algebra $A^{\mathrm{coord}}_b = \widehat{\mathcal{O}}_{\mathbb{C}^2,b} \cong \mathbb{C}[[z_1, z_2]]$ acts by the substitution $z_i \mapsto \phi_i$, and the Koszul field $\psi \in \mathfrak{gl}_N[1]$ imposes $Q\psi = [\phi_1, \phi_2]$. Thus the classical boundary algebra is

$$A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}\big(\mathfrak{gl}_N,\ \mathrm{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])\big),$$

the Chevalley–Eilenberg/Koszul algebra of the derived commuting stack $[\mu^{-1}_{\mathrm{der}}(0)/\mathrm{GL}_N]$. For $f \in \bar A = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ the brane evaluation is the trace

$$J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)} \in \mathcal{O}(\mu^{-1}_{\mathrm{der}}(0))^{GL_N}.$$

Trace cyclicity kills ordinary commutators, and the Koszul differential records the derived moment-map relation; adjacent orderings of the matrix substitution are therefore cohomologous on the derived commuting stack. For $f \in \mathfrak{h}$, the same formula is interpreted in the completed formal brane algebra.

The theorem constructs the closed-to-open dictionary on the formal-Darboux trace sector,

$$\Phi_N : C^\bullet_{\mathrm{CE,cont}}(\widehat{\mathfrak{g}}_{\mathrm{cent}}) \dashrightarrow C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}), \qquad (c_f, u_f, K) \mapsto (\theta_f, J(f), N),$$

where $\widehat{\mathfrak{g}}_{\mathrm{cent}}$ is the Capelli projective central extension and θ_f is a Koszul lift of the Hamiltonian Schouten derivation. After scalar reduction, stable trace passage, and the stated Tate/pro-Matlis admissible completion, this is the Chevalley–Eilenberg/polyvector coordinate isomorphism on $\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$. It is the trace-sector formal-Darboux stalk of the mixed holomorphic–topological E_2-Deligne conjecture; Beilinson–Drinfeld proves the corresponding chiral centre theorem in complex dimension 1 [28]. At fixed finite N it is not an isomorphism from the whole bulk to the whole

Hochschild centre; Procesi–Razmyslov trace relations, stack stabilizer classes, and unreduced P_0-centrality homotopies are additional finite-N data.

The constant Hamiltonian line removed from $\mathfrak{h}$ survives on the brane as the trace $U(1)$. It produces the finite-N Capelli class

$$\omega(f,g) = [\{f,g\}]_0, \qquad \{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N, \qquad [\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_{\mathrm{proj}} = \hbar N \cdot \mathbf{1}.$$

These formulas represent $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})[[\hbar]]$. When a determinant or modular line with connection exists and its Atiyah class pulls back to $[\bar{c}]$, this class is the projective curvature at the trace generator.

The holomorphic singular product is the two-complex-dimensional diagonal local-cohomology kernel

$$\mathsf{J}_x(z)\mathsf{J}_y(w) \sim \frac{d\zeta_1\, d\zeta_2}{\zeta_1\zeta_2}\, \mathsf{J}_{[x,y]_{\widehat{\mathfrak{g}}_{\mathrm{cent}}}}(w), \qquad \zeta_i = z_i - w_i.$$

It is the residue-pairing extension of the Heisenberg curve kernel; a vertex algebra appears only after choosing a holomorphic curve restriction and matching the pushed-forward bracket, BV pairing, brane image, and anomaly. On the cyclic framed stable branch $\mathbb{C}[\phi_1, \phi_2] \cdot v = \mathbb{C}^N$, the same formal-Darboux chart gives the ADHM presentation of $\mathrm{Hilb}^N(\mathbb{C}^2)$, and its Weyl/Moyal deformation is the corresponding noncommutative Hilbert-scheme normalization with centre-of-mass charge $\hbar N$.

The global mixed holomorphic–topological Deligne identification is the descent problem for this local fibre over the brane-link boundary. Its obstruction classes are the all-arity chiral E_2-Deligne class, the Costello–Gwilliam descent class, the all-order quantum master equation class, and the class of the open–closed comparison cone. The formal calculation gives the trace-sector CE/PV coordinate map, the arity-two diagonal kernel, and the Capelli projective anomaly. Compact Calabi–Yau enumerative theory, Hall–Drinfeld–Borcherds comparison, and the operator lift of Φ_{10} enter through their matched-conventions comparison theorems.

Summary

The B-model topological string of Bershadsky–Cecotti–Ooguri–Vafa [1] is the topological B-twist of the $\mathcal{N} = (2,2)$ worldsheet sigma model on a Calabi–Yau target. In the Costello–Li BV renormalisation [12][13][14], its local Costello holomorphic–topological sector [25] lives on $X = \mathbb{R}^2_{t,s} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$: the theory is topological in (t,s), holomorphic in (z_1, z_2), and near a holomorphic-symplectic Dirac brane vacuum its closed fields form Hamiltonian BF theory. The closed-string fields $\mathcal{A}$ take values in the reduced Hamiltonian Lie algebra of the formal symplectic polydisk, $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ with Poisson bracket $\{f,g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$; the antifields $\mathcal{B}$ take values in the continuous Grothendieck–Matlis residue dual $\mathfrak{h}^\vee_{\mathrm{cont}} = \mathrm{Ann}_{\mathbf{Res}}(\mathbb{C} \cdot 1) \subset H^2_{(z_1,z_2)}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2$ paired by $\langle f, \omega\rangle = \mathbf{Res}_0 f\,\omega$. Together they form the shifted-cotangent BF Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ on the formal symplectic polydisk $\widehat{\mathbb{C}^2}_0$, with classical action

$$S = \int_X \langle \mathcal{B}, (d + \bar{\partial})\mathcal{A} + \tfrac{1}{2}[\mathcal{A}, \mathcal{A}]_{\mathfrak{h}} \rangle.$$

The classical observables form the holomorphic factorization algebra $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ on the holomorphic surface in the sense of Costello–Gwilliam [15].

The open boundary is a stack of N Dirac branes on the worldline $\mathbb{R}_t \subset X$. Its two holomorphic transverse coordinates become Chan–Paton matrices $\phi_1, \phi_2 \in \mathfrak{gl}_N$ under the substitution $z_i \mapsto \phi_i$, subject to the moment-map constraint $[\phi_1, \phi_2] = 0$. The antifield $\psi \in \mathfrak{gl}_N[1]$ with $Q\psi = [\phi_1, \phi_2]$ gives the Koszul resolution of the moment-map ideal, equivalently the derived zero fibre of $\mu(\phi_1, \phi_2) = [\phi_1, \phi_2]$. The classical open-string algebra on the worldline is the Chevalley–Eilenberg cochain complex

$$A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}\big(\mathfrak{gl}_N,\ \mathrm{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])\big),$$

with Koszul differential $Q\psi = [\phi_1, \phi_2]$. A polynomial closed Hamiltonian $f \in \bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$ acts on this brane sector through the trace map

$$J : \bar{A} \longrightarrow \mathcal{O}(\mu^{-1}_{\mathrm{der}}(0))^{GL_N}, \qquad J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)},$$

the GL_N-invariant function obtained by substituting the matrices ϕ_i into f and taking the trace on the derived commuting variety $\mu^{-1}_{\mathrm{der}}(0)$ at the brane vacuum $b \in \mathbb{C}^2$. For $f \in \mathfrak{h}$, $J(f)$ denotes the continuous extension to the completed formal brane algebra.

The Mixed Holomorphic–Topological Deligne conjecture asserts that the physical bulk is the chiral derived E_2-centre of the primitive boundary open factorisation category $\mathcal{C}^{\mathrm{op}}_\partial$ on the Costello–Gwilliam brane-link boundary ∂X of the Dirac brane defect $L = \mathbb{R}_t \times \{b\} \subset X$. The theorem proved here is the formal-Darboux stalk of this assertion in the trace sector:

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \dashrightarrow C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}), \qquad c_f \mapsto \theta_f, \quad u_f \mapsto J(f).$$

After scalar reduction, stable trace passage, and the stated Tate/pro-Matlis admissible completion, this map is the continuous CE/PV coordinate isomorphism on the trace algebra. It is the local Darboux coordinate expression of the derived chiral centre in the part detected by Hamiltonian traces. At fixed finite N, finite trace identities, stack stabilizer classes, and unreduced centrality homotopies remain outside this trace-sector isomorphism. Away from the stalk, the corresponding global assertion is the open chiral E_d-Deligne conjecture at $d = 2$.

The classical BV pair is the field $\mathcal{A}$ valued in $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\mathfrak{h}$ and antifield $\mathcal{B}$ valued in $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\mathfrak{h}^\vee_{\mathrm{cont}}[1]$, paired by the (-1)-shifted symplectic form $\int_X \langle \mathcal{B}, \mathcal{A}\rangle$ obtained from the residue pairing on $\mathfrak{h} \otimes \mathfrak{h}^\vee_{\mathrm{cont}}$ and integration of mixed-HT forms against $dz_1\, dz_2$. The BV differential is $Q = d + \bar\partial + d_{\mathrm{CE}}$, with d_{CE} the Chevalley–Eilenberg differential of $\mathfrak{g}$; the master equation $\{S, S\} = 0$ holds because the BF interaction $I = \frac{1}{2}\int_X \langle \mathcal{B}, [\mathcal{A}, \mathcal{A}]_{\mathfrak{h}}\rangle$ encodes the Lie bracket on $\mathfrak{h}$. Renormalisation proceeds in the heat-kernel scheme of Costello–Gwilliam: a gauge-fixing operator $Q^{\mathrm{GF}} = d^* + \bar\partial^*$ for chosen flat and Hermitian metrics gives the Laplacian $D = [Q, Q^{\mathrm{GF}}]$, heat kernel $K_t = e^{-tD}$, and propagator $P_{\varepsilon,L} = \int_\varepsilon^L \frac{1}{2}(Q^{\mathrm{GF}}\otimes 1 + 1\otimes Q^{\mathrm{GF}})\, K_t\, dt$. At RG scale L the effective interaction is the Costello transform $I[L] = W(P_{\varepsilon,L}, I)$, and the quantum master equation

$$QI[L] + \tfrac{1}{2}\{I[L], I[L]\}_L + \hbar\Delta_L\, I[L] = 0$$

is the consistency condition for renormalised quantisation. Costello deformation cohomology of $I[L]$ is the obstruction theory governing the closed-to-open coupling Φ_N (Theorem 1.26); its reduced non-scalar obstruction is the class $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]$ (Theorem E.38).

Tate setup and the residue pairing. Throughout, $\mathfrak{h}$ carries the linear topology defined by the images of $\mathfrak{m}^k \subset A$, with continuous dual $\mathfrak{h}^\vee_{\mathrm{cont}} = \mathrm{Hom}^{\mathrm{cts}}_{\mathbb{C}}(\mathfrak{h}, \mathbb{C})$ in the category of Tate $\mathbb{C}$-vector spaces (linearly compact / pro-finite). The residue pairing identifies $\mathfrak{h}^\vee_{\mathrm{cont}}$ with the Matlis principal-part module

$$\mathcal{P} = \mathrm{Ann}_{\mathbf{Res}}(\mathbb{C}\cdot 1) \subset H^2_{\mathfrak{m}}(A) \otimes \omega_A, \quad \mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\,\rho_{a,b}, \quad \omega_A = dz_1 \wedge dz_2, \quad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2,$$

the holomorphic-surface two-variable refinement of the Heisenberg current on the formal disk. Inverse limits over Matlis windows $\mathcal{P}_{\le N} = \bigoplus_{a+b\le N} \mathbb{C}\rho_{a,b}$ are read as derived limits $\mathbf{R}\varprojlim_N \mathcal{P}_{\le N}$, with $\varprojlim^1$-data tracked.

The moment-map equation $[\phi_1, \phi_2] = 0$ defines a singular variety on $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$: for $N \ge 2$ the entries of $[\phi_1, \phi_2]$ have syzygies and define the derived commuting-pair stack. The Koszul complex $\mathbb{C}[\phi_1, \phi_2, \psi]$ with $Q\psi = [\phi_1, \phi_2]$ produces a derived enhancement: the ghost-zero brane moduli is the derived commuting-pair stack $[\mu_{\mathrm{der}}^{-1}(0)/GL_N]$. The classical Hilbert scheme $\mathcal{H}\mathrm{ilb}^N(\mathbb{C}^2)$ is recovered on the cyclic framed stable branch obtained by adjoining a framing vector $v \in \mathbb{C}^N$ with $\mathbb{C}[\phi_1, \phi_2] \cdot v = \mathbb{C}^N$. The Nekrasov–Schwarz noncommutative Hilbert scheme is obtained from the framed ADHM/noncommutative deformation; its central parameter is carried by the framing or by a projective central extension, not by the finite-dimensional equation $[\phi_1, \phi_2] = \hbar I_N$ in $\mathfrak{gl}_N$ [47].

Uniqueness of J. Procesi–Razmyslov [17][18] identifies the polynomial GL_N-invariants on $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$ with the algebra generated by mixed traces of words in ϕ_1, ϕ_2; Loday–Quillen–Tsygan [2][3] identify the primitive part with the cyclic-word trace classes. Combined with Dirac descent through the moment-map ideal, formal-Darboux naturality, and the linear normalisation $J(z_i) = \overline{\mathrm{Tr}\,\phi_i}$, J is forced monomial-by-monomial (Theorem 5.42); among admissible protected Hamiltonian sectors at a holomorphic-symplectic brane point, J is the formal-Darboux normal-form datum (Theorem 5.147).

Concretely the closed observables are the Chevalley–Eilenberg chains of the local Lie algebra $\Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}$ on $U \subset \mathbb{R}^2_{\mathrm{top}}$, $B \subset \mathbb{C}^2_{\mathrm{hol}}$ (Definition 1.12, Proposition 1.36); pushforward along $\pi\colon X \to \mathbb{C}^2_{\mathrm{hol}}$ integrates out the topological factor and gives $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}(B) = \mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}(D^2 \times B) \in \mathrm{Alg}_{E_2}$ (Theorem 1.22, Corollary 1.23); the E_2-valuation arises from the locally constant factorization-algebra structure on $\mathbb{R}^2_{\mathrm{top}}$ (Lurie–Costello–Gwilliam). The native holomorphic singular product of $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ is a two-variable expansion on the diagonal $\Delta \subset \mathbb{C}^2_z \times \mathbb{C}^2_w$: the singular coefficient module is the Grothendieck local-cohomology principal-part module $H^2_\Delta(\mathbb{C}[[z_1, z_2, w_1, w_2]])\, d\zeta_1 d\zeta_2$, with arity-two Cauchy kernel $K^{(2)}_\Delta(z, w) = d\zeta_1\, d\zeta_2/(\zeta_1\zeta_2)$ in formal diagonal coordinates $\zeta_i = z_i - w_i$ (Theorem 1.17). A one-dimensional chiral algebra is obtained after constructing a restriction or pushforward to a curve $C \hookrightarrow \mathbb{C}^2$. Such a construction must specify the transverse modes, the pushed-forward bracket, the BV pairing, the image of the brane chart, and the anomaly normalisation. Heisenberg, affine Kac–Moody, $\beta\gamma$, Virasoro, and W-algebra vertex algebras enter as curve-chiral comparison objects with these extra hypotheses.

The closed-to-open dictionary on Chevalley–Eilenberg generators of $\mathfrak{g}$ assigns to each generator-dual coordinate c_f the Hamiltonian Schouten derivation $\theta_f \in C^1(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ and to each cotangent-dual coordinate u_f the brane observable $O_f = J(f) \in C^0(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ (Theorem 1.26). A lift to factorisation algebras is precisely a chain map

$$\Phi_N : \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk},0} \longrightarrow Z^{\mathrm{der}}_{E_1}(A^{\mathrm{cl}}_{\partial,N})$$

whose existence at the chain level requires vanishing of the obstruction vector $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$ (Theorem E.53) with null-homotopies constructed in Appendix G.

Capelli scalar anomaly. Closed Hamiltonian reduction quotients the constant Hamiltonian line $\mathbb{C}\cdot 1$, on which $\mathrm{Tr}\ 1 = N$; the brane retains it as the trace-$U(1)$ centre-of-mass cocycle of N coincident D-branes on noncommutative $\mathbb{C}^2_\hbar$. The class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})[[\hbar]]$, $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$, has three faces:

the Lie 2-cocycle $\omega(f, g) = [\{f, g\}]_0$ on $\bar{A}$ (Lemma 6.2), the Poisson bracket of trace coordinates $\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$, and the projective Moyal commutator of the Weyl-ordered trace representation,

$$[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_{\mathrm{proj}} = \hbar N \cdot \mathbf{1}$$

(Theorem E.13). Here $[\ ,\ ]_{\mathrm{proj}}$ denotes the central component of the Weyl-ordered commutator in the Capelli central extension $\mathfrak{h}_{\hbar,N}$. The number N is the trace-$U(1)$ central charge of the noncommutative D-brane stack on $\mathbb{C}^2_\hbar$. A determinant-line curvature interpretation of this class requires the determinant-line datum of Theorem 6.8. The strict scalar-reduced Hamiltonian action lifts to a projective action on the Capelli central extension $0 \to \mathbb{C} \to \mathfrak{h}_{\hbar,N} \to \bar{A} \to 0$.

Obstruction calculus. Write Θ_{loc} for the local trace-substitution datum $(J, \phi_1, \phi_2, \psi, c_f \mapsto \theta_f, u_f \mapsto J(f))$. Let $d^L_{\mathrm{QME}} = Q + \{I[L], -\}_L + \hbar\Delta_L$ denote Costello's scale-L quantum master differential on local functionals; its associated-graded classical part is $d_{\mathrm{cl}} = Q + \{I_0, -\}$. Globalization, quantization, all-bidegree Weyl/Moyal compatibility, and chain-level centre maps place this datum in different native deformation complexes. A single curvature equation is defined only after a common deformation complex and comparison morphisms have been constructed (Theorem 7.1). All-bidegree Weyl/trace compatibility is controlled by the radial Weyl cokernel $\Omega^{\mathrm{rad}}_{a,b} \in \operatorname{coker} B_{a,b}$ on the decorated Poincaré–Birkhoff–Witt (PBW) Stokes operator $B_{a,b}$ (Theorems 5.218, F.44), with $\Omega^{\mathrm{rad}}_{a,b} = 0$ for all (a, b) under the Harish-Chandra/Levasseur–Stafford radial-image hypothesis of Statement F.8; the chain-level P_0-factorization-centre lift of Φ_N by the factorization-centre vector $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$ (Theorem E.53); the reduced non-scalar Costello QME by the class $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1\big(\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I),\ d^L_{\mathrm{QME}}\big)$ in the filtered weighted brane-defect local-functional complex, whose order-three CE-source row θ_3 gives the prototype $H^1_{\mathrm{bare}} = \mathbb{C} \cdot \mathsf{E}_{\theta_3,M}$. After a scalar-zero local counterterm $C_{\theta_3,M}$ with $d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$ is adjoined to the finite-window complex, the coefficient $c = 1$ kills the row in any finite window $M \ni \theta_3$ (Theorem E.38); and the attempted strict all-window Bochner–Martinelli current transfer meets $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$, nonzero on every finite Matlis window $\mathcal{P}_{\le N}$ (Theorem 8.40), with coherent action on the pro-Matlis substituted module $\widehat{\mathcal{P}}_\Pi = \varprojlim_N \mathcal{P}_{\le N}$ (Theorem 8.42). These four classes are the obstructions identified for the centre, quantum, radial, and current-transfer comparisons considered here; each comparison carries its null-homotopy in its native complex. Minimal-model classification and an exact count of nonzero higher brackets are independent classification theorems.

The local model is the holomorphic factorization algebra of Hamiltonian currents on $\mathbb{C}^2$. Its coefficient dual is the Matlis principal-part module $\mathcal{P}$. Under a curve restriction $C \hookrightarrow \mathbb{C}^2$, $\mathcal{P}$ restricts to the Kac–Moody central-pairing module Ω_K and $\mathfrak{h}$ to the regular functions on the curve. Costello's twisted M-theory family [25] motivates the source conventions; no identification with that theory is used in the proof. On the cyclic framed stable branch it gives the trace chart on $\mathcal{H}\mathrm{ilb}^N(\mathbb{C}^2)$; the Moyal deformation is the corresponding framed noncommutative deformation, not an ordinary finite-rank scalar commutator equation.

The global formulation has a source category C and the composite functor $\Phi_d = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi^{\mathrm{FA}}_d$. The proof below uses only the local formal stalk. Its native object is the local holomorphic E_2 factorization algebra $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ on the formal holomorphic surface. Conventions match a global theorem through the matched-conventions obstruction class $\mathrm{Ob}_{\mathrm{comparison}}$ of Theorem 11.13: a nonzero coordinate of $\mathrm{Ob}_{\mathrm{comparison}}$ is the obstruction to using the local Hamiltonian BF theorem inside a compact target carrying a BCOV/Kodaira–Spencer propagator, a Calabi–Yau-to-chiral specialization, an Igusa/Borcherds–Kac–Moody normalization, a Gromov–Witten/Donaldson–Thomas/Pandharipande–Thomas virtual trace, or an Ooguri–Strominger–Vafa attractor comparison.

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LOCAL DICTIONARY

Linear-algebraic category: Tate $\mathbb{C}$-vector spaces. All linear-algebraic constructions take place in the symmetric monoidal category $\mathrm{Tate}_{\mathbb{C}}$ of Tate $\mathbb{C}$-vector spaces: ind-pro finite-dimensional $\mathbb{C}$-vector spaces equipped with linearly compact lattices. A linearly compact module is a lattice object, not the whole Tate object. The completed tensor product $\widehat{\otimes}$ is the topological tensor product on $\mathrm{Tate}_{\mathbb{C}}$; continuous duality is the strict functor $V \mapsto V^{\vee}_{\mathrm{cont}} = \mathrm{Hom}^{\mathrm{cts}}_{\mathbb{C}}(V, \mathbb{C})$. The $\mathbb{C}$-algebra $A = \mathbb{C}[[z_1, z_2]]$ is a complete linearly topologized $\mathbb{C}$-algebra with the $\mathfrak{m}$-adic filtration $\mathfrak{m}^k = (z_1, z_2)^k A$; $\mathfrak{h} = A/\mathbb{C}\cdot 1$ is pro-finite-dimensional in this filtration, and its strict continuous dual is the Grothendieck–Matlis principal-part module

$$\mathfrak{h}^{\vee}_{\mathrm{cont}} = \mathcal{P} = \mathrm{Ann}_{\mathbf{Res}}(\mathbb{C}\cdot 1) \subset H^2_{\mathfrak{m}}(A) \otimes \omega_A, \qquad \mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\,\rho_{a,b}, \quad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2,$$

ind-finite-dimensional, with the residue pairing $\langle f, \omega\rangle = \mathrm{Res}_0\, f\omega$ a perfect pairing of Tate $\mathbb{C}$-modules. Inverse limits over Matlis windows $\mathcal{P}_{\le N} = \bigoplus_{a+b\le N} \mathbb{C}\rho_{a,b}$ are read as derived limits $\mathbf{R}\varprojlim_N \mathcal{P}_{\le N}$, with $\varprojlim^1$-data tracked. Polynomial calculations use the dense ind-finite-dimensional sub-Lie algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$; completed CE/PV calculations use $\mathfrak{h}$ itself. Factorization algebras throughout are taken in the Costello–Gwilliam sense (precosheaf on opens, Weiss-cover descent), refined to mixed-HT opens on $\mathbb{R}^2 \times \mathbb{C}^2$. This is the analytic incarnation of the factorisable-$\mathcal{D}$-module-on-$\mathrm{Ran}(\mathbb{C}^2)$ description — the two-complex-dimensional extension of the $\mathcal{D}$-module-on-Ran presentation of curve chiral algebras formalised in [28, §3], equivalent to the analytic form via the de Rham complex; the BV / heat-kernel formalism selects Costello–Gwilliam as native.

$$X = \mathbb{R}^2_{t,s} \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}, \qquad \omega = dz_1 \wedge dz_2.$$

Unimodularity datum.

Datum	**Definition**
Ω_{loc}	The fixed holomorphic volume and symplectic form on the formal polydisk $$\widehat{\mathbb{C}^2}_0 = \operatorname{Spf}\mathbb{C}[[z_1,z_2]], \qquad \Omega_{\text{loc}} = dz_1\wedge dz_2.$$ It trivializes the local canonical line and fixes the residue orientation.
Poisson tensor and bracket	The inverse bivector is $$P = \partial_{z_1}\otimes\partial_{z_2} - \partial_{z_2}\otimes\partial_{z_1}, \qquad \{f,g\} = \partial_{z_1}f\,\partial_{z_2}g - \partial_{z_2}f\,\partial_{z_1}g.$$ Polynomial calculations use $\bar{A} = \mathbb{C}[z_1,z_2]/\mathbb{C}\cdot 1$; completed CE/PV calculations use $\mathfrak{h} = \mathbb{C}[[z_1,z_2]]/\mathbb{C}\cdot 1$. Constants are removed because they generate the zero Hamiltonian vector field.
Hamiltonian vector field	The convention is $$\iota_{X_f}\Omega_{\text{loc}} = -df, \qquad X_f = (\partial_{z_1}f)\partial_{z_2} - (\partial_{z_2}f)\partial_{z_1}.$$ Thus $X_f(g) = \{f,g\}$ and $[X_f,X_g] = X_{\{f,g\}}$ after projecting constants.
Unimodularity / divergence-free condition	For a holomorphic vector field $X = a_1\partial_{z_1} + a_2\partial_{z_2}$, $$\mathcal{L}_X\Omega_{\text{loc}} = (\operatorname{div}_{\Omega_{\text{loc}}} X)\Omega_{\text{loc}}, \qquad \operatorname{div}_{\Omega_{\text{loc}}} X = \partial_{z_1}a_1 + \partial_{z_2}a_2.$$ Every Hamiltonian vector field is divergence-free: $$\operatorname{div}_{\Omega_{\text{loc}}} X_f = -\partial_{z_1}\partial_{z_2}f + \partial_{z_2}\partial_{z_1}f = 0.$$ Conversely, on the formal polydisk the holomorphic Poincare lemma identifies $\mathbf{Ker}(\partial_{\Omega_{\text{loc}}}: \mathrm{PV}^1 \to \mathrm{PV}^0)$ with Hamiltonians modulo constants.
Closed BV pairing	For the Hamiltonian BF fields $$\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\mathfrak{h}[1]\oplus\mathfrak{h}^\vee_{\text{cont}}[2]\big),$$ the BV pairing is the compactly supported local pairing obtained by wedging the form factors, integrating against Ω_{loc}, and applying the continuous evaluation $\mathfrak{h}^\vee_{\text{cont}}\otimes\mathfrak{h}\to\mathbb{C}$. It has degree -1. A Costello coefficient coevaluation kernel additionally requires a Tate/weighted category in which the Casimir converges.
Residue / cyclic trace density	The same orientation fixes the principal-part residue pairing $$\rho_{a,b} = z_1^{-a-1}z_2^{-b-1}\,dz_1dz_2, \qquad \operatorname{Res}_0(\rho_{a,b}z_1^m z_2^n) = \delta_{a,m}\delta_{b,n},$$ with $a+b>0$ in the reduced cotangent module. Hamiltonian vector fields preserve dz_1dz_2, so residue integration by parts gives the coadjoint action. On the open brane, the top Ext class is the residue orientation dual to $dz_1\wedge dz_2$, and the cyclic trace is $$\operatorname{Tr}_{\text{open}}(\eta\otimes u\otimes M) = \int_{\mathbb{R}}\eta\,[\varepsilon_1\varepsilon_2]u\,\operatorname{Tr}_N(M).$$

Kinematic locality.

Theorem 0.1 *(Kinematic locality).* On the local pair (X, L) with unimodularity datum $\omega = dz_1 \wedge dz_2$ and reduced Hamiltonian Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$:

(i) The equal-time bracket kernel $\delta(t - t')$ on two brane-time points, smeared against test functions $a(t), b(t')$, gives the support-local P_0-bracket $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$ with ab the pointwise product on $\mathbb{R}_t$; disjoint supports bracket to zero.

(ii) Every topological translation $v \in T\mathbb{R}^2_{\mathrm{top}}$ is Q-exact, $L_v = [Q, \iota_v]$, so $H^\bullet(\mathrm{Obs}^{\mathrm{cl}})$ is locally constant along $\mathbb{R}_t$ and inclusions of disks act by quasi-isomorphism.

(iii) Every antiholomorphic translation $\partial/\partial\bar{z}_i$ is Q-exact, $L_{\partial/\partial\bar{z}_i} = [Q, \iota_{\partial/\partial\bar{z}_i}]$, so the surviving $\bar\partial$-cohomology at the brane point is the formal jet algebra $\mathfrak{h}$, Grothendieck-dually paired with $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2$.

Items (ii) and (iii) hold on Q-cohomology; (i) is a chain-level identity after distributional pairing.

Trace observables $\mathrm{Tr}\, f(\phi_1, \phi_2)$ smeared as $B_f(a)$ satisfy (i)–(iii): support locality from the equal-time bracket of the matrix-model fields, topological locality from the Q-exact t-direction, formal-holomorphic locality from the $\bar\partial$-exact transverse direction. Shifted-cotangent currents $\Theta_\rho(B)$ satisfy (i) and (ii), and their formal-holomorphic content is the residue current $\rho \in \mathcal{P}$ at the brane point.

The dg Lie algebra of an acceptable factorization algebra is therefore locally constant in $\mathbb{R}^2_{\mathrm{top}}$ and holomorphic in $B \subset \mathbb{C}^2_{\mathrm{hol}}$,

$$\Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1],$$

with P_0-bracket on smeared currents fixed by (i). This produces the $\mathbb{C}^2$ holomorphic factorization algebra $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$ and its mixed topological-current enhancement $\mathcal{F}^{\mathrm{mix}}_{\mathrm{Ham}}$.

Holomorphic factorization on $\mathbb{C}^2$ and its curve restrictions. The holomorphic object forced by Theorem 0.1 is a two-complex-dimensional factorization algebra on $\mathbb{C}^2_{\mathrm{hol}}$. The taxonomy below records each object on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ and the restriction data required to descend to a one-complex-dimensional factorization algebra on a holomorphic curve.

Object	**Definition and range**
Formal Hamiltonian holomorphic factorization algebra on $\mathbb{C}^2$	The $\mathbb{C}^2$ Hamiltonian holomorphic factorization algebra is $$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}},$$ the formal-Darboux Hamiltonian BF object on $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$. For a holomorphic polydisk B its dg Lie algebra is $$\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1], \quad \mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1,$$ with differential $\bar\partial$ plus the Hamiltonian BF differential. The formal stalk of observables is the completed CE/PV object $$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \simeq \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$$ in the coordinate/admissible sense recorded below. This object is the two-complex-dimensional holomorphic factorization algebra; a one-complex-dimensional vertex algebra appears only after specifying a curve restriction.

Object	Definition and range
Native two-dimensional holomorphic singular product	For two points $z, w \in \mathbb{C}^2$, set $\zeta_i = z_i - w_i$. The singular coefficient module is $$H^2_\Delta(\mathbb{C}[[z,w]])\,d\zeta_1 d\zeta_2 = \bigoplus_{a,b\geq 0} \mathbb{C}[[w_1,w_2]]\,\rho^\Delta_{a,b},$$ where $$\rho^\Delta_{a,b} = \zeta_1^{-a-1}\zeta_2^{-b-1}d\zeta_1 d\zeta_2.$$ The normalized two-dimensional Cauchy kernel is $$K^{(2)}_\Delta = \rho^\Delta_{0,0} = \frac{d\zeta_1 d\zeta_2}{\zeta_1\zeta_2}.$$ For $x, y \in \mathfrak{g}$ the native current singular product is $$\mathsf{J}_x(z)\mathsf{J}_y(w) \sim K^{(2)}_\Delta(z,w)\mathsf{J}_{[x,y]}(w),$$ which gives $\mathsf{B}_f\mathsf{B}_g \sim K^{(2)}_\Delta \mathsf{B}_{\{f,g\}}$, $\mathsf{B}_f\Theta_\rho \sim K^{(2)}_\Delta \Theta_{f\cdot\rho}$, and $\Theta_\rho\Theta_\sigma \sim 0$. Arity-$n$ extension and operadic associativity: Proposition 5.7. The quantum brane trace adds $\hbar N\,\omega(f,g)K^{(2)}_\Delta\mathbf{1}$, with $\omega(f,g) = [\{f,g\}]_0$. This is the formal-coordinate two-complex-dimensional OPE. A one-variable Laurent expansion is the additional curve-restriction datum recorded below.
Mixed topological-current enhancement over $\mathbb{R}^2$	For a topological open $U \subset \mathbb{R}^2_{\mathrm{top}}$ and holomorphic polydisk $B \subset \mathbb{C}^2_{\mathrm{hol}}$, the mixed current dg Lie algebra is $$\mathfrak{g}^{\mathrm{mix}}_{U,B} = \Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]).$$ Its observables are locally constant in U and holomorphic in B. The brane-line current algebras below are its support-local one-real-dimensional restrictions.
Weyl/Moyal open-string brane algebra and cyclic realisations	The degree-zero quantum brane target is the Weyl/Moyal deformation of the formal symplectic polydisk. Put $$P = \partial_{z_1}\otimes\partial_{z_2} - \partial_{z_2}\otimes\partial_{z_1}.$$ Then $$f * g = m \circ \exp\!\left(\frac{\hbar}{2}P\right)(f\otimes g),$$ $$\{f,g\}_\hbar = \hbar^{-1}(f*g - g*f),$$ with constants quotiented in the Hamiltonian Lie algebra $\bar{A}_\hbar$. At finite N the open model is the matrix Weyl algebra generated by $(\phi_1)^i{}_j, (\phi_2)^k{}_l$ with $$[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar\,\delta^i_l\delta^k_j,$$ followed by the quantum moment-map reduction and the connected trace projection. Cyclic and supercyclic bar-cobar realisations are trace or supertrace word complexes for matrix or supermatrix branes.

Object	Definition and range
Defect/principal-part current algebra	For an interval $I \subset \mathbb{R}_{\mathrm{brane}}$, $$\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}, \qquad \mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}, \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$ where $$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2.$$ The reduced target is $$A^{\mathrm{pp}}_\partial(I) = \widehat{\mathbf{S}}(\Omega^0_c(I)\widehat{\otimes}\bar{A})\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1]),$$ with brackets $$\begin{aligned}\{B_f(a), B_g(b)\} &= B_{\{f,g\}}(ab),\\ \{B_f(a), \Theta_\rho(B)\} &= \Theta_{f\cdot\rho}(aB),\\ \{\Theta_\rho(B), \Theta_\sigma(C)\} &= 0.\end{aligned}$$ This is a support-local real-current P_0 algebra with Matlis principal parts in two holomorphic variables.
One-dimensional vertex/OPE restriction boundary	A restriction of the $\mathbb{C}^2$ Hamiltonian holomorphic factorization algebra to a one-complex-dimensional vertex/OPE algebra starts from the native diagonal singular product above and requires the extra datum $$\mathfrak{R}_L = (\iota_L, \mathcal{B}_\perp, \mathsf{R}_{L,\mathcal{B}_\perp}, V_L, \mathbf{1}_L, T_L, Y_L, \zeta_{L,\hbar}),$$ where $\iota_L\colon L = \mathrm{Spf}\,\mathbb{C}[[w]] \hookrightarrow \widehat{\mathbb{C}^2}_0$ is a chosen holomorphic line or defect germ, $\mathcal{B}_\perp$ is a transverse boundary condition or normal vacuum, and $\mathsf{R}_{L,\mathcal{B}_\perp}$ is a restriction functor producing a one-complex-dimensional holomorphic factorization algebra $$\mathcal{F}_{L,\mathcal{B}_\perp}(D) = \mathsf{R}_{L,\mathcal{B}_\perp}(\mathcal{F}_{\mathbb{C}^2})(D)$$ for disks $D \subset L$. The functor must be built from a specified tubular-neighbourhood, shriek/star restriction, or defect-coupling construction, and must prove independence of the auxiliary transverse radius, functoriality for disjoint disks, and translation covariance along the coordinate w. The tuple also includes the restricted state space V_L, vacuum, translation, state-field map, and Zhu/zero-mode map. In the $\mathbb{C}\times\mathbb{R}$ interface, such a V_L datum factors through $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$ and is identified with V_{red} in $\mathfrak{I}_{\mathrm{ch}}$.

Object	Definition and range
OPE data for a restricted curve object	After such a restriction one must construct $$V_L = H^\bullet(\mathcal{F}_{L,\mathcal{B}_\perp}(\Delta_0)), \qquad Y_L(a,w)b = \sum_{n\in\mathbb{Z}} (a_{(n)}b)\, w^{-n-1}.$$ The factorization product for two disks centered at w and 0 must have a finite-pole expansion $$\mu_{w,0}(a \boxtimes b) \sim \sum_{n\geq 0} (a_{(n)}b)\, w^{-n-1} + \mu^{\mathrm{reg}}_{w,0}(a,b),$$ with $a_{(n)}b = 0$ for $n \gg 0$, a vacuum $\mathbf{1}_L$, translation $[T_L, Y_L(a,w)] = \partial_w Y_L(a,w)$, and locality $$(w_1 - w_2)^N [Y_L(a,w_1), Y_L(b,w_2)] = 0$$ for $N \gg 0$. These Laurent modes and finite singular orders belong to the restricted curve object. The two-complex-dimensional holomorphic factorization algebra uses the singular coefficient $H^2_\Delta(\mathbb{C}[[z,w]])\, d\zeta_1 d\zeta_2$; after curve restriction the one-variable coefficient module is $\mathbb{C}((w))/\mathbb{C}[[w]]\, dw$.
Compatibility with the Weyl/Moyal brane algebra	A one-dimensional vertex/OPE restriction must also specify how the degree-zero brane algebra acts on the restricted object. Equivalently it requires a continuous Zhu/zero-mode map $$\zeta_{L,\hbar} \colon (\bar{A}_\hbar, *) \longrightarrow \mathrm{Zhu}(V_L)$$ or fields $J_f(w)$ satisfying $$J_f(w) J_g(0) \sim \frac{\kappa_L(f,g)}{w^2} + \frac{J_{\{f,g\}_\hbar}(0)}{w},$$ with the central level κ_L specified, and on the cotangent current sector $$[J_{f,(0)}, \Theta_\rho] = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho},$$ together with $\zeta_{L,\hbar}(f * g) = \zeta_{L,\hbar}(f)\zeta_{L,\hbar}(g)$. The Weyl/Moyal calculation gives $f * g$ and the reduced interval-current brackets; Theorem 1.17 gives the native two-dimensional singular expansion. The data $\zeta_{L,\hbar}$, the fields $J_f(w)$, the central level, and the finite Laurent curve expansion are separate one-dimensional restriction constructions.

Object	Definition and range
Chiral algebra interface tuple	An externally constructed chiral algebra on a curve enters as the reduced vertex object V_{red} in $$\mathfrak{I}_{\mathrm{ch}} = (\mathcal{R}_{\mathbb{C}\times\mathbb{R}}, \mathcal{F}_{\mathrm{red}}, V_{\mathrm{red}}, Q_{\mathrm{BRST}}, \mathbf{1}, T, Y, \mathrm{wt}, \zeta_{\hbar}, H_{\mathrm{anom}}).$$ Here $\mathcal{R}_{\mathbb{C}\times\mathbb{R}}$ is a controlled pushforward to $\mathbb{R}_t \times \mathbb{C}_{z_1}$ retaining the z_2-mode or the full two-index principal-part coefficient system $$\mathfrak{h}_{\mathrm{red}} = \mathbb{C}[[z_1]][[z_2]]/\mathbb{C}\cdot 1, \qquad \mathcal{P}_{\mathrm{red}} \subset H^2_{(z_1,z_2)}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2.$$ $\mathcal{F}_{\mathrm{red}}$ is the pushed-forward factorization algebra on $\mathbb{R}_t \times \mathbb{C}_{z_1}$. The interface must prove the pushed-forward bracket, BV pairing, brane image, factorization-to-vertex comparison, BRST nilpotence, Zhu/zero-mode map, Capelli normalization, Moyal multiplicativity, the completed stable Hochschild/HKR/CE-PV comparison, and anomaly transport. The obstruction vector is $$\mathrm{Ob}_{\mathfrak{I}_{\mathrm{ch}}} = (\mathrm{Ob}_{\mathrm{red}}, \mathrm{Ob}_{\mathrm{vert}}, \mathrm{Ob}_{\mathrm{Zhu}}, \mathrm{Ob}_{\mathrm{nat}}).$$ V_{red} is a reduced shadow of $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$ on a holomorphic curve; the two-complex-dimensional algebra remains $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$.

Normal Ω-background. The brane-preserving equivariant deformation scales the normal bundle $N_L X = \mathbb{R}_s \oplus \mathbb{C}_{z_1} \oplus \mathbb{C}_{z_2}$ with t fixed; the fixed locus condition singles out this normal scaling. The objects and identities below define the deformation in the mixed de Rham/Dolbeault coefficient category: the Cartan differential $Q_\Omega = Q + \iota_{V_\Omega}$, the localized normal contraction, the equivariant CE/PV bracket, and the residue-versus-Euler normalization.

Object or convention	Definition
Ambient pair and fixed stratum	The local pair is $$X = \mathbb{R}^2_{t,s} \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}.$$ The normal bundle of the brane line is $$N_L X = \mathbb{R}_s \oplus \mathbb{C}_{z_1} \oplus \mathbb{C}_{z_2}.$$ The Ω-background scales this bundle with t fixed.
Equivariant parameters	Write $$T_\Omega = \mathbb{C}^*_{\epsilon_s} \times \mathbb{C}^*_{\epsilon_1} \times \mathbb{C}^*_{\epsilon_2}.$$ The parameters $\epsilon_s, \epsilon_1, \epsilon_2$ are equivariant weights. The open trace orientation uses the separate Ext generators $\varepsilon_1, \varepsilon_2$. The action is $$(s, z_1, z_2) \mapsto (\lambda_s s, \lambda_1 z_1, \lambda_2 z_2), \qquad t \mapsto t.$$ Its infinitesimal vector field is $$V_\Omega = \epsilon_s s \partial_s + \epsilon_1 z_1 \partial_{z_1} + \epsilon_2 z_2 \partial_{z_2}.$$

Object or convention	Definition
Equivariant differential	The Cartan differential on the local mixed complex is $$Q_\Omega = Q + \iota_{V_\Omega}, \qquad Q_\Omega^2 = L_{V_\Omega}.$$ Thus Q_Ω is an honest differential on the T_Ω-invariant subcomplex. Away from invariants the correct object is the T_Ω-equivariant curved Cartan complex with curvature L_{V_Ω}.
Normal-weight localization ring	For a finite (N, M)-window let $\mathsf{X}_{N,M}$ be the finite set of nonzero moving characters that occur in that window. It contains the normal Hamiltonian and residue-dual characters $$n\epsilon_s + a\epsilon_1 + b\epsilon_2, \qquad n\epsilon_s - a\epsilon_1 - b\epsilon_2$$ whenever the corresponding moving summands occur. The localized coefficient ring is $$R_\Omega^{N,M} = \mathbb{C}[\epsilon_s, \epsilon_1, \epsilon_2, \hbar_\omega][\chi^{-1} \mid \chi \in \mathsf{X}_{N,M}].$$ Equivalently, one inverts only the nonzero weights present in the finite normal window. This is the algebra in which moving normal modes may be contracted. The factor $\epsilon_s\epsilon_1\epsilon_2$ is inverted in the Euler-localization branch after that branch has been declared. The self-dual branch is a specialization with its own allowed inversions: base-change first, then invert only the surviving nonzero characters, leaving $\epsilon_1 + \epsilon_2$ as a zero character on the self-dual branch.
Normal contraction datum	A localized deformation retract $$\mathsf{C}_\Omega = (\pi_L, j_L, h_\Omega) \colon \mathcal{E}_\Omega(X) \rightleftarrows \mathcal{E}_\Omega(L)$$ for the mixed de Rham/Dolbeault coefficient complex, with $$\pi_L j_L = \mathrm{id}, \qquad Q_\Omega h_\Omega + h_\Omega Q_\Omega = \mathrm{id} - j_L\pi_L$$ on the localized invariant normal complex. On a homogeneous normal summand of nonzero weight λ, h_Ω is the Euler/Koszul homotopy divided by λ, with signs fixed by the de Rham degree in the s-direction and the Dolbeault degree in the z_i-directions.

<table>
<tr><th>Object or convention</th><th>Definition</th></tr>
<tr><td>Hamiltonian and cotangent weights</td><td>For $H_{a,b} = z_1^a z_2^b$, $a + b > 0$,
$$w(H_{a,b}) = a\epsilon_1 + b\epsilon_2.$$
For the principal-part residue basis,
$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2, \qquad w(\rho_{a,b}) = -a\epsilon_1 - b\epsilon_2.$$
The holomorphic symplectic character is
$$\chi_\omega = \epsilon_1 + \epsilon_2, \qquad w(dz_1 \wedge dz_2) = \chi_\omega, \qquad w(P) = -\chi_\omega.$$
The Weyl/Moyal deformation parameter, when present, has
$$w(\hbar_{\mathrm{W}}) = \chi_\omega.$$</td></tr>
<tr><td>Three $\hbar$-parameters</td><td>The Omega convention distinguishes three formal parameters:
$$\hbar_{\mathrm{QME}}, \qquad \hbar_{\mathrm{W}}, \qquad \hbar_\omega.$$
Here $\hbar_{\mathrm{QME}}$ is the Costello/BV loop-counting parameter in graph and curvature equations; $\hbar_{\mathrm{W}}$ is the Weyl/Moyal deformation parameter in the open brane algebra; and $\hbar_\omega$ is the equivariant line parameter of weight χ_ω used to scalarize the Poisson bracket. A Weyl branch may identify $\hbar_{\mathrm{W}} = \hbar_\omega$. Identification of $\hbar_{\mathrm{QME}}$ with another parameter is an explicit theorem convention.</td></tr>
<tr><td>Self-dual bracket</td><td>On the self-dual locus
$$\epsilon_1 + \epsilon_2 = 0$$
the Poisson tensor has weight 0. The Hamiltonian bracket $\{f, g\}$ is then a scalar T_Ω-equivariant Lie bracket on $\mathfrak{h}$ after quotienting constants.</td></tr>
<tr><td>Refined bracket off the self-dual locus</td><td>Off the self-dual locus the ordinary Poisson bracket has weight
$$w(\{f, g\}) = w(f) + w(g) - (\epsilon_1 + \epsilon_2).$$
It is therefore a bracket valued in the inverse symplectic-character line. The scalar refined bracket is obtained only after adjoining the line parameter $\hbar_\omega$ of weight $w(\hbar_\omega) = \epsilon_1 + \epsilon_2$:
$$[f, g]_\Omega = \hbar_\omega \{f, g\}.$$
In the Weyl/Moyal specialization $\hbar_\omega = \hbar_{\mathrm{W}}$. Before this scalarization, the refined Hamiltonian algebra is valued in the inverse symplectic-character line; after adjoining $\hbar_\omega$ it becomes a scalar equivariant Hamiltonian Lie algebra.</td></tr>
</table>

Object or convention	Definition
Equivariant CE/PV generator rule	The equivariant CE/PV map keeps the same generator labels $$c_{a,b} \mapsto \theta_{a,b}, \qquad u_{a,b} \mapsto O_{a,b},$$ but all differentials and brackets are read in the T_Ω-graded completed coordinate algebra. In the refined convention the contraction bracket and the d_π-terms carry the factor $\hbar_\omega$. In the self-dual convention this factor has weight 0 and may be suppressed.
Equivariant quantum brane algebra	The degree-zero open quantum brane target is the projective central-character reduction $$A^q_{N,\Omega} = \Big(\mathrm{Weyl}_{\hbar_{\mathrm{W}}}(\mathrm{Mat}^2_N)/\mathcal{W}_N\widehat{\mu}_{\chi_N}(\mathfrak{sl}_N)\Big)^{SL_N}, \qquad YX - XY + \hbar_{\mathrm{W}} NI \equiv 0 \pmod{\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)}$$ with $$w(X) = \epsilon_1, \qquad w(Y) = \epsilon_2, \qquad w(\hbar_{\mathrm{W}}) = \epsilon_1 + \epsilon_2.$$ The central-character relation is weight homogeneous. Its scalar $U(1)$ contact term is the same Capelli line as in the non-equivariant dictionary, now graded by $\epsilon_1 + \epsilon_2$.
Residue normalization	The residue convention uses the formal residue orientation $$\mathrm{Res}_{z_1=z_2=0}\big(z_1^{-1}z_2^{-1}dz_1dz_2\big) = 1$$ and the open top Ext coefficient $[\varepsilon_1\varepsilon_2]$. Its normalization constant is $$\nu_\Omega^{\mathrm{res}} = 1.$$ In this convention the brane trace is normalized by the residue/cyclic density. Any $\epsilon_s\epsilon_1\epsilon_2$ factor is written explicitly as a comparison constant.
Euler-localization normalization	The smooth equivariant localization convention pushes from X to L with the inverse normal Euler class $$e_{T_\Omega}(N_LX)^{-1} = (\epsilon_s\epsilon_1\epsilon_2)^{-1},$$ up to the orientation sign σ_s fixed by the chosen real s-density and the complex orientation $dz_1 \wedge dz_2$. Its normalization constant is $$\nu_\Omega^{\mathrm{Eu}} = \sigma_s(\epsilon_s\epsilon_1\epsilon_2)^{-1}.$$ A theorem comparing this convention with the residue convention must state the sign and the exact factor. Until then the two normalizations are separate branches.

Object or convention	Definition
Stratified factorization datum	The brane-localized object is a stratified mixed HT factorization datum on the pair (X, L), $$\mathcal{F}^{\mathrm{str}}_{\Omega}(X, L) = \big(\mathcal{F}_{\Omega,X\setminus L}, \mathcal{F}_{\Omega,L}, \mathcal{M}_{\Omega,L}, j^*, i^!, \mu_{\mathrm{str}}\big),$$ where the closed bulk stratum is the T_Ω-equivariant mixed bulk factorization algebra, the open defect stratum is the brane-line current object, and $\mathcal{M}_{\Omega,L}$ is the defect module. Protected observables become stratified factorization-homology traces in the presence of the brane-defect QME and the trace-state normalization.
Open obstruction vector	The residual obligations are $$\begin{aligned}\mathrm{Ob}_\Omega = \big(&\mathrm{ob}_{\Omega,\mathrm{Cartan}}, \mathrm{ob}_{\Omega,\mathrm{contr}},\\ &\mathrm{ob}_{\Omega,\mathrm{CE/PV}}, \mathrm{ob}_{\Omega,\mathrm{norm}},\\ &\mathrm{ob}_{\Omega,\mathrm{strat}}, \mathrm{ob}_{\Omega,\mathrm{QME}},\\ &\mathrm{ob}_{\Omega,N\to\infty}\big).\end{aligned}$$ These name, respectively, the mixed Cartan model, the localized normal contraction, the equivariant CE/PV bracket comparison, the residue-versus-Euler normalization theorem, the stratified factorization structure, the Costello brane-defect graph/QME complex, and the physical large-N trace-state theorem.

Canonical maps and obstruction symbols.

Symbol	Meaning
$\Phi_{\mathfrak{l}}$	Finite-dimensional cotangent CE/PV map $$C^\bullet_{\mathrm{CE}}(\mathfrak{l} \ltimes \mathfrak{l}^\vee[1]) \to \mathrm{PV}(\mathbf{S}(\mathfrak{l})), \qquad c \mapsto \theta, \quad u \mapsto O.$$
Φ_{coord}	Formal-polydisk coordinate CE/PV map reconstructed from admissible finite coordinate tests and pro-window coordinate cochains for $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ and $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$. It is a completed coordinate dg commutative algebra isomorphism before any P_0 bracket is asserted.
$\Phi^B_{P_0}$	Restriction of Φ_{coord} to a bracket-admissible subalgebra $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$: $$\Phi^B_{P_0} : \Phi^{-1}_{\mathrm{coord}}(B) \xrightarrow{\sim} B.$$ The completed P_0 comparison in $\mathcal{B}^{\mathrm{adm}}$ (Theorem 5.6, axioms (i)–(iv)): B carries a complete Hausdorff topology, continuous inclusion in the coordinate product, continuous multiplication, continuous d^{coord}_π, and a continuous closed contraction bracket; the criterion is Köthe bracket continuity (Remark 5.9, check (6)). Boundary Hamiltonians inside this P_0 comparison require the relevant O_f and $d^{\mathrm{coord}}_\pi O_f$ to lie in B.

Symbol	Meaning
Θ_B	Kernel-admissible Maurer–Cartan tensor in a specified completed convolution dg Lie algebra $\mathcal{K}_B$, $$\Theta_B = \sum_I H_I \otimes \theta^I + \sum_I \eta^I \otimes O_I,$$ attached to B. Asserted only when the displayed diagonal sums converge and the convolution differential and bracket are continuous in $\mathcal{K}_B$; the kernel-admissibility test is the Maurer–Cartan identity $d\Theta_B + \frac{1}{2}[\Theta_B, \Theta_B] = 0$ (Theorem 5.6, axiom (v); Remark 5.9, check (7)).
η^I	Shifted cotangent generator $\eta^I = (H_I)^\vee[1] \in \mathfrak{h}^\vee_{\mathrm{cont}}[1]$, for $I = (a, b)$ and $a + b > 0$. It is the source of the O_I coordinate in the CE/PV map.
$\mathfrak{U}_{\mathfrak{h}}$	Admissible formal-local interface datum in the category $\mathcal{B}^{\mathrm{adm}}$ of Theorem 5.6 (seven admissibility axioms (i)–(vii); host details in Appendix G §1). The categorical universal object is $$\mathfrak{U}_{\mathfrak{h}} = (\Phi_{\mathrm{coord}}, \{\Phi^B_{P_0}\}_B, \{\Theta_B\}_B),$$ where B ranges over bracket-admissible subalgebras in the middle term and kernel-admissible data in the last term.
B_{fin}, E_q	Model admissibility tests. B_{fin} is the finite-support coordinate subalgebra on a finite window. E_q, for $q > 1$, is the weighted Köthe test target with seminorms $$p_n(x) = \sup_{a,b} \lvert x_{a,b}\rvert q^{a+b}(1 + a + b)^n.$$ It is bracket-admissible only after the finite-word tensor convention and the corresponding convolution estimate are fixed; it is kernel-admissible only if the diagonal tensor converges.
$\mathfrak{o}^{\mathrm{br}}, \mathfrak{o}^{\mathrm{coad}}$	Transition obstruction terms for bracket and coadjoint compatibility in the pro/mixed coordinate CE/PV comparison.
$\mathfrak{o}^{\mathrm{Cas}}$	Strict product/direct-sum kernel obstruction for the Casimir, propagator, and Costello endpoint problem. It belongs to the endpoint coefficient problem; the coordinate CE/PV identity is the separate coordinate comparison.
$\mathfrak{o}_{\mathrm{BV,end}}$	Endpoint obstruction tuple $(\mathfrak{o}_{\mathrm{Cas}}, \mathfrak{o}_{\mathrm{def}}, \mathfrak{o}_{\mathrm{RG}}, \mathfrak{o}_{\mathrm{QME}})$ controlling a Costello-compatible coefficient endpoint.
$\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}}$	Matched-conventions descent vector $(\mathrm{per}_{\mathrm{tar}}, \mathrm{ob}_{\mathrm{WR}}, \mathrm{ob}_{\mathrm{anom}}, \mathrm{ob}_{\mathrm{QME}}, \mathrm{ob}_{\mathrm{conv}}, \mathrm{ob}_{\mathrm{hyp}}, \mathrm{ob}_{6r}, \mathrm{ob}_{\mathrm{MD}})$, indexed by the comparison target $\mathfrak{C}_{\mathrm{tar}}$.

Non-scalar Costello/QME labels. The local obstruction complex named in Problem 5.220 is recorded below. The scalar projection σ_{sc} is associated-graded data; the projection $\widehat{\sigma}_{\mathrm{sc}}$ is a filtered chain map existing exactly when the successive scalar-lift obstructions vanish; the reduced principal-part brackets live after de Rham-current contraction; the unreduced non-scalar QME is the curvature equation $\mathrm{Curv}(K_{\hbar,w,I}) = 0$ of the bulk-to-defect kernel.

Symbol or datum	Meaning
$\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$	Filtered brane-defect local-functional complex $$\varprojlim_M \Big(O^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}^M_w\|_I; A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)), d_M\Big), \qquad d_M = Q + \{I^w_{0,M}, -\}_{\hbar,M}.$$ The inverse limit is taken over finite Hamiltonian windows with the $\hbar$-adic, scalar-contact, and finite-window filtrations. The differential has degree +1. This is the target complex for the bulk-to-defect kernel and QME obstruction; the reduced current algebra is $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$.
$F^\bullet\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$	Complete decreasing closed scalar-contact filtration preserved by d_{QME}. In the ordinary scalar-reduced branch take $F^r = J^r_{\mathrm{sc}}$, where J_{sc} is the closed ideal of functionals with zero associated-graded scalar-contact symbol. In the balanced branch it is the closed filtration generated by identity-endomorphism contractions inside the balanced scalar-contact complex. Its role is to define the associated-graded scalar symbol and the obstruction tower for lifting that symbol.
σ_{sc}	Associated-graded scalar-symbol extractor $$\sigma_{\mathrm{sc}}:\ \mathrm{gr}_F\,\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]].$$ It has cochain degree 0 as an associated-graded map. A projection of the filtered QME complex is the lifted map $\widehat{\sigma}_{\mathrm{sc}}$. On a degree-one scalar contact with two Hamiltonian arguments it records the Lie two-cocycle coefficient of γ_1. In the ordinary branch this leading symbol is $\hbar N[\bar{c}]$.
$\widehat{\sigma}_{\mathrm{sc}}$	Actual scalar-contact projection: a continuous filtered degree-0 cochain map $$\widehat{\sigma}_{\mathrm{sc}}: \mathfrak{Q}^\bullet_{w,\partial,\hbar}(I) \to C^\bullet_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]]$$ whose associated graded is σ_{sc}. It exists exactly when the scalar-lift obstruction classes $o^{(r)}_{\sigma,w}(I)$ vanish successively with compatible finite-window choices. The residual non-scalar complex is defined only after this map has been chosen. On a balanced scalar-contact complex the proved projection is the zero chain map $\widehat{\sigma}_{\mathrm{bal,sc}} = 0$.
$o^{(r)}_{\sigma,w}(I)$	Successive scalar-contact lift obstruction. For a filtered linear lift s of σ_{sc}, the chain defect $\partial_s = d_{\mathrm{CE}}s - s d_{\mathrm{QME}}$ lowers $F^\bullet$. Its first nonzero part gives $$o^{(r)}_{\sigma,w}(I) \in H^1\Big(\mathrm{Hom}\big(\mathrm{gr}_F\,\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I), C^\bullet_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]]\big)_{-r}\Big).$$ In the ordinary scalar-reduced $\mathfrak{gl}_N$ branch the first two-argument evaluation is $\hbar N\,\omega(f,g)\gamma_1$, nonzero on $(f,g)=(z_1,z_2)$.
$\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}$	Current source for the unreduced non-scalar lift, $$\big(\Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}_{w,\hbar}\big) \ltimes \big(\mathcal{D}'_c(I)\widehat{\otimes}\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}\big)[1].$$ The topology is the completed current topology with the weighted finite-window and $\hbar$-adic completions. On CE coordinates, $c_{B,\rho}$ has degree 1 and $u_{a(t)dt\otimes f}$ has degree 0. Before scalar reduction one may replace the Hamiltonian source by the central-character source.

Symbol or datum	Meaning
$\varkappa_{\hbar,w,I}$	Continuous bulk-to-defect kernel, equivalently a degree-0 Maurer–Cartan element in the completed convolution Lie algebra, $$\varkappa_{\hbar,w,I}\colon C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}) \longrightarrow \mathfrak{Q}^\bullet_{w,\partial,\hbar}(I),$$ with generator values $$u_{a(t)dt\otimes f} \mapsto B^{\mathrm{BV}}_f(a), \qquad c_{B,\rho} \mapsto \Theta^{\mathrm{BV}}_\rho(B).$$ Its reduced image must recover $B_f(a)$ and $\Theta_\rho(B)$, but the kernel itself lives before scalar reduction, before de Rham-current contraction, and before replacing unreduced open observables by freely adjoined current generators.
$\mathrm{Ob}_{w,\partial,\hbar}$	Curvature cocycle of the bulk-to-defect kernel, $$\mathrm{Ob}_{w,\partial,\hbar} = d_{\mathrm{QME}}\varkappa_{\hbar,w,I} + \frac{1}{2}[\varkappa_{\hbar,w,I}, \varkappa_{\hbar,w,I}]_\hbar.$$ It has QME degree 1. The scalar branch first requires $\widehat{\sigma}_{\mathrm{sc}}(\mathrm{Ob}_{w,\partial,\hbar}) = 0$. Only then does its residual class lie in the non-scalar complex.
$\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}$	Residual non-scalar obstruction after the scalar-contact projection has been constructed and the scalar component has been killed: $$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1\big(\ker\widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{QME}}\big).$$ This class is defined after choosing the scalar-contact projection; the reduced principal-part bracket formulas leave it as a residual non-scalar obstruction.
$\boldsymbol{Q}^\bullet_{w,M}(I)$	Finite-window residual complex $$\boldsymbol{Q}^\bullet_{w,M}(I) = \ker(\widehat{\sigma}_{\mathrm{sc},M}) \subset \Big(\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}^M_w\|_I; A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)), d_M\Big).$$ The bracket, interaction, and differential are the transported finite-window structures given by the admissibility datum. The unweighted Hamiltonian bracket enters only through that transport.
$C_{n,M}$, $C^{w_\circ}_{\Gamma,w}(\epsilon)$	Finite-window counterterms. At order $\hbar^n$, $C_{n,M} \in \boldsymbol{Q}^0_{w,M}(I)$ must satisfy $$d_M C_{n,M} = -\mathfrak{o}^{\mathrm{ns}}_{n,M}, \qquad \pi_{M,N}C_{n,M} = C_{n,N}.$$ Graph representatives $C^{w_\circ}_{\Gamma,w}(\epsilon)$ must also commute with window truncation and weight transport. They are degree-zero local functionals in the filtered QME complex.
$\mathfrak{O}^{\mathrm{ns}}_n$	Order-n finite-window obstruction vector $$\mathfrak{O}^{\mathrm{ns}}_n = \big(([\mathfrak{o}^{\mathrm{ns}}_{n,M}])_M, \lambda_n\big),$$ where the first component lies in $\varprojlim_M H^1(\boldsymbol{Q}^\bullet_{w,M}(I))$, and $\lambda_n \in \varprojlim^1_M H^0(\boldsymbol{Q}^\bullet_{w,M}(I))$ is the primitive-compatibility class. Its vanishing is exactly the finite-window criterion for compatible order-n counterterms.

Symbol or datum	Meaning
$\mathsf{E}_{\theta_3,M}$, $C_{\theta_3,M}$	First proved larger-row non-scalar obstruction and its permitted primitive. In the one-row complex, $$Q^0_{\theta_3,M} = 0, \qquad Q^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_M = 0,$$ and the residual is $\mathfrak{o}^{\mathrm{ns}}_{\theta_3,M} = \mathsf{E}_{\theta_3,M}$. The covector $\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$ proves that this class is nonzero in the given subcomplex. A valid $C_{\theta_3,M}$ is one of three explicit witnesses, $$K^M_{\Theta_3}(\eta_{\theta_3,M}), \qquad C^{\mathrm{ct}}_{\theta_3,M}, \qquad \text{or a companion-face primitive,}$$ with differential $d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$, scalar-contact value zero, and vanishing Roos compatibility class under finite-window projection.
I_N, η, χ_N	Moment-ideal and scalar primitive data. The quantum Hamiltonian reduction uses the left moment-map ideal $I_N = \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$ and then removes the scalar empty trace. The scalar primitive lives on the unreduced Hamiltonian source $A = \mathbb{C}[z_1, z_2]$: $$\eta(f) = -[f]_0, \qquad d_{\mathrm{CE}}\eta = \omega, \qquad \chi_N(f) = N[f]_0 = -N\eta(f).$$ Thus $\hbar N\omega + d_{\mathrm{CE}}(\hbar\chi_N) = 0$ before scalar reduction. This primitive lives on A; after passage to $\bar{A} = A/\mathbb{C}\cdot 1$, the scalar-reduced QME complex requires its own counterterm data.
$E_{a,b,N}$, $A^j{}_i(a,b,N)$	Quantum shear moment-ideal primitive datum in the matrix Weyl algebra. Put $$E_{a,b,N} = \sum_{j=0}^{b} \mathrm{Tr}(Y^j X^a Y^{b-j}) - J_N\Big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\Big).$$ The all-N shear congruence required for the radial/Weyl image is equivalent to $$E_{a,b,N} \in I_N \quad\Longleftrightarrow\quad E_{a,b,N} = \sum_{i,j} A^j{}_i(a,b,N)\widehat{\mu}^i{}_j$$ for coefficients $A^j{}_i(a,b,N) \in \mathcal{W}_N$. This is a degree-zero left-ideal primitive problem in the quantum moment-map ideal. The scalar CE primitive η, scalar-contact projection, and Costello QME counterterms are independent finite-row data.
Reduced principal-part brackets	Target-side brackets in $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$ after scalar reduction and de Rham-current contraction: $$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \quad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$ $$\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0.$$ They use multiplication of a distribution by a smooth test function. They prove the reduced current interface. Costello wavefront estimates, unreduced centrality homotopies, and the non-scalar QME are additional analytic and homotopical data.

Symbol or datum	Meaning
Unreduced non-scalar QME realization	The full graph theorem is the construction of $\kappa_{\hbar,w,I}$ and compatible counterterms $C_{\hbar,w}$. Put $K_{\hbar,w,I} = \kappa_{\hbar,w,I} + C_{\hbar,w}$. The equation is $$\mathrm{Curv}(K_{\hbar,w,I}) = d_{\mathrm{QME}} K_{\hbar,w,I} + \frac{1}{2}[K_{\hbar,w,I}, K_{\hbar,w,I}]_{\hbar} = 0.$$ It requires the Costello finite-scale BV propagator and Laplacian, locality and counterterm estimates, scalar-contact chain projection, vanishing of $\mathfrak{O}_n^{\mathrm{ns}}$ for every order, and unreduced product and P_0-bracket homotopies against arbitrary open observables.

Common local rings and quotients. $\mathfrak{h}$ labels Hamiltonian observables; $\mathfrak{g}_{\mathrm{str}}$ labels CE/PV closed cochains; $\bar{A}$ labels the polynomial trace sector; N labels the Dirac brane stack rank.

Symbol	Meaning
$R = \mathbb{C}[[z_1, z_2]]$	Complete local ring of the formal symplectic polydisk.
$\mathfrak{m} = (z_1, z_2)$	Maximal ideal of functions vanishing at the brane point.
$\mathfrak{h} = R/\mathbb{C} \cdot 1$	Reduced Hamiltonian Lie algebra of the formal polydisk. Constants are removed because they generate the zero Hamiltonian vector field.
$\mathfrak{g}_{\mathrm{str}}$	Strict shifted cotangent Hamiltonian Lie algebra $$\mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1].$$ Its Hamiltonian side is a product completion and its cotangent side is continuous dual/direct-sum in residue coordinates. This is the formal-local CE/PV object. A Costello coefficient kernel additionally requires the endpoint Casimir data.
$\mathfrak{g}_w$	Weighted finite-window Tate replacement used as a regulator. It carries the same coordinate-window CE/PV formulas after passage through the admissible tower. Its weight topology is auxiliary data for the admissible tower; the strict cotangent completion is $\mathfrak{g}_{\mathrm{str}}$.
$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$	Polynomial Hamiltonian sector used in explicit trace and Moyal calculations.
$\bar{A}_{\mathrm{desc}}$	Abstract polynomial label space for PBW one-ψ descendants; it carries a different $\mathfrak{h}$-action from the Matlis principal-part module $\mathcal{P}$ at the same bidegree labels.
$[\bar{c}]$	Distinguished scalar $U(1)$ anomaly class in $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$, realized by the constant term of the Poisson bracket and by the Capelli shift.
N	Rank of the Dirac brane stack. Degreewise stable trace statements are formed at fixed total word degree and then for N in the stable range.
D	Taylor truncation degree when a finite quotient $\mathfrak{m}/\mathfrak{m}^{D+1}$ is used. The brane rank is denoted N.

Large-N and quotient conventions. An algebraic primitive projection is defined inside the stable range; a physical cumulant requires a state $\omega_{N,\Omega}$.

Symbol or operation	Meaning
$\mathrm{Prim}_{\mathrm{st}}$	Degreewise stable primitive invariant sector. In total word degree d, the Procesi–Razmyslov stable range is $N \geq d$.
$\mathrm{Conn}_{N\to\infty}$	Connected primitive/indecomposable projection in the algebraic stable trace sector after finite-N normalization and passage to the stable range. Physical cumulants require a state $\omega_{N,\Omega}$.
Hamiltonian scalar quotient	The closed scalar $1 \in \mathbb{C}[z_1, z_2]$ and the open empty trace $\mathrm{Tr}_N(1) = N$ are omitted as primitive Hamiltonian generators. The unit of the symmetric algebra remains.

Separation rule. PBW special-fibre labels $H_{a,b}$, $O_{a,b}$, $\Psi_{a,b}$, CE/PV labels c^I, u_I, θ^I, O_I, and Matlis labels $\rho_{a,b}$ carry different $\mathfrak{h}$-actions and different polarizations. Shared indices are labels for separate modules.
The Fourier–Rees theorem (Theorem 8.12) is the only comparison morphism between PBW and Matlis labels: it matches associated-graded labels through the Fourier delta Rees lattice $\mathcal{D}_\hbar/(z_1, z_2)$, with Čech realization carrying the factor $(-1)^{a+b}\hbar^{a+b}a!b!$. Before $\hbar$ is inverted this is a Rees sublattice; as a module statement it uses a Fourier-twisted polarization. The Hamiltonian coadjoint action is the action on $\mathcal{P}$. Beyond the Fourier–Rees theorem the spans of $\Psi_{a,b}$ and $\rho_{a,b}$ carry distinct $\mathfrak{h}$-module structures.

Source–comparison table.

Source (algebraic)	Target (physical)	Comparison morphism	Scope
coordinate-window tower / mixed-continuous replacement attached to $\mathfrak{g}_{\mathrm{str}}$	$\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$	coordinate dg identity; bracketed comparison only after choosing a bracket-admissible B	full unreduced factorization centre requires centrality and descent data
bracket-admissible $B \subset \mathrm{PV}_{\mathrm{coord}}$	P_0 realization $\Phi^B_{P_0}$	restriction of Φ_{coord} when B gives completeness, d_π^{coord}-stability, product stability, and a continuous contraction bracket	bracket-admissible subalgebra inside the chosen completion
kernel-admissible B	Maurer–Cartan kernel Θ_B	diagonal-convergent kernel in the chosen completed tensor product	strict ambient product completion requires separate Casimir convergence
$C^\bullet_{\mathrm{CE}}(\mathfrak{g})$ / $\mathrm{PV}(A)$	reduced P_0 central-operation model	local Poisson cochain model	unreduced $Z^{P_0}_{\mathrm{fact}}$ requires centrality homotopies
$U_\hbar(\mathfrak{g})/(\hbar)$ PBW special fibre	$\mathbf{S}(\mathfrak{h} \oplus \mathfrak{h}^\vee[1])$ and then selected $\mathbf{S}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1])$ labels	associated-graded label comparison	PBW and Matlis labels carry different actions

Source (algebraic)	Target (physical)	Comparison morphism	Scope
admissible Tate / relative bar-cobar envelope	$\widehat{U}(\mathfrak{g}_{\mathrm{adm}})$	PBW envelope under conilpotent CE chains, exact continuous duality, PBW completeness, and strong convergence	available after specifying admissibility data for the formal polydisk
c^I	θ^I	generator rule $c^I \mapsto \theta^I$ of Φ_{coord}	CE ghost coordinate
u_I	O_I and brane-line Hamiltonian coordinates $B_f(a)$	generator rule $u_I \mapsto O_I$ of Φ_{coord}	boundary Hamiltonian coordinate
$C^\bullet_{\mathrm{CE}}(\mathfrak{g}_I)$ on intervals	reduced brane-line factorization target	cofactorization/ opposite-interval source; factorization only after the CE coproduct is paired with target multiplication	variance reversal handled by the CE coproduct and target multiplication
$\Psi_{a,b}$	stable primitive one-ψ PBW special-fibre class	stable-trace identification in the trace complex	PBW trace label with polynomial action
$\rho_{a,b}$	$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$	residue/Matlis pairing $f \otimes \rho \mapsto \mathrm{Res}_0(f\rho)$	Matlis principal-part label with Hamiltonian coadjoint action
$\mathsf{F}_{\mathrm{Rees}}$	$\bigoplus \hbar^{a+b}\mathbb{C}[[\hbar]]\rho_{a,b}$	Fourier-polarized Rees label bridge	polarization-changing Rees comparison
$\Omega^\bullet_c(I) \otimes \mathfrak{h}$	brane-line Hamiltonian currents	reduced current model	$C^\infty_c(I)$ denotes only the degree-zero summand
$\mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1]$	principal-part current sector $\Theta_\rho(B)$	post-de-Rham-current contraction	uses multiplication of a distribution by a smooth test function
formal Moyal algebra $\bar{A}_\hbar$	degree-zero Weyl coefficient target	reduced algebraic Moyal map with raw weights 1 and 1/24	finite-N radial-parts theorem requires the radial image data

Source (algebraic)	Target (physical)	Comparison morphism	Scope
finite-N radial-parts target	stable connected trace realization of the Moyal coefficients	finite verified relative datum: formal coefficients, Capelli normalization, free normal-diagram obstruction $\mathfrak{o}^{\mathrm{rad}}_{a,b}$, edge PBW strip, and the remaining uniform splitting or left-cokernel theorem	requires formal Moyal coefficients, the radial-parts hypothesis, and the normal-diagram obstruction calculation; Costello analytic graph/QME realization is a separate construction
Costello brane-defect kernel	closed-open BV/QME specialization	obstruction datum $\mathrm{Ob}_{w,\partial,\hbar}$ of bulk-to-defect curvature	requires bulk-to-defect curvature and counterterm data
scalar cocycle $[\bar{c}]$	$U(1)$ anomaly / Capelli contact line	distinguished class in $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$	one scalar contact class inside the unreduced QME problem

PBW special-fibre open trace labels. Algebraic side: the formal-stalk chart at the $\hbar = 0$ special fibre. Physical side: the trace observable $\overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ on the matrix Weyl brane after moment-map reduction. Comparison morphism: the PBW limit of that trace observable, an isomorphism of stable trace complexes in the Procesi–Razmyslov stable range $N \geq d$.

Symbol	Meaning
$H_{a,b} = z_1^a z_2^b$	Polynomial Hamiltonian label in $\bar{A}$, with $a + b > 0$. In the PBW special fibre it labels the scalar-reduced trace observable $O_{a,b}$.
$O_{a,b}$	Open ghost-zero trace class $\overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}$, with $a + b > 0$ in the reduced Hamiltonian label set.
$\Upsilon_{a,b}$	Primitive one-ψ stable open class, represented by the symmetrised trace with one ψ, a copies of ϕ_1, and b copies of ϕ_2, with $a + b > 0$ after removing the scalar descendant line. It lies in the PBW special-fibre open shadow and carries the corresponding polynomial action. In the reduced comparison it gives the primitive indecomposable P_0-shadow label.
$\mathsf{F}_{\mathrm{Rees}}$	Fourier–Rees label bridge of Theorem 8.12. It sends the polynomial label $z_1^a z_2^b$ to the Fourier-delta label $\partial_1^a \partial_2^b e_0$, whose Čech realization is $(-1)^{a+b} \hbar^{a+b} a! b! \rho_{a,b}$. After inverting $\hbar$, division by this Rees factor gives the principal-part label $\rho_{a,b}$. It changes polarization and is the comparison between the PBW label $\Upsilon_{a,b}$ and the Matlis label $\rho_{a,b}$.

The shared indices on $H_{a,b}$, $\Upsilon_{a,b}$, and $\rho_{a,b}$ are labels for separate modules: the span of the $\Upsilon_{a,b}$ is the PBW special-fibre trace label space, while $\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P}$ is the principal-part current module. The Fourier–Rees bridge $\mathsf{F}_{\mathrm{Rees}}$ is the only comparison morphism between PBW and Matlis sectors.

Closed CE/PV labels. Algebraic side: the closed-side complex of CE/PV cochains. Physical side: the closed-string P_0-central operation on brane observables. Comparison morphism: Φ_{coord}, the coordinate dg CE/PV isomorphism on generators $c^I \mapsto \theta^I$, $u_I \mapsto O_I$; its restriction to a bracket-admissible B is the P_0 realization $\Phi^B_{P_0}$.

Symbol	Meaning
$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1]$	Shifted cotangent Lie algebra of Hamiltonian canonical transformations.
H_I	Hamiltonian generator in $\mathfrak{h}$.
η^I	Shifted cotangent generator $(H_I)^{\vee}[1]$, with $I = (a, b)$ and $a + b > 0$.
c^I	Chevalley–Eilenberg coordinate dual to a Hamiltonian generator $H_I \in \mathfrak{h}$. Under Φ_{coord}, $c^I \mapsto \theta^I$.
u_I	Chevalley–Eilenberg coordinate dual to the shifted cotangent generator. Under Φ_{coord}, $u_I \mapsto O_I$. Boundary Hamiltonian observables come from u_I.
θ^I	Polyvector coordinate corresponding to c^I under the coordinate CE/PV isomorphism.
O_I	Hamiltonian observable coordinate corresponding to u_I under the coordinate CE/PV isomorphism.
Φ_{coord}	Coordinate dg CE/PV map determined on generators by $c^I \mapsto \theta^I$ and $u_I \mapsto O_I$. Its P_0 restriction is $\Phi^B_{P_0}$ only after a bracket-admissible B is chosen.

Matlis principal-part labels. Algebraic side: the Grothendieck-dual of the formal jet algebra $\mathfrak{h}$, namely the residue/Matlis principal-part module $\mathcal{P} \cong \mathfrak{h}^{\vee}_{\mathrm{cont}}$. Physical side: the principal-part current observable $\Theta_\rho(B)$ on a brane interval. Comparison morphism: the residue pairing $\mathfrak{h} \otimes \mathcal{P} \to \mathbb{C}$, $f \otimes \rho \mapsto \mathrm{Res}_0(f\rho)$, with Hamiltonian coadjoint action on $\mathcal{P}$ its image. This is the formal-holomorphic locality (iii) of Theorem 0.1.

Symbol	Meaning
$\mathcal{P}$	Reduced principal-part module $\mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) \subset H^2_{\mathfrak{m}}(R)\, dz_1 dz_2$, identified with $\mathfrak{h}^{\vee}_{\mathrm{cont}}$.
$\rho_{a,b}$	Principal-part residue current $z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$, $a + b > 0$, with the Matlis coadjoint action.
$z_1^p z_2^q \cdot \rho_{a,b}$	Coadjoint action $-(pb - qa + p - q)\rho_{a-p+1,b-q+1}$. This is the Hamiltonian coadjoint principal-part action. The Matlis R-module action and the PBW action on $\Psi_{a,b}$ are separate structures.
$\mathcal{P}_I$	$\mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}$, the compactly supported distributional principal-part currents on an interval I.

As stated in the separation rule above,

$$\Psi_{a,b} \in \text{PBW special-fibre open homology}, \qquad \rho_{a,b} \in \mathcal{P} \cong \mathfrak{h}^{\vee}_{\mathrm{cont}}$$

live in different modules. The only comparison morphism between them is the Fourier–Rees label bridge $\mathsf{F}_{\mathrm{Rees}}$ of Theorem 8.12, which acts on associated-graded labels through Fourier-twisted polarization. The Hamiltonian coadjoint action remains the action on the Matlis module.

Interval-current and boundary labels. Algebraic side: the brane-line objects produced by the support-locality differential of Theorem 0.1 (i). Physical side: the smeared brane measurement $B_f(a)$ and the principal-part current observable $\Theta_\rho(B)$. Comparison morphism: the de Rham-current contraction $u_{a(t)dt\otimes f} \mapsto B_f(a)$, $c_{B,\rho} \mapsto \Theta_\rho(B)$; a smooth test function multiplies a distribution.

Symbol	Meaning
$\mathfrak{h}_I$	$\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}$, the compactly supported de Rham Hamiltonian currents on an interval $I \subset \mathbb{R}$. Degree-zero test functions appear only after taking the explicit measurement notation.
$\mathfrak{g}_I$	$\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$, the shifted cotangent current algebra on I, with principal-part currents in the shifted cotangent direction.
$B_f(a)$	Smeared boundary Hamiltonian observable $\int_I a(t)\overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt$. Its CE source is $u_{a(t)dt\otimes f}$.
$A^{\mathrm{Ham}}_\partial(I)$	Reduced Hamiltonian brane-line P_0 factorization algebra $\widehat{\mathbf{S}}(\mathfrak{h}_I)$.
$A^{\mathrm{pp}}_\partial(I)$	Reduced principal-part defect-current model after de Rham-current contraction $\widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1])$. An unreduced BV observable algebra maps to it in the presence of the centrality homotopies, coefficient kernel, brane-defect propagator, descent data, and BV/QME solution.
$\Theta_\rho(B)$	Principal-part current observable with $B \in \mathcal{D}'_c(I)$ and $\rho \in \mathcal{P}$. It is the boundary notation for $B \otimes \rho[1]$ in the reduced model and pairs with smooth test functions by $B(a)\,\mathrm{Res}(\rho f)$.
aB	Multiplication of a compactly supported distribution B by a smooth test function $a \in \Omega^0_c(I)$. The reduced brackets use this operation and never require multiplying two distributions.

Conventions and notation for the local theorem

The conventions below bind throughout the body, the appendices, and every cross-target statement; deviation requires an explicit matched-conventions theorem (Theorem 11.13, §11.4).

Geometric and algebraic conventions. The complex target dimension is $d = \dim_{\mathbb{C}} X$; the formal-Darboux trace theorem proves the trace-sector map

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \dashrightarrow C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}), \qquad c_f \mapsto \theta_f, \quad u_f \mapsto J(f),$$

with stable scalar-reduced CE/PV equivalence after the admissible completion (Definition 1.1, Theorem 5.6) at the Dirac brane stack on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$, with $d = \dim_{\mathbb{C}} \widehat{\mathbb{C}^2}_0 = 2$. The local geometry is $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ at a brane: the only Calabi–Yau datum is the trivial holomorphic symplectic $\Omega_{\mathrm{loc}} = dz_1 \wedge dz_2$ on $\widehat{\mathbb{C}^2}_0$, used for the BV pairing, divergence-free Hamiltonian fields, and cyclic trace densities of Definition C.1. Compact CY_3, BCOV, quintic, Ooguri–Strominger–Vafa (OSV), Gopakumar–Vafa (GV), Maulik–Nekrasov–Okounkov–Pandharipande (MNOP), Abel–Jacobi, cohomological Hall algebra (CoHA), Igusa, and Borcherds–Kac–Moody (BKM) belong to compact or automorphic comparison theorems governed by Theorem 11.4 and the matched-conventions obstruction class $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}})$. The native bulk object is the holomorphic E_2 / factorization algebra on $\mathbb{C}^2$, $\Omega^{0,\bullet}_c(B) \widehat{\otimes} \mathfrak{g}$, with shifted-cotangent BF Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$, $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ (§2, Theorem 2.11); it is two-complex-dimensional. A curve vertex operator algebra (VOA) is a one-dimensional chiral object. It can be compared with the native factorization algebra only after a controlled reduction gives explicit z_2-mode or principal-part data, pushed-forward bracket, BV pairing, brane image, and anomaly matching. Only then do the Zhu algebra and the Vol. II $\mathbb{C} \times \mathbb{R}$ chiral-topological theorems apply to the restricted curve theory (Remark 1.27). The worldsheet Σ is a complex curve, possibly with boundary or marked points; framing on S^3 is stated when invoked, with deviation flagged.

Sign, degree, and renormalization conventions. BV degree, ghost number, form degree, Koszul / CE / PV / Gerstenhaber bracket signs, cyclic trace, CE/PV cochain comparison, and the heat-kernel propagator $(K_L - K_\epsilon)$ are fixed in Appendix C and Appendix B. The QFT loop $\hbar$ controls the local BV/Moyal expansion; the string-genus g_s belongs to the ambient genus expansion. The local theorem uses $\hbar$. Compatibility with the BCOV holomorphic-anomaly equation is the matched-conventions statement of Theorem 11.21.

Homological taxonomy. Hochschild homology $HH_\bullet$, cyclic homology, and negative-cyclic homology $HC^-_\bullet$ are distinguished at every reference (Appendix C): the cyclic trace symbol Tr is the graded cyclic quotient, its Hochschild shadow is obtained by forgetting the S^1-equivariance in the cyclic complex, and the Connes B-operator is the connecting operator in the SBI sequence. The negative-cyclic class is the Calabi–Yau pairing on the target category. Drinfeld centre $Z(C)$, derived centre Z^{der}, and chiral derived centre $Z^{\mathrm{der}}_{\mathrm{ch}}$ are distinguished: the bar construction $\mathrm{Bar}(A)$ is the twisting coalgebra, while the bulk is the derived chiral centre. For a chiral algebra A on a smooth curve C the $d = 1$ chiral derived centre is $Z^{\mathrm{der}}_{\mathrm{ch}}(A) \simeq C^\bullet_{\mathrm{ch}}(A, A)$ by Beilinson–Drinfeld [28, §4]. At $d = 2$ the Mixed Holomorphic–Topological Deligne conjecture asserts the equivalence of the bulk with $Z^{\mathrm{der}}_{E_2^{\mathrm{ch}}}(C_\partial^{\mathrm{op}})$, the chiral derived E_2-centre of the boundary open category. The formal-Darboux trace theorem gives the trace-sector map at the brane vacuum and identifies the obstruction classes for the full derived-centre comparison. Three distinct algebraic structures appear:

(i) the bulk chiral E_2 factorization algebra $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ on $\mathbb{C}^2$;

(ii) the formal-Darboux local-observable algebra $\mathrm{Obs}^{\mathrm{cl}}_b(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}})$, obtained by continuous duality from the current stalk at the brane vacuum and carrying the chiral structure restricted to the formal neighbourhood of b;

(iii) the boundary E_1-brane algebra $A^{\mathrm{cl}}_{\partial,N}$ along the worldline $\mathbb{R}_t$, whose derived E_1-centre $Z^{\mathrm{der}}_{E_1}(A^{\mathrm{cl}}_{\partial,N}) \simeq HH^\bullet(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ is an E_2-algebra by Deligne's E_1-Hochschild theorem.

The formal coordinate algebra $A^{\mathrm{coord}}_b = \widehat{\mathcal{O}}_{\mathbb{C}^2,b} \cong \mathbb{C}[[z_1, z_2]]$ gives Hamiltonians on the Darboux normal disk. The brane algebra whose Hochschild cochains occur below is the boundary algebra $A^{\mathrm{cl}}_{\partial,N}$. When the notation $C^\bullet_{\mathrm{ch}}(A_b, A_b)$ is used for the Deligne target, A_b denotes the chosen boundary algebra with the restricted chiral structure, not the coordinate ring A^{coord}_b. The local theorem of §5.3 constructs the trace-sector Hochschild generator assignment

$$c_f \mapsto \theta_f, \qquad u_f \mapsto J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

for polynomial f, with formal f interpreted after completion. After the stable scalar-reduced trace quotient and admissible completion this is the CE/PV coordinate isomorphism. At fixed finite N the full derived E_1-centre contains trace relations, stabilizer classes, and centrality homotopies outside that trace sector. The Mixed Holomorphic–Topological Deligne conjecture is the global identification of the bulk chiral E_2 algebra $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ with the chiral derived E_2-centre of C_∂^{op}; this formal-Darboux theorem gives its trace-sector formal coordinate and leaves the global derived-centre identification as the obstruction-and-cone problem. Where $C^\bullet_{\mathrm{ch}}(A_b, A_b)$ appears below, A_b carries the restricted chiral structure and the symbol denotes the local Hochschild target of this trace-sector map.

Cross-volume objects anchored to the present theorem. The global boundary category lives on the Kato–Nakayama log boundary ∂X^{KN}. The local calculation uses the Costello–Gwilliam brane-link boundary

$\partial_{\mathrm{link}} X$ of the Dirac brane defect $L \subset X$. A comparison between these two boundaries is part of the matched-conventions datum. The open category $\mathcal{C}_\partial^{\mathrm{op}}$ is primitive boundary data equipped with a bulk–boundary action; it is not the restriction of the bulk factorization algebra. Factorisation-homology specialisation along a $(d-1)$-manifold Σ_{d-1} and a curve C is $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}(\mathcal{F}) = \int_{\Sigma_{d-1}} \mathcal{F}|_C$ (Lurie–Ayala–Francis). The Dirac brane stack at N branes is the derived commuting variety $[\mu_{\mathrm{der}}^{-1}(0)/\mathrm{GL}_N]$ with Koszul differential $Q\phi_i = 0$, $Q\psi = [\phi_1, \phi_2]$ resolving the moment-map ideal of conjugation on $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$. The Capelli scalar $\hbar N[\bar{c}]$ is the projective Lie anomaly at the trace generator (Theorem 6.8). The compact Igusa comparison identifies the modular line bundle on $\overline{\mathcal{A}}_g$ with the central projective-curvature line after matched conventions; the Igusa cusp form Φ_{10} is the level-2 scalar section, and $\Delta_5 = \Phi_{10}^{1/2}$ denotes a chosen square-root branch on the paramodular cover. Its operator-level lift on $K3 \times E$ is the obstruction question treated in §11.3. The CE/PV dictionary $c_f \mapsto \theta_f$, $u_f \mapsto J(f)$ of Lemma C.2 is the trace-sector Koszul assignment from the bulk formal stalk to the brane-side E_1-Hochschild cochain complex $C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ at the chosen Dirac brane vacuum $b_p \in \mathcal{C}_\partial^{\mathrm{op}}$, with formal coordinate algebra $A^{\mathrm{coord}}_{b_p} = \widehat{\mathcal{O}}_{\mathbb{C}^2,p}$ (Theorem 5.147). The full Mixed Holomorphic–Topological Deligne identification is the derived-centre problem controlled below.

Theorem-control predicates. Every theorem is stated under the following hypotheses:

(i) Native $\mathbb{C}^2$ holomorphic E_2 / factorization-algebra taxonomy is retained before any curve-chiral reduction; a curve VOA result is invoked only after a controlled reduction with pushed-forward bracket, brane image, and anomaly matching.

(ii) BMK comparison: the one-pair analytic pro-Matlis retract of Theorem 8.42 is the pro-Matlis replacement for strict all-window support-local current transfer; the obstruction on every fixed finite Matlis window is $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$ (Theorem 8.40), and the pro-tower $\widehat{\mathcal{P}}_\Pi = \varprojlim_N \mathcal{P}_{\le N}$ is the target replacement in which that obstruction is represented.

(iii) Larger non-scalar θ_3 row is evidence only with one of: a Chevalley–Eilenberg ancestor, a scalar-zero Costello local counterterm, or a complete companion-face table (Theorem E.38); tower compatibility passes through $\Delta^1_{M,N} = -\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N$ together with the secondary $\varprojlim^1 H^0$ primitive class.

(iv) The radial / Weyl statement is $\Omega^{\mathrm{rad}}_{a,b} \in \operatorname{coker} B_{a,b}$, equivalently the decorated PBW Stokes obstruction for $D^{\square}_{a,b} = C^+_{a,b}\partial_2$, with obstruction exactly a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ such that $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$ (Theorem F.44).

(v) Larger non-scalar Costello / QME theorem requires the data tuple: filtered scalar projection $\widehat{\sigma}_{\mathrm{sc}}$, finite row arrays, primitive matrix $A^M c = -r^M$, transition matrices, Roos compatibility, centrality homotopies, and the curved bulk-to-defect kernel (Theorem E.53).

(vi) The brane-preserving Ω-background uses the normal scaling on $N_L X = \mathbb{R}_s \oplus \mathbb{C}_{z_1} \oplus \mathbb{C}_{z_2}$ with the brane time t fixed, torus $T_\Omega = \mathbb{C}^*_{\epsilon_s} \times \mathbb{C}^*_{\epsilon_1} \times \mathbb{C}^*_{\epsilon_2}$, twisted differential $Q_\Omega = Q + \iota_{V_\Omega}$ with $Q_\Omega^2 = L_{V_\Omega}$, inverted normal weights, residue-vs-Euler normalization, and stratified factorization data; literal (t, s)-rotation changes the brane fixed locus and belongs to a different fixed-locus problem.

Compact comparison hypotheses. A compact comparison theorem contains the local formal-Darboux stalk together with the compact object to which it is compared. According to the case, this object is the BCOV/Kodaira–Spencer propagator with its anomaly sign, the Calabi–Yau-to-chiral specialization functor, the Igusa/Borcherds–Kac–Moody normalization, the Gromov–Witten/Donaldson–Thomas/Pandharipande–Thomas virtual trace, the Ooguri–Strominger–Vafa attractor variables, the mixed-Dunn six-relation structure, or the global Hamiltonian descent object on a compact holomorphic-symplectic surface. The comparison is the matched-conventions theorem (Theorem 11.4, Theorem 11.13) with Koszul-descent obstruction class $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}})$; a nonzero coordinate of $\mathrm{Ob}_{\mathrm{comparison}}$ is the obstruction to matched-conventions transport.

The formal-Darboux theorem and its boundary

The formal-Darboux theorem identifies the coordinate CE/PV dictionary

$$c_f \longmapsto \theta_f, \qquad u_f \longmapsto J(f)$$

inside the admissible filtered nuclear Tate, pro-Matlis, and Capelli-projective trace category $\mathcal{B}^{\mathrm{adm}}$. The strict ambient product does not carry this theorem: the degree-lowering Hamiltonian bracket has a Jacobi obstruction (Lemma 3.37, Theorem 3.38), and the diagonal Casimir becomes a Maurer–Cartan kernel only after the weighted Tate completion of Theorem 5.58.

In this completed category the bracket-admissible polyvector sector has the P_0-lift of Theorem 5.5; the bar–cobar and open A_∞-Koszul comparison is controlled by the filtered twisting-cochain and $\varprojlim^1$-obstruction of Lemma 4.25. The local trace-sector map

$$\mathrm{Obs}^{\mathrm{cl}}_b(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}) \dashrightarrow_{\Phi_N} C^\bullet_{\mathrm{Hoch,adm}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$$

is Theorem 5.6; the Capelli scalar is the projective curvature class $\hbar N[\bar{c}]$ of Theorem 6.8.

The global open–closed comparison on the Costello–Gwilliam brane-link boundary ∂X requires the vanishing and chosen null-homotopies of the chiral E_2-Deligne, descent, and all-order QME obstruction classes, together with a contraction of $\mathrm{Cone}(\mathrm{OCA}_N)$ (Theorem 5.13). Operator lifts of the modular line on $K3 \times E$ and Hall–Drinfeld–Borcherds comparison on compact non-toric Calabi–Yau threefolds are compact comparison theorems governed by $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}})$, not consequences of the formal stalk alone.

The strict-product obstructions of row two force the admissible completion. Remark 1.6 records the three obstruction mechanisms (degree-lowering bracket; divergent Casimir; $\varprojlim^1$ defect); the admissible model carries their specified null-homotopies inside $\mathcal{B}^{\mathrm{adm}}$.

1 THE SETUP

Mixed holomorphic–topological string theory on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ has, at each formal holomorphic-symplectic brane point, a trace-sector closed-to-open map into the chiral Hochschild cochains of the boundary algebra; this formal-Darboux coordinate expression is computed at N Dirac branes.

1.1 The Dirac brane formal-stalk chart in one calculation

Definition 1.1 *(Formal-Darboux stalk chart at a brane vacuum).* A *brane vacuum at a closed point* $p \in \mathbb{C}^2$ is an object b_p of the primitive open boundary factorization category C^{op}_∂ on the Costello–Gwilliam brane-link boundary ∂X of the Dirac brane defect $L_p = \mathbb{R}_t \times \{p\} \subset X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ (link sphere bundle over the worldline). Its formal coordinate algebra is the completed local ring

$$A^{\mathrm{coord}}_{b_p} = \widehat{\mathcal{O}}_{\mathbb{C}^2,p} \cong \mathbb{C}[[z_1, z_2]].$$

The *formal-Darboux stalk chart at b_p* is the quadruple $\big(\widehat{\mathbb{C}^2}_p,\ \omega_p,\ A^{\mathrm{coord}}_{b_p},\ (z_1, z_2)\big)$ with formal completion $\widehat{\mathbb{C}^2}_p = \operatorname{Spf} \widehat{\mathcal{O}}_{\mathbb{C}^2,p}$, holomorphic symplectic form $\omega_p = dz_1 \wedge dz_2$, chart algebra $A^{\mathrm{coord}}_{b_p}$, and formal Darboux coordinates (z_1, z_2) from the formal holomorphic Darboux theorem. The *Dirac brane formal-Darboux stalk chart at N branes over b_p* adjoins the Chan–Paton coordinate assignment

$$z_i \mapsto \phi_i \in \mathfrak{gl}_N, \qquad \psi \in \mathfrak{gl}_N[1], \qquad Q\psi = [\phi_1, \phi_2]\ :$$

the $\mathfrak{gl}_N$-valued pair $(\phi_1, \phi_2) \in \mathfrak{gl}_N \oplus \mathfrak{gl}_N$ is the matrix coordinate pair of the N-brane stack on the holomorphic-symplectic factor (its transverse position), and Dirac/moment-map reduction is the Koszul derived zero fibre of the commutator moment map. Affine Kac–Moody, free $\beta\gamma$, Virasoro, and W_N vertex algebras are curve chiral algebras which may be compared with the Hamiltonian formal disk only after a curve-reduction datum and anomaly matching. For a factorization algebra $\mathcal{X}$ on $\mathbb{R}^2_t \times \mathbb{C}^2_{z_1,z_2}$, the *formal-Darboux stalk of $\mathcal{X}$ at p* is the restriction $\mathcal{X}|_{\mathbb{R}^2_t \times \widehat{\mathbb{C}^2}_p}$ in these coordinates.

A polynomial Hamiltonian $f \in \bar{A}$ on the formal symplectic normal polydisk has trace character

$$J(f) = \overline{\operatorname{Tr} f(\phi_1, \phi_2)},$$

a GL_N-invariant trace function on the brane stack; formal Hamiltonians are evaluated only in the completed brane algebra. The supporting geometry is the product

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1,z_2},$$

formally completed at the origin of $\mathbb{C}^2$. Each factor carries one ingredient of the calculation: the worldline $\mathbb{R}_t$ carries the open-line observable algebra, the transverse direction $\mathbb{R}_s$ carries the bulk-to-boundary propagator, and the formal holomorphic-symplectic normal disk $\widehat{\mathbb{C}^2}_0$ carries the Hamiltonian algebra $\mathfrak{h}$. Replacing the formal completion $\widehat{\mathbb{C}^2}_0$ by a global holomorphic-symplectic surface $Y \supset \widehat{Y}_p$ gives only the formal restriction of a global target theorem. Matched-conventions descent, anomaly cancellation, and the comparison morphism are additional data (§11.4).

A stack of N Dirac branes on $L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\} \subset X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2$ is a Chan–Paton matrix coordinate assignment

$$z_i \longmapsto \phi_i \in \mathfrak{gl}_N,$$

equipped with the Koszul field $\psi \in \mathfrak{gl}_N[1]$ resolving the commutator moment map; on H^0 of the derived zero fibre it becomes the usual representation of the commutative formal coordinate algebra by commuting matrices. The holomorphic symplectic form $dz_1 \wedge dz_2$ pulls back along this substitution and traces to

$$\theta_N = \operatorname{Tr}(\phi_1\, d\phi_2), \qquad \Omega_N = \operatorname{Tr}(d\phi_1 \wedge d\phi_2),$$

and GL_N-conjugation acts symplectically with moment map $\mu(\epsilon) = -\operatorname{Tr} \epsilon[\phi_1, \phi_2]$, so Dirac reduction imposes the first-class constraint $[\phi_1, \phi_2] = 0$. On H^0 the Chan–Paton pair lies in the commuting variety; on its reduced semisimple locus the simultaneous spectrum of (ϕ_1, ϕ_2) is a length-N configuration in the polydisk. The trace character map at this derived Chan–Paton datum is

$$J : \mathfrak{h} \to \operatorname{Im}(J)^{\wedge}_{\mathrm{Tate}} \subset A^{\mathrm{cl}}_{\partial,N}, \qquad J(f) = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}, \qquad \mathfrak{h} = A/\mathbb{C} \cdot 1,$$

the completed gauge-invariant trace character; on polynomial Hamiltonians it is the ordinary trace on the finite-N brane algebra. The Chan–Paton pair defines a representation of the commutative coordinate ring A on H° of the Koszul derived zero fibre. Before this passage the brane datum is $(\phi_1, \phi_2, \psi, Q\psi = [\phi_1, \phi_2])$. The Hamiltonian Lie algebra $\mathfrak{h}$ acts on the underived commuting representation functor by Hamiltonian vector fields and on the derived zero fibre by the induced Q-commuting derivations.

For polynomial $f \in A_{\text{poly}} = \mathbb{C}[z_1, z_2]$ the substitution $f(\phi_1, \phi_2) \in \operatorname{End}(V)$ is defined on the commuting variety $\operatorname{Rep}_N(A_{\text{poly}}) = \{(\phi_1, \phi_2) \in \mathfrak{gl}_N^2 : [\phi_1, \phi_2] = 0\}$ and gives the polynomial trace map $J_{\text{poly}} \colon A_{\text{poly}}/\mathbb{C} \cdot 1 \to \mathcal{O}(\operatorname{Rep}_N(A_{\text{poly}}))^{GL_N}$. For formal $f \in A = \mathbb{C}[[z_1, z_2]]$ the substitution lives on the underived formal representation functor

$$\operatorname{Rep}_N^\wedge(A)(R) = \big\{(\phi_1, \phi_2) \in (\mathfrak{m}_R \otimes \operatorname{End}V)^2 : [\phi_1, \phi_2] = 0\big\}, \qquad R \text{ Artin local over } \mathbb{C},$$

on which every formal power series in ϕ_1, ϕ_2 terminates modulo $\mathfrak{m}_R^k$; the formal trace map is $J \colon \mathfrak{h} \to \mathcal{O}(\operatorname{Rep}_N^\wedge(A))^{GL_N}$, and the Tate completion $A_\partial = \widehat{\mathbf{S}}_{\text{Tate}}(\mathfrak{h})$ records its image at the universal Artin base. The polynomial commuting variety $\operatorname{Rep}_N(A_{\text{poly}}) \subset \mathfrak{gl}_N^2$ is the uncompleted algebraic model; the formal functor $\operatorname{Rep}_N^\wedge(A)$ is its formal completion at the closed point $\{\phi_1 = \phi_2 = 0\}$ inside the commuting variety, and on polynomial Hamiltonians the two evaluations agree. The derived formal brane is obtained from this functor by adjoining the odd variable ψ with Koszul differential $Q\psi = [\phi_1, \phi_2]$.

An odd matrix antifield $\psi \in \mathfrak{gl}_N[1]$ and BV differential Q with $Q\phi_i = 0$, $Q\psi = [\phi_1, \phi_2]$ realize the moment-map ideal as the Koszul derived zero fibre, and the matrix entries of $[\phi_1, \phi_2]$ have syzygies for $N \geq 2$. The open sector is the derived zero fibre of the conjugation moment map (§3.4). The closed BV sector takes its fields in the shifted-cotangent extension

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}_{\text{cont}}^\vee[1].$$

A Hamiltonian $f \in \mathfrak{h}$ carries two Chevalley–Eilenberg (CE) coordinates on $\mathfrak{g}$: the generator-dual c_f acting as the Hamiltonian Schouten polyvector (PV) field θ_f on the underived commuting representation functor, and the cotangent-dual u_f mapping to the boundary observable $O_f = J(f)$ on the brane; the CE/PV cochain dictionary on generators is

$$c_f \longmapsto \theta_f, \qquad u_f \longmapsto O_f$$

(Theorem 4.20). The residue pairing realizes the cotangent module $\mathfrak{h}_{\text{cont}}^\vee$ as the Grothendieck–Matlis principal-part module

$$\mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2,$$

with Taylor Hamiltonians paired by the residue $\langle f, \rho_{a,b}\rangle = \mathbf{Res}_0(f\,\rho_{a,b})$ detecting the $z_1^a z_2^b$-coefficient of f (Theorem 5.179); the residues $\rho_{a,b}$ are the cotangent momenta conjugate to the trace-map evaluation of Taylor coefficients, distinct from the polynomial PBW shadows of the open one-ψ sector.

Example 1.2 *(Single-point evaluation, $N = 1$).* For $N = 1$ the matrices commute automatically, and $J(f) = f(\phi_1, \phi_2)$ is the evaluation map $\mathbb{C}[[z_1, z_2]] \to \mathbb{C}$ at the point $(\phi_1, \phi_2) \in \mathbb{C}^2$. This is the underived support of the single brane. The derived zero fibre also contains the closed scalar Koszul class $[\psi]$, since $Q\psi = 0$ for $\mathfrak{gl}_1$. The trace-sector map J forgets this decoupled scalar antifield. Procesi–Razmyslov degenerates to the polynomial ring on two scalar variables, and J is the classical evaluation homomorphism.

Example 1.3 *(Two Dirac branes, generic commuting pair, $N = 2$).* For $N = 2$ the moment-map constraint $[\phi_1, \phi_2] = 0$ cuts the generic stratum of $\mathfrak{gl}_2 \oplus \mathfrak{gl}_2$ to simultaneously diagonalisable pairs $\phi_1 \sim \operatorname{diag}(a_1, a_2)$, $\phi_2 \sim \operatorname{diag}(b_1, b_2)$. The trace map evaluates on monomial Hamiltonians as the two-variable power sum of the eigenvalue pairs:

$$J(z_1^p z_2^q) = \operatorname{Tr}\big(\phi_1^p \phi_2^q\big) = a_1^p b_1^q + a_2^p b_2^q, \qquad p + q \geq 1.$$

In particular $J(z_1) = a_1 + a_2$, $J(z_2) = b_1 + b_2$, and $J(z_1 z_2) = a_1 b_1 + a_2 b_2$ are the two-brane trace values of the trace generators $O_{1,0}, O_{0,1}, O_{1,1}$. The cyclic trace invariants $\{J(z_1^p z_2^q)\}_{p+q\geq 1}$ generate the GL_2-invariant ring on the commuting variety $\mathfrak{C}_2 = \{(\phi_1, \phi_2) \in \mathfrak{gl}_2^2 : [\phi_1, \phi_2] = 0\}$ by Procesi–Razmyslov [17][18]; the dictionary $u_{p,q} \mapsto J(z_1^p z_2^q)$ identifies the boundary observables with the single-trace generators. Outside the generic stratum the commuting variety has singular strata at coincident eigenvalue pairs; the trace map descends to a single-valued expression on each stratum.

Example 1.4 *(Linear Hamiltonians and the scalar anomaly).* For $f = z_1$, $g = z_2$, the unreduced brane Poisson bracket is

$$\{\operatorname{Tr}\phi_1, \operatorname{Tr}\phi_2\} = N.$$

This scalar is the open-side image of the closed–open mismatch Lie 2-cocycle $\bar{c} \in Z^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$, the cohomology class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ attached to the constant Hamiltonian line that vanishes closed-side and survives open-side as the brane centre of mass (Lemma 6.2, Theorem 6.3). Moyal quantization with Weyl-symmetric quantization map $\Phi_\hbar$ and Kontsevich–Moyal star $\star$ returns the same scalar in the projective trace representation, $[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_{\mathrm{proj}} = \hbar N \cdot \mathbf{1}$; the first scalar-reduced $\mathfrak{gl}_N$ quantum-master-equation obstruction is the same class shifted by $\hbar$, $W_1^{\mathrm{sc,ord}} = \hbar N[\bar{c}]$ (Theorem E.13).

Example 1.5 *(A single Matlis residue).* The Grothendieck–Matlis principal-part

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2 \in \mathcal{P} \subset H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2$$

in the local-cohomology module $H^2_{\mathfrak{m}} = H^2_{\{0\}}$ at the maximal ideal $\mathfrak{m} = (z_1, z_2)$ pairs with $f \in \mathfrak{h}$ by $\langle \rho_{a,b}, f\rangle = \operatorname{Res}_0(\rho_{a,b} f)$ and detects the Taylor coefficient of $z_1^a z_2^b$ in f. The residue-coadjoint formula $z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}$ of Theorem 5.179 gives a non-locally-nilpotent action on $\mathcal{P}$. A polynomial PBW class attached to the same (a, b)-bidegree exists in the open one-ψ sector with a strictly different $\mathfrak{h}$-module structure (Corollary 5.173); the residue mode here and that polynomial class are separated by the Fourier-polarized Rees bridge.

Remark 1.6 *(Strict realization and admissible completion).* Examples 1.2–1.5 already display three obstructions to a strict realization of the local open–closed dictionary. First, the bracket $[H_p, H_q] \subset H_{p+q-2}$ on the Hamiltonian polydisk Lie algebra $\mathfrak{h}$ is *degree-lowering*: a linear Hamiltonian $z_i \in H_1$ does not preserve any finite Taylor truncation $\mathfrak{h}_{\leq N}$, so the plain projected bracket fails Jacobi in finite windows (Lemma 3.37, Theorem 3.38). Second, the diagonal Casimir $\sum_i e_i \otimes e^i$ over the principal-part basis is a *divergent* sum on $\mathfrak{h} \otimes \mathfrak{h}^\vee_{\mathrm{cont}}$: the unweighted Tate basis $\{z_1^a z_2^b\}$ and its residue dual $\{\rho_{a,b}\}$ of Example 1.5 are mutually paired but the sum has no convergent meaning before admissible weighting. Third, the inverse-limit reconstruction along the Taylor tower carries a *Milnor* $\lim^1$ *defect* (Lemma 4.25) that blocks the strict bar–cobar identification.

The local theorem (§5.3, Theorem 5.6) isolates the trace-sector completion. The weighted-Tate completion of Theorem 5.58 makes the Casimir converge; the pro-Matlis completion of Theorem 5.179 stores the principal-part module without a degree-lowering pathology; the Capelli-projective central extension

$\widehat{\mathfrak{g}}_{\mathrm{cent}}$ of §6 records the finite-N scalar of Example 1.4 as the projective Lie anomaly $\hbar N\,[\bar{c}]$. These choices give the completed CE/PV trace-sector map. They do not turn the finite-N brane algebra into the whole derived centre; the strict-product and unreduced P_0-centrality extensions remain the obstruction problem.

1.2 Mixed holomorphic-topological strings

In the local theory, a mixed-HT string-field-theoretic sector is a classical BV theory whose local observables form a factorization algebra in the Costello–Gwilliam sense [15], together with a cyclic open BV theory on a brane and a bulk-to-boundary operation. The *topological* qualifier names a factorization algebra satisfying the topological locality of Proposition 1.36: every translation acts null-homotopically. The *holomorphic* qualifier names a factorization algebra satisfying the holomorphic locality: every antiholomorphic translation acts null-homotopically. The protected algebraic content under Theorem 1.37 is the classical and quantum master equations, locality, and factorization associativity; the topological and holomorphic qualifiers refine this content by constraining how local operators vary across the chosen spacetime directions. The word *string* appears here in this local string-field-theoretic sense, attached to the BV / factorization data; global theories of worldsheet moduli or amplitudes require additional compact field-theory data.

Gwilliam–Williams [5] formulate a holomorphic string in the Costello BV/factorization formalism, with Dolbeault fields (γ, β, c, b), one-loop quantization in the critical complex dimension, and a factorization algebra recovering BRST cohomology. The model here is a formal local target-space sector with de Rham/Floer directions, a holomorphic-symplectic formal polydisk, Hamiltonian cotangent BF fields, and a brane large-N operator algebra.

Definition 1.7 *(Local topological factorization sector; forced by item (2) of Proposition 1.36).* Let M be a smooth manifold. A local topological closed-string sector on M is a classical BV theory whose local observables form a locally constant factorization algebra on M: for inclusions of sufficiently small disks $U \subset V$, the structure map

$$\mathrm{Obs}^{\mathrm{cl}}(U) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(V)$$

is a quasi-isomorphism, equivalently every translation $v \in TM$ acts on observables by a homotopy $L_v = [Q, \iota_v]$. Attaching branes means specifying a cyclic open BV theory and the bulk-to-boundary central-operation data it carries. In a de Rham model the closed fields carry d_M plus the BV linearised differential; for pairwise disjoint disks $U_1, \ldots, U_k \subset V$, the locally constant factorization product $\bigotimes_i \mathrm{Obs}^{\mathrm{cl}}(U_i) \to \mathrm{Obs}^{\mathrm{cl}}(V)$ is the operator product on disjoint supports, depending only on the isotopy class of the embedding by the translation homotopy $L_v = [Q, \iota_v]$; equivalently the Lurie–Costello–Gwilliam E_n-algebra structure on the value at a single disk.

Remark 1.8 *(Canonical completion).* Locally constant factorization is exactly the data of a topological sector in the Lurie–Costello–Gwilliam sense, and the homotopy $L_v = [Q, \iota_v]$ is the constructive content of locality item (2). Strict translation invariance, or imposing a metric, breaks compatibility with that homotopy and is therefore excluded. The exact A-model factor used below is the self-Floer model for $\mathbb{R} \subset T^*\mathbb{R}$, represented by the de Rham complex $\Omega^\bullet(\mathbb{R})$; its local observables are insensitive to the metric of the interval.

Definition 1.9 *(Local holomorphic factorization sector; forced by item (3) of Proposition 1.36).* Let Y be a complex manifold. A local holomorphic sector on Y is a classical BV theory whose fields are Dolbeault complexes on Y, whose differential has the form $\bar{\partial} + Q_0$ with Q_0 holomorphic, and whose interactions are holomorphic polydifferential operators. Its local observables form a holomorphic factorization algebra: on coordinate polydisks the factorization products and single-open structure maps are represented by holomorphic polydifferential operations, and every antiholomorphic translation $\partial/\partial\bar{z}_i$ acts null-homotopically through $L_{\partial/\partial\bar{z}_i} = [Q, \iota_{\partial/\partial\bar{z}_i}]$. Branes are cyclic open BV theories with bulk-to-boundary operations.

Remark 1.10 *(Canonical completion).* The continuous-dual data attached to a brane point is the principal-part module $\mathcal{P}$ of §1.6, forced by Theorem 5.179; the Dolbeault presentation, with the homotopy $L_{\partial/\partial\bar{z}_i} = [Q, \iota_{\partial/\partial\bar{z}_i}]$, is what locality item (3) constructs. The active geometry below is $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$. BCOV/Kodaira–Spencer [1] gives the source conventions: Dolbeault polyvectors, cotangent BV notation, the divergence determined by a holomorphic volume form. Here the holomorphic-symplectic form $dz_1 \wedge dz_2$ and the volume form $dz_1 dz_2$ provide the BV pairing, divergence operator, and cyclic trace densities; divergence-free vector fields are locally Hamiltonians modulo constants.

Remark 1.11 *(Gwilliam–Williams sector vs. the local target-space sector).* The classical local observable theory of Gwilliam–Williams is a pure holomorphic BV/factorization sector on a Riemann surface, with worldsheet ghost, BRST, and critical-dimension quantum structure. Its critical-dimension and BRST-cohomology theorems live on the worldsheet; the model below is the local target-space sector $\mathbb{R}^2 \times \mathbb{C}^2$, with holomorphic factor Hamiltonian BF (the holomorphic-symplectic Kodaira–Spencer analogue) and topological factor de Rham/Floer. The critical complex dimension is a worldsheet ghost-anomaly statement for the holomorphic bosonic string; the local $\mathbb{R}^2 \times \mathbb{C}^2$ Hamiltonian sector is governed by Definition 1.12.

Definition 1.12 *(Classical local mixed holomorphic-topological sector; the constructed factorization algebra carrying all three locality differentials).* Let M be a smooth manifold and Y a complex manifold. A classical local mixed holomorphic-topological closed sector on $M \times Y$ consists of:

(i) a sheaf of cochain complexes of fields $\mathcal{E}$ which, on product coordinate opens $D \times B$, has the form

$$\mathcal{E}(D \times B) = \Omega^\bullet(D) \widehat{\otimes} \Omega^{0,\bullet}(B, \mathcal{V})$$

for a graded holomorphic vector bundle or formal holomorphic vector space $\mathcal{V}$;

(ii) a degree-(-1) local BV pairing on $\mathcal{E}$ and a differential

$$Q = d_M + \bar{\partial}_Y + Q_0,$$

with Q_0 holomorphic over Y and linear over de Rham forms on M;

(iii) a degree-zero local functional $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E})$, built from differential operators in the M directions and holomorphic polydifferential operators in the Y directions, satisfying $QI + \frac{1}{2}\{I, I\} = 0$;

(iv) the associated classical observables

$$\mathrm{Obs}^{\mathrm{cl}}(U) = \big(\widehat{\mathrm{Sym}}(\mathcal{E}^!_c(U)), Q_{\mathrm{obs}}\big), \qquad Q_{\mathrm{obs}} = Q^! + \{I, -\},$$

with the standard factorization product for disjoint supports. Here $\mathcal{E}^!_c(U)$ is the density-dual cosheaf of fields, and extension by zero gives the covariant structure maps used by the factorization algebra.

These data must satisfy the mixed locality condition: for every inclusion of small disks $D \subset D'$ in M and every holomorphic polydisk $B \subset Y$, the structure map

$$\mathrm{Obs}^{\mathrm{cl}}(D \times B) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(D' \times B)$$

is a quasi-isomorphism, and on B each antiholomorphic translation $\partial/\partial\bar{z}_i$ acts null-homotopically through a degree-(-1) derivation η_i with $[Q_{\mathrm{obs}}, \eta_i] = \mathcal{L}_{\partial/\partial\bar{z}_i}$.

A mixed brane is supported on $N \times Z$, with $N \subset M$ a smooth submanifold and $Z \subset Y$ a holomorphic submanifold or formal holomorphic neighbourhood of one, specified by an object $\mathcal{F}$ of the local holomorphic brane category on Z. Its open fields have the derived form

$$\Omega^\bullet(N)\widehat{\otimes}\mathrm{RHom}_Y(\mathcal{F}, \mathcal{F}) \otimes \mathrm{End}(V)[1],$$

with a cyclic trace. For a smooth brane this recovers the Dolbeault model $\Omega^\bullet(N)\widehat{\otimes}\Omega^{0,\bullet}(Z, \mathcal{W})$; for a skyscraper brane the Ext algebra. A full bulk-to-boundary mixed datum requires a morphism of factorization algebras

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed}}\big|_{N\times Z} \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}}),$$

compatible with BV brackets and factorization products. The Hamiltonian model below constructs the degree-zero Hamiltonian current part of this map; the full closed CE/cotangent factorization-centre lift is the obstruction problem Problem 5.49.

Remark 1.13 *(Canonical completion).* Item (1) fixes the field sheaf as $\Omega^\bullet(D)\widehat{\otimes}\Omega^{0,\bullet}(B)$: de Rham in the topological direction is required by item (2) of Proposition 1.36, Dolbeault in the holomorphic direction by item (3). Item (2) fixes $Q = d_M + \bar{\partial}_Y + Q_0$: the first two summands are the differentials whose contraction homotopies build the topological and holomorphic localities, the third encodes the BV interaction. Item (3) imposes the classical master equation, the protected content of Theorem 1.37. Item (4) is the Costello–Gwilliam observable functor, the unique factorization assignment compatible with disjoint-union locality and the BV pairing. Replacing any item violates a named locality differential or a protected algebraic condition.

Remark 1.14 *(Quantum extension).* A full quantum mixed holomorphic-topological string datum is the corresponding $\hbar$-adic factorization algebra obtained from a renormalized BV action satisfying the quantum master equation on the canonical weighted-Tate completion B_w (Theorem 5.58). The componentwise quantum coefficient statement is weighted BV kernel convergence, the balanced scalar-contact zero projection on its completion, reduced weighted Moyal principal-part current brackets, and a degree-zero Hamiltonian trace comparison controlled by the radial/Weyl normal-form criterion. The finite normal-form computations close the balanced diagonal through $(6, 6)$, all non-edge bidegrees of total degree at most 13, and every total-degree-14 case with one component at most 6:

$$(3, 11), (11, 3), (4, 10), (10, 4), (5, 9), (9, 5), (6, 8), (8, 6).$$

The balanced $(7, 7)$ is the first undecided case in the exact kernel-correction window. Proposition F.32 closes the one-moving-letter edge strips $\mathfrak{o}_{2,b} = \mathfrak{o}_{a,2} = 0$. The all-interior statement is the obstruction problem $\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}$, whose positive form is the decorated PBW Stokes identity $D^{\square}_{a,b} = C^+_{a,b}\partial_2$; obstruction is exactly a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. The finite-window non-scalar closed-open BV criterion is the vanishing problem in $\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$ of Problem 5.220.

A full renormalized quantum mixed h-t string requires Costello graph data outside the local Hamiltonian theorem: the analytic graph-realization datum of Problem 5.213, the quantum master equation in the filtered local-functional complex $\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$ of Problem 5.220, and the all-bidegree radial vanishing $\Omega^{\mathrm{rad}}_{a,b} = 0$ of the preceding remark.

A pure topological sector is locally constant in all real directions; examples include the A-model local/Floer direction and the Chern–Simons-type open field theory on a topological brane. A pure holomorphic sector is Dolbeault in its complex directions and produces a holomorphic factorization algebra; Gwilliam–Williams' holomorphic bosonic string is a worldsheet example. A mixed holomorphic-topological sector lives on $M \times Y$ with locally constant factorization in M and holomorphic factorization in Y. The A/B terminology specialises this: the A-side gives the de Rham direction and the B-side gives the Dolbeault or formal holomorphic direction.

1.3 The local mixed model

The local geometric datum is

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}, \qquad \omega = dz_1 \wedge dz_2,$$

completed at L and replaced by its formal Darboux normal polydisk $(\widehat{\mathbb{C}^2}_0, \omega)$. Each factor is required.

The RG-stable completion. $\mathbb{R}_t$ is the brane worldline: the cyclic open BV theory and its trace observables $J(f)(t)$ live as currents on it. $\mathbb{R}_s$ is the topological normal/propagator direction along which BV ghosts move and the moment-map multiplier becomes a one-form. This transverse topological direction gives the propagator and the bulk-to-brane locality mechanism, since the mixed-locality homotopy $L_v = [Q, \iota_v]$ of Proposition 1.36 (b) requires a topological direction transverse to the worldline. $\mathbb{C}^2$ is the formal holomorphic-symplectic normal disk: two complex coordinates give the Hamiltonian bracket and a Darboux pair. Formal completion at L is dictated by the trace substitution: the brane carries only the formal holomorphic transverse geometry, because ϕ_1, ϕ_2 live in $\mathfrak{gl}^2_N$ and the substitution $z_i \mapsto \phi_i$ is formal substitution into a power series. Removing the formal completion forces the brane to see irrelevant global geometry and breaks the scalar-reduced trace identification of Lemma 3.1.

The Dirac brane is the formal-stalk chart, fixing the open BV sector and its first-class boundary action. Closed insertions act through central operations in Witten's source convention, fixing the closed-side target as a derived P_0-centre with reduced Hamiltonian model $Z^{P_0}_{\mathrm{red,Ham}}$ and dictionary $c^I \mapsto \theta^I, u_I \mapsto O_I$. Continuous duality is residue duality (Grothendieck), fixing the cotangent coefficient module as the Matlis principal-part module $\mathcal{P}$, separated from the polynomial PBW special fibre. Together with the three locality differentials of Proposition 1.36 – support, topological, formal holomorphic – these structures determine the constructed factorization-algebra model on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$: its dg Lie algebra is $\Omega^{\bullet}_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}(\mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1])$ on every product open $U \times B$, locally constant in U and holomorphic in B. Definition 1.15 records this object as $\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}$ with mixed enhancement $\mathcal{F}^{\mathrm{mix}}_{\mathrm{Ham}}$. Compact and global comparisons are governed by obstruction classes not contained in the formal-Darboux theorem.

Definition 1.15 *(Local Hamiltonian holomorphic factorization algebra).* The local Hamiltonian holomorphic factorization algebra for the formal-Darboux theorem, with Dirac's source convention, Grothendieck residue duality, and the three locality differentials of Proposition 1.36 on the formal Darboux polydisk, is

$$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}.$$

On a holomorphic polydisk $B \subset \mathbb{C}^2_{\text{hol}}$ its dg Lie algebra is

$$\Omega_c^{0,\bullet}(B)\widehat{\otimes}\mathfrak{g}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\text{cont}}[1], \qquad \mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1,$$

and the formal stalk of observables is the continuous Chevalley–Eilenberg cochain algebra $C^\bullet_{\text{CE,cont}}(\mathfrak{g})$. Its identification with Schouten polyvectors is the coordinate-window CE/PV theorem of §5.2; the strict-product P_0 structure requires the admissible completion stated there.

The mixed topological-current enhancement is

$$\mathcal{F}^{\text{mix}}_{\text{Ham}}(U \times B), \qquad \mathfrak{g}^{\text{mix}}_{U,B} = \Omega^\bullet_c(U)\widehat{\otimes}\Omega_c^{0,\bullet}(B)\widehat{\otimes}\mathfrak{g},$$

for $U \subset \mathbb{R}^2_{\text{top}}$. This object is locally constant in U and holomorphic in B, realising the topological and formal holomorphic locality differentials of Proposition 1.36.

The open quantum brane model is the Weyl/Moyal algebra of the formal symplectic polydisk, together with its finite-N matrix Weyl Hamiltonian reduction and matrix or supermatrix cyclic trace bar-cobar models. The support-local defect/cotangent model on an interval $I \subset \mathbb{R}_{\text{brane}}$ is the principal-part current algebra

$$\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}, \qquad \mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}, \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$

with reduced target

$$A^{\text{pp}}_\partial(I) = \widehat{\mathbf{S}}(\Omega^0_c(I)\widehat{\otimes}\bar{A})\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1]).$$

Its quantum version replaces the classical coadjoint action by the Moyal coadjoint action.

"Chiral" here means this two-complex-dimensional holomorphic factorization algebra and its mixed topological-current extensions. Its native singular product is the two-variable diagonal local-cohomology expansion of Theorem 1.17. A one-dimensional vertex operator algebra (VOA) / one-variable OPE statement requires a restriction theorem choosing a complex line or holomorphic defect, a transverse boundary condition or restriction functor, and the finite Laurent OPE expansion on that restricted object.

PROPOSITION 1.16 *(Constructed Hamiltonian CE/factorization observables).* In the formal local mixed model $\mathbb{R}^2_{\text{top}} \times \widehat{\mathbb{C}^2}_{0,\text{hol}}$, the Hamiltonian BF fields of Definition 1.12 reduce at a formal holomorphic insertion to the shifted cotangent dg Lie algebra

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\text{cont}}[1], \qquad \mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1.$$

For a holomorphic polydisk B, the constructed holomorphic observable algebra is represented by the continuous CE observables of the compactly supported Dolbeault dg Lie algebra

$$\mathcal{F}^{\text{hol}}_{\text{Ham}}(B) = C^\bullet_{\text{CE,cont}}\big(\Omega_c^{0,\bullet}(B)\widehat{\otimes}\mathfrak{g}\big).$$

For a product open $U \times B$, its mixed topological-current observable enhancement is represented by

$$\mathcal{F}^{\text{mix}}_{\text{Ham}}(U \times B) = C^\bullet_{\text{CE,cont}}\big(\Omega^\bullet_c(U)\widehat{\otimes}\Omega_c^{0,\bullet}(B)\widehat{\otimes}\mathfrak{g}\big).$$

These assignments are the local holomorphic observable algebras of the mixed holomorphic-topological BF theory. Their formal stalk is $C^\bullet_{\text{CE,cont}}(\mathfrak{g})$, and its identification with the PV model is only the coordinate/admissible theorem cited below.

Proof. Proposition 2.7 identifies the divergence-free holomorphic vector fields in the formal polydisk with Hamiltonians modulo constants. The Hamiltonian BF action then has connection field valued in $\mathfrak{h}[1]$ and cotangent field valued in $\mathfrak{h}^\vee_{\mathrm{cont}}[2]$. After the de Rham and Dolbeault Poincare contractions at the formal insertion point, Proposition 6.20 identifies the residual closed local operators with $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$, where the CE differential is dual to the semidirect bracket $[f,g]=\{f,g\}$, $[f,\eta_\lambda]=\eta_{\mathrm{ad}^\vee_f\lambda}$, and $[\eta_\lambda,\eta_\mu]=0$.

On a polydisk one keeps the compactly supported Dolbeault factor; on a mixed product open one also keeps the compactly supported de Rham factor. Extension by zero gives the factorization structure maps, while the local Poincare homotopies give holomorphic factorization in B and local constancy in U, exactly as in Definitions 1.9 and 1.12. The PV, P_0, kernel, and bar-cobar refinements are the admissible consequences of the coordinate CE/PV theorem and the reduced central-operation comparison. The native two-dimensional singular product is the diagonal local-cohomology theorem inserted next; the one-dimensional vertex/OPE structure requires the restriction data of Theorem 0.1. □

1.3.1 The formal two-dimensional holomorphic singular product

The singular product of the local model is a diagonal expansion on $\mathbb{C}^2\times\mathbb{C}^2$. A Laurent expansion on a curve arises after choosing a curve restriction. Let

$$\Delta\subset\mathbb{C}^2_z\times\mathbb{C}^2_w,\qquad \zeta_i=z_i-w_i,$$

and let $j\colon(\mathbb{C}^2\times\mathbb{C}^2)\setminus\Delta\hookrightarrow\mathbb{C}^2\times\mathbb{C}^2$ be the open complement. The singular coefficient module of a two-dimensional holomorphic singular product is the local-cohomology module

$$\mathcal{P}_\Delta=H^2_\Delta\big(\mathbb{C}[[z_1,z_2,w_1,w_2]]\big)\,d\zeta_1 d\zeta_2=\bigoplus_{a,b\geq 0}\mathbb{C}[[w_1,w_2]]\,\rho^\Delta_{a,b},$$

with

$$\rho^\Delta_{a,b}=\zeta_1^{-a-1}\zeta_2^{-b-1}\,d\zeta_1 d\zeta_2,\qquad \mathrm{Res}_\Delta\big(\rho^\Delta_{a,b}\,\zeta_1^m\zeta_2^n\big)=\delta_{a,m}\delta_{b,n}.$$

The two-dimensional Cauchy kernel is the generator

$$K^{(2)}_\Delta(z,w)=\rho^\Delta_{0,0}=\frac{d\zeta_1 d\zeta_2}{\zeta_1\zeta_2}.$$

In the region $|z_i|>|w_i|$ it has the formal expansion

$$K^{(2)}_\Delta(z,w)=\sum_{a,b\geq 0}z_1^{-a-1}z_2^{-b-1}w_1^a w_2^b\,dz_1 dz_2.$$

This is the two-variable local-cohomology expansion used below. It is the local-cohomology singular expansion of the surface theory. A one-dimensional chiral algebra on a curve $C\hookrightarrow\mathbb{C}^2$ requires a separate restriction theorem (Theorem 0.1) replacing $\mathcal{P}_\Delta$ by the one-variable principal parts of C, together with the pushforward bracket, BV pairing, brane image, and anomaly matching.

For $x\in\mathfrak{g}=\mathfrak{h}\ltimes\mathfrak{h}^\vee_{\mathrm{cont}}[1]$, write $\mathsf{J}_x(z)$ for the linear holomorphic current determined by x in the CE factorization algebra

$$B\longmapsto C^\bullet_{\mathrm{CE,cont}}\big(\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g}\big).$$

For $f\in\mathfrak{h}$ and $\rho\in\mathfrak{h}^\vee_{\mathrm{cont}}$ we also write

$$\mathsf{B}_f=\mathsf{J}_f,\qquad \Theta_\rho=\mathsf{J}_{\rho[1]}.$$

THEOREM 1.17 *(Formal two-dimensional holomorphic singular product).* The singular part of the formal holomorphic product of the Hamiltonian BF factorization algebra on $\widehat{\mathbb{C}^2}_0$ is

$$\mathsf{J}_x(z)\mathsf{J}_y(w) \sim K_\Delta^{(2)}(z,w)\,\mathsf{J}_{[x,y]_\mathfrak{g}}(w), \qquad x, y \in \mathfrak{g}.$$

Equivalently, for Hamiltonians $f, g \in \mathfrak{h}$ and principal parts $\rho, \sigma \in \mathfrak{h}^\vee_{\mathrm{cont}}$,

$$\begin{aligned}
&\mathsf{B}_f(z)\mathsf{B}_g(w) \sim K_\Delta^{(2)}(z,w)\,\mathsf{B}_{\{f,g\}}(w),\\
&\mathsf{B}_f(z)\Theta_\rho(w) \sim K_\Delta^{(2)}(z,w)\,\Theta_{f\cdot\rho}(w),\\
&\Theta_\rho(z)\Theta_\sigma(w) \sim 0,
\end{aligned}$$

where $(f\cdot\rho)(g) = -\rho(\{f,g\})$. The multi-index singular products are

$$\big(\mathsf{J}_x\big)_{(0,0)}\mathsf{J}_y = \mathsf{J}_{[x,y]_\mathfrak{g}}, \qquad \big(\mathsf{J}_x\big)_{(a,b)}\mathsf{J}_y = 0 \quad \text{for } (a,b) \neq (0,0),$$

with descendant products obtained by applying the holomorphic translation operators:

$$\partial_z^\alpha \mathsf{J}_x(z)\,\mathsf{J}_y(w) \sim (-1)^{|\alpha|}\alpha!\,\rho^\Delta_{\alpha_1,\alpha_2}(z,w)\,\mathsf{J}_{[x,y]_\mathfrak{g}}(w).$$

Proof. The CE factorization algebra is generated by the compactly supported Dolbeault local Lie algebra

$$\Omega_c^{0,\bullet}(B)\widehat{\otimes}\mathfrak{g}.$$

For disjoint polydisks the product is extension by zero. When two insertions collide, the only singular operation in the CE differential is the local Lie bracket $[x,y]_\mathfrak{g}$; this is the singular binary operation of the shifted-cotangent Lie algebra. The Dolbeault resolution of the diagonal identifies the singular quotient of the product of two holomorphic insertions with

$$H^2_\Delta(\mathcal{O}_{\mathbb{C}^2\times\mathbb{C}^2})\,d\zeta_1 d\zeta_2,$$

and the normalized generator is $K_\Delta^{(2)}$, since its residue on 1 is 1. Therefore the collision of the two currents is $K_\Delta^{(2)}\mathsf{J}_{[x,y]}$.

Substituting

$$[f,g] = \{f,g\}, \qquad [f,\rho[1]] = (f\cdot\rho)[1], \qquad [\rho[1],\sigma[1]] = 0$$

gives the three displayed singular products. The coefficient extraction formula follows from

$$\mathrm{Res}_\Delta(\rho^\Delta_{a,b}\zeta_1^m\zeta_2^n) = \delta_{a,m}\delta_{b,n}.$$

The descendant identity is

$$\partial_{z_1}^{\alpha_1}\partial_{z_2}^{\alpha_2}\frac{1}{\zeta_1\zeta_2} = (-1)^{|\alpha|}\alpha_1!\alpha_2!\,\zeta_1^{-\alpha_1-1}\zeta_2^{-\alpha_2-1}.$$

Associativity of all iterated two-dimensional singular products is the Jacobi identity in $\mathfrak{g}$, and the compatibility with Q is the Chevalley–Eilenberg identity $d^2_{\mathrm{CE}} = 0$. $\square$

Corollary 1.18 *(Binary generation of higher singular products).* The iterated n-point singular product in the strict CE current model is generated by Theorem 1.17, the regular commutative product, and the Chevalley–Eilenberg differential. All arity-n singular products with $n \geq 3$ are iterated binary collision products.

Proof. The coefficient algebra is the CE algebra of the strict dg Lie algebra $\Omega_c^{0,\bullet}(B)\widehat{\otimes}\mathfrak{g}$. Its only nonlinear structure map is the binary Lie bracket of $\mathfrak{g}$; the arity-n collision maps are therefore the iterated binary collision maps with the Koszul signs of the CE differential. Jacobi is the equality of the two binary collision orders along the small diagonal. This proves both associativity and the absence of independent higher singular coefficients. □

Corollary 1.19 *(Quantum and brane trace contact term).* After Moyal quantization of the Hamiltonian pair and scalar reduction on the brane,

$$\mathsf{B}_f(z)\mathsf{B}_g(w) \sim K_\Delta^{(2)}(z,w)\Big(\mathsf{B}_{\{f,g\}_\hbar}(w) + \hbar N\,\omega(f,g)\mathbf{1}\Big),$$

where

$$\{f,g\}_\hbar = \hbar^{-1}(f * g - g * f) \bmod \mathbb{C}\cdot 1, \qquad \omega(f,g) = [\{f,g\}]_0.$$

For $f = z_1$ and $g = z_2$,

$$\mathsf{B}_{z_1}(z)\mathsf{B}_{z_2}(w) \sim \hbar N\, K_\Delta^{(2)}(z,w)\mathbf{1}.$$

This is the Capelli scalar as a two-dimensional holomorphic contact term.

Proof. The degree-zero quantum brane algebra replaces the Poisson bracket by the Moyal commutator. The scalar-reduced Hamiltonian quotient removes the constant term on the closed side, while the Chan–Paton trace sends 1 to N. Hence the constant Taylor coefficient of the classical bracket contributes the central cocycle $\omega(f,g)$, and the quantum normalization multiplies it by $\hbar N$. For $(f,g) = (z_1,z_2)$, $\omega(z_1,z_2) = 1$, giving the displayed contact term. □

Remark 1.20 *(Physical reading).* The free BF contraction is

$$\alpha_f(z)\,\beta_\rho(w) \sim \langle \rho, f\rangle K_\Delta^{(2)}(z,w),$$

with α the Hamiltonian field and β the shifted cotangent field. The cubic interaction

$$I_{\mathrm{BF}} = \frac{1}{2}\int_{\mathbb{C}^2} \langle \beta, \{\alpha,\alpha\}\rangle\, dz_1 dz_2$$

turns the free contraction into the current singular product of Theorem 1.17. The quantum brane trace adds precisely the centre-of-mass scalar of Corollary 1.19.

1.3.2 The closed–open factorization triple: E_2 bulk, E_1 brane, derived-centre map

The local-string-theoretic content of the model is exhausted by three factorization objects and the morphism between them. All three are constructed from the same dg Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ of §2 together with the Dirac brane-stack data $\mathrm{Kosz}_N = \big(\mathbb{C}[\mathfrak{gl}_N^{\oplus 2}] \otimes \Lambda^\bullet \mathfrak{gl}_N[1],\ Q\psi = [\phi_1,\phi_2]\big)$ of Construction 3.20.

Definition 1.21 (Closed mixed-HT current factorization algebra). For a product open $O = U \times B \subset \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$, the compactly supported local dg Lie algebra is

$$\mathcal{L}_{\mathrm{Ham},c}(O) = \Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{g},$$

with differential $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2} + d_{\mathfrak{g}}$ and bracket $[\alpha \otimes x, \beta \otimes y] = (\alpha \wedge \beta) \otimes [x, y]_{\mathfrak{g}}$. The closed classical mixed-HT factorization algebra is its CE chain complex,

$$\mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}(O) = C^{\mathrm{CE}}_\bullet(\mathcal{L}_{\mathrm{Ham},c}(O)),$$

with factorization product induced by extension by zero $\bigoplus_i \mathcal{L}_{\mathrm{Ham},c}(O_i) \to \mathcal{L}_{\mathrm{Ham},c}(O)$ for disjoint $O_i \subset O$. The local operator algebra is the continuous dual of the formal current stalk after the de Rham–Dolbeault contraction and the residue pairing:

$$\mathrm{Obs}^{\mathrm{cl}}_b(\mathcal{F}_{\mathrm{Ham}}) := \mathrm{Hom}_{\mathrm{cont}}\big(C^{\mathrm{CE}}_\bullet(\mathcal{L}_{\mathrm{Ham},c,b}), \mathbb{C}\big) \simeq C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}).$$

This identification uses exact continuous duality in the admissible Tate envelope.

THEOREM 1.22 *(Mixed-HT factorization theorem).* The assignment $O \mapsto \mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}(O)$ is a prefactorization algebra on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ satisfying:

(i) (*Topological locality.*) For every $v \in T\mathbb{R}^2$, $L_v = [D, \iota_v]$ on $\mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}$; hence the assignment $U \mapsto \mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}(U \times V)$ is locally constant in the topological directions for every fixed holomorphic open V.

(ii) (*Holomorphic locality.*) $L_{\partial/\partial\bar{z}_i} = [D, \iota_{\partial/\partial\bar{z}_i}]$ on $\mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}$; hence the assignment $V \mapsto \mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}(U \times V)$ is a holomorphic factorization algebra on $\mathbb{C}^2$ via Dolbeault descent.

(iii) (*Factorization product.*) For pairwise disjoint $O_1, \dots, O_k \subset O$, $\otimes_i \mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}(O_i) \to \mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}(O)$ is a chain map and Weiss-cosheaf descent over hypercovers holds.

Equivalently, $\mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}} \in \mathrm{Fact}_{\mathrm{HT}}(\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}})$ in the sense of Definition 1.12.

Proof. Topological locality. Cartan's homotopy $[d_{\mathbb{R}^2}, \iota_v] = L_v$ on $\Omega^\bullet_c(U)$ lifts to $[D, \iota_v] = L_v$ on $\mathcal{L}_{\mathrm{Ham},c}(O)$, since ι_v commutes with $\bar{\partial}_{\mathbb{C}^2}$ and $d_{\mathfrak{g}}$. The CE functor carries this homotopy to $\mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}$. *Holomorphic locality.* Identical argument with ι_v replaced by $\iota_{\partial/\partial\bar{z}_i}$ and $d_{\mathbb{R}^2}$ by $\bar{\partial}_{\mathbb{C}^2}$. *Factorization product.* Extension-by-zero $\Omega^\bullet_c(O_i) \hookrightarrow \Omega^\bullet_c(O)$ on each factor of $\mathcal{L}_{\mathrm{Ham},c}$ is a dg Lie injection for disjoint supports; CE chains of a coproduct of dg Lie algebras is the tensor product of CE chains, giving the factorization product. The Weiss-cosheaf condition is the standard de Rham/Dolbeault descent for compactly supported forms. □

COROLLARY 1.23 *(E_2-bulk algebra).* Fix a holomorphic polydisk $B \subset \mathbb{C}^2$ and a topological disk $D^2 \subset \mathbb{R}^2$. The bulk classical algebra

$$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}(B) = \mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}(D^2 \times B)$$

is an E_2-algebra under the Lurie–Costello–Gwilliam locally constant factorization structure on $\mathbb{R}^2$; the assignment $B \mapsto \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}(B)$ is a holomorphic factorization algebra on $\mathbb{C}^2$ valued in E_2-algebras.

Proof. Theorem 1.22 (1) says that $U \mapsto \mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}(U \times B)$ is a locally constant factorization algebra on $\mathbb{R}^2$; the value on a topological disk is therefore an E_2-algebra by the Lurie–Costello–Gwilliam classification of locally constant factorization algebras on $\mathbb{R}^n$. Holomorphic factorization in B is item (2) of the same theorem. □

Definition 1.24 *(E_1-brane algebra).* Let Kosz_N be the Dirac BV complex of Construction 3.20. The classical brane factorization algebra on $L = \mathbb{R}_t$ assigns to an interval $I \subset \mathbb{R}_t$ the CE complex

$$\mathrm{Obs}^{\mathrm{cl}}_{\partial,N}(I) = C^\bullet_{\mathrm{CE,cont}}\big(\Omega^\bullet_c(I) \otimes \mathfrak{gl}_N,\ \Omega^\bullet_c(I) \otimes \mathrm{Kosz}_N\big),$$

with factorization product extension by zero on $\Omega^\bullet_c$ and multiplication of Kosz_N-valued cochains. Local constancy on $\mathbb{R}_t$ makes the value on a small interval an E_1-algebra:

$$A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}\big(\mathfrak{gl}_N,\ \mathrm{Kosz}_N\big) \in \mathrm{Alg}_{E_1}.$$

Its ghost-zero Q-cohomology is the stable trace ring of Theorem 5.42: $H^0(A^{\mathrm{cl}}_{\partial,N}) \simeq H^0(\mathcal{O}(\mu^{-1}_{\mathrm{der}}(0))^{GL_N}, Q)$.

A polynomial Hamiltonian $f \in \bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ acts on the underived commuting representation functor through a Schouten vector field θ^0_f, dual on the open side to the trace character map $u_f \mapsto J_f$. For $f \in \mathfrak{h}$, the formula extends only after the brane coordinate algebra is completed at the formal Darboux point. A lift to the Koszul derived zero fibre is a choice of homotopy on the moment-map equation.

LEMMA 1.25 *(Hamiltonian vector field on the representation stack).* For $f \in \bar{A}$, and continuously for $f \in \mathfrak{h} = A/\mathbb{C}\cdot 1$ after formal completion at the Darboux point, the assignment

$$\theta_f \phi_1 = -\partial_{z_2} f\big|_{(\phi_1,\phi_2)}, \qquad \theta_f \phi_2 = +\partial_{z_1} f\big|_{(\phi_1,\phi_2)}$$

defines a canonical derivation $\theta^0_f \in \mathrm{Der}\big(\mathcal{O}(\mathrm{Rep}^\wedge_N(A))\big)$ of the underived commuting formal representation functor. A Q-commuting derivation of the Koszul cdga $\mathcal{O}(\mu^{-1}_{\mathrm{der}}(0))$ lifting θ^0_f consists of a noncommutative lift $\widetilde{\theta}_f$ and an odd component $h_f = \widetilde{\theta}_f \psi \in \mathfrak{gl}_N[1]$ satisfying

$$Qh_f = \widetilde{\theta}_f\big([\phi_1, \phi_2]\big) = [\widetilde{\theta}_f \phi_1, \phi_2] + [\phi_1, \widetilde{\theta}_f \phi_2].$$

The pair $(\widetilde{\theta}_f, h_f)$ is determined only up to a Q-closed vertical derivation. Its action on H^0 trace characters is independent of that choice and is

$$\theta^0_f(J_g) = J_{\{f,g\}_{\bar{A}}} \qquad (f, g \in \mathfrak{h}),$$

with $\{-,-\}_{\bar{A}}$ the Hamiltonian Poisson bracket on $\mathfrak{h}$.

Proof. The Hamiltonian derivation $\delta_f(z_1) = -\partial_{z_2} f$, $\delta_f(z_2) = \partial_{z_1} f$ of the commutative algebra A induces θ^0_f by functoriality of the representation functor of A. Since $\delta_f(g) = \{f, g\}_{\bar{A}}$, cyclicity of trace gives $\theta^0_f(J_g) = J_{\{f,g\}_{\bar{A}}}$.

The derived coordinate algebra is the Koszul resolution of the ideal generated by $\mu = [\phi_1, \phi_2]$. A derivation of the quotient lifts to this Koszul cdga after choosing a null-homotopy for its action on μ. The displayed equation is precisely the condition $[Q, \widetilde{\theta}_f]\psi = 0$. Adding a Q-closed vertical term to h_f changes the lift, but not the induced derivation of the underived quotient or the displayed trace action on H^0. □

THEOREM 1.26 *(Trace-sector action and CE/PV reduction).* There is a generator-level trace-sector action by Hochschild cochains

$$\Phi_N : \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk},0} \dashrightarrow C^\bullet(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}), \qquad Z^{\mathrm{der}}_{E_1}(A^{\mathrm{cl}}_{\partial,N}) \simeq HH^\bullet(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}),$$

whose value on Hamiltonian CE generators is the explicit action

$$c_f \longmapsto \theta_f \in C^1(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}), \quad u_f \longmapsto O_f = J_f \cdot (\,\cdot\,) \in C^0(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}) \quad (f \in \bar{A}),$$

where $J_f = J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ acts by multiplication and θ_f is a chosen Koszul lift of the Hamiltonian vector field of Lemma 1.25. For formal $f \in \mathfrak{h}$, the assignment is defined in the completed brane trace algebra. The displayed target is the Hochschild target of Deligne's E_1-theorem; it is not an assertion that the bulk is equivalent to the whole derived centre, and it is not by itself a cochain map from Chevalley–Eilenberg cochains to ordinary Hochschild cochains. The proved dg identity is the Chevalley–Eilenberg/polyvector identity of Theorem 4.20 on $c^{a,b}$, $u_{a,b}$, with Schouten differential $d_\pi = [\pi, -]_S$. If compatible Koszul lifts $(\widetilde{\theta}_f, h_f)$ and Hochschild equivariance homotopies are fixed, the assignment extends to the trace Hochschild subcomplex generated by $J(f)$ and θ_f. The full chain-level lift to the strict factorization-centre morphism $\Phi^{\mathrm{cl}} \colon \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}|_L \to Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N})$ is governed by the four-component obstruction vector $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$ of Theorem E.53. The CE/PV coordinate map $c_f \mapsto \theta_f$, $u_f \mapsto O_f$ is the dg coordinate identity in the reduced Hamiltonian P_0 model; its Hochschild image is a generator action. A Hochschild-cochain map Φ_N, and hence the chain-level P_0-centre map Φ^{cl}, exists only after compatible null-homotopies of $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$ have been chosen.

Proof. The boundary algebra $A^{\mathrm{cl}}_{\partial,N}$ is an E_1-algebra, and the bulk algebra $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk},0}$ is an E_2-algebra of Chevalley–Eilenberg coordinates on $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$. The Hochschild complex $C^\bullet(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ carries the Gerstenhaber bracket of degree -1; its cohomology is the derived E_1-centre, with HH^0 the centre (multiplications) and HH^1 the outer derivations.

The assignment is defined on CE generators as Hochschild cochains. For $f \in \bar{A}$, the Hamiltonian-dual coordinate c_f is sent to the Hamiltonian vector field $\theta_f \in C^1(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ of Lemma 1.25 (the classical brane algebra is graded-commutative with odd antifield generator ψ, and a Koszul lift is part of the chosen datum); the cotangent-dual coordinate u_f is sent to the Hochschild 0-cochain $J_f \cdot (\,\cdot\,) \in C^0(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$, multiplication by the trace character $J_f = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$. For $f \in \mathfrak{h}$ the same formula has meaning only in the completed formal brane algebra. The action identities in Hochschild cochains are

$$[\theta_f, \theta_g]_G = \theta_{\{f,g\}_{\bar{A}}}, \qquad \theta_f(J_g) = J_{\{f,g\}_{\bar{A}}},$$

by Lemma 1.25 and Proposition 5.171. They express the Hamiltonian action on open observables. The Chevalley–Eilenberg differential itself is matched in the reduced Hamiltonian P_0 model:

$$d_\pi \theta^K = -\tfrac{1}{2} C^K_{IJ} \theta^I \theta^J, \qquad d_\pi O_K = C^L_{KJ} O_L \theta^J,$$

which is Theorem 4.20. Thus $c \mapsto \theta$ and $u \mapsto O$ are a dg substitution in (PV, d_π), while their Hochschild realisation is an action datum. It becomes a Hochschild cochain map only after the Koszul lifts satisfy $[Q, \theta_f] = 0$ and the centre-lift homotopies have been supplied. Continuous extension across the graded-commutative coordinate algebra is Theorem 4.20. Deligne's E_1-Hochschild theorem identifies the target $Z^{\mathrm{der}}_{E_1}(A^{\mathrm{cl}}_{\partial,N})$ with the Hochschild cochain E_2-algebra $C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$, whose cohomology is $HH^\bullet(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$. The displayed generator formulas define the Hochschild trace action. The Costello–Gwilliam factorization-algebra lift is the strict centre-map problem governed by $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$. Uniqueness of the boundary values $u_f \mapsto J(f)$ is Theorem 5.42.

An explicit Hochschild-cochain model for Φ_N at every cochain degree is the lift Φ^{cl} to the strict P_0-factorization-centre, governed by the obstruction $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$ of Theorem E.53; the curvature class is the difference between the derived E_1-centre E_2-structure of Deligne's theorem and the strict P_0-factorization-centre. □

Remark 1.27 *(Native bulk object on $\mathbb{C}^2$).* The native object is the mixed holomorphic-topological factorization algebra $\mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}$ on $X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$. The bulk $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} = \pi_* \mathcal{F}^{\mathrm{cl}}_{\mathrm{Ham}}$ is its pushforward along $\pi : X \to \mathbb{C}^2_{\mathrm{hol}}$, the holomorphic factorization algebra $B \mapsto \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}(B)$ on $\mathbb{C}^2_{\mathrm{hol}}$ valued in E_2-algebras — the E_2-valuation encoding the locally constant $\mathbb{R}^2_{\mathrm{top}}$ factor by the Lurie–Costello–Gwilliam classification. A one-dimensional vertex algebra is a derived object, recovered after a restriction datum—a holomorphic curve $C \hookrightarrow \mathbb{C}^2$ with transverse boundary condition, or a factorization-homology pushforward—which singles out a one-complex-dimensional holomorphic factorization restriction on which Zhu / finite-Laurent OPE applies. Before any such restriction, the native singular product is two-complex-dimensional: Theorem 1.17 identifies the diagonal local-cohomology coefficient module with $H^2_\Delta(\mathbb{C}[[z,w]])\, d\zeta_1 d\zeta_2$ and computes

$$\mathsf{J}_x(z)\mathsf{J}_y(w) \sim K^{(2)}_\Delta(z,w)\mathsf{J}_{[x,y]_{\mathfrak{g}}}(w).$$

Proposition 1.28 *(Bochner–Martinelli current data and finite-window obstruction for E_2 transfer).* Let $B_0 \Subset B_1 \Subset B_2 \Subset B \subset \mathbb{C}^2$ be nested polydisks around the collision point and choose $\chi \in C^\infty_c(B_2)$ with $\chi = 1$ on B_1. Put

$$\mathcal{P} = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

and

$$\mathcal{P}_{\leq N} = \bigoplus_{0<a+b\leq N} \mathbb{C}\rho_{a,b}.$$

The coordinate window $\mathfrak{m}/\mathfrak{m}^{N+1}$, dual to $\mathcal{P}_{\leq N}$, is the Hamiltonian counterpart of the retained principal parts; the full Hamiltonian Lie bracket requires the completed algebra $\mathfrak{h}$.

On $(B_\zeta \times B_z) \setminus \Delta$, set

$$\eta = \zeta - z, \qquad r^2 = |\eta_1|^2 + |\eta_2|^2, \qquad d\bar\eta_j = d\bar\zeta_j - d\bar z_j.$$

The Dolbeault homotopy kernel is the Bochner–Martinelli–Koppelman current [37, Ch. III, Def. 1.2, Lem. 1.3]

$$K_{\mathrm{BM}}(\zeta,z) = \frac{1}{(2\pi i)^2} \frac{\overline{\eta_1}\, d\bar\eta_2 - \overline{\eta_2}\, d\bar\eta_1}{r^4} \wedge d\zeta_1 \wedge d\zeta_2.$$

The several-complex-variables sources are used here only for this pre-cutoff kernel, its component $\bar\partial$-identity away from the diagonal, and the classical Koppelman homotopy formula on bounded C^1 domains for C^1 forms [38, §5, Def. 5.1, Thm. 5.2, Eq. (10)]. The cutoff-current limit H_χ, the annular diagonal-current sign, the finite moment projection p_N, the residue-current section i_N, the analytic/pro-Matlis quotient, the Hamiltonian/Matlis arity-two action, and the Costello/QME or factorization transfer below are additional constructions. The $d\bar z_j$-terms in $d\bar\eta_j$ are part of the kernel; the $d\bar z = 0$ component is only the scalar integral kernel. With a radial cutoff $\vartheta_\epsilon(r)$, equal to 0 for $r \leq \epsilon$ and to 1 for $r \geq 2\epsilon$, put

$$K_{\chi,\epsilon} = \chi(\zeta)\chi(z)\vartheta_\epsilon(|\zeta - z|)K_{\mathrm{BM}}(\zeta,z).$$

The current limit

$$H_\chi \alpha(z) = \lim_{\epsilon\to 0} \int_{\zeta\in B} K_{\chi,\epsilon}(\zeta,z) \wedge \alpha(\zeta)$$

is well-defined on each fixed finite configuration stratum. The singularity is locally integrable in real dimension four ($|K_{\mathrm{BM}}| \sim r^{-3}$), and for separated compact supports its derivatives obey estimates

$$|\nabla^k K_{\mathrm{BM}}(\zeta, z)| \le C_k |\zeta - z|^{-3-k}.$$

Let $i_N(\rho_{a,b})$ be the support-at-zero current

$$\Delta_{a,b} = \frac{(-1)^{a+b}}{a!\,b!}\partial_{z_1}^a \partial_{z_2}^b \delta_0\, d\bar z_1 \wedge d\bar z_2, \qquad \langle \Delta_{a,b}, f\rangle = \mathrm{Res}_0(\rho_{a,b} f),$$

and define the finite principal-part moment projection on top Dolbeault degree by

$$p_N(\alpha) = \sum_{0<a+b\le N} \left(\int_B z_1^a z_2^b\, \alpha^{0,2} \wedge dz_1 \wedge dz_2 \right) \rho_{a,b}.$$

The data above construct the current-limit operator H_χ, the finite moment map p_N, and the residue-current section i_N. A finite-window homotopy $H_{\chi,N}$ on the full compact-support/current complex additionally requires the identity

$$\bar\partial H_{\chi,N} + H_{\chi,N}\bar\partial = \mathrm{id} - i_N p_N + E_{\chi,N} \tag{$*_N$}$$

is blocked, in the full-current completion, by the finite-window obstruction vector

$$\mathrm{Ob}_N^{\text{BMK-curr}} = \left(\mathrm{ob}_{\mathrm{diag}}, \mathrm{ob}_{\mathrm{an/fin},N}, \mathrm{ob}_{\mathfrak{h},N}, \mathrm{ob}_{\text{collar-fact},N} \right).$$

Here $\mathrm{ob}_{\mathrm{diag}}$ is the sign and scalar comparison between the annular diagonal current $\bar\partial\vartheta_\epsilon(|\zeta - z|) \wedge K_{\mathrm{BM}}$ and the Matlis derivative-delta convention; with the displayed kernel and real outward boundary orientation its angular mass is -1, so a BMK diagonal-current convention must be declared before the sign in $i_N p_N$ is fixed. The component $\mathrm{ob}_{\mathrm{an/fin},N}$ is the quotient between the full analytic moment kernel $\ker(\mathcal{O}(B_1)^\vee \to \mathcal{P}_{\le N})$ and the smaller support-at-zero span generated by $\Delta_{0,0}$ and the $\Delta_{a,b}$ with $a + b > N$. The component $\mathrm{ob}_{\mathfrak{h},N}$ is the Hamiltonian-invariance defect

$$\pi_N(\mathfrak{h} \cdot \ker \pi_N),$$

already nonzero because

$$z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1},$$

while $\rho_{N+1,0}$ is killed by the window and $\rho_{0,1}$ is retained. Finally $\mathrm{ob}_{\text{collar-fact},N}$ is the still-open obstruction, away from one nested pair, to simultaneous preservation of the same relation subspace by the collar quotient, extension-by-zero maps, factorization products, and the BMK homotopy.

Consequently strict finite support-local current quotient retaining $\mathcal{P}_{\le N}$, killing the scalar and high support-at-zero currents, and carrying the full Hamiltonian factorization action is obstructed. The projected finite-window L_∞ extension is obstructed as well: the forced return block already has

$$\kappa_N(z_1^{N+2}, \rho_{N+1,0}) = -(N+2)\rho_{0,1},$$

and the projected Jacobiator on $(z_1^{N+2}, z_2, \rho_{N,0})$ is

$$(N+1)(N+2)\rho_{0,1}.$$

Since the escape-return image spans the retained finite Matlis window, the unary-boundary closure of these returns coincides with the full collapse of $\mathcal{P}_{\leq N}$ onto the trivial module; every nontrivial finite Matlis window therefore retains the Bochner–Martinelli leak. A one-pair derived analytic quotient does give a differential identity, after the full analytic BMK contraction and diagonal orientation are fixed: quotient the relative current complex by representatives of $\ker(\mathcal{O}(B_1)^\vee \to \mathcal{P}_{\leq N})$, normalize the homotopy by

$$(1 - I_{\mathrm{an}} P_{\mathrm{an}}) H_\chi (1 - I_{\mathrm{an}} P_{\mathrm{an}}),$$

and then

$$\bar{\partial} H_{\mathrm{BM},N} + H_{\mathrm{BM},N} \bar{\partial} = \mathrm{id} - i_N p_N.$$

This quotient is analytic and one-pair relative. The source theorem used here is only the pre-cutoff Koppelman identity; the representative quotient, the normalized contraction, and the compatible pro-Matlis moment map are local constructions of this proposition.

Keeping the entire Taylor tower gives the positive one-pair pro-Matlis replacement. Let

$$\widehat{\mathcal{P}}_\Pi = \prod_{\substack{a,b \geq 0 \\ a+b>0}} \mathbb{C}\rho_{a,b}, \qquad \mathcal{P}_{\mathrm{an}}(B_1) = \tau\big(\mathcal{O}(B_1)^\vee_{\mathrm{red}}\big) \subset \widehat{\mathcal{P}}_\Pi,$$

where

$$\tau(\lambda) = \big(\lambda(z_1^a z_2^b)\big)_{a+b>0}.$$

On the single-pair analytic BMK completion, after the diagonal orientation is fixed and the collar error is quotiented, the normalized contraction $H_\chi^\circ = (1 - I_{\mathrm{an}} P_{\mathrm{an}}) H_\chi (1 - I_{\mathrm{an}} P_{\mathrm{an}})$ satisfies

$$\bar{\partial} H_\chi^\circ + H_\chi^\circ \bar{\partial} = \mathrm{id} - I_{\mathrm{an}} P_{\mathrm{an}}.$$

Hence the pro-analytic moment map $P_\Pi^{\mathrm{an}} = \tau P_{\mathrm{an}}$ has compatible finite projections

$$\pi_N P_\Pi^{\mathrm{an}} = p_N, \qquad \pi_{M,N} p_M = p_N.$$

This is a one-pair analytic pro-Matlis retract.

The binary operation

$$\ell_2^{\mathrm{BM},N}(x, y) = p_N[ix, iy]$$

is therefore retained only as an arity-two Hamiltonian/Matlis coefficient comparison induced by p_N and i_N. A finite-window homotopy requires an additional construction. It agrees, coefficient by coefficient and in every fixed finite window, with the Hamiltonian semidirect bracket. Thus for Hamiltonians $f, g \in \mathfrak{h}$,

$$\ell_2^{\mathrm{BM},N}(f, g) = \mathrm{pr}_{\leq N}\big(\partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g\big) \pmod{\mathbb{C}},$$

while for $f = z_1^p z_2^q$ and $\rho_{i,j} = z_1^{-i-1} z_2^{-j-1} dz_1 dz_2$,

$$\ell_2^{\mathrm{BM},N}(f, \rho_{i,j}) = -(pj - qi + p - q)\, \mathrm{pr}_{\leq N}\, \rho_{i-p+1,j-q+1},$$

with negative-index terms equal to zero, and

$$\ell_2^{\mathrm{BM},N}(\rho, \sigma) = 0.$$

This is the Matlis coadjoint action of Theorem 5.179.

Only after $\mathrm{Ob}_N^{\text{BMK-curr}}$ is killed, or after the one-pair analytic quotient is replaced by a factorization-current completion, does the first possible obstruction to a strict all-arity E_2 collision operation become the ternary cutoff term

$$\mathrm{Ob}_3^{\chi}(x,y,z) = p[H_{\mathrm{BM}}([ix,iy] - i\ell_2^{\mathrm{BM}}(x,y)), iz] + \text{cyclic Koszul terms}.$$

It vanishes on separated support strata by the estimates above, and its coefficient Jacobiator on the collision stratum is the Jacobiator of $\mathfrak{h} \ltimes \mathcal{P}[1]$, hence zero. Any surviving term is therefore a cutoff/collar contribution supported on the small diagonal. The attempted strict all-window current E_2 transfer is controlled by the six-entry obstruction vector

$$\mathrm{Ob}_{\mathrm{BM}}^{\Pi} = \left(\mathrm{ob}_{\oplus}, \mathrm{ob}_{I_\Pi}, \mathrm{ob}_{\mathfrak{h},\Pi}, \mathrm{ob}_{\mathrm{collar},\Pi}, \mathrm{ob}_{3,\Pi}, \mathrm{ob}_{\mathrm{unif},\Pi}\right).$$

Here $\mathrm{ob}_{\oplus}$ is the nonzero class of the all-window moment sequence in $\widehat{\mathcal{P}}_\Pi/\mathcal{P}$; ob_{I_Π} is the nonexistence of a compact-current section $i_\Pi : \widehat{\mathcal{P}}_\Pi \to \mathcal{D}_0'^{0,2}(B)$ extending the finite derivative-delta sections; and $\mathrm{ob}_{\mathfrak{h},\Pi}$ is the obstruction to the full formal Hamiltonian action on the pro-product module, detected by the divergent $\rho_{0,1}$-coefficient for $f = \sum_{p\geq 2} z_1^p$ acting on $\lambda = \sum_{p\geq 2} \rho_{p-1,0}$. The remaining entries are the compatible collar primitive class, the first ternary small-diagonal transfer class, and the all-window uniform estimate class. The first, second, third, and sixth entries are nonzero in the current topology; the collar and ternary entries vanish after the corresponding primitive classes are constructed.

The principal-part face of the transfer is genuinely two-variable:

$$\frac{1}{(2\pi i)^2}\frac{d\zeta_1 \wedge d\zeta_2}{(\zeta_1 - z_1)(\zeta_2 - z_2)} = \sum_{a,b\geq 0} z_1^a z_2^b \rho_{a,b}(\zeta).$$

Killing z_2 kills the Hamiltonian bracket, and the $b=0$ principal-part line is unstable because $z_1 \cdot \rho_{a,0} = -\rho_{a,1}$.

Proof. The Koppelman identity for K_{BM} gives the smooth-domain, pre-cutoff Dolbeault homotopy [38, §5, Thm. 5.2]. The cutoff derivative, annular diagonal current, finite moments, and Matlis orientation below are checked locally. After multiplication by the cutoff, the derivative controlling the boundary term is

$$\bar\partial(\chi(\zeta)\chi(z)\vartheta_\epsilon K_{\mathrm{BM}}) = \chi\chi\,\vartheta_\epsilon\,\bar\partial K_{\mathrm{BM}} + \bar\partial(\chi\chi)\,\vartheta_\epsilon K_{\mathrm{BM}} + \chi\chi\,\bar\partial\vartheta_\epsilon \wedge K_{\mathrm{BM}}.$$

The second summand is collar-supported. The third summand is an annular diagonal current; its sign and scalar are the first entry of $\mathrm{Ob}_N^{\text{BMK-curr}}$. Applying finite moments gives the formula for p_N. The normalization of i_N follows by integration by parts against a holomorphic test function, giving the residue functional represented by $\rho_{a,b}$, once the diagonal-current orientation is tied to the Matlis convention.

The full-current finite-window identity has a Stokes obstruction on the current complex. Such an identity makes every closed top current whose analytic Dolbeault cohomology class lies in $\ker \pi_N$ exact after quotienting by the finite selected moments, while pairing with a holomorphic function detects that class. In particular, $T = \Delta_{N+1,0}$ has $p_N(T) = 0$, but $\langle T, z_1^{N+1}\rangle = 1$, whereas $\langle \bar\partial S, z_1^{N+1}\rangle = 0$ for every current S. The scalar current $\Delta_{0,0}$ gives the same obstruction because $\rho_{0,0}$ is excluded from the reduced principal-part module.

The stronger support-local quotient obstruction is Hamiltonian invariance. A quotient relation must be an $\mathfrak{h}$-submodule relation. The Matlis coadjoint formula gives $z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}$; hence killing the high mode $\rho_{N+1,0}$ while retaining the finite mode $\rho_{0,1}$ is incompatible with the full Hamiltonian action: high transverse holomorphic modes leak back into low brane-momentum modes. This proves the nonexistence of the strict finite support-local quotient stated above; the all-window form is Theorem 8.40, with the pro-Matlis retract as the target replacement representing this leak (Corollary 8.41).

The one-pair analytic quotient is formal once the full analytic BMK contraction exists locally: the normalized homotopy annihilates the chosen representatives of $\ker \pi_N$, so it descends and the analytic projector becomes $i_N p_N$ in the quotient. On the unquotiented Taylor tail, the same normalized contraction gives $\bar\partial H^{\circ}_{\chi} + H^{\circ}_{\chi}\bar\partial = \mathrm{id} - I_{\mathrm{an}} P_{\mathrm{an}}$, and Taylor moments give $P^{\mathrm{an}}_{\Pi} = \tau P_{\mathrm{an}}$ with $\pi_N P^{\mathrm{an}}_{\Pi} = p_N$. Because this pro-Matlis retract depends on the analytic moment space, a representative map, and a chosen polydisk pair, a factorization-current theorem requires additional support-local data.

The arity-two formula is the residue transpose calculation. On holomorphic coefficient labels this is the Poisson bracket modulo constants, projected after the coefficient is computed. On cotangent labels the residue identity

$$\mathrm{Res}_0((f \cdot \rho)g) = -\mathrm{Res}_0(\rho\{f, g\})$$

gives exactly the principal-part coadjoint formula already proved in Theorem 5.179. The cotangent summand is abelian, so the (ρ, σ)-bracket is zero.

The displayed ternary expression is therefore only the next obstruction after the finite-current condition is crossed. Away from collisions the kernel is smooth and the separated-support estimates force it to vanish in the factorization product. On the coefficient collision stratum the semidirect Lie algebra satisfies Jacobi. The remaining work is the collar, diagonal, and completed all-window transfer data. □

Proposition 1.29 *(Darboux polydisk constructions).* For

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\},$$

with holomorphic Darboux form $dz_1 \wedge dz_2$, the local theorem has the following five constructions.

(i) The closed Hamiltonian BF coefficient algebra is

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1, \qquad P = \mathfrak{h}^{\vee}_{\mathrm{cont}}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes P[1].$$

The quotient by constants comes from the kernel of the Hamiltonian vector-field map.

(ii) The finite-N open chart is the Dirac/Koszul matrix model

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0,$$

obtained from the first-order action $\int_{\mathbb{R}_t} \mathrm{Tr}(\phi_1\, d\phi_2 + A[\phi_1, \phi_2])$. Thus the open algebra is the derived zero-fibre of the conjugation moment map, with commutativity imposed homologically.

(iii) After killing the scalar Hamiltonian and the empty trace, the reduced CE/PV central-operation comparison is the coordinate map

$$\Phi_{\mathrm{coord}}(c^I) = \theta^I, \qquad \Phi_{\mathrm{coord}}(u_I) = O_I.$$

The c-coordinate gives the Schouten vector-field operation; the u-coordinate gives the boundary Hamiltonian observable.

(iv) The open one-ψ classes are PBW special-fibre labels. The deformation-level cotangent dual is P, realized by Matlis principal parts and acted on by the coadjoint action. The PBW/Rees label bridge is extra Fourier-polarized structure relating two label systems with distinct $\mathfrak{h}$-actions.

(v) The Bochner–Martinelli–Koppelman current limit, finite Matlis moment maps, arity-two coefficient comparison, and one-pair analytic pro-Matlis retract $P_\Pi^{\mathrm{an}} = \tau P_{\mathrm{an}}$, $\pi_N P_\Pi^{\mathrm{an}} = p_N$, are constructed in Proposition 1.28. A strict support-local finite-window current quotient is obstructed by $\mathrm{Ob}_N^{\text{BMK-curr}}$. A strict all-arity E_2 transfer requires first crossing this finite-current condition and then solving the six-entry all-window obstruction problem $\mathrm{Ob}_{\mathrm{BM}}^{\Pi}$.

Proof. The formal Darboux polydisk is fixed in the opening of this section. Item (1) is Proposition 2.7 and Proposition 1.16. Item (2) is the constrained-mechanics computation of Lemma 3.15, Lemma 3.14, and Proposition 3.16. Item (3) is Theorem 4.20, with the scalar convention of Remark 2.3 and Theorems 6.3–6.6. Item (4) is Theorem 5.181 together with the principal-part uniqueness theorem 5.179. Item (5) is Proposition 1.28. □

Remark 1.30 *(Formal stalk restriction).* The Hamiltonian identification is formal local. On a holomorphic symplectic surface X, a holomorphic vector field v is symplectic when

$$\alpha_v = -\iota_v \omega$$

is a closed holomorphic one-form, and Hamiltonian only after choosing f with $df = \alpha_v$, in the convention $\iota_{X_f}\omega = -df$. A global Hamiltonian sector requires a sheaf of local Hamiltonians and the vanishing of the corresponding holomorphic de Rham obstruction. The holomorphic Poincare lemma makes this obstruction vanish on the formal polydisk.

PROPOSITION 1.31 *(Formal-stalk Hamiltonian restriction).* Let (X, ω) be a connected holomorphic symplectic surface and let $p \in X$. Put

$$\mathcal{H}_X = \mathcal{O}_X/\mathbb{C}_X, \qquad \widehat{\mathfrak{h}}_{X,p} = \widehat{\mathcal{O}}_{X,p}/\mathbb{C}.$$

The differential identifies $\mathcal{H}_X$ with the sheaf of closed holomorphic one-forms, hence with the sheaf of holomorphic symplectic vector fields by $\alpha = -\iota_v\omega$. For every $p \in X$ there is a Lie algebra restriction

$$H^0(X, \mathcal{H}_X) \longrightarrow \widehat{\mathfrak{h}}_{X,p}.$$

After choosing a formal Darboux coordinate system, the formal neighbourhood $(\widehat{X}_p, \omega)$ has the same completed Hamiltonian Lie algebra as the formal polydisk used in the local model. The resulting stalk-level boundary map

$$\widehat{\mathfrak{h}}_{X,p} \longrightarrow H^0_{\mathrm{Ham}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{brane}})$$

is independent of this choice up to the natural action of the formal symplectomorphism group. The subspace coming from global Hamiltonian functions is the image of

$$H^0(X, \mathcal{O}_X)/\mathbb{C} \longrightarrow H^0(X, \mathcal{H}_X).$$

The obstruction for a locally Hamiltonian symplectic vector field to come from a global Hamiltonian function is its period class in $H^1(X, \mathbb{C}_X)$.

Proof. The holomorphic Poincare lemma gives an isomorphism

$$d : \mathcal{O}_X/\mathbb{C}_X \xrightarrow{\sim} \Omega^1_{X,\mathrm{cl}}.$$

The holomorphic symplectic form identifies a symplectic vector field v with the closed one-form $\alpha_v = -\iota_v\omega$. Thus a section of $\mathcal{H}_X$ is exactly a local Hamiltonian primitive, modulo local constants, and the Poisson bracket is independent of the local constants. Taking the germ at p and completing gives the displayed restriction to $\widehat{\mathfrak{h}}_{X,p}$.

The formal holomorphic Darboux theorem identifies the completed local symplectic algebra at p with $\mathbb{C}[[z_1, z_2]]$ equipped with $dz_1 \wedge dz_2$. Under this identification Hamiltonians modulo constants become $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$, and the point-brane Ext algebra is the coordinate-free exterior algebra $\mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda^* T_p X$, with cyclic trace fixed by ω_p. Hence the open BV reduction and the Hamiltonian boundary evaluation are exactly the formal local construction after choosing Darboux coordinates. A different Darboux coordinate system differs by a formal symplectomorphism, which acts on Hamiltonians and on the normal coordinate algebra; the substitution map $z_i \mapsto \phi_i$ is equivariant for this action.

The exact sequence

$$0 \to \mathbb{C}_X \to \mathcal{O}_X \to \mathcal{H}_X \to 0$$

gives

$$0 \to H^0(X, \mathcal{O}_X)/\mathbb{C} \to H^0(X, \mathcal{H}_X) \xrightarrow{\delta} H^1(X, \mathbb{C}_X) \to H^1(X, \mathcal{O}_X).$$

For a locally Hamiltonian vector field, δ is the period class of the closed one-form $-\iota_v\omega$. Vanishing of this class is exactly the existence of a single global holomorphic Hamiltonian primitive, unique modulo constants. □

Remark 1.32 *(Hamiltonian period class and global descent).* The vanishing $H^1(X, \mathbb{C}_X) = 0$ kills the Hamiltonian period obstruction in Proposition 1.31. It is one matched-conventions hypothesis for globalization. Descent, QME, anomaly cancellation, locality, and the OCA comparison-cone contraction are the remaining data.

The closed mixed BF sector is the consequence of Witten's source convention applied to canonical transformations of the Darboux polydisk. The closed-field target is the shifted cotangent dg Lie algebra $\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1]$; on product opens

$$\Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big), \qquad Q = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}, \qquad S = \tfrac{1}{2}\int \langle \beta, \{\alpha, \alpha\}\rangle\, dz_1 dz_2,$$

which Proposition 1.34 verifies as a mixed holomorphic-topological BV sector in the sense of Definition 1.12. The constants kernel of the Hamiltonian vector-field map forces the quotient $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$; the U(1) centre class $[\bar{c}]$ of §6.1 is the scalar obstruction to lifting this reduced trace action to the unreduced polynomial Hamiltonian Lie algebra.

The open brane is the product of the exact local A-brane $\mathbb{R} \subset T^*\mathbb{R}$ and the B-brane $\mathcal{O}_p$ at the formal holomorphic point. Cyclic open string field theory [4], in the AKSZ form [6], gives the formal action $S(a) = \sum_{n\geq 1} \frac{1}{n+1} \mathrm{Tr}(a\, m_n(a, \ldots, a))$ on $\mathrm{RHom}(\mathcal{F}, \mathcal{F})[1]$; the affine skyscraper Yoneda algebra $\mathrm{RHom}_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda(\varepsilon_1, \varepsilon_2)$ of Proposition 3.9 reduces the ghost-zero piece $\alpha_0 = A_{\mathrm{CS}} + \chi_1\varepsilon_1 + \chi_2\varepsilon_2$ to the Dirac first-order matrix model (7) under $\phi_1 = -\chi_1$, $\phi_2 = \chi_2$, $A = -A_{\mathrm{CS}}$. The local unimodular datum is the residue/cyclic pairing on this exterior algebra: the holomorphic volume form on the Darboux polydisk.

A topological line carrying a stack of $\mathfrak{gl}_N$ Dirac branes meets a formal holomorphic-symplectic polydisk transverse to it. The brane line selects a Lagrangian inside that polydisk; the polydisk gives the Hamiltonian deformations as conjugate matrix coordinates on the brane.

1.4 Local setup at one brane stack

A stack of N branes carries the $\mathfrak{gl}_N$-valued Heisenberg pair

$$(\phi_1, \phi_2) \in \mathfrak{gl}_N \oplus \mathfrak{gl}_N, \qquad \Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2),$$

of holomorphic conjugate matrix coordinates on the brane worldvolume. The trace functional of a polynomial Hamiltonian $f \in \bar{A}$ on the brane is $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$; for $f \in \mathfrak{h}$ it is the completed formal trace.

The pair (ϕ_1, ϕ_2) is the degree-zero part of the derived Chan–Paton coordinate datum. After H^0 it is a representation of the two formal holomorphic coordinates by commuting matrices. Its scalar reduction is one holomorphic symplectic pair. Thus the transverse local geometry is a single holomorphic symplectic surface (Y, ω) at the brane point.

The formal-stalk chart uses the germ of Y at that point. Holomorphic Darboux trivialises the germ to

$$(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2),$$

so the target contributes only its formal symplectic polydisk.

The brane stack carries an open trace observable $\mathrm{Tr}\, f(\phi_1, \phi_2)$ for every Hamiltonian f. The trace is supported on the one-dimensional brane worldline. A second topological direction gives the transverse propagator direction for the BV ghosts, and the moment-map multiplier becomes a one-form there. Thus a real two-dimensional topological factor is the minimal completion in which the Dirac moment-map evaluation and the bulk-to-brane locality coexist.

Under the formal-stalk-chart hypotheses — a brane stack carrying the $\mathfrak{gl}_N$-valued holomorphic-symplectic pair (ϕ_1, ϕ_2), open trace observables along a one-dimensional brane worldline, a propagator direction transverse to that worldline inside the topological factor, and a single transverse holomorphic symplectic surface at the brane point — the three independent locality differentials of Proposition 1.36 fix the geometry as:

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1, z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}, \qquad \omega = dz_1 \wedge dz_2, \tag{1}$$

with $\mathbb{R}_t$ the brane worldline, $\mathbb{R}_s$ the transverse topological direction, and $\mathbb{C}^2_{z_1, z_2}$ the formal Darboux polydisk. The holomorphic factor is read through its formal symplectic polydisk $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$; compact or global features of an ambient holomorphic symplectic surface contribute this local Darboux datum to the theorem.

The brane has topological worldvolume direction $\mathbb{R}_t$ and holomorphic factor the point $p = 0 \in \widehat{\mathbb{C}^2}_0$, realised in the brane category as the skyscraper sheaf $\mathcal{O}_p$. The mixed brane is therefore

$$L = \mathbb{R}_t \times \{0\} \times \{p\} \hookrightarrow X,$$

the mixed brane used in the local Darboux calculation.

A brane stack is the derived Chan–Paton datum

$$\phi_1, \phi_2 \in \mathfrak{gl}_N, \qquad \psi \in \mathfrak{gl}_N[1], \qquad Q\psi = [\phi_1, \phi_2],$$

for the formal coordinate algebra $A = \mathbb{C}[[z_1, z_2]]$. On H^0 this becomes an N-dimensional representation of A by commuting matrices, hence a class in the commuting-pair quotient. The Hamiltonian Lie algebra $\mathfrak{h} = A/\mathbb{C}$ acts on the underived commuting representation functor by Hamiltonian vector fields; the $c_f \mapsto \theta_f$ coordinate records its infinitesimal motion on the moduli. The trace character of a bulk Hamiltonian $f \in \bar{A}$ at the shifted-cotangent coordinate u_f is

$$J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}, \tag{2}$$

the gauge-invariant trace of f evaluated on the Chan–Paton pair, projected to the stable scalar-reduced quotient; for $f \in \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ this is read in the completed formal brane algebra. Under the formal-stalk-chart hypotheses of Proposition 1.34 together with GL_N-invariance, formal Darboux naturality, and stable scalar reduction, three independent structures force J. First, Procesi–Razmyslov 1976/1980 [17][18] identify the stable connected GL_N-invariants on $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$ as cyclic words in two letters; under the stated hypotheses these are the only possible gauge-invariant scalar functionals (Proposition 3.22). Second, the moment-map zero-fibre $[\phi_1, \phi_2] = 0$ makes the cyclic words commute pairwise on cohomology and collapses every cyclic trace to the commutative monomial trace $\mathrm{Tr}(\phi_1^k \phi_2^l)$ (Lemma 3.30, Corollary 3.32). Third, stable GL_N-invariance together with Darboux equivariance under the formal symplectomorphism group of $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$ and linear normalization on z_1, z_2 forces the scalar correction to vanish at every monomial $z_1^a z_2^b$: the naturality lemma gives $J(z_1^a z_2^b) = \mathrm{Tr}(\phi_1^a \phi_2^b)$ directly, and the formal extension $f = \sum_{a,b} f_{a,b} z_1^a z_2^b \mapsto J(f) = \sum_{a,b} f_{a,b} \mathrm{Tr}(\phi_1^a \phi_2^b)$ follows by linearity and $\mathfrak{m}$-adic continuity. Under those three hypotheses J is the unique GL_N-invariant, Darboux-natural, scalar-reduced bulk-to-brane evaluation (Theorem 5.42; Corollary 5.40; Theorem 5.147 for the formal-Darboux normal-form statement among admissible protected ambient data).

§§3.3–3.4 build the matrix-Chan–Paton BV theory whose ghost-zero classical observables are the cyclic Procesi–Razmyslov traces on the moment-map zero-fibre, with Koszul antifield ψ resolving the constraint $[\phi_1, \phi_2] = 0$. §2.1 builds the closed shifted-cotangent BV theory whose Hamiltonian factor is $\widehat{\mathfrak{h}}$ and whose cotangent factor is the Grothendieck–Matlis principal-part module $\mathcal{P} = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$. §2.2–4.3 assemble the Hamiltonian BF action and the bulk-to-brane coupling $f \mapsto B_f(a)$ that smears $J(f)$ along the brane line and matches the reduced Hamiltonian Lie bracket on the closed side with the P_0-bracket on the brane line (Theorem 4.5, Theorem 5.191). §7.5 records the obstruction calculus blocking the unreduced Tate-coefficient cotangent lift of this matched bracket: vector obstruction Ob_T of Theorem E.53 on the attempted strict all-window side, with $\mathfrak{o}_{\mathrm{Cas}}$ the Casimir-kernel block and BMK obstruction Theorem 8.40 excluding one-pair analytic strict transfer and selecting the pro-Matlis retract as the replacement target (Corollary 8.41). §3.5 closes the section with the operator algebra of the open brane and the stable large-N reduction in which J becomes a connected single-trace primitive generator and the Capelli/Moyal scalar shift identifies with the closed-side anomaly class $[\bar{c}]$.

Theorem 5.147 is the fixed-frame normal-form statement. For an admissible protected Hamiltonian sector at a holomorphic-symplectic brane point, a chosen formal Darboux frame and protected comparison datum produce the formal Darboux model $(\widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1],\ \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1],\ [\bar{c}])$ with the bulk-to-brane evaluation (2). The Darboux-frame torsor records only the coordinate freedom. Membership of an ambient datum in the protected formal-stalk class requires the four-component protected twist class $\mathrm{Ob}^{\mathrm{univ}}_{p,X}$ to be represented and killed.

The cochain-level comparison between the closed Hamiltonian sector and the open brane observables has five parts. (1) Finite-dimensional cotangent CE/PV identification on every truncation $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ (Theorem 5.102). (2) Coordinate formal-polydisk envelope: the same generator dictionary $c^{a,b} \mapsto \theta^{a,b}$, $u_{a,b} \mapsto O_{a,b}$ extends to the completed coordinate algebras of Definition 4.17 (Theorem 4.20). (3) P_0-bracket enhancement on a chosen bracket-admissible target $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ (Definition 4.18); the strict ambient product is foreclosed by Lemma 5.212.

(4) Maurer–Cartan diagonal kernel: the formal Casimir $C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_i e_{d,i}\otimes e^i_d$ converges in the chosen completion — realised by the weighted Tate envelope of Corollary 5.71. (5) Bar-cobar/Koszul Quillen equivalence (Theorem 4.24), under a Tate/nilpotent/weighted completion with bounded inverse-limit hypotheses. The eight-component vector Ob_T of §7.5 records which level is reached and which obstruction blocks the next.

Remark 1.33 *(c_f versus u_f).* The dictionary distinguishes two CE coordinates of a Hamiltonian $f \in \widehat{\mathfrak{h}}$ on the closed side. The Hamiltonian-dual CE coordinate c^f maps under the cotangent CE/PV comparison to the Schouten polyvector coordinate θ^f on $\widehat{\mathbf{S}}(\bar{A})$, the generator of an infinitesimal canonical motion as a vector field on the brane phase space. The shifted-cotangent-dual CE coordinate u_f maps to the boundary observable $O_f = J(f)$, the gauge-invariant scalar reading of f on the Chan–Paton pair. The closed Hamiltonian generator enters the boundary algebra through u_f (Theorem 4.20, Proposition 4.15): a vector field acts on observables, and perfection of the Hamiltonian Lie algebra kills closed length-one CE cochains in the Hamiltonian direction, so u_f is the CE coordinate whose CE/PV image O_f is itself an observable.

PROPOSITION 1.34 *(Geometry determined by the formal-stalk chart).* The geometry (1) is the formal local mixed model for the formal-stalk-chart datum (Ω_N, Tr) of the brane stack. The open polynomial observable family equals $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$, with completed extension for $f \in \mathfrak{h}$, by the GL_N-invariant Darboux-natural scalar-reduced criterion (Theorem 5.147), and the mixed factorization-algebra shape is fixed by the three independent locality differentials of Proposition 1.36, with the trace observables exhibiting all three localities and the shifted-cotangent currents only the first two (Theorem 0.1). Its closed observable theory is the mixed holomorphic-topological Hamiltonian BF sector of Definition 1.12, with classical master equation; its mixed locality is the Cartan/Dolbeault homotopy. The brane open observable theory is the cyclic Dirac formal-stalk-chart theory of §3.3 and §3.4; the constructed bulk-to-brane operation is the degree-zero Hamiltonian boundary evaluation (Construction 3.25, Theorem 5.191). The unreduced cotangent factorization centre is the obstruction problem Problem 5.49; the open-side centrality-homotopy obstruction is Remark 5.190.

Proof. Closed BV data: on a product open $D \times B$ in the completed formal neighbourhood of $\mathbb{R}^2 \times \{p\} \subset \mathbb{R}^2 \times \mathbb{C}^2$, Proposition 2.7 identifies divergence-free polyvectors-one with Hamiltonians modulo constants, so Hamiltonian BF in the Kodaira-Spencer convention has fields

$$\mathcal{E}_{\mathrm{cl}}(D\times B) = \Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B)\widehat{\otimes}(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]),$$

with $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ the polynomial dense subspace. The differential is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ (so $Q_0 = 0$ in Definition 1.12). The BV pairing is the local pairing on compactly supported fields, obtained by wedge product of forms, integration over $D \times B$ against $dz_1 dz_2$, and the continuous evaluation $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}} \otimes \widehat{\mathfrak{h}} \to \mathbb{C}$; for $\alpha \in \Omega^p\widehat{\otimes}\widehat{\mathfrak{h}}[1]$ and $\beta \in \Omega^q\widehat{\otimes}\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]$ one has $|\alpha| = p-1$, $|\beta| = q-2$, and the top-form condition $p+q=4$ makes the pairing degree -1.

Master equation: with

$$F_\alpha = D\alpha + \tfrac{1}{2}\{\alpha,\alpha\}, \qquad D_\alpha = D + \{\alpha, -\}, \qquad S = \int_{D\times B}\langle\beta, F_\alpha\rangle dz_1 dz_2,$$

Jacobi for the holomorphic Poisson bracket and the derivation property of D give $D_\alpha F_\alpha = 0$; compact support and coadjoint-invariance of the evaluation pairing fix the BV transpose

$$\int \langle D^\vee_\alpha \beta, \xi\rangle dz_1 dz_2 = -(-1)^{|\beta|}\int \langle \beta, D_\alpha \xi\rangle dz_1 dz_2,$$

whence $\frac{1}{2}\{S,S\}_{\mathrm{BV}} = \int \langle D_\alpha^\vee \beta, F_\alpha\rangle dz_1 dz_2 = -(-1)^{|\beta|}\int \langle \beta, D_\alpha F_\alpha\rangle dz_1 dz_2 = 0$. The free and BF pieces

$$S_{\mathrm{free}} = \int \langle \beta, D\alpha\rangle dz_1 dz_2, \qquad I_{\mathrm{BF}} = \frac{1}{2}\int \langle \beta, \{\alpha,\alpha\}\rangle dz_1 dz_2,$$

contain the differential once and the Hamiltonian interaction once.

Mixed locality: for a constant translation v in $\mathbb{R}^2$ and an anti-holomorphic coordinate field $\partial/\partial \bar{z}_i$, $[D, \iota_v] = \mathcal{L}_v$ and $[D, \iota_{\partial/\partial \bar{z}_i}] = \mathcal{L}_{\partial/\partial \bar{z}_i}$. Both contractions are derivations of the local observable algebra and commute with $\{I_{\mathrm{BF}}, -\}$ up to those Lie derivatives, so $[Q_{\mathrm{obs}}, \iota_v] = \mathcal{L}_v$ and $[Q_{\mathrm{obs}}, \iota_{\partial/\partial \bar{z}_i}] = \mathcal{L}_{\partial/\partial \bar{z}_i}$. The de Rham Poincare lemma on D and the Dolbeault Poincare lemma on B give the quasi-isomorphism for small inclusions and the holomorphic dependence on B, as required by Definition 1.12.

Open BV data: the brane is a point inside the holomorphic factor, so its derived endomorphism algebra is the Yoneda algebra of the skyscraper $\mathcal{O}_p$ on $\widehat{\mathbb{C}^2}_0$, identified with $\Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2)$ by Proposition 3.9; the brane open fields are

$$\Omega^*(\mathbb{R}) \otimes \mathbf{Ext}^*_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p, \mathcal{O}_p) \otimes \mathfrak{gl}_N[1],$$

the derived $\mathrm{RHom}_Y(\mathcal{O}_p, \mathcal{O}_p)$-data of Definition 1.12, the derived endomorphism algebra of the skyscraper point. The cyclic structure is the trace on $\mathfrak{gl}_N$ tensored with the Berezin trace on the exterior algebra; §3.4 derives both the action and the Koszul differential of this open theory directly from the formal-stalk-chart premise. The bulk-to-brane evaluation $f \mapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is Construction 3.25 and its bracket compatibility is Theorem 5.191; the unreduced cotangent factorization-centre lift is the obstruction problem Problem 5.49. □

1.5 *Notation for the local theorem*

The local model uses three disjoint alphabets. Letters from one alphabet are operations; letters from another are evaluations; the remaining letters are distributional cotangent modes. Coincident (a,b)-labels across alphabets are an indexing convention, never an identification.

Symbol	**Meaning**
$R = \mathbb{C}[[z_1, z_2]]$	Complete local ring of the formal symplectic polydisk.
$\mathfrak{h} = R/\mathbb{C}\cdot 1$	Reduced Hamiltonian Lie algebra.
$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$	Polynomial Hamiltonian subalgebra used in trace and Moyal computations.
$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$	Shifted cotangent Lie algebra of Hamiltonian canonical transformations.
c^I (operation)	CE coordinate dual to a Hamiltonian generator $H_I \in \mathfrak{h}$; produces a Schouten vector field.
u_I (boundary observable)	CE coordinate dual to a shifted-cotangent generator $\eta_I \in \mathfrak{h}^\vee_{\mathrm{cont}}[1]$; produces a boundary Hamiltonian observable.
θ^I, O_I	PV coordinates; the CE/PV map sends $c^I \mapsto \theta^I$, $u_I \mapsto O_I$.
$O_{a,b}$ (trace observable)	Open trace observable $\overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}$, $a+b>0$.
$\Psi_{a,b}$ (PBW class)	Primitive one-ψ PBW special-fibre descendant in bidegree (a,b).
$\rho_{a,b}$ (cotangent mode)	Principal part $z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$, basis element of $\mathfrak{h}^\vee_{\mathrm{cont}}$.
$B_f(a)$	Smeared boundary Hamiltonian $\int a(t)\overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt$.

Remark 1.35 *(Two CE coordinates of a Hamiltonian: operation and evaluation). (i) The two CE coordinates of a Hamiltonian.* The boundary Hamiltonian observable on the brane is the image of the shifted-cotangent coordinate u_f; the Hamiltonian CE coordinate c_f acts on observables as the Schouten vector field θ^I. Their bracket compatibility $[\theta^I, O_J]_S = \delta^I_J$ is the Schouten–observable pairing organising the P_0-centrality homotopies of Problem 5.49; the dictionary is Lemma 4.1.

(ii) Bidegree coincidence $\Upsilon_{a,b} \leftrightarrow \rho_{a,b}$. The polynomial PBW descendant $\Upsilon_{a,b}$ and the principal part $\rho_{a,b}$ share the bidegree (a, b) and carry distinct $\mathfrak{h}$-module structures (Corollary 5.173), connected by the Fourier-polarized Rees bridge of Theorem 8.12. See Example 2.2 for the lowest non-trivial worked case.

1.6 Three notions of locality

The local model carries three independent locality differentials: contact in the brane direction, cohomological triviality of topological translations on $\mathbb{R}^2_{\mathrm{top}}$, and residue support of antiholomorphic translations on $\widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$. The three are independent coordinates of the locality structure.

(a) Support locality along the brane. For an open interval $I \subset \mathbb{R}$ on the world-line, define the compactly supported Hamiltonian current algebra $\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\widehat{\mathfrak{h}}$ with bracket induced from the de Rham product and the Hamiltonian bracket. In the degree-zero test-function notation used for explicit smeared trace evaluations this reads

$$[a \otimes f,\ b \otimes g] = ab \otimes \{f, g\}_{\widehat{\mathfrak{h}}}.$$

The reduced Hamiltonian open target on I is $A^{\mathrm{Ham}}_\partial(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$, generated by the scalar-quotiented smeared trace functionals

$$B_f(a) = \int_I a(t)\, \overline{\operatorname{Tr} f\big(\phi_1(t), \phi_2(t)\big)}\, dt.$$

Their bracket in the reduced Hamiltonian target is the canonical equal-time bracket:

$$\big\{B_f(a), B_g(b)\big\}_{A^{\mathrm{Ham}}_\partial} = B_{\{f,g\}_{\widehat{\mathfrak{h}}}}(ab).$$

The bracket is the reduced one; before scalar quotient the unreduced bracket carries the central term $N\omega(f, g)\int_I ab\, dt$, with ω the closed–open mismatch cocycle of Lemma 6.2. Disjoint supports multiply, since $ab = 0$ outside the intersection; overlapping supports give a delta-function contact term. Disjoint smeared trace evaluations multiply, and Poisson brackets are supported where the test functions meet.

(b) Topological locality. In the topological direction (the $\mathbb{R}^2$ factor), translations are Q-exact:

$$L_v = [Q, \iota_v].$$

Equivalently, positive jets in the topological direction vanish in local-operator cohomology by the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$ (Lemma 3.21). The precise position of an insertion is cohomologically immaterial; the collision pattern of insertions determines the product, and the factorisation algebra is locally constant in the topological direction.

(c) Formal holomorphic locality. In the holomorphic direction (the $\mathbb{C}^2$ factor), the model has been completed at the brane point $p \in \mathbb{C}^2$. It is a theory of holomorphic jets at p. Antiholomorphic translations are Q-exact:

$$L_{\partial/\partial\bar{z}_i} = [Q, \iota_{\partial/\partial\bar{z}_i}].$$

The kinematic differential of the closed sector is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, and on the formal-jet model $\bar{\partial}$-cohomology is concentrated in degree zero. The cotangent boundary sector is *distributionally local*: a coadjoint mode is a transverse principal part $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2$, paired with a brane Hamiltonian by residue,

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}.$$

The momentum dual field attached to the defect has support on the defect with transverse behaviour encoded by residues. The polynomial one-ψ descendants $\Psi_{k,l}$ realise the PBW special fibre of $\mathfrak{h}^\vee[1]$ (Proposition 5.172); Corollary 5.173 records the deformation-level mismatch with the coadjoint module, exactly the distinction between polynomial and distributional realisations.

Proposition 1.36 *(Locality forced by the three differentials).* In the reduced Hamiltonian model the three locality statements above are consequences of three explicit operators, with (t, t') two brane-time points and s the transverse $\mathbb{R}_s$-coordinate reserved for the geometry of X:

$$\delta(t - t'), \qquad L_v = [Q, \iota_v] \quad (v \in T\mathbb{R}^2_{\mathrm{top}}), \qquad L_{\partial/\partial\bar{z}_i} = [Q, \iota_{\partial/\partial\bar{z}_i}].$$

Support locality gives strict vanishing of P_0-brackets for disjoint smeared Hamiltonian currents. Topological locality gives the locally constant brane-line factorization structure after the compact-support density shift. Formal holomorphic locality gives Taylor Hamiltonians and residue-dual principal parts at the brane point. These are independent pieces of structure. The unreduced cotangent centrality homotopies require additional data. The trace observables exhibit all three localities; the shifted-cotangent currents exhibit only the first two (Theorem 0.1).

Proof. Support locality is the calculation of Remark 4.10: after smearing, the factor $\delta(t - t')$ becomes the product ab, so disjoint supports give zero bracket. This proves the strict support-local factorization identity used by Proposition 4.9. Topological locality follows because translation along a topological direction is Q-exact; moving an insertion changes it by a Q-boundary. On compactly supported interval currents the surviving cohomology class is the density line of Lemma 4.8. Formal holomorphic locality follows in the same way for antiholomorphic translations, while holomorphic dependence is completed at the brane point and paired with cotangent modes by the residue pairing. The three operators live in different directions. Centrality homotopies for arbitrary unreduced open observables are additional structure. □

1.7 Unitarity

The protected local algebraic question is the BV consistency calculation on a twisted sector: classical and quantum master equations on the ghost/antifield complex with cohomologically trivialised topological time. Positive-definite inner products, Wightman positivity, and reflection positivity belong to a parent unitary theory. The protected sector studied here is its cohomological algebra of observables.

The replacements for unitarity are the consistency conditions of the BV formalism: the classical master equation

$$QI + \tfrac{1}{2}\{I, I\} = 0$$

and the quantum master equation

$$QI + \tfrac{1}{2}\{I, I\} + \hbar\Delta I = 0,$$

together with associativity of the resulting factorization algebra. The native two-dimensional holomorphic singular product is the diagonal local-cohomology expansion of Theorem 1.17; a one-dimensional VOA/OPE algebra is present only after a one-complex-dimensional reduction datum retaining the required transverse coefficient system. If the local model arises as a protected sector of a larger unitary theory (for example, the topological twist of a supersymmetric quantum field theory whose local observables in the twisted sector arise as cohomology with respect to a supercharge), then unitarity is carried by the larger theory, and the local sector retains only the cohomological algebra of protected observables. Lorentzian unitarity belongs to such a parent theory.

THEOREM 1.37 *(BV replacement for unitarity in the protected sector).* For the local mixed holomorphic-topological sector, the consistency conditions are

$$QI + \frac{1}{2}\{I, I\} = 0, \qquad QI + \frac{1}{2}\{I, I\} + \hbar\Delta I = 0$$

where the second equation is imposed only after a BV Laplacian and a renormalization scheme have been constructed, together with associativity and locality of the resulting factorization products. These conditions imply that the cohomology of local observables carries the stated P_0 or quantum factorization algebra structure and that closed insertions act through the central-operation complex of that structure — the P_0-bracket complex classically, the chiral E_2-Hochschild centre quantum-mechanically. Positivity, a Hilbert-space inner product, and a Lorentzian unitary evolution operator belong to the parent theory. If the sector is obtained as Q-cohomology of a unitary parent theory, the unitarity datum belongs to that parent together with the choice of Q-compatible adjoint.

Proof. The classical master equation says that the interacting differential $Q_I = Q + \{I, -\}$ squares to zero, since $Q_I^2 = \{QI + \tfrac{1}{2}\{I, I\}, -\}$. The quantum master equation is the corresponding statement for the quantum BV differential $Q + \{I, -\} + \hbar\Delta$. Factorization associativity is the compatibility of local products with disjoint inclusions and collision limits; locality says these products are supported on the collision loci allowed by the factorization algebra. These are exactly the algebraic structures used below to form P_0-brackets, central operations, and Moyal deformations. The construction never introduces a positive sesquilinear pairing on the BRST complex; ghosts and antifields are cohomological variables. Conversely, if a larger physical theory gives a Hilbert space, a positive form, and a supercharge Q compatible with adjoints, then the local theory is its protected cohomological algebra. That parent datum is additional structure beyond the reduced Hamiltonian CE/PV theorem. □

The configuration is fixed: the formal-Darboux geometry $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$, the brane line L, the Hamiltonian Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, and the three locality differentials. Closure of the BV theory on this configuration forces the shifted-cotangent extension of $\mathfrak{h}$ by its continuous dual, with a nondegenerate equivariant pairing of cohomological degree -1: the next chapter constructs and characterises that extension.

2 THE SHIFTED-COTANGENT BF LIE ALGEBRA

The Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ constructed below is the closed-side dg Lie algebra of Theorem 5.6; its Capelli-projective central extension $\widehat{\mathfrak{g}}_{\mathrm{cent}}$ enters as the bulk Chevalley–Eilenberg target.

The holomorphic-symplectic Darboux normal form on $(\widehat{\mathbb{C}^2}_0, \omega = dz_1 \wedge dz_2)$ determines the closed BV sector by three uniqueness statements. The Lie algebra of Hamiltonian vector fields modulo constants is $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ (Proposition 2.7). The shifted-cotangent extension $\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ is the unique extension among Lie extensions of $\widehat{\mathfrak{h}}$ by an abelian odd ideal carrying a non-degenerate equivariant pairing of cohomological degree -1 (Theorem 2.1). The Hamiltonian BF action is the unique compatible local classical master equation (CME) action on this BV pair (Theorem 2.11). These statements reduce the construction to Cartan's identity and divergence, the BV cotangent extension, and Grothendieck–Matlis residue duality.

Hamiltonian vector fields from Cartan's identity. An infinitesimal canonical transformation of $(\widehat{\mathbb{C}^2}_0, \omega = dz_1 \wedge dz_2)$ is a vector field X annihilating ω under the Lie derivative. Cartan's identity reduces this to closedness of $\iota_X\omega$, which on the formal polydisk is exactness: $\iota_X\omega = -df$ for a unique f mod $\mathbb{C}$. Hence the symmetry vector field is the Hamiltonian vector field X_f, and the kernel of $f \mapsto X_f$ is the constant line. Equivalently, by Proposition 2.7, the polyvector divergence $\partial_\Omega \colon \mathbf{PV}^1 \to \mathbf{PV}^0$ on the polyvector-field complex $\mathbf{PV}^\bullet$ of the polydisk identifies its kernel with Hamiltonian vector fields modulo constants. The closed Lie algebra is therefore

$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1, \qquad \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1, \tag{3}$$

with the constant-Poisson bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g \pmod{\mathbb{C}}$. Constants vanish because their Hamiltonian vector field is zero. This is the kernel of the divergence-Hamiltonian identification. Polynomial trace and Moyal calculations use the polynomial subalgebra $\bar{A}$; completed CE/Koszul constructions use the formal completion $\widehat{\mathfrak{h}}$.

The cotangent extension from canonical-transformation BV. A BV theory of canonical transformations extends $\widehat{\mathfrak{h}}$ to a Lie algebra of fields by adjoining a degree-shifted cotangent partner: each Hamiltonian ghost c acquires its antifield u in the dual direction. The Hamiltonian sector of BCOV/Kodaira–Spencer theory [1] fixes this language. The closed BV Lie algebra is the shifted cotangent extension

$$\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1], \tag{4}$$

with the Hamiltonian bracket on $\widehat{\mathfrak{h}}$ and the coadjoint action on $\mathfrak{h}^\vee_{\mathrm{cont}}[1]$, and the closed observables are the continuous Chevalley–Eilenberg cochains

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{Ham}} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big).$$

The cochain CE/PV identity proved later (Theorem 4.20) lives on the coordinate envelope $C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g})$—the product over finite graded-commutative words in the CE coordinates—and the continuous CE cochains are its continuity-controlled subspace.

The cotangent module from residue duality. The shifted cotangent partner carries an antifield labelled by the continuous dual $\mathfrak{h}^\vee_{\mathrm{cont}}$. This dual is the principal-part module selected by residue duality. The natural pairing $\widehat{\mathfrak{h}} \otimes \mathfrak{h}^\vee_{\mathrm{cont}} \to \mathbb{C}$ must be continuous for the formal-disk filtration and equivariant for the Hamiltonian action. Both requirements name a unique module.

THEOREM 2.1 *(Defect momentum is the principal-part module).* Let $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ carry the $\mathfrak{m}$-adic topology, $\mathfrak{m} = (z_1, z_2)$. The continuous $\mathfrak{h}$-equivariant dual is, up to the residue scalar of Theorem 5.177, uniquely the Grothendieck–Matlis principal-part module

$$\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\, \rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

with the residue pairing $\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}$ and the residue-coadjoint action $z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}$. Equivalently, this is the unique continuous total-degree-zero $\mathfrak{h}$-module structure on a continuous dual of $\widehat{\mathfrak{h}}$ compatible with the residue pairing on the formal polydisk.

Proof. This is Theorem 5.179: local cohomology [19] of the maximal ideal $\mathfrak{m} = (z_1, z_2)$ identifies the continuous dual of the formal completion with the principal-part module $\mathcal{P}$; the residue pairing produces the displayed coadjoint formula on monomials, and any other continuous total-degree-zero $\mathfrak{h}$-equivariant pairing is a scalar multiple of the residue pairing. After the residue scalar is fixed, the isomorphism is canonical. □

The PBW Rees algebra of $\mathfrak{g}$ has special fibre

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]),$$

where $U_\hbar(\mathfrak{g})$ denotes the Rees deformation of the universal enveloping algebra and $\hbar$ the formal deformation parameter. The bidegree-(a, b) generators of the one-ψ sector inside this commutative special-fibre algebra are denoted $\Psi_{a,b}$; the deformation-level cotangent module is $P = \mathfrak{h}^\vee_{\mathrm{cont}}$, realised by the principal parts $\rho_{a,b}$ of Theorem 2.1. Theorem 8.12 is the Fourier-polarized Rees bridge between these label systems; PBW descendants and principal parts have distinct $\mathfrak{h}$-module structures.

A Taylor Hamiltonian is what the brane samples. A principal part is the dual distributional momentum concentrated at the brane: the residue functional that detects a Taylor coefficient. Cotangent fields on a formal brane are residues. Each $\rho_{a,b}$ is a defect-supported transverse mode: a distributional object on the brane line whose holomorphic transverse behaviour is encoded by a single principal-part coefficient. An abstract copy $\bar{A}_{\mathrm{desc}}$ of $\bar{A}$ sees the same labels through the polynomial PBW special-fibre descendants $\Psi_{k,l}$ introduced in Construction 3.25; they carry the same (a, b)-bidegree as the principal parts but a different $\mathfrak{h}$-module structure (Corollary 5.173); see §1.5 for the alphabet separation.

Example 2.2 (The lowest non-trivial residue mode). Take the linear Hamiltonian z_1 and act on the lowest reduced residue $\rho_{1,0}$. The coadjoint formula of Theorem 2.1 with $(p, q, a, b) = (1, 0, 1, 0)$ gives

$$z_1 \cdot \rho_{1,0} = -(0 - 0 + 1 - 0)\rho_{1,1} = -\rho_{1,1}.$$

The polynomial PBW class $\Psi_{1,0}$, labelled identically, carries a different action: under the open one-ψ descendant module of Proposition 5.172 the linear Hamiltonian shifts $\Psi_{k,l}$ by the bracket on the open-trace ring. Same label, two $\mathfrak{h}$-module structures: the PBW descendant action and the residue-coadjoint action. The residue mode is what the defect momentum sees; the polynomial PBW class is what the open trace ring records. The Fourier-polarized Rees bridge of Theorem 8.12 is the intermodule passage between the two. This worked case computes the residue-side action of $\rho_{a,b}$ explicitly, complementing Remark 1.35.

Remark 2.3 *(Scalar Hamiltonian and centre-of-mass term).* In $\bar{A}$, $\{z_1, z_2\} = 1 \equiv 0$ by quotient. On the unreduced brane phase space, $\{\mathrm{Tr}(\phi_1), \mathrm{Tr}(\phi_2)\} = N$. Constants vanish closed-side; the same constant survives open-side as the brane centre-of-mass symplectic pair. Every bracket involving linear Hamiltonians is read after projection to the Hamiltonian quotient by scalar trace classes. Quantisation returns this asymmetry as the Capelli shift (Theorem 6.6 below; cf. Remark 5.196).

Remark 2.4 *(Cotangent obstruction).* Theorem 2.1 resolves the identification of $\mathfrak{h}^\vee_{\mathrm{cont}}$. The unreduced lift of the cotangent factorisation centre is the problem of promoting the closed cochain CE/PV identity (Theorem 4.20) to a chain-level morphism $\Phi_{\mathrm{cl}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}|_L \to Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N})$ has centrality homotopies on the cotangent (one-antifield) sector as its obstruction-theoretic datum. The exact obstruction datum is named in Problem 5.49; the strict all-window Bochner–Martinelli current transfer obstruction is $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$. A global central extension to the bracket-admissible $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ of Definition 4.18 requires the vanishing of $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$ on the strict ambient product completion.

2.1 Closed mixed Hamiltonian sector

Closed insertions on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ are infinitesimal canonical transformations of the formal holomorphic symplectic polydisk $\big(\widehat{\mathbb{C}^2}_0,\ \omega = dz_1 \wedge dz_2\big)$. Divergence-freeness under the holomorphic volume $\Omega = dz_1 \wedge dz_2$ identifies the infinitesimal symmetries with the reduced Hamiltonian Lie algebra $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ (Proposition 2.7). The BV cotangent extension of a classical deformation problem then attaches a degree-shifted dual partner; continuity of the polydisk topology fixes that dual to be the Grothendieck–Matlis principal-part module $\mathcal{P} = \mathfrak{h}^\vee_{\mathrm{cont}}$, with the residue basis

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2 \quad (a, b \geq 0,\ a + b > 0)$$

dual to the Taylor monomial $H_{a,b} = z_1^a z_2^b$ under the residue pairing of Proposition 5.176. The same labels carry distinct $\widehat{\mathfrak{h}}$-module structures: the polynomial one-ψ descendant $\Upsilon_{a,b}$ of Construction 3.25 carries the open-trace bracket; the Matlis residue class $\rho_{a,b}$ carries the residue coadjoint action; the two are separated by Lemma 3.37. The polarization theorem 8.1 forces the residue-side choice over the polynomial PBW class: the Moyal coadjoint action acts on the residue module, and its image on the polynomial substitute lies inside the same-action obstruction class. Both factors are forced: the Hamiltonian factor by Proposition 2.7, the cotangent factor by the residue-pairing rigidity Theorem 5.177 and the polarization Theorem 8.1. The closed BV field complex is the unique local realization of the resulting shifted-cotangent Lie algebra $\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ (Theorem 2.8).

Remark 2.5 *(Polyvector and divergence conventions).* Holomorphic polyvector fields $\mathbf{PV}^{*,*}(\mathbb{C}^2) \cong \Omega^{0,*}(\mathbb{C}^2)[\partial_{z_1}, \partial_{z_2}]$ carry the divergence operator $\partial_\Omega = \iota_\Omega^{-1} \circ \partial_{\mathrm{hol}} \circ \iota_\Omega$ determined by $\Omega = dz_1 \wedge dz_2$. In the formal polynomial model Dolbeault cohomology is represented by Taylor polynomials. ∂_Ω vanishes on functions; on poly-1-vectors $X = a_1\partial_{z_1} + a_2\partial_{z_2}$, $\partial_\Omega X = \partial_{z_1} a_1 + \partial_{z_2} a_2$. The image of ∂_Ω from $\mathbf{PV}^0$ is zero, since ∂_Ω vanishes on functions. Constants are removed by $f \mapsto X_f$ to give the reduced polynomial algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$.

Lemma 2.6 *(Polynomial Poincare primitive).* Let

$$\theta = a(z_1, z_2)\, dz_1 + b(z_1, z_2)\, dz_2$$

be a polynomial one-form on $\mathbb{C}^2$. If $\partial_{z_2} a = \partial_{z_1} b$, then there is a polynomial $f \in \mathbb{C}[z_1, z_2]$, unique modulo constants, such that $\theta = df$.

Proof. Define

$$f(z_1, z_2) = \int_0^1 \big(z_1 a(tz_1, tz_2) + z_2 b(tz_1, tz_2)\big)\, dt.$$

Since a and b are polynomials, the integral is termwise division of each homogeneous term by its degree plus one, hence f is polynomial. Differentiate under the integral sign. Closedness says $\partial_{z_2} a = \partial_{z_1} b$, so

$$\begin{aligned}
\partial_{z_1} f &= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_1} b(tz)\big)\, dt \\
&= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_2} a(tz)\big)\, dt \\
&= \int_0^1 \frac{d}{dt}\big(t\, a(tz)\big)\, dt = a(z).
\end{aligned}$$

The same calculation, with a and b interchanged, gives $\partial_{z_2} f = b(z)$. Therefore $df = \theta$. If $df = dg$, then $d(f - g) = 0$, hence $f - g$ is constant. $\square$

PROPOSITION 2.7 *(Hamiltonian polyvector reduction).* In the formal polynomial model on $(\mathbb{C}^2, \omega = dz_1 \wedge dz_2)$, the divergence-closed polyvector-one fields reduce to Hamiltonian vector fields:

$$\mathbf{Ker}(\partial_\Omega : \mathbf{PV}^1 \to \mathbf{PV}^0) \cong \{df : f \in \mathbb{C}[z_1, z_2]\} \cong \bar{A}.$$

Under the convention

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df,$$

the commutator of vector fields is

$$[X_f, X_g] = X_{\{f,g\}_{\bar{A}}}.$$

Proof. Let $X = a_1 \partial_{z_1} + a_2 \partial_{z_2}$. The divergence operator is $\partial_\Omega X = \partial_{z_1} a_1 + \partial_{z_2} a_2$. Contracting with ω gives

$$\iota_X \omega = a_1\, dz_2 - a_2\, dz_1.$$

The equation $\partial_\Omega X = 0$ is exactly the closedness of this one-form:

$$d(\iota_X \omega) = (\partial_{z_1} a_1 + \partial_{z_2} a_2)\, dz_1 \wedge dz_2 = 0.$$

By Lemma 2.6, there is a polynomial f, unique modulo constants, such that $\iota_X \omega = -df$. Therefore

$$a_1 = -\partial_{z_2} f, \qquad a_2 = \partial_{z_1} f,$$

so $X = X_f$. Constants are exactly the kernel of $f \mapsto X_f$, giving $\bar{A}$.

Finally, for any polynomial h,

$$\begin{aligned}
[X_f, X_g](h) &= \{f, \{g, h\}\} - \{g, \{f, h\}\} \\
&= \{\{f, g\}, h\}
\end{aligned}$$

by the Jacobi identity for the constant Poisson bracket. Thus $[X_f, X_g] = X_{\{f,g\}}$, and projecting constants gives the bracket on $\bar{A}$. $\square$

Replacing $\bar{A}$ by $\Omega^{0,*}(\mathbb{C}^2)\,\widehat{\otimes}\,\bar{A}$ extends the same statement to Dolbeault coefficients: the defining equations are coefficient-wise, constants drop by the same argument, and the bracket is the Dolbeault-coefficient extension of $\{-,-\}_{\bar{A}}$.

Theorem 2.8 *(Closed BV field complex of canonical transformations).* Let $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ be the kinematic differential on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$. The BV cotangent extension of $\widehat{\mathfrak{h}}$, applied to the formal local data of $(\widehat{\mathbb{C}^2}_0, \omega)$, is the field complex

$$\mathcal{E}_{\mathrm{cl}} = \Omega^*(\mathbb{R}^2)\,\widehat{\otimes}\,\Omega^{0,*}(\mathbb{C}^2)\,\widehat{\otimes}\,\big(\widehat{\mathfrak{h}}[1] \oplus \mathfrak{h}^\vee_{\mathrm{cont}}[2]\big), \tag{5}$$

with $\widehat{\mathfrak{h}}[1]$ the Hamiltonian connection sector α and $\mathfrak{h}^\vee_{\mathrm{cont}}[2]$ the BV cotangent partner β. This complex is the unique local realization, in the Dolbeault–de Rham category, of the shifted-cotangent Lie algebra $\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ carrying:

(i) the BV pairing $\int_{\mathbb{R}^2\times\mathbb{C}^2} \langle \beta, \alpha\rangle\, dz_1 dz_2$ of degree -1, with $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}} \otimes \widehat{\mathfrak{h}} \to \mathbb{C}$ the residue pairing of Proposition 5.176;

(ii) the kinematic differential D acting coefficient-wise;

(iii) the bracket on the Lie factor $\widehat{\mathfrak{h}}$ given by the holomorphic Poisson bracket of ω extended to Dolbeault coefficients, and the coadjoint action on the cotangent factor.

Proof. The Hamiltonian factor $\widehat{\mathfrak{h}}$ is given by Proposition 2.7: the divergence-free polyvector-one fields, modulo constants, are exactly the Hamiltonian vector fields of $\bar{A}$, and the Dolbeault extension is coefficient-wise. The cotangent factor $\mathfrak{h}^\vee_{\mathrm{cont}} = \mathcal{P}$ is fixed by Proposition 5.176 and the residue-pairing rigidity Theorem 5.177: among continuous total-degree-zero $\widehat{\mathfrak{h}}$-equivariant pairings the Matlis residue pairing is unique up to a single scalar. The BV pairing on $\mathcal{E}_{\mathrm{cl}}$ is then the integrated form of this residue pairing against the de Rham/Dolbeault top form. The degree shifts $[1]$ on $\widehat{\mathfrak{h}}$ and $[2]$ on $\mathfrak{h}^\vee_{\mathrm{cont}}$ are fixed by the BV cotangent principle: a Maurer–Cartan element of $\mathfrak{g} = L \ltimes L^\vee[1]$ in a degree-zero local field theory on a four-real-dimensional spacetime sits in $\Omega^*\widehat{\otimes}\Omega^{0,*}\widehat{\otimes}L[1]$, and the partner sits in $\Omega^*\widehat{\otimes}\Omega^{0,*}\widehat{\otimes}L^\vee[2]$ so that the integrated pairing has degree -1. The computation in Proposition 1.34 verifies these degrees on the four-form factor. $\square$

The boundary image of β, under the bulk-to-brane evaluation of Section 4.3, is the Matlis principal-part defect current of Theorem 5.186: cotangent fields are measured by that current and the cotangent module is the Matlis residue module. The BF action of Section 2.2 is the unique cyclic local functional whose classical master equation says that α is a Maurer–Cartan element of $\mathfrak{g}$.

Remark 2.9 *(Cotangent module as Matlis dualizing object).* The cotangent factor $\mathcal{P}$ is the local-cohomology module $H^2_{\mathfrak{m}}(R)\, dz_1 dz_2$ on $R = \mathbb{C}[[z_1, z_2]]$ with the constant Hamiltonian line removed: the Matlis dualizing object on R, paired with $\bar{A} = R/\mathbb{C}\cdot 1$ by residue.

Remark 2.10 *(Casimir/Tate obstruction to a strict cotangent kernel).* The pairing of Theorem 2.8 is non-degenerate but the diagonal coevaluation tensor $C_{\widehat{\mathfrak{h}}} = \sum_{a+b>0} H_{a,b} \otimes \rho_{a,b}$ diverges in the strict tensor product $\widehat{\mathfrak{h}}\widehat{\otimes}\mathfrak{h}^\vee_{\mathrm{cont}}$: this is the Tate-Casimir obstruction Lemma 5.212. The closed BV complex is therefore well posed as a continuous bilinear pairing. A strict kernel requires passage to a kernel-admissible model through the weighted-Tate envelope of Definition 5.55; in the strict ambient product completion, the P_0-algebra structure is obstructed (Definition 4.18). Every unreduced cotangent-kernel construction must clear this obstruction.

2.2 Hamiltonian BF action

The Hamiltonian BF action is forced by three data: the holomorphic-symplectic Darboux normal form on $(\widehat{\mathbb{C}^2}_0, \omega = dz_1 \wedge dz_2)$, the cyclic BV trace on $\mathcal{E}_{\mathrm{cl}}$ of (5), and linearity in the cotangent partner. Theorem 2.8 fixes the BV pair $(\mathcal{E}_{\mathrm{cl}}, \omega_{\mathrm{BV}})$ and the shifted-cotangent Lie algebra $\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1]$; the canonical degree-$(-1)$ evaluation pairing $\langle \beta, f\rangle = \beta(f)$ between cotangent and Hamiltonian factors, wedged with the form factors and integrated over $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ against $dz_1 \wedge dz_2$, extends to the cyclic BV pairing ω_{BV}. Under cyclicity, locality, and linearity in the cotangent partner, the classical master equation $\frac{1}{2}\{S, S\}_{\mathrm{BV}} = 0$ has a unique cyclic local solution modulo D-exact additions: the BF action of (6) below. Equivalently, the Hamiltonian BF action is the unique compatible local CME action on the BV pair $(\mathcal{E}_{\mathrm{cl}}, \omega_{\mathrm{BV}})$, forced by holomorphic-symplectic Darboux normal form, cyclic BV trace, and linearity in cotangent partner.

THEOREM 2.11 *(Cotangent BF action: unique cyclic Maurer–Cartan functional).* Let $\langle -, - \rangle$ be the integrated residue pairing between $\Omega^*(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,*}(\mathbb{C}^2)\widehat{\otimes}\mathfrak{h}^{\vee}_{\mathrm{cont}}[2]$ and $\Omega^*(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,*}(\mathbb{C}^2)\widehat{\otimes}\widehat{\mathfrak{h}}[1]$, contracted by wedging form factors and integrating against $dz_1 \wedge dz_2$. The cyclic local functional

$$S = \int_{\mathbb{R}^2\times\mathbb{C}^2} \Big\langle \beta,\ D\alpha + \tfrac{1}{2}\{\alpha, \alpha\} \Big\rangle\, dz_1\, dz_2 \tag{6}$$

is, up to a D-exact form, the unique cyclic local functional on $\mathcal{E}_{\mathrm{cl}}$ satisfying:

(i) S is local in the de Rham/Dolbeault sense and cyclic for the integrated residue pairing;

(ii) S is linear in the cotangent partner β;

(iii) the classical BV master equation $\frac{1}{2}\{S, S\}_{\mathrm{BV}} = 0$ holds and encodes the Maurer–Cartan equation $F_\alpha = D\alpha + \frac{1}{2}\{\alpha, \alpha\} = 0$ on $\widehat{\mathfrak{h}}$ together with the dual transport $D^{\vee}_{\alpha}\beta = 0$ on $\mathfrak{h}^{\vee}_{\mathrm{cont}}$.

The form-valued Hamiltonian bracket on $\widehat{\mathfrak{h}}$ is

$$[\eta \otimes f, \theta \otimes g] = (-1)^{|f||\theta|}(\eta \wedge \theta) \otimes \{f, g\},$$

with Koszul sign reducing to +1 on the ghost-zero (even-coefficient) sector.

Proof. Existence. The action (6) is constructed in Proposition 1.34, where the classical master equation $\frac{1}{2}\{S, S\}_{\mathrm{BV}} = 0$ is verified by the coadjoint invariance of the residue pairing and the Cartan identity $D_\alpha F_\alpha = 0$ on $\widehat{\mathfrak{h}}$.

Uniqueness. Cyclicity, locality, and linearity in β restrict S to a sum

$$S = \int \langle \beta, P\alpha \rangle + \int \langle \beta, Q(\alpha, \alpha) \rangle \quad (\mathrm{mod}\ D\text{-exact}),$$

with P a degree-one $\mathbb{C}$-linear local operator on $\Omega^*\widehat{\otimes}\Omega^{0,*}\widehat{\otimes}\widehat{\mathfrak{h}}[1]$ and Q a degree-zero local bilinear bracket on the same space. Higher powers β^k, $k \geq 2$, are excluded by linearity in the cotangent factor; pure α-polynomials are excluded because cyclicity for the (β, α) pairing pairs every term to a single β. The classical master equation on this ansatz reads

$$\tfrac{1}{2}\{S, S\}_{\mathrm{BV}} = \int \Big\langle \beta, P^2\alpha + \big(PQ(\alpha, \alpha) + 2Q(\alpha, P\alpha)\big) + 2Q(\alpha, Q(\alpha, \alpha))\Big\rangle,$$

where the three displayed pieces are homogeneous in α of degrees 1, 2, 3. Vanishing of the degree-1 piece forces $P^2 = 0$, and locality plus the kinematic axioms (i),(iii) then identify P with the differential

$D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ acting coefficient-wise. Vanishing of the degree-2 piece forces D to be a derivation of Q; vanishing of the degree-3 piece forces Q to satisfy Jacobi. The unique local bilinear D-derivation on $\widehat{\mathfrak{h}}$ that satisfies Jacobi and extends coefficient-wise the holomorphic Poisson bracket of ω is $\{-,-\}$ itself. Therefore S is given by (6) modulo a D-exact term. □

The classical equation of motion is the flatness statement $F_\alpha = 0$: α is a $\mathfrak{g}$-Maurer–Cartan element, equivalently a connection on a bundle of formal holomorphic-symplectic surfaces over $\mathbb{R}^2$. The cotangent partner β is the Lagrange multiplier enforcing this equation; the BV gauge symmetries are

$$\delta_\varepsilon \alpha = D\varepsilon + \{\alpha, \varepsilon\}, \qquad \delta_\varepsilon \beta = \mathrm{ad}_\varepsilon^\vee \beta,$$

and the cotangent gauge symmetry

$$\delta_\lambda \alpha = 0, \qquad \delta_\lambda \beta = D_\alpha \lambda,$$

with $D_\alpha = D + \{\alpha, -\}$ and $\lambda \in \Omega^*\widehat{\otimes}\Omega^{0,*}\widehat{\otimes}\mathfrak{h}^\vee_{\mathrm{cont}}[1]$. The second symmetry is reducible on shell: $\lambda = D_\alpha \mu$ gives $D_\alpha \lambda = \{F_\alpha, \mu\}$, which vanishes when $F_\alpha = 0$.

Remark 2.12 *(Component reading).* In ghost-zero coordinates on $\mathbb{R}^2_{x_i} \times \mathbb{C}^2_{z_i}$,

$$\alpha = \alpha_{x_i}\, dx_i + \alpha_{\bar{z}_j}\, d\bar{z}_j, \qquad \beta = \beta_{x_1 x_2}\, dx_1 dx_2 + \beta_{x_i \bar{z}_j}\, dx_i\, d\bar{z}_j + \beta_{\bar{z}_1\bar{z}_2}\, d\bar{z}_1 d\bar{z}_2.$$

The component $\alpha_{\bar{z}_j} d\bar{z}_j$ is the Hamiltonian potential whose vector field is the Beltrami differential of a complex-structure deformation of $\mathbb{C}^2$; $\alpha_{x_i} dx_i$ is the connection on $\mathbb{R}^2$ valued in Hamiltonian vector fields, organizing the bundle of $\mathbb{C}^2$s over $\mathbb{R}^2$. $F_\alpha = 0$ asserts joint integrability and flat holonomy. The bulk theory is defined on the full mixed space-time $\mathbb{R}^2 \times \mathbb{C}^2$; the brane line $L = \mathbb{R} \times \{0\} \times \{p\}$ enters only via the boundary evaluation of Section 4.3.

Remark 2.13 *(Named obstruction to a renormalized BF theory).* Theorem 2.11 produces a formal-local cyclic functional satisfying the classical BV master equation. Its lift to a renormalized field theory requires the analytic data of Problem 5.213: an elliptic gauge fixing, a Costello propagator $P_{\epsilon,L}$ on $\mathbb{R}^2 \times \mathbb{C}^2$, a finite-scale graph effective action with graphwise renormalization, and the QME $(d + \hbar\Delta)e^{I/\hbar} = 0$ to all orders. The two named obstructions are: (a) the Tate-Casimir obstruction Lemma 5.212, which prevents the strict diagonal cotangent kernel from converging in $\widehat{\mathfrak{h}}\widehat{\otimes}\mathfrak{h}^\vee_{\mathrm{cont}}$ and hence forces a weighted-Tate envelope on the propagator coefficient module; (b) the QME obstruction class of Problem 5.73, whose vanishing in the weighted-Tate envelope is the unreduced one-antifield quantum lift. Before (a) and (b) are fixed, (6) remains a formal-local generating functional, and the perturbative phase space is the derived MC stack of Remark 2.14.

Remark 2.14 *(Phase space).* Circle-regularizing one topological direction gives $\mathbb{R} \times S^1 \times Y_{\mathrm{hol}}$ with Y_{hol} holomorphic symplectic, and $F_\alpha = 0$ becomes the Maurer–Cartan problem for Hamiltonian vector fields on $S^1 \times Y_{\mathrm{hol}}$. The perturbative phase space is the derived cotangent stack

$$T^*_{\mathrm{der}}\mathrm{MC}\big(\Omega^*(S^1)\widehat{\otimes}\Omega^{0,*}(Y_{\mathrm{hol}}, \mathrm{Ham}(Y_{\mathrm{hol}})), D, \{-,-\}\big),$$

collapsing on the underived tangent at the trivial solution to the ordinary cotangent of the linearized deformation complex. The analytic data of Remark 2.13 fixes this regularized model to a propagator-grade theory.

The closed BV theory of the local model is the Hamiltonian BF theory on $\widehat{\mathbb{C}^2}_0$ with fields valued in the shifted-cotangent Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ and master action $S = \int \langle \beta, D\alpha + \frac{1}{2}\{\alpha, \alpha\}\rangle dz_1 dz_2$, forced by Darboux + cyclic trace + linearity in the cotangent partner. The open boundary at a stack of N Dirac branes fixes the Chan–Paton substitution $z_i \mapsto \phi_i \in \mathfrak{gl}_N$ with moment map $\mu(\phi_1, \phi_2) = [\phi_1, \phi_2]$; the open BV theory is the derived commuting variety $[\mu^{-1}_{\mathrm{der}}(0)/\mathrm{GL}_N]$, constructed in the section below.

3 The derived commuting variety stack at N Dirac branes

The Dirac brane stack constructed here is the open E_1-algebra $A^{\mathrm{cl}}_{\partial,N}$ of Theorem 5.6; its $\mathfrak{gl}_N$-equivariant trace Hochschild subalgebra is Theorem 5.10.

3.1 *The Dirac brane stack*

The primitive open object is the open category C^{op}_∂ on the Costello–Gwilliam brane-link boundary ∂X; a chosen vacuum b gives the boundary algebra $A_b = \mathrm{End}_{C^{\mathrm{op}}_\partial}(b)$. The formal Darboux coordinate algebra $A^{\mathrm{coord}}_b = \widehat{\mathcal{O}}_{\mathbb{C}^2,b}$ contains the Hamiltonians, and at a stack of N Dirac branes the boundary algebra used in Hochschild cochains is $A^{\mathrm{cl}}_{\partial,N}$. The formal-Darboux trace theorem is stated in this coordinate chart of the primitive open object.

A holomorphic Hamiltonian $f \in \mathbb{C}[[z_1, z_2]]$ on the formal symplectic polydisk $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$ is infinite-dimensional. The finite-rank reduction used here is the open boundary that names the two normal coordinates: a stack of N Dirac branes on the topological line

$$L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\} \subset X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2_{z_1,z_2}$$

realises the Dirac brane stack as the derived commuting variety stack $[\mu^{-1}_{\mathrm{der}}(0)/GL_N]$ with moment map $\mu(\phi_1, \phi_2) = [\phi_1, \phi_2]$. The transverse position is the Chan–Paton pair

$$(\phi_1, \phi_2) \in \mathfrak{gl}_N \oplus \mathfrak{gl}_N$$

on the holomorphic-symplectic factor (Pantev–Toën–Vaquié–Vezzosi shifted symplectic structures [44]; the brane realization is Calaque [45]; the BV/Koszul realisation of the moment-map zero fibre is Costello [12, Ch. 5]). A polynomial Hamiltonian f has trace evaluation

$$J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$$

on this chart; a formal $f \in \mathfrak{h}$ has the same expression only in the completed brane algebra. This is the formal Darboux coordinate of the trace-sector map from the bulk $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_{b_p}$ to the brane-side E_1-Hochschild cochain complex $C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$. The full derived-centre identification is the Costello–Gwilliam/Deligne theorem under its hypotheses; the trace calculation uses only the formal Darboux polydisk at the brane point and the Chan–Paton substitution

$$z_1 \rightsquigarrow \phi_1, \qquad z_2 \rightsquigarrow \phi_2, \qquad \phi_i \in \mathfrak{gl}_N,$$

sending holomorphic coordinates to transverse Chan–Paton matrices. The symplectic potential, moment map, Dirac constraint, BV differential, and gauge-invariant operator algebra are all determined by this substitution.

The substitution forces the open BV theory. The pullback of $dz_1 \wedge dz_2$ along the substitution, traced over Chan–Paton, is

$$\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2), \qquad \theta_N = \mathrm{Tr}(\phi_1\, d\phi_2).$$

GL_N-conjugation acts symplectically. Its moment map is

$$\mu(\epsilon) = -\,\mathrm{Tr}(\epsilon[\phi_1, \phi_2]), \qquad \epsilon \in \mathfrak{gl}_N,$$

so the first-class constraint is

$$[\phi_1, \phi_2] = 0.$$

On this locus the Chan–Paton pair commutes and the N branes sit at N points of the polydisk. With multiplier $A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N$ the Dirac first-order action is

$$S_\partial = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big), \tag{7}$$

and the constraint is realized by the Koszul derived zero-fibre model for the moment-map equation [12, Ch. 5]. Introduce the odd matrix-valued antifield

$$\psi \in \mathfrak{gl}_N[1].$$

It is the BV antifield $\psi = A^*$ of the Lagrange multiplier A. It is independent of ϕ_1, ϕ_2. Let Q be the derivation defined on generators by

$$Q\phi_1 = Q\phi_2 = 0, \qquad Q\psi = [\phi_1, \phi_2]. \tag{8}$$

The displayed equation is the Koszul resolution of $\mu^{-1}(0)$ on the BV side: every matrix entry of the commutator equals $Q\psi^i{}_j$ and is therefore Q-exact, so the moment-map ideal vanishes in cohomology. The matrix entries of $[\phi_1, \phi_2]$ have syzygies in $\mathbb{C}[\phi_1{}^i{}_j, \phi_2{}^i{}_j]$ for $N \geq 2$; the construction is the derived zero fibre of the moment map. Its ghost-zero invariant cohomology

$$\mathcal{O}\big([\mu_{\mathrm{der}}^{-1}(0)/GL_N]\big) = H^0\big((\mathbf{S}^\bullet \mathfrak{gl}_N^{\oplus 2} \otimes \Lambda^\bullet \mathfrak{gl}_N[1])^{GL_N}, Q\big)$$

identifies the structure sheaf of the derived commuting variety stack with the GL_N-invariant cohomology of the Koszul complex; the degree-zero classical invariant ring of the commuting-pair quotient is the bottom row of this identification. The degree-zero Hamiltonian trace comparison ignores higher Koszul homology; that homology enters only through explicit invocation. The ghost-zero invariant ring is

$$\mathcal{R}_N = \mathbb{C}[\phi_1{}^i{}_j, \phi_2{}^i{}_j] \otimes \Lambda(\psi^i{}_j), \qquad H^0(\mathcal{R}_N^{GL_N}, Q).$$

Lemma 3.14 produces (7) from the shifted open Chern–Simons coefficients; Lemma 3.15 verifies the moment-map content; Proposition 3.16 verifies that ghost-zero truncation is consistent. The action (7), the differential (8), and $\mathcal{R}_N^{GL_N}$ all come from the Chan–Paton substitution and moment-map reduction.

Koszul independence of word order for J. $J(f)$ depends only on the symbol f. Adjacent letters in any trace word commute up to a Q-exact correction,

$$Q\ \mathrm{Tr}(\psi\, v\, u) = \mathrm{Tr}([\phi_1, \phi_2]\, v\, u),$$

so the difference of the two adjacent orderings is the BRST image of an antifield insertion. Cyclic invariance and this Koszul homotopy reduce every cyclic word in ϕ_1, ϕ_2 to a symmetric symbol $z_1^k z_2^l$: the trace family $\{J(f)\}$ is the connected single-trace primitive content of the Chan–Paton pair, the full stable GL_N-invariant ring is the polynomial algebra on this primitive (Procesi–Razmyslov first fundamental theorem), and the closed sector $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ acts on the Hamiltonian trace sector through J.

LEMMA 3.1 *(Trace realisation of the formal polydisk).* In the stable connected ghost-zero open observables $H^0(\mathcal{R}_N^{GL_N}, Q)$, the substitution map

$$J: \mathbb{C}[z_1, z_2] \longrightarrow H^0(\mathcal{R}_N^{GL_N}, Q)\big|_{\text{stable conn.}}, \qquad f \longmapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)},$$

is surjective with kernel exactly the constant line $\mathbb{C}\cdot 1 \subset \mathbb{C}[z_1, z_2]$. Procesi–Razmyslov first fundamental theorem identifies the full stable GL_N-invariant ring with the polynomial algebra on the trace family $\{J(f) : f \in \bar{A}\}$, so every gauge-invariant operator is a polynomial in trace observables, while the primitive Hamiltonian sector $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ acts through J.

Proof. Procesi–Razmyslov [17][18] identifies $\mathbb{C}[\mathfrak{gl}_N \oplus \mathfrak{gl}_N]^{GL_N}$ with the algebra generated by traces of noncommutative monomials in two letters. Cyclic invariance equates traces of cyclically equivalent words; the Q-exactness of (8) equates traces differing by one adjacent transposition. Iterating, every cyclic word $\mathrm{Tr}(w(\phi_1, \phi_2))$ reduces to a symmetric symbol $\mathrm{Tr}(\phi_1^k \phi_2^l) = J(z_1^k z_2^l)$ in $H^0(\mathcal{R}_N^{GL_N}, Q)$. Stable connected linear independence over N of these symmetric symbols is standard. The constant Hamiltonian substitutes to $N \cdot I$, a scalar multiple of the identity, killed by passage to the connected stable class. $\square$

The substitution-realised Lie algebra image of $\bar{A}$ inside the open trace cohomology generates the stable open trace algebra.

Example 3.2 *($N = 1$: the geometric prototype).* At $N = 1$, $[\phi_1, \phi_2] = 0$ holds tautologically, so the Dirac constraint imposes nothing. The Koszul field ψ is closed, not contractible; it gives the decoupled scalar derived direction excluded from the Hamiltonian trace sector. The Chan–Paton pair $(\phi_1, \phi_2) \in \mathbb{C}^2$ is a chosen point of the formal polydisk, and

$$J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)} = f(\phi_1, \phi_2) \in \mathbb{C}$$

is ordinary scalar evaluation modulo the constant for polynomial f. Formal f are evaluated after completion at the chosen point. J reduces to the closed-string evaluation map $f \mapsto f(p)$ at a chosen polydisk point p, and the Procesi–Razmyslov invariant theory reduces to scalars. Every later non-commutative refinement of J restricts on the zero-ψ trace sector to this scalar evaluation when $N = 1$; the matrix structure enters with the brane stack.

LEMMA 3.3 *(Cyclic framing branch and the framed noncommutative deformation).* Let

$$X_N^{\mathrm{cyc}} = \{(\phi_1, \phi_2, v) \in \mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathbb{C}^N : [\phi_1, \phi_2] = 0,\ \mathbb{C}[\phi_1, \phi_2]\cdot v = \mathbb{C}^N\}$$

be the open locus of cyclic stable framed pairs. The GL_N-action on X_N^{cyc} is free, and the underived quotient

$$X_N^{\mathrm{cyc}}/GL_N \;\cong\; \mathcal{H}\mathrm{ilb}^N(\mathbb{C}^2)$$

is the Nakajima ADHM presentation of the Hilbert scheme of N points on $\mathbb{C}^2$ [46]. The unframed derived commuting stack does not become underived on this locus: the cokernel of the unframed moment-map differential is dual, under the trace pairing, to the commutant of (ϕ_1, ϕ_2), and on a cyclic point this commutant is $\mathbb{C}[z_1, z_2]/I$, not zero in general. Thus the Koszul antifield $\psi \in \mathfrak{gl}_N[1]$ records the excess derived structure of the unframed moment-map presentation. The smoothness of $\mathcal{H}\mathrm{ilb}^N(\mathbb{C}^2)$ belongs to the cyclic framed quotient, or equivalently to the full framed ADHM presentation.

The Nekrasov–Schwarz noncommutative Hilbert scheme is the framed ADHM/noncommutative deformation [47]. It is not the finite-dimensional fibre $[\phi_1, \phi_2] = \hbar I_N$, since $\mathrm{Tr}[\phi_1, \phi_2] = 0$ in $\mathfrak{gl}_N$. Its central parameter is represented by the framing term in the ADHM moment map, or by the corresponding projective central extension of the Weyl action. The Capelli scalar $\hbar N[\bar{c}]$ of Example 1.4 is this projective Lie anomaly of the trace representation, not the trace of an ordinary finite-rank commutator.

Proof. Freeness of the GL_N-action on X_N^{cyc}: if $g \in GL_N$ fixes (ϕ_1, ϕ_2, v), then $g\phi_i v = \phi_i v$ and $gv = v$, so by cyclicity g fixes a spanning set of $\mathbb{C}^N$, forcing $g = I$ [46, Theorem 1.9]. The identification $X_N^{\mathrm{cyc}}/GL_N = \mathcal{H}\mathrm{ilb}^N(\mathbb{C}^2)$ sends $[(\phi_1, \phi_2, v)]$ to the ideal $I_{(\phi,v)} = \{f \in \mathbb{C}[z_1, z_2] : f(\phi_1, \phi_2)v = 0\}$, which has colength N by cyclicity and is a point of the Hilbert scheme; inversely, an ideal I gives (ϕ_1, ϕ_2, v) on $\mathbb{C}[z_1, z_2]/I$ by multiplication and the class of 1 [46, Theorem 1.14].

For the unframed moment map the differential $d\mu\colon \mathfrak{gl}_N^{\oplus 2} \to \mathfrak{gl}_N$ is

$$(\delta\phi_1, \delta\phi_2) \longmapsto [\delta\phi_1, \phi_2] + [\phi_1, \delta\phi_2].$$

Its adjoint for the trace pairing sends $T \in \mathfrak{gl}_N$ to $([\phi_2, T], [T, \phi_1])$. Hence the cokernel of $d\mu$ is dual to the commutant of the pair. If v is cyclic and I is the corresponding colength-N ideal, this commutant is the image of $\mathbb{C}[z_1, z_2]/I$. It has dimension N, so cyclicity cannot contract the Koszul field ψ in the unframed derived commuting stack. The smooth Hilbert scheme is the quotient of the cyclic framed commuting-pair variety; the full Nakajima ADHM presentation gives the regular framed moment map in which the framing variables carry the trace direction.

The finite-rank equation $[\phi_1, \phi_2] = \hbar I_N$ has no solution for $\hbar \neq 0$. Nekrasov–Schwarz replaces it by the noncommutative/framed ADHM relation, where the scalar term is cancelled by the framing contribution. On the trace sector this same scalar appears as the projective cocycle $[\bar{c}] = [\omega(z_1, z_2)c_1 \wedge c_2] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})[[\hbar]]$ of Theorem 6.3. Theorem 6.6 therefore computes the projective Lie anomaly of the Weyl-ordered trace representation. After multiplication by N, this anomaly is the $U(1)$ centre-of-mass charge, not a classical matrix commutator. $\square$

J is exact at the level of the scalar trace algebra. The full P_0-factorization centre requires additional cotangent centrality homotopies: the chain-level morphism from closed observables on L to the open factorization centre has a structural obstruction.

Remark 3.4 *(Cotangent obstruction beyond J).* Lemma 3.1 resolves the scalar-trace generator question: scalar closed insertions $F = J(f)$ commute with every open trace class up to the moment-map operation $\{J(f), J(g)\} = J(\{f, g\})$ of Theorem 4.4 below. The closed Hamiltonian BF sector on the polydisk and the open brane stack on L generate factorization algebras of classical observables, denoted $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}$ and $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}$ (constructed in §2.1, §3.4); the P_0-factorization centre $\mathrm{Z}^{P_0}_{\mathrm{fact}}$ of a factorization algebra is its centralizer for the shifted Poisson bracket and the factorization product (Problem 5.49). The strengthening sought beyond Lemma 3.1 is a chain-level morphism

$$\Phi_{\mathrm{cl}} \colon \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow Z^{P_0}_{\mathrm{fact}}\big(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}\big),$$

asking for centrality homotopies linking closed insertions to *arbitrary* open insertions, in particular to the principal-part cotangent currents Θ_ρ (the boundary current attached to a residue $\rho \in \mathfrak{h}^\vee_{\mathrm{cont}}$ by smearing, defined in Theorem 5.186 below). Substitution into the Chan–Paton pair encodes the symbol-level information of f. The cotangent-direction antifield content of the closed BV algebra is controlled by the unreduced cotangent centrality homotopy data of Problem 5.49 together with the Bochner–Martinelli obstruction vector $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$, a six-component cohomology vector blocking strict all-window support-local current transfer to the brane line (Theorem 8.40). J is the scalar part of Φ_{cl}; the cotangent part is controlled by the obstruction classes above.

3.2 Hamiltonian sector and stable large N

The trace substitution J produces an open trace algebra at every finite N; the closed Hamiltonian algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ is its image under J only after a stable trace stabilization is performed. Four hypotheses control that stabilization: stable invariant theory, BRST symmetrization, primitive cyclic projection, and a finite-N window in which the projection is a chain map. Inside that window the Procesi–Razmyslov trace algebra collapses to commutative monomials, the connected primitive class is read by Loday–Quillen–Tsygan, and the resulting completed coefficient ring is the formal Hamiltonian envelope of $\mathfrak{h}$. Outside the window finite-N trace identities re-enter and break the projection. Strict identification of the raw finite-N operator algebras with the formal envelope requires an open-trace A_∞-Koszul homotopy datum (level 5 of §4.2), which is an additional hypothesis beyond the stable hypotheses.

Coefficient algebras: $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ on the polynomial side and $\mathfrak{h} = \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ on the completed side, with $\mathfrak{h}^\vee = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ its continuous dual; $\widehat{\mathbf{S}}$ denotes the completed symmetric algebra and $C^\bullet_{\mathrm{CE,cont}}$ the matched continuous CE complex. Constants are quotiented out per §1.5: a constant Hamiltonian has zero Hamiltonian vector field. On the open side $\mathrm{Tr}_N(1) = N$ is the empty trace, a central scalar tracked through scalar reduction while retaining the operator unit; for a positive word-degree trace expression T, $\overline{T}$ denotes its connected class. Shifts follow the BV convention: an element of $V[1]$ has parity reversed and cohomological degree one lower than the corresponding element of V. The CE/PV cotangent block below uses its own coordinate grading: the odd BV cotangent generator $\lambda[1]$ has BV degree -1, while the dual CE coordinate u_λ is assigned PV function degree 0; "dual" in that block means the CE-suspended cotangent coordinate dual in this grading convention.

At a fixed brane point the reduced BV algebra is generated by two even matrices ϕ_1, ϕ_2 and the odd antifield $\psi = A^*$, with

$$Q\phi_1 = Q\phi_2 = 0, \qquad Q\psi = [\phi_1, \phi_2].$$

The moment-map constraint is imposed homologically; for $N \geq 2$ the commuting-pair equations form a nonregular sequence, so the construction is the derived zero fibre. On compactly supported brane variations the boundary terms vanish, so local operators in the interior are computed from the Koszul zero-fibre complex.

Stable large-N means the degreewise stable value of the invariant algebras under block restriction, accessed only after four hypotheses. Procesi–Razmyslov stable invariant theory [17][18]: stable invariants of two matrices are freely generated by cyclic trace words. In the connected sector the Koszul differential

$Q\psi = [\phi_1, \phi_2]$ makes adjacent transpositions Q-exact (Lemma 3.30), so ghost-zero cyclic words collapse on H^0 to commutative monomials $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$, $k + l > 0$ (Proposition 3.22, Corollary 3.32). Connected-trace projection: the Loday–Quillen–Tsygan cyclic-homology calculation [2][3][24] gives the primitive single-cycle class; same-rank deletion has a nonzero commutator coderivation, and the finite-rank bridge is the separated-block zigzag $P^{\mathrm{sep}}_{M,K,L} = \beta_{M,K,L} P_{1\mathrm{cyc}} \alpha_{M,K,L}$ of Remark 3.7 below. Stable-window finite-N bound: the LQT projection is a chain map only for $N \geq \max(K, L + 2)$ (Theorem 3.6); outside this window finite-N trace identities re-enter and break the projection.

Identifying the finite-N operator algebras with the strict formal Hamiltonian envelope requires the open-trace A_∞-Koszul homotopy datum $(A_{\mathrm{op}}, C_{\mathrm{op}}, B_{\mathrm{CE/PV}}, \tau_{\mathrm{op}})$ of Definition 4.26, proved by Theorem 4.27; its cohomology controls the bar-cobar quasi-isomorphism $\Omega\mathrm{Bar}(A_{\mathrm{op}}) \xrightarrow{\sim} B_{\mathrm{CE/PV}}$ and the dual CE-side enveloping comparison $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \simeq A^!_{\mathrm{op}}$ of clause (3) of Theorem 5.5. Thus the connected-trace calculation of Theorem 5.6 has an HC_{*-1}-normalization precisely after the displayed finite-window and Milnor conditions have been imposed.

THEOREM 3.5 *(Loday–Quillen–Tsygan stable trace theorem).* Let A be a unital associative algebra over a field of characteristic zero and let $\mathfrak{gl}_\infty(A) = \varinjlim_N \mathfrak{gl}_N(A)$. As connected graded-commutative Hopf algebras,

$$H^{\mathrm{CE}}_*\big(\mathfrak{gl}_\infty(A); \mathbb{C}\big) \cong \mathbf{S}\big(HC_{*-1}(A)\big).$$

Equivalently,

$$\mathrm{Prim}\, H^{\mathrm{CE}}_*\big(\mathfrak{gl}_\infty(A); \mathbb{C}\big) \cong HC_{*-1}(A).$$

THEOREM 3.6 *(Stable Eulerian finite-window LQT projection).* Let

$$A_\psi = T(x, y, \psi), \qquad d\psi = xy - yx,$$

with

$$\mathrm{wt}(x) = (1, 0), \qquad \mathrm{wt}(y) = (0, 1), \qquad \mathrm{wt}(\psi) = (1, 1).$$

For a Koszul-weight bound K, let $A_{\psi,K} = A_\psi / F^{>K} A_\psi$, and let L bound CE length. In the stable block-sum Hopf CE window $\mathsf{H}^{\mathrm{st}}_{K,L}(A_\psi)$, with product induced by block diagonal sum and coproduct the CE coproduct, put

$$P_{1\mathrm{cyc}} = \log^*(\mathrm{id}) = \sum_{m=1}^{L} \frac{(-1)^{m-1}}{m} (\mathrm{id} - u\epsilon)^{*m}.$$

Then $P_{1\mathrm{cyc}}$ is a chain idempotent whose image is the stable single-cycle trace span $\mathrm{im}\,\Theta_{N,K,L}$. For

$$N \geq \max(K, L + 2),$$

the finite-window LQT projection is the chain map

$$\lambda_{\mathrm{LQT},K,L} = \Theta^{-1}_{N,K,L} P_{1\mathrm{cyc}} : \mathsf{H}^{\mathrm{st}}_{K,L}(A_\psi) \longrightarrow CC_{\mathrm{red},\leq L}(A_{\psi,K})[1].$$

If one integer W controls both $K \leq W$ and $L \leq W$, this means $N \geq W + 2$. The scalar one-ψ line is a chain quotient only after quotienting the two-term scalar-Koszul subcomplex

$$\Theta_1(\psi) \longmapsto \Theta_1(xy - yx), \qquad [\psi] \longmapsto [xy - yx],$$

or after passing to homology.

Proof. The quotient $A_{\psi,K}$ is a dg quotient because $d\psi = xy - yx$ preserves the Koszul bidegree $(1,1)$. The trace cycle map $\Theta_{N,K,L}$ sends a reduced cyclic word $a_0|\cdots|a_r$, $r+1 \le L$, to the usual cyclic matrix tensor

$$\sum_{i_0,\dots,i_r} (E_{i_0 i_1} a_0) \wedge \cdots \wedge (E_{i_r i_0} a_r),$$

and the matrix bracket gives the reduced cyclic differential with the standard graded cyclic signs. In particular $d_{\mathrm{CE}}\Theta_1(\psi) = \Theta_1(xy - yx)$.

The stable CE window is formed before projecting. Block diagonal sum is a chain product, the CE coproduct is a chain coproduct, and $\mathrm{id} - u\epsilon$ is a chain map; hence every convolution power $(\mathrm{id} - u\epsilon)^{*m}$ and therefore $P_{1\mathrm{cyc}}$ is a chain map. Stable invariant CE tensors are indexed by permutations of the CE factors, and a permutation decomposes uniquely into cycles. A single cycle is exactly a trace-cycle tensor in $\operatorname{im}\Theta_{N,K,L}$; a multicycle tensor is a block-sum product of such single cycles. The Hopf logarithm kills decomposable products because on a product of n positive single-cycle generators its coefficient is

$$\sum_{m=1}^{n} (-1)^{m-1}(m-1)!\,S(n,m),$$

the coefficient of $t^n/n!$ in $\log(1 + (e^t - 1)) = t$. This is 1 for $n = 1$ and 0 for $n > 1$, proving idempotence and the stated image. The bound $N \ge K$ removes trace identities in the displayed Koszul window, and $N \ge L + 2$ is the sufficient CE stable range in length at most L. □

Remark 3.7 *(Same-rank deletion vs. the LQT projection).* The LQT projection $P_{1\mathrm{cyc}}$ is the stable block-sum Hopf primitive projection, distinct from the fixed-rank operation P_{del} that keeps one-cycle permutation tensors and deletes multicycle tensors inside one exterior CE complex. For homogeneous a, b

$$P_{\mathrm{del}} d_{\mathrm{CE}}\big(\Theta_1(a) \wedge \Theta_1(b)\big) = \pm\Theta_1(ab - (-1)^{|a||b|}ba), \qquad d_{\mathrm{CE}} P_{\mathrm{del}}\big(\Theta_1(a) \wedge \Theta_1(b)\big) = 0.$$

Thus same-rank deletion has a nonzero chain defect when the graded commutator is nonzero. The theorem uses the stable block-sum Hopf primitive projection before returning to any finite rank. More precisely, set $B(K,L) = \max(K, L+2)$ and $M = L\,B(K,L)$. Let

$$\beta_{M,K,L} \colon \mathsf{H}^{\mathrm{st}}_{K,L}(A_\psi) \to C^{\mathrm{CE}}_*(\mathfrak{gl}_M(A_{\psi,K}))^{\mathrm{sep}}_{\le L}$$

be the normalized signed average over injections of the connected Hopf factors into the L labeled diagonal $B(K,L)$-blocks, and let

$$\alpha_{M,K,L} \colon C^{\mathrm{CE}}_*(\mathfrak{gl}_M(A_{\psi,K}))^{\mathrm{sep}}_{\le L} \to \mathsf{H}^{\mathrm{st}}_{K,L}(A_\psi)$$

forget the block labels and read back the stable block-sum Hopf word. Thus $\alpha_{M,K,L}\beta_{M,K,L} = \mathrm{id}$. Then

$$P^{\mathrm{sep}}_{M,K,L} = \beta_{M,K,L} P_{1\mathrm{cyc}} \alpha_{M,K,L}$$

is the strict finite-rank chain idempotent on the separated-block subcomplex. This zigzag is the finite-rank bridge. A same-rank chain-homotopy bridge has obstruction the collision coderivation

$$\Gamma_N = d_{\mathrm{same}} F_N - F_N d_\oplus,$$

whose first nonzero component is

$$\Gamma_N(\Theta_1^N(a) \wedge \Theta_1^N(b)) = \pm\Theta_1^N(ab - (-1)^{|a||b|}ba).$$

3.3 Open mixed brane states

The open state space is fixed by the trace substitution: every brane observable is a polynomial in (ϕ_1, ϕ_2) traced over the Chan-Paton index, modulo the moment-map first-class constraint $[\phi_1, \phi_2] = 0$. The Koszul derived zero-fibre model for that constraint, the Procesi–Razmyslov invariant theory of the surviving traces, and the local geometry of the brane line and skyscraper point split the open states into a topological factor along the line, a $\mathfrak{gl}_N$ Chan-Paton factor, and a Koszul-dual exterior factor on two odd derived normal coordinates. The matrix entries of $[\phi_1, \phi_2]$ have syzygies for $N \geq 2$; this is the derived zero fibre.

Koszul reduction of the formal-stalk chart

The Dirac brane stack carries the polynomial coordinate ring

$$\mathcal{R}_N^{\text{poly}} = \mathbb{C}[(\phi_1)^i{}_j, (\phi_2)^i{}_j], \qquad 1 \leq i, j \leq N,$$

together with the holomorphic symplectic form $\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2)$ and the GL_N-action by simultaneous conjugation. Conjugation is Hamiltonian with moment map $\mu_\epsilon = -\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$ (Lemma 3.15); its zero locus is the variety $\{[\phi_1, \phi_2] = 0\}$ of commuting matrix pairs.

Proposition 3.8 *(Derived zero-fibre and Procesi–Razmyslov trace ring).* The ghost-zero open observable algebra of the Dirac brane stack is the classical BV cohomology of the Koszul complex

$$\mathcal{R}_N = \mathcal{R}_N^{\text{poly}} \otimes \Lambda_{\mathbb{C}}(\psi^i{}_j), \qquad Q\psi^i{}_j = [\phi_1, \phi_2]^i{}_j, \qquad Q\phi_1 = Q\phi_2 = 0,$$

on GL_N-invariants:

$$\mathcal{O}^{\text{cl,gho}}_{\text{brane},N} = H^0(\mathcal{R}_N^{GL_N}, Q).$$

The GL_N-invariant subring $(\mathcal{R}_N^{\text{poly}})^{GL_N}$ is generated, by the Procesi–Razmyslov first fundamental theorem, by cyclic traces $\mathrm{Tr}\ w(\phi_1, \phi_2)$ of words w in two letters; passage to H^0 imposes the relations $Q\psi = [\phi_1, \phi_2] = 0$.

Proof. Since GL_N is reductive in characteristic zero, invariants is exact and commutes with cohomology, so $H^0(\mathcal{R}_N^{GL_N}, Q) = H^0(\mathcal{R}_N, Q)^{GL_N}$. The fields ϕ_1, ϕ_2 sit in ghost number zero and ψ in ghost number -1; $Q\psi = [\phi_1, \phi_2]$ is the standard Koszul differential of the $\mathfrak{gl}_N$-valued moment map, so classical BV theory identifies $H^0(\mathcal{R}_N, Q)$ with the polynomial functions on the scheme $\{[\phi_1, \phi_2] = 0\} \subset \mathfrak{gl}_N \oplus \mathfrak{gl}_N$. Procesi (1976) and Razmyslov (1974) prove that $(\mathcal{R}_N^{\text{poly}})^{GL_N}$ is generated by cyclic traces of words in (ϕ_1, ϕ_2); restriction to the zero-fibre imposes the bracket-vanishing relations. □

The Koszul complex $\mathcal{R}_N$ is the open ghost-zero state space at a single brane point. Two further data assemble it into a sheaf along the brane line L and over the formal holomorphic point: the topological factor along $\mathbb{R}_t$ and the holomorphic factor transverse to L inside $\widehat{\mathbb{C}^2}_0$.

Topological factor along the brane line

The brane line has trivial curvature and holonomy: its self-Floer complex is exact and wrapped generators are excluded by the local model. For a compact interval $I \subset \mathbb{R}_t$ and a compactly supported Hamiltonian perturbation, the local Fukaya complex $CF^*(\mathbb{R}_t, \mathbb{R}_t; I)$ is the Morse complex on I, and passage to local operators in the interior identifies it with the de Rham complex $\Omega^*(\mathbb{R}_t)$ acting by compactly supported variations. Wrapped generators at infinity are excluded by the local-model hypothesis.

Holomorphic factor at the brane point

The brane is a point inside $\widehat{\mathbb{C}^2}_0$; its holomorphic state space is the derived endomorphism algebra of the skyscraper $\mathcal{O}_p$. Koszul duality identifies it with the exterior algebra on the two derived normal coordinates whose ghost-zero shadows are the matrix coordinates ϕ_1, ϕ_2.

PROPOSITION 3.9 *(Affine skyscraper self-Ext).* Let $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $k = R/\mathfrak{m}$. The Yoneda algebra of the skyscraper at the origin is

$$\mathrm{Ext}^*_R(k, k) \cong \Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2), \qquad |\varepsilon_i| = 1,$$

the free graded-commutative algebra on two odd generators.

Proof. The pair (z_1, z_2) is a regular sequence in R; the affine Koszul resolution

$$K = \big(0 \to R\, e_{12} \xrightarrow{d_2} R\, e_1 \oplus R\, e_2 \xrightarrow{d_1} R \to 0\big), \quad d_1(ae_1 + be_2) = az_1 + bz_2, \quad d_2(e_{12}) = z_2 e_1 - z_1 e_2,$$

is a free R-resolution of k. Applying $\mathrm{Hom}_R(-, k)$ gives

$$0 \to k \xrightarrow{0} k\, \varepsilon_1 \oplus k\, \varepsilon_2 \xrightarrow{0} k\, \varepsilon_{12} \to 0,$$

because z_1, z_2 act by zero on k; hence $\mathrm{Ext}^p_R(k, k)$ is one-dimensional in degree 0, two-dimensional in degree 1, and one-dimensional in degree 2. The Yoneda product is computed by dualising the comultiplication of the Koszul dg-algebra $R \otimes \Lambda(e_1, e_2)$ with $d(e_i) = z_i$, under which $e_i \mapsto e_i \otimes 1 + 1 \otimes e_i$; duality forces $\varepsilon_i^2 = 0$, $\varepsilon_1\varepsilon_2 = -\varepsilon_2\varepsilon_1$, with $\varepsilon_{12} = \varepsilon_1\varepsilon_2$ spanning the top Ext group. □

Remark 3.10 *(Koszul duality and derived normal coordinates).* $\Lambda(\varepsilon_1, \varepsilon_2)$ is Koszul dual to the polynomial ring $\mathbb{C}[\phi_1, \phi_2]$ on two even generators in the point-brane normal model. For a stack of N branes the Chan–Paton coordinate cdga is the matrix-valued moment-map Koszul complex with $Q\psi = [\phi_1, \phi_2]$; the skyscraper Ext algebra gives the formal normal directions, not the full derived commuting-stack coordinate ring. The ghost-zero scalar coefficients χ_1, χ_2 of $\varepsilon_1, \varepsilon_2$ in §3.4 are, after Dirac normalization, the matrix coordinates ϕ_1, ϕ_2; the Procesi–Razmyslov trace ring of Proposition 3.8 reappears there as the ghost-zero observable algebra of the open BV theory.

Open-state factorisation along L

Combining the three factors, the local open state space at the brane is

$$\mathcal{S}_L^{\mathrm{open}} = \Omega^*(\mathbb{R}_t) \otimes \mathrm{Ext}^*_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p, \mathcal{O}_p) \otimes \mathfrak{gl}_N[1] = \Omega^*(\mathbb{R}_t) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1].$$

The shift [1] is the BV ghost-shift of Definition 1.12: a form of de Rham degree p and Grassmann degree q sits in ghost number $p + q - 1$; the brane gauge field has $p = 1, q = 0$ (ghost 0), the two scalar fields have $p = 0, q = 1$ (ghost 0), and the remaining components carry nonzero ghost number, encoding the BV antifields and ghosts of the GL_N-conjugation gauge symmetry of the Chan–Paton pair.

3.4 Open mixed brane field theory

Under the Dirac brane stack hypotheses of §1.4 — the $\mathfrak{gl}_N$-valued holomorphic-symplectic pair (ϕ_1, ϕ_2) carrying the form $\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2)$ and its Liouville one-form $\theta_N = \mathrm{Tr}(\phi_1\, d\phi_2)$ (Lemma 3.15), the moment-map zero-fibre $[\phi_1, \phi_2] = 0$ coupled by a Lagrange multiplier $A \in \Omega^1(\mathbb{R}_t) \otimes \mathfrak{gl}_N$ along the brane line, and the GL_N-conjugation gauge symmetry generated by that moment map — the open

BV action, BV differential, and ghost-zero classical observable algebra are forced. The action is the Dirac first-class functional (9); the BV differential is the Koszul derived zero-fibre model for the moment-map equation (10) with antifield $\psi = A^*$; the ghost-zero classical observables are the Procesi–Razmyslov derived zero-fibre of Proposition 3.8. Together they assemble into the formal first-order Chern–Simons theory (11) on $\mathbb{R}_t \times \mathbb{C}^{0|2}$, with Dirac normalization fixing the signs.

Action: Chan–Paton Liouville plus Dirac multiplier

The Chan–Paton symplectic form $\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2)$ admits the Liouville one-form $\theta_N = \mathrm{Tr}(\phi_1\, d\phi_2)$. The brane moment map for simultaneous GL_N-conjugation is $\mu_\epsilon = -\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$ (Lemma 3.15); coupling its zero-fibre constraint by a Lagrange multiplier $A \in \Omega^1(\mathbb{R}_t) \otimes \mathfrak{gl}_N$ along the brane line gives the Dirac first-class action

$$S_\partial = \int_{\mathbb{R}_t} \theta_N - \mu_A = \int_{\mathbb{R}_t} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big). \tag{9}$$

The variation of A imposes the moment-map zero-fibre, and the variation of ϕ_i imposes the Hamiltonian flow generated by μ_A along the brane line; this is the Dirac constraint reduction of the formal-stalk chart.

BV differential: Koszul derived zero-fibre model for the moment-map equation

The first-class constraint $[\phi_1, \phi_2] = 0$ is realized by the derived zero-fibre model with an antifield $\psi = A^* \in \mathfrak{gl}_N[1]$ and the Koszul differential

$$Q\psi^i{}_j = [\phi_1, \phi_2]^i{}_j, \qquad Q\phi_1 = Q\phi_2 = 0. \tag{10}$$

The matrix entries of $[\phi_1, \phi_2]$ have syzygies for $N \geq 2$; this is the derived zero fibre of the moment map. Its ghost-zero invariant cohomology identifies the open classical observable algebra with $H^0(\mathcal{R}_N^{GL_N}, Q)$ (Proposition 3.8): the cotangent direction of the moment-map zero-fibre is exactly the BV antifield, the Koszul differential is exactly the BV differential.

Cyclic trace and BV pairing

The cyclic trace pairing comes from two ingredients: the Berezin integral on the exterior factor $\Lambda(\varepsilon_1, \varepsilon_2)$, which projects onto the top class $\varepsilon_1\varepsilon_2$ (residue dual to $dz_1 \wedge dz_2$); the de Rham integral on the brane line, which projects onto the top form. Together they fix the open cyclic trace

$$\mathrm{Tr}_{\mathrm{open}}(\eta \otimes u \otimes M) = \int_{\mathbb{R}_t} \eta\, [\varepsilon_1\varepsilon_2]u\ \mathrm{Tr}_N(M),$$

where $[\varepsilon_1\varepsilon_2]u$ is the coefficient of the top Ext class. The de Rham integral carries degree -1 and the Ext residue degree -2; after the field shift $[1]$, the induced BV pairing has degree -1, as required by Definition 1.12.

The total open BV field is therefore $\alpha \in \Omega^*(\mathbb{R}_t) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1]$, and a single first-order Chern-Simons action records all three pieces at once: the Liouville term, the moment-map multiplier, and the Koszul antifield.

Proposition 3.11 *(Open BV action from the formal-stalk chart).* The open BV theory of the mixed brane is the formal first-order Chern-Simons theory

$$S(\alpha) = \int_{\mathbb{R}_t \times \mathbb{C}^{0|2}} \tfrac{1}{2}\,\mathrm{Tr}(\alpha\, d\alpha) + \tfrac{1}{3}\,\mathrm{Tr}\ \alpha^3, \qquad d = d_{\mathrm{dR}}, \tag{11}$$

on the open BV field $\alpha \in \Omega^*(\mathbb{R}_t) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1]$, with the Berezin convention $\int_{\mathbb{C}^{0|2}} d\varepsilon_1 d\varepsilon_2 : \varepsilon_1\varepsilon_2 \mapsto 1$ and zero differential on the exterior factor. Its restriction to the ghost-zero coefficient fields, after the linear Dirac normalization $(A_{\mathrm{CS}}, \chi_1, \chi_2) \mapsto (-A, -\phi_1, \phi_2)$, is the Dirac action S_∂ of (9); its BV differential Q restricted to the antifield $\psi = A^*$ is the moment-map Koszul differential (10); its ghost-zero classical observable algebra is the Procesi–Razmyslov derived zero-fibre $H^0(\mathcal{R}_N^{GL_N}, Q)$ of Proposition 3.8.

Proof. The reduction of (11) to the ghost-zero coefficient sector is Lemma 3.14; the identification of the BV differential with the moment-map Koszul differential is Lemma 3.15; the identification of the ghost-zero classical observables with the derived zero-fibre is Proposition 3.16 combined with Proposition 3.8. □

Construction 3.12 *(The open field from the Ext basis).* Write $\varepsilon_{12} = \varepsilon_1\varepsilon_2$. Before imposing ghost number, an open field is a finite sum

$$\alpha = a_0 + a_1\varepsilon_1 + a_2\varepsilon_2 + a_{12}\varepsilon_{12},$$

where each a_I is a de Rham form on $\mathbb{R}_t$ with values in $\mathfrak{gl}_N$, shifted by $[1]$. The ghost-number-zero components are

$$A_{\mathrm{CS}} \in \Omega^1(\mathbb{R}_t) \otimes \mathfrak{gl}_N, \qquad \chi_1, \chi_2 \in \Omega^0(\mathbb{R}_t) \otimes \mathfrak{gl}_N,$$

with coefficient form $\alpha_0^{\mathrm{coef}} = A_{\mathrm{CS}} + \chi_1\varepsilon_1 + \chi_2\varepsilon_2$. The Dirac variables of the trace-sector chart are

$$\phi_1 = -\chi_1, \qquad \phi_2 = \chi_2, \qquad A = -A_{\mathrm{CS}},$$

fixed once and for all so that the Liouville one-form is $\mathrm{Tr}(\phi_1 d\phi_2)$ and the moment-map multiplier term is $A[\phi_1, \phi_2]$. The Berezin integral kills every monomial except the coefficient of ε_{12}, so each ghost-zero monomial of (11) is obtained by multiplying the three displayed summands and keeping exactly one real one-form and the product $\varepsilon_1\varepsilon_2$.

Remark 3.13 *(Ghost shift).* Total degree of an $(\Omega^p, \varepsilon\text{-degree } q)$ monomial is $p + q$; after the BV shift $[1]$ it sits in ghost number $p + q - 1$. The brane gauge field $(p = 1, q = 0)$ and the two scalar fields $(p = 0, q = 1)$ sit in ghost zero; the components proportional to ε_i record the motion of the brane in the two derived normal directions of the skyscraper.

3.4.1 Ghost-zero open field content

Lemma 3.14. With the Berezin convention $\int_{\mathbb{C}^{0|2}} \varepsilon_1\varepsilon_2 = 1$, the restriction of the BV action to the ghost-zero coefficient fields, written in the Dirac variables of Construction 3.12, is, up to a boundary term on $\mathbb{R}$,

$$S(\alpha_0) = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Proof. Only monomials containing one real one-form and the factor $\varepsilon_1\varepsilon_2$ survive integration over $\mathbb{R} \times \mathbb{C}^{0|2}$. In coefficient coordinates $\alpha_0^{\mathrm{coef}} = A_{\mathrm{CS}} + \chi_1\varepsilon_1 + \chi_2\varepsilon_2$, the tensor-product Koszul sign gives

$$\frac{1}{2}\int_{\mathbb{R}} \mathrm{Tr}\big(-\chi_1\, d\chi_2 + \chi_2\, d\chi_1\big) \equiv -\int_{\mathbb{R}} \mathrm{Tr}(\chi_1\, d\chi_2),$$

modulo an exact one-form. In the cubic term the six orderings of $A_{\mathrm{CS}}, \chi_1\varepsilon_1, \chi_2\varepsilon_2$ reduce, by cyclicity of the trace and $\varepsilon_2\varepsilon_1 = -\varepsilon_1\varepsilon_2$, to

$$3\,\mathrm{Tr}(A_{\mathrm{CS}}\chi_1\chi_2 - A_{\mathrm{CS}}\chi_2\chi_1).$$

The coefficient $\frac{1}{3}$ in the Chern-Simons action therefore produces $\mathrm{Tr}(A_{\mathrm{CS}}[\chi_1, \chi_2])$. Substituting $\phi_1 = -\chi_1$, $\phi_2 = \chi_2$, and $A = -A_{\mathrm{CS}}$ gives the displayed Dirac-normalized action. □

Lemma 3.15 *(Dirac reduction of the brane phase space).* Let

$$\mathcal{P}_N = \mathfrak{gl}_N \oplus \mathfrak{gl}_N$$

with coordinates (ϕ_1, ϕ_2) and symplectic form

$$\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2).$$

With the convention $\iota_{X_f}\Omega_N = -df$, simultaneous conjugation by $\mathfrak{gl}_N$ is Hamiltonian with moment map components

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

They satisfy the first-class identity

$$\{\mu_\epsilon, \mu_\eta\} = -\mu_{[\epsilon,\eta]}.$$

Thus this comoment is anti-equivariant for the displayed Poisson convention, equivalently equivariant for the opposite Lie algebra; the zero locus and the generated constraint ideal are unchanged. The action of Lemma 3.14 is the corresponding Dirac first-order action

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2) - \mu_A,$$

and varying A imposes the constrained phase space

$$\mu^{-1}(0)/GL_N = \{[\phi_1, \phi_2] = 0\}/GL_N.$$

Proof. The symplectic form gives

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j.$$

For $v_\epsilon(\phi_i) = [\epsilon, \phi_i]$, cyclicity of the trace gives

$$d\mu_\epsilon(\delta\phi_1, \delta\phi_2) = -\mathrm{Tr}\big(\epsilon([\delta\phi_1, \phi_2] + [\phi_1, \delta\phi_2])\big) = -\Omega_N(v_\epsilon, (\delta\phi_1, \delta\phi_2)).$$

Thus $X_{\mu_\epsilon} = v_\epsilon$. The moment-map bracket is therefore the anti-equivariance identity $\{\mu_\epsilon, \mu_\eta\} = -\mu_{[\epsilon,\eta]}$. Substituting $\epsilon = A(t)$ gives

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2) - \mu_A = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big),$$

and the A-equation is $[\phi_1, \phi_2] = 0$. □

Proposition 3.16 *(Classical BV truncation).* The subspace of ghost-number-zero coefficient fields

$$\alpha_0^{\mathrm{coef}} = A_{\mathrm{CS}} + \chi_1\varepsilon_1 + \chi_2\varepsilon_2$$

with all nonzero ghost-number components set to zero is a consistent classical truncation of the open BV theory. After the linear Dirac-normalization of Construction 3.12, the induced classical action is the first-order gauged matrix model of Lemma 3.14.

Proof. Decompose the BV field as $\alpha = \sum_r \alpha^{(r)}$ by ghost number. The differential $d_{\mathbb{R}}$ and the product on $\Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N$ preserve total degree, and the BV action has ghost number zero. Hence every monomial in the Euler-Lagrange expression for a component of ghost number $r \neq 0$ contains at least one field component of nonzero ghost number when evaluated at $\alpha = \alpha_0^{\mathrm{coef}}$. Setting all such components to zero therefore makes the discarded equations vanish. The remaining equation is exactly the variation of the ghost-zero action computed in Lemma 3.14.

Concretely, the full Euler-Lagrange expression is the curvature

$$F(\alpha) = d\alpha + \alpha^2.$$

For $\alpha_0^{\mathrm{coef}} = A_{\mathrm{CS}} + \chi_1\varepsilon_1 + \chi_2\varepsilon_2$ on a line,

$$F(\alpha_0^{\mathrm{coef}}) = (d\chi_1 + [A_{\mathrm{CS}}, \chi_1])\varepsilon_1 + (d\chi_2 + [A_{\mathrm{CS}}, \chi_2])\varepsilon_2 + [\chi_1, \chi_2]\varepsilon_1\varepsilon_2.$$

The BV pairing keeps exactly real-form degree one and $\varepsilon_1\varepsilon_2$. Hence the displayed three components pair exactly with the coefficient variations of $A_{\mathrm{CS}}, \chi_2, \chi_1$ respectively, while variations of nonzero ghost number see complementary curvature components that are zero for α_0^{coef} by real-form degree or ε-degree. □

The truncated action is gauge-invariant under

$$\delta_\varepsilon \phi_i = [\varepsilon, \phi_i], \qquad \delta_\varepsilon A = d\varepsilon + [A, \varepsilon], \qquad \varepsilon \in \Omega^0(\mathbb{R}_t) \otimes \mathfrak{gl}_N :$$

the $d\varepsilon$ variation of $\mathrm{Tr}(\phi_1 d\phi_2)$ produces $-\mathrm{Tr}(d\varepsilon[\phi_1, \phi_2])$, which cancels the $d\varepsilon$ variation of $\mathrm{Tr}(A[\phi_1, \phi_2])$; the remaining terms cancel by invariance of the trace pairing. This is the gauge identity that identifies the brane gauge field as the Lagrange multiplier for the moment-map zero-fibre.

Remark 3.17 *(Skyscraper–matrix duality).* The same open theory arises from the affine $\mathbb{C}^2$ point-brane Yoneda calculation by tensoring $\mathcal{O}_p$ with $\mathbb{C}^N$ and passing to derived endomorphisms; the Berezin integral on the Ext factor is the residue dual to $dz_1 \wedge dz_2$. This is the open Koszul duality between the Chan–Paton coordinate ring $\mathbb{C}[\phi_1, \phi_2]$ and the skyscraper Ext algebra $\Lambda(\varepsilon_1, \varepsilon_2)$ (Remark 3.10).

Remark 3.18 *(AKSZ form).* In AKSZ language the open theory is the one-dimensional gauged sigma model with Hamiltonian target $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$ and Liouville potential $\theta_N = \mathrm{Tr}(\phi_1 d\phi_2)$. The Poisson bivector

$$\Pi = \sum_{i,j} \partial_{(\phi_1)^i{}_j} \wedge \partial_{(\phi_2)^j{}_i}, \qquad \{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j,$$

is dual to Ω_N under $\iota_{X_f}\omega = -df$; the Hamiltonian $\mu_\epsilon = -\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$ generates $\delta_\epsilon \phi_i = [\epsilon, \phi_i]$; its zero-locus quotient is the stack of commuting matrix pairs modulo conjugation. Equivalently the same theory descends from three-dimensional Chern–Simons on $\mathbb{R} \times T^2$ restricted to harmonic forms $1, \eta_1, \eta_2, \eta_1\eta_2$ on T^2, with $A_{3d} = A_t\, dt + \phi_1\eta_1 + \phi_2\eta_2$.

The P_0-algebra structure on the unreduced open observables, with the cotangent direction realised at the chain level, requires data beyond the reduced construction; its data and the obstruction it must overcome are the following.

Problem 3.19 *(Unreduced cotangent lift of the open BV theory).* Construct an unreduced Tate-coefficient cotangent enlargement of the open BV observables of Proposition 3.11 realising the cotangent direction

$\mathcal{P} = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ of the closed Hamiltonian Lie algebra at the chain level on the brane line. The required data are: an unreduced enlargement

$$\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \mathbin{\widehat{\otimes}} \widehat{\mathbf{S}}\big(\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I)\big),$$

with $\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I) = C^{\mathrm{dR}}_{c,\mathrm{cur}}(I) \widehat{\otimes} \operatorname{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)[1]$; a chain-level primitive projection π_{prim} compatible with Q_{open}, constants, decomposable multi-traces, and factorization products; centrality homotopies for the cotangent vertex Θ_ρ against arbitrary unreduced open observables; P_0-bracket homotopies

$$\{B_f, \Theta_\rho\} \simeq \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\} \simeq 0,$$

before primitive projection. The open-side data is the restriction to the brane line of the full unreduced cotangent lift recorded as Problem 5.49.

The obstruction mode of the centrality homotopies and Tate-cotangent module of Problem 3.19 is the obstruction class of Remark 5.190: the same Tate-coefficient cotangent module governs both the open-side centrality homotopies and the unreduced bulk-to-brane factorization-centre lift.

3.5 Open-brane operators and the large N limit

A stack of N Dirac branes carries a finite-matrix trace map of the formal symplectic polydisk: its operator algebra is generated by cyclic words in two $\mathfrak{gl}_N$-valued Chan–Paton matrices, and its Weyl quantisation produces a scalar shift $\hbar N$ whose class survives connected single-trace projection. Procesi–Razmyslov 1976 ([17][18]; Proposition 3.22) fixes the classical content of the formal-stalk-chart operator side; Capelli 1890 ([26][27]; Remark 5.196) forces the quantum scalar. Theorem 5.6 reads its operator side off the constructions below.

Three large-N operations of Definition 3.41 apply in fixed order: degreewise invariant stabilisation $\mathrm{Prim}_{\mathrm{st}}$, connected single-trace extraction $\mathrm{Conn}_{N\to\infty}$, and Hamiltonian scalar quotient π_{Ham}. Their composite image is the stable brane Hamiltonian algebra $\mathbf{S}(V_H)$ with $V_H = \bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]$ (Proposition 3.40), whose connected indecomposable quotient $\mathfrak{m}/\mathfrak{m}^2 \cong V_H$ is matched by the closed side of §6.2 as continuous CE cochains on $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$. The four-window scalar-anomaly transgression (Theorem E.13) is the bridge between the classical Lie-cohomology class $[\bar{c}]$ and the Weyl/Capelli quantum shift; cancellation on one window forces cancellation on the other three. The all-bidegree residual after Capelli renormalisation is $\Omega^{\mathrm{rad}}_{a,b} \in \operatorname{coker} B_{a,b}$, the decorated PBW Stokes obstruction of Theorem 5.217.

The Dirac brane sector (Lemma 3.15, Proposition 3.16) is the BV theory of two $\mathfrak{gl}_N$-valued matrices and one Koszul antifield, with $Q\psi = [\phi_1, \phi_2]$. In characteristic zero, GL_N is linearly reductive, so degreewise stable invariants are exact in the function-level calculation; invariant theory controls this step, while the unreduced Lie-CE BRST cohomology of $\mathfrak{gl}_N$ has scalar stabilizer classes outside the Hamiltonian trace sector. Procesi–Razmyslov identifies the stable invariant algebra with the free commutative algebra on nonempty cyclic words in two letters (Proposition 3.22). The Koszul homotopy for adjacent transpositions collapses the zero-ψ sector to commutative monomials on the formal symplectic plane (Lemma 3.30, Corollary 3.32). The cyclic-word homotopy (Proposition 3.27) computes the primitive one-ψ Koszul homology in nonconstant bidegree (k, l) as the line spanned by the symmetrised trace

$$\Psi_{k,l} = \sum_{w\in \mathsf{W}_{k,l}} \operatorname{Tr}\big(\psi\, w(\phi_1, \phi_2)\big).$$

The Hamiltonian descendant algebra is therefore $\widehat{\mathbf{S}}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1])$ (Proposition 3.40); its connected indecomposable quotient is $V_H = \bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]$.

Quantization tests the compatibility of Weyl ordering with the brane trace map. On the unreduced brane the classical phase-space mismatch $\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$ survives Weyl quantisation as a projective Capelli contact term. In the matrix Weyl algebra, after quantum Hamiltonian reduction by $\mathfrak{sl}_N$, the normal-ordered moment relation reads

$$YX - XY + \hbar N I \equiv 0 \pmod{\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)},$$

and therefore $\mathrm{Tr}(A\,XY) - \mathrm{Tr}(A\,YX) = \hbar N\,\mathrm{Tr}(A)$ on every covariant matrix word A. The right-hand side is a nonempty connected trace whenever A is nonconstant: the single-trace primitive projection removes the empty trace and disconnected products while retaining this lower-degree trace insertion. The Capelli shift is the quantum mirror of the closed-side scalar class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ of Theorem 6.4: the same scalar appears classically as the central-extension obstruction and quantum mechanically as the Weyl-ordering anomaly, and Theorem 6.6 identifies them. The four-window identification — Lie cohomology, classical trace algebra, Capelli quantum shift, Moyal star product — is the scalar-anomaly transgression Theorem E.13: each window detects the same cocycle, so cancelling it on one window forces cancellation on the other three.

The operator-side comparison closes at finite N through the Capelli-renormalised generator

$$J_N(z_1^a z_2^b) = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(z_1^a z_2^b)(\phi_1, \phi_2)$$

of Lemma 5.198: the change of basis from unrenormalized normal-ordered traces $T_{a,b}$ to J_N is unipotent over $\mathbb{C}[[\hbar N]]$, and the displayed Capelli shift is exactly the $r = 1$ tail of that triangular system. In the J_N-basis the Capelli term is represented by the triangular basis change; the residual all-bidegree obstruction is the quantum shear primitive

$$E_{a,b,N} \in \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N), \qquad \Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}$$

of Theorem 5.217, equivalently the decorated PBW Stokes identity $D^{\square}_{a,b} = C^+_{a,b}\partial_2$; an obstruction is exactly a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. This is the open-trace A_∞-Koszul homotopy datum of Remark 3.44.

Construction 3.20 *(The reduced BV algebra at the insertion point).* At a fixed point of the brane line, introduce three matrix variables

$$\phi_1, \phi_2 \in \mathfrak{gl}_N, \qquad \psi \in \mathfrak{gl}_N[1].$$

The reduced local polynomial algebra is

$$\mathcal{R}_N = \mathbb{C}[(\phi_1)^i{}_j, (\phi_2)^i{}_j] \otimes \Lambda\big((\psi)^i{}_j\big).$$

Its reduced Koszul/BV differential is the derivation determined on generators by

$$Q(\phi_1)^i{}_j = 0, \qquad Q(\phi_2)^i{}_j = 0, \qquad Q(\psi)^i{}_j = ([\phi_1, \phi_2])^i{}_j.$$

This formula is the algebraic encoding of the A-equation of motion: varying

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives $[\phi_1, \phi_2] = 0$, and the antifield $\psi = A^*$ records that equation as a Koszul differential.

Lemma 3.21 *(De Rham jets).* Positive jets in the topological direction vanish in local operator cohomology. Equivalently, every cohomology class has a representative determined by the values of the fields and antifields at the insertion point.

Proof. For a coordinate t on the line, the de Rham part of the BRST operator is $d_{\mathbb{R}}$. On the jet algebra one has the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$. Hence a local expression containing a positive t-derivative is cohomologous to zero after applying the usual Euler homotopy on jet degree. This kills topological-direction jets and leaves the holomorphic z_i-derivatives used later on the closed side. □

Proposition 3.22 *(Stable trace invariants).* Let $F = \mathbb{C}\langle x_1, x_2\rangle$ and let $\mathrm{Cyc}(F) = F/[F,F]$ be the vector space of cyclic words. For each N, GL_N acts on $\mathfrak{gl}_N^2$ by simultaneous conjugation. The degreewise stable invariant algebra

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} := \bigoplus_{d\geq 0} \varprojlim_N \big(\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}\big)_d$$

with respect to block-restriction maps in each word-degree is the free commutative algebra generated by the nonempty cyclic words:

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} \cong \mathbf{S}\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}}\big),$$

where a cyclic word w maps to the trace function $(\phi_1, \phi_2) \mapsto \mathrm{Tr}\, w(\phi_1, \phi_2)$. More precisely, in total positive word-degree d this map is an isomorphism for every $N \geq d$, after omitting the scalar empty trace as a primitive generator.

Proof. Procesi–Razmyslov invariant theory [17][18] gives two facts. First, $\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}$ is generated by trace functions of words in the two generic matrices. Second, all polynomial relations among these trace functions are consequences of the fundamental trace identities of matrix size N; in particular every homogeneous trace relation has total word-degree at least $N + 1$. Fix a total word-degree d. For every $N \geq d$ the degree-d relations are exactly cyclicity of the trace. Passing degree by degree to the stable inverse limit removes all finite-N trace identities. The surviving identification is $\mathrm{Tr}(uv) = \mathrm{Tr}(vu)$. This is exactly the symmetric algebra on cyclic words. □

Lemma 3.23 *(Stable marked trace invariants).* In the stable invariant theory used below, the primitive part linear or quadratic in an odd marked matrix ψ is obtained from the corresponding even three-matrix trace invariant by polarization and skew-symmetrisation in the marked letters. The only stable single-trace relation is graded cyclicity: moving one ψ through the trace has sign $+1$, while moving one ψ past another ψ gives a minus sign.

Proof. Apply Proposition 3.22 to three even generic matrices and take the homogeneous summand linear, respectively quadratic, in the third matrix. The Procesi–Razmyslov trace identities vanish in the fixed total word-degree once N is large enough. Declaring the marked matrix odd is the usual super-polarization: the linear part is unchanged, and the quadratic part is projected to the alternating component under exchange of the two marked occurrences. The trace relation therefore becomes exactly graded cyclicity. These are the signs used in the finite complex for one- and two-ψ words. □

Remark 3.24. The empty trace is the finite-rank scalar $\mathrm{Tr}_N(1) = N$, depending on N and omitted from the stable primitive generators. The stable symmetric algebra still has its ordinary unit; what is omitted is the empty trace as a generator of the connected primitive part. The class $\mathrm{Tr}(\psi)$ is a genuine stable odd primitive class. It is excluded from the Hamiltonian comparison because constants have been removed from the closed Hamiltonian algebra.

Varying

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives the moment-map equation $[\phi_1, \phi_2] = 0$. In the reduced BV complex,

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0$$

on invariant trace functions.

Construction 3.25 *(Boundary evaluation of a closed Hamiltonian).* Let $f = \sum_{k,l} c_{k,l} z_1^k z_2^l \in \mathbb{C}[z_1, z_2]$. The brane replaces the two normal coordinates z_1, z_2 by the two adjoint scalar fields ϕ_1, ϕ_2. Choosing the ordered lift with all ϕ_1's to the left of all ϕ_2's gives the invariant function

$$\widetilde{\partial}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l}\, \mathrm{Tr}(\phi_1^k \phi_2^l) \in \mathcal{R}_N^{GL_N}.$$

Let

$$H^0_{\mathrm{Ham}}(\mathcal{R}_N^{GL_N}, Q) := H^0(\mathcal{R}_N^{GL_N}, Q)/\mathbb{C}\cdot \mathrm{Tr}(1)$$

as a vector space, and as a Lie algebra for the Poisson bracket. The associative operator algebra retains its unit. Constants give the scalar $c_{0,0}N$ and therefore vanish in this Hamiltonian quotient. Thus the construction descends to a linear map

$$\partial_{\mathrm{bb},N} : \bar{A} \to H^0_{\mathrm{Ham}}(\mathcal{R}_N^{GL_N}, Q), \qquad [f] \longmapsto \overline{\widetilde{\partial}_{\mathrm{bb},N}(f)}.$$

Let $\mathsf{W}_{k,l}$ be the set of words in two letters containing k copies of the first letter and l copies of the second. The first descendant is obtained by inserting one Koszul variable and summing over all ordered lifts: For $k + l > 0$ the descendant part of the Hamiltonian comparison is

$$\partial^{(1)}_{\mathrm{bb},N}(e^{\mathrm{PBW}}_{k,l}) = \Psi_{k,l}, \qquad \Psi_{k,l} := \sum_{w\in \mathsf{W}_{k,l}} \mathrm{Tr}\big(\psi\, w(\phi_1, \phi_2)\big).$$

Here $e^{\mathrm{PBW}}_{k,l}$ is an abstract special-fibre PBW label in bidegree (k, l), distinct from the Taylor-dual coadjoint functional $\delta_{k,l}$ and from the Matlis principal part $\rho_{k,l}$. The same formula at $(k, l) = (0, 0)$ gives the decoupled scalar antifield $\mathrm{Tr}(\psi)$, which is excluded from the Hamiltonian pairing because constants have been removed from $\bar{A}$.

LEMMA 3.26 *(First descendants are cycles).* For every $k, l \geq 0$, the symmetrised trace $\Psi_{k,l}$ satisfies $Q\Psi_{k,l} = 0$.

Proof. Write $x = \phi_1$ and $y = \phi_2$. Since x and y are even and $Q\psi = [x, y]$, one has

$$Q\Psi_{k,l} = \sum_{w\in \mathsf{W}_{k,l}} \mathrm{Tr}\big((xy - yx)w(x, y)\big).$$

Group the resulting traces by cyclic word of multidegree $(k + 1, l + 1)$. The coefficient of a cyclic word in the positive sum is the number of cyclic adjacent occurrences of the block xy; the coefficient in the negative sum is the number of cyclic adjacent occurrences of the block yx. On a circle the number of transitions from an x-run to a y-run equals the number of transitions from a y-run to an x-run. The two grouped sums are therefore equal, so $Q\Psi_{k,l} = 0$. □

PROPOSITION 3.27 *(Stable primitive one-antifield Koszul homology).* Fix $k, l \geq 0$. The middle homology of the primitive single-trace complex with one displayed ψ and k, l displayed x, y letters is one-dimensional, generated by the class of $\Upsilon_{k,l}$. The nonconstant part relevant to the Hamiltonian comparison is the direct sum over $k + l > 0$.

Proof. Write $x = \phi_1$ and $y = \phi_2$. Let $\mathsf{W}_{k,l}$ be the set of ordinary words with k letters x and l letters y, and let $\mathsf{N}_{a,b}$ be the set of cyclic words with a letters x and b letters y. By Lemma 3.23, primitive cyclic words are taken modulo the graded cyclic rule

$$\mathrm{Tr}(ab) = (-1)^{|a||b|}\mathrm{Tr}(ba), \qquad |x| = |y| = 0, \quad |\psi| \equiv 1 \pmod 2.$$

Graded cyclicity gives a unique representative $\mathrm{Tr}(\psi w)$ for every one-ψ cyclic word. Thus the relevant one-ψ chain group is

$$C_1(k, l) = \mathbb{C}\{\mathsf{W}_{k,l}\}.$$

The zero-ψ target is

$$C_0(k, l) = \mathbb{C}\{\mathsf{N}_{k+1,l+1}\},$$

and the differential is the incidence map

$$\partial_1(w) = [xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}}.$$

The two-ψ source is the alternating quotient generated by traces $\mathrm{Tr}(\psi u \psi v)$ with u and v ordinary words whose total bidegree is $(k - 1, l - 1)$:

$$\mathrm{Tr}(\psi u \psi v) = -\mathrm{Tr}(\psi v \psi u).$$

Hence diagonal pairs vanish, and if $k = 0$ or $l = 0$ this source is zero. Applying $Q\psi = xy - yx$ gives

$$\begin{aligned}\partial_2(u, v) &= Q\,\mathrm{Tr}(\psi u \psi v)\\ &= \mathrm{Tr}\big(\psi vxyu - \psi vyxu - \psi uxyv + \psi uyxv\big),\end{aligned}$$

so, after removing the common front ψ,

$$\partial_2(u, v) = vxyu - vyxu - uxyv + uyxv.$$

The displayed formula changes sign under $(u, v) \leftrightarrow (v, u)$, so it is well defined on the alternating quotient. A direct substitution shows $\partial_1\partial_2 = 0$.

If $k = 0$ or $l = 0$, then $C_1(k, l)$ is spanned by x^k or y^l, $\partial_1 = 0$, and $C_2(k, l) = 0$. Hence the middle homology is already the line spanned by $\Upsilon_{k,l}$. Assume from now on that $k, l > 0$.

The first homology is computed by the following finite complex. Form a graph $\Gamma_{k,l}$ whose vertices are the necklaces in $\mathsf{N}_{k+1,l+1}$ and whose oriented edges are the words $w \in \mathsf{W}_{k,l}$, with

$$w : [yxw]_{\mathrm{cyc}} \longrightarrow [xyw]_{\mathrm{cyc}}.$$

Then ∂_1 is the cellular boundary of this graph. For each pair (u, v) as above attach the square whose oriented boundary is $vxyu - vyxu - uxyv + uyxv$. Denote the resulting two-complex by $X_{k,l}$.

Equivalently, $X_{k,l}$ is the two-skeleton of the cyclic-word cellular complex $Y_{k,l}$. The full complex $Y_{k,l}$ has an r-cell for r disjoint edge-interior point choices on a cycle with $l + 1$ oriented edges and

$k+1$ indistinguishable vertex points, modulo cyclic relabelling of the edge labels. A vertex is a cyclic distribution of the points among the vertices of this cycle, hence a necklace in $\mathsf{N}_{k+1,l+1}$. A one-cell moves one chosen point across one oriented edge; cutting the cycle at this moving edge records exactly a word $w \in \mathsf{W}_{k,l}$, and its two endpoints are $[yxw]_{\mathrm{cyc}}$ and $[xyw]_{\mathrm{cyc}}$. A two-cell records two independent edge-interior points, and its oriented boundary is precisely $vxyu - vyxu - uxyv + uyxv$. Therefore the cellular chain complex of $Y_{k,l}$ through degree two is exactly

$$C_2(k,l) \longrightarrow C_1(k,l) \longrightarrow C_0(k,l),$$

and the inclusion $X_{k,l} = Y_{k,l}^{(2)} \hookrightarrow Y_{k,l}$ induces an isomorphism on H_1, since cells of dimension at least three leave first homology unchanged.

The homology of $Y_{k,l}$ is the homology of the cycle. The Abel-map argument runs as follows. Before quotienting by cyclic relabelling, send a point configuration to the sum of the point positions in $\mathbb{R}/(l+1)\mathbb{Z}$, and extend linearly over each cell. In the universal cover of the target circle, a fibre is described by weakly ordered lifted point positions with fixed total sum; it is a convex polytope, subdivided by the same cells. Hence the Abel map is a cellular homology equivalence to S^1 by Vietoris–Begle. The finite cyclic relabelling of the y-gaps is equivariant for this Abel map: it rotates the target circle, and every stabilizer acts affinely on a convex fibre, whose quotient is again contractible. Thus the quotient target is still a circle and the quotient complex $Y_{k,l}$ has the homology of S^1. Consequently

$$H_1(X_{k,l};\mathbb{C}) \cong H_1(S^1;\mathbb{C}) \cong \mathbb{C}.$$

Lemma 3.26 identifies this representative as a cycle in the original trace complex. The functional

$$\epsilon\big(\mathrm{Tr}(\psi w)\big) = 1, \qquad w \in \mathsf{W}_{k,l},$$

kills every displayed two-ψ boundary because each boundary has two positive and two negative summands, while

$$\epsilon(\Psi_{k,l}) = \binom{k+l}{k} \neq 0.$$

Thus the one-ψ homology in bidegree (k,l) is the one-dimensional line generated by $[\Psi_{k,l}]$. □

Remark 3.28. A finite exact homology check constructs the finite complex $C_2(k,l) \to C_1(k,l) \to C_0(k,l)$ over $\mathbb{Q}$, with the alternating two-ψ source above, and confirms $\dim\ker\partial_1/\mathrm{im}\,\partial_2 = 1$ for $0 \le k,l \le 5$. It also checks the open Hamiltonian descendant action of Proposition 5.172 (below) for exponents at most 3, and the closed coadjoint Taylor-dual formula together with the principal-part realization of Proposition 5.176 (below) for exponents at most 5. The finite exact check verifies the displayed finite models directly.

COROLLARY 3.29 *(PBW special-fibre one-antifield pairing).* Let $\bar{A}_{\mathrm{desc}}[1]$ be the abstract shifted PBW label copy with basis $e_{k,l}^{\mathrm{PBW}}$ in nonconstant bidegree (k,l). The assignment

$$e_{k,l}^{\mathrm{PBW}} \longmapsto [\Psi_{k,l}]$$

extends to an isomorphism from $\bar{A}_{\mathrm{desc}}[1]$ to the nonconstant primitive one-ψ brane homology.

Proof. The monomials $z_1^k z_2^l$ with $k + l > 0$ form the polynomial basis of $\bar{A}$ and therefore index the PBW special-fibre label copy $\bar{A}_{\mathrm{desc}}$. By Proposition 3.27, the primitive one-ψ homology has one generator in each bidegree, namely $[\Psi_{k,l}]$. Discarding the scalar bidegree $(0, 0)$ and sending the abstract PBW label in every nonconstant bidegree to that generator gives a filtered vector-space isomorphism. This is the boundary pairing used in the PBW special fibre. The continuous Taylor-dual/coadjoint module and the Matlis principal-part realization are separate deformation-level objects. □

Lemma 3.30 *(Koszul homotopy for adjacent transpositions).* In the invariant reduced BV cohomology,

$$\mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v)$$

is Q-exact for all ordinary words u, v in ϕ_1, ϕ_2.

Proof. By cyclicity of the trace,

$$\begin{aligned} Q\,\mathrm{Tr}(\psi v u) &= \mathrm{Tr}([\phi_1, \phi_2] v u) \\ &= \mathrm{Tr}(\phi_1\phi_2 v u - \phi_2\phi_1 v u) \\ &= \mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v). \end{aligned}$$

□

Theorem 3.31 *(Normal-ordering homotopy for cyclic words).* Write $x = \phi_1$ and $y = \phi_2$. For every ordinary necklace $[w]$ in x, y, choose once and for all a linear representative $c([w])$. Suppose $c([w])$ has a occurrences of x and b occurrences of y, and choose a sequence of adjacent transpositions

$$c([w]) = w_0,\ w_1, \ldots,\ w_r = x^a y^b,$$

where each step has the form

$$w_s = A_s\, yx\, B_s, \qquad w_{s+1} = A_s\, xy\, B_s.$$

Define the one-ψ chain

$$H_c([w]) := -\sum_{s=0}^{r-1} \mathrm{Tr}(\psi B_s A_s).$$

Then

$$QH_c([w]) = \mathrm{Tr}(c([w])) - \mathrm{Tr}(x^a y^b).$$

Equivalently, if $N([w]) = z_1^a z_2^b$ and $i(z_1^a z_2^b) = \mathrm{Tr}(x^a y^b)$, then on this ordinary-word sector with the chosen cuts

$$QH_c = \mathrm{id} - iN.$$

The homotopy depends on the chosen cyclic cut. If a second cut c' is chosen, then $H_{c'}([w]) - H_c([w])$ is a closed one-ψ chain; its homology class is the corresponding scalar multiple, determined by the change of cut, of the primitive descendant $\Psi_{a-1,b-1}$ when $a, b > 0$. Thus this theorem gives a homotopy on the ordinary-word sector for the chosen cyclic cut; contraction of the full cyclic open complex requires additional data.

Proof. For a single step $w_s = A_s yxB_s \to w_{s+1} = A_s xyB_s$, Lemma 3.30 gives

$$Q\,\mathrm{Tr}(\psi B_s A_s) = \mathrm{Tr}(A_s xyB_s) - \mathrm{Tr}(A_s yxB_s) = \mathrm{Tr}(w_{s+1}) - \mathrm{Tr}(w_s).$$

Summing over s and inserting the minus sign in the definition of $H_c([w])$ telescopes:

$$QH_c([w]) = -\sum_{s=0}^{r-1}\big(\mathrm{Tr}(w_{s+1}) - \mathrm{Tr}(w_s)\big) = \mathrm{Tr}(w_0) - \mathrm{Tr}(w_r).$$

Since $w_0 = c([w])$ and $w_r = x^a y^b$, this is the displayed identity. A fixed sorting sequence chooses one path for each chosen cut, so the formula extends linearly to the selected zero-ψ necklace span.

If c' is a cyclic rotation of c, the two telescoping formulas have the same boundary because $\mathrm{Tr}(c([w])) = \mathrm{Tr}(c'([w]))$. Therefore $H_{c'}([w]) - H_c([w])$ is Q-closed. The one-ψ homology computation of Proposition 3.27 identifies its class with a scalar multiple of $\Upsilon_{a-1,b-1}$ determined by the change of cut; for example $[yxx] = [xxy]$, but the cuts yxx and xxy give homotopies differing by $-2[\psi x]$. This is the dependence on the cyclic cut: H_c records the chosen cut datum. A cyclic contraction requires additional data. □

COROLLARY 3.32 *(Matrix evaluation is commutative in cohomology).* If two words in ϕ_1, ϕ_2 have the same numbers of ϕ_1 and ϕ_2, then their traces define the same class in the reduced BV cohomology. Consequently $\partial_{\mathrm{bb},N}(f)$ is determined by the commutative polynomial $f \in \mathbb{C}[z_1, z_2]$ and is independent of the chosen ordered lift.

Proof. Apply Theorem 3.31 to each word. Both traces are Q-homologous to the same normal-ordered trace $\mathrm{Tr}(\phi_1^a \phi_2^b)$, where a, b are their common numbers of letters. Hence they define the same reduced BV cohomology class. □

Definition 3.33 *(Unital Hamiltonian cyclic Koszul Lie model).* Let

$$E_{\mathrm{K}} = \mathbb{C}x \oplus \mathbb{C}y \oplus \mathbb{C}\psi, \qquad |x| = |y| = 0, \quad |\psi| = -1,$$

and let

$$\mathfrak{p}_{\mathrm{K}} = \mathrm{Cyc}\big(T(E_{\mathrm{K}})\big) = T(E_{\mathrm{K}})/[T(E_{\mathrm{K}}), T(E_{\mathrm{K}})]_{\mathrm{gr}}$$

be the graded cyclic words, including the scalar word $[1]$. Let $\bar{\mathfrak{p}}_{\mathrm{K}} := \mathfrak{p}_{\mathrm{K}}/\mathbb{C}[1]$ denote its scalar-reduced quotient. The differential is induced by

$$Qx = Qy = 0, \qquad Q\psi = xy - yx.$$

With cyclic derivatives ∂_x, ∂_y, define the constant necklace bracket by

$$\{F, G\}_{\mathrm{neck}} = \big[\partial_x F\, \partial_y G - \partial_y F\, \partial_x G\big]_{\mathrm{cyc}},$$

with the standard Koszul sign from cyclic rotation. Since x, y are even, the displayed formula carries sign $+1$ on the ordinary Hamiltonian words.

THEOREM 3.34 *(Unital cyclic Koszul P_0 model).* The complex $(\mathfrak{p}_{\mathrm{K}}, Q, \{-,-\}_{\mathrm{neck}})$ is a dg Lie algebra. For every interval $I \subset \mathbb{R}$,

$$\mathfrak{p}_{\mathrm{K}}(I) := \Omega_c^\bullet(I) \otimes \mathfrak{p}_{\mathrm{K}}$$

with differential $d_{\mathbb{R}} + Q$ and bracket

$$[\alpha \otimes F, \beta \otimes G] = (-1)^{|F||\beta|}(\alpha \wedge \beta) \otimes \{F, G\}_{\mathrm{neck}}$$

is a compactly supported current dg Lie algebra. The graded-commutative algebra

$$\mathrm{Obs}^{\mathrm{cyc}}_{\mathrm{open}}(I) := \mathbf{S}(\mathfrak{p}_{\mathrm{K}}(I))$$

with the biderivation extending this bracket is a P_0 factorization algebra on the brane line, with structure maps given by extension by zero and multiplication on disjoint intervals.

The CE/PV comparison used later is the degree-zero Hamiltonian identity of Theorem 5.2; extending it to all ψ-words requires a graded coordinate theorem with the full signs and degree shifts specified.

Proof. The necklace bracket is the standard constant-symplectic necklace bracket for the even pair x, y. Its graded antisymmetry and Jacobi identity are the usual cancellation of the cyclic sum of commutators in $T(E_{\mathrm{K}})$. The scalar word must be present: for example $\{[x], [y]\}_{\mathrm{neck}} = [1]$. The derivation Q replaces only uncontracted occurrences of ψ by $xy - yx$ and fixes the paired variables x, y. Expanding the bracket as the signed sum over cut pairs $(x \in F, y \in G)$ and $(y \in F, x \in G)$, Q never hits a contracted letter and carries the same super-cyclic cut sign as applying Q before bracketing; the terms where Q hits G acquire the usual factor $(-1)^{|F|}$. Hence

$$Q\{F, G\}_{\mathrm{neck}} = \{QF, G\}_{\mathrm{neck}} + (-1)^{|F|}\{F, QG\}_{\mathrm{neck}}.$$

Finally $Q^2 = 0$ is immediate from $Qx = Qy = 0$.

Tensoring with $\Omega^\bullet_c(I)$ gives the displayed current dg Lie algebra by the standard tensor-product sign rule; $d_{\mathbb{R}}$ is a derivation of $\wedge$, and Q is a derivation of the necklace bracket. For disjoint intervals $I_1, \dots, I_n \subset J$, extension by zero sends each current into J, and the factorization product is the product in the symmetric algebra. If two currents have disjoint support, their current bracket has zero de Rham product, so the P_0-bracket is local for the factorization structure.

The final assertion gives the stated boundary: an all-degree cotangent-coordinate comparison is additional data beyond this theorem. □

PROPOSITION 3.35 *(Normal-ordering quotient of the unital cyclic Koszul model).* Define

$$N : \mathfrak{p}_{\mathrm{K}} \longrightarrow \mathbb{C}[z_1, z_2]$$

on a cyclic word by

$$N([w]) = \begin{cases} z_1^{\#_x(w)} z_2^{\#_y(w)}, & w \text{ is an ordinary } x, y\text{-word}, \\ 0, & w \text{ contains at least one } \psi. \end{cases}$$

Then N is a dg Lie morphism from $(\mathfrak{p}_{\mathrm{K}}, Q, \{-,-\}_{\mathrm{neck}})$ to the unreduced polynomial Hamiltonian Lie algebra $\mathbb{C}[z_1, z_2]$ with zero differential and Poisson bracket $\partial_{z_1} \wedge \partial_{z_2}$. Composing with $\mathbb{C}[z_1, z_2] \to \bar{A}$ gives the reduced Hamiltonian quotient $N_{\mathrm{red}} : \bar{\mathfrak{p}}_{\mathrm{K}} \to \bar{A}$. The section $i(z_1^a z_2^b) = [x^a y^b]$ and the homotopy H_c for the chosen cyclic cut of Theorem 3.31 satisfy $QH_c = \mathrm{id} - iN$ on the chosen zero-ψ necklace sector. The full cyclic complex is governed by the cyclic homotopy-transfer construction.

Proof. If a word contains ψ, every term of its necklace bracket with any word still contains ψ, because the bracket only differentiates in x and y. Hence both sides of the Lie identity vanish after N whenever one word contains ψ. For ordinary words F, G, cyclic derivative counting gives

$$N(\partial_x F) = \partial_{z_1} N(F), \qquad N(\partial_y F) = \partial_{z_2} N(F),$$

and similarly for G. Therefore

$$N\{F, G\}_{\text{neck}} = \partial_{z_1} N(F)\, \partial_{z_2} N(G) - \partial_{z_2} N(F)\, \partial_{z_1} N(G) = \{N(F), N(G)\}.$$

The chain-map property follows generatorwise and then by the derivation rule. If a word contains one ψ, each Q-summand replaces that ψ by $xy - yx$, and the two resulting ordinary commutative monomials agree after N. If a word contains two or more ψ's, every Q-summand still contains a ψ, hence maps to zero. On ordinary words $Q = 0$. Thus $NQ = 0$. The final displayed homotopy identity is precisely Theorem 3.31 with the chosen cuts. □

LEMMA 3.36 *(Obstruction to contraction onto the Hamiltonian quotient).* A degree -1 operator K on

$$\mathfrak{p}_{\mathrm{K}} = \mathrm{Cyc}\big(T(x, y, \psi)\big)$$

satisfying

$$QK + KQ = \mathrm{id} - iN$$

with N killing every ψ-containing word would force the primitive one-ψ classes to vanish.

Proof. For each nonconstant bidegree (k, l), Proposition 3.27 gives a closed one-ψ primitive $\Psi_{k,l}$ whose class is nonzero. Since $N(\Psi_{k,l}) = 0$, the displayed identity would imply

$$\Psi_{k,l} = QK(\Psi_{k,l}) + KQ(\Psi_{k,l}) = QK(\Psi_{k,l}),$$

contradicting $[\Psi_{k,l}] \neq 0$. Hence any unreduced normal-ordering transfer must either retain the descendant module and higher coherence homotopies, or restrict to the zero-ψ sector with a chosen cyclic cut. □

LEMMA 3.37 *(Gauge-paired necklace obstruction to the plain Hamiltonian quotient).* Enlarge E_{K} to $E_{\mathrm{BV}} = \mathbb{C}x \oplus \mathbb{C}y \oplus \mathbb{C}a \oplus \mathbb{C}\psi$ with $|a| = 1$, $|\psi| = -1$, and add the constant pairing $\langle a, \psi\rangle = 1$ to the necklace bracket. The map that sends an x, y-word to its commutative monomial and kills every word containing a or ψ has a nonzero Lie defect.

Proof. With left cyclic derivatives and the displayed pairing convention,

$$\{[a], [\psi x]\}_{\text{neck}} = [x]$$

exactly. The plain projection sends the left-hand bracket to z_1, while it sends both $[a]$ and $[\psi x]$ individually to zero, so the bracket of their projected images is zero. Thus the full gauge-paired cyclic BV algebra requires an additional quotient, homotopy transfer, or explicit centrality data before it can map to the Hamiltonian formal-disk quotient. □

Theorem 3.38 *(Gauge-BRST cyclic algebra and the normal-ordering obstruction).* Let

$$E_{\mathrm{BRST}} = \mathbb{C}x \oplus \mathbb{C}y \oplus \mathbb{C}\gamma \oplus \mathbb{C}\psi, \qquad |x| = |y| = 0, \quad |\gamma| = 1, \quad |\psi| = -1,$$

and put

$$\mathfrak{p}_{\mathrm{BRST}} = \mathrm{Cyc}\big(T(E_{\mathrm{BRST}})\big).$$

Equip E_{BRST} with the even constant pairing

$$\langle x, y\rangle = 1, \qquad \langle y, x\rangle = -1, \qquad \langle \gamma, \psi\rangle = \langle \psi, \gamma\rangle = 1,$$

and the corresponding graded necklace bracket. The cyclic Hamiltonian

$$S = [\gamma(xy - yx)] + [\psi\gamma\gamma]$$

has degree 1 and satisfies $\{S, S\}_{\mathrm{neck}} = 0$. Hence $Q = \{S, -\}_{\mathrm{neck}}$ makes $\mathfrak{p}_{\mathrm{BRST}}$ a dg Lie algebra, with

$$Q\gamma = \gamma^2, \quad Qx = \gamma x - x\gamma, \quad Qy = \gamma y - y\gamma, \quad Q\psi = xy - yx + \gamma\psi + \psi\gamma.$$

The normal-ordering map

$$N_{\mathrm{BRST}} \colon \mathfrak{p}_{\mathrm{BRST}} \longrightarrow \mathbb{C}[z_1, z_2], \qquad [w] \longmapsto \begin{cases} z_1^{\#_x(w)} z_2^{\#_y(w)}, & w \text{ contains only } x, y, \\ 0, & w \text{ contains } \gamma \text{ or } \psi, \end{cases}$$

is a chain map with a nonzero Lie defect. A contraction $QK + KQ = \mathrm{id} - iN_{\mathrm{BRST}}$ on the full cyclic complex is obstructed by the length-one class $[\psi]$.

Proof. The constant-pairing necklace bracket has graded Jacobi by the standard two-cut cancellation in cyclic words. The displayed Hamiltonian gives the four formulas for Q by direct cyclic differentiation. A direct check of the generators gives $Q^2 = 0$: $Q\gamma^2 = 0$, $Q^2x = Q(\gamma x - x\gamma) = 0$, $Q^2 y = 0$, and, for $\mu = xy - yx$, $Q\mu = \gamma\mu - \mu\gamma$, whence

$$Q^2\psi = Q\mu + Q(\gamma\psi) + Q(\psi\gamma) = (\gamma\mu - \mu\gamma) + (-\gamma\mu - \gamma\psi\gamma) + (\mu\gamma + \gamma\psi\gamma) = 0.$$

Therefore $Q = \{S, -\}$ is a square-zero derivation of the necklace bracket.

The chain-map statement for N_{BRST} follows as in Proposition 3.35: ordinary words are Q-closed after projection; a single ψ contributes $xy - yx$, whose two normal orderings agree, while the $\gamma\psi$ and $\psi\gamma$ terms are killed; every summand containing γ is killed. Its Lie defect is the gauge-paired bracket $\{[\gamma], [\psi x]\}_{\mathrm{neck}} = [x]$, whose image is z_1, whereas both factors have zero image. Finally $[\psi]$ is closed:

$$Q[\psi] = [xy - yx] + [\gamma\psi + \psi\gamma] = 0,$$

since $[xy] = [yx]$ and $[\psi\gamma] = -[\gamma\psi]$ in the cyclic quotient. The image of Q misses $[\psi]$. With respect to cyclic word length, Q raises length by one: Qx, Qy, $Q\gamma$, and $Q\psi$ are quadratic words. Hence the length-one class $[\psi]$ represents nonzero homology. Equivalently, the derivation rule never produces a lone ψ-letter. Thus a full contraction would force $[\psi] = QK[\psi]$, contradiction. The extended BRST cyclic algebra is the correct unreduced primitive object; the Hamiltonian normal-ordering sector is only a quotient/chain-map sector and still requires separate centrality and factorization-center data. □

PROPOSITION 3.39 *(Stalk central multiplication).* Let

$$A_N^{\mathrm{pt}} = (\mathcal{R}_N^{GL_N}, Q)$$

be the invariant reduced open Koszul algebra at a brane insertion point. For $f \in \mathbb{C}[z_1, z_2]$, set

$$B_f = \widetilde{\partial}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l} \mathrm{Tr}(\phi_1^k \phi_2^l).$$

Then $QB_f = 0$. Regard B_f as a Hochschild 0-cochain; its associated central operation is multiplication

$$L_{B_f} : A_N^{\mathrm{pt}} \to A_N^{\mathrm{pt}}, \qquad a \longmapsto B_f a.$$

This Hochschild 0-cochain is closed. The same statement holds for multiplication by a closed one-antifield class $\Upsilon_{k,l}$. After projection to the Hamiltonian quotient and stabilization in N, the degree-zero classes satisfy

$$\{B_f, B_g\}_{\mathrm{open}} = B_{\{f,g\}_{\bar{A}}}.$$

Proof. Since $Q\phi_1 = Q\phi_2 = 0$, every B_f is Q-closed. The algebra A_N^{pt} is graded-commutative, so the Hochschild differential of a 0-cochain b is

$$d_{\mathrm{Hoch}} b(a) = Qb\, a + ba - (-1)^{|b||a|} ab.$$

This vanishes for every closed graded-central element b. The classes $\Upsilon_{k,l}$ are closed by Lemma 3.26 and are central for the same graded-commutative reason. This is only a statement about multiplication by a closed polynomial descendant in the stalk algebra: $\Upsilon_{k,l}$ is a PBW special-fibre descendant distinct from the principal-part cotangent boundary field $\rho_{k,l}$. The continuous Matlis coadjoint action is the separate action of Theorem 5.179. Ordered-lift independence for B_f is cohomological and is exactly Corollary 3.32. The Poisson identity follows from Theorem 5.191 (below). □

Let $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The removed constant corresponds on the gauge side to the scalar $\mathrm{Tr}(1) = N$ and has zero Hamiltonian vector field.

PROPOSITION 3.40 *(Stable brane Hamiltonian algebra and connected quotient).* Put

$$V_H = \bar{A} \oplus \bar{A}_{\mathrm{desc}}[1].$$

In the stable large-N open BV model, the Hamiltonian descendant algebra generated by the nonempty primitive ghost-zero traces and the nonconstant primitive one-ψ Koszul homology is

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane},H,\mathrm{st}} \cong \mathbf{S}(V_H).$$

Its connected single-trace, or indecomposable, quotient is

$$Q\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane},H,\mathrm{st}} = \mathfrak{m}/\mathfrak{m}^2 \cong V_H.$$

Under J, the class $z_1^k z_2^l$ with $k + l > 0$ maps to $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$; its descendant basis vector maps to the symmetrised trace with one ψ and k, l copies of ϕ_1, ϕ_2. All Lie comparisons on single traces use V_H after scalar reduction; CE/PV polyvector functions use the commutative algebra $\mathbf{S}(V_H)$.

Proof. Proposition 3.22 gives the free commutative algebra on cyclic words. Lemma 3.30 identifies any two cyclic words with the same numbers of ϕ_1 and ϕ_2, because adjacent transpositions generate all reorderings. Thus the connected ghost-zero cyclic words reduce to the monomials $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$ with $k + l > 0$. Proposition 3.27 computes the primitive one-ψ homology in each multidegree; the nonconstant generators are $\Upsilon_{k,l}$ with $k + l > 0$. Products of the connected ghost-zero primitive classes and these one-ψ primitive classes give the free graded-commutative multi-trace algebra $\mathbf{S}(V_H)$ in the Hamiltonian sector. Projection to indecomposables $\mathfrak{m}/\mathfrak{m}^2$ recovers the connected single-trace quotient V_H. The full primitive calculation proves that stable primitive homology vanishes for ψ-degree $p \geq 2$; decomposable products of one-ψ classes remain only as products in the symmetric algebra. □

Definition 3.41 *(Three large-N operations).* The three operations have different domains and are applied in the order below.

(i) *Degreewise invariant stabilisation* $\mathrm{Prim}_{\mathrm{st}}$. At fixed word degree d, take $N \geq d$ so that Procesi–Razmyslov trace invariants lie beyond finite-N trace identities. This is the operation used in Proposition 3.22.
(ii) *Connected single-trace extraction* $\mathrm{Conn}_{N\to\infty}$. After passing to the stable invariant algebra, project the symmetric algebra on cyclic-word generators to its primitive/indecomposable summand. Products of two nonempty traces are decomposable and are killed by this projection.
(iii) *Hamiltonian scalar quotient* π_{Ham}. Remove the empty trace $\mathrm{Tr}(1) = N$ on the open side and the constant Hamiltonian 1 on the closed side. The removed line is the scalar $U(1)$ line whose obstruction class is $[\bar{c}]$.

The reduced Hamiltonian sector used below is

$$\pi_{\mathrm{Ham}}\,\mathrm{Conn}_{N\to\infty}\,\mathrm{Prim}_{\mathrm{st}}\big(\mathcal{R}_N^{GL_N}, Q\big).$$

The formula is an ordered construction: first stabilize invariant theory degree by degree, then extract connected primitive trace classes, then quotient the scalar Hamiltonian line. The unprojected $N \to \infty$ algebra and the permuted composites are different objects.

Remark 3.42. The word "sector" is literal. The calculation above presents the stable algebra generated by the primitive zero-ψ Hamiltonians and the nonconstant primitive one-ψ homology generated by $\Upsilon_{k,l}$ for $k + l > 0$. The full invariant Koszul homology in higher ψ-count lies outside this sector, as does the decoupled scalar class $\mathrm{Tr}(\psi)$.

COROLLARY 3.43 *(Full primitive Koszul homology).* Let

$$A_\psi = \mathbb{C}\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| \equiv 1 \pmod 2,$$

with differential $d\psi = xy - yx$ and $dx = dy = 0$. Put

$$\mathrm{Cyc}(A_\psi) = A_\psi/[A_\psi, A_\psi]_{\mathrm{gr}}, \qquad [uv] = (-1)^{|u||v|}[vu].$$

Give x, y, ψ weights

$$\mathrm{wt}(x) = (1, 0), \qquad \mathrm{wt}(y) = (0, 1), \qquad \mathrm{wt}(\psi) = (1, 1).$$

For fixed (R, S) define

$$K_{R,S}^p = \mathbb{C}\{[w]_{\mathrm{cyc}} : \#\psi(w) = p,\ \#x(w) = R - p,\ \#y(w) = S - p\},$$

with $0 \leq p \leq \min(R, S)$. The differential is

$$\partial[w] = \sum_{i:a_i=\psi} (-1)^{\#\{\psi \text{ before } i\}} [a_1 \cdots a_{i-1} xy\, a_{i+1} \cdots a_n - a_1 \cdots a_{i-1} yx\, a_{i+1} \cdots a_n]_{\mathrm{cyc}}.$$

The primitive homology is

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C} \cdot [x^R y^S], & p = 0, \\ \mathbb{C} \cdot [\Psi_{R-1,S-1}], & p = 1,\ R, S \geq 1, \\ 0, & p \geq 2. \end{cases}$$

Equivalently, the primitive stable Koszul homology is $\mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta$ with $|\theta| = 1$ and $\mathrm{wt}(\theta) = (1, 1)$.

This is a stable primitive cyclic-word computation. The proof identifies the graded cyclic complex with the cellular chains of the cyclic-word cellular complex and uses the Abel map to compute the state complex as $H_\bullet(S^1; \mathbb{C})$; see Theorem 5.129 and Theorem D.12. The result identifies the polynomial one-ψ span as a PBW special-fibre object; Matlis principal parts and the Moyal coadjoint module are distinct deformation-level objects.

Remark 3.44 *(Open-trace A_∞-Koszul homotopy datum).* After the Capelli renormalisation (Lemma 5.198, Corollary 5.199) the stable connected single-trace target is the Lie-algebra image of the Moyal Hamiltonian source on the degree-zero generator restriction (Corollary 5.201); the residual obstruction to the all-bidegree ordinary Weyl-trace bracket comparison is the quantum shear primitive

$$E_{a,b,N} \in \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N), \qquad \mathfrak{o}_{a,b} = [R^{\mathrm{free}}_{a,b}] \in \operatorname{coker} C_{a,b},$$

equivalently the dual-potential class $\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}$ of Theorem 5.217; obstruction is exactly a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. The edge strips $\mathfrak{o}_{2,b}$ and $\mathfrak{o}_{a,2}$ vanish (Proposition F.32); the finite-window interior identities are recorded in Proposition F.33; the all-interior assertion is the decorated PBW Stokes signed-row vanishing theorem. The decoupled scalar class $\mathrm{Tr}(\psi)$ and the higher ψ-count primitive homology lie outside the sector and outside this homotopy datum. Stable large-N brane-trace asymptotics enter only through Definition 4.26.

The open BV brane sector is the derived commuting variety $[\mu^{-1}_{\mathrm{der}}(0)/\mathrm{GL}_N]$ with Koszul antifield ψ and BV differential $Q\psi = [\phi_1, \phi_2]$, with stable trace observables generating the open-string sector classical observables factorization algebra on the brane line. The trace functional of a polynomial closed Hamiltonian $f \in \bar{A}$ on this brane sector is the GL_N-invariant $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$, extended to $\mathfrak{h}$ only after completion; its universal property and CE-coordinate expression are the content of the section below.

4 BOUNDARY ALGEBRA AND TRACE MAP

The trace map $J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2)$ is forced by Procesi–Razmyslov stable invariants, Loday–Quillen–Tsygan primitivity, and Darboux naturality on the moment-map zero fibre; the Koszul homotopy for adjacent transpositions makes it well defined on the derived stack (Theorem 5.6, step (3) of the proof). The same identity is isolated in Remark 5.9.

4.1 The reduced CE/PV trace map

The trace map at a Dirac brane vacuum is

$$J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)} \qquad (f \in \bar{A}),$$

the formal Darboux coordinate expression at $b_p \in C_\partial^{\mathrm{op}}$ of the trace-sector map from the bulk stalk to the brane-side E_1-Hochschild complex $C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ (Deligne's E_1-Hochschild theorem gives the E_2-structure; see Conventions and notation for the local theorem). Here $b_p \in C_\partial^{\mathrm{op}}$ is the chosen vacuum at $p = 0 \in \mathbb{C}^2$ (Definition 1.1), with chart algebra $A^{\mathrm{coord}}_{b_p} = \widehat{\mathcal{O}}_{\mathbb{C}^2,p} \cong \mathbb{C}[[z_1, z_2]]$ on the brane line $L = \mathbb{R}_t \times \{z_1 = z_2 = 0\}$ of §3; J is the stable scalar-reduced cyclic-trace class of the unreduced $J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2)$. For $f \in \mathfrak{h}$, the same symbol denotes the continuous extension to the completed brane algebra. The closed BV theory carries two CE coordinate types and the chart carries only one of them: Hamiltonian ghosts c^I—dual to generators $H_I \in \widehat{\mathfrak{h}}$—generate the closed motion on the polydisk; cotangent coordinates u_I—dual to the shifted antifield generators in $\mathfrak{h}^\vee_{\mathrm{cont}}[1]$—are the boundary observables. The map J is the unique $\mathbb{C}$-linear, GL_N-equivariant, Darboux-natural primitive single-trace map $\mathfrak{h} \to \mathcal{O}(\mu_{\mathrm{der}}^{-1}(0))^{GL_N}$ satisfying scalar reduction, Dirac descent, and $J(z_i) = \mathrm{Tr}\,\phi_i$ (Theorem 5.147). Centrality is the moment-map identity $\{J(f), J(g)\}_\partial = J(\{f, g\}_{\mathfrak{h}})$ on the trace sector of $A^{\mathrm{cl}}_{\partial,N}$. Closed Hamiltonians act through c_f; J records u_f; below, the symbols c_f, u_f, θ_f, O_f refer back to Lemma 4.1.

The CE/PV comparison is graded into five levels at separate admissibility hypotheses.

level	statement	admissibility
1	coordinate cochain identity $\Phi_{\mathrm{coord}}(c^I) = \theta^I$, $\Phi_{\mathrm{coord}}(u_I) = O_I$	no extra datum
2	P_0-bracket enhancement on the chosen sub-PV model	bracket-admissibility of B
3	Maurer–Cartan kernel compatibility	kernel datum $(B, \mathcal{K}_B)$
4	bar-cobar / enveloping equivalence $\Omega C^\bullet_{\mathrm{CE}} \xrightarrow{\sim} A$	Tate / nilpotent / weighted datum
5	open-trace A_∞-Koszul homotopy	A_∞-homotopy datum $(A_{\mathrm{op}}, C_{\mathrm{op}}, B_{\mathrm{CE/PV}}, \tau_{\mathrm{op}})$

The datum proved without extra structure is level 1. Levels 2–5 hold under the separate admissibility hypotheses stated in Theorem 5.5: B is a graded sub-Lie algebra of the polyvector coordinate ring $\mathrm{PV}_{\mathrm{coord}}$ closed under the Schouten bracket (bracket-admissibility, Definition 4.18); $\mathcal{K}_B$ is a target completion of the diagonal tensor product in which the Casimir converges (kernel-admissibility); the Tate / nilpotent / weighted datum specifies a bar-cobar replacement of the closed CE complex with a conilpotent endpoint and $\varprojlim^1$-vanishing; and the A_∞-homotopy datum $(A_{\mathrm{op}}, C_{\mathrm{op}}, B_{\mathrm{CE/PV}}, \tau_{\mathrm{op}})$ collects an open-trace A_∞ algebra A_{op}, a Koszul-dual coalgebra C_{op}, the matching CE/PV bracket model $B_{\mathrm{CE/PV}}$, and a twisting cochain τ_{op}. References below to "the CE/PV comparison" identify their level explicitly.

Lemma 4.1 *(Two CE coordinates of a Hamiltonian).* A Hamiltonian $f \in \mathfrak{h}$ carries two CE coordinates on the closed shifted-cotangent BV side.

- As generator of canonical transformations: $c_f \in \mathfrak{h}^\vee[-1]$ is the Hamiltonian-dual CE coordinate, and $\Phi_{\mathrm{coord}}(c_f) = \theta_f$, the Schouten vector field on the boundary observable algebra.
- As closed cotangent insertion: $u_f \in \mathfrak{h} \subset \mathfrak{g}[1]$ is the shifted-cotangent-dual coordinate, and $\Phi_{\mathrm{coord}}(u_f) = O_f$, the boundary observable $J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2)$ through the coordinate algebra A^{coord}_b after stable scalar reduction.

Closed Hamiltonians act through c_f; the trace map J records u_f; the CE/PV theorem (Theorem 5.5, Theorem 4.20) implements the duality.

Proof. Theorem 5.2 fixes the generator signs in the finite-dimensional case: $\Phi_1(c^\xi) = \theta^\xi$, $\Phi_1(u_x) = O_x$. Theorem 5.3 extends this generator assignment to the formal-polydisk coordinate envelope. Equating c_f with O_f breaks the coordinate bracket-compatibility $[\theta^I, O_J]_S = \delta^I_J$, which is what every P_0-centrality homotopy uses (clause (2) of Theorem 5.5). The collision is already at the bracket on generators, before any central extension or kernel datum can enter: the obstruction is the generator-level bracket defect. □

Remark 4.2 *(Generator table).*

symbol	CE coordinate on	Φ_{coord}-image	physical role
c_f	$\mathfrak{h}^\vee[-1]$	$\theta_f \in \mathrm{Vect}(\widehat{\mathbf{S}}(\bar{A}))$	generator of canonical motion
u_f	$(\mathfrak{h}^\vee_{\mathrm{cont}}[1])^\vee$	$O_f = J(f) \in \widehat{\mathbf{S}}(\bar{A})$	coordinate trace map J through A_b^{coord}

LEMMA 4.3 *(Bracket on observables).* For $f, g \in \mathfrak{h}$,

$$\theta_f(O_g) = O_{\{f,g\}}.$$

Equivalently, the Schouten field $\theta_f = \Phi_{\mathrm{coord}}(c_f)$ acts on the boundary observable $O_g = \Phi_{\mathrm{coord}}(u_g) = J(g)$ as the Hamiltonian derivation $g \mapsto \{f, g\}$.

Proof. On generators c_f, u_g of $C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g})$ the CE differential gives $d_{\mathrm{CE}} u_g(c_f) = u_{\{f,g\}}$. The CE/PV identification Theorem 5.2 sends $c_f \mapsto \theta_f$ and $u_g \mapsto O_g$, and the Schouten pairing $[\theta_f, O_g]_S = \theta_f(O_g)$ on $\mathrm{PV}(\widehat{\mathbf{S}}(\bar{A}))$ carries the same derivation on linear functions; the equality on the brane side is the moment-map identity $\{J(f), J(g)\}_\partial = J(\{f,g\}_{\mathfrak{h}})$ of Theorem 5.191. □

The CE/PV dictionary. On generators the cochain CE/PV identification of Theorem 4.20 is

$$c^I \longmapsto \theta^I, \qquad u_I \longmapsto O_I.$$

This is Lemma 4.1 written on a basis: the Hamiltonian-dual CE coordinate produces a Schouten vector field, the shifted-cotangent-dual CE coordinate produces a boundary Hamiltonian observable. The brane substitution is the image of u_f; c_f is the Hamiltonian vector-field coordinate.

Word ordering is BRST-exact, the bracket records the symbol. At finite N the trace map $J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2)$ (equivalently $\overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ after stable scalar reduction) requires a word ordering of the noncommuting matrices in $f(\phi_1, \phi_2)$. The choice is irrelevant in $H^0(\mathcal{R}_N^{GL_N}, Q)$: adjacent transpositions are exact via $Q\ \mathrm{Tr}(\psi v u) = \mathrm{Tr}([\phi_1, \phi_2] v u)$ of (8), so $J(f)$ depends only on the symbol f. The matrix Poisson bracket induced by $\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$ is

$$\{F, G\} = \mathrm{Tr}\big(\partial_{\phi_1} F \cdot \partial_{\phi_2} G - \partial_{\phi_2} F \cdot \partial_{\phi_1} G\big)$$

with cyclic matrix derivatives. On monomials,

$$\big\{\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}\big\} = \overline{(kn - lm)\,\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1})}, \tag{12}$$

and assembling over the symbol algebra after scalar reduction,

$$\big\{J(f), J(g)\big\}_\partial = J\big(\{f,g\}_{\mathfrak{h}}\big).$$

J is a Lie algebra morphism $\bar{A} \to H^0(\mathcal{R}_N^{GL_N}, Q)$ modulo the central direction §6.1 diagnoses.

Finite Taylor truncation obstruction. The plain finite-rank reduction of $\widehat{\mathfrak{h}}$ is its Taylor-degree truncation $\widehat{\mathfrak{h}}/\mathfrak{m}^{N+1}$. The Lie-quotient obstruction is explicit: linear Hamiltonians z_1, z_2 generate brackets of strictly lower Taylor degree, the projected brackets have a Jacobi obstruction after truncation, and the diagonal cotangent Casimir $C_{\widehat{\mathfrak{h}}} = \sum_{d\geq 1} e_{d,i} \otimes e^i_d$ misses the strict product/direct-sum endpoint. Finite coordinate projections exist only as coefficient tests. They assemble to the completed coordinate CE/PV isomorphism

$$\Phi_{\mathrm{coord}} : C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$$

of Theorem 4.20. A P_0 comparison is asserted only after a bracket-admissible $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ is chosen (Definition 4.18); a Maurer–Cartan kernel is asserted only after the diagonal tensor converges in a kernel-admissible tensor product. Both are part of the reduced Hamiltonian central-operation model; the strict ambient product has a separate convergence obstruction.

Theorem 4.4 *(Witten centrality on the Dirac brane: the local moment-map trace identity).* In the reduced classical P_0 factorization algebra A_∂ on the brane line, the trace map $J : \bar{A} \to A_\partial$, $J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2)$, with smeared form $f \mapsto B_f(a) = \int_L a(t)\, J(f)(t)\, dt$, realises every closed Hamiltonian $f \in \bar{A}$ as a boundary observable whose ordered products and P_0-bracket with smeared Hamiltonian insertions $B_g(b)$ are the moment-map operations

$$\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab), \qquad F \cdot G \simeq (-1)^{|F||G|} G \cdot F \text{ (scalar } F).$$

In particular J lands inside the reduced P_0-central-operation complex of A_∂, $J(f)$ is null-homotopic for every scalar (constant) f, and J is Witten's classical centrality computation [4] read in the local algebraic model $Z^{P_0}_{\mathrm{red,Ham}}(A_{\partial,\mathrm{coord}}; B) = B$. The unreduced P_0-factorization centre lift, including centrality homotopies against arbitrary open observables and principal-part cotangent currents, is exactly Problem 5.49.

Proof. The brane equal-time Poisson bracket induced by the symplectic potential $\theta_N = \mathrm{Tr}(\phi_1\, d\phi_2)$ of Lemma 3.15 is, on two brane-time points (t, t'),

$$\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(t')\} = \delta^i_l \delta^k_j\, \delta(t - t').$$

Smearing by $a(t)b(t')$ replaces $\delta(t - t')$ by the product ab; contraction of the matrix derivatives of $\mathrm{Tr}\, f(\phi_1, \phi_2)$ and $\mathrm{Tr}\, g(\phi_1, \phi_2)$ returns the constant Poisson bracket on the symbols, evaluated again on the Chan–Paton pair. This produces the moment-map identity $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$; on monomials the explicit coefficient $kn - lm$ is the open trace bracket (12). Cyclic invariance and the Q-exact swaps reduce every monomial bracket to its symbol-level Hamiltonian bracket on $\bar{A}$.

Constants in $\bar{A}$ have zero Hamiltonian vector field, so the scalar bracket component is null-homotopic; the central direction $\mathbb{C} \cdot J(1)$ is exactly the U(1) centre treated in §6.1. The locally constant brane-line factorisation product compares two ordered products of a closed and an open insertion; the homotopy connecting them is the Hochschild centrality datum of the associated E_1-algebra stalk, and the P_0 refinement imposes the same compatibility for the matrix Poisson bracket. Both are produced by the moment-map identity above.

The unreduced cotangent statement requires centrality homotopies for arbitrary open observables, including the principal-part cotangent currents Θ_ρ of Theorem 5.186. The trace substitution J provides the scalar bracket sector; it leaves these cotangent homotopies as the obstruction of Problem 5.49. □

4.2 The two algebras

The trace substitution J produces the open algebra. The Dirac brane stack is the Dirac-reduced first-order action (7); its ghost-zero invariant ring is $H^0(\mathcal{R}_N^{GL_N}, Q)$. By Lemma 3.1 the stable connected GL_N-invariants are exactly the trace family $\{J(f) = \overline{\operatorname{Tr} f(\phi_1, \phi_2)} : f \in \bar{A}\}$, with kernel the constants; any polynomial GL_N-invariant Darboux-natural scalar-reduced map agreeing with the moment-map action on linear generators must equal J (Theorem 5.147). On a brane interval I the smeared trace observables generate the open-string sector classical observables factorization algebra

$$\mathcal{A}_N(I) = \widehat{\mathbf{S}}\big(\Omega_c^0(I)\widehat{\otimes}\bar{A}\big), \qquad B_f(a) = \int_I a(t)\, J(f)(t)\, dt,$$

with smearing test function $a \in \Omega_c^0(I)$, boundary observable $B_f(a)$, and bracket $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$ (Remark 4.10); the support-locality variant $A_\partial^{\mathrm{Ham}}(I)$ on the full de Rham complex appears in §1.6.

Hamiltonian ghosts move; cotangent coordinates measure. The closed BV field content is therefore (4), $\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}_{\mathrm{cont}}^\vee[1]$, and the closed observables are *by definition* the continuous CE cochains

$$\mathcal{B} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}).$$

On a holomorphic polydisk $B \subset \mathbb{C}^2_{\mathrm{hol}}$ and a topological open $U \subset \mathbb{R}^2_{\mathrm{top}}$, the constructed holomorphic and mixed factorization algebras are

$$\mathcal{F}^{\mathrm{hol}}_{\mathrm{Ham}}(B) = C^\bullet_{\mathrm{CE,cont}}\big(\Omega_c^{0,\bullet}(B)\widehat{\otimes}\mathfrak{g}\big), \qquad \mathcal{F}^{\mathrm{mix}}_{\mathrm{Ham}}(U \times B) = C^\bullet_{\mathrm{CE,cont}}\big(\Omega_c^\bullet(U)\widehat{\otimes}\Omega_c^{0,\bullet}(B)\widehat{\otimes}\mathfrak{g}\big),$$

of Definition 1.15.

The comparison between $\mathcal{A}_N$ and $\mathcal{B}$ is the cochain-level CE/PV identity on generators

$$c^f \longmapsto \theta^f, \qquad u_f \longmapsto O_f,$$

of Lemma 4.1: closed Hamiltonians act through c, boundary observables are read off u. The cotangent module is the Grothendieck–Matlis principal-part $\mathcal{P}$, and the Moyal coadjoint action acts on the residue side as a deformation-level structure distinct from the polynomial PBW class (Theorem 8.1).

The recognition criterion has five levels (Theorem 5.5). Each level adjoins one extra structure to the level beneath; obstruction at level k lives in a single obstruction class.

Level	Comparison	Hypothesis
1	$C^\bullet_{\mathrm{CE}}(T^*[1]\mathfrak{l}) \cong \mathrm{PV}(\mathbf{S}\mathfrak{l})$ on a finite-dim Lie algebra $\mathfrak{l}$ (Theorem 5.2)	characteristic zero
2	$\Phi_{\mathrm{coord}}\colon C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ on the formal-polydisk coordinate envelope (Theorem 5.3)	coordinatewise finiteness
3	P_0-bracket lift on a sub-PV model $B \subset \mathrm{PV}_{\mathrm{coord}}$	B bracket-admissible (Definition 4.18)
4	Maurer–Cartan diagonal kernel $\Theta_B \in \mathcal{K}_B$	$(B, \mathcal{K}_B)$ kernel-admissible; diagonal converges
5	Bar-cobar/enveloping $(C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}))^! \simeq \widehat{U}(\mathfrak{g}_{\mathrm{adm}})$ and matched A_∞-Koszul on $\mathcal{A}_N$-side	admissible Tate/nilpotent/relative/weighted replacement; conilpotent endpoint; $\varprojlim^1$-vanishing; open-trace homotopy datum

Level 1 is independent. Level 2 is the formal-polydisk lift of level 1 by coefficientwise finite tests. Level 3 chooses a topologised subalgebra in which the Schouten bracket converges; the strict ambient product completion has the P_0-bracket obstruction. Level 4 chooses a kernel-admissible target in which the diagonal Casimir converges to a Maurer–Cartan element. Level 5 adds an admissible bar-cobar replacement on the closed Lie side and a matched A_∞-Koszul homotopy datum on the open trace side; the five obstructions of clause (4) of Theorem 5.5 block the strict ambient product from satisfying it. The minimal RG-stable analytic completion forced by J — closed under the Schouten bracket and pointwise product, with heat-kernel propagator extension and Casimir convergence — is the weighted-Tate completion B_w (Theorem 5.58); it satisfies levels 3 and 4, with level 5 under the open-trace A_∞-Koszul homotopy datum.

The brane-line factorization-current comparison (Theorem 4.11) is the level-3 application: it identifies the reduced Hamiltonian sector of $\mathcal{F}^{\mathrm{mix}}_{\mathrm{Ham}}|_L$ with $\mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(\mathcal{A}_N)$ on each interval.

Identification of the raw finite-N operator algebras with the strict formal Hamiltonian envelope requires an open-trace A_∞-Koszul homotopy datum $(A_{\mathrm{op}}, C_{\mathrm{op}}, B_{\mathrm{CE/PV}}, \tau_{\mathrm{op}})$ with exact obstruction complex inside an admissible Tate/stable category, conilpotent endpoint, and $\varprojlim^1$-vanishing (Theorem 5.5, item (3)). This datum is exactly what extends the cochain CE/PV identity to bar-cobar/enveloping conclusions and large-N operator-algebra identifications. The unreduced cotangent-included lift to a chain morphism of factorization algebras

$$\Phi_{\mathrm{cl}}\colon \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed}}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N})$$

is the obstruction problem of Problem 5.49. The five named classes — the strict Casimir support obstruction $C_{\mathfrak{h}}$, the linear translation term $\{z_1, z_2^{N+1}\} = (N+1)z_2^N$, the $H_2 \cong \mathfrak{sl}_2$ lower-central obstruction to conilpotence, the associated-graded cone $H^\bullet(\mathrm{Cone}(\mathrm{gr}\,\kappa_\tau))$, and the Milnor $\varprojlim^1$ defect (Theorem E.53; Theorem 5.5, item (4)) — are each independently necessary. Removing any one collapses the comparison; a chain morphism requires all five. These classes block the unreduced cotangent centrality homotopies of Theorem 4.4.

Costello's protected sector of perturbative twisted M-theory on $\mathbb{R} \times X$, with X a holomorphic-symplectic surface and a stack of N topological branes on $\mathbb{R} \times p$, gives the physical motivation for a Hamiltonian formal sector at the symplectic polydisk $\widehat{X}_p$. Once an ambient background gives an admissible protected Hamiltonian restriction, the local theorem of §5.3 applies to that restriction.

4.3 *Reduced Hamiltonian brane coupling*

The brane is a constrained N-Chan–Paton chart of $(\widehat{\mathbb{C}^2}_0, \omega)$ with normal coordinates $\phi_1, \phi_2 \in \mathfrak{gl}_N$ and equal-time Poisson bracket $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(t')\} = \delta^i_l \delta^k_j \delta(t - t')$ on two brane-time points (t, t') (Lemma 3.15, Remark 4.10). A closed Hamiltonian $f \in \bar{A}$ acts on the brane by Dirac substitution $z_i \rightsquigarrow \phi_i$ followed by trace and stable scalar reduction:

$$J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

Smearing along the brane line $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ by a compactly supported test density $a(t)\, dt$ gives the boundary observable

$$B_f(a) = \int_L a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt \in A_\partial^{\mathrm{Ham}}(I). \tag{13}$$

The bulk Hamiltonian bracket equals the brane Poisson bracket of these observables (Theorem 4.5); the moment-map operation comes from the Dirac equal-time bracket of Lemma 3.15 together with Remark 4.10.

The matrix-Chan–Paton BV theory of §§3.3–3.4 gives the brane-line trace observables; the closed shifted-cotangent BV theory of §2.1 gives the Hamiltonian generators $f \in \bar{A}$; the smeared substitution $B_f(a)$ of (13) matches their brackets at the level of the reduced P_0-bracket (Theorem 4.5, Theorem 4.11). The all-window strict cotangent extension of this matched bracket — the unreduced Tate-coefficient cotangent lift — is the obstruction calculus of §7.5.

THEOREM 4.5 *(Brane trace-map principle: bulk Hamiltonians as moment-map currents).* For $f, g \in \bar{A}$ and $a, b \in \Omega^0_c(\mathbb{R})$, the boundary observables (13) satisfy the strict P_0-bracket identity

$$\{B_f(a), B_g(b)\}_{A_\partial^{\mathrm{Ham}}} = B_{\{f,g\}_{\bar{A}}}(ab), \tag{14}$$

and the bracket vanishes whenever the supports of a and b are disjoint. The ghost-zero specialisation of (14) is the unsmeared closed-side Lie homomorphism $\{J(f), J(g)\}_\partial = J(\{f, g\}_{\bar{A}})$ of Theorem 5.191. Before scalar reduction the same bracket carries the U(1) centre anomaly of Corollary 5.192: the Capelli/Moyal shift $N \cdot \omega(f, g) \int_{\mathbb{R}} a(t)b(t)\, dt$ obstructs the Chan–Paton-line lift.

Proof. Smearing the equal-time Dirac bracket $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(t')\} = \delta^i_l \delta^k_j \delta(t - t')$ against $a(t)b(t')$ turns $\delta(t - t')$ into $a(t)b(t)$. The matrix symplectic form contracts the Hamiltonian derivatives of f and g to the holomorphic Poisson bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ evaluated at (ϕ_1, ϕ_2). Tracing and projecting to the stable scalar-reduced quotient gives (14); disjoint-support vanishing follows because the smeared product ab is pointwise zero. This is the level-one component of Theorem 4.11. The center anomaly is the unreduced refinement Corollary 5.192 (ii); after scalar reduction it is killed and (14) survives. □

The shifted-cotangent partner β on the brane is the principal-part defect current of Theorem 5.186. Together with Theorem 4.5 the trace map J and the principal-part defect current $\rho \mapsto \Theta_\rho$ form a cosheaf source datum on intervals $I \subset \mathbb{R}$.

Construction 4.6 *(Reduced line-defect current kernel).* Let

$$X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}, \qquad L = \mathbb{R}_{x_1} \times \{x_2 = 0\} \times \{p\} \subset X,$$

and let $i : L \hookrightarrow X$ be the inclusion. This is the line defect supporting the brane currents. In the reduced formal local model, a compactly supported test density $a(t)\, dt$ on L defines the defect-supported current $i_*(a(t)\, dt)$, locally written as $a(x_1)\delta(x_2)\delta_p\, dx_1 dx_2$ together with the formal holomorphic delta-current at p.

The degree-zero Hamiltonian part of the reduced PBW/Rees defect-current vertex is the smeared boundary current (13):

$$K_L^{\mathrm{cl}}(a \otimes f) = B_f(a), \qquad f \in \bar{A}.$$

For the cotangent principal-part source, the reduced vertex sends

$$B \otimes \rho[1] \longmapsto \Theta_\rho(B), \qquad B \in \mathcal{D}'_c(L), \quad \rho \in \mathcal{P},$$

using Theorem 5.186. These formulas define a reduced PBW/Rees defect-current kernel. A cochain-level map from closed CE observables, and the quantum Costello brane-defect kernel, are separate objects: they require the Ward identity, anomaly cancellation, finite-scale propagator, and quantum moment-map compatibility.

Construction 4.7 *(Interval Hamiltonian and factorization algebras).* For an interval $I \subset \mathbb{R}$, define

$$\begin{aligned} \mathfrak{h}_I &= \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}, \\ \mathcal{P}_I &= \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}, \\ \mathfrak{g}_I &= \mathfrak{h}_I \ltimes \mathcal{P}_I[1], \end{aligned}$$

with brackets

$$\begin{aligned} [a \otimes f, b \otimes g]_{\mathfrak{h}_I} &= ab \otimes \{f, g\}_{\bar{A}}, \\ [a \otimes f, B \otimes \rho[1]]_{\mathfrak{g}_I} &= aB \otimes (f \cdot \rho)[1], \\ [B \otimes \rho[1], C \otimes \sigma[1]]_{\mathfrak{g}_I} &= 0, \end{aligned}$$

where products in $\Omega^\bullet_c(I)$ are de Rham products and, in the displayed degree-zero formulas, aB is multiplication of a compactly supported current by a smooth compactly supported function and $f \cdot \rho = \pi_{\mathrm{pp}}\{f, \rho\}$ is the principal-part coadjoint action of 5.176. Inclusions $I' \subset I$ act by extension by zero on compactly supported de Rham forms and currents; disjoint unions act by sum. This makes $I \mapsto \mathfrak{h}_I, \mathfrak{g}_I$ cosheaves of (graded) Lie algebras on $\mathbb{R}$. For $j : I \hookrightarrow J$, the map $j_!$ is defined on $\Omega^\bullet_c(I)$ because supports are compactly contained in I; equivalently, smooth representatives vanish near the boundary of I inside J. It satisfies

$$j_!(\alpha \wedge \beta) = j_!\alpha \wedge j_!\beta, \qquad j_!(aB) = (j_!a)(j_!B),$$

in the degree-zero current notation. On CE cochains the same extension-by-zero maps induce pullback/coproduct maps; the CE cochain source is contravariant in this convention.

Define the boundary P_0 factorization algebras on $\mathbb{R}$ in the formal stable Hamiltonian sector by

$$\begin{aligned} A_\partial^{\mathrm{Ham}}(I) &= \widehat{\mathbf{S}}(\mathfrak{h}_I), \\ A_\partial^{\mathrm{pp}}(I) &= \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1]), \end{aligned}$$

with degree-zero reduced Hamiltonian P_0 bracket extended as a biderivation from

$$\begin{aligned} \{a \otimes f, b \otimes g\}_{A_\partial} &= ab \otimes \{f, g\}_{\bar{A}}, \\ \{a \otimes f, B \otimes \rho[1]\}_{A_\partial} &= aB \otimes (f \cdot \rho)[1], \\ \{B \otimes \rho[1], C \otimes \sigma[1]\}_{A_\partial} &= 0. \end{aligned}$$

The factorization product on disjoint intervals is the multiplication in $\widehat{\mathbf{S}}$ after extension by zero. These are the reduced objects after de Rham-current contraction for the reduced Hamiltonian current central-operation map. The Hamiltonian summand $A_\partial^{\mathrm{Ham}}$ is the connected component of the formal Hamiltonian sector of the open brane factorization algebra; the principal-part summand $A_\partial^{\mathrm{pp}}/A_\partial^{\mathrm{Ham}}$ is the reduced current object of 5.186.

LEMMA 4.8 *(Compact-support interval contraction).* Let $I \subset \mathbb{R}$ be a nonempty open interval, oriented by the coordinate t. Choose a compactly supported one-form $\lambda_I(t)\,dt \in \Omega_c^1(I)$ with $\int_I \lambda_I(t)\,dt = 1$. Define

$$p_I : \Omega_c^\bullet(I) \longrightarrow \mathbb{C}[-1], \qquad p_I(\alpha) = \int_I \alpha,$$

with p_I zero on degree 0, and

$$i_I : \mathbb{C}[-1] \longrightarrow \Omega_c^\bullet(I), \qquad i_I(1) = \lambda_I(t)\,dt.$$

Extend compactly supported forms by zero to $\mathbb{R}$. For a one-form $\alpha = f(t)\,dt$, set

$$K_I(\alpha)(t) = \int_{-\infty}^t \left(f(s) - \lambda_I(s) \int_I f(r)\,dr \right) ds, \qquad K_I(\varphi) = 0 \quad (\varphi \in \Omega_c^0(I)).$$

Then $K_I : \Omega_c^1(I) \to \Omega_c^0(I)$ is well-defined and compactly supported, and

$$dK_I + K_I d = \mathrm{id}_{\Omega_c^\bullet(I)} - i_I p_I.$$

Consequently $\Omega_c^\bullet(I)$ contracts onto its compact-support cohomology line $\mathbb{C}[-1]$. Acyclicity of $\Omega_c^\bullet(I)$ is replaced by the following exact statement: a compactly supported primitive of a compactly supported one-form exists precisely on the kernel of the integral map, and the integral class is the surviving compact-support cohomology generator.

Proof. The one-form inside the integral defining $K_I(\alpha)$ has compact support and total integral zero. Its primitive therefore vanishes near both ends of I, after extension by zero to $\mathbb{R}$, so $K_I(\alpha) \in \Omega_c^0(I)$.

For $\alpha = f(t)\,dt$,

$$dK_I(\alpha) = \left(f(t) - \lambda_I(t) \int_I f(r)\,dr \right) dt = \alpha - i_I p_I(\alpha).$$

For a compactly supported function φ,

$$K_I(d\varphi)(t) = \int_{-\infty}^t \varphi'(s)\,ds = \varphi(t),$$

because φ vanishes near the left end, and $p_I(\varphi) = 0$. These two identities are exactly $dK_I + K_I d = \mathrm{id} - i_I p_I$ in degrees 1 and 0. $\square$

Proposition 4.9 *(Locality of the brane P_0 bracket).* Let $a \in \Omega^0_c(I_1)$, $b \in \Omega^0_c(I_2)$ with $I_1 \cap I_2 = \emptyset$. For all $f, g \in \bar{A}$,

$$\{B_f(a), B_g(b)\}_{A_\partial^{\mathrm{Ham}}} = 0.$$

In particular, $A_\partial^{\mathrm{Ham}}$ has the strict factorization product property: the bracket between sections supported on disjoint intervals vanishes.

Proof. By 4.7, $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$. The pointwise product $ab \equiv 0$ since the supports are disjoint, so $B_{\{f,g\}}(ab) = 0$. □

Remark 4.10 *(Locality from the canonical equal-time bracket).* Locality is the canonical equal-time bracket of the first-order open BV theory, written on two brane-time points (t, t') (with s reserved throughout for the transverse $\mathbb{R}_s$ coordinate of $X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2$):

$$\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(t')\} = \delta^i_l \delta^k_j\, \delta(t - t'), \qquad \{(\phi_a)^i{}_j(t), (\phi_a)^k{}_l(t')\} = 0 \ \ (a = 1, 2).$$

Tracing $f(\phi_1, \phi_2)(t)$ and $g(\phi_1, \phi_2)(t')$, applying the Leibniz rule, and integrating against $a(t)b(t')\delta(t - t')$ produces exactly the smeared identity $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$ of 5.192 in unsmeared form. In the classical reduced, de Rham-contracted sector this delta-function locality produces a strict support-local P_0 factorization product. The unreduced BV current complex still requires its own homotopies, contact-term choices, and QME data.

Theorem 4.11 *(Interval-wise reduced Hamiltonian CE/PV central-operation map, Hamiltonian sector).* Let $\mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N}$ be the reduced open-brane factorization algebra of 4.14. Let $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H,\mathrm{Ham}}|_L$ denote the formal classical Hamiltonian-mode subfactorization algebra of the cotangent BF theory on the formal neighbourhood of L obtained by restricting to the connected Hamiltonian central-operation sector. Define the interval-wise CE cochain cofactorization object, equivalently a factorization object on the opposite interval category, by

$$\mathcal{F}_{\mathrm{cl}}(I) = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_I), \qquad \mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$

where extension by zero on $\Omega^\bullet_c$ and on compactly supported currents induces the CE coproduct. Then:

(i) there is a continuous interval-wise morphism of P_0 cofactorization source data to the reduced Hamiltonian central-operation target

$$\Phi^{\mathrm{Ham}}_{\mathrm{fact}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H,\mathrm{Ham}}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A_\partial^{\mathrm{Ham}}),$$

whose underlying CE/PV map on each interval I is given on length-one coordinates by

$$B_f(a) := \Phi_I(u_{a(t)dt\otimes f}) = \int_I a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt,$$

and

$$\Phi_I(c_{B\otimes\rho}) = \theta_{B\otimes\rho}.$$

The reduced principal-part current representative $\Theta_\rho(B)$ belongs to the boundary local-dual model of Theorem 5.186; its comparison with the CE/PV vector-field coordinate is made after that reduction.

(ii) this morphism is an isomorphism of P_0-cdg algebras in each fixed interval, after the stated reduced current model is chosen, and is compatible with the CE coproduct on the source and disjoint factorization products on the target;

(iii) after applying the compact-support de Rham contraction $p_I : \Omega_c^\bullet(I) \to \mathbb{C}[-1]$, the induced Hamiltonian density sector is locally constant in the topological direction $\mathbb{R}$ in the sense of Lurie, *Higher Algebra*, §5.5; Costello–Gwilliam [15, Vol. I §6.4] gives the published factorization-algebra terminology. Extension by zero $\mathcal{D}'_c(I') \to \mathcal{D}'_c(I)$ for arbitrary compactly supported distributional currents requires separate analysis;

(iv) the factorization product $A_\partial^{\mathrm{Ham}}(I_1) \otimes A_\partial^{\mathrm{Ham}}(I_2) \to A_\partial^{\mathrm{Ham}}(V)$ for disjoint $I_1 \sqcup I_2 \subset V$ corresponds under $\Phi_{\mathrm{fact}}^{\mathrm{Ham}}$ to the CE coproduct on the closed side induced by extension by zero $\Omega_c^\bullet(I_j) \hookrightarrow \Omega_c^\bullet(V)$;

(v) at the stalk $I \to \{t_0\}$, $\Phi_{\mathrm{fact}}^{\mathrm{Ham}}$ restricts to the stalkwise central operation

$$f \longmapsto L_{B_f} \in C^0_{\mathrm{Hoch}}\big(A_N^{\mathrm{pt}}, A_N^{\mathrm{pt}}\big)$$

of 3.39.

Proof. Step 1. Identify the interval-wise reduced P_0 model. In the formal stable Hamiltonian sector, the boundary factorization algebra $A_\partial^{\mathrm{Ham}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$ is graded-commutative, with strict P_0 bracket inherited from $\mathfrak{h}_I$ as a biderivation. Here $\mathrm{Z}^{P_0}_{\mathrm{red,Ham}}$ means the reduced interval-wise Poisson-CE model $\mathrm{PV}^\bullet\big(\widehat{\mathbf{S}}(\mathfrak{h}_I)\big) = \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathfrak{h}^\vee_{I,\mathrm{cont}}[-1])$ with the Schouten bracket. A genuine Hochschild/factorization-centre identification for the completed Tate algebra is a continuous Hochschild–Kostant–Rosenberg theorem with interval functoriality; the proof here uses the explicit PV model. Restricting to the polynomial subalgebra $\mathbf{S}(\mathfrak{h}_I) \subset \widehat{\mathbf{S}}(\mathfrak{h}_I)$ and to the constant polyvectors gives the central multiplication operations, whose boundary names are the Hamiltonian currents $B_f(a)$. The CE coordinates c dual to Hamiltonian generators map instead to constant polyvectors. This is the interval version of the generator dictionary $c^I \mapsto \theta^I, u_I \mapsto O_I$ in 4.20.

Step 2. Construct the CE source. By 5.166, applied to $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$ in the stated completed category,

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_I) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{I,\mathrm{cont}}[-1]\big),$$

and its CE differential is the dual of the semidirect Lie bracket. The Hamiltonian part is controlled by the de Rham-current pairing on the $\Omega_c^\bullet(I)$ factor and the residue-Taylor pairing on $\bar{A}$. In degree-zero notation this is the familiar pairing between compactly supported test functions and compactly supported currents; the polynomial subspace $\bar{A} \subset \widehat{\bar{A}}$ pairs with the discrete Taylor dual generated by the $\delta_{a,b}$. The shifted-cotangent CE coordinate represented by $a(t)dt \otimes f$ is dual to the cotangent generator determined by that current. Its boundary image is $B_f(a)$.

Step 3. Define $\Phi_{\mathrm{fact}}^{\mathrm{Ham}}(I)$ on generators. By 6.20, the closed Hamiltonian cotangent BF observables on the formal neighbourhood of $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ in the Hamiltonian sector are $\widehat{\mathbf{S}}(\mathfrak{h}^\vee_{I,\mathrm{cont}}[-1])$, with CE differential dual to the Hamiltonian Lie bracket on $\mathfrak{g}_I$. The interval-wise map sends a length-one shifted-cotangent CE coordinate represented by $a(t)dt \otimes f$ to the element of the central operations factorization algebra given by the multiplication operator

$$L_{B_f(a)} \in \mathrm{Hoch}^0\big(A_\partial^{\mathrm{Ham}}(I)\big) \subset \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}\big(A_\partial^{\mathrm{Ham}}\big)(I),$$

with $B_f(a) = \Phi_I(u_{a(t)dt\otimes f})$ as displayed. By 3.39, this operator is closed (Hochschild 0-cocycle on the graded-commutative open stalk). Extend $\Phi_{\mathrm{fact}}^{\mathrm{Ham}}(I)$ as a continuous algebra morphism on $\widehat{\mathbf{S}}$.

Step 4. Verify CE differential compatibility. On length two, the CE differential of a shifted-cotangent generator represented by $a(t)dt \otimes f$ is the coadjoint term dual to the Lie bracket on $\mathfrak{h}_I$. By 5.192 (its sharpened form below),

$$\{L_{B_f(a)}, L_{B_g(b)}\}_{A_\partial^{\mathrm{Ham}}} = L_{B_{\{f,g\}}(ab)}.$$

Pulling back through $\Phi_{\mathrm{fact}}^{\mathrm{Ham}}$ matches this with the shifted-cotangent CE differential dual to $[a \otimes f, b \otimes g]_{\mathfrak{h}_I} = ab \otimes \{f, g\}$. Inductive extension by length and the biderivation property propagates this to all CE lengths.

Step 5. Quasi-isomorphism on each interval. Both sides are free graded-commutative algebras on the same chosen compact-support current reduced Tate vector space after the generator dictionary $c \mapsto \theta$, $u \mapsto O$, with the same CE differential induced from the semidirect Lie bracket of $\mathfrak{g}_I$ (closed side: by definition; brane side: by the reduced interval-wise PV model of Step 1 and the Schouten bracket). The map respects the symmetric-algebra generators by construction. Thus it is an isomorphism of underlying graded vector spaces and intertwines the differentials. The map is therefore an isomorphism of P_0-cdg algebras on each interval.

Step 6. Factorization product compatibility. For disjoint $I_1, I_2 \subset V$, the CE source extension by zero gives $\mathfrak{h}_{I_1} \oplus \mathfrak{h}_{I_2} \hookrightarrow \mathfrak{h}_V$. Thus the CE source is cofactorization: if $j_{I_k,V,!} : \mathfrak{g}_{I_k} \to \mathfrak{g}_V$ denotes extension by zero, the source structure map is

$$\Delta_{I_1,I_2;V} : C^\bullet_{\mathrm{CE}}(\mathfrak{g}_V) \to C^\bullet_{\mathrm{CE}}(\mathfrak{g}_{I_1}) \widehat{\otimes} C^\bullet_{\mathrm{CE}}(\mathfrak{g}_{I_2}).$$

On the brane side, $A_\partial^{\mathrm{Ham}}(I_1) \otimes A_\partial^{\mathrm{Ham}}(I_2) \to A_\partial^{\mathrm{Ham}}(V)$ is the multiplication map under the inclusion of $\mathfrak{h}_{I_1} \oplus \mathfrak{h}_{I_2} \hookrightarrow \mathfrak{h}_V$. By 4.9, the bracket between the two factors vanishes on disjoint supports, so the product is strictly multiplicative. The induced map between central-operation factorization algebras is the corresponding tensor product of multiplication operators. The compatibility is the mixed-variance equation

$$\Phi_V = m_{I_1,I_2;V} \circ \left(\Phi_{I_1} \otimes \Phi_{I_2}\right) \circ \Delta_{I_1,I_2;V},$$

for the operations whose support lies in $I_1 \sqcup I_2 \subset V$. Equivalently, the diagram

$$\begin{tikzcd}
\mathcal{F}_{\mathrm{cl}}(I_1) \otimes \mathcal{F}_{\mathrm{cl}}(I_2) \arrow[r, "\Phi(I_1)\otimes\Phi(I_2)"] & \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A_\partial^{\mathrm{Ham}})(I_1) \otimes \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A_\partial^{\mathrm{Ham}})(I_2) \arrow[d] \\
\mathcal{F}_{\mathrm{cl}}(V) \arrow[u] \arrow[r, "\Phi(V)"] & \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A_\partial^{\mathrm{Ham}})(V)
\end{tikzcd}$$

commutes after vertical reversal of the left arrow (CE coproduct = dual of extension by zero).

Step 7. Locally constant structure. For each interval I, Lemma 4.8 gives the explicit contraction

$$\Omega_c^\bullet(I) \underset{i_I}{\overset{p_I}{\rightleftarrows}} \mathbb{C}[-1], \qquad i_I(1) = \lambda_I(t)\,dt, \quad p_I(\alpha) = \int_I \alpha,$$

with $dK_I + K_I d = \mathrm{id} - i_I p_I$. The maps p_I are natural for extension by zero; the choice of i_I and K_I is a choice of density representative for the compact-support cohomology generator. The degree-zero boundary notation $B_f(a)$ used above already includes the shifted-cotangent degree, so the density representative appears as a degree-zero CE coordinate after the BV shift. Translation is homotopically trivial by the same de Rham contraction, hence the interval-wise objects are locally constant in the topological direction after this density contraction.

Step 8. Stalk recovery. Under the density contraction, a length-one shifted-cotangent CE coordinate represented by $a(t)dt \otimes f$ maps to the boundary current $B_f(a)$ and hence to the multiplication operator $L_{B_f(a)}$. Stalk recovery is obtained by any cofinal family of compactly supported unit densities supported near t_0. This represents the compact-support cohomology generator under the locally constant factorization-algebra quasi-isomorphism; degree-zero point evaluation is replaced by compact-support density contraction. □

Remark 4.12 *(Locally constant equivalence and E_1-algebra translation).* In the locally constant setting on $\mathbb{R}$, factorization algebras are equivalent to homotopy associative algebras up to translation invariance by Lurie, *Higher Algebra*, §5.5.4, in the Costello–Gwilliam vocabulary of locally constant factorization algebras [15, Vol. I §6.4]. The Hochschild cochain complex of the E_1 algebra computes the factorization-center stalk. The reduced Hamiltonian theorem here uses the explicit PV model $\widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathfrak{h}_I^\vee[-1])$ and its strict P_0 subalgebra of multiplication operators. Identifying this PV model with continuous Hochschild cochains for the completed Tate algebra is a separate HKR/topology theorem. The CE source identification of 4.11 is exactly the closed dual of this PV picture in the connected Hamiltonian sector.

Remark 4.13 *(Presentations of one-dimensional factorization algebras).* The precise equivalence used here is Lurie's identification of locally constant factorization algebras on the line with E_1-algebras [41, Thm. A.3.7.5]. The Costello–Gwilliam model-categorical language is used only as a compatible presentation convention unless the cited corollary is separately anchored.

For a factorization algebra $\mathcal{A}$ on the line, write $\mathrm{Z}_{\mathrm{fact}}(\mathcal{A})$ for the factorization algebra of E_1 central operations; in the locally constant one-dimensional case its stalk is the Hochschild cochain complex of the corresponding E_1 algebra. The notation $\mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathcal{A})$ means the same central operations equipped with classical Poisson/BV compatibility homotopies. The reduced Hamiltonian proof uses the explicit local central-operation model.

THEOREM 4.14 *(Hamiltonian-current central operation, smeared).* 4.11 restricted to the length-one CE component on each interval recovers the degree-zero Hamiltonian current: the assignment

$$\Phi_{I,N}(u_{a(t)dt\otimes f})(u) = B_f(a)\cdot u = \Big(\int_I a(t)\,\overline{\mathrm{Tr}\, f(\phi_1(t),\phi_2(t))}\,dt\Big)u$$

is a closed Hochschild 0-cochain, hence a degree-zero central operation, in $\mathrm{Z}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N})(I)$, the assignments are compatible with interval inclusions and disjoint factorization products, and the induced operations satisfy

$$\{\Phi(u_{a(t)dt\otimes f}),\Phi(u_{b(t)dt\otimes g})\}_{A_\partial^{\mathrm{Ham}}} = \Phi(u_{a(t)b(t)dt\otimes\{f,g\}_{\bar{A}}}).$$

The degree-zero current is the length-one component of the reduced CE/PV central-operation comparison: its content is the strict P_0 identity, one of the structure equations defining $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ of 4.11.

Proof. In the reduced open model $Q\phi_1 = Q\phi_2 = 0$. Hence $\overline{\mathrm{Tr}\, f(\phi_1,\phi_2)}$ is Q-closed for every Hamiltonian f, and the compactly supported integral is Q-closed. Since the reduced open observable algebra is graded-commutative, multiplication by this closed element is a Hochschild 0-cocycle; this is the interval-wise version of 3.39. Extension by zero of the compactly supported test density is compatible with enlargement to a larger interval through the extension map $j_!$. For disjoint intervals 4.9 gives vanishing of the P_0 bracket between the two factors, so the factorization product of two such operations is the strict multiplication by $B_f(a)B_g(b)$; this is the multiplicative extension from $\mathfrak{h}_I$ to $A_\partial^{\mathrm{Ham}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$; the linear current bracket is its indecomposable part.

The Poisson statement is local on the line. For compactly supported test functions a, b, 5.192 (the strict P_0 identity on the brane factorization algebra) gives

$$\{B_f(a), B_g(b)\}_{A_\partial^{\mathrm{Ham}}} = B_{\{f,g\}_{\hat{A}}}(ab).$$

Passing to the connected stable Hamiltonian quotient removes only the scalar Hamiltonian line, which is central for the bracket. Thus the current bracket is preserved. Taking a and b supported in a small coordinate interval and then passing to the locally constant stalk recovers 5.191. □

Proposition 4.15 *(CE-source obstruction for perfect Hamiltonian Lie algebras).* Let $\mathfrak{h}$ be a perfect Lie algebra in **TateVec** (i.e. $[\mathfrak{h}, \mathfrak{h}] = \mathfrak{h}$). Then every length-one CE coordinate c^I dual to a Hamiltonian generator $H_I \in \mathfrak{h}$ has nonzero CE differential:

$$d_{\mathrm{CE}} c^K = -\tfrac{1}{2}\, C^K_{IJ}\, c^I c^J \neq 0.$$

Equivalently, $H^1(\mathfrak{h}; \mathbb{C}) = 0$.

This obstruction is the structural source of the cotangent direction: under 4.20, the boundary observable $O_I = J(H_I) \in A_\partial$ is

$$O_I = \Phi_{\mathrm{coord}}(u_I),$$

the image of the CE coordinate dual to the shifted cotangent generator $\eta_{\partial_I} \in \mathfrak{h}^\vee[1]$. The boundary observable lives in the cotangent direction of the CE cochain algebra. Its CE differential is $d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J$, mapping under Φ_{coord} to the Hamiltonian vector field $d_\pi^{\mathrm{coord}} O_K = C^L_{KJ} O_L \theta^J$ of O_K in the Schouten algebra.

This is exactly what the bracket-admissible P_0-map produces: a function whose Schouten differential is its Hamiltonian vector field.

Proof. $\mathfrak{h}$ perfect means each basis vector has nonzero projection from some bracket combination; equivalently, for every length-one Hamiltonian CE coordinate c^K there are indices I, J with $C^K_{IJ} c^I c^J \neq 0$. Hence every c^K has nonzero differential, and H^1 vanishes. Identification $O_I = \Phi_{\mathrm{coord}}(u_I)$ is the content of 5.40 combined with $\Phi_{\mathrm{coord}}(u_I) = O_I$ from 4.20. Differentials match on generators by 5.168. □

Remark 4.16. The reduced Hamiltonian Lie algebra $\mathfrak{h} = \widehat{A}/\mathbb{C}\cdot 1 = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ of the formal symplectic polydisk in dimension two is perfect: for every monomial one has, before quotienting constants, $z_1^a z_2^b = \left\{z_1, \frac{1}{b+1} z_1^a z_2^{b+1}\right\}$, so $[\mathfrak{h}, \mathfrak{h}] = \mathfrak{h}$. In particular $[H_{1,0}, H_{m,n}] = n\, H_{m,n-1}$ and $[H_{0,1}, H_{m,n}] = -m\, H_{m-1,n}$. The hypothesis of 4.15 is met. The CE-length-one cochain $\xi_{a,b} = c^{a,b}$ dual to $H_{a,b} = z_1^a z_2^b$ has nonzero differential; closed CE cochains have zero linear Hamiltonian part. This is the source obstruction.

The CE/PV identification therefore proceeds at the level of the coordinate completion of the cochain algebra. Finite coordinate words carry the bracket and differential, and convergence of brackets and Casimir kernels on the ambient product is an admissibility hypothesis.

Definition 4.17 *(Formal-polydisk coordinate CE/PV completion).* The coordinate CE/PV completion is the product over finite graded-commutative words in the Hamiltonian CE coordinates c^I, shifted-cotangent CE coordinates u_I, Schouten coordinates θ^I, and boundary Hamiltonian coordinates O_I.

All differentials and brackets are read coefficientwise in finite coordinate windows and then passed to the compatible product completion.

Definition 4.18 *(Bracket-admissible CE/PV realizations).* A coordinate realization $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ is bracket-admissible when the Schouten bracket and the Hamiltonian boundary brackets of the generators extend continuously from every finite coordinate window to B. The worked nontrivial class is the weighted-Tate coefficient pair of Definition 5.55, with bracket-admissibility established by Corollary 5.71 and the diagonal kernel-admissibility refinement by Theorem 5.67. The strict ambient product completion sits below admissibility at both levels: the continuous Schouten biderivation extends from finite words to the full product only through the weighted-Tate replacement, and the diagonal Casimir $C_{\mathfrak{h}}$ converges only in coefficient categories with a regulator (Lemma 5.212); item (4) of Theorem 5.5 records the combined obstruction datum.

LEMMA 4.19 *(Formal-polydisk coordinate differential formulas).* In the coordinate completion,

$$d_{\mathrm{CE}}c^K = -\frac{1}{2}C^K_{IJ}c^I c^J, \qquad d_{\mathrm{CE}}u_K = C^L_{KJ}u_L c^J,$$

and

$$d_\pi^{\mathrm{coord}}\theta^K = -\frac{1}{2}C^K_{IJ}\theta^I\theta^J, \qquad d_\pi^{\mathrm{coord}}O_K = C^L_{KJ}O_L\theta^J.$$

THEOREM 4.20 *(Admissible reduced Hamiltonian CE/PV central-operation map, cochain level).* Let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$$

be the reduced Hamiltonian Lie algebra of the formal symplectic polydisk, equipped with the coordinate completion of Definition 4.17. Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ be the shifted-cotangent extension. Let $A_{\partial,\mathrm{coord}}$ be the coordinate completed symmetric algebra on the stable scalar-reduced Hamiltonian trace labels.

Then there is a canonical isomorphism of completed dg graded-commutative coordinate algebras

$$\Phi_{\mathrm{coord}} : C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}}),$$

determined on generators by

$$\Phi_{\mathrm{coord}}(c^I) = \theta^I, \qquad \Phi_{\mathrm{coord}}(u_I) = O_I,$$

where c^I is the CE coordinate dual to $H_I \in \mathfrak{h}$ and u_I is the CE coordinate dual to the shifted cotangent generator $\eta_{\partial_I} \in \mathfrak{h}^\vee[1]$.

If $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ is a bracket-admissible realization of the reduced Hamiltonian central-operation model, then the restriction

$$\Phi^B_{P_0} : \Phi^{-1}_{\mathrm{coord}}(B) \xrightarrow{\sim} B$$

is the cochain-level open-closed reduced Hamiltonian central-operation isomorphism in the formal/local sector. The strict ambient product completion has a separate P_0-algebra obstruction. Tate bar-cobar and enveloping conclusions require the separate admissible conilpotent hypotheses of Lemma 5.166.

Proof. Generators and coordinate extension. By Definition 4.17, $C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g})$ is the product over finite graded-commutative words in the letters u_I, c^I, with $|u_I| = 0$ and $|c^I| = 1$. The complex $\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ is the identical coordinate product with O_I, θ^I in place of u_I, c^I. The assignment

$c^I \mapsto \theta^I, u_I \mapsto O_I$ therefore extends uniquely to a continuous bijection of completed coordinate dg-commutative algebras. This is a coordinate envelope statement. Literal continuous CE cochains form the continuous subspace inside this product when the coefficient support is continuous for the chosen topology; the whole coordinate product is the formal coordinate envelope.

Cochain identity on generators. The CE differential is

$$d_{\mathrm{CE}} c^K = -\tfrac{1}{2} C^K_{IJ} c^I c^J, \qquad d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J.$$

By Lemma 4.19,

$$d_\pi^{\mathrm{coord}} \theta^K = -\tfrac{1}{2} C^K_{IJ} \theta^I \theta^J, \qquad d_\pi^{\mathrm{coord}} O_K = C^L_{KJ} O_L \theta^J.$$

Therefore

$$\Phi_{\mathrm{coord}}(d_{\mathrm{CE}} c^K) = -\tfrac{1}{2} C^K_{IJ} \theta^I \theta^J = d_\pi^{\mathrm{coord}} \theta^K = d_\pi^{\mathrm{coord}} \Phi_{\mathrm{coord}}(c^K),$$

$$\Phi_{\mathrm{coord}}(d_{\mathrm{CE}} u_K) = C^L_{KJ} O_L \theta^J = d_\pi^{\mathrm{coord}} O_K = d_\pi^{\mathrm{coord}} \Phi_{\mathrm{coord}}(u_K).$$

Cochain identity on the full coordinate algebra. Both d_{CE} and d_π^{coord} are degree-1 derivations of the completed graded-commutative algebra structures on each side. They satisfy the graded Leibniz rule with the same Koszul sign convention. Φ_{coord} is an algebra homomorphism, so

$$\begin{aligned}\Phi_{\mathrm{coord}}(d_{\mathrm{CE}}(xy)) &= \Phi_{\mathrm{coord}}\big((d_{\mathrm{CE}} x) y + (-1)^{|x|} x\, d_{\mathrm{CE}} y\big) \\ &= \Phi_{\mathrm{coord}}(d_{\mathrm{CE}} x)\, \Phi_{\mathrm{coord}}(y) + (-1)^{|x|} \Phi_{\mathrm{coord}}(x)\, \Phi_{\mathrm{coord}}(d_{\mathrm{CE}} y),\end{aligned}$$

while

$$\begin{aligned}d_\pi^{\mathrm{coord}} \Phi_{\mathrm{coord}}(xy) &= d_\pi^{\mathrm{coord}}\big(\Phi_{\mathrm{coord}}(x) \Phi_{\mathrm{coord}}(y)\big) \\ &= (d_\pi^{\mathrm{coord}} \Phi_{\mathrm{coord}}(x))\, \Phi_{\mathrm{coord}}(y) + (-1)^{|x|} \Phi_{\mathrm{coord}}(x)\, d_\pi^{\mathrm{coord}} \Phi_{\mathrm{coord}}(y).\end{aligned}$$

Generator-level agreement plus Leibniz/algebra-homomorphism gives agreement on every word in $\{c^I, u_J\}$ at fixed total length; coordinate-product completeness of both sides plus continuity of $d_{\mathrm{CE}}, d_\pi^{\mathrm{coord}}, \Phi_{\mathrm{coord}}$ for finite coordinate projections extends the equality to the completed coordinate algebras.

Inverse. The assignment $\theta^I \mapsto c^I, O_I \mapsto u_I$ extends to a continuous algebra homomorphism in the opposite direction. By the same Leibniz argument, it intertwines d_π^{coord} and d_{CE}. The two compositions fix generators, hence fix the completed algebras. So Φ_{coord} is an isomorphism of complete dg graded-commutative algebras.

Cotangent/Schouten bracket compatibility. On finite words the CE side carries the canonical (-1)-shifted Poisson structure on $C^\bullet_{\mathrm{CE,coord}}(T^*[1]\mathfrak{h})$ with

$$\{c^I, u_J\}_{\mathrm{CE}} = \delta^I_J, \quad \{c^I, c^J\}_{\mathrm{CE}} = 0, \quad \{u_I, u_J\}_{\mathrm{CE}} = 0.$$

These are the degree-(-1) cotangent brackets on the cochain algebra of a shifted cotangent Lie algebra. The bracket compatibility used here is the displayed generator computation; it is separate from the degree-zero reduced Hamiltonian P_0 bracket on boundary currents. The finite-word Schouten bracket on $\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ satisfies

$$[\theta^I, O_J]_S = \delta^I_J, \quad [\theta^I, \theta^J]_S = 0, \quad [O_I, O_J]_S = 0,$$

the last because O_I, O_J are functions on the formal affine space $\mathfrak{h}^\vee$ and the Schouten bracket of two functions is zero. Generator-level matching gives $\Phi_{\mathrm{coord}}(\{c^I, u_J\}_{\mathrm{CE}}) = \Phi_{\mathrm{coord}}(\delta^I_J) = \delta^I_J = [\theta^I, O_J]_S = [\Phi_{\mathrm{coord}}(c^I), \Phi_{\mathrm{coord}}(u_J)]_S$, and similarly for the other two pairs. This proves bracket compatibility on finite words. On a bracket-admissible B, continuity and the biderivation rule extend the equality to $\Phi^{-1}_{\mathrm{coord}}(B)$ and B. Bracket statements on the strict ambient product completion require a bracket-admissible target.

Identification with the reduced P_0 Hamiltonian central-operation model. For the reduced constant-stalk convention used here, the reduced P_0 Hamiltonian central-operation model of a Poisson algebra A is by definition represented by the Poisson cochain complex $\mathrm{PV}(A)$ with the Lichnerowicz differential $[\pi, -]_S$. This is the local Poisson-CE model; an unreduced Costello–Gwilliam central-operations theorem is the corresponding unreduced theorem. Therefore any bracket-admissible B representing this local Poisson-CE model identifies with $Z^{P_0}_{\mathrm{red,Ham}}(A_{\partial,\mathrm{coord}}; B)$. Therefore $\Phi^B_{P_0}$ is the reduced Hamiltonian central-operation isomorphism in this local stalk sense; the ambient product coordinate map requires bracket-admissibility first. □

PROPOSITION 4.21 *(Reduced coordinate coupling equations).* Let A be a bracket-admissible dg P_0-target and let

$$\kappa\colon C^\bullet_{\mathrm{CE,coord}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]) \longrightarrow A$$

be a continuous graded-commutative algebra map. Write $X^I = \kappa(c^I)$ and $\mu_I = \kappa(u_I)$. Then κ is a dg map exactly when

$$d_A X^K + \frac{1}{2} C^K_{IJ} X^I X^J = 0, \qquad d_A \mu_K - C^L_{KJ} \mu_L X^J = 0.$$

It is a P_0-map exactly when, in addition, the generator contraction identities are satisfied:

$$\{X^I, \mu_J\}_A = \delta^I_J, \qquad \{X^I, X^J\}_A = 0, \qquad \{\mu_I, \mu_J\}_A = 0.$$

The Dirac CE/PV solution is $X^I = \theta^I$, $\mu_I = O_I$. For an unreduced factorization target the same equations are replaced by the curvature condition

$$\mathrm{Curv}(\kappa) = d\kappa + \frac{1}{2}[\kappa, \kappa] = 0$$

in the completed convolution complex, together with the descent and QME data required for such an unreduced lift.

Proof. The source is freely generated, as a completed graded-commutative coordinate algebra, by c^I and u_I. A continuous algebra map is therefore determined by the two coordinate families X^I and μ_I. Intertwining the differentials is exactly the two displayed equations, because $d_{\mathrm{CE}} c^K = -\frac{1}{2} C^K_{IJ} c^I c^J$ and $d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J$. The P_0 condition is likewise the generator-level cotangent/Schouten contraction identity; bracket-admissibility is the hypothesis that extends this identity from generators to the completed target. The last sentence is the same condition written in the convolution dg Lie algebra where an unreduced factorization lift lives. □

THEOREM 4.22 *(Abstract bracket-compatible CE/PV recognition criterion).* Let $L = \varprojlim_\alpha L_\alpha$ be a complete Hausdorff filtered dg Lie algebra over $\mathbb{C}$, with finite-dimensional bracket-compatible dg Lie quotients L_α, and let $D = L^\vee_{\mathrm{cont}} = \varinjlim_\alpha L^\vee_\alpha$. Put $G = L \ltimes D[1]$. Form the completed coordinate CE and PV algebras by taking products over finite words and finite coordinate windows. Assume:

(U1) the internal differential, Lie bracket on L, and coadjoint action on D are continuous, equivalently every finite coordinate projection of $d_L x$, $[x, y]$, and $x \cdot \lambda$ factors through finite Lie quotients of the arguments;

(U2) the linear Poisson bivector
$$\pi_L = \frac{1}{2} C_{ij}^k O_k \theta^i \theta^j$$
determined by the structure constants of L is a continuous Schouten bivector on $\widehat{\mathbf{S}}_{\mathrm{coord}}(L)$;

(U3) the source cotangent bracket $\{c^i, u_j\}_{\mathrm{CE}} = \delta^i_j$ and the target Schouten contraction bracket $[\theta^i, O_j]_S = \delta^i_j$ are continuous biderivations on the chosen mixed or weighted completion. This is a summability hypothesis beyond the generator formula on the ambient product.

Then the generator assignment

$$\Phi_L \colon C^\bullet_{\mathrm{CE,cont}}(G) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{coord}}(L)\big), \qquad c^i \mapsto \theta^i, \quad u_i \mapsto O_i,$$

is an isomorphism of completed dg algebras with the degree-(-1) cotangent/Schouten contraction bracket. The summability condition (U3) extends the compatible pro finite-window dg-commutative CE/PV identity to a global contraction bracket. This is a formal-local cochain theorem, with lower-central Tate-pronilpotence and a strict kernel $C_L \in L \widehat{\otimes} D$ as separate stronger hypotheses. The strict formal Hamiltonian polydisk $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ lies outside this criterion: $\mathfrak{h}$ lacks an exhaustion by finite-dimensional bracket-compatible Lie quotients (Remark 4.16), and the formal-disk case is covered by the coordinate finite-coefficient theorem 4.20.

Proof. The completed CE algebra of $G = L \ltimes D[1]$ and the completed PV algebra of the linear formal space L are products over the same finite-word coordinate monomials after $c^i \leftrightarrow \theta^i$ and $u_i \leftrightarrow O_i$. Hence Φ_L is a continuous graded-commutative algebra isomorphism with continuous inverse.

Assumption (U1) makes the CE differential continuous: every finite coordinate of the internal differential, of $d_{\mathrm{CE}} c^k$, and of $d_{\mathrm{CE}} u_i$ is computed from finitely many source coordinates. The internal differential is carried to the identical linear vector field on the PV side by the coordinate substitution. Assumption (U2) makes $d_{\pi_L} = [\pi_L, -]_S$ a continuous differential on the PV side. On generators the bracket terms are the same structure-constant formulas:

$$d_{\mathrm{CE}} c^k = -\tfrac{1}{2} C_{ij}^k c^i c^j, \qquad d_{\pi_L} \theta^k = -\tfrac{1}{2} C_{ij}^k \theta^i \theta^j,$$

and

$$d_{\mathrm{CE}} u_i = C_{ij}^\ell u_\ell c^j, \qquad d_{\pi_L} O_i = C_{ij}^\ell O_\ell \theta^j.$$

Jacobi for L gives $d_{\mathrm{CE}}^2 = 0$ and $[\pi_L, \pi_L]_S = 0$, hence $d_{\pi_L}^2 = 0$. Both differentials are continuous derivations, so generator-level compatibility propagates to $\Phi_L d_{\mathrm{CE}} = d_{\pi_L} \Phi_L$ on the completed algebras.

Assumption (U3) gives the same conclusion for the contraction brackets:

$$\{c^i, u_j\}_{\mathrm{CE}} = \delta^i_j \quad \text{and} \quad [\theta^i, O_j]_S = \delta^i_j$$

on generators, with all same-type brackets zero. Continuity and the biderivation rule extend the equality to finite words and then to the chosen mixed or weighted completion. A strict Casimir kernel is stronger than the coordinate biderivation; the unweighted ambient product can carry a global biderivation obstruction. This boundary is the strict-Casimir obstruction of Lemma 5.212. □

PROPOSITION 4.23 *(Low-degree obstruction to a pronilpotent full-polydisk splitting).* The subspace $\mathfrak{m}^3 \subset \mathfrak{h} = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ has a nonzero ideal-defect in the full reduced Hamiltonian Lie algebra $\mathfrak{h}$. Consequently the vector-space decomposition

$$\mathfrak{h} = (H_1 \oplus H_2) \oplus \mathfrak{m}^3, \qquad H_d = \operatorname{Span}\{z_1^a z_2^b : a + b = d\},$$

lacks the semidirect-product structure of a low-degree sector with a pronilpotent ideal.

Proof. The linear Hamiltonian $z_1 \in H_1$ lowers $z_2^3 \in \mathfrak{m}^3$:

$$\{z_1, z_2^3\} = 3z_2^2 \in H_2.$$

Since $H_2 \cap \mathfrak{m}^3 = 0$, the bracket leaves $\mathfrak{m}^3$, giving the ideal-defect of $\mathfrak{m}^3$ in $\mathfrak{h}$. A semidirect-product presentation with pronilpotent radical $\mathfrak{m}^3$ forces that radical to be Lie-stable under the low-degree sector; the displayed bracket contradicts this condition. □

THEOREM 4.24 *(Relative filtered CE/PV reconstruction criterion).* Let $\mathfrak{h}_{\mathrm{rel}}$ be a complete filtered dg Lie algebra with a finite-dimensional reductive degree-zero subalgebra $\mathfrak{a}$, a complete pronilpotent dg ideal $\mathfrak{n}$, and a filtered splitting

$$\mathfrak{h}_{\mathrm{rel}} \cong \mathfrak{a} \ltimes \mathfrak{n}.$$

Let D_{rel} be a continuous $\mathfrak{h}_{\mathrm{rel}}$-module with a perfect filtered pairing with $\mathfrak{h}_{\mathrm{rel}}$ on every finite quotient, and assume the coevaluation tensor belongs to $\mathfrak{h}_{\mathrm{rel}} \widehat{\otimes} D_{\mathrm{rel}}$. Put

$$\mathfrak{g}_{\mathrm{rel}} = \mathfrak{h}_{\mathrm{rel}} \ltimes D_{\mathrm{rel}}[1], \qquad A_{\mathrm{rel}} = \widehat{\mathbf{S}}(\mathfrak{h}_{\mathrm{rel}}).$$

Then the generator assignment $c \mapsto \theta, u \mapsto O$ defines a filtered isomorphism

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{rel}}) \xrightarrow{\sim} \mathrm{PV}(A_{\mathrm{rel}})$$

of completed dg algebras with the degree-(-1) cotangent/Schouten contraction bracket. If, in addition, the relative bar-cobar functor is taken in the category of $\mathfrak{a}$-modules and the $\mathfrak{n}$-coalgebra is conilpotent, this cochain identity gives relative Lie-Koszul duality:

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{rel}}, \mathfrak{a}) \simeq \big(\widehat{U}_{\mathfrak{a}}(\mathfrak{g}_{\mathrm{rel}})\big)^!.$$

This criterion applies under the displayed hypotheses. Proposition 4.23 shows that the natural decomposition $\mathfrak{h} = (H_1 \oplus H_2) \oplus \mathfrak{m}^3$ has an ideal-defect under the full bracket, so the criterion applies only after an independent relative model gives the displayed semidirect-product hypotheses.

Proof. The cochain-level statement is the completed version of Theorem 5.2: finite quotients of the filtration carry the finite CE/PV identity, the transition maps preserve $c \mapsto \theta, u \mapsto O$, and completeness identifies the inverse limit with the displayed completed complexes. The coevaluation hypothesis is the exact condition needed to keep the cotangent generators inside the same completed tensor category.

For the Koszul-dual statement, work relative to the reductive subalgebra $\mathfrak{a}$. The coalgebra generated by the pronilpotent ideal $\mathfrak{n}$ is conilpotent, so the Lie/commutative bar-cobar equivalence applies to the relative CE chains. Reductivity of $\mathfrak{a}$ makes passage to $\mathfrak{a}$-modules exact on finite quotients; in the filtered inverse limit this gives the completed relative equivalence. Dualizing yields the displayed cochain Koszul-dual form. □

Lemma 4.25 *(Filtered cobar admissibility criterion).* Let C be a coaugmented conilpotent dg coalgebra and B a complete augmented filtered A_∞-algebra in the same complete linear category. For a continuous degree-one map $\tau \colon C \to B$, define

$$\mathrm{MC}(\tau) = d_B\tau + \tau d_C + \sum_{r\geq 2} m_r^B(\tau^{\otimes r})\Delta_C^{(r)} \in \mathrm{Hom}^2_{\mathrm{cont}}(C, B),$$

where $\Delta_C^{(r)}$ is the r-fold reduced coproduct. The equation $\mathrm{MC}(\tau) = 0$ is equivalent to the existence of the adjoint complete cobar morphism

$$\kappa_\tau \colon \Omega C \longrightarrow B.$$

Assume the filtrations on ΩC and B are bounded below, complete, separated, and strongly convergent, with finite-dimensional associated graded pieces in every total degree. Then κ_τ is an admissible filtered quasi-isomorphism if and only if

$$H^\bullet\big(\mathrm{Cone}(\mathrm{gr}\,\kappa_\tau)\big) = 0$$

and the two finite-window Milnor defects

$$\varprojlim\nolimits^1_M H^{n-1}((\Omega C)_{\leq M}), \qquad \varprojlim\nolimits^1_M H^{n-1}(B_{\leq M})$$

vanish for every n.

Proof. The Maurer–Cartan formula is the bar-cobar adjunction written in the continuous convolution complex. For a dg codomain it reduces to $d_B\tau + \tau d_C + \mu_B(\tau\otimes\tau)\Delta_C = 0$, which says exactly that $\Omega C \to B$ commutes with differentials and multiplication. The displayed A_∞ formula is the same statement with all codomain multiplications retained.

Filter domain and codomain by tensor length and by finite coordinate windows. Exactness of the associated-graded cone is equivalent to $\mathrm{gr}\,\kappa_\tau$ being a quasi-isomorphism. The two $\varprojlim^1$-terms are the Milnor errors in computing the cohomology of the completed domain and codomain from their finite windows. With those errors zero, strong convergence gives the spectral-sequence comparison theorem in both directions. Necessity is the definition of admissible filtered quasi-isomorphism in this envelope: it is detected on associated graded and on the exact finite-window towers. □

Definition 4.26 (Admissible open-trace A_∞-Koszul homotopy datum). An admissible open-trace A_∞-Koszul homotopy datum consists of:

(K1) a finite-window tower of one-object cyclic open dg categories whose endomorphism complexes are $\Omega_c^\bullet(I) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathrm{End}(\mathbb{C}^N)$, with cyclic trace $\int_I \eta\, [\varepsilon_1\varepsilon_2]\, \mathrm{Tr}_N(M)$ and Dirac/Koszul reduction $Q\psi = [\phi_1, \phi_2]$;

(K2) the reduced cyclic dg algebra $A_\psi = T(x, y, \psi)$, $d\psi = xy - yx$, with $CC_{\mathrm{red}}(A_\psi)$ and the necklace H_0-quotient $\mathrm{Cyc}(A_\psi)$;

(K3) a finite-window Loday–Quillen–Tsygan comparison: with Koszul weights $\mathrm{wt}(x) = (1, 0)$, $\mathrm{wt}(y) = (0, 1)$, $\mathrm{wt}(\psi) = (1, 1)$ and $A_{\psi,K} = A_\psi/F^{>K}A_\psi$, the stable single-cycle map

$$\lambda_{\mathrm{LQT},K,L} \colon \mathrm{Prim}^{1\mathrm{cyc}}_{N,K,L} \longrightarrow CC_{\mathrm{red},\leq L}(A_{\psi,K})[1]$$

is a quasi-isomorphism in the stable range $N \geq \max(K, L+2)$; the same-rank deletion of multicycle tensors has a nonzero chain defect, and the separated-block stabilization zigzag $P^{\mathrm{sep}}_{M,K,L} = \beta_{M,K,L} P_{1\mathrm{cyc}} \alpha_{M,K,L}$ gives the finite-rank bridge;

(K4) a completed augmented filtered A_∞-algebra A_{op} in the scalar-reduced stable trace/Koszul/BV sector, with associated graded $\operatorname{gr} A_{\mathrm{op}} \cong \widehat{\mathbf{S}}(V_H)$ and $\operatorname{Indec}\operatorname{gr} A_{\mathrm{op}} = \mathfrak{m}/\mathfrak{m}^2 \cong V_H$; the indecomposables module is either $\bar{A} \oplus \mathcal{P}[1]$ or $\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]$ together with a proved Fourier–Rees/Moyal bridge to $\mathcal{P}[1]$; the raw same-label map $\Upsilon_{a,b} \mapsto \rho_{a,b}$ is replaced by the proved bridge;

(K5) the completed closed CE/PV complex

$$B_{\mathrm{CE/PV}} = C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \cong \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{coord}}(\mathfrak{h}_{\mathrm{adm}})\big),$$

with the generator dictionary $c^f \mapsto \theta^f$, $u_f \mapsto O_f$;

(K6) the open bar coalgebra $C_{\mathrm{op}} = \mathrm{Bar}(A_{\mathrm{op}})$, equipped with a complete bounded-below filtration compatible with the bar differential, coproduct, and finite coordinate windows;

(K7) a continuous degree-one map $\tau_{\mathrm{op}} \colon C_{\mathrm{op}} \to B_{\mathrm{CE/PV}}$ satisfying the explicit Maurer–Cartan equation

$$d_{B_{\mathrm{CE/PV}}}\tau_{\mathrm{op}} + \tau_{\mathrm{op}} b_{\mathrm{op}} + \mu_{B_{\mathrm{CE/PV}}}(\tau_{\mathrm{op}} \otimes \tau_{\mathrm{op}})\Delta = 0;$$

(K8) the induced completed cobar morphism $\kappa_{\tau_{\mathrm{op}}} \colon \Omega C_{\mathrm{op}} \to B_{\mathrm{CE/PV}}$ whose associated-graded linear symbol is $B_f \mapsto O_f$, $\Theta_{\rho_{a,b}} \mapsto \theta^{a,b}$;

(K9) the admissibility obstruction complex

$$\mathfrak{C}_{\tau_{\mathrm{op}}} = \mathrm{Cone}(\operatorname{gr}\kappa_{\tau_{\mathrm{op}}}) \oplus \bigoplus_n \varprojlim\nolimits^1_M H^{n-1}\big((\Omega C_{\mathrm{op}})_{\le M}\big)[-n] \oplus \bigoplus_n \varprojlim\nolimits^1_M H^{n-1}\big((B_{\mathrm{CE/PV}})_{\le M}\big)[-n];$$

(K10) exact continuous dualization on the same finite-window tower; and cyclicity/BV-pairing data: the scalar word is retained before scalar reduction, graded cyclicity uses $\mathrm{Tr}(ab) = (-1)^{|a||b|}\,\mathrm{Tr}(ba)$, two ψ-letters skew, the open trace is continuous, and the BV pairing has degree -1 after the de Rham, Ext, and field-shift conventions.

THEOREM 4.27 *(Admissible open-side filtered A_∞-Koszul homotopy).* For a datum in the sense of Definition 4.26, the following are equivalent:

(a) τ_{op} is an admissible open-side twisting cochain, i.e. $\kappa_{\tau_{\mathrm{op}}}$ is an admissible filtered quasi-isomorphism of complete augmented A_∞-algebras with the displayed associated-graded symbol;

(b) the Maurer–Cartan equation in (K7) holds and the obstruction complex $\mathfrak{C}_{\tau_{\mathrm{op}}}$ from (K9) is exact.

Under these equivalent conditions,

$$\Omega\mathrm{Bar}(A_{\mathrm{op}}) \xrightarrow{\sim} B_{\mathrm{CE/PV}}$$

and hence $A_{\mathrm{op}} \simeq B_{\mathrm{CE/PV}}$ in the completed filtered A_∞ homotopy category. If the CE-side cobar map $\Omega C^{\mathrm{CE}}_{*}(\mathfrak{g}_{\mathrm{adm}}) \to A_{\mathrm{op}}$ satisfies the same exact cone and $\varprojlim^1$ criterion, exact continuous dualization gives the Koszul-dual cochain form

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \simeq A^{!}_{\mathrm{op}}.$$

Proof. Apply Lemma 4.25 with $C = C_{\mathrm{op}}$ and $B = B_{\mathrm{CE/PV}}$. The Maurer–Cartan equation in (K7) is precisely the condition that $\kappa_{\tau_{\mathrm{op}}}$ is a morphism. Exactness of $\mathfrak{C}_{\tau_{\mathrm{op}}}$ says simultaneously that $\operatorname{gr}\kappa_{\tau_{\mathrm{op}}}$ is a quasi-isomorphism and that both finite-window towers compute the completed cohomology. The filtered comparison theorem then gives the admissible quasi-isomorphism, and the converse is the admissible-filtered definition.

The bar-cobar counit $\Omega\mathrm{Bar}(A_{\mathrm{op}}) \to A_{\mathrm{op}}$ is a filtered quasi-isomorphism under the same boundedness and convergence hypotheses, so the admissible $\kappa_{\tau_{\mathrm{op}}}$ identifies A_{op} with $B_{\mathrm{CE/PV}}$ in the homotopy category. The final displayed equivalence is the CE-side specialization of the same criterion, followed by exact continuous dualization. □

5 CE/PV dictionary as Koszul resolution

The dictionary $c_f \mapsto \theta_f, u_f \mapsto J(f) \cdot (-)$ is the Koszul model for the trace-sector map to the chiral Hochschild complex at the brane vacuum; the admissible criterion for its bracket-admissible and kernel-admissible enhancements is Theorem 5.5. Theorem 5.6 identifies this trace sector after scalar reduction and completion, not the whole finite-N Hochschild complex.

5.1 *The concrete dictionary*

Two type distinctions on three disjoint alphabets fix the dictionary. Witten's source convention forces the closed-CE-coordinate distinction $c \mapsto \theta$ versus $u \mapsto O$: the Hamiltonian-dual coordinate c carries vector-field operations, while the cotangent-dual coordinate u carries boundary Hamiltonians. Grothendieck residue duality forces the cotangent-realization distinction Ψ versus ρ: the polynomial one-ψ trace classes $\Psi_{k,l}$ are PBW special-fibre labels on the open side; the principal parts $\rho_{a,b}$ are the deformation-level cotangent module on the closed side. Coincident bidegrees $(k, l) = (a, b)$ are an indexing convention; the underlying modules are distinct.

Open PBW special fibre. For $k, l \geq 0$, $k + l > 0$, the open Hamiltonian trace label is

$$O_{k,l} = \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}.$$

The primitive one-antifield PBW special-fibre label is the symmetrised trace

$$\Psi_{k,l} = \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}\big(\psi\, w(\phi_1, \phi_2)\big),$$

with $\mathsf{W}_{k,l}$ the set of words containing k copies of ϕ_1 and l copies of ϕ_2. Both $O_{k,l}$ and $\Psi_{k,l}$ are open brane classes on the connected stable PBW special fibre. CE coordinates and principal parts belong to the closed-side alphabets below.

Closed CE/PV coordinates. For $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, write $H_{k,l} = z_1^k z_2^l$ for the Hamiltonian basis element and $\eta_{k,l} = \delta_{k,l}[1] \in \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ for the shifted cotangent Lie generator, with $\delta_{k,l}(z_1^a z_2^b) = \delta_{k,a}\delta_{l,b}$. The CE/PV comparison uses the dual coordinate alphabet

$$c^{k,l} \quad \text{dual to } H_{k,l}, \qquad u_{k,l} \quad \text{dual to } \eta_{k,l}.$$

The CE/PV theorem maps $c^{k,l} \mapsto \theta^{k,l}$ (Schouten constant vector field) and $u_{k,l} \mapsto O_{k,l}$ (boundary Hamiltonian on the brane), with the assignment $u \mapsto O_{k,l}$ forced by the uniqueness of J as the $\mathbb{C}$-linear, GL_N-equivariant, Darboux-natural primitive single-trace map satisfying scalar reduction and Dirac descent on linear generators (Theorem 5.147). Witten's source convention takes the boundary observable through the cotangent coordinate u and the Schouten action on observables through the Hamiltonian coordinate c. The dictionary $c_f \mapsto \theta_f$, $u_f \mapsto J(f)$ is the Koszul resolution of the trace summand in $C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ at the coordinate chart $A^{\mathrm{coord}}_{b_p}$; the cotangent CE/PV reduced central-operation construction of §5.2 is the formal-Darboux stalk expression of that trace-sector map. The Mixed Holomorphic–Topological Deligne conjecture is the further derived-centre identification with the bulk $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_{b_p}$.

Matlis principal parts. Continuous duality on the Hamiltonian polydisk realizes $\mathfrak{h}^\vee_{\mathrm{cont}}$ as the reduced Matlis principal-part module

$$\mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2, \qquad \langle \rho_{a,b}, H_{m,n}\rangle = \delta_{a,m}\delta_{b,n},$$

with the residue-coadjoint action of Proposition 5.176 and uniqueness given by Theorem 5.179.

Label collision. The vector-space pairing $e^{\mathrm{PBW}}_{k,l} \mapsto [\Psi_{k,l}]$ on the PBW special fibre is the open PBW realization. The principal-part realization and the $\mathfrak{h}$-module isomorphism to $\mathcal{P}$ require the deformation-level bridge: the polynomial one-ψ class $\Psi_{k,l}$ and the principal part $\rho_{a,b}$ share a bidegree label and live in different $\mathfrak{h}$-modules. The available passage from the PBW label carried by $\Psi_{k,l}$ to the principal-part label $\rho_{k,l}$ is the Fourier-polarized Rees $D_\hbar$-module bridge of Theorem 8.12; it is extra polarized structure relating different $\mathfrak{h}$-module actions. The unreduced quantum boundary observable is the unreduced boundary object. Corollary 5.173 records the exact deformation-level mismatch.

5.2 *The CE/PV Koszul resolution of the brane E_1-Hochschild*

The dictionary $c_f \mapsto \theta_f$, $u_f \mapsto J(f)$ is the Koszul resolution of the trace summand in the brane-side derived E_1-Hochschild complex $C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ at the formal-Darboux stalk of the Dirac brane vacuum b_p: closed CE cochains of the shifted cotangent Hamiltonian Lie algebra are Schouten polyvectors of the boundary Hamiltonian polynomial algebra, on the coordinate window, and a closed Hamiltonian becomes a brane observable through the cotangent coordinate u_f; the Hamiltonian coordinate c_f gives the Schouten vector field. The Mixed Holomorphic–Topological Deligne conjecture identifies this trace sector with the bulk in the presence of the derived-centre, descent, and QME data. The Schouten/cotangent bracket carrying the comparison has degree -1; the reduced Hamiltonian boundary bracket on O_I is the degree-zero Poisson bracket induced by the matrix Poisson form. Completion at the formal polydisk is coefficientwise: each chosen coefficient of $d_{\mathrm{CE}}c$, $d_{\mathrm{CE}}u$, $d^{\mathrm{coord}}_\pi\theta$, or $d^{\mathrm{coord}}_\pi O$ is a finite monomial coefficient, so the finite-dimensional Schouten identity assembles to the formal polydisk identity on the coordinate envelope. Bar-cobar/Koszul duality is the separate admissible enhancement, governed by conilpotence, exact continuous duality, filtered cobar convergence, and the open-trace A_∞-Koszul homotopy datum named in §4.2. The minimal RG-stable analytic completion in which the heat-kernel propagator extends and the Casimir converges is the weighted-Tate completion B_w, generated by $\{O_f, \theta_f\}_{f\in\bar{A}}$ under the Schouten bracket and pointwise product (Theorem 5.58).

Definition 5.1 *(Reduced Hamiltonian P_0 central-operation complex).* Let A be a complete graded-commutative Poisson algebra with a chosen topology in which the Poisson bivector π and the Schouten contraction bracket act continuously. Its reduced Hamiltonian P_0 central-operation complex is the Poisson cochain algebra

$$Z^{P_0}_{\mathrm{red,Ham}}(A) := \mathrm{PV}(A), \qquad d_\pi = [\pi, -]_S,$$

with the completed Schouten bracket of degree -1. This is a local reduced model for central operations. The ordinary Poisson centre, the strict ambient coordinate product, the unreduced factorization-centre object, and the degree-zero reduced Hamiltonian Poisson bracket on boundary currents are distinct structures. For the formal polydisk this definition is used on explicitly topologized bracket-admissible completions B. An HKR or Hochschild-center identification is a continuous Hochschild–Kostant–Rosenberg theorem compatible with the chosen completion and interval functoriality.

THEOREM 5.2 *(Coordinate-free cotangent CE/PV identity).* Let $\mathfrak{l}$ be a finite-dimensional Lie algebra over $\mathbb{C}$, and put

$$T^*[1]\mathfrak{l} = \mathfrak{l} \ltimes \mathfrak{l}^\vee[1].$$

The dual module is the coadjoint module:

$$(x \cdot \lambda)(y) = -\lambda([x, y]).$$

Let $A_\mathfrak{l} = \mathbf{S}(\mathfrak{l})$, regarded as functions on $\mathfrak{l}^\vee$, with the Kirillov Poisson bracket. Then there is a canonical isomorphism of dg graded-commutative algebras with their degree-(-1) cotangent/Schouten brackets

$$\Phi_\mathfrak{l} : C^\bullet_{\mathrm{CE}}(T^*[1]\mathfrak{l}) \xrightarrow{\sim} Z^{P_0}_{\mathrm{red,Ham}}(A_\mathfrak{l}) = \mathrm{PV}(A_\mathfrak{l}).$$

For $x \in \mathfrak{l}$, let O_x be the linear function on $\mathfrak{l}^\vee$ and let u_x be the CE coordinate obtained by pairing the shifted cotangent generator $\mathfrak{l}^\vee[1]$ with x. For $\xi \in \mathfrak{l}^\vee$, let c^ξ be the Hamiltonian CE coordinate and θ^ξ the constant vector field on $\mathfrak{l}^\vee$ in the direction ξ. Then

$$\Phi_\mathfrak{l}(u_x) = O_x, \qquad \Phi_\mathfrak{l}(c^\xi) = \theta^\xi.$$

Proof. Choose a basis e_i of $\mathfrak{l}$, dual basis ϵ^i, and structure constants $[e_i, e_j] = C^k_{ij} e_k$. Write $O_i = O_{e_i}$, $u_i = u_{e_i}$, $c^i = c^{\epsilon^i}$, and $\theta^i = \theta^{\epsilon^i}$. The CE differential of the shifted cotangent Lie algebra is

$$d_{\mathrm{CE}} c^k = -\frac{1}{2} C^k_{ij} c^i c^j, \qquad d_{\mathrm{CE}} u_k = C^\ell_{kj} u_\ell c^j.$$

The Kirillov Poisson tensor on $A_\mathfrak{l}$ is

$$\pi = \frac{1}{2} C^k_{ij} O_k \theta^i \theta^j.$$

The Schouten differential $d_\pi = [\pi, -]_S$ satisfies

$$d_\pi \theta^k = -\frac{1}{2} C^k_{ij} \theta^i \theta^j, \qquad d_\pi O_k = C^\ell_{kj} O_\ell \theta^j.$$

Hence the substitution $c^i \mapsto \theta^i$, $u_i \mapsto O_i$ intertwines the differentials on generators. Both differentials are derivations, so the equality extends to all CE words. The Schouten and degree-(-1) cotangent/Schouten brackets are also identified on generators:

$$\{c^i, u_j\}_{\mathrm{CE}} = \delta^i_j, \qquad [\theta^i, O_j]_S = \delta^i_j,$$

with the same zero brackets among two c's, two u's, two θ's, or two O's. The biderivation rule gives bracket compatibility on the whole algebra. A change of basis acts by the same linear substitution on both sides, so the isomorphism is coordinate-free. □

THEOREM 5.3 *(Formal-polydisk source-to-boundary CE/PV coordinate reconstruction).* Let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$$

be the reduced Hamiltonian Lie algebra of the formal symplectic polydisk, and let $A^{\mathrm{st}}_{\partial,H}$ be the stable scalar-reduced Hamiltonian trace algebra. Under the stable trace identification

$$A^{\mathrm{st}}_{\partial,H} \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}),$$

whose connected/indecomposable quotient is $\mathfrak{h}$, the coordinate finite-coefficient construction gives

$$C^{\bullet}_{\mathrm{CE,coord}}(\mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1]) \dashrightarrow Z^{P_0}_{\mathrm{red,Ham}}(A^{\mathrm{st}}_{\partial,H}).$$

This is a coordinate dg-algebra comparison. After a mixed or weighted completion makes the coordinate contraction bracket continuous, the dashed generator map extends to the completed bracketed comparison on that bracket-admissible realization. In the strict ambient coordinate product the coordinate contraction bracket has a global continuity obstruction; a global P_0-bracket requires a bracket-admissible completion. The boundary evaluation

$$J(f) = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$$

is the image of the shifted-cotangent coordinate u_f. The Hamiltonian CE coordinate c^f maps to the Hamiltonian vector-field operation $\{J(f), -\}$. The coordinate map Φ_{coord} is functorial under formal Darboux coordinate changes. The P_0, kernel, and principal-part enhancements are functorial only for Darboux changes that transport the chosen admissible completion and preserve the diagonal convergence data.

Proof. The finite theorem fixes the generator signs. The formal-disk passage is coefficientwise through finite coefficient projections. For each chosen coefficient of $d_{\mathrm{CE}}c$, $d_{\mathrm{CE}}u$, $d^{\mathrm{coord}}_{\pi}\theta$, or $d^{\mathrm{coord}}_{\pi}O$, only finitely many monomial labels can contribute after coordinate projection. These finite coefficient identities are compatible under restriction of coordinate sets and determine Φ_{coord}. A completed bracketed limit requires the mixed or weighted continuity condition recorded in Theorem 4.20.

Proposition 3.22, Lemma 3.30, and Corollary 3.32 identify the stable scalar-reduced ghost-zero brane Hamiltonians with the completed symmetric algebra on $\mathfrak{h}$. The matrix Poisson calculation gives

$$\{J(f), J(g)\}_{\partial} = J(\{f, g\}_{\mathfrak{h}}),$$

so this identification is an isomorphism of reduced Hamiltonian Poisson algebras. Under Theorem 5.2, the linear function coordinate O_f is precisely $\Phi_{\mathrm{coord}}(u_f)$, while $\Phi_{\mathrm{coord}}(c^f) = \theta^f$ acts as the constant vector-field generator whose Schouten differential on O_g is the Hamiltonian action. After choosing a bracket-admissible realization, the closed CE cochains of the shifted cotangent Hamiltonian Lie algebra give the reduced Hamiltonian P_0-central operations on the stable brane algebra.

A formal Darboux change is a formal symplectomorphism. It acts by Lie algebra automorphisms on $\mathfrak{h}$, by Poisson automorphisms on $\widehat{\mathbf{S}}(\mathfrak{h})$, and by substitution on the matrix normal-coordinate evaluation after passing to the scalar-reduced stable trace quotient. The coordinate CE/PV construction is functorial for these Lie algebra automorphisms, so Φ_{coord} is Darboux-coordinate independent. A bracket-admissible B, a kernel-admissible tensor product, or a divisor-dependent principal-part completion is extra structure; functoriality of those enhancements requires preservation or explicit transport of that structure. □

Remark 5.4 *(Obstruction to the strict-product CE/PV comparison).* The strict ambient product completion has the following obstruction to the coordinate CE/PV identity: linear Hamiltonians lower Taylor degree, projected brackets have a low-order Jacobi obstruction, and the diagonal Casimir sum is divergent. Proposition 4.23 records the precise low-degree class ($\{z_1, z_2^{N+1}\} = (N+1)z_2^N$) and the $H_2 \cong \mathfrak{sl}_2$ lower-central obstruction to conilpotence; Theorem 5.108 and Theorem 5.111 state the colimit and conilpotence obstructions; the Milnor $\varprojlim^1$ defect of Lemma 4.25 blocks strict inverse-limit reconstruction. Strict-product realization of the P_0-bracket and Maurer–Cartan kernel is replaced by admissible Tate, nilpotent, relative, or weighted completions, the mechanism for satisfying the criterion of Theorem 5.5.

5.3 The local theorem

Theorem 5.6 is the formal-Darboux answer to the mixed-HT Deligne problem at the Dirac brane vacuum b_p. The closed local operator algebra

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$$

acts on the boundary algebra by Hamiltonian Schouten derivations and trace functions:

$$c_f \mapsto \theta_f, \qquad u_f \mapsto J(f)\cdot(-), \qquad J(f) = \operatorname{Tr} f(\phi_1,\phi_2) \quad (f \in \bar{A}).$$

For $f \in \mathfrak{h}$, $J(f)$ is evaluated in the completed formal brane algebra. After scalar reduction, passage to the stable trace algebra $\widehat{\mathbf{S}}(\mathfrak{h})$, and the stated Tate/pro-Matlis admissible completion, this action is the continuous CE/PV coordinate isomorphism. At fixed finite N, the Procesi–Razmyslov trace ideal, the Chevalley–Eilenberg stabilizer classes, and the unreduced P_0-centrality homotopies measure the part of the Hochschild centre not seen by the trace-sector map. The finite-N scalar anomaly is the projective Lie class $\hbar N[\bar{c}]$ (Theorem 6.8). The global identification on ∂X (Theorem 5.13) remains governed by the vanishing of

$$\mathrm{Ob}^{\mathrm{global}} = \bigl(\mathrm{Ob}^{\mathrm{ch}\,E_2}_{\mathrm{Deligne}}, \mathrm{Ob}^{\mathrm{desc}}_{\partial}, \mathrm{Ob}^{\mathrm{QME}}_{\text{all-order}}\bigr).$$

It also requires a contracting homotopy for $\mathrm{Cone}(\mathrm{OCA}_N)$.

The arithmetic kernel the local theorem must recover is the first bracket calculation (12), $\{O_{k,l}, O_{m,n}\}_\partial = (kn - lm)O_{k+m-1,l+n-1}$. The minimal RG-stable analytic completion forced by J is the weighted-Tate completion B_w, generated by $\{O_f, \theta_f\}_{f\in\bar{A}}$ under the Schouten bracket and pointwise product, in which the heat-kernel propagator extends and the Casimir converges (Theorem 5.58).

THEOREM 5.5 *(Coordinate CE/PV theorem and admissible Koszul criterion).* In the formal local Hamiltonian polydisk model the cotangent CE/PV construction extends through the following exact ladder.

(i) For every finite-dimensional Lie algebra $\mathfrak{l}$, the assignment $c \mapsto \theta,\ u \mapsto O$ gives

$$C^\bullet_{\mathrm{CE}}\bigl(T^*[1]\mathfrak{l}\bigr) \cong \mathrm{PV}\bigl(\mathbf{S}(\mathfrak{l})\bigr)$$

as dg graded-commutative algebras with the degree-(-1) cotangent/Schouten bracket.

(ii) For the completed formal symplectic polydisk $\mathfrak{h} = \mathbb{C}[[z_1,z_2]]/\mathbb{C}\cdot 1$, the same generator assignment gives the pro coordinate-window cochain isomorphism

$$\Phi_{\mathrm{coord}}\colon C^\bullet_{\mathrm{CE,coord}}\bigl(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]\bigr) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}}).$$

After fixing the residue scalar, $\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P}$ by Theorem 5.179. If $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ is bracket-admissible, then

$$\Phi^B_{P_0}\colon \Phi^{-1}_{\mathrm{coord}}(B) \xrightarrow{\sim} B$$

is a dg P_0-isomorphism. If B is Hamiltonian-spanning for a class $S \subset \mathfrak{h}$, then for every $f \in S$ the boundary Hamiltonian B_f is the image of the shifted cotangent coordinate u_f, while the Hamiltonian CE coordinate c^f maps to the vector-field operation. If a kernel-admissible datum $(B,\mathcal{K}_B)$ is fixed, the associated diagonal tensor Θ_B satisfies the Maurer–Cartan equation in $\mathcal{K}_B$. The strict product/direct-sum endpoint carries the obstruction to the P_0, boundary-Hamiltonian, and kernel conclusions; these conclusions require the stated admissibility hypotheses.

(iii) If an admissible Tate, nilpotent, relative, or weighted replacement gives conilpotent CE chains, exact continuous duality, a filtered twisting cochain, strong convergence, and exact finite-window inverse limits, then the closed-side cochain identity extends only to the admissible bar-cobar/enveloping statement

$$\left(C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}})\right)^! \simeq \widehat{U}(\mathfrak{g}_{\mathrm{adm}}).$$

If, in addition, an open-trace A_∞-Koszul homotopy datum $(A_{\mathrm{op}}, C_{\mathrm{op}}, B_{\mathrm{CE/PV}}, \tau_{\mathrm{op}})$ exists and its obstruction complex is exact, then

$$\Omega\mathrm{Bar}(A_{\mathrm{op}}) \xrightarrow{\sim} B_{\mathrm{CE/PV}}, \qquad A_{\mathrm{op}} \simeq B_{\mathrm{CE/PV}}.$$

If the CE-side cobar comparison also passes the same finite-window and $\varprojlim^1$ tests, then

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \simeq A^!_{\mathrm{op}}.$$

(iv) The obstruction to promoting the strict full formal-disk coefficient pair to this bar-cobar statement decomposes as: the strict Casimir support obstruction $C_{\mathfrak{h}}$, the low-degree linear translation term $\{z_1, z_2^{N+1}\} = (N+1)z_2^N$, the $H_2 \cong \mathfrak{sl}_2$ lower-central obstruction to conilpotence, the associated-graded cone $H^\bullet(\mathrm{Cone}(\mathrm{gr}\,\kappa_\tau))$, and the Milnor $\varprojlim^1$ defect. Quantum or graph-level extensions add the mixed brane-defect QME obstruction class.

Proof. The first item is the finite-dimensional cotangent CE/PV identity Theorem 5.2. The coordinate formal-disk identity in the second item is the cochain-level theorem Theorem 4.20, whose generator clause already records $\Phi_{\mathrm{coord}}(u_f) = O_f$ and $\Phi_{\mathrm{coord}}(c^f) = \theta^f$; the boundary-Hamiltonian evaluation $O_f = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is Theorem 5.3; the diagonal element $\Theta_B = \sum_I H_I \otimes \theta^I + \sum_I \eta^I \otimes O_I$ on a kernel-admissible target satisfies the Maurer–Cartan equation by Theorem 5.67 for the explicit weighted-Tate realization. The third item separates two structures. The closed Lie bar-cobar/enveloping comparison is the relative filtered CE/PV reconstruction criterion Theorem 4.24, whose admissible cobar quasi-isomorphism is detected by Lemma 4.25 on associated graded together with the two finite-window Milnor defects; Lemma 5.166 gives the Tate-pronilpotent case. Proposition 4.23 places the unmodified formal polydisk outside the criterion's domain. The additional open A_∞-Koszul comparison requires the open-side homotopy datum of Definition 4.26, with admissibility proved by Theorem 4.27. The obstruction terms in the fourth item are Theorem 5.108, Theorem 5.111, and the finite-window inverse-limit obstruction clause; the QME class is the residual class of Problem 5.73. Thus the formal-Darboux pro coordinate-window cochain identity is a theorem, bracket and kernel realizations are exactly the admissible B clauses of that theorem, and the enveloping/open-Koszul extensions are precisely the two stated admissibility criteria. □

THEOREM 5.6 *(Formal-Darboux trace-sector theorem at N Dirac branes).* Fix the formal Darboux local model

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \widehat{\mathbb{C}}^2_{z_1,z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}, \qquad R = \mathbb{C}[[z_1, z_2]], \qquad \mathfrak{h} = R/\mathbb{C}\cdot 1,$$

with Poisson bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ and Grothendieck–Matlis residue dual

$$P = \mathfrak{h}^\vee_{\mathrm{cont}} = \mathrm{Ann}_{\mathbf{Res}}(\mathbb{C}\cdot 1) \subset H^2_{\mathfrak{m}}(R)\, dz_1 dz_2, \qquad P = \bigoplus_{a+b>0} \mathbb{C}\, \rho_{a,b}, \quad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2, \quad \mathfrak{m} = (z_1, z_2),$$

$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}$. The shifted-cotangent Hamiltonian Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes P[1]$ carries, at a stack of N Dirac branes, the Capelli-projective central extension whose central cocycle is supported on the Hamiltonian summand:

$$0 \longrightarrow \mathbb{C}\cdot K \longrightarrow \widehat{\mathfrak{g}}_{\mathrm{cent}} \longrightarrow \mathfrak{g} \longrightarrow 0, \tag{15}$$

$$[(f,\eta,aK),(g,\xi,bK)] = \big(\{f,g\},\ \mathrm{ad}_f^\vee \xi - \mathrm{ad}_g^\vee \eta,\ \hbar N\,\omega(f,g)K\big), \quad \omega(f,g) = [\{f,g\}]_0,$$

with zero central term whenever one argument lies in $P[1]$. Here $[\ ]_0$ is the coefficient of 1 in the Poisson bracket of the unique representatives with zero constant term. The cocycle is the extension class of $0 \to \mathbb{C}\cdot 1 \to R \to \mathfrak{h} \to 0$; hence its Chevalley–Eilenberg differential vanishes by the Jacobi identity in R, and the semidirect $P[1]$-summand contributes no central term. The cocycle is continuous on $\mathfrak{h}$ and restricts on $\bar{A} = \mathbb{C}[z_1,z_2]/\mathbb{C}\cdot 1$ to the projective Lie class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A},\mathbb{C})[[\hbar]]$ (Theorem 6.8). The corresponding Chevalley–Eilenberg generator is the degree-one dual central coordinate c_K; the evaluation $K \mapsto \mathrm{Tr}\,\mathbf{1} = N$ is a central character imposed only after passage to the projective current/enveloping algebra. Let

$$A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}\big(\mathfrak{gl}_N,\ \mathrm{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])\big), \qquad Q\phi_i = 0, \quad Q\psi = [\phi_1,\phi_2],$$

be the finite-N brane E_1-algebra retaining $\mathrm{Tr}\,\mathbf{1} = N$, and $\mathcal{B}^{\mathrm{adm}}$ the admissible filtered nuclear Tate / pro-Matlis / Capelli-projective category of Appendix G. The base admissible structure for the trace-sector theorem consists of the first five conditions below. The last two entries are additional centre and quantum data; they are hypotheses for the corresponding extensions, not consequences of the trace-sector calculation.

(i) continuous duality is exact;
(ii) the residue pairing $\mathfrak{h} \otimes P \to \mathbb{C}$ is perfect;
(iii) the diagonal Casimir $C = \sum_I H_I \otimes \eta_I \in \mathfrak{h}\widehat{\otimes} P$ converges in the weighted nuclear topology;
(iv) the Hamiltonian bracket on $\mathfrak{h}$ and the coadjoint action on P are continuous, with weighted Köthe seminorms $p_n(\{x,y\}) \le C_n\, p_{n+2}(x)\, p_{n+2}(y)$;
(v) the diagonal local-cohomology kernel $K^{(2)}_\Delta = d\zeta_1 d\zeta_2/(\zeta_1\zeta_2)$ on $\Delta \subset \mathbb{C}^2_z \times \mathbb{C}^2_w$ is bracket- and kernel-admissible;
(vi) the Hochschild-centrality convolution Lie algebra is completed; a centre lift, when chosen, is a Maurer–Cartan element with components $\{h_{\alpha,a}, h_{\alpha,a,b}, \dots\}$;
(vii) quantum and unreduced cotangent extensions use scalar-contact and QME counterterm towers $\{C_{n,M}\}$ satisfying Milnor/Roos compatibility $\pi_{M,N} C_{n,M} = C_{n,N}$.

Then the continuous local-observable algebra at b of the formal current stalk of the bulk mixed-HT Hamiltonian BF factorization algebra $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} = \pi_* C^{\mathrm{CE}}_\bullet(\Omega^\bullet_c(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}_c(\mathbb{C}^2)\widehat{\otimes}\widehat{\mathfrak{g}}_{\mathrm{cent}})$ is identified by exact continuous Tate duality with $C^\bullet_{\mathrm{CE,cont}}(\widehat{\mathfrak{g}}_{\mathrm{cent}})$. There is a trace-sector generator assignment

$$\mathrm{Obs}^{\mathrm{cl}}_b(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}) \ \simeq\ C^\bullet_{\mathrm{CE,cont}}(\widehat{\mathfrak{g}}_{\mathrm{cent}}) \dashrightarrow_{\Phi_N} C^\bullet_{\mathrm{Hoch,adm}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}), \tag{16}$$

determined on Chevalley–Eilenberg generators by the two compatible target components

$$(c_f, u_f) \ \longmapsto\ \Big(\theta_f,\ J(f)\cdot(-)\Big), \qquad c_K \longmapsto \gamma_1 \text{ in } C^\bullet_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1]. \tag{17}$$

Here θ_f is a chosen Koszul lift, $J(f)\cdot(-)$ is a Hochschild 0-cochain, c_K is the Chevalley–Eilenberg dual of the central Lie generator, $\gamma_{\mathbf{1}}$ is the scalar-contact ghost for the omitted constant Hamiltonian line, and the projective central character later evaluates K by $K \mapsto \operatorname{Tr}\mathbf{1} = N$. The trace map is $J(f) = \operatorname{Tr} f(\phi_1,\phi_2)$ for $f \in \bar{A}$, with formal extension in the completed brane algebra. The Schouten derivation satisfies

$$\theta_f(\phi_1) = -\partial_{z_2} f(\phi_1,\phi_2), \qquad \theta_f(\phi_2) = \partial_{z_1} f(\phi_1,\phi_2), \qquad Q\theta_f(\psi) = \theta_f([\phi_1,\phi_2]). \tag{18}$$

If the admissible Hochschild-centre lift of Definition 5.8 exists, the action extends to a closed-to-open Hochschild cochain morphism with equivariance homotopies $H^{\mathrm{prod}}_{f,-}, H^{P_0}_{f,-}, H^3_{f,-,-}, \ldots$. After applying the scalar-reduction functor, passing to the stable trace algebra $A^{\mathrm{st}}_{\partial,H} \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$, and keeping the admissible P_0-bracket, Φ_N becomes the continuous Chevalley–Eilenberg/polyvector chosen-coordinate isomorphism

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \;\simeq\; \big(\mathrm{PV}_{\mathrm{adm}}(A^{\mathrm{st}}_{\partial,H}), d_\pi\big) \;\simeq\; HH^\bullet_{\mathrm{adm,HKR}}\big(A^{\mathrm{st}}_{\partial,H}, A^{\mathrm{st}}_{\partial,H}\big).$$

The last isomorphism is the completed Hochschild–Kostant–Rosenberg identification for the stable scalar-reduced trace algebra; it is not an assertion about the finite-N Hochschild complex. For fixed finite N, equation (16) is not an equivalence onto the full Hochschild cochain complex. Its finite-N kernel and cokernel are governed by the trace-identity ideal, the stack-level Chevalley–Eilenberg stabilizer classes, and the unreduced P_0-centrality obstruction vector of Theorem E.53. The arity-two holomorphic singular product on linear currents J_x, $x \in \widehat{\mathfrak{g}}_{\mathrm{cent}}$, is

$$\mathsf{J}_x(z)\mathsf{J}_y(w) \;\sim\; K^{(2)}_\Delta(z,w)\,\mathsf{J}_{[x,y]_{\widehat{\mathfrak{g}}_{\mathrm{cent}}}}(w), \qquad \zeta_i = z_i - w_i; \tag{19}$$

on Hamiltonian generators J_f, $f \in \mathfrak{h}$, the finite-N projective correction is

$$\mathsf{J}_f(z)\mathsf{J}_g(w) \;\sim\; K^{(2)}_\Delta(z,w)\big(\mathsf{J}_{\{f,g\}}(w) + \hbar N[\{f,g\}]_0 \cdot \mathbf{1}\big). \tag{20}$$

The forgetful functor $U_{\mathrm{sc}}\colon \mathcal{B}^{\mathrm{adm}} \to \mathcal{B}^{\mathrm{red}}$ sending $K \mapsto 0$, $\hbar N[\bar{c}] \mapsto 0$, the pro-Matlis tower $\mathcal{P}^{\Pi}_q = \varprojlim P_{\le N} \mapsto P_{\mathrm{coord}}$, the equivariance homotopy hierarchy to its cohomology class, and the QME counterterm tower to the classical CE/PV differential recovers the scalar-reduced trace-sector map

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \dashrightarrow C^\bullet_{\mathrm{Hoch,red}}(A^{\mathrm{cl,red}}_{\partial,N}, A^{\mathrm{cl,red}}_{\partial,N}), \qquad J(f) \rightsquigarrow \overline{\operatorname{Tr} f(\phi_1,\phi_2)},$$

and this map is an equivalence exactly after replacement of the finite-N algebra by the stable scalar-reduced trace algebra above.

Proof. (1) BV locality. The local Lie algebra

$$\Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}\widehat{\mathfrak{g}}_{\mathrm{cent}}$$

with differential $d + \bar{\partial} + d_{\mathrm{CE}}$ is a mixed-HT local Lie algebra. Topological translations and antiholomorphic translations are Q-exact. Hence the bulk observables are locally constant in $\mathbb{R}^2_{\mathrm{top}}$ and holomorphic in $\mathbb{C}^2_{\mathrm{hol}}$.

(2) Formal stalk contraction. The disk contraction in the topological directions and the formal Dolbeault contraction at b identify the compactly supported current stalk with $C^{\mathrm{CE}}_\bullet(\widehat{\mathfrak{g}}_{\mathrm{cent}})$. Applying the continuous local-observable dual gives

$$C^\bullet_{\mathrm{CE,cont}}(\widehat{\mathfrak{g}}_{\mathrm{cent}}).$$

(3) Trace descends through the Koszul complex. Lemma 3.30 gives

$$\mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v) = Q\,\mathrm{Tr}(\psi v u).$$

Thus $J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2)$ is a function on the derived commuting stack for $f \in \bar{A}$, and for $f \in \mathfrak{h}$ after completion at the formal Darboux point. In the stable trace range the primitive single-trace classes are generated by the Procesi– Razmyslov traces; at fixed N their relations remain in the trace-identity ideal.

(4) Hamiltonian action preserves the derived constraint. The derivation θ_f of (18) preserves the moment-map ideal and extends to ψ with $[Q, \theta_f] = 0$ once the Koszul homotopy h_f has been chosen. It therefore defines a Hochschild 1-cochain, and multiplication by $J(f)$ defines a Hochschild 0-cochain. This gives the generator action; the Hochschild cochain map requires the equivariance homotopies named in the theorem.

(5) The bracket is Hamiltonian plus the Capelli scalar. The matrix Poisson calculation gives

$$\{J(f), J(g)\}_\partial = J(\{f, g\}_{\mathfrak{h}}) + N[\{f, g\}]_0$$

on the trace sector. The first term is the Hamiltonian bracket. The second term is the pullback of the central extension $0 \to \mathbb{C} \cdot 1 \to R \to R/\mathbb{C} \cdot 1 \to 0$, represented by the central generator K in $\widehat{\mathfrak{g}}_{\mathrm{cent}}$.

(6) CE/PV comparison. The finite-dimensional shifted-cotangent identity $C^\bullet_{\mathrm{CE}}(\mathfrak{l} \ltimes \mathfrak{l}^\vee[1]) \cong \mathrm{PV}(\mathbf{S}(\mathfrak{l}))$ extends coefficientwise to the formal disk under the admissible continuity estimates of $\mathcal{B}^{\mathrm{adm}}$. It gives a chosen-coordinate isomorphism on the stable scalar-reduced trace algebra $A^{\mathrm{st}}_{\partial,H} \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$. The same generator formula gives a map to the finite-N Hochschild complex, but the Procesi–Razmyslov ideal and stack BRST classes prevent the finite-N target from being identified with the stable symmetric algebra.

(7) E_2-structure and singular product. Local constancy in the two topological directions gives the bulk E_2-structure. Hochschild cochains of the brane E_1-algebra carry the Deligne E_2-structure. Under the admissible Hochschild-centre lift, the induced map Φ_N respects the Gerstenhaber operations on the completed trace sector, since

$$[\theta_f, \theta_g] = \theta_{\{f,g\}}, \qquad \theta_f(J(g)) = J(\{f, g\}), \qquad [J(f), J(g)]_{\mathrm{proj}} = \hbar N[\{f, g\}]_0.$$

The diagonal local-cohomology class $d\zeta_1 d\zeta_2/(\zeta_1\zeta_2)$ gives the arity-two holomorphic singular product (19), and the central cocycle gives (20).

The forgetful functor U_{sc} removes the central generator, the pro-Matlis tail, and the higher centrality data. Its image is the scalar-reduced trace-sector CE/PV comparison. A full finite-N derived-centre equivalence is precisely the additional centrality and descent problem measured by the obstruction vector named in the theorem. □

PROPOSITION 5.7 *(Diagonal local-cohomology kernel operad on $\mathbb{C}^2$).* Let

$$K_n = H^{2(n-1)}_{\Delta_n}(\mathbb{C}[[z^{(1)}, \ldots, z^{(n)}]]) \bigotimes_{i=1}^{n-1} d\zeta_{i,1}\, d\zeta_{i,2}, \qquad \Delta_n = \{z^{(1)} = \cdots = z^{(n)}\}, \quad \zeta_i = z^{(i)} - z^{(n)},$$

the arity-n diagonal local-cohomology kernel on $\mathbb{C}^{2n}$. The collection $\{K_n\}_{n\ge 1}$ with residue composition $\gamma\colon K_n \otimes K_{m_1} \otimes \cdots \otimes K_{m_n} \to K_{m_1+\cdots+m_n}$ defined by pullback of singular principal parts to nested diagonals followed by iterated Grothendieck residues is a formal coloured dg kernel operad in nuclear Tate vector spaces; on the Hamiltonian BF model it acts on $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ by

$$\mu_n\colon K_n \otimes \left(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}\right)^{\boxtimes n} \longrightarrow \Delta_{n,*}\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, \quad \mu_n(K_n; \mathsf{J}_{x_1}, \ldots, \mathsf{J}_{x_n}) = K_n \cdot \mathsf{J}_{\ell_n(x_1,\ldots,x_n)},$$

with $\ell_2 = [-,-]_{\widehat{\mathfrak{g}}_{\mathrm{cent}}}$ and the higher L_∞-brackets $\ell_{n\geq 3}$ transferred from the CE factorization structure ($\ell_{n\geq 3} = 0$ in the strict Hamiltonian BF model, modulo quantum counterterm transfer). The arity-two component is $K_2 = K_\Delta^{(2)} = d\zeta_1 d\zeta_2/(\zeta_1\zeta_2)$ of (19).

Proof. Arity-two: $K_2 = H^2_\Delta(\mathbb{C}[[z,w]])\, d\zeta_1 d\zeta_2$ is the residue class of Theorem 5.186. Arity-three associativity: the iterated Grothendieck residue on the nested diagonal $\Delta_3 \subset \Delta_2 \subset \mathbb{C}^6$ factors as $\mathrm{Res}_{z^{(2)}=z^{(3)}} \circ \mathrm{Res}_{z^{(1)}=z^{(3)}} = \mathrm{Res}_{z^{(1)}=z^{(2)}=z^{(3)}}$, matching $\gamma(K_2 \otimes K_2) = K_3$; signs agree with Koszul signs of CE cochains by the standard sign rule for nested residues. Higher arities $n \geq 4$ inherit the kernel associativity from the factorization-algebra composition law [15] applied to disjoint and nested polydisk factorisations of $\mathbb{C}^{2n}$; continuity in $\mathcal{B}^{\mathrm{adm}}$ is the kernel-admissibility condition (v) of Theorem 5.6. Curve restriction to a smooth $C \hookrightarrow \mathbb{C}^2$ with transverse residue datum $B_\perp\colon \Omega_c^{0,1}(\mathbb{C}) \to \mathbb{C}$ specialises $K_\Delta^{(2)}$ to the one-dimensional Cauchy kernel $K_C^{(1)}(z,w) = dw/(w-w')$ by direct residue computation $\mathrm{Res}_{u=u'}\, d(w-w')d(u-u')/((w-w')(u-u')) = d(w-w')/(w-w')$. The all-arity chiral E_2-Deligne action on the derived centre is a separate Hochschild-centre lift problem. □

Definition 5.8 *(Admissible Hochschild-centre lift).* An admissible lift of the closed-to-open coupling $\Phi_N\colon \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}\big|_b \to Z^{\mathrm{der,adm}}_{E_1}\big(A^{\mathrm{cl}}_{\partial,N}\big)$ is a Maurer–Cartan element

$$C_{\Phi_N} = \Phi_N + H^{\mathrm{prod}} + H^{P_0} + H^3 + \cdots$$

in the completed Hochschild-centrality convolution L_∞-algebra $\mathcal{K}^{\mathrm{cent}}_{\mathcal{B}^{\mathrm{adm}}}$ of clause (vi) of Theorem 5.6 satisfying

$$dC_{\Phi_N} + \tfrac{1}{2}\big[C_{\Phi_N}, C_{\Phi_N}\big] + \tfrac{1}{3!}\ell_3\big(C_{\Phi_N}^{\otimes 3}\big) + \cdots = 0,$$

with H^{prod} the product equivariance homotopy

$$[\Phi_N(x), M_a]_G - M_{\rho_x(a)} = dH^{\mathrm{prod}}_{x,a},$$

where ρ_x is the closed-on-open Hamiltonian action and M_a denotes multiplication by the open observable a. The P_0 homotopy satisfies

$$\{\Phi_N(x), M_a\}_{P_0} - M_{\rho_x^{P_0}(a)} = dH^{P_0}_{x,a}.$$

Null-centrality is the special case $\rho_x = \rho_x^{P_0} = 0$, hence applies only to invariant centre classes. The term H^3 is the order-three Jacobiator/associator $dH^3_{x,a,b} = \mathrm{Jac}\big(H^{\mathrm{prod}}, H^{P_0}\big)$, and higher H^n the Stasheff coherences. The Maurer–Cartan equation is the homotopy-equivariance condition for extending the trace-sector action of closed currents on open observables to a derived E_1-centre morphism.

Remark 5.9 *(Low-degree consequences).* Theorem 5.6 has the following consequences inside $\mathcal{B}^{\mathrm{adm}}$.

(1) $N = 1$. $\phi_i = z_i$, $J(f) = f$, and the formal one-pair Weyl algebra has $[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_\star = \hbar$; the projective curvature reduces to $\hbar[\bar{c}]$.

(2) *Finite-N trace bracket.* $\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$; setting this datum to zero projects the finite-N trace bracket to the scalar-reduced shadow.

(3) *Koszul homotopy.* $\mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v) = Q\,\mathrm{Tr}(\psi v u)$ (Lemma 3.30).

(4) *Monomial bracket.* $\{J(z_1^a z_2^b), J(z_1^c z_2^d)\} = (ad-bc)J(z_1^{a+c-1}z_2^{b+d-1}) + N[\{f,g\}]_0$.

(5) *CE/PV differential.* Under $c^k \mapsto \partial_{\xi_k}$, $u_k \mapsto \xi_k$, $d_{\mathrm{CE}}c^k = -\frac{1}{2}f^k_{ij}c^i c^j$ matches $d_\pi(\partial_{\xi_k})$ and $d_{\mathrm{CE}}u_k = f^j_{ik}u_j c^i$ matches $d_\pi(\xi_k)$ (Theorem 4.20).

(6) *Köthe bracket continuity.* $p_n(\{x,y\}) \le C_n\, p_{n+2}(x)\, p_{n+2}(y)$ in the weighted seminorms of clause (iv).
(7) *Kernel Maurer–Cartan.* $d\Theta_B + \frac{1}{2}[\Theta_B, \Theta_B] = 0$ (Theorem 5.5).
(8) *Hochschild equivariance.* $[\Phi_N(x), M_a]_G - M_{\rho_x(a)} = dH^{\mathrm{prod}}_{x,a}$ and $\{\Phi_N(x), M_a\}_{P_0} - M_{\rho_x^{P_0}(a)} = dH^{P_0}_{x,a}$ for every closed generator x and open observable a; invariant centre classes are the case $\rho_x = \rho_x^{P_0} = 0$ (Definition 5.8).
(9) *QME recursion.* At order $\hbar^n$ and window M, $d_M C_{n,M} = -o^{\mathrm{ns}}_{n,M}$ and $\pi_{M,N} C_{n,M} = C_{n,N}$.
(10) *Global descent first obstruction.* The Čech–Weiss class in $H^2(\partial X, \underline{\mathrm{Hom}}^{-1}(\mathcal{H}^\bullet, \mathcal{H}^\bullet))$ from Theorem 5.13.
(11) *Curve restriction.* $R_{L,B_\perp}(K^{(2)}_\Delta) = dw/(w - w')$ (Proposition 5.7).

THEOREM 5.10 *(Hamiltonian trace generators define the admissible trace Hochschild subalgebra).* Inside $\mathcal{B}^{\mathrm{adm}}$,

$$C^\bullet_{\mathrm{Hoch,tr,adm}}\big(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}\big) \;=\; \big\langle J(f),\, \theta_f : f \in \mathfrak{h} \big\rangle_{\smile,\{-,-\}},$$

where the left-hand side denotes the trace Hochschild subalgebra generated under cup product and Gerstenhaber bracket by the trace coordinates $J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2)$ for polynomial f, its completed formal extension, and their Schouten derivations θ_f. The trace-U(1) central character belongs to the projective current algebra; its Chevalley–Eilenberg dual c_K maps to the scalar-contact ghost γ_1, and K evaluates as $\mathrm{Tr}\, \mathbf{1} = N$ only after the projective central extension has been chosen. The scalar-reduced shadow at $K = 0$ gives the identity for the scalar-reduced trace subalgebra. The theorem makes no assertion that this subalgebra is the whole finite-N Hochschild centre.

Proof. The finite-window CE/PV calculation (Theorem 4.20; Theorem 5.24) identifies the admissible Hamiltonian trace summand with the corresponding polyvector summand in $PV^\bullet_{\mathrm{adm}}(\mathrm{Sym}\,\mathfrak{h})^{GL_N}$. Its interpretation as a Hochschild-centre summand uses the admissible Hochschild-centre lift of Definition 5.8; it is not a consequence of associative bar-cobar duality alone. Procesi–Razmyslov stable trace invariant theory [17][18] (Proposition 3.22) generates the GL_N-invariant polynomial functions on $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$ by the cyclic trace words $\mathrm{Tr}\, f(\phi_1, \phi_2)$, $f \in \mathbb{C}[z_1, z_2]$; Koszul resolution by ψ with $Q\psi = [\phi_1, \phi_2]$ extends this generation to the stable trace subalgebra of $\mathcal{O}(\mu^{-1}_{\mathrm{der}}(0))^{GL_N}$. The corresponding trace polyvector summand is generated as a graded-commutative algebra over this stable function ring by the Schouten derivatives θ_f along the symplectic structure on $\widehat{\mathbb{C}}^2$. The projective Capelli completion is a separate central extension adjoining K with the cocycle $[(f, aK), (g, bK)] = (\{f,g\}, \hbar N\, \omega(f,g)\, K)$. The bracket-admissible Hamiltonian-spanning image of Theorem 5.5 contains every $J(f)$ and every θ_f, and is closed under cup product and Gerstenhaber bracket. □

PROPOSITION 5.11 *(Forgetful diagram for compact, boundary, and formal data).* The compact target data, the global boundary data, the formal Capelli data, the bracket-admissible P_0 data, the coordinate associated-graded data, and the scalar-reduced data form a diagram $T = (C, \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, A^{\mathrm{cl}}_\partial, Z, \Phi, \mathcal{O}, \mathcal{H})$ connected by forgetful morphisms

$$T_{\mathrm{compact}} \xrightarrow{U_{\mathrm{target}}} T_{\partial,\mathrm{global}} \xrightarrow{\mathrm{Stalk}_b} T_{\mathrm{adm,Capelli}} \xrightarrow{H_Q} T_{P_0^B} \xrightarrow{\mathrm{gr}_{\hbar,F}} T_{\mathrm{coord}} \xrightarrow{U_B} T_{\mathrm{red}}.$$

A theorem formulated at a point of the diagram implies its image under every arrow. The converse requires the data omitted by the corresponding forgetful morphism: the Capelli projective Lie anomaly

$\hbar N[\bar{c}]$; the pro-Matlis residue tower $\mathcal{P}_q^{\Pi} = \varprojlim P_{\leq N}$; the diagonal local-cohomology kernel $K_{\Delta}^{(2)}$; the Hochschild centrality homotopy hierarchy $\{h_{\alpha,a}, h_{\alpha,a,b}, \dots\}$; the all-order QME counterterm tower $\{C_{n,M}\} \in \mathfrak{Q}^{\bullet}_{w,\partial,\hbar}$. Theorem 5.6 is the formal Capelli datum $T_{\mathrm{adm,Capelli}}$.

Proof. U_{target} forgets compact-target comparison data of Theorem 11.13. Stalk_b restricts to the formal Darboux neighbourhood: the descent class $\mathrm{Ob}_{\partial}^{\mathrm{desc}}$ vanishes on a contractible polydisk, and the global chiral E_2-Deligne datum restricts to the arity-two kernel $K_{\Delta}^{(2)}$. H_Q takes cohomology of the QME differential, retaining the induced projective E_2-central action on $H^{\bullet}$. $\mathrm{gr}_{\hbar,F}$ takes the associated graded for the $\hbar$-adic and scalar-contact filtrations, turning the projective quantum centre into its classical P_0-shadow with the cocycle as leading scalar class. U_B forgets the topology, convergence, and kernel admissibility of $\mathcal{B}^{\mathrm{adm}}$, retaining the coordinate CE/PV identity $c_f \mapsto \theta_f$, $u_f \mapsto O_f$. The last arrow to T_{red} quotients constants, killing K and the pro-Matlis tail. These are exactly the components omitted by the displayed arrows. □

Remark 5.12 *(Admissible CE/PV criterion and specialization).* The CE/PV cochain identity in 5.5 is the formal-Darboux coordinate statement in the reduced Hamiltonian sector. Its enveloping/Koszul enhancement is the admissible criterion: one must construct the coefficient category in which CE chains are conilpotent, continuous duality is exact, and the filtered cobar spectral sequence converges. Nilpotent, relative, and weighted Tate replacements are mechanisms for satisfying that criterion; the strict product/direct-sum endpoint is governed by the obstruction datum. The other proved statements are dimension-two specific (they use the residue pairing on $\mathbb{C}^2$, the matrix Weyl algebra of pairs (X, Y), and the U(1) centre of $\mathfrak{gl}_N$). Higher-dimensional analogues require the same cochain identity together with the appropriate residue pairing, matrix Weyl data, and a separate verification of the required conilpotent, relative, weighted, or completed coefficient hypotheses.

The admissible category $\mathcal{B}^{\mathrm{adm}}$ records the local-to-strict obstruction classes structurally. The factorization-centre lift datum $\mathrm{Ob}_{P_0}^{\mathrm{fact}} \in H^1(\mathrm{Cone}(\mathrm{gr}\,\kappa_{\tau})) \oplus \varprojlim^1 H^0$ of Problem 5.49 (strict Casimir support, the low-degree term $\{z_1, z_2^{N+1}\} = (N+1)z_2^N$, the $H_2 \cong \mathfrak{sl}_2$ lower-central obstruction, and the Milnor $\varprojlim^1$ defect) requires a null-homotopy before the bracket-admissible kernel image of Theorem 5.5 is obtained. The non-scalar QME class $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{QME}})$ of Problem 5.220, with first exposed row θ_3 and one-row covector $\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$, requires a scalar-zero Costello counterterm system. The radial PBW Stokes class $[(T_{a,b}, E^+_{a,b})] \in \mathrm{coker}\, B_{a,b}$ of Proposition F.25, equivalently the decorated Stokes obstruction for $D^{\square}_{a,b} = C^+_{a,b}\partial_2$, requires radial signed-row vanishing or a contracting homotopy in the stated bidegree. The six-entry Bochner–Martinelli–Koppelman vector $(\mathrm{ob}_{\oplus}, \mathrm{ob}_{I_{\Pi}}, \mathrm{ob}_{\mathfrak{h},\Pi}, \mathrm{ob}_{\mathrm{collar},\Pi}, \mathrm{ob}_{3,\Pi}, \mathrm{ob}_{\mathrm{unif},\Pi})$ of Proposition 1.28 is recorded by the pro-Matlis principal-part current target and requires pro-Matlis coherence. The trace-$U(1)$ scalar anomaly $\hbar N[\bar{c}]$ is represented by the projective Capelli central extension $\widehat{\mathfrak{g}}_{\mathrm{cent}}$ and reappears in $\mathcal{B}^{\mathrm{adm}}$ as the central charge at the trace generator. A modular-line curvature interpretation requires the determinant-line datum and connection.

The Hamiltonian on the formal symplectic normal disk is evaluated on the brane's matrix-valued transverse coordinate by trace; Dirac reduction forces the commuting-matrix algebra; the shifted-cotangent BF Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1]$ separates canonical motion from evaluation; the Chevalley–Eilenberg/polyvector dictionary sends Hamiltonian ghosts c_f to Schouten derivations θ_f and cotangent coordinates u_f to brane observables $J(f)$; continuous duality realises cotangent modes as Matlis principal parts; the projective central charge $\hbar N[\bar{c}]$ is the projective Lie anomaly at the trace generator. It becomes determinant-line curvature only after a $(\det_J^{1/2}, \nabla_J)$ whose Atiyah class pulls back to $[\bar{c}]$.

THEOREM 5.13 *(Global open–closed comparison criterion on the Costello–Gwilliam brane-link boundary).* Let a pre-open–closed comparison datum preOCA_N of Definition 9.5 be given in the admissible category $\mathcal{B}^{\mathrm{adm}}$ on the brane-link boundary ∂X, with source the bulk mixed-HT factorization algebra $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ of Definition 1.21 and target

$$\underline{\mathrm{End}}_{\mathrm{SC}^{\mathrm{HT}}_{(1,2)}}(\mathcal{C}^{\mathrm{op}}_{\partial}).$$

The centre target after the centrality lift is

$$Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}^{\mathrm{op}}_{\partial})$$

of Definition 9.1 (C3). Assume the formal trace-sector fibre of preOCA_N at a Dirac brane Darboux chart is the morphism of Theorem 9.6, equivalently Theorem 5.6. If the three-component obstruction $\mathrm{Ob}^{\mathrm{global}}$ below admits a null-homotopy (with typed five-component refinement (38)), then preOCA_N is promoted to an open–closed morphism OCA_N. If the cone of the resulting OCA_N has a contracting homotopy, then OCA_N is a global open–closed quasi-isomorphism in $\mathcal{B}^{\mathrm{adm}}$. An equivalence of mixed-HT SC-algebras requires the corresponding operadic homotopy-inverse datum. The typed refinement is Theorem 9.16.

Let $X = \mathbb{R}^2_{\mathrm{top}} \times Y_{\mathrm{hol}}$ with Y_{hol} a holomorphic-symplectic target (locally $\mathbb{C}^2_{\mathrm{hol}}$ in the formal-Darboux chart), $L \subset X$ the Dirac brane defect along the worldline, and ∂X its Costello–Gwilliam brane-link boundary [15, Vol. II] — the link sphere bundle over L, equivalently the boundary of an excised tubular neighbourhood of L in X. Let $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ be the bulk mixed-HT factorization algebra of Definition 1.21 and $\mathcal{C}^{\mathrm{op}}_{\partial}$ the open boundary factorisation category of Dirac branes on ∂X. The global comparison

$$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \longrightarrow Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}^{\mathrm{op}}_{\partial})$$

is a quasi-isomorphism in $\mathcal{B}^{\mathrm{adm}}$ after a null-homotopy of the three-component global obstruction tuple together with a contracting homotopy of $\mathrm{Cone}(\mathrm{OCA}_N)$:

$$\mathrm{Ob}^{\mathrm{global}} := \left(\mathrm{Ob}^{\mathrm{ch}\,E_2}_{\mathrm{Deligne}}, \mathrm{Ob}^{\mathrm{desc}}_{\partial}, \mathrm{Ob}^{\mathrm{QME}}_{\text{all-order}}\right)$$

in the corresponding chiral-operadic, descent, and QME obstruction complexes. A nonzero component, or a non-acyclic comparison cone, obstructs the global open–closed quasi-isomorphism. The three components are:

- $\mathrm{Ob}^{\mathrm{ch}\,E_2}_{\mathrm{Deligne}}$ is the obstruction to extending the arity-two diagonal product $K^{(2)}_{\Delta} = d\zeta_1 d\zeta_2/(\zeta_1\zeta_2)$ to a chiral E_2-operad with operadic associativity on all arities — the chiral E_d-Deligne conjecture at $d = 2$ (Conjecture 11.1);
- $\mathrm{Ob}^{\mathrm{desc}}_{\partial} = [\delta_{\check{C}}] \in \check{H}^2(\mathfrak{U}; \mathcal{E}^{-1}_{\mathrm{desc}})$ is the Čech obstruction class in the Weiss descent complex of a brane-link cover $\mathfrak{U}$ trivialising the local holomorphic-symplectic Darboux data;
- $\mathrm{Ob}^{\mathrm{QME}}_{\text{all-order}}$ is the all-order quantum master-equation obstruction tower of the closed-to-open lift of Φ_N at every Costello renormalisation scale, in the renormalisation-group filtered complex constructed by the Costello–Li theory of Statement 5.208 and Statement 5.207.

Remark 5.14 *(Compact modular-line addendum).* The operator-level lift of the modular line bundle on the compact target $K3 \times E$ belongs to the external compact comparison datum. If that comparison is imposed, one adds the obstruction $\mathrm{Ob}^{V}_{\mathrm{mod}}$ to lifting $\Omega_{\mathrm{central}}$ to an operator-valued cocycle in its own compact-target complex. Its scalar curvature section is the Igusa cusp form Φ_{10}, with Borcherds square-root branch $\Delta_5 = \Phi_{10}^{1/2}$.

Remark 5.15 *(Formal representatives and the global obstruction).* Restricted to a formal-Darboux neighbourhood of a brane vacuum $b \in X$, the local representatives are: the arity-two diagonal kernel $K_\Delta^{(2)}$, the trace-sector CE/PV map $c_f \mapsto \theta_f$, $u_f \mapsto J(f)$, and the Capelli central class. They are the arity-two and trace-sector boundary values whose extension, gluing, renormalisation, anomaly trivialisation, locality homotopy, and comparison-cone acyclicity are measured by $\mathrm{Ob}^{\mathrm{global},5}$ and $\mathrm{Cone}(\mathrm{OCA}_N)$ along ∂X.

5.4 Tate bar-cobar admissibility and the ∞-categorical comparison of the cochain bar-cobar identity

Inside an admissible Tate envelope satisfying Statement 5.18, the bar-cobar adjunction $\Omega \dashv \mathrm{Bar}$ is required to satisfy the admissible unit and counit weak-equivalence hypotheses (Theorem 5.31). The CE/PV map $c_f \mapsto \theta_f$, $u_f \mapsto J(f)$ then gives an admissible filtered equivalence between the closed CE cochain algebra and the Schouten polyvector algebra of the Hamiltonian formal disk. This is a Koszul comparison for the trace sector. It is not the construction of the full Hochschild cochain complex, nor of the derived chiral centre; those require the Hochschild-centre lift and its centrality homotopies. After restriction to a holomorphic curve, the $d = 1$ matching to the bulk side is the chiral-algebra theorem of [28]. The arity-two holomorphic collision is the two-dimensional diagonal singular product of Theorem 1.17; the Tate bar-cobar comparison controls only the Koszul algebraic coordinate of that stalk: $\Omega C_*^{\mathrm{CE}}(\mathfrak{g}) \xrightarrow{\sim} \widehat{U}(\mathfrak{g})$ on the Lie side, dualized through exact continuous duality to

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \simeq A^!_{\mathrm{op}},$$

with the linear symbol $B_f \mapsto O_f$, $\Theta_{\rho_{a,b}} \mapsto \theta^{a,b}$ of Definition 5.22.

For zero-differential finite-dimensional pieces $\{V_d\}_{d\geq 1}$, the inclusion

$$V_\oplus = \bigoplus_{d\geq 1} V_d \hookrightarrow V_\Pi = \prod_{d\geq 1} V_d,$$

filtered by degree tails, is an isomorphism on every finite quotient and on associated graded, yet $H^0(V_\Pi/V_\oplus) = V_\Pi/V_\oplus \neq 0$ (Lemma 5.27). The tail-quotient class is invisible to admissible filtered weak equivalence and constitutes one summand of the Koszul obstruction datum (the five-summand obstruction complex collecting the window, topology, conilpotence, Maurer–Cartan, and cobar obstruction components for the twist τ): $\mathcal{O}_{\mathrm{Kos}} = \mathcal{O}_{\mathrm{win}} \oplus \mathcal{O}_{\mathrm{top}} \oplus \mathcal{O}_{\mathrm{conilp}} \oplus \mathcal{O}_{\mathrm{MC}}(\tau) \oplus \mathcal{O}_{\mathrm{cobar}}(\tau)$, of Definition 5.23. The recognition criterion

$$\omega_{\mathrm{win}} = 0, \quad \mathcal{O}_{\mathrm{conilp}} = 0, \quad \mathcal{O}_{\mathrm{MC}}(\tau) = 0, \quad H^\bullet\big(\mathcal{O}_{\mathrm{top}} \oplus \mathcal{O}_{\mathrm{cobar}}(\tau)\big) = 0$$

is necessary and sufficient for filtered-cobar admissibility (Theorem 5.24). It is not, by itself, an ordinary cohomology equivalence of the chiral Hochschild target, since Lemma 5.27 exhibits tail cohomology invisible to admissible weak equivalence. An ordinary Hochschild-cohomology statement requires a tail-vanishing or Roos-acyclicity hypothesis. Theorem 5.31 descends to an equivalence of associated ∞-categories (Corollary 5.32). The Weiss-cosheaf condition on the unreduced factorization algebra over $\mathbb{R}^2 \times \mathbb{C}^2$ remains separate (Proposition 5.39); it requires the brane-defect heat-kernel propagator extension of Problem 5.49 item (2).

5.4.1 The Tate chain category

Let **TateVec** be the closed symmetric monoidal category of complete filtered Tate $\mathbb{C}$-vector spaces of 5.166. The filtration on every object is taken *complete and Hausdorff*: $V = \varprojlim_w V_{\le w}$ and $\bigcap_w F^w V = 0$. Concrete morphisms are filtration-preserving continuous chain maps. The working ambient category is the bounded-below filtered subcategory $\mathbf{TateChain}_{\ge 0}$ of non-negatively filtered chain complexes in **TateVec**, large enough to contain every CE chain coalgebra and completed enveloping algebra used here. The exact finite-window Mittag-Leffler category is constructed first. The model-categorical statements add transfer, smallness, and monoid axioms to this exact category; their scope is the exact category constructed here.

Definition 5.16 *(Strict finite-window Mittag-Leffler Tate category).* Let $\mathbf{MLTateChain}^{\mathrm{str}}_{\ge 0}$ be the category whose objects are inverse systems

$$V = (V_{\le M}, \pi^V_{M,N})_{M \ge N \ge 1}$$

of bounded-below finite-dimensional chain complexes over $\mathbb{C}$, with degreewise surjective transition maps. The represented complete object is

$$|V| = \varprojlim_M V_{\le M}, \qquad F^M|V| = \ker(|V| \to V_{\le M-1}).$$

Morphisms are compatible families of chain maps $f_{\le M} : V_{\le M} \to W_{\le M}$. The completed tensor product is defined on paired windows by

$$V \widehat{\otimes}_{\mathrm{ML}} W = (V_{\le M} \otimes W_{\le M})_{M \ge 1},$$

with the induced surjective transition maps, and

$$|V \widehat{\otimes}_{\mathrm{ML}} W| = \varprojlim_M (V_{\le M} \otimes W_{\le M}).$$

The continuous dual attached to this strict tower is

$$V^\vee_{\mathrm{ML}} = \varinjlim_M V^\vee_{\le M}.$$

A morphism is an admissible filtered weak equivalence when each $f_{\le M}$ and the induced associated-graded map are quasi-isomorphisms.

The only universal property asserted for this category is the ordinary inverse-limit one: a compatible family $x_M \in V_{\le M}$ determines a unique element $x \in |V|$, and every map out of a test strict tower is exactly a compatible family of finite-window maps. Initial and terminal properties among all completions of a Tate vector space lie outside this assertion.

PROPOSITION 5.17 *(Exactness and kernels in the strict ML category).* In $\mathbf{MLTateChain}^{\mathrm{str}}_{\ge 0}$:

(i) the filtration on $|V|$ is complete and Hausdorff;

(ii) inverse limits of strict surjective towers are exact, so $H^\bullet(|V|) = \varprojlim_M H^\bullet(V_{\le M})$ and $\varprojlim^1_M H^{\bullet-1}(V_{\le M}) = 0$;

(iii) completed tensor products and continuous duals are computed by the displayed finite-window formulas;

(iv) if $H = (H_{\le M})$ and $D = (D_{\le M})$ are paired strict ML systems and finite tensors $C_M \in H_{\le M} \otimes D_{\le M}$ are compatible under transition, then

$$C = \varprojlim_M C_M \in |H|\widehat{\otimes}_{\mathrm{ML}}|D|$$

exists and is the unique strict ML kernel with those finite-window projections.

Proof. Completeness is the definition of $|V|$ as an inverse limit; the intersection of the kernels $F^M|V|$ is zero because an element whose image vanishes in every finite quotient is the zero compatible family. Surjectivity of the transition maps gives the Mittag-Leffler condition. The Milnor exact sequence then has zero $\varprojlim^1$-term, proving exactness and the cohomology formula. The tensor and dual formulas are definitions inside the strict category and are finite-dimensional on each window. Finally, a compatible tensor family $(C_M)_M$ is precisely an element of the inverse limit $\varprojlim_M (H_{\le M} \otimes D_{\le M})$, which is the completed tensor product by definition. The uniqueness is the inverse-limit universal property of Definition 5.16. □

Statement 5.18 *(Admissible Tate model envelope).* An admissible Tate model envelope is a bounded-below, locally presentable symmetric monoidal model category $\mathbf{TateChain}^{\mathrm{adm}}_{\ge 0}$ equipped with a fully faithful exact symmetric-monoidal embedding of the strict category

$$\mathbf{MLTateChain}^{\mathrm{str}}_{\ge 0} \hookrightarrow \mathbf{TateChain}^{\mathrm{adm}}_{\ge 0},$$

and containing the CE chains, completed enveloping algebras, completed symmetric algebras, and finite-window inverse systems used below. Weak equivalences are admissible filtered quasi-isomorphisms: quasi-isomorphisms on every finite quotient and on associated graded. The tensor product restricts to $\widehat{\otimes}_{\mathrm{ML}}$ on strict Mittag-Leffler systems, and filtered inverse limits with surjective transition maps are exact by Proposition 5.17. The additional hypotheses are transfer and smallness for the bar-cobar adjunction, the monoid axiom for completed tensor products, and compatibility of $\widehat{\otimes}$ with the finite-window model structures. All Quillen statements below are relative to such an envelope. The strict Mittag-Leffler category gives exact inverse limits; it does not by itself give local presentability, Hovey recognition, the monoid axiom, or operadic transfer.

LEMMA 5.19 *(Mittag-Leffler dictionary for closed Tate windows).* Let

$$\{V_{\le M}\}_{M\ge 1} \quad \text{and} \quad \{W_{\le M}\}_{M\ge 1}$$

be inverse systems of finite-dimensional complexes with surjective transition maps, and put

$$V = \varprojlim_M V_{\le M}, \qquad W = \varprojlim_M W_{\le M}.$$

In the strict ML category, and hence in any admissible envelope containing it, the following assertions hold:

(i) $\varprojlim$ is exact on these systems, so $H^\bullet(V) = \varprojlim_M H^\bullet(V_{\le M})$;

(ii) continuous dualization changes the inverse system into the direct system of finite duals: $V^\vee_{\mathrm{cont}} = \varinjlim_M V^\vee_{\le M}$;

(iii) completed tensor products are computed windowwise: $V\widehat{\otimes}W = \varprojlim_M (V_{\le M} \otimes W_{\le M})$ whenever the transition maps are the paired finite-window truncations;

(iv) a map $f : V \to W$ is an admissible filtered quasi-isomorphism iff each finite-window map $f_{\le M} : V_{\le M} \to W_{\le M}$ is a quasi-isomorphism and the induced associated-graded maps are quasi-isomorphisms.

Proof. Surjectivity of the transition maps gives the Mittag-Leffler condition, hence $\varprojlim^1 = 0$. The cohomology assertion is the standard Milnor exact sequence with the $\varprojlim^1$-term killed. Continuous linear functionals on V factor through a finite quotient $V_{\le M}$, giving the displayed direct limit of duals. The completed tensor product in Definition 5.16 is defined by the same finite-window inverse system, so item (iii) is tautological inside the strict category. Item (iv) is the definition of weak equivalence in Definition 5.16 and Statement 5.18. □

Definition 5.20 *(Closed-window obstruction terms).* Let $\mathfrak{h} = \varprojlim_M \mathfrak{h}_{\le M}$ be a complete filtered Lie algebra with finite-dimensional windows and surjective transition maps $\pi_{M,N}$. Write $D_{\le M}$ for a finite dual module paired with $\mathfrak{h}_{\le M}$, with surjective transition maps $\pi^\vee_{M,N}$, and put $D_{\mathrm{cont}} = \varprojlim_M D_{\le M}$. For $M \ge N$ define:

$$\mathfrak{o}^{\mathrm{br}}_{M,N}(x, y) = \pi_{M,N}[x, y]_M - [\pi_{M,N}x, \pi_{M,N}y]_N,$$

$$\mathfrak{o}^{\mathrm{coad}}_{M,N}(x, \lambda) = \pi^\vee_{M,N}(x \cdot_M \lambda) - (\pi_{M,N}x) \cdot_N (\pi^\vee_{M,N}\lambda),$$

and

$$\mathfrak{o}^{\mathrm{Cas}}_{M,N} = (\pi_{M,N} \otimes \pi^\vee_{M,N})C_M - C_N.$$

The closed window system is *cochain-CE/PV-admissible* if $\mathfrak{o}^{\mathrm{br}}$ and $\mathfrak{o}^{\mathrm{coad}}$ vanish for all $M \ge N$. It is *kernel-admissible* only after the additional Casimir term $\mathfrak{o}^{\mathrm{Cas}}$ vanishes in the chosen completed tensor category. The coordinate CE/PV cochain identity contains the bracket and coadjoint transition equations; kernel-admissibility adds the Casimir equation.

LEMMA 5.21 *(Closed CE/PV finite-window compatibility and variance boundary).* Assume the closed window system of Definition 5.20 is cochain-CE/PV-admissible and each finite window has the finite cotangent CE/PV identity

$$\Phi_M : C^\bullet_{\mathrm{CE}}\big(\mathfrak{h}_{\le M} \ltimes D_{\le M}[1]\big) \xrightarrow{\sim} \mathrm{PV}\big(\mathbf{S}(\mathfrak{h}_{\le M})\big).$$

Then the Φ_M are compatible as finite coordinate tests under the contravariant CE pullbacks and the corresponding PV coordinate maps. These maps form compatible finite coordinate tests. A completed admissible filtered quasi-isomorphism

$$\Phi : C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h} \ltimes D_{\mathrm{cont}}[1]\big) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big)$$

follows only after one gives either a chain-side inverse limit with exact continuous dualization or a topology whose cochain transition maps are dg maps. If the kernel-admissible Casimir condition is also given, the same finite windows carry compatible P_0-pairings and kernels in that chosen tensor category.

Proof. The finite map Φ_M is determined on generators by $c \mapsto \theta$ and $u \mapsto O$. Vanishing of $\mathfrak{o}^{\mathrm{br}}_{M,N}$ says the CE differential on the Hamiltonian ghosts commutes with transition maps. Vanishing of $\mathfrak{o}^{\mathrm{coad}}_{M,N}$ says the cotangent-coordinate part of the CE differential, and equivalently the Lichnerowicz differential on O-coordinates, commutes with transition maps. Therefore the finite squares with Φ_M and Φ_N commute in the variance appropriate to cochains. A quotient $\mathfrak{h}_{\le M} \twoheadrightarrow \mathfrak{h}_{\le N}$ induces CE cochain maps by pullback along the quotient. Thus inverse-limit language on cochains requires the additional continuous-dualization or topology hypothesis stated above. Vanishing of $\mathfrak{o}^{\mathrm{Cas}}_{M,N}$ is the separate statement that the finite P_0-brackets and BV coefficient pairings are compatible under transition in the chosen kernel-admissible tensor category. □

5.4.2 Twisting cochains and the filtered-cobar admissibility criterion

Definition 5.22 *(Open-side twisting cochain datum).* Fix an admissible filtered-cobar datum

$$\mathfrak{h} = \varprojlim_{M} \mathfrak{h}_{\le M}, \qquad D = \varinjlim_{M} \mathfrak{h}^{\vee}_{\le M}, \qquad \mathfrak{g} = \mathfrak{h} \ltimes D[1],$$

a complete augmented filtered open A_∞-algebra A_{op}, and the completed CE/PV target

$$B_{\mathrm{cl}} = C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) \cong \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{coord}}(\mathfrak{h})\big).$$

Put $C_{\mathrm{op}} = \mathrm{Bar}(A_{\mathrm{op}})$, with reduced bar differential b_{op}, coproduct Δ, and iterated reduced coproducts $\Delta^{(r)}$. The bar coalgebra is required to be locally conilpotent in the completed sense: for every finite codomain window of B_{cl} and every finite coaugmentation quotient of C_{op}, the iterated reduced coproducts $\Delta^{(r)}$ vanish for all sufficiently large r. Equivalently, the conilpotence obstruction in Definition 5.23 is zero. This condition makes the Maurer–Cartan curvature below a continuous map and gives the twisting-cochain datum.

An open-side twisting cochain is a continuous degree-one map

$$\tau\colon C_{\mathrm{op}} \longrightarrow B_{\mathrm{cl}}$$

vanishing on the coaugmentation and satisfying the Maurer–Cartan equation

$$d_{B_{\mathrm{cl}}}\tau + \tau b_{\mathrm{op}} + \sum_{r\ge 2} m_r^{B_{\mathrm{cl}}}\big(\tau^{\otimes r}\big)\Delta^{(r)} = 0.$$

For the dg P_0-algebra B_{cl}, the higher multiplications vanish and this reads

$$d_{B_{\mathrm{cl}}}\tau + \tau b_{\mathrm{op}} + \mu_{B_{\mathrm{cl}}}(\tau\otimes\tau)\Delta = 0.$$

Equivalently,

$$d_{\mathrm{conv}}\tau + \frac{1}{2}[\tau,\tau]_{\mathrm{conv}} = 0, \qquad d_{\mathrm{conv}} f = d_{B_{\mathrm{cl}}} f - (-1)^{|f|} f\, b_{\mathrm{op}},$$

in the continuous convolution dg Lie algebra $\mathrm{Hom}_{\mathrm{cont}}(C_{\mathrm{op}}, B_{\mathrm{cl}})$.

The twisting cochain induces the completed cobar morphism

$$\kappa_\tau\colon \Omega C_{\mathrm{op}} \longrightarrow B_{\mathrm{cl}}.$$

Its associated-graded symbol is the map

$$\mathrm{gr}\,\kappa_\tau\colon\ \mathrm{gr}\,\Omega\mathrm{Bar}(A_{\mathrm{op}}) \longrightarrow \mathrm{gr}\,B_{\mathrm{cl}}.$$

In the Hamiltonian trace presentation the required linear symbol is

$$\mathrm{gr}\,\kappa_\tau(B_f) = O_f, \qquad \mathrm{gr}\,\kappa_\tau(\Theta_{\rho_{a,b}}) = \theta^{a,b},$$

after the scalar trace is removed and the cotangent generator is read in the principal-part local-dual basis.

Definition 5.23 *(Open-side Koszul obstruction complex).* In the notation of Definition 5.22, put $D_{\le M} = \mathfrak{h}^{\vee}_{\le M}$, $\mathrm{PV}_{\le M} = \mathrm{PV}(\mathbf{S}(\mathfrak{h}_{\le M}))$, and $B_{\mathrm{cl},\le M}$ for the corresponding finite CE/PV window. The obstruction datum is

$$\mathcal{O}_{\mathrm{Kos}}(\mathfrak{h}, D, A_{\mathrm{op}}, \tau) = \mathcal{O}_{\mathrm{win}} \oplus \mathcal{O}_{\mathrm{top}} \oplus \mathcal{O}_{\mathrm{conilp}} \oplus \mathcal{O}_{\mathrm{MC}}(\tau) \oplus \mathcal{O}_{\mathrm{cobar}}(\tau).$$

The window and conilpotence summands are degree-zero:

$$\mathcal{O}_{\mathrm{win}} = \prod_{M \ge N} \big(\mathrm{Hom}(\Lambda^2 \mathfrak{h}_{\le M}, \mathfrak{h}_{\le N}) \oplus \mathrm{Hom}(\mathfrak{h}_{\le M} \otimes D_{\le M}, D_{\le N})\big) \oplus Q_{\mathrm{Cas}},$$

where

$$Q_{\mathrm{Cas}} = \frac{\varprojlim_M (\mathfrak{h}_{\le M} \otimes D_{\le M})}{\mathrm{im}(\mathfrak{h}\widehat{\otimes} D)}.$$

The distinguished window class is

$$\omega_{\mathrm{win}} = \big(\mathfrak{o}^{\mathrm{br}}_{M,N}, \mathfrak{o}^{\mathrm{coad}}_{M,N}, [(C_{\le M})_M]\big),$$

where the first two terms are those of Definition 5.20 and the last term is the compatible finite-window Casimir family modulo strict completed tensors. The topology summand is

$$\mathcal{O}^n_{\mathrm{top}} = {\varprojlim}^1_M H^{n-1}\big(C^\bullet_{\mathrm{CE}}(\mathfrak{g}_{\le M})\big) \oplus {\varprojlim}^1_M H^{n-1}(\mathrm{PV}_{\le M}).$$

The conilpotence summand is

$$\mathcal{O}_{\mathrm{conilp}} = \Big(C^{\mathrm{CE}}_*(\mathfrak{g}) / \bigcup_r F^{(r)}_{\mathrm{coaug}} C^{\mathrm{CE}}_*(\mathfrak{g})\Big) \oplus \bigcap_{r \ge 1} \mathfrak{g}^{(r)}.$$

The Maurer–Cartan summand is the explicit curvature

$$\mathcal{O}_{\mathrm{MC}}(\tau) = \mathrm{MC}(\tau) = d_{\mathrm{conv}}\tau + \frac{1}{2}[\tau, \tau]_{\mathrm{conv}} \in \mathrm{Hom}^2_{\mathrm{cont}}(C_{\mathrm{op}}, B_{\mathrm{cl}}).$$

The cobar obstruction complex is

$$\mathcal{O}_{\mathrm{cobar}}(\tau) = \mathrm{Cone}(\mathrm{gr}\,\kappa_\tau) \oplus \bigoplus_n {\varprojlim}^1_M H^{n-1}\big((\Omega C_{\mathrm{op}})_{\le M}\big)[-n] \oplus \bigoplus_n {\varprojlim}^1_M H^{n-1}\big(B_{\mathrm{cl},\le M}\big)[-n].$$

THEOREM 5.24 *(Filtered-cobar admissibility criterion).* Fix a datum as in Definitions 5.22 and 5.23. Assume the filtrations on ΩC_{op} and B_{cl} are bounded below, complete, separated, and strongly convergent, with finite-dimensional associated graded pieces in every total degree. Assume moreover that the local conilpotence condition in Definition 5.22 holds and continuous dualization is exact on the finite-window towers. Then the following conditions are equivalent.

(A1) The datum is admissible: the generator substitution $c \mapsto \theta$, $u \mapsto \mathcal{O}$ is the completed CE/PV identity, τ is a twisting cochain, and

$$\kappa_\tau \colon \Omega\mathrm{Bar}(A_{\mathrm{op}}) \longrightarrow B_{\mathrm{cl}}$$

is an admissible filtered quasi-isomorphism with the linear symbol of Definition 5.22.

(A2) The obstruction equations are exact:

$$\omega_{\mathrm{win}} = 0, \qquad O_{\mathrm{conilp}} = 0, \qquad O_{\mathrm{MC}}(\tau) = 0, \qquad H^{\bullet}\big(O_{\mathrm{top}} \oplus O_{\mathrm{cobar}}(\tau)\big) = 0.$$

When these equivalent conditions hold,

$$B_{\mathrm{cl}} \simeq A_{\mathrm{op}}$$

in the homotopy category of complete augmented filtered A_∞-algebras. If the CE coalgebra is also used as the closed Koszul source, exact continuous duality identifies the dual form as

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) \simeq A^{!}_{\mathrm{op}}$$

precisely when the CE-side cobar map $\Omega C^{\mathrm{CE}}_{*}(\mathfrak{g}) \to A_{\mathrm{op}}$ has the same exact associated-graded cone and $\varprojlim^1$ obstruction.

Proof. The equality $\omega_{\mathrm{win}} = 0$ is equivalent to strict finite-window compatibility of the CE differential and the Schouten differential, with P_0-pairings added only in the kernel-admissible tensor category; this is exactly Lemma 5.21 and Theorem 4.22. The vanishing of O_{top} is the Milnor and variance condition identifying cohomology of the completed CE/PV objects with the admissible finite-window calculation.

The local conilpotence hypothesis makes the Maurer–Cartan sum finite on each finite codomain window, hence a continuous curvature. The equation $O_{\mathrm{MC}}(\tau) = 0$ is then the Maurer–Cartan equation in the continuous convolution dg Lie algebra. It is equivalent to the statement that the adjoint map κ_τ commutes with the completed cobar differential and the A_∞-operations. This equality gives the morphism tested below.

After the Maurer–Cartan equation holds, filter ΩC_{op} and B_{cl} by tensor length and by the finite coordinate windows. The mapping-cone summand in $O_{\mathrm{cobar}}(\tau)$ is acyclic exactly when $\mathrm{gr}\,\kappa_\tau$ is a quasi-isomorphism. The two $\varprojlim^1$-summands are exactly the Milnor defects for the completed source and target towers. Strong convergence and completeness then make the spectral-sequence comparison theorem equivalent to admissible filtered quasi-isomorphism of κ_τ. Necessity follows from the definition of admissible filtered quasi-isomorphism in the envelope: it is detected on the associated graded and on the exact finite-window towers.

The bar-cobar counit $\Omega\mathrm{Bar}(A_{\mathrm{op}}) \to A_{\mathrm{op}}$ is an admissible filtered quasi-isomorphism under the same boundedness and convergence hypotheses. Composing its inverse in the homotopy category with κ_τ gives $A_{\mathrm{op}} \simeq B_{\mathrm{cl}}$. The final statement is the same argument applied to the CE coalgebra $C^{\mathrm{CE}}_{*}(\mathfrak{g})$, followed by exact continuous dualization. □

Definition 5.25 *(Tate cofibrantly generated model structure in the envelope).* Define generating cofibrations I and acyclic cofibrations J by

$$I = \big\{F^w S^{n-1} \hookrightarrow F^w D^n \mid n \in \mathbb{Z}_{\geq 0},\ w \in \mathbb{N}\big\}, \qquad J = \big\{0 \hookrightarrow F^w D^n \mid n \in \mathbb{Z}_{\geq 0},\ w \in \mathbb{N}\big\},$$

where S^{n-1}, D^n are the standard sphere and disk chain complexes in **Vec** and the Tate window $F^w(-)$ takes the w-th piece of the filtration. A morphism $f : V \to W$ in $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ is:

(i) a *weak equivalence* if it is an admissible filtered quasi-isomorphism in the sense of 5.166: a quasi-isomorphism on every quotient $V/F^wV \to W/F^wW$ and on the associated graded;

(ii) a *cofibration* if it is in the class generated by I under transfinite composition, retracts, and pushouts;

(iii) a *fibration* if it has the right lifting property against J.

Theorem 5.26 *(Admissible-envelope model axiom).* Under Statement 5.18, the category $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ is a cofibrantly generated symmetric monoidal model category whose weak equivalences are admissible filtered quasi-isomorphisms. The sets I and J of Definition 5.25 are the chosen generating cofibrations and acyclic cofibrations. The obstruction to replacing this axiom by a construction is the simultaneous verification of local presentability, smallness, Hovey recognition, the pushout-product axiom, the monoid axiom, and compatibility with finite-window truncations.

Proof. The assertion is the model-category part of Statement 5.18. The strict Mittag-Leffler category proves exactness of inverse limits with surjective transition maps, but it does not prove smallness for transfinite cell complexes, stability of acyclic cofibrations under pushout, or the monoid axiom. Hovey recognition gives the displayed model category once those hypotheses are included in the envelope. □

Lemma 5.27 *(Tail quotient obstruction invisible to finite-window weak equivalence).* Let $\{V_d\}_{d\geq 1}$ be nonzero finite-dimensional vector spaces with zero differential, and put

$$V_\oplus = \bigoplus_{d\geq 1} V_d, \qquad V_\Pi = \prod_{d\geq 1} V_d.$$

Filter both complexes by degree tails. The inclusion $V_\oplus \hookrightarrow V_\Pi$ is an admissible filtered weak equivalence in the sense of Statement 5.18: it is an isomorphism on every finite quotient and on associated graded pieces. Its cofiber has

$$H^0(V_\Pi/V_\oplus) = V_\Pi/V_\oplus \neq 0.$$

Proof. The quotient by the tail $d > w_0$ is

$$V_\oplus / \bigoplus_{d>w_0} V_d \cong \bigoplus_{d\leq w_0} V_d \cong \prod_{d\leq w_0} V_d \cong V_\Pi / \prod_{d>w_0} V_d.$$

Thus the inclusion is an isomorphism on every finite quotient. The associated graded in degree d is V_d on both sides, so the associated graded map is also an isomorphism. This is precisely an admissible filtered weak equivalence.

Since the differential is zero, ordinary cohomology is the underlying vector space. A sequence with one nonzero component in every degree defines a nonzero class of $V_\Pi/V_\oplus$. Hence the cofiber has nonzero H^0. An ordinary quasi-isomorphism of chain complexes would have acyclic cofiber, so the inclusion is only a filtered weak equivalence in the admissible envelope. □

5.4.3 Operad-algebra transfer

Let $\mathrm{Lie}_{\mathbf{TateVec}}$ be the Lie operad in **TateVec**, regarded as a Σ-cofibrant Σ-module (by the standard Koszul resolution of the Σ-cofibrant replacement), and similarly $\mathrm{Ass}^+_{\mathbf{TateVec}}$ for augmented associative. A $\mathrm{Lie}_{\mathbf{TateVec}}$-algebra is a Tate Lie algebra in **TateVec**; an $\mathrm{Ass}^+_{\mathbf{TateVec}}$-algebra is a Tate complete augmented associative algebra. Write

$$\mathrm{TateLie} = \mathrm{Lie}_{\mathbf{TateVec}}\text{-alg}_{\geq 0}, \qquad \mathrm{TateAssocAlg}^+ = \mathrm{Ass}^+_{\mathbf{TateVec}}\text{-alg}_{\geq 0}$$

in $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$.

Lemma 5.28 *(Transferred model structures on Tate operad-algebras).* Suppose the admissible envelope also satisfies operadic admissibility for $\mathcal{P} \in \{\mathrm{Lie}_{\mathbf{TateVec}}, \mathrm{Ass}^+_{\mathbf{TateVec}}\}$: path objects exist for free $\mathcal{P}$-algebras, pushouts along generating acyclic cofibrations are weak equivalences, and filtered assembly preserves these weak equivalences. Then the free-forget adjunction $\mathrm{Free}_{\mathcal{P}} \dashv \mathrm{Forget}$ for $\mathcal{P}$ transfers the cofibrantly generated model structure of 5.26 to $\mathcal{P}\text{-alg}_{\geq 0}$ in the sense of Hinich [35, §2.2]: the weak equivalences and fibrations in $\mathcal{P}\text{-alg}_{\geq 0}$ are the maps whose underlying chain map in $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ is one, and the cofibrations are determined by lifting.

Proof. Hinich's transfer theorem requires exactly the hypotheses listed above:

(1) a path object on every algebra of the form $\mathrm{Free}_{\mathcal{P}}(F^w D^n)$;
(2) that pushouts along generating acyclic cofibrations be weak equivalences.

Item (1) is the path-object part of operadic admissibility. Item (2) is the pushout-stability part. Window-wise finite-dimensional arguments are necessary tests, but they do not replace the transfer hypothesis in the Tate envelope. □

The dual coalgebra side requires more care. Hinich's transfer applies to algebra categories; the coalgebra case is governed instead by conilpotence and the bar-cobar adjunction. The substitute is the Aubry-Lemaire-Smirnov [24] §11.4 model structure on conilpotent dg coalgebras transferred through the bar-cobar adjunction itself.

Definition 5.29 *(Conilpotent Tate coalgebras).* Let

$$\begin{aligned}\mathrm{TateConilpCoAlg} = \Big\{\, (C, \Delta_C, d_C) \ : & \ C \in \mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}, \\ & \Delta_C \colon C \to C \widehat{\otimes} C \text{ continuous coproduct}, \\ & \text{coaugmentation filtration exhaustive} \,\Big\}.\end{aligned}$$

with morphisms continuous coalgebra maps. Conilpotency means the coaugmentation filtration $F^{(n)}_{\mathrm{coaug}} C = \ker(\overline{\Delta}^{(n)})$ is exhaustive: $C = \bigcup_n F^{(n)}_{\mathrm{coaug}} C$.

LEMMA 5.30 *(Transferred model structure on Tate conilpotent coalgebras).* Suppose the conilpotent coalgebra transfer of Lefèvre–Hasegawa is valid in the chosen admissible Tate envelope: Ω preserves coalgebra cofibrations, detects the declared weak equivalences, and the unit $C \to \mathrm{Bar}\,\Omega C$ is an admissible filtered weak equivalence for every conilpotent Tate coalgebra. Then TateConilpCoAlg admits a model structure with

- weak equivalences = maps $f\colon C \to C'$ such that $\Omega(f)\colon \Omega(C) \to \Omega(C')$ is a weak equivalence in $\mathrm{TateAssocAlg}^{+}$ (5.28);
- cofibrations = monomorphisms of underlying filtered chain complexes;
- fibrations = right lifting against acyclic cofibrations.

Equivalently, this is the Lefevre-Hasegawa [40, Th. 1.3.1.2] transferred model structure in the admissible Tate envelope.

Proof. Lefevre-Hasegawa, with proof sketched in Loday-Vallette §11.4.10, constructs the conilpotent-coalgebra transferred model structure in the symmetric monoidal category of $\mathbb{Q}$- or $\mathbb{C}$-vector spaces. The proof passes through the cobar-bar adjunction $\Omega\colon \mathrm{ConilpCoAlg} \rightleftarrows \mathrm{AugAssocAlg}\colon \mathrm{Bar}$ and uses that

(i) Ω takes coalgebra cofibrations (= mono with filtered cokernel) to algebra cofibrations;
(ii) Ω inverts coalgebra weak equivalences;
(iii) the unit $\eta_C\colon C \to \mathrm{Bar}\,\Omega(C)$ is a weak equivalence for every conilpotent dg coalgebra.

Items (i)–(iii) are precisely the transfer hypotheses in the Tate envelope. Each finite window is a necessary check against the ordinary Lefèvre–Hasegawa proof, but the passage to the completed object uses the admissible filtered weak equivalence of the envelope. □

5.4.4 Bar-cobar comparison in the admissible envelope

THEOREM 5.31 *(Tate bar-cobar comparison in the admissible envelope).* Assume Statement 5.18, Lemma 5.28, and Lemma 5.30, and assume that the bar-cobar unit and counit are admissible filtered weak equivalences on the conilpotent and augmented objects under consideration. Then The bar-cobar adjunction

$$\Omega : \mathrm{TateConilpCoAlg} \rightleftarrows \mathrm{TateAssocAlg}^{+} : \mathrm{Bar}$$

gives a derived equivalence on the full subcategory generated by those admissible cofibrant-fibrant objects. A Quillen equivalence of the entire Tate model categories requires the unit and counit weak-equivalence hypotheses for all cofibrant and fibrant objects. In particular:

(i) Ω is left Quillen: it preserves cofibrations and acyclic cofibrations.

(ii) Bar is right Quillen: it preserves fibrations and acyclic fibrations.

(iii) For every admissible cofibrant conilpotent coalgebra C and every admissible fibrant augmented associative algebra A in the named subcategories, the unit $\eta_C : C \to \mathrm{Bar}\,\Omega(C)$ and the counit $\varepsilon_A : \Omega\,\mathrm{Bar}(A) \to A$ are weak equivalences.

(iv) The induced derived adjunction restricts to an equivalence on the homotopy subcategories generated by the admissible objects satisfying the displayed unit and counit hypotheses.

Proof. Items (1) and (2) are immediate from 5.30: weak equivalences in TateConilpCoAlg are defined as maps whose Ω is a weak equivalence, and fibrations on the coalgebra side are defined by lifting.

For item (3), the unit η_C: filter both sides by symmetric / tensor length on the underlying **TateVec** object, plus the Tate window filtration F^w. The associated graded gives, windowwise, a finite-dimensional dg coalgebra and the standard bar-cobar resolution $C^{\mathrm{gr}} \to \mathrm{Bar}\,\Omega(C^{\mathrm{gr}})$. By Loday-Vallette [24, §2.2.5, §11.1.5], this is a quasi-isomorphism on each window. Conilpotency (5.29) ensures the coaugmentation filtration on C is exhaustive and Hausdorff, so the abutment of the spectral sequence of the filtration computes the global qiso. Mittag-Leffler exactness in the Tate filtration (5.166 item 1) lifts the windowwise qiso to a global filtered qiso.

For the counit $\varepsilon_A : \Omega\,\mathrm{Bar}(A) \to A$, filter both sides by augmentation/tensor length and by the Tate window. On every finite window and every finite length cap this is the ordinary bar-cobar counit for an augmented dg associative algebra over $\mathbb{C}$, hence a quasi-isomorphism by the standard normalized bar resolution [24, §2.2.5, §11.1.5]. The filtrations are bounded below, complete, and Hausdorff in the admissible envelope, so the spectral sequence comparison and Mittag-Leffler exactness lift the windowwise quasi-isomorphism to the completed Tate object.

In the Lie specialization $A = \widehat{U}(\mathfrak{g})$, the same counit can also be read through the PBW filtration: $\mathrm{gr}_{\mathrm{PBW}}\widehat{U}(\mathfrak{g}) \cong \widehat{S}_{\mathrm{Tate}}(\mathfrak{g})$, and the associated graded comparison is the symmetric Koszul resolution of Loday-Vallette [24, §3.4.13]. This is the closed-side PBW form used later, but the Quillen counit above is the associative bar-cobar counit for general A.

Item (4) is formal from (1)-(3) on the stated admissible cofibrant-fibrant subcategory. On the whole model categories, Quillen's adjoint-functor criterion is the assertion that the same unit and counit are weak equivalences for every cofibrant and fibrant object. □

5.4.5 Promotion to the associated ∞-categories

COROLLARY 5.32 *(Infinity-categorical Tate bar-cobar comparison).* Apply the Dwyer-Kan / Hammock-localization functor to the admissible subcategory comparison of 5.31:

$$N_\Delta\big(\mathrm{TateConilpCoAlg}^{\mathrm{adm,cf}}, W\big) \simeq N_\Delta\big(\mathrm{TateAssocAlg}^{+,\mathrm{adm,cf}}, W\big)$$

as the ∞-categories associated to the admissible model subcategories in the sense of Lurie's *Higher Algebra* §A.3, where cf denotes the cofibrant-fibrant objects satisfying the unit and counit hypotheses, and W is the inherited class of weak equivalences. The equivalence is induced by $L\Omega$ and $R\,\mathrm{Bar}$ on this admissible subcategory.

Proof. Theorem 5.31 gives inverse derived functors on the admissible cofibrant-fibrant subcategories. The Dwyer–Kan localization of a relative equivalence of such subcategories is an equivalence of the associated ∞-categories. □

5.4.6 Identification with the Lie operadic Koszul duality

The bar-cobar comparison of 5.31 is *associative* bar-cobar. Specialization to the Lie operad gives the cochain-level comparison used in the CE/PV theorem.

COROLLARY 5.33 *(Tate Lie bar-cobar / CE Koszul duality).* For $\mathfrak{g}$ a lower-central Tate-pronilpotent graded Lie algebra in the sense of 5.166, the admissible bar-cobar comparison applied to $A = \widehat{U}(\mathfrak{g}) \in \mathrm{TateAssocAlg}^+$ yields the equivalence in the homotopy category

$$L\Omega\big(C_*^{\mathrm{CE}}(\mathfrak{g})\big) \xrightarrow{\sim} \widehat{U}(\mathfrak{g}), \qquad R\,\mathrm{Bar}\big(\widehat{U}(\mathfrak{g})\big) \xrightarrow{\sim} C_*^{\mathrm{CE}}(\mathfrak{g}),$$

and hence, after continuous dualization, a Lie-Koszul-dual equivalence of cochain algebras

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \simeq \big(\widehat{U}(\mathfrak{g})\big)^!$$

in the corresponding homotopy category of complete augmented Tate algebras.

No symmetric monoidal ∞-categorical lift is asserted from this corollary. Such a lift requires a weak monoidal Quillen equivalence for the chosen Tate operadic model and a separate proof that the Lie-operadic base change preserves the completed Koszul duality.

Proof. Apply 5.31 to the Lie-operadic Koszul-dual pair (Lie, CoComm) in **TateVec**. Loday-Vallette [24, Th. 13.4.6] identifies the operadic bar-cobar with the Lie-bar-cobar of CE chains and completed enveloping algebra. The statement is therefore an equivalence in the homotopy category determined by the admissible envelope. Monoidal functoriality is a further structure, not a consequence of the associative bar-cobar counit. □

LEMMA 5.34 *(Closed-side Tate formalism).* Let $\mathfrak{h} = \varprojlim_M \mathfrak{h}_{\le M}$ be a closed Hamiltonian Tate Lie algebra whose finite windows are cochain-CE/PV-admissible, and kernel-admissible when a P_0 kernel is used, in the sense of Definition 5.20. Assume:

(K1) the shifted cotangent extension $\mathfrak{g} = \mathfrak{h} \ltimes D_{\mathrm{cont}}[1]$ is an object of $\mathbf{TateChain}^{\mathrm{adm}}_{\ge 0}$;

(K2) the CE coalgebra $C_*^{\mathrm{CE}}(\mathfrak{g})$ is conilpotent for the coaugmentation filtration;

(K3) the completed enveloping algebra $\widehat{U}(\mathfrak{g}) = \varprojlim_M \widehat{U}(\mathfrak{g}_{\le M})$ is complete and separated for the PBW filtration;

(K4) the finite-window PBW associated graded maps assemble to

$$\mathrm{gr}_{\mathrm{PBW}}\widehat{U}(\mathfrak{g}) \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{g}).$$

Then the closed-side formalism has two compatible consequences:

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\Phi} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big)$$

is the admissible completed CE/PV identity obtained from the finite-window family of Lemma 5.21 together with the exact continuous-dualization and variance data in (K1)–(K4), and

$$L\Omega C^{\mathrm{CE}}_{*}(\mathfrak{g}) \simeq \widehat{U}(\mathfrak{g}), \qquad R\,\mathrm{Bar}\,\widehat{U}(\mathfrak{g}) \simeq C^{\mathrm{CE}}_{*}(\mathfrak{g})$$

is the Tate Lie bar-cobar / CE Koszul duality of Corollary 5.33. The two constructions are compatible with the same finite-window transition maps.

Proof. The first assertion is Lemma 5.21 plus the exact continuous-dualization and variance hypotheses included above. The obstruction terms $\mathfrak{o}^{\mathrm{br}}$ and $\mathfrak{o}^{\mathrm{coad}}$ are the cochain CE/PV transition obstructions; $\mathfrak{o}^{\mathrm{Cas}}$ is the additional kernel/P_0-pairing obstruction. Their vanishing gives the finite-window compatibility; the admissible envelope gives the cochain inverse-limit datum for $\Phi = \varprojlim_M \Phi_M$.

For the Koszul assertion, hypotheses (K1)–(K4) put $C^{\mathrm{CE}}_{*}(\mathfrak{g})$ and $\widehat{U}(\mathfrak{g})$ in the admissible model envelope with complete filtrations. Conilpotency is the coalgebra hypothesis needed for Bar Ω, and PBW completeness identifies the associated graded of the enveloping algebra with the completed symmetric algebra. Corollary 5.33 therefore applies. Both constructions are obtained by first applying the finite-dimensional CE/PV and bar-cobar statements on $\mathfrak{g}_{\le M}$, then passing through the same Mittag-Leffler inverse limit; hence the transition maps are the same. □

5.4.7 Module reconstruction from the CE/PV identity

THEOREM 5.35 *(Admissible-envelope module reconstruction from the CE/PV identity).* Let $\mathfrak{h}_{\mathrm{adm}}$ be a pro-finite-dimensional Lie algebra in **TateVec** whose shifted-cotangent extension $\mathfrak{g}_{\mathrm{adm}} = \mathfrak{h}_{\mathrm{adm}} \ltimes \mathfrak{h}^{\vee}_{\mathrm{adm,cont}}[1]$ satisfies the lower-central Tate-pronilpotent admissible-envelope hypotheses of 5.31. Assume further that:

(M1) the strict cochain map

$$\Phi\colon C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) \longrightarrow \mathrm{PV}\big(\widehat{S}_{\mathrm{Tate}}(\mathfrak{h}_{\mathrm{adm}})\big)$$

is the admissible CE/PV quasi-isomorphism obtained from the finite-window family of Lemma 5.21 after the required variance and exact continuous-dualization data are given;

(M2) the completed dg P_0-algebras on both sides lie in an admissible transferred P_0-algebra model structure inside $\mathbf{TateChain}^{\mathrm{adm}}_{\ge 0}$, with weak equivalences detected on underlying admissible filtered quasi-isomorphisms;

(M3) the module categories are formed in the same locally presentable monoidal envelope, and the two algebras are replaced by cofibrant-fibrant representatives when needed.

Then Φ induces an equivalence, inside that admissible Tate model envelope, of module ∞-categories over the two identified completed dg P_0-algebras

$$\widetilde{\Phi}\colon N_{\Delta}\big(C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}})\text{-mod}, W\big) \xrightarrow{\sim} N_{\Delta}\big(\mathrm{PV}\big(\widehat{S}_{\mathrm{Tate}}(\mathfrak{h}_{\mathrm{adm}})\big)\text{-mod}, W\big)$$

in the ∞-categorical setting of Lurie's *Higher Algebra* §5.5. The hypotheses (M1)–(M3) presuppose an already constructed admissible P_0-algebra envelope. The conclusion is the algebraic ∞-categorical equivalence inside that envelope. The completed dg P_0 model structure and the factorization-centre theorem for the full untruncated formal Hamiltonian algebra are separate constructions.

Proof. Assumption (M1) gives a strict admissible filtered quasi-isomorphism of completed dg P_0-algebras. The strictness is the generator-level identity $c \mapsto \theta,\ u \mapsto O$, together with $d_\pi \Phi = \Phi d_{\mathrm{CE}}$ and bracket compatibility from the finite-window CE/PV calculation.

By (M2), this strict weak equivalence is a weak equivalence in the transferred P_0-algebra model structure. Replace the two P_0-algebras by cofibrant-fibrant representatives if necessary. Extension and restriction of scalars along a weak equivalence of admissible algebra objects give a Quillen equivalence of module model categories; this is the standard Schwede–Shipley/Lurie mechanism for module categories over weakly equivalent algebra objects [43] and [41, Th. 7.1.4.6]. Applying Dwyer–Kan localization gives the displayed equivalence of module ∞-categories.

Functoriality in the Lie algebra gives interval-wise compatibility for admissible interval objects $\mathfrak{h}_{\mathrm{adm},I} = \Omega^\bullet_c(I) \otimes \mathfrak{h}_{\mathrm{adm}}$ when these interval objects remain in the same admissible envelope. This gives interval-wise categorical compatibility. Weiss descent and unreduced open-BV centrality are governed by their own descent and centrality criteria. □

Remark 5.36 *(Relation to Lurie's higher-algebraic Koszul duality).* Lurie [41, §5.5.5] constructs a higher-algebraic Koszul duality between augmented ∞-operad-algebras and their Koszul-dual coalgebras. For the operad $\mathrm{Lie}_{\mathbf{TateVec}}$, Koszul self-duality (with shift) gives the equivalence of 5.33 in the lower-central Tate-pronilpotent setting. The two constructions agree: weak equivalences in the model-categorical presentation are admissible filtered quasi-isomorphisms (5.166), and Lurie A.3.7 with the Hinich transfer of 5.28 identifies them with the Hammock-localized weak equivalences of the ∞-categorical presentation.

5.4.8 Conilpotence and Casimir convergence

PROPOSITION 5.37 *(Bar-cobar conilpotence separated from Casimir convergence).* The admissible bar-cobar comparison of 5.31 and the resulting ∞-categorical equivalence of 5.35, when their admissible-envelope hypotheses hold, are independent of the convergence of the Tate Casimir element. The cochain complexes $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$ and $\mathrm{PV}\big(\widehat{S}_{\mathrm{Tate}}(\mathfrak{h})\big)$ are equivalent in the model category of 5.28 after imposing the same conilpotent admissibility hypotheses. The required hypothesis here is conilpotence; Casimir convergence belongs to the analytic kernel problem.

Proof. The bar-cobar comparison uses $\widehat{S}^c(\mathfrak{g}[1])$ and $T(\mathfrak{g})$ as underlying graded objects. Both are well-defined and conilpotent in the admissible envelope under 5.166 items 1, 2 (continuous dual exact, conilpotent CE coalgebra). The formal Casimir $\sum_d e_{d,i} \otimes e_d^{i,\vee}$ is a kernel hypothesis only when it converges in the chosen coefficient category; in the strict product/direct-sum Tate endpoint it is precisely the obstruction isolated in Lemma 5.212. Such a kernel is required for the Costello [12] heat-kernel propagator construction at the analytic factorization-algebra level where the bracket is not strict. The algebraic bar-cobar comparison uses only the conilpotent admissibility hypotheses: the cochain-level Φ extends, in the admissible model envelope, to an ∞-categorical equivalence. □

PROPOSITION 5.38 *(Strict product/direct-sum kernel obstruction).* Let $\mathcal{C}_{\mathrm{ker}}$ be the category whose objects are complete coefficient pairs (H, D), equipped with finite quotients $(H_{\leq M}, D_{\leq M})$, compatible transition maps, and a tensor $K \in H \widehat{\otimes} D$ whose image in $H_{\leq M} \otimes D_{\leq M}$ is the finite Casimir C_M for every M. Morphisms preserve the finite quotients and carry K to K'. The original strict product/direct-sum pair

$$H_\Pi = \prod_{d\geq 1} H_d, \qquad D_\oplus = \bigoplus_{d\geq 1} H_d^\vee$$

lies outside $\mathcal{C}_{\mathrm{ker}}$. More strongly, any object of $\mathcal{C}_{\mathrm{ker}}$ mapping to the strict pair by a finite-window identity morphism produces the obstructed infinite-support tensor in $H_\Pi \widehat{\otimes} D_\oplus$.

Thus initial, terminal, and representing universal properties for kernel-admissible completions pass through a genuine kernel object. A weighted product completion is a constructed regulator for tail-sensitive ordinary cohomology classes.

Proof. If the strict pair were an object, it would contain a tensor $K \in H_\Pi \widehat{\otimes} D_\oplus$ whose image in every finite window is C_M. If a kernel-admissible object mapped to the strict pair by a finite-window identity morphism, the image of its kernel would give the same obstructed tensor. But $D_\oplus$ has finite support in the degree direction, while the compatible family $(C_M)_M$ has a nonzero $H_d^\vee$-component for every d. This is exactly the support obstruction of Theorem 5.94. Therefore the strict pair lies outside $\mathcal{C}_{\mathrm{ker}}$, and universal-property assertions must pass through a kernel-admissible completion. □

PROPOSITION 5.39 *(Residual Weiss-cosheaf condition).* The ∞-categorical equivalence of 5.35 on stalks and on the locally-constant interval-factorization $\mathcal{F}_{\mathrm{cl}}$ of 4.11 gives the cochain-level categorical comparison. The Weiss-cosheaf condition for the *unreduced* factorization algebra $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}$ on $\mathbb{R}^2 \times \mathbb{C}^2$ requires the additional descent check below. The Weiss / descent condition for an unreduced factorization algebra on a higher-dimensional manifold requires checking Čech descent for Weiss covers, an independent cosheaf-theoretic statement. Costello–Gwilliam Vol. I §§6.1, 6.5, 7.2 provide the factorization, cosheaf, and basis-extension vocabulary; the present argument uses them as vocabulary and records the unreduced descent obligation as a residual class. The admissible Tate Quillen equivalence closes the cochain-level ∞-categorical comparison; the Weiss descent for the unreduced factorization algebra remains a separate cosheaf-theoretic obligation requiring the heat-kernel propagator extension of Problem 5.49. The two-dimensional formal singular product calculation is the separate result of Theorem 1.17.

Proof. Lurie [41, §5.5.4], with the same terminology as Costello-Gwilliam [15, Vol. I §6.4]: a locally constant factorization algebra on $\mathbb{R}$ determines an E_1-algebra up to the standard contractible choice; the bar-cobar comparison of 5.31 realizes this on the admissible subcategory. On a higher-dimensional manifold, the Weiss cover $\{U_\alpha\}$ has structure-map descent independent of the algebraic bar-cobar; in particular the heat-kernel propagator on $\mathbb{R}^2 \times \mathbb{C}^2$ enters the analytic check. This heat-kernel propagator is the analytic datum relating the bar-cobar identity to Weiss descent. □

COROLLARY 5.40 *(Moment-map specialisation).* In the situation of 4.20, the degree-zero specialization of Φ_{coord} on the cotangent direction is the moment-map / Kirillov assignment

$$J : \mathfrak{h} \longrightarrow A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}), \qquad J(f) = O_f.$$

The closed-side Lie bracket $\{f, g\}_{\mathfrak{h}}$ matches the Kirillov–Poisson bracket on A_∂:

$$\{J(f), J(g)\}_\partial = J\big(\{f, g\}_{\mathfrak{h}}\big).$$

Proof. J is the restriction of Φ_{coord} to the cotangent direction $u_I \mapsto O_I$ followed by linear extension along $\mathfrak{h} \hookrightarrow A_\partial$ (the canonical embedding of $\mathfrak{h}$ as the linear functions on $\mathfrak{h}^\vee$). Bracket compatibility is the (c^J, u_K)-versus-(θ^J, O_K) identity from 4.20 read off in degree zero. Equivalently, this is the standard Kirillov–Kostant–Souriau theorem for linear Poisson structures. □

LEMMA 5.41 *(Natural-correction obstruction).* Let $\widetilde{J}\colon \mathfrak{h} \to A_\partial$ be a linear map satisfying stable polynomial GL_∞-invariance (Proposition 3.22), descent through the Dirac ideal $[\phi_1, \phi_2] = 0$, the scalar quotient $\overline{\mathrm{Tr}(1)} = 0$, and primitive block-sum additivity:

$$\widetilde{J}_{N_1+N_2}(f)\big|_{\phi_i=\phi_i'\oplus\phi_i''} = \widetilde{J}_{N_1}(f) + \widetilde{J}_{N_2}(f),$$

the value at the block-diagonal stacking $(\phi_1, \phi_2) = (\phi_1', \phi_2') \oplus (\phi_1'', \phi_2'')$ on $\mathfrak{gl}_{N_1} \oplus \mathfrak{gl}_{N_2} \subset \mathfrak{gl}_{N_1+N_2}$ reading on the right as a vector-space sum in A_∂. Equivalently, $\widetilde{J}(f)$ is primitive for the block-sum coproduct $\Delta_\oplus$ on A_∂ sending each single-trace generator $\overline{\mathrm{Tr}}$ to $\overline{\mathrm{Tr}} \otimes 1 + 1 \otimes \overline{\mathrm{Tr}}$. Then for every $a + b > 0$,

$$\widetilde{J}(z_1^a z_2^b) = c_{a,b}\,\overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}, \qquad c_{a,b} \in \mathbb{C}.$$

Proof. Stable invariant theory (Proposition 3.22) writes $\widetilde{J}(z_1^a z_2^b)$ as a polynomial in nonempty cyclic trace words on the free pair, of total bidegree (a, b); scalar reduction kills empty traces. Restriction to the moment-map zero-fibre with the Koszul homotopy for adjacent transpositions (Lemma 3.30) collapses each cyclic word at bidegree (p, q) to the commutative monomial trace $\overline{\mathrm{Tr}(\phi_1^p \phi_2^q)}$ (Corollary 3.32), so

$$\widetilde{J}(z_1^a z_2^b) = \sum_{\lambda \vdash (a,b)} c_\lambda \prod_{i=1}^{\ell(\lambda)} \overline{\mathrm{Tr}(\phi_1^{p_i} \phi_2^{q_i})},$$

summed over partitions of (a, b) into $\ell(\lambda)$ nonempty parts (p_i, q_i). Block-sum stacking sends the single-trace generator $\overline{\mathrm{Tr}(\phi_1^p \phi_2^q)}$ to its block-diagonal sum $\overline{\mathrm{Tr}((\phi_1')^p(\phi_2')^q)} + \overline{\mathrm{Tr}((\phi_1'')^p(\phi_2'')^q)}$ — this is the Loday–Quillen–Tsygan primitive on $H_*^{\mathrm{CE}}(\mathfrak{gl}_\infty(A_{\bar f}))$ — and a length-ℓ product to a sum of 2^ℓ terms with $2^\ell - 2$ cross-block products outside the LQT primitive. The primitive block-sum identity therefore vanishes the $\ell \geq 2$ coefficients, leaving the single-part partition $\lambda = ((a, b))$. □

THEOREM 5.42 *(Uniqueness of J).* Let $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ be the reduced Hamiltonian Lie algebra of the formal symplectic polydisk and let $A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$ be the stable scalar-reduced Hamiltonian trace algebra of Proposition 3.40. Among linear maps

$$\widetilde{J}\colon \mathfrak{h} \longrightarrow A_\partial$$

satisfying:

(U1) (*Polynomial GL_N-invariance, stable*) $\widetilde{J}(f)$ is a stable GL_N-invariant trace polynomial in (ϕ_1, ϕ_2) in the sense of Proposition 3.22;

(U2) (*Dirac constraint*) $\widetilde{J}(f)$ descends modulo the moment-map ideal $[\phi_1, \phi_2] = 0$;

(U3) (*Scalar reduction*) $\widetilde{J}$ factors through the stable scalar-reduced quotient $A_\partial = A_\infty^{GL_\infty}/\mathbb{C}\cdot \mathrm{Tr}(1)$;

(U4) (*Darboux naturality*) for every formal canonical transformation $\Phi\colon \widehat{\mathbb{C}^2}_0 \to \widehat{\mathbb{C}^2}_0$ preserving $\omega = dz_1 \wedge dz_2$, $\widetilde{J}(\Phi^* f) = \Phi^* \widetilde{J}(f)$ under the induced action on A_∂;

(U5) (*Linear normalization*) $\widetilde{J}(z_1) = \overline{\mathrm{Tr}\,\phi_1}$, $\widetilde{J}(z_2) = \overline{\mathrm{Tr}\,\phi_2}$;

(U6) (*Primitive block-sum additivity*) under the block-diagonal stacking $(\phi_1, \phi_2) = (\phi_1', \phi_2') \oplus (\phi_1'', \phi_2'')$ on $\mathfrak{gl}_{N_1} \oplus \mathfrak{gl}_{N_2} \subset \mathfrak{gl}_{N_1+N_2}$,

$$\widetilde{J}_{N_1+N_2}(f)\big|_{\phi_i=\phi_i'\oplus\phi_i''} = \widetilde{J}_{N_1}(f) + \widetilde{J}_{N_2}(f) \qquad \text{in } A_\partial,$$

the brane substitution $J(f) = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$ is the unique solution. Lie-algebra morphism $\{\widetilde{J}(f), \widetilde{J}(g)\}_\partial = \widetilde{J}(\{f, g\}_{\hat{A}})$ holds for every solution (Theorem 5.191); uniqueness is the content of (U1)–(U6). The categorical normal-form refinement, including protected admissibility, is Theorem 5.147 of §7.5.

Remark 5.43 *(Worked instances).* The three running examples of §1.1 determine the theorem concretely. Example 1.2 is the $N = 1$ case: the matrices commute automatically, scalar reduction is trivial, and $J(f) = f(\phi_1, \phi_2)$ is the classical evaluation map at a point of $\mathbb{C}^2$. Example 1.4 shows the linear Hamiltonian normalisation $z_i \mapsto \overline{\operatorname{Tr} \phi_i}$ at work and exhibits the scalar reduction killing the constant-Hamiltonian line. Example 1.5 contrasts the Taylor side (what J samples) with the residue side (the Matlis dual that pairs with Taylor coefficients), which is why the natural-correction obstruction works monomial-by-monomial through finite Hamiltonian flows.

Proof. Lemma 5.41, drawing on (U1), (U2), (U3), (U6), gives $\widetilde{J}(z_1^a z_2^b) = c_{a,b} \overline{\operatorname{Tr}(\phi_1^a \phi_2^b)}$ on every bidegree $a + b > 0$ with scalars $c_{a,b} \in \mathbb{C}$. Linear normalization (U5) gives $c_{1,0} = c_{0,1} = 1$.

Darboux naturality (U4) propagates the scale to every bidegree. The formal Darboux symplectomorphism group $G_{\mathrm{Darb}} = \mathrm{Symp}_{\mathrm{form}}(\widehat{\mathbb{C}^2}_0)$ acts on A_∂ through the algebra-homomorphism matrix substitution $\Phi_* \overline{\operatorname{Tr} f(\phi_1, \phi_2)} = \overline{\operatorname{Tr}\,(\Phi^* f)(\phi_1, \phi_2)}$. The Hamiltonian flow generated by $H_1^{(2)} = -z_1^2/2$ acts by $\Phi^*_{H_1^{(2)}} z_2 = z_2 + t z_1$, and on z_2^k gives

$$\widetilde{J}\big((z_2 + t z_1)^k\big) = \sum_{a+b=k} \binom{k}{a} t^a c_{a,b} \overline{\operatorname{Tr}(\phi_1^a \phi_2^b)}, \qquad \Phi_* \widetilde{J}(z_2^k) = c_{0,k} \overline{\operatorname{Tr}(\phi_2 + t\phi_1)^k} = \sum_{a+b=k} \binom{k}{a} t^a c_{0,k} \overline{\operatorname{Tr}(\phi_1^a \phi_2^b)}.$$

Equality at each t^a gives $c_{a,b} = c_{0,k}$ for $a + b = k$; the symmetric linear flow $z_1 \mapsto z_1 - t z_2$ on z_1^k gives $c_{a,b} = c_{k,0}$ and hence $c_{0,k} = c_{k,0}$ within total degree k. Linear shears preserve total degree, so they are insufficient by themselves to reach the linear normalisation $c_{1,0} = c_{0,1} = 1$. The required cross-degree morphism is the nonlinear flow generated by $H_k^{(1)} = -z_2^{k+1}/(k+1)$,

$$\Phi^*_{H_k^{(1)}} z_1 = z_1 - t z_2^k, \qquad \Phi^*_{H_k^{(1)}} z_2 = z_2,$$

which raises the Taylor degree of z_1 from 1 to k and acts on the matrix substitution as $(\phi_1, \phi_2) \mapsto (\phi_1 - t\phi_2^k, \phi_2)$. Naturality (U4) on the linear monomial z_1 reads

$$\widetilde{J}(z_1 - t z_2^k) = \widetilde{J}(z_1) - t\,\widetilde{J}(z_2^k) = \overline{\operatorname{Tr} \phi_1} - t c_{0,k} \overline{\operatorname{Tr} \phi_2^k}$$

on the source against

$$\Phi_* \widetilde{J}(z_1) = \overline{\operatorname{Tr}(\phi_1 - t\phi_2^k)} = \overline{\operatorname{Tr} \phi_1} - t\,\overline{\operatorname{Tr} \phi_2^k}$$

on the matrix side. Equality at t^1 forces $c_{0,k} = 1$ for every $k \geq 1$; the linear-shear identity then propagates $c_{a,b} = 1$ on every bidegree of total degree k. Continuity in the $\hat{\mathfrak{h}}$-topology extends to formal power series, so $\widetilde{J} = J$. □

THEOREM 5.44 *(Hypothesis rigidity for J).* The substantive hypotheses are (U4) and (U5): removing either admits a positive-dimensional family of solutions $\widetilde{J} \neq J$. Primitive block-sum additivity (U6) is the Loday–Quillen–Tsygan-explicit form of single-trace forcing; given (U1)–(U5), it is automatic, but separating it from (U4) rebalances the proof, with (U6) carrying shape (single-trace) and (U4)+(U5) carrying scale (Lemma 5.41 monomial-by-monomial). Hypotheses (U1), (U2), (U3) are codomain specifications, each implied by (U4)+(U5) on an enlarged ambient: on $\mathbb{C}[\mathfrak{gl}_N^2]/\langle[\phi_1, \phi_2]\rangle$, $\widetilde{J}$ takes values in the trace algebra; on $\mathbf{S}(\mathrm{Cyc}(F)_{\text{nonempty}})$, $\widetilde{J}$ equals the symmetrized cyclic trace, which Dirac collapse identifies with J; on $A_\partial \oplus \mathbb{C} \cdot \overline{\mathrm{Tr}(1)}$, the $\overline{\mathrm{Tr}(1)}$-component is zero.

Proof. Removing (U4). Choose any sequence $\{c_{a,b}\}_{a+b\geq 2} \subset \mathbb{C}$ and set $\widetilde{J}(z_1^a z_2^b) = c_{a,b}\,\overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}$ for $a + b \geq 2$, $\widetilde{J}(z_i) = \overline{\mathrm{Tr}\,\phi_i}$. The family satisfies (U1)–(U3), (U5), (U6); Darboux flow $z_2 \mapsto z_2 + t z_1$ on z_2^k forces $c_{a,k-a} = c_{0,k}$ and the reverse forces $c_{0,k} = 1$, hence every $c_{a,b} = 1$.

Removing (U5). The family $\widetilde{J}_\lambda^{\mathrm{rs}} = \lambda J$, $\lambda \in \mathbb{C}^\times$, satisfies (U1)–(U4) and (U6); (U5) requires $\lambda = 1$.

Removing (U6) with (U4) intact. The two-trace correction $\widetilde{J}_\alpha^{\mathrm{mt}}(z_1 z_2) = \overline{\mathrm{Tr}(\phi_1\phi_2)} + \alpha\,\overline{\mathrm{Tr}\,\phi_1}\,\overline{\mathrm{Tr}\,\phi_2}$, $\widetilde{J}_\alpha^{\mathrm{mt}}(z_1^a z_2^b) = \overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}$ elsewhere, satisfies (U1), (U2), (U3), (U5) for every α; Darboux flow $z_2 \mapsto z_2 + t z_1$ on z_2^2 reads source-side t^1-coefficient $2\,\overline{\mathrm{Tr}(\phi_1\phi_2)} + 2\alpha\,\overline{\mathrm{Tr}\,\phi_1}\,\overline{\mathrm{Tr}\,\phi_2}$ against matrix-side $2\,\overline{\mathrm{Tr}(\phi_1\phi_2)}$, forcing $\alpha = 0$. The same Darboux propagation kills every higher multi-trace correction inductively: any $M : V_H \to \mathbf{S}^{\geq 2}(V_H)$ compatible with the full Hamiltonian flow algebra reduces to $M = 0$. Hence (U6) is implied by (U4)+(U5)+(U1)–(U3); inside the full hypothesis set $\widetilde{J} = J$.

(U1) reduction. On $\mathbb{C}[\mathfrak{gl}_N^2]/\langle[\phi_1, \phi_2]\rangle$, (U5) gives $\widetilde{J}(z_i) = \overline{\mathrm{Tr}\,\phi_i}$ (single trace, hence GL_N-invariant). The flow $(\Phi_t^{H_k^{(1)}})^* z_1 = z_1 - t z_2^k$ under (U4) gives

$$\widetilde{J}(z_1) - t\widetilde{J}(z_2^k) = \overline{\mathrm{Tr}\,\phi_1} - t\overline{\mathrm{Tr}\,\phi_2^k},$$

hence $\widetilde{J}(z_2^k) = \overline{\mathrm{Tr}\,\phi_2^k}$. The reverse flow gives $\widetilde{J}(z_1^k)$; composite flows give $\widetilde{J}(z_1^a z_2^b) = \overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}$ at every bidegree.

(U2) reduction. On $\mathbf{S}(\mathrm{Cyc}(F)_{\text{nonempty}})$ (Proposition 3.22), (U5) gives $\widetilde{J}(z_i) = \overline{\mathrm{Tr}\,\phi_i}$; matrix substitution sends $\overline{\mathrm{Tr}\ P(\phi)}$ to $\overline{\mathrm{Tr}\ P(\Phi(\phi))}$. The t^j-coefficient of $(z_2 + t z_1)^k$ under (U4), matched against $\overline{\mathrm{Tr}(\phi_2 + t\phi_1)^k}$, gives

$$\widetilde{J}(z_1^j z_2^{k-j}) = \frac{1}{\binom{k}{j}} \sum_{w \in \mathsf{W}_{j,k-j}} \overline{\mathrm{Tr}\ w(\phi_1, \phi_2)}.$$

At bidegree $(2, 2)$: $\frac{2}{3}\overline{\mathrm{Tr}(\phi_1\phi_1\phi_2\phi_2)} + \frac{1}{3}\overline{\mathrm{Tr}(\phi_1\phi_2\phi_1\phi_2)}$. On commuting matrices both cyclic classes equal $\overline{\mathrm{Tr}(\phi_1^2\phi_2^2)} = J(z_1^2 z_2^2)$; Lemma 3.30 extends this to every bidegree.

(U3) reduction. On $A_\partial \oplus \mathbb{C} \cdot \overline{\mathrm{Tr}(1)}$, write $\widetilde{J}(f) = \widetilde{J}_{\mathrm{red}}(f) + L(f)\,\overline{\mathrm{Tr}(1)}$ with $L : \mathfrak{h} \to \mathbb{C}$ linear. Matrix substitution fixes $\overline{\mathrm{Tr}(1)}$, so (U4) gives $L(\Phi^* z_i) = L(z_i)$; the Hamiltonian flows give $L(z_i^k) = 0$ for every $k \geq 1$, and composite flows give $L \equiv 0$. □

COROLLARY 5.45 *(CE/PV generator identity, formal symplectic polydisk).* Let $\mathfrak{h} = \widehat{A}/\mathbb{C} \cdot 1 = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ be the reduced Hamiltonian Lie algebra of the formal symplectic polydisk on $\mathbb{C}^2$, with bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ modulo constants. Choose the Taylor basis $H_{a,b} = z_1^a z_2^b$ for $a, b \geq 0$, $a + b > 0$. Structure constants:

$$[H_{a,b}, H_{c,d}] = (ad - bc)\, H_{a+c-1,b+d-1},$$

with the convention that $H_{m,n} = 0$ if $\min(m, n) < 0$ or $m + n = 0$.

Set $A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$, with linear coordinates $O_{a,b}$ on $\mathfrak{h}^\vee$ and Kirillov–Poisson bracket $\{O_{a,b}, O_{c,d}\}_\partial = (ad - bc)O_{a+c-1,b+d-1}$. The generator calculation underlying 4.20 gives the cochain-level coordinate CE/PV identity, read in the pro coordinate-window sense,

$$\Phi_{\mathrm{coord}} : C^\bullet_{\mathrm{CE,coord}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]) \xrightarrow{\sim} \mathrm{PV}_{\mathrm{coord}}(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})), \qquad c^{a,b} \mapsto \theta^{a,b}, \quad u_{a,b} \mapsto O_{a,b}.$$

Its moment-map specialisation is the boundary evaluation map $J(f) = O_f = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ of 5.191 after identifying A_∂ with the stable scalar-reduced Hamiltonian trace algebra of 3.40. The P_0 and factorization central-operation interpretation is imposed only after the coordinate-window or mixed/weighted continuity hypotheses in the theorem-data table; the strict product coordinate completion has a separate global P_0-algebra obstruction.

Proof. The structure constants are an instance of 2.7 read on monomials. The CE/PV differential identity is checked directly on the monomial generators $c^{a,b}$, $u_{a,b}$ and then extended by the completed Leibniz rule; lower-central Tate-pronilpotence of the full Hamiltonian algebra is a bar-cobar hypothesis. The moment-map specialisation follows from 5.40 together with 5.191. □

Example 5.46 *(Low-degree monomial verification).* Direct verification of Φ_{coord} on small monomials of the formal symplectic polydisk.

- *Degree-1 generators.* Take $H_{1,0} = z_1$, $H_{0,1} = z_2$. The bracket $[H_{1,0}, H_{0,1}] = 1 \cdot 1 - 0 \cdot 0 = 1$ lifts to the constant $H_{0,0}$, which is set to zero in $\mathfrak{h}$. Hence $C^{0,0}_{(1,0),(0,1)} = 0$ in $\mathfrak{h}$. Consequently the generator $u_{0,0}$ is omitted, and the $u_{0,0}$ degree of freedom is omitted, consistent with constants being removed.
- *Degree-2 calculation.* Take $K = (1, 1)$. The differential of u_K uses outgoing brackets with H_K:

  $$d_{\mathrm{CE}} u_{1,1} = \sum_{c,d} (d - c)\, u_{c,d} c^{c,d}.$$

  Hence the first low-degree terms are

  $$d_{\mathrm{CE}} u_{1,1} = 2u_{0,2}c^{0,2} - 2u_{2,0}c^{2,0} - u_{1,0}c^{1,0} + u_{0,1}c^{0,1} + \cdots .$$

  Each term $C^L_{(1,1),J}\, u_L\, c^J$ arises from $\{H_{1,1}, H_J\} = \sum_L C^L_{(1,1),J}\, H_L$. For example, $\{H_{1,1}, H_{1,0}\} = \partial_{z_1}(z_1 z_2) \cdot 0 - \partial_{z_2}(z_1 z_2) \cdot 1 = -z_1 = -H_{1,0}$, so $C^{1,0}_{(1,1),(1,0)} = -1$, contributing $-u_{1,0}\, c^{1,0}$ to $d_{\mathrm{CE}} u_{1,1}$. On the open/PV side,

  $$d_\pi^{\mathrm{coord}} O_{1,1} = 2O_{0,2}\theta^{0,2} - 2O_{2,0}\theta^{2,0} - O_{1,0}\theta^{1,0} + O_{0,1}\theta^{0,1} + \cdots ,$$

  the same expression with $u \to O$, $c \to \theta$. Hence $\Phi_{\mathrm{coord}}(d_{\mathrm{CE}} u_{1,1}) = d_\pi^{\mathrm{coord}} \Phi_{\mathrm{coord}}(u_{1,1})$. An independent exact arithmetic calculation verifies the same structure constants on every monomial pair with exponents up to 6 via the first-order Moyal coefficient $kn - lm$ at $\hbar^1$, which equals C^K_{IJ} for these indices.
- *Higher-order word.* For $K = (1, 1)$, $J = (1, 0)$, $J' = (0, 1)$, the cochain element $u_{1,1}c^{1,0}c^{0,1}$ has degree $0 + 1 + 1 = 2$. Its CE differential, by graded Leibniz, is

  $$d_{\mathrm{CE}}\big(u_{1,1}c^{1,0}c^{0,1}\big) = (d_{\mathrm{CE}} u_{1,1})c^{1,0}c^{0,1} - u_{1,1}(d_{\mathrm{CE}} c^{1,0})c^{0,1} + u_{1,1}c^{1,0}(d_{\mathrm{CE}} c^{0,1}),$$

  with each summand applying the generator formulas above. The same word on the PV side is $O_{1,1}\theta^{1,0}\theta^{0,1}$, and d_π^{coord} applied with the same Leibniz convention produces the corresponding triple sum with $u \to O$ and $c \to \theta$. Term-by-term equality follows from the generator identities; the full word equality is the algebra-homomorphism Leibniz argument from 4.20.

Construction 5.47 *(Compact current PBW/Rees source).* Let $\mathcal{P}$ denote the principal-part module

$$\mathcal{P} = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\,\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

identified with $\mathfrak{h}^\vee_{\mathrm{cont}}$ by the residue pairing of Proposition 5.176. Put

$$\mathfrak{g}_{\mathrm{pp}} = \mathfrak{h} \ltimes \mathcal{P}[1],$$

with zero differential and bracket

$$[f, g] = \{f, g\}, \qquad [f, \rho[1]] = (f \cdot \rho)[1], \qquad [\rho[1], \sigma[1]] = 0,$$

where $f \cdot \rho = \pi_{\mathrm{pp}}\{f, \rho\}$. On monomials this action is

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with negative-index terms zero.

For an interval $I \subset \mathbb{R}$, define the compactly supported current Lie algebra

$$\mathfrak{J}_{\mathfrak{g}}(I) = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{g}_{\mathrm{pp}}, \qquad [\alpha \otimes x, \beta \otimes y] = \alpha\beta \otimes [x, y].$$

Its PBW/Rees algebra is

$$U_\hbar\big(\mathfrak{J}_{\mathfrak{g}}(I)\big) = \widehat{T}\big(\mathfrak{J}_{\mathfrak{g}}(I)\big)[[\hbar]] / \langle xy - (-1)^{|x||y|} yx - \hbar[x, y]\rangle.$$

The PBW filtration is the tensor-word filtration. Its special fibre is

$$U_\hbar\big(\mathfrak{J}_{\mathfrak{g}}(I)\big)/(\hbar) \cong \widehat{\mathbf{S}}\big(\mathfrak{J}_{\mathfrak{g}}(I)\big),$$

equipped with the semiclassical P_0 bracket

$$\{\bar{X}, \bar{Y}\}_{P_0} = \big(\hbar^{-1}(XY - (-1)^{|X||Y|}YX)\big) \bmod \hbar.$$

This is the PBW/Rees source for the Hamiltonian and principal-part comparison. A closed CE cochain algebra map requires a bar-cobar twisting construction or a completed pairing.

Problem 5.48 *(PBW-source P_0-centre action criterion and CE comparison datum).* Construct a filtered factorization-center action of the compactly supported current PBW/Rees source $U_\hbar(\mathfrak{J}_{\mathfrak{g}})$ of Construction 5.47. With the cotangent generators included, the target is the enlarged principal-part boundary object of Theorem 5.186. In the classical special fibre the action must send

$$f \longmapsto B_f = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}, \qquad \rho_{a,b}[1] \longmapsto \Theta_{a,b},$$

and satisfy the stalkwise bracket identities

$$\{B_f, B_g\}_{\mathrm{open}} = B_{\{f,g\}}, \qquad \{B_f, \Theta_\rho\}_{\mathrm{open}} = \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\}_{\mathrm{open}} = 0.$$

This special-fibre statement is the natural P_0 theorem. A strict action of the full filtered Rees algebra on the same commutative reduced target requires additional quantized boundary data: in the source

$$XY - (-1)^{|X||Y|}YX = \hbar[X, Y],$$

while multiplication operators on a commutative classical target commute. Thus the Rees-level theorem requires either a deformed boundary target or an explicit homotopical quantization of the central operations. Consequently the compact current PBW/Rees source gives a P_0-central-operation action only after passing to the classical special fibre, or after replacing the boundary target by a genuinely deformed/homotopical central-operation object. The primitive one-ψ projection is an indecomposable-label projection; the filtered Rees action datum is additional. The boundary data must include differential compatibility, factorization centrality homotopies, coherent P_0 bracket homotopies, and compact-support naturality for interval inclusions and disjoint unions before the de Rham contraction.

Separately compare this PBW/Koszul-dual source with the closed CE observable algebra by an explicit twisting or pairing construction. This comparison is the extra datum required for a cochain-level map from closed CE observables to extend a PBW/Koszul-dual boundary action.

Problem 5.49 *(Unreduced Tate-coefficient cotangent lift datum).* Construct a Tate or weighted-coefficient extension of the BV heat-kernel calculus on $\mathbb{R}^2 \times \mathbb{C}^2$. The native formal-coordinate holomorphic singular product has already been computed: its diagonal coefficient module, local-cohomology expansion, and current singular product are Theorem 1.17. The required extension lifts that binary singular product, together with the trace map J, through the unreduced cotangent sector, Costello heat-kernel coefficients, and the quantum master equation tower. Thus the coefficient category must satisfy:

(i) the formal Casimir $C_{\mathfrak{h}} = \sum_{d\geq 1} \sum_i e_{d,i} \otimes e^i_d$ converges as an element of $\widehat{\mathfrak{h}} \,\widehat{\otimes}\, \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ this is proved by 5.71(a) via the weighted Tate completion of 5.55;

(ii) the Costello propagator $P_{\epsilon,L}$ extends to that coefficient category this is proved by 5.71(b);

(iii) the primitive indecomposable projection of the principal-part current sector

$$\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \,\widehat{\otimes}\, \widehat{\mathbf{S}}\big(\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I)\big)$$

admits a chain-level primitive projection π_{prim} with verified Q-compatibility and factorization compatibility on the indecomposable quotient the primitive projection is constructed in 5.133; the cotangent centrality homotopies remain part of item (4);

(iv) the reduced principal-part defect-current theorem of 5.6 lifts to the unreduced factorization-algebra model and the degree-zero Moyal/Weyl theorem extends to the cotangent (one-antifield) sector at all orders in $\hbar$ after the primitive projection of 5.140; unreduced cotangent centrality and the one-antifield quantum lift require vanishing of the QME obstruction class of 5.73; the truncated nilpotent sector $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ closed classically by 5.102 and obstructed quantum-mechanically by 5.103.

Equivalently, lift the cochain-level 4.20 and the Hamiltonian-sector factorization-algebra 4.11 to a cochain-level morphism of factorization algebras

$$\Phi_{\mathrm{cl}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow Z^{P_0}_{\mathrm{fact}}\big(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}\big)$$

on the unreduced (cotangent-included) sector, with all P_0-center homotopies, factorization centrality homotopies, the stable connected quotient, and a boundary realization of the shifted cotangent generators. Any global version must also pass through the local-Hamiltonian descent and period obstruction of 1.31; a stalkwise formal disk map gives the local stalk, while Weiss/Ran descent for factorization centers requires the descent datum.

Weighted Tate completion of the coefficient pair

The Hamiltonian Lie algebra on the formal symplectic plane $\mathbb{C}^2$,

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1 = \prod_{d\geq 1} H_d, \qquad H_d = \mathrm{Span}\{z_1^a z_2^b : a + b = d\},$$

carries the Hamiltonian bracket $\{H_p, H_q\} \subset H_{p+q-2}$ and the formal Casimir

$$C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_i e_{d,i} \otimes e_d^i$$

in any homogeneous basis $\{e_{d,i}\}$ of H_d with dual basis $\{e_d^i\}$ fixed by $\langle e_d^j, e_{d,i}\rangle = \delta_i^j$. Costello's heat-kernel BV calculus on the bulk-plus-defect field complex

$$\mathcal{E} = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\mathfrak{h}[1] \oplus \mathfrak{h}^\vee_{\mathrm{cont}}[2]\big)$$

factors the propagator as

$$P_{\epsilon,L} = P_{\epsilon,L}^{\mathrm{base}}\widehat{\otimes} C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1]},$$

the bulk piece being the standard heat-kernel inverse and the coefficient piece $C_{\mathfrak{h}}$ extended to the cotangent direction; the propagator exists in $\mathcal{E}\widehat{\otimes}\mathcal{E}$ exactly when $C_{\mathfrak{h}}$ exists as a coefficient kernel. In the strict product/direct-sum Tate pair $(\mathfrak{h}_\Pi, D_\oplus) = \big(\prod_{d\geq 1} H_d, \bigoplus_{d\geq 1} H_d^\vee\big)$ the second factor has support in finitely many degrees, while $C_{\mathfrak{h}}$ has nonzero $H_d^\vee$-component in every degree, so $C_{\mathfrak{h}} \notin \mathfrak{h}_\Pi\widehat{\otimes}D_\oplus$ (Lemma 5.212; Theorem 5.94 gives the finite-window form). The coefficient category itself must change: the second factor must be a product, dualized through an intermediate Mittag–Leffler inverse-limit weight system. The smallest complete filtered subspace of $\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ containing $\{O_f, \theta_f\}_{f\in\bar{A}}$, closed under d_π, the Schouten bracket $[-,-]_{\mathrm{Sch}}$, and the pointwise product, and complete in such a weighted topology is the weighted Tate envelope B_w of Theorem 5.58.

Strict Tate pair: Casimir-kernel obstruction

Remark 5.50 *(Strict Tate continuous duality).* In the strict Tate pair

$$\mathfrak{h}_\Pi = \prod_{d\geq 1} H_d, \qquad D_\oplus = \bigoplus_{d\geq 1} H_d^\vee, \tag{21}$$

Lemma 5.166 (1) realizes $D_\oplus$ as the continuous dual of $\mathfrak{h}_\Pi$ as Tate vector spaces: the continuous dual of a Tate product is the strict direct sum of duals, and *vice versa*. The second tensor factor of the formal Casimir $C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_i e_{d,i} \otimes e_d^i$ has nonzero $H_d^\vee$-component in every d, hence lies in $\prod_d H_d^\vee$. The direct sum $\bigoplus_d H_d^\vee = D_\oplus$ contains only finitely supported dual-degree expansions.

Remark 5.51 *(Costello's coefficient requirements).* Costello's heat-kernel BV calculus [12, Ch. 5] writes the BV propagator on a free elliptic complex $(\mathcal{E}, Q, \langle\ ,\ \rangle)$ as the kernel of $\int_\epsilon^L Q^{\mathrm{GF}} e^{-tD}\,dt$ in $\mathcal{E}\widehat{\otimes}\mathcal{E}$. When the field complex factors as

$$\mathcal{E} = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\mathfrak{h}[1] \oplus \mathfrak{h}^\vee_{\mathrm{cont}}[2]\big),$$

with the Tate coefficient pair acting passively on the spacetime, the propagator factors as

$$P_{\epsilon,L} = P_{\epsilon,L}^{\mathrm{base}}\widehat{\otimes} C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1]}. \tag{22}$$

The spacetime piece $P^{\text{base}}_{\epsilon,L}$ on $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$ is the standard bulk BV inverse, continuous as an operator by Remark 5.211. The coefficient piece is the BV inverse pairing on $\mathfrak{h} \oplus \mathfrak{h}^\vee_{\text{cont}}[1]$, i.e. the Casimir $C_{\mathfrak{h}}$ together with its dual coevaluation. The full kernel therefore exists in $\mathcal{E}\widehat{\otimes}\mathcal{E}$ exactly when $C_{\mathfrak{h}}$ exists as a coefficient kernel.

Costello's heat-kernel BV requires three properties of the coefficient module:

(C1) a distributional inverse pairing, i.e. a coefficient coevaluation $C_{\mathfrak{h}} \in \mathfrak{h}\widehat{\otimes}\mathfrak{h}^\vee_{\text{cont}}$ realizing the bilinear form with kernel zero in the cohomology of the heat operator;

(C2) regularity in the cutoff parameters ϵ and L, in the sense that the heat-kernel propagator $P_{\epsilon,L}$ is a continuous family in the coefficient topology and converges as $L \to \infty$, $\epsilon \to 0^+$ to a distributional kernel after counterterm subtraction;

(C3) compatibility with the Costello RG flow $W(P_{\epsilon,L}, I)$: the weight or topology defining the coefficient module is preserved by the renormalization flow on effective interactions induced by varying (ϵ, L), in the precise sense that the source of the flow lies in the same completed coefficient category as its target.

The strict Tate pair (21) has the obstruction to (C1). Requirement (C3) is the residual locality/QME problem attached to coefficient enlargement: Problem 5.73.

Degree weights and the canonical completion B_w

A degree weight on $\mathfrak{h}$ selects a coefficient category closed under the Schouten differential d_π, the Schouten bracket $[-,-]_{\text{Sch}}$, and the pointwise product, while hosting the Casimir kernel: bracket-tame for the Schouten bracket, windowwise coadjoint-continuous for the cotangent action, and Mittag–Leffler-coevaluating for the heat-kernel propagator.

Definition 5.52 *(Polynomial Hamiltonian degree weight).* A *degree weight* is a function $w\colon \mathbb{Z}_{\geq 1} \to \mathbb{R}_{>0}$ satisfying:

(W1) *Bracket-tame:* there exists a constant $C_w \geq 0$ such that $w(p+q-2) \leq C_w\, w(p)w(q)$ for every $p, q \geq 2$;

(W2) *Windowwise coadjoint-continuity:* for every Hamiltonian degree bound R and every resulting Tate window w_0, the principal-part coadjoint shifts $\rho_{a,b} \mapsto \rho_{a-p+1,b-q+1}$ with $p+q \leq R$ and $a+b-p-q+2 \leq w_0$ involve only the finite source window $a+b \leq w_0 + R - 2$, and the finitely many corresponding weight ratios have a maximum;

(W3) *Mittag-Leffler coevaluation:* for each w_0 the finite-window Casimir $\sum_{d\leq w_0}\sum_i w(d)e_{d,i} \otimes w(d)^{-1}e^i_d$ is compatible under the surjective truncation maps $w_0' \to w_0$. Equivalently, the coefficient kernel is a strict Mittag-Leffler inverse system of finite-dimensional kernels.

LEMMA 5.53 *(Existence of weights).* Every exponential weight $w(d) = q^d$ with $q > 1$ satisfies (W1)–(W3). The proof uses only finite-window maxima for the coadjoint action; the unbounded principal-part degree prevents a uniform bound.

Proof. For $w(d) = q^d$, (W1) reads $q^{p+q-2} \leq C_w q^{p+q}$, i.e. $C_w = q^{-2}$ suffices. For (W2), fix R and w_0. If $p+q \leq R$ and the resulting degree is $a+b-p-q+2 \leq w_0$, then $a+b \leq w_0 + R - 2$. Hence the set of possible triples $(p, q, a+b)$ is finite, and the corresponding ratios of powers of q admit a finite maximum; the bound depends on the finite-window restriction. Condition (W3) is the strict Mittag-Leffler compatibility of finite-window Casimirs under truncation. The transition maps are surjective, hence the system is Mittag-Leffler. □

Remark 5.54 *(Choice of exponential weight).* The exponential weight $w(d) = q^d$ is the working choice, with $q > 1$ formally an element of $\mathbb{R}_{>0}$ and concretely set to $q = 2$ for the proofs below. The choice is a grading weight on the coefficient module. It acts on coefficients, with the formal polydisk fixed. Thus the Poisson bracket and the counterterm representatives are compared between two weights only after transport by the diagonal map $R_{w,w'}$ below. In coordinates, the bracket constants in the basis $w(d)e_{d,i}$ are conjugated by the finite ratios $w(p)w(q)/w(p+q-2)$, and condition (W1) is exactly the boundedness condition needed for this transported bracket on each finite graph. The weight is therefore a controlled coefficient regulator; rescaling the variables z_1, z_2 is a different operation on the original Hamiltonian bracket.

Definition 5.55 *(Weighted Tate coefficient pair).* Given a degree weight w as in Definition 5.52, write $H_d^{(w)}$ for the rescaled copy of the finite-dimensional degree piece H_d with basis $w(d)e_{d,i}$, and $(H_d^\vee)^{(w^{-1})}$ for the rescaled dual basis $w(d)^{-1}e_d^i$. Define

$$\mathfrak{h}_w = \varprojlim_{w_0 \in \mathbb{Z}_{\geq 1}} \prod_{d=1}^{w_0} H_d^{(w)}$$

with the Tate filtration $F^{w_0}\mathfrak{h}_w = \ker(\mathfrak{h}_w \to \prod_{d \leq w_0} H_d^{(w)})$, and define the *weighted coefficient dual*

$$\mathfrak{h}_{\mathrm{cont}}^{\vee,w} = \varprojlim_{w_0 \in \mathbb{Z}_{\geq 1}} \prod_{d=1}^{w_0} (H_d^\vee)^{(w^{-1})}.$$

The pairing is fixed degreewise by $\langle w(d)^{-1}e_d^i, w(d)e_{d,j}\rangle = \delta_j^i$. The weighted Tate coefficient pair is $(\mathfrak{h}_w, \mathfrak{h}_{\mathrm{cont}}^{\vee,w})$. The second factor is a product coefficient module replacing the strict continuous dual of the first product Tate space.

Remark 5.56 *(Topological content).* Both $\mathfrak{h}_w$ and $\mathfrak{h}_{\mathrm{cont}}^{\vee,w}$ are Mittag-Leffler inverse limits of finite-dimensional Tate windows, and both are products in the degree direction. The construction replaces the direct-sum dual by a symmetric Mittag-Leffler product coefficient module. This is an additional coefficient category for the BV kernel; the ordinary continuous dual of $\mathfrak{h}_w$ remains the strict Tate dual.

Definition 5.57 *(The weighted-Tate envelope B_w).* Inside the bracket-admissible polyvector subspace $B \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ of Definition 4.18, the weighted Tate envelope is

$$B_w := \overline{\left\langle \{O_f, \theta_f\}_{f \in \bar{A}} \right\rangle_{d_\pi, [-,-]_{\mathrm{Sch}}, \cdot}}^{\mathfrak{h}_w \widehat{\otimes} \mathfrak{h}_{\mathrm{cont}}^{\vee,w}},$$

the smallest complete filtered subspace of $\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ containing the brane-side generators $\{O_f, \theta_f\}_{f \in \bar{A}}$, closed under the Schouten differential d_π, the Schouten bracket $[-,-]_{\mathrm{Sch}}$, and the pointwise product, and complete in the weighted Tate topology of $\mathfrak{h}_w \widehat{\otimes} \mathfrak{h}_{\mathrm{cont}}^{\vee,w}$. The polyvector grading on B_w is the Schouten degree; the coefficient grading is the polynomial degree carried by the weight w.

THEOREM 5.58 *(Canonical weighted-Tate envelope).* Let w be a degree weight as in Definition 5.52 and let B_w be the envelope of Definition 5.57. Then B_w is the minimal complete filtered subspace of $\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ in which Costello's heat-kernel BV calculus and the brane-side Schouten algebra on $\{O_f, \theta_f\}_{f \in \bar{A}}$ both close.

(a) *Closure under the Schouten algebra.* B_w is closed under multiplication, Schouten differential, and Schouten bracket:

$$B_w \cdot B_w \subset B_w, \qquad d_\pi B_w \subset B_w, \qquad [B_w, B_w]_{\mathrm{Sch}} \subset B_w.$$

(b) *Casimir convergence.* The diagonal-rescaled Casimir

$$C_{\mathfrak{h},w} = \sum_{d\geq 1}\sum_{i}\big(w(d)e_{d,i}\big)\otimes\big(w(d)^{-1}e^{i}_{d}\big)$$

converges in the weighted continuous tensor product $\mathfrak{h}_w\widehat{\otimes}\mathfrak{h}^{\vee,w}_{\mathrm{cont}}$. Under the P_0-algebra map $e_{d,i}\mapsto O_{e_{d,i}}$, $e^{i}_{d}\mapsto\theta^{e^{i}_{d}}$ of Theorem 4.20, the kernel lies in $B_w\widehat{\otimes}B_w$ and realizes Costello requirement (C1) as the BV inverse pairing on the cotangent extension.

(c) *Heat-kernel propagator extension.* The Costello heat-kernel propagator

$$P^{w}_{\epsilon,L} = P^{\mathrm{base}}_{\epsilon,L}\widehat{\otimes}C_{\mathfrak{h}\oplus\mathfrak{h}^{\vee}[1],w}$$

extends from the strict Tate field complex to the weighted complex $\mathcal{E}_w$ as a continuous family for $0<\epsilon\leq L<\infty$, satisfying the BV identity $[Q,P^{w}_{\epsilon,L}] = K^{w}_{\epsilon}-K^{w}_{L}$ and Costello regularity (C2): continuity in (ϵ,L) in the weighted Tate topology, and existence of the $L\to\infty$, $\epsilon\to 0^{+}$ limit as a distributional kernel after the standard bulk counterterm subtraction. The coefficient piece $C_{\mathfrak{h},w}$ is counterterm-free; it is the fixed Mittag-Leffler kernel of (b).

(d) *Graphwise RG stability.* For every fixed Costello graph Γ and every $0<\epsilon\leq L<\infty$, the contribution $W_{\Gamma}(P^{w}_{\epsilon,L},I)$ of any vertexwise polynomial-degree- bounded interaction $I\in\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ stays inside B_w. Each fixed graph is bounded by a finite constant times the corresponding spacetime graph weight, with constants given by (W1) and (W2).

(e) *Cochain-level CE/PV extension.* The reduced central-operation cochain Φ of Theorem 4.20 extends to the weighted cotangent extension $\mathfrak{g}_w = \mathfrak{h}_w\ltimes\mathfrak{h}^{\vee,w}_{\mathrm{cont}}[1]$ as a continuous cochain map landing in B_w:

$$\Phi^{w}\colon C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_w)\longrightarrow B_w\subset \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}_w)\big).$$

Conversely, any complete filtered subspace $B'\subset\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ satisfying (a), containing $\{O_f,\theta_f\}_{f\in\bar{A}}$, and containing $C_{\mathfrak{h},w}\in B'\widehat{\otimes}B'$ as a continuous element of the weighted Tate topology, contains B_w.

Proof. (a): with the linear Kirillov Poisson tensor $\pi = \frac{1}{2}C^{k}_{ij}O_k\theta^{i}\theta^{j}$ of Theorem 4.20, the Schouten differential $d_\pi = [\pi,-]_S$ acts on generators by

$$d_\pi\theta^{k} = -\tfrac{1}{2}C^{k}_{ij}\theta^{i}\theta^{j},\qquad d_\pi O_k = C^{\ell}_{kj}O_\ell\theta^{j},$$

with C^{k}_{ij} the Hamiltonian structure constants on $\mathfrak{h}$. The bracket-tame estimate (W1) bounds, on each finite Tate window, the polynomial-degree shift produced by a single d_π or a single Schouten bracket by a finite constant; the windowwise coadjoint-continuity (W2) bounds the cotangent-leg shift. Multiplication is closed by definition of the envelope. The closure $\overline{(\cdot)}$ in Definition 5.57 adjoins all d_π-, $[-,-]_{\mathrm{Sch}}$-, and product-completions, and that closure is itself stable under the same operations because each is continuous in the weighted Tate topology by (W1)–(W2).

(b): this is Proposition 5.62 below.

(c): this is Proposition 5.64.

(d): this is Proposition 5.69.

(e): the cochain identity $\Phi d_{\mathrm{CE}} = d_\pi\Phi$ is degreewise algebraic on $\mathfrak{h}\oplus\mathfrak{h}^{\vee}[1]$ and is preserved by the diagonal weight rescaling. The image of generators $c_f,u_f\in C^{\bullet}_{\mathrm{CE}}(\mathfrak{g}_w)$ is, by the P_0-algebra map convention, $\theta_f, O_f \in B_w$; closure of B_w under d_π, $[-,-]_{\mathrm{Sch}}$, and product (item (a)) makes Φ^w land in B_w. The continuity for the weighted Tate topology is the same windowwise estimate as in (d).

Minimality: any B' satisfying (a) and containing $\{O_f, \theta_f\}$ contains all d_π-, bracket-, and product-iterates of those generators, hence the algebraic envelope. Containing $C_{\mathfrak{h},w} \in B' \widehat{\otimes} B'$ as a continuous element forces B' to be complete in the weighted Tate topology, hence to contain the closure B_w. □

Definition 5.59 *(Category of admissible weighted-Tate completions).* Fix a degree weight w as in Definition 5.52. The category Adm_w has objects: complete filtered subspaces $B' \subset \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$ satisfying

(A1) *(generation)* B' contains the brane-side generators $\{O_f, \theta_f\}_{f \in \bar{A}}$;

(A2) *(Schouten closure)* $B' \cdot B' \subset B'$, $d_\pi B' \subset B'$, and $[B', B']_{\mathrm{Sch}} \subset B'$;

(A3) *(Casimir convergence)* the weighted Casimir $C_{\mathfrak{h},w}$ of (23) converges in $B' \widehat{\otimes} B'$ in the weighted Tate topology;

(A4) *(heat-kernel propagator extension)* the Costello heat-kernel propagator $P^w_{\epsilon,L} = P^{\mathrm{base}}_{\epsilon,L} \widehat{\otimes} C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$ extends to a continuous family for $0 < \epsilon \le L < \infty$ on B' satisfying $[Q, P^w_{\epsilon,L}] = K^w_\epsilon - K^w_L$;

(A5) *(graphwise RG boundedness)* for every Costello graph Γ and every vertexwise polynomial-degree bounded interaction $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$, the contribution $W_\Gamma(P^w_{\epsilon,L}, I)$ stays in B' with constants given by (W1) and (W2).

Morphisms $B' \hookrightarrow B''$ are continuous filtered inclusions containing the generators on the nose and intertwining d_π, $[-,-]_{\mathrm{Sch}}$, the pointwise product, the weighted Casimir, and the heat-kernel propagator.

COROLLARY 5.60 *(B_w is initial in Adm_w).* B_w of Definition 5.57 is the initial object of Adm_w: every $B' \in \mathsf{Adm}_w$ admits a unique morphism $\iota_{B'} : B_w \hookrightarrow B'$.

Proof. By Theorem 5.58, $B_w \subset B'$ for every $B' \in \mathsf{Adm}_w$, and the inclusion is an Adm_w-morphism. A morphism $B_w \to B'$ is determined on $\{O_f, \theta_f\}_{f \in \bar{A}}$, hence on the Schouten-bracket-product envelope, hence by Tate continuity on B_w. □

Remark 5.61. Adm_w depends on w; its admissibility conditions are intrinsic to $(\mathfrak{h}_w, \mathfrak{h}^{\vee,w}_{\mathrm{cont}}, P^w_{\epsilon,L})$.

PROPOSITION 5.62 *(Casimir convergence).* In the weighted Tate coefficient pair $(\mathfrak{h}_w, \mathfrak{h}^{\vee,w}_{\mathrm{cont}})$ of Definition 5.55, the formal Casimir

$$C_{\mathfrak{h},w} = \sum_{d \ge 1} \sum_i \left(w(d) e_{d,i}\right) \otimes \left(w(d)^{-1} e^i_d\right) \tag{23}$$

converges as an element of $\mathfrak{h}_w \widehat{\otimes} \mathfrak{h}^{\vee,w}_{\mathrm{cont}}$. The pairing of $C_{\mathfrak{h},w}$ with any test functional in the dual is the identity on the corresponding finite-dimensional Tate window, so $C_{\mathfrak{h},w}$ is the BV coefficient inverse pairing in the weighted category.

Proof. The displayed Casimir, restricted to the Tate window $d \le w_0$, is the finite sum $\sum_{d \le w_0} \sum_i (w(d) e_{d,i}) \otimes (w(d)^{-1} e^i_d)$, an element of $\prod_{d \le w_0} w(d) H_d \otimes \prod_{d \le w_0} w(d)^{-1} H^\vee_d$. This is a tensor in a finite-dimensional vector space because each window is finite-dimensional. The tower of finite-window Casimirs is compatible: enlarging w_0 extends the sum by the terms in degrees $w_0 + 1, \dots, w_0'$. The inverse limit in **TateVec** exists by Definition 5.55 and (W3) of Definition 5.52.

Pairing the Casimir with a test functional $\phi \in \mathfrak{h}^{\vee,w}_{\mathrm{cont}}$, supported in finitely many Tate windows, gives a finite sum of identity pairings, so the result is the canonical evaluation. Hence $C_{\mathfrak{h},w}$ realizes the inverse pairing on the weighted Tate pair, satisfying Costello requirement (C1). □

Remark 5.63 *(Independence of basis).* Definition (23) is written in a basis $\{e_{d,i}\}$ of H_d, and the tensor is basis-independent. The dual basis $\{e^i_d\}$ is fixed by $\langle e_{d,i}, e^j_d\rangle = \delta^j_i$. A change of basis $e_{d,i} \mapsto \sum_j O^j_i e_{d,j}$ with $O \in GL(H_d)$ shifts the dual by the inverse transpose, yielding the same Casimir. Thus $C_{\mathfrak{h},w} = \sum_d \mathrm{id}_{H_d \otimes H_d^\vee}$ is the basis-independent identity element of $\bigoplus_d H_d \otimes H_d^\vee$ extended to the weighted product.

Heat-kernel propagator on the weighted pair

PROPOSITION 5.64 *(Heat-kernel propagator extension to the weighted pair).* Let w be a degree weight as in Definition 5.52. The Costello heat-kernel propagator $P^{\mathrm{op}}_{\epsilon,L}$ of Remark 5.211 extends from the strict Tate field complex to the weighted field complex

$$\mathcal{E}_w = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\mathfrak{h}_w[1] \oplus \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[2]\big), \tag{24}$$

to give a continuous family of BV propagators $P^w_{\epsilon,L} \in \mathcal{E}_w\widehat{\otimes}\mathcal{E}_w$ for $0 < \epsilon \le L < \infty$, satisfying:

(1) the BV identity $[Q, P^w_{\epsilon,L}] = K^w_\epsilon - K^w_L$, where $K^w_t = K^{\mathrm{base}}_t\widehat{\otimes}C_{\mathfrak{h},w}$ is the weighted heat kernel of the BV inverse pairing;
(2) Costello regularity (C2): the family is continuous in (ϵ, L) in the weighted Tate topology;
(3) the limit $L \to \infty$, $\epsilon \to 0^+$ exists as a distributional kernel after the standard spacetime counterterm subtraction in Costello's heat-kernel renormalization-group construction [12, Chs. 5–7]; the coefficient direction leaves the spacetime singular support unchanged. Locality and QME counterterm compatibility for the mixed brane-defect theory remain the content of Problem 5.73.

Proof. By the factorization (22), the propagator splits as a tensor product of the spacetime kernel $P^{\mathrm{base}}_{\epsilon,L}$ on $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$ and the coefficient Casimir $C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$ on the weighted Tate pair. The spacetime piece is unchanged by the weight on the coefficient direction; it is the Costello heat-kernel inverse on the bulk free elliptic complex $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$, constructed in [12, Ch. 5]. The BV identity for $P^{\mathrm{base}}_{\epsilon,L}$ is standard: with $Q^{\mathrm{base}} = d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}$, gauge-fixing $Q^{\mathrm{GF}} = d^*_{\mathbb{R}^2} + \bar\partial^*_{\mathbb{C}^2}$, and the associated Laplacian $\Delta = [Q^{\mathrm{base}}, Q^{\mathrm{GF}}]$, one has

$$[Q^{\mathrm{base}}, P^{\mathrm{base}}_{\epsilon,L}] = \int_\epsilon^L [Q^{\mathrm{base}}, Q^{\mathrm{GF}}e^{-t\Delta}]\, dt = \int_\epsilon^L \partial_t(-e^{-t\Delta})\, dt = e^{-\epsilon\Delta} - e^{-L\Delta} = K^{\mathrm{base}}_\epsilon - K^{\mathrm{base}}_L.$$

The coefficient piece is the Casimir of Proposition 5.62, which is annihilated by the trivial differential on $\mathfrak{h}_w[1] \oplus \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[2]$. Therefore $[Q, P^w_{\epsilon,L}] = K^w_\epsilon - K^w_L$, and the family is continuous in (ϵ, L) because $C_{\mathfrak{h},w}$ is a fixed element of the weighted pair and the bulk family is continuous. Item (3), at the level asserted here, is the Costello bulk regularity results [12, Chs. 5–7] tensored with a fixed coefficient kernel. This proves finite-scale kernel regularity and absence of new coefficient wave-front directions. The brane-defect QME obstruction classes are separate local-functional classes. □

Remark 5.65 *(Brane defect).* The brane defect $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\} \subset \mathbb{R}^2 \times \mathbb{C}^2$ pulls the weighted field complex $\mathcal{E}_w$ to its restriction along the defect:

$$\mathcal{E}^\partial_w = \Omega^\bullet(\mathbb{R})\widehat{\otimes}\big(\mathfrak{h}_w[1] \oplus \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[2]\big).$$

The boundary propagator $P^{w,\partial}_{\epsilon,L} \in \mathcal{E}^\partial_w\widehat{\otimes}\mathcal{E}^\partial_w$ is the bulk-to-boundary restriction of $P^w_{\epsilon,L}$ followed by the zero-mode-removed midpoint propagator $G(t,s) = \frac{1}{2}\,\mathrm{sgn}(t-s)$ of Proposition 5.223. The weighted Casimir $C_{\mathfrak{h},w}$ is the same on the boundary as in the bulk; the weight w is independent of the coordinate t.

Normal Ω-weights and equivariant kernel admissibility

The brane-preserving equivariant deformation rescales the normal bundle $N_L X = \mathbb{R}_s \oplus \mathbb{C}_{z_1} \oplus \mathbb{C}_{z_2}$ by the torus $T_\Omega = \mathbb{C}^*_{\epsilon_s} \times \mathbb{C}^*_{\epsilon_1} \times \mathbb{C}^*_{\epsilon_2}$, with t fixed. The equivariant differential is $Q_\Omega = Q + \iota_{V_\Omega}$ for the generating vector field $V_\Omega = \epsilon_s s \partial_s + \epsilon_1 z_1 \partial_{z_1} + \epsilon_2 z_2 \partial_{z_2}$, with $Q_\Omega^2 = L_{V_\Omega}$. The weighted-Tate envelope B_w extends to a stratified factorization-algebra over (X, L) by a diagonal equivariant Casimir; the residue-versus-Euler normalization enters as a one-dimensional scalar comparison map, and resonance among the normal characters is the exact obstruction to localizing the contraction.

Definition 5.66 *(Equivariant finite normal window).* Fix normal equivariant parameters $\epsilon_s, \epsilon_1, \epsilon_2$, with t fixed and s, z_1, z_2 carrying weights $\epsilon_s, \epsilon_1, \epsilon_2$. For a finite normal/Hamiltonian window (N, M), write

$$\mathsf{W}_{N,M} = \{(n, a, b) \mid 0 \leq n \leq N,\ a, b \geq 0,\ 1 \leq a + b \leq M\}.$$

The integer n is the total s-weight of the homogeneous normal summand; if a de Rham ds-factor is present, its weight is included in n. For the Hamiltonian monomial $s^n z_1^a z_2^b$ the normal character is

$$\chi^H_{n,a,b} = n\epsilon_s + a\epsilon_1 + b\epsilon_2 \tag{25}$$

For the residue-dual principal-part label with the same s-weight, the holomorphic part has the opposite character:

$$\chi^\vee_{n,a,b} = n\epsilon_s - a\epsilon_1 - b\epsilon_2 \tag{26}$$

Let $\mathsf{X}_{N,M}$ be the finite set of nonzero formal characters among $\chi^H_{n,a,b}$ and $\chi^\vee_{n,a,b}$ for those homogeneous summands which the normal Koszul contraction is required to move. The finite-window equivariant coefficient ring is

$$R_\Omega^{N,M} = \mathbb{C}[\epsilon_s, \epsilon_1, \epsilon_2, \hbar_\omega][\chi^{-1} \mid \chi \in \mathsf{X}_{N,M}], \qquad \mathrm{wt}(\hbar_\omega) = \epsilon_1 + \epsilon_2$$

A scalar specialization of $(\epsilon_s, \epsilon_1, \epsilon_2)$ is (N, M)-nonresonant if every $\chi \in \mathsf{X}_{N,M}$ remains nonzero. It is all-window nonresonant if this holds for every (N, M). An analytic Köthe norm in place of the algebraic Mittag–Leffler product topology requires the stronger q-moderate nonresonance condition: there are constants A, Q such that

$$|\chi_{n,a,b}|^{-1} \leq A\, Q^{n+a+b}$$

for every inverted character. This quantitative bound lifts the finite-window algebraic kernels from nonresonance to an analytic all-window bounded homotopy in a fixed normed scale.

THEOREM 5.67 *(Equivariant weighted-kernel admissibility).* Let w be a degree weight satisfying Definition 5.52. Work in a finite normal window (N, M) over $R_\Omega^{N,M}$, and let $\mathsf{A}_{N,M}$ be a homogeneous basis of the corresponding Hamiltonian-plus-cotangent coefficient pair. For $\alpha = (n, a, b, \tau) \in \mathsf{A}_{N,M}$, put $d(\alpha) = a + b$, let $e_{\alpha,w} = w(d(\alpha))e_\alpha$, let $e^\alpha_{w^{-1}} = \big(w(d(\alpha))\big)^{-1} e^\alpha$, and let χ_α be the character (25) for Hamiltonian labels and (26) for cotangent labels.

Then the equivariant diagonal coefficient Casimir

$$C_{\Omega,w}^{N,M} = \sum_{\alpha \in \mathsf{A}_{N,M}} e_{\alpha,w} \otimes e^\alpha_{w^{-1}}$$

is a finite-window kernel, and these kernels are compatible under the normal/Hamiltonian truncation maps. Thus the inverse system of $C_{\Omega,w}^{N,M}$'s is a strict Mittag–Leffler diagonal Casimir in the weighted coefficient category.

Suppose moreover that on each moving homogeneous normal summand there is a finite-dimensional Koszul numerator k_α^{Kos} with

$$Q_\Omega k_\alpha^{\mathrm{Kos}} + k_\alpha^{\mathrm{Kos}} Q_\Omega = \chi_\alpha \ \mathrm{id}_\alpha$$

before division by the character. Then the normal contracting kernel

$$H_{\Omega,w}^{N,M} = \sum_{\alpha \in \mathsf{A}_{N,M}^{\mathrm{mov}}} \chi_\alpha^{-1} \, e_{\alpha,w} \otimes e_{w^{-1}}^{\alpha} \otimes k_\alpha^{\mathrm{Kos}} \tag{27}$$

is a well-defined element of the finite-window weighted kernel module and satisfies

$$Q_\Omega H_{\Omega,w}^{N,M} + H_{\Omega,w}^{N,M} Q_\Omega = \mathrm{id} - \mathrm{pr}_{\chi=0}$$

on that window. A character with specialized value zero is a genuine resonant summand. It is retained in $\mathrm{pr}_{\chi=0}$, and the contraction is defined on the moving summand.

The residue and Euler normalizations leave the diagonal Casimir fixed. They change the brane pushforward/scalar-contact normalization. With orientation sign $\sigma_s = \pm 1$,

$$\nu_\Omega^{\mathrm{res}} = 1, \qquad \nu_\Omega^{\mathrm{Eu}} = \sigma_s \big(\epsilon_s \epsilon_1 \epsilon_2\big)^{-1}.$$

A normalized equivariant kernel row is therefore $K_{\Omega,w,\nu}^{N,M} = \nu_\Omega K_{\Omega,w}^{N,M}$. The comparison between residue and Euler conventions is the scalar map

$$K_{\Omega,w,\mathrm{res}}^{N,M} \longmapsto \nu_\Omega^{\mathrm{Eu}} K_{\Omega,w,\mathrm{res}}^{N,M}.$$

This factor is required when both conventions appear in one scalar-contact or trace statement; omitting it is exactly the normalization obstruction.

Proof. The Casimir is finite in the window (N, M). The two diagonal weights $w(d(\alpha))$ and $\big(w(d(\alpha))\big)^{-1}$ cancel in the evaluation pairing, so the finite sum is the identity tensor on the corresponding finite-dimensional coefficient quotient. Truncating from (N, M) to a smaller window deletes the omitted homogeneous labels, sending $C_{\Omega,w}^{N,M}$ to the smaller-window Casimir. This proves strict Mittag–Leffler compatibility.

For a moving homogeneous summand, the displayed Koszul identity gives

$$Q_\Omega (\chi_\alpha^{-1} k_\alpha^{\mathrm{Kos}}) + (\chi_\alpha^{-1} k_\alpha^{\mathrm{Kos}}) Q_\Omega = \mathrm{id}_\alpha$$

after adjoining χ_α^{-1} to the coefficient ring. Summing over the finite set of moving labels gives (27) and the homotopy identity with the unmoved zero-character projection. If a specialized character vanishes, χ_α^{-1} lies outside the specialized coefficient ring, so resonance is precisely the obstruction to moving that summand.

The residue convention fixes the formal residue density; the Euler convention pushes along the normal bundle with inverse equivariant Euler class $\sigma_s(\epsilon_s \epsilon_1 \epsilon_2)^{-1}$. The coefficient Casimir pairs dual bases before either pushforward is applied, so it is unchanged. The two conventions differ only in the one-dimensional scalar localization line: scalar-contact rows and trace rows are multiplied by the chosen ν_Ω, which is the displayed comparison map. $\square$

COROLLARY 5.68 *(Equivariant Costello QME obstruction vector).* Theorem 5.67 gives the coefficient Casimir and the localized normal contracting kernel in the weighted finite-window category, the datum needed to form an equivariant Costello kernel complex. Once equivariant brane-defect Costello data are given, in the sense of Definition E.41, the exact obstruction vector to the Costello QME is

$$\mathrm{Ob}^{\mathrm{QME}}_{\Omega,w} = \Big(\mathcal{R}_\Omega, \mathrm{ob}_{\Omega,\nu}, \{\mathfrak{s}_{n,\Omega}\}_{n\geq 1}, \{(([\mathfrak{o}^{\mathrm{ns}}_{n,\Omega,M}])_M, \lambda_{n,\Omega})\}_{n\geq 1}\Big).$$

Here

$$\mathcal{R}^{N,M}_\Omega = \{\alpha \in \mathsf{A}^{\mathrm{mov}}_{N,M} \mid \chi_\alpha = 0\}$$

is the resonance obstruction to the normal contraction, $\mathrm{ob}_{\Omega,\nu}$ is the residue/Euler normalization obstruction, $\mathfrak{s}_{n,\Omega,M}$ is the scalar condition

$$\widehat{\sigma}_{\mathrm{sc},\Omega,M}(R^{\mathrm{ns}}_{n,\Omega,M}),$$

and

$$([\mathfrak{o}^{\mathrm{ns}}_{n,\Omega,M}])_M \in \varprojlim_M H^1(\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M}(I)), \qquad \lambda_{n,\Omega} \in {\varprojlim}^1_M H^0(\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M}(I))$$

are the non-scalar cohomology and primitive-compatibility obstructions. The equivariant weighted kernel data determine a Costello brane-defect QME solution exactly on the common zero locus of this vector, and only there.

Proof. Theorem 5.67 constructs $C_{\Omega,w}$, the normal denominators, and the normalization line. Costello's QME is the Maurer–Cartan equation for the Hom-valued brane-defect graph kernel plus local counterterms. Therefore, after the equivariant datum is present, the scalar projection must first be a chain projection and its scalar condition must vanish. Only then is the residual a class in the non-scalar kernel complex, and the finite-window counterterm criterion is the same inverse-limit H^1 and ${\varprojlim}^1 H^0$ criterion as Theorem 5.84, with Q, d_M, $\widehat{\sigma}_{\mathrm{sc}}$ replaced by their equivariant versions. Resonant normal modes and normalization mismatch occur before this cohomology class is even formed. Hence the displayed vector is necessary and sufficient for the equivariant weighted-kernel construction to become a QME solution. □

Renormalization-flow compatibility

The Costello RG flow on effective interactions is the assignment $I_L = W(P_{\epsilon,L}, I)$, where W is the homotopy renormalization group action on functionals [12, Ch. 6]. Costello requirement (C3) demands that the RG flow on $\mathcal{E}_w$-valued interactions preserve the weighted coefficient module: a polynomial-degree-bounded interaction in $\mathcal{O}(\mathcal{E}_w)$ must produce I_L in the same weighted category.

PROPOSITION 5.69 *(Graphwise RG-tame interactions).* Let $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ be a local interaction polynomial in the fields, of total polynomial-degree bounded by D_0 at each vertex in the coefficient direction. For every fixed Costello graph Γ and every $0 < \epsilon \leq L < \infty$, the graph contribution $W_\Gamma(P^w_{\epsilon,L}, I)$ has finite coefficient degree bounded by $D_0 V(\Gamma) - 2E(\Gamma)$ and lies in the weighted completed coefficient category. The bracket-tame estimate (W1), together with the finite-window coadjoint condition (W2), bounds this single graph by a finite constant times the corresponding spacetime graph weight.

Proof. The interaction I has bounded polynomial degree D_0 at each vertex in the coefficient direction by hypothesis. A connected Costello graph Γ with V vertices, each of polynomial degree $\leq D_0$, and E

internal edges, contributes a coefficient pairing of external length at most $V \cdot D_0 - 2E$ in the source factor and a matching length in the target. By the bracket-tame estimate (W1), each internal edge contraction $C_{\mathfrak{h},w} : \mathfrak{h}_w \widehat{\otimes} \mathfrak{h}^{\vee,w}_{\mathrm{cont}} \to \mathbb{C}$ changes the total weight by at most the finite constant attached to the degrees present in that graph. Coadjoint legs are controlled windowwise by (W2). Hence the graph contribution is a well-defined element of the weighted completion. The assertion is graphwise; the full effective interaction is Costello's completed sum over graph degrees. □

THEOREM 5.70 *(Weighted RG flow, graphwise compatibility).* Let w be a degree weight as in Definition 5.52 and let $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ be a local interaction satisfying a vertexwise polynomial-degree bound D_0. Each fixed graph contribution to the Costello RG flow

$$I_L = W(P^w_{\epsilon,L}, I)$$

lies in the weighted Tate completion and is continuous in (ϵ, L). The completed sum over counterterms and the QME obstruction class require the locality datum of Problem 5.73.

Proof. By Proposition 5.69, every Feynman diagram considered individually remains in the weighted Tate category. Its finite coefficient-degree bound is $D_0 V(\Gamma) - 2E(\Gamma)$, and continuity in (ϵ, L) follows from Proposition 5.64 (2). Passing from graphwise well-definedness to a renormalized QME solution is precisely the residual analytic problem. □

Cotangent-lift inside B_w

COROLLARY 5.71 *(Cotangent-lift inside the weighted coefficient category).* Take $w(d) = q^d$ with $q > 1$ as in Remark 5.54. Then

(a) the Casimir $C_{\mathfrak{h},w}$ converges in $\mathfrak{h}_w \widehat{\otimes} \mathfrak{h}^{\vee,w}_{\mathrm{cont}}$;
(b) the Costello heat-kernel propagator $P^w_{\epsilon,L}$ extends to the weighted field complex $\mathcal{E}_w$ as a continuous BV family;
(c) each fixed Costello graph contribution from a vertexwise polynomial-degree-bounded interaction stays inside B_w;
(d) the central-operation cochain Φ of Theorem 4.20 extends to the weighted cotangent extension $\mathfrak{g}_w = \mathfrak{h}_w \ltimes \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[1]$ as a continuous cochain map

$$\Phi^w : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_w) \longrightarrow B_w \subset \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}_w)\big).$$

Relative to Problem 5.73, the all-order lift $\Phi^w_{\hbar,\mathrm{loc}}$ is Proposition 5.72.

Proof. This is Theorem 5.58 parts (b)–(e), evaluated at the exponential weight $w(d) = q^d$, $q > 1$. The P_0-algebra reading of Φ^w uses the bracket-admissible envelope of Definition 4.18. The strict continuous duality statement of Lemma 5.166 applies to the product/direct-sum Tate pair. The needed dual generators are inserted by the symmetric Mittag-Leffler coefficient pairing of Remark 5.56. □

PROPOSITION 5.72 *(Weighted QME lift under obstruction vanishing).* Assume the Costello-locality/QME compatibility identity of Problem 5.73: the finite-window counterterm system for the mixed brane-defect interaction has vanishing obstruction class and is compatible with the weighted coefficient category. Under this assumption, the formal quantum CE/PV central-operation map $\Phi^{(0)}_\hbar$ governed by Theorem 5.217 has a weighted-continuous cotangent extension

$$\Phi^w_{\hbar,\mathrm{loc}} : C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]\big)[[\hbar]] \xrightarrow[\text{QME obstruction}=0]{\sim} Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^\hbar_{\mathrm{open}}\big)|_{\mathfrak{g}_w}.$$

The displayed isomorphism is a theorem in the weighted-continuous sector with vanishing QME obstruction. The original unweighted Tate pair is governed by the endpoint obstruction of Theorem 5.94.

Proof. The algebraic cochain identity is given by Corollary 5.71. The BV graph realization decomposes the cotangent action $\mathrm{ad}^\vee$ into Costello graphs in $\mathcal{E}_w$, and Theorem 5.70 keeps each fixed graph inside the weighted category. The nonformal point is the cancellation between Costello-locality counterterms and the bulk-boundary kernel at the brane defect. The proposition assumes exactly that weighted QME compatibility and the vanishing of the obstruction class named in Problem 5.73. With that assumption in force, the formal $\hbar$-adic construction of Theorem 5.217 applies in the weighted coefficient envelope. The graphwise finite-scale assertions of Theorem 5.70 are anterior to this obstruction and to the principal-part current extension; that extension is the additional datum for compatibility with $\mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1]$. □

Problem 5.73 *(Costello-locality compatibility for the weighted RG flow). Counterterm datum.* Find a compatible counterterm system $\{C_{n,M}\}_{n\geq 1, M\geq 1}$ for the mixed brane-defect interaction at the defect $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ in the weighted field complex $\mathcal{E}_w$ such that:

(L1) each $C_{n,M}$ is a local functional in $\mathcal{O}^{\hbar^n}_{\mathrm{loc}}(\mathcal{E}^{\leq M}_w)$ of bounded vertex polynomial degree;

(L2) the truncations are compatible: $\pi_{M,N} C_{n,M} = C_{n,N}$ for $M \geq N$;

(L3) the recursion solves the QME order by order.

Named obstruction class. The class

$$[\mathrm{Ob}^{\mathrm{loc}}_{w,\partial}] \in H^1\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\}\big)$$

obstructing (L1)–(L3); equivalently, by Theorem 5.84, the pair $\big(([\mathfrak{o}^{\mathrm{ns}}_{n,M}])_M, \lambda_n\big)$ in $\varprojlim_M H^1(\boldsymbol{Q}^\bullet_{w,M}(I)) \oplus \varprojlim^1_M H^0(\boldsymbol{Q}^\bullet_{w,M}(I))$. Estimate (W1) bounds vertex weights; (W2) bounds coadjoint legs graph by graph; Theorem 8.17 controls wave-front and finite-window locality estimates. The remaining datum is the counterterm system that turns Hadamard control into anomaly cancellation.

COROLLARY 5.74 *(Descendant cochain comparison under weighted QME vanishing).* Assume the obstruction class in Problem 5.73 vanishes, so that Proposition 5.72 applies. Then the Tate-coefficient lift of the descendant cochain comparison of Problem 5.222 exists in the weighted Tate completion only under this hypothesis: the deformation

$$\Phi^{(0,1)}_{\hbar,\mathrm{loc}} \colon C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]\big)^{(0)+(1)}[[\hbar]] \xrightarrow[\text{QME obstruction}=0]{\sim} Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^{\hbar}_{\mathrm{open}}\big)^{(0)+(1)}|_{\mathfrak{g}_w}$$

is an isomorphism on the degree-zero plus one-antifield sector in the weighted theory with vanishing QME obstruction.

Remark 5.75 *(Sector restriction).* The weighted mechanism restricts the cotangent direction from the ambient polynomial Hamiltonian dual to the weighted dual $\mathfrak{h}^{\vee,w}_{\mathrm{cont}} = \prod_d w(d)^{-1} H^\vee_d$. The strict polynomial Hamiltonian dual $\mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_d H^\vee_d$ is a *dense subspace* of $\mathfrak{h}^{\vee,w}_{\mathrm{cont}}$ for every w with $w(d) \to \infty$, since the strict direct sum is finite-degree truncations and the weighted product is the weighted Mittag-Leffler inverse-limit completion. Hence the weighted lift extends the cochain map continuously from the strict polynomial sector to the weighted product completion. This recovers the weighted-continuous Hamiltonian identities. Unweighted regulator-independence and identification of the weighted product with the original strict continuous dual are separate statements.

Regulator independence and its boundary

Definition 5.76 *(Regulator-admissible finite-window sector).* A local functional, interaction, obstruction cocycle, or graph coefficient in the weighted theory is *regulator-admissible* if it is represented by a strict Mittag-Leffler family on the finite Tate windows

$$\{\mathcal{E}_w^{w_0}\}_{w_0\geq 1}$$

and satisfies the following four conditions.

(A1) The transition maps in the coefficient and local-functional directions are the truncations to degrees $d \leq w_0$, or the induced restriction obtained by setting higher-degree coefficient fields to zero. They are degreewise surjective maps of complexes.

(A2) At each fixed $\hbar$-order and for each fixed Costello graph, the coefficient expression uses only vertex-wise polynomial-degree-bounded Hamiltonian and cotangent legs.

(A3) If w, w' are two degree weights satisfying Definition 5.52, the diagonal finite-window rescaling

$$R^{w_0}_{w,w'}(w(d)e_{d,i}) = w'(d)e_{d,i}, \qquad R^{w_0}_{w,w'}(w(d)^{-1}e^i_d) = w'(d)^{-1}e^i_d$$

transports the interaction, differential, bracket, propagator contraction, and counterterm representative from the w-window to the w'-window. The bracket is the transported bracket; comparison of two weights is made after this transport, while raw unweighted structure constants are held outside the comparison.

(A4) The cohomology being computed is the admissible filtered cohomology detected on finite quotients and associated graded pieces in the envelope of Statement 5.18; observables are restricted to data visible in finite quotients or associated graded pieces.

Definition 5.77 *(Exponential finite-window regulator tower).* For $q > 1$ put $w_q(d) = q^d$. For a finite Hamiltonian window M set

$$\mathfrak{h}_{q,\leq M} = \prod_{1\leq d\leq M} q^d H_d, \qquad D_{q,\leq M} = \prod_{1\leq d\leq M} q^{-d} H_d^\vee,$$

and

$$\mathcal{E}_q^M = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\mathfrak{h}_{q,\leq M}[1] \oplus D_{q,\leq M}[2]\big).$$

For $M \geq N$, let

$$\pi^q_{M,N} \colon (\mathfrak{h}_{q,\leq M}, D_{q,\leq M}) \twoheadrightarrow (\mathfrak{h}_{q,\leq N}, D_{q,\leq N})$$

be degree truncation. The weight transport and the unweighting map on the window M are

$$R^M_{q,q'}(q^d e_{d,i}) = (q')^d e_{d,i}, \qquad R^M_{q,q'}(q^{-d}e^i_d) = (q')^{-d}e^i_d,$$

and

$$U^M_q(q^d e_{d,i}) = e_{d,i}, \qquad U^M_q(q^{-d}e^i_d) = e^i_d.$$

The finite-window unweighted endpoint is the finite-dimensional pair

$$(\mathfrak{h}_{\leq M}, D_{\leq M}) = \Big(\prod_{d\leq M} H_d, \prod_{d\leq M} H_d^\vee\Big).$$

Its transition map for $M \geq N$ is denoted $\pi^\circ_{M,N}$.

LEMMA 5.78 *(Finite-window transport squares).* For $q, q' > 1$ and $M \geq N$, the maps of Definition 5.77 satisfy

$$\pi^{q'}_{M,N} R^M_{q,q'} = R^N_{q,q'} \pi^q_{M,N}, \qquad \pi^\circ_{M,N} U^M_q = U^N_q \pi^q_{M,N},$$

and

$$U^M_{q'} R^M_{q,q'} = U^M_q, \qquad R^M_{q',q''} R^M_{q,q'} = R^M_{q,q''}.$$

The truncated Casimir

$$C^M_q = \sum_{d \leq M} \sum_i (q^d e_{d,i}) \otimes (q^{-d} e^i_d)$$

is transported to $C^M_{q'}$ by $R^M_{q,q'}$ and to the ordinary finite Casimir $C^M_\circ$ by U^M_q. Hence every Costello graph whose coefficient legs lie in $d \leq M$ has the same unweighted coefficient after applying U^M_q, independently of q.

Proof. All four identities are degreewise identities on the two bases $q^d e_{d,i}, q^{-d} e^i_d$. Truncation forgets degrees $d > N$, while $R^M_{q,q'}$ and U^M_q multiply only the surviving degree-d basis vector by a scalar. Thus truncation and transport commute.

The Casimir statement follows term by term:

$$R^M_{q,q'}\big((q^d e_{d,i}) \otimes (q^{-d} e^i_d)\big) = (q')^d e_{d,i} \otimes (q')^{-d} e^i_d,$$

and U^M_q sends the same term to $e_{d,i} \otimes e^i_d$. A fixed graph has finitely many coefficient contractions and external coefficient legs. If all of them lie in the window M, applying U^M_q to each leg removes every diagonal weight and each internal Casimir becomes $C^M_\circ$. □

Statement 5.79 *(Weighted RG transport criterion).* A weighted mixed brane-defect interaction enters the regulator theorem only with the following data.

(G1) Its coefficient and local-functional restrictions to $d \leq w_\circ$ form strict Mittag-Leffler towers with degreewise surjective transition maps.

(G2) At every fixed $\hbar$-order and for every fixed Costello graph, the coefficient expression is vertexwise bounded in polynomial Hamiltonian and cotangent degree.

(G3) For any admissible degree weights w, w', the diagonal finite-window maps $R^{w_\circ}_{w,w'}$ transport the differential, bracket, propagator contraction, interaction, and counterterm representatives.

(G4) The cohomology theory is the admissible filtered cohomology of Statement 5.18.

(G5) The Costello RG recursion $W(P_{\epsilon,L}, I)$ preserves the same strict Mittag-Leffler tower: every RG step has source and target in the weighted coefficient envelope.

Proof. Conditions (G1)–(G4) are the four clauses of Definition 5.76. Condition (G5) is the coefficient part of Costello requirement (C3). Proposition 5.69 proves (G2) for a fixed graph. Theorem 8.17 proves that the finite-window Hadamard estimates pass through the Mittag-Leffler limit when the windowwise counterterms are compatible. The all-order datum beyond this graphwise statement is the obstruction class of Problem 5.73. □

THEOREM 5.80 *(Transport groupoid of admissible weights).* On the class of interactions satisfying 5.79, admissible degree weights form a connected transport groupoid. Its objects are the weighted finite-window local-functional complexes

$$C^{w_\circ}_w = \big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w_\circ}_w), Q + \{I^w_\circ, -\}\big),$$

and its arrows are the diagonal maps $R^{w_0}_{w,w'}$. For every w, w', w'' and every finite window w_0,

$$R^{w_0}_{w',w''} R^{w_0}_{w,w'} = R^{w_0}_{w,w''}, \qquad R^{w_0}_{w,w} = \mathrm{id},$$

and these arrows commute with truncation in w_0, Costello graph contractions, and RG counterterm representatives.

Proof. The formula defining $R^{w_0}_{w,w'}$ is diagonal:

$$w(d)e_{d,i} \mapsto w'(d)e_{d,i}, \qquad w(d)^{-1}e^i_d \mapsto w'(d)^{-1}e^i_d.$$

Composition multiplies the scalars degree by degree, hence gives the displayed groupoid law. Since the maps are diagonal in d, they commute with truncation to $d \leq w_0$. By 5.79(G3), the transported bracket, interaction, propagator, and counterterm representatives are the corresponding w'-objects. Therefore $R^{w_0}_{w,w'}$ is an isomorphism of the finite-window RG complexes and intertwines every fixed graph contraction. □

THEOREM 5.81 *(Canonical equivalence of finite-degree local observables).* Let $w(d) = q^d$ and $w'(d) = (q')^d$, with $q, q' > 1$. Let O^w be a regulator-admissible local observable, graph coefficient, or obstruction cocycle represented on a finite window $d \leq M$. Then $R^M_{q,q'}$ is the canonical identification of the q-weighted and q'-weighted representatives on that finite window:

$$O^{q'}_M = R^M_{q,q'} O^q_M.$$

After writing both representatives in the common unweighted basis,

$$U^M_q(O^q_M) = U^M_{q'}(O^{q'}_M).$$

If O^q_M is a cocycle for $Q + \{I^q_M, -\}$, then the common unweighted representative is a cocycle for the unweighted finite-window transported differential. If $\mathrm{Ob}_{q,M}$ is the finite-window obstruction cocycle produced by the Costello recursion, then

$$U^M_q(\mathrm{Ob}_{q,M}) = U^M_{q'}(\mathrm{Ob}_{q',M}), \qquad \pi^\circ_{M,N} U^M_q(\mathrm{Ob}_{q,M}) = U^N_q(\mathrm{Ob}_{q,N})$$

for $M \geq N$. Thus a finite-degree local observable, fixed graph coefficient, or obstruction class is a single object of the finite-window transported complex, presented in different weighted bases by the choices $q > 1$.

Proof. The observable lies in the finite complex C^M_q. By Theorem 5.80, $R^M_{q,q'}$ is an isomorphism from C^M_q to $C^M_{q'}$, carrying the differential, bracket, propagator contraction, interaction, counterterm representative, and obstruction cocycle. Lemma 5.78 gives $U^M_{q'} R^M_{q,q'} = U^M_q$. Therefore both weighted representatives have the same coefficient array after unweighting.

Since $R^M_{q,q'}$ is a cochain isomorphism, cocycles go to cocycles and coboundaries go to coboundaries. Applying U^M_q identifies the same finite complex in the common unweighted basis; the transported differential is $U^M_q(Q + \{I^q_M, -\})(U^M_q)^{-1}$. Hence the cocycle condition is preserved. The obstruction equalities are the same statement for the obstruction cochain produced at the relevant RG step. The transition identity follows from $\pi^\circ_{M,N} U^M_q = U^N_q \pi^q_{M,N}$ and strict Mittag-Leffler compatibility of the finite-window obstruction system. □

THEOREM 5.82 *(Regulator independence inside the admissible finite-window sector).* Let w, w' be degree weights satisfying Definition 5.52, and let $I_\circ^w$ be a regulator-admissible mixed brane-defect interaction satisfying 5.79. Put $I_\circ^{w'} = R_{w,w'} I_\circ^w$. Then the diagonal rescalings $R_{w,w'}^{w_\circ}$ assemble to an isomorphism of strict Mittag-Leffler local-functional complexes

$$R_{w,w'}: \varprojlim_{w_\circ}\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_\circ}), Q + \{I_\circ^w, -\}\big) \xrightarrow{\cong} \varprojlim_{w_\circ}\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_{w'}^{w_\circ}), Q + \{I_\circ^{w'}, -\}\big).$$

Under this isomorphism:

(i) truncated Casimirs, heat-kernel propagators, and fixed graphwise Costello contractions are identified;

(ii) the obstruction class
$$\mathrm{Ob}_w \in H^1_{\mathrm{adm}}(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_\circ^w, -\})$$
maps to $\mathrm{Ob}_{w'}$;

(iii) vanishing or nonvanishing of the mixed brane-defect QME obstruction is independent of the admissible weight.

These isomorphisms are the arrows of the transport groupoid of Theorem 5.80; they are functorial in triples of admissible weights.

Proof. On the finite window $d \le w_\circ$, the map $R_{w,w'}^{w_\circ}$ is an invertible linear map on the coefficient space and the identity on the spacetime factor. The bracket and the interaction in the w'-window are, by admissibility, the transported bracket and transported interaction. The map sends

$$\sum_{d \le w_\circ} \sum_i (w(d) e_{d,i}) \otimes (w(d)^{-1} e_d^i)$$

to the corresponding w'-weighted truncated Casimir. Hence it sends $P_{\epsilon,L}^{w,w_\circ}$ to $P_{\epsilon,L}^{w',w_\circ}$, commutes with the differential Q, and carries the transported BV bracket and every finite-window graph contraction to the w'-window contraction. Condition (A3) gives the same statement for $I_\circ$, the counterterm representatives, and the obstruction cocycle.

The maps $R_{w,w'}^{w_\circ}$ commute with truncation in $w_\circ$. The finite-window complexes therefore form an isomorphic pair of strict Mittag-Leffler inverse systems. By (A1), these are degreewise surjective towers of complexes. Hence the inverse limit is exact in the admissible envelope of Statement 5.18; equivalently $\varprojlim^1 = 0$ for this tower. Passing to the inverse limit gives the displayed isomorphism of local-functional complexes and identifies their admissible filtered cohomology classes. In particular the obstruction class vanishes for w if and only if the transported obstruction class vanishes for w'.

Functoriality in triples is the groupoid identity proved in Theorem 5.80. □

Definition 5.83 *(Finite-window non-scalar QME tower).* Fix an interval I, a degree weight w, and a regulator-admissible interaction satisfying Statement 5.79. For a finite Hamiltonian window M, put

$$D_{w,\le M} := \prod_{d \le M} w(d)^{-1} H_d^\vee,$$

and define the truncated principal-part target

$$A_{\partial,\hbar}^{\mathrm{pp},w,M}(I) = \widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}\big(\Omega_c^\bullet(I) \otimes \prod_{d \le M} w(d) H_d[[\hbar]]\big) \widehat{\otimes} \widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}\big((\mathcal{D}_c'(I) \widehat{\otimes} D_{w,\le M}[[\hbar]])[1]\big).$$

Assume that the scalar-contact projection of Problem 5.220 has been chosen windowwise, commutes with truncation, and is transported by $R^M_{w,w'}$. The finite-window non-scalar obstruction complex is

$$Q^\bullet_{w,M}(I) = \ker(\widehat{\sigma}_{\mathrm{sc},M}) \subset \Big(\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}\big(\mathcal{E}^M_w|_I; A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)\big), d_M\Big),$$

where $d_M = Q + \{I^w_{0,M}, -\}_{\hbar,M}$. The bracket, interaction, and differential are the transported finite-window structures given by the weighted RG criterion; plain projected Hamiltonian brackets are replaced by this transported structure. The inverse system $\{Q^\bullet_{w,M}(I)\}_{M\geq 1}$ uses degree truncation $\pi_{M,N}$ for $M \geq N$, together with the $\hbar$-adic filtration, the scalar-contact filtration, and the finite-window coefficient filtration.

THEOREM 5.84 *(Finite-window counterterm criterion for the non-scalar QME).* Work in the tower of Definition 5.83. Suppose that a counterterm system has been chosen through order $\hbar^{n-1}$, and let

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} \in Z^1\big(Q^\bullet_{w,M}(I)\big)$$

be the order-n non-scalar residual after those lower-order choices. Equivalently, in the curved Maurer–Cartan form,

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} = [\hbar^n]\Big(\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar,M} + d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big),$$

with the quadratic term omitted for the linearized equation $\mathrm{Ob}^{\mathrm{red}} + d_M C = 0$. Compatible order-$n$ finite-window counterterms

$$C_{n,M} \in Q^0_{w,M}(I), \qquad d_M C_{n,M} = -\mathfrak{o}^{\mathrm{ns}}_{n,M}, \qquad \pi_{M,N} C_{n,M} = C_{n,N},$$

exist if and only if the following two obstructions vanish:

$$\big([\mathfrak{o}^{\mathrm{ns}}_{n,M}]\big)_M \in \varprojlim_M H^1\big(Q^\bullet_{w,M}(I)\big),$$

and, after choosing any windowwise primitives $c_{n,M}$ with $d_M c_{n,M} = -\mathfrak{o}^{\mathrm{ns}}_{n,M}$, the Milnor primitive-compatibility class

$$\lambda_n \in {\varprojlim}^1_M H^0\big(Q^\bullet_{w,M}(I)\big)$$

represented, after passing to cohomology, by

$$\big[\pi_{M,N} c_{n,M} - c_{n,N}\big] \in H^0\big(Q^\bullet_{w,N}(I)\big)$$

is zero. If the inverse system $\{H^0(Q^\bullet_{w,M}(I))\}_M$ is Mittag–Leffler, then $\lambda_n = 0$ automatically, so windowwise vanishing of the H^1-classes is sufficient.

Iterating this criterion for all $n \geq 1$ constructs an $\hbar$-adic compatible counterterm $C_{\hbar,w} = \sum_{n\geq 1} \hbar^n C_{n,w}$ in the admissible finite-window inverse limit. The transport maps $R^M_{w,w'}$ identify the obstruction pair

$$\mathfrak{O}^{\mathrm{ns}}_n = \big(([\mathfrak{o}^{\mathrm{ns}}_{n,M}])_M, \lambda_n\big)$$

for all admissible weights. Therefore a nonzero $\mathfrak{O}^{\mathrm{ns}}_n$, at the first order where it occurs, is the precise finite-window obstruction to killing the non-scalar QME class by compatible weighted counterterms.

Proof. The lower-order counterterms solve the QME modulo $\hbar^n$. Applying d_M to the displayed order-n residual and using $d_M^2 = 0$, the graded Jacobi identity, and the lower-order equations gives $d_M \mathfrak{o}^{\mathrm{ns}}_{n,M} = 0$. The scalar-contact projection is a chain map by Definition 5.83, and the balanced scalar branch has zero scalar symbol, so the residual lies in $\mathcal{Q}^{\bullet}_{w,M}(I)$.

Necessity is immediate. A compatible counterterm $C_{n,M}$ is a primitive of $-\mathfrak{o}^{\mathrm{ns}}_{n,M}$, hence each H^1-class vanishes. Its truncation identities make $\pi_{M,N} C_{n,M} - C_{n,N} = 0$, so the associated Milnor $\varprojlim^1 H^0$-class is also zero.

Conversely, assume the two displayed obstructions vanish. The first condition gives primitives $c_{n,M}$ window by window. For $M \geq N$, the difference $\pi_{M,N} c_{n,M} - c_{n,N}$ is d_N-closed because the residuals commute with truncation. The cohomology classes of these closed degree-zero differences form the standard Roos 1-cocycle computing $\varprojlim^1_M H^0(\mathcal{Q}^{\bullet}_{w,M}(I))$. Vanishing of λ_n, together with the degreewise Mittag–Leffler property of the finite-window truncation tower, means that the primitives can be adjusted by degree-zero cocycles and degree-zero boundaries so that the adjusted primitives commute with every truncation map. Those adjusted primitives are $C_{n,M}$. Passing to the inverse limit gives $C_{n,w}$.

The assertion about the H^0-Mittag–Leffler hypothesis is the standard vanishing of $\varprojlim^1$ for a Mittag–Leffler inverse system. Weight invariance follows from Theorem 5.82: the maps $R^M_{w,w'}$ are cochain isomorphisms, commute with truncation, and carry residuals, primitives, and their difference cocycles to the corresponding data for w'. □

Definition 5.85 *(Regular-density closed-exchange finite-window complex).* Work in the tower of Definition 5.83, and assume the regular-density Costello graph kernel of Proposition B.17 on every finite Hamiltonian window. Thus the cotangent current labels are restricted to

$$\mathcal{D}^{\mathrm{reg}}_c(I) = \{\, b(t)dt \mid b \in C^\infty_c(I) \,\},$$

and the finite-window graph weights

$$W^{R,\mathrm{reg},M}_{\Gamma,L}(a_\bullet, b_\bullet; f_\bullet, \rho_\bullet)$$

are renormalized brane-defect local functionals compatible with interval extension, degree truncation, and admissible weight transport. Let $\mathcal{G}^{\bullet,\mathrm{reg}}_{w,M}(I)$ be the complete filtered $\mathbb{C}[[\hbar]]$-span, inside $\mathcal{Q}^{\bullet}_{w,M}(I)$, of the scalar-zero members among these graph weights and their regular-density local counterterms. The zero-internal-edge generators are the coefficient-current terms

$$\iota_I(B^w_f(a)), \qquad \iota_I(\Theta^w_\rho(b(t)dt)),$$

whenever they have zero scalar-contact projection.

Define

$$\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I) \subset \mathcal{Q}^{\bullet}_{w,M}(I)$$

to be the smallest complete filtered d_M-stable subspace containing $\mathcal{G}^{\bullet,\mathrm{reg}}_{w,M}(I)$. Its differential is

$$d^{X,\mathrm{reg}}_M = d_M|_{\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)}.$$

This is a finite-window graph complex. The plain finite-dimensional Hamiltonian Lie quotient is the obstruction of Lemma 5.210. The window is a coefficient-support and graph-locality condition on already constructed regular-density local functionals.

PROPOSITION 5.86 *(Regular-density construction of X and Ξ).* The complexes of Definition 5.85 form an inverse system

$$\rho^{X,\mathrm{reg}}_{M,N}\colon \mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I) \longrightarrow \mathcal{X}^{\bullet,\mathrm{reg}}_{w,N}(I), \qquad M \geq N,$$

by restricting the finite-window truncation maps $\pi_{M,N}$ of $\boldsymbol{Q}^{\bullet}_{w,M}(I)$. The closed-exchange insertion map is the degree-zero inclusion

$$\Xi^{\mathrm{reg}}_{M}\colon \mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I) \hookrightarrow \boldsymbol{Q}^{\bullet}_{w,M}(I).$$

It is a cochain map and satisfies

$$\pi_{M,N}\Xi^{\mathrm{reg}}_{M} = \Xi^{\mathrm{reg}}_{N}\rho^{X,\mathrm{reg}}_{M,N}.$$

For admissible weights w, w', the diagonal transports restrict to cochain isomorphisms

$$R^{M,X}_{w,w'}\colon \mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I) \xrightarrow{\cong} \mathcal{X}^{\bullet,\mathrm{reg}}_{w',M}(I)$$

and intertwine the inclusions:

$$R^{M}_{w,w'}\Xi^{\mathrm{reg}}_{M} = \Xi^{\mathrm{reg}}_{M,w'}R^{M,X}_{w,w'}$$

Consequently the abstract closed-exchange criterion of Theorem 5.88 applies to the regular-density tower $\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$ with $\Xi_M = \Xi^{\mathrm{reg}}_{M}$.

Proof. Proposition B.17 gives the finite-window identities

$$\pi_{M,N}W^{R,\mathrm{reg},M}_{\Gamma,L} = W^{R,\mathrm{reg},N}_{\Gamma,L}\pi^{\mathrm{CE}}_{M,N}, \qquad R^{M}_{w,w'}W^{R,\mathrm{reg},M}_{\Gamma,L} = W^{R,\mathrm{reg},M}_{\Gamma,L,w'}R^{M,\mathrm{CE}}_{w,w'}$$

and the same compatibility for the regular-density counterterms. Since $\widehat{\sigma}_{\mathrm{sc},M}$ is a chain map in Definition 5.83, its kernel $\boldsymbol{Q}^{\bullet}_{w,M}(I)$ is a subcomplex. Therefore scalar-zero graph weights and their d_M-closures stay in $\boldsymbol{Q}^{\bullet}_{w,M}(I)$.

Truncation $\pi_{M,N}$ and weight transport $R^{M}_{w,w'}$ are cochain maps on the non-scalar tower. They carry the generating regular-density graph weights and counterterms to the corresponding generators in the lower window or transported weight. By minimality of the complete d_M-stable subcomplex generated by these elements, they restrict to the displayed maps on $\mathcal{X}^{\bullet,\mathrm{reg}}$.

The map Ξ^{reg}_{M} is inclusion of a subcomplex of $\boldsymbol{Q}^{\bullet}_{w,M}(I)$, hence $d_M\Xi^{\mathrm{reg}}_{M} = \Xi^{\mathrm{reg}}_{M}d^{X,\mathrm{reg}}_{M}$. The compatibility with $\rho^{X,\mathrm{reg}}_{M,N}$ and $R^{M,X}_{w,w'}$ is precisely the restriction of the same identities on $\boldsymbol{Q}^{\bullet}$. □

PROPOSITION 5.87 *(Raw kernel obstruction to a stronger cochain map).* Let $\kappa^{\mathrm{graph},\mathrm{reg},M}_{\hbar,w,I}$ be the finite-window regular-density graph kernel of Proposition B.17, viewed as a degree-zero element of the convolution Hom from the regular-density CE source to the target $\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$. If one tries to take the regular-density CE source itself as the closed-exchange complex and $\Xi_M = \kappa^{\mathrm{graph},\mathrm{reg},M}_{\hbar,w,I}$, then the exact obstruction to an ordinary cochain map into $\boldsymbol{Q}^{\bullet}_{w,M}(I)$ is the pair

$$\widehat{\sigma}_{\mathrm{sc},M}\kappa^{\mathrm{graph},\mathrm{reg},M}_{\hbar,w,I}, \qquad d_{\mathbf{K}}\kappa^{\mathrm{graph},\mathrm{reg},M}_{\hbar,w,I} = d_M\kappa^{\mathrm{graph},\mathrm{reg},M}_{\hbar,w,I} - \kappa^{\mathrm{graph},\mathrm{reg},M}_{\hbar,w,I}d_{\mathrm{CE}}$$

The first term is the obstruction to landing in $\ker \widehat{\sigma}_{\mathrm{sc},M}$; the second is the obstruction to commuting with differentials. If the closed-exchange datum is required to be a $P_{0,\hbar}$-central-operation kernel, the ordinary cochain obstruction is strengthened to the Maurer–Cartan curvature

$$\mathrm{Curv}_{\mathbf{K}}(\kappa^{\mathrm{graph},\mathrm{reg},M}_{\hbar,w,I}) = d_{\mathbf{K}}\kappa^{\mathrm{graph},\mathrm{reg},M}_{\hbar,w,I} + \frac{1}{2}[\kappa^{\mathrm{graph},\mathrm{reg},M}_{\hbar,w,I}, \kappa^{\mathrm{graph},\mathrm{reg},M}_{\hbar,w,I}]_{\mathbf{K}}.$$

Thus the regular-density graph complex gives a proved closed-exchange tower by the subcomplex construction above. The stronger assertion that the raw bulk-to-defect CE kernel itself is the cochain map is equivalent to vanishing of the displayed scalar and convolution defects, with finite-window and weight-transport compatibility.

Proof. The convolution differential is $d_{\mathbf{K}}T = d_M T - (-1)^{|T|} T d_{\mathrm{CE}}$. Since $\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}$ has degree zero, the displayed formula is exactly the commutator of the proposed map with the two differentials. Hence it vanishes if and only if the raw kernel is a cochain map into the ambient target. It lands in the non-scalar complex precisely when its scalar-contact projection vanishes.

The bracket term is the completed convolution bracket of Definition B.15. A $P_{0,\hbar}$-central-operation kernel is a Maurer–Cartan element in that convolution dg Lie algebra, so its exact obstruction is $d_{\mathbf{K}}\kappa + \frac{1}{2}[\kappa,\kappa]_{\mathbf{K}}$. The generator-level form of this statement is the obstruction list in Proposition B.21. The compatibility clauses follow from the same truncation and weight-transport identities used in Proposition 5.86. □

THEOREM 5.88 *(Closed-exchange plus counterterm criterion).* Work in the non-scalar tower of Definition 5.83. Let

$$\rho^X_{M,N}\colon \mathcal{X}^\bullet_{w,M}(I) \longrightarrow \mathcal{X}^\bullet_{w,N}(I), \qquad M \geq N,$$

be an inverse system of closed-exchange local-functional complexes, and let

$$\Xi_M\colon \mathcal{X}^\bullet_{w,M}(I) \longrightarrow \mathcal{Q}^\bullet_{w,M}(I)$$

be degree-zero cochain maps satisfying

$$\pi_{M,N}\Xi_M = \Xi_N \rho^X_{M,N}.$$

Assume also that the closed-exchange tower and the maps Ξ_M are transported by $R^M_{w,w'}$, in the same sense as the non-scalar tower. Let

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} \in Z^1(\mathcal{Q}^\bullet_{w,M}(I)), \qquad \pi_{M,N}\mathfrak{o}^{\mathrm{ns}}_{n,M} = \mathfrak{o}^{\mathrm{ns}}_{n,N},$$

be the order-n residual after the lower-order choices.

For a fixed window M, the branch equation

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} + \Xi_M(W_{n,M}) + d_M C_{n,M} = 0, \qquad W_{n,M} \in Z^1(\mathcal{X}^\bullet_{w,M}(I)), \quad C_{n,M} \in \mathcal{Q}^0_{w,M}(I),$$

has a solution if and only if

$$[\mathfrak{o}^{\mathrm{ns}}_{n,M}] \in -\operatorname{im}\Big(H^1(\Xi_M)\colon H^1(\mathcal{X}^\bullet_{w,M}(I)) \to H^1(\mathcal{Q}^\bullet_{w,M}(I))\Big).$$

In the inverse limit, put

$$\beta^{\mathrm{cl}}_n = \Big[\big([\mathfrak{o}^{\mathrm{ns}}_{n,M}]\big)_M\Big] \in \operatorname{coker}\left(\varprojlim_M H^1(\mathcal{X}^\bullet_{w,M}(I)) \xrightarrow{\varprojlim H^1(\Xi_M)} \varprojlim_M H^1(\mathcal{Q}^\bullet_{w,M}(I))\right).$$

Compatible closed-exchange cocycles and compatible counterterms solving

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} + \Xi_M(W_{n,M}) + d_M C_{n,M} = 0, \qquad \rho^X_{M,N}W_{n,M} = W_{n,N}, \quad \pi_{M,N}C_{n,M} = C_{n,N},$$

exist if and only if $\beta_n^{\mathrm{cl}} = 0$ and one can choose the following secondary data with zero Roos classes. Vanishing of β_n^{cl} is equivalent to the existence of a compatible cohomology class

$$\alpha = (\alpha_M)_M \in \varprojlim_M H^1(\mathcal{X}^\bullet_{w,M}(I))$$

with

$$H^1(\Xi_M)(\alpha_M) = -[\mathfrak{o}^{\mathrm{ns}}_{n,M}] \quad \text{for every } M.$$

For such a choice of α, the representative-compatibility class

$$\mu_n^{\mathrm{cl}}(\alpha) \in \varprojlim\nolimits^1_M H^0(\mathcal{X}^\bullet_{w,M}(I))$$

vanishes. Explicitly, choose cocycles $w_M \in Z^1(\mathcal{X}^\bullet_{w,M}(I))$ representing α_M. Since α is compatible, for the cofinal successor tower one can choose $u_M \in \mathcal{X}^0_{w,M}(I)$ with

$$\rho^X_{M+1,M} w_{M+1} - w_M = d^X_M u_M.$$

With the Roos convention

$$\Delta_X((a_M)_M) = \big(\rho^X_{M+1,M} a_{M+1} - a_M\big)_M,$$

the class of $([u_M])_M$ in $\operatorname{coker} \Delta_X = \varprojlim\nolimits^1_M H^0(\mathcal{X}^\bullet_{w,M}(I))$ is $\mu_n^{\mathrm{cl}}(\alpha)$. Its vanishing is exactly the condition that the representatives w_M can be changed by degree-zero coboundaries to compatible cocycles $W_{n,M}$.

After choosing such compatible $W_{n,M}$, the counterterm compatibility class

$$\lambda_n^{\mathrm{cl}}(W) \in \varprojlim\nolimits^1_M H^0(\mathcal{Q}^\bullet_{w,M}(I))$$

vanishes. Namely, set

$$r_M = \mathfrak{o}^{\mathrm{ns}}_{n,M} + \Xi_M(W_{n,M}).$$

The class of r_M in $H^1(\mathcal{Q}^\bullet_{w,M}(I))$ is zero, so choose $c_M \in \mathcal{Q}^0_{w,M}(I)$ with $d_M c_M = -r_M$. The differences

$$\pi_{M+1,M} c_{M+1} - c_M \in Z^0(\mathcal{Q}^\bullet_{w,M}(I))$$

define, with the analogous Roos operator Δ_Q, the class

$$\lambda_n^{\mathrm{cl}}(W) = \big[\big([\pi_{M+1,M} c_{M+1} - c_M]\big)_M\big] \in \operatorname{coker} \Delta_Q$$

This class vanishes exactly when the primitives c_M can be changed by degree-zero cocycles and coboundaries to compatible counterterms $C_{n,M}$.

If the inverse system $\{H^0(\mathcal{X}^\bullet_{w,M}(I))\}_M$ is Mittag–Leffler, then $\mu_n^{\mathrm{cl}}(\alpha) = 0$ for every admissible lift α. If $\{H^0(\mathcal{Q}^\bullet_{w,M}(I))\}_M$ is Mittag–Leffler, then $\lambda_n^{\mathrm{cl}}(W) = 0$ for every compatible closed-exchange representative W. Under both Mittag–Leffler hypotheses, the compatible closed-exchange branch is obstructed only by β_n^{cl}. The counterterm-only criterion of Theorem 5.84 is the special case $\mathcal{X}^\bullet_{w,M}(I) = 0$. The admissible weight transports $R^M_{w,w'}$ identify the fixed-window image condition, β_n^{cl}, and the secondary Roos classes for w with the corresponding data for w'.

Proof. The fixed-window statement is the cohomology calculation. If the branch equation holds, then passing to H^1 gives $[\mathfrak{o}^{\mathrm{ns}}_{n,M}] = -H^1(\Xi_M)[W_{n,M}]$. Conversely, if the displayed image condition holds, choose a cocycle $W_{n,M}$ in the closed-exchange complex whose class maps to $-[\mathfrak{o}^{\mathrm{ns}}_{n,M}]$. Then $\mathfrak{o}^{\mathrm{ns}}_{n,M} + \Xi_M(W_{n,M})$ is a d_M-boundary, so a degree-zero $C_{n,M}$ gives the branch equation.

The compatibility of the residuals and the identity $\pi_{M,N}\Xi_M = \Xi_N\rho^X_{M,N}$ make $([\mathfrak{o}^{\mathrm{ns}}_{n,M}])_M$ a point of $\varprojlim_M H^1(Q^\bullet_{w,M}(I))$. Its cokernel class is zero exactly when a compatible cohomology class α in the closed-exchange tower maps to its negative. This proves the first inverse-limit obstruction.

Fix such an α. The differences $\rho^X_{M+1,M}w_{M+1} - w_M$ are closed degree-one elements with zero cohomology class, hence have the form $d^X_M u_M$. Changing the representatives w_M by degree-zero coboundaries changes $([u_M])_M$ by the Roos coboundary Δ_X. Thus the displayed class $\mu^{\mathrm{cl}}_n(\alpha)$ is independent of the choices. In the two-term Roos complex computing the first derived inverse limit of the tower $H^0(X^\bullet_{w,M}(I))$, its vanishing is exactly the equation making these differences exact after changing w_M. This is the exact condition for compatible cocycles $W_{n,M}$.

With compatible $W_{n,M}$, the cochains $r_M = \mathfrak{o}^{\mathrm{ns}}_{n,M} + \Xi_M(W_{n,M})$ are compatible d_M-cocycles whose windowwise H^1-classes vanish. Choose primitives c_M. Since $\pi_{M+1,M}r_{M+1} = r_M$, the differences $\pi_{M+1,M}c_{M+1} - c_M$ are d_M-closed of degree zero. Changing the primitives by closed degree-zero cochains, and by boundaries, changes their cohomology classes by the Roos coboundary Δ_Q. Hence the displayed $\lambda^{\mathrm{cl}}_n(W)$ is independent of primitive choices and vanishes exactly when the primitives can be adjusted to a compatible system $C_{n,M}$. This system is precisely the required compatible local counterterm family.

The last paragraph is the standard vanishing of $\varprojlim^1$ for Mittag–Leffler inverse systems of vector spaces. Therefore an H^0-Mittag–Leffler hypothesis on the closed-exchange tower kills the representative obstruction μ^{cl}_n, and the corresponding hypothesis on the non-scalar counterterm tower kills λ^{cl}_n. If both hold, only the cokernel H^1-class remains. If the closed-exchange tower is zero, the cokernel condition reduces to vanishing of $([\mathfrak{o}^{\mathrm{ns}}_{n,M}])_M$, and the only secondary class is the λ_n of Theorem 5.84. The weight-transport assertion follows because $R^M_{w,w'}$ is a compatible cochain isomorphism on both towers and intertwines Ξ_M, residuals, truncation maps, cocycle representatives, and primitive-difference cocycles. □

PROPOSITION 5.89 *(Regular-density quotient obstruction for the first residual).* Work in the regular-density tower of Definition 5.85 and Proposition 5.86. Let n_0 be the first $\hbar$-order at which the lower-order choices leave a compatible non-scalar residual

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} \in Z^1\big(Q^\bullet_{w,M}(I)\big), \qquad \pi_{M,N}\mathfrak{o}^{\mathrm{ns}}_{n_0,M} = \mathfrak{o}^{\mathrm{ns}}_{n_0,N}.$$

Put

$$\begin{aligned} C^{\bullet,\mathrm{reg}}_{w,M}(I) &= Q^\bullet_{w,M}(I)/X^{\bullet,\mathrm{reg}}_{w,M}(I), \\ \bar{d}_M &= d_M \bmod X^{\bullet,\mathrm{reg}}_{w,M}(I), \\ q_M \colon Q^\bullet_{w,M}(I) &\twoheadrightarrow C^{\bullet,\mathrm{reg}}_{w,M}(I). \end{aligned}$$

The maps $\pi_{M,N}$ and $R^M_{w,w'}$ descend to quotient cochain maps on $C^{\bullet,\mathrm{reg}}$. Define the finite-window quotient class

$$\eta^{\mathrm{reg}}_{n_0,M} = \big[q_M(\mathfrak{o}^{\mathrm{ns}}_{n_0,M})\big] \in H^1\big(C^{\bullet,\mathrm{reg}}_{w,M}(I)\big).$$

Then, for a fixed M,

$$[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}] \in -\operatorname{im} H^1(\Xi^{\mathrm{reg}}_M) \iff \eta^{\mathrm{reg}}_{n_0,M} = 0.$$

Equivalently, the regular-density closed-exchange branch at window M is solvable exactly when the image of the residual is exact in $Q^{\bullet}_{w,M}(I)/X^{\bullet,\mathrm{reg}}_{w,M}(I)$.

The compatible classes

$$\eta^{\mathrm{reg}}_{n_0} = \big(\eta^{\mathrm{reg}}_{n_0,M}\big)_M \in \varprojlim_M H^1\big(C^{\bullet,\mathrm{reg}}_{w,M}(I)\big)$$

are the quotient shadow of the closed-exchange obstruction

$$\beta^{\mathrm{cl,reg}}_{n_0} \in \operatorname{coker}\left(\varprojlim_M H^1(X^{\bullet,\mathrm{reg}}_{w,M}(I)) \xrightarrow{\varprojlim H^1(\Xi^{\mathrm{reg}}_M)} \varprojlim_M H^1(Q^{\bullet}_{w,M}(I))\right).$$

If $\eta^{\mathrm{reg}}_{n_0} \neq 0$, then $\beta^{\mathrm{cl,reg}}_{n_0} \neq 0$. On the zero locus of $\eta^{\mathrm{reg}}_{n_0}$, choose windowwise classes

$$\alpha_M \in H^1(X^{\bullet,\mathrm{reg}}_{w,M}(I)), \qquad H^1(\Xi^{\mathrm{reg}}_M)(\alpha_M) = -[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}].$$

Let

$$K^{\mathrm{reg}}_M = \ker\Big(H^1(\Xi^{\mathrm{reg}}_M) : H^1(X^{\bullet,\mathrm{reg}}_{w,M}(I)) \to H^1(Q^{\bullet}_{w,M}(I))\Big).$$

For the cofinal successor tower the differences

$$\rho^{X,\mathrm{reg}}_{M+1,M}\alpha_{M+1} - \alpha_M \in K^{\mathrm{reg}}_M$$

define a Roos class

$$\kappa^{\mathrm{reg}}_{n_0} \in \varprojlim{}^1_M K^{\mathrm{reg}}_M.$$

It is independent of the chosen windowwise lifts up to Roos coboundary, and

$$\beta^{\mathrm{cl,reg}}_{n_0} = 0 \iff \eta^{\mathrm{reg}}_{n_0} = 0 \text{ and } \kappa^{\mathrm{reg}}_{n_0} = 0.$$

In particular, suppose the branch is restricted to a complete regular-density subcomplex

$$Q^{\bullet,\mathrm{rd}}_{w,M}(I) \subset Q^{\bullet}_{w,M}(I)$$

on which

$$Q^{\bullet,\mathrm{rd}}_{w,M}(I) = X^{\bullet,\mathrm{reg}}_{w,M}(I)$$

compatibly with truncation and admissible weight transport, and assume $\mathfrak{o}^{\mathrm{ns}}_{n_0,M} \in Z^1(Q^{\bullet,\mathrm{rd}}_{w,M}(I))$ for all M. Then

$$\beta^{\mathrm{cl,reg}}_{n_0} = 0.$$

The canonical compatible choice $W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns}}_{n_0,M}$ and $C_{n_0,M} = 0$ has

$$\mu^{\mathrm{cl}}_{n_0} = 0, \qquad \lambda^{\mathrm{cl}}_{n_0} = 0.$$

Thus a regular-density equality $X^{\bullet,\mathrm{reg}} = Q^{\bullet,\mathrm{rd}}$ kills the first residual only for residuals lying in that specified subcomplex. Outside it, the exact obstruction is the quotient/Roos pair $(\eta^{\mathrm{reg}}_{n_0}, \kappa^{\mathrm{reg}}_{n_0})$ above.

Proof. Proposition 5.86 says that $\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$ is a complete d_M-stable subcomplex of $\mathcal{Q}^{\bullet}_{w,M}(I)$, and that truncation and admissible weight transport preserve this subcomplex. Therefore the quotient complexes and their transition maps are defined. Since the residuals commute with truncation, the classes $\eta^{\mathrm{reg}}_{n_0,M}$ form an inverse-system element; weight transport gives the same statement for w'.

For a fixed window, the short exact sequence of complexes

$$0 \longrightarrow \mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I) \xrightarrow{\Xi^{\mathrm{reg}}_M} \mathcal{Q}^{\bullet}_{w,M}(I) \xrightarrow{q_M} C^{\bullet,\mathrm{reg}}_{w,M}(I) \longrightarrow 0$$

gives the long exact cohomology sequence. Exactness at $H^1(\mathcal{Q}^{\bullet}_{w,M}(I))$ gives

$$\operatorname{im} H^1(\Xi^{\mathrm{reg}}_M) = \ker H^1(q_M).$$

The sign is irrelevant over $\mathbb{C}$. Hence the image criterion is exactly the vanishing of $[q_M(\mathfrak{o}^{\mathrm{ns}}_{n_0,M})]$. Equivalently, if $q_M(\mathfrak{o}^{\mathrm{ns}}_{n_0,M}) = -\bar{d}_M \bar{C}_M$, choose any lift C_M of $\bar{C}_M$. Then $\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + d_M C_M$ lies in $\mathcal{X}^{1,\mathrm{reg}}_{w,M}(I)$ and is closed, so $W_{n_0,M} = -(\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + d_M C_M)$ solves the branch equation. The converse follows by applying q_M to the branch equation.

Passing to the inverse limit, the quotient maps induce a map from the displayed cokernel to $\varprojlim_M H^1(C^{\bullet,\mathrm{reg}}_{w,M}(I))$, and the image of $\beta^{\mathrm{cl,reg}}_{n_0}$ is precisely $\eta^{\mathrm{reg}}_{n_0}$. Thus nonvanishing of $\eta^{\mathrm{reg}}_{n_0}$ forces nonvanishing of $\beta^{\mathrm{cl,reg}}_{n_0}$.

Assume now $\eta^{\mathrm{reg}}_{n_0} = 0$. Windowwise exactness gives the classes α_M. Since both $\rho^{X,\mathrm{reg}}_{M+1,M}\alpha_{M+1}$ and α_M map to $-[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}]$, their difference lies in K^{reg}_M. Changing the windowwise lift α_M by an element of K^{reg}_M changes the difference family by the Roos coboundary. Hence the class $\kappa^{\mathrm{reg}}_{n_0}$ is well defined. It vanishes exactly when the α_M can be adjusted to a compatible element of $\varprojlim_M H^1(\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I))$ mapping to $-([\mathfrak{o}^{\mathrm{ns}}_{n_0,M}])_M$. This is exactly the vanishing of the cokernel class $\beta^{\mathrm{cl,reg}}_{n_0}$.

On a subcomplex $\mathcal{Q}^{\bullet,\mathrm{rd}} = \mathcal{X}^{\bullet,\mathrm{reg}}$, the inclusion is the identity. If the residual lies in that subcomplex, then $W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns}}_{n_0,M}$ is a compatible d_M-cocycle in $\mathcal{X}^{1,\mathrm{reg}}_{w,M}(I)$, and $C_{n_0,M} = 0$ gives

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + \Xi^{\mathrm{reg}}_M(W_{n_0,M}) + d_M C_{n_0,M} = 0.$$

The representative differences defining $\mu^{\mathrm{cl}}_{n_0}$ are zero because the residuals are compatible, and the primitive differences defining $\lambda^{\mathrm{cl}}_{n_0}$ are zero because every primitive is chosen to be 0. This proves the saturated regular-density assertion. $\square$

PROPOSITION 5.90 *(First regular-density residual criterion).* The first non-scalar residual used in the closed-exchange criterion is the curvature coefficient determined by the chosen scalar-contact chain projection, the chosen lower-order counterterms, and the finite window M:

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} = [\hbar^{n_0}]\Big(\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar,M} + d_M C_{<n_0,M} + \frac{1}{2}\{C_{<n_0,M}, C_{<n_0,M}\}_{\hbar,M}\Big) \in Z^1\big(\mathcal{Q}^{\bullet}_{w,M}(I)\big),$$

where n_0 is the first order at which this compatible family survives the lower-order choices. Thus the residual lies in the saturated regular-density subcomplex

$$\mathcal{Q}^{\bullet,\mathrm{rd}}_{w,M}(I) = \mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$$

if and only if the following regular-density residual criterion is given for every finite window M, compatibly with $\pi_{M,N}$ and $R^M_{w,w'}$.

(R1) The displayed curvature coefficient is represented, in $Q^{1}_{w,M}(I)$, by a complete filtered sum whose image in every finite filtration quotient is finite:

$$\sum_{\Gamma,\ell} a_{\Gamma,\ell}\, W^{R,\mathrm{reg},M}_{\Gamma,L,w}(a_{\Gamma,\ell,\bullet}, b_{\Gamma,\ell,\bullet}dt; f_{\Gamma,\ell,\bullet}, \rho_{\Gamma,\ell,\bullet}) \;+\; d_M B^{\mathrm{reg}}_{n_0,M},$$

where all defect cotangent labels are smooth compactly supported densities $b_{\Gamma,\ell,j}(t)dt$, all coefficient labels lie in degree $\leq M$, and $B^{\mathrm{reg}}_{n_0,M}$ is a regular-density local counterterm in the graph/counterterm complex.

(R2) The representative has zero scalar-contact projection:

$$\widehat{\sigma}_{\mathrm{sc},M}(\mathfrak{o}^{\mathrm{ns}}_{n_0,M}) = 0$$

(R3) The equality with the curvature coefficient above holds in the local-functional complex $\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}^{M}_{w}|_{I}; A^{\mathrm{pp},w,M}_{\partial,\hbar}(I))$ before cohomology and before scalar reduction.

(R4) The representatives commute with finite-window truncation and admissible weight transport:

$$\pi_{M,N}\mathfrak{o}^{\mathrm{ns}}_{n_0,M} = \mathfrak{o}^{\mathrm{ns}}_{n_0,N}, \qquad R^{M}_{w,w'}\mathfrak{o}^{\mathrm{ns}}_{n_0,M,w} = \mathfrak{o}^{\mathrm{ns}}_{n_0,M,w'}$$

If this criterion holds, then

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} \in Z^{1}\big(X^{\bullet,\mathrm{reg}}_{w,M}(I)\big)$$

for every M. The closed-exchange equation is solved by the explicit compatible representatives

$$W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns}}_{n_0,M}, \qquad C_{n_0,M} = 0,$$

and hence

$$\eta^{\mathrm{reg}}_{n_0} = 0, \qquad \kappa^{\mathrm{reg}}_{n_0} = 0, \qquad \beta^{\mathrm{cl,reg}}_{n_0} = 0, \qquad \mu^{\mathrm{cl}}_{n_0} = 0, \qquad \lambda^{\mathrm{cl}}_{n_0} = 0$$

for this saturated branch.

When the criterion is unmet, the regular-density branch remains undecided. The exact computable obstruction is

$$\eta^{\mathrm{reg}}_{n_0,M} = \Big[q_M[\hbar^{n_0}]\Big(\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar,M} + d_M C_{<n_0,M} + \frac{1}{2}\{C_{<n_0,M}, C_{<n_0,M}\}_{\hbar,M}\Big)\Big] \in H^{1}\big(C^{\bullet,\mathrm{reg}}_{w,M}(I), \bar{d}_M\big).$$

Computing this class requires the finite-window coefficient array of $\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar,M}$, the lower-order counterterms $C_{<n_0,M}$, the scalar-contact chain projection $\widehat{\sigma}_{\mathrm{sc},M}$, and the reduction of the displayed degree-one cochain modulo the regular-density graph/counterterm subcomplex. If all these quotient classes vanish, choose quotient primitives $\bar{C}_{n_0,M}$ and lifts $C_{n_0,M} \in Q^{0}_{w,M}(I)$ with

$$q_M(\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + d_M C_{n_0,M}) = 0.$$

Then

$$x_{n_0,M} := \mathfrak{o}^{\mathrm{ns}}_{n_0,M} + d_M C_{n_0,M} \in Z^{1}\big(X^{\bullet,\mathrm{reg}}_{w,M}(I)\big), \qquad \alpha_M = [-x_{n_0,M}]$$

is a windowwise lift of $-[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}]$. The remaining cohomology-lift obstruction is precisely

$$\kappa^{\mathrm{reg}}_{n_0} = \Big[\big(\rho^{X,\mathrm{reg}}_{M+1,M}\alpha_{M+1} - \alpha_M\big)_M\Big] \in \varprojlim\nolimits^{1}_{M} \ker H^{1}(\Xi^{\mathrm{reg}}_{M}).$$

Only after this class vanishes does the regular-density branch have a compatible cohomology lift; the representative and primitive compatibility classes $\mu^{\mathrm{cl}}_{n_0}$ and $\lambda^{\mathrm{cl}}_{n_0}$ are then the secondary Roos classes of Theorem 5.88.

Proof. The formula for $\mathfrak{o}^{\mathrm{ns}}_{n_0,M}$ is the order n_0 specialization of Theorem 5.84. The Bianchi identity in the QME complex gives $d_M\mathfrak{o}^{\mathrm{ns}}_{n_0,M} = 0$, and the chosen scalar-contact chain projection places it in $\mathcal{Q}^{\bullet}_{w,M}(I)$.

Conditions (R1)–(R4) are exactly the definition of membership in the complete regular-density graph/counterterm subcomplex constructed in Definition 5.85 and Proposition 5.86. Hence the criterion is equivalent to $\mathfrak{o}^{\mathrm{ns}}_{n_0,M} \in Z^1(\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I))$ compatibly in the finite-window tower. Since Ξ^{reg}_M is inclusion on the saturated subcomplex, $W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns}}_{n_0,M}$ and $C_{n_0,M} = 0$ give

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + \Xi^{\mathrm{reg}}_M(W_{n_0,M}) + d_M C_{n_0,M} = 0$$

Compatibility of the residual family makes the Roos differences for W vanish, and the zero primitives make the counterterm Roos differences vanish. This proves the displayed vanishing of η, κ, β, μ, and λ.

This criterion gives the invariant operation beyond quotienting by $\mathcal{X}^{\bullet,\mathrm{reg}}$. Applying Proposition 5.89 to the explicit curvature coefficient gives the displayed $\eta^{\mathrm{reg}}_{n_0,M}$. Its computation needs precisely the coefficient array of the curvature coefficient, the lower-order counterterms entering it, the chain projection defining $\mathcal{Q}$, and the quotient reduction modulo $\mathcal{X}^{\mathrm{reg}}$. If $\eta^{\mathrm{reg}}_{n_0,M} = 0$, a quotient primitive gives $q_M(\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + d_M C_{n_0,M}) = 0$, hence $x_{n_0,M} \in \mathcal{X}^{1,\mathrm{reg}}_{w,M}(I)$. Closedness follows from $d_M\mathfrak{o}^{\mathrm{ns}}_{n_0,M} = 0$ and $d^2_M = 0$. The Roos differences of the resulting cohomology classes lie in $\ker H^1(\Xi^{\mathrm{reg}}_M)$ because both terms map to $-[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}]$. This is exactly the $\kappa^{\mathrm{reg}}_{n_0}$ obstruction of Proposition 5.89. □

PROPOSITION 5.91 *(Regular-density criterion for the evaluated curvature).* Work at a fixed finite window M. Assume the finite-window graph/QME system of Definition 8.21 is the regular-density system: every cotangent defect label is $B_j = b_j(t)dt \in \mathcal{D}^{\mathrm{reg}}_c(I)$, the finite graph list is closed under the d_M-boundary and BV-bracket terms appearing in the system, and the scalar-contact projection is zero on the chosen subcomplex. Let

$$\mathrm{Ob}^{\mathrm{red,reg}}_{w,\partial,\hbar,M} = \mathrm{ev}_{\mathcal{G}_M}\,\mathrm{Curv}_{\mathbf{K}}\big(\kappa^{\mathrm{graph,reg},M}_{\hbar,w,I}\big) \in \mathcal{Q}^1_{w,M}(I)$$

be the evaluated reduced curvature of the regular-density graph kernel. Suppose the lower-order counterterms are compatible and regular-density:

$$C^{\mathrm{reg}}_{<n_0,M} \in \mathcal{Q}^{0,\mathrm{rd}}_{w,M}(I) = \mathcal{X}^{0,\mathrm{reg}}_{w,M}(I), \qquad \pi_{M,N}C^{\mathrm{reg}}_{<n_0,M} = C^{\mathrm{reg}}_{<n_0,N}.$$

Put

$$\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M} = [\hbar^{n_0}]\Big(\mathrm{Ob}^{\mathrm{red,reg}}_{w,\partial,\hbar,M} + d_M C^{\mathrm{reg}}_{<n_0,M} + \frac{1}{2}\{C^{\mathrm{reg}}_{<n_0,M}, C^{\mathrm{reg}}_{<n_0,M}\}_{\hbar,M}\Big).$$

Then

$$\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M} \in Z^1\big(\mathcal{Q}^{\bullet,\mathrm{rd}}_{w,M}(I)\big) = Z^1\big(\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)\big)$$

compatibly with finite-window truncation and admissible weight transport.

More explicitly, after expanding the finite graph system at the coefficient of $\hbar^{n_0}$, there is a finite indexing set $\mathcal{A}^{\mathrm{reg}}_{n_0,M}$ and coefficients $\varepsilon_{\alpha,M} \in \mathbb{C}$ such that

$$\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M} = \sum_{\alpha\in\mathcal{A}^{\mathrm{reg}}_{n_0,M}} \varepsilon_{\alpha,M}\, G^M_{\alpha,w} + [\hbar^{n_0}]d_M C^{\mathrm{reg}}_{<n_0,M} + \frac{1}{2}[\hbar^{n_0}]\{C^{\mathrm{reg}}_{<n_0,M}, C^{\mathrm{reg}}_{<n_0,M}\}_{\hbar,M},$$

where each graph generator is of the form

$$G^M_{\alpha,w} = W^{R,\mathrm{reg},M}_{\Gamma_\alpha,L,w}(a_{\alpha,\bullet}, b_{\alpha,\bullet}dt; f_{\alpha,\bullet}, \rho_{\alpha,\bullet}), \qquad b_{\alpha,j} \in C^\infty_c(I),$$

or is one of the regular-density local counterterm generators attached to such a labelled graph. The last two displayed terms also lie in $\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$: the first by d_M-stability, the second by the bracket-closure clause in the finite-window graph/QME system.

Hence the regular-density residual criterion of Proposition 5.90 is given for this evaluated curvature. The closed-exchange solution is

$$W_{n_0,M} = -\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M}, \qquad C_{n_0,M} = 0,$$

and

$$\eta^{\mathrm{reg}}_{n_0} = \kappa^{\mathrm{reg}}_{n_0} = \beta^{\mathrm{cl,reg}}_{n_0} = \mu^{\mathrm{cl}}_{n_0} = \lambda^{\mathrm{cl}}_{n_0} = 0$$

on this regular-density branch.

If the chosen curvature contains non-regular labelled data, choose any linear splitting modulo $\mathcal{Q}^{\bullet,\mathrm{rd}}_{w,M}(I)$ and write

$$\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar,M} = \mathrm{Ob}^{\mathrm{red,reg}}_{w,\partial,\hbar,M} + \mathrm{Ob}^{\mathrm{red},\neg\mathrm{rd}}_{w,\partial,\hbar,M}, \qquad C_{<n_0,M} = C^{\mathrm{reg}}_{<n_0,M} + C^{\neg\mathrm{rd}}_{<n_0,M}.$$

The resulting quotient class is independent of the splitting. The non-regular summand is generated exactly by graph weights

$$W^{R,\mathrm{WF},M}_{\Gamma,L,w}(a_\bullet, B_\bullet; f_\bullet, \rho_\bullet) \quad \text{with some } B_j \notin \mathcal{D}^{\mathrm{reg}}_c(I),$$

by the corresponding distributional current counterterms, and by d_M- or BV-bracket terms involving such counterterms. Its quotient obstruction is the explicit component

$$\begin{aligned}\eta^{\mathrm{reg}}_{n_0,M} = \Big[q_M[\hbar^{n_0}]\Big(&\mathrm{Ob}^{\mathrm{red},\neg\mathrm{rd}}_{w,\partial,\hbar,M} + d_M C^{\neg\mathrm{rd}}_{<n_0,M}\\ &+ \{C^{\mathrm{reg}}_{<n_0,M}, C^{\neg\mathrm{rd}}_{<n_0,M}\}_{\hbar,M} + \frac{1}{2}\{C^{\neg\mathrm{rd}}_{<n_0,M}, C^{\neg\mathrm{rd}}_{<n_0,M}\}_{\hbar,M}\Big)\Big]\end{aligned}$$

in

$$H^1\big(\mathcal{Q}^{\bullet}_{w,M}(I)/\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)\big).$$

Thus a single wavefront-admissible but non-smooth cotangent label, or a distributional current counterterm forced by such a label, is the exact datum that leaves $Q^{\mathrm{rd}} = X^{\mathrm{reg}}$.

Proof. Proposition B.17 constructs $W^{R,\mathrm{reg},M}_{\Gamma,L,w}$ with smooth compactly supported density labels and proves compatibility with interval extension, finite-window truncation, and admissible weight transport. Definition 5.85 then puts all scalar-zero regular graph weights and their regular local counterterms in $\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$, and closes this space under d_M. The finite-window graph/QME system includes the BV-bracket terms forced by the finite graph differential, so the quadratic counterterm term remains in the same complete regular-density graph/counterterm complex. Hence every summand in the displayed $\hbar^{n_0}$-coefficient lies in $\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I)$.

The Bianchi identity in the convolution dg Lie algebra gives closedness of the first unsolved coefficient after the lower-order equations have been imposed. The scalar projection is zero on the chosen regular-density system, so this closed coefficient lies in $\mathcal{Q}^{\bullet}_{w,M}(I)$. The same truncation and weight-transport identities hold term by term for the graph weights, counterterms, differential, and bracket; therefore the residuals are compatible in the finite-window tower.

Since $Q^{\bullet,\mathrm{rd}}_{w,M}(I) = X^{\bullet,\mathrm{reg}}_{w,M}(I)$ on this branch and Ξ^{reg}_M is inclusion, the displayed $W_{n_0,M}$ and $C_{n_0,M} = 0$ give

$$\mathfrak{o}^{\mathrm{ns,reg}}_{n_0,M} + \Xi^{\mathrm{reg}}_M(W_{n_0,M}) + d_M C_{n_0,M} = 0.$$

Compatibility of the residual family makes the closed-exchange Roos differences vanish, and the primitive family is identically zero. The quotient class $\eta^{\mathrm{reg}}_{n_0}$, the lift class $\kappa^{\mathrm{reg}}_{n_0}$, and the three closed-exchange and counterterm obstruction classes therefore vanish.

For a non-regular label set, the quotient map q_M kills exactly the regular-density subcomplex. Expanding the same curvature expression into regular and non-regular summands therefore leaves only the four non-regular terms displayed in the formula for $\eta^{\mathrm{reg}}_{n_0,M}$: the non-regular graph curvature, the d_M-image of non-regular lower counterterms, the mixed bracket with a regular counterterm, and the purely non-regular bracket. The wavefront-admissible graph complex of Proposition B.19 is proved as a distributional graph complex. When some $B_j \notin \mathcal{D}^{\mathrm{reg}}_c(I)$, the term can enter the regular-density branch only if its displayed quotient class is killed by an explicit degree-zero primitive in the ambient distributional quotient complex. This proves the obstruction formula. □

THEOREM 5.92 *(Wavefront singular-current obstruction).* Fix a finite window M and a wavefront-admissible finite graph/QME system as in Proposition B.19. Put

$$\mathcal{S}_c(I) = \mathcal{D}'_c(I)/\mathcal{D}^{\mathrm{reg}}_c(I).$$

A non-smooth label $B \in \mathcal{D}'_c(I)$ with non-empty wavefront has nonzero class in $\mathcal{S}_c(I)$. This quotient class obstructs a canonical regular-density representative of B modulo $\mathcal{D}^{\mathrm{reg}}_c(I)$: if $B - B^{\mathrm{sm}} \in \mathcal{D}^{\mathrm{reg}}_c(I)$ for some $B^{\mathrm{sm}} \in \mathcal{D}^{\mathrm{reg}}_c(I)$, then B was already regular.

Let the first non-scalar wavefront residual be written as

$$\mathfrak{o}^{\mathrm{ns,WF}}_{n_0,M} = [\hbar^{n_0}]\Big(\mathrm{Ob}^{\mathrm{red,WF}}_{w,\partial,\hbar,M} + d_M C^{\mathrm{WF}}_{<n_0,M} + \frac{1}{2}\{C^{\mathrm{WF}}_{<n_0,M}, C^{\mathrm{WF}}_{<n_0,M}\}_{\hbar,M}\Big) \in Z^1\big(Q^{\bullet}_{w,M}(I)\big),$$

with at least one cotangent label in $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M) \setminus \mathcal{D}^{\mathrm{reg}}_c(I)$ or at least one distributional current counterterm. Its singular-current component is

$$\begin{aligned}\mathfrak{s}^{\mathrm{WF}}_{n_0,M} = q_M[\hbar^{n_0}]\Big(&\mathrm{Ob}^{\mathrm{red},\neg\mathrm{rd}}_{w,\partial,\hbar,M} + d_M C^{\neg\mathrm{rd}}_{<n_0,M} \\ &+ \{C^{\mathrm{reg}}_{<n_0,M}, C^{\neg\mathrm{rd}}_{<n_0,M}\}_{\hbar,M} + \frac{1}{2}\{C^{\neg\mathrm{rd}}_{<n_0,M}, C^{\neg\mathrm{rd}}_{<n_0,M}\}_{\hbar,M}\Big) \in C^{1,\mathrm{reg}}_{w,M}(I),\end{aligned} \tag{28}$$

where

$$C^{\bullet,\mathrm{reg}}_{w,M}(I) = Q^{\bullet}_{w,M}(I)/X^{\bullet,\mathrm{reg}}_{w,M}(I).$$

The obstruction class is exactly

$$\eta^{\mathrm{reg}}_{n_0,M} = [\mathfrak{s}^{\mathrm{WF}}_{n_0,M}] \in H^1\big(C^{\bullet,\mathrm{reg}}_{w,M}(I)\big).$$

More concretely, for a graph summand $W^{R,\mathrm{WF},M}_{\Gamma,L,w}(a_\bullet, B_\bullet; f_\bullet, \rho_\bullet)$, choose arbitrary smooth densities $B^{\mathrm{sm}}_j \in \mathcal{D}^{\mathrm{reg}}_c(I)$ and put $S_j = B_j - B^{\mathrm{sm}}_j$. By multilinearity,

$$q_M W^{R,\mathrm{WF},M}_{\Gamma,L,w}(a_\bullet, B_\bullet; f_\bullet, \rho_\bullet) = q_M \sum_{\emptyset \neq J \subset \{1,\dots,r\}} W^{R,\mathrm{WF},M}_{\Gamma,L,w}(a_\bullet, B^J_\bullet; f_\bullet, \rho_\bullet),$$

where $B^J_j = S_j$ for $j \in J$ and $B^J_j = B^{\mathrm{sm}}_j$ for $j \notin J$. This expression is independent of the chosen smooth densities after passing to $\mathcal{C}^{\bullet,\mathrm{reg}}_{w,M}(I)$. The zero-internal-edge case, whenever the generator lies in the non-scalar kernel, is the current class

$$q_M \iota_I(\Theta^w_\rho(B)) = [B] \otimes \rho \in \mathcal{S}_c(I) \widehat{\otimes} D_{w,\le M},$$

so a singular cotangent current with nonzero coefficient is already a nonzero regular-density quotient generator.

The wavefront of $\mathfrak{s}^{\mathrm{WF}}_{n_0,M}$ is contained in the finite union obtained from the graph pushforward of

$$\mathrm{WF}(K^M_{\Gamma,\alpha}) + \mathrm{WF}(B_\Gamma)$$

over the Hadamard coefficients appearing at order n_0, together with the local counterterm wavefronts supported on

$$\bigcup_{\Delta \in \mathrm{Diag}(\Gamma)} \Delta \cap \mathrm{supp}(B_\Gamma).$$

Thus the component in (28) is the exact wavefront/counterterm component left after quotienting by the regular-density graph complex.

A finite-window counterterm kills the non-regular branch if and only if there is a degree-zero distributional counterterm $C^{\mathrm{sing}}_{n_0,M} \in \mathcal{Q}^0_{w,M}(I)$ such that

$$\bar{d}_M\, q_M(C^{\mathrm{sing}}_{n_0,M}) = -\,\mathfrak{s}^{\mathrm{WF}}_{n_0,M}.$$

In that case $\eta^{\mathrm{reg}}_{n_0,M} = 0$, and the remaining compatibility obstruction is the Roos class $\kappa^{\mathrm{reg}}_{n_0}$ of Proposition 5.89. If every such distributional primitive is obstructed, then

$$\eta^{\mathrm{reg}}_{n_0,M} \neq 0,$$

and the regular-density closed-exchange branch has a smoothing obstruction. The honest enlargement is a new wavefront-labelled closed-exchange complex distinct from $\mathcal{X}^{\bullet,\mathrm{reg}}$.

Proof. A compactly supported smooth density has empty wavefront set. Therefore a compactly supported distribution with non-empty wavefront has nonzero class modulo smooth densities. This proves the label-level obstruction to canonical smoothing. The zero-internal-edge generator shows that this obstruction is already visible in the current target: $\Theta^w_\rho(B)$ records the distribution B linearly in $\mathcal{D}'_c(I)\widehat{\otimes} D_{w,\le M}$, and the quotient by $\mathcal{X}^{\bullet,\mathrm{reg}}$ kills only the smooth-density subspace.

Proposition B.19 constructs the finite-window wavefront graph weights and the corresponding distributional current counterterms. The regular-density subcomplex of Definition 5.85 contains exactly the smooth-label graph weights, their regular counterterms, and their d_M-closures. Hence applying q_M to the first residual removes precisely the regular-density terms and leaves the four non-regular terms displayed in Formula (28). Changing the auxiliary smooth densities B^{sm}_j changes the expansion only by graph terms with all labels smooth; these lie in $\mathcal{X}^{\bullet,\mathrm{reg}}$, so the quotient expression is independent of the splitting.

Closedness follows from the same Bianchi identity used in Proposition 5.91, after passing to the quotient complex. The wavefront and support containment is exactly the Hörmander product and extension statement in Proposition B.19: graph terms have wavefront contained in the pushforward of the product wavefronts, and local extension ambiguities are supported on the collision diagonals meeting $\mathrm{supp}(B_\Gamma)$.

Finally, $H^1(C^{\bullet,\mathrm{reg}}_{w,M}(I), \bar{d}_M)$ is the cohomology of the quotient complex. Therefore the class represented by $\mathfrak{s}^{\mathrm{WF}}_{n_0,M}$ vanishes exactly when it is a $\bar{d}_M$-boundary, equivalently exactly when a degree-zero distributional counterterm $C^{\mathrm{sing}}_{n_0,M}$ with the displayed equation exists. For a non-boundary class, the quotient obstruction $\eta^{\mathrm{reg}}_{n_0,M}$ is nonzero and the regular-density branch is obstructed. □

THEOREM 5.93 *(Distributional primitive cone for the wavefront quotient).* Keep the hypotheses and notation of Theorem 5.92, and assume that the chosen wavefront graph/QME systems are compatible under finite-window truncation and admissible weight transport. Thus

$$\pi_{M,N}\mathfrak{o}^{\mathrm{ns,WF}}_{n_0,M} = \mathfrak{o}^{\mathrm{ns,WF}}_{n_0,N}, \qquad R^M_{w,w'}\mathfrak{o}^{\mathrm{ns,WF}}_{n_0,M,w} = \mathfrak{o}^{\mathrm{ns,WF}}_{n_0,M,w'}.$$

Let

$$\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M} := \mathfrak{o}^{\mathrm{ns,WF}}_{n_0,M} \in Z^1\big(\boldsymbol{Q}^\bullet_{w,M}(I)\big)$$

be the closed residual lift of $\mathfrak{s}^{\mathrm{WF}}_{n_0,M}$, so that $q_M\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M} = \mathfrak{s}^{\mathrm{WF}}_{n_0,M}$.

Define the local wavefront-primitive extension by

$$\widetilde{\boldsymbol{Q}}^{\bullet,\mathrm{WFpr}}_{w,M}(I) = \boldsymbol{Q}^\bullet_{w,M}(I)\widehat{\oplus}\mathbb{C}[[\hbar]]\cdot p^{\mathrm{WF}}_{n_0,M}, \qquad |p^{\mathrm{WF}}_{n_0,M}| = 0,$$

completed in the same $\hbar$-adic, scalar-contact, and finite-window filtrations as $\boldsymbol{Q}^\bullet_{w,M}(I)$. The new generator is placed in $\hbar$-order n_0, in the scalar-contact filtration of $\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M}$, and with local support and wavefront label equal to the support and wavefront cone displayed in Theorem 5.92. Set

$$\widetilde{d}_M|_Q = d_M, \qquad \widetilde{d}_M\, p^{\mathrm{WF}}_{n_0,M} = -\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M}.$$

Let

$$\widetilde{C}^{\bullet,\mathrm{reg}}_{w,M}(I) = \widetilde{\boldsymbol{Q}}^{\bullet,\mathrm{WFpr}}_{w,M}(I)/\mathcal{X}^{\bullet,\mathrm{reg}}_{w,M}(I),$$

and write again q_M for the quotient map and $\bar{d}_M$ for the induced differential. Then $\widetilde{\bar{d}}^2_M = 0$, the inclusion $\boldsymbol{Q}^\bullet_{w,M}(I) \hookrightarrow \widetilde{\boldsymbol{Q}}^{\bullet,\mathrm{WFpr}}_{w,M}(I)$ is a filtered cochain map, and the element

$$C^{\mathrm{sing}}_{n_0,M} := p^{\mathrm{WF}}_{n_0,M}$$

satisfies the quotient primitive equation

$$\bar{d}_M q_M(C^{\mathrm{sing}}_{n_0,M}) = -\mathfrak{s}^{\mathrm{WF}}_{n_0,M}.$$

The extension is strict Mittag–Leffler on compatible finite-window towers: define

$$\widetilde{\pi}_{M,N}\, p^{\mathrm{WF}}_{n_0,M} = p^{\mathrm{WF}}_{n_0,N}$$

and let $\widetilde{\pi}_{M,N}$ be $\pi_{M,N}$ on $\boldsymbol{Q}^\bullet_{w,M}(I)$. For admissible weights set

$$\widetilde{R}^M_{w,w'}\, p^{\mathrm{WF}}_{n_0,M,w} = p^{\mathrm{WF}}_{n_0,M,w'}$$

and let $\widetilde{R}^M_{w,w'}$ be $R^M_{w,w'}$ on $\boldsymbol{Q}^\bullet_{w,M}(I)$. These maps are continuous filtered cochain maps and commute with the quotient to $\widetilde{C}^{\bullet,\mathrm{reg}}$.

The new generator $p^{\mathrm{WF}}_{n_0,M}$ lives in the extended complex, outside the original Costello counterterm complex $\boldsymbol{Q}^\bullet_{w,M}(I)$, and the construction keeps a singular current distinct from a smooth density. A primitive inside $\boldsymbol{Q}^\bullet_{w,M}(I)$ exists exactly under the vanishing condition $\eta^{\mathrm{reg}}_{n_0,M} = 0$ of Theorem 5.92.

Proof. The residual $\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M}$ is d_M-closed by the Bianchi identity used in Theorem 5.92. Hence

$$\widetilde{d}^2_M\, p^{\mathrm{WF}}_{n_0,M} = -\, d_M \widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M} = 0,$$

and $\widetilde{d}^2_M = 0$ on the old summand because $d^2_M = 0$. The image of the new generator has the same local support, wavefront cone, $\hbar$-order, scalar-contact order, and finite coefficient window assigned to the generator. Therefore the differential is continuous for the completed filtration topology.

Since $q_M \widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M} = \mathfrak{s}^{\mathrm{WF}}_{n_0,M}$, applying the quotient map to the defining differential gives

$$\bar{d}_M q_M (p^{\mathrm{WF}}_{n_0,M}) = q_M \widetilde{d}_M\, p^{\mathrm{WF}}_{n_0,M} = -\,\mathfrak{s}^{\mathrm{WF}}_{n_0,M}.$$

This is the displayed primitive equation.

The truncation and weight-transport formulas are cochain identities because the closed residual lifts are compatible:

$$\widetilde{\pi}_{M,N} \widetilde{d}_M\, p^{\mathrm{WF}}_{n_0,M} = -\,\pi_{M,N} \widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M} = -\,\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,N} = \widetilde{d}_N \widetilde{\pi}_{M,N}\, p^{\mathrm{WF}}_{n_0,M},$$

and similarly

$$\widetilde{R}^M_{w,w'} \widetilde{d}_M\, p^{\mathrm{WF}}_{n_0,M,w} = -\, R^M_{w,w'} \widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M,w} = -\,\widehat{\mathfrak{s}}^{\mathrm{WF}}_{n_0,M,w'} = \widetilde{d}_M \widetilde{R}^M_{w,w'}\, p^{\mathrm{WF}}_{n_0,M,w}.$$

On the old summand these are the already proved identities for $\mathcal{Q}^\bullet_{w,M}(I)$. The transition maps are surjective on the single new generator and are the old strict Mittag–Leffler maps on $\mathcal{Q}$, so the extended tower is again strict Mittag–Leffler. The same formulas descend to the quotient because $\mathcal{X}^{\bullet,\mathrm{reg}}$ is preserved by truncation and weight transport.

The last assertion is formal. The construction adjoins a new degree-zero generator whose differential is the closed wavefront residual. It leaves the quotient $\mathcal{D}'_c(I)/\mathcal{D}^{\mathrm{reg}}_c(I)$ unchanged on the original current labels, and therefore leaves singular distributions singular. In the original complex alone, the primitive equation is precisely the statement that $\mathfrak{s}^{\mathrm{WF}}_{n_0,M}$ is a boundary in the original quotient complex, i.e. that $\eta^{\mathrm{reg}}_{n_0,M}$ vanishes. □

THEOREM 5.94 *(Strict unweighted endpoint obstruction).* Let

$$\mathfrak{h}_\Pi = \prod_{d\geq 1} H_d, \qquad D_\oplus = \bigoplus_{d\geq 1} H_d^\vee$$

be the original product/direct-sum Tate pair. The support obstruction excludes a tensor $K \in \mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$, and likewise a tensor in $D_\oplus \widehat{\otimes} \mathfrak{h}_\Pi$, whose projection to every finite window $d \leq w_0$ is the finite Casimir

$$C^{w_0}_{\mathfrak{h}} = \sum_{d\leq w_0} \sum_i e_{d,i} \otimes e^i_d.$$

Consequently the strict unweighted pair has the endpoint obstruction for a finite-window Costello regulator theory satisfying the coefficient kernel requirement (C1) and the windowwise compatibility used in Theorem 5.82. Equivalently, the support obstruction persists for every $q \to 1^+$ family of weighted finite-window kernels in $\mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$ with projections $C^{w_0}_{\mathfrak{h}}$.

Moreover, for any admissible weight w, the inclusion $D_\oplus \hookrightarrow D_{\Pi,w} := \prod_{d\geq 1} w(d)^{-1} H_d^\vee$ is only an admissible filtered weak equivalence. Its cofiber has nonzero H^0, so any observable or obstruction class detecting the quotient $D_{\Pi,w}/D_\oplus$ is outside the proved regulator-independent sector.

Proof. The direct-sum factor $D_\oplus$ is coefficient-finite by Lemma 5.166 item (1): every element of a completed tensor product with $D_\oplus$ has support in only finitely many $H_d^\vee$-degrees. Projection of K to $C_{\mathfrak{h}}^{w_0}$ for all w_0 forces a nonzero $H_d^\vee$-component of K in every degree d. This is infinite support in the direct-sum direction, a contradiction. The same argument applies after reversing the tensor factors.

Costello's heat-kernel BV propagator for the cotangent coefficient field is $P_{\epsilon,L}^{\mathrm{base}}\widehat{\otimes}K$. Since the support obstruction removes K from the strict product/direct-sum pair, the strict pair has an operator-level heat homotopy while the coefficient distributional inverse pairing and graphwise regulator endpoint are obstructed.

Suppose a $q \to 1^+$ endpoint existed in the strict pair. Its finite-window projection is forced to be the limit of the weighted finite-window kernels. After the weighted bases are identified with the unweighted basis, every such finite-window kernel is exactly $C_{\mathfrak{h}}^{w_0}$, independent of q. Such an endpoint therefore gives a tensor K with the obstructed projections above. This proves the endpoint assertion.

Finally, with zero differential,

$$H^0(D_{\Pi,w}/D_\oplus) = D_{\Pi,w}/D_\oplus.$$

The class of any sequence with one nonzero covector in every degree is nonzero in this quotient. Therefore the completion changes ordinary cohomology even though finite quotients and associated graded pieces have zero image of the tail. This proves the asserted boundary of regulator independence. □

Definition 5.95 *(Endpoint-admissible coefficient module).* An endpoint for the weighted regulators $w_q(d) = q^d$, $q > 1$, is *admissible* if it consists of a complete Hausdorff coefficient pair $(\mathfrak{h}_*, D_*)$, finite quotients $(\mathfrak{h}_*, D_*)^{w_0}$, and diagonal transport maps from every weighted window to the endpoint window, satisfying:

(E1) the finite quotients form strict Mittag-Leffler towers with surjective transition maps and endpoint quotients $H_{\le w_0} \oplus H_{\le w_0}^\vee$;

(E2) there is a coefficient kernel $C_* \in \mathfrak{h}_*\widehat{\otimes}D_*$ whose projection to $d \le w_0$ is $C_{\mathfrak{h}}^{w_0}$ for every w_0;

(E3) the endpoint transport maps carry differential, Poisson bracket, coadjoint action, propagator contraction, interaction, counterterm representatives, and the CE/PV central-operation generator assignment $c \mapsto \theta$, $u \mapsto O$ to their endpoint versions;

(E4) local observables compute the admissible filtered cohomology of Statement 5.18, so exactness of the strict Mittag-Leffler limit applies;

(E5) the Costello RG recursion $W(P_{\epsilon,L}, I)$ preserves the endpoint tower and the endpoint counterterms.

THEOREM 5.96 *(Endpoint stabilization criterion).* Let $w_q(d) = q^d$, $q > 1$. If there is an endpoint-admissible coefficient module in the sense of Definition 5.95, then the transport maps assemble to isomorphisms

$$R_{q,*}\colon \varprojlim_{w_0}\left(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_{w_q}^{w_0}), Q + \{I_0^{w_q}, -\}\right) \xrightarrow{\cong} \varprojlim_{w_0}\left(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_*^{w_0}), Q + \{I_0^*, -\}\right)$$

for all $q > 1$, functorially in q. Hence finite-degree local observables, fixed-graph Costello contractions, RG counterterm representatives, and the mixed brane-defect QME obstruction class stabilize at the endpoint. More precisely, if O_M^q is represented in the finite window $d \le M$, then its endpoint representative is

$$O_M^* = R_{q,*}^M O_M^q$$

and is independent of q after the common finite-window identification with $H_{\le M} \oplus H^{\vee}_{\le M}$. The notation $q \to 1^+$ is therefore stabilization of this finite-window representative inside the endpoint module. It is a finite-window stabilization in place of strict product/direct-sum convergence. In particular

$$R_{q,*}(\mathrm{Ob}_q) = \mathrm{Ob}_*, \qquad \mathrm{Ob}_q = 0 \iff \mathrm{Ob}_* = 0.$$

Proof. Replace the second admissible weight w' in Theorem 5.82 by the endpoint object. Conditions (E1)–(E5) are exactly the hypotheses used there: strict Mittag-Leffler exactness, existence of the coefficient kernel, transport of the algebraic and graphwise structures, admissible filtered cohomology, and preservation by RG recursion. The same finite-window diagonal maps therefore give isomorphisms of strict inverse systems. Passing to the exact inverse limit identifies cohomology and sends the obstruction cocycle to the endpoint obstruction cocycle. Functoriality follows from the composition law of the transport maps on each finite window.

For a representative supported in $d \le M$, the endpoint statement is already finite-dimensional: $R^M_{q,*}$ is the diagonal map from the q-basis to the endpoint basis, and the endpoint transport commutes with the unweighting maps by condition (E3). Thus the finite-window coefficient array is q-independent. The inverse-limit assertion adds exactly the strict Mittag-Leffler compatibility required by (E1) and exactness in the admissible envelope. □

THEOREM 5.97 *(Finite-window $q \to 1^+$ stabilization of the CE/PV generator map).* Let $w_q(d) = q^d$, $q > 1$, and let

$$\Phi^q_M : C^\bullet_{\mathrm{CE}}\big(\mathfrak{h}_{q,\le M} \ltimes D_{q,\le M}[1]\big) \longrightarrow \mathrm{PV}\big(\mathbf{S}(\mathfrak{h}_{q,\le M})\big)$$

be the finite-window restriction of the weighted CE/PV generator map. Let $\Phi^{0,M}$ be the finite-dimensional unweighted CE/PV map on $(\mathfrak{h}_{\le M}, D_{\le M})$. Then

$$U^M_q\, \Phi^q_M = \Phi^{0,M}\, U^M_q$$

as cochain P_0-algebra maps. Consequently the unweighted finite-window representative $U^M_q \Phi^q_M (U^M_q)^{-1}$ is independent of q and has $q \to 1^+$ limit $\Phi^{0,M}$.

If an endpoint-admissible coefficient module exists, the same identity passes to the strict Mittag-Leffler inverse limit: $R_{q,*}\Phi^q = \Phi^* R_{q,*}$. For the original product/direct-sum endpoint the obstruction term is precisely

$$\mathfrak{o}_{\mathrm{Cas}} = \big(C^M_{\mathfrak{h}}\big)_{M\ge 1} \notin \mathfrak{h}_\Pi \widehat{\otimes} D_\oplus,$$

equivalently the nonexistence of a tensor whose finite-window projections are all $C^M_{\mathfrak{h}}$.

Proof. On generators, Φ^q_M sends the CE coordinate dual to $q^d e_{d,i}$ to the Schouten vector coordinate dual to $q^d e_{d,i}$, and sends the CE coordinate dual to $q^{-d} e^i_d[1]$ to the Hamiltonian coordinate $O^q_{d,i}$. The map U^M_q removes the same diagonal weight on both the CE and PV sides. Therefore the square commutes on the generators c, u, θ, O. Since both CE and Schouten differentials are derivations and their structure constants are transported by the same diagonal conjugation, the generator check gives the equality of dg P_0-algebra maps.

The displayed equality says that the finite-window map, after being written in the unweighted basis, is exactly the finite CE/PV map $\Phi^{0,M}$; it is q-independent. Hence the $q \to 1^+$ limit exists on every finite window.

Under Definition 5.95, the endpoint maps commute with finite-window transition maps and carry the CE/PV generator assignment to the endpoint assignment. Exactness of the strict Mittag-Leffler inverse limit then passes the finite-window identity to the endpoint. In the original product/direct-sum pair, condition (E2) is violated by Theorem 5.94; the obstruction is the compatible family of finite Casimirs whose global tensor is obstructed in $\mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$. □

COROLLARY 5.98 *(Strict direct-sum endpoint obstruction criterion).* The strict product/direct-sum pair $(\mathfrak{h}_\Pi, D_\oplus)$ violates the endpoint-admissibility criterion exactly at condition (E2): the compatible family of finite Casimirs has an obstructed representing coefficient kernel in $\mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$.

Proof. Condition (E2) is exactly the coefficient-kernel condition ruled out by Theorem 5.94. □

PROPOSITION 5.99 *(The completion changes ordinary cohomology).* Let

$$D_\oplus = \bigoplus_{d \geq 1} H_d^\vee, \qquad D_{\Pi,w} = \prod_{d \geq 1} w(d)^{-1} H_d^\vee$$

with zero differential. The finite-window inclusion $D_\oplus \hookrightarrow D_{\Pi,w}$ induces an isomorphism on every quotient by the tail $d > w_0$, and therefore is a weak equivalence in the admissible filtered sense of Statement 5.18. Its ordinary cofiber has nonzero degree-zero cohomology:

$$H^0(D_{\Pi,w}/D_\oplus) = D_{\Pi,w}/D_\oplus \neq 0,$$

for example the class of any sequence with one nonzero covector in every degree is nonzero. Thus the weighted product contains tail states omitted by the strict continuous dual. Any observable detecting such a tail is outside the regulator-admissible sector of Definition 5.76.

Example 5.100 *(Super-exponential bracket obstruction).* The super-exponential function $w(d) = \exp(d^2)$ violates the degree weight condition: the ratio

$$\frac{w(p+q-2)}{w(p)w(q)} = \exp\big((p+q-2)^2 - p^2 - q^2\big)$$

is unbounded along $p = q \to \infty$, so condition (W1) is violated. The Poisson bracket violates bracket-tameness in that coefficient topology, and the graphwise constants in Proposition 5.69 can be unbounded in the degrees used by the interaction. Theorem 5.82 applies to the admissible weighted theory.

Nilpotent truncation $\mathfrak{o}_{\mathrm{nilp}}$. The truncation $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ is finite-dimensional, nilpotent of class $\leq N-2$, and supports the finite classical CE/PV identification of 5.102. The recovery of the full Hamiltonian source from this nilpotent inverse system $\{\mathfrak{h}_{\leq N}\}_N$ has its obstruction in the low-degree sector $\mathfrak{m}/\mathfrak{m}^3 = H_1 \oplus H_2$; the obstruction $\mathfrak{o}_{\mathrm{nilp}}$ is the triple

$$\mathfrak{o}_{\mathrm{nilp}} = \big(\mathfrak{o}_N^{\mathrm{lin}},\ \mathfrak{o}^{\mathrm{ss}},\ \mathfrak{o}_{\leq 2}^{\mathrm{Cas}}\big)$$

of Definition 5.106: the linear-Hamiltonian tail-noninvariance defect $\mathfrak{o}_N^{\mathrm{lin}}(z_1; z_2^{N+1}) = (N+1)z_2^N$, the semisimple bracket im $\mathfrak{o}^{\mathrm{ss}} = H_2 \cong \mathfrak{sl}_2$ preventing lower-central Tate-pronilpotence, and the missing low-degree Casimir component $\mathfrak{o}_{\leq 2}^{\mathrm{Cas}}$ outside any finite $\mathfrak{m}$-adic window. By Lemma 5.107 a finite $\mathfrak{m}$-adic tower extending $\{\mathfrak{h}_{\leq N}\}_N$ to include $H_1 \oplus H_2$ exists exactly when these three components vanish. Since the displayed components are nonzero, the nilpotent window cannot recover the full source. The replacement datum is a coefficient category in which the full Casimir family of 5.109 converges, together with a relative or reductive replacement for the lower-central Tate-pronilpotent hypothesis treating the semisimple $\mathfrak{sl}_2 \subset H_2$ directly. This is separate from the completed formal-disk CE/PV map of Theorem 5.6. At quantum level the same finite truncation is obstructed by 5.103: the Moyal bracket of z_1^4, z_2^4 carries the $\hbar^2$-residue $24\hbar^2 z_1 z_2 \notin \mathfrak{m}^3$, so a finite quantum CE source on $\mathfrak{h}_{\leq N}$ requires a Moyal-flat truncation or a projected bracket with new Jacobi proof.

Nilpotent truncation of the cotangent module

The full CE/PV trace-sector map of Theorem 5.6 is a completed formal-Darboux statement for the reduced Hamiltonian Lie algebra $\mathfrak{h} = \mathfrak{m} = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$. This subsection studies a different construction: the lower-central pronilpotent finite-window route. That route starts with the cubic Lie subalgebra $\mathfrak{m}^3 \subset \mathfrak{h}$. It does not prove the full formal-Darboux CE/PV map, and it does not obstruct that map. It records the obstruction to recovering the full Hamiltonian source from nilpotent $\mathfrak{m}$-adic quotients. The obstruction to this nilpotent recovery is the explicit triple $\left(\mathfrak{o}_N^{\mathrm{lin}}, \mathfrak{o}^{\mathrm{ss}}, \mathfrak{o}_{\leq 2}^{\mathrm{Cas}}\right)$.

For every $N \geq 1$,

$$\{z_1, z_2^{N+1}\} = (N+1)z_2^N \notin \mathfrak{m}^{N+1}, \qquad \{z_1^2, z_2^2\} = 4z_1z_2, \qquad [H_2, H_2] = H_2 \cong \mathfrak{sl}_2.$$

The first identity is the linear-tail obstruction $\mathfrak{o}_N^{\mathrm{lin}}(z_1; z_2^{N+1}) = (N+1)z_2^N$, ruling out $\mathfrak{m}^{N+1}$ as a Lie ideal once $z_1 \in H_1$ is admitted. The last two encode the semisimple obstruction $\mathfrak{o}^{\mathrm{ss}} \colon \Lambda^2 H_2 \to H_2$ with image $\mathfrak{sl}_2$, ruling out nilpotence of H_2. Together with the low-degree Casimir component $\mathfrak{o}_{\leq 2}^{\mathrm{Cas}} = \sum_i e_{1,i} \otimes e_1^i + \sum_i e_{2,i} \otimes e_2^i$, these form the obstruction triple of Definition 5.106.

On the cubic ideal $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$, where the obstruction triple is silent, the finite cotangent CE/PV identity $\Phi_{\leq N} \colon C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\leq N,I}) \xrightarrow{\sim} \mathrm{PV}(\widehat{\mathbf{S}}(\mathfrak{h}_{\leq N,I}))$ is the cubic-truncation trace-sector model for $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_b$, proved in the finite classical sector (Theorem 5.102). Theorem 5.103 excludes the all-order Moyal lift on the same finite truncation; Theorem 5.108 shows that the inverse limit of this nilpotent tower recovers the proper subalgebra $\mathfrak{m}^3 \subsetneq \mathfrak{h}$. The full formal-Darboux CE/PV map is given elsewhere by the completed coordinate construction. Problem 5.110 is the narrower problem of replacing the nilpotent finite-window route by a relative or reductive route that includes $H_1 \oplus H_2$ without using false pronilpotence.

The truncation

Let $\mathfrak{h} = (z_1, z_2)\mathbb{C}[[z_1, z_2]] = \mathfrak{m}$ be the reduced Hamiltonian Lie algebra of the formal symplectic polydisk on $\mathbb{C}^2$, with bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ modulo constants. Let $\mathfrak{m}^k = (z_1, z_2)^k\mathbb{C}[[z_1, z_2]]$ for $k \geq 1$ be the k-th power of the maximal ideal. Fix integers $N \geq k \geq 3$ and define the *nilpotent truncation*

$$\mathfrak{h}_{\leq N}^{(k)} := \mathfrak{m}^k/\mathfrak{m}^{N+1},$$

with $\mathfrak{m}$-adic Poisson bracket inherited from the ambient $\mathfrak{h}_{\mathrm{poly}}$. The standard choice is $k = 3$:

$$\mathfrak{h}_{\leq N} := \mathfrak{h}_{\leq N}^{(3)} = \mathfrak{m}^3/\mathfrak{m}^{N+1}.$$

Concretely, $\mathfrak{h}_{\leq N}$ is the finite-dimensional vector space spanned by monomials $z_1^a z_2^b$ with $3 \leq a + b \leq N$.

The cutoff lies at $k = 3$. At $k = 2$, the degree-2 piece $H_2 = \mathrm{span}\{z_1^2, z_1z_2, z_2^2\}$ closes under the Poisson bracket and carries the $\mathfrak{sl}_2$-relations

$$\{z_1^2, z_2^2\} = 4z_1z_2, \quad \{z_1^2, z_1z_2\} = 2z_1^2, \quad \{z_1z_2, z_2^2\} = 2z_2^2,$$

so $\mathfrak{m}^2/\mathfrak{m}^{N+1}$ contains a nonabelian semisimple subalgebra and violates nilpotence, hence the lower-central Tate-pronilpotence hypothesis of 5.166. At $k = 3$ the bracket raises $\mathfrak{m}$-adic degree strictly.

LEMMA 5.101 *(Truncation is a finite-dimensional nilpotent Lie algebra).* For every $N \geq 3$:

(i) $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ is a finite-dimensional Lie algebra of dimension $\binom{N+2}{2} - 6$ (the number of monomials of degree between 3 and N inclusive).

(ii) $\mathfrak{h}_{\leq N}$ is nilpotent of class at most $N-2$ in the lower central series.

(iii) $\mathfrak{h}^{\vee}_{\leq N}$ is finite-dimensional, hence the formal Casimir $C_{\mathfrak{h}_{\leq N}} = \sum_{3 \leq d \leq N} \sum_i e_{d,i} \otimes e^i_d$ is a convergent element of $\mathfrak{h}_{\leq N} \otimes \mathfrak{h}^{\vee}_{\leq N}$ as a finite sum in the ordinary tensor product.

(iv) The linear Hamiltonians $z_1, z_2 \in \mathfrak{m} \setminus \mathfrak{m}^3$, the conformal-symplectic generators $H_2 = \mathrm{span}\{z_1^2, z_1 z_2, z_2^2\} \cong \mathfrak{sl}_2$, and the constant Hamiltonian $1 \notin \mathfrak{m}$ are omitted from $\mathfrak{h}_{\leq N}$ by construction.

Proof. (i) The space $\mathfrak{m}^3$ is a Lie subalgebra of the polynomial Hamiltonian Lie algebra $\mathfrak{h}_{\mathrm{poly}}$, and $\mathfrak{m}^{N+1}$ is a Lie ideal inside $\mathfrak{m}^3$: for $f \in \mathfrak{m}^p$ and $g \in \mathfrak{m}^q$ one has $\{f, g\} \in \mathfrak{m}^{p+q-2}$, since the Poisson bracket on $\mathbb{C}[z_1, z_2]$ lowers total polynomial degree by exactly two, and each partial derivative lowers $\mathfrak{m}$-adic degree by exactly one. For $p, q \geq 3$ the bracket lands in $\mathfrak{m}^{p+q-2} \subset \mathfrak{m}^4 \subset \mathfrak{m}^3$, making $\mathfrak{m}^3$ a Lie subalgebra; for $p \geq 3$ and $q \geq N+1$ the bracket lands in $\mathfrak{m}^{p+q-2} \subset \mathfrak{m}^{N+2} \subset \mathfrak{m}^{N+1}$, making $\mathfrak{m}^{N+1}$ a Lie ideal of $\mathfrak{m}^3$. The quotient $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ inherits a well-defined Lie bracket. Dimension count: monomials $z_1^a z_2^b$ with $3 \leq a+b \leq N$ number $\binom{N+2}{2} - 6$ (the total number with $0 \leq a+b \leq N$ minus the six of degrees 0, 1, 2).

(ii) The Poisson bracket of two elements of $\mathfrak{m}^3$ raises $\mathfrak{m}$-adic degree by at least 1: $\{H_p, H_q\} \subset H_{p+q-2} \subset \mathfrak{m}^{p+q-2}$, and for $p, q \geq 3$, $p+q-2 \geq 4 > p$ (since $q \geq 3$). Thus the descending Lie series

$$\mathfrak{h}^{(\geq m)}_{\leq N} := \underbrace{[\mathfrak{h}_{\leq N}, [\mathfrak{h}_{\leq N}, \ldots]]}_{m \text{ brackets}} \subset \left(\mathfrak{m}^{m+3}/\mathfrak{m}^{N+1}\right)$$

satisfies $\mathfrak{h}^{(\geq m)}_{\leq N} = 0$ for $m+3 \geq N+1$, i.e. for $m \geq N-2$. Hence $\mathfrak{h}_{\leq N}$ is nilpotent of class at most $N-2$.

(iii) Immediate from finite-dimensionality.

(iv) Immediate from the $\mathfrak{m}^3$-adic cutoff and the exclusion $a+b \leq 2$. □

The truncation kills the linear and conformal-symplectic sectors: the obstruction $\{z_1, z_2^{N+1}\} = (N+1)z_2^N \notin \mathfrak{m}^{N+1}$ of 5.210 requires $z_1 \in H_1$, which the cubic cutoff omits. The Casimir is the finite sum $C_{\mathfrak{h}_{\leq N}} = \sum_{d=3}^N \sum_i e_{d,i} \otimes e^i_d \in \mathfrak{h}_{\leq N} \otimes \mathfrak{h}^{\vee}_{\leq N}$; the product/direct-sum support obstruction of 5.212, which depends on infinite degree expansion, vanishes.

The classical truncated cotangent CE/PV identification

THEOREM 5.102 *(Classical truncated CE/PV identification: Koszul resolution on the cubic ideal).* Set $\mathfrak{g}_{\leq N} = \mathfrak{h}_{\leq N} \ltimes \mathfrak{h}^{\vee}_{\leq N}[1]$. For every interval $I \subset \mathbb{R}$ let $\mathfrak{g}_{\leq N, I} = \Omega^\bullet_c(I) \widehat{\otimes} \mathfrak{h}_{\leq N} \ltimes \mathcal{D}'_c(I) \widehat{\otimes} \mathfrak{h}^{\vee}_{\leq N}[1]$. Then, after the Poisson, residue, and interval-smearing conventions fixed above, the dictionary $c^I \mapsto \theta^I$, $u_I \mapsto O_I$ is the Koszul resolution of the chiral Hochschild cohomology $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_b$ on the cubic truncation: a filtered cochain isomorphism of completed dg graded-commutative P_0-algebras

$$\Phi_{\leq N} : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\leq N, I}) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}(\mathfrak{h}_{\leq N, I})\big)$$

on every interval $I \subset \mathbb{R}$, with the same Schouten/Lichnerowicz convention as in 4.20. It is compatible with extension by zero and with disjoint-interval products in this reduced finite model, and on the

Hamiltonian generators it recovers the stalkwise operation $f \mapsto L_{B_f}$. The cotangent generators are finite principal-part labels; their unreduced boundary centrality homotopies remain part of Problem 5.49.

Proof. The Lie algebra $\mathfrak{g}_{\le N}$ is finite-dimensional ($\dim \mathfrak{g}_{\le N} = 2 \dim \mathfrak{h}_{\le N} = 2\binom{N+2}{2} - 12$), hence pro-finite-dimensional in the trivial constant filtration. It is lower-central Tate-pronilpotent in the sense of 5.166: by 5.101(ii) the descending Lie powers of $\mathfrak{h}_{\le N}$ vanish after at most $N-2$ brackets. The cotangent summand is abelian, and nilpotence of the semidirect product uses the degree-lowering coadjoint action: since $[H_p, H_q] \subset H_{p+q-2}$, an element $f \in H_p$, $p \ge 3$, sends $H_d^\vee$ to $H_{d-p+2}^\vee$, hence lowers dual degree by at least one. Repeated action therefore kills every finite quotient $\mathfrak{h}_{\le N}^\vee$. Together with nilpotence of $\mathfrak{h}_{\le N}$, the descending Lie powers of $\mathfrak{g}_{\le N} = \mathfrak{h}_{\le N} \ltimes \mathfrak{h}_{\le N}^\vee[1]$ also terminate. Hence $\bigcap_m \mathfrak{g}_{\le N}^{(\ge m)} = 0$ in any Tate filtration, and the lower-central Tate-pronilpotence hypothesis of 5.166 is satisfied trivially.

Apply 4.20 (the cochain-level reduced Hamiltonian CE/PV central-operation theorem) to $\mathfrak{h}_{\le N}$. The conclusion is a canonical isomorphism of completed dg graded-commutative P_0-algebras

$$\Phi_{\le N}^{\mathrm{stalk}} : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\le N}) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}(\mathfrak{h}_{\le N})\big),$$

with the explicit generator-level data of 4.20.

To pass from the stalk identification to the factorization-algebra identification on intervals $I \subset \mathbb{R}$, the Hamiltonian-sector factorization extension of 4.11 applies verbatim with $\mathfrak{h}$ replaced by $\mathfrak{h}_{\le N}$: the locality bracket $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(t')\} = \delta^i_l \delta^k_j \delta(t-t')$ on two brane-time points is unchanged, the smearing $\mathfrak{h}_{\le N,I} = \Omega^\bullet_c(I) \mathbin{\widehat{\otimes}} \mathfrak{h}_{\le N}$ is canonical because $\mathfrak{h}_{\le N}$ is finite-dimensional in the ordinary tensor category, and the strict P_0 bracket follows from the delta-function locality.

In this finite truncated classical sector, the cotangent factor has a finite Tate-Casimir kernel. The principal-part sub-factorization algebra

$$\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,\le N}(I) = \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},\le N}(I) \mathbin{\widehat{\otimes}} \widehat{\mathbf{S}}\big(\mathfrak{h}^\vee_{\le N,I}[1]\big)$$

admits a chain-level primitive projection $\pi^{\le N}_{\mathrm{prim}}$ because the Tate Casimir $C_{\mathfrak{h}_{\le N}} = \sum_{d=3}^N \sum_i e_{d,i} \otimes e^i_d$ is a finite sum, hence a literal element of the (uncompleted) tensor product $\mathfrak{h}_{\le N} \otimes \mathfrak{h}^\vee_{\le N}$. The Q-compatibility, factorization compatibility, and P_0-bracket homotopies are then formal, since every step uses finite-support tensors in the direct-sum dual.

The bracket compatibility and graded Leibniz extension to the completed finite CE/PV algebra is the same argument as in 4.20, with the additional simplification that all completions are trivial because $\mathfrak{h}_{\le N}$ is finite-dimensional. Compatibility with extension by zero, disjoint-interval factorization, and the stalkwise limit follow as in 4.11. □

Quantum obstruction for the finite truncation

THEOREM 5.103 *(Quantum obstruction for the nilpotent truncation).* The classical finite truncation 5.102 is proved in the finite classical sector. The same finite Lie algebra $\mathfrak{h}_{\le N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ has an obstruction to an all-order Moyal deformation for $N \ge 4$: the reduced Moyal bracket has a closure obstruction. Consequently an all-order quantum descendant theorem on the strict finite truncation requires an additional Moyal-flat replacement or a projected bracket with Jacobi and cochain compatibility.

What remains true at quantum level is the restriction of the full-sector degree-zero map 5.217 to the finite set of generators of $\mathfrak{h}_{\le N}$, viewed inside the full $\mathfrak{h}_\hbar$. This is a restriction of a proved map on the full

Hamiltonian source. A closed CE theorem for a finite quantum Lie subalgebra requires the additional data above.

Proof. The Moyal bracket is

$$\{f, g\}_\hbar = \sum_{j\geq 0} \frac{\hbar^{2j}}{(2j+1)!\,2^{2j}} P^{2j+1}(f, g), \qquad P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

If $f \in \mathfrak{m}^p$ and $g \in \mathfrak{m}^q$, the order-$\hbar^{2j}$ summand lies in $\mathfrak{m}^{p+q-2(2j+1)}$ when nonzero. The Poisson term $j = 0$ preserves $\mathfrak{m}^3$ as a Lie subalgebra, but the higher Moyal terms can leave the ideal. For $N \geq 4$,

$$P^3(z_1^4, z_2^4) = 24z_1 \cdot 24z_2 = 576z_1z_2,$$

so

$$\{z_1^4, z_2^4\}_\hbar = 16z_1^3z_2^3 + 24\hbar^2 z_1z_2 + \cdots.$$

The term $24\hbar^2 z_1z_2$ has $\mathfrak{m}$-adic degree 2. It has degree 2 and lies outside both the cubic ideal $\mathfrak{m}^3$ and the constants killed by the reduction modulo $\mathbb{C}\cdot 1$. Thus $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ has a Moyal-closure obstruction for $N \geq 4$.

Therefore the CE differential for a hypothetical finite quantum source $\mathfrak{h}_{\leq N,\hbar} \ltimes \mathfrak{h}^\vee_{\leq N,\hbar}[1]$ is incompatible with simple restriction of the Moyal bracket. A projected bracket carries its own Jacobi class; the projected-degree warning of 5.210 shows that this is a genuine Jacobi condition. The all-order theorem 5.217 remains valid on the full Hamiltonian algebra $\mathfrak{h}_\hbar$, and its map can be evaluated on generators of degrees $3, \ldots, N$. That operation yields restricted values; a finite quantum CE source requires the additional Moyal-flat structure above. The descendant cotangent sector at $\hbar = 0$ is exactly 5.102; at $\hbar \neq 0$ it is the named Moyal-flat truncation problem. □

PROPOSITION 5.104 *(Sectors removed by truncation).* The truncation $\mathfrak{h} \rightsquigarrow \mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$ removes the following five data from the local theorem 5.6:

(i) The linear Hamiltonians $z_1, z_2 \in \mathfrak{m} \setminus \mathfrak{m}^2$, which generate translations in the symplectic plane and the center-of-mass mode of the brane.

(ii) The conformal-symplectic $\mathfrak{sl}_2$ generators $H_2 = \mathrm{span}\{z_1^2, z_1z_2, z_2^2\}$, which generate rotations and dilations of the symplectic plane. This is the semisimple sector that prevents $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ from being nilpotent.

(iii) The constant Hamiltonian $1 \notin \mathfrak{m}$, on which the $\hbar$-normalization of the Moyal product depends. The central element $\mathbb{C}\cdot 1 \subset \mathfrak{h}_{\mathrm{poly}}$ is omitted from $\mathfrak{m}^3$ a fortiori, so the $\hbar$-normalization in the truncated theorem is conventional.

(iv) The U(1) center anomaly cohomology class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ of 5.6(f), since the cocycle

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = (kn - lm)\,\mathbf{1}[(k+m, l+n) = (1,1)]$$

is concentrated on multidegree $(1,1)$, with both arguments of total $\mathfrak{m}$-adic degree 1, hence the cocycle pairs only elements outside $\mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$. The central anomaly has support outside the truncation.

(v) The stable connected limit $N \to \infty$ on the full $\mathfrak{h}$. The inverse system $\{\mathfrak{h}_{\leq N}\}_{N\geq 3}$ has structure maps the natural surjections $\mathfrak{h}_{\leq N+1} \twoheadrightarrow \mathfrak{h}_{\leq N}$ induced by the $\mathfrak{m}$-adic projection $\mathfrak{m}^{N+1} \subset \mathfrak{m}^{N+2}$. The inverse limit is $\varprojlim_N \mathfrak{h}_{\leq N} = \mathfrak{m}^3$, the $\mathfrak{m}^3$-truncation of the Hamiltonian Lie algebra.

What remains classically: the third-order and higher canonical transformations, which form a nilpotent Lie subalgebra $\mathfrak{m}^3 \subset \mathfrak{h}$, a Lie ideal in $\mathfrak{m}^2$, admitting the classical finite cotangent CE/PV identification of Theorem 5.102. The unreduced open BV factorization-centre morphism requires additional data beyond this truncation. The all-order Moyal descendant lift is excluded by 5.103 unless extra quantum truncation data are added.

Proof. (i)–(iv) are immediate from the definitions and from the $\mathfrak{m}$-adic concentration of ω. (v) is the explicit inverse limit calculation: a compatible family $\{f_N \in \mathfrak{m}^3/\mathfrak{m}^{N+1}\}_N$ assembles to a unique formal series in $\mathfrak{m}^3 \subset \mathbb{C}[[z_1, z_2]]$, and the $\mathfrak{m}$-adic Lie structure is preserved at every level. □

Cubic inverse-limit survivor

PROPOSITION 5.105 *(Cubic coordinate-window survivor and variance boundary).* The finite Lie quotients

$$\mathfrak{h}_{\le N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$$

recover the complete cubic Lie algebra $\varprojlim_N \mathfrak{h}_{\le N} = \mathfrak{m}^3$. The finite cotangent maps $\Phi_{\le N}$ of 5.102 form a compatible family of coordinate tests. Their functoriality is contravariant: it is pullback along Lie quotients, with additional data required for an inverse system of dg CE cochain projections. A Lie quotient $\mathfrak{h}_{\le M} \twoheadrightarrow \mathfrak{h}_{\le N}$ induces CE cochain maps by pullback in the opposite direction. Consequently $\Phi_{\mathfrak{m}^3}$ denotes the coordinate-window family $\{\Phi_{\le N}\}_N$ together with the chain-side inverse limit and an exact continuous-dualization datum; the expression $\varprojlim_N \Phi_{\le N}$ on CE cochains requires those additional data.

This is the maximal survivor of the nilpotent tower only in that precise sense: the Lie-side tower recovers $\mathfrak{m}^3$, while a completed cochain P_0-statement requires the additional variance and bracket-admissibility data named above.

Proof. The Lie-side inverse limit is the usual $\mathfrak{m}$-adic completion of $\mathfrak{m}^3$. The variance obstruction appears already at $M = 4$, $N = 3$. In $\mathfrak{h}_{\le 4}$,

$$\{z_1^3, z_2^3\} = 9z_1^2 z_2^2, \qquad d_{\mathrm{CE}} c^{2,2} \supset -9c^{3,0}c^{0,3}.$$

The coordinate projection to the $N = 3$ cochain window kills $c^{2,2}$ and leaves $c^{3,0}c^{0,3}$, so

$$p(d_{\mathrm{CE}} c^{2,2}) = -9c^{3,0}c^{0,3} \neq 0 = d_{\mathrm{CE}}\, p(c^{2,2}).$$

Thus projection of CE cochains has a nonzero dg defect. The correct finite functoriality is contravariant pullback along Lie quotients, while a completed cochain theorem must be obtained either from a chain-side inverse limit followed by exact continuous dualization or from a separately given topology whose cochain transition maps are dg maps.

The Lie algebra $\mathfrak{m}^3$ is lower-central Tate-pronilpotent in the $\mathfrak{m}$-adic Tate filtration: its descending Lie powers satisfy $(\mathfrak{m}^3)^{(\ge m)} \subset \mathfrak{m}^{m+3}$ because each iterated bracket raises the $\mathfrak{m}$-adic degree by at least 1 when both arguments lie in $\mathfrak{m}^3$ (since $\{H_p, H_q\} \subset \mathfrak{m}^{p+q-2}$ and $p, q \ge 3$ implies $p + q - 2 \ge p + 1$). Hence $\bigcap_m (\mathfrak{m}^3)^{(\ge m)} = 0$, and the lower-central Tate-pronilpotence hypothesis of 5.166 holds. The Tate Casimir $C_{\mathfrak{m}^3} = \sum_{d\ge 3} \sum_i e_{d,i} \otimes e_d^i$ is governed by the same product/direct-sum convergence question as in 5.212, but on the smaller direct-sum dual $\bigoplus_{d\ge 3} H_d^\vee \subset \bigoplus_{d\ge 1} H_d^\vee$. The finite maps $\Phi_{\le N}$ at every finite N use only the finite Casimir $\sum_{3\le d\le N} \sum_i e_{d,i} \otimes e_d^i$. Passing from these finite identities to a completed cochain identity is exactly the continuous-dualization and variance datum isolated in the statement, rather than a formal inverse-limit consequence. □

Definition 5.106 *(Low-degree obstruction datum for the nilpotent tower).* For $N \geq 3$ let

$$\pi_N : \mathfrak{m}^3 \twoheadrightarrow \mathfrak{h}_{\leq N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$$

be the nilpotent window projection. The two obstruction terms for adjoining the removed low-degree sector are:

$$\mathfrak{o}_N^{\mathrm{lin}}(x;\tau) = \{x, \tau\} \bmod \mathfrak{m}^{N+1}, \qquad x \in H_1,\ \tau \in \mathfrak{m}^{N+1}/\mathfrak{m}^{N+2},$$

and

$$\mathfrak{o}^{\mathrm{ss}} : \Lambda^2 H_2 \to H_2, \qquad \mathfrak{o}^{\mathrm{ss}}(x,y) = \{x,y\}.$$

Concretely

$$\mathfrak{o}_N^{\mathrm{lin}}(z_1; z_2^{N+1}) = (N+1)z_2^N \neq 0 \quad \text{in } \mathfrak{m}^N/\mathfrak{m}^{N+1},$$

and

$$\{z_1^2, z_2^2\} = 4z_1z_2, \qquad \{z_1^2, z_1z_2\} = 2z_1^2, \qquad \{z_1z_2, z_2^2\} = 2z_2^2.$$

The compatible Casimir missing from the nilpotent tower is

$$\mathfrak{o}_{\leq 2}^{\mathrm{Cas}} = \sum_i e_{1,i} \otimes e_1^i + \sum_i e_{2,i} \otimes e_2^i.$$

LEMMA 5.107 *(Obstruction-datum criterion for enlarging the tower).* A finite $\mathfrak{m}$-adic tower extending $\{\mathfrak{h}_{\leq N}\}_{N\geq 3}$ to a Lie tower containing $H_1 \oplus H_2$ and using the same quotient maps exists only if $\mathfrak{o}_N^{\mathrm{lin}} = 0$ for all N and $\mathfrak{o}^{\mathrm{ss}}$ is nilpotent in the lower-central filtration. A cotangent CE/PV extension of the inverse limit to the full $\mathfrak{h} = \mathfrak{m}$ also requires a coefficient category in which the compatible tensor $\mathfrak{o}_{\leq 2}^{\mathrm{Cas}} + \sum_{d\geq 3}\sum_i e_{d,i} \otimes e_d^i$ converges.

Proof. If H_1 is included and the window N is still the quotient by $\mathfrak{m}^{N+1}$, then the tail $\mathfrak{m}^{N+1}$ must be a Lie ideal for the enlarged source. The displayed value $\{z_1, z_2^{N+1}\} = (N+1)z_2^N \notin \mathfrak{m}^{N+1}$ shows that this ideal condition is exactly the vanishing of $\mathfrak{o}_N^{\mathrm{lin}}$, and the displayed value is nonzero.

If H_2 is included, then its bracket remains inside H_2 and equals the nonabelian $\mathfrak{sl}_2$ bracket displayed in Definition 5.106. The image of $\mathfrak{o}^{\mathrm{ss}}$ is all of H_2, so the descending Lie powers never leave H_2. This is exactly the obstruction to nilpotence and to the conilpotent bar-cobar hypothesis.

The finite cotangent windows for $\mathfrak{h}_{\leq N}$ use the finite Casimir $\sum_{3\leq d\leq N}\sum_i e_{d,i} \otimes e_d^i$. Extending to $\mathfrak{h}$ adds the missing $d = 1, 2$ components and then the whole compatible family over all $d \geq 1$. Thus the full cotangent lift requires precisely the convergence of the displayed full Casimir family. □

THEOREM 5.108 *(Low-degree obstruction to nilpotent-window recovery of the full Hamiltonian source).* The coordinate-window family $\Phi_{\mathfrak{m}^3}$ of 5.105 is the classical cotangent lift datum on $\mathfrak{m}^3$. Recovery of all of $\mathfrak{h} = \mathfrak{m}$ must include the low-degree sectors $\mathfrak{m}/\mathfrak{m}^3 = H_1 \oplus H_2 = \mathbb{C} \cdot z_1 \oplus \mathbb{C} \cdot z_2 \oplus \operatorname{span}\{z_1^2, z_1z_2, z_2^2\}$, which by 5.210(iii) and 5.107 has a stability obstruction under any $\mathfrak{m}$-adic cutoff: the bracket $\{z_1, z_2^{N+1}\} = (N+1)z_2^N$ is the explicit obstruction. Equivalently, the obstruction datum is

$$(\mathfrak{o}_N^{\mathrm{lin}},\ \mathfrak{o}^{\mathrm{ss}},\ C_{\mathfrak{h}}),$$

where

$$\mathfrak{o}_N^{\mathrm{lin}}(z_1; z_2^{N+1}) = (N+1)z_2^N \neq 0, \qquad \operatorname{im}\mathfrak{o}^{\mathrm{ss}} = H_2 \cong \mathfrak{sl}_2,$$

and $C_{\mathfrak{h}}$ is the full infinite Casimir family below. Equivalently, the Tate Casimir $C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_i e_{d,i} \otimes e_d^i$ has low-degree components $\sum_i e_{1,i} \otimes e_1^i \in H_1 \otimes H_1^\vee$ and $\sum_i e_{2,i} \otimes e_2^i \in H_2 \otimes H_2^\vee$ that the truncation removes; the degree-1 component is the source of the linear-Hamiltonian translation, and the degree-2 component is the source of the conformal-symplectic $\mathfrak{sl}_2$ direction, where semisimplicity replaces nilpotence. Recovering the full $\mathfrak{h}$ requires a coefficient category in which $C_{\mathfrak{h}}$ converges *including* both low-degree components, which is exactly the Casimir support obstruction of 5.212. The truncation route leaves that obstruction intact; it removes the sector supporting the obstruction. A recovery theorem using this nilpotent-window route must therefore give both a coefficient category in which $C_{\mathfrak{h}}$ converges and a relative or reductive model for the H_2-sector.

Proof. The tower used in 5.105 is $\mathfrak{h}_{\le N} = \mathfrak{m}^3/\mathfrak{m}^{N+1}$. Since the transition maps are the usual quotient maps,

$$\varprojlim_N \mathfrak{h}_{\le N} = \varprojlim_N \mathfrak{m}^3/\mathfrak{m}^{N+1} = \mathfrak{m}^3$$

as a complete filtered Lie algebra. The homogeneous pieces H_1 and H_2 have zero image in every term of this tower; hence the inverse limit misses them.

The obstruction is intrinsic to the chosen finite windows. Including H_1 while keeping the same $\mathfrak{m}$-adic finite windows gives a Lie-ideal obstruction in the tails: for every N,

$$z_2^{N+1} \in \mathfrak{m}^{N+1}, \qquad \{z_1, z_2^{N+1}\} = (N+1)z_2^N \notin \mathfrak{m}^{N+1}.$$

Therefore the quotient by the tail carries a full-Hamiltonian bracket obstruction. If one includes H_2, the degree-two Hamiltonians contain the $\mathfrak{sl}_2$ bracket $[H_2, H_2] = H_2$, so the finite quotient violates nilpotence. Thus a finite nilpotent $\mathfrak{m}$-adic tower is incompatible with the low-degree sector $H_1 \oplus H_2$.

Finally, the full Casimir $C_{\mathfrak{h}}$ has nonzero components in $H_1 \otimes H_1^\vee$, $H_2 \otimes H_2^\vee$, and in every higher $H_d \otimes H_d^\vee$. The nilpotent tower sees only the $d \ge 3$ part. Hence the inverse-limit cotangent lift on $\mathfrak{m}^3$ reaches the full Hamiltonian sector only after an additional coefficient category makes the full Casimir converge. $\square$

COROLLARY 5.109 *(Full maximal-ideal obstruction class).* For the full reduced formal Hamiltonian algebra $\mathfrak{h} = \mathfrak{m} = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$, the direct formal-local CE/PV recognition theorem remains valid in the finite-coordinate product completion. The filtered-cobar Koszul recognition criterion of Theorem 5.24, however, has a nonzero obstruction class in O_{Kos} of Definition 5.23. Its visible components are

$$\mathfrak{o}_N^{\mathrm{lin}}(z_1; z_2^{N+1}) = (N+1)z_2^N \neq 0, \qquad \operatorname{im} \mathfrak{o}^{\mathrm{ss}} = H_2 \cong \mathfrak{sl}_2,$$

$$[C_{\mathfrak{h}}] = \left[\sum_{d\ge 1}\sum_i e_{d,i} \otimes e_d^i\right] \in Q_{\mathrm{Cas}}, \qquad H_2 \subset \bigcap_{r\ge 1} \mathfrak{h}^{(r)}.$$

Therefore a full-$\mathfrak{h}$ bar-cobar/enveloping theorem requires, before any open-side A_∞ comparison, all of the following: a coefficient topology representing $C_{\mathfrak{h}}$, continuity of the finite-window bracket and coadjoint transition maps, vanishing $\varprojlim^1$ for the completed CE/PV and cobar towers, a relative or reductive treatment of the H_2-sector, and a twisting cochain whose associated-graded cobar map has acyclic cone.

Proof. The formal-Darboux CE/PV part is Theorem 4.20, equivalently the formal-Darboux instance of Theorem 4.22: the bracket and coadjoint formulas are continuous for finite coordinate projections, so the generator substitution $c \mapsto \theta$, $u \mapsto O$ is a completed dg commutative cochain isomorphism. The P_0-algebra reading is finite-level or bracket-admissible data; an unqualified inverse limit of ambient coordinate products gives a different object.

The filtered-cobar recognition theorem has an additional linear obstruction, the same computation as in Definition 5.106: adjoining H_1 to the $\mathfrak{m}$-adic nilpotent windows makes the tail non-invariant because $\{z_1, z_2^{N+1}\} = (N+1)z_2^N \notin \mathfrak{m}^{N+1}$. The semisimple obstruction is the displayed $\mathfrak{sl}_2$ bracket on H_2, hence $H_2 \subset \bigcap_r \mathfrak{h}^{(r)}$. This gives a nonzero lower-central component in O_{conilp}.

The Casimir component is nonzero in the strict product/direct-sum topology by the support argument of Lemma 5.212: the compatible finite-window Casimirs have nonzero identity component in every $H_d \otimes H_d^\vee$, while a strict tensor with a direct-sum dual has only finitely many dual-degree components. Finally, even after changing topology and handling H_2 relatively, the filtered-cobar criterion still demands the cobar cone and the Milnor $\varprojlim^1$ terms in Definition 5.23 to vanish. These are precisely the listed remaining requirements. □

Problem 5.110 *(Relative replacement of the nilpotent* $\mathfrak{m}^3$*-window).* Replace the coordinate-window Koszul-resolution datum $\Phi_{\mathfrak{m}^3}$ of 5.105 from $\mathfrak{m}^3$ by a relative construction on all of $\mathfrak{h} = \mathfrak{m}$ by giving: (D1) a Tate or weighted-coefficient extension in which the degree-1 and degree-2 Casimir components converge as kernels for the BV propagator; (D2) a relative or reductive replacement for the lower-central Tate-pronilpotence hypothesis of 5.166 in which the conformal-symplectic $\mathfrak{sl}_2$ direction H_2 is included as semisimple data. By 5.106 the obstruction terms are the triple $\big(\mathfrak{o}_N^{\mathrm{lin}}, \mathfrak{o}^{\mathrm{ss}}, \mathfrak{o}_{\le 2}^{\mathrm{Cas}}\big)$; by 5.212 (D1) is the same coefficient-category problem as 5.49. The nilpotent-truncation route closes the Koszul resolution on the cubic ideal $\mathfrak{m}^3$; the displayed triple is the precise obstruction to using the same pronilpotent route for the full $\mathfrak{h}$. It is not an obstruction to the completed formal-Darboux CE/PV construction of Theorem 5.6.

THEOREM 5.111 *(Semisimple obstruction to lower-central Tate-pronilpotence).* A second obstruction to extending the nilpotent $\mathfrak{m}^3$ bar-cobar route to all of $\mathfrak{h}$ is structural: the inclusion of $H_2 = \mathfrak{sl}_2$ violates the lower-central Tate-pronilpotence route used in Lemma 5.166 because $[H_2, H_2] = H_2$ is its own descending Lie series. Even if the degree-1 and degree-2 Casimir components converged in some weighted Tate completion, direct application of the pronilpotent bar-cobar route would still be obstructed. Ordinary finite-dimensional bar-cobar and PBW formalism remain valid for $\mathfrak{sl}_2$; the lower-central Tate-pronilpotent criterion must be replaced, on a L containing H_2, by a relative or reductive model that builds in semisimple subalgebras separately.

Proof. Let L be any filtered Lie algebra containing the degree-two Hamiltonians H_2 as a Lie subalgebra with the usual Poisson bracket. The descending Lie powers of L satisfy $L^{(n)} \cap H_2 \supset H_2$ for every n, because $[H_2, H_2] = H_2$. Hence

$$H_2 \subset \bigcap_{n \ge 1} L^{(n)}.$$

The intersection of the descending Lie powers is therefore nonzero. This contradicts the lower-central Tate-pronilpotence condition used in Lemma 5.166. The bar-cobar argument applies to the nilpotent radical $\mathfrak{m}^3$. A Lie source containing the semisimple H_2-sector requires a relative or reductive replacement. □

Costello–Li precedent

Remark 5.112 *(Costello–Li balanced matrix truncation as structural model).* Costello and Li [13] prove the quantum perturbative BCOV theorem on flat holomorphic-volume space by an $N \to \infty$ stable connected limit through the open gauge algebra $\mathfrak{gl}(N \mid N)$, with the directed Lie embedding $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N+k \mid N+k)$ and the BV-cohomology vanishing $H^*_{\mathrm{BV}}\big(\mathfrak{gl}(N \mid N)\big) = 0$

(supertrace identity $\mathrm{str}(I) = N - N = 0$ kills the putative U(1)-anomaly). The structural analogy with the truncation here:

(i) Each window $\mathfrak{gl}(N \mid N)$ is finite-dimensional, BV-cohomology vanishes, the universal CE/Schouten identification and the BV-propagator coefficient kernel are well-defined on each window.
(ii) The inverse / direct system is the Lie embedding $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N+1 \mid N+1)$ on the Costello–Li side, the Lie surjection $\mathfrak{h}_{\le N+1} \twoheadrightarrow \mathfrak{h}_{\le N}$ here.
(iii) The U(1)-anomaly is killed by the structure of each window on the Costello–Li side (supertrace), and is removed by the $\mathfrak{m}^3$-truncation here (the cocycle ω pairs only multidegree-$(1,0)$ with multidegree-$(0,1)$, both in $H_1 \subset \mathfrak{m} \setminus \mathfrak{m}^2$, hence outside $\mathfrak{m}^3$).
(iv) The colimit $N \to \infty$ converges on the Costello–Li side to the full $\mathfrak{gl}(\infty \mid \infty)$, providing the stable connected single-trace algebra. Here the inverse limit converges to the proper subalgebra $\mathfrak{m}^3 \subsetneq \mathfrak{h}$ by Theorem 5.108.

The structural difference: the $\mathfrak{gl}(N \mid N)$ system is graded by matrix size, with trivial action of one window on the next-window orthogonal complement; the $\mathfrak{h} = \mathfrak{m}$ system is graded by $\mathfrak{m}$-adic degree, with the linear and quadratic sectors acting non-trivially on every higher-degree window, which is precisely the finite-window obstruction of 5.210(iii) plus the $\mathfrak{sl}_2 \subset H_2$ semisimple obstruction to lower-central Tate-pronilpotence.

Tate bar–cobar admissibility $\mathfrak{o}_{\mathrm{Quil}}$. The cochain CE/PV identification of 4.20 lifts to a bar–cobar Quillen statement only after the admissible model-envelope hypotheses in $\mathbf{TateChain}^{\mathrm{adm}}_{\ge 0}$ of Statement 5.18. The component $\mathfrak{o}_{\mathrm{Quil}}$ is the open-side Koszul obstruction class in the Koszul obstruction complex $\mathcal{O}_{\mathrm{Kos}}$ of Definition 5.23, recording the conilpotent-coalgebra hypothesis on the bar source, the bounded filtration, the strong-convergence criterion, and the cone of the filtered cobar map. By 5.109 the visible components on the full Hamiltonian source $\mathfrak{h} = \mathfrak{m}$ are

$$\mathfrak{o}^{\mathrm{lin}}_N(z_1; z_2^{N+1}) \neq 0, \qquad \mathrm{im}\,\mathfrak{o}^{\mathrm{ss}} = H_2, \qquad [C_{\mathfrak{h}}] \in Q_{\mathrm{Cas}}, \qquad H_2 \subset \bigcap_{r \ge 1} \mathfrak{h}^{(r)},$$

together with the Milnor $\varprojlim^1$ terms required by the filtered cobar admissibility criterion of 4.25. The required datum is the joint vanishing inside $\mathcal{O}_{\mathrm{Kos}}$: give a coefficient topology representing $C_{\mathfrak{h}}$, continuity of finite-window bracket and coadjoint transition maps, vanishing $\varprojlim^1$ for the completed CE/PV and cobar towers, a relative/reductive treatment of the H_2-sector, and a twisting cochain whose associated-graded cobar map has acyclic cone. Inside the weighted regulator-admissible sector and after the substitution of the finite-window scalar projection of 5.84, the Tate Quillen equivalence becomes an admissible filtered quasi-isomorphism in the sense of 5.16. The Quillen statement itself is 5.31 of 5.4; the obstruction class $\mathfrak{o}_{\mathrm{Quil}}$ is the cone of its visible obstruction components on the full Hamiltonian source.

BV-cohomology endpoint $\mathfrak{o}_{\mathrm{BV,end}}$. A Costello–Li type endpoint for the mixed brane-defect coefficient module is admissible only after the four-component tuple

$$\mathfrak{o}_{\mathrm{BV,end}} = \big(\mathfrak{o}_{\mathrm{Cas}}, \mathfrak{o}_{\mathrm{def}}, \mathfrak{o}_{\mathrm{RG}}, \mathfrak{o}_{\mathrm{QME}}\big)$$

of Definition 6.16 vanishes simultaneously: the coefficient kernel class $\mathfrak{o}_{\mathrm{Cas}}$ is the residual already named under T1; $\mathfrak{o}_{\mathrm{def}}$ is the obstruction to extending the bulk propagator, bracket transport, interaction, and counterterm representatives across the topological line defect with ordinary $\mathfrak{gl}_N$ open observables; $\mathfrak{o}_{\mathrm{RG}}$ is the obstruction to preserving the strict finite-window Mittag-Leffler tower under the Costello recursion $W(P_{\epsilon,L}, I)$ of [12]; and $\mathfrak{o}_{\mathrm{QME}}$ is the mixed brane-defect local-functional class

$$\mathfrak{o}_{\mathrm{QME}} \in H^1\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w),\ Q + \{I_0, -\}\big)$$

of Problem 5.73. The class $\mathfrak{o}_{\mathrm{BV,end}}$ vanishes exactly under the four-datum Costello–Li extension criterion of 6.18: non-compact/Omega-background BV vanishing on $\mathbb{R}^2 \times \mathbb{C}^2$; mixed topological–holomorphic kinematics on $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$; brane-defect transport with $\mathfrak{gl}_N$ gauge data deciding the scalar contact class of E.12; and the coefficient-system Tate completion already fixed by T1. Casimir-balanced equivariant data in the sense of Definition 5.149 kill the scalar component $\mathfrak{o}^{\mathrm{sc}}_{\mathrm{QME}}$ by 5.150; the non-scalar residual is the obstruction class of 5.84 in $H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{QME}})$ together with the secondary $\varprojlim^1 H^0$ class.

Chain-level primitive projection $\mathfrak{o}_{\mathrm{prim}}$. The chain-level primitive projection $\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \to \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$ of 5.115 is a strict Q-chain map (Theorem 5.117), strictly factorization-compatible after the unit-retaining linear projection (Theorem 5.119), and has strictly compatible P_0-bracket (Theorem 5.120, with the homotopy $h_{P_0} \equiv 0$ on every stratum). It is the indecomposable projection on the unreduced open complex. The unreduced cotangent factorization-centre morphism is the additional target. The component $\mathfrak{o}_{\mathrm{prim}}$ is the residual cotangent-label class: a centrality-homotopy obstruction in the cotangent target $\widehat{\mathbf{S}}(\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I))$, and a one-antifield boundary realization datum. The class $\mathfrak{o}_{\mathrm{prim}}$ is killed by the chain-level primitive cotangent projection of 5.132 together with its single-antifield homology comparison 5.123, lifted to a strict cochain P_0-centrality homotopy with verified bulk-to-boundary kernel 5.140 and a chosen null-homotopy of the positive-ψ primitive summands killed by $\pi^{\mathrm{Ham}}_{\leq 1}$. The primitive projection closes the cosheaf and factorization compatibilities of the indecomposable projection algebraically; the analytic residual is exactly the cotangent-label data and its centrality.

5.5 *Primitive indecomposable P_0-shadow: the Koszul resolution at F_1/F_0*

The chain-level primitive P_0-shadow is the cocycle representing the trace summand of the map from the bulk stalk $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_b$ to the brane-side derived E_1-Hochschild complex $HH^\bullet(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ in Theorem 5.6, at the graded piece F_1/F_0 of the trace-count filtration on the open observables. The two-dimensional holomorphic singular product on this stalk is the diagonal OPE of Theorem 1.17; this file constructs the primitive open-side shadow of the same stalk as a chain-level representative in the primitive trace filtration.

Item (3) of Problem 5.49 is the chain-level projection

$$\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \longrightarrow \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}, \qquad \bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1,$$

that exists strictly at the algebraic chain level and realizes the Koszul-resolution cocycle representing $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_b$ at F_1/F_0. Its binary operation is read through the $K^{(2)}_\Delta$-OPE before the brane-line contraction to interval currents.

The chain-level primitive component $\mathfrak{o}_{\mathrm{prim}}$ of the unreduced cotangent lift obstruction vector Ob_T of Definition 7.3 is the class of the residual cotangent label shadow as a centrality-homotopy obstruction in the cotangent target $\widehat{\mathbf{S}}(\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I))$, together with a one-antifield boundary realization datum. The construction below kills the indecomposable-shadow part of $\mathfrak{o}_{\mathrm{prim}}$ on the graded piece F_1/F_0; the residue is the secondary Koszul obstruction, the pair $(\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}, \mathrm{ob}^{P_0}_{\mathrm{cent}})$ of centrality homotopies on the multi-trace strata, plus the cotangent-target Moyal-coadjoint intertwiner controlled by Theorem 5.138. The Roos secondary class $\lambda_n \in \varprojlim^1_M H^0\big(\boldsymbol{Q}^\bullet_{w,M}(I)\big)$ of Theorem 5.84 is a separate, analytic-side companion of this obstruction, controlling the QME-tower primitive-compatibility for the brane-defect QME endpoint of $\mathfrak{o}_{\mathrm{BV,end}}$ (item T4); its vanishing is needed for the Costello-renormalized one-loop and higher-loop counterterms. The chain-level construction below uses the algebraic primitive splitting.

For the cyclic-word Koszul complex $K^\bullet_{R,S}$ of Definition 5.124,

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C}\cdot[x^R y^S], & p = 0, \\ \mathbb{C}\cdot[\Psi_{R-1,S-1}], & p = 1 \text{ and } R, S \geq 1, \\ 0, & p \geq 2, \end{cases}$$

where $\Psi_{a,b} = \sum_{w\in \mathsf{W}_{a,b}} \mathrm{Tr}(\psi w)$ is the symmetrised one-ψ trace (Theorem 5.123, Lemmas 5.127, 5.128). The displayed homology is the Abel-map Vietoris–Begle equivalence $\widetilde{Y}_{R,S} \simeq_{\mathrm{H}} S^1$ of the cellular cyclic-word complex $\widetilde{Y}_{R,S}$, passed to cyclic coinvariants by $\mathbb{C}[C_S]$-semisimplicity. The chain-level primitive therefore lives exactly on the Hamiltonian sector $\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]$, the closed Hamiltonian algebra $\bar{A}$ summed with its odd-shifted one-ψ descendant copy $\bar{A}_{\mathrm{desc}} = \bigoplus_{a+b>0} \mathbb{C}\cdot[\Psi_{a,b}]$; the homology contains exactly these two Hamiltonian summands.

For the unreduced cotangent factorization-centre morphism

$$\Phi^{\mathrm{unred}}_{\mathrm{fact}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}\big(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}\big)$$

of Problem 5.49, modulo the analytic and convention sublattice $(\mathfrak{o}_{\mathrm{Cas}}, \mathfrak{o}_{\mathrm{nilp}}, \mathfrak{o}_{\mathrm{Quil}}, \mathfrak{o}_{\mathrm{Had}}, \mathfrak{o}_{\mathrm{univ}}) = 0$ of Ob_T, the first obstruction coordinates are given by the joint centrality / QME / descent tuple

$$\mathrm{Ob}_{\mathrm{cent/QME/desc}} = \big(\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}, \mathrm{ob}^{P_0}_{\mathrm{cent}}, \mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}, \delta^{\mathrm{Weiss}}_{\check{C}/R}\big)$$

whose vanishing is necessary, and sufficient only after the full filtered obstruction complex, all higher descent classes, and compatible null-homotopies have been constructed. Its entries are the factorization-product centrality class $\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}$, the P_0-bracket centrality class $\mathrm{ob}^{P_0}_{\mathrm{cent}}$, the brane-defect QME endpoint class $\mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}$, and the stratified Weiss Čech–Roos descent class $\delta^{\mathrm{Weiss}}_{\check{C}/R}$ of Remark 5.162. This tuple is the restriction of Theorem 5.163 to the two components $\mathfrak{o}_{\mathrm{prim}}$ and $\mathfrak{o}_{\mathrm{BV,end}}$ of Ob_T, expanded into their centrality, QME, and descent constituents, under the displayed sublattice constraint. $\mathfrak{o}_{\mathrm{cv}}$ is not set to zero here; its local Weiss component is the descent coordinate displayed above, while its compact comparison component belongs to the matched-conventions obstruction class. $\mathfrak{o}_{\mathrm{prim}}$ gives $\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}$ and $\mathrm{ob}^{P_0}_{\mathrm{cent}}$; the QME endpoint component of $\mathfrak{o}_{\mathrm{BV,end}}$ gives $\mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}$; the descent class $\delta^{\mathrm{Weiss}}_{\check{C}/R}$ is the local stratification of the descent component of $\mathfrak{o}_{\mathrm{cv}}$ of Remark 5.162, whose local form is forced by the Weiss-cosheaf condition on the unreduced factorization centre even when the cross-target part of $\mathfrak{o}_{\mathrm{cv}}$ vanishes.

In this filtration the chain-level primitive shadow is the *associated graded* of the unreduced lift problem on its zero-ψ Hamiltonian and one-ψ descendant generator subspaces. Strict Q-, factorization-, and P_0-bracket compatibilities are separate lift obstructions until a filtered obstruction complex and its spectral sequence are constructed.

Theorem 5.133 extracts the algebraic primitive/indecomposable projection on the free symmetric algebra $\mathbf{S}(\mathrm{Cyc}(F)_{\mathrm{nonempty}})$, with $F = \mathbb{C}\langle x_1, x_2\rangle$ the free associative algebra on the two Chan–Paton matrix generators and $\mathrm{Cyc}(F)_{\mathrm{nonempty}} = F/[F,F] \setminus \mathbb{C}\cdot 1$ its space of nontrivial cyclic-word generators, composed with the zero-ψ Hamiltonian abelianization $\mathrm{Cyc}(F)_{\mathrm{nonempty}} \twoheadrightarrow \bar{A}$; the result is a strict Q-chain map, factorization-compatible after unit retention, and P_0-bracket-compatible with explicit zero homotopy.

The construction draws on the chain structure of the reduced BV algebra (Construction 3.20), the de Rham contraction $H^\bullet(\Omega^\bullet_c(I)) \cong H^\bullet_c(I;\mathbb{C})$ (Lemma 3.21), and the stable trace invariant theory $\mathcal{R}^{GL_\infty,\mathrm{st}}_\infty \cong \mathbf{S}(\mathrm{Cyc}(F)_{\mathrm{nonempty}})$ (Proposition 3.22). Analytic propagators and Tate-completion convergence enter the remaining obstruction groups.

Beyond the indecomposable shadow, the quantum descendant label obstruction Proposition 5.134 shows the Moyal coadjoint obstruction to intertwining the classical primitive map with the principal-part target. The exact same-action Rees obstruction Theorem 5.138 identifies this as a local-nilpotence mismatch, which the Fourier-twisted Rees construction Theorem 5.137 sidesteps by changing polarization with the module action fixed.

Construction 5.113 *(Trace-count filtration on the open observables).* Let $\mathcal{R}_N$ be as in Construction 3.20. In the stable connected limit $N \to \infty$, Proposition 3.22 identifies

$$\mathcal{R}^{GL_\infty,\mathrm{st}}_\infty \cong \mathbf{S}\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}}\big)$$

with $F = \mathbb{C}\langle x_1, x_2\rangle$ and the cyclic words being primitive generators (Lemma 3.23 extends this to the super-cyclic case with one displayed odd ψ). The *trace-count filtration* on $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) = \Omega^\bullet_c(I)\widehat{\otimes}\mathcal{R}^{GL_N}_N$ is the symmetric-algebra filtration $F_k Obs$ generated by products of at most k primitive trace classes (cyclic words):

$$F_0 \subset F_1 \subset F_2 \subset \cdots \subset \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I).$$

F_0 is the algebra of compactly supported de Rham forms on I with scalar coefficients (the empty-trace stratum); F_1/F_0 is the *primitive* or *single-trace* component, isomorphic as a graded vector space to $\Omega^\bullet_c(I)\widehat{\otimes}\mathrm{Cyc}(F)_{\mathrm{nonempty}}$ in the stable limit.

LEMMA 5.114 *(Trace-count filtration is preserved by Q and by the open P_0 bracket).* In the stable connected limit, the BV differential Q and the open P_0 bracket $\{-,-\}_{\mathrm{open}}$ both preserve the trace-count filtration of Construction 5.113.

Proof. By Construction 3.20, $Q\psi = [\phi_1, \phi_2]$ and $Q\phi_i = 0$. As an interior derivation on $\mathcal{R}_N$, Q replaces a single ψ inside a trace by $[\phi_1, \phi_2]$ in the same trace, preserving the trace count; smearing by a de Rham form on I preserves it as well, so $QF_k \subset F_k$.

The equal-time bracket $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(t')\} = \delta^i_l\delta^k_j\delta(t-t')$ of Remark 4.10 is the Procesi–Razmyslov cycle-merging operation: it sends a pair $(\mathrm{Tr}\, w_1, \mathrm{Tr}\, w_2)$ to a single trace by Wick-contracting one ϕ_1 in w_1 with one ϕ_2 in w_2 and joining the cycles, hence the bracket of two single traces is a single trace (Theorem 5.191). By the Leibniz rule, the bracket of a k_1-trace product with a k_2-trace product has total trace count $k_1 + k_2 - 1$, lying in $F_{k_1+k_2-1} \subset F_{k_1+k_2}$. □

Construction 5.115 *(Primitive indecomposable P_0-shadow).* In the stable connected limit, the symmetric-algebra structure $\mathcal{R}^{GL_\infty,\mathrm{st}}_\infty \cong \mathbf{S}(\mathrm{Cyc}(F)_{\mathrm{nonempty}})$ has the natural projection onto the linear span of generators (the Hopf-algebraic primitive/indecomposable part). Tensored with the identity on $\Omega^\bullet_c(I)$, this gives

$$\pi_{\mathrm{prim}} : \Omega^\bullet_c(I)\widehat{\otimes}\mathbf{S}(\mathrm{Cyc}(F)_{\mathrm{nonempty}}) \twoheadrightarrow \Omega^\bullet_c(I)\widehat{\otimes}\mathrm{Cyc}(F)_{\mathrm{nonempty}}.$$

Composing this shadow map with the identification $\mathrm{Cyc}(F)_{\mathrm{nonempty}} \twoheadrightarrow \bar{A}$ on the Hamiltonian-trace subspace (the abelianization of cyclic words: the trace of a cyclic word in the commuting variables

ϕ_1, ϕ_2, taken modulo Q-exact reordering, is exactly the polynomial Hamiltonian $f \in \bar{A}$, a stable cyclic-word/polynomial dictionary recorded by Construction 3.25 and Lemma 5.170), gives

$$\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \twoheadrightarrow \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}.$$

π_{prim} is the zero-ψ Hamiltonian shadow: primitive traces containing at least one displayed ψ are killed here and are treated separately by the one-ψ label shadow of Construction 5.132.

For factorization products one must retain the empty-trace stratum until the final linear projection. Define the unit-retaining shadow

$$\pi^{\mathrm{Ham}}_{\le 1} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \longrightarrow \Omega^\bullet_c(I)\widehat{\otimes}(\mathbb{C}\cdot 1 \oplus \bar{A})$$

as projection to the scalar stratum F_0 together with the zero-ψ Hamiltonian summand of F_1/F_0, killing $F_{\ge 2}$ and the positive-ψ primitive summands. Then π_{prim} is $\pi^{\mathrm{Ham}}_{\le 1}$ followed by projection to the $\bar{A}$-summand. At finite N this is a degreewise stable construction. For a fixed word-degree d and $N \ge d$, Proposition 3.22 identifies the degree-d trace invariants with the stable cyclic-word span, and the above shadow map is applied on that stable range. Outside the stable range the finite-N trace identities remain part of the finite-N algebra; the statement is made after passing to the stable connected limit.

LEMMA 5.116 *(Kernel of the Hamiltonian primitive shadow).* On the zero-ψ Hamiltonian subalgebra the kernel of π_{prim} is the closed ideal $F_{\ge 2}$ generated by products of at least two primitive trace classes, together with the empty-trace stratum F_0, and the primitive abelianization kernel

$$\ker\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}} \to \bar{A}\big) \subset F_1/F_0.$$

On the full reduced BV algebra before the cotangent label shadow is added, the kernel also contains the positive ψ-primitive summands. Equivalently, the unit-retaining map $\pi^{\mathrm{Ham}}_{\le 1}$ has kernel $F_{\ge 2}$, the primitive abelianization kernel, and the non-Hamiltonian primitive summands, while π_{prim} further kills F_0.

Proof. By construction π_{prim} is the projection onto zero-ψ Hamiltonian generators of the symmetric algebra, which is the standard projection to primitive indecomposable generators followed by the abelianization $\mathrm{Cyc}(F)_{\mathrm{nonempty}} \to \bar{A}$. The primitive projection kills the augmentation ideal squared $(\mathbf{S}^{\ge 1})^2 = \mathbf{S}^{\ge 2}$. The abelianization then kills primitive reordering differences, for instance the class of $[xyxy]_{\mathrm{cyc}} - [xxyy]_{\mathrm{cyc}}$, and the final projection to $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ kills the constants $\mathbb{C}\cdot 1 \subset F_0$. Primitive classes with positive ψ-count lie in the kernel because the Hamiltonian target has zero ψ-count. Tensoring with the de Rham cosheaf $\Omega^\bullet_c(I)$ preserves these direct summands because the symmetric-algebra split is over the discrete coefficient ring on cyclic-word generators. □

THEOREM 5.117 *(Primitive indecomposable P_0-shadow is a Q-chain map).* With total differential $Q_{\mathrm{tot}} = d_{\mathrm{dR}} + Q_{\mathrm{BV}}$ on the source and the de Rham differential on $\Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$, π_{prim} commutes strictly with Q_{tot}:

$$\pi_{\mathrm{prim}} \circ Q_{\mathrm{tot}} = Q_{\mathrm{tot}} \circ \pi_{\mathrm{prim}}.$$

Proof. The de Rham part commutes with π_{prim} because the projection acts only on the trace label. For the BV part, Lemma 5.114 gives that Q_{BV} preserves the trace-count filtration $F_\bullet$. On F_0 the BV differential is zero on the matrix factor, and π_{prim} kills the scalar Hamiltonian line. On a zero-ψ primitive trace B_f, $Q_{\mathrm{BV}}B_f = 0$ because $Q_{\mathrm{BV}}\phi_i = 0$, while primitive reordering differences remain in the abelianization kernel. On a primitive one-ψ trace $\mathrm{Tr}(\psi w(x,y))$,

$$Q_{\mathrm{BV}}\ \mathrm{Tr}(\psi w) = \mathrm{Tr}((xy - yx)w).$$

Its image in $\bar{A}$ is zero, since the two cyclic words become the same commutative monomial after abelianization. Primitive terms with at least two ψ's remain outside the zero-ψ Hamiltonian target after one application of Q_{BV}, and multi-trace terms remain multi-trace by Lemma 5.114. Thus the two composites agree on every summand. □

Remark 5.118 *(The primitive target as the indecomposable part).* The target $\Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$ is the *indecomposable* or *linear primitive* part of the brane P_0 factorization algebra $A^{\mathrm{Ham}}_\partial(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$ of Construction 4.7: it is the generating Lie subspace $\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$. Equivalently, it is the quotient $\widehat{\mathbf{S}}(\mathfrak{h}_I)/\widehat{\mathbf{S}}^{\geq 2}(\mathfrak{h}_I) \cong \mathfrak{h}_I$ in the augmentation-ideal grading, after removal of constants. The factorization product on the indecomposable target side is therefore the linear (Lie) bracket of $\mathfrak{h}_I$; the multiplication of two distinct primitive elements lands in $\mathbf{S}^{\geq 2}$, which is the zero space at the indecomposable level.

THEOREM 5.119 *(Linearized factorization compatibility of the indecomposable shadow).* For disjoint intervals $I_1 \sqcup I_2 \subset V$ in $\mathbb{R}$, let

$$\lambda_{V;I_1,I_2} : \big(\Omega^\bullet_c(I_1)\widehat{\otimes}(\mathbb{C}\cdot 1 \oplus \bar{A})\big) \otimes \big(\Omega^\bullet_c(I_2)\widehat{\otimes}(\mathbb{C}\cdot 1 \oplus \bar{A})\big) \longrightarrow \Omega^\bullet_c(V)\widehat{\otimes}\bar{A}$$

be the exactly-one-primitive part of the brane factorization product after extension to V: it keeps the scalar–primitive and primitive–scalar summands and kills the scalar–scalar and primitive–primitive summands. Then

$$\pi_{\mathrm{prim}} \circ \mu = \lambda_{V;I_1,I_2} \circ \big(\pi^{\mathrm{Ham}}_{\leq 1} \otimes \pi^{\mathrm{Ham}}_{\leq 1}\big).$$

Thus the indecomposable shadow is compatible with the factorization product through the unit-retaining linear projection; it is a linearized factorization map to $\bar{A}$.

Proof. By trace-count stratum. If both factors have nonzero primitive trace count, their factorization product has at least two nonempty traces and is killed by π_{prim}. The right-hand side is also zero because $\lambda_{V;I_1,I_2}$ kills the primitive–primitive summand. If both factors lie in F_0, the product remains scalar and is killed by the final projection to $\bar{A}$; $\lambda_{V;I_1,I_2}$ kills the scalar–scalar summand.

The only surviving cases are scalar–primitive and primitive–scalar. The unit-retaining projections remember precisely those two summands. Applying the brane factorization product and then projecting to the exactly-one-primitive Hamiltonian component gives the same class as first applying $\pi^{\mathrm{Ham}}_{\leq 1}$ on both factors and then applying $\lambda_{V;I_1,I_2}$. Positive-ψ primitive factors are killed on both sides of this zero-ψ Hamiltonian statement; their separate label shadow is treated below. Hence the displayed identity holds strictly. □

THEOREM 5.120 *(P_0 bracket compatibility, with explicit homotopy).* For $F, G \in \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I)$ in the stable connected limit,

$$\pi_{\mathrm{prim}}\big(\{F,G\}_{\mathrm{open}}\big) - \big\{\pi_{\mathrm{prim}}F, \pi_{\mathrm{prim}}G\big\}_{\bar{A}} = (Q \circ h_{P_0} + h_{P_0} \circ Q)(F \otimes G),$$

with the explicit homotopy $h_{P_0} \equiv 0$ on every stratum $F_{k_1} \otimes F_{k_2}$. The chain-level P_0-bracket compatibility square commutes strictly.

Proof. On $(k_1, k_2) = (1, 1)$, the open bracket of two single traces is a single trace by Procesi–Razmyslov cycle merging, and Theorem 5.191 gives $\{B_f(a), B_g(b)\}_{\mathrm{open}} = B_{\{f,g\}_{\bar{A}}}(ab)$; π_{prim} sends this to $ab \otimes \{f,g\}_{\bar{A}}$, which agrees with the $\bar{A}$-bracket of $\pi_{\mathrm{prim}}B_f(a) = a \otimes f$ and $\pi_{\mathrm{prim}}B_g(b) = b \otimes g$ by the construction of Construction 4.7. Off the $(1,1)$-stratum both sides are zero: on (k_1, k_2) with $\max(k_1, k_2) \geq 2$

the Leibniz rule sends the bracket into $F_{\geq 2} = \ker \pi_{\mathrm{prim}}$ (Lemma 5.116); on $(0, k)$ or $(k, 0)$ the factor in F_0 is a scalar de Rham form, on which the equal-time bracket acts trivially. The defect is identically zero, so $h_{P_0} \equiv 0$. □

Construction 5.121 *(Primitive one-antifield chain complex).* The trace-count filtration places the one-ψ primitive generators on the graded piece F_1/F_0 below the multi-trace strata, and Theorem 5.117 reduces strict Q-compatibility there to a finite combinatorial homology calculation: identify every closed one-ψ primitive cycle modulo Q of a two-ψ primitive.

Put $x = \phi_1$, $y = \phi_2$, and let $\mathsf{W}_{k,l}$ be the set of ordinary words with k letters x and l letters y. Let $\mathsf{N}_{a,b}$ be the set of cyclic words with a letters x and b letters y. The primitive one-ψ chain model in bidegree (k, l) is

$$C_2(k,l) \xrightarrow{\partial_2} C_1(k,l) \xrightarrow{\partial_1} C_0(k,l),$$

where

$$C_1(k,l) = \mathbb{C}\{\mathsf{W}_{k,l}\}, \qquad C_0(k,l) = \mathbb{C}\{\mathsf{N}_{k+1,l+1}\}.$$

The basis vector $w \in \mathsf{W}_{k,l}$ represents $\mathrm{Tr}(\psi w)$, with the graded cyclic rule $\mathrm{Tr}(ab) = (-1)^{|a||b|}\,\mathrm{Tr}(ba)$, $|x| = |y| = 0$, $|\psi| \equiv 1 \pmod 2$. The first boundary is

$$\partial_1(w) = [xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}}.$$

The two-ψ source is the alternating quotient generated by pairs of ordinary words u, v, of total bidegree $(k-1, l-1)$, with

$$[u,v] = -[v,u], \qquad [u,u] = 0.$$

In this quotient

$$C_2(k,l) = \big(\mathbb{C}\{(u,v) : \deg u + \deg v = (k-1, l-1)\}\big)_{\mathrm{alt}},$$

and $C_2(k,l) = 0$ if $k = 0$ or $l = 0$. Thus the displayed complex is literally the cellular chain complex in dimensions 2, 1, 0: C_0 is the necklace vertex set, C_1 is the one-edge-interior-point edge set, and C_2 is the oriented two-edge-interior-point square set.

LEMMA 5.122 *(Two-antifield boundary sign).* In the chain model of Construction 5.121, the boundary of the generator represented by $\mathrm{Tr}(\psi u \psi v)$ is

$$\partial_2(u,v) = vxyu - vyxu - uxyv + uyxv.$$

This formula is well-defined on the alternating quotient and satisfies $\partial_1\partial_2 = 0$.

Proof. The differential is the odd derivation determined by $Q\psi = xy - yx$, $Qx = Qy = 0$. Since the second ψ is preceded by one odd letter, replacing it carries the Koszul sign -1:

$$Q\ \mathrm{Tr}(\psi u \psi v) = \mathrm{Tr}\big((xy - yx)u\psi v\big) - \mathrm{Tr}\big(\psi u(xy - yx)v\big).$$

In the first term the even word $(xy - yx)u$ is moved cyclically past the final displayed ψv, producing sign $+1$. Removing the common front ψ gives $vxyu - vyxu - uxyv + uyxv$. Interchanging u and v reverses this expression:

$$uxyv - uyxv - vxyu + vyxu = -(vxyu - vyxu - uxyv + uyxv),$$

so the formula descends to the alternating quotient $C_2(k,l)$.

Applying ∂_1 to the four summands gives the signed sum of cyclic words obtained by inserting two adjacent commutators. Each final cyclic word occurs twice with opposite signs, according to the order in which the two displayed ψ's are replaced. Equivalently $Q^2 = 0$ for the odd derivation Q. Hence $\partial_1\partial_2 = 0$. □

THEOREM 5.123 *(Primitive one-antifield homology).* For every $k, l \geq 0$,

$$H_1\big(C_2(k,l) \to C_1(k,l) \to C_0(k,l)\big) \cong \mathbb{C} \cdot [\Upsilon_{k,l}].$$

The Hamiltonian comparison uses the nonconstant sum $k + l > 0$; the scalar class $\Upsilon_{0,0} = \mathrm{Tr}(\psi)$ lies in the removed constant-Hamiltonian sector; $\bar{A}_{\mathrm{desc}}$ contains the nonconstant classes.

Proof. If $k = 0$ or $l = 0$, then $C_1(k,l)$ is spanned by the single word x^k or y^l, $\partial_1 = 0$, and $C_2(k,l) = 0$. The class is represented by $\Upsilon_{k,l}$.

Assume $k, l > 0$. Form the graph $\Gamma_{k,l}$ with vertices $\mathsf{N}_{k+1,l+1}$ and oriented edges $w \in \mathsf{W}_{k,l}$,

$$w : [yxw]_{\mathrm{cyc}} \longrightarrow [xyw]_{\mathrm{cyc}}.$$

Its cellular boundary is ∂_1. Attach one square for each alternating pair (u, v) with boundary $vxyu - vyxu - uxyv + uyxv$. By Lemma 5.122, this square boundary is exactly ∂_2; denote the resulting two-complex by $X_{k,l}$.

Put $R = k + 1$ and $S = l + 1$. $X_{k,l}$ is the two-skeleton of the cyclic-word cellular complex $Y_{R,S}$. Before cyclic relabelling, a p-cell is a configuration of $R - p$ stationary indistinguishable points on the vertices of an S-edge oriented circle and p points in distinct edge interiors. The orientation is the increasing order of the occupied edge labels. Traversing the labelled circle writes x for stationary points, y for an empty edge, and ψ for an occupied edge interior. Hence vertices are $\mathsf{N}_{R,S}$, one-cells are the words $w \in \mathsf{W}_{k,l}$, and two-cells are exactly the alternating pairs (u, v) above.

The Abel map sends a labelled configuration to the sum of its point positions in $\mathbb{R}/S\mathbb{Z}$ and is affine on every cell. After lifting the target to $\mathbb{R}$ and fixing a cyclic order of the lifted points, a fibre is the convex polytope cut out by

$$t_1 \leq \cdots \leq t_R \leq t_1 + S, \qquad \sum_i t_i = \text{constant},$$

together with the cell inequalities. The cells subdivide this polytope. Thus the labelled Abel map is a homology equivalence with S^1.

Cyclic relabelling by $r \in C_S$ adds r to every point position, so the Abel coordinate is translated by Rr. The action on the target circle is orientation-preserving and therefore trivial on H_0 and H_1. Since $\mathbb{C}[C_S]$ is semisimple, homology commutes with coinvariants. If a stabilizer $G \subset C_S$ fixes an Abel value, it acts affinely on each lifted convex fibre; averaging gives a G-fixed point and straight-line contraction to it descends to a contraction of the quotient fibre. Therefore the cyclic quotient still has the homology of S^1. Finally, attaching cells of dimension at least three preserves H_1, because H_1 only sees $C_2 \to C_1 \to C_0$. Hence $H_1(X_{k,l}; \mathbb{C}) \cong \mathbb{C}$.

The symmetrised trace

$$\Upsilon_{k,l} = \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}(\psi w)$$

is a cycle: the coefficient of any zero-ψ necklace in $Q\Upsilon_{k,l}$ counts cyclic xy-transitions minus cyclic yx-transitions, and these two numbers are equal on a circle. The functional $\epsilon(\mathrm{Tr}(\psi w)) = 1$ kills every two-ψ boundary, because the formula in Lemma 5.122 has two positive and two negative summands, while

$$\epsilon(\Upsilon_{k,l}) = \binom{k+l}{k} \neq 0.$$

Each one-cell occurring in $\Upsilon_{k,l}$ is oriented in the positive Abel direction. Hence its Abel image is $\binom{k+l}{k}$ times the fundamental circle class. The unique H_1-line is generated by $[\Upsilon_{k,l}]$. □

Definition 5.124 *(Full primitive ψ-complex).* Let

$$A_\psi = \mathbb{C}\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| \equiv 1 \pmod 2,$$

with $d\psi = xy - yx$ and $dx = dy = 0$. In the graded cyclic quotient $\mathrm{Cyc}(A_\psi) = A_\psi/[A_\psi, A_\psi]_{\mathrm{gr}}$, put

$$\mathrm{wt}(x) = (1,0), \qquad \mathrm{wt}(y) = (0,1), \qquad \mathrm{wt}(\psi) = (1,1).$$

For fixed (R,S), define

$$K^p_{R,S} = \mathbb{C}\{[w]_{\mathrm{cyc}} : \#\psi(w) = p,\ \#x(w) = R - p,\ \#y(w) = S - p\},$$

with $0 \le p \le \min(R,S)$. The boundary is

$$\partial[w] = \sum_{i:a_i=\psi} (-1)^{\#\{\psi \text{ before } i\}} [a_1 \cdots a_{i-1} xy\, a_{i+1} \cdots a_n - a_1 \cdots a_{i-1} yx\, a_{i+1} \cdots a_n]_{\mathrm{cyc}}.$$

The sign is the Koszul sign for replacing the i-th odd letter by the even commutator $xy - yx$; with this sign $\partial^2 = 0$.

LEMMA 5.125 *(All-ψ face sign).* Let $u_1, \dots, u_p$ be ordinary words in x, y. In $K^{p-1}_{R,S}$,

$$\partial[\psi u_1 \psi u_2 \cdots \psi u_p] = \sum_{j=1}^{p} (-1)^{j-1} [\psi u_1 \cdots \psi u_{j-1} xy\, u_j \psi u_{j+1} \cdots \psi u_p - \psi u_1 \cdots \psi u_{j-1} yx\, u_j \psi u_{j+1} \cdots \psi u_p]_{\mathrm{cyc}}.$$

Under the cyclic-word cellular model below, this is exactly the oriented cubical boundary of the corresponding p-cell.

Proof. The j-th displayed ψ has precisely $j-1$ odd letters before it. Replacing it by the even commutator $xy - yx$ therefore contributes the Koszul sign $(-1)^{j-1}$, with sign $+1$ from the remaining letters. The cellular p-cell is oriented by the increasing order of its p occupied ψ-edges. Its j-th coordinate has terminal face xy and initial face yx, with cubical sign $(-1)^{j-1}$. These two formulae agree term by term. If a cyclic rotation moves a block containing a odd letters past a block containing b odd letters, the cellular orientation changes by $(-1)^{ab}$, the graded cyclic sign. Hence the formula descends to the cyclic quotient. □

Construction 5.126 *(Cyclic-word cellular model).* Assume $R, S > 0$. Let $\widetilde{Y}_{R,S}$ be the labelled state complex on an oriented S-edge circle. A p-cell consists of $R - p$ stationary indistinguishable points on vertices and p points in the interiors of p distinct edges. Traversing the labelled circle writes x for each stationary point at a vertex and writes y, respectively ψ, for an empty edge, respectively an edge containing an interior point. Thus p-cells are labelled words with $R-p$ x's, $S-p$ y's, and p ψ's.

The cyclic group C_S rotates the labelled edge set. The quotient chain complex is taken with the orientation character already encoded by the graded cyclic rule $[uv]=(-1)^{|u||v|}[vu]$.

LEMMA 5.127 *(Cellular chain identification).* For $R,S>0$, the cellular chains of Construction 5.126, after cyclic relabelling, identify with the full primitive cyclic Koszul complex:

$$C_p(\widetilde{Y}_{R,S};\mathbb{C})_{C_S}\cong K^p_{R,S}.$$

Under this identification the cellular boundary is the boundary of Definition 5.124.

Proof. A labelled p-cell has a unique expression $x^{a_1}m_1\cdots x^{a_S}m_S$, where $m_i\in\{y,\psi\}$, exactly p markers are ψ, and $\sum a_i=R-p$. Every graded cyclic word admits such a representative after moving the cut to an edge marker; choosing the labelled starting edge makes the representative unique. Forgetting the labelled starting edge is exactly passage to graded cyclic words, modulo cyclic relabelling only. The sign check is Lemma 5.125; cyclic relabelling by a rotation that moves a moving-edge coordinates past b moving-edge coordinates changes orientation by $(-1)^{ab}$, which is the same sign as moving the corresponding odd ψ-blocks in the graded cyclic quotient. □

LEMMA 5.128 *(Abel map and cyclic stabilizers).* The quotient state complex $Y_{R,S}=\widetilde{Y}_{R,S}/C_S$, with the orientation local system of Lemma 5.127, has the homology of S^1. The surviving H_1-line is represented by the Abel-positive cycle $\Psi_{R-1,S-1}$.

Proof. Before cyclic relabelling, in the universal cover of the S-edge circle, a lifted configuration is described by weakly ordered point positions

$$t_1\le\cdots\le t_R\le t_1+S.$$

The Abel map sends the configuration to $\sum_i t_i\in\mathbb{R}/S\mathbb{Z}$ and is affine on every cell. After lifting the target circle to $\mathbb{R}$, each fibre is the convex polytope cut out by the displayed inequalities and by the fixed-sum equation. The cells subdivide this polytope. Every lifted fibre is therefore a contractible finite polyhedral complex, and the labelled Abel map is a proper PL map with acyclic fibres. Vietoris–Begle gives a cellular homology equivalence

$$\widetilde{Y}_{R,S}\simeq_{\mathrm{H}} S^1.$$

A cyclic relabelling by $r\in C_S$ adds r to each point position, so the Abel coordinate is translated by Rr on $\mathbb{R}/S\mathbb{Z}$; relabelling acts on the target by orientation-preserving rotations and trivially on H_0 and H_1. Since $\mathbb{C}[C_S]$ is semisimple, coinvariants commute with homology:

$$H_\bullet(C_\bullet(\widetilde{Y}_{R,S})_{C_S})\cong H_\bullet(\widetilde{Y}_{R,S})_{C_S}\cong H_\bullet(S^1).$$

The fibre statement survives stabilizers. If a subgroup $G\subset C_S$ fixes a target Abel value, choose the lifted fibre before decomposing it into cells. It is a convex polytope preserved by the affine G-action. Averaging any point of the fibre gives a G-fixed point, and the straight-line contraction to that fixed point is G-equivariant and respects the cellular subdivision. The stabilizer quotient of the fibre is therefore contractible.

The cycle $\Psi_{R-1,S-1}=\sum_w[\psi w]_{\mathrm{cyc}}$ is closed by the same transition count used in Theorem 5.123. Each of its one-cells is oriented in the positive Abel direction, so its image is a nonzero multiple of the fundamental class of the target circle, and it spans the unique H_1-line. □

THEOREM 5.129 *(Full primitive higher-ψ Koszul homology).* For every $R, S \geq 0$, the full primitive cyclic Koszul complex of Definition 5.124 has homology

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C} \cdot [x^R y^S], & p = 0, \\ \mathbb{C} \cdot [\Psi_{R-1,S-1}], & p = 1,\ R, S \geq 1, \\ 0, & p \geq 2. \end{cases}$$

Equivalently, the stable primitive single-trace Koszul homology is

$$\mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta, \qquad |\theta| = 1, \qquad \mathrm{wt}(\theta) = (1, 1),$$

with $x^a y^b \theta$ represented by $\Psi_{a,b}$.

Proof. If $R = 0$ or $S = 0$, the complex is concentrated in positive ψ-degree zero and $K^0_{R,S}$ is the one-dimensional span of the necklace x^R or y^S, including the empty word when $R = S = 0$.

Assume $R, S > 0$. Lemma 5.127 identifies $K^\bullet_{R,S}$ with the cyclic-coinvariant cellular chains of $\widetilde{Y}_{R,S}$. Lemma 5.128 computes their homology as $H_\bullet(S^1; \mathbb{C})$. Thus only degrees 0 and 1 survive. The H_0-line is the commutative necklace class $[x^R y^S]$. The H_1-line is the Abel-positive fundamental class, represented by $\Psi_{R-1,S-1}$. The vanishing for $p \geq 2$ is the vanishing of $H_{\geq 2}(S^1; \mathbb{C})$; the H_1 calculation is already visible in the three-term skeleton, since higher cells leave $\ker(C_1 \to C_0)/\mathrm{im}(C_2 \to C_1)$. □

THEOREM 5.130 *(Hamiltonian sector extracted from full psi homology).* The Hamiltonian comparison uses exactly the nonconstant summands

$$\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1] = \bigoplus_{R+S>0} \mathbb{C} \cdot [x^R y^S] \oplus \bigoplus_{a+b>0} \mathbb{C} \cdot [\Psi_{a,b}].$$

The sectors outside that comparison are precisely: the scalar degree-zero class $[1]$, the scalar odd class $\Psi_{0,0} = \mathrm{Tr}(\psi)$, and the zero higher primitive ψ-sectors $H_p(K_{R,S}) = 0$ for $p \geq 2$. Identifying the remaining $\Psi_{a,b}$ line with principal parts $\rho_{a,b}$ as an $\mathfrak{h}$-module, or after Moyal quantization, belongs to the descendant-target comparison.

Proof. Constants are removed from the closed Hamiltonian algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$, and the open empty trace is the scalar $\mathrm{Tr}(1) = N$. This removes the $H_0(K_{0,0})$ line. The class $\Psi_{0,0}$ is a genuine odd primitive class, but it is the descendant of the removed constant Hamiltonian and is discarded with that scalar sector. By Theorem 5.129, all primitive $p \geq 2$ homology groups vanish. The only surviving positive Hamiltonian labels are therefore the displayed nonconstant H_0 and H_1 lines. The final sentence is exactly the module-structure separation proved later by Proposition 5.134 and Theorem 5.136. □

THEOREM 5.131 *(Primitive descendant homology).* The stable primitive cyclic Koszul complex generated by x, y, ψ, with $Q\psi = xy - yx$ and only graded-cyclic trace relations, has homology

$$\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1] = \bigoplus_{R+S>0} \mathbb{C} \cdot [x^R y^S] \oplus \bigoplus_{a+b>0} \mathbb{C} \cdot [\Psi_{a,b}].$$

Thus the descendant side computed in this statement is exactly the one-ψ line $\bar{A}_{\mathrm{desc}}[1]$.

After the Matlis normalization

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m} \delta_{b,n},$$

the only flat $D_\hbar$-bridge constructed from the polynomial descendant labels is the Fourier-twisted Rees bridge of Theorem 5.137. If the original Hamiltonian action of z_1, z_2 is required to be preserved, the exact obstruction is the local-nilpotence mismatch of Theorem 5.138.

Proof. The first assertion is Theorem 5.129 together with the scalar removal in Theorem 5.130. The cellular proof identifies this stable primitive cyclic Koszul complex with the cyclic-word cellular complex and proves $H_p(K_{R,S}) = 0$ for $p \geq 2$. The calculation computes the homology of this complex. It does not classify other primitive chain-level constructions unless a specified chain equivalence identifies them with this complex.

The Matlis pairing fixes the dual coefficient normalization up to the single scalar allowed by residue-pairing rigidity; the displayed formula chooses that scalar. With this normalization, the positive flat Rees construction is exactly the Fourier-twisted cyclic $D_\hbar$-module $\mathcal{D}_\hbar/(z_1, z_2)$ embedded into the principal-part Čech lattice with the explicit factor $(-1)^{a+b}\hbar^{a+b}a!b!$. If one removes the Fourier twist and asks for the same Hamiltonian action, local nilpotence of the linear Hamiltonians is preserved by flat generic-fibre isomorphism but differs on the two sides. This is precisely the obstruction theorem cited in the statement. □

Construction 5.132 *(Primitive one-ψ label shadow).* Extend π_{prim} to the one-ψ primitive shadow by recording the labels of the symmetrised traces $\Upsilon_{a,b}$ of Construction 3.25, with target the reduced principal-part label space $\mathcal{P}$, the $\mathbb{C}$-vector space spanned by the basis vectors $\rho_{a,b} = z_1^{-a-1}z_2^{-b-1}dz_1dz_2$ of the local-cohomology Matlis dual of $\bar{A}$ at the origin (Theorem 5.179), shifted by [1] for the odd ψ-degree. The domain here is the enlarged principal-part shadow object $\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I)$: the Hamiltonian primitive sector is smeared by $\Omega^\bullet_c(I)$, and the principal-part label sector is smeared by $\mathcal{D}'_c(I)$. Define

$$\pi^{\mathrm{cot}}_{\mathrm{prim}} : \widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) \longrightarrow \big(\Omega^\bullet_c(I)\widehat{\otimes}\bar{A}\big) \oplus \big(\mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1]\big)$$

by:

- on the zero-ψ Hamiltonian sector, $\pi^{\mathrm{cot}}_{\mathrm{prim}} = \pi_{\mathrm{prim}}$ composed with the inclusion into the $\Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$ summand;
- on a primitive one-ψ word $\mathrm{Tr}(\psi w)$ of bidegree (a, b), $a + b > 0$, smeared by a distributional test current $B \in \mathcal{D}'_c(I)$,
$$\pi^{\mathrm{cot}}_{\mathrm{prim}}\big(B \otimes \mathrm{Tr}(\psi w)\big) = \binom{a+b}{a}^{-1} B \otimes \rho_{a,b}[1].$$
Hence
$$\pi^{\mathrm{cot}}_{\mathrm{prim}}\big(B \otimes \Upsilon_{a,b}\big) = B \otimes \rho_{a,b}[1] = \Theta_{a,b}(B),$$
where $\Theta_\rho(B) = B \otimes \rho[1]$ is the shifted cotangent current of label $\rho \in \mathcal{P}$ smeared by $B \in \mathcal{D}'_c(I)$, in the convention of Construction 4.7.
- the scalar class $\Upsilon_{0,0} = \mathrm{Tr}(\psi)$ maps to the removed odd scalar line, equivalently to zero in the Hamiltonian principal-part target $\mathcal{P}[1]$;
- on primitive single-trace chains with at least two displayed ψ's, $\pi^{\mathrm{cot}}_{\mathrm{prim}} \equiv 0$;
- on multi-trace products of one-ψ classes (or mixed products of zero-ψ and one-ψ traces), $\pi^{\mathrm{cot}}_{\mathrm{prim}} \equiv 0$ (projection away from the multi-trace stratum).

This is a shadow map to the reduced principal-part label target. The polynomial-realization obstruction theorem 5.183 says precisely that the open polynomial one-ψ descendants and the principal-part currents carry distinct $\mathfrak{h}$-module structures. The map above is therefore an indecomposable shadow map after imposing the reduced current target $\mathcal{P}[1]$; the unreduced boundary-observable realization of $\mathcal{P}$ inside the polynomial BV/Koszul algebra is governed by an unreduced target.

THEOREM 5.133 *(Primitive indecomposable P_0-shadow and one-ψ label shadow).* On every interval I, in the stable connected limit, the Hamiltonian zero-ψ primitive indecomposable P_0-shadow π_{prim} of Construction 5.115 and the one-ψ shadow $\pi^{\mathrm{cot}}_{\mathrm{prim}}$ of Construction 5.132 give a surjection to the generator space, equivalently the indecomposable quotient, of the boundary factorization algebra

$$\pi^{\mathrm{cot}}_{\mathrm{prim}} : \widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) \twoheadrightarrow \mathfrak{h}_I \oplus \mathcal{P}_I[1]$$

in the sense of Construction 4.7, with the following compatibilities:

(i) *(BV chain map.)* $\pi^{\mathrm{cot}}_{\mathrm{prim}} \circ Q_{\mathrm{tot}} = Q_{\mathrm{tot}} \circ \pi^{\mathrm{cot}}_{\mathrm{prim}}$ strictly at the chain level (Theorem 5.117 and Theorem 5.123 extending it to the cotangent sector via the closed cocycle property of $\Upsilon_{a,b}$);

(ii) *(Factorization product.)* The zero-ψ Hamiltonian-current compatibility is the unit-retaining linearized statement of Theorem 5.119. The one-ψ label shadow satisfies the analogous unit-retaining linearized statement: scalar–one-ψ and one-ψ–scalar products extend the one-ψ label by zero to the larger interval, while products with two positive primitive factors are killed in the indecomposable quotient;

(iii) *(P_0 bracket.)* The P_0-bracket compatibility square commutes strictly modulo a zero homotopy on the Hamiltonian zero-ψ indecomposable quotient (Theorem 5.120). On the one-ψ shadow the map records labels only; the intertwining of the open polynomial descendant PBW action with the principal-part coadjoint action has the obstruction of Proposition 5.134. The reduced target bracket $\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ is given separately by Theorem 5.186;

(iv) *(Cosheaf.)* $\pi^{\mathrm{cot}}_{\mathrm{prim}}$ is compatible with extension by zero: on the Hamiltonian summand this is extension by zero in $\Omega^\bullet_c$, and on the principal-part cotangent summand it is extension by zero in $\mathcal{D}'_c$.

The target is the nonunital indecomposable quotient: multiplication of two positive primitive generators is zero after projection, while the Lie/P_0 bracket remains the Hamiltonian bracket. Thus the statement is a chain map to the nonunital indecomposable P_0-quotient of $\bar{A}$.

Equivalently, $\pi^{\mathrm{cot}}_{\mathrm{prim}}$ is the chain-level realization of the unreduced-to-reduced primitive shadow required by Problem 5.49, item (3). The unreduced cotangent factorization-center lift remains the higher problem.

Proof. Items (1), (2), (3) on the Hamiltonian zero-ψ sector are Theorems 5.117, 5.119, 5.120 above.

One-ψ shadow. The functional on one-ψ words used in Construction 5.132 sends every word of bidegree (a, b) to the same scalar $\binom{a+b}{a}^{-1}\rho_{a,b}$. It kills the two-$\psi$ boundary because the boundary formula of Lemma 5.122 has two positive and two negative one-ψ summands of the same bidegree. It sends the indecomposable label represented by $\Upsilon_{a,b}$ to the imposed reduced target basis vector $\rho_{a,b}$; the $\mathfrak{h}$-equivariance obstruction for the polynomial one-ψ module is Proposition 5.134. It kills the scalar $\Upsilon_{0,0}$. Primitive chains with $p \geq 2$ displayed ψ's are killed by definition; after one application of Q_{BV} they land in the kernel described above, since the all-ψ boundary is a signed sum of one fewer ψ-chains and the equal-weight functional cancels the two-ψ faces while higher faces still have positive ψ-count. With the current differential on $\mathcal{D}'_c(I)$, this proves strict Q-compatibility on the one-ψ label summand.

The factorization-product compatibility on one-ψ labels is linearized in the same sense as Theorem 5.119. A scalar factor acts as the unit and extends the one-ψ current by zero to the larger interval. A product of a one-ψ primitive with any positive primitive trace has at least two primitive factors and is killed in the indecomposable quotient. Thus the unit-retaining exactly-one-positive projection agrees on the two routes. The P_0-bracket compatibility asserted here is the zero-ψ Hamiltonian statement already proved above. The one-ψ shadow records labels; compatibility between the open descendant PBW action and the principal-part coadjoint action has difference Corollary 5.173. The bracket of two cotangent generators is zero on both sides because $\mathcal{P}[1]$ is an abelian odd ideal in $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$.

Cosheaf compatibility is automatic on each summand: extension by zero on the $\Omega^\bullet_c$ Hamiltonian summand and on the $\mathcal{D}'_c$ principal-part current factor commute with the fixed trace-label projections.

The chain-level surjectivity is the surjectivity of the projection onto primitive indecomposable generators in the stable symmetric algebra (each generator is hit by itself), tensored with the identity on the compactly supported de Rham or distributional current factor on the same interval. □

PROPOSITION 5.134 *(Quantum obstruction to the primitive one-ψ label map).* Let $\mathfrak{h}_\hbar = \bar{A}_\hbar$ carry the Weyl/Moyal bracket

$$\{f,g\}_\hbar = \sum_{j\geq 0} \frac{\hbar^{2j}}{(2j+1)!\,2^{2j}} P^{2j+1}(f,g), \qquad P = \partial_{z_1}\otimes\partial_{z_2} - \partial_{z_2}\otimes\partial_{z_1}.$$

The chain-level one-ψ primitive label map $\Upsilon_{a,b} \mapsto \rho_{a,b}$ has a nonzero compatibility obstruction for the $\mathfrak{h}_\hbar$-coadjoint action. The obstruction already occurs in the indecomposable target and survives projection away from decomposable multi-trace terms.

Proof. Compatibility is equality of the polynomial PBW descendant action of Proposition 5.172, after the label replacement $\widetilde{\Upsilon}_{a,b} \mapsto \rho_{a,b}$, with the Moyal coadjoint action

$$(\mathrm{ad}^\vee_{f,\hbar}\,\delta)(g) = -\delta(\{f,g\}_\hbar).$$

At the single Hamiltonian $f = z_1^4$ and the single descendant label $(a,b) = (1,1)$ the two routes diverge.

On the polynomial one-ψ side, Proposition 5.172 gives

$$z_1^4 : \widetilde{\Upsilon}_{1,1} \longmapsto 4\,\widetilde{\Upsilon}_{4,0}.$$

On the Moyal coadjoint side the classical Poisson term has negative first target index and vanishes, while the P^3 term is nonzero:

$$P^3(z_1^4, z_2^4) = 576\, z_1 z_2, \qquad \frac{1}{3!\,2^2} P^3(z_1^4, z_2^4) = 24\, z_1 z_2.$$

Hence

$$\mathrm{ad}^\vee_{z_1^4,\hbar}\,\delta_{1,1} = -24\,\hbar^2 \delta_{0,4} + \cdots,$$

equivalently $z_1^4 \cdot_\hbar \rho_{1,1} = -24\,\hbar^2 \rho_{0,4} + \cdots$ in the principal-part realization. The two answers have different $\hbar$-order and different primitive label. Since both sides have already been projected to the linear indecomposable summand, adding decomposable multi-trace terms have zero image under $\pi^{\mathrm{cot}}_{\mathrm{prim}}$ and leave this square unchanged.

The finite exact-arithmetic computation verifies the same obstruction through exponents ≤ 5 and Moyal order ≤ 5: 1200 of 1225 nonzero instances disagree under the bidegree label map, and 958 cases have genuine higher Moyal coadjoint terms. The same finite computation also checks the corrected principal-part target below in 3375 finite representation identities with exponents ≤ 3, Moyal order ≤ 5, and $\hbar$-power ≤ 4, and checks the linear-generator leading-term obstruction to every $\hbar$-adic deformation of the classical label shadow in 70 tests with labels ≤ 5. $\square$

Construction 5.135 *(Reduced Moyal principal-part descendant target).* The corrected quantum descendant target is obtained by changing the target, with the polynomial one-ψ representatives fixed. For an interval $I \subset \mathbb{R}$ put

$$\mathfrak{h}_{\hbar,I} = \Omega_c^\bullet(I)\widehat{\otimes}\bar{A}_\hbar, \qquad \mathcal{P}_{\hbar,I} = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[[\hbar]].$$

Define the reduced quantum principal-part descendant complex

$$A^{\mathrm{pp}}_{\partial,\hbar}(I) = \widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}(\mathfrak{h}_{\hbar,I})\widehat{\otimes}\widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}(\mathcal{P}_{\hbar,I}[1]).$$

The differential is the de Rham/current differential on the interval factors and zero on the coefficient modules $\bar{A}_\hbar$ and $\mathcal{P}[[\hbar]]$. The bracket is the Moyal semidirect bracket

$$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \qquad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

with

$$\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0.$$

Here $a, b \in \Omega_c^0(I)$, $B, C \in \mathcal{D}'_c(I)$, and aB denotes multiplication of a compactly supported distribution by a smooth test function.

In monomial coordinates the full coadjoint action is as follows. Put

$$C_r(p,q;m,n) = \sum_{\alpha=0}^{r}(-1)^\alpha\binom{r}{\alpha}(p)_{r-\alpha}(q)_\alpha(m)_\alpha(n)_{r-\alpha}.$$

Then

$$z_1^p z_2^q \cdot_\hbar \rho_{a,b} = -\sum_{\substack{r\geq 1\\ r\ \mathrm{odd}}} \frac{\hbar^{r-1}}{2^{r-1}r!}\, C_r(p,q;a-p+r,b-q+r)\,\rho_{a-p+r,b-q+r},$$

with the convention that negative indices and the scalar target $(0,0)$ are zero. This is a finite sum for each monomial $z_1^p z_2^q$, because the falling factorials vanish for $r > p+q$. The $r=1$ term is the classical Matlis coadjoint action of Theorem 5.179; the $r \geq 3$ terms are the genuine Moyal corrections.

This construction gives a reduced distributional principal-part target for the quantum descendant sector. The polynomial PBW classes $\Upsilon_{a,b}$ remain on the source side.

THEOREM 5.136 *($\hbar$-adic obstruction for the one-ψ label shadow).* Let $M_{\Upsilon,\hbar}$ be a $\mathbb{C}[[\hbar]]$-flat complex generated over $\mathbb{C}[[\hbar]]$ by lifts of the primitive polynomial one-ψ classes $\widetilde{\Upsilon}_{a,b}$, $a+b>0$. Assume:

(i) the source action of the linear Hamiltonians reduces modulo $\hbar$ to the PBW descendant action

$$z_1 \cdot_0 \widetilde{\Upsilon}_{a,b} = b\,\widetilde{\Upsilon}_{a,b-1}, \qquad z_2 \cdot_0 \widetilde{\Upsilon}_{a,b} = -a\,\widetilde{\Upsilon}_{a-1,b};$$

(ii) the target coefficient module is $\mathcal{P}[[\hbar]][1]$, equivalently any fixed interval-current summand of $\mathcal{P}_{\hbar,I}[1]$ from Construction 5.135, with the full Moyal coadjoint action;

(iii) a comparison map $T_\hbar : M_{\Psi,\hbar} \to \mathcal{P}[[\hbar]][1]$ has the classical one-ψ label shadow as leading term:

$$T_\hbar(\widetilde{\Psi}_{a,b}) = \rho_{a,b}[1] + O(\hbar).$$

Then the intertwining equation for $T_\hbar$ has a nonzero obstruction for the $\mathfrak{h}_\hbar$-action. The obstruction already appears after reduction modulo $\hbar$ and for the linear Hamiltonian z_1. Consequently the strict quantum descendant lift of the classical primitive label shadow is obstructed under these assumptions. The surviving quantum target in this statement is the reduced principal-part complex itself, with the polynomial one-ψ classes remaining on the source side.

Proof. Reduction modulo $\hbar$ of a hypothetical intertwiner gives, for the class $\widetilde{\Psi}_{1,0}$ and PBW source action,

$$z_1 \cdot_0 \widetilde{\Psi}_{1,0} = 0,$$

so the source-then-target route gives

$$T_0(z_1 \cdot_0 \widetilde{\Psi}_{1,0}) = 0.$$

The target-then-action route gives, by the $r = 1$ term of Construction 5.135,

$$z_1 \cdot_0 T_0(\widetilde{\Psi}_{1,0}) = z_1 \cdot \rho_{1,0}[1] = -\rho_{1,1}[1].$$

The principal-part basis vector $\rho_{1,1}$ is nonzero in $\mathcal{P}$. This contradiction is independent of all higher $\hbar$-corrections to $T_\hbar$, because those corrections vanish after taking the leading term. It is also independent of the higher Moyal terms: the linear Hamiltonian has only the $r = 1$ Moyal term.

The same computation with z_2 gives $z_2 \cdot_0 \widetilde{\Psi}_{0,1} = 0$ on the PBW side but $z_2 \cdot \rho_{0,1} = \rho_{1,1}$ on the principal-part side. Thus the same mismatch occurs for both coordinates. Since the coefficient differential on $\mathcal{P}[[\hbar]]$ is zero, this leading coefficient represents a nonzero homology class in the reduced target. A higher homotopy can remove it only after changing the target complex or the classical source action, which is outside the stated assumptions. □

THEOREM 5.137 *(Universal Fourier-twisted Rees D-hbar construction).* Let

$$D_\hbar = \mathbb{C}[[\hbar]]\langle z_1, z_2, \partial_1, \partial_2\rangle / ([\partial_i, z_j] = \hbar\delta_{ij},\ [z_i, z_j] = [\partial_i, \partial_j] = 0)$$

be the Rees Weyl algebra. Let

$$O_\hbar = D_\hbar/(\partial_1, \partial_2), \qquad \Delta^{\mathsf{F}}_\hbar = D_\hbar/(z_1, z_2),$$

and let e_0 be the image of 1 in $\Delta^{\mathsf{F}}_\hbar$. The quotients are left cyclic $D_\hbar$-modules, with left ideals generated by ∂_1, ∂_2 and by z_1, z_2. PBW normal form gives $O_\hbar$ the free $\mathbb{C}[[\hbar]]$-basis $z_1^a z_2^b$ and gives $\Delta^{\mathsf{F}}_\hbar$ the free basis $\partial_1^a \partial_2^b e_0$, so both modules are $\mathbb{C}[[\hbar]]$-flat. The Fourier automorphism

$$\mathsf{F}(z_i) = \partial_i, \qquad \mathsf{F}(\partial_i) = -z_i$$

identifies the Fourier twist of $O_\hbar$ with $\Delta^{\mathsf{F}}_\hbar$. In the Čech local-cohomology lattice generated by $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$, the map $e_0 \mapsto \rho_{0,0}$ sends

$$\partial_1^a \partial_2^b e_0 \longmapsto (-1)^{a+b} \hbar^{a+b} a! b!\, \rho_{a,b}.$$

This is a flat $D_\hbar$-module bridge in the Fourier-conjugated sense: $P \in D_\hbar$ acts on the polynomial source as P, and on the delta target as $\mathsf{F}(P)$. After tensoring with $\mathbb{C}((\hbar))$, it gives a filtered Fourier–Rees label bijection after this polarization change between the nonconstant polynomial label span and $\mathcal{P}((\hbar))$. For the original Hamiltonian action it is a Fourier-conjugated $D_\hbar$-module bridge. Before $\hbar$ is inverted, the image is the explicit Rees sublattice

$$\bigoplus_{a,b\geq 0} \mathbb{C}[[\hbar]]\,(-1)^{a+b}\hbar^{a+b}a!b!\,\rho_{a,b}.$$

Proof. The formulas for F preserve the Rees Weyl relation:

$$[\mathsf{F}(\partial_i),\mathsf{F}(z_j)] = [-z_i,\partial_j] = \hbar\delta_{ij}.$$

The polynomial module $\mathcal{O}_\hbar$ is generated by 1 killed by ∂_1,∂_2. After applying F, the cyclic generator is killed by z_1, z_2, which is exactly the defining cyclic vector of $\Delta_\hbar^{\mathsf{F}}$.

In the Čech model ∂_i acts as $\hbar\partial_{z_i}$. Hence

$$\begin{aligned}\partial_1^a\partial_2^b\rho_{0,0} &= \hbar^{a+b}\partial_{z_1}^a\partial_{z_2}^b(z_1^{-1}z_2^{-1}dz_1dz_2)\\ &= (-1)^{a+b}\hbar^{a+b}a!b!\rho_{a,b}.\end{aligned}$$

The factorials and signs are units over $\mathbb{C}$; the nontrivial Rees datum is the power $\hbar^{a+b}$. Inverting $\hbar$ turns this sublattice into the whole principal-part label span. □

Theorem 5.138 *(Exact same-action Rees obstruction).* The following three requirements on a $\mathbb{C}[[\hbar]]$-flat Rees module $M_\hbar$ have a nonzero local-nilpotence obstruction:

(i) its special fibre is the polynomial primitive descendant module generated by the $\widetilde{\Psi}_{a,b}$, $a+b>0$, with linear Hamiltonian action

$$z_1\cdot\widetilde{\Psi}_{a,b} = b\,\widetilde{\Psi}_{a,b-1}, \qquad z_2\cdot\widetilde{\Psi}_{a,b} = -a\,\widetilde{\Psi}_{a-1,b};$$

(ii) on the generic fibre $M_\hbar((\hbar))$, the same linear Hamiltonians z_1, z_2 act locally nilpotently, as they do in the polynomial Rees Weyl module;

(iii) the generic fibre is isomorphic, as a module for the same Hamiltonian Lie algebra, to $\mathcal{P}((\hbar))$ with the Matlis coadjoint action

$$z_1\cdot\rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2\cdot\rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Equivalently, matching the local-nilpotence profile of z_1, z_2 on the generic fibre is a necessary condition for any flat same-action Rees bridge from polynomial one-ψ descendants to principal parts, and that necessary condition has a nonzero obstruction. The polynomial-to-principal-part bridge considered here exists after the Fourier change of polarization in Theorem 5.137; the same-action Rees family is obstructed.

Proof. Local nilpotence of a fixed operator is invariant under module isomorphism over $\mathbb{C}((\hbar))$. On the polynomial descendant side, the displayed formulas lower the b-index for z_1 and the a-index for z_2; hence every finite vector is killed by a high enough power of each linear Hamiltonian. The same is true for the polynomial Rees Weyl module after inverting $\hbar$, because multiplying the operators by the invertible scalar $\hbar$ preserves local nilpotence.

On $\mathcal{P}((\hbar))$, every nonzero finite Laurent vector has an infinite orbit under the linear Hamiltonians. For $v = \sum c_{a,b}\rho_{a,b} \neq 0$,

$$z_1^N v = \sum c_{a,b}(-1)^N (b+1)(b+2)\cdots(b+N)\rho_{a,b+N}.$$

The basis vectors $\rho_{a,b+N}$ remain distinct for distinct (a,b), and every rising factorial displayed above is a nonzero element of $\mathbb{C}((\hbar))$. Thus $z_1^N v \neq 0$ for all N. The same argument using z_2 shifts a upward and proves non-local-nilpotence for z_2. Hence $M_\hbar((\hbar))$ is incompatible with $\mathcal{P}((\hbar))$ under the same Hamiltonian action. □

Remark 5.139. The chain-level primitive shadow uses the symmetric-algebra splitting in place of a heat-kernel propagator on $\mathbb{R}^2 \times \mathbb{C}^2$: the Costello–Gwilliam role of $G_L = \int_0^L K_t\, dt$ is played by the symmetric-algebra splitting, and Lemma 3.21 (positive jets are Q-exact) identifies the E_1-stalk with its C^∞-smearing. The two routes agree on classical cocycles by uniqueness of the primitive projection on a symmetric algebra; the renormalized one-loop lift remains the Costello brane-defect propagator problem of Problem 5.49, items (1)–(2).

COROLLARY 5.140 *(Primitive indecomposable shadow of the reduced cotangent comparison).* Combining Theorem 5.133 with the cochain-level reduced CE/PV central-operation comparison (Theorem 4.20) and the Hamiltonian factorization-algebra equivalence (Theorem 4.11), one obtains a chain-level primitive indecomposable shadow of the reduced cotangent comparison:

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \xrightarrow{\Phi_{\mathrm{cl}}} \mathrm{Z}^{P_0}_{\mathrm{red,Ham}}(A^{\mathrm{pp}}_\partial)$$
$$\xrightarrow{\pi^{\mathrm{cot}*}_{\mathrm{prim}}} \text{primitive indecomposable shadow of the open boundary sector.}$$

The unreduced cotangent factorization-center morphism lifts the composite once the missing data of Problem 5.49 are fixed: the centrality homotopies for principal-part currents against arbitrary unreduced open observables, the Tate/weighted coefficient kernel, and the brane-defect propagator extension. These three data are the named obstruction to the unreduced lift.

Proof. Φ_{cl} is constructed at the cochain level by Theorem 4.20 on the reduced sector after de Rham-current contraction. Theorem 4.11 extends this to a P_0 factorization algebra equivalence in the Hamiltonian sector. Theorem 5.133 gives the primitive indecomposable P_0-shadow and one-ψ label shadow. The composition therefore exists only at the indecomposable-shadow level. Since the one-ψ projection carries the descendant PBW / principal-part coadjoint obstruction, the asserted result is exactly the shadow statement. □

Remark 5.141. Theorem 5.133 is the E_1-page of the obstruction filtration on the zero-ψ and one-ψ primitive generator subspaces. Higher differentials $d_2, d_3, \ldots$ are the unreduced P_0- and product-centrality homotopies ($\mathrm{ob}^{P_0}_{\mathrm{cent}}, \mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}$); the Tate/weighted Casimir kernel $\mathfrak{o}_{\mathrm{Cas}}$ and stratified Weiss Čech–Roos descent class $\delta^{\mathrm{Weiss}}_{\check{C}/R}$; the Costello renormalized brane-defect QME endpoint $\mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}$ carried by $\mathfrak{o}_{\mathrm{BV,end}}$; and the one-$\psi$ Moyal-coadjoint intertwiner whose nonexistence is Proposition 5.134 together with Theorem 5.138. The chain-level construction uses algebraic data (5.139); the renormalized BV homotopies remain Problem 5.49 (1)–(2).

Remark 5.142. The de Rham contraction of 3.21 enters Construction 5.115 only through the compactly supported density class: $H^\bullet(\Omega_c^\bullet(I)) = H_c^\bullet(I;\mathbb{C})$ is one-dimensional in degree 1 for an open interval I, represented by $\lambda_I(t)\,dt$ with integral 1; the degree-zero notation in the primitive shadow is the shifted BV/cotangent coordinate paired against this density in place of point-evaluation in $H_c^0(I)$.

Hadamard/Mittag-Leffler endpoint $\mathfrak{o}_{\mathrm{Had}}$. The propagator factorization $P^w_{\epsilon,L} = P^{\mathrm{base}}_{\epsilon,L}\widehat{\otimes}C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$ is exact; the bulk Hadamard parametrix is universal in the bulk Riemannian/Hermitian data and the coefficient factor is algebraic. By Lemma 8.16 every truncation $P^{w,w_0}_{\epsilon,L}$ admits Costello's Hadamard expansion, and the wave-front bound is w_0-uniform. The Mittag-Leffler limit exists and Theorem 8.17 produces the Hadamard parametrix on the inverse limit, with finite-window locality preserved graph by graph. The component $\mathfrak{o}_{\mathrm{Had}}$ is the endpoint class

$$\mathfrak{o}_{\mathrm{Had}} = [C_n[\Gamma]] \in H^1\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w),\ Q + \{I_0, -\}\big)$$

of 8.17 (iii) for the mixed brane-defect interaction. This class is the QME companion of $\mathfrak{o}_{\mathrm{QME}}$: Hadamard control gives the finite-window locality regulator and the wave-front compatibility, and isolates the QME class as the precise residual. The class $\mathfrak{o}_{\mathrm{Had}}$ vanishes after one produces a compatible graphwise counterterm family $\{C_n[\Gamma]^{w_0}\}_{w_0}$ whose inverse limit $C_n[\Gamma] \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ represents the zero class in the displayed cohomology, equivalently the Hadamard coefficient criterion of 8.19: the all-order quantum source-to-boundary CE/PV comparison reduces to a single brane-defect QME vanishing inside the regulator-admissible sector of 5.76. Hadamard control gives the analytic factor; anomaly cancellation is the non-scalar QME cohomology class of 5.84.

Protected universality $\mathfrak{o}_{\mathrm{univ}}$. The protected Hamiltonian sector at a brane point of Definition 5.143 is the restriction of perturbative twisted-M-theory data $\mathcal{T}_{\mathrm{tw}M}(\mathbb{R}\times X;\mathbb{R}\times p)$ to the open BV theory on the brane line, the closed BV theory at the formal symplectic polydisk $\widehat{X}_p$, and the bulk-to-boundary evaluation. Its admissibility for the local theorem requires formal Darboux identification with $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1,z_2]]/\mathbb{C}\cdot 1$ and $\mathbf{Ext}^*_X(\mathcal{O}_p,\mathcal{O}_p)$, brane-defect locality through the weighted Tate field complex, and scalar anomaly compatibility with the class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A},\mathbb{C})$ of Lemma 6.2. The component $\mathfrak{o}_{\mathrm{univ}}$ is the four-component protected twist obstruction tuple of Definition 5.145,

$$\mathfrak{o}_{\mathrm{univ}} = \big(\mathrm{per}(v),\ C_{\mathrm{cl},p},\ \mathrm{ob}_{\mathrm{scal}},\ \mathrm{ob}^{\mathrm{prot}}_{\mathrm{QME}}\big),$$

with the period class $\mathrm{per}(v) = \partial(\alpha_v) \in H^1(X,\mathbb{C}_X)$ for every holomorphic symplectic vector field used by the comparison; the closed-field comparison cone $C_{\mathrm{cl},p} = \mathrm{Cone}(\chi_{\mathrm{cl}})$; the scalar anomaly mismatch $\mathrm{ob}_{\mathrm{scal}} = \chi_{\mathrm{anom}}([\mathrm{anom}_{\mathrm{prot}}]) - [\bar{c}]$; and the protected brane-defect QME class $[QI_{\mathrm{prot}} + \frac{1}{2}\{I_{\mathrm{prot}}, I_{\mathrm{prot}}\} + \hbar\Delta I_{\mathrm{prot}}]$. The vanishing condition for $\mathfrak{o}_{\mathrm{univ}}$ is the joint vanishing of these four classes with chosen null-homotopies, equivalently the admissible restriction of Definition 5.144; under that vanishing the protected Hamiltonian sector identifies with the formal-local Hamiltonian model and the Casimir-balanced equivariant scalar QME criterion 5.150 sets the scalar component to zero exactly on the locus $\kappa_{\mathfrak{S}} = 0$.

5.6 Formal-Darboux normal form of the local Hamiltonian sector

The CE/PV Koszul resolution gives the trace summand of the map from the bulk stalk $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_b$ to the brane-side derived E_1-Hochschild complex $HH^\bullet(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ in Theorem 5.6 of §5.2. Its universal property is the formal-Darboux normal-form property of the trace sector. The local theorem of §5.3

lives at one brane point of the formal symplectic polydisk $\widehat{\mathbb{C}^2}_0$. An ambient theory on a connected holomorphic-symplectic surface (X, ω) with brane point p, including any comparison with Costello's perturbative twisted M-theory [25], requires its own open fields, closed fields, bulk-to-boundary evaluation, and anomaly matching before it restricts to this formal Hamiltonian model. The admissible cases form the category $\mathsf{ProtHam}$ defined below. The formal Darboux model

$$\mathfrak{M}^{\mathrm{Ham}}_{\mathrm{loc}} = \big(\widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[1],\ \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1],\ [\bar{c}]\big), \quad \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1,$$

is the formal-Darboux normal form in $\mathsf{ProtHam}$ (Theorem 5.147); equivalently, it gives the normal form of the trace-sector Koszul map from $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_b$ at the formal-Darboux stalk. A pre-admissible datum is promoted to $\mathsf{ProtHam}$ after the four-coordinate class

$$\mathrm{Ob}^{\mathrm{univ}}_{p,X} = \big(\mathrm{per}(v),\ [C_{\mathrm{cl},p}],\ \mathrm{ob}_{\mathrm{scal}},\ \mathrm{ob}_{\mathrm{QME}}\big)$$

of Definition 5.145 is represented and killed by chosen null-homotopies; the four coordinates lie in independent cohomology groups (the de Rham extension on X, the closed-comparison cone, $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$, and the weighted protected local-functional complex), and a single nonzero coordinate obstructs the corresponding admissibility clause. In Koszul language, $\mathrm{Ob}^{\mathrm{univ}}_{p,X}$ is the obstruction table for extending the formal-Darboux trace-sector map along the ambient datum at (X, ω, p).

A compact theorem using this formal restriction contains a second geometric or automorphic object: the BCOV/Kodaira–Spencer propagator, the Maulik–Nekrasov–Okounkov–Pandharipande virtual trace, a Gopakumar–Vafa or Ooguri–Strominger–Vafa scalar, a Donaldson–Thomas/Pandharipande–Thomas wall-crossing trace, a Calabi–Yau-to-chiral specialization, or an Igusa/Borcherds–Kac–Moody normalization. The comparison with that object is governed by the matched-conventions descent class of Corollary 5.161 and §11.4. Vanishing of $\mathrm{Ob}^{\mathrm{univ}}_{p,X}$ proves only the formal Hamiltonian restriction at (X, ω, p).

Definition 5.143 *(Protected Hamiltonian sector at a brane point).* Let (X, ω) be a connected holomorphic-symplectic surface, let $p \in X$, and let $\mathcal{T}_{\mathrm{tw}M}(\mathbb{R} \times X; \mathbb{R} \times p)$ denote perturbative twisted-M-theory data on $\mathbb{R} \times X$ in the presence of a stack of N one-dimensional topological branes supported on $\mathbb{R} \times p$ (Costello [25]). A *protected Hamiltonian sector at* $\widehat{p}$ is a restricted datum

$$\mathcal{T}_{\mathrm{tw}M}(\mathbb{R} \times X; \mathbb{R} \times p)\big|_{\mathrm{Ham},\widehat{p}} = \big(\mathcal{E}_{\mathrm{open,prot}}, \mathcal{E}_{\mathrm{cl,prot}}, \partial^{\mathrm{prot}}_{\mathrm{bb}}\big)$$

consisting of the protected open BV theory on the brane line, the protected closed BV theory coming from formal Hamiltonian operators at the symplectic polydisk $\widehat{X}_p$, and the protected bulk-to-boundary evaluation on the brane algebra.

The realization datum is the construction of this triple from the ambient BV fields and to prove the vanishing of the obstruction datum below.

Definition 5.144 *(Admissible protected Hamiltonian restriction).* A protected Hamiltonian sector is *admissible for the local theorem* if it comes with the following data.

(i) *Formal Darboux Hamiltonian reduction.* The completed closed Hamiltonian algebra at p is identified, up to formal symplectomorphism, with $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, and the point brane Ext algebra is identified with $\mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda^* T_p X$, with cyclic trace induced by ω_p.

(ii) *Brane-defect locality.* The protected operator restriction to $\mathbb{R} \times p$ factors through the weighted Tate field complex and is compatible with the interval-wise factorization map $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$, the chain-level primitive projection, and the weighted Hadamard parametrix used in the local theorem.

(iii) *Scalar anomaly compatibility.* The centre-of-mass scalar anomaly of the protected brane stack maps to the class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ used in the local scalar-anomaly comparison; here $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$ is the reduced polynomial Hamiltonian algebra at the brane point.

Definition 5.145 *(Protected twist obstruction datum).* Fix a protected twisted-M background near $\mathbb{R} \times p$ and a comparison with the Hamiltonian formal-disk model. The comparison carries four obstruction classes.

First, every holomorphic symplectic vector field v determines the closed one-form

$$\alpha_v = -\iota_v \omega.$$

The holomorphic de Rham exact sequence

$$0 \longrightarrow \mathbb{C}_X \longrightarrow \mathcal{O}_X \xrightarrow{d} \Omega^1_{X,\mathrm{cl}} \longrightarrow 0$$

gives the period obstruction

$$\mathrm{per}(v) = \delta(\alpha_v) \in H^1(X, \mathbb{C}_X).$$

This is the obstruction to choosing a single global holomorphic Hamiltonian primitive for v.

Second, a chosen local comparison of closed protected fields with the Hamiltonian BF fields gives a cone

$$C_{\mathrm{cl},p} = \mathrm{Cone}\Big(\mathcal{E}^!_{\mathrm{cl,prot},c}(\mathbb{R} \times \widehat{X}_p) \xrightarrow{\chi_{\mathrm{cl}}} \Omega^\bullet_c(\mathbb{R})\widehat{\otimes}(\widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1])^!\Big).$$

The comparison is a field-level equivalence only after this cone is acyclic in the protected summand.

Third, the scalar anomaly mismatch is

$$\mathrm{ob}_{\mathrm{scal}} = \chi_{\mathrm{anom}}\big([\mathrm{anom}_{\mathrm{prot}}]\big) - [\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}),$$

where χ_{anom} is the anomaly map induced by the comparison and $[\bar{c}]$ is the scalar class used in the local theorem.

Fourth, after choosing the weighted coefficient model used by the local theorem, the brane-defect quantum obstruction is the class

$$\mathrm{ob}_{\mathrm{QME}} = \Big[QI_{\mathrm{prot}} + \frac{1}{2}\{I_{\mathrm{prot}}, I_{\mathrm{prot}}\} + \hbar\Delta I_{\mathrm{prot}}\Big] \in H^1\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_{\mathrm{prot},w}), Q + \{I_{0,\mathrm{prot}}, -\}\big).$$

Its vanishing is the precise statement that the protected brane-defect interaction satisfies the QME in the same normalization as Problem 5.73.

LEMMA 5.146 *(Coordinate-free objects of* $\mathsf{ProtHam}$*).* Definition 5.144 imposes on an object (X, ω, p, Θ) of $\mathsf{ProtHam}$ coordinate-free data for Θ and treats the Darboux expression and the bulk-to-boundary evaluation by trace substitution. An admissible protected Hamiltonian restriction Θ at $\widehat{p}$ is a quadruple of independent data: a sheaf of dgla brackets on $\widehat{p}$, a brane-defect open BV theory, a bulk-to-boundary evaluation $\mathrm{ev}_\Theta \colon \Theta \to \mathcal{A}_\partial(L)$, and a scalar anomaly representative $[\bar{c}]_\Theta$; the morphism conditions preserve each piece. Examples:

(1) $(X, \omega) = (T^*\Sigma, d\theta_\Sigma)$ for Σ a Riemann surface, evaluated at a point of the zero section, with Θ given by the cotangent BV theory of Σ;

(2) (X, ω) any K3 surface at a smooth point, with Θ the Hochschild restriction of a curved A_∞-deformation of the structure sheaf;

(3) (X, ω) a holomorphic-symplectic surface with coordinate-free symplectic form, Θ the protected Hamiltonian sector of an arbitrary Darboux choice.

The objects (1)–(3) are presented coordinate-free; the Darboux symplectomorphism, the trace-substitution evaluation, and the identification of $[\bar{c}]_\Theta$ with $[\bar{c}]$ are conclusions of Theorem 5.147; ProtHam membership is coordinate-free.

Proof. The four pieces of an admissible protected Hamiltonian restriction are listed in Definition 5.144 independent of a coordinate choice. The functorial datum ev_Θ may be given by other evaluations: the cotangent example (1) admits the de Rham residue evaluation as ev_Θ; the Hochschild example (2) admits the Connes–Tsygan evaluation; the symplectic-surface example (3) admits any ω-compatible polynomial evaluation that satisfies (U1)–(U5). Each receives the trace-substitution form as the conclusion of the formal Darboux reduction, the brane-line factorization restriction $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$, and the scalar-anomaly identification used in the proof of Theorem 5.147. □

THEOREM 5.147 *(Formal-Darboux normal form of the Hamiltonian trace sector; corollary of the CE/PV Koszul resolution identification).* Let ProtHam denote the category whose objects are triples (X, ω, p) with (X, ω) a connected holomorphic-symplectic surface and $p \in X$ a brane point, together with an admissible protected Hamiltonian sector at $\widehat{p}$ in the sense of Definition 5.144, and whose morphisms are formal Darboux symplectomorphisms at the brane point that intertwine the protected open BV theory, the protected closed BV theory, the protected bulk-to-boundary evaluation, and the scalar anomaly class $[\bar{c}]$. Let $\mathfrak{M}^{\mathrm{Ham}}_{\mathrm{loc}}$ denote the formal Darboux model

$$\mathfrak{M}^{\mathrm{Ham}}_{\mathrm{loc}} = \big(\widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[1],\ \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1],\ [\bar{c}]\big), \qquad \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1,$$

with bulk-to-boundary evaluation $f \mapsto \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$ on the brane line. Let $\mathcal{D}_{X,p}$ be the torsor of formal Darboux frames

$$\eta\colon (\widehat{X}_p, \omega) \xrightarrow{\sim} (\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2).$$

For every object (X, ω, p, Θ) of ProtHam, every $\eta \in \mathcal{D}_{X,p}$, and every fixed protected comparison datum entering the definition of ProtHam, there is a normal-form morphism

$$\mu_\eta\colon (X, \omega, p, \Theta) \longrightarrow \mathfrak{M}^{\mathrm{Ham}}_{\mathrm{loc}}$$

in ProtHam. Its coordinate part is the formal Darboux reduction

$$\mathfrak{h}_{X,p} \xrightarrow{\sim} \widehat{\mathfrak{h}}, \qquad \mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p) \xrightarrow{\sim} \Lambda^* T_p X, \qquad [\bar{c}]_\Theta \mapsto [\bar{c}],$$

composed with the brane-line factorization restriction $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ of Theorem 4.11 and the protected bulk-to-boundary evaluation. Darboux frames act on the coordinate part; this statement does not classify the full mapping groupoid of protected open and closed BV comparison data.

A pre-admissible ambient datum equipped with the obstruction class $\mathrm{Ob}^{\mathrm{univ}}_{p,X}$ of Definition 5.145 is promoted to an object of ProtHam after those obstruction coordinates are represented and killed by chosen null-homotopies. A nonzero coordinate obstructs the corresponding admissibility clause, not a universal Hom-set terminality statement.

Proof. The normal-form assertion for $\mathfrak{M}^{\mathrm{Ham}}_{\mathrm{loc}}$ is the Darboux-frame form of the trace-sector CE/PV Koszul map of §5.2: any protected datum on a connected holomorphic-symplectic surface restricts to that Koszul resolution at the formal-Darboux point $p \in \partial X$ of the brane-link boundary once the four coordinates of $\mathrm{Ob}^{\mathrm{univ}}_{p,X}$ vanish, since the Koszul-resolution map is canonical on the formal-Darboux stalk and frame dependence is the Darboux–Weinstein rigidity of the stalk. The detailed verification follows.

Existence of μ_η is read off the three clauses of Definition 5.144. Formal Darboux Hamiltonian reduction gives a symplectomorphism $\mathfrak{h}_{X,p} \xrightarrow{\sim} \widehat{\mathfrak{h}}$ of the completed closed Hamiltonian algebra and an identification $\mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p) \xrightarrow{\sim} \Lambda^* T_p X$ of the point-brane Ext algebra carrying the cyclic trace induced by ω_p; this fixes the closed and open underlying objects of $\mathfrak{M}^{\mathrm{Ham}}_{\mathrm{loc}}$. Brane-defect locality factors the protected operator restriction through the weighted Tate field complex and the brane-line factorization map $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ of Theorem 4.11, with target $f \mapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ by Theorem 5.42. Scalar anomaly compatibility identifies $[\bar{c}]_\Theta$ with $[\bar{c}]$ inside $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ through χ_{anom}. Their composition is μ_η.

Formal Darboux frames form a torsor under G_{Darb}, and changing the frame changes only the coordinate expression of the displayed normal form. The open BV theory, closed BV theory, bulk-to-boundary map, scalar anomaly, and chosen null-homotopies are part of the protected comparison datum and are not classified by the Darboux torsor.

Lemma 5.153 below says that a pre-admissible datum is promoted after each of the four classes $\mathrm{per}(v)$, $[C_{\mathrm{cl},p}]$, $\mathrm{ob}_{\mathrm{scal}}$, $\mathrm{ob}_{\mathrm{QME}}$ is represented and killed by chosen null-homotopies. These classes measure the corresponding admissibility clauses (Darboux Hamiltonian reduction, closed-field protected comparison, scalar anomaly compatibility, brane-defect QME locality). □

COROLLARY 5.148 *(Restriction to the formal-Darboux trace-sector map).* Any protected datum $(X, \omega, p, \Theta) \in \mathsf{ProtHam}$ restricts at the formal-Darboux point $p \in \partial X$ of the Costello–Gwilliam brane-link boundary to the trace-sector formal-Darboux CE/PV Koszul map Φ_{coord} of §5.2, landing in the derived Hochschild complex of the chosen brane algebra, with bulk-to-boundary evaluation $u_f \mapsto J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ for polynomial f, extended to formal f only after completion, and $c_f \mapsto \theta_f$, uniquely up to inner formal Darboux symplectomorphism. Equivalently, $\mathrm{Ob}^{\mathrm{univ}}_{p,X}$ is the obstruction to convergence of that trace-sector map along the ambient datum at (X, ω, p): a single nonzero coordinate is the precise obstruction (period, closed-field comparison cone, scalar anomaly, brane-defect QME) to extending the formal-Darboux trace-sector map along Θ.

Proof. Theorem 5.147 gives, for each formal Darboux frame and protected comparison datum, the normal-form morphism μ_η; §5.2 identifies the formal-Darboux stalk of $\mathfrak{M}^{\mathrm{Ham}}_{\mathrm{loc}}$ with the trace-sector CE/PV Koszul map under $c_f \mapsto \theta_f$, $u_f \mapsto J(f)$. Composing gives the corollary; the obstruction reading is Lemma 5.153 restated for the trace-sector map. □

The QME coordinate of $\mathrm{Ob}^{\mathrm{univ}}_{p,X}$*: Casimir-balance and the queer counterexample.* The first three coordinates of $\mathrm{Ob}^{\mathrm{univ}}_{p,X}$ – the period class, the closed-field comparison cone, and the scalar anomaly mismatch – are classical: they are read off the holomorphic-symplectic geometry, the protected closed sector, and the Lie cohomology of $\bar{A}$ respectively, and their vanishing is a structural statement about the ambient datum. The fourth coordinate $\mathrm{ob}_{\mathrm{QME}}$ is a quantum BV class that depends on the Chan-Paton matrix factor. The Casimir-balanced equivariant scalar QME criterion (Theorem 5.150 below) identifies an explicit locus $\kappa_{\mathfrak{s}} = 0$ on which the scalar component of the QME obstruction vanishes identically, and exhibits $\mathfrak{gl}(N|N)$ as an instance. The queer-centre theorem (Theorem 5.152 below) exhibits a complementary locus where unsigned parity-equivariance breaks down and a nonzero odd scalar class $\hbar N\, [\bar{c}]^{\mathrm{otr}}$ survives, showing that the algebraic spin-Hecke coefficient datum governs the cohomology class, while chain-level QME cancellation requires the additional homotopy data. Together these two theorems give the precise dichotomy controlling the QME coordinate: scalar QME-vanishing is ruled by a single matrix coefficient $\kappa_{\mathfrak{s}}$, and a target null-homotopy of the surviving odd class is the residual ingredient for the queer brane stack.

Definition 5.149 *(Casimir-balanced equivariant QME datum).* A Casimir-balanced equivariant brane-defect datum consists of:

(i) a matrix Lie superalgebra $\mathfrak{s}$ acting on the Chan-Paton factor, together with an even nondegenerate invariant form $B_{\mathfrak{s}}$ used to build the BV heat-kernel contraction;

(ii) a trace character $\mathrm{tr}_{\mathfrak{s}}$ on the matrix factor. Let $C_{B_{\mathfrak{s}}} = \sum_\alpha e_\alpha \otimes e^\alpha$ be the Casimir for $B_{\mathfrak{s}}$, put $K_{\mathfrak{s}} = m(C_{B_{\mathfrak{s}}})$, and define the scalar contraction coefficient $\kappa_{\mathfrak{s}}$ on the scalar trace line by

$$\mathrm{tr}_{\mathfrak{s}}(AK_{\mathfrak{s}}) = \kappa_{\mathfrak{s}}\, \mathrm{tr}_{\mathfrak{s}}(A).$$

The datum is Casimir-balanced when $\kappa_{\mathfrak{s}} = 0$;

(iii) the weighted brane-defect local-functional complex

$$\mathfrak{Q}_{\mathfrak{s}} = \Big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{\mathfrak{s}}_{\mathrm{bd},w}),\, d_{\mathrm{QME}} = Q + \{I_{0,\mathfrak{s}}, -\}\Big);$$

(iv) a d_{QME}-stable scalar filtration and a filtered chain projection

$$\widehat{\sigma}_{\mathrm{sc}} \colon \mathfrak{Q}_{\mathfrak{s}} \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1]$$

whose associated graded is the scalar-symbol map σ_{sc};

(v) a $P_{(z_1)}$-equivariant coefficient topology on the formal symplectic polydisk, where

$$P_{(z_1)} = \{\varphi \in \mathrm{Symp}_{\mathrm{form}}(\widehat{\mathbb{C}^2}_0) : \varphi^*(z_1) \in (z_1)\}.$$

The scalar associated graded of $\mathfrak{Q}_{\mathfrak{s}}$ is the Chevalley–Eilenberg complex of the reduced Hamiltonian algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ with coefficients in the central boundary ghost line $\mathbb{C}\gamma_1$.

THEOREM 5.150 *(Casimir-balanced equivariant scalar QME vanishing).* Let $(\mathfrak{s}, B_{\mathfrak{s}}, \mathrm{tr}_{\mathfrak{s}})$ be an equivariant brane-defect datum in the sense of Definition 5.149. Write $\omega(f, g) = [\{f, g\}]_0 \in \mathbb{C}$ for the constant Taylor coefficient of the Poisson bracket on $\mathbb{C}[z_1, z_2]$ (Lemma 6.2). For $a, b \in \Omega^0_c(\mathbb{R})$ and $f, g \in \mathbb{C}[z_1, z_2]$, the scalar part of the mixed brane-defect QME obstruction has the explicit representative

$$\mathrm{Ob}^{\mathfrak{s}}_{\mathrm{sc}}(\gamma_1; a, f; b, g) = \hbar\, \kappa_{\mathfrak{s}}\, \omega(f, g) \int_{\mathbb{R}} a(t) b(t) \gamma_1(t)\, dt \in \mathfrak{Q}_{\mathfrak{s}},$$

with associated graded scalar class

$$[\mathrm{Ob}^{\mathfrak{s}}_{\mathrm{sc}}] = \hbar\, \kappa_{\mathfrak{s}}\, [\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}\gamma_1).$$

Two consequences follow. First, the scalar QME obstruction vanishes identically, $P_{(z_1)}$-equivariantly, on the Casimir-balanced locus $\kappa_{\mathfrak{s}} = 0$; for the standard $\mathfrak{gl}(N|N)$ supertrace pairing $K_{\mathfrak{s}} = \mathrm{sdim}(V) I = 0$, so $\kappa_{\mathfrak{s}} = 0$ and Casimir-balance holds. Second, off the Casimir-balanced locus the scalar channel carries the precise nonzero class $\hbar\, \kappa_{\mathfrak{s}} [\bar{c}]$, which is the exact obstruction to scalar QME-flatness of the brane-defect interaction.

If, in addition, the reduced non-scalar summand

$$\mathfrak{Q}^{\mathrm{nsc}}_{\mathfrak{s}} = \mathrm{Cone}\big(\widehat{\sigma}_{\mathrm{sc}} \colon \mathfrak{Q}_{\mathfrak{s}} \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1]\big)[-1]$$

has zero degree-one cohomology after the chosen weighted regulator and interval-factorization null-homotopies are imposed, then the complete degree-one brane-defect QME obstruction class in $H^1(\mathfrak{Q}_\mathfrak{s})$ vanishes. The non-scalar vanishing hypothesis is the additional condition extending the result beyond the scalar channel.

Proof. The scalar contraction in the BV bracket is obtained by closing the matrix index in the single boundary loop. With the invariant form $B_\mathfrak{s}$ this contraction is the multiplication $K_\mathfrak{s} = m(C_{B_\mathfrak{s}})$ of the Casimir tensor, and its scalar trace-line coefficient is $\kappa_\mathfrak{s}$. The target part of the same local bracket is the constant Taylor coefficient cocycle

$$\omega(f, g) = [\{f, g\}]_0$$

of Lemma 6.2, and the brane-line locality gives the compactly supported factor $\int_\mathbb{R} a(t)b(t)\gamma_1(t)\,dt$. This proves the displayed representative.

Passing to the scalar associated graded keeps only the Hamiltonian cocycle $[\bar{c}]$ and the matrix coefficient $\kappa_\mathfrak{s}$, giving the displayed cohomology class. Casimir-balance sets this coefficient to zero, so the degree-one representative is the zero cochain itself. The $P_{(z_1)}$-equivariance follows because $\kappa_\mathfrak{s}$ is matrix-factor data and ω is the symplectic Poisson cocycle on the target factor. The completion at the brane axis is functorial only under $P_{(z_1)}$, which is exactly the equivariance group in the hypothesis.

The last assertion uses the long exact sequence of cohomology for the displayed cone, which is defined only for the filtered chain projection $\widehat{\sigma}_{\mathrm{sc}}$ included in the datum. The scalar quotient class is zero by the first part, and the reduced degree-one cohomology is zero by hypothesis. Therefore the total degree-one obstruction class in $H^1(\mathfrak{Q}_\mathfrak{s})$ vanishes. □

Definition 5.151 *(Spin-Hecke queer coefficient datum).* Let $N \geq 1$ and

$$\mathfrak{q}(N) = \left\{ \begin{pmatrix} A & B \\ B & A \end{pmatrix} : A, B \in \mathfrak{gl}_N \right\} \subset \mathfrak{gl}(N|N).$$

Put

$$J = \begin{pmatrix} 0 & I_N \\ -I_N & 0 \end{pmatrix}, \qquad P^\mathfrak{q} = \begin{pmatrix} I_N & 0 \\ 0 & -I_N \end{pmatrix},$$

and define the queer trace, valued in the odd line $\Pi\mathbb{C}$ (parity reversal of $\mathbb{C}$; Π shifts even degree to odd):

$$\mathrm{otr}\begin{pmatrix} A & B \\ B & A \end{pmatrix} = \mathrm{Tr}(B) \in \Pi\mathbb{C}.$$

A spin-Hecke datum for the queer brane is the Sergeev Hecke–Clifford action on $V^{\otimes n}$, $V = \mathbb{C}^{N|N}$, whose central-character comparison identifies

$$\mathrm{Tr}_{\mathfrak{gl}_N}(I_N) = \mathrm{otr}_{\mathfrak{q}(N)}(J) = N.$$

THEOREM 5.152 *(Queer-centre anomaly from the spin-Hecke datum).* Assume the spin-Hecke datum of Definition 5.151. For the queer-trace boundary evaluation $f \mapsto \mathrm{otr}\, f(\phi_1, \phi_2)$, the odd scalar QME representative on the queer brane stack is

$$\mathrm{Ob}^{\mathrm{otr}}_{\mathrm{sc}}(\gamma_1; a, f; b, g) = \hbar N\, \omega(f, g) \int_\mathbb{R} a(t)b(t)\gamma_1(t)\,dt \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}^\mathfrak{q}_{\mathrm{bd},w})_{\bar{1}},$$

with associated graded CE class

$$\hbar N[\bar{c}]^{\mathrm{otr}} \in H^2_{\mathrm{Lie}}(\bar{A}, \Pi\mathbb{C})_{\bar{1}}.$$

Three statements follow. First, the odd class $\hbar N[\bar{c}]^{\mathrm{otr}}$ is nonzero for every $N \geq 1$ and is independent of the even class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})_{\bar{0}}$, because they sit in the two direct summands $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}) \oplus H^2_{\mathrm{Lie}}(\bar{A}, \Pi\mathbb{C})$ of the even-odd decomposition. Second, the odd class survives over $\mathbb{C}[[\hbar]]$ and admits a null-homotopy only after either an odd scalar quotient is imposed on the target or a target null-homotopy of the odd line $\Pi\mathbb{C} \cdot [\bar{c}]^{\mathrm{otr}}$ is given; the class itself is the precise residual obstruction. Third, the spin-Hecke datum fixes the coefficient identity on cohomology; chain-level QME cancellation has the parity obstruction

$$P^{\mathfrak{q}} J (P^{\mathfrak{q}})^{-1} = -J$$

obstructs the unsigned parity-equivariant regulator used in Theorem 5.150.

Consequently, the queer brane stack is a named ambient datum on which the QME coordinate $\mathrm{ob}_{\mathrm{QME}}$ of $\mathrm{Ob}^{\mathrm{univ}}_{p,X}$ is nonzero in the cohomology; admissibility of the queer datum requires either an odd-scalar quotient at the target or a chosen null-homotopy of the odd class, given as part of the obstruction-vanishing data of Definition 5.145.

Proof. The queer brane has two central coefficient lines: the even line generated by the identity and the odd line generated by J. The queer trace kills the even identity and evaluates the odd central element by

$$\mathrm{otr}(J) = \mathrm{Tr}(I_N) = N.$$

The spin-Hecke datum asserts that this coefficient is preserved by the Sergeev central-character comparison; the Hamiltonian cocycle on the target remains the same $\omega(f, g) = [\{f, g\}]_0$. Therefore the scalar boundary loop gives exactly the displayed odd representative.

Passing to associated graded removes the brane-line smearing and keeps the Lie two-cocycle $\Pi\omega$, hence the class $\hbar N[\bar{c}]^{\mathrm{otr}}$. Nontriviality is the same calculation as Lemma 6.2: a primitive $\bar{\eta} : \bar{A} \to \Pi\mathbb{C}$ would lift with $\bar{\eta}(1) = 0$, but then

$$\partial\bar{\eta}(z_1, z_2) = -\bar{\eta}(\{z_1, z_2\}) = -\bar{\eta}(1) = 0$$

while $\omega(z_1, z_2) = 1$. The even and odd classes are independent because they lie in the two direct summands $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}) \oplus H^2_{\mathrm{Lie}}(\bar{A}, \Pi\mathbb{C})$. Since $N \geq 1$ in the datum and $\hbar$ is formal, the displayed odd class survives inside $H^2_{\mathrm{Lie}}(\bar{A}, \Pi\mathbb{C})[[\hbar]]$. Killing it requires an additional target quotient or null-homotopy beyond the spin-Hecke coefficient comparison.

Finally, $P^{\mathfrak{q}} J (P^{\mathfrak{q}})^{-1} = -J$ is an explicit matrix computation. Thus the queer channel has signed parity; the super-balanced cancellation uses the unsigned parity condition. The algebraic spin-Hecke datum gives the coefficient class; a null-homotopy of this odd obstruction is an additional datum. □

LEMMA 5.153 *(Primary obstruction test).* A protected twisted-M background gives an admissible protected Hamiltonian restriction in the sense of Definition 5.144 if and only if the four classes of Definition 5.145 vanish in the senses stated there: the period classes $\mathrm{per}(v)$ vanish for the Hamiltonian vector fields used by the sector, the closed-field comparison cone $C_{\mathrm{cl},p}$ is acyclic on the protected summand, the scalar mismatch $\mathrm{ob}_{\mathrm{scal}}$ is zero, and a solution of the QME class $\mathrm{ob}_{\mathrm{QME}}$ has been chosen compatibly with compactly supported interval factorization.

Proof. The period statement is the connecting morphism of $0 \to \mathbb{C}_X \to O_X \xrightarrow{d} \Omega^1_{X,\mathrm{cl}} \to 0$: local Hamiltonian primitives h_i for α_v differ by constants on overlaps, and that Čech cocycle is exactly $\delta(\alpha_v)$; vanishing of $\mathrm{per}(v) = \delta(\alpha_v) \in H^1(X, \mathbb{C}_X)$ is precisely the statement that the local primitives glue to a global Hamiltonian, modulo one global constant.

The closed-field comparison is a quasi-isomorphism of protected closed fields with Hamiltonian BF fields exactly when its mapping cone $C_{\mathrm{cl},p} = \mathrm{Cone}(\chi_{\mathrm{cl}})$ is acyclic on the protected summand: a nonzero cohomology class in the cone is either a protected closed field invisible to the formal Hamiltonian model, or a Hamiltonian BF field outside the protected-sector image, and either witness blocks the field-level identification.

Scalar compatibility is the equality $\chi_{\mathrm{anom}}([\mathrm{anom}_{\mathrm{prot}}]) = [\bar{c}]$ in $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$, where the comparison map χ_{anom} is the anomaly-comparison morphism of Definition 5.145; both classes are central Lie cocycles on $\bar{A}$ by construction, so their equality is the assertion $\mathrm{ob}_{\mathrm{scal}} = \chi_{\mathrm{anom}}([\mathrm{anom}_{\mathrm{prot}}]) - [\bar{c}] = 0$.

Finally, the displayed QME expression is the standard degree-one obstruction class for deforming a classical BV interaction to a quantum BV interaction in the fixed weighted coefficient category; vanishing of this class with chosen interval-factorization null-homotopies is exactly the brane-defect locality clause of Definition 5.144. The four vanishings therefore give the four admissibility clauses, and conversely each admissibility clause records the vanishing of its class. □

Definition 5.154 *(Ambient-to-protected realization map).* An ambient-to-protected realization map is a morphism from the ambient twisted-M BV fields near $\mathbb{R} \times p$ to the triple

$$(\mathcal{E}_{\mathrm{open,prot}}, \mathcal{E}_{\mathrm{cl,prot}}, \partial^{\mathrm{prot}}_{\mathrm{bb}})$$

whose closed-field cone is $C_{\mathrm{cl},p}$, whose scalar anomaly map is χ_{anom}, and whose brane interaction has QME obstruction $\mathrm{ob}_{\mathrm{QME}}$.

PROPOSITION 5.155 *(Admissible protected-sector reduction).* Let (X, ω) be a connected holomorphic-symplectic surface and let $p \in X$. Suppose that a perturbative twisted-M background with branes on $\mathbb{R} \times p$ gives an admissible protected Hamiltonian sector at $\widehat{p}$. Then, after the Darboux coordinate and scalar-anomaly normalizations included in the admissibility data, that restricted sector is identified with the formal local reduced Hamiltonian CE/PV brane model of §5.3.

The open side is

$$\Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1],$$

the closed side is

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}), \qquad \mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1], \qquad \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1,$$

and the bulk-to-boundary evaluation is

$$f \longmapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

Hence the proved Hamiltonian, factorization, and scalar-anomaly statements of the local theorem apply to the protected sector. The all-order one-antifield quantum extension requires vanishing of the mixed brane-defect QME obstruction of Problem 5.73 for that sector.

Proof. The formal Darboux part of admissibility identifies the completed Hamiltonian algebra at the brane point with $\mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, after choosing formal Darboux coordinates. The point-brane Ext algebra and its cyclic trace give the open coefficient data of the Dirac brane formal-stalk model.

Brane-defect locality gives the factorization-algebraic restriction to the brane line and matches it with the local factorization map, primitive projection, and weighted Hadamard reduction already used in the local theorem. These hypotheses put the protected sector in the same algebraic category as the local model. QME-flatness for the one-antifield quantum extension is the vanishing condition $\mathrm{ob}_{\mathrm{QME}} = 0$ in Theorem 5.156.

Scalar anomaly compatibility identifies the brane centre-of-mass anomaly with the class $[\bar{c}]$ appearing in the local anomaly comparison. The three admissibility hypotheses therefore give exactly the data on which the local theorem acts. Applying that theorem gives the asserted protected-sector reduction. □

THEOREM 5.156 *(Protected twist-to-Hamiltonian comparison criterion).* Let (X, ω) be a connected holomorphic-symplectic surface and $p \in X$. Suppose a perturbative twisted-M background with branes on $\mathbb{R} \times p$ gives a protected sector together with the obstruction datum of Definition 5.145. Assume:

(i) the period classes $\mathrm{per}(v)$ vanish for the locally Hamiltonian symplectic vector fields used by the sector;
(ii) the closed-field comparison cone $C_{\mathrm{cl},p}$ is acyclic on the protected summand;
(iii) $\mathrm{ob}_{\mathrm{scal}} = 0$;
(iv) $\mathrm{ob}_{\mathrm{QME}} = 0$, with a chosen solution compatible with compactly supported interval factorization.

Then the protected sector determines an admissible protected Hamiltonian restriction. Consequently the reduced brane-line Hamiltonian CE/PV, factorization-current, and scalar-anomaly statements of the formal local theorem apply to that protected sector. If, in addition, the QME solution includes the one-antifield weighted Tate lift of Problem 5.73, then the corresponding one-antifield quantum Hamiltonian comparison is valid for that protected sector and for that sector only.

Proof. By Lemma 5.153, the four hypotheses give the Darboux Hamiltonian reduction, the field-level protected comparison, scalar anomaly compatibility, and brane-defect QME locality required in Definition 5.144. Applying Proposition 1.31 at the formal polydisk and then the admissible protected-sector reduction gives the local Hamiltonian BF model and its bulk-to-boundary evaluation. The last assertion is exactly the weighted Tate/QME extension isolated in Problem 5.73, transported through the chosen protected-sector equivalence. □

LEMMA 5.157 *(Formal restriction and compact comparison).* An admissible protected Hamiltonian restriction contributes precisely the formal Hamiltonian polydisk, the Dirac brane Ext algebra, the brane-line factorization restriction, and the scalar anomaly class $[\bar{c}]$. Let T be a compact BCOV, MNOP/Abel–Jacobi, Calabi–Yau-to-chiral, Igusa, Borcherds, or Borcherds–Kac–Moody comparison target. A theorem for T uses the formal restriction through a convention object $\mathfrak{C}_T$, a morphism $\mathfrak{C}^{\mathrm{Ham}}_{\widehat{p}} \to \mathfrak{C}_T$, and null-homotopies for the target-specific obstruction classes

$$\mathrm{Ob}_T^{\mathrm{cmp}} = \left(\mathrm{per}_T, \mathrm{ob}_T^{\mathrm{desc}}, \mathrm{ob}_T^{\mathrm{BCOV}}, \mathrm{ob}_T^{\mathrm{QME}}, \mathrm{ob}_T^{\mathrm{conv}}, \mathrm{ob}_T^{\mathrm{hyp}}, \mathrm{ob}_T^{\mathrm{CY\text{-}ch}}, \mathrm{ob}_T^{\mathrm{Igusa/BKM}}, \mathrm{ob}_T^{6r/\mathrm{MD}}\right),$$

with a component present exactly when the corresponding structure occurs in T. A single class in $H^1(\mathfrak{K}_T)$ exists only after a deformation complex $\mathfrak{K}_T$ and comparison maps from the native complexes have been constructed. A nonzero component is the obstruction to the corresponding compact theorem.

Proof. The three admissibility clauses record Darboux Hamiltonians at $\widehat{X}_p$, the point-brane Ext algebra, brane-defect locality on $\mathbb{R} \times p$, and scalar anomaly compatibility. The global data of any target theorem

— compact Calabi–Yau threefold, negative-cyclic Calabi–Yau class, the compact CY-to-chiral target specialization datum $\Phi_d = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi^{\mathrm{FA}}_d$, compact worldsheet or S^3 framing, BCOV propagator and holomorphic anomaly equation, Borcherds Jacobi form, Weyl chamber, compact Hall orientation — enter as independent components of $\mathrm{Ob}^{\mathrm{cmp}}_T$ in their native target complexes. Two ambient theories with the same admissible formal restriction differ exactly at these components. Vanishing with specified null-homotopies is the admissibility datum for using the formal restriction in T; a nonzero component is the named target obstruction. □

Definition 5.158 *(Protected-sector matched-conventions contribution).* The protected-sector reduction contributes the following datum to any matched-conventions target theorem:

$$\mathfrak{C}^{\mathrm{Ham}}_{\widehat{p}} = \big(d = 2, \widehat{X}_p, \omega_p, \mathbb{R}_{\mathrm{brane}}, \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}, \mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p), [\bar{c}], \Phi^{\mathrm{Ham}}_{\mathrm{fact}}\big).$$

The remaining datum for a compact BCOV, CY-to-chiral, or Igusa/BKM theorem consists of the compact target, its Calabi–Yau or modular class, the worldsheet or specialization curve, the framing, the global propagator or Hall orientation, and the target obstruction complex.

COROLLARY 5.159 *(Formal uniformity after admissibility).* For every connected holomorphic-symplectic surface (X, ω), every brane point $p \in X$, and every protected twisted-M realization whose obstruction datum vanishes, the restricted sector is identified with the same formal Hamiltonian brane model

$$\big(\widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[1], \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1], [\bar{c}]\big)$$

after choosing a formal Darboux coordinate at p.

Proof. The period vanishing gives global Hamiltonian primitives for the vector fields used in the sector. The acyclic closed comparison cone identifies the protected closed fields with $\widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[1]$. The Ext-algebra identification gives the Dirac brane formal-stalk chart, and scalar compatibility fixes the same class $[\bar{c}]$. These are exactly the hypotheses of Proposition 1.31. □

Remark 5.160 *(Dirac brane formal-stalk chart).* In an admissible protected Hamiltonian sector, the brane is the constrained formal-stalk chart whose stable scalar-reduced trace map is the GL_N-invariant Darboux-natural extension of the moment-map action on linear generators (Theorem 5.147). The Dirac brane stack reads the formal symplectic normal polydisk through trace. Dirac reduction gives the open trace algebra, the closed Hamiltonian BF theory is the shifted cotangent theory of canonical transformations of that polydisk, and the closed observables identify with derived Poisson-central operations on the stable open Hamiltonian brane algebra in the reduced local sense proved by Theorem 5.6. The scalar anomaly is the same class $[\bar{c}]$ on the closed and open sides. The corresponding ambient centrality statement is the matched-conventions interface of Corollary 5.161.

COROLLARY 5.161 *(Protected-sector matched-conventions criterion).* A theorem in a compact BCOV, CY-to-chiral, or Igusa/BKM target uses the protected-sector reduction as a matched-conventions descent datum: it defines a morphism $\mathfrak{C}^{\mathrm{Ham}}_{\widehat{p}} \to \mathfrak{C}_T$ and null-homotopies for every asserted component of $\mathrm{Ob}^{\mathrm{cmp}}_T$.

Proof. Lemma 5.157 identifies the target-specific obstruction components $\mathrm{Ob}^{\mathrm{cmp}}_T$. A morphism of convention data and null-homotopies for the asserted components give exactly the compact, worldsheet, modular, orientation, and descent data needed by the compact theorem. Corollary 5.159 then gives the $d = 2$ formal-Darboux normalization functorially in the chosen chart. □

Matched-conventions component $\mathfrak{o}_{\mathrm{cv}}$. The local convention base $\mathfrak{C}^{(2)}_{\mathrm{loc}}$ of Definition 11.5 is the formal holomorphic dimension $d = 2$ brane-line stalk; compact CY_3 BCOV, the compact Calabi–Yau-to-chiral comparison, and the Borcherds / Borcherds–Kac–Moody (BKM)–Igusa bridge carry their own convention, descent, and modular data. By Lemma 11.6 the formal-Darboux theorem contributes that stalk to a compact comparison $\mathfrak{T}$ through a matched-conventions datum $\mathfrak{C}_{\mathrm{tar}}$ of Definition 11.7 and a null-homotopy of the Koszul-descent obstruction class $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{T}) = (\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}}, \mathrm{Ob}_{\mathrm{glob}}, \mathrm{ob}_{6r}, \mathrm{ob}_{\mathrm{MD}})$. The component $\mathfrak{o}_{\mathrm{cv}}$ is therefore

$$\mathfrak{o}_{\mathrm{cv}} = \mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}}) \in H^1(\mathcal{K}_{\mathrm{obs}}) \oplus H^1(\mathfrak{K}_{\mathrm{KD}}),$$

with named subclasses the period class $\mathrm{per}_X(v) = \partial(-\iota_v\omega) \in H^1(X, \mathbb{C}_X)$ of Definition 11.8, the Weiss/Ran descent class of Definition 11.9, the scalar anomaly compatibility class $\mathrm{ob}_{\mathrm{anom}}$, the target QME class $\mathrm{ob}_{\mathrm{QME}}$, the convergence and hyperdescent classes $\mathrm{ob}_{\mathrm{conv}}$, $\mathrm{ob}_{\mathrm{hyp}}$, the CY-to-chiral/Igusa/BKM matched-conventions classes $\mathrm{ob}_{\mathrm{CY\text{-}ch}}$, $\mathrm{ob}_{\mathrm{Igusa/BKM}}$, the six-relation factorization class ob_{6r}, and the mixed-Dunn class $\mathrm{ob}_{\mathrm{MD}}$, together with the curvature $[d\Theta_{\mathrm{KD}} + \frac{1}{2}[\Theta_{\mathrm{KD}}, \Theta_{\mathrm{KD}}]]$. The class $\mathfrak{o}_{\mathrm{cv}}$ vanishes precisely when these coordinates vanish with chosen null-homotopies, equivalently under the hypotheses of Theorem 11.13; under that vanishing the target inherits the formal-stalk component $\chi_{\mathrm{op}}^{-1}\Phi^{\mathrm{Ham}}_{\mathrm{fact}}\chi_{\mathrm{cl}}$ of 11.6. A divergence of conventions is a nonzero coordinate of $\mathfrak{o}_{\mathrm{cv}}$, which records exactly the structures captured by $\mathfrak{C}_{\mathrm{tar}}$. The full compact-comparison theorem 11.4, the matched-conventions criterion 11.13, and the per-target divergence theorems for compact-CY_3 BCOV, the compact Calabi–Yau-to-chiral comparison, the Borcherds/BKM–Igusa bridge, and the Gromov–Witten (GW) / Donaldson–Thomas (DT) / Pandharipande–Thomas (PT) / Maulik–Nekrasov–Okounkov–Pandharipande (MNOP) comparison are stated in the appendix (§11.4); their statements are deferred to keep the local theorem from being read as a global theorem.

Remark 5.162 *(Stratified Weiss/Cech–Roos descent obstruction).* Inside the local stratified construction the descent component of $\mathfrak{o}_{\mathrm{cv}}$ is sharpened by the Cech cone

$$\Delta^{N,M}_{V,\mathcal{U}} = \mathrm{Cone}\Big(\mathcal{F}^{\mathrm{str}}_{\Omega,N,M}(V) \to \mathrm{Tot}\, \check{C}^{\bullet}(\mathcal{U}; \mathcal{F}^{\mathrm{str}}_{\Omega,N,M})\Big)$$

for collared stratified Weiss covers $\mathcal{U}$ of V, together with the Roos compatibility class $\mathfrak{w}^{\check{C}/R}_{V,\Omega,N,M}$ under refinements, finite windows, interval inclusions, and admissible weight transports. The internal descent entry is

$$\partial^{\check{C}/R}_{\mathrm{Weiss}} = \Big(\{H^{\bullet}(\Delta^{N,M}_{V,\mathcal{U}})\}_{V,\mathcal{U},N,M},\ \mathfrak{w}^{\check{C}/R}_{V,\Omega,N,M}\Big),$$

the datum used by the local factorization-center morphism. Choosing Darboux coordinates pointwise and gluing the resulting stalk maps produces a global morphism after this descent datum is made acyclic; the displayed cone class is the obstruction.

THEOREM 5.163 *(Iff-vanishing of the Tate-coefficient cotangent lift obstruction vector).* Let Ob_T be the obstruction vector of Definition 7.3.

- *Lift.* An unreduced Tate-coefficient cotangent factorization-centre morphism

$$\Phi_{\mathrm{cl}} \colon \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N})$$

  with all P_0-centrality homotopies, factorization centrality homotopies, stable connected quotient, and a boundary realization of the shifted cotangent generators in the sense of Problem 5.49.

- *Hypotheses.* The reduced bulk-to-brane evaluation $J : \bar{A} \to A_\partial^{\mathrm{Ham}}$ of Theorem 5.147; the brane line L and gauge group $\mathfrak{gl}_N$; a coefficient category together with chosen compatible null-homotopies for each component class.
- *Obstruction class.* The eight-component vector $\mathrm{Ob}_T = (\mathfrak{o}_{\mathrm{Cas}}, \mathfrak{o}_{\mathrm{nilp}}, \mathfrak{o}_{\mathrm{Quil}}, \mathfrak{o}_{\mathrm{BV,end}}, \mathfrak{o}_{\mathrm{prim}}, \mathfrak{o}_{\mathrm{Had}}, \mathfrak{o}_{\mathrm{univ}}, \mathfrak{o}_{\mathrm{cv}})$ of Definition 7.3.
- *Vanishing criterion.* Φ_{cl} exists precisely when Ob_T is represented and killed by chosen compatible null-homotopies. The all-order one-antifield quantum extension additionally requires a null-homotopy of $\mathfrak{o}_{\mathrm{QME}}$. The compact or cross-target version additionally requires a null-homotopy of $\mathfrak{o}_{\mathrm{cv}}$.
- *First nonzero example.* The Casimir kernel $\mathfrak{o}_{\mathrm{Cas}}$ is nonzero on the strict product/direct-sum Tate pair $(\mathfrak{h}_\Pi, D_\oplus)$: the compatible family $(C_{\mathfrak{h}}^M)_M$ lies in the cokernel of the represented-kernel map into $\mathfrak{h}\widehat{\otimes}D$ (Lemma 5.212, Theorem 5.94). This is the canonical construction defect blocking the strict-unweighted endpoint and forcing the weighted Tate completion of Definition 5.55.
- *Physical meaning.* The matched-bracket open-closed identity that holds at degree zero (Theorem 4.11) lifts to the cotangent direction after installing the regulator data for the strict ambient product completion; the residue cotangent module $\mathcal{P} = \mathfrak{h}_{\mathrm{cont}}^\vee$ is genuinely Tate, with finite-window approximations, and the diagonal Casimir $C_{\mathfrak{h}}$ converges in $\mathfrak{h}\widehat{\otimes}\mathcal{P}$ precisely after the regulator is installed. Vanishing of Ob_T names exactly the regulator data — weight system, propagator extension, truncation, primitive projection, parametrix expansion, protected anomaly cancellation, and matched-conventions descent — under which the unreduced lift is constructed.

Proof. The classical assertion assembles the component criteria. Component $\mathfrak{o}_{\mathrm{Cas}} = 0$ is the weighted Casimir kernel of 5.62 together with the heat-kernel propagator extension of 5.64, giving the classical cochain CE/PV cotangent extension 5.71 (d). Component $\mathfrak{o}_{\mathrm{nilp}} = 0$ is the joint vanishing of the linear, semisimple, and low-degree Casimir terms of Lemma 5.107, closing the truncation tower. Component $\mathfrak{o}_{\mathrm{Quil}} = 0$ is the open-side Koszul-descent vanishing of 5.109 in $\mathcal{O}_{\mathrm{Kos}}$; inside the admissible Tate model envelope of 5.18, this gives the Tate Quillen equivalence refining the cochain identification. Component $\mathfrak{o}_{\mathrm{BV,end}} = 0$ gives the endpoint-admissible coefficient module of 5.95, and component $\mathfrak{o}_{\mathrm{prim}} = 0$ gives the strict cotangent centrality homotopy and the cotangent-label boundary realization. Component $\mathfrak{o}_{\mathrm{Had}} = 0$ is the Hadamard endpoint vanishing of 8.19. Component $\mathfrak{o}_{\mathrm{univ}} = 0$ gives the protected Hamiltonian comparison through Theorem 5.150 and identifies the formal-local model with its protected restriction. Together these give the cochain-level CE/PV map, the Tate Quillen envelope, the analytic kernel, the descent datum, and the protected realization, which is exactly Problem 5.49. Conversely, an existing Φ_{cl} with the listed homotopies exhibits each component class through its definition: the Casimir kernel through the convergent BV pairing, the truncation classes through the finite $\mathfrak{m}^3$-adic cotangent, the Koszul cone through the bar-cobar comparison, the BV endpoint through the coefficient and RG transport, the primitive class through the factorization projection, the Hadamard class through the parametrix expansion of the propagator, the protected class through the comparison morphisms, and the cross-target class through the matched-conventions descent.

The quantum and cross-target statements are Proposition 5.72 and Theorem 11.13, respectively. $\mathfrak{o}_{\mathrm{QME}} = 0$ gives the all-order one-antifield extension of the degree-zero Moyal/Weyl theorem inside the weighted regulator-admissible sector; $\mathfrak{o}_{\mathrm{cv}} = 0$ with the matched-conventions datum gives the global or cross-target factorization-centre morphism whose formal stalk is the local Hamiltonian map. □

COROLLARY 5.164 *(Proved sublattice and residual).* The proved sublattice of Ob_T consists of: the weighted-completion classical lift (5.71 (a)–(d) inside the regulator-admissible sector); the classical

$\mathfrak{m}^3$ truncation (5.102); the chain-level primitive projection with strict Q-, factorization-, and P_0-bracket compatibility (5.117, 5.119, 5.120); the Hadamard finite-window parametrix (8.17); the scalar Casimir-balanced equivariant QME criterion (5.150); and the matched-conventions transfer under hypotheses of 11.13. The residual obstruction is

$$\mathrm{Ob}_T^{\mathrm{res}} = \left(\mathfrak{o}^{\mathrm{ns}}_{\mathrm{QME}}, \mathfrak{o}^{\mathrm{low}}_{\mathrm{nilp}}, \mathfrak{o}^{\mathrm{univ}}_{\mathrm{Quil}}, \mathfrak{o}^{\mathrm{cot}}_{\mathrm{prim}}, \mathfrak{o}^{\mathrm{prot}}_{\mathrm{univ}}, \mathfrak{o}^{\mathrm{glob}}_{\mathrm{cv}}\right),$$

with components: the non-scalar mixed brane-defect QME class $[(\mathfrak{o}^{\mathrm{ns}}_{n,M})_M] \in \varprojlim_M H^1(\mathcal{K}^\bullet_{\mathrm{ns},M}(I))$ together with its secondary $\varprojlim^1 H^0$ primitive class (5.84 and 5.68); the low-degree truncation triple $(\mathfrak{o}^{\mathrm{lin}}_N, \mathfrak{o}^{\mathrm{ss}}, \mathfrak{o}^{\mathrm{Cas}}_{\leq 2})$ of 5.106; the open-side Koszul-descent class in O_{Kos} of 5.23; the cotangent centrality homotopy and one-antifield boundary realization residue represented by $\mathfrak{o}_{\mathrm{prim}}$; the protected twist-obstruction tuple of 5.145; and the matched-conventions obstruction class $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}})$ of 11.13. Each is a precise obstruction class with an explicit vanishing datum.

Proof. The proved sublattice is the disjoint union of the cited theorems; each gives the corresponding component. The residual is the complement, with vanishing datum given by the corresponding paragraph above. Iff-vanishing is Theorem 5.163. □

5.7 Stratified CE/PV and admissible Koszul duality

The closed local operator algebra $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$ of §6.2 and the open brane Hamiltonian descendant algebra $\widehat{\mathbf{S}}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1])$ of §3.5 are two E_1-algebras built from the same Lie/coComm operadic pair, on different coefficient categories. Koszul duality (Priddy 1970, [23]; Loday–Vallette 2012, [24]) exchanges commutative cochains for associative enveloping algebras whenever its convergence hypotheses are met. On the strict formal-polydisk product/direct-sum pair those hypotheses have one obstruction class — the Tate Casimir $\mathfrak{o}_{\mathrm{Cas}}$; on the weighted Tate replacement of §7.5 they are met. The stratification below names the coefficient complex and the precise obstruction class at each boundary.

(i) *Operadic data.* The closed BV side has dg Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$; the open Hamiltonian side has a coCommutative comodule on $\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]$, namely brane traces and primitive Koszul descendants. The Lie/coComm operadic pair is the precondition of Koszul duality. The cotangent-dual CE coordinate u_f is the CE coordinate whose CE/PV image is the boundary observable O_f (Theorem 4.20); the Hamiltonian-dual coordinate c_f records the Schouten action on observables.

(ii) *Admissible Tate CE/bar-cobar.* The cochain complex is the completed cotangent CE/PV identity of Theorem 5.5: valid for the formal polydisk on finite coordinate identities, with a kernel-admissible Maurer–Cartan tensor appearing only after the diagonal converges. The enveloping algebra comparison is a bar-cobar criterion that applies precisely when the chosen Tate or weighted coefficient category makes the CE coalgebra conilpotent and the filtered inverse limits exact. The strict product/direct-sum full-polydisk coefficient pair $\left(\prod_d H_d\right) \oplus \left(\bigoplus_d H_d^\vee\right)$ has the bar-cobar obstruction (Lemma 5.212); the obstruction tuple $\mathfrak{o}_{\mathrm{Cas}}$ is killed by weighted Tate completion (Corollary 5.71) or by nilpotent truncation, never discarded. The minimal RG-stable analytic completion forced by J — closed under the Schouten bracket and pointwise product, with heat-kernel propagator extension and Casimir convergence — is the weighted-Tate completion B_w (Theorem 5.58).

(iii) *Tate bar-cobar admissibility (Lefèvre–Hasegawa, Hinich, Loday–Vallette, Hovey, Lurie).* Under admissible lower-central Tate-pronilpotence the completed Lie bar-cobar map

$\Omega C^{\mathrm{CE}}_{*}(\mathfrak{g}_{\mathrm{adm}}) \to \widehat{U}(\mathfrak{g}_{\mathrm{adm}})$ is a filtered quasi-isomorphism in **TateVec** (Lemma 5.166, Proposition 5.169). The lower-central Tate-pronilpotence hypothesis is an additional condition on the strict full polydisk. Weighted completion gives the topology and Casimir convergence; lower-central pronilpotence requires nilpotent truncation or a relative reductive replacement for the $H_2 \cong \mathfrak{sl}_2$ sector. The strict Beilinson–Feigin–Mazur [22] product pair carries the same obstruction.

(iv) *Coordinate CE/PV reduced P_0 central operation.* On the coordinate window the CE/PV identity matches Hamiltonian-dual CE coordinates to Schouten polyvector coordinates, and shifted-cotangent-dual CE coordinates to boundary Hamiltonian functions on the Chan–Paton pair (Theorem 4.20). This is a cochain-level reduced morphism. A strict factorisation P_0-centre morphism on the unreduced cotangent side requires the unreduced lift obstruction Remark 5.190, recorded in Problem 5.49.

(v) *PBW/Rees special fibre.* The Rees enveloping algebra $U_\hbar(\mathfrak{g})$ has PBW special fibre at $\hbar = 0$ isomorphic to $\widehat{\mathbf{S}}(\mathfrak{g})$, giving the label-level identification of the brane Hamiltonian descendants $\Psi_{a,b}$ with the abstract bidegree $\bar{A}_{\mathrm{desc}}$ (Corollary 3.29). The symmetric PBW action on $\Psi_{a,b}$ and the coadjoint action on the cotangent module $\mathcal{P}$ are distinct (Theorem 5.181, Corollary 5.173, Theorem 8.1); the two formulas have opposite shift directions and differ by $-(pb - qa + p - q)$ versus $(pb - qa)$. The PBW special fibre and the coadjoint module have distinct Hamiltonian actions.

(vi) *Matlis/Grothendieck cotangent module.* The continuous dual $\mathfrak{h}^{\vee}_{\mathrm{cont}}$ is the Grothendieck–Matlis principal-part module $\mathcal{P}$, the residue annihilator of the constants quotient (Matlis 1958, [19]; Hartshorne local cohomology, [20]; Theorem 5.179). The cotangent module $\mathcal{P}$ is the residue principal-part module, distinguished from the polynomial open BV/Koszul matrix model (Proposition 5.182, Theorem 5.183) by the residue-coadjoint pole-raising action: linear Hamiltonians act locally nilpotently on polynomial observables and pole-raisingly on residues.

(vii) *Brane-line factorisation-current comparison.* Hamiltonians smeared on a brane interval lift to factorisation currents whose binary collisions reproduce the Hamiltonian Poisson bracket and the residue-coadjoint action on $\mathcal{P}$ (Theorem 4.11, Theorem 5.186); de Rham-current contraction along the brane line returns the open trace algebra (Theorem 5.191); the Moyal quantum coefficient comparison is the componentwise Theorem 5.218. Each remaining obstruction is a finite-window obstruction theorem: the attempted strict all-window Bochner–Martinelli transfer is obstructed by $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$, with the strict obstruction proved for every N by the Hamiltonian-flow leak $z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}$ (Theorem 8.40, Corollary 8.41); the unreduced factorisation-centre lift by the four-component obstruction vector of Theorem E.53 and Problem 5.49; the non-scalar Costello QME by Problem 5.220, with the smallest non-scalar window θ_3 carrying both the bare value $H^1 = \mathbb{C} \cdot E_{\theta_3} \neq 0$ and the counterterm-adjoined row value $H^1 = 0$ (Theorem E.38); the all-bidegree decorated PBW Stokes identity by $\Omega^{\mathrm{rad}}_{a,b}$, with the radial bicomplex interior-acyclic under the radial-parts statement (Theorem F.28).

Definition 5.165. For an augmented complete dg algebra B, its Koszul dual is

$$B^! = \mathrm{RHom}_B(\mathbb{C}, \mathbb{C})$$

[23]. Commutative Chevalley-Eilenberg cochains are regarded as E_1 algebras by restriction. All bar and cobar constructions below are taken in the category of complete filtered Tate vector spaces, with continuous maps. The completions are by symmetric or tensor length together with the coefficient

topology coming from Taylor degree. For a Tate space V, the notation $V^\vee_{\mathrm{cont}}$ means the continuous dual with its dual Tate topology, $\widehat{\otimes}$ means the completed Tate tensor product, and

$$\widehat{\mathbf{S}}^n(V) = (V^{\widehat{\otimes} n})_{\mathbb{S}_n}.$$

Chevalley-Eilenberg chains below are direct sums in CE length with completed coefficients in each length; CE cochains, completed cobar algebras, and completed enveloping algebras are products in the corresponding length filtration.

LEMMA 5.166 *(Tate CE chains, completed PBW, and Koszul duality).* Work in the symmetric monoidal category **TateVec** of complete filtered Tate $\mathbb{C}$-vector spaces:

$$V = \varprojlim_{w\in\mathbb{N}} V_{\le w}, \qquad V_{\le w} \text{ finite-dimensional}, \qquad F^w V = \mathbf{Ker}\big(V \to V_{\le w-1}\big),$$

with $\widehat{\otimes}$ the completed tensor product $V\widehat{\otimes}W = \varprojlim_w (V_{\le w} \otimes W_{\le w})$ and $V^\vee_{\mathrm{cont}} = \varinjlim_w V^*_{\le w}$ the continuous dual. Admissible quasi-isomorphisms are the filtered ones: chain maps inducing a quasi-isomorphism on every quotient V/F^wV and on the associated graded.

Let $\mathfrak{g}$ be a Tate graded Lie algebra whose filtration is by closed dg Lie ideals, so that every $\mathfrak{g}_{\le w}$ used below is a finite-dimensional dg Lie quotient. Assume $\mathfrak{g}$ is lower-central Tate-pronilpotent in the sense that the descending Lie powers $\mathfrak{g}^{(\ge m)} = [\mathfrak{g}, \mathfrak{g}^{(\ge m-1)}]$ satisfy $\bigcap_m \big(\mathfrak{g}^{(\ge m)} + F^w\mathfrak{g}\big) = F^w\mathfrak{g}$ for every w. Then:

(i) For $V = \prod_d V_d$ with all V_d finite-dimensional and $W = \bigoplus_d W_d$ with all W_d finite-dimensional,

$$(V \oplus W)^\vee_{\mathrm{cont}} = \bigoplus_d V_d^\vee \oplus \prod_d W_d^\vee,$$

and the dualization $V \mapsto V^\vee_{\mathrm{cont}}$ is exact and continuous in the filtration F^w.

(ii) The continuous Chevalley–Eilenberg cochains

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \mathrm{Hom}_{\mathrm{cont}}\big(\widehat{\mathbf{S}}^c(\mathfrak{g}[1]), \mathbb{C}\big)$$

identify with $\widehat{\mathbf{S}}_{\mathrm{Tate}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big)$. The CE coalgebra $C^{\mathrm{CE}}_*(\mathfrak{g}) = \widehat{\mathbf{S}}^c(\mathfrak{g}[1])$ is conilpotent in CE length: every homogeneous CE chain has finite symmetric length and is killed by sufficiently high iterated reduced coproduct. Lower-central Tate-pronilpotence is a separate inverse-limit/bar-cobar convergence hypothesis, used below for the completed cobar and PBW comparison.

(iii) The continuous bar resolution of the augmentation $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \to \mathbb{C}$ has dual cobar algebra $\Omega C^{\mathrm{CE}}_*(\mathfrak{g})$, and the canonical Lie bar–cobar map

$$\Omega C^{\mathrm{CE}}_*(\mathfrak{g}) \longrightarrow \widehat{U}(\mathfrak{g})$$

is an admissible (filtered) quasi-isomorphism in **TateVec**.

(iv) The PBW filtration on $\widehat{U}(\mathfrak{g})$ by tensor length is complete and separated. Its associated graded is the completed symmetric algebra

$$\mathrm{gr}^\bullet_F \widehat{U}(\mathfrak{g}) \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{g}).$$

Proof. Item (1). A continuous functional on $V = \prod V_d$ depends on finitely many coordinates: the topology on $\mathbb{C}$ is discrete, so the kernel of any continuous map contains some $F^w V$. Therefore $V^\vee_{\mathrm{cont}} = \bigoplus_d V_d^*$. Conversely, a continuous functional on the discrete sum $W = \bigoplus W_d$ is given by an arbitrary product of degreewise functionals, hence $W^\vee_{\mathrm{cont}} = \prod_d W_d^*$. Direct sum of these gives the displayed formula. Exactness of dualization in **TateVec** is preservation of finite limits; continuity in the Tate filtration is direct.

Item (2). The completed CE coalgebra $\widehat{\mathbf{S}}^c(\mathfrak{g}[1]) = \bigoplus_n \widehat{\mathbf{S}}^n(\mathfrak{g}[1])$ has coaugmentation coideal $\widehat{\mathbf{S}}^{\geq 1}(\mathfrak{g}[1])$. An element in CE length n is killed by the $(n+1)$-fold reduced coproduct; this is the conilpotence used in the coalgebra. Passage to completed cobar algebras and PBW enveloping algebras after the Tate inverse limit uses the lower-central Tate-pronilpotence hypothesis: modulo every finite filtration step the descending Lie powers have empty intersection, so the completed bar-cobar spectral sequence is separated. The displayed equality identifies cochains as continuous functionals on the CE coalgebra, using item (1) windowwise.

Item (3). Filter the CE coalgebra by symmetric coalgebra length and by the chosen Tate quotient system. Project to a fixed bracket-compatible finite dg Lie quotient $\mathfrak{g}_{\leq w}$ and bound symmetric/tensor length to a fixed n. The resulting finite-dimensional diagram is the ordinary Lie bar–cobar comparison [24]: the bar resolution computes $\mathrm{RHom}_{C^\bullet_{\mathrm{CE}}(\mathfrak{g}_{\leq w}, \leq n)}(\mathbb{C}, \mathbb{C})$, and the cobar algebra maps to the universal enveloping algebra of the same finite Lie quotient. These finite comparisons are compatible with enlargement of the quotient and increase of the length cap because the transition maps are Lie-algebra quotient maps. Passing to the Tate limit and completing in tensor length gives the asserted filtered quasi-isomorphism in **TateVec**. The PBW filtration is, quotientwise, the standard PBW filtration on $U(\mathfrak{g}_{\leq w})$, hence preserved under inverse limits. Arbitrary Taylor-degree coordinate windows of the full formal Hamiltonian polydisk lie outside this hypothesis and outside this lemma.

Item (4). Tensor length is a separated, descending filtration on $\widehat{U}(\mathfrak{g})$ because the Lie relations $xy - (-1)^{|x||y|}yx - [x, y] = 0$ have leading tensor-length term $xy - (-1)^{|x||y|}yx$ of length 2 and lower-order term $-[x, y]$ of length 1; so the associated graded retains only the symmetric (length-fixed) part. Completeness is clear from the Tate construction. The associated graded is therefore the completed symmetric algebra $\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{g})$. □

Remark 5.167 *(Bar-cobar obstruction datum).* The lower-central Tate-pronilpotence hypothesis in item (2) is the exact bar-cobar criterion used here. It is satisfied by the nilpotent truncations and by the weighted admissible replacements used later. The full reduced formal Hamiltonian algebra with linear modes is already covered at cochain level by the pro coordinate-window/mixed-continuous CE/PV calculation; its enveloping/Koszul lift is governed by the obstruction terms of Remark 4.16 and Theorem 5.5. The Beilinson–Feigin–Mazur explicit product/direct-sum Tate convention [22] is the special case $\mathfrak{g} = \left(\prod P_d\right) \oplus \left(\bigoplus Q_d\right)$ with the bracket bilinear in the two displayed pieces.

LEMMA 5.168 *(Linear Poisson Schouten differential).* Let $\mathfrak{h}$ be a pro-finite-dimensional Lie algebra in **TateVec**, and choose a mixed or weighted completion on which the linear Kirillov–Kostant–Souriau bracket is continuous. The bracket is $\{O_I, O_J\}_\partial = O_{[H_I, H_J]} = C^K_{IJ}\, O_K$ on a chosen homogeneous basis $\{H_I\}$ of $\mathfrak{h}$, where $O_I \in \mathfrak{h}^*$ is the linear coordinate. The associated Poisson bivector is the linear bivector

$$\pi_\partial = \tfrac{1}{2}\, C^K_{IJ}\, O_K\, \theta^I \wedge \theta^J, \qquad \theta^I = \partial/\partial O_I \in \mathrm{Der}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big).$$

Assume that this linear bivector and the Schouten contraction bracket are continuous on the chosen mixed or weighted completion. Then the Lichnerowicz/Schouten differential $d_\pi = [\pi_\partial, -]_S$ on

$$\mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big) = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\widehat{\otimes}\widehat{\Lambda}(\theta^I)$$

acts on generators by

$$d_\pi\theta^K = -\tfrac{1}{2}\,C^K_{IJ}\,\theta^I\theta^J, \qquad d_\pi O_K = C^L_{KJ}\,O_L\,\theta^J.$$

Proof. Direct calculation. The Schouten Leibniz rule with $[\theta^I, O_J]_S = \delta^I_J$, $[\theta^I, \theta^J]_S = 0$, $[O_I, O_J]_S = 0$ gives

$$\begin{aligned}
[\pi_\partial, O_K]_S &= \tfrac{1}{2}C^L_{IJ}\,O_L\,[\theta^I\theta^J, O_K]_S \\
&= \tfrac{1}{2}C^L_{IJ}\,O_L\,\big(\delta^I_K\theta^J - \delta^J_K\theta^I\big) \\
&= \tfrac{1}{2}\big(C^L_{KJ} - C^L_{JK}\big)O_L\theta^J = C^L_{KJ}O_L\theta^J,
\end{aligned}$$

using the antisymmetry $C^L_{IJ} = -C^L_{JI}$ in the lower indices. For $[\pi_\partial, \theta^K]_S$, the Schouten bracket with θ^K uses $[\theta^I\theta^J, \theta^K]_S = 0$ on the bivector factor and $[O_L, \theta^K]_S = -\delta^K_L$; the result is $-\frac{1}{2}C^K_{IJ}\theta^I\theta^J$. □

PROPOSITION 5.169 *(Admissible Tate Koszul-duality criterion).* Let $\mathfrak{g}_{\mathrm{adm}}$ be a Tate graded Lie algebra satisfying the lower-central Tate-pronilpotence hypothesis of Lemma 5.166 — for the formal-local setting, a nilpotent bracket-compatible finite-window replacement, the weighted admissible envelope in which the hypothesis is explicitly imposed, or another stated admissible replacement. Put

$$B_{\mathrm{adm}} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_{\mathrm{adm}}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{adm,cont}}[-1]\big).$$

Then

$$B^!_{\mathrm{adm}} \simeq \widehat{U}(\mathfrak{g}_{\mathrm{adm}})$$

in the completed Tate category. For the strict full formal Hamiltonian algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ with its linear and conformal-symplectic low-degree modes, the same conclusion is a criterion: it follows after giving a coefficient replacement or relative model killing the obstruction tuple of Theorem 5.5.

Proof. This is exactly Lemma 5.166 applied to $\mathfrak{g}_{\mathrm{adm}}$. The lower-central Tate-pronilpotence condition is the decisive hypothesis: it ensures that the coefficient-completed CE coalgebra is conilpotent in every finite Tate window, that the continuous bar resolution is admissible, and that the completed Lie bar-cobar map

$$\Omega C^{\mathrm{CE}}_*(\mathfrak{g}_{\mathrm{adm}}) \longrightarrow \widehat{U}(\mathfrak{g}_{\mathrm{adm}})$$

is a filtered quasi-isomorphism.

For the full formal Hamiltonian algebra, the direct CE/PV generator calculation gives the cochain identity before any bar-cobar hypothesis is imposed. To turn that identity into an enveloping/Koszul-duality theorem one must construct an admissible replacement, relative semisimple model, or weighted coefficient category satisfying the displayed lower-central Tate-pronilpotence and convergence hypotheses. The required data are exactly the Casimir, low-degree, semisimple, cone, and inverse-limit datum of Theorem 5.5. □

The completed PBW/order Rees enveloping algebra

$$U_\hbar(\mathfrak{g}) = \widehat{T}(\mathfrak{g})[[\hbar]] \Big/ \overline{\langle xy - (-1)^{|x||y|} yx - \hbar[x, y] \rangle},$$

with closure in tensor length and the Tate coefficient topology, makes the PBW special fibre precise in every admissible replacement to which Proposition 5.169 applies. The Rees parameter $\hbar$ records word length in the enveloping algebra and is independent of Taylor degree. At $\hbar = 0$ the completed PBW special fibre is

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]\big).$$

After choosing the monomial-dual Taylor basis, the underlying label set of $\mathfrak{h}^\vee$ is identified with an abstract copy $\bar{A}_{\mathrm{desc}}$; this identification transports labels only, with the $\mathfrak{h}$-module structure specified separately. The degreewise polynomial subalgebra $\mathbf{S}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1])$ is the stable classical brane Hamiltonian descendant algebra of Proposition 3.40: the degreewise polynomial part of the PBW associated graded gives the brane Hamiltonian descendant labels. For the full formal Hamiltonian algebra this is the PBW/Rees label level. A filtered enveloping/Koszul-duality theorem is available under the admissible or relative replacement data of Proposition 5.169. The interacting closed Hamiltonian bracket deforms the special fibre inside that admissible enveloping algebra; on the brane side the matching deformation at order one is the Hamiltonian trace bracket proved below.

5.7.1 First-order Poisson identification

The formal-stalk chart evaluates a closed Hamiltonian on the matrix-valued transverse coordinate (ϕ_1, ϕ_2); Dirac reduction forces $[\phi_1, \phi_2] = 0$; the surviving algebra of cyclic single traces is the first non-trivial coefficient identity. The question is whether the closed Poisson bracket on $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ and the matrix trace bracket give equal coefficients on the trace generators.

The PBW special-fibre label correspondence used in the reduced comparison reads

$$z_1^k z_2^l \leftrightarrow \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}.$$

The reduced Poisson bracket on the closed side is

$$\{z_1^k z_2^l,\ z_1^m z_2^n\}_{\bar{A}} = \overline{(kn - lm)\, z_1^{k+m-1} z_2^{l+n-1}}.$$

If an exponent in the displayed monomial is negative then the coefficient $kn - lm$ is already zero; if the displayed monomial is constant, its class in $\bar{A}$ is zero. The matching bracket on the open side is the canonical bracket induced by the first-order kinetic term $\int \mathrm{Tr}(\phi_1\, d\phi_2)$, with the sign convention $\iota_{X_f}\omega = -df$ fixed in the AKSZ remark above:

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j, \qquad \{(\phi_1)^i{}_j, (\phi_1)^k{}_l\} = 0 = \{(\phi_2)^i{}_j, (\phi_2)^k{}_l\}.$$

LEMMA 5.170 *(Hamiltonian trace bracket).* The canonical matrix Poisson bracket descends to the invariant reduced BV cohomology and then to the Hamiltonian Lie quotient by constant trace classes.

Proof. For $\epsilon \in \mathfrak{gl}_N$, the component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$ of the moment map generates the infinitesimal conjugation action, with the sign fixed by the convention for the bracket. Hence $\{F, \mu_\epsilon\} = 0$ for every invariant function F. If I_μ is the ideal generated by all μ_ϵ, then $\{F, \mu_\epsilon K\} = \mu_\epsilon\{F, K\}$ lies again in I_μ. Therefore the bracket of invariant functions descends to $(\mathcal{R}_N/I_\mu)^{GL_N}$, equivalently to the degree-zero reduced BV cohomology. The central line $\mathbb{C}\cdot\mathrm{Tr}(1)$ is central for this Lie bracket, so the Hamiltonian vector-space quotient by constants is compatible with the bracket. The associative quotient by the unit is a different object. $\square$

PROPOSITION 5.171 *(First-order trace bracket).* In the stable connected Hamiltonian trace cohomology,

$$\{\overline{\mathrm{Tr}(\phi_1^k\phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m\phi_2^n)}\} = \overline{(kn-lm)\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}.$$

Proof. Write $F_{k,l} = \mathrm{Tr}(\phi_1^k\phi_2^l)$. The matrix Poisson bracket induced by

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l\delta^k_j$$

is

$$\{F, G\} = \mathrm{Tr}\left(\frac{\partial F}{\partial\phi_1}\frac{\partial G}{\partial\phi_2} - \frac{\partial F}{\partial\phi_2}\frac{\partial G}{\partial\phi_1}\right),$$

where the derivatives are cyclic matrix derivatives. Varying one occurrence at a time gives

$$\frac{\partial F_{k,l}}{\partial\phi_1} = \sum_{a=0}^{k-1}\phi_1^{k-1-a}\phi_2^l\phi_1^a,$$
$$\frac{\partial F_{k,l}}{\partial\phi_2} = \sum_{b=0}^{l-1}\phi_2^{l-1-b}\phi_1^k\phi_2^b.$$

Therefore the first summand in the bracket is a sum of kn traces, each obtained by removing one ϕ_1 from the first word and one ϕ_2 from the second word. The second summand is a sum of lm traces obtained by removing one ϕ_2 from the first word and one ϕ_1 from the second. Every trace appearing in either nonzero sum has multidegree $(k+m-1, l+n-1)$. Lemma 3.30 identifies all cyclic words with this multidegree in reduced Koszul cohomology. The coefficient is therefore $kn-lm$, and the common class is $\overline{\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}$. When the common word has multidegree $(0,0)$, this class is the scalar trace and is zero in the Hamiltonian quotient by constants. □

PROPOSITION 5.172 *(Open action on primitive descendants).* Let

$$\widetilde{\Psi}_{a,b} = \binom{a+b}{a}^{-1}\Psi_{a,b}$$

be the averaged primitive one-ψ class. In the stable primitive one-ψ homology,

$$\{\overline{\mathrm{Tr}(\phi_1^p\phi_2^q)}, \widetilde{\Psi}_{a,b}\} = (pb-qa)\,\widetilde{\Psi}_{a+p-1,b+q-1},$$

with the convention that a term with a negative exponent is zero. Thus the unreduced polynomial descendant classes carry the adjoint polynomial Hamiltonian action. The Hamiltonian descendant sector is the quotient by the scalar line $\mathbb{C}\widetilde{\Psi}_{0,0}$, so the displayed action is projected further by killing the scalar target.

Proof. Put $x=\phi_1$ and $y=\phi_2$. The cyclic derivatives of $F_{p,q} = \mathrm{Tr}(x^py^q)$ are

$$\frac{\partial F_{p,q}}{\partial x} = \sum_{i=0}^{p-1}x^{p-1-i}y^qx^i, \qquad \frac{\partial F_{p,q}}{\partial y} = \sum_{j=0}^{q-1}y^{q-1-j}x^py^j.$$

For a one-ψ cyclic word $\mathrm{Tr}(\psi w)$ of bidegree (a,b), differentiating with respect to y gives one summand for each of the b occurrences of y in w, and differentiating with respect to x gives one summand for each of the a occurrences of x. Multiplying by the displayed cyclic derivatives of $F_{p,q}$ therefore produces pb positive one-ψ words and qa negative one-ψ words for each $w \in \mathsf{W}_{a,b}$, all of total bidegree $(a+p-1, b+q-1)$.

The bracket is a cycle because $F_{p,q}$ is invariant, hence $\{F_{p,q}, \mu_\epsilon\} = 0$ for every moment-map component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[x,y])$. Equivalently, the Hamiltonian vector field of $F_{p,q}$ preserves the Koszul differential $Q\psi = [x,y]$ and acts on Koszul homology. By Proposition 3.27, the target primitive one-ψ homology in that bidegree is one-dimensional. The functional $\epsilon(\mathrm{Tr}(\psi w)) = 1$ detects its coefficient. Since $\epsilon(\widetilde{\Psi}_{a,b}) = 1$, the averaged cycle gives

$$\epsilon\big(\{F_{p,q}, \widetilde{\Psi}_{a,b}\}\big) = pb - qa.$$

Also $\epsilon(\widetilde{\Psi}_{a+p-1,b+q-1}) = 1$. Hence the displayed formula follows. If the target bidegree is $(0,0)$, the same formula is still the unreduced primitive one-ψ action; the Hamiltonian descendant quotient sends that scalar class to zero. □

COROLLARY 5.173 *(Descendant action and coadjoint action: structural separation).* Under the PBW special-fibre pairing of Corollary 3.29, the open Hamiltonian descendant action on the averaged polynomial $\widetilde{\Psi}_{a,b}$-basis is the symmetric PBW action

$$F_{p,q} : \widetilde{\Psi}_{a,b} \longmapsto (pb-qa)\, \widetilde{\Psi}_{a+p-1,b+q-1},$$

while the closed coadjoint action on the principal-part basis $\rho_{a,b} \leftrightarrow \delta_{a,b} \in \mathfrak{h}^\vee_{\mathrm{cont}}$ is the negative-transpose coadjoint action

$$\mathrm{ad}^\vee_{z_1^p z_2^q}\, \delta_{a,b} = -(pb - qa + p - q)\, \delta_{a-p+1,b-q+1},$$

with negative-index terms zero. The two actions agree only on the trivial Hamiltonian $p = q = 0$. The deformation quantisation of the cotangent factor sees the second action; the special-fibre PBW symmetric algebra sees the first. The structural reason is the PBW special-fibre versus coadjoint-module separation of Theorem 5.181 below: the $\widetilde{\Psi}_{a,b}$ and the $\rho_{a,b}$ have the same index set but different $\mathfrak{h}$-module structures.

Proof. Proposition 5.172 gives the symmetric PBW formula on $\widetilde{\Psi}_{a,b}$. The coadjoint action is dual to the Poisson bracket: for $f = z_1^p z_2^q$ and the Taylor-dual basis vector $\delta_{a,b}$,

$$(\mathrm{ad}^\vee_f\, \delta_{a,b})(z_1^m z_2^n) = -\delta_{a,b}\big((pn-qm) z_1^{p+m-1} z_2^{q+n-1}\big),$$

giving the displayed coadjoint formula. For example, z_1 acts on open descendants by $\widetilde{\Psi}_{a,b} \mapsto b\, \widetilde{\Psi}_{a,b-1}$, whereas it acts coadjointly by $\delta_{a,b} \mapsto -(b+1)\, \delta_{a,b+1}$. The scalar edge $\{\, \mathrm{Tr}(\phi_1), \widetilde{\Psi}_{0,1}\, \} = \Psi_{0,0}$ is removed with the scalar descendant. The two formulas have opposite index shift directions and differ by $-(pb - qa + p - q)\rho_{a-p+1,b-q+1} - (pb-qa)\rho_{a+p-1,b+q-1}$ after any bidegree-only vector-space relabelling between the two alphabets, which is non-zero unless both terms vanish identically; this happens exactly on the trivial $p = q = 0$ action. The PBW special-fibre product identification is Corollary 3.29; the coadjoint half is Theorem 5.179; the Laurent/distributional principal-part realization is Proposition 5.176 and the separation Theorem 5.181. □

The coadjoint identification on the principal-part side requires three notions of locality on the brane line $\mathbb{R}$ to remain distinct.

LEMMA 5.174 *(Three notions of locality).* On the brane line $\mathbb{R}$ with formal holomorphic transverse coordinates z_1, z_2 at the brane point $p \in \mathbb{C}^2$, the following three notions of locality are independent.

(i) *Support locality:* distributional support on the brane: a section $T \otimes m$ of $\mathcal{D}'_c(I) \widehat{\otimes} M$ has support locality whenever $\operatorname{supp} T \subset I \subset \mathbb{R}$.

(ii) *Topological locality:* locally constant in the $\mathbb{R}$-direction up to Q-exact translations: every section is identified with its translate by a parameter on $\mathbb{R}$ modulo the open BV differential Q_{open}.

(iii) *Formal holomorphic locality:* polynomial, or Taylor-series, dependence on the transverse coordinates z_1, z_2.

The polynomial trace observables $O_f(a) = \int_{\mathbb{R}} a(t)\, \overline{\operatorname{Tr} f(\phi_1(t), \phi_2(t))}\, dt$ have all three. The shifted cotangent currents $\Theta_\rho(B)$ of Construction 5.185 have support locality and topological locality. Their transverse content is momentum data attached to the brane, living in the $\mathbb{C}$-linear residue annihilator of scalar Hamiltonians inside the top local-cohomology principal-part module, $\operatorname{Ann}_{\text{res}}(\mathbb{C} \cdot 1)$.

Proof. Independence is a category-theoretic statement: a polynomial in z_1, z_2 with point support is constant in $\mathbb{R}$, separating support locality from formal holomorphic locality; a delta-current at one point of $\mathbb{R}$ has compact support and only trivial Q-exact translation, separating support locality from topological locality. The cotangent currents $\Theta_\rho(B)$ are smooth or distributional sections of the line $\mathbb{R}$ (hence (1)) and translation-invariant up to Q_{open}-exact cocycles (hence (2)); their transverse content $\rho \in \mathcal{P}$ is a top-local-cohomology principal part, hence a residue-dual distributional mode. $\square$

Remark 5.175. In canonical quantization of a constrained system, the configuration variables are locally polynomial in spatial coordinates and the conjugate momenta are dual distributions. This is exactly the separation Lemma 5.174 records: configuration observables on a brane have formal holomorphic locality (3), and momentum observables have only (1) and (2). The principal-part sector is the algebraic image of this separation in the local Hamiltonian BF model.

PROPOSITION 5.176 *(Principal-part realization of the coadjoint module).* Let

$$\mathcal{P} = \bigoplus_{\substack{a,b \geq 0 \\ a+b>0}} \mathbb{C}\, \rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

with residue pairing

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \operatorname{Res}_{z_1=z_2=0} z_1^{m-a-1} z_2^{n-b-1}\, dz_1 dz_2 = \delta_{a,m}\delta_{b,n}.$$

Let π_{pp} denote projection of Laurent two-forms to the span of the principal parts $\rho_{a,b}$. The action

$$f \cdot \rho = \pi_{\text{pp}}(\{f, \rho\})$$

identifies $\mathcal{P}$ with the continuous coadjoint module $\mathfrak{h}^\vee_{\text{cont}}$. On monomials,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1, b-q+1},$$

with the convention that a term with a negative index is zero.

Proof. The residue pairing identifies $\rho_{a,b}$ with the Taylor-dual functional $\delta_{a,b}$. Hamiltonian vector fields preserve the holomorphic volume form $dz_1 dz_2$, so integration by parts for the residue pairing gives

$$\langle \{f, \rho\}, g \rangle = -\langle \rho, \{f, g\} \rangle.$$

Hence the Laurent principal-part action realizes the coadjoint action. Directly, for $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$,

$$\begin{aligned}\{f, z_1^{-a-1} z_2^{-b-1}\} &= p z_1^{p-1} z_2^q (-(b+1)) z_1^{-a-1} z_2^{-b-2} \\ &\quad - q z_1^p z_2^{q-1} (-(a+1)) z_1^{-a-2} z_2^{-b-1} \\ &= -(pb - qa + p - q) z_1^{p-a-2} z_2^{q-b-2}.\end{aligned}$$

Since $z_1^{p-a-2} z_2^{q-b-2} dz_1 dz_2 = \rho_{a-p+1,b-q+1}$ when both indices are nonnegative, principal-part projection gives the displayed formula. □

Two rigidity statements determine the principal-part identification after the residue scalar is fixed: among continuous total-degree-zero pairings the residue pairing is unique up to a single nonzero scalar, and among continuous total-degree-zero $\mathfrak{h}$-module isomorphisms the coadjoint identification $\mathcal{P} \cong \mathfrak{h}^\vee_{\mathrm{cont}}$ is unique modulo that scalar.

THEOREM 5.177 *(Residue pairing, normalized up to scalar).* Up to a nonzero scalar, the residue pairing

$$\langle \cdot, \cdot \rangle : \mathcal{P} \times (\mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1) \longrightarrow \mathbb{C}, \qquad \langle \rho_{a,b}, [z_1^m z_2^n] \rangle = \delta_{a,m} \delta_{b,n},$$

is the unique non-degenerate continuous $\mathfrak{h}$-equivariant pairing graded of total degree zero with the property that every element of $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1)$ annihilates the constant line.

Proof. Existence. The residue pairing is non-degenerate by the displayed Kronecker relation; it is continuous because $\mathcal{P}$ is graded discrete and $\mathbb{C}[[z_1, z_2]]$ carries the Taylor filtration; equivariance is integration by parts for the Poisson bracket, since Hamiltonian vector fields preserve the holomorphic two-form $dz_1 dz_2$:

$$\langle \{f, \rho\}, g \rangle = \mathrm{Res}(\{f, \rho\} g) = -\,\mathrm{Res}(\rho \{f, g\}) = -\langle \rho, \{f, g\} \rangle.$$

Uniqueness. Let $\beta : \mathcal{P} \times (\mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1) \to \mathbb{C}$ be a second pairing satisfying the same conditions, and write $\beta_{a,b;m,n} = \beta(\rho_{a,b}, z_1^m z_2^n)$. Since β is graded of total degree zero, $\beta_{a,b;m,n} = 0$ unless $a + b = m + n$.

Diagonal weight constraint. The Hamiltonian $z_1 z_2$ acts diagonally on monomials by $\{z_1 z_2, z_1^m z_2^n\} = (n - m) z_1^m z_2^n$. Direct Poisson computation on principal parts uses the Hamiltonian vector field $X_{z_1 z_2} = z_2 \partial_{z_2} - z_1 \partial_{z_1}$ (with $X_f = (\partial_{z_1} f) \partial_{z_2} - (\partial_{z_2} f) \partial_{z_1}$), giving $X_{z_1 z_2}(z_1^{-a-1} z_2^{-b-1}) = (a - b)\, z_1^{-a-1} z_2^{-b-1}$, so $\{z_1 z_2, \rho_{a,b}\} = (a - b) \rho_{a,b} = -(b - a) \rho_{a,b}$. Equivariance $\beta(\{z_1 z_2, \rho_{a,b}\}, z_1^m z_2^n) = -\beta(\rho_{a,b}, \{z_1 z_2, z_1^m z_2^n\})$ gives $-(b - a) \beta_{a,b;m,n} = -(n - m) \beta_{a,b;m,n}$, hence $\beta_{a,b;m,n} = 0$ unless $a - b = m - n$. Combined with the weight constraint $a + b = m + n$, this gives $\beta_{a,b;m,n} = 0$ unless $(a, b) = (m, n)$. So β is diagonal in the residue basis: $\beta_{a,b;m,n} = c_{a,b}\, \delta_{a,m} \delta_{b,n}$ for some scalars $c_{a,b} \in \mathbb{C}$.

Coefficients are constant. The Hamiltonian generator $f = z_1$ has $X_{z_1} = \partial_{z_2}$, so $\{z_1, z_1^m z_2^n\} = n\, z_1^m z_2^{n-1}$ and a direct Poisson computation gives $\{z_1, \rho_{a,b}\} = -(b + 1) \rho_{a,b+1}$. Equivariance reads $\beta(\{z_1, \rho_{a,b}\}, z_1^m z_2^n) = -\beta(\rho_{a,b}, \{z_1, z_1^m z_2^n\})$, i.e., $-(b + 1) \beta_{a,b+1;m,n} = -n\, \beta_{a,b;m,n-1}$. Both sides vanish identically unless the diagonality condition is met on each pair. The right-hand side is non-zero only if $(a, b) = (m, n - 1)$, i.e., $m = a$, $n = b + 1$. The left-hand side is non-zero only if $(a, b + 1) = (m, n)$, i.e., $m = a$, $n = b + 1$. Both indices coincide, giving $(b + 1)\, c_{a,b+1} = (b + 1)\, c_{a,b}$, hence $c_{a,b+1} = c_{a,b}$ when both coefficients occur in the reduced index set; that is, for $a + b > 0$. The reduced module $\mathcal{P}$ starts at $a + b > 0$, so the case $a = b = 0$ is omitted. Symmetrically from

$f = z_2$ (using $X_{z_2} = -\partial_{z_1}$, so $\{z_2, z_1^m z_2^n\} = -m\, z_1^{m-1} z_2^n$ and $\{z_2, \rho_{a,b}\} = (a+1)\rho_{a+1,b}$) the equality $c_{a+1,b} = c_{a,b}$ holds when both coefficients occur in the reduced index set. Connecting these chains through $c_{1,1}$, all $c_{a,b}$ for $a+b \geq 1$ equal a single constant $c \in \mathbb{C}$. Non-degeneracy forces $c \neq 0$.

Therefore $\beta = c\,\langle\cdot,\cdot\rangle$ is the residue pairing scaled by c. The condition on constants holds because $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$; equivalently $\rho_{0,0} \notin \mathcal{P}$. □

Remark 5.178. Theorem 5.177 is better understood as Matlis-local-cohomology rigidity. Let $R = \mathbb{C}[[z_1, z_2]]$ and $\mathfrak{m} = (z_1, z_2)$. The top local cohomology module, in the Čech-Laurent residue model of Grothendieck–Hartshorne local cohomology and Kunz–Cox–Dickenstein power-series residues [20][21],

$$H^2_{\mathfrak{m}}(R)\, dz_1 dz_2 = \bigoplus_{a,b\geq 0} \mathbb{C}\, z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$$

is, after the trivialization $\omega_R = R\, dz_1 dz_2$, the injective hull $E_R(\mathbb{C})$ given by Matlis duality [19]. The reduced principal-part module $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$ is the $\mathbb{C}$-linear residue annihilator of constants. The R-submodule structure belongs to $E_R(\mathbb{C})$, with the scalar line treated in the ambient Matlis module. The reduced module is the $\mathbb{C}$-linear annihilator of the scalar Hamiltonian under the Matlis residue evaluation pairing. Hamiltonian vector fields preserve $dz_1 dz_2$, so this restricted pairing transports their action to the coadjoint action. The explicit coefficient proof above is the coordinate form of this local-duality statement.

THEOREM 5.179 *(Principal-part coadjoint isomorphism, Matlis form).* After fixing the residue scalar in Theorem 5.177, the residue pairing induces an isomorphism of $\mathfrak{h}$-modules

$$\Phi_{\mathrm{coad}} : \mathcal{P} \xrightarrow{\sim} \mathfrak{h}^\vee_{\mathrm{cont}}, \qquad \rho_{a,b} \longmapsto \partial_{a,b},$$

characterised uniquely (up to the residue scalar c of Theorem 5.177) by the property that for every $f \in \mathfrak{h}$ and every $g \in \mathfrak{h}$,

$$\langle \Phi^{-1}_{\mathrm{coad}}(\mathrm{ad}^\vee_f\, \partial), g\rangle = -\langle \Phi^{-1}_{\mathrm{coad}}(\partial), \{f, g\}\rangle.$$

In monomial form,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\, \rho_{a-p+1,b-q+1},$$

with the convention that any term with a negative index is zero. The isomorphism is continuous, of total degree zero, and the unique continuous total-degree-zero $\mathfrak{h}$-module isomorphism $\mathcal{P} \to \mathfrak{h}^\vee_{\mathrm{cont}}$ after the residue scalar is fixed; before that normalization it is unique modulo scalar.

Proof. Existence on monomials. Let $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$. Since $\{z_1, z_2\} = 1$,

$$\{f, z_1^{-a-1} z_2^{-b-1}\} = p\, z_1^{p-1} z_2^q \cdot (-(b+1)) z_1^{-a-1} z_2^{-b-2} - q\, z_1^p z_2^{q-1} \cdot (-(a+1)) z_1^{-a-2} z_2^{-b-1},$$

which simplifies to $-(pb - qa + p - q)\, z_1^{p-a-2} z_2^{q-b-2}$. Multiplying by $dz_1 dz_2$ identifies this with $\rho_{a-p+1,b-q+1}$ when both indices are ≥ 0; otherwise the product lies outside $\mathcal{P}$ and the principal-part projection π_{pp} of Proposition 5.176 gives zero. The integration by parts formula $\langle\{f,\rho\}, g\rangle = -\langle \rho, \{f, g\}\rangle$ identifies this with the coadjoint action.

Continuity. The dual filtration on $\mathfrak{h}^\vee_{\mathrm{cont}}$ has $F^N \mathfrak{h}^\vee_{\mathrm{cont}}$ equal to the functionals vanishing on $\mathfrak{m}^{N+1} \subset \mathfrak{h}$, where $\mathfrak{m}$ is the maximal ideal. The grading of $\mathcal{P}$ by total weight $a+b$ corresponds under Φ_{coad} to the dual filtration: $F^N = \bigoplus_{a+b\leq N} \mathbb{C}\, \partial_{a,b}$ is the Φ_{coad}-image of $\bigoplus_{a+b\leq N} \mathbb{C}\, \rho_{a,b}$.

Verification on three instances. With the convention $\{z_1, z_2\} = 1$, the Hamiltonian vector field is $X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}$.

(i) $(p, q, a, b) = (1, 0, 0, 1)$: the formula gives $z_1 \cdot \rho_{0,1} = -(1 \cdot 1 - 0 \cdot 0 + 1 - 0)\rho_{0-1+1,1-0+1} = -2\rho_{0,2}$. Direct: $X_{z_1} = \partial_{z_2}$, so $X_{z_1}(z_1^{-1}z_2^{-2}) = -2z_1^{-1}z_2^{-3}$, and multiplying by $dz_1 dz_2$ gives $-2\rho_{0,2}$. Match.

(ii) $(p, q, a, b) = (0, 1, 1, 0)$: the formula gives $z_2 \cdot \rho_{1,0} = -(0 \cdot 0 - 1 \cdot 1 + 0 - 1)\rho_{1-0+1,0-1+1} = 2\rho_{2,0}$. Direct: $X_{z_2} = -\partial_{z_1}$, so $X_{z_2}(z_1^{-2}z_2^{-1}) = 2z_1^{-3}z_2^{-1}$, multiplied by $dz_1 dz_2$ gives $2\rho_{2,0}$. Match.

(iii) $(p, q, a, b) = (2, 1, 1, 1)$: the formula gives $z_1^2 z_2 \cdot \rho_{1,1} = -(2 \cdot 1 - 1 \cdot 1 + 2 - 1)\rho_{0,1} = -2\rho_{0,1}$. Direct: $X_{z_1^2 z_2} = 2z_1 z_2\,\partial_{z_2} - z_1^2\,\partial_{z_1}$; $X_{z_1^2 z_2}(z_1^{-2}z_2^{-2}) = -4z_1^{-1}z_2^{-2} + 2z_1^{-1}z_2^{-2} = -2z_1^{-1}z_2^{-2}$, multiplied by $dz_1 dz_2$ gives $-2\rho_{0,1}$. Match.

These three cases are confirmed as part of the 1225-case exact check, which verifies the formula for $0 \le p, q, a, b \le 5$, $p + q > 0$, $a + b > 0$.

Uniqueness. Suppose $\Phi' : \mathcal{P} \to \mathfrak{h}^\vee_{\mathrm{cont}}$ is another continuous $\mathfrak{h}$-equivariant isomorphism of total degree zero. Define

$$\beta_{\Phi'}(\rho, g) = \Phi'(\rho)(g).$$

This is a non-degenerate continuous total-degree-zero pairing $\mathcal{P} \times \mathfrak{h} \to \mathbb{C}$, it kills the scalar line by definition of $\mathfrak{h}$, and $\mathfrak{h}$-equivariance of Φ' gives the coadjoint equivariance condition of Theorem 5.177. Therefore $\beta_{\Phi'} = c'\langle \cdot, \cdot \rangle$ for a unique $c' \in \mathbb{C}^\times$, and hence $\Phi' = c'\Phi_{\mathrm{coad}}$. The scalar is exactly the residue scalar of Theorem 5.177. □

Remark 5.180 *(Sign convention).* The sign $-(pb - qa + p - q)$ on the principal-part side and the sign $(pb - qa)$ on the averaged polynomial $\widetilde{\Upsilon}_{a,b}$ side of Proposition 5.172 are both correct in their own conventions; they differ because the $\Upsilon_{a,b}$ are special-fibre PBW representatives and the $\rho_{a,b}$ are deformation-compatible coadjoint representatives. See Theorem 5.181 below for the precise separation.

THEOREM 5.181 *(PBW special fibre versus coadjoint deformation module).* Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ be the semidirect Lie algebra of Proposition 2.7. The polynomial one-ψ basis $\Upsilon_{a,b} \in \bar{A}_{\mathrm{desc}}$ and the principal-part basis $\rho_{a,b} \in \mathcal{P} \cong \mathfrak{h}^\vee_{\mathrm{cont}}$ are related as follows.

(i) *Special-fibre PBW basis.* The $\widetilde{\Upsilon}_{a,b}$ form the primitive one-ψ PBW label basis in the special-fibre symmetric algebra $\mathbf{S}(\mathfrak{h} \oplus \mathfrak{h}^\vee[1])\big|_{\hbar=0}$, and the open Hamiltonian Lie action is the *symmetric PBW action* on this generator span:

$$F_{p,q} : \widetilde{\Upsilon}_{a,b} \longmapsto (pb - qa)\,\widetilde{\Upsilon}_{a+p-1,b+q-1}.$$

The shift $(a + p - 1, b + q - 1)$ is determined by multiplication by the Hamiltonian monomial and by the scalar Hamiltonian quotient.

(ii) *Deformation-level coadjoint basis.* The $\rho_{a,b}$ form a basis of the deformation-compatible continuous coadjoint module $\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P}$ of Theorem 5.179, and the $\mathfrak{h}$-action is the *principal-part coadjoint action* on this basis:

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\,\rho_{a-p+1,b-q+1}.$$

The displayed shift is the negative-transpose residue action. Linear Hamiltonians raise pole order, while higher monomials have the mixed direction prescribed by the formula.

(iii) *Module separation.* The bidegree relabelling between the PBW special-fibre basis $\widetilde{\Upsilon}_{a,b}$ and the coadjoint basis $\rho_{a,b} \leftrightarrow \delta_{a,b}$ is an isomorphism of graded vector spaces and label sets. Pulling the coadjoint $\mathfrak{h}$-action across this relabelling gives $-(pb - qa + p - q)\rho_{a-p+1,b-q+1}$, while the symmetric PBW action displayed in (1) gives $(pb - qa)\rho_{a+p-1,b+q-1}$. The formal Moyal/Weyl

Rees deformation controls the Weyl algebra and its continuous dual $\mathfrak{h}^\vee_{\hbar,\mathrm{cont}}$; the polynomial one-$\psi$ descendant module and $\mathcal{P}$ remain distinct $\mathfrak{h}$-modules. The Fourier-polarized Rees label bridge is constructed in Theorem 8.12; the same-action and unreduced quantum-observable realizations remain distinct deformation problems.

Thus the two formulas define distinct $\mathfrak{h}$-modules in the formal-local category: the PBW special fibre and the continuous coadjoint cotangent module, respectively. The structural separation recorded in Corollary 5.173 follows.

Proof. (1) is Proposition 5.172, with the rephrasing in the PBW special-fibre basis. (2) is Theorem 5.179. For (3), the symmetric PBW action on the nonconstant $\widetilde{\Psi}_{a,b}$ labels is the PBW special-fibre action after the Hamiltonian scalar quotient. The coadjoint action on $\rho_{a,b}$ is the negative-transpose principal-part action. These two module behaviours are separated by the displayed formulas already at the linear Hamiltonians z_1, z_2; hence any bidegree-only relabelling between $\widetilde{\Psi}_{a,b}$ and $\partial_{a,b}$ deviates from an $\mathfrak{h}$-module map already at the linear Hamiltonians. The Rees-deformation $A_\hbar = (\mathbb{C}[[z_1, z_2]][[\hbar]], *)$ has the property that the formal Moyal product $f * g = fg + \frac{\hbar}{2}\{f, g\} + O(\hbar^2)$ acts on its dual by the coadjoint formula at *all* orders of $\hbar$: the dual representation of the Moyal Lie algebra $(\mathfrak{h}_\hbar, [-,-]_{\mathfrak{h}_\hbar})$ on $\mathfrak{h}^\vee_\hbar$ is the residue pairing extended $\mathbb{C}[[\hbar]]$-linearly. Specialising to $\hbar = 0$ recovers the Poisson coadjoint action of the completed Hamiltonian algebra $\mathfrak{h}$ on $\mathfrak{h}^\vee = \mathcal{P}$, and its restriction to the dense polynomial subalgebra $\bar{A} \subset \mathfrak{h}$. The PBW filtration is the Rees filtration, and the special-fibre $\hbar = 0$ symmetric algebra is the associated graded; the symmetric PBW action on $\widetilde{\Psi}_{a,b}$ is exactly the $\hbar = 0$ *associated graded* image of the classical Hamiltonian vector-field action on the polynomial PBW descendant labels; the adjoint and coadjoint actions of the Lie algebra live on the deformation-level module. It therefore separates the descendant module from the continuous coadjoint module, and identifies the Rees/Fourier bridge as additional Fourier-polarized structure. □

PROPOSITION 5.182 *(Polynomial boundary obstruction for principal parts).* In the polynomial open BV/Koszul matrix model studied here, any realization of the principal-part coadjoint module requires a boundary sector beyond ordinary polynomial boundary local cohomology classes

$$\Theta_{a,b}, \qquad a, b \geq 0, \quad a + b > 0,$$

of Proposition 5.176.

Proof. The linear Hamiltonians

$$O_{1,0} = \overline{\mathrm{Tr}(\phi_1)}, \qquad O_{0,1} = \overline{\mathrm{Tr}(\phi_2)}$$

act on polynomial boundary observables by the constant Hamiltonian vector fields ∂_{ϕ_2} and $-\partial_{\phi_1}$, respectively. On every ordinary polynomial observable these operators are locally nilpotent: after enough iterations the polynomial degree in the differentiated variable is exhausted.

On the principal-part coadjoint module, however, Proposition 5.176 gives

$$O_{1,0} \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad O_{0,1} \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Both operators have non-locally-nilpotent action on every nonzero principal-part generator. Local nilpotence is preserved under isomorphism of modules. Therefore any class $\Theta_{a,b}$ representing $\rho_{a,b}$ requires a non-polynomial boundary realization; the residue principal-part module is the unique such realization. □

The categorical refinement records the same local-nilpotence obstruction at the level of polynomial sub-bimodules. The statement is restricted to this local-nilpotence obstruction; classification of all $\bar{A}$-module morphisms out of $\mathcal{P}$ is a separate question.

THEOREM 5.183 *(Categorical polynomial-realization obstruction for principal parts).* Let $\mathsf{P}_{\mathrm{pol}}$ denote the abelian category of $\mathbb{Z}$-graded bimodules $M = \bigoplus_{n \geq 0} M_n$ over the polynomial Hamiltonian Lie algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$, on which:

(i) every element acts as a Hamiltonian vector field;
(ii) every linear Hamiltonian acts locally nilpotently: for every $m \in M$ and every $i \in \{1, 2\}$, there exists N with $(z_i \cdot)^N m = 0$.

This category models polynomial boundary observables of the open BV theory with respect to the polynomial sub-bimodule structure preserved by all morphisms of the boundary algebra into itself. Then $\mathcal{P}$ lies outside $\mathrm{Ob}(\mathsf{P}_{\mathrm{pol}})$. More generally, an injective $\bar{A}$-module map $\mathcal{P} \hookrightarrow M$ with $M \in \mathrm{Ob}(\mathsf{P}_{\mathrm{pol}})$ is obstructed by local nilpotence. The unique realization of $\mathcal{P}$ is the residue principal-part realization, with the residue-coadjoint pole-raising Hamiltonian action.

Proof. By Theorem 5.179, $z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}$ and $z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}$ on every generator. Iteration gives $(z_1\cdot)^N \rho_{a,b} = (-1)^N (b+1)(b+2)\cdots(b+N)\, \rho_{a,b+N} \neq 0$ for every N, so z_1 acts non-locally-nilpotently on every nonzero element of $\mathcal{P}$. The same argument applies to z_2.

Hence $\mathcal{P}$ itself violates condition (2) and lies outside of $\mathsf{P}_{\mathrm{pol}}$. If $\Phi : \mathcal{P} \hookrightarrow M$ were an injective $\bar{A}$-module map into an object of $\mathsf{P}_{\mathrm{pol}}$, then $m = \Phi(\rho_{a,b}) \neq 0$ for some generator. Equivariance gives

$$z_1^N \cdot m = \Phi(z_1^N \cdot \rho_{a,b}) = (-1)^N (b+1)\cdots(b+N)\Phi(\rho_{a,b+N}),$$

which is nonzero for every N by injectivity. This contradicts local nilpotence of z_1 on M. The same argument applies with z_2. □

Remark 5.184 *(Polynomial obstruction).* The categorical obstruction applies to every polynomial sub-bimodule of every boundary observable algebra with locally nilpotent z_i-action; it is uniform in the boundary algebra and independent of the particular choice of classes $\Theta_{a,b}$. The distributional/Laurent principal-part construction of Construction 5.185 is therefore required for this realization.

Construction 5.185 *(Stalkwise boundary local-dual principal-part sector).* Let M_∂^{pp} be the reduced continuous dual of the stable connected Hamiltonian trace sector, with basis $\Theta_{a,b}$ dual to

$$O_{m,n} = \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}, \qquad m + n > 0,$$

by

$$\langle \Theta_{a,b}, O_{m,n} \rangle = \delta_{a,m}\delta_{b,n}.$$

Equivalently,

$$M_\partial^{\mathrm{pp}} \cong \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) \subset H^2_{(z_1,z_2)}\big(\mathbb{C}[z_1, z_2]\big)\, dz_1 dz_2 \quad \text{after completing at } (z_1, z_2),$$

the continuous dual functionals vanishing on constants, with basis $\rho_{a,b}$ for $a + b > 0$. Explicitly,

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \qquad a + b > 0,$$

with

$$\Theta_{a,b} \longleftrightarrow \rho_{a,b}.$$

This is a $\mathbb{C}$-linear residue annihilator stable for the Hamiltonian coadjoint action. The ambient R-submodule is $E_R(\mathbb{C})$, and quotienting by the non-invariant line $\mathbb{C}\,\rho_{0,0}$ is a different operation. Define the Hamiltonian action by negative transpose of the proved boundary Hamiltonian bracket:

$$\langle \{O_f, \Theta\}, O_g \rangle = -\langle \Theta, \{O_f, O_g\}_{\mathrm{open}} \rangle.$$

Then this stalkwise local-dual sector realizes the principal-part coadjoint action:

$$\{O_{p,q}, \Theta_{a,b}\}_{\mathrm{open}} = -(pb - qa + p - q)\Theta_{a-p+1,b-q+1},$$

with negative-index terms zero. Declare the odd local-dual ideal abelian:

$$\{\Theta_{a,b}, \Theta_{c,d}\}_{\mathrm{open}} = 0.$$

Equivalently, the assignment

$$z_1^p z_2^q \longmapsto O_{p,q}, \qquad \rho_{a,b}[1] \longmapsto \Theta_{a,b}$$

realizes stalkwise the Lie algebra $\bar{A} \ltimes \mathcal{P}[1]$.

THEOREM 5.186 *(Support-local principal-part current P_0 prefactorization datum).* After the de Rham contraction to the reduced local model on the brane line, the local-dual sector of Construction 5.185 extends to a support-local P_0 prefactorization datum on $\mathbb{R}$. In this theorem the de Rham differential has already been contracted; the Hamiltonian smearing uses degree-zero test functions, and the cotangent summand uses compactly supported distributions. Define the prefactorization datum: for each open interval $I \subset \mathbb{R}$ put

$$\mathfrak{J}_\partial^{\mathrm{pp}}(I) = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1], \qquad \mathfrak{J}_\partial^{\mathrm{Ham}}(I) = \Omega_c^0(I)\widehat{\otimes}\bar{A},$$

and

$$A_\partial^{\mathrm{Ham}}(I) = \widehat{\mathbf{S}}\big(\mathfrak{J}_\partial^{\mathrm{Ham}}(I)\big), \qquad A_\partial^{\mathrm{pp}}(I) = A_\partial^{\mathrm{Ham}}(I)\widehat{\otimes}\widehat{\mathbf{S}}\big(\mathfrak{J}_\partial^{\mathrm{pp}}(I)\big).$$

In this theorem the open factor is this reduced connected Hamiltonian current algebra $A_\partial^{\mathrm{Ham}}(I)$. The displayed brackets define the reduced Hamiltonian/principal-part current bracket. Brackets with arbitrary reduced or unreduced open BV observables require the unreduced lift of Remark 5.190. Write

$$\mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(I) = A_\partial^{\mathrm{pp}}(I),$$

where $\widehat{\mathbf{S}}$ is the degreewise completed symmetric algebra in the chosen current category: each finite symmetric degree is formed before completion, and the P_0 biderivation is extended continuously from generators. The construction uses this degreewise completed symmetric-algebra functor. The datum is equipped with extension by zero for inclusions of intervals and with the following structures:

(i) the differential Q_{open} on the open factor and zero on $\mathfrak{J}_\partial^{\mathrm{pp}}(I)$;

(ii) the factorization product on disjoint intervals $I_1 \sqcup I_2 \subset J$ given by extension by zero followed by graded-commutative multiplication;

(iii) the cosheaf structure on $I \mapsto \Omega_c^0(I)$ and $I \mapsto \mathcal{D}'_c(I)$ given by extension by zero;

(iv) the smearing maps

$$B_f(a) = \int_I a(t) B_f(t)\, dt, \qquad O_f(a) = a \otimes f, \qquad \Theta_\rho(B) = B \otimes \rho[1],$$

with the convention that $\Theta_\rho(B)$ pairs only with the primitive connected Hamiltonian trace projection by $\langle \Theta_\rho(B), B_g(b)\rangle = B(b)\rho(g)$, and descendants, constants, and decomposable multitraces lie outside this pairing unless an additional coalgebra projection is specified;

(v) the P_0 Poisson bracket

$$\begin{aligned}
\{B_f(a), B_g(b)\} &= B_{\{f,g\}}(ab),\\
\{B_f(a), \Theta_\rho(B)\} &= \Theta_{f\cdot\rho}(aB),\\
\{\Theta_\rho(B), \Theta_\sigma(C)\} &= 0,
\end{aligned}$$

where $a, b \in \Omega_c^0$, $B, C \in \mathcal{D}_c'$, and aB is the multiplication of a current by a smooth degree-zero form;

(vi) the smeared current morphism

$$\widetilde{\partial}_{\mathrm{bb},I} : \Omega_c^0(I) \otimes \bar{A} \oplus \mathcal{D}_c'(I) \otimes \mathcal{P}[1] \longrightarrow A_\partial^{\mathrm{pp}}(I),$$

$$a \otimes f \longmapsto B_f(a), \qquad B \otimes \rho[1] \longmapsto \Theta_\rho(B).$$

This degreewise datum defines a support-local P_0 prefactorization datum on $\mathbb{R}$. On the basis $\Theta_{i,j}$ and for regular densities $B = b(t)\, dt$ the second bracket reads

$$\{O_{p,q}(a), \Theta_{i,j}(b)\} = -(pj - qi + p - q)\, \Theta_{i-p+1, j-q+1}(ab),$$

again with negative-index terms zero. The smeared map $\widetilde{\partial}_{\mathrm{bb},I}$ is the interval-level morphism into the reduced connected Hamiltonian P_0 current sector. An unsmeared canonical map on an interval requires a chosen compactly supported unit or density.

Proof. Cosheaf axioms. Both Ω_c^0 and $\mathcal{D}_c'$ are cosheaves on $\mathbb{R}$ for the extension-by-zero structure. Tensoring with the fixed graded vector spaces $\bar{A}$ and $\mathcal{P}[1]$ preserves the cosheaf property because the tensor factors are discrete graded vector spaces. Taking the degreewise completed symmetric algebra gives the prefactorization products used here: on disjoint intervals the structure map is extension by zero followed by the graded-commutative product, and the P_0 bracket is the continuous biderivation generated by the displayed current brackets. The odd generators Θ_ρ therefore generate the completed graded-symmetric, equivalently exterior, principal-part factor.

Disjoint-support locality. For disjoint $I_1, I_2 \subset J$, the multiplication map

$$\mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(I_1) \otimes \mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(I_2) \longrightarrow \mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}(J)$$

is the graded-commutative product after extending by zero. This is the standard prefactorization product. Because the bracket $\{O_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ involves multiplication of a smooth function by a distribution and is supported on $\operatorname{supp} a \cap \operatorname{supp} B$, disjoint I_1, I_2 have disjoint such supports and the corresponding bracket vanishes; the factorization product is therefore Poisson-commutative on disjoint intervals, as required for the P_0 structure.

Q-compatibility. $Q_{\mathrm{open}}B_f = 0$ on Hamiltonian trace classes by Proposition 3.39. By declaration $Q_{\mathrm{open}}\Theta_\rho = 0$ on the Θ_ρ-factor. Therefore the differential is zero on the Θ_ρ-factor and acts trivially on the bracket structure displayed above. This is consistent with the reduced model after de Rham-current contraction.

P_0 *identities.* The first bracket $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$ is Theorem 5.191 smeared by test functions, as in Corollary 5.192. The second bracket $\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ is the negative transpose under the residue pairing of Theorem 5.177 applied to the first bracket; in coordinates with $\rho = \rho_{a',b'}$ and $f = z_1^p z_2^q$ this reads

$$\{O_{p,q}(a), \Theta_{a',b'}(B)\} = -(pb' - qa' + p - q)\, \Theta_{a'-p+1,b'-q+1}(aB),$$

exactly the principal-part coadjoint formula of Theorem 5.179. The third bracket $\{\Theta_\rho(B), \Theta_\sigma(C)\} = 0$ is by declaration of the abelian odd ideal.

Jacobi identity. The semidirect Lie algebra $\bar{A} \ltimes \mathcal{P}[1]$ has Jacobi identity $[\mathrm{ad}_f^\vee, \mathrm{ad}_g^\vee] = \mathrm{ad}_{\{f,g\}}^\vee$ for $f, g \in \bar{A}$ on the continuous dual. Smearing by test functions and distributions preserves Jacobi because the $C^\infty(\mathbb{R})$-module action on distributions is associative. Therefore the P_0 bracket on the smeared sector also satisfies Jacobi.

Bulk-boundary morphism. The map $\widetilde{\partial}_{\mathrm{bb},I}$ sends $a \otimes f$ to the smeared polynomial trace observable $B_f(a)$ by Theorem 5.191, and sends $B \otimes \rho[1]$, with $\mathcal{P}[1] \cong \mathfrak{h}_{\mathrm{cont}}^\vee[1]$ via Theorem 5.179, to the $\Theta_\rho(B)$-sector. The brackets verified above are exactly the smeared semidirect brackets on the degree-zero line-current version of $\bar{A} \ltimes \mathcal{P}[1]$. Therefore $\widetilde{\partial}_{\mathrm{bb},I}$ is a morphism into the reduced connected Hamiltonian P_0 current sector. □

LEMMA 5.187 *(Distribution-locality criterion for principal-part currents).* Let $a \in C_c^\infty(I)$, $B \in \mathcal{D}_c'(I)$, and $\rho, \sigma \in \mathcal{P}$. The product aB used in Theorem 5.186 is the standard multiplication of a distribution by a smooth function,

$$(aB)(\varphi) = B(a\varphi), \qquad \varphi \in C_c^\infty(I).$$

With this convention the reduced principal-part brackets are well-defined by

$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0.$$

Multiplication of two distributions is avoided. This is support-locality in the reduced line-current model. Costello-local functional locality and wavefront/RG-admissibility are separate analytic data.

Proof. Multiplication $B \mapsto aB$ is a continuous endomorphism of $\mathcal{D}_c'(I)$, because $a\varphi \in C_c^\infty(I)$ for every test function φ. The first displayed bracket therefore lands again in $\mathcal{D}_c'(I)\widehat{\otimes}\mathcal{P}[1]$. Its support is contained in $\mathrm{supp}(a) \cap \mathrm{supp}(B)$, so it is local on the line. The second bracket is zero by the abelian odd-ideal definition of the reduced principal-part sector; the product BC is omitted. Jacobi with two Hamiltonian entries and one principal-part entry follows from associativity $a(bB) = (ab)B$ and from the coadjoint identity $[f\cdot, g\cdot]\rho = \{f, g\} \cdot \rho$. Jacobi with two principal-part entries is immediate from the zero bracket. □

Remark 5.188 *(Reduced principal-part currents).* Theorem 5.186 is a reduced support-local P_0 prefactorization datum after de Rham-current contraction, with $\Omega_c^0(I)$ in the Hamiltonian current factor and $\mathcal{D}_c'(I)$ in the cotangent factor. The raw distributional target has additional degree-zero classes under interval enlargement: if $I \subsetneq J$, a delta-current supported in $J \setminus I$ lies in $\mathcal{D}'_c(J)$ and outside the image of extension from $\mathcal{D}'_c(I)$. The unreduced BV/factorization algebra is separate. In particular, the equation $Q\Theta_\rho = 0$ is part of the added reduced local-dual factor; an unreduced derivation requires an unreduced de Rham current complex, a chain-level primitive projection, and Hochschild centrality homotopies against arbitrary open observables.

THEOREM 5.189 *(Reduced prefactorization realization of the boundary local dual).* After de Rham contraction in the reduced setting, the assignment

$$\rho_{a,b}[1] \longmapsto \Theta_{a,b}$$

is a chain-level realization of the boundary local dual within the support-local P_0 prefactorization datum $\mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}$ of Theorem 5.186. The intervalwise smeared current morphism $\widetilde{\partial}_{\mathrm{bb},I}$ extends, in this reduced principal-part target, the Hamiltonian bulk-to-boundary Lie algebra map of Theorem 5.191 to the closed-coadjoint sector $\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$.

Proof. Theorem 5.186 gives the support-local P_0 prefactorization datum and verifies that $\widetilde{\partial}_{\mathrm{bb},I}$ is a smeared current Lie algebra morphism for each interval I. The identification $\rho_{a,b}[1] \leftrightarrow \Theta_{a,b}$ is by the basis of the local-dual factor, consistent with Construction 5.185. The closed source side $\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ is the semidirect product computed in Proposition 2.7, and its image in the support-local P_0 prefactorization datum is the displayed Lie subalgebra $\bar{A} \oplus \mathcal{P}[1]$. □

Remark 5.190 *(Unreduced lift obstruction).* Theorem 5.189 resolves the boundary local-dual prefactorization realization at the reduced level after de Rham-current contraction, with the cotangent current differential already contracted. An unreduced boundary BV/factorization lift requires: define the compactly supported de Rham functoriality before contraction, the full BV differential and homotopies, the pairing with ordinary open observables before primitive trace projection, and the coherent P_0-center homotopies.

The unreduced enlargement has the form

$$\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I)\widehat{\otimes}\widehat{\mathbf{S}}\big(\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I)\big),$$

where

$$\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I) = C^{\mathrm{dR}}_{c,\mathrm{cur}}(I)\widehat{\otimes}\,\mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)[1].$$

Here $C^{\mathrm{dR}}_{c,\mathrm{cur}}(I)$ is the compactly supported de Rham current cosheaf on the line, with differential dual to $d_{\mathbb{R}}$, and $\mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$ is the principal-part residue annihilator of constants. One must also construct a chain-level primitive/indecomposable projection

$$\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \longrightarrow \Omega^\bullet_c(I)\widehat{\otimes}\bar{A}$$

compatible with Q_{open}, constants, decomposable multi-traces, and factorization products. The required homotopies are the centrality homotopies for Θ_ρ against arbitrary unreduced open observables and the P_0 bracket homotopies

$$\{B_f, \Theta_\rho\} \simeq \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\} \simeq 0$$

before primitive projection. Only after this enlargement does the full cotangent factorization-center lift extend the reduced realization $\rho_{a,b}[1] \mapsto \Theta_{a,b}$ to a chain-level morphism on the unreduced sector. The unreduced lift is the quantum-level homotopical lift beyond the classical reduced theorem above; it sits in the residual Problem 5.222.

THEOREM 5.191 *(Classical bulk-to-boundary Hamiltonian map).* In the stable connected Hamiltonian trace sector of the formal local mixed holomorphic-topological theory, the boundary evaluation maps $\partial_{\mathrm{bb},N}$ of Construction 3.25 stabilize to a linear map

$$\partial_{\mathrm{bb}} : \bar{A} \longrightarrow H^0_{\mathrm{Ham}}(\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane}}), \qquad f \longmapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

This map is a morphism of Lie algebras from the closed Hamiltonian Poisson algebra $\bar{A}$ to the connected open Hamiltonian trace Poisson algebra:

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \partial_{\mathrm{bb}}\big(\{f, g\}_{\bar{A}}\big).$$

This is the degree-zero Hamiltonian bracket component of the reduced CE/PV central-operation map to the brane observables. The full Poisson factorization-center morphism belongs to the Poisson factorization-center construction.

Proof. The map is well-defined on $\bar{A}$ by Corollary 3.32; constants map to the scalar N and vanish in the Hamiltonian quotient by constants. Lemma 5.170 gives the open Hamiltonian trace Poisson bracket. For $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, Proposition 5.171 gives

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}.$$

The closed Poisson bracket is

$$\{f, g\}_{\bar{A}} = \overline{(kn - lm) z_1^{k+m-1} z_2^{l+n-1}},$$

and applying ∂_{bb} gives the same expression. Bilinearity proves the identity for all polynomials. □

COROLLARY 5.192 *(Classical local bulk-boundary coupling with explicit U(1) centre anomaly).* Let $a, b \in \Omega^0_c(\mathbb{R})$ be scalar compactly supported test functions on the brane line, and for $f \in \bar{A}$ define the smeared boundary Hamiltonian

$$I_\partial(a, f) = B_f(a) = \int_{\mathbb{R}} a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt \in A^{\mathrm{Ham}}_\partial(\mathbb{R}).$$

Then the following two strengthenings of the bracket identity hold jointly.

(i) Strict P_0 identity on disjoint supports. In the P_0 bracket of $A^{\mathrm{Ham}}_\partial$,

$$\{I_\partial(a, f), I_\partial(b, g)\}_{A^{\mathrm{Ham}}_\partial} = I_\partial(ab, \{f, g\}_{\bar{A}}),$$

and the bracket vanishes whenever the supports of a and b are disjoint. The locality bracket is the canonical equal-time bracket $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(t')\} = \delta^i_l \delta^k_j \delta(t - t')$ of 4.10 on two brane-time points; smearing against $a(t)b(t')$ produces the pointwise product ab after integration of the delta function. This is the level-one component of 4.11.

(ii) Explicit U(1) centre anomaly constant. Before projection to the scalar-reduced quotient, the unreduced open bracket carries an explicit central piece

$$\{I_\partial(a, f), I_\partial(b, g)\}_{\mathrm{open}} = I_\partial(ab, \{f, g\}_{\bar{A}}) + N \cdot \omega(f, g) \int_{\mathbb{R}} a(t)b(t)\, dt,$$

where ω is the cocycle of 6.2 and the prefactor N is the rank of the brane stack and is the value of the U(1) centre character $\mathrm{Tr} : \mathbb{C} \cdot I \to \mathbb{C}, I \mapsto N$. After projection to $H^0_{\mathrm{red}} = H^0/\mathbb{C} \cdot \mathrm{Tr}(1)$ the second term vanishes, and one recovers

$$\overline{\{I_\partial(a,f), I_\partial(b,g)\}_{\mathrm{open}}} = \overline{I_\partial(ab, \{f,g\}_{\bar{A}})}.$$

Statements (i) and (ii) are compatible: (ii) is the unreduced refinement, (i) is the level-one factorization-algebra restatement on the scalar-reduced sector. The equal-time delta locality of (i) produces a strict P_0 factorization algebra, while (ii) records the homotopical U(1) centre anomaly that obstructs lifting (i) to the unreduced sector.

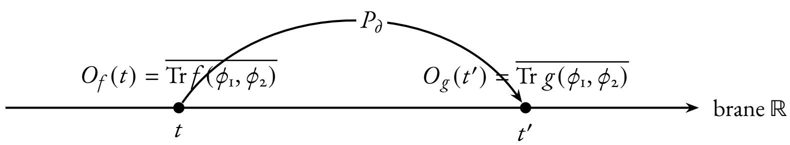
Proof. Proof of (i). By 4.7, the P_0 bracket on $A_\partial^{\mathrm{Ham}}$ is the biderivation with $\{a \otimes f, b \otimes g\}_{A_\partial} = ab \otimes \{f,g\}_{\bar{A}}$ on length-one generators, so $\{B_f(a), B_g(b)\}_{A_\partial^{\mathrm{Ham}}} = B_{\{f,g\}}(ab) = I_\partial(ab, \{f,g\}_{\bar{A}})$. 4.9 then gives vanishing on disjoint supports because $ab \equiv 0$ pointwise. The same identity is the smeared content of the equal-time delta-function bracket of 4.10: tracing $f(\phi_1, \phi_2)(t)$ and $g(\phi_1, \phi_2)(t')$, applying the Leibniz rule, and integrating against $a(t)b(t')\delta(t-t')$ produces the pointwise product ab after the delta-function integration.

Proof of (ii). The unreduced bracket of two integrals is, by locality of the canonical bracket and 5.191,

$$\begin{aligned}\{I_\partial(a,f), I_\partial(b,g)\} &= \iint_{\mathbb{R}^2} a(t)b(t')\,\{\mathrm{Tr}\, f(\phi(t)), \mathrm{Tr}\, g(\phi(t'))\}\,dt\,dt' \\ &= \int_{\mathbb{R}} a(t)b(t)\,\mathrm{Tr}\big(\{f,g\}_{\bar{A}}\big)\,dt + \int_{\mathbb{R}} a(t)b(t)\,\omega(f,g)\,N\,dt,\end{aligned}$$

where the splitting is by the central decomposition of 6.4: the central scalar $\omega(f,g) \cdot N$ is removed by the projection to H^0_{red}, and the remaining piece is $I_\partial(ab, \{f,g\}_{\bar{A}})$ exactly. The constants are explicit: the prefactor of the central piece is $N \cdot \omega(f,g)$, and the smeared coefficient is $\int ab\,dt$. The classical-to-quantum reduction of 6.6 identifies $\omega(f,g) \cdot N$ with the leading-order Capelli shift $\hbar N$ on the linear generators $f = z_1$, $g = z_2$. Compact support of a, b kills boundary terms. $\square$

$$P_\partial = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$$



$$|\operatorname{Aut}(\Gamma_1)| = 1, \quad \frac{1}{|\operatorname{Aut}(\Gamma_1)|} = 1, \quad \text{one edge}$$

$$\text{ordered: } \hbar P_\partial/2; \quad \text{commutator: } \hbar P_\partial$$

$$\text{bracket:} \quad \hbar(kn - lm)\,\overline{\operatorname{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})} \quad \text{on } f = z_1^k z_2^l,\ g = z_1^m z_2^n$$

FIGURE 0.1: Notation diagram for the reduced open-line order-$\hbar$ raw commutator graph Γ_1. Two ordered boundary insertions sit on the brane line $\mathbb{R}$ at distinct times t, t'; the unique propagator P_∂ contracts one z_1-leg on one side against one z_2-leg on the other (and the antisymmetric partner). The automorphism group is trivial. In the Weyl ordered product the ordered one-edge weight is $\hbar P_\partial/2$; after subtracting the reverse order the commutator weight is $\hbar P_\partial$. The bivector contraction on monomials reproduces the cocycle coefficient $(kn - lm)$ of Proposition 5.171, the order-$\hbar$ Moyal coefficient of Proposition 5.216, and the U(1) centre cocycle $\omega(f, g)$ of Lemma 6.2 realised at multidegree $(k + m, l + n) = (1, 1)$.

The graph Γ_1 of Figure 0.1 records the single canonical contraction producing the coefficient $(kn - lm)$ in Proposition 5.171. An independent Feynman-integral proof has Costello graph weights as its input.

5.7.2 Quantization and the brane trace map

Three operations are at stake: Moyal star-quantisation of Hamiltonians on the formal symplectic polydisk, Weyl quantisation of the matrix entries of (ϕ_1, ϕ_2), and the brane trace map $J : f \mapsto \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$ extended to a quantum trace map $\Phi_\hbar$. The question is whether they commute up to identifiable obstruction classes. Theorem 5.218 names the four components on which compatibility is proved: weighted BV kernel convergence, balanced-supertrace scalar QME cancellation on its completion, weighted Moyal principal-part current brackets, and a explicit degree-zero Hamiltonian trace comparison. The obstruction coordinates are explicit: the all-bidegree radial/Weyl normal-form question is the single cokernel class $\Omega^{\mathrm{rad}}_{a,b} \in \operatorname{coker} B_{a,b}$ (Proposition F.25); the Costello QME question is the filtered obstruction tower

$$\left(\{o^{(r)}_{\sigma,w}\}_r,\ [\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}],\ \varprojlim{}^1 H^0\right),$$

where the displayed $H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{QME}})$ class exists only after the scalar projection has been lifted and the finite-window tower has been made compatible (Problem 5.220); the smallest non-scalar window θ_3 splits into a bare $H^1 = \mathbb{C} \cdot E_{\theta_3} \neq 0$ and a counterterm-adjoined row $H^1 = 0$ with coefficient $c = 1$ (Theorem E.38).

The U(1) centre-of-mass cocycle. On the closed side, the Hamiltonian vector field of every constant is zero; the further quotient $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$ removes the unit from the source. On the brane side, the unit $1 \in \mathbb{C}[z_1, z_2]$ lifts to $\operatorname{Tr}(I_N) = N$, the rank of the brane stack: the center-of-mass mode of the matrix coordinate, invisible to Hamiltonian reduction. The two linear generators $\mathrm{Tr}(\phi_1)$ and $\mathrm{Tr}(\phi_2)$ are the open-side images of z_1 and z_2, and on them the unreduced matrix Poisson bracket reads

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = \mathrm{Tr}(I_N) = N,$$

while $\{z_1, z_2\}_{\bar{A}} = 0$ modulo constants. Quantising the same linear generators by the Moyal star and reading the Weyl-ordered trace representation gives the projective commutator

$$[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_{\mathrm{proj}} = \hbar N \cdot \mathbf{1}.$$

This detects on the closed side the Lie two-cocycle $\bar{c} \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ with $\bar{c}(f, g) = [\{f, g\}]_0$, and on the brane side reproduces the Capelli contact term in the reduced ordinary scalar $\mathfrak{gl}_N$ branch [26][27]. The three faces of the same scalar obstruction — Lie 2-cocycle on $\bar{A}$, Poisson bracket of trace coordinates, and projective Moyal–Weyl commutator — realise the same class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})[[\hbar]]$ (Theorems 6.3, 6.4, 6.6, E.13). The cancellation branches are catalogued in Theorem 5.214: the unreduced source, the central-character source, the opposite central extension, the balanced supertrace, the algebraic central-contact cone, and the Costello-local closed-exchange criterion. The all-order Moyal/Weyl identity extending the linear-generator computation is the constant-Poisson deformation quantisation [7].

The formal Moyal target is the unique translation-invariant deformation of the constant Poisson tensor on the formal holomorphic polydisk. It fixes the algebraic coefficients that any quantum closed-open BV construction must recover. Construction of that BV theory is separate. The classical degree-zero Hamiltonian bracket is the proposed special fibre; the cotangent/descendant part is recorded as Problem 5.222. Put

$$A_\hbar = \big(\mathbb{C}[[z_1, z_2]][[\hbar]], *\big),$$

where $*$ is the Weyl/Moyal product

$$f * g = m \circ \exp\!\Big(\tfrac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\Big)(f \otimes g).$$

The symbol $\bar{A}_\hbar$ denotes the central quotient of the underlying Moyal Lie algebra by $\mathbb{C}[[\hbar]]$; the associative algebra quotient is a different object. Set

$$[f, g]_{\mathrm{raw}} = f * g - g * f, \qquad \{f, g\}_\hbar = \hbar^{-1}[f, g]_{\mathrm{raw}}.$$

The degree-zero quantum Hamiltonian Lie algebra is

$$\mathfrak{h}_\hbar = \bar{A}_\hbar, \qquad [f, g]_{\mathfrak{h}_\hbar} = \{f, g\}_\hbar.$$

The specialization at $\hbar = 0$ is the Poisson Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ used above.

On the brane side, the open action

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2)$$

makes the matrix entries of ϕ_1 and ϕ_2 conjugate variables. The canonical algebraic quantization of the unconstrained first-order matrix phase space is the completed Weyl algebra $\mathcal{W}_N$ generated by those entries with

$$[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar\, \delta^i_l \delta^k_j, \qquad [(\phi_a)^i{}_j, (\phi_a)^k{}_l] = 0 \quad (a = 1, 2).$$

The gauge field imposes the conjugation moment-map constraint. With the commutator convention above, the normal-ordered quantum comoment is

$$\widehat{\mu}_\epsilon = -\mathrm{Tr}\big([\epsilon, \phi_1]\phi_2\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

This formula is kept in normal order: for $\epsilon = 1$ it vanishes, as the scalar subgroup acts trivially. The induced infinitesimal conjugation action on the Weyl algebra is fixed by

$$\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a;$$

changing to the opposite commutator convention changes the displayed comoment by the corresponding sign. Since the scalar comoment is zero, the effective constraint algebra is $\mathfrak{pgl}_N$, represented by $\mathfrak{sl}_N$ after choosing the trace pairing. The quantum brane local algebra used here is the reduced BRST, or degree-zero normalizer, quantum Hamiltonian reduction

$$\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N} = C^\bullet_{\mathrm{BRST}}\big(\mathfrak{sl}_N, \mathcal{W}_N, \widehat{\mu}\big),$$

whose degree-zero cohomology is the normalizer quotient

$$H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N} = \mathrm{N}_{\mathcal{W}_N}\big(\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)\big)/\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N).$$

The constant Hamiltonian trace is projected out in degree zero as before. The classical special fibre is the reduced BV algebra of Construction 3.20.

The quantum boundary map is Weyl-ordered trace evaluation:

$$\partial_{\mathrm{bb},\hbar,N}(f) = \overline{\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)}, \qquad f \in \bar{A}_\hbar,$$

where Op_{W} denotes total symmetrisation before passing to the trace. At $\hbar = 0$ this recovers Construction 3.25. After dividing the commutator by $\hbar$ and reducing modulo $\hbar$, the raw commutator identity recovers Theorem 5.191.

PROPOSITION 5.193 *(Reduced BRST descent of Weyl-ordered Hamiltonian traces).* For every $f \in \bar{A}_\hbar$, the Weyl-ordered trace

$$F_{f,N} = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$$

is invariant under the quantum comoment:

$$[\widehat{\mu}_\epsilon, F_{f,N}] = 0, \qquad \epsilon \in \mathfrak{gl}_N.$$

Hence $F_{f,N}$ lies in the normalizer of the quantum moment-map ideal and defines a degree-zero class in $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ after removing the scalar Hamiltonian.

Proof. Total Weyl symmetrisation is equivariant for linear changes of the matrix variables, and simultaneous conjugation acts linearly on (ϕ_1, ϕ_2). Therefore $\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$ is invariant under infinitesimal conjugation. By the defining property of the quantum comoment,

$$\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a,$$

this invariance is exactly $[\widehat{\mu}_\epsilon, F_{f,N}] = 0$. If $I = \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$ is the left quantum moment-map ideal, then for every generator $\widehat{\mu}_\epsilon$, $\widehat{\mu}_\epsilon F_{f,N} = F_{f,N}\widehat{\mu}_\epsilon \in I$. Thus $F_{f,N}$ normalizes I and descends to the normalizer quotient describing $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$. $\square$

PROPOSITION 5.194 *(Reduced BRST descent of arbitrary Weyl-ordered Hamiltonian products).* For every finite collection $f_1, \ldots, f_r \in \bar{A}_\hbar$, the ordered product

$$F = \prod_{i=1}^{r} F_{f_i,N}, \qquad F_{f_i,N} = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f_i)(\phi_1, \phi_2),$$

is invariant under the quantum comoment:

$$[\widehat{\mu}_\epsilon, F] = 0, \qquad \epsilon \in \mathfrak{gl}_N.$$

Equivalently, every monomial in the Weyl-ordered single-trace generators normalizes the quantum moment-map ideal and defines a degree-zero class in $H^0 \mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ after removing the empty trace.

Proof. By the convention $\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a$, the operator $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ acts on the associative algebra $\mathcal{W}_N$ by infinitesimal GL_N-conjugation, and is therefore a derivation of the associative product. By Proposition 5.193, each individual factor $F_{f_i,N}$ is annihilated by this derivation. Derivations of an associative algebra preserve products: the Leibniz expansion of $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ on the ordered product $\prod_{i=1}^{r} F_{f_i,N}$ is a sum of summands each containing a factor $\hbar^{-1}[\widehat{\mu}_\epsilon, F_{f_i,N}] = 0$. Hence $[\widehat{\mu}_\epsilon, F] = 0$, and as in the proof of Proposition 5.193, $\widehat{\mu}_\epsilon F = F\widehat{\mu}_\epsilon \in \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$, so F normalizes the left moment-map ideal and descends. □

Remark 5.195. The argument uses only that $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ is a derivation. The Capelli shift obstructs the bracket identity between traces while their joint reduced BRST descent follows from derivation invariance.

Remark 5.196 *(Normal-ordering obstruction).* Proposition 5.193 proves reduced BRST descent. The quantum Moyal theorem requires the bracket identity. Put $X = \phi_1$ and $Y = \phi_2$. With the normal ordering above,

$$\widehat{\mu}_\epsilon = -\mathrm{Tr}([\epsilon, X]Y) = \mathrm{Tr}\big(\epsilon(YX - XY + \hbar N I)\big).$$

For $\epsilon = I$ this comoment is identically zero; equivalently the displayed matrix has trace zero inside the Weyl algebra. Thus the nontrivial quantum moment equations are represented in the quantum Hamiltonian reduction by the traceless matrix congruence

$$YX - XY + \hbar N I \equiv 0 \pmod{\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)},$$

whose special fibre is the classical commuting-pair equation. Consequently, for a covariant matrix word A,

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\,\mathrm{Tr}(A) \quad \mathrm{mod}\ \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N).$$

This Capelli-type shift can feed lower-degree traces into the connected sector. A stable quantum theorem must prove that the chosen connected extraction removes or renormalizes this shift, or else change the sector or normalization.

Remark 5.197 *(Capelli-renormalized trace generators).* Let

$$T_{a,b} = \mathrm{Tr}(X^a Y^b)$$

denote the normal-ordered trace. The Weyl-ordered trace

$$J_N(z_1^a z_2^b) = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(z_1^a z_2^b)(X, Y)$$

is triangular over the normal-ordered traces after imposing the quantum moment relation:

$$J_N(z_1^a z_2^b) \equiv \sum_{r=0}^{\min(a,b)} \left(-\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} T_{a-r,b-r}.$$

For example,

$$J_N(z_1 z_2) = T_{1,1} - \frac{\hbar N^2}{2}, \qquad J_N(z_1^2 z_2) = T_{2,1} - \hbar N\, T_{1,0},$$

and

$$J_N(z_1 z_2^2) = T_{1,2} - \hbar N\, T_{0,1}, \qquad J_N(z_1^2 z_2^2) = T_{2,2} - 2\hbar N\, T_{1,1} + \frac{\hbar^2 N^3}{2}.$$

The normal-ordered basis carries the Capelli-shifted commutator already in low degree; a direct calculation gives

$$\hbar^{-1}[T_{2,1}, T_{0,2}] = 4T_{1,2} - 2\hbar N\, T_{0,1} \quad \bmod\ \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N),$$

while the scalar Moyal bracket records the leading $4T_{1,2}$. The connected, or primitive trace, projection retains the Capelli shift: it removes disconnected products and the empty trace, leaving the nonempty primitive trace $T_{0,1}$ in the residual. The stable connected quantum theorem therefore uses normal-ordered primitive traces with the triangular $\hbar N$ correction built into the definition. The natural finite-N generators are the Capelli-renormalized Weyl traces $J_N(f)$, equivalently

$$J_N(f) = T_N\left(\exp\left(-\frac{\hbar N}{2}\partial_{z_1}\partial_{z_2}\right) f\right)$$

in the normal-ordered trace notation $T_N(z_1^a z_2^b) = T_{a,b}$.

LEMMA 5.198 *(Capelli-renormalized stable connected trace).* Let $\mathrm{Tr}_N^{\mathrm{ren}}(f) = J_N(f)$ for $f \in \mathbb{C}[z_1, z_2][[\hbar]]$. Write $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$ for the left moment-map ideal. Then:

(i) the map $\mathrm{Tr}_N^{\mathrm{ren}}\colon \bar{A}_\hbar \longrightarrow \mathcal{W}_N / I_N$ is well defined modulo the empty (constant) trace;

(ii) the inverse triangular system reads

$$T_{a,b} = \sum_{r=0}^{\min(a,b)} \left(\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} J_N(z_1^{a-r} z_2^{b-r}),$$

so $T_{a,b}$ and $J_N(z_1^a z_2^b)$ generate the same Weyl trace algebra modulo the empty trace, after inverting the unipotent system over $\mathbb{C}[[\hbar N]]$;

(iii) the Capelli contact term of Remark 5.196, $\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\,\mathrm{Tr}(A) \bmod I_N$, reproduces, on J_N generators, exactly the $r = 1$ tail of the triangular formula. Consequently the displayed shift is *represented by the change of basis* between $T_{a,b}$ and $J_N(z_1^a z_2^b)$; the J_N-commutator has the scalar contact term already built into the basis;

(iv) the leading symbol of $J_N(f)$ is $\mathrm{Tr}\, f(X, Y)$ in associated graded order, so $\mathrm{Tr}_N^{\mathrm{ren}}$ stabilises in the large-N limit and its stable connected projection coincides with the leading primitive trace.

Proof. The triangular formula of Remark 5.197 is total Weyl symmetrisation of $z_1^a z_2^b$ inside the matrix Weyl algebra, in which one normal-ordered swap of $z_1 z_2$ commutes through $\hbar N$. This is the local bivariate Weyl-normal-ordering formula proved by the contraction count in Lemma F.7; the classical Capelli source context is [26][27]. The exact bivariate triangular trace identity is proved here by the displayed contraction count. A finite exact check verifies $J_N \circ T = \mathrm{id}$ on exponents $a, b \leq 10$. The system is unipotent over $\mathbb{C}[[\hbar N]]$ with $r = 0$ diagonal, hence invertible there; the sign in the inverse is $(+\hbar N/2)^r$, again verified by the same finite computation.

For (iii), expand

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \mathrm{Tr}(A[X,Y]) = \hbar N\,\mathrm{Tr}(A) - \mathrm{Tr}\big(A(YX - XY + \hbar NI)\big)$$

using the Weyl moment relation $YX - XY + \hbar NI = 0 \bmod I_N$ from Remark 5.196; the subtracted term lies in I_N and the displayed image is $\hbar N\,\mathrm{Tr}(A)$. On J_N generators this is the $r = 1$ tail term of the triangular expansion, by direct comparison with Remark 5.197. Consequently, in the J_N-basis the Capelli contact term is the change-of-basis matrix from T to J.

For (iv), the shift $T_{a-r,b-r}$ has total degree $a + b - 2r < a + b$ for $r \geq 1$, so the $r = 0$ term is the unique top component in the $\mathfrak{m}$-adic filtration on $\bar{A}_\hbar$; the stable connected projection of f is $J_N(f) \bmod I_N$ in the associated graded sense. □

COROLLARY 5.199 *(Renormalized stable connected trace map).* The map

$$\mathrm{Tr}^{\mathrm{ren}} \colon \bar{A}_\hbar \longrightarrow H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty}), \qquad f \longmapsto \big[\overline{J_N(f)}\big]_{N\to\infty}$$

is well defined and is a $\mathbb{C}[[\hbar]]$-linear injection of $\bar{A}_\hbar$ onto the stable Capelli-renormalized single-trace generators. The bar $\overline{(-)}$ denotes the connected single-trace primitive projection: it removes the empty trace and disconnected products, and it is consistent with the J_N-basis at every finite N by Lemma 5.198(iii). The notation $\big[\overline{J_N(f)}\big]_{N\to\infty}$ means this renormalized J-basis transition, or equivalently the associated-graded stable class determined by the leading primitive trace. The literal block-restriction inverse-limit element in the unrenormalized normal-ordered trace basis over plain $\mathbb{C}[[\hbar]]$ is a different object.

THEOREM 5.200 *(Reduced Moyal commutator: radial hypothesis and obstruction criterion).* Let $\mathcal{W}_N$ be the Weyl algebra of $T^*\,\mathrm{Mat}_N$ and let $\widehat{\mu}$ be the normal-ordered conjugation comoment above. For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, put

$$J_N(f) = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(X, Y).$$

Let $\mathfrak{t} \subset \mathfrak{gl}_N$ be the diagonal Cartan, with coordinates $\lambda_1, \ldots, \lambda_N$, and let

$$\Delta = \prod_{i<j}(\lambda_i - \lambda_j).$$

Put

$$S_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

The ordinary-trace comparison is proved under the following hypotheses. Assume the radial-parts hypotheses of Statement F.8: after Vandermonde conjugation, the Harish-Chandra map on invariants has the quantum moment-map ideal as kernel and therefore injects the reduced invariant quotient into Weyl-invariant differential operators on the regular Cartan. Under this hypothesis the radial-part map satisfies

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1} S_N(f) \Delta.$$

Consequently, for every $N \geq 1$ and all f, g the defect

$$D_N(f, g) = \hbar^{-1}[J_N(f), J_N(g)] - J_N(\{f, g\}_\hbar)$$

lies in the quantum moment-map ideal $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$ and therefore vanishes in the degree-zero quantum Hamiltonian reduction

$$\mathrm{N}_{\mathcal{W}_N}(I_N)/I_N.$$

After the connected single-trace projection $\overline{(-)}\colon \mathcal{W}_N/I_N \twoheadrightarrow \mathcal{W}_N/I_N/(\text{empty trace} \oplus \text{disconnected products})$, the descended commutator on the J_N-images is the finite-rank image of the scalar Moyal bracket:

$$\left[\eth_{\mathrm{bb},\hbar,N}(f), \eth_{\mathrm{bb},\hbar,N}(g)\right]_{\mathrm{conn,raw}} = \hbar\, \eth_{\mathrm{bb},\hbar,N}(\{f,g\}_\hbar).$$

The identification with the scalar Moyal algebra itself holds only after passing to the degreewise stable range and then to the large-N connected sector. Here the large-N limit is the renormalized associated-graded connected transition system of Corollary 5.199. Finite-N trace identities remain in each rank, and the scalar character $\hbar N$ has already been recorded in the J_N-basis and then quotiented along the Hamiltonian scalar line. The identity stabilises in this large-N sense, defining $\eth_{\mathrm{bb},\hbar}\colon \mathfrak{h}_\hbar \longrightarrow H^{\mathrm{o}}_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})$ satisfying

$$\hbar^{-1}[\eth_{\mathrm{bb},\hbar}(f), \eth_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \eth_{\mathrm{bb},\hbar}(\{f,g\}_\hbar),$$

which is the bracket equation recorded in Remark 5.203.

The image-normalization clause of Statement F.8 is the quantum shear primitive: for every a, b, N,

$$E_{a,b,N} = \sum_{j=0}^{b} \mathrm{Tr}(Y^j X^a Y^{b-j}) - J_N\Big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\Big) \in \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N).$$

Equivalently one must kill the free normal-diagram obstruction

$$\mathfrak{o}_{a,b} = [R^{\mathrm{free}}_{a,b}] \in \operatorname{coker} C_{a,b}$$

by functorial lifts $K_{a,b} \in \ker \partial$. In the dual-potential formulation this is the vanishing of

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b},$$

equivalently the decorated PBW Stokes condition $D^{\square}_{a,b} = C^+_{a,b}\partial_2$; obstruction is exactly a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. The complete edge families $\mathfrak{o}_{2,b}$ and $\mathfrak{o}_{a,2}$ vanish by Proposition F.32. The finite free normal-form computations prove the displayed interior windows, including the $K_{4,4}$ lift and the non-edge total-degree checks recorded in Proposition F.33 and the exact normal-form calculation. For $a, b \geq 3$ in all degrees, all-interior closure is equivalent to constructing the associated-graded two-colour necklace homotopy, or proving the dual-potential theorem: every signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ satisfies $\lambda(E^+_{a,b}) = \phi(T_{a,b})$. The obstruction is exactly such a row with nonzero $\lambda(E^+_{a,b}) - \phi(T_{a,b})$.

Proof. Identify $\mathcal{W}_N$ with $\mathcal{D}_\hbar(\mathfrak{gl}_N)$, where X is multiplication and $Y = -\hbar\partial^t_X$. The quantum Hamiltonian reduction for conjugation is the normalizer quotient by the left ideal generated by the traceless comoment. The scalar matrix has zero comoment, as in Remark 5.196; equivalently the degree-zero function calculation is the $\mathfrak{sl}_N$ reduction together with rational GL_N-invariants. Since GL_N is linearly reductive in characteristic zero, the functor of rational algebraic invariants is exact on this function-level calculation. The unreduced Lie-CE cohomology of $\mathfrak{gl}_N$ is outside this argument.

The Harish-Chandra radial-parts theorem, whose source lineage is Harish-Chandra, Wallach, and Levasseur–Stafford [8][9][10][11], is exactly the assertion that the Vandermonde-conjugated radial map on invariants has the moment-map ideal as kernel, injects this quantum Hamiltonian reduction, and has the generator image normalization recorded in Statement F.8. The nonlinear shear obstruction in Lemma F.13 explains why the mixed Weyl-trace image is kept as part of those hypotheses; pure-power radial images give only the special cases. Therefore the displayed radial formula for $J_N(f)$ is valid only under the exact hypotheses named there.

Distinct one-particle Weyl summands commute, hence

$$\hbar^{-1}[S_N(f), S_N(g)] = S_N(\{f, g\}_\hbar).$$

Conjugation by Δ preserves commutators, so $\mathrm{Rad}_0(D_N(f, g)) = 0$. Injectivity of the Harish-Chandra reduction map gives

$$D_N(f, g) \in \big(\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)\big)^{GL_N},$$

hence $D_N(f, g)$ is zero in the degree-zero quantum Hamiltonian reduction. The Cartan-rank-2 case has been checked by exact arithmetic: the direct $N = 2$ radial restriction of $J_2(f)$ is checked on 80 operator/polynomial pairs, and the $N = 3$ finite-rank sweep checks 80 corresponding pairs. The same computation also checks

$$[S_2(f), S_2(g)] = S_2([f, g]_*)$$

in the normal-form convention $z_2 = -\hbar\partial_\lambda$ on the primary pairs, and on the higher-order pairs that exhibit nonzero P^3 and P^5 coefficients. These are finite-rank coefficient checks; the all-N source theorem remains separate.

For the connected-projection statement, the projection $\overline{(-)}$ acts on the basis $T_{a,b} \mapsto \overline{T_{a,b}} \equiv \overline{J_N(z_1^a z_2^b)}$ mod (Capelli triangular). By Lemma 5.198(iii) the Capelli contact term is represented by the change of basis between $T_{a,b}$ and J_N, so on $\overline{J_N(f)}$-classes the descended commutator is the image of $\hbar J_N(\{f, g\}_\hbar)$ mod I_N under the projection, i.e. $\hbar\, \partial_{\mathrm{bb},\hbar,N}(\{f, g\}_\hbar)$.

Stability in N: after passage to the renormalized associated-graded J_N-basis transition maps, the radial identity is compatible with increasing N. Enlarging $\mathfrak{t}_N \subset \mathfrak{gl}_N$ to $\mathfrak{t}_{N+k}$ adjoins k mutually commuting one-particle Weyl summands, and the bracket $[S_N(f), S_N(g)] = S_N(\{f, g\}_\hbar)$ is preserved at every N. By Lemma 5.198(iv), the connected extraction depends only on the leading symbol filtration, which the Capelli triangular system preserves. Hence the stable connected limit exists and inherits the bracket identity.

The final obstruction statement is Lemma F.15 together with Definition F.21 and Statement F.22. The edge vanishing is Proposition F.32. The finite interior identities are the free normal-diagram identities of Proposition F.20, Proposition F.33, and the exact-arithmetic normal-form computation in the free trace-diagram basis. These data give finite-window identities and edge theorems; the functorial all-interior splitting is the corresponding interior theorem. □

COROLLARY 5.201 *(Degree-zero quantum comparison and decorated radial obstruction).* Under the same radial-parts hypotheses as Theorem 5.200, the map

$$\partial_{\mathrm{bb},\hbar}\colon \mathfrak{h}_\hbar = \bar{A}_\hbar \longrightarrow H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty}), \qquad f \longmapsto \big[\overline{J_N(f)}\big]_{N\to\infty},$$

is a Lie-algebra map for $\{\ ,\ \}_\hbar$ on the source and $\hbar^{-1}[-,-]_{\mathrm{conn,raw}}$ on the target. Its associated graded at $\hbar = 0$ is the degree-zero part of Theorem 5.6. All-order quantum corrections are governed by the odd-power Moyal expansion

$$\{f, g\}_\hbar = \sum_{s \geq 0} \frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g),$$

so its second-order normalized coefficient is $\frac{1}{24} P^3(f, g)$. This term can be nonzero.

Under the stated finite-window and radial hypotheses, the formal Moyal coefficients and the Capelli triangular trace basis remain proved. The quantum shear primitive needed for the ordinary Weyl-trace comparison is proved on the edge strips by Proposition F.32 and in the finite verified interior windows. Its all-interior extension is exactly the dual-potential obstruction theorem

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}$$

of Proposition F.25: it vanishes if and only if the decorated PBW Stokes identity $D^{\square}_{a,b} = C^+_{a,b}\partial_2$ holds, with obstruction represented exactly by a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$.

Proof. This is Theorem 5.200, combined with the Moyal odd-symmetry of Proposition 5.216: P^r changes by $(-1)^r$ under interchange of f and g, so only odd r survive in the raw star commutator. Dividing by $\hbar$ shifts these odd raw powers to even powers in the normalized Hamiltonian bracket. Hence the normalized $\hbar^2$ coefficient is $P^3(f, g)/24$, and the normalized $\hbar^4$ coefficient is $P^5(f, g)/1920$. □

Remark 5.202 *(Explicit second- and fourth-order corrections).* For higher-order instances that exercise nontrivial P^3 and P^5:

$$[z_1^3 z_2, z_1 z_2^3]_* = 8\hbar\, z_1^3 z_2^3 - 3\hbar^3\, z_1 z_2 + O(\hbar^5),$$
$$[z_1^4 z_2^2, z_1^2 z_2^4]_* = 12\hbar\, z_1^5 z_2^5 - 40\hbar^3\, z_1^3 z_2^3 + 6\hbar^5\, z_1 z_2 + O(\hbar^7).$$

After normalization by $\hbar^{-1}$, the first example gives

$$\{z_1^3 z_2, z_1 z_2^3\}_\hbar = 8 z_1^3 z_2^3 - 3\hbar^2 z_1 z_2 + O(\hbar^4).$$

The $\hbar^3$ raw coefficient is therefore the normalized $\hbar^2$ correction $\frac{1}{24} P^3(f, g)$, in agreement with Proposition 5.216. The non-affine bracket correction reduces, in this constant Poisson tensor case, to exactly the higher Moyal coefficients.

Remark 5.203 *(Degree-zero quantum Hamiltonian sector).* Assume the radial image datum of Statement F.8. Then Corollary 5.201 identifies the stable connected single-trace quotient of $\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ and proves that $\partial_{\mathrm{bb},\hbar,N}$ stabilizes to a map

$$\partial_{\mathrm{bb},\hbar} : \mathfrak{h}_\hbar \longrightarrow H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})$$

satisfying

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f, g\}_\hbar).$$

Thus the connected quantum brane algebra is the Weyl/Moyal deformation of the degree-zero Hamiltonian sector. Its associated graded at $\hbar = 0$ is the degree-zero part of Theorem 5.6. The J_N-basis change records the finite-N Capelli contact and its triangular $\hbar N$ lower-trace terms; the connected projection removes only the empty trace and decomposable products. Interior bidegrees reduce to the exact Weyl-trace image clause of Statement F.8, equivalently to quantum shear primitives $E_{a,b,N} \in \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$. In the free indexed normal-diagram model this is the vanishing of $\mathfrak{o}_{a,b}$ together with functorial lifts $K_{a,b}$. In the sharper dual-potential form the obstruction is $\Omega^{\mathrm{rad}}_{a,b}$, equivalently decorated PBW Stokes for $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$; an obstruction is a signed $B^{*}_{a,b}$-row with nonzero $\lambda(E^{+}_{a,b}) - \phi(T_{a,b})$. The edge strips are proved; the interior all-bidegree theorem is the associated-graded two-colour necklace homotopy, or this equivalent dual-potential signed-row vanishing theorem.

Remark 5.204. The degree-zero Hamiltonian target is the Weyl/Moyal algebra of the formal symplectic polydisk, the constant-Poisson case of deformation quantization [7]. Costello's BV/RG construction [12] and the flat open-closed BCOV/HCS quantization of Costello–Li [13][14] apply under their elliptic, gauge-fixing, renormalization, and obstruction-vanishing hypotheses; the mixed ordinary-$\mathfrak{gl}_N$ Hamiltonian BF defect problem here lies outside their hypothesis class and requires the specialization datum of Problem 5.213.

Remark 5.205. Corollary 5.173 gives the Hamiltonian action on the polynomial representatives $\Psi_{k,l}$: it is the PBW special-fibre action. The cotangent field carries the coadjoint action. A quantum descendant theorem requires a boundary model for the continuous dual $\mathfrak{h}_{\hbar}^{\vee,\mathrm{cont}}$, for example a Laurent or distributional boundary realization.

Remark 5.206. The full cotangent deformation replaces $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}[1]$ by

$$\mathfrak{g}_{\hbar} = \mathfrak{h}_{\hbar} \ltimes \mathfrak{h}_{\hbar}^{\vee,\mathrm{cont}}[1]$$

and requires $C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}_{\hbar})$ as the closed quantum observable algebra. The polynomial boundary representatives carry the PBW special-fibre action; the deformation-level coadjoint action requires the principal-part target.

5.7.3 Graph formalism and specialization datum

The Moyal coefficients $\hbar P$, $\hbar^3 P^3/24$, $\hbar^5 P^5/1920$, $\hbar^7 P^7/322560$ of the previous subsection form the algebraic model. Costello's heat-kernel BV calculus produces coefficients of the same operad-theoretic shape from different data: an elliptic parent theory, a propagator, a counterterm scheme, an anomaly class, and a trace normalisation. The question is whether these two calculations land on the same coefficients on the brane-defect complex.

Statement 5.207 *(Costello perturbative BV construction).* For an elliptic BV theory with a gauge fixing, heat kernel K_t, propagator

$$P_{\epsilon,L} = \int_{\epsilon}^{L} Q^{\mathrm{GF}} K_t \, dt,$$

and a renormalization scheme, Costello constructs finite-scale effective interactions by the RG operator

$$W(P_{\epsilon,L}, I[\epsilon]) = \hbar \log\bigl(\exp(\hbar \partial_{P_{\epsilon,L}}) \exp(I[\epsilon]/\hbar)\bigr).$$

The renormalized interaction is obtained by adding local counterterms and taking the $\epsilon \to 0$ limit. It has the connected-graph expansion

$$I[L] = \sum_{\Gamma \text{ connected}} \frac{\hbar^{b_1(\Gamma)+\sum_v g_v}}{|\operatorname{Aut}\Gamma|} W_{\Gamma}(P_{\epsilon,L}; I_{g_v}[\epsilon] + I^{\mathrm{CT}}_{g_v}(\epsilon)),$$

satisfies RG flow and locality, and satisfies the quantum master equation when the obstruction classes vanish [12]. The resulting quantum observables form a factorization algebra [15][16].

Statement 5.208 *(Costello–Li flat open-closed BCOV theorem).* Costello–Li prove a perturbative quantization of open-closed BCOV theory on flat $\mathbb{C}^d$, for d odd, unique up to the equivalences in their renormalization scheme, with open gauge algebra $\mathfrak{gl}(N \mid N)$ and compatibility under $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N + k \mid N + k)$. In that flat open-closed BCOV/HCS theory they compute the anomaly cancellation needed for the graph construction [14]. Ordinary $\mathfrak{gl}_N$ and the mixed $\mathbb{R}^2 \times \mathbb{C}^2$ Hamiltonian brane model require an additional reduction or summand argument.

Remark 5.209. Theorem 5.6 uses the formal Hamiltonian BF calculation. The brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ is a defect in the mixed space; a spacetime-boundary interpretation requires a chosen half-space model.

LEMMA 5.210 *(Finite-window obstruction for the full Hamiltonian sector).* The polynomial Hamiltonian Lie algebra of the formal symplectic plane has the following finite-window obstructions to a finite-dimensional Lie truncation compatible with the full Hamiltonian sector:

(i) finite degree windows have a Lie-subalgebra obstruction;
(ii) projected degree-window brackets can violate Jacobi;
(iii) $\mathfrak{m}$-adic jet quotients have a Lie-quotient obstruction in the presence of linear Hamiltonians;
(iv) directed finite-dimensional Lie-subalgebra exhaustion is obstructed containing the full polynomial Hamiltonian algebra.

Proof. Let H_d be the span of homogeneous polynomials of degree d, and let $\mathfrak{m} = (z_1, z_2)$. The Poisson bracket satisfies $\{H_p, H_q\} \subset H_{p+q-2}$. Thus

$$\{z_1^N, z_2^N\} = N^2 z_1^{N-1} z_2^{N-1},$$

which has degree $2N - 2 > N$ for $N > 2$, so degree $\leq N$ windows have a Lie-subalgebra obstruction.

Projecting the bracket back to a degree window creates a Jacobi obstruction. For the bracket $\pi_{\leq 3}\{-,-\}$, take

$$f = z_2, \qquad g = z_2^3, \qquad h = z_1^2 z_2.$$

A direct calculation gives the projected Jacobiator

$$[[f, g]_3, h]_3 + [[g, h]_3, f]_3 + [[h, f]_3, g]_3 = 6z_2^3.$$

The $\mathfrak{m}$-adic quotients have a stability obstruction under translations generated by linear Hamiltonians:

$$\{z_1^{N+1}, z_2\} = (N + 1) z_1^N \notin \mathfrak{m}^{N+1}.$$

Hence $\mathfrak{h}/\mathfrak{m}^{N+1}$ has a Lie-quotient obstruction for the full Hamiltonian algebra.

Any finite-dimensional Lie subalgebra containing

$$a = z_1^2 z_2^2, \qquad b = z_1^3 z_2$$

contains all iterated brackets

$$(\mathrm{ad}_a)^n(b) = (-4)^n z_1^{n+3} z_2^{n+1}, \qquad n \geq 0,$$

an infinite linearly independent set. Thus directed finite-dimensional Lie-subalgebra exhaustion is obstructed for the full polynomial Hamiltonian algebra. □

Remark 5.211 *(Linear heat operator versus BV kernel).* On the linear field complex

$$\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1]\oplus\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big),$$

the flat gauge fixing

$$Q^{\mathrm{GF}} = d^*_{\mathbb{R}^2} + \bar{\partial}^*_{\mathbb{C}^2}, \qquad \Delta = [d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}, Q^{\mathrm{GF}}]$$

gives finite-scale heat operators

$$K_t = K_t^{\mathrm{base}}\widehat{\otimes}\mathrm{id}, \qquad P^{\mathrm{op}}_{\epsilon,L} = \int_\epsilon^L Q^{\mathrm{GF}}K_t^{\mathrm{base}}\,dt\widehat{\otimes}\mathrm{id}$$

satisfying

$$[d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}, P^{\mathrm{op}}_{\epsilon,L}] = K_\epsilon - K_L$$

as continuous operators on the completed coefficient spaces. This is an operator-level homotopy. Costello's BV propagator also requires a coefficient coevaluation kernel for the Tate pair $\widehat{\mathfrak{h}}\oplus\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$. In the product/direct-sum topology used here, the formal sum of homogeneous coefficient coevaluations diverges in the completed tensor product. Thus the analytic graph problem requires a genuine Tate or weighted coefficient extension of the heat-kernel calculus, or a declared restriction to a smaller sector.

LEMMA 5.212 *(Tate Casimir obstruction for the full Hamiltonian sector).* In the product/direct-sum Tate topology

$$\mathfrak{h} = \prod_{d\ge 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d\ge 1} H_d^\vee,$$

the coefficient coevaluation

$$C_{\mathfrak{h}} = \sum_{d\ge 1}\sum_i e_{d,i}\otimes e^i_d$$

lies in neither $\mathfrak{h}\widehat{\otimes}\mathfrak{h}^\vee_{\mathrm{cont}}$ nor $\mathfrak{h}^\vee_{\mathrm{cont}}\widehat{\otimes}\mathfrak{h}$. Consequently the full Hamiltonian sector lacks a strict BV pairing in this topology; a Costello coefficient kernel here requires a different coefficient category. The operator-level homotopy of Remark 5.211 gives a BV propagator, BV Laplacian, and renormalized graph weights only after a coefficient category in which the Casimir converges has been chosen.

Proof. For each d, choose a basis $\{e_{d,i}\}_i$ of H_d and the dual basis $\{e^i_d\}_i$ of $H_d^\vee$. An element of a completed tensor product with the direct-sum factor $\bigoplus_d H_d^\vee$ has finite support in that direct-sum direction in the Tate convention used here. The displayed formal Casimir has a nonzero component in every $H_d^\vee$. Hence the tensor diverges. The same support obstruction holds with the tensor factors reversed. Costello's heat-kernel BV formalism requires the inverse pairing as a distributional kernel on the field complex; an abstract continuous operator is insufficient. The weighted Tate completion of Corollary 5.71 resolves this Casimir obstruction by changing the coefficient category; the unweighted diagnosis here remains the structural reason that change is required. □

Problem 5.213 *(Hamiltonian specialization datum for Costello perturbative BV).* • *Theory:* a Costello open–closed BV theory in the heat-kernel renormalization formalism whose restriction is the formal-local Hamiltonian BF sector $\mathbb{R}^2\times\mathbb{C}^2$ of §1.4.

- *Hypotheses:* an elliptic parent or stated dimensional reduction; field topology, BV pairing, gauge fixing, heat-kernel propagator; weighted Tate coefficient kernel (Corollary 5.71); brane-defect condition; ribbon graph data; Capelli/scalar normalisation; bulk-to-defect kernel.

- *Obstruction complex:* the Tate Casimir $\mathfrak{o}_{\mathrm{Cas}}$ (Lemma 5.212); the scalar-symbol class $\hbar N[\bar{c}]$ (Theorem 5.214); the reduced non-scalar QME class $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]$ (Problem 5.220); the θ_3 finite-window class (Theorem E.38).
- *Vanishing criterion:* weighted Tate completion closes $\mathfrak{o}_{\mathrm{Cas}}$; a balanced or unreduced source replacement closes $\hbar N[\bar{c}]$; the reduced non-scalar class is closed exactly on the locus where the filtered scalar-contact tower of Theorem 5.84 closes.
- *First nonzero example:* the strict full-polydisk product/direct-sum coefficient pair carries a nonzero $\mathfrak{o}_{\mathrm{Cas}}$; the ordinary scalar-reduced $\mathfrak{gl}_N$ branch carries nonzero $\hbar N[\bar{c}]$ at order $\hbar^1$ on the linear pair (z_1, z_2).
- *Physical meaning:* the analytic specialization datum is the elliptic parent and the brane-defect renormalisation under which Costello's heat-kernel BV calculus and the Hamiltonian trace map is simultaneously controllable.

Statement. Costello gives the general renormalized BV graph calculus. For the Hamiltonian sector here, the required theorem data begin with an elliptic closed-open parent theory, or a stated dimensional reduction of one, whose restriction is this Hamiltonian BF sector and whose obstruction class vanishes. The required data are:

(i) choose the mechanism: either prove that the Hamiltonian cotangent BF complex

$$\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1]\oplus\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big)$$

is literally a closed sector of a Costello open–closed BV theory, or prove that it is obtained by a specified dimensional reduction. The chosen mechanism must determine the field topology, degree -1 BV pairing, differential, local interaction, bracket degree, and a homotopy retract closed under the propagator and RG flow. Since $\widehat{\mathfrak{h}}$ is a completed Tate coefficient system, the proof goes through a Tate or weighted-coefficient extension; finite-degree and $\mathfrak{m}$-adic truncations carry the obstruction of Lemma 5.210. It must construct a genuine Tate or weighted-coefficient extension of Costello's heat-kernel calculus [12] compatible with the Poisson bracket and RG flow, or restrict to a smaller Lie sector such as the classical $\mathfrak{m}^3/\mathfrak{m}^{N+1}$ truncation and explicitly declare that translations, the conformal $\mathfrak{sl}_2$ sector, and the all-order Moyal closure have been removed. By Lemma 5.212, a damping factor inside the direct-sum dual fixed above is insufficient; the coefficient category itself must be replaced by one in which the Casimir converges, or the sector must be restricted;

(ii) specify the asymptotic conditions on $\mathbb{C}^2$ and the brane-defect condition on $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$;

(iii) choose the induced finite-scale gauge fixing, heat-kernel propagator, and BV Laplacian in this sector. The infrared limit $L \to \infty$ may be used only after the zero modes and harmonic projection have been fixed;

(iv) compute the obstruction/anomaly class for this mixed Hamiltonian sector. Concretely, represent the obstruction as a local functional class in the chosen bulk-defect local-functional complex. If the defect cotangent sector is represented by principal-part currents, the target must either be replaced by a current-compatible complex

$$\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}_w;\mathfrak{J}^{\mathrm{pp}}_\partial),$$

with wavefront and counterterm rules for $\mathcal{D}'_c$-coefficients, or the theorem must restrict B to regular compactly supported densities. In the weighted Tate category the obstruction is a class

$$\mathrm{Ob}_{w,n} \in H^1_{\mathrm{adm}}\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I^w_0, -\}\big)$$

coming from the n-loop obstruction to the quantum master equation after counterterms. Costello–Li's flat $\mathfrak{gl}(N \mid N)$ anomaly cancellation gives a source theorem and a model; transfer requires proof. To execute it one must prove that constant-Hamiltonian projection and connected trace extraction kill the ordinary $\mathfrak{gl}_N$ anomaly, or else replace the open sector by the appropriate super or special-linear variant. The first defect-local obstruction is the Capelli contact term

$$\omega_{\mathrm{Cap}}(A) = N\,\overline{\mathrm{Tr}(A)},$$

represented by

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\,\mathrm{Tr}(A) \quad \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N).$$

It survives the ordinary connected trace projection, since the right-hand side is a nonempty connected trace when A is nonconstant. The anomaly datum must therefore be a filtered local class $\mathrm{Ob}_{\mathrm{bd}}$ whose scalar-symbol projection is

$$\sigma_{\mathrm{sc}}(\mathrm{Ob}_{\mathrm{bd}}) = \hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]].$$

A closed-exchange cancellation is the datum of $W_{\mathrm{closed\,exch}}$ and a counterterm C with scalar leading class

$$[w_{\mathrm{add}}] = -\hbar N[\bar{c}] \qquad \text{in } H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]]$$

and with

$$\omega_{\mathrm{Cap}} + W_{\mathrm{closed\,exch}} + (Q + \{I_0, -\})C = 0.$$

The proved local mechanisms here are the specified supertrace and central-character normalizations; any exhaustive classification is controlled by the displayed obstruction complex;

(v) construct the bulk-to-defect kernel as a curved/cochain-level map κ to the Poisson central-operation object of the brane observables. After counterterms it must satisfy

$$\mathrm{Curv}(\kappa) = d\kappa + \frac{1}{2}[\kappa,\kappa] = 0.$$

On degree-zero Hamiltonians it must send $f(z_1, z_2)$ to the Weyl-ordered boundary observable $\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$, and it must satisfy the Ward identity and the quantum moment-map constraint. Its cotangent component must either restrict the principal-part current B to regular defect densities, or prove that the chosen Costello bulk-defect observable category admits the required compactly supported distributions with wavefront, counterterm, finite-window, and RG compatibility;

(vi) define the relevant boundary ribbon graphs: cyclic order, boundary components, automorphism factors, propagator labels, and the relation between Costello's loop-counting $\hbar$ and the Weyl contraction parameter. Equivalently, one must prove that a single boundary cross-contraction is $P/2$ in the Weyl convention;

(vii) fix the one-propagator boundary graph normalization. For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the raw first boundary commutator must have coefficient

$$\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1},$$

equivalently the normalized commutator must have coefficient

$$(kn - lm)z_1^{k+m-1}z_2^{l+n-1};$$

(viii) after the first normalization, compute the three-propagator boundary graph normalization and prove that the connected order-$\hbar^3$ commutator contributes exactly

$$\frac{\hbar^3}{24}P^3(f,g)$$

with $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$;

(ix) give the convention for extracting connected single-trace and multi-trace coefficients as primitive trace classes, and prove that this extraction cancels, renormalizes, or otherwise accounts for the normal-ordering shift of Remark 5.196;

(x) construct the renormalization scheme and counterterm locality statement in this mixed bulk-defect sector. For every divergent graph whose $\epsilon \to 0$ limit is subtracted, choose compatible representatives

$$C^{w_\circ}_{\Gamma,w}(\epsilon) \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w_\circ}_w)$$

with

$$\pi_{w'_\circ w_\circ} C^{w'_\circ}_{\Gamma,w} = C^{w_\circ}_{\Gamma,w}, \qquad R^{w_\circ}_{w,w'} C^{w_\circ}_{\Gamma,w} = C^{w_\circ}_{\Gamma,w'},$$

prove support on the relevant bulk diagonal or brane-collision stratum, and check that RG flow carries these representatives inside the same local-functional subcomplex;

(xi) extend the bulk-to-defect kernel to the cotangent boundary-current target, or record the obstruction. In the reduced setting the extension must land in the principal-part current object of Theorem 5.186; before contraction it must contain the unreduced data of Remark 5.190. On the cotangent summand it must realize the principal-part coadjoint action of Theorem 5.179, and at the Moyal level the $\mathfrak{h}^\vee_\hbar$-action induced by the residue pairing. The allowed substitute for polynomial one-ψ representatives is a completed continuous-dual boundary realization.

THEOREM 5.214 *(Scalar QME obstruction and cancellation branches).* • *Obstruction:* cancellation of the scalar-symbol QME obstruction in the ordinary scalar-reduced $\mathfrak{gl}_N$ brane-defect model.

- *Hypotheses:* on the source side, an $\bar{A}$-cochain J with $d_{\mathrm{CE}}J = -\hbar N\bar{c}$, or an alternative source category in which $\bar{c}$ is cohomologous to zero; on the Costello side, finite-window complexes $\mathcal{X}^\bullet_{\mathrm{sc},w,M}$, cochain maps $\Xi^{\mathrm{Cost,sc}}_M$, a filtered scalar chain projection $\widehat{\sigma}_{\mathrm{sc},M}$, and compatible local counterterms.
- *Obstruction complex:* the Lie cohomology class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]]$, and its filtered Costello image $\beta^{\mathrm{Cost,sc}}_M = [\hbar N[\bar{c}_M]] \in \operatorname{coker} H^1(\widehat{\sigma}_{\mathrm{sc},M}\Xi^{\mathrm{Cost,sc}}_M)$.
- *Vanishing criterion:* either replace the scalar-reduced source by the unreduced source, the central-character source, the opposite central extension, the balanced $\mathfrak{gl}(N|N)$-supertrace source, or the algebraic central-contact cone — each cancellation is a source replacement, made explicit below; alternatively, give a Costello-local closed-exchange triple $(W_M, C_M, \widehat{\sigma}_{\mathrm{sc},M})$ with $\beta^{\mathrm{Cost,sc}}_M = 0$ and compatible truncation.
- *First nonzero example:* on the linear pair (z_1, z_2), $\omega(z_1,z_2) = 1$ and $\bar{c} \neq 0$ in $H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})$; the projective Moyal commutator gives the Capelli term $\hbar N \cdot \mathbf{1}$ at order $\hbar^1$.
- *Physical meaning:* the scalar-symbol QME obstruction records the brane center-of-mass anomaly that survives ordinary $\mathfrak{gl}_N$ Hamiltonian reduction; cancelling it requires changing what one means by "Hamiltonian" (source replacement) or giving a closed-exchange counterterm with a matched scalar leading class.

Statement. Let $A = \mathbb{C}[z_1, z_2]$, $\bar{A} = A/\mathbb{C}\cdot 1$, and

$$\omega(f, g) = [\{f, g\}]_0.$$

Write $\bar{c}$ for the induced Lie two-cocycle on $\bar{A}$. In the ordinary scalar-reduced $\mathfrak{gl}_N$ brane-defect model, the scalar-symbol part of the mixed QME obstruction is represented by

$$\mathrm{Ob}_{\mathrm{sc}}(\gamma_1; a, f; b, g) = \hbar N\, \omega(f, g) \int_{\mathbb{R}} a(t)b(t)\gamma_1(t)\, dt,$$

and its associated-graded CE class is

$$\hbar N[\bar{c}]\, \gamma_1 \in H^2_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]].$$

Forgetting the central ghost coefficient gives $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}; \mathbb{C})[[\hbar]]$. For $N > 0$, this class is nonzero in the scalar-reduced source. Hence the leading scalar symbol obstructs internal scalar-reduced local counterterms.

The cancellation mechanisms have distinct types.

(i) On the unreduced source A, the functional $\eta(f) = -[f]_0$ satisfies $d_{\mathrm{CE}}\eta = \omega$. Thus

$$\hbar N \omega + d_{\mathrm{CE}}(-\hbar N \eta) = 0$$

before quotienting constants.

(ii) Equivalently, after replacing the scalar-reduced source by the unreduced source with central character $\chi_N(f) = N[f]_0$, the scalar term becomes exact before it becomes a reduced obstruction class.

(iii) Equivalently in the scalar-replacement category, define the opposite scalar central extension

$$\bar{A}^{\mathrm{cl}}_{-\hbar N \bar{c}} = \bar{A} \oplus \mathbb{C}[[\hbar]]K_{\mathrm{cl}},$$

with

$$[(f, aK_{\mathrm{cl}}), (g, bK_{\mathrm{cl}})] = \big(\{f, g\}_{\bar{A}}, -\hbar N\, \bar{c}(f, g)K_{\mathrm{cl}}\big), \qquad [K_{\mathrm{cl}}, -] = 0.$$

If $k^\vee(K_{\mathrm{cl}}) = 1$ and $k^\vee(\bar{A}) = 0$, then for $J_{\mathrm{op}} = -k^\vee$

$$d_{\mathrm{CE}} J_{\mathrm{op}} = -\hbar N\, \bar{c}, \qquad \mathrm{Ob}_{\mathrm{sc}} + d_{\mathrm{CE}} J_{\mathrm{op}} = 0.$$

The cancellation is compatible with scalar finite-window projections retaining z_1, z_2. It is a source replacement; after forgetting K_{cl}, the class $\hbar N[\bar{c}]$ on $\bar{A}$ reappears.

(iv) In the balanced supertrace model $V = \mathbb{C}^{N|N}$, the scalar coefficient is $\mathrm{sdim}\, V = 0$; the scalar brane-defect QME representative vanishes in that model.

(v) The finite-window algebraic central-contact cone is a genuine changed source branch. After adjoining a defect-supported central closed source line $\mathbb{C}[[\hbar]]e_M$ with

$$\Xi^{\mathrm{alg,sc}}_M(e_M) = -\hbar N\, \bar{c}_M$$

on the scalar-contact line, the choice $W_M = e_M$ and $C_M = 0$ cancels the finite-window scalar row, and the corresponding β, μ, λ_C classes vanish in that enlarged algebraic cone.

(vi) In the original scalar-reduced ordinary $\mathfrak{gl}_N$ branch, a same-branch closed-exchange or brane-defect contribution can cancel the scalar symbol only if its leading scalar class obeys

$$[w_{\mathrm{add}}] = -\hbar N[\bar{c}] \qquad \text{in } H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]].$$

Equivalently, a Costello-local closed-exchange solution must contain finite-window closed-source complexes $\mathcal{X}^\bullet_{\mathrm{sc},w,M}(I)$, cochain maps

$$\Xi^{\mathrm{Cost,sc}}_M : \mathcal{X}^\bullet_{\mathrm{sc},w,M}(I) \to \mathfrak{Q}^\bullet_{\mathrm{sc},w,M}(I),$$

a filtered scalar chain projection $\widehat{\sigma}_{\mathrm{sc},M}$, cocycles W_M, and counterterms C_M such that

$$\mathrm{Ob}_{\mathrm{sc},M} + \Xi^{\mathrm{Cost,sc}}_M(W_M) + d_M C_M = 0, \qquad H^1(\widehat{\sigma}_{\mathrm{sc},M}\Xi^{\mathrm{Cost,sc}}_M)[W_M] = -\hbar N[\bar{c}_M],$$

with compatible truncation and vanishing inverse-limit β, μ, λ_C classes.

The algebraic cone in (4) becomes a Costello-local closed-exchange theorem only after its formal generator is replaced by a closed source complex, finite-scale propagator, bulk-to-defect graph map, source-face Hom differential, local counterterm support, scalar chain projection, and compatible finite-window representatives. For current genuine Costello-local towers the ordinary scalar-reduced branch is blocked first by

$$o^{(1),\mathrm{sc}}_{\sigma,M} = \hbar N[\bar{c}_M] \neq 0,$$

and, after any additional scalar condition, by the image obstruction

$$\beta^{\mathrm{Cost,sc}}_M = [\hbar N[\bar{c}_M]] \in \operatorname{coker} H^1(\widehat{\sigma}_{\mathrm{sc},M}\Xi^{\mathrm{Cost,sc}}_M).$$

A complete classification of scalar QME cancellations is a compact theorem. Any further mechanism must be an explicitly constructed null-homotopy in the filtered QME obstruction complex whose scalar-symbol class cancels $\hbar N[\bar{c}]$.

Proof. The representative and its nonzero associated-graded class are exactly Theorem E.12. Non-exactness in the reduced source is the last part of that theorem: a primitive on $\bar{A}$ would lift to a primitive on A vanishing on 1, but $\omega(z_1, z_2) = 1$ and $\{z_1, z_2\} = 1$.

The first branch is Lemma E.11, multiplied by $\hbar N$. The central-character branch is Theorem E.15; it is a source replacement before scalar reduction. A counterterm inside the original reduced obstruction complex is a different datum. The opposite central-extension branch is the same CE sign calculation written as an honest central extension of $\bar{A}$: with $(d_{\mathrm{CE}}\lambda)(x, y) = -\lambda([x, y])$, the central bracket $-\hbar N\bar{c}(f, g)K_{\mathrm{cl}}$ gives $d_{\mathrm{CE}}k^\vee = \hbar N\bar{c}$, hence $d_{\mathrm{CE}}(-k^\vee) = -\hbar N\bar{c}$. Jacobi is exactly the cocycle identity for $\bar{c}$, and finite-window compatibility follows because $\bar{c}$ is detected on the retained pair z_1, z_2. The balanced-supertrace branch is Theorem E.16. The algebraic central-contact cone is the finite-window source extension recorded in the scalar QME source extension; its differential is zero and its scalar image is declared to be $-\hbar N\bar{c}_M$, so the displayed cancellation is immediate in that enlarged source. The same declaration records the missing Costello-local closed-exchange data. The necessary condition for any same-branch ordinary cancellation is Theorem E.52. The additional Costello-local image and scalar-chain obstructions are the filtered Hom conditions in Problem 5.220. □

Remark 5.215 *(Parabolic functoriality of the local model).* The completion along the brane divisor $z_1 = 0$ used here breaks the formal symplectomorphism group $\mathrm{Symp}_{\mathrm{form}}(\mathbb{C}^2, 0)$ to the parabolic stabiliser

$$P_{(z_1)} = \{\varphi \in \mathrm{Symp}_{\mathrm{form}}(\mathbb{C}^2, 0) : \varphi^*(z_1) \in (z_1)\}.$$

Any compact comparison therefore requires a parabolic-equivariant comparison. Full $\mathrm{Symp}_{\mathrm{form}}$-equivariance is additional compact comparison data.

5.7.4 Formal Weyl/Moyal algebra, all orders

The order-$\hbar$ projective coefficient

$$[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_{\mathrm{proj}} = \hbar N \cdot \mathbf{1}$$

identifies the linear-generator anomaly on the Weyl-ordered trace representation; the all-order question is whether the same central class persists at orders $\hbar^3, \hbar^5, \dots$. The unique translation-invariant deformation of the constant Poisson tensor on the formal holomorphic polydisk [7] fixes the all-order target on the closed side.

The Weyl/Moyal Rees convention is

$$f * g = m \circ \exp\left(\frac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\right)(f \otimes g).$$

The factor $1/2$ is already determined by linear Hamiltonians in this Weyl convention: if $u = az_1 + bz_2$ and $v = cz_1 + dz_2$, then $[\widehat{u}, \widehat{v}] = \hbar(ad - bc)$, so the Baker-Campbell-Hausdorff formula for Weyl-ordered exponentials contributes $\exp(\hbar P(u, v)/2)$. Differentiating these exponentials gives the cross-contraction $P/2$ in the formal Weyl convention. Then the star commutator is odd in $\hbar$:

$$[f, g]_* = \hbar\{f, g\} + \frac{\hbar^3}{24} P^3(f, g) + O(\hbar^5),$$

where $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$. Thus the $\hbar^2$ coefficient of the raw star commutator vanishes in this quantization convention. The normalized quantum Hamiltonian bracket is instead

$$\{f, g\}_\hbar = \hbar^{-1}[f, g]_* = \sum_{s \geq 0} \frac{\hbar^{2s}}{2^{2s}(2s + 1)!} P^{2s+1}(f, g).$$

Proposition 5.216 *(Moyal coefficients on monomials).* Let $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$. For every $r \geq 0$,

$$P^r(f, g) = \sum_{a=0}^{r} (-1)^a \binom{r}{a} (k)_{r-a} (l)_a (m)_a (n)_{r-a} z_1^{k+m-r} z_2^{l+n-r},$$

where $(s)_q = s(s - 1) \cdots (s - q + 1)$ and $(s)_0 = 1$. Terms with a negative displayed exponent are understood to be zero; equivalently the corresponding falling factorial already vanishes. Consequently

$$[f, g]_* = \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{2}{2^r r!} \hbar^r P^r(f, g),$$

so the second-order coefficient of the raw commutator vanishes. The first four odd raw coefficients are

$$\hbar P(f, g), \qquad \frac{\hbar^3}{24} P^3(f, g), \qquad \frac{\hbar^5}{1920} P^5(f, g), \qquad \frac{\hbar^7}{322560} P^7(f, g).$$

Proof. Since

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1},$$

the two summands commute as differential operators on the tensor product. The binomial theorem gives

$$P^r = \sum_{a=0}^{r} (-1)^a \binom{r}{a} \partial_{z_1}^{r-a} \partial_{z_2}^{a} \otimes \partial_{z_1}^{a} \partial_{z_2}^{r-a}.$$

Applying this to $z_1^k z_2^l \otimes z_1^m z_2^n$ gives the displayed falling factorial formula. Interchanging f and g changes $P^r(f, g)$ by the sign $(-1)^r$, so the star commutator keeps exactly odd r and contributes the factor $2/(2^r r!)$ in order r. □

THEOREM 5.217 *(Degree-zero Moyal target and decorated radial obstruction).* Let $\mathfrak{h}_\hbar = \bar{A}_\hbar$ be the Moyal Hamiltonian Lie algebra, and let $\mathfrak{h}_{\hbar,u}$ be its shifted-cotangent coordinate copy, with generator u_f labelled by $f \in \mathfrak{h}_\hbar$. Let

$$H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})_{\mathrm{ren}}$$

denote the stable connected Hamiltonian trace sector generated by the Capelli-renormalized J-basis transition classes $\big[\overline{J_N(f)}\big]_{N\to\infty}$. These are associated-graded stable classes in the renormalized transition system. Naive normal-ordered block-restriction limits are different objects.

The following data are proved under the stated finite-window and radial obstruction hypotheses.

(i) The Weyl/Moyal coefficients of Proposition 5.216 give the all-order formal Hamiltonian source. The odd raw coefficients are $2\hbar^r P^r/(2^r r!)$, and the first four are $1, 1/24, 1/1920, 1/322560$.

(ii) The Capelli-renormalized J_N-basis of Lemma 5.198 gives the stable connected trace target as a $\mathbb{C}[[\hbar]]$-module, after removing the empty trace and decomposable products.

(iii) The ordinary Weyl-trace image clause is equivalent, after the homomorphism, injectivity, and pure-power parts of the radial hypotheses, to the moment-ideal primitive $E_{a,b,N} \in \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$ of Lemma F.15. In the free normal-diagram model this is the vanishing of

$$\mathfrak{o}_{a,b} = [R^{\mathrm{free}}_{a,b}] \in \operatorname{coker} C_{a,b}$$

with functorial lifts $K_{a,b} \in \ker \partial$. Equivalently, in the dual-potential obstruction theorem,

$$\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}$$

must vanish, or equivalently the decorated PBW Stokes identity $D^{\square}_{a,b} = C^+_{a,b}\partial_2$ must hold; an obstruction is a signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$. Proposition F.32 proves this primitive for every edge bidegree $(2, b)$ and $(a, 2)$. Proposition F.20, Proposition F.33, and the exact-arithmetic normal-form calculation prove finite interior identities. For all $a, b \geq 3$, the all-interior assertion is the associated-graded two-colour necklace homotopy, equivalently the dual-potential signed-row theorem above.

If the radial image hypotheses of Statement F.8 hold, or if the equivalent quantum shear primitive construction above is constructed together with the homomorphism, injectivity, and pure-power clauses, then

$$\Phi^{(0)}_\hbar : \mathfrak{h}_{\hbar,u} \xrightarrow{\sim} H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})_{\mathrm{ren}}, \qquad u_f \longmapsto \partial_{\mathrm{bb},\hbar}(f) = [\overline{J_N(f)}]_{N\to\infty}$$

is the u_f-coordinate component of the quantum P_0-operation: it is a $\mathbb{C}[[\hbar]]$-linear isomorphism onto the renormalized degree-zero Hamiltonian sector. It is a Lie-algebra map on these Hamiltonian coordinates,

$$\Phi^{(0)}_{\hbar}(u_{\{f,g\}_\hbar}) = \hbar^{-1}[\Phi^{(0)}_{\hbar}(u_f), \Phi^{(0)}_{\hbar}(u_g)]_{\mathrm{conn,raw}}.$$

Under this assumption, the associated graded at $\hbar = 0$ is the degree-zero Hamiltonian restriction of Φ in Theorem 5.6. The descendant ($\mathfrak{h}^\vee$, one-antifield Koszul) sector remains governed by Problem 5.49; see Problem 5.222 below for the corresponding obstruction.

Proof. Item (1) is Proposition 5.216. Item (2) is Corollary 5.199 and Lemma 5.198. Item (3) is the chain Proposition F.14, Lemma F.15, Definition F.21, and Statement F.22, sharpened by Proposition F.25 to $\Omega^{\mathrm{rad}}_{a,b} = 0$, equivalently $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, with the signed-row obstruction criterion. The edge theorem is Proposition F.32. The finite interior statements cited there are proved exact free normal-diagram identities; by Lemma F.19 they are all-rank identities in their displayed windows.

Under the radial image hypotheses, the displayed $\mathbb{C}[[\hbar]]$-linear bijection follows from the stable Capelli-renormalized generators. Flatness over $\mathbb{C}[[\hbar]]$ follows from the unipotent Capelli triangular system over $\mathbb{C}[[\hbar N]]$ (Lemma 5.198(ii)); the associated graded is the classical trace map because the $r = 0$ triangular term is the leading symbol (Lemma 5.198(iv)). The Lie identity is precisely Theorem 5.200, passed to the connected stable limit. The cotangent cochain summand is governed by the Tate-coefficient obstruction datum. □

THEOREM 5.218 *(Componentwise quantum coefficient comparison).* Moyal-quantised Hamiltonians, Weyl-quantised brane operators, and the brane trace map commute on a four-component product target, with the obstruction of any other component recorded as a separate class on its own completion.

- *Components proved compatible:* weighted BV kernel convergence $\mathcal{K}^{w}_{\mathrm{BV}}$; balanced $\mathfrak{gl}(N|N)$-supertrace zero scalar QME representative 0_{sc} on its scalar-contact completion; reduced weighted Moyal principal-part current target $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$; verified-range degree-zero Hamiltonian trace target $T_{\mathrm{Ham},\hbar}$.
- *Hypotheses:* weighted Tate completion $\mathfrak{h}_w$ (Theorem 5.58), Costello propagator and weighted RG calculus (Corollary 5.71), balanced supertrace replacement (Theorem E.16), Capelli-renormalised J_N-basis transitions (Lemma 5.198), and the explicit enumeration cited in clause (4).
- *Obstruction classes outside this product:* the ordinary-$\mathfrak{gl}_N$ scalar QME class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})[[\hbar]]$ (Theorem 5.214); the all-bidegree radial/Weyl normal-form class $\Omega^{\mathrm{rad}}_{a,b} \in \operatorname{coker} B_{a,b}$ (Proposition F.25); the reduced non-scalar Costello QME class $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d^{L}_{\mathrm{QME}})$ (Problem 5.220); the θ_3 bare / counterterm-adjoined row dichotomy (Theorem E.38).
- *Vanishing criterion for the four named components:* the displayed equalities of items (1)–(4) hold on the scalar-contact balanced completion and on the verified range listed in Proposition F.32, Proposition F.31, Proposition F.33, and Theorem 5.217.
- *First nonzero example outside the proved components:* the ordinary scalar branch is blocked at order $\hbar^1$ on linear generators by the projective scalar $\hbar N \cdot \mathbf{1}$; the reduced non-scalar Costello QME branch is blocked at the smallest non-scalar window θ_3, where the bare $H^1 = \mathbb{C}\cdot E_{\theta_3} \neq 0$.
- *Physical meaning:* the Moyal star, the matrix Weyl algebra, and the brane trace map agree componentwise on the proved completion; mismatch on any of the four obstruction classes is the precise location at which compatibility breaks and a counterterm, source replacement, or scalar-contact lift is required.

Statement. Fix an exponential degree weight $w(d) = q^d$, $q > 1$, and work over $\mathbb{C}[[\hbar]]$. Write

$$\mathfrak{h}_{w,\hbar} = \mathfrak{h}_w[[\hbar]], \qquad \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}} = \mathfrak{h}^{\vee,w}_{\mathrm{cont}}[[\hbar]].$$

The Moyal bracket on $\mathfrak{h}_{w,\hbar}$ is transported from $\bar{A}_\hbar$ by the diagonal weight rescaling of Definition 5.55. Let

$$\mathcal{K}^w_{\mathrm{BV}} = (C_{\mathfrak{h},w}, P^w_{\epsilon,L}, \mathcal{E}_w)$$

be the weighted coefficient-kernel datum of Corollary 5.71. For an interval $I \subset \mathbb{R}$, let $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$ be the weighted reduced Moyal principal-part current envelope

$$\mathfrak{h}^w_{\hbar,I} = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}_{w,\hbar}, \qquad \mathcal{P}^w_{\hbar,I} = \mathcal{D}'_c(I)\widehat{\otimes}\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}},$$

$$A^{\mathrm{pp},w}_{\partial,\hbar}(I) = \widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}(\mathfrak{h}^w_{\hbar,I})\widehat{\otimes}\widehat{\mathbf{S}}_{\mathbb{C}[[\hbar]]}(\mathcal{P}^w_{\hbar,I}[1]).$$

Its dense strict Matlis subtarget is the reduced Moyal principal-part target $A^{\mathrm{pp}}_{\partial,\hbar}(I)$ of Construction 5.135, obtained by replacing $\mathfrak{h}_{w,\hbar}$ with $\bar{A}_\hbar$ and $\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}$ with $\mathcal{P}[[\hbar]]$. The Moyal coadjoint action on the weighted envelope is the finite-window continuous extension of the action in that construction; the assertion is restricted to the vertexwise polynomial-degree-bounded weighted sector. Let $T_{\mathrm{Ham},\hbar} = H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})_{\mathrm{ren}}$ be the stable connected Hamiltonian trace target of Theorem 5.217. Its formal Moyal and Capelli pieces are proved there; the ordinary-trace Lie comparison is reduced to the dual-potential radial obstruction $\Omega^{\mathrm{rad}}_{a,b}$, equivalently the decorated PBW Stokes identity $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, with all edge strips proved and all-interior closure requiring the signed-row vanishing theorem. In the balanced $\mathfrak{gl}(N \mid N)$ supertrace branch put 0_{sc} for the zero scalar QME representative of Theorem E.16; on a balanced scalar-contact completion this is the zero filtered scalar projection of Proposition E.18.

The componentwise quantum coefficient comparison is the product prefactorization target

$$\mathfrak{S}_{\hbar,w}(I) = \mathcal{K}^w_{\mathrm{BV}} \times \{0_{\mathrm{sc}}\} \times A^{\mathrm{pp},w}_{\partial,\hbar}(I) \times T_{\mathrm{Ham},\hbar},$$

where the first, second, and fourth factors are constant in I, and the principal-part current factor uses extension by zero and the factorization product of Theorem 5.186. This product target carries the following proved data.

(i) The weighted Casimir $C_{\mathfrak{h},w}$ and the finite-scale BV propagator $P^w_{\epsilon,L}$ converge in the weighted Tate coefficient category. Each fixed Costello graph with vertexwise polynomial-degree-bounded Hamiltonian labels remains in that category.

(ii) On every balanced scalar-contact completion of Definition E.17, the balanced $\mathfrak{gl}(N \mid N)$ supertrace model has zero filtered scalar projection $\widehat{\sigma}_{\mathrm{bal,sc}} = 0$, zero chain-map defect, and vanishing scalar-projection obstruction classes $o^{(r)}_{\sigma,w,\mathrm{bal}}(I)$. In the ordinary scalar-reduced $\mathfrak{gl}_N$ branch the scalar class is instead $\hbar N[\bar{c}]$ and requires the alternative mechanisms of Theorem 5.214.

(iii) The reduced weighted quantum cotangent current brackets are

$$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \qquad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

$$\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0.$$

Here $\mathrm{ad}^\vee_{f,\hbar}$ is the Moyal coadjoint action on $\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}$, whose restriction to $\mathcal{P}[[\hbar]]$ is the action of Construction 5.135. This is the Moyal coadjoint action, distinct from the polynomial one-ψ PBW action. Thus the classical principal-part current construction is transported to the quantum target only after replacing $f \cdot \rho$ by the full Moyal coadjoint action and after imposing weighted finite-window continuity.

(iv) Under the radial image hypotheses, equivalently after solving the quantum shear primitive construction of Theorem 5.217, the degree-zero Hamiltonian generator map

$$\Phi_{\hbar}^{(0)} : \mathfrak{h}_{\hbar,u} \xrightarrow{\sim} T_{\mathrm{Ham},\hbar}$$

is the $\mathbb{C}[[\hbar]]$-linear Lie comparison of Theorem 5.217.

Consequently the weighted coefficient/BV kernel, the balanced scalar-contact zero projection, the reduced weighted Moyal principal-part current target, and the radial-constrained degree-zero Moyal coefficient comparison give the componentwise theorem. The full non-scalar Costello graph realization, unreduced factorization centre morphism, and all-interior radial theorem are separate constructions governed by the obstruction formulation above.

Proof. Item (1) is Corollary 5.71: its parts (a)–(c) give Casimir convergence, heat-kernel propagator extension, and graphwise weighted RG compatibility. Item (2) is Theorem E.16 together with Proposition E.18; the ordinary scalar alternatives and obstruction criterion are assembled in Theorem 5.214.

At $\hbar = 0$, the current brackets are the reduced principal-part brackets of Theorem 5.186 and Corollary B.8. The quantum target is obtained from the completed continuous-dual coadjoint action, with the polynomial one-ψ labels fixed. On the dense strict Matlis subtarget it is the semidirect Moyal current complex of Construction 5.135, whose action is by $(\mathrm{ad}^{\vee}_{f,\hbar}\,\rho)(g) = -\rho(\{f,g\}_{\hbar})$. The extension from $\mathcal{P}[[\hbar]]$ to $\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}$ is windowwise continuous: the Moyal coadjoint formula is finite for each polynomial Hamiltonian, and condition (W2) in Definition 5.52 gives the required finite-window bounds for vertexwise polynomial-degree-bounded arguments. Proposition 5.134 and Theorem 5.136 show why this hypothesis is necessary: the PBW one-ψ label map has a nonzero intertwining obstruction for the Moyal coadjoint action. Hence item (3) is a cotangent-side reduced current construction with the correct quantum action; transport from polynomial descendants has its own obstruction class.

Item (4) is Theorem 5.217: its formal Moyal, Capelli, and edge-strip pieces are proved, while the all-interior ordinary-trace comparison is exactly Proposition F.25, namely $\Omega^{\mathrm{rad}}_{a,b} = 0$, equivalently decorated PBW Stokes $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$, with signed $B^{*}_{a,b}$-rows as the obstruction functionals. The product prefactorization structure is componentwise: the weighted BV kernel, scalar zero representative, and stable trace factor are constant in the interval, while the principal-part current factor is functorial by extension by zero and has the disjoint-interval product of Theorem 5.186. These components are independent factors, so their product carries exactly the displayed product comparison; Costello graph/QME assertions require the separate obstruction complex. □

Remark 5.219 *(Exact finite-window verification).* The explicit clause of Theorem 5.218 is verified by exact rational arithmetic. The Procesi–Razmyslov–Capelli inverse identity and the prequotient scalar Moyal expansion on the rectangular window $0 \le k, l, m, n \le 10$ to Moyal order ≤ 11, confirming the identities of Proposition F.32 and Theorem 5.217. The same finite-window calculation verifies the descendant action and the residue-coadjoint formula $z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}$ of Theorem 5.179 on $0 \le k, l \le 5$. The radial Stokes boundary $B_{a,b}$ and decorated weight $C^{+}_{a,b}$ are built as rational matrices; $\ker B^{*}_{a,b}$ and $\operatorname{coker} B_{a,b}$ follow by exact Gaussian elimination. Outside the window, the all-bidegree obstruction $\Omega^{\mathrm{rad}}_{a,b} \in \operatorname{coker} B_{a,b}$ is the cohomology class blocking extension (Proposition F.25).

Problem 5.220 *(Non-scalar quantum P_0-operation obstruction complex).* After the four proved components of Theorem 5.218, the remaining non-scalar Costello QME comparison is the filtered obstruction triple $(o^{(r)}_{\sigma,w}, [\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}], \lambda)$. Single-class language applies only after the scalar projection and tower-compatibility data have been fixed.

- *Morphism:* a chain-level Costello-local $P_{0,\hbar}$-factorisation-centre map $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]) \to Z^{P_{0,\hbar}}_{\mathrm{fact}}(\mathrm{Obs}^{\hbar}_{\mathrm{open}})|_{\mathfrak{S}_{\hbar,w}}$, curved bulk-to-defect kernel $\kappa_{\hbar,w,I}$ included.
- *Hypotheses:* a complete d^L_{QME}-stable scalar-contact filtration with vanishing successive scalar-symbol extension classes $o^{(r)}_{\sigma,w}(I)$; a filtered chain projection $\widehat{\sigma}_{\mathrm{sc}}$; a curved bulk-to-defect kernel $\kappa_{\hbar,w,I}$ on the current source of Definition B.14; compatible local counterterms $C^{w_0}_{\Gamma,w}(\epsilon)$ with finite-window transport.
- *Obstruction class:* the reduced non-scalar QME cohomology class $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker\widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{QME}})$, together with its finite-window $\varprojlim^1 H^0$-primitive-compatibility class.
- *Vanishing criterion:* successive vanishing of $o^{(r)}_{\sigma,w}(I)$ for $r > 0$ builds $\widehat{\sigma}_{\mathrm{sc}}$; on a fixed window the row equation $A^M c = -r^M$ is solvable iff $[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}]$ lies in $-\operatorname{im} H^1(\Xi_M)$; in the inverse-limit tower vanishing of the closed-exchange triple $(\beta^{\mathrm{cl}}_{n_0}, \mu^{\mathrm{cl}}_{n_0}, \lambda^{\mathrm{cl}}_{n_0})$ is the criterion in that tower.
- *First nonzero example or known obstruction:* the smallest non-scalar window θ_3 carries the bare $H^1 = \mathbb{C}\cdot\mathsf{E}_{\theta_3,M} \neq 0$ with covector $\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$; a scalar-zero counterterm $C_{\theta_3,M}$ with $d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$ forces $c = 1$ unique and $H^1 = 0$ (Theorem E.38); the lower source-coordinate Bianchi route of Proposition 5.221 exhibits a nontrivial cokernel $\widetilde{\lambda}_{02,\mathfrak{b}}$ surviving the current Bianchi-exposed completion.
- *Physical meaning:* the unreduced Costello-local closed-exchange or local-counterterm cancellation of the non-scalar QME residual is the bulk-to-defect Ward identity compatible with the quantum moment-map constraint.

Statement. The non-scalar Costello graph/QME obstruction criterion is the vanishing problem in the filtered local-functional complex

$$\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I) = \Big(\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}_w|_I;\ A^{\mathrm{pp},w}_{\partial,\hbar}(I)),\ d^L_{\mathrm{QME},w} = Q + \{I^w[L], -\}_{\hbar,L} + \hbar\Delta_L\Big),$$

where the local functionals have compact support on the brane line, coefficients in the weighted field complex $\mathcal{E}_w$, and target currents in $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$. The scalar-symbol projection is only associated-graded data until it is lifted through the scalar-contact filtration. The first obstruction is the filtered lift of the scalar projection; the reduced non-scalar class is reached after this lift. Choose a complete d^L_{QME}-stable scalar-contact filtration $F^\bullet\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$. For any filtered linear lift s of the scalar-symbol map, the successive chain-map defects define classes

$$o^{(r)}_{\sigma,w}(I) \in H^1\Big(\mathrm{Hom}\big(\mathrm{gr}_F\,\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I), C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]\big)_{-r}\Big), \qquad r > 0,$$

with the Hom differential of Lemma E.4. The first condition is the successive vanishing of these $o^{(r)}_{\sigma,w}(I)$, with compatible choices in the completed finite-window inverse limit. Only then is there a filtered chain projection

$$\widehat{\sigma}_{\mathrm{sc}} : \mathfrak{Q}^\bullet_{w,\partial,\hbar}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]$$

whose associated graded is the scalar-symbol projection of Definition E.3.

The balanced scalar-contact branch has a vanishing subcomplex: on $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I) \subset \mathfrak{Q}^{\bullet}_{w,\partial,\hbar,\mathrm{str}}(I)$, Proposition E.18 gives the zero filtered chain projection $\widehat{\sigma}_{\mathrm{bal,sc}} = 0$ and kills every $o^{(r)}_{\sigma,w,\mathrm{bal}}(I)$. Extending this result to the full graph/counterterm completion requires an additional construction: either prove that the relevant scalar-contact symbols still factor through sdim $\mathbb{C}^{N|N}$, or compute and kill the extension classes of Remark E.19.

After this lift has been constructed, the balanced supertrace branch requires a continuous bulk-to-defect kernel

$$\kappa_{\hbar,w,I} \colon C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}) \longrightarrow \mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$$

for the current source of Definition B.14. Its curvature

$$\mathrm{Ob}_{w,\partial,\hbar} = d^{L}_{\mathrm{QME}}\kappa_{\hbar,w,I} + \frac{1}{2}[\kappa_{\hbar,w,I}, \kappa_{\hbar,w,I}]_{\hbar}$$

must be a cocycle satisfying

$$\widehat{\sigma}_{\mathrm{sc}}(\mathrm{Ob}_{w,\partial,\hbar}) = 0$$

and the remaining reduced non-scalar class

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^{1}\Big(\ker \widehat{\sigma}_{\mathrm{sc}}, d^{L}_{\mathrm{QME}}\Big)$$

must vanish. At finite windows this is the tower $Q^{\bullet}_{w,M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M}$ of Definition 5.83. The compatible counterterm recursion is exactly the criterion of Theorem 5.84: at each $\hbar$-order the inverse-limit H^1-class and the $\varprojlim^{1} H^{0}$-primitive-compatibility class must vanish.

More explicitly, let n_0 be the first order where lower-order choices leave a residual

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} \in Z^{1}(Q^{\bullet}_{w,M}(I)).$$

Counterterms alone close this order exactly on the vanishing locus of the obstruction vector

$$\mathfrak{O}^{\mathrm{ns}}_{n_0} = \big(([\mathfrak{o}^{\mathrm{ns}}_{n_0,M}])_M, \lambda_{n_0}\big)$$

of Theorem 5.84. A closed-exchange cancellation is additional cochain data: finite-window complexes $X^{\bullet}_{w,M}(I)$, compatible cochain maps

$$\Xi_M \colon X^{\bullet}_{w,M}(I) \longrightarrow Q^{\bullet}_{w,M}(I),$$

and cochains satisfying the branch equation

$$\mathfrak{o}^{\mathrm{ns}}_{n_0,M} + \Xi_M(W_{n_0,M}) + d_M C_{n_0,M} = 0, \qquad W_{n_0,M} \in Z^{1}(X^{\bullet}_{w,M}(I)), \quad C_{n_0,M} \in Q^{0}_{w,M}(I).$$

For a fixed window this equation is solvable if and only if

$$[\mathfrak{o}^{\mathrm{ns}}_{n_0,M}] \in -\operatorname{im}\Big(H^{1}(\Xi_M) \colon H^{1}(X^{\bullet}_{w,M}(I)) \to H^{1}(Q^{\bullet}_{w,M}(I))\Big).$$

In the inverse-limit tower the exact closed-exchange obstruction is the triple

$$\Big(\beta^{\mathrm{cl}}_{n_0}, \mu^{\mathrm{cl}}_{n_0}, \lambda^{\mathrm{cl}}_{n_0}\Big),$$

where

$$\beta^{\mathrm{cl}}_{n_0} \in \operatorname{coker}\left(\varprojlim_M H^1(\mathcal{X}^\bullet_{w,M}(I)) \xrightarrow{\varprojlim H^1(\Xi_M)} \varprojlim_M H^1(Q^\bullet_{w,M}(I))\right),$$

$$\mu^{\mathrm{cl}}_{n_0} \in {\varprojlim}^1_M H^0(\mathcal{X}^\bullet_{w,M}(I)), \qquad \lambda^{\mathrm{cl}}_{n_0} \in {\varprojlim}^1_M H^0(Q^\bullet_{w,M}(I)).$$

Vanishing of this triple is the exact typed criterion for a compatible closed-exchange kernel plus local counterterm to kill the first non-scalar residual. Constructing $\mathcal{X}^\bullet$, Ξ, and such a class $[W]$ is the remaining finite-row obstruction.

For a fixed binary generator pair x, y, the row form of the centrality obstruction is the same elementary finite-window matrix problem. Once the finite graph data give a scalar-zero basis ρ_i^M for the degree $|x|+|y|$ residual rows and homotopy rows η_j^M of degree $|x|+|y|-1$, write

$$R^{\mathrm{cent}}_{x,y,M} = \sum_i r_i^M \rho_i^M, \qquad d_M \eta_j^M = \sum_i A_{ij}^M \rho_i^M.$$

Then $H^M_{x,y} = \sum_j h_j^M \eta_j^M$ is a binary centrality homotopy in the chosen finite row span if and only if

$$A^M h^M = -r^M.$$

The compatible inverse-limit homotopy additionally requires

$$[P^{M,N} h^M - h^N_{\pi x, \pi y}] = 0 \quad \text{in} \quad H^{|x|+|y|-1}(Q^\bullet_{w,N}(I)).$$

Thus the coefficient-current zero row gives evidence only for the coefficient-current row; the enlarged Costello row is the displayed matrix obstruction.

Equivalently, one must construct compatible finite-window local counterterms $C^{w_0}_{\Gamma,w}(\epsilon)$ and the bulk-to-defect kernel $\kappa_{\hbar,w,I}$ such that

$$\operatorname{Curv}(\kappa_{\hbar,w,I} + C_{\hbar,w}) = d^L_{\mathrm{QME}}(\kappa_{\hbar,w,I} + C_{\hbar,w}) + \frac{1}{2}[\kappa_{\hbar,w,I} + C_{\hbar,w}, \kappa_{\hbar,w,I} + C_{\hbar,w}]_\hbar = 0,$$

with finite-window compatibility

$$\pi_{w_0' w_0} C^{w_0'}_{\Gamma,w} = C^{w_0}_{\Gamma,w}, \qquad R^{w_0}_{w,w'} C^{w_0}_{\Gamma,w} = C^{w_0}_{\Gamma,w'}.$$

This is the precise obstruction complex for the

$$C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]\big) \longrightarrow Z^{P_0,\hbar}_{\mathrm{fact}}\big(\mathrm{Obs}^\hbar_{\mathrm{open}}\big)|_{\mathfrak{S}_{\hbar,w}}.$$

PROPOSITION 5.221 *(Finite-window criterion for the first θ_3 row).* In a larger finite-window graph system, let Θ_3 be a genuine marked source-face row of $\hbar$-order 3. The exposed one-row scalar-zero summand is

$$Q^0_{\theta_3,M} = 0, \qquad Q^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_M = 0,$$

with

$$\mathsf{E}_{\theta_3,M} = -K^M_{\Theta_3}(\zeta_{M,\nu_3}), \qquad \mathfrak{o}^{\mathrm{ns}}_{\theta_3,M} = \mathsf{E}_{\theta_3,M}, \qquad \widehat{\sigma}_{\mathrm{sc},M}(\mathsf{E}_{\theta_3,M}) = 0.$$

In this finite subcomplex the class $[\mathsf{E}_{\theta_3,M}]$ is nonzero. It is killed in an enlarged finite-window system precisely by one of the following data, together with scalar-zero and finite-window transport compatibility.

First, a CE-source ancestor

$$\eta_{\theta_3,M}, \qquad d_{\mathrm{CE}}\eta_{\theta_3,M} = \zeta_{M,\nu_3},$$

whose induced degree-zero row $C^{\mathrm{CE}}_{\theta_3,M} = K^M_{\Theta_3}(\eta_{\theta_3,M})$ satisfies

$$d_M C^{\mathrm{CE}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}.$$

Second, a Costello local counterterm primitive $C^{\mathrm{ct}}_{\theta_3,M} \in \mathcal{Q}^0_{w,M}(I)$ satisfying

$$d_M C^{\mathrm{ct}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}, \qquad \widehat{\sigma}_{\mathrm{sc},M}(C^{\mathrm{ct}}_{\theta_3,M}) = 0.$$

Third, a complete companion-face table $F_{\Theta_3,M}$ with signs $\epsilon^{\mathrm{CE}}_{\Theta_3,\nu}$ and rows $E^M_{\Theta_3,\nu}$ such that

$$\sum_{\nu \in F_{\Theta_3,M}} \epsilon^{\mathrm{CE}}_{\Theta_3,\nu} E^M_{\Theta_3,\nu} = 0, \qquad E^M_{\Theta_3,\nu_3} = \mathsf{E}_{\theta_3,M}.$$

In the first two cases the finite-window tower must also satisfy

$$[\pi_{M,N} C_{\theta_3,M} - C_{\theta_3,N}] = 0 \quad \text{in} \quad H^0(\mathcal{Q}^\bullet_{w,N}(I)).$$

In the companion-face case the source-face transport matrices must satisfy

$$\pi_{M,N} E^M_{\Theta_3,\nu} = \sum_{\nu'} v^{M,N}_{\Theta_3,\nu;\Theta_3,\nu'} E^N_{\Theta_3,\nu'}, \qquad v^{M,P} = v^{N,P} v^{M,N},$$

and must preserve the displayed signed zero sum. A formal symbol C_{θ_3} given apart from the differential $d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$, scalar-zero value, and tower compatibility is not a counterterm datum; a counterterm consists of all three data.

The currently tested finite source routes are obstructed in a sharper sense. In the ordinary pure two-u CE source envelope, in every finite Hamiltonian degree, the covector

$$q_{2u} = e^*_{((H_{0,1},H_{2,0}),c_{\rho_{2,1}})} - \frac{1}{2} e^*_{((H_{0,1},H_{0,1}),c_{\rho_{0,2}})}$$

satisfies

$$q_{2u} d_{\mathrm{CE}} = 0, \qquad q_{2u}(\zeta_{M,\nu_3}) = 1.$$

Hence the ordinary pure two-u degree enlargement has an obstruction to giving a CE ancestor for ζ_{M,ν_3}. This is an exact obstruction theorem for that source envelope; marked-source, Hom-valued companion, and local-functional counterterm completions are separate source categories. The tested eight-face companion row has coordinates

$$(2, -2, 3, 2, -1, 1, -2, -3)$$

on

$$E_{\nu_{02}}, E_{\nu_{20}}, E_{\nu_{03}}, E_{\nu^{(1)}_{12}}, E_{\nu^{(2)}_{12}}, E_{\theta_3}, E_{\nu^{(2)}_{21}}, E_{\nu_{30}},$$

and diagonal transport to the $N = 2$ window leaves $2E^2_{\nu_{02}} - 2E^2_{\nu_{20}}$. The covector q_{2u} evaluates to zero on this eight-face row, so the row is incompatible with the required marked source column. The normalized lower-window cokernel functional $\lambda^{(2)}_{\nu_{02}} = \frac{1}{2}(E^2_{\nu_{02}})^*$ evaluates to 1 on the transported residual. One must therefore give either a scalar-zero column $A^M_{\theta_3,C} = -1$, a marked source generator outside the ordinary pure two-u algebra with q_{2u}-value 1, or a companion relation whose lower-window transport kills this residual by a derived non-diagonal transport, lower primitive, or relation $E^2_{\nu_{02}} = E^2_{\nu_{20}}$. The canonical lower source-coordinate column is

$$\zeta_2^0 = u_{a,H_{1,0}} u_{b,H_{0,1}}, \qquad D^2_{02,20} = K^2_{\Theta_3}(\zeta_2^0),$$

with source differential

$$d_{\mathrm{CE}}\zeta_2^0 = 2\zeta_{\nu_{02}} - 2\zeta_{\nu_{20}}.$$

This proves source exactness of the transported lower row. It proves local-functional exactness only after the Hom-valued Bianchi defect

$$\mathfrak{b}^2_{02,20} = d_{\mathrm{ns},2} K^2_{\Theta_3}(\zeta_2^0) - K^2_{\Theta_3}(d_{\mathrm{CE}}\zeta_2^0)$$

is killed:

$$d_{\mathrm{ns},2} D^2_{02,20} = -2E^2_{\nu_{02}} + 2E^2_{\nu_{20}} + \mathfrak{b}^2_{02,20}.$$

In the Bianchi-exposed lower basis

$$(E^2_{\nu_{02}}, E^2_{\nu_{20}}, \mathfrak{b}^2_{02,20})$$

the column of $D^2_{02,20}$ and the residual are

$$A^2_D = (-2, 2, 1)^T, \qquad r_2 = (2, -2, 0)^T.$$

The equation $A^2_D c = -r_2$ is obstructed in this free presentation. The cokernel

$$\widetilde{\lambda}_{02,\mathfrak{b}} = \frac{1}{2}(E^2_{\nu_{02}})^* + (\mathfrak{b}^2_{02,20})^*$$

satisfies

$$\widetilde{\lambda}_{02,\mathfrak{b}} A^2_D = 0, \qquad \widetilde{\lambda}_{02,\mathfrak{b}}(r_2) = 1.$$

Thus the lower source primitive is a genuine local-functional primitive only in the quotient where $\mathfrak{b}^2_{02,20} = 0$, or after a scalar-zero local counterterm $B^2_{02,20}$ with $d_{\mathrm{ns},2} B^2_{02,20} = -\mathfrak{b}^2_{02,20}$ exists, with locality and transport compatibility.

In the current Bianchi-exposed finite-row completion the $B^2_{02,20}$ column is omitted. The only degree-zero lower column is $D^2_{02,20}$, so the Bianchi counterterm equation has right-hand side vector

$$-\mathfrak{b}^2_{02,20} = (0, 0, -1)^T.$$

It lies outside the image of A^2_D, since

$$\widetilde{\lambda}_{02,\mathfrak{b}} A^2_D = 0, \qquad \widetilde{\lambda}_{02,\mathfrak{b}}((0, 0, -1)^T) = -1.$$

The minimal controlled enlargement must therefore add a genuine scalar-zero local-functional column

$$A^2_B = (0, 0, -1)^T, \qquad d_{\mathrm{ns},2} B^2_{02,20} = -\mathfrak{b}^2_{02,20},$$

with regular-density or wavefront-admissible locality and compatible finite-window transport. In a tower this requires

$$\Delta^1_{M,N} := -\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N = d_{\mathrm{ns},N}(\pi_{M,N} B^M - B^N)$$

together with vanishing of the H^1 Bianchi transport class on the right-hand side and then of the resulting secondary $\varprojlim^1 H^0$-class after any transport correction.

Proof. The one-row complex used here has zero degree-zero term and has one degree-one basis vector. Hence $H^1(Q^\bullet_{\theta_3,M}) \cong \mathbb{C}\mathsf{E}_{\theta_3,M}$. The covector

$$\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$$

satisfies $\ell_{\theta_3} d_M = 0$ and evaluates nontrivially on the residual, so it is a left-cokernel functional for the nonzero fixed-window H^1-class.

After adjoining any finite collection of degree-zero rows, the fixed-window question is the elementary linear system

$$A^M c^M = -r^M,$$

where A^M is the matrix of d_M from degree-zero rows to residual rows and r^M is the coordinate vector of $\mathfrak{o}^{\mathrm{ns}}_{\theta_3,M}$. A CE ancestor or local counterterm with differential coefficient -1 is exactly a column of A^M equal to $-\mathsf{E}_{\theta_3,M}$, so $c^M = 1$ solves the equation. If the column is named while this differential entry is omitted, the old covector ℓ_{θ_3} still annihilates the image of A^M and still evaluates to 1 on r^M; the obstruction survives.

The lower source-coordinate column illustrates the distinction between source exactness and local-functional exactness. The source calculation gives $d_{\mathrm{CE}}\zeta_2^0 = 2\zeta_{\nu_{02}} - 2\zeta_{\nu_{20}}$, but $K^2_{\Theta_3}$ becomes a chain map after its Hom-valued Bianchi defect is killed. In the Bianchi-exposed basis the matrix equation is $A^2_D c = -r_2$, with $A^2_D = (-2, 2, 1)^T$ and $r_2 = (2, -2, 0)^T$. The first two coordinates force $c = 1$, while the Bianchi coordinate forces $c = 0$. Equivalently $\widetilde{\lambda}_{02,\mathfrak{b}} A^2_D = 0$ and $\widetilde{\lambda}_{02,\mathfrak{b}}(r_2) = 1$. Hence this computation gives an exact obstruction class. A completed local-functional counterterm requires the additional column displayed below.

The same cokernel proves absence of the $B^2_{02,20}$ column in the current completion. The target vector $(0, 0, -1)^T$ forces a scalar c with $c(-2, 2, 1) = (0, 0, -1)$; the first two coordinates force $c = 0$, while the third forces $c = -1$. Equivalently $\widetilde{\lambda}_{02,\mathfrak{b}}$ annihilates every current boundary column and evaluates to -1 on the target. A formal new column $A^2_B = (0, 0, -1)^T$ closes this fixed matrix equation. The corresponding local-functional theorem consists of the local functional, scalar-zero value, locality or wavefront data, and finite-window transport data.

A companion-face table is a signed source-face relation. The primitive is a row datum. It changes the residual row by replacing the one-face CE source expression with the full signed CE boundary. It is admissible exactly when the signed sum is zero and the truncation matrices preserve that zero sum in the inverse system. The displayed Roos condition is precisely the obstruction to choosing the fixed-window primitives compatibly. These are necessary and sufficient finite-window conditions; the analytic Costello graph construction and all-graph QME solution require additional data. □

Problem 5.222 *(Tate-coefficient lift of the descendant cochain isomorphism).* In the unweighted Tate pair, the diagnosis of Lemma 5.212 identifies the obstruction as the coefficient-kernel obstruction; the resolution requires a category-level change of coefficient module. The classical Tate obstruction is closed by the weighted Tate completion of Corollary 5.71(a)–(d); the all-order quantum lift on the descendant sector is the quantum specialization of 5.49 under the hypothesis of 5.73.

PROPOSITION 5.223 *(Open-line Weyl boundary product from midpoint kernel).* For Weyl-ordered Hamiltonian boundary observables placed on the open line $\mathbb{R}$ in the reduced first-order open model $S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2)$, on two brane-time points $(t, t') \in \mathbb{R}^2_{\mathrm{brane}}$ (with s reserved for the transverse $\mathbb{R}_s$ coordinate of X) and the zero-mode-removed midpoint propagator

$$G(t, t') = \tfrac{1}{2}\,\mathrm{sgn}(t - t'), \qquad \langle \phi_1(t)\phi_2(t')\rangle = \hbar G(t, t'), \qquad \langle \phi_2(t)\phi_1(t')\rangle = -\hbar G(t, t'),$$

the boundary product on $\bar{A}_\hbar$ is

$$F \star G = m \circ \exp\left(\frac{\hbar}{2} P_\partial\right)(F \otimes G), \qquad P_\partial = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

Equivalently, every odd raw commutator coefficient is $\frac{2}{2^r r!}\hbar^r P^r(f,g)$. In particular the first and third coefficients are $\hbar P(f,g)$ and $\frac{\hbar^3}{24}P^3(f,g)$. This reduced open-line computation is independent of Costello specialization. With these conventions, the Weyl normalization is determined by:

(i) the matrix Weyl algebra commutator $[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar\delta^i_l\delta^k_j$;

(ii) the zero-mode-removed midpoint propagator $G(\pm,\mp) = \pm\frac{1}{2}$;

(iii) the Baker–Campbell–Hausdorff determination of the cross-contraction weight $P/2$ on linear Hamiltonians;

(iv) the linear-Hamiltonian commutator $[\widehat{u},\widehat{v}] = \hbar(ad - bc)$ for $u = az_1 + bz_2, v = cz_1 + dz_2$.

Proof. Items (i)–(iv) form a normalization system. By (i)–(ii) the midpoint propagator at $t \neq t'$ has weight $\pm\hbar/2$. By (iv) the linear-Hamiltonian Weyl commutator is $\hbar(ad - bc)$, so the cross-contraction weight on linear Hamiltonians is exactly $1/2$, giving $P/2$. Hence by (iii) the BCH formula on the Heisenberg algebra of linear Hamiltonians fixes the cross-contraction kernel on the boundary as $P_\partial = P$ with weight $\hbar/2$.

The all-order boundary product on $\bar{A}_\hbar$ is then determined by: (a) the local Weyl/BCH computation for the constant Poisson tensor, with the constant-Poisson case of Kontsevich deformation quantization [7]; (b) Proposition 5.216 together with item (iii) above; (c) the antisymmetrisation argument of Theorem 5.224.

The scalar-normalization clause is the content of Lemma 5.198(iii): the Capelli contact term is represented by the $T \leftrightarrow J_N$ change of basis, and the connected single-trace projection is the standard primitive projection. Consequently all odd raw commutator coefficients are $\frac{2}{2^r r!}\hbar^r P^r(f,g)$, and for $r = 1, 3$ they are $\hbar P(f,g)$ and $\frac{\hbar^3}{24}P^3(f,g)$. The four primary cases (a)–(d) and a sweep of 14641 monomial pairs through order $\hbar^{11}$ give exact-arithmetic verification of this coefficient formula. The finite-N radial-parts hypothesis remains the independent radial-parts hypothesis. □

THEOREM 5.224 *(Reduced open-line Wick theorem).* In the reduced open first-order line model

$$S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2),$$

on two brane-time points $(t,t') \in \mathbb{R}^2_{\text{brane}}$ (with s reserved for the transverse $\mathbb{R}_s$ coordinate of X), choose the zero-mode-removed midpoint propagator

$$G(t,t') = \frac{1}{2}\,\mathrm{sgn}(t-t'), \qquad \langle\phi_1(t)\phi_2(t')\rangle = \hbar G(t,t'), \quad \langle\phi_2(t)\phi_1(t')\rangle = -\hbar G(t,t').$$

For ordered insertions $t > t'$, let $W_r^>(f,g)$ be the contribution of the graph with exactly r cross-contractions between the two Weyl-ordered boundary vertices. Let $W_r^<(f,g)$ be the reversed ordering contribution. Then

$$W_r^>(f,g) = \frac{1}{r!}\left(\frac{\hbar}{2}\right)^r P^r(f,g), \qquad W_r^<(f,g) = \frac{1}{r!}\left(-\frac{\hbar}{2}\right)^r P^r(f,g).$$

Consequently the reduced open-line commutator is

$$[f,g]_{\text{Wick}} = \sum_{\substack{r\geq 1\\ r \text{ odd}}} \frac{2}{2^r r!}\hbar^r P^r(f,g),$$

and all even orders vanish. For monomials $f = z_1^k z_2^l$, $g = z_1^m z_2^n$,

$$P^r(f,g) = \sum_{a=0}^{r}(-1)^a \binom{r}{a}(k)_{r-a}(l)_a(m)_a(n)_{r-a} z_1^{k+m-r} z_2^{l+n-r}.$$

In particular the one-edge and three-edge commutator contributions are

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24}P^3(f,g).$$

Proof. The midpoint value $G(+,-) = 1/2$ assigns one copy of

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$$

to each cross-contraction, with weight $\hbar/2$. With Weyl vertices self-contractions vanish. The graph with r identical cross-edges has automorphism group S_r, so the ordered contribution is

$$W_r^{>}(f,g) = \frac{1}{r!}\left(\frac{\hbar}{2}\right)^r P^r(f,g).$$

Reversing the order changes the midpoint propagator from $1/2$ to $-1/2$, giving the displayed formula for $W_r^{<}$. Hence

$$W_r^{>}(f,g) - W_r^{<}(f,g) = \frac{1-(-1)^r}{2^r r!}\hbar^r P^r(f,g),$$

which is zero for even r and $2\hbar^r P^r(f,g)/(2^r r!)$ for odd r.

The monomial formula is the expansion of $P^r = (\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})^r$ by the binomial theorem: the summand with a copies of $-\partial_{z_2} \otimes \partial_{z_1}$ contributes the sign $(-1)^a$, the binomial coefficient $\binom{r}{a}$, and the four falling factorials shown above. The values at $r = 1$ and $r = 3$ give $\hbar P(f,g)$ and $\hbar^3 P^3(f,g)/24$. □

Remark 5.225. Theorem 5.224 is a computation in the reduced open-line Weyl model. The Costello closed-open graph theorem still requires an elliptic parent or dimensional reduction, a brane-defect condition, finite-scale gauge fixing, the induced bulk-to-defect propagator, orientations and measures on configuration spaces, counterterms, anomaly cancellation, and compatibility with the quantum Hamiltonian reduction.

5.7.5 Third-order Costello graph class

The order-$\hbar^1$ Moyal coefficient P is fixed by the linear-Hamiltonian Heisenberg commutator and gives the target $\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$. At order $\hbar^3$ the Moyal target is $P^3/24$; a Costello-graph realisation of the same coefficient requires a separate trivalent boundary contraction diagram. The equality of the two coefficients is the content of this subsection.

For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the third-order Moyal coefficient from Proposition 5.216 is

$$P^3(f,g) = \sum_{a=0}^{3}(-1)^a \binom{3}{a}(k)_{3-a}(l)_a(m)_a(n)_{3-a} z_1^{k+m-3} z_2^{l+n-3},$$

where $(r)_s = r(r-1)\cdots(r-s+1)$ and $(r)_0 = 1$. Therefore

$$[f,g]_* = \hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1} + \frac{\hbar^3}{24}P^3(f,g) + O(\hbar^5).$$

The exact-arithmetic verification checks this first-order coefficient, the vanishing of the second-order coefficient in the raw commutator, the displayed third-order formula, and the star-commutator normalization on every monomial pair with exponents between 0 and 10 and commutator orders through 11.

This algebraic computation is an observable-product target, distinct from an effective action computation. In Costello's formalism the connected effective action and the product/OPE of quantum observables are related but distinct. The formal holomorphic singular coefficient of the product is the $K_\Delta^{(2)}$-kernel of Theorem 1.17; the finite-scale calculation below controls the brane-defect observable product that quantizes it. For two boundary observables F, G, the finite-scale free-field model has the Wick product

$$F \star_L G = m \circ \exp(\hbar \partial_{P_{\partial,L}})(F \otimes G),$$

where $P_{\partial,L}$ is the boundary or brane-defect propagator induced by the chosen kernel. The connected single-trace part is the primitive trace component extracted from this product. In this convention the order in $\hbar$ counts boundary cross-contractions; the loop order of the connected effective interaction is a different grading.

Thus there are two equivalent normalizations to keep separate. In Costello's observable-product notation the required boundary contraction kernel is

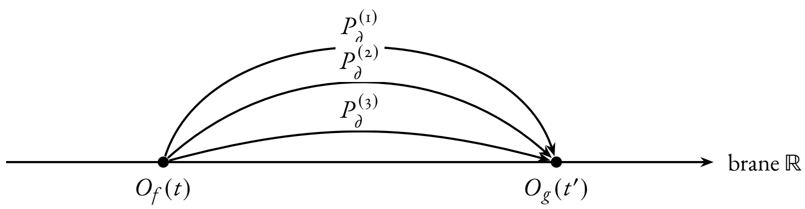
$$P_{\partial,L} \rightsquigarrow \frac{1}{2}P$$

on degree-zero Hamiltonians, so that

$$m \circ \exp(\hbar \partial_{P_{\partial,L}}) \rightsquigarrow m \circ \exp\left(\frac{\hbar}{2}P\right),$$

the Weyl/Moyal product used above. The ordered product has labelled arguments (f,g); its r identical cross-contractions contribute the denominator $r!$. The ordered product uses exactly this denominator. The reverse order is the separate product $g * f$; their difference forms the commutator.

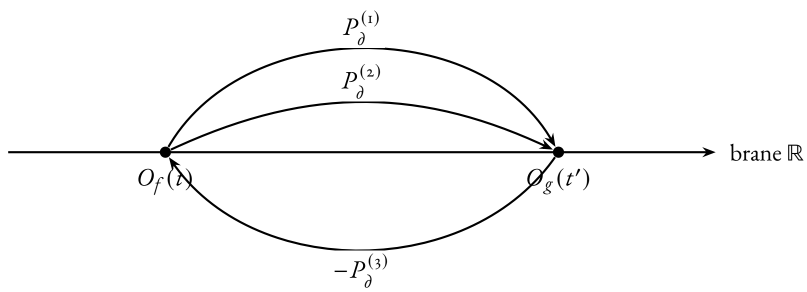
Γ_{3a}: three propagators, common orientation



$$\mathrm{Aut}(\Gamma_{3a}) = S_3, \quad |\mathrm{Aut}(\Gamma_{3a})| = 3! = 6, \quad \frac{1}{|\mathrm{Aut}(\Gamma_{3a})|} = \frac{1}{6}$$

orientations $a = 0, 3$: contributions $(k)_3(n)_3$ and $-(l)_3(m)_3$

Γ_{3b}: two propagators forward, one reversed



$$\mathrm{Aut}(\Gamma_{3b}) = S_2, \quad |\mathrm{Aut}(\Gamma_{3b})| = 2! = 2, \quad \frac{1}{|\mathrm{Aut}(\Gamma_{3b})|} = \frac{1}{2}$$

orientations $a = 1, 2$: contributions $-3(k)_2 lm(n)_2$ and $+3k(l)_2(m)_2 n$

FIGURE 0.2: The third-order graphs Γ_{3a} (top) and Γ_{3b} (bottom) for the order-$\hbar^3$ boundary contractions whose tensor target is $P^3(f, g)$. Each cross-contraction contributes one copy of the Poisson bivector P_∂; reversing the orientation of a propagator contributes a relative sign. The orientation multiplicities are the binomial coefficients $\binom{3}{a}$ of Proposition 5.216: Γ_{3a} realises $a \in \{0, 3\}$ with automorphism group S_3 permuting the three identical propagators (symmetry factor $1/6$), and Γ_{3b} realises $a \in \{1, 2\}$ with automorphism group S_2 permuting the two same-orientation propagators (symmetry factor $1/2$). These are two descriptions of the same count: either use the unexpanded aggregate P-edge count with automorphism S_3, or expand into orientation classes with residual automorphism $S_a \times S_{3-a}$; the equality is $\binom{3}{a}/3! = 1/(a!(3-a)!)$: the unexpanded S_3 denominator and the residual $S_a \times S_{3-a}$ denominator are alternative descriptions of the same total automorphism count. Summing all four orientation classes, $P^3(f, g) = \sum_{a=0}^{3} (-1)^a \binom{3}{a} (k)_{3-a}(l)_a(m)_a(n)_{3-a}\, z_1^{k+m-3} z_2^{l+n-3}$. The third-order raw commutator weight after antisymmetrising is $\hbar^3/24$. The reduced open-line weights are computed in Theorem 5.224; the Costello counterterm and anomaly checks remain the analytic obligations of Problem 5.228.

The graphs Γ_{3a} and Γ_{3b} of Figure 0.2 record the displayed order-$\hbar^3$ boundary contraction types whose tensor target is $P^3(f, g)$. Each cross-contraction contributes one copy of the constant Poisson bivector P. Their reduced open-line weights and automorphism factors are computed above. Their realization as Costello brane-defect graphs still requires the sector specialization, propagator, counterterm, and anomaly checks in the problem below.

THEOREM 5.226 *(Costello first/third normalization recognition criterion).* Assume the Hamiltonian specialization data of Problem 5.213 have been fixed for a Costello open–closed BV theory, and assume that the induced brane-defect observable product restricts on degree-zero Hamiltonians to the midpoint kernel $P/2$ of Theorem 5.224. Assume further that at order $\hbar^3$ every connected local term outside the three-parallel-edge cross-contraction has been shown to vanish, to be counterterm-exact, or to land in the discarded scalar trace. Then the first- and third-order Costello boundary commutator coefficients are determined by the Weyl/Moyal coefficients:

(i) the connected one-propagator brane-defect ribbon graph carries coefficient $\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$ for monomial arguments $f = z_1^k z_2^l$, $g = z_1^m z_2^n$;
(ii) the surviving three-propagator contribution carries coefficient $\frac{\hbar^3}{24}P^3(f,g)$;
(iii) any finite counterterm ambiguity that changes the first or third coefficient is fixed by matching the Moyal coefficients 1 and 1/24 at these orders. All-order Costello uniqueness is a separate statement.

Equivalently, the Weyl/Moyal coefficient test for a Costello specialization at these orders holds precisely when the specialization gives the above data and the normalization defects

$$\mathfrak{n}_1 = P_{\partial,L}|_{\mathrm{Ham}^\circ} - \frac{1}{2}P,$$

$$\mathfrak{n}_3 = \left[\mathrm{ObsProd}^{(3)}_{\mathrm{Costello}}(f,g) - \frac{\hbar^3}{24}P^3(f,g)\right]_{\mathrm{Ham}^\circ,\mathrm{conn},\mathrm{red}}$$

vanish, together with the scalar-contact, reduced-kernel, and θ_3 graph-row obstructions. The obstruction tuple is

$$\mathfrak{O}^{1,3}_{\mathrm{Cost}} = (\mathfrak{o}_{\mathrm{Cas}}, \hbar N[\bar{c}] \text{ or } o^{(r)}_{\sigma,w}, [\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}], \mathfrak{O}_{\theta_3}, \mathfrak{n}_1, \mathfrak{n}_3).$$

The reduced datum gives Theorem 5.224 and the reduced open-line Wick normalization. Costello heat-kernel graph realization requires the full specialization data.

Proof. Under the stated hypotheses, the degree-zero boundary observable product is the reduced open-line Weyl product of Proposition 5.223. Thus each boundary cross-contraction contributes the operator P with weight $\hbar/2$. Here the S_3 count treats the three edges as copies of the aggregate bivector $P = A - B$. After expanding P^3, a mixed orientation class has residual automorphism $S_a \times S_{3-a}$ and binomial coefficient $\binom{3}{a}$. These are equivalent normalizations: $\binom{3}{a}/3! = 1/(a!(3-a)!)$; the aggregate S_3 denominator and the residual orientation-class denominator are alternative descriptions of the same total automorphism count.

Item (i): one edge, trivial edge automorphism group, weight $\hbar P/2$ for the ordered product (Theorem 5.224); antisymmetrisation gives

$$\hbar P(f,g) = \hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$$

on monomials.

Item (ii): for three parallel edges between the two labelled boundary observables, the edge automorphism group is S_3. The ordered weight is

$$\hbar^3 \frac{1}{3!}\left(\frac{1}{2}\right)^3 P^3(f,g) = \frac{\hbar^3}{48}P^3(f,g),$$

and antisymmetrisation doubles it to $\hbar^3 P^3(f,g)/24$. The hypotheses exclude the remaining connected local terms at the same order.

Item (iii): a finite local counterterm that changes the one- or three-edge boundary product changes the coefficient of P or P^3. The linear-Hamiltonian Heisenberg commutator fixes the coefficient of P to

1, and Proposition 5.216 fixes the coefficient of P^3 to $1/24$ in the raw commutator. Thus matching these two coefficients removes precisely the counterterm freedom visible at these orders. The displayed defect tuple is a restatement of the hypotheses needed to promote this reduced computation to a Costello graph realization: the first defect fixes the brane-restricted propagator, the third-order defect fixes the connected reduced product, and the remaining entries are the scalar, reduced-kernel, and θ_3 graph-row conditions developed in Problem 5.220. □

Remark 5.227. In the reduced open-line model the Weyl parameter $\hbar$ counts boundary cross-contractions. If a Costello specialization restricts to this free boundary kernel, then its observable-product expansion must be read with this edge-counting convention on degree-zero Hamiltonians. The specialization data of Problem 5.213 identify the formal Weyl/Moyal normalization above with a Costello graph statement.

Problem 5.228 *(Costello first/third graph-normalization obstruction datum).* Work inside the observable product constructed by Statement 5.207, after solving the Hamiltonian specialization data of Problem 5.213. The first graph fixes the normalization; the third graph tests whether the Costello specialization lands on the Weyl/Moyal product, fixing the deformation-quantization convention. The analytic obstruction is reduced to the following checks:

(i) identify the connected one-propagator brane-defect ribbon graph whose tensor contraction is $P(f,g)$, and prove that its antisymmetrised boundary product gives the raw coefficient $\hbar P(f,g)$;
(ii) fix the sign, automorphism factor, boundary orientation, and trace normalization in this first-order graph so that the coefficient is $\hbar(kn-lm)z_1^{k+m-1}z_2^{l+n-1}$ on monomials;
(iii) identify the connected three-propagator brane-defect ribbon graphs whose tensor contraction is $P^3(f,g)$;
(iv) prove that every other connected order-$\hbar^3$ contribution in the Hamiltonian sector vanishes, is counterterm-exact, or lands in the discarded scalar trace;
(v) for every term declared counterterm-exact in the preceding item, exhibit compatible local counterterm representatives $C^{w_\circ}_{\Gamma,w}(\epsilon)\in\mathcal{O}_{\mathrm{loc}}(\mathcal{E}^{w_\circ}_w)$, prove Costello locality and RG compatibility, and verify that their restriction to degree-zero Hamiltonians preserves the fixed one-edge coefficient except by the allowed Moyal normalization choice;
(vi) compute the graph weight and automorphism factor after the first-order normalization and show that the coefficient is $\frac{1}{24}$;
(vii) check that connected large-N extraction preserves the factor of N, sign, and scalar moment-map normalization;
(viii) verify that the same observable product, counterterm subtraction, and connected extraction commute with the principal-part current extension of Theorem 5.186 on the reduced target. If the assertion is made before de Rham-current contraction, identify the additional unreduced homotopies of Remark 5.190. In particular the degree-zero Moyal product must act on the cotangent current summand through the principal-part coadjoint action; polynomial one-ψ descendants carry the PBW special-fibre action.

The target normalization is the pair of algebraic coefficients

$$\hbar P(f,g),\qquad \frac{\hbar^3}{24}P^3(f,g)$$

in the Weyl/Moyal commutator. Theorem 5.226 records the implication once the Costello specialization gives these checks.

Remark 5.229. In characteristic zero, GL_N is linearly reductive on rational representations. Therefore taking GL_N-invariants is exact, and

$$H\big((\mathcal{R}_N)^{GL_N}, Q\big) \cong H(\mathcal{R}_N, Q)^{GL_N}.$$

This justifies the degree-zero rational invariant calculation used above. The full Lie-algebra Chevalley–Eilenberg BRST cohomology of $\mathfrak{gl}_N$ is a separate stack-level calculation. A stack-level presentation remembers stabilizers, including the trivially acting scalar $B\mathbb{G}_m$; those ghost and stabilizer classes are outside the Hamiltonian trace sector computed here. Linearly reductive exactness identifies derived algebraic invariants of functions with ordinary invariants in this setting.

The dictionary $c_f \mapsto \theta_f, u_f \mapsto J(f)$ is the trace-sector CE/PV coordinate map at the formal-Darboux stalk of the Dirac brane vacuum b_p (Theorem 5.6). After the stable scalar-reduced trace quotient and the admissible completion it is the CE/PV isomorphism. At fixed finite N it is a Hochschild generator assignment; a cochain map requires the centrality homotopies and is not the whole derived E_1-centre. Scalar reduction kills the constant Hamiltonian on the closed side and retains a U(1) centre-of-mass anomaly on the open side: a single Lie 2-cocycle $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ classified in the section below.

6 The Capelli scalar

The Capelli scalar $\hbar N[\bar{c}]$ is the projective Lie anomaly of the trace representation, realised by the Capelli central extension $0 \to \mathbb{C} \cdot K \to \widehat{\mathfrak{g}}_{\mathrm{cent}} \to \mathfrak{g} \to 0$ of (15); the defining bracket value is $\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$ (Remark 5.9, check (2)). A determinant-line curvature interpretation is a further theorem after the determinant line and its connection are constructed.

6.1 The U(1) centre-of-mass cocycle

A constant Hamiltonian $c \in \mathbb{C} \subset \mathbb{C}[z_1, z_2]$ acts trivially on the polydisk: its Hamiltonian vector field vanishes, so it sits in the kernel of the canonical-transformation reduction (3). The same constant pulled back along the substitution $z_i \mapsto \phi_i$ becomes $c \cdot I$, and the matrix Poisson partner of the symplectic potential is

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = \mathrm{Tr}(I \cdot I) = N,$$

the brane centre-of-mass charge. Closed-side: invisible. Open-side: visible, equal to N. The asymmetry is a single Lie cohomology class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$, realised both as an abelian-extension obstruction on $\bar{A}$ and as the U(1) trace centre of $\mathfrak{gl}_N$, and it survives Moyal quantisation as the order-$\hbar$ Capelli shift. The cocycle has three concrete faces — Lie 2-cocycle on $\bar{A}$, Poisson bracket of trace coordinates, and projective Moyal–Weyl commutator

$$[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_{\mathrm{proj}} = \hbar N \cdot \mathbf{1}$$

— packaging the four equivalent representatives of Theorem E.13 (the abstract central extension and the explicit Lie central extension reduce to one face on $\bar{A}$).

The closed–open mismatch cocycle. Write $\mathfrak{h}_{\mathrm{poly}} = \mathbb{C}[z_1, z_2]$ for the unreduced polynomial Hamiltonian Lie algebra, constants kept; the bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ again takes values in $\mathfrak{h}_{\mathrm{poly}}$. Define

$$\omega\colon \mathfrak{h}_{\mathrm{poly}} \otimes \mathfrak{h}_{\mathrm{poly}} \longrightarrow \mathbb{C}, \qquad \omega(f, g) = [\,\{f, g\}\,]_0,$$

the constant Taylor coefficient of the bracket. On monomials,

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = \begin{cases} kn - lm, & (k+m, l+n) = (1,1), \\ 0, & \text{otherwise}, \end{cases}$$

so ω is concentrated in the unique bidegree where the bracket multidegree $(k+m-1, l+n-1)$ lands at $(0,0)$: the constant-Taylor direction the canonical-transformation reduction (3) discards. This is the direction on which the closed and open sides disagree.

Example 6.1 *(Linear Hamiltonians and the brane centre of mass).* The two linear Hamiltonians $f_1 = z_1$ and $f_2 = z_2$ measure

$$J(z_1) = \overline{\operatorname{Tr}\phi_1}, \qquad J(z_2) = \overline{\operatorname{Tr}\phi_2},$$

the brane centre of mass in the two transverse directions. On the unreduced phase space the matrix Poisson bracket of these two observables is

$$\{\operatorname{Tr}\phi_1, \operatorname{Tr}\phi_2\} = \operatorname{Tr}(I \cdot I) = N,$$

so the brane centre-of-mass pair is a Heisenberg pair with charge N: the open-side image of the unique cocycle bidegree $\omega(z_1, z_2) = 1$. Closed-side, $\{z_1, z_2\} = 1 \equiv 0$ in $\bar{A}$ by quotient. Moyal quantisation returns the same projective scalar at order $\hbar$:

$$[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_{\mathrm{proj}} = \hbar N \cdot \mathbf{1}.$$

The linear computation truncates exactly at first order because the Moyal expansion needs at least one second derivative on each tensor factor and the linear Hamiltonian has none (Theorem 6.6). The three faces of the anomalous scalar — Lie 2-cocycle on $\bar{A}$, Poisson bracket of trace coordinates, Moyal–Weyl commutator — realise the same class. The trace-$U(1)$ central extension and the Capelli shift are the two constructed interpretations of the multiplier N (Theorem 6.8).

Lemma 6.2 *(Closed–open mismatch lemma).* The form ω is a Lie 2-cocycle on $\mathfrak{h}_{\mathrm{poly}}$: exact on the unreduced algebra and obstructed after the constant-quotient. Equivalently:

- on $\mathfrak{h}_{\mathrm{poly}}$, $\omega = \delta\eta$ with $\eta(f) = -[f]_0$;
- on $\bar{A} = \mathfrak{h}_{\mathrm{poly}}/\mathbb{C}\cdot 1$, the restricted cocycle represents a nonzero class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$.

The class $[\bar{c}]$ is the closed-side obstruction blocking the descent of η past the constant line. Up to the multiplier $\operatorname{Tr} I = N$ it is the open-side scalar $\{\operatorname{Tr}\phi_1, \operatorname{Tr}\phi_2\} = N$ pulled back through the boundary evaluation map of Theorem 6.4: the same scalar mismatch, read on each side.

Proof. Cocycle identity. Project each term of the Jacobi identity on $\mathfrak{h}_{\mathrm{poly}}$ to its constant Taylor coefficient:

$$\omega(f, \{g, h\}) + \omega(g, \{h, f\}) + \omega(h, \{f, g\}) = 0.$$

Antisymmetry of ω is inherited from antisymmetry of the bracket. Hence $\omega \in Z^2_{\mathrm{Lie}}(\mathfrak{h}_{\mathrm{poly}}, \mathbb{C})$.

Exactness on the unreduced algebra. Set $\eta(f) = -[f]_0$. Then for all $f, g \in \mathfrak{h}_{\mathrm{poly}}$, $\delta\eta(f,g) = -\eta(\{f,g\}) = [\{f,g\}]_0 = \omega(f,g)$.

Obstruction after constant-quotient. The cocycle ω descends to $\bar{A}$ because it already vanishes on the constant line. The primitive η has value $\eta(1) = -1$ on the constant line. Suppose some $\bar{\eta}\colon \bar{A} \to \mathbb{C}$ primitive of $[\bar{c}]$ existed; lift to $\tilde{\eta}\colon \mathfrak{h}_{\mathrm{poly}} \to \mathbb{C}$ by setting $\tilde{\eta}(1) = 0$. Then $\tilde{\eta}(\{z_1, z_2\}) = \tilde{\eta}(1) = 0$, while $\omega(z_1, z_2) = 1$, contradicting $\delta\tilde{\eta} = \omega$.

Open-side image. The closed cocycle on the unit bidegree is $\omega(z_1, z_2) = 1$; the boundary evaluation $\partial^{\mathrm{full}}_{\mathrm{bb},N}$ of Theorem 6.4 multiplies by $\mathrm{Tr}\, I = N$ to give $\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = \mathrm{Tr}(I \cdot I) = N$. Same axis, two faces. $\square$

The class $[\bar{c}]$ is Chevalley–Eilenberg cohomology on the closed side and the $\mathfrak{gl}_N$ trace centre on the open side.

THEOREM 6.3 *(Closed-side face: the scalar abelian extension).* The class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ of Lemma 6.2 is the obstruction class to splitting the abelian Lie extension

$$0 \longrightarrow \mathbb{C} \cdot 1 \longrightarrow \mathfrak{h}_{\mathrm{poly}} \longrightarrow \bar{A} \longrightarrow 0.$$

Equivalently, $\mathfrak{h}_{\mathrm{poly}} \cong \bar{A}_{[\bar{c}]} = \bar{A} \oplus \mathbb{C}K$ as Lie algebras, with bracket

$$[\,(f, aK), (g, bK)\,] = (\{f,g\}_{\bar{A}},\ \bar{c}(f,g)\,K),$$

and $\mathbb{C} \cdot [\bar{c}] \subset H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ is the distinguished anomaly line attached to the constant-Hamiltonian central extension.

Proof. The constant Hamiltonian $1 \in \mathfrak{h}_{\mathrm{poly}}$ has zero Hamiltonian vector field, so $\mathbb{C} \cdot 1$ is central; the displayed sequence is a central extension and its class is its standard Chevalley–Eilenberg obstruction. Computing on monomial generators,

$$[z_1^k z_2^l, z_1^m z_2^n]_{\mathfrak{h}_{\mathrm{poly}}} = (kn - lm) z_1^{k+m-1} z_2^{l+n-1},$$

and projecting to the central line $\mathbb{C} \cdot 1$ keeps the $(0,0)$-Taylor coefficient. This is the cocycle ω, hence the extension class is $[\bar{c}]$. Non-triviality is Lemma 6.2; the scalar-extension line $\mathbb{C} \cdot [\bar{c}] \subset H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ is therefore nonzero, and splitting the corresponding single extension is the problem of lifting back to $\mathfrak{h}_{\mathrm{poly}}$. $\square$

THEOREM 6.4 *(Open-side face: the U(1) trace centre at finite N).* At finite N the matrix Poisson bracket $\{F, G\} = \mathrm{Tr}(\partial_{\phi_1} F\, \partial_{\phi_2} G - \partial_{\phi_2} F\, \partial_{\phi_1} G)$ on $\mathbb{C}[\mathfrak{gl}_N \oplus \mathfrak{gl}_N]^{GL_N}$ satisfies

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$$

in the unreduced ring; this scalar vanishes in the reduced quotient $H^0_{\mathrm{red}} = H^0/\mathbb{C} \cdot \mathrm{Tr}(1)$. Under the boundary evaluation map

$$\partial^{\mathrm{full}}_{\mathrm{bb},N}\colon \mathbb{C}[z_1, z_2] \longrightarrow \mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}, \qquad f \longmapsto \mathrm{Tr}\, f(\phi_1, \phi_2),$$

the closed cocycle ω on $\bar{A}$ multiplies by $\mathrm{Tr}\, I = N$ to give the open trace bracket on the central direction; the resulting central extension of the open trace algebra is therefore the image of $[\bar{c}]$. Since $\mathbb{C} \cdot I \subset \mathfrak{gl}_N$ is the U(1) centre with character Tr, the brane realisation of $[\bar{c}]$ is exactly the U(1) centre of $\mathfrak{gl}_N$ with charge N.

Proof. On the linear generators $\partial_{\phi_1}\operatorname{Tr}\phi_1 = I$ and $\partial_{\phi_2}\operatorname{Tr}\phi_2 = I$; the cross derivatives vanish. The matrix Poisson bracket is therefore $\operatorname{Tr}(I \cdot I) = N$ in the unreduced ring and zero modulo $\mathbb{C} \cdot \operatorname{Tr}(1)$. The boundary evaluation map sends $1 \in \mathbb{C}[z_1, z_2]$ to $\operatorname{Tr} I = N$, and on the cocycle bidegree $(k+m, l+n) = (1,1)$,

$$\partial^{\mathrm{full}}_{\mathrm{bb},N}\big(\{z_1^k z_2^l, z_1^m z_2^n\}\big) = (kn - lm) \cdot N = N \cdot \omega(z_1^k z_2^l, z_1^m z_2^n),$$

which is the closed cocycle of Lemma 6.2 times the multiplier $\operatorname{Tr} I = N$. The central extension class on the open side is therefore the image of $[\bar{c}]$, and the central generator $I \in \mathfrak{gl}_N$ is detected by the trace character. $\square$

Remark 6.5 *(Reduced quotient).* Theorems below using $\mathfrak{h}$, $\bar{A}$, or $\mathcal{A}_\partial = H^\circ_{\mathrm{red}}$ live *after* killing the scalar; the constant central line is removed on each side. The unreduced statement keeps $\mathbb{C} \cdot 1 \subset \mathfrak{h}_{\mathrm{poly}}$ closed-side and $\mathbb{C} \cdot \operatorname{Tr}(1)$ open-side, and the gap is $[\bar{c}]$.

Quantisation tests trace-map compatibility. Two operations close in on the brane: Weyl/Moyal quantization $\Phi_\hbar$ of the formal symplectic polydisk, and the matrix-trace map J of the quantized symbol on the Chan–Paton pair. In the projective trace representation the Moyal commutator has the same central cocycle as the trace commutator $[\overline{\operatorname{Tr}\phi_1^\hbar}, \overline{\operatorname{Tr}\phi_2^\hbar}]$. Linear Hamiltonians expose the first quantum obstruction. The Moyal expansion truncates because each higher-order term P^r requires r derivatives in each tensor factor and the second derivative of a linear function vanishes; the surviving leading term is the projective scalar $\hbar N \cdot \mathbf{1}$, exactly the order-$\hbar$ image of $\omega(z_1, z_2) = 1$ on the unreduced algebra. The classical class $[\bar{c}]$ and its order-$\hbar$ Capelli shift are one Lie 2-cocycle, read on the closed Lie side, on the unreduced trace algebra side, and on the Moyal/Weyl side; this is the quantum face of the brane trace comparison. Linear trace generators have only the order-$\hbar$ correction.

THEOREM 6.6 *(Moyal survival of $\bar{c}$: the order-$\hbar$ Capelli shift).* Let $\Phi = (\phi_1, \phi_2)$ be the classical brane field and $\Phi_\hbar = (\phi_1^\hbar, \phi_2^\hbar)$ its Weyl quantisation at finite N, with the Moyal product [7] $f * g = m \circ \exp\big(\frac{\hbar}{2}P\big)(f \otimes g)$, $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$. In the degree-zero quantum Hamiltonian reduction the normal-ordered comoment imposes the Capelli/canonical relation

$$Y_\hbar X_\hbar - X_\hbar Y_\hbar + \hbar N\, I \equiv 0, \qquad X_\hbar = \phi_1^\hbar,\ Y_\hbar = \phi_2^\hbar,$$

modulo the quantum moment-map ideal. This is a matrix Weyl-algebra congruence in the projective Hamiltonian reduction, not an equality in a finite-dimensional matrix representation. On the linear trace generators $X = \operatorname{Tr}\phi_1$, $Y = \operatorname{Tr}\phi_2$ this is the classical bracket $\{\operatorname{Tr}\phi_1, \operatorname{Tr}\phi_2\} = N$ of Theorem 6.4 times $\hbar$; higher-order Moyal corrections vanish. Equivalently, the Rees quantisation $U_\hbar(\bar{A}_{[\bar{c}]})$ of the closed-side central extension carries the cocycle $[\bar{c}]$ as its leading-order PBW splitting obstruction, and the trace character $K \mapsto N\mathbf{1}$ on the central generator returns the scalar Capelli contact $\hbar N$.

Proof. Linear-generator collapse of the Moyal expansion. The Weyl convention gives $[X_\hbar, Y_\hbar]_* = \hbar\{X, Y\} + O(\hbar^3)$ in general. On the linear trace generators $X = \operatorname{Tr}\phi_1$, $Y = \operatorname{Tr}\phi_2$, every higher-order term P^r requires at least r derivatives in each tensor factor, but the second derivative of a linear function vanishes. The Moyal expansion truncates at first order, so

$$X_\hbar Y_\hbar - Y_\hbar X_\hbar = \hbar\{X, Y\} = \hbar N$$

exactly. Rearranging gives the displayed normal-ordered moment relation in the quantum Hamiltonian reduction. The order-one Capelli identity [26][27] is the same relation read on the universal-enveloping side; the Razmyslov–Procesi description [18][17] of the full trace-relation ideal at finite N is separate from this calculation.

Central-extension matching. The Rees algebra $U_\hbar(\bar{A}_{[\bar{c}]})$ has special fibre $\widehat{\mathbf{S}}(\bar{A}_{[\bar{c}]})$ at $\hbar = 0$ and contains $\bar{A}_{[\bar{c}]}$ as a Lie subalgebra at $\hbar = 1$. PBW associated-graded carries the cocycle ω as the primary obstruction to splitting the deformation: ω is the leading Moyal commutator on the central direction. Evaluating the central generator K by the finite-N trace character $K \mapsto NI$ converts the order-$\hbar$ central term into the scalar Capelli contact $\hbar N$.

Conclusion. Classical $[\bar{c}]$ and order-$\hbar$ quantum Capelli shift carry the same Lie 2-cocycle on $\bar{A}$. They are the same datum, read at $\hbar = 0$ and at first order in $\hbar$. □

Remark 6.7 *(Three faces, one class).* The mismatch class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ has three named faces:

- closed side, abelian-extension obstruction (Theorem 6.3);
- open side, U(1) trace centre with charge N (Theorem 6.4);
- Moyal side, order-$\hbar$ Capelli contact (Theorem 6.6).

Scalar reduction kills the empty-trace scalar $\mathrm{Tr}(1) = N$; the linear trace observables $\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2$, whose unreduced bracket is N and whose reduced bracket is zero, remain. The obstruction beyond this scalar sector is the unreduced factorisation centre of Problem 5.49 and the higher-bidegree QME obstruction of Problem 5.220: $[\bar{c}]$ is the scalar component, those problems carry the higher-bidegree components.

THEOREM 6.8 *(Capelli scalar and determinant-line condition).* Let $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$, with the completed formal meaning for $f \in \mathfrak{h}$, be the trace generator of §4.2. The trace representation of the central extension $0 \to \mathbb{C}K \to \bar{A}_{[\bar{c}]} \to \bar{A} \to 0$, with $K \mapsto N$, has projective Lie anomaly

$$\kappa_{\mathrm{Cap}}(J) = \hbar N [\bar{c}] \qquad \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})[[\hbar]],$$

where $[\bar{c}]$ is the Lie 2-cocycle of Lemma 6.2. If a determinant line $\det_J^{1/2}$ and a connection ∇_J are constructed for this projective trace representation, and if their Atiyah class pulls back to $[\bar{c}]$, then the curvature of ∇_J restricts to the same class $\hbar N [\bar{c}]$. The determinant-line construction supplies the line and connection; the local theorem supplies the Lie cocycle.

Proof. The cocycle identity is the content of Theorem E.13: the four representatives of $[\bar{c}]$ coincide under the canonical scalar projection, with the multiplier $\hbar N$ coming from the open Chan–Paton trace pairing. The brane E_1-algebra $A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}(\mathfrak{gl}_N, \mathrm{Kosz}([\phi_1, \phi_2]))$ on the worldline $\mathbb{R}_t$ has trace coordinate $u_f \mapsto J(f)$ and CE coordinate $c_f \mapsto \theta_f$ by §4.1; under this identification the projective Capelli normal-ordered shift $Y_\hbar X_\hbar - X_\hbar Y_\hbar + \hbar N\, I \equiv 0$ (Theorem 6.6) is the formal-Darboux expression of the trace representation of the central generator $K \mapsto N$. The linear Hamiltonian generators z_1, z_2 act on the trace representation with classical Capelli identity [26][27]; its order-$\hbar$ deformation is the central extension with cocycle $\hbar N \omega$, where $\omega(z_1, z_2) = 1$ by Corollary C.8. Hence the Lie cohomology class is $\hbar N [\bar{c}]$. A determinant-line curvature formula is a statement about the determinant line and connection named in the theorem. □

Remark 6.9 *(Determinant line and scheme dependence).* The local construction gives a projective Lie central extension, not a determinant line by itself. A square-root determinant line $\det_J^{1/2}$, a connection on it, and a pullback of its Atiyah class to $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ are additional data. When those data exist and the pullback is $[\bar{c}]$, the Capelli shift $\hbar N\, I$ is the curvature value at the linear trace generators. The numerical coefficient $\hbar N$ is scheme-dependent in the sense of the factorization-current convention table §B: the canonical scheme is the symmetric Weyl ordering $z_1 z_2 \mapsto \frac{1}{2}(\phi_1\phi_2 + \phi_2\phi_1)$ with Moyal parameter $\hbar$ and Chan–Paton trace $\operatorname{Tr} I = N$; changing scheme by an inner automorphism shifts $[\bar{c}]$ by a coboundary and leaves the class invariant. Reordering by an outer automorphism (e.g. a triangular Capelli splitting versus normal ordering) shifts the trace coefficient by a central character. Preservation of a modular-line interpretation is a statement about the determinant-line datum.

BV-cohomology vanishing and the unreduced cotangent lift

The linearized BV differential $d_{\mathrm{BV}} = Q + \{I_0, -\}$ of degree +1 acts on $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ for the mixed geometry $X = \mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ with brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ and ordinary $\mathfrak{gl}_N$ gauge data. An endpoint-admissible coefficient module for Problem 5.49 extends the already computed native holomorphic singular product of Theorem 1.17 to the unreduced Costello/QME endpoint; it requires the four-component class

$$\mathfrak{o}_{\mathrm{BV,end}} = \left(\mathfrak{o}_{\mathrm{Cas}},\ \mathfrak{o}_{\mathrm{def}},\ \mathfrak{o}_{\mathrm{RG}},\ \mathfrak{o}_{\mathrm{QME}}\right)$$

of Definition 6.16 to vanish (Proposition 6.17): the strict-pair Casimir cokernel $\mathfrak{o}_{\mathrm{Cas}}$ (5.212), the brane-line propagator extension obstruction $\mathfrak{o}_{\mathrm{def}}$ (8.17 (2)), the strict-window Mittag–Leffler preservation class $\mathfrak{o}_{\mathrm{RG}}$ (5.16), and the local-functional QME anomaly class $\mathfrak{o}_{\mathrm{QME}} \in H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ of Problem 5.73.

The BV grading is the cotangent-coordinate grading of E.1 together with the signs of C.1, C.8: $|\Delta| = +1$, the reduced boundary P_0-bracket has degree 0, the Schouten bracket has degree −1.

The closed BCOV vanishing of [13] and the open-closed extension of [14] differ from the present geometry along five independent axes: compact Hodge geometry against the formal jet $\widehat{\mathbb{C}^2}_0$; pure holomorphic-volume kinematics $\bar{\partial} + t\,\partial_\Omega$ against mixed $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$; empty bulk against the transverse line defect L; the $\mathfrak{gl}(N \mid N)$ supertrace cancellation $\operatorname{str}(I) = N - N = 0$ against the ordinary $\mathfrak{gl}_N$ scalar contact $\hbar N[\bar{c}]$ of E.12; the worldsheet t-Tate direction against the coefficient-system Tate direction $\widehat{\mathfrak{h}}\widehat{\otimes}\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ carrying $\mathfrak{o}_{\mathrm{Cas}}$. Each axis names a distinct endpoint component of $\mathfrak{o}_{\mathrm{BV,end}}$ under the cited hypotheses (Theorem 6.15); the data needed to close them is enumerated in Theorem 6.18. The chain-level primitive shadow 5.133 closes the classical indecomposable shadow by an algebraic argument inside the admissible Tate envelope of 5.31.

BV degree convention and Costello renormalization datum

Remark 6.10 *(BV grading and Δ for the brane-defect endpoint).* The BV grading is fixed by E.1: $|\phi_1| = |\phi_2| = 0$, $|\psi = A^*| = -1$, $Q\psi = [\phi_1, \phi_2]$, $|Q| = +1$; a Hamiltonian generator H_f has degree 0, a shifted cotangent generator $\eta_\rho = \rho[1]$ is odd of BV degree −1, and the CE cotangent–coordinate grading is $|c_f| = +1$, $|u_\rho| = 0$. In this convention the linearized BV differential

$$d_{\mathrm{BV}} = Q + \{I_0, -\}$$

acting on $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ has degree +1, and a quantum master equation *anomaly representative* is a degree-+1 cocycle in $\left(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), d_{\mathrm{BV}}\right)$. The Schouten/cotangent–coordinate antibracket has degree −1; the reduced Hamiltonian boundary P_0-bracket has degree 0. The scalar Hamiltonian cocycle $\omega(z_1, z_2) = 1$ of C.6 controls the central contribution to the symbol, and the local Poisson sign convention $\{z_1, z_2\} = 1$ is the one of C.1.

Remark 6.11 *(Costello (2011) heat-kernel renormalization datum invoked).* The final obstruction comparison invokes the renormalization formalism of [12, Ch. 2 §2, Ch. 2 §6, Ch. 5 §6]: a parametrix $P_{\epsilon,L} = \int_\epsilon^L ds\, K_s$ for a generalized Laplacian, the homotopy RG flow

$$W(P_{\epsilon,L}, I) = \hbar \log\bigl(\exp(\hbar\partial_{P_{\epsilon,L}}) \exp(I/\hbar)\bigr) = \sum_{\Gamma \text{ connected}} \frac{\hbar^{\text{loops}(\Gamma)}}{|\text{Aut}(\Gamma)|} W_\Gamma(P_{\epsilon,L}, I),$$

the scale-zero limit $\lim_{\epsilon\to 0} W(P_{\epsilon,L}, I)$ constructed by counterterm subtraction, and the QME $QI + \hbar\Delta_L I + \frac{1}{2}\{I, I\}_L = 0$ at scale L. In the present brane-defect geometry the propagator factorizes (when it exists) as $P^w_{\epsilon,L} = P^{\text{base}}_{\epsilon,L} \widehat{\otimes} C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$ (8.17); the bulk Hadamard parametrix is taken from [12, Ch. 3] on $\mathbb{R}^2_{\text{top}} \times \mathbb{C}^2_{\text{hol}}$ and the coefficient factor is the algebraic kernel of T1. The components $\mathfrak{o}_{\text{def}}$ and $\mathfrak{o}_{\text{RG}}$ of $\mathfrak{o}_{\text{BV,end}}$ measure the brane-line extension class of $P^{\text{base}}_{\epsilon,L}$ and the strict-window preservation class of $W(P_{\epsilon,L}, -)$ inside this convention.

Costello–Li 2012 hypotheses, restated

Statement 6.12 *(Costello–Li 2012 BCOV construction, restated).* Let X satisfy the compact Calabi–Yau holomorphic-volume hypotheses of [13], with volume form Ω, $\dim_{\mathbb{C}} X = d$. Let $\text{PV}(X)[[t]] = \bigoplus_{i,j} \text{PV}^{i,j}(X)[[t]]$ be the polyvector-field complex with the Tate parameter t for the worldsheet S^1 rotation, BV differential

$$Q_{\text{BCOV}} = \bar{\partial} + t\,\partial_\Omega, \qquad |Q_{\text{BCOV}}| = +1,$$

with $\partial_\Omega = \Omega^{-1} \circ \partial \circ \Omega$ the divergence operator, and the (-1)-shifted symplectic pairing

$$\langle \alpha, \beta \rangle = \int_X (\alpha \wedge \beta)\big|_{\text{PV}^{d,d}} \cdot \Omega^2.$$

The classical BCOV interaction I_{BCOV} is the cubic vertex of [13, §3]. The cited construction gives the generalized classical BCOV complex, its local-functional obstruction complex, and the renormalization formalism of [12] adapted to compact Calabi–Yau X. The 2015 open-closed sequel [14] extends the construction to flat $\mathbb{C}^d$ (d odd) coupled to open gauge data $\mathfrak{gl}(N \mid N)$, with the U(1)-anomaly cancellation $\text{str}(I) = N - N = 0$. Each is used here only under its stated geometric and coefficient hypotheses.

Hypothesis separation on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $L = \mathbb{R}_{\text{brane}} \times \{0\} \times \{p\}$ and $\mathfrak{gl}_N$

PROPOSITION 6.13 *(Costello–Li 2012/2015 hypothesis separation for the brane-defect endpoint).* Let $\mathfrak{o}_{\text{BV,end}} = (\mathfrak{o}_{\text{Cas}}, \mathfrak{o}_{\text{def}}, \mathfrak{o}_{\text{RG}}, \mathfrak{o}_{\text{QME}})$ be the four-component endpoint obstruction class of 6.16 for the formal local geometry $X = \mathbb{R}^2_{\text{top}} \times \widehat{\mathbb{C}^2}_{0,\text{hol}}$ with brane $L = \mathbb{R}_{\text{brane}} \times \{0\} \times \{p\}$ and ordinary $\mathfrak{gl}_N$ gauge data. The hypotheses of 6.12 differ from this local model along five independent axes. In each axis the corresponding endpoint obstruction is named at the level of a cocycle representative:

(S1) *Compact Hodge-theoretic geometry vs. formal jet.* The 2012 theorem requires compact Calabi–Yau X with Hodge-theoretic harmonic projection $\Pi_{\text{harm}} : \Omega^{0,\bullet} \to \mathcal{H}^{0,\bullet}$. On the formal jet $\widehat{\mathbb{C}^2}_0$, the corresponding datum is the one-dimensional cohomology $H^*_{\text{Dol}}(\widehat{\mathbb{C}^2}_0) = \mathbb{C}$ in degree zero. Hence the 2012 BV-cohomology vanishing $H^*_{\text{BV,BCOV}}(X) = 0$ above degree zero is a statement about the compact BCOV complex. The present $\bigl(\mathcal{O}_{\text{loc}}(\mathcal{E}_w), d_{\text{qBV}}\bigr)$ on $X = \mathbb{R}^2_{\text{top}} \times \widehat{\mathbb{C}^2}_{0,\text{hol}}$ has its own cocycle representative of $\mathfrak{o}_{\text{QME}}$. The 2015 open-closed theorem replaces compact X by flat $\mathbb{C}^d$ (d odd) but keeps the holomorphic-volume kinematics and the $\mathfrak{gl}(N \mid N)$ anomaly cancellation; it is a different geometric and coefficient setting.

(S2) *Pure holomorphic-volume vs. mixed topological–holomorphic kinematics.* The 2012 BV differential is $Q_{\mathrm{BCOV}} = \bar{\partial} + t\,\partial_\Omega$ on $\mathrm{PV}(X)[[t]]$. The kinematic differential of the present model is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ on $\Omega^{\bullet}_{\mathbb{R}^2}(I)\widehat{\otimes}\Omega^{0,\bullet}_{\mathbb{C}^2}\widehat{\otimes}(\mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1])$, the holomorphic E_2 factorization algebra on $\mathbb{C}^2$. The two complexes have distinct generators and differentials, so transfer of a Costello–Li class on $(\mathrm{PV}(X)[[t]], Q_{\mathrm{BCOV}})$ to a cocycle of Δ on the mixed complex requires a kinematic comparison map. The protected sector of twisted M-theory [25] carries this mixed kinematic shape as physical motivation.

(S3) *Brane and topological line defect.* The 2012 theorem concerns the closed sector; the propagator $P^{2012}_{\epsilon,L}$ on $\mathrm{PV}(X)[[t]]$ is a closed-sector propagator. The 2015 theorem has an open-closed sector on flat $\mathbb{C}^d$ with open gauge $\mathfrak{gl}(N \mid N)$, where the open boundary is the boundary of a worldsheet disk in $\mathbb{C}^d$. The present endpoint uses a transverse line $L \subset \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2$ point-supported in the holomorphic direction. Extension of $P^{\mathrm{base}}_{\epsilon,L}$ across L with ordinary $\mathfrak{gl}_N$ brackets is encoded by the obstruction $\mathfrak{o}_{\mathrm{def}}$ of 6.16.

(S4) *Supertrace cancellation vs. ordinary $\mathfrak{gl}_N$ scalar contact.* On $\mathfrak{gl}(N \mid N)$ the [14] cancellation $\mathrm{str}(I) = N - N = 0$ kills the putative U(1)-anomaly at every loop order. On ordinary $\mathfrak{gl}_N$ the same closed exchange gives the scalar contact term $\hbar N\,\omega(f,g)\gamma_1$ of E.12, with cohomology class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A},\mathbb{C})$. If a determinant line with connection exists and its Atiyah class pulls back to $[\bar{c}]$, its curvature restricts to this class at the trace generator $J(f)$ (Theorem 6.8). Costello–Li graph signs leave this class as the local trace-line representative of the cancelled supertrace anomaly; a graph-level comparison with the Costello–Li cancellation is additional data. Hence $\mathfrak{o}_{\mathrm{QME}}$ carries the residual $\hbar N[\bar{c}]$ summand for ordinary $\mathfrak{gl}_N$ at the U(1) centre-of-mass slice, and that summand is a separate endpoint cocycle in the stated Costello–Li BV complex.

(S5) *Worldsheet t-Tate vs. coefficient-system Tate.* The 2012 Tate parameter t is a worldsheet S^1 rotation completion of the polyvector field complex. The Tate direction required by Problem 5.49 is along the coefficient pair

$$\widehat{\mathfrak{h}}\,\widehat{\otimes}\,\widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}, \qquad C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_{i} e_{d,i}\otimes e^{i}_{d}.$$

By 5.212 the strict pair $(\mathfrak{h}_{\Pi}, D_{\oplus})$ has the represented-kernel obstruction for simultaneous projection to every finite Casimir $C^{M}_{\mathfrak{h}}$: the dual side admits only finitely many degree components, and the compatible kernel has nonzero identity component in every $H^{\vee}_{d}$. Hence the cocycle representative $\mathfrak{o}_{\mathrm{Cas}} = [(C^{M}_{\mathfrak{h}})_M]$ is nonzero in the cokernel. This coefficient direction is governed by the endpoint Tate completion; the Costello–Li t-Tate completion is the worldsheet completion.

Each axis identifies a distinct endpoint component of $\mathfrak{o}_{\mathrm{BV,end}}$: (S1)+(S2) give the kinematic comparison problem for $\mathfrak{o}_{\mathrm{QME}}$; (S3) gives $\mathfrak{o}_{\mathrm{def}}$; (S4) names the residual $\hbar N[\bar{c}]$ summand of $\mathfrak{o}_{\mathrm{QME}}$; (S5) names the cokernel class of $\mathfrak{o}_{\mathrm{Cas}}$. The remaining component $\mathfrak{o}_{\mathrm{RG}}$ is the strict finite-window Mittag-Leffler preservation class for the recursion $W(P_{\epsilon,L}, I)$ of 6.11; this preservation is an endpoint coefficient condition in addition to the compact-target Costello–Li theorems.

Proof. Each axis is a direct comparison of the hypotheses of 6.12 (and its 2015 sequel) with the data of 5.6, the linear-heat / Casimir analysis (5.211, 5.212), the scalar contact theorem E.12, and the strict-window category 5.16.

(S1). Compactness enters [13] through the Hodge-theoretic harmonic projection used to construct the BV propagator and to verify the QME at the parametrix limit $\epsilon \to 0$, $L \to \infty$. On the formal jet $\widehat{\mathbb{C}^2}_0$, the corresponding datum is the one-dimensional Dolbeault cohomology $H^{*}_{\mathrm{Dol}}(\widehat{\mathbb{C}^2}_0)$.

(S2). The 2012 BV differential is purely Dolbeault plus the divergence $t\,\partial_\Omega$; the present $D = d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}$ carries an additional de Rham factor on the topological line. A symmetric-monoidal comparison map $(\mathrm{PV}(X)[[t]], Q_{\mathrm{BCOV}}) \to (\Omega^\bullet_{\mathbb{R}^2}(I)\widehat{\otimes}\Omega^{0,\bullet}_{\mathbb{C}^2}\widehat{\otimes}\cdots, D)$ is additional endpoint data.

(S3). The brane L is transverse to the topological factor and point-supported in the holomorphic factor; the open boundary in [14] is a worldsheet boundary on flat $\mathbb{C}^d$. The present endpoint requires the Wilson line on a transverse topological direction. The bulk Hadamard parametrix $P^{\mathrm{base}}_{\epsilon,L}$ of [12, Ch. 3] on $\mathbb{R}^2 \times \mathbb{C}^2_{\mathrm{hol}}$ gives the bulk kernel; the brane-line trace is a further endpoint datum.

(S4). By E.12 the closed exchange between two Hamiltonian sources B_f, B_g on the brane line produces the universal scalar contact density $\hbar N\,\omega(f,g)\,\gamma_1(t)\,a(t)b(t)$. Its associated-graded scalar symbol is the Lie 2-cocycle $\omega(f,g)\,\gamma_1$ on $\bar A$, with $\omega(z_1,z_2) = 1$ by C.8; this is a nonzero class in $H^2_{\mathrm{Lie}}(\bar A,\mathbb{C})$ by 5.6 part (f). On $\mathfrak{gl}(N \mid N)$ the same scalar is multiplied by $\mathrm{sdim}(\mathbb{C}^{N|N}) = N - N = 0$, killing the representative. On ordinary $\mathfrak{gl}_N$ it is $\hbar N$, and absorption of this term requires an endpoint counterterm.

(S5). The compatible Casimir family $(C^M_{\mathfrak{h}})_M$ projects nonzero into every $H_d \otimes H_d^\vee$ for $d \le M$, while the strict direct-sum dual $D_\oplus = \bigoplus_d H_d^\vee$ has only finitely many nonzero degree components per element. Thus 5.212 identifies a nonzero cokernel class for the map

$$\mathfrak{h}_\Pi\widehat{\otimes}D_\oplus \longrightarrow (\mathfrak{h}_\Pi\widehat{\otimes}D_\oplus)^{\wedge\bullet}.$$

Equivalently the cocycle representative of $\mathfrak{o}_{\mathrm{Cas}}$ is nonzero, and the worldsheet t-Tate completion of [13] commutes with this coefficient direction trivially.

Finally, $\mathfrak{o}_{\mathrm{RG}}$ is the assertion that $W(P_{\epsilon,L},-)$ sends the strict ML category $\mathbf{MLTateChain}^{\mathrm{str}}_{\ge 0}$ of 5.16 into itself with surjective finite-window transition maps; this is a statement about the algebraic-coefficient direction of the propagator, while the Costello–Li theorems address the worldsheet direction. □

COROLLARY 6.14 *(Costello–Li construction and the unreduced coefficient endpoint).* The Costello–Li 2012 BCOV construction gives a pure closed BCOV field theory; the 2015 sequel gives the open-closed extension on flat $\mathbb{C}^d$ with $\mathfrak{gl}(N \mid N)$ gauge data. The present $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ brane-defect endpoint is governed by the four cocycle classes in $\mathfrak{o}_{\mathrm{BV,end}}$. The unreduced Tate-coefficient cotangent lift therefore additionally requires the endpoint coefficient data of Definition 5.95. The classical primitive indecomposable shadow on the open observables is closed by the algebraic argument of 5.133; the admissible Tate envelope on which the obstruction tuple is computed is provided by the model-categorical envelope of 5.18. The unreduced cotangent factorization-centre morphism and the quantum one-antifield extension of part (e) of 5.6 require the weighted coefficient category of T1 and the mixed brane-defect locality/QME datum of Problem 5.73 together with 8.17.

Proof. By Proposition 6.13, each Costello–Li hypothesis selects a distinct final BV comparison problem for a cocycle representative in $\mathfrak{o}_{\mathrm{BV,end}}$. Vanishing of the tuple is therefore the additional final BV datum specified in the corollary. The remaining clauses are the classical chain-level lift of 5.133, the model-categorical envelope of 5.18, and the weighted finite-scale coefficient kernel of T1; the endpoint data are exactly those listed in Definition 5.95. □

THEOREM 6.15 *(Costello–Li endpoint coefficient criterion).* The Costello–Li 2012 BCOV construction [13] and its 2015 open-closed extension [14], in the forms restated in 6.12, give an endpoint-admissible coefficient module of Definition 5.95 for the formal local brane-defect geometry only after three independent classical data are added:

(E2) coefficient Casimir kernel, (E3) mixed brane-defect transport, (E5) endpoint RG preservation, with (E1) and (E4) the inherited finite-window and Mittag-Leffler envelope conditions. A quantum endpoint additionally requires vanishing of the QME component $\mathfrak{o}_{\mathrm{QME}}$ of Definition 6.16. Equivalently: the four cocycle representatives in $\mathfrak{o}_{\mathrm{BV,end}}$ require independent endpoint primitives on this geometry.

Proof. An endpoint-admissible coefficient module requires a coefficient kernel $C_* \in \mathfrak{h}_* \widehat{\otimes} D_*$ whose projection to every finite Hamiltonian window is the finite Casimir $C_{\mathfrak{h}}^M$, together with mixed brane-defect transport of the bracket, propagator, interaction, counterterms, and RG recursion (Definition 5.95). By Proposition 6.13, each axis (S1)–(S5) names a distinct cocycle in $\mathfrak{o}_{\mathrm{BV,end}}$: (S5) gives $\mathfrak{o}_{\mathrm{Cas}}$, (S3) gives $\mathfrak{o}_{\mathrm{def}}$, the algebraic-coefficient direction gives $\mathfrak{o}_{\mathrm{RG}}$, and (S1)+(S2)+(S4) give $\mathfrak{o}_{\mathrm{QME}}$. Hence (E2), (E3), (E5), and the QME vanishing are the additional endpoint data. □

Definition 6.16 *(BV endpoint obstruction tuple, with explicit cocycle representatives).* For the formal mixed local geometry $X = \mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ with brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$ and ordinary $\mathfrak{gl}_N$ gauge data, the BV endpoint obstruction tuple is

$$\mathfrak{o}_{\mathrm{BV,end}} = \big(\mathfrak{o}_{\mathrm{Cas}},\ \mathfrak{o}_{\mathrm{def}},\ \mathfrak{o}_{\mathrm{RG}},\ \mathfrak{o}_{\mathrm{QME}}\big).$$

Each component is named with its ambient cohomology, BV degree, and explicit cocycle representative:

(O1) *Casimir cokernel class* $\mathfrak{o}_{\mathrm{Cas}}$ — in $\mathrm{coker}\big(\mathfrak{h}_\Pi \widehat{\otimes} D_\oplus \to \varprojlim_M (\mathfrak{h} \otimes \mathfrak{h}^\vee)_{\le M}\big)$, degree 0; cocycle representative the compatible family

$$\mathfrak{o}_{\mathrm{Cas}} = \big[(C_{\mathfrak{h}}^M)_{M \ge 1}\big], \qquad C_{\mathfrak{h}}^M = \sum_{d \le M} \sum_i e_{d,i} \otimes e^i_d,$$

with transition maps the standard truncations $C_{\mathfrak{h}}^{M+1} \mapsto C_{\mathfrak{h}}^M$; nonzero in the strict pair by 5.212.

(O2) *Brane-defect transport class* $\mathfrak{o}_{\mathrm{def}}$ — in $H^0\big(\mathrm{Hom}^\bullet_{\mathrm{loc}}(P^{\mathrm{base}}_{\epsilon,L}, P^{\mathrm{brane}}_{\epsilon,L}),\ \partial_L\big)$, degree 0; cocycle representative the brane-line restriction datum

$$\mathfrak{o}_{\mathrm{def}} = \big[\partial_L^* P^{\mathrm{base}}_{\epsilon,L} - P^{\mathrm{brane}}_{\epsilon,L}\big]$$

together with the bracket-transport obstruction for $\{\cdot,\cdot\}_{\mathrm{open}}$ of C.9 across L with ordinary $\mathfrak{gl}_N$ brackets, equivalently the Hadamard locality class of 8.17 item (2).

(O3) *RG-window preservation class* $\mathfrak{o}_{\mathrm{RG}}$ — in the Mittag-Leffler $\varprojlim^1 H^0$ of the strict-window functional complex $\mathbf{MLTateChain}^{\mathrm{str}}_{\ge 0}$ of 5.16, degree 0; cocycle representative the compatible family of recursion-induced transition defects

$$\mathfrak{o}_{\mathrm{RG}} = \big[\mathrm{T}^{M,M-1} \circ W(P_{\epsilon,L}, -)|_{\le M} - W(P_{\epsilon,L}, -)|_{\le M-1} \circ \mathrm{T}^{M,M-1}\big] \in \varprojlim\nolimits^1_M H^0,$$

where $\mathrm{T}^{M,M-1}$ is the strict ML truncation of 5.16 and $W(P_{\epsilon,L}, -)$ is the Costello recursion of 6.11.

(O4) *QME anomaly class* $\mathfrak{o}_{\mathrm{QME}}$ — in the local-functional cohomology

$$H^1\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w),\ d_{\mathrm{qBV}}\big), \qquad d_{\mathrm{qBV}} = Q + \{I_0, -\} + \hbar\Delta_L,\ \ |d_{\mathrm{qBV}}| = +1,$$

degree +1; cocycle representative the local functional

$$\mathfrak{o}_{\mathrm{QME}} = \big[QI_0 + \hbar\Delta_L I_0 + \tfrac{1}{2}\{I_0, I_0\}_L\big]$$

produced by the quantum master equation of [12, Ch. 5] at scale L, restricted to the mixed brane-defect interaction I_0 of Problem 5.73. In the ordinary $\mathfrak{gl}_N$ branch this class carries the residual scalar contact summand $\hbar N\,\omega(f,g)\,\gamma_1$ of E.12, whose scalar projection at the U(1) centre-of-mass slice is the Capelli class $\hbar N[\bar{c}]$ of Chapter 6 (Theorem 6.8); in the balanced $\mathbb{C}^{N|N}$ branch the same summand vanishes by E.16. The scalar projection is therefore the local-stalk value of the projective Lie anomaly. It is the value of $\Omega_{\mathrm{central}}\big|_{J(f)}$ only after a determinant line with connection has been fixed as in Theorem 6.8.

PROPOSITION 6.17 *(Endpoint obstruction tuple).* An endpoint-admissible coefficient module of Definition 5.95 for the brane-defect geometry of 6.16 exists only if all four cocycle classes $\mathfrak{o}_{\mathrm{Cas}}$, $\mathfrak{o}_{\mathrm{def}}$, $\mathfrak{o}_{\mathrm{RG}}$, $\mathfrak{o}_{\mathrm{QME}}$ of 6.16 vanish. These vanishings are sufficient after the compatible primitives assemble in the strict Mittag–Leffler envelope of 5.16. In particular, a Costello–Li type BV-cohomology theorem closes the present coefficient endpoint only when it proves vanishing of all four classes simultaneously, with primitives compatible with the strict finite-window Mittag-Leffler envelope of 5.16.

Proof. (O1)⇔(E2). Definition 5.95(E2) is the existence of a coefficient kernel $C_* \in \mathfrak{h}_*\widehat{\otimes}D_*$ whose finite-window projections are the $C^M_{\mathfrak{h}}$; equivalently a primitive for the cocycle representative of $\mathfrak{o}_{\mathrm{Cas}}$ under the represented-kernel map.

(O2)⇔(E3). Condition (E3) requires transport of bracket $\{\cdot,\cdot\}_{\mathrm{open}}$, propagator $P^{\mathrm{base}}_{\epsilon,L}$, interaction I_0, and counterterm representatives across the brane line L with ordinary $\mathfrak{gl}_N$ brackets; equivalently a primitive for the Hom-complex cocycle representative of $\mathfrak{o}_{\mathrm{def}}$.

(O3)⇔(E5). Condition (E5) is preservation of the strict finite-window Mittag-Leffler tower under the Costello recursion $W(P_{\epsilon,L},-)$; equivalently vanishing of the $\varprojlim^1 H^0$ class of $\mathfrak{o}_{\mathrm{RG}}$.

(O4) and the quantum endpoint. The endpoint QME is the vanishing of the degree-+1 local-functional class $\mathfrak{o}_{\mathrm{QME}}$ in $H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), d_{\mathrm{qBV}})$, where $d_{\mathrm{qBV}} = Q + \{I_0,-\} + \hbar\Delta_L$. The operator Δ_L enters as the second-order summand of the quantum BV differential; it is not the differential whose cohomology carries the anomaly class.

Necessity is the same chain of equivalences in reverse: a nonzero component removes one of the endpoint data listed above. The primitive-compatibility clause is the requirement that the chosen cobounding primitives be morphisms in the strict ML category $\mathbf{MLTateChain}^{\mathrm{str}}_{\geq 0}$, so that the endpoint module sits in $\mathbf{TateChain}^{\mathrm{adm}}_{\geq 0}$ (5.18). □

Five-datum extension criterion

THEOREM 6.18 *(Costello–Li extension criterion for the mixed coefficient endpoint).* Let $\mathfrak{o}_{\mathrm{BV,end}}$ be the obstruction tuple of Definition 6.16. A Costello–Li type BV-cohomology theorem proves vanishing of $\mathfrak{o}_{\mathrm{BV,end}}$ on the formal local geometry $X = \mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$, brane $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{p\}$, gauge $\mathfrak{gl}_N$, only if it gives the following four classical and one quantum data simultaneously, each addressing a distinct cocycle representative in $\mathfrak{o}_{\mathrm{BV,end}}$:

- *(I1) Non-compact / Ω-background BV-cohomology vanishing* replacing (S1). A vanishing theorem for the linearized BV differential d_{BV} on $(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), d_{\mathrm{BV}})$ over the non-compact base $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$, either via a separately anchored Ω-background route in the normal bundle $N_L X = \mathbb{R}_s \oplus \mathbb{C}_{z_1} \oplus \mathbb{C}_{z_2}$, with the vector field, equivariant differential $Q_\Omega = Q + \iota_{V_\Omega}$, $Q^2_\Omega = L_{V_\Omega}$ mechanism, inverted normal weights, residue/Euler normalization, and stratified factorization on (X, L), or via compactification with Lagrangian boundary at infinity preserving the brane L as a fixed locus.

(I2) Topological–holomorphic mixed kinematics replacing (S2). The vanishing must hold for $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ on $\Omega^{\bullet}_{\mathbb{R}^2}(I)\widehat{\otimes}\Omega^{0,\bullet}_{\mathbb{C}^2}\widehat{\otimes}(\mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1])$, together with its linearized BV-kernel complex of 5.211 and the cotangent [1]-extension of the Hamiltonian sector (E.1).

(I3) Brane-defect extension with ordinary $\mathfrak{gl}_N$ scalar contact decision replacing (S3)–(S4). The vanishing must allow the transverse one-real-dimensional defect L carrying open gauge $\mathfrak{gl}_N$, with comparison to the flat-$\mathbb{C}^d$ $\mathfrak{gl}(N \mid N)$ setup of [14], and must decide the scalar contact class

$$\hbar N\, \omega(f, g)\, \gamma_{\mathbf{1}} \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}), \qquad \omega(z_1, z_2) = 1,$$

of E.12. After a determinant line with connection has been fixed, this is the value of $\Omega_{\mathrm{central}}\big|_{J(f)}$ at the trace generator (Theorem 6.8). The class is then decided by one of the four named cancellations: closed-exchange cancellation, $\mathbb{C}^{N|N}$ supertrace removal (E.16), central-character normalization (E.15), or renormalized boundary generators.

(I4) Coefficient-system Tate completion replacing (S5). The coefficient direction $\widehat{\mathfrak{h}}\,\widehat{\otimes}\,\widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}$ must support the Casimir $C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_i e_{d,i} \otimes e^i_d$ as a convergent represented kernel; by 5.212 this requires a coefficient category change, given for example by the weighted Tate pair $(\mathfrak{h}_w, \mathfrak{h}^{\vee,w}_{\mathrm{cont}})$ of T1.

(I5) Quantum endpoint vanishing. A primitive for the degree-+1 cocycle $\mathfrak{o}_{\mathrm{QME}} = \big[QI_0 + \hbar\Delta_L I_0 + \tfrac{1}{2}\{I_0, I_0\}_L\big]$ in $H^1\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), d_{\mathrm{qBV}}\big)$, with the Costello-2011 counterterm subtraction of 6.11 compatible with the strict-window Mittag-Leffler tower of 5.16. Casimir-balanced equivariant data of 5.149 kill the scalar component $\mathfrak{o}^{\mathrm{sc}}_{\mathrm{QME}}$; the non-scalar residual is the obstruction class of 5.84 in $H^1(\ker\widehat{\sigma}_{\mathrm{sc}}, d_{\mathrm{qBV}})$ together with the secondary $\varprojlim^1 H^0$ primitive class.

Equivalently, (I1)–(I5) are exactly the data of Problem 5.49 for the present brane-defect endpoint, after the arity-two formal OPE of Theorem 1.17 has been fixed.

Proof. By Proposition 6.13, axes (S1)–(S5) name pairwise distinct cocycle representatives in $\mathfrak{o}_{\mathrm{BV,end}}$: (S1)+(S2) give the kinematic side of $\mathfrak{o}_{\mathrm{QME}}$, (S3) for $\mathfrak{o}_{\mathrm{def}}$, (S4) for the residual $\hbar N[\bar{c}]$ summand of $\mathfrak{o}_{\mathrm{QME}}$, and (S5) for $\mathfrak{o}_{\mathrm{Cas}}$; the algebraic-coefficient direction governs $\mathfrak{o}_{\mathrm{RG}}$. Components (I1)–(I4) give primitives, in turn, for the cocycle representatives forced by (S1)+(S2), (S3)+(S4)$_{\mathrm{transport}}$, the strict-window Mittag-Leffler preservation of $W(P_{\epsilon,L}, -)$, and the Casimir convergence; (I5) gives the quantum primitive. By Proposition 6.17 this is exactly vanishing of $\mathfrak{o}_{\mathrm{BV,end}}$, with primitives compatible with $\mathbf{MLTateChain}^{\mathrm{str}}_{\geq 0}$ of 5.16.

For necessity: dropping (I1) leaves the kinematic representative of $\mathfrak{o}_{\mathrm{QME}}$ unprimitivated; dropping (I2) leaves D requiring its endpoint vanishing theorem; dropping (I3) retains $\mathfrak{o}_{\mathrm{def}}$ and the scalar contact summand of $\mathfrak{o}_{\mathrm{QME}}$; dropping (I4) retains $\mathfrak{o}_{\mathrm{Cas}}$ as a nonzero cokernel class by 5.212; dropping (I5) retains the degree-+1 anomaly primitive as endpoint data. The chain-level primitive shadow of 5.133 closes the classical indecomposable shadow by an algebraic argument; (I3) and (I5) give the brane-defect scalar and quantum endpoint data. The Quillen envelope of 5.31 embeds the strict category of (I4) in an admissible Tate model envelope. □

6.2 Operators in the closed Hamiltonian sector

On the closed side of the local mixed model, Hamiltonian-dual CE coordinates produce Schouten vector-field operations on the brane polynomial algebra, while shifted-cotangent-dual CE coordinates produce boundary Hamiltonian observables on the Chan–Paton pair. The closed local operator algebra is, after one kinematic contraction and one BRST identification, the continuous CE cochains

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{Ham}} \cong C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}), \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1],$$

with $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ the constants-quotiented Hamiltonian polydisk algebra and $\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P}$ its continuous residue dual (Theorem 5.179); this is Proposition 6.20.

The kinematic BRST operator $Q_0 = d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}$ contracts every positive topological and antiholomorphic jet by the Euler homotopy in the real direction (Lemma 3.21) and the formal Dolbeault Poincaré lemma in the holomorphic normal directions; Q_0-cohomology retains only the holomorphic Taylor coefficients of the Hamiltonian field α and the cotangent field β. The residual interaction differential reproduces the Lie brackets of $\mathfrak{g}$: two Hamiltonian α-fields contribute the closed Poisson bracket $\{f, g\}$, one Hamiltonian field and one cotangent field contribute the coadjoint action, and two cotangent fields contribute zero. The residual differential is the Chevalley–Eilenberg differential of $\mathfrak{g}$, and the closed local operator algebra is its continuous cochain algebra.

The open side of §3.5 gives the ghost-zero connected trace monomials $\overline{\mathrm{Tr}(\phi_1^k\phi_2^l)}$, the primitive one-$\psi$ Koszul descendants $\Psi_{a,b}$, and the open-trace A_∞-Koszul homotopy datum of Remark 3.44. The closed side matches these as continuous CE cochains via the coordinate CE/PV identity of Theorem 5.5: on the coordinate window, Hamiltonian-dual CE coordinates c^I map to Schouten polyvector coordinates θ^I on $\widehat{\mathbf{S}}(\bar A)$, and shifted-cotangent-dual CE coordinates u_I map to boundary Hamiltonian observables O_I on the Chan–Paton pair. The assignment $c_f \mapsto O_f$ is a type error (Theorem 4.20). The full stratified comparison is §5.7 below.

Let $\mathfrak{h} = \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ with the projected Poisson bracket, and let $\mathfrak{h}^\vee$ be its continuous Taylor dual. The cotangent field is a coadjoint module, so the local Hamiltonian Lie algebra is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1].$$

For $f, g \in \mathfrak{h}$ and $\lambda, \mu \in \mathfrak{h}^\vee$, the brackets are

$$[f, g] = \{f, g\}, \qquad [f, \eta_\lambda] = \eta_{\mathrm{ad}_f^\vee \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0,$$

where $(\mathrm{ad}_f^\vee \lambda)(g) = -\lambda(\{f, g\})$. On polynomial monomials in the dense subalgebra $\bar A$ this reads

$$[z_1^k z_2^l, z_1^m z_2^n] = \overline{(kn - lm) z_1^{k+m-1} z_2^{l+n-1}},$$

where the bar means projection to $\bar A$; in particular the constant bracket of z_1 and z_2 is zero in the reduced Hamiltonian Lie algebra.

LEMMA 6.19 *(Shifted coadjoint signs).* Let η_λ denote the suspension of $\lambda \in \mathfrak{h}^\vee$ into $\mathfrak{h}^\vee[1]$. Since the Hamiltonians in $\mathfrak{h}$ are even,

$$[f, \eta_\lambda] = \eta_{\mathrm{ad}_f^\vee \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0.$$

Consequently the enveloping algebra relations involving two shifted coadjoint generators are odd-commutative:

$$\eta_\lambda \eta_\mu + \eta_\mu \eta_\lambda = 0.$$

Proof. The semidirect product bracket before shifting is the ordinary coadjoint action of the even Lie algebra $\mathfrak{h}$ on its continuous dual. Suspending the module raises the degree of every λ by one. The mixed-bracket sign is $+1$ because $|f| = 0$. The shifted module is abelian as a Lie ideal, so its bracket with itself is zero; in the associative enveloping algebra this zero Lie bracket gives $\eta_\lambda\eta_\mu - (-1)^{1\cdot 1}\eta_\mu\eta_\lambda = 0$. □

The kinematic BRST operator is

$$Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

The same jet contraction as in Lemma 3.21, together with the Dolbeault Poincare lemma in the formal local model, leaves precisely the holomorphic Taylor coefficients. Thus the linear local coordinates are the functionals

$$\partial_{z_1}^k \partial_{z_2}^l \alpha|_0, \qquad \partial_{z_1}^k \partial_{z_2}^l \beta|_0.$$

With the coefficient-dual convention fixed above, these are represented by $(k!l!)^{-1}\partial_{z_1}^k \partial_{z_2}^l|_0$. Since zero-form components of α and β have negative field ghost numbers, the corresponding linear functions have positive ghost numbers.

Proposition 6.20 *(Closed local operators are continuous CE cochains).* The classical local operators of the formal Hamiltonian cotangent BF sector are the continuous Chevalley-Eilenberg cochains

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{Ham}} \cong C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]),$$

with CE differential dual to the Lie brackets above.

Proof. Work at the formal insertion point. The jet algebra of the fields is the completed free graded-commutative algebra on the Taylor coefficients of α and β. The kinematic differential $Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ contracts all positive real and anti-holomorphic jets: in the real directions the homotopy is $[d_{\mathbb{R}^2}, \iota_{\partial_{x_i}}] = \partial_{x_i}$, and in the anti-holomorphic directions it is the formal Dolbeault Poincare homotopy. The Q_0-cohomology of the jet algebra is therefore generated by the holomorphic Taylor coefficients alone.

The remaining α-coefficients are Taylor Hamiltonians modulo constants, hence elements of $\mathfrak{h} = \widehat{\mathfrak{h}}$; degreewise polynomial coefficients form the dense subalgebra $\bar{A}$. The cotangent field β gives the odd shifted continuous dual $\mathfrak{h}^\vee_{\mathrm{cont}}[1]$. Thus the linear space of residual local modes is $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$.

The nonlinear BRST differential is determined by the BV Hamiltonian

$$S = \int \Big\langle \beta, D\alpha + \frac{1}{2}\{\alpha, \alpha\} \Big\rangle.$$

The linear part is exactly Q_0, already killed in cohomology. The quadratic term sends two Hamiltonian α-modes f, g to their reduced Poisson bracket $\{f, g\}$, and the action on the cotangent variable is the coadjoint action displayed above. The quadratic term has zero component with two cotangent fields, so $[\eta_\lambda, \eta_\mu] = 0$. Hence the residual BRST differential on functions is precisely the derivation dual to the Lie bracket on $\mathfrak{g}$, which is the Chevalley-Eilenberg differential. Continuity records the completion over polynomial degree. □

Remark 6.21 *(Strict-product P_0 comparison block).* The cochain-level CE/PV identity of Theorem 5.5 matches the generators of $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$ to Schouten polyvectors of the boundary polynomial algebra $\widehat{\mathbf{S}}(\bar{A})$. The match is exact only on the coordinate-window CE generators; promoting it to a strict product P_0-comparison on the full Hamiltonian polydisk has a Tate-Casimir obstruction. The full disk algebra $\mathfrak{h}$ carries the product/direct-sum Tate pair $\mathfrak{h} \oplus \mathfrak{h}^\vee_{\mathrm{cont}}$ of Lemma 5.212: the coefficient coevaluation $C_{\mathfrak{h}} = \sum_d \sum_i e_{d,i} \otimes e^i_d$ diverges in each completed tensor factor; the Casimir obstruction $\mathfrak{o}_{\mathrm{Cas}}$ is the block, killed only by the weighted Tate completion (Corollary 5.71) or by the nilpotent truncation $\mathfrak{m}^3/\mathfrak{m}^{N+1}$. Promotion of the generator-level CE/PV identity to a strict factorisation P_0-centre morphism on the unreduced cotangent side is governed by the obstruction tuple Problem 5.49, with classical

Tate-coefficient closure reduced to Problem 5.222, and Roos/Cech-descent compatibility recorded in Remark 5.162. The PBW special-fibre/coadjoint deformation separation (Theorem 5.181, Corollary 5.173) shows the polynomial one-ψ descendants of §3.5 carry a different $\mathfrak{h}$-module structure than the residue cotangent module $\mathcal{P}$; the cotangent side admits a unique residue realization, separated from any polynomial substitute by Theorem 5.183. Three obstructions remain: $\mathfrak{o}_{\mathrm{Cas}}$ (BFM Casimir), the unreduced lift obstruction $\mathrm{Ob}_{\mathrm{P}_0,\mathrm{unred}}$ of Remark 5.190, and the PBW/coadjoint module-structure mismatch. They are killed successively in §5.7.

The Capelli scalar $\hbar N[\bar{c}]$ is the projective Lie anomaly evaluated at the trace generator (Theorem 6.8). It lives in Lie cohomology and becomes a component of a larger obstruction vector only after a common target deformation complex has been constructed.

All-bidegree Capelli through the radial-parts identity. The linear-generator collapse of Theorem 6.6 extends to all bidegrees (a,b) through a per-bidegree exact rational reduction: at each bidegree (a,b) the radial bicomplex carries a finite matrix $B_{a,b} \in M_{r_0\times r_1}(\mathbb{Q})$ and a comparison vector $v_{a,b} = (T_{a,b}, E^+_{a,b})$, and $\Omega^{\mathrm{rad}}_{a,b} \in \operatorname{coker} B_{a,b}$ classifies whether $v_{a,b}$ lies in $\operatorname{im} B_{a,b}$ (Bareiss elimination over $\mathbb{Q}$ decides the question in exact rational arithmetic; §F, Theorem F.44). When $v_{a,b} \in \operatorname{im} B_{a,b}$ the elimination returns the unique correction $C^+_{a,b}$ with $B_{a,b}C^+_{a,b} = v_{a,b}$ and $\Omega^{\mathrm{rad}}_{a,b} = 0$; otherwise the obstruction covector $(\phi, -\lambda) \in \ker B^*_{a,b}$ realises the equivalence class $[(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}$. On the listed range of Corollary F.45 all classes vanish and the Capelli identification extends; outside the listed range $\Omega^{\mathrm{rad}}_{a,b}$ is the radial obstruction class in its native decorated PBW–Stokes bicomplex. A counterterm interpretation requires a target deformation complex and comparison map. The Razmyslov–Procesi radical structure of the trace ideal at finite N lies outside the radial reduction: every step is exact rational and basis-canonical.

7 The obstruction calculus: native complexes and local-to-global coherence

The obstruction taxonomy records the global obstructions to the formal-Darboux trace-sector stalk map of Theorem 5.6. The six formal-Darboux obstruction classes $\Omega^{\mathrm{rad}}_{a,b}$, $\hbar N[\bar{c}]$, $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]$, $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$, $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$, $\mathrm{per}_X(v)$ live in different native complexes and in different cohomological degrees. They become coordinates of one curvature only after a common filtered deformation complex $\mathfrak{K}_{\mathcal{T}}$, comparison maps, and projection maps have been constructed. The conventions of Conventions and notation for the local theorem fix the symbols: $d = \dim_{\mathbb{C}} X$, $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ as local geometry, $\mathfrak{h} = \mathbb{C}[[z_1,z_2]]/\mathbb{C}\cdot 1$ and $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ as the native shifted-cotangent BF Lie algebra, the holomorphic E_2 factorization-algebra surface, the BV / Koszul / Gerstenhaber sign rules of Appendix C, the explicit Costello propagator weight in heat-kernel renormalization, the $\hbar$ versus g_s discipline, the Hochschild / cyclic / negative-cyclic distinction, the Capelli projective Lie class $\hbar N[\bar{c}]$ of Theorem 6.8, whose determinant-line curvature interpretation requires additional determinant data, and the matched-conventions obstruction class $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}})$ of Theorem 11.13. Every class named below is first stated in its native complex.

The shifted-cotangent BF Lie algebra of §2, $\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$, is the native shifted-cotangent Lie algebra of the local closed fields. Deformations of the BV action are controlled by the corresponding local-functional deformation complex. When a target $\mathcal{T}$ gives a filtered L_∞-deformation complex $(\mathfrak{K}_{\mathcal{T}}, F^\bullet, d, \ell_2, \ldots)$, the transported trace substitution J is a local datum Θ_{loc}. Image under quantization, globalization, factorization-centre lift, modular clutching, and CY-categorical specialisation produces the obstruction classes $\hbar N[\bar{c}]$, $\Omega^{\mathrm{rad}}_{a,b}$, $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]$, $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$, $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$, and $\mathrm{per}_X(v)$ as native obstruction classes. A single curvature decomposition requires an additional common deformation complex.

7.1 Filtered L_∞ comparison deformation complex

The data of a target deformation problem are $(\mathfrak{K}^\bullet_{\mathcal{T}}, F^\bullet, d, \ell_2, \ell_3, \dots)$: a filtered L_∞-algebra in cohomological degree, with filtration $\mathfrak{K}_{\mathcal{T}} = F^0 \supseteq F^1 \supseteq F^2 \supseteq \cdots$ that is exhaustive and complete in the sense that $\mathfrak{K}_{\mathcal{T}} \xrightarrow{\sim} \varprojlim_n \mathfrak{K}_{\mathcal{T}}/F^n$, with each ℓ_k of weight at least $k-1$ for the filtration so that the curvature series of an element of F^1 converges in the filtered topology.

Given a degree-one element $\Theta \in F^1\mathfrak{K}^1_{\mathcal{T}}$, the curvature is the formal series

$$F_{\mathcal{T}}(\Theta) = d\Theta + \tfrac{1}{2}\ell_2(\Theta,\Theta) + \tfrac{1}{3!}\ell_3(\Theta,\Theta,\Theta) + \cdots,$$

convergent in the filtered topology. The Maurer–Cartan equation $F_{\mathcal{T}}(\Theta) = 0$ cuts out the strict $\mathcal{T}$-coherences. Twists are filtered cochains $\tau \in F^1\mathfrak{K}^1_{\mathcal{T}}$ (or, when the necessary extension is structural and higher than first order, central extensions of $\mathfrak{K}_{\mathcal{T}}$ by an abelian filtered cochain complex):

$$\mathrm{TwGlob}_{\mathcal{T}}(\Theta) = \left\{ \tau \in F^1\mathfrak{K}^1_{\mathcal{T}} \;:\; F_{\mathcal{T}}(\Theta+\tau) = 0 \right\} / \mathrm{gauge}.$$

A morphism of target deformation complexes $\mathfrak{K}_{\mathcal{T}} \to \mathfrak{K}_{\mathcal{T}'}$ is a filtered L_∞-morphism, sending Maurer–Cartan elements to Maurer–Cartan elements.

7.2 The transported local datum

The formal-stalk-chart hypotheses of Theorem 5.147 – Chan–Paton substitution $z_i \rightsquigarrow \phi_i$, the Dirac constraint $[\phi_1, \phi_2] = 0$, and the trace map $J(f) = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$ – define the underlying datum; the chosen comparison object determines its controlling deformation algebra.

Global Hamiltonian descent. For a holomorphic-symplectic ambient (X, ω) with a brane L and a Darboux cover $\{U_i\}$ symplectomorphic to $\widehat{\mathbb{C}^2}_0$, the local data are charts $\Phi_i : U_i \to \widehat{\mathbb{C}^2}_0$ with local Hamiltonians h_i; the deformation complex is $\mathfrak{K}_{\mathrm{glob}} = \operatorname{Tot} C^\bullet_{\check{C}/\mathrm{WR}}(\{U_i\}, \mathfrak{K}_{\mathrm{loc}})$ – the totalization of the Čech–Weiss–Ran complex of local formal-stalk-chart coefficients. The transported element is $\Theta_{\mathrm{glob}} = (\Theta_i, \alpha_{ij}, \beta_{ijk}, \dots)$, with curvature $F_{\mathrm{glob}}(\Theta_{\mathrm{glob}})$ carrying the Hamiltonian period $\mathrm{per}_X(v) = [h_i - h_j] \in H^1(X, \mathbb{C}_X)$ as one component.

Costello quantum master equation. The Costello deformation complex is the filtered local-functional complex $\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$ with scale-L quantum BV differential

$$d^L_{\mathrm{qBV}} = Q + \{I[L], -\}_L + \hbar\Delta_L.$$

The transported element is the classical interaction I_{loc} of the open Hamiltonian BF theory, completed in the weighted-Tate envelope of Theorem 5.58. If a filtered chain scalar projection $\widehat{\sigma}_{\mathrm{sc}}$ exists and its scalar-lift obstruction tower vanishes, the reduced non-scalar curvature is $[F_{\mathrm{QME}}(\Theta_{\mathrm{loc}})] = [\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{qBV}})$.

Unreduced P_0-factorization-centre lift. The deformation complex is the mapping complex $\mathfrak{I}_{\mathrm{fact}} = \mathrm{RHom}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}|_L, Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}))$, augmented by BV/QME and stratified Weiss–Čech–Roos descent directions. The transported element is the reduced CE/PV cochain morphism of Theorem 4.20 restricted to the brane. The curvature decomposes as the four-component vector $\mathrm{Ob}_{\mathrm{cent/QME/desc}} = (\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}, \mathrm{ob}^{P_0}_{\mathrm{cent}}, \mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}, \partial^{\mathrm{Weiss}}_{\check{C}/R})$ of Theorem E.53.

Decorated radial Stokes / Weyl normal form. The deformation complex is the finite-dimensional radial bicomplex of Definition F.27 with decoration generators for PBW/Stokes corrections. At each bidegree (a,b) the transported element is the radial comparison vector $v_{a,b} = (T_{a,b}, E^+_{a,b})$; the curvature is the per-bidegree class $\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}$, and the obstruction dual is $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$.

Strict finite-window Bochner–Martinelli current transfer. The transfer problem uses the tower of finite Matlis windows $P_{\le N} = \bigoplus_{a+b\le N} \mathbb{C}\rho_{a,b}$ with action of $\mathfrak{h}$ by Hamiltonian vector fields. The compatibility of this action with strict native all-window support-local transfer is exactly what the BMK leak obstructs; the transported element is the brane current pair $(P_{\le N}, \rho_{0,1})$. The curvature is the residue-coadjoint leak $z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}$, encoded as the obstruction $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$. The pro-Matlis target is the replacement after this obstruction, not a strict native all-window transfer.

Modular open-closed clutching. Modular coherence on the brane-line cyclic factorization dg category C^{op} is a target deformation problem of the same shape as the previous six, controlled by the strict law $d\Theta_C + \frac{1}{2}[\Theta_C, \Theta_C] = 0$ in $Z^{\mathrm{ch}}_{\mathrm{der}}(C^{\mathrm{op}})$ and its Ω_{mod}-twisted version carrying the determinant-line twist of the projective clutching law $\mathrm{Cl}^*\mathrm{Tr}_{g+1,n-2} = \mathrm{contract}_C(\mathrm{Tr}_{g,n}) \otimes e^{\Omega_{\mathrm{node}}}$. The data $(b, \mathrm{Tr}_C, \Theta^{oc,\log}_C, \Omega_{\mathrm{mod}})$ belong to the companion compact comparison theorem in Vol II 3d HT bulk-boundary and Vol III CY cyclic trace; any comparison with the present local theorem factors through §11.4.

7.3 *Twists: three mechanisms*

A nonvanishing curvature class is killed in three ways: null-homotopy, central extension, and target replacement.

Null-homotopy twist. The class is exact in the original target: $F_{\mathcal{T}}(\Theta) = d_\Theta \tau$. The corrected element $\Theta + \tau$ satisfies the strict Maurer–Cartan equation in the same target. This is the language of QME counterterms, homotopy-centrality witnesses, decorated radial Stokes corrections, Weiss/Ran descent homotopies, and many finite-window radial adjustments. The model example is the θ_3 finite-window H^1: bare $H^1 = \mathbb{C}\cdot E_{\theta_3} \neq 0$ under the cleared QME differential, while the counterterm-adjoined row complex has $H^1 = 0$ after a given scalar-zero local functional satisfying $dC = -E$, $c = 1$ (Theorem E.38).

Structural / central twist. The class is closed and central: $F_{\mathcal{T}}(\Theta) = \omega$, $d\omega = 0$, ω central. One enlarges $\mathfrak{K}_{\mathcal{T}}$ by a central extension carrying ω on the trace line. The corrected element satisfies the master equation in the extended target, and Θ becomes a section of a projective structure. The scalar anomaly $\hbar N[\bar c] \in H^2_{\mathrm{Lie}}(\bar A, \mathbb{C})[[\hbar]]$ is the prototype: the ordinary scalar-reduced Costello obstruction in the $\mathfrak{gl}_N$ branch obstructs internal counterterms, and the admissible object is the central extension $0 \to \mathbb{C}\cdot\mathbf{1} \to \mathfrak{h}_{\hbar,N} \to \bar A \to 0$ under which the brane carries a projective Hamiltonian action $[\Phi_\hbar(f), \Phi_\hbar(g)]_\star = \hbar\Phi_\hbar(\{f,g\}) + \hbar N \bar c(f,g)\mathbf{1}$. A global Hamiltonian primitive has instead a descent torsor with Čech 1-class $\mathrm{per}_X(v)$. It becomes gerbe-valued only after a degree-two central extension class has been constructed.

Target-replacement twist. The class is closed, inexact, and noncentral: every admissible extension of $\mathfrak{K}_{\mathcal{T}}$ for which the corrected curvature is zero is obstructed. Coherence requires a target substitution $\mathcal{T} \rightsquigarrow \mathcal{T}'$ along a forced embedding. The attempted strict all-window Bochner–Martinelli current transfer is the prototype: in every fixed finite Matlis window $\mathcal{P}_{\le N}$, the residue-coadjoint Jacobiator $(N+1)(N+2)\rho_{0,1} \neq 0$ is non-absorptive in the strict multiplication structure, so internal counterterm restoration is obstructed. The pro-Matlis tower $\widehat{\mathcal{P}}_\Pi = \varprojlim_N \mathcal{P}_{\le N}$ (Corollary 8.41) is the replacement target selected by the leak: the modified coherence law is $\pi_N(f \cdot \xi) = \pi_N(f \cdot \pi_{N+r(f)}\xi)$, Hamiltonian action is coherent only in the pro-tower, and any fixed finite current window is a measurement truncation of the closed dynamical subsystem.

7.4 Four obstruction mechanisms

Theorem 7.1 *(Four obstruction mechanisms for the formal-Darboux stalk).* Let $\mathcal{T}$ be a target deformation problem with filtered L_∞ algebra $(\mathfrak{K}_\mathcal{T}, F^\bullet, d, \ell_k)$, and let $\Theta_{\mathrm{loc}} \in F^1\mathfrak{K}^1_\mathcal{T}$ be the transported local formal-stalk-chart datum. Call an *admissible extension* of $\mathfrak{K}_\mathcal{T}$ any L_∞ central extension by a closed degree-2 cocycle. The following four cases are the possible native mechanisms. They are mutually exclusive only in a single obstruction complex with projection maps and vanishing stabilizer cohomology.

(i) Strict coherence: $F_\mathcal{T}(\Theta_{\mathrm{loc}}) = 0$ in $\mathfrak{K}^2_\mathcal{T}$.
(ii) Counterterm coherence: the class $[F_\mathcal{T}(\Theta_{\mathrm{loc}})]$ is exact in $H^2(\mathfrak{K}_\mathcal{T}, \Theta_{\mathrm{loc}})$, with $\tau \in F^1\mathfrak{K}^1_\mathcal{T}$ satisfying $F_\mathcal{T}(\Theta_{\mathrm{loc}} + \tau) = 0$ in $\mathfrak{K}_\mathcal{T}$; the corrected element is the τ-twisted $\mathcal{T}$-coherence in the same target.
(iii) Projective coherence: the class is central and inexact, $[F_\mathcal{T}(\Theta_{\mathrm{loc}})] = [\omega]$ with ω central in $\mathfrak{K}_\mathcal{T}$, and the corrected element solves the master equation in a central extension by ω; the global structure is projective.
(iv) Target replacement: outside a chosen realization of cases (1)–(3), $[F_\mathcal{T}(\Theta_{\mathrm{loc}})]$ is nonzero in $H^2(\mathfrak{K}_\mathcal{T}, \Theta_{\mathrm{loc}})$, and every admissible extension has an obstruction to a central representative; coherence requires a replacement target $\mathcal{T}'$ given as additional mathematical data.

The local formal-Darboux theorems decompose as follows.

Class	Curvature meaning	Mechanism
$\hbar N[\bar{c}]$	scalar-reduced Hamiltonian action	central extension, projective
$\Omega^{\mathrm{rad}}_{a,b}$	Weyl/radial all-bidegree normal form	decorated PBW/Stokes counterterm
$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]$	reduced non-scalar QME	Costello local counterterm
$\mathrm{Ob}_{\mathrm{cent/QME/desc}}$	unreduced factorization-centre lift	derived centre, homotopy-centrality
$\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$	strict finite-current transfer	target replacement, pro-Matlis
$\mathrm{per}_X(v)$	global Hamiltonian descent	Hamiltonian descent torsor

The table records the native mechanism in which each class is proved.

Theorem 7.2 *(Six obstruction coordinates in a common target).* The six formal-Darboux obstruction classes

$$\Omega^{\mathrm{rad}}_{a,b}, \quad \hbar N[\bar{c}], \quad [\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}], \quad \mathrm{Ob}^{\Pi}_{\mathrm{BM}}, \quad \mathrm{Ob}_{\mathrm{cent/QME/desc}}, \quad \mathrm{per}_X(v)$$

are defined in six native obstruction complexes. If a target $\mathcal{T}$ gives a common filtered deformation complex with comparison maps from these six complexes, their images form a six-component obstruction vector. Without that additional structure there is no single curvature class whose projections are these six objects.

Proof. Each class is established in its native complex: $\Omega^{\mathrm{rad}}_{a,b}$ in the radial bicomplex (Appendix F, Theorem F.44; counterterm cell $d_\Theta \tau$); $\hbar N[\bar{c}]$ in scalar Lie cohomology (Lemma 6.2, Theorem 6.8; projective cell $\Omega_{\mathrm{central}}$); $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]$ in the weighted Tate complex (Theorem E.38; counterterm cell); $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$ on finite Matlis windows (Theorem 8.40; genuine-obstruction cell $\notin$ im d_Θ); $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$ in the unreduced cotangent lift (Theorem E.53; counterterm cell); $\mathrm{per}_X(v)$ as the global period class on X (Theorem 5.147; descent-torsor cell). The final assertion is precisely the absence of an unproved common curvature complex. □

7.5 Tate-coefficient cotangent lift: native obstruction classes

The unreduced cotangent extension of the bracket match $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ of §4.3 (Theorem 4.5, Theorem 4.11) produces six native obstruction classes:

$$\Omega^{\mathrm{rad}}_{a,b}, \quad \hbar N[\bar{c}], \quad [\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}], \quad \mathrm{Ob}^{\Pi}_{\mathrm{BM}}, \quad \mathrm{Ob}_{\mathrm{cent/QME/desc}}, \quad \mathrm{per}_X(v)$$

in the radial bicomplex, Lie cohomology, the weighted QME complex, the finite-window Bochner–Martinelli transfer problem, the unreduced centrality/QME/descent complex, and Hamiltonian period cohomology. They are not projections of a single curvature until a common target deformation complex and comparison maps are constructed. The strict cochain-level morphism of factorization algebras

$$\Phi_{\mathrm{cl}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N})$$

realising the cotangent direction $\mathfrak{h}^\vee_{\mathrm{cont}}[1]$ with all P_0-centrality homotopies and a boundary realisation of the shifted-cotangent generators is the existence problem Problem 5.49.

The four-component obstruction vector $(\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}, \mathrm{ob}^{P_0}_{\mathrm{cent}}, \mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}, \delta^{\mathrm{Weiss}}_{\check{C}/R})$ of Theorem E.53 resolves the Tate-completed lift inside the $(d_\Theta \tau)$-cell: joint vanishing with chosen null-homotopies constructs the morphism, and a nonzero entry is the precise construction defect. The chain-level primitive projection of Theorem 5.133 computes the associated graded primitive complex. A spectral sequence converging to $\mathrm{ob}^{P_0}_{\mathrm{cent}}$ is an additional construction once the full filtered obstruction complex has been specified.

The first proved nonzero entry is the Casimir-kernel block $\mathfrak{o}_{\mathrm{Cas}}$: the strict product/direct-sum Tate pair $(\mathfrak{h}_\Pi, D_\oplus)$ has an obstruction to a coefficient kernel projecting to every finite Casimir $C^M_{\mathfrak{h}} = \sum_{d \le M} \sum_i e_{d,i} \otimes e^i_d$ (Lemma 5.212, Theorem 5.94), so the strict-unweighted endpoint is foreclosed. Theorem 8.40 records the finite-window Bochner–Martinelli obstruction; strict all-window support-local transfer is the separate BMK ladder problem with obstruction vector $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$. By Corollary 8.41 the pro-Matlis retract is the inverse-limit receiver for the fixed finite-window projections and BMK moment data. The unreduced cotangent lift therefore has its obstruction on the strict ambient product completion and is realized only inside a supplied refinement: the weighted Tate completion (Corollary 5.71), the nilpotent truncation $\mathfrak{m}^3/\mathfrak{m}^{N+1}$, or the pro-Matlis retract.

Definition 7.3 (Tate-coefficient cotangent lift obstruction vector). The *Tate-coefficient cotangent lift obstruction vector* for the formal local model $X = \mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ with brane line $L = \mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{0\}$ and ordinary $\mathfrak{gl}_N$ gauge data is

$$\mathrm{Ob}_T = \big(\mathfrak{o}_{\mathrm{Cas}}, \mathfrak{o}_{\mathrm{nilp}}, \mathfrak{o}_{\mathrm{Quil}}, \mathfrak{o}_{\mathrm{BV,end}}, \mathfrak{o}_{\mathrm{prim}}, \mathfrak{o}_{\mathrm{Had}}, \mathfrak{o}_{\mathrm{univ}}, \mathfrak{o}_{\mathrm{cv}}\big),$$

with components: the weighted Casimir kernel class $\mathfrak{o}_{\mathrm{Cas}}$, the nilpotent truncation class $\mathfrak{o}_{\mathrm{nilp}}$, the Quillen/Tate-bar-cobar class $\mathfrak{o}_{\mathrm{Quil}}$, the BV endpoint tuple $\mathfrak{o}_{\mathrm{BV,end}}$, the chain-level primitive class $\mathfrak{o}_{\mathrm{prim}}$, the Hadamard locality class $\mathfrak{o}_{\mathrm{Had}}$, the protected universality class $\mathfrak{o}_{\mathrm{univ}}$, and the cross-target matched-conventions class $\mathfrak{o}_{\mathrm{cv}}$. The Costello–Li 2012 BCOV/HCS theorem belongs to a different hypothesis class from the present brane-defect geometry; the hypothesis-separation lemma below forces the eight components.

Weighted Casimir kernel $\mathfrak{o}_{\mathrm{Cas}}$. The strict product/direct-sum Tate pair $(\mathfrak{h}_\Pi, D_\oplus)$ has an obstruction to a coefficient kernel projecting to every finite Casimir $C^M_{\mathfrak{h}} = \sum_{d \le M} \sum_i e_{d,i} \otimes e^i_d$, so $\mathfrak{o}_{\mathrm{Cas}}$ is the class of the compatible family $(C^M_{\mathfrak{h}})_M$ in the cokernel of the represented-kernel map into $\mathfrak{h}\widehat{\otimes}D$. Its strict-pair nonvanishing is the unweighted endpoint obstruction $\mathfrak{o}_{\mathrm{Cas}} \notin \mathfrak{h}_\Pi \widehat{\otimes} D_\oplus$ of 5.94. The vanishing datum for $\mathfrak{o}_{\mathrm{Cas}}$ is a weight w, an exponential $w_q(d) = q^d$ with $q > 1$ being the working choice, satisfying the bracket-tame, finite-window coadjoint-continuity, and Mittag-Leffler coevaluation conditions of Definition 5.52. Within the weighted Tate pair $(\mathfrak{h}_w, \mathfrak{h}^{\vee,w}_{\mathrm{cont}})$ of Definition 5.55, the same compatible family converges by Proposition 5.62, and the heat-kernel BV propagator $P^w_{\epsilon,L}$ extends to a continuous family by Proposition 5.64. The classical cochain CE/PV cotangent extension to $\mathfrak{g}_w$ is then 5.71 (d). The unweighted endpoint is closed only under 5.95; in the weighted sector the residue defect is the regulator-independence quotient of 5.82, and Proposition 5.99 records that the inclusion $D_\oplus \hookrightarrow D_{\Pi,w}$ lies outside the class of ordinary quasi-isomorphisms, so any class detecting the tail $D_{\Pi,w}/D_\oplus$ is outside the regulator-admissible sector of 5.76.

The four-case trace-sector obstruction table and the six native obstruction classes (Theorems 7.1, 7.2) live in their native complexes. The inverse-limit coefficient system $\widehat{\mathcal{P}}_\Pi = \varprojlim_N \mathcal{P}_{\le N}$ records the finite-Matlis-window components; the Bochner–Martinelli obstruction on each finite window selects the pro-Matlis replacement target for the coefficient side of the bulk-boundary comparison, constructed in the section below.

8 The pro-Matlis envelope: Matlis principal parts and the continuous cotangent module

Configuration observables on the brane are the polynomial trace classes $O_f = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$: Taylor in the matrix-valued normal coordinates, forming the PBW special-fibre alphabet. Their conjugate cotangent momenta, by Grothendieck local-cohomology duality, are the residues $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$, top local-cohomology generators of $R = \mathbb{C}[[z_1, z_2]]$ at the maximal ideal (z_1, z_2). The brane measures positions polynomially; its cotangent momenta live as residues. The same local-cohomology object also gives the two-point diagonal coefficients of the native holomorphic singular product: after replacing the origin support $\mathfrak{m} = (z_1, z_2)$ by the diagonal $\Delta \subset \mathbb{C}^2_z \times \mathbb{C}^2_w$, the basis becomes $\rho^\Delta_{a,b} = (z_1 - w_1)^{-a-1}(z_2 - w_2)^{-b-1} d\zeta_1 d\zeta_2$, and Theorem 1.17 uses $\rho^\Delta_{0,0}$ as the Cauchy kernel.

The $\mathfrak{m}$-adic continuous dual of the reduced Hamiltonian Lie algebra $\mathfrak{h} = R/\mathbb{C} \cdot 1$ sits inside $H^2_{\mathfrak{m}}(R)\, dz_1 dz_2$ as the scalar annihilator of the constant line; the residue pairing identifies it canonically with the direct sum $\mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\, \rho_{a,b}$. The Hamiltonian coadjoint action, transported through this residue identification, is the principal-part residue formula $z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\, \rho_{a-p+1, b-q+1}$, with negative-index targets zero. The polynomial PBW one-ψ descendants $\widetilde{\Psi}_{a,b}$ carry the same labels (a, b) and a strictly different $\mathfrak{h}$-module structure: linear Hamiltonians act locally nilpotently on the polynomial side and never locally nilpotently on $\mathcal{P}$, giving the basis-relabeling obstruction. Fourier–Rees conjugation gives the only universal label bridge, and only after a polarization change: it sends $\mathcal{O}_\hbar = \mathcal{D}_\hbar/(\partial_1, \partial_2)$ to $\Delta^{\mathsf{F}}_\hbar = \mathcal{D}_\hbar/(z_1, z_2)$, whose Čech realization sits inside the standard local-cohomology lattice as $\bigoplus \hbar^{a+b}\mathbb{C}[[\hbar]]\, \rho_{a,b}$; the full lattice $\bigoplus \mathbb{C}[[\hbar]]\, \rho_{a,b}$ is recovered only after $\hbar$ is inverted.

Source conventions: Matlis duality [19], Grothendieck–Hartshorne local cohomology [20], and the power-series residue model of Kunz–Cox–Dickenstein, Chapter 5 [21].

8.1 *The defect polarization*

A canonical-transformation BV theory on the formal symplectic polydisk $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$ carries two coordinate types: Hamiltonian-dual coordinates c^I, which generate canonical motion, and shifted-cotangent coordinates u_I, which carry antifields. The brane line $L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}$ reads only the second alphabet, through the trace substitution of Lemma 3.1: a polynomial Hamiltonian becomes the brane configuration observable $O_f = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$. A formal Hamiltonian $f \in \mathfrak{h}$ gives a completed Taylor function of the Chan–Paton pair. The cotangent momentum dual to O_f is constrained by three properties: it pairs with Hamiltonian observables to a scalar; it is continuous for the $\mathfrak{m}$-adic topology on $\mathfrak{h}$, where $\mathfrak{m} = (z_1, z_2)$; it intertwines the Hamiltonian coadjoint action coming from the Poisson bracket on the polydisk. Those three constraints together select the residue-valued cotangent polarization.

THEOREM 8.1 *(Defect Polarization).* On the formal symplectic polydisk $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$, the brane defect line $L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\}$ carries a canonical polarization with the following two sides.

(i) *Configuration side: polynomial / Taylor.* Hamiltonian configuration observables on the brane are the polynomial trace family

$$O_f = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}, \qquad f \in \mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1.$$

The Lie algebra acting through these observables is the reduced polynomial Hamiltonian sub-Lie algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1 \subset \mathfrak{h}$, each generator represented by a constant Hamiltonian vector field on the polynomial PBW special-fibre alphabet. These observables form the PBW one-ψ shadow span: $\widetilde{\Psi}_{a,b} \in \bar{A}_{\mathrm{desc}}$, bidegree (a, b), with linear Hamiltonians acting locally nilpotently.

(ii) *Cotangent side: distributional / residual.* Cotangent momenta on the same brane are the Grothendieck–Matlis principal parts

$$\mathcal{P} = \bigoplus_{\substack{a,b \geq 0 \\ a+b>0}} \mathbb{C}\, \rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

basis-paired with the Taylor monomial dual by the Grothendieck residue $\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}$, and acted on by Hamiltonians through the principal-part coadjoint formula

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1, b-q+1},$$

with negative-index targets zero. Linear Hamiltonians act on $\mathcal{P}$ by raising pole order, never locally nilpotently. The two-point diagonal version replaces $\rho_{a,b}$ by $\rho^{\Delta}_{a,b}$; its $(0, 0)$ generator is the kernel $K^{(2)}_{\Delta}$ in the native singular product.

The two sides are distinct $\mathfrak{h}$-modules. The bidegree relabelling $\widetilde{\Psi}_{a,b} \leftrightarrow \rho_{a,b}$ carries the same index set (a, b) and has the local-nilpotence obstruction to intertwining the two Hamiltonian actions: linear-Hamiltonian local nilpotence holds on the polynomial side, while the residual side has non-locally-nilpotent action (Theorem 8.11). The Moyal coadjoint deformation acts on the residual side; the polynomial one-ψ shadow has the same obstruction after inverting $\hbar$, and the Fourier–Rees label bridge of Theorem 8.12 is the unique universal interpolation, available only after a Fourier polarization change of the action itself.

In particular, the continuous dual of the Hamiltonian Lie algebra of the defect is the residual side:

$$\mathfrak{h}^{\vee}_{\mathrm{cont}} \cong \mathcal{P} \qquad \text{(Proposition 8.6)},$$

and the residue-coadjoint pairing is the unique perfect continuous pairing intertwining the two sides (Theorem 8.10).

Proof. The configuration side is Lemma 3.1: the gauge-invariant content of the Chan–Paton pair on the brane is the trace family $\{O_f\}_{f\in\bar{A}}$, with $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$ acting by Hamiltonian vector fields on the polynomial PBW special-fibre alphabet $\widetilde{\Psi}_{a,b}$; local nilpotence of the linear Hamiltonians z_1, z_2 on this side is Theorem 8.11.

On the cotangent side, the basis identity $\mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\,\rho_{a,b}$ and the residue Kronecker pairing $\langle \rho_{a,b}, z_1^m z_2^n\rangle = \delta_{a,m}\delta_{b,n}$ are Definition 8.5. The continuous-dual identification $\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P}$ is Proposition 8.6. The principal-part coadjoint action with monomial coefficient $-(pb - qa + p - q)$ is Proposition 8.7 and Lemma 8.8; iteration on linear Hamiltonians produces the rising-factorial $(-1)^N(b+1)\cdots(b+N)\rho_{a,b+N}$ in Theorem 8.11, excluding local nilpotence.

Linear Hamiltonians act locally nilpotently on the polynomial side and have infinite orbits on the residual side, by the same theorem, so the same-action module isomorphism is obstructed. The unique universal label bridge between the two sides is Theorem 8.12; that the bridge requires a Fourier polarization change of action is the obstruction clause of the same theorem and Corollary 8.13. The residue-coadjoint pairing is unique up to one scalar by Theorem 8.10. ☐

Remark 8.2 *(Finite-window verification of the polarization).* The exact rational calculation of $\mathrm{ad}^\vee_{z_1^p z_2^q}(\delta_{a,b})$ on Taylor-monomial test labels gives the residual-side coefficient $-(pb - qa + p - q)$, identifies the one-functional support, and confirms $-(pb - qa + p - q)\rho_{a-p+1,b-q+1}$ on every quadruple (p, q, a, b) with exponents ≤ 5 and $p + q, a + b > 0$: 1225 cases, all match. The same calculation evaluates the bidegree-only label map $\widetilde{\Psi}_{a,b} \mapsto \rho_{a,b}$ by the exact Moyal coadjoint action: 1200 of 1225 cases register a PBW-vs-Moyal mismatch, and 958 cases carry genuine higher Moyal coadjoint terms which the polynomial one-ψ action lacks at every order in $\hbar$. The smallest explicit obstruction is the pair

$$\mathrm{ad}^\vee_{z_1^4}(\delta_{1,1}) = -24\,\hbar^2\,\delta_{0,4}, \qquad \widetilde{\Psi}_{1,1} \mapsto 4\,\widetilde{\Psi}_{4,0} \quad \text{(polynomial side)};$$

the two functionals are supported at different bidegrees and at different powers of $\hbar$. The Moyal star-bracket coefficients on monomials up to exponents ≤ 10 and orders ≤ 11 have antisymmetric odd-order tower $1, \frac{1}{24}, \frac{1}{1920}, \frac{1}{322560}, \ldots$ matching the closed-form $\frac{1}{2^{r-1}r!}$ for odd r.

8.2 Construction of the residual side

Definition 8.3 *(Topology and reduced Hamiltonians).* Put

$$R = \mathbb{C}[[z_1, z_2]], \qquad \mathfrak{m} = (z_1, z_2), \qquad \mathfrak{h} = R/\mathbb{C}\cdot 1.$$

The topology on R is the $\mathfrak{m}$-adic topology. The topology on $\mathfrak{h}$ is the quotient topology, equivalently the topology transported through the zero-constant section

$$\mathfrak{h} \simeq \mathfrak{m}, \qquad \overline{f} \longmapsto f - f(0).$$

A fundamental system of neighbourhoods of zero is

$$F^N\mathfrak{h} = \mathfrak{m}^N, \qquad N \ge 1,$$

via this splitting. The Lie bracket is the Poisson bracket

$$\{f, g\} = \partial_{z_1} f\,\partial_{z_2} g - \partial_{z_2} f\,\partial_{z_1} g$$

followed by projection modulo constants. The adjoint maps are continuous for this topology: if $f \in \mathfrak{m}$, then $\{f, \mathfrak{m}^{N+1}\} \subset \mathfrak{m}^N$. The quotient $\mathfrak{h}$ removes the constant class, which acts by zero.

LEMMA 8.4 *(Continuous dual).* The continuous dual of $\mathfrak{h}$, with $\mathbb{C}$ discrete, is

$$\mathfrak{h}^\vee_{\mathrm{cont}} = \bigcup_{N \geq 1} (\mathfrak{h}/F^N\mathfrak{h})^\vee = \bigoplus_{\substack{a,b \geq 0 \\ a+b>0}} \mathbb{C}\, \delta_{a,b},$$

where

$$\delta_{a,b}(z_1^m z_2^n) = \delta_{a,m}\delta_{b,n}.$$

Proof. A linear functional on the $\mathfrak{m}$-adic vector space $\mathfrak{h} \simeq \mathfrak{m}$ is continuous exactly when its kernel contains $F^N\mathfrak{h} = \mathfrak{m}^N$ for some N. Such a functional factors through the finite-dimensional quotient $\mathfrak{m}/\mathfrak{m}^N$, hence is a finite linear combination of Taylor coefficient functionals. Conversely every finite linear combination of the displayed $\delta_{a,b}$ kills all sufficiently high powers of $\mathfrak{m}$, hence is continuous. □

Definition 8.5 *(Reduced Matlis principal parts).* The top local cohomology module of R is written in Čech Laurent form as

$$H^2_{\mathfrak{m}}(R)\, dz_1 dz_2 = \bigoplus_{a,b \geq 0} \mathbb{C}\, z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

Equivalently this is the Čech quotient

$$R[z_1^{-1}, z_2^{-1}]/(R[z_1^{-1}] + R[z_2^{-1}])$$

after multiplication by $dz_1 dz_2$; the displayed direct sum is the standard representative basis in which both exponents are strictly negative. The basis vector $\rho_{0,0} = z_1^{-1} z_2^{-1} dz_1 dz_2$ is the nonzero functional in $H^2_{\mathfrak{m}}(R)\, dz_1 dz_2$ which evaluates the constant coefficient. The reduced principal-part module is the $\mathbb{C}$-linear residue annihilator of constants,

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1) = \bigoplus_{\substack{a,b \geq 0 \\ a+b>0}} \mathbb{C}\, \rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

Equip $\mathcal{P}$ with the locally finite direct-sum topology

$$\mathcal{P} = \varinjlim_N \mathcal{P}_{\leq N}, \qquad \mathcal{P}_{\leq N} = \bigoplus_{\substack{a,b \geq 0 \\ 0<a+b \leq N}} \mathbb{C}\, \rho_{a,b},$$

and give $\mathfrak{h}^\vee_{\mathrm{cont}}$ the corresponding inductive-limit topology through the basis $\delta_{a,b}$. Thus the residue identification below is a continuous map of the declared locally finite topological vector spaces. The space $\mathcal{P}$ is a Hamiltonian-coadjoint module; the R-submodule structure of $H^2_{\mathfrak{m}}(R)\, dz_1 dz_2$ is obstructed on $\mathcal{P}$, since multiplication by z_i can hit the excluded scalar-residue line. The residue pairing is

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \mathrm{Res}_{z_1=z_2=0}\, z_1^{m-a-1} z_2^{n-b-1}\, dz_1 dz_2 = \delta_{a,m}\delta_{b,n}.$$

PROPOSITION 8.6 *(Matlis realization of the continuous dual).* The residue pairing induces a continuous vector-space isomorphism

$$\mathcal{P} \xrightarrow{\sim} \mathfrak{h}^\vee_{\mathrm{cont}}, \qquad \rho_{a,b} \longmapsto \delta_{a,b}.$$

It is the scalar-reduced restriction of the Matlis evaluation pairing for the complete regular local ring R: after the trivialization $\omega_R = R\,dz_1dz_2$, $H^2_{\mathfrak{m}}(R)\,dz_1dz_2 \cong E_R(\mathbb{C})$. The subspace $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1)$ is the $\mathbb{C}$-linear scalar-annihilator dual to $\mathfrak{h} = R/\mathbb{C}\cdot 1$. The R-module structure on $E_R(\mathbb{C})$ can hit the excluded scalar residue line under multiplication. The Hamiltonian coadjoint action is transported separately by the residue pairing (Proposition 8.7) and is the structure used by the brane cotangent sector.

Proof. The displayed residue formula identifies each basis vector $\rho_{a,b}$, $a+b>0$, with the continuous Taylor coefficient functional $\delta_{a,b}$. Lemma 8.4 shows that these coefficient functionals form the whole continuous dual. The exclusion of $\rho_{0,0}$ is exactly the condition that the functional vanish on the scalar line $\mathbb{C}\cdot 1\subset R$; this is the continuous dual of the quotient $R/\mathbb{C}\cdot 1$. On finite truncations the map sends $\mathcal{P}_{\le N}$ isomorphically to the functionals vanishing on $\mathfrak{m}^{N+1}$, so both the map and its inverse are continuous for the declared locally finite topologies. □

Proposition 8.7 *(Residue coadjoint action).* Under the isomorphism of Proposition 8.6, the coadjoint action of $f\in\mathfrak{h}$ is the principal-part residue action

$$f\cdot\rho = \pi_{\mathrm{pp}}\{f,\rho\},$$

where π_{pp} projects Laurent two-forms to the span of the principal parts. On monomials,

$$z_1^p z_2^q\cdot\rho_{a,b} = -(pb-qa+p-q)\,\rho_{a-p+1,b-q+1},$$

with the convention that any term with a negative index is zero. If the displayed target is $\rho_{0,0}$, then the coefficient $pb-qa+p-q$ is zero; hence the action preserves $\mathcal{P}$ and avoids the scalar residue line, and the action preserves the scalar-annihilator subspace. The coefficient $-(pb-qa+p-q)$ is the residue side of the defect polarization (Theorem 8.1); on the polynomial PBW one-ψ shadow the analogous coefficient is the symmetric $pb-qa$, and the two formulas differ already at the linear Hamiltonians.

Proof. Let $X_f = (\partial_{z_1}f)\partial_{z_2} - (\partial_{z_2}f)\partial_{z_1}$. Hamiltonian vector fields preserve the volume form dz_1dz_2. For a Laurent coefficient ϕ and $g\in R$, the residue of a total z_i-derivative is zero, hence

$$0 = \mathrm{Res}\big(L_{X_f}(\phi g\,dz_1dz_2)\big) = \mathrm{Res}\big((\{f,\phi\}g + \phi\{f,g\})\,dz_1dz_2\big).$$

Therefore the residue pairing satisfies integration by parts,

$$\langle\{f,\rho\},g\rangle = -\langle\rho,\{f,g\}\rangle,$$

which is exactly the coadjoint sign convention $(\mathrm{ad}_f^\vee\lambda)(g) = -\lambda(\{f,g\})$.

For $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1}z_2^{-b-1}\,dz_1dz_2$, expand the bracket. The two terms come from $\partial_{z_1}f\cdot\partial_{z_2}\phi$ and $-\partial_{z_2}f\cdot\partial_{z_1}\phi$:

$$\begin{aligned}\{f, z_1^{-a-1}z_2^{-b-1}\} &= pz_1^{p-1}z_2^q\,(-(b+1))z_1^{-a-1}z_2^{-b-2}\\ &\quad - qz_1^p z_2^{q-1}\,(-(a+1))z_1^{-a-2}z_2^{-b-1}\\ &= \big(-p(b+1)+q(a+1)\big)z_1^{p-a-2}z_2^{q-b-2}\\ &= -(pb-qa+p-q)z_1^{p-a-2}z_2^{q-b-2}.\end{aligned}$$

Multiplication by $dz_1 dz_2$ gives $\rho_{a-p+1,b-q+1}$ precisely when the two new indices are nonnegative; otherwise the principal-part projection gives zero. If $a - p + 1 = b - q + 1 = 0$, then $a = p - 1$ and $b = q - 1$, and the coefficient $pb - qa + p - q = p(q-1) - q(p-1) + p - q$ reduces to zero by direct expansion. Thus the action preserves the scalar-annihilator $\mathcal{P}$. The signed coefficient is verified by direct computation on 1225 cases with exponents ≤ 5. $\square$

LEMMA 8.8 *(Continuity of the coadjoint action).* The formula of Proposition 8.7 extends from polynomial Hamiltonians to all of $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$. The bilinear action

$$\mathfrak{h} \times \mathcal{P} \longrightarrow \mathcal{P}$$

is continuous for the $\mathfrak{m}$-adic topology on $\mathfrak{h}$ and the locally finite direct-sum topology on $\mathcal{P}$.

Proof. Fix $\rho_{a,b}$. In the monomial formula a term $z_1^p z_2^q \cdot \rho_{a,b}$ can be nonzero only if $p \leq a + 1$ and $q \leq b + 1$. Thus an arbitrary formal Hamiltonian acts on $\rho_{a,b}$ through a finite jet, so the action is well-defined and continuous in the Hamiltonian variable. Conversely, if $a + b \leq N$, every nonzero target has total index

$$(a - p + 1) + (b - q + 1) = a + b - (p + q) + 2 \leq N + 1$$

because $p + q > 0$ for a reduced Hamiltonian. Hence each fixed Hamiltonian sends $\mathcal{P}_{\leq N}$ into $\mathcal{P}_{\leq N+1}$. This is exactly continuity for the locally finite direct limit. $\square$

LEMMA 8.9 *(Coadjoint centralizer).* Every $\mathfrak{h}$-module endomorphism of $\mathcal{P}$, with the coadjoint action above, is multiplication by one scalar.

Proof. The vector $\rho_{1,0}$ is cyclic. Indeed $z_1^2 \cdot \rho_{1,0} = -2\rho_{0,1}$; repeated action by the linear Hamiltonians z_1 and z_2 then reaches every $\rho_{a,b}$, $a + b > 0$, with nonzero rising-factorial coefficient.

Let T be an $\mathfrak{h}$-module endomorphism. Since $z_1 z_2 \cdot \rho_{a,b} = (a - b)\rho_{a,b}$, the vector $T\rho_{1,0}$ lies in the eigenvalue-one subspace

$$\bigoplus_{r \geq 0} \mathbb{C}\, \rho_{r+1,r}.$$

It is a finite sum because $\mathcal{P}$ is a direct sum: $T\rho_{1,0} = \sum_{r=0}^{R} c_r \rho_{r+1,r}$. The Hamiltonian z_2^2 kills $\rho_{1,0}$, hence it kills $T\rho_{1,0}$. But for $r \geq 1$,

$$z_2^2 \cdot \rho_{r+1,r} = (2r + 4)\rho_{r+2,r-1},$$

while the $r = 0$ target has negative second index and is zero. The displayed nonzero targets are linearly independent, so $c_r = 0$ for all $r \geq 1$. Therefore $T\rho_{1,0} = c_0 \rho_{1,0}$, and cyclicity gives $T = c_0\, \mathrm{id}_{\mathcal{P}}$. $\square$

8.3 Rigidity of the residue pairing

THEOREM 8.10 *(Residue pairing rigidity).* Up to multiplication by one scalar in $\mathbb{C}^\times$, the residue pairing is the unique perfect continuous pairing

$$\mathcal{P} \times \mathfrak{h} \longrightarrow \mathbb{C}$$

which is equivariant for the coadjoint action above. Here *perfect* means that the adjoint map $\mathcal{P} \to \mathfrak{h}^\vee_{\mathrm{cont}}$ is a topological vector-space isomorphism. In particular the same uniqueness holds in the narrower class of total-degree-zero pairings. Fixing the scalar by

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}$$

is therefore the universal dual-module normalization for the principal-part cotangent sector.

Proof. Let β be another perfect continuous equivariant pairing. It gives a topological vector-space isomorphism

$$\Phi_\beta : \mathcal{P} \longrightarrow \mathfrak{h}^\vee_{\mathrm{cont}}, \qquad \Phi_\beta(\rho)(g) = \beta(\rho, g).$$

Equivariance of β says

$$\beta(f \cdot \rho, g) = -\beta(\rho, \{f, g\}),$$

so Φ_β is an $\mathfrak{h}$-module map to the coadjoint module. Let Φ_{res} be the residue isomorphism of Proposition 8.6. Then $T = \Phi_{\mathrm{res}}^{-1}\Phi_\beta$ is an $\mathfrak{h}$-module automorphism of $\mathcal{P}$. By Lemma 8.9, $T = c\,\mathrm{id}_{\mathcal{P}}$. Since Φ_β is an isomorphism, $c \neq 0$, and $\beta = c\,\langle\cdot\,,\,\cdot\rangle_{\mathrm{res}}$. The displayed Kronecker normalization sets $c = 1$. □

8.4 Local-nilpotence obstruction to polynomial realization

Theorem 8.11 *(Polynomial descendant obstruction for principal parts).* Let M be a polynomial boundary descendant module on which the linear Hamiltonians z_1, z_2 act locally nilpotently. Then every $\mathfrak{h}$-module map from $\mathcal{P}$ to M has zero image. In particular the polynomial PBW one-ψ descendants $\widetilde{\Psi}_{a,b}$ and the principal parts $\rho_{a,b}$ have a nonzero ordinary $\mathfrak{h}$-module identification obstruction.

Proof. In any ordinary polynomial boundary model, the linear Hamiltonians act as constant coefficient Hamiltonian vector fields; iterating either z_1 or z_2 eventually exhausts the polynomial degree. Thus these actions are locally nilpotent: for every $m \in M$ and every $i \in \{1, 2\}$ there exists $N \geq 1$ with $(z_i\cdot)^N m = 0$.

On $\mathcal{P}$ the situation is rigorously the opposite. Proposition 8.7 specialises to

$$z_1 \cdot \rho_{a,b} = -(b+1)\,\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\,\rho_{a+1,b},$$

and iteration produces the rising-factorial coefficient

$$z_1^N \cdot \rho_{a,b} = (-1)^N (b+1)(b+2)\cdots(b+N)\,\rho_{a,b+N} \neq 0$$

for every $N \geq 0$; the same holds for z_2. Local nilpotence is preserved by module isomorphism and by restriction to submodules. Therefore a polynomial descendant module with locally nilpotent linear Hamiltonians has the local-nilpotence obstruction to realizing $\mathcal{P}$, or any nonzero submodule of $\mathcal{P}$. The obstruction is independent of $\hbar$ and survives every label relabelling between the two alphabets.

For the primitive PBW labels $\widetilde{\Psi}_{a,b}$ the obstruction is already visible at the boundary of the index quadrant. After the scalar descendant is removed, the special-fibre action of the linear Hamiltonians on the polynomial one-ψ shadow is

$$z_1 \cdot \widetilde{\Psi}_{a,b} = b\,\widetilde{\Psi}_{a,b-1}, \qquad z_2 \cdot \widetilde{\Psi}_{a,b} = -a\,\widetilde{\Psi}_{a-1,b}.$$

Thus z_1 kills $\widetilde{\Psi}_{a,0}$ for every $a > 0$, whereas on the residual side $z_1 \cdot \rho_{a,0} = -\rho_{a,1} \neq 0$. Every rescaling $\widetilde{\Psi}_{a,b} \mapsto \lambda_{a,b}\rho_{a,b}$ with $\lambda_{a,b} \neq 0$ preserves the obstruction: the linear Hamiltonian z_1 lowers the second index on the polynomial side and raises it on the residual side. □

8.5 Universal Fourier–Rees label bridge

Theorem 8.12 *(Universal Fourier–Rees construction and exact obstruction).* Let

$$\mathcal{D}_\hbar = \mathbb{C}[[\hbar]]\langle z_1, z_2, \partial_1, \partial_2\rangle \big/ \big([\partial_i, z_j] = \hbar\delta_{ij}, [z_i, z_j] = [\partial_i, \partial_j] = 0\big)$$

be the Rees Weyl algebra. Let

$$\mathcal{O}_\hbar = \mathbb{C}[z_1, z_2][[\hbar]] \cong \mathcal{D}_\hbar/(\partial_1, \partial_2)$$

be the polynomial Rees module, and let

$$\Delta^{\mathsf{F}}_\hbar = \mathcal{D}_\hbar/(z_1, z_2)$$

be the Fourier delta Rees module, generated by a vector e_0 killed by z_1, z_2. These quotients are left cyclic $\mathcal{D}_\hbar$-modules, with the indicated left ideals generated by ∂_1, ∂_2 and by z_1, z_2. PBW normal form gives $\mathcal{O}_\hbar$ the free $\mathbb{C}[[\hbar]]$-basis $z_1^a z_2^b$ and gives $\Delta^{\mathsf{F}}_\hbar$ the free basis $\partial_1^a \partial_2^b e_0$; in particular both Rees modules are $\mathbb{C}[[\hbar]]$-flat. The Čech local-cohomology lattice

$$\Delta^{\mathrm{std}}_\hbar = H^2_{\mathfrak{m}}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2[[\hbar]]$$

carries the standard action in which z_i multiplies and $\partial_i = \hbar\partial_{z_i}$. Sending $e_0 \mapsto \rho_{0,0} = z_1^{-1}z_2^{-1}dz_1dz_2$ realizes $\Delta^{\mathsf{F}}_\hbar$ as the Rees sublattice

$$\iota_{\mathsf{F}}(\Delta^{\mathsf{F}}_\hbar) = \bigoplus_{a,b\geq 0} \mathbb{C}[[\hbar]]\,(-1)^{a+b}\hbar^{a+b}a!b!\,\rho_{a,b} \subset \Delta^{\mathrm{std}}_\hbar.$$

This inclusion becomes an isomorphism after tensoring with $\mathbb{C}((\hbar))$. At the $\mathbb{C}[[\hbar]]$-lattice level it is the displayed Rees sublattice. The Rees Fourier automorphism

$$\mathsf{F}(z_i) = \partial_i, \qquad \mathsf{F}(\partial_i) = -z_i$$

identifies the Fourier twist $\mathsf{F}(\mathcal{O}_\hbar)$ with $\Delta^{\mathsf{F}}_\hbar$. This is the flat positive bridge: the action of $P \in \mathcal{D}_\hbar$ on the polynomial source is carried to the action of $\mathsf{F}(P)$ on the Fourier delta target. On basis labels in the Čech realization,

$$z_1^a z_2^b \longmapsto \partial_1^a \partial_2^b e_0 \xrightarrow{\iota_{\mathsf{F}}} \partial_1^a \partial_2^b \rho_{0,0} = (-1)^{a+b}\hbar^{a+b}a!b!\,\rho_{a,b}.$$

After inverting $\hbar$, this gives a filtered Fourier–Rees identification of label spaces between nonconstant polynomial labels and the reduced principal parts $\rho_{a,b}$, $a + b > 0$, but only after the Fourier twist of action displayed above. Over the Rees lattice before inverting $\hbar$, the construction records the degree-$a + b$ shift by the explicit factor $\hbar^{a+b}$. The special fibre of the abstract Rees module keeps the labels $\partial_1^a \partial_2^b e_0$; the Čech inclusion into $\Delta^{\mathrm{std}}_\hbar$ sends the positive-degree labels into $\hbar\Delta^{\mathrm{std}}_\hbar$. Thus the comparison is an associated-graded label match; ordinary local-cohomology lattices retain the displayed Rees shift.

The obstruction for the original Hamiltonian action is exact. The Hamiltonian action on polynomial descendants is by vector fields $X_f = \{f, -\}$ on $\mathcal{O}_\hbar((\hbar))$. In the Rees Weyl algebra the normalized linear operators are $\hbar X_{z_1} = \partial_2$ and $\hbar X_{z_2} = -\partial_1$. Their Fourier transforms are $\mathsf{F}(\hbar X_f)$, whereas the Matlis cotangent action on $\mathcal{P}((\hbar))$ is still $\hbar X_f$, acting on Laurent principal parts and projected to the reduced principal-part span. Already for the linear Hamiltonians,

$$\hbar X_{z_1} = \partial_2, \qquad \hbar X_{z_2} = -\partial_1,$$

but

$$\mathsf{F}(\hbar X_{z_1}) = -z_2, \qquad \mathsf{F}(\hbar X_{z_2}) = z_1.$$

The transformed linear operators are locally nilpotent on $\Delta^{\mathsf{F}}_\hbar$; the Rees-normalized coadjoint operators

$$\hbar X_{z_1}\rho_{a,b} = -(b+1)\hbar\rho_{a,b+1}, \qquad \hbar X_{z_2}\rho_{a,b} = (a+1)\hbar\rho_{a+1,b}$$

have infinite orbits on every nonzero reduced principal part after inverting $\hbar$. Dividing by $\hbar$ gives the same conclusion for the unnormalized Hamiltonian vector fields. The Fourier–Rees construction therefore identifies the polynomial descendant module with $\mathcal{P}$ as a module over a Fourier-conjugate copy of the reduced Hamiltonian Lie algebra; identification over the unconjugated action carries the same-action obstruction recorded below. Any universal bridge selects one of two routes: Fourier-conjugate the action as above, or carry the same-action obstruction in the original Matlis coadjoint frame.

Proof. The displayed formulas for F preserve the Rees Weyl relations, since

$$[\mathsf{F}(\partial_i), \mathsf{F}(z_j)] = [-z_i, \partial_j] = \hbar\delta_{ij}.$$

The polynomial module $\mathcal{O}_\hbar$ is cyclic with generator 1 annihilated by ∂_1, ∂_2. After Fourier twist the cyclic generator is annihilated by z_1, z_2, so the twisted module is $\mathcal{D}_\hbar/(z_1, z_2) = \Delta^{\mathsf{F}}_\hbar$. In the Čech model, $z_i\rho_{0,0} = 0$, and applying $\partial_i = \hbar\partial_{z_i}$ to $\rho_{0,0}$ generates the Rees lattice displayed in the theorem. It generates the full standard local-cohomology lattice only after $\hbar$ is inverted. Therefore $\mathsf{F}(\mathcal{O}_\hbar) \cong \Delta^{\mathsf{F}}_\hbar$, with the Čech realization ι_{F}.

Since ∂_i acts as $\hbar\partial_{z_i}$ on Laurent representatives,

$$\partial_1^a\partial_2^b\rho_{0,0} = \hbar^{a+b}\partial_{z_1}^a\partial_{z_2}^b(z_1^{-1}z_2^{-1}dz_1dz_2) = (-1)^{a+b}\hbar^{a+b}a!b!\rho_{a,b}.$$

Removing the scalar label $a = b = 0$ gives the reduced label comparison after the stated Rees normalization. Since the factors $a!b!$ and $(-1)^{a+b}$ are units over $\mathbb{C}$, the only genuine Rees-lattice shift is the power $\hbar^{a+b}$.

For the obstruction, compare the two linear actions. On polynomial descendants the Rees-normalized operators $\hbar X_{z_1} = \partial_2$ and $\hbar X_{z_2} = -\partial_1$ are locally nilpotent; after inverting $\hbar$, the same is true for X_{z_1} and X_{z_2}. Fourier sends the Rees-normalized operators to multiplication by $-z_2$ and z_1 on $\Delta^{\mathsf{F}}_\hbar$; multiplication by z_i lowers pole order and is again locally nilpotent. The coadjoint Matlis action is distinct from the Fourier-transformed action. It keeps the same vector fields, now acting on Laurent principal parts:

$$\hbar X_{z_1}\rho_{a,b} = \hbar\partial_{z_2}\rho_{a,b} = -(b+1)\hbar\rho_{a,b+1}, \qquad \hbar X_{z_2}\rho_{a,b} = -\hbar\partial_{z_1}\rho_{a,b} = (a+1)\hbar\rho_{a+1,b}.$$

After tensoring with $\mathbb{C}((\hbar))$, every nonzero vector in $\mathcal{P}((\hbar))$ has an infinite orbit under either linear Hamiltonian: for a finite nonzero linear combination, every power of X_{z_1} shifts each nonzero basis summand to a distinct $\rho_{a,b+N}$, with nonzero rising-factorial coefficient, and the same argument applies to X_{z_2}. Local nilpotence is invariant under module isomorphism. Thus a Fourier–Rees comparison preserving the original Hamiltonian action has the same-action obstruction; the positive construction changes the polarization and is a flat $\mathcal{D}_\hbar$-module bridge for the Fourier-conjugated action stated above. □

Corollary 8.13 *(Same-action obstruction criterion).* Let $M_\hbar$ be a $\mathbb{C}[[\hbar]]$-flat family whose generic fibre carries the polynomial descendant action of the linear Hamiltonians z_1, z_2, so that X_{z_1} and X_{z_2} act locally nilpotently after tensoring with $\mathbb{C}((\hbar))$. Then $M_\hbar((\hbar))$ and $\mathcal{P}((\hbar))$ with the Matlis coadjoint action have a same-action module-isomorphism obstruction. The same statement holds for the Rees-normalized operators $\hbar X_{z_i}$, since $\hbar$ is invertible on the generic fibre. Equivalently, local nilpotence of the two linear Hamiltonians on the generic fibre is a necessary condition for a flat same-action Rees bridge, and $\mathcal{P}((\hbar))$ violates that condition on every nonzero vector. Any same-action comparison must therefore either change the Hamiltonian action by a Fourier twist, enlarge the boundary target to a distributional principal-part sector, or state a different deformation problem.

Proof. This is the local-nilpotence obstruction of Theorem 8.11 after extension of scalars to $\mathbb{C}((\hbar))$. The two linear Hamiltonians remain locally nilpotent on the polynomial descendant generic fibre and remain non-locally-nilpotent on every nonzero element of $\mathcal{P}((\hbar))$. Multiplication by the invertible scalar $\hbar$ preserves local nilpotence. □

Remark 8.14 *(Lattice content of the Fourier–Rees bridge).* The shared index set (a, b) is the Fourier–Rees associated-graded convention, with explicit factorial and $\hbar^{a+b}$ normalization. The Fourier twist gives $\mathcal{D}_\hbar/(z_1, z_2)$, whose Čech realization is the Rees sublattice $\bigoplus \hbar^{a+b}\mathbb{C}[[\hbar]]\rho_{a,b}$. The full $\bigoplus \mathbb{C}[[\hbar]]\rho_{a,b}$ appears after $\hbar$ is inverted. Fourier sends the Hamiltonian vector fields to Fourier-transformed operators on $\Delta^{\mathsf{F}}_\hbar$; the Matlis cotangent module uses the original coadjoint vector fields on principal parts.

8.6 Pro-Matlis envelope and bulk-boundary duality

The obstruction classes of §7.5 use the pro-Matlis target

$$\widehat{\mathcal{P}}_\Pi = \varprojlim_N \mathcal{P}_{\leq N}, \qquad \mathcal{P}_{\leq N} = \bigoplus_{0<a+b\leq N} \mathbb{C}\,\rho_{a,b}.$$

The reduced Hamiltonian Lie algebra $\mathfrak{h} = R/\mathbb{C}\cdot 1$, $R = \mathbb{C}[[z_1, z_2]]$, carries the $\mathfrak{m}$-adic topology with maximal ideal $\mathfrak{m} = (z_1, z_2)$; its continuous dual is the Grothendieck–Matlis principal-part module

$$\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\,\rho_{a,b} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) \subset H^2_{\mathfrak{m}}(R)\otimes\omega_R, \quad \rho_{a,b} = z_1^{-a-1}z_2^{-b-1}\,dz_1dz_2,$$

the scalar-annihilator subspace of the top local cohomology of R (Matlis [19], Grothendieck–Hartshorne local cohomology [20], Kunz–Cox–Dickenstein [21]; Theorem 2.1, Proposition 5.176, Section 8). The residue Kronecker pairing $\langle\rho_{a,b}, z_1^m z_2^n\rangle = \delta_{a,m}\delta_{b,n}$ identifies each $\rho_{a,b}$ with the continuous Taylor coefficient functional, and the Hamiltonian coadjoint action transports through this residue identification to the principal-part residue formula

$$z_1^p z_2^q\cdot\rho_{a,b} = -(pb - qa + p - q)\,\rho_{a-p+1,b-q+1}$$

(negative-index terms zero; Proposition 5.176). The finite Matlis window $\mathcal{P}_{\leq N}$ is the truncation at pole order $a + b \leq N$; the inverse limit $\widehat{\mathcal{P}}_\Pi$ is the pro-Matlis envelope used for the N-tower coefficient comparison. Matlis duality at finite presentation gives the finite-window Tate dual: a finitely generated R-module M of finite length supported at $\mathfrak{m}$ admits the Matlis dual $M^\vee = \mathrm{Hom}_R(M, E_R(\mathbb{C}))$ with $M^{\vee\vee} = M$, realised inside $\mathcal{P}_{\leq N}$ for window size N bounding the pole order of $M^\vee$ (Matlis [19]; Definition 8.5, Lemma 8.8).

Leakage out of the finite Matlis window $\mathcal{P}_{\leq N}$ for a class is the finite-rank Matlis duality obstruction on the corresponding window; the obstruction theorem at level N is Theorem 8.40, the Tate-cohomology identity

$$z_1^{N+2}\cdot\rho_{N+1,0} = -(N+2)\,\rho_{0,1}$$

exhibiting a high transverse holomorphic mode at degree $(N + 1, 0)$ transported under Hamiltonian flow into a low brane-momentum mode at degree $(0, 1)$. Every finite-Matlis cutoff retains this leakage. The pro-Matlis replacement theorem (Theorem 8.42) selects $\widehat{\mathcal{P}}_\Pi$ as the completed receiver for the fixed finite-window moment maps. The full $\mathfrak{h}$-coadjoint action lives on the untruncated pro-object; the finite-window maps are compatible projections, not strict equivariant quotients (Corollary 8.41).

Bulk-boundary duality. The bulk side of the formal-Darboux stalk is the Chevalley–Eilenberg cochain complex $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}, M)$ of the shifted-cotangent Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ (Theorem 2.8); the boundary trace side is the stable scalar-reduced trace algebra

$$A^{\mathrm{st}}_{\partial,H} \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}), \qquad z_1^a z_2^b \longmapsto J(z_1^a z_2^b).$$

The full finite-N brane algebra is instead

$$A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}(\mathfrak{gl}_N, \mathrm{Kosz}([\phi_1, \phi_2])).$$

The first object is the stable trace shadow of the second, not its definition. The duality with the bulk CE complex lives in the pro-Matlis target. At each finite Hamiltonian window the truncated cotangent CE/PV coordinate identification holds at finite presentation (Theorem 5.102), and the residue pairing $\mathcal{P}_{\le N} \times (\mathfrak{h}/\mathfrak{m}^{N+1}) \to \mathbb{C}$ is finite-rank perfect. The duality pro-extends through $\widehat{\mathcal{P}}_\Pi = \varprojlim_N \mathcal{P}_{\le N}$ by the pro-Matlis receiver property (Theorem 8.42); coherence on the inverse limit is the $\widehat{\mathcal{P}}_\Pi$-coherent extension of the finite-window dualities. The discarded strict all-window support-local current transfer is governed by $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$; this obstruction blocks that strict target and motivates the pro-Matlis replacement (Theorem 8.40).

The $\varprojlim^1$ secondary class. Lifting the finite-window Matlis duality to the inverse-limit target requires choosing a coherent counterterm tower; the obstruction is the secondary $\varprojlim^1 H^0$ primitive class in the inverse system of finite Hamiltonian windows; this is the Roos $\varprojlim^1$-cocycle (Roos, derived functors of inverse limits)

$$\lambda^{\mathrm{fw}}_{n_0} \in \varprojlim\nolimits^1_M H^0\big(Q^\bullet_{\mathcal{G},M}(I)\big)$$

of the Hadamard / Mittag-Leffler residual, measuring the obstruction to compatible windowwise primitives at the lowest unsolved order n_0 (§7.5 on $\mathfrak{o}_{\mathrm{Had}}$; the $\mathrm{Ob}^{\mathrm{HML}}_w = (\mathrm{Ob}_w, \lambda^{\mathrm{fw}}_{n_0})$ of Theorem 8.17). The two coordinates lie in independent cohomology groups: a vanishing graphwise QME class is independent of tower-compatibility of windowwise primitives, and a Mittag-Leffler tower can produce zero $\lambda^{\mathrm{fw}}_{n_0}$ while the underlying cohomology obstruction is nonzero. Coherent primitives on $\widehat{\mathcal{P}}_\Pi$ require $\lambda^{\mathrm{fw}}_{n_0} = 0$. The class $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$ remains the obstruction to the discarded strict all-window finite-current transfer.

The pro-Matlis target is the coefficient home for the finite-window Matlis components of the obstruction vector along the N-tower. The component of the obstruction vector Ob_T of Definition 7.3 that lands on $\widehat{\mathcal{P}}_\Pi$ is a coherence class for the inverse-limit extension of finite-window Matlis duality, with the Roos $\varprojlim^1$ secondary tracking the obstruction to compatible windowwise primitives. The Casimir extension of §6.1 and its Capelli scalar $\hbar N[\bar{c}]$ sit at the trace generator inside the same pro-Matlis envelope; the construction problem each component inscribes is the choice of pro-Matlis-coherent counterterm tower that null-homotopes the corresponding class.

Hadamard parametrix on the Mittag-Leffler coefficient module

Costello's heat-kernel BV calculus [12, Chs. 5–7] requires a free elliptic complex with finite-rank coefficient bundle; its Hadamard parametrix expansion together with Hörmander wave-front tracking [36, Ch. 7] locates the singular support of the propagator on the bulk diagonal. The weighted Tate field complex (24),

$$\mathcal{E}_w = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\Big(\prod_{d\ge 1} w(d) H_d[1] \oplus \prod_{d\ge 1} w(d)^{-1} H_d^\vee[2]\Big),$$

carries an infinite tower of coefficient bundles indexed by Hamiltonian degree, so the coefficient direction has infinite rank. The question is whether that finite-rank machinery commutes, graphwise and at every order in $\hbar$, with the strict Mittag-Leffler inverse limit in the coefficient direction; equivalently, whether universal summation of the windowwise counterterms survives passage to the inverse-limit weighted Tate envelope, and what class measures the obstruction.

The Hadamard/Mittag-Leffler (H/ML) commutation residual measuring this obstruction is a pair of independent cohomology classes: a primary QME-obstruction class in the local-functional cohomology of the weighted complex, and a secondary Roos $\varprojlim^1$-cocycle on the inverse system of finite Hamiltonian windows. Concretely,

$$\mathrm{Ob}_w^{\mathrm{HML}} = \big(\mathrm{Ob}_w,\ \lambda_{n_0}^{\mathrm{fw}}\big) \in H^1\big(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\}\big) \,\oplus\, \varprojlim\nolimits^1_M H^0\big(\boldsymbol{Q}^\bullet_{\mathcal{G},M}(I)\big).$$

The first coordinate Ob_w is the QME obstruction class in local-functional cohomology with differential $Q + \{I_0, -\}$; the second coordinate $\lambda_{n_0}^{\mathrm{fw}}$ is the Roos $\varprojlim^1$-cocycle measuring the obstruction to compatible windowwise primitives at the lowest unsolved order n_0. The two coordinates lie in independent cohomology groups and detect independent obstruction modes: graphwise QME vanishing and tower-compatibility of windowwise primitives are separate conditions, and a Mittag-Leffler tower can produce zero $\lambda_{n_0}^{\mathrm{fw}}$ while the underlying cohomology obstruction is nonzero. Theorem 8.17 reduces the analytic content of Problem 5.73 – exact propagator factorization, empty coefficient-Casimir wave front, strict ML transition surjectivity – to the algebraic remainder: graphwise QME in the local-functional class Ob_w and an inverse-limit Roos primitive class $\lambda_{n_0}^{\mathrm{fw}}$ controlled by Theorem 8.22. For the fixed graphwise extension problem under these hypotheses, vanishing of both coordinates is necessary and sufficient for compatible graphwise quantum extension on $\mathcal{E}_w$; a single nonzero coordinate is the precise nonzero obstruction mechanism. The θ_3 row of Proposition 8.32 exhibits a local-functional row that passes the full-equivariant scalar-condition vanishing and presents its missing primitive as the named datum (CE-ancestor, scalar-zero counterterm, or complete companion-face table).

The construction completing the residual on a given graph system gives four ingredients: the finite Hamiltonian window M and graph list $\mathcal{G}_M$; a chain projection $\widehat{\sigma}_{\mathrm{sc},M}$ onto the scalar-symbol sector with vanishing image of the residual; primitives $c_{n,M}$ of the non-scalar residual at every window; and Roos-compatible truncation coefficients across windows. A violated ingredient names the corresponding nonzero coordinate of $\mathrm{Ob}_w^{\mathrm{HML}}$. The finite-row complex – graph arrays, truncation matrices, scalar projection, and primitives – yields a finite cochain whose vanishing is tested at every window M and every order $\hbar^n$.

The principal-part current sector carries the parallel finiteness condition. Theorem 8.40 below proves, by the single Tate-cohomology identity $z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\,\rho_{0,1}$, the obstruction to Hamiltonian-equivariant principal-part transfer into a finite Matlis target: high transverse holomorphic modes leak back into low brane momentum modes under Hamiltonian flow. The replacement structure used below is the one-pair analytic pro-Matlis retract (Corollary 8.41), with its fixed finite-window projections and Bochner–Martinelli moment map (Theorem 8.42). The Hadamard / QME residual $\mathrm{Ob}_w^{\mathrm{HML}}$ and the BMK finite-window obstruction live in independent sectors of the local theorem, with separate formulas.

Remark 8.15 *(Spacetime/coefficient factorization of the Hadamard parametrix).* In Costello's heat-kernel BV calculus [12, Ch. 5], the Hadamard parametrix enters as the asymptotic expansion of the heat kernel K_t^{base} on the bulk free elliptic complex $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$, combined with Hörmander wave-front-set tracking [36, Ch. 7] to control the singular support of the propagator $P_{\epsilon,L}^{\mathrm{base}}$ along the diagonal. Costello's heat-kernel locality formalism [12, Chs. 5–7] is used here only as the heat-kernel/RG vocabulary for free elliptic complexes: local functionals admit Hadamard-parametrix expansions whose counterterms stay in the local-functional class $\mathcal{O}_{\mathrm{loc}}$ under the hypotheses of that formalism. The local datum here is this heat-kernel/RG formalism.

In the weighted setting of 22, the propagator factors as $P^{w}_{\epsilon,L} = P^{\mathrm{base}}_{\epsilon,L}\widehat{\otimes}C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$. The bulk piece $P^{\mathrm{base}}_{\epsilon,L}$ is independent of the coefficient weight; the coefficient piece $C_{\mathfrak{h},w}$ is purely algebraic, with zero scale parameter and zero Laplacian. The Hadamard parametrix construction therefore lives entirely on the bulk factor; the coefficient factor enters only by tensor.

Lemma 8.16 *(Windowwise parametrix compatibility).* For each $w_0 \in \mathbb{Z}_{\geq 1}$, let

$$C^{w_0}_{\mathfrak{h},w} = \sum_{d\leq w_0}\sum_i (w(d)e_{d,i}) \otimes (w(d)^{-1}e^i_d) \in \Big(\prod_{d\leq w_0} w(d)H_d\Big) \otimes \Big(\prod_{d\leq w_0} w(d)^{-1}H^\vee_d\Big)$$

denote the truncated weighted Casimir. Define the windowwise field complex

$$\mathcal{E}^{w_0}_w = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\bigg(\Big(\prod_{d\leq w_0} w(d)H_d\Big)[1] \oplus \Big(\prod_{d\leq w_0} w(d)^{-1}H^\vee_d\Big)[2]\bigg)$$

and the windowwise propagator $P^{w,w_0}_{\epsilon,L} = P^{\mathrm{base}}_{\epsilon,L}\widehat{\otimes}C^{w_0}_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}$. Then:

(1) For each w_0, $\mathcal{E}^{w_0}_w$ is a free elliptic complex with finite-rank coefficient bundle, and Costello's Hadamard parametrix construction [12, Ch. 5] applies verbatim to $P^{w,w_0}_{\epsilon,L}$.

(2) The truncations $\{P^{w,w_0}_{\epsilon,L}\}_{w_0\geq 1}$ form a strict Mittag-Leffler inverse system in the coefficient direction, with surjective transition maps $P^{w,w_0'}_{\epsilon,L} \twoheadrightarrow P^{w,w_0}_{\epsilon,L}$ for $w_0' \geq w_0$ given by truncation to degrees $d \leq w_0$.

(3) The wave-front set $\mathrm{WF}(P^{w,w_0}_{\epsilon,L})$ is contained in the bulk wave-front set $\mathrm{WF}(P^{\mathrm{base}}_{\epsilon,L}) \subset T^*(\mathbb{R}^2 \times \mathbb{C}^2) \setminus 0$, independently of w_0, since the coefficient factor $C^{w_0}_{\mathfrak{h},w}$ is a finite-rank algebraic tensor with empty spacetime singular support.

Proof. (1) For fixed w_0, the coefficient bundle $\big(\prod_{d\leq w_0} w(d)H_d\big)[1] \oplus \big(\prod_{d\leq w_0} w(d)^{-1}H^\vee_d\big)[2]$ is a finite-rank $\mathbb{Z}$-graded vector space (finite product of finite-dimensional H_d's). Costello's Hadamard parametrix construction [12, Ch. 5] applies verbatim to any free elliptic complex with finite-rank coefficient bundle, so it applies to $\mathcal{E}^{w_0}_w$. The bulk propagator $P^{\mathrm{base}}_{\epsilon,L}$ is unchanged; the coefficient factor $C^{w_0}_{\mathfrak{h},w}$ is a fixed finite-rank tensor.

(2) Surjectivity of the transition maps is the truncation $\prod_{d\leq w_0'} w(d)H_d \twoheadrightarrow \prod_{d\leq w_0} w(d)H_d$ on the coefficient direction; the bulk factor is unchanged. An inverse system with surjective transition maps satisfies the Mittag-Leffler condition trivially, so the inverse limit is exact in the strict finite-window category of Definition 5.16, and Proposition 5.17 gives $\varprojlim^1 = 0$ for this tower. Compatibility with $P^{w,w_0'}_{\epsilon,L} \mapsto P^{w,w_0}_{\epsilon,L}$ follows from the tensor factorization and the truncation of the Casimir.

(3) The Hörmander wave-front set of a tensor product distribution $T_1 \otimes T_2$ on $X_1 \times X_2$ satisfies $\mathrm{WF}(T_1 \otimes T_2) \subseteq \big(\mathrm{WF}(T_1) \times \{(x_2, 0) : x_2 \in X_2\}\big) \cup \big(\{(x_1, 0) : x_1 \in X_1\} \times \mathrm{WF}(T_2)\big) \cup \big(\mathrm{WF}(T_1) \times \mathrm{WF}(T_2)\big)$ ([36, Thm. 8.2.9]). Here $T_1 = P^{\mathrm{base}}_{\epsilon,L}$ has wave-front set in the conormal of the diagonal of $\mathbb{R}^2 \times \mathbb{C}^2$, and $T_2 = C^{w_0}_{\mathfrak{h},w}$ is a finite-rank algebraic tensor with constant point dependence, hence smooth, so $\mathrm{WF}(T_2) = \emptyset$. Hence $\mathrm{WF}(P^{w,w_0}_{\epsilon,L}) = \mathrm{WF}(P^{\mathrm{base}}_{\epsilon,L}) \times \{0\}$, identified with $\mathrm{WF}(P^{\mathrm{base}}_{\epsilon,L})$ on the spacetime, independent of w_0. □

Theorem 8.17 *(Hadamard control on the Mittag-Leffler coefficient module).* Let $w(d) = q^d$ with $q > 1$ be the exponential degree weight of Remark 5.54, and let $\mathcal{E}_w$ be the weighted field complex

(24). The Hadamard parametrix construction [12, Ch. 5] extends from each finite-window restriction $\mathcal{E}_w^{w_0}$ (Lemma 8.16) to the Mittag-Leffler limit $\mathcal{E}_w = \varprojlim_{w_0} \mathcal{E}_w^{w_0}$, preserving the Hadamard wave-front and finite-window locality estimates graph by graph. Specifically:

(i) The propagator $P^w_{\epsilon,L}$ admits a Hadamard parametrix expansion as the inverse limit

$$P^w_{\epsilon,L} = \varprojlim_{w_0} P^{w,w_0}_{\epsilon,L},$$

with bulk wave-front set independent of w_0.

(ii) For every local interaction $I \in \mathcal{O}_{\text{loc}}(\mathcal{E}_w)$ of bounded polynomial-degree D_0 at each vertex in the coefficient direction, every fixed Costello graph Γ of order $\hbar^n$ contributes a compatible finite-window counterterm system whose weighted coefficient size is bounded, on every finite codomain window, by a finite graph-dependent constant times $|C_n^{\text{base}}[\Gamma]|$, where $|C_n^{\text{base}}[\Gamma]|$ is the bulk spacetime counterterm of Costello.

(iii) For every fixed graph whose finite-window counterterms are degreewise surjective and compatible under the Mittag-Leffler transition maps, the corresponding inverse-limit representative lies in the local-functional class $\mathcal{O}_{\text{loc}}(\mathcal{E}_w)$. The associated endpoint obstruction is the class in $H^1(\mathcal{O}_{\text{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ for the mixed brane-defect interaction.

(iv) The construction has the ordinary strict inverse-limit universal property: a compatible graphwise counterterm family $\{C_n[\Gamma]^{w_0}\}_{w_0}$ determines a unique element $C_n[\Gamma] \in \varprojlim_{w_0} \mathcal{O}_{\text{loc}}(\mathcal{E}_w^{w_0})$, and every finite-window projection of this element is the given counterterm.

Thus the Hadamard and wave-front part of Problem 5.73 is exactly the finite-window locality plus Mittag-Leffler compatibility statement above. The theorem's hypotheses are graphwise; summability over graph topologies and QME vanishing are the separate obstruction data recorded in the displayed cohomology group.

Proof. (i) Each truncation $P^{w,w_0}_{\epsilon,L}$ admits Costello's Hadamard parametrix expansion by Lemma 8.16 (1). The Mittag-Leffler inverse system of (2) of that lemma is strict (surjective transition maps), so the inverse limit

$$P^w_{\epsilon,L} = \varprojlim_{w_0} P^{w,w_0}_{\epsilon,L}$$

exists in the weighted Tate category and inherits the wave-front bound (3) of that lemma uniformly in w_0. The inverse limit of the Hadamard expansions is itself a Hadamard expansion: the asymptotic Hadamard coefficients $a_k(x,y)$ on the bulk diagonal are independent of the coefficient direction, so the windowwise Hadamard expansion of $P^{w,w_0}_{\epsilon,L}$ is

$$\Big(\sum_{k\geq 0} a_k(x,y)\, t^k\Big)\widehat{\otimes} C^{w_0}_{\mathfrak{h},w},$$

and the inverse limit in the coefficient direction commutes with the asymptotic expansion in the bulk parameter t.

(ii) Fix a graph Γ with $V(\Gamma)$ vertices and $E(\Gamma)$ internal edges. Each vertex carries an interaction of polynomial degree $\leq D_0$ in the coefficient direction (hypothesis on I). Each internal edge contracts one source leg with one target leg via the Casimir $C_{\mathfrak{h},w}$. By (W1), each such contraction multiplies the weight by a factor $\leq C_w$ on Hamiltonian legs, and the cotangent legs are bounded on the finite windows appearing in the graph by (W2). The total coefficient size after $E(\Gamma)$ contractions is bounded by a finite graph-dependent constant times the bulk counterterm norm. The bulk counterterm norm $|C_n^{\mathrm{base}}[\Gamma]|$ is finite by Costello's bulk locality theorem [12, Chs. 5–7].

This graph-dependent constant is finite for the fixed graph Γ and the fixed finite codomain window under consideration. A summability statement over all graph topologies at a fixed $\hbar$-order belongs to the renormalized QME problem, together with the usual stability, counterterm, and locality hypotheses of Costello's graph expansion. The present theorem needs only the graphwise bound: for a fixed graph, the coefficient degrees and the corresponding weight ratios lie in a finite set, so the compatible finite-window counterterm system has a Mittag-Leffler limit when the windowwise representatives are themselves compatible.

(iii) The bulk Costello locality theorem at fixed window w_0 is used only through a finite-window theorem: the windowwise counterterm $C_n[\Gamma]^{w_0} \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_0})$ is a local functional with kernel supported on the diagonal of $\mathbb{R}^2 \times \mathbb{C}^2$. By the wave-front bound of Lemma 8.16 (3), the support is the bulk diagonal independently of w_0. The transition map $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_0'}) \to \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_0})$ for $w_0' \geq w_0$ is the restriction of polynomial functionals to truncated coefficient indices, equivalently setting higher coefficient fields to zero; in the regulator-admissible sector these maps are degreewise surjective. This preserves locality (the support of the truncated kernel is the same diagonal). The inverse limit $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w) = \varprojlim_{w_0} \mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w^{w_0})$ exists in the weighted Tate category and is the local-functional class on $\mathcal{E}_w$. The windowwise counterterms $\{C_n[\Gamma]^{w_0}\}_{w_0}$ form a compatible inverse system, and their inverse limit $C_n[\Gamma] = \varprojlim_{w_0} C_n[\Gamma]^{w_0}$ lies in $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w)$ with bound (ii). Its universal property is precisely the one proved in Proposition 5.17: it is the unique compatible strict ML element with these finite-window projections.

The endpoint obstruction class $[C_n[\Gamma]] \in H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ is represented by the inverse limit of the compatible windowwise classes when that compatibility holds. The theorem identifies its analytic support and coefficient topology. Its vanishing is an additional algebraic/BV condition on the mixed brane-defect interaction. □

Remark 8.18. The Mittag-Leffler limit in the coefficient direction commutes with Hörmander's wave-front analysis because the propagator factorization (22) is exact (the bulk wave-front set is computed once and tensored), the coefficient factor $C_{\mathfrak{h},w}$ is an algebraic tensor in the weighted Tate category with $\mathrm{WF} = \emptyset$, and the Hadamard parametrix coefficients $a_k(x, y)$ depend only on the bulk Riemannian/Hermitian data of $\mathbb{R}^2 \times \mathbb{C}^2$. The wave-front bound of Lemma 8.16 (3) is therefore exact and w_0-uniform; the Mittag-Leffler limit preserves the local-functional class graph by graph at every order $\hbar^n$ whenever the windowwise counterterm system is compatible.

COROLLARY 8.19 *(Hadamard coefficient criterion for the weighted QME lift).* In the graphwise Mittag-Leffler sector of Theorem 8.17, Proposition 5.72 is equivalent to the vanishing of the mixed brane-defect QME obstruction class. The all-order quantum source-to-boundary CE/PV comparison

$$\begin{aligned}\Phi_\hbar^w \colon C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h}_{w,\hbar} \ltimes \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}[1]\big)[[\hbar]] \\ \longrightarrow Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^\hbar_{\mathrm{open}}\big)\big|_{\mathfrak{g}_w}\end{aligned}$$

extends to the weighted cotangent direction at all orders in $\hbar$ as a continuous $\mathbb{C}[[\hbar]]$-linear cochain map precisely when the obstruction class in $H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\})$ vanishes. Equivalently, this vanishing statement is the datum required by Corollary 5.74.

Proof. Proposition 5.72 assumes the Costello-locality/QME compatibility identity of Problem 5.73. By Theorem 8.17, the Hadamard wave-front-set and Mittag-Leffler coefficient-topology pieces of that identity are controlled. The proposition is exactly the vanishing statement for the mixed brane-defect QME obstruction class. □

THEOREM 8.20 *(Weighted RG locality obstruction criterion).* Before adjoining the principal-part current extension, in the graphwise finite-window sector of Theorem 8.17, the analytic part of Problem 5.73 is exactly the following finite-window coefficient criterion:

$$\begin{aligned} P^{w}_{\epsilon,L} &= P^{\mathrm{base}}_{\epsilon,L}\widehat{\otimes}C_{\mathfrak{h}\oplus\mathfrak{h}^\vee[1],w}, \\ \mathrm{WF}(C^{w_0}_{\mathfrak{h},w}) &= \emptyset, \\ \varprojlim_{w_0}{}^{1} P^{w,w_0}_{\epsilon,L} &= 0. \end{aligned}$$

These conditions hold by Lemma 8.16. After they are imposed, the endpoint datum in the weighted QME lift is the class

$$\mathrm{Ob}_w \in H^1(\mathcal{O}_{\mathrm{loc}}(\mathcal{E}_w), Q + \{I_0, -\}).$$

Proof. The propagator factorization is (22); its finite-window form is the definition of $P^{w,w_0}_{\epsilon,L}$ in Lemma 8.16. The same lemma proves that the coefficient Casimir is an algebraic tensor with empty spacetime wave-front set and that the finite-window propagators form a strict Mittag-Leffler system. Thus the Hadamard singular support, asymptotic coefficients, and locality estimates are exactly the bulk Costello data tensored with a compatible coefficient kernel. The only term of Problem 5.73 remaining after these analytic statements is the local-functional QME class displayed above. The $\mathcal{D}'_c(I)\otimes\mathcal{P}[1]$ defect-current compatibility clause is the additional datum of Problem 5.213. □

Finite-window graph/QME assembly

The QME obstruction class isolated by Theorem 8.17 is a sum over Costello graphs at every $\hbar$-order, each graph carrying its renormalized weight, source-face, BV-bracket, and counterterm contributions. Extracting it requires a finite graph list $\mathcal{G}_M$ on a finite Hamiltonian window M, with labels and incidence signs declared, and the order-n residual written as a finite cochain.

Definition 8.21 *(Finite-window graph/QME system).* Fix an interval I, an admissible weight w, a finite Hamiltonian window M, and a finite set $\mathcal{G}_M$ of connected Costello graphs. A finite-window graph/QME system over $(I, w, M, \mathcal{G}_M)$ consists of the following data.

(F1) For every $\Gamma \in \mathcal{G}_M$ with r_Γ cotangent external labels, a tuple $B^\Gamma_\bullet = (B^\Gamma_1, \dots, B^\Gamma_{r_\Gamma})$ satisfying one of the two proved label conditions:

$$B^\Gamma_\bullet \in \big(\mathcal{D}^{\mathrm{reg}}_c(I)\big)^{r_\Gamma} \quad \text{or} \quad B^\Gamma_\bullet \in \mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M).$$

The second condition is the graphwise relation of Definition B.18. If one wants an honest CE source from a linear subspace $\mathcal{V}\subset\mathcal{D}'_c(I)$, every finite tuple from $\mathcal{V}$ must satisfy the corresponding graphwise relation. Otherwise the source is only the finite CE span of the specified labelled arguments. Denote the resulting source CE object by $C^\bullet_{\mathrm{lab}}(\mathcal{G}_M)$.

(F2) The finite-scale regulator data

$$P^{w,M}_{\epsilon,L}, \qquad \Delta^{w,M}_{\epsilon,L}, \qquad I^w_{0,M},$$

together with graph weights $W^M_{\Gamma,\epsilon,L}(B^\Gamma_\bullet)$, local counterterms $C^M_{\Gamma,w}(\epsilon; B^\Gamma_\bullet)$, and renormalized weights

$$W^{R,M}_{\Gamma,L,w} = \lim_{\epsilon\to 0}\big(W^M_{\Gamma,\epsilon,L} - C^M_{\Gamma,w}(\epsilon; B^\Gamma_\bullet)\big).$$

These are the regular-density weights of Proposition B.17, or the wavefront-admissible weights of Proposition B.19.

(F3) A complete filtered local-functional subcomplex

$$\mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I) = \Big(\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc},\mathcal{G},M}(\mathcal{E}^{M}_{w}|_{I};\, A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)), d_M\Big), \qquad d_M = Q + \{I^{w}_{0,M}, -\}_{\hbar,M},$$

generated, and then completed in the $\hbar$-adic and scalar-contact filtrations, by the displayed renormalized graph weights, their local counterterms, and the d_M-boundaries and BV brackets forced by the finite graph differential on $\mathcal{G}_M$. This is a subcomplex only after those boundary and bracket terms have been included.

(F4) A filtered chain projection on this subcomplex,

$$\widehat{\sigma}_{\mathrm{sc},M} : \mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I) \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{1})[1][[\hbar]],$$

whose associated graded is the scalar-symbol projection. Equivalently, either the graph data lie in the balanced graph/counterterm subcomplex of Definition E.20, or the scalar-extension obstructions

$$\partial^{(0)}_{\mathrm{Cost},\sigma}(I), \qquad \mathfrak{o}^{(r)}_{\mathrm{Cost},\sigma}(I), \quad r > 0,$$

of Theorem E.22 have been killed on the chosen finite-window graph/counterterm subcomplex.

(F5) A degree-zero graph kernel

$$\kappa^{M}_{\mathcal{G},w,I} \in \mathrm{Hom}_{\mathrm{cont}}\Big(C^{\bullet}_{\mathrm{lab}}(\mathcal{G}_M), \mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I)\Big),$$

restricted to the chosen label set, whose zero-internal-edge part is $\kappa^{\mathrm{coef},M}_{\hbar,w,I}$ of Proposition B.16 and whose positive graph components are the renormalized weights $W^{R,M}_{\Gamma,L,w}$.

Summation over graph topologies is part of the datum exactly when a graph-summability hypothesis holds.

Theorem 8.22 *(Finite-window graph/QME assembly).* Let $(I, w, M, \mathcal{G}_M)$ carry a finite-window graph/QME system in the sense of Definition 8.21. Put

$$S^{\bullet} = C^{\bullet}_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{1})[1][[\hbar]], \qquad Q^{\bullet}_{\mathcal{G},M}(I) = \ker(\widehat{\sigma}_{\mathrm{sc},M}) \subset \mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I).$$

Let

$$\mathrm{Ob}^{\mathrm{graph}}_{\mathcal{G},M} = \mathrm{ev}_{\mathcal{G}}\, \mathrm{Curv}_{\mathbf{K}}(\kappa^{M}_{\mathcal{G},w,I}) \in \mathfrak{Q}^{1}_{\mathcal{G},M}(I)$$

be the graph curvature evaluated on the chosen finite labelled CE cycles; before evaluation it is the corresponding Hom-valued curvature. The source-differential term is included in $\mathrm{Curv}_{\mathbf{K}}$. Here $\mathrm{Curv}_{\mathbf{K}} = d_{\mathbf{K}} + \frac{1}{2}[-,-]_{\mathbf{K}}$ as in Definition B.15. Suppose counterterms have been chosen through order $\hbar^{n-1}$, and write $C_{<n,M} \in \mathfrak{Q}^{0}_{\mathcal{G},M}(I)$ for their sum. The order-n residual is

$$\mathfrak{o}^{\mathrm{fw}}_{n,M} = [\hbar^{n}]\Big(\mathrm{Ob}^{\mathrm{graph}}_{\mathcal{G},M} + d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big).$$

Then the following statements hold.

(1) The analytic graph weights in the system are finite-window local functionals satisfying the stated locality hypotheses. For regular density labels this is Proposition B.17; for wavefront-admissible labels this is Proposition B.19. The statement applies to these regular-density and wavefront-admissible label classes.

(2) The first obstruction is scalar. If

$$\mathfrak{s}_{n,M} := [\hbar^n]\,\widehat{\sigma}_{\mathrm{sc},M}\Big(\mathrm{Ob}^{\mathrm{graph}}_{\mathcal{G},M} + d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big)$$

is nonzero in S^1, the order-n graph/QME equation has a scalar obstruction before passage to the non-scalar kernel. In the balanced graph/counterterm subcomplex this scalar term is zero by Proposition E.21.

(3) If the scalar term vanishes through order n and the lower-order QME equations hold, then

$$\mathfrak{o}^{\mathrm{fw}}_{n,M} \in Z^1(\mathcal{Q}^\bullet_{\mathcal{G},M}(I)).$$

An order-n finite-window counterterm $C_{n,M} \in \mathcal{Q}^0_{\mathcal{G},M}(I)$ satisfying

$$d_M C_{n,M} = -\mathfrak{o}^{\mathrm{fw}}_{n,M}$$

exists if and only if

$$[\mathfrak{o}^{\mathrm{fw}}_{n,M}] = 0 \quad \text{in} \quad H^1(\mathcal{Q}^\bullet_{\mathcal{G},M}(I), d_M).$$

If this class is nonzero, it is the precise finite-window non-scalar obstruction to the graph/QME realization at order $\hbar^n$.

Proof. The first assertion is exactly the analytic datum given by the regular-density and wavefront-admissible kernel propositions. The finite-window hypothesis makes every coefficient factor finite rank, so Theorem 8.17 gives the Hadamard counterterm control for each fixed graph. The label restriction is essential: outside $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M)$, the product of the graph Hadamard coefficient with the pushed-forward current label requires the wavefront-admissible product condition to define a canonical distribution.

The scalar statement is formal. A filtered chain projection $\widehat{\sigma}_{\mathrm{sc},M}$ is part of the system. If the displayed scalar projection of the residual is nonzero, then the residual has nonzero scalar component outside $\ker \widehat{\sigma}_{\mathrm{sc},M}$, so every counterterm inside that kernel leaves the scalar component unchanged. In the balanced graph/counterterm subcomplex the projection is the zero chain map, hence the scalar branch vanishes before the non-scalar class is formed.

Finally, the completed convolution Hom is a filtered dg Lie algebra. The Bianchi identity

$$d_{\mathbf{K}}\,\mathrm{Curv}_{\mathbf{K}}(\kappa) + [\kappa, \mathrm{Curv}_{\mathbf{K}}(\kappa)]_{\mathbf{K}} = 0$$

and the lower-order QME equations imply that the first unsolved $\hbar^n$-coefficient is closed in the convolution differential. After evaluation on labelled CE cycles this is d_M-closed. Since the scalar projection is a chain map and its order-n value is zero, this coefficient lies in the kernel complex $\mathcal{Q}^\bullet_{\mathcal{G},M}(I)$. Adding a degree-zero counterterm changes the first unsolved coefficient by the d_M-boundary of that counterterm; quadratic counterterm terms are already included in the displayed residual. Therefore the finite-window equation is solvable exactly when the displayed degree-one cohomology class vanishes. □

Proposition 8.23 *(Finite-window-to-limit obstruction)*. Suppose that finite-window graph/QME systems $\{(I, w, M, \mathcal{G}_M)\}_{M\geq 1}$ are given and satisfy the following additional hypotheses.

(L1) The graph system is compatible under truncation: $\pi_{M,N}\mathfrak{Q}^{\bullet}_{\mathcal{G},M}(I) \subset \mathfrak{Q}^{\bullet}_{\mathcal{G},N}(I)$, the kernels, counterterms, and scalar projections commute with $\pi_{M,N}$, and the same compatibility holds under admissible weight transport $R^{M}_{w,w'}$.

(L2) At the fixed $\hbar$-order under consideration the graph sum is finite in each window, or a Costello graph-summability hypothesis holds. This graph-summability hypothesis is separate from the finite-window theorem.

(L3) The residuals $\mathfrak{o}^{\mathrm{fw}}_{n,M}$ of Theorem 8.22 are compatible:

$$\pi_{M,N}\mathfrak{o}^{\mathrm{fw}}_{n,M} = \mathfrak{o}^{\mathrm{fw}}_{n,N}, \qquad M \geq N.$$

Put

$$\mathfrak{O}^{\mathrm{fw}}_{n} = \Big(\big([\mathfrak{o}^{\mathrm{fw}}_{n,M}]\big)_M,\, \lambda^{\mathrm{fw}}_n\Big),$$

where

$$\big([\mathfrak{o}^{\mathrm{fw}}_{n,M}]\big)_M \in \varprojlim_{M} H^1\big(Q^{\bullet}_{\mathcal{G},M}(I)\big),$$

and, after choosing windowwise primitives $c_{n,M}$ with $d_M c_{n,M} = -\mathfrak{o}^{\mathrm{fw}}_{n,M}$,

$$\lambda^{\mathrm{fw}}_n \in \varprojlim{}^1_M\, H^0\big(Q^{\bullet}_{\mathcal{G},M}(I)\big)$$

is represented by the Roos cocycle

$$\big[\pi_{M,N}c_{n,M} - c_{n,N}\big] \in H^0\big(Q^{\bullet}_{\mathcal{G},N}(I)\big).$$

Then compatible order-n counterterms $C_{n,M}$ with $\pi_{M,N}C_{n,M} = C_{n,N}$ exist if and only if $\mathfrak{O}^{\mathrm{fw}}_n = 0$. If the tower $\{H^0(Q^{\bullet}_{\mathcal{G},M}(I))\}_M$ is Mittag–Leffler, then $\lambda^{\mathrm{fw}}_n = 0$, and the inverse-limit obstruction is only the compatible H^1-class above.

Hypotheses (L1)–(L3) are the scope of the inverse-limit graph/QME theorem. When (L2) is unmet, the graph sum itself lies outside the tower, so the obstruction class is unformed.

Proof. Windowwise, Theorem 8.22 says that the only non-scalar obstruction is the class of $\mathfrak{o}^{\mathrm{fw}}_{n,M}$ in H^1. Necessity of the two displayed inverse-limit conditions is immediate from any compatible primitive $C_{n,M}$. Conversely, vanishing of the compatible H^1-class gives primitives $c_{n,M}$ window by window. The differences $\pi_{M,N}c_{n,M} - c_{n,N}$ are closed degree-zero elements, and their cohomology classes are the standard Roos 1-cocycle computing $\varprojlim{}^1_M H^0(Q^{\bullet}_{\mathcal{G},M}(I))$. The vanishing of λ^{fw}_n is exactly the statement that the $c_{n,M}$'s can be adjusted by closed degree-zero terms and boundaries to become compatible under all truncation maps. The Mittag–Leffler sufficient condition is the standard vanishing of $\varprojlim{}^1$ for a Mittag–Leffler tower. The weight-transport compatibility follows from the same diagonal transport squares used in Theorem 5.82. □

The order-n residual of Proposition 8.23 is a sum of graph-differential, CE-source, BV-bracket, and lower-counterterm contributions, each carrying its own incidence sign, scalar projection, and truncation matrix. Computing, testing, and propagating the residual window by window requires a finite array of rows recording every contribution in normal form.

Definition 8.24 *(Finite-window graph array).* Fix a finite-window graph/QME system $(I, w, M, \mathcal{G}_M)$, a codimension-two closed marked Costello list $\mathcal{G}_M^{\mathrm{mk}}$, and a labelled CE cycle ζ_M. A finite-window graph array at order n is a finite set

$$\mathsf{A}_{n,M}^{\mathrm{fw}} = \{\alpha\}$$

together with the following row data.

(A1) *Graph type and order.* A row has a type

$$\tau_\alpha \in \{\mathrm{dK}, \mathrm{CE}, \mathrm{bracket}, \text{lower-}d, \text{lower-bracket}\}.$$

For dK, CE, and bracket rows it specifies, respectively, a marked graph Γ_α, a CE-source face of such a graph, or an ordered pair $(\Gamma_{\alpha,1}, \Gamma_{\alpha,2})$, all in the finite marked list. The order is

τ_α	$\lvert\alpha\rvert_\hbar$
dK, CE	$\lvert\Gamma_\alpha\rvert_\hbar$
bracket	$\lvert\Gamma_{\alpha,1}\rvert_\hbar + \lvert\Gamma_{\alpha,2}\rvert_\hbar$
lower-d, lower-bracket	n,

and only rows with $|\alpha|_\hbar = n$ enter $\mathsf{A}_{n,M}^{\mathrm{fw}}$.

(A2) *Edge labels.* Each internal edge of the graph carries one of the finite-window edge kernels

$$P_{\epsilon,L}^{w,M}, \qquad \Delta_{\epsilon,L}^{w,M}, \qquad E_{\mathrm{Weyl}},$$

where E_{Weyl} is the boundary Weyl edge whose scalar two-Hamiltonian part is $(f, g) \mapsto \omega(f, g)\gamma_1$. The two half-edge coefficient labels lie in

$$\mathfrak{h}_{w,\le M} \cup D_{\le M}^{w}, \qquad D_{\le M}^{w} = \prod_{d \le M} w(d)^{-1} H_d^{\vee},$$

and the edge record includes the source/target orientation used by the BV bracket sign.

(A3) *Vertex labels.* Each vertex is labelled by one of the finite system labels

$$u_{a(t)dt\otimes f}, \quad c_{B,\rho}, \quad I_{0,M}^{w}, \quad W_{\Gamma,L,w}^{R,M}(B_\bullet^\Gamma), \quad C_{\Gamma,w}^{M}(\epsilon; B_\bullet^\Gamma), \quad \gamma_1,$$

with f, ρ in degree $\le M$ and $B_\bullet^\Gamma$ either regular-density or graphwise wavefront-admissible as in Definition 8.21.

(A4) *Markers and incidence coefficient.* The row contains a marker tensor

$$\mathfrak{m}_\alpha = (\mathfrak{m}_{\alpha,L})_L, \qquad \mathfrak{m}_{\alpha,L} \in \mathcal{M}_{m_L}^{\mathrm{full}}(\mathbb{C}^{N|N})$$

or the explicitly chosen smaller marker complex. If the row comes from an elementary boundary operation $\beta_i \in \mathsf{B}_M(\Gamma_\alpha)$, its incidence coefficient is

$$\iota_{\alpha,M} = (-1)^{i-1}\varepsilon_{\beta_i}.$$

For CE-source and convolution-bracket rows, $\iota_{\alpha,M}$ is the corresponding CE or BV Koszul sign, including the factor $1/2$ in the bracket row.

(A5) *Weight and filtrations.* The row records

$$d_{\max}(\alpha) = \max\{d : H_d \text{ or } H_d^\vee \text{ occurs in the row}\} \leq M,$$

its scalar-contact filtration $p_\alpha = \#\{\text{scalar-contact open loops}\}$, and, for a boundary operation β_i,

$$r_\alpha = p_\alpha - p(\partial_{\beta_i}\Gamma_\alpha, \mu_{\beta_i}(\mathfrak{m}_\alpha)).$$

(A6) *Curvature value and scalar condition.* The row evaluates to a local functional

$$R^{\mathrm{row}}_{\alpha,M} \in \mathfrak{Q}^1_{\mathcal{G},M}(I).$$

For the graph-curvature part these are

τ_α	$R^{\mathrm{row}}_{\alpha,M}$
dK	$\iota_{\alpha,M}\, d_M K^M_{\Gamma_\alpha}$
CE	$\iota_{\alpha,M}\, K^M_{\Gamma_\alpha}$
bracket	$\iota_{\alpha,M}\{K^M_{\Gamma_{\alpha,1}}, K^M_{\Gamma_{\alpha,2}}\}_{\hbar,M}.$

The two lower-counterterm row types are the corresponding homogeneous summands of $[\hbar^n]d_M C_{<n,M}$ and $\frac{1}{2}[\hbar^n]\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}$. The scalar-condition entry is

$$S_{\alpha,M} = \widehat{\sigma}_{\mathrm{sc},M}(R^{\mathrm{row}}_{\alpha,M}).$$

If α is a first filtration-lowering two-Hamiltonian scalar contact row, this specializes to

$$S_{\alpha,M} = \iota_{\alpha,M}\ \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m}_\alpha)\big)\ \omega(f_{\partial_{\beta_i}\Gamma_\alpha}, g_{\partial_{\beta_i}\Gamma_\alpha})\gamma_1.$$

The non-scalar reduced-curvature contribution is the scalar-zero sum

$$R^{\mathrm{ns}}_{n,M} = \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} R^{\mathrm{row}}_{\alpha,M} \quad \text{provided} \quad \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} S_{\alpha,M} = 0.$$

An entrywise non-scalar summand requires an additional chain splitting of $\widehat{\sigma}_{\mathrm{sc},M}$; the displayed scalar-zero sum is the canonical datum in the present array.

(A7) *Truncation in M.* For $M \geq N$ the array contains truncation coefficients $t^{M,N}_{\alpha\alpha'} \in \mathbb{C}$ with $\alpha' \in \mathsf{A}^{\mathrm{fw}}_{n,N}$, defined by

$$\pi_{M,N} R^{\mathrm{row}}_{\alpha,M} = \sum_{\alpha'} t^{M,N}_{\alpha\alpha'} R^{\mathrm{row}}_{\alpha',N}, \qquad \pi_{M,N} S_{\alpha,M} = \sum_{\alpha'} t^{M,N}_{\alpha\alpha'} S_{\alpha',N}.$$

A row whose coefficient labels lie entirely above degree N has all $t^{M,N}_{\alpha\alpha'} = 0$. A filling of this equation is exactly the truncation entry required by the lower-window theorem.

(A8) *Order-two boundary compositions.* For an order-2 closure check the array also records the ordered pairs of distinct first scalar-contact boundary operations

$$(\beta_i, \beta_j), \qquad r(\beta_i) = r(\beta_j) = 1, \qquad |\beta_i|_\hbar = |\beta_j|_\hbar = 1,$$

whose two composites occur in the codimension-two closure table of Definition E.27. If $\mathrm{pos}_i(j)$ denotes the position of the induced face $\beta_j|\beta_i$ after deleting β_i, the composite incidence coefficient is

$$\iota_{\beta_j|\beta_i,\beta_i} = (-1)^{i-1}(-1)^{\mathrm{pos}_i(j)-1}\varepsilon_{\beta_i}\varepsilon_{\beta_j|\beta_i}.$$

The corresponding two-contact scalar condition is the row

$$S^{(2)}_{\beta_j|\beta_i,M} = \iota_{\beta_j|\beta_i,\beta_i}\left(\prod_{L\in\mathcal{L}_{\mathrm{sc}}(\partial_{\beta_j|\beta_i}\partial_{\beta_i}\Gamma)} \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}'_L)\right)\omega(f_{\beta_i}, g_{\beta_i})\omega(f_{\beta_j|\beta_i}, g_{\beta_j|\beta_i})\gamma_{\mathbf{1}}^{(2)}.$$

Here $\mathfrak{m}'_L$ is the transported marker on the resulting scalar loop L, and $\gamma_{\mathbf{1}}^{(2)}$ denotes the ordered two-contact associated-graded scalar-contact symbol. A surviving product $\gamma_{\mathbf{1}}^2$ in the target algebra is a separate datum. The equality of resulting graphs, coefficients, and marker transports in (E.27) is exactly the order-2 composite-face datum.

The array is *valid* if the row sum reproduces the order-n residual:

$$R_{n,M} = \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} R^{\mathrm{row}}_{\alpha,M}, \qquad \mathfrak{s}_{n,M} = \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} S_{\alpha,M},$$

and if the truncation coefficients satisfy the displayed equations for every $M \geq N$.

PROPOSITION 8.25 *(First finite-window array rows).* The universal computed rows are the reduced current interface row in order 0 and the two-Hamiltonian scalar-contact row in order 1.

On the length-one reduced current interface, Theorem B.7 gives strict CE/PV compatibility. Thus the order-0 row has

$$R^{\mathrm{row}}_{0,M} = 0, \qquad S_{0,M} = 0.$$

For the ordered Hamiltonian arguments (a, z_1), (b, z_2), the one-cross boundary row has graph type dK, one boundary Weyl edge E_{Weyl}, two external Hamiltonian vertices, one scalar-contact vertex $\gamma_{\mathbf{1}}$, and

$$\omega(z_1, z_2) = 1, \qquad |\alpha_{\mathrm{sc}}|_{\hbar} = 1.$$

Its scalar condition is

$$S_{\alpha_{\mathrm{sc}},M} = \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_{\mathrm{sc}}}(\mathfrak{m})\big)\gamma_{\mathbf{1}},$$

after extracting the $\hbar^1$-coefficient and using the orientation convention in which the ordered pair (z_1, z_2) has positive sign. Therefore:

branch	$S_{\alpha_{\mathrm{sc}},M}$	$R^{\mathrm{ns}}_{1,M}$ from this row
$\mathfrak{gl}_N$ ordinary	$N\gamma_{\mathbf{1}}$	scalar branch
even block marker $\lambda_0 P_{\bar{0}} + \lambda_1 P_{\bar{1}}$	$N(\lambda_0 - \lambda_1)\gamma_{\mathbf{1}}$	formed when $\lambda_0 = \lambda_1$
$\mathfrak{gl}(N\|N)$ full-equivariant	0	0

Consequently, for the minimal full-equivariant closed array containing only the strict current-interface rows and this scalar-contact row, the first order residual is zero and

$$C_{1,M} = 0$$

is the compatible order-one non-scalar counterterm. For any larger graph system, the first non-scalar residual is

$$R^{\mathrm{ns}}_{1,M} = \sum_{\alpha \in \mathsf{A}^{\mathrm{fw}}_{1,M} \setminus \{\alpha_{\mathrm{sc}}\}} R^{\mathrm{row}}_{\alpha,M}$$

after the scalar condition has vanished. Its primitive is decided only after giving the remaining order-one rows of Definition 8.24.

Proof. The order-0 assertion is exactly the strict current-source/target identity of Theorem B.7 on the length-one generators: the CE differential and target P_0 bracket use the same semidirect product constants.

The order-1 scalar-contact row is the computation of Proposition E.23, written in the row notation of Definition 8.24. The monomial value is $\omega(z_1, z_2) = 1$. In the ordinary branch the open index loop contributes N. In the even block-diagonal marker branch the cyclic supertrace is $N(\lambda_0 - \lambda_1)$. In the full-equivariant branch, Lemma E.31 kills the cyclic supertrace of every transported marker tensor in $\mathbb{C}^{N|N}$; the empty marker word is killed by the same $\operatorname{sdim} \mathbb{C}^{N|N} = 0$ calculation. Thus the row is already zero as a local functional in the full-equivariant branch.

If the array contains only the displayed order-one row, the order-one residual is zero, so the zero counterterm is a primitive and is compatible under all truncation maps. If further order-one rows are present, the displayed sum is the definition of the remaining scalar-zero residual. A primitive for it is a theorem exactly when the missing row values, signs, scalar conditions, and truncation coefficients are given and the resulting H^1-class vanishes. □

Proposition 8.26 *(Universal scalar-contact bracket rows).* Work in the local full-equivariant marked branch with $V = \mathbb{C}^{N|N}$, and let α_{sc} be the one-cross two-Hamiltonian scalar-contact row of Proposition 8.25. Write $K^M_{\alpha_{\mathrm{sc}}}$ for the local functional carried by this row. Then

$$K^M_{\alpha_{\mathrm{sc}}} = 0$$

in the finite-window local-functional complex.

Consequently every order-2 convolution-bracket row containing α_{sc} has the following entries. For any order-1 component β for which the bracket row is present,

row	$R^{\mathrm{row}}_{b,M}$
$b(\alpha_{\mathrm{sc}}, \beta)$	$\frac{1}{2}\varepsilon_{\alpha_{\mathrm{sc}},\beta}\{K^M_{\alpha_{\mathrm{sc}}}, K^M_{\beta}\}_{\hbar,M} = 0$
$b(\beta, \alpha_{\mathrm{sc}})$	$\frac{1}{2}\varepsilon_{\beta,\alpha_{\mathrm{sc}}}\{K^M_{\beta}, K^M_{\alpha_{\mathrm{sc}}}\}_{\hbar,M} = 0$
$b(\alpha_{\mathrm{sc}}, \alpha_{\mathrm{sc}})$	$\frac{1}{2}\varepsilon_{\alpha_{\mathrm{sc}},\alpha_{\mathrm{sc}}}\{K^M_{\alpha_{\mathrm{sc}}}, K^M_{\alpha_{\mathrm{sc}}}\}_{\hbar,M} = 0$

Their scalar conditions are

$$S_{b,M} = 0.$$

For every $M \geq N$, the truncation row may be chosen as

$$t^{M,N}_{bb'} = 0 \qquad \text{for every lower-window bracket row } b',$$

since $\pi_{M,N} R^{\mathrm{row}}_{b,M} = 0$ and $\pi_{M,N} S_{b,M} = 0$. These rows therefore have zero order-2 obstruction and zero counterterm.

This proposition closes only the bracket family containing α_{sc}. Bracket rows $b(\beta_1, \beta_2)$ with $\beta_1, \beta_2 \neq \alpha_{\mathrm{sc}}$ remain part of the chosen graph system and must be fixed row by row.

Proof. Proposition 8.25 proves the stronger full-equivariant fact that the scalar-contact row is zero as a local functional, because every transported scalar-loop marker has zero cyclic supertrace on $\mathbb{C}^{N|N}$. The BV/convolution bracket is bilinear and continuous on the finite-window completed local-functional complex, hence any bracket with $K^M_{\alpha_{\rm sc}} = 0$ is zero before passing to cohomology. Applying the chain projection $\widehat{\sigma}_{{\rm sc},M}$ to zero gives the displayed scalar conditions. Truncation sends zero to zero, so the zero truncation matrix row is compatible with the defining equations of Definition 8.24. □

PROPOSITION 8.27 *(Genuine order-two graph-weight rows).* Work in the local $\mathbb{R}^2_{\rm top} \times \mathbb{C}^2_{\rm hol}$ full-equivariant marked branch with $V = \mathbb{C}^{N|N}$. Let $\mathcal{G}^{\rm gen}_{2,M}$ be the part of a finite marked Costello list consisting of genuine seed graph weights of $\hbar$-order 2: these are genuine seed rows, distinct from the codimension-two composites of two first scalar-contact boundary faces and from the bracket rows containing $\alpha_{\rm sc}$ closed in Proposition 8.26. For $\Gamma \in \mathcal{G}^{\rm gen}_{2,M}$, write

$$K^M_\Gamma = W^{R,M}_{\Gamma,L,w}(B^\Gamma_\bullet)$$

for the normalized renormalized finite-window graph weight, and let ε_Γ be the coefficient with which this component occurs in $d_M \kappa^M_{\mathcal{G},w,I}$, including the orientation, Koszul, and symmetry factors given by the finite graph system.

The genuine order-two rows have the following form.

(G1) The graph-differential row $d\Gamma$ has

$$\tau_{d\Gamma} = {\rm dK}, \qquad |d\Gamma|_\hbar = 2,$$

inherits its edge labels, vertex labels, marker tensor, and coefficient degree from Γ, and has

$$R^{\rm row}_{d\Gamma,M} = \varepsilon_\Gamma\, d_M K^M_\Gamma, \qquad S_{d\Gamma,M} = \varepsilon_\Gamma\, \widehat{\sigma}_{{\rm sc},M}(d_M K^M_\Gamma).$$

(G2) Before evaluation on a strict CE cycle, or when the Hom-valued kernel table is retained, each source face is a separate row. Write the relevant part of the source differential as

$$d_{\rm CE}\zeta_M = \sum_\nu \varepsilon^{\rm CE}_{\Gamma,\nu}\zeta_{M,\nu},$$

where $\varepsilon^{\rm CE}_{\Gamma,\nu}$ includes the CE Koszul sign and the graph coefficient attached to the Γ-component. The source row is

$$R^{\rm row}_{{\rm CE}(\Gamma,\nu),M} = -\varepsilon^{\rm CE}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu}),$$

and its scalar condition is

$$S_{{\rm CE}(\Gamma,\nu),M} = -\varepsilon^{\rm CE}_{\Gamma,\nu}\widehat{\sigma}_{{\rm sc},M}\big(K^M_\Gamma(\zeta_{M,\nu})\big).$$

If the curvature has already been evaluated on a strict labelled CE cycle, this source-face family is empty.

If these row values and all their d_M-faces lie in the full-equivariant marked complex of Theorem E.32, then

$$S_{d\Gamma,M} = 0, \qquad S_{{\rm CE}(\Gamma,\nu),M} = 0.$$

This is a scalar-condition statement. The local-functional values K^M_Γ and $d_M K^M_\Gamma$ require their own row data.

After scalar-condition vanishing, the genuine seed-graph contribution to the order-2 reduced curvature is

$$R^{\mathrm{ns,gen}}_{2,M} = \sum_{\Gamma\in\mathcal{G}^{\mathrm{gen}}_{2,M}} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{\Gamma,\nu} \varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu}) \in \mathcal{K}^1_{\mathrm{ns},M}(I).$$

The second sum is zero after strict CE-cycle evaluation. A genuine order-2 counterterm exists precisely when

$$[R^{\mathrm{ns,gen}}_{2,M}] = 0 \quad \text{in} \quad H^1(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}).$$

If this class is nonzero, it is the explicit finite-window obstruction carried by the genuine seed graphs.

Truncation is part of the row data. For $M \geq N$ one must choose normalized coefficients

$$\pi_{M,N}(\varepsilon_\Gamma K^M_\Gamma) = \sum_{\Gamma'} u^{M,N}_{\Gamma\Gamma'}\, \varepsilon_{\Gamma'} K^N_{\Gamma'},$$

and then the graph-differential truncation entries are

$$t^{M,N}_{d\Gamma,d\Gamma'} = u^{M,N}_{\Gamma\Gamma'}.$$

For source rows one must likewise give

$$\pi_{M,N}\big(-\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu})\big) = \sum_{\Gamma',\nu'} v^{M,N}_{\Gamma,\nu;\Gamma',\nu'}\, \big(-\varepsilon^{\mathrm{CE}}_{\Gamma',\nu'} K^N_{\Gamma'}(\zeta_{N,\nu'})\big),$$

and set $t^{M,N}_{\mathrm{CE}(\Gamma,\nu),\mathrm{CE}(\Gamma',\nu')} = v^{M,N}_{\Gamma,\nu;\Gamma',\nu'}$. Rows whose coefficient labels lie entirely above degree N have zero truncation coefficients. The displayed truncation identities are exactly the row entries required by the lower-window theorem.

In the minimal full-equivariant local system of Proposition 8.28, $\mathcal{G}^{\mathrm{gen}}_{2,M} = \emptyset$ and the source-face set is empty. Hence

$$R^{\mathrm{ns,gen}}_{2,M} = 0, \qquad C^{\mathrm{gen}}_{2,M} = 0.$$

In any larger system, vanishing, counterterms, and obstruction are decided by the following data: the finite generator set $\mathcal{G}^{\mathrm{gen}}_{2,M}$; the renormalized local weights K^M_Γ with regular-density or wavefront-admissible current labels; the full $d_M K^M_\Gamma$ expansion with orientation signs; the CE-source decomposition and signs unless strict CE-cycle evaluation has removed them; proof that every row lies in the full-equivariant marked complex, or else its scalar projection; and the truncation matrices displayed above.

Proof. This is the order-2 part of the convolution-curvature formula in Theorem 8.36, with the scalar-contact bracket family removed by Proposition 8.26. The differential term contributes $\varepsilon_\Gamma d_M K^M_\Gamma$. The source term in $d_{\mathbf{K}} = d_M - (-1)^{|\cdot|}(\cdot)d_{\mathrm{CE}}$ contributes the displayed minus sign for a degree-zero kernel. After evaluation on a strict CE cycle, that source term is zero.

The scalar-condition assertion follows from Theorem E.32: on the full-equivariant marked complex the filtered scalar projection is the zero chain map. Since this projection only kills the scalar shadow, it gives a scalar-condition theorem for the non-scalar local functional K^M_Γ.

The counterterm criterion is exactly Definition E.33: once the scalar condition is zero, the row sum is a degree-one element of the non-scalar kernel complex, and a degree-zero counterterm exists exactly when this element is a $d_{\mathrm{ns},M}$-boundary. The truncation formulas are the row identities required in Definition 8.24, written for normalized graph and source rows. In the minimal system the genuine seed-row set is empty, so the sum is empty. □

PROPOSITION 8.28 *(Order-two finite-window boundary rows).* Work in the local $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ full-equivariant marked branch with $V = \mathbb{C}^{N|N}$. Let the minimal array be generated by the strict current-interface rows and the one-cross scalar-contact row of Proposition 8.25. Assume the finite marked Costello list contains exactly these minimal rows: the genuine $|\Gamma|_\hbar = 2$ seed-graph set, the order-2 CE-source-face set, and the order-2 graph-counterterm-face set are empty. Use $C_{1,M} = 0$.

Then the only order-2 boundary data forced by codimension-two closure are the two ordered composites of two first scalar-contact boundary operations. For $o = \beta_1 \wedge \beta_2$ they have incidence signs

$$\iota_{\beta_2|\beta_1,\beta_1} = \varepsilon_{\beta_1}\varepsilon_{\beta_2|\beta_1}, \qquad \iota_{\beta_1|\beta_2,\beta_2} = -\varepsilon_{\beta_2}\varepsilon_{\beta_1|\beta_2}.$$

Their scalar conditions are

$$\begin{aligned} S^{(2)}_{\beta_2|\beta_1,M} &= \varepsilon_{\beta_1}\varepsilon_{\beta_2|\beta_1}\prod_L \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}'_L)\,\omega(f_{\beta_1}, g_{\beta_1})\omega(f_{\beta_2|\beta_1}, g_{\beta_2|\beta_1})\gamma_1^{(2)}, \\ S^{(2)}_{\beta_1|\beta_2,M} &= -\varepsilon_{\beta_2}\varepsilon_{\beta_1|\beta_2}\prod_L \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}''_L)\,\omega(f_{\beta_2}, g_{\beta_2})\omega(f_{\beta_1|\beta_2}, g_{\beta_1|\beta_2})\gamma_1^{(2)}. \end{aligned}$$

The codimension-two closure equations identify the two resulting graphs, the two products of coefficients, and the transported marker tensors. Therefore the pair sum is zero. In the full-equivariant branch it is stronger: every factor $\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}'_L)$ and $\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}''_L)$ is zero by Lemma E.31; hence each two-contact scalar row already vanishes as a local scalar row.

Under these minimal hypotheses the order-2 reduced non-scalar row sum is

$$R^{\mathrm{ns}}_{2,M} = 0, \qquad C_{2,M} = 0$$

is a compatible order-2 counterterm.

For any larger finite-window system, the order-2 residual is decided by the added row data. After scalar-condition vanishing its exact row sum is

$$\begin{aligned} R^{\mathrm{ns}}_{2,M} = \sum_{|\Gamma|_\hbar=2} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{|\Gamma'|_\hbar=2} \varepsilon^{\mathrm{CE}}_{\Gamma'} K^M_{\Gamma'} \\ + \frac{1}{2}\sum_{|\Gamma_1|_\hbar=|\Gamma_2|_\hbar=1} \varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M} \\ + [\hbar^2]\Big(d_M C_{<2,M} + \frac{1}{2}\{C_{<2,M}, C_{<2,M}\}_{\hbar,M}\Big). \end{aligned}$$

Thus a nonzero order-2 obstruction or its counterterm can be asserted only after giving the genuine two-cross/contact graph weights K^M_Γ, the CE-source faces, the BV bracket rows between order-1 graph components distinct from those containing α_{sc} or another retained zero local-functional row such as α_{11}, the lower-counterterm rows if a nonminimal $C_{1,M}$ was chosen, the scalar projection of any non-full-equivariant two-contact symbol, and the truncation coefficients $t^{M,N}_{\alpha\alpha'}$.

Proof. The signs are those of Lemma E.29. Starting with $o = \beta_1 \wedge \beta_2$, the face β_1 followed by $\beta_2|\beta_1$ contributes $+1$, while the face β_2 followed by $\beta_1|\beta_2$ contributes -1. The codimension-two closure equations Equation (E.27) identifies the resulting graph, coefficient, and marker transport in the two composites, so the two boundary-composition rows cancel pairwise.

The scalar-condition formula is the order-2 instance of the last row datum in Definition 8.24: one scalar-contact factor is contributed by each first-lowering boundary face. In the full-equivariant marked complex, marker transport preserves the signed Brauer span. Lemma E.31 gives zero cyclic supertrace for every transported scalar-loop marker, including the empty marker word. Hence the two-contact scalar rows vanish before passing to cohomology.

Under the stated minimal hypotheses the genuine order-2 graph-row set, CE-source-row set, and order-1 non-scalar primitive set are empty, and the full-equivariant trace calculation kills the remaining scalar-contact row. The possible bracket row between two one-cross scalar-contact components is also zero, because each such component has zero local-functional value in the full-equivariant branch by Proposition 8.25. The reduced non-scalar order-2 row sum is therefore zero, and the zero counterterm is compatible under finite-window truncation.

Removing any minimal hypothesis reintroduces exactly the displayed row types from the curvature formula in Theorem 8.36. The bracket rows containing α_{sc} are the universal zero rows of Proposition 8.26; bracket rows containing a retained α_{11}-row vanish by the same bilinearity argument and Proposition E.45. The remaining graph weights, signs, scalar projections, and truncation coefficients are the data defining $R^{\mathrm{ns}}_{2,M}$, the order-2 obstruction class, and its counterterm equation. □

PROPOSITION 8.29 *(Truncation matrices for the computed finite-window rows).* Work with the finite-window graph arrays of Definition 8.24. For $M \geq N$, use the following label-retaining convention: if a labelled row has $d_{\max} \leq N$, the same label denotes its lower-window row; if $d_{\max} > N$, the row is truncated away and all entries from it are zero. With this convention the computed rows have the following truncation matrices.

(T1) The one-cross scalar-contact row α_{sc} of Proposition 8.25 has $d_{\max}(\alpha_{\mathrm{sc}}) = 1$. For $M \geq N \geq 1$,

$$t^{M,N}_{\alpha_{\mathrm{sc}}\alpha_{\mathrm{sc}}} = 1, \qquad t^{M,N}_{\alpha_{\mathrm{sc}}\alpha'} = 0 \quad (\alpha' \neq \alpha_{\mathrm{sc}}).$$

This applies to the ordinary, even-block, and full-equivariant branches. In the full-equivariant branch the row value is already zero, so the zero matrix would also satisfy the row equations; the displayed matrix is the chosen transport of the retained row label.

(T2) The labelled scalar-zero row α_{11} of Proposition E.45, with ordered arguments (z_1, z_1), has

$$R^{\mathrm{row}}_{\alpha_{11},M} = 0, \qquad S_{\alpha_{11},M} = 0, \qquad d_{\max}(\alpha_{11}) = 1.$$

If the row convention retains this labelled zero row, then for $M \geq N \geq 1$

$$t^{M,N}_{\alpha_{11}\alpha_{11}} = 1, \qquad t^{M,N}_{\alpha_{11}\alpha'} = 0 \quad (\alpha' \neq \alpha_{11}).$$

If zero rows are suppressed, the α_{11}-row and its truncation matrix are empty.

(T3) Let $\gamma_{21} = (\beta_2|\beta_1, \beta_1)$ and $\gamma_{12} = (\beta_1|\beta_2, \beta_2)$ denote the two ordered codimension-two scalar-contact composites in Proposition 8.28. When the degree-$\leq N$ window contains the two Hamiltonian labels, the composite rows are transported diagonally:

$$t^{M,N}_{\gamma_{21}\gamma_{21}} = 1, \qquad t^{M,N}_{\gamma_{12}\gamma_{12}} = 1, \qquad t^{M,N}_{\gamma_{ij}\gamma'} = 0 \quad (\gamma' \neq \gamma_{ij}).$$

The codimension-two pair cancellation is therefore stable under truncation. In the full-equivariant branch the stronger rowwise vanishing is also stable.

(T4) Let b be any order-2 bracket row containing a retained zero local-functional row

$$\alpha_0 \in \{\alpha_{\mathrm{sc}}, \alpha_{11}\}.$$

Thus $b = b(\alpha_0, \beta)$, $b = b(\beta, \alpha_0)$, or $b = b(\alpha_0, \alpha_0)$, whenever the displayed row belongs to the chosen finite marked list. Then

$$R^{\mathrm{row}}_{b,M} = 0, \qquad S_{b,M} = 0,$$

and the zero truncation row is

$$t^{M,N}_{bb'} = 0 \qquad \text{for every lower-window bracket row } b'.$$

(T5) When genuine order-2 seed rows are zero, as in the minimal full-equivariant system, their truncation matrices are empty. If a larger system gives genuine seeds $\Gamma \in \mathcal{G}^{\mathrm{gen}}_{2,M}$, then the graph-differential rows are transported by the normalized matrix $u^{M,N}$:

$$\pi_{M,N}(\varepsilon_\Gamma K^M_\Gamma) = \sum_{\Gamma'} u^{M,N}_{\Gamma\Gamma'}\, \varepsilon_{\Gamma'} K^N_{\Gamma'}, \qquad t^{M,N}_{d\Gamma,d\Gamma'} = u^{M,N}_{\Gamma\Gamma'}.$$

Source-face rows, when present, are transported by the normalized matrix $v^{M,N}$:

$$\pi_{M,N}\big(-\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu})\big) = \sum_{\Gamma',\nu'} v^{M,N}_{\Gamma,\nu;\Gamma',\nu'}\, \big(-\varepsilon^{\mathrm{CE}}_{\Gamma',\nu'} K^N_{\Gamma'}(\zeta_{N,\nu'})\big),$$

$$t^{M,N}_{\mathrm{CE}(\Gamma,\nu),\mathrm{CE}(\Gamma',\nu')} = v^{M,N}_{\Gamma,\nu;\Gamma',\nu'}.$$

Under the projection-closed row-basis hypotheses of Proposition 8.30, these are the coordinate entries of the finite-window projection. These hypotheses give exactly the lower-window entries $u^{M,N}_{\Gamma\Gamma'}$ and $v^{M,N}_{\Gamma,\nu;\Gamma',\nu'}$ for those given genuine rows.

(T6) Bracket rows $b(\beta_1, \beta_2)$ with $\beta_i \notin \{\alpha_{\mathrm{sc}}, \alpha_{\mathrm{II}}\}$ require data beyond the present local computation. After a lower-window bracket row basis, or an equivalent row presentation with chosen coordinates, is fixed, their required truncation matrix is the coordinate family $q^{M,N}$ satisfying

$$\pi_{M,N}\Big(\frac{1}{2}\varepsilon_{\beta_1,\beta_2}\{K^M_{\beta_1}, K^M_{\beta_2}\}_{\hbar,M}\Big) = \sum_{\beta_1',\beta_2'} q^{M,N}_{\beta_1,\beta_2;\beta_1',\beta_2'}\, \frac{1}{2}\varepsilon_{\beta_1',\beta_2'}\{K^N_{\beta_1'}, K^N_{\beta_2'}\}_{\hbar,N}.$$

These q-entries, together with the underlying order-1 row values, signs, and lower-window bracket-basis reduction, are still missing data for the enlarged system.

The nonempty displayed matrices are stable under further truncation: for $M \geq N \geq P$,

$$t^{M,P} = t^{M,N} t^{N,P}$$

on the scalar-contact, scalar-zero, codimension-two, and zero-bracket families. For genuine seed rows and nonzero added bracket rows, stability is exactly the cocycle condition

$$u^{M,P}_{\Gamma\Gamma''} = \sum_{\Gamma'} u^{M,N}_{\Gamma\Gamma'} u^{N,P}_{\Gamma'\Gamma''},$$

$$v^{M,P}_{\Gamma,\nu;\Gamma'',\nu''} = \sum_{\Gamma',\nu'} v^{M,N}_{\Gamma,\nu;\Gamma',\nu'} v^{N,P}_{\Gamma',\nu';\Gamma'',\nu''},$$

and the analogous equation for q.

Proof. For the one-cross scalar-contact rows the coefficient labels are z_1, z_2, and for α_{11} the labels are z_1, z_1. These all have window degree 1. Therefore $\pi_{M,N}$ preserves the labelled row for $N \geq 1$, and kills it only when the lower window omits the degree-one labels. The diagonal matrices in (T1) and (T2) are exactly this statement. The row α_{11} is zero because $\omega(z_1, z_1) = 0$.

The codimension-two rows use the same two degree-one Hamiltonian labels and the same transported marker words before and after projection. The finite-window marked-complex compatibility preserves the resulting graph, coefficient product, marker transport, and incidence signs. Hence the two ordered composites truncate to the same labelled composites in the lower window, and their pair cancellation survives.

If a bracket row contains α_{sc} or α_{11}, one bracket argument is zero as a local functional in the retained full-equivariant row family. Bilinearity of the BV/convolution bracket gives $R^{\mathrm{row}}_{b,M} = 0$, and applying the scalar projection gives $S_{b,M} = 0$. The zero truncation row is then compatible with the defining equations of Definition 8.24.

For genuine seeds and for nonzero added bracket rows, the displayed formulas are additional data beyond the cutoff; they are the required decompositions of the projected normalized row values in the chosen lower-window row basis. Applying $\pi_{N,P}\pi_{M,N} = \pi_{M,P}$ gives the displayed cocycle equations. Conversely, those equations are exactly the statement that the given matrices define a compatible finite-window row system. $\square$

Proposition 8.30 *(Projection formulas for the u, v, and q rows).* Work in a larger finite-window graph system in the local $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ full-equivariant marked branch. Fix $M \geq N$. Suppose the system gives ordered lower-window row bases, or equivalently row presentations with chosen coordinate maps, for the following three finite-dimensional spans:

$$\begin{aligned}
\mathcal{U}_M &= \mathrm{span}\{\mathbf{K}^M_\Gamma\}_{\Gamma \in \mathcal{G}^{\mathrm{gen}}_{2,M}}, \qquad \mathbf{K}^M_\Gamma = \varepsilon_\Gamma K^M_\Gamma,\\
\mathcal{V}_M &= \mathrm{span}\{\mathbf{E}^M_{\Gamma,\nu}\}_{\Gamma,\nu}, \qquad \mathbf{E}^M_{\Gamma,\nu} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu}),\\
\mathcal{B}_M &= \mathrm{span}\{\mathbf{B}^M_{\beta_1,\beta_2}\}_{\beta_i \notin \{\alpha_{\mathrm{sc}}, \alpha_{11}\}}, \qquad \mathbf{B}^M_{\beta_1,\beta_2} = \frac{1}{2}\varepsilon^M_{\beta_1,\beta_2}\{K^M_{\beta_1}, K^M_{\beta_2}\}_{\hbar,M}.
\end{aligned}$$

Assume these spans are projection-closed:

$$\pi_{M,N}\mathcal{U}_M \subset \mathcal{U}_N, \qquad \pi_{M,N}\mathcal{V}_M \subset \mathcal{V}_N, \qquad \pi_{M,N}\mathcal{B}_M \subset \mathcal{B}_N,$$

and that $\pi_{M,N}$ is the chain and BV-bracket projection on the chosen finite-window subcomplex:

$$\pi_{M,N} d_M = d_N \pi_{M,N}, \qquad \pi_{M,N}\{x, y\}_{\hbar,M} = \{\pi_{M,N}x, \pi_{M,N}y\}_{\hbar,N}$$

for the rows under consideration.

Then the missing entries isolated above are defined by coordinates of finite-window projection:

$$\pi_{M,N}\mathbf{K}^M_\Gamma = \sum_{\Gamma'} u^{M,N}_{\Gamma,\Gamma'} \mathbf{K}^N_{\Gamma'},$$

$$\pi_{M,N}\mathbf{E}^M_{\Gamma,\nu} = \sum_{\Gamma',\nu'} v^{M,N}_{\Gamma,\nu;\Gamma',\nu'} \mathbf{E}^N_{\Gamma',\nu'},$$

and

$$\pi_{M,N}\mathbf{B}^M_{\beta_1,\beta_2} = \sum_{\beta_1',\beta_2'} q^{M,N}_{\beta_1,\beta_2;\beta_1',\beta_2'} \mathbf{B}^N_{\beta_1',\beta_2'}.$$

These equations are the definitions of $u^{M,N}$, $v^{M,N}$, and $q^{M,N}$ relative to the chosen lower-window coordinates. If the displayed lower-window families are honest bases, the entries are unique. If the system has a spanning family with relations, the coordinates are part of the presentation that fixes the canonical matrix entry.

The u-entries also transport the differential rows:

$$\pi_{M,N}(\varepsilon_\Gamma d_M K^M_\Gamma) = \sum_{\Gamma'} u^{M,N}_{\Gamma,\Gamma'} \,\varepsilon_{\Gamma'} d_N K^N_{\Gamma'}.$$

This is the first coordinate equation after applying $d_N \pi_{M,N} = \pi_{M,N} d_M$.

For the bracket rows there is a useful factorized form. Suppose the added nonzero order-1 rows have transport

$$\pi_{M,N} K^M_\beta = \sum_\eta r^{M,N}_{\beta,\eta} K^N_\eta, \qquad \eta \notin \{\alpha_{\mathrm{sc}}, \alpha_{\mathrm{II}}\},$$

and write the lower-window ordered raw bracket reduction as

$$\frac{1}{2}\{K^N_{\eta_1}, K^N_{\eta_2}\}_{\hbar,N} = \sum_{\beta'_1,\beta'_2} \Theta^N_{\eta_1,\eta_2;\beta'_1,\beta'_2} \mathbf{B}^N_{\beta'_1,\beta'_2}.$$

Then

$$q^{M,N}_{\beta_1,\beta_2;\beta'_1,\beta'_2} = \varepsilon^M_{\beta_1,\beta_2} \sum_{\eta_1,\eta_2} r^{M,N}_{\beta_1,\eta_1} r^{M,N}_{\beta_2,\eta_2} \Theta^N_{\eta_1,\eta_2;\beta'_1,\beta'_2}.$$

If the lower system retains all ordered normalized bracket rows with trivial row relations, then

$$\Theta^N_{\eta_1,\eta_2;\beta'_1,\beta'_2} = \frac{\delta_{\eta_1,\beta'_1}\delta_{\eta_2,\beta'_2}}{\varepsilon^N_{\beta'_1,\beta'_2}},$$

and hence

$$q^{M,N}_{\beta_1,\beta_2;\beta'_1,\beta'_2} = \frac{\varepsilon^M_{\beta_1,\beta_2}}{\varepsilon^N_{\beta'_1,\beta'_2}} r^{M,N}_{\beta_1,\beta'_1} r^{M,N}_{\beta_2,\beta'_2}.$$

For $M \geq N \geq P$, these coordinate matrices satisfy

$$u^{M,P}_{\Gamma,\Gamma''} = \sum_{\Gamma'} u^{M,N}_{\Gamma,\Gamma'} u^{N,P}_{\Gamma',\Gamma''},$$

$$v^{M,P}_{\Gamma,\nu;\Gamma'',\nu''} = \sum_{\Gamma',\nu'} v^{M,N}_{\Gamma,\nu;\Gamma',\nu'} v^{N,P}_{\Gamma',\nu';\Gamma'',\nu''},$$

and

$$q^{M,P}_{\beta_1,\beta_2;\theta_1,\theta_2} = \sum_{\beta'_1,\beta'_2} q^{M,N}_{\beta_1,\beta_2;\beta'_1,\beta'_2} q^{N,P}_{\beta'_1,\beta'_2;\theta_1,\theta_2}.$$

Consequently the enlarged-system order-2 non-scalar residual belongs to the compatible finite-window tower only after the following data are fixed: the genuine seed weights, source-face values, nonzero order-1 rows, all orientation/Koszul/symmetry signs, CE-source transport, lower-window row bases or presentations, scalar-condition vanishing or scalar projections, and the projection-closure identities above. With this system the obstruction is exactly the class of the resulting row sum in $H^1(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M})$, together with the Roos compatibility class in $\varprojlim^1 H^0(\mathcal{K}^\bullet_{\mathrm{ns},M}(I))$. This system is the data defining the non-scalar QME obstruction class.

Proof. The first three displayed equations are the coordinate expansion of the projected normalized row values in the chosen lower-window row bases. Projection-closure is exactly the hypothesis that these coordinates exist. If the lower rows are linearly independent, the coordinate expansion is unique. If there are row relations, uniqueness is lost unless the system gives a presentation or a coordinate splitting.

The $d\Gamma$-row formula follows by applying the chain-map identity to $\pi_{M,N}\mathbf{K}^M_\Gamma$. The q-formula follows by applying the BV-bracket identity:

$$\begin{aligned}\pi_{M,N}\mathbf{B}^M_{\beta_1,\beta_2} &= \frac{1}{2}\varepsilon^M_{\beta_1,\beta_2}\{\pi_{M,N}K^M_{\beta_1}, \pi_{M,N}K^M_{\beta_2}\}_{\hbar,N} \\ &= \frac{1}{2}\varepsilon^M_{\beta_1,\beta_2}\sum_{\eta_1,\eta_2} r^{M,N}_{\beta_1,\eta_1} r^{M,N}_{\beta_2,\eta_2}\{K^N_{\eta_1}, K^N_{\eta_2}\}_{\hbar,N},\end{aligned}$$

and then reducing each lower raw bracket by the matrix Θ^N. The ordered normalized-row specialization is the case where the lower raw bracket is already one basis row after division by its incidence coefficient.

The cocycle identities are forced by $\pi_{N,P}\pi_{M,N} = \pi_{M,P}$. Expanding first directly from M to P, and then through N, gives two coordinate expansions of the same projected row in the same lower-window coordinates. Equality of coordinates gives the three displayed matrix product formulas. These compatibility identities are precisely the truncation part of Definition 8.24. Once they are fixed, the obstruction criterion is the non-scalar kernel criterion of Definition E.33 and Proposition 8.23. □

Proposition 8.31 *(Finite-row primitive search interface).* Let $\mathsf{A}^{\mathrm{fw}}_{n,M}$ be a valid finite-window graph array at order n, and suppose its scalar condition has vanished:

$$\sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} S_{\alpha,M} = 0.$$

Choose an ordered row presentation

$$\rho^M_1, \dots, \rho^M_a$$

for the degree-one non-scalar rows retained at this order: graph differentials, CE-source faces, nonzero bracket rows, and the lower counterterm rows. Write the residual as

$$R_{n,M} = \sum_i r^M_i \rho^M_i.$$

Choose also an ordered finite list of degree-zero counterterm primitives

$$\eta^M_1, \dots, \eta^M_b \in \mathcal{K}^0_{\mathrm{ns},M}(I)$$

given by the same finite system, and write

$$d_{\mathrm{ns},M}\eta^M_j = \sum_i A^M_{ij}\rho^M_i.$$

Then a counterterm in the given primitive span, $C_{n,M} = \sum_j c_j\eta^M_j$, exists if and only if the finite linear system

$$A^M c = -r^M$$

is solvable. When it is obstructed, any covector $\ell \in (\mathbb{C}^a)^\vee$ satisfying

$$\ell A^M = 0, \qquad \ell(r^M) \neq 0$$

separates r^M from im A^M, hence the class of $R_{n,M}$ is nonzero in the cohomology of the displayed finite row complex.

For a tower $M \geq N$, let $T^{M,N}$ be the row-truncation matrix on the ρ-presentation and $P^{M,N}$ the chosen transport matrix on the primitive columns. The row block $T^{M,N}$ is the direct sum of the already computed identity or zero blocks, the u-block for genuine seed rows, the v-block for CE-source faces, and the q-block for nonzero bracket rows. A compatible primitive search is defined only when

$$T^{M,N} r^M = r^N, \qquad T^{M,N} A^M = A^N P^{M,N}.$$

After windowwise solutions c_M have been chosen, the remaining compatibility obstruction is the Roos class represented by

$$\left[P^{M,N} c_M - c_N\right] \in H^0(\mathcal{K}^\bullet_{\mathrm{ns},N}(I)).$$

Thus the primitive search has three conditions, in this order: scalar condition, fixed-window linear solvability, and Roos $\varprojlim^1$ compatibility. The last condition is reached only after fixed-window H^1-obstructions detected by such covectors ℓ vanish; in that case the fixed-window obstruction is decisive.

Proof. Once the scalar condition vanishes, the residual lies in the non-scalar kernel complex. Restricting the possible counterterm to the given degree-zero span turns the equation $d_{\mathrm{ns},M} C_{n,M} = -R_{n,M}$ into the displayed matrix equation. Solvability is equivalent to $-r^M \in \operatorname{im} A^M$. In finite dimension this is detected by a linear functional annihilating im A^M and pairing nontrivially with r^M, exactly as written.

The truncation identities say that the row residual and the boundaries of primitive columns commute with finite-window projection. The computed blocks are those of Propositions 8.29 and 8.30. If c_M and c_N solve the two fixed-window equations, then $P^{M,N} c_M - c_N$ is closed in degree zero by $T^{M,N} A^M = A^N P^{M,N}$ and $T^{M,N} r^M = r^N$. Its cohomology class is the Roos 1-cocycle of Proposition 8.23. □

In the minimal full-equivariant array all order-one and order-two rows close to zero (Propositions 8.25, 8.26, 8.28); the first non-trivial obstruction therefore appears at order three. The smallest enlargement exhibiting a non-scalar row with nonzero local-functional value is a single CE-source face attached to a genuine marked Costello seed at $\hbar^3$, the row θ_3 of Proposition 8.32, with obstruction class $\mathrm{Ob}^{\Pi}_{\theta_3}$ and companion-face construction tracked through Theorem 8.35.

PROPOSITION 8.32 *(Concrete order-three row in an enlarged system).* Work in the full-equivariant marked branch with $V = \mathbb{C}^{N|N}$, and let $M \geq 3$. Enlarge the minimal finite-window system by one Hom-valued CE-source face row

$$\theta_3 = \mathrm{CE}(\Theta_3, \nu_3), \qquad |\Theta_3|_\hbar = 3,$$

where Θ_3 is a genuine marked Costello seed, distinct from a scalar-contact closure composite and from a bracket row. The row data are as follows.

The graph type and order are

$$\tau_{\theta_3} = \mathrm{CE}, \qquad |\theta_3|_\hbar = 3.$$

Its finite labels are

$$\begin{aligned} e_{\mathrm{pp}} :&\quad P^{w,M}_{\epsilon,L}, \quad h_{2,1} = z_1^2 z_2,\ \rho_{2,1}, \\ e_{\mathrm{ct}} :&\quad \Delta^{w,M}_{\epsilon,L}, \quad h_{1,1} = z_1 z_2,\ \rho_{1,1}, \\ e_{\mathrm{W}} :&\quad E_{\mathrm{Weyl}}, \quad z_1^2,\ z_2 \end{aligned}$$

with $B_\theta \in \mathcal{D}_c^{\mathrm{reg}}(I)$ on the cotangent external leg. The vertex labels are

$$u_{a(t)dt\otimes z_1^2}, \qquad u_{b(t)dt\otimes z_2}, \qquad c_{B_\theta,\rho_{2,1}}, \qquad I^w_{0,M}, \qquad W^{R,M}_{\Theta_3,L,w}(B^\Theta_\bullet).$$

The marker tensor is a full $\mathfrak{gl}(N|N)$-equivariant signed Brauer marker

$$\mathfrak{m}_{\theta_3} \in \mathcal{M}^{\mathrm{full}}(\mathbb{C}^{N|N}), \qquad \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}_{\theta_3}) = 0.$$

Choose the source-face coefficient

$$d_{\mathrm{CE}}\zeta_M = +\zeta_{M,\nu_3} + \text{other faces}$$

on the Θ_3-component. Since the kernel has degree zero, the incidence sign of this CE row in $-\kappa^M_{\mathcal{G},w,I}(d_{\mathrm{CE}}\zeta_M)$ is -1.

Write

$$K^M_{\Theta_3} = W^{R,M}_{\Theta_3,L,w}(B^\Theta_\bullet).$$

The row value is the nonzero symbolic local functional

$$R^{\mathrm{row}}_{\theta_3,M} = -K^M_{\Theta_3}(\zeta_{M,\nu_3}) =: \mathsf{E}_{\theta_3,M} \neq 0.$$

Its scalar condition is

$$S_{\theta_3,M} = \widehat{\sigma}_{\mathrm{sc},M}(\mathsf{E}_{\theta_3,M}) = 0.$$

Indeed, the row lies in the full-equivariant marked complex, so Theorem E.32 kills its scalar projection; also the displayed Weyl half-edge has $\omega(z_1^2, z_2) = 0$.

The class-forming closure check is part of the row datum:

$$d_{\mathrm{ns},M}\mathsf{E}_{\theta_3,M} = 0.$$

This equation is part of the valid obstruction-row data; with it given, the row defines

$$\mathfrak{o}^{\mathrm{ns}}_{\theta_3,M} = [\mathsf{E}_{\theta_3,M}] \in H^1\Big(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}\Big).$$

The truncation matrix is explicit. For $M \geq N$,

$$\pi_{M,N}\mathsf{E}_{\theta_3,M} = \begin{cases} \mathsf{E}_{\theta_3,N}, & N \geq 3, \\ 0, & N < 3, \end{cases}$$

hence

$$t^{M,N}_{\theta_3\theta_3} = 1 \quad (N \geq 3), \qquad t^{M,N}_{\theta_3\alpha'} = 0 \quad (\alpha' \neq \theta_3),$$

and every entry from θ_3 is zero for $N < 3$. These entries satisfy $t^{M,P} = t^{M,N}t^{N,P}$.

Therefore this enlarged-system row is a nonzero local-functional cochain with vanishing scalar condition. It is an explicit symbolic obstruction datum. A fixed-window order-three counterterm exists precisely when

$$\mathfrak{o}^{\mathrm{ns}}_{\theta_3,M} = 0, \qquad d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$$

for some $C_{\theta_3,M} \in \mathcal{K}^0_{\mathrm{ns},M}(I)$. This row names the class; solving the larger order-three QME branch requires an added primitive or companion rows.

PROPOSITION 8.33 *(θ_3 finite-row obstruction criterion).* Keep the one-row order-three system of Proposition 8.32. In the finite row complex given there, set

$$\mathcal{K}^0_{\theta_3,M} = 0, \qquad \mathcal{K}^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_{\theta_3,M} = 0,$$

and take residual $R_{\theta_3,M} = \mathsf{E}_{\theta_3,M}$. The scalar condition has already vanished:

$$\widehat{\sigma}_{\mathrm{sc},M}(\mathsf{E}_{\theta_3,M}) = 0.$$

The primitive-search matrix is the 1×0 zero matrix and the target vector is $-\mathsf{E}_{\theta_3,M}$. Hence the equation $A^M c = -r^M$ has a nonzero obstruction in the displayed finite row complex. The covector

$$\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$$

annihilates im $d_{\theta_3,M} = 0$ and satisfies $\ell_{\theta_3}(R_{\theta_3,M}) = 1$. Therefore

$$[\mathsf{E}_{\theta_3,M}] \neq 0 \quad \text{in} \quad H^1(\mathcal{K}^\bullet_{\theta_3,M}).$$

This is an exact obstruction statement for the given finite row subcomplex. A primitive in a larger local-functional complex requires extra data. To kill the row one must add either a degree-zero primitive $C_{\theta_3,M}$ with $d_{\mathrm{ns},M} C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$, or additional Hom-valued companion rows whose sum changes the residual vector.

If such a primitive $C_{\theta_3,M}$ is given and truncates by

$$\pi_{M,N} C_{\theta_3,M} = C_{\theta_3,N} \quad (N \geq 3), \qquad \pi_{M,N} C_{\theta_3,M} = 0 \quad (N < 3),$$

then the fixed-window search is solved and the Roos $\varprojlim^1$-class of this one-row tower is zero. A nonzero truncation defect leaves zero fixed-window obstruction but represents the Roos class by

$$[\pi_{M,N} C_{\theta_3,M} - C_{\theta_3,N}] \in H^0(\mathcal{K}^\bullet_{\mathrm{ns},N}(I)).$$

For the minimal order-two full-equivariant rows and for bracket rows containing α_{sc} or α_{II}, the same primitive search has residual vector 0, so the zero counterterm is the unique given solution and its Roos class is zero.

Proof. This is Proposition 8.31 applied to the one-row presentation. The degree-zero row set is empty, so the image of the finite differential is zero. The residual vector is the basis vector $\mathsf{E}_{\theta_3,M}$, so the displayed covector is a cokernel separating covector. The Roos statement is the same proposition applied after a hypothetical primitive has been added. The minimal order-two and zero-bracket rows have zero residual before cohomology, hence the zero primitive is compatible under truncation. □

PROPOSITION 8.34 *(θ_3 primitive-entry criterion).* Keep the row $\theta_3 = \mathrm{CE}(\Theta_3, \nu_3)$ and the notation of Propositions 8.32 and 8.33. A proposed degree-zero primitive for this row is admissible for the finite-row theorem only after the row array gives one of the following two primitive entries.

(P1) A CE-ancestor entry

$$\eta_{\theta_3,M}, \qquad d_{\mathrm{CE}} \eta_{\theta_3,M} = \zeta_{M,\nu_3},$$

together with the Hom-valued Bianchi and companion-face terms needed to prove

$$C^{\mathrm{CE}}_{\theta_3,M} = K^M_{\Theta_3}(\eta_{\theta_3,M}) \in \mathcal{K}^0_{\mathrm{ns},M}(I), \qquad d_{\mathrm{ns},M} C^{\mathrm{CE}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}.$$

(P2) A local counterterm entry $C^{\mathrm{ct}}_{\theta_3,M} \in \mathcal{K}^{\circ}_{\mathrm{ns},M}(I)$, with its locality proof, scalar projection, and differential expansion, satisfying

$$d_{\mathrm{ns},M} C^{\mathrm{ct}}_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}.$$

Equivalently, in the ordered one-row presentation

$$\rho^M_1 = \mathsf{E}_{\theta_3,M}, \qquad r^M = (1),$$

the given primitive column $C_{\theta_3,M}$ must have boundary-matrix coefficient

$$A^M_{\theta_3,C} = -1, \qquad d_{\mathrm{ns},M} C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}.$$

Admissible row data give the displayed differential entry with coefficient -1 for the primitive. Any other datum must first be replaced by an explicitly given primitive whose differential column is the one above.

The scalar and tower conditions are part of the same datum:

$$\widehat{\sigma}_{\mathrm{sc},M}(C_{\theta_3,M}) = 0, \qquad \pi_{M,N} C_{\theta_3,M} = \begin{cases} C_{\theta_3,N}, & N \geq 3, \\ 0, & N < 3. \end{cases}$$

With the row transport $T^{M,N}_{\theta_3\theta_3} = 1$ for $N \geq 3$ and 0 for $N < 3$, these equations give

$$T^{M,N} A^M = A^N P^{M,N}, \qquad [P^{M,N} c_M - c_N] = 0 \quad \text{for the solution } c_M = 1.$$

Thus such a primitive kills the fixed-window row and has zero Roos compatibility class. The one-row complex here has $K^{\circ}_{\theta_3,M} = 0$, so Proposition 8.33 remains the active obstruction statement.

A complete companion-face table is a different condition. It replaces the one-face residual by a full CE-source-face row sum

$$\sum_{\nu \in F_{\Theta_3,M}} R^{\mathrm{row}}_{\mathrm{CE}(\Theta_\nu,\nu),M}, \qquad \nu_3 \in F_{\Theta_3,M},$$

whose ν_3-summand is $\mathsf{E}_{\theta_3,M}$. Such a table is admissible only if it gives every face, every CE coefficient, every scalar condition, the $v^{M,N}$-truncation matrices with their cocycle identity, and the codimension-two closure table, and if

$$\sum_{\nu \in F_{\Theta_3,M}} R^{\mathrm{row}}_{\mathrm{CE}(\Theta_\nu,\nu),M} = 0, \qquad \sum_{\nu \in F_{\Theta_3,M}} S_{\mathrm{CE}(\Theta_\nu,\nu),M} = 0.$$

The current one-face table is only an interface for this possibility: its residual is $\mathsf{E}_{\theta_3,M}$. Hence it leaves the θ_3 row active and leaves the primitive Roos class unformed. The first source-algebraic companion table containing ν_3 is recorded in Theorem 8.35; it produces an obstruction datum.

Proof. For a primitive column, apply Proposition 8.31 with degree-one basis $\rho^M_1 = \mathsf{E}_{\theta_3,M}$ and residual vector $r^M = (1)$. The equation $A^M c = -r^M$ is solved by the given primitive with $c = 1$ exactly when the primitive column is $A^M_{\theta_3,C} = -1$. The CE-ancestor and local-counterterm alternatives are precisely two ways of producing such a degree-zero column. The displayed target differential, scalar projection, and support proof give the boundary matrix entry used by the finite row complex.

The truncation assertion is the identity/zero transport already attached to the row $\mathsf{E}_{\theta_3,M}$. With the same identity/zero rule for $C_{\theta_3,M}$, the matrix equation $TA = AP$ is tautological: above degree 3 it reads $(-1) = (-1)$, and below degree 3 both sides are 0. The chosen coefficients $c_M = 1$ are therefore compatible, so the Roos representative vanishes.

For a companion-face table the residual vector itself is replaced by the signed row sum of all CE-source faces. The finite-row theorem can test that new row sum only after the table gives its scalar conditions, truncation coordinates, and codimension-two closure data. A nonzero signed sum remains a residual; a zero signed sum cancels before the primitive search and leaves the primitive Roos class unformed. □

THEOREM 8.35 *(θ_3 companion-face construction obstruction).* Work in the order-three full-equivariant marked branch of Proposition 8.32, and put $H_{p,q} = z_1^p z_2^q$. On the standard symplectic $\mathbb{C}^2$ with $\omega = dz_1 \wedge dz_2$ the Hamiltonian Poisson rule is $\{H_{a,b}, H_{r,s}\} = (as - br)H_{a+r-1,b+s-1}$; in particular $\{H_{1,0}, H_{2,1}\} = H_{2,0}$. Consider the source-algebra ancestor

$$\zeta_M^0 = u_{a,H_{1,0}}\, u_{b,H_{0,1}}\, c_{B_\theta,\rho_{2,1}}.$$

Its monomial CE differential is the first source-algebraic table of the same external shape whose source faces contain the displayed ν_3-row $u_{a,H_{2,0}} u_{b,H_{0,1}} c_{B_\theta,\rho_{2,1}}$. After truncation to Hamiltonian degree ≤ 3, removal of constants, and cancellation of the two $\rho_{1,1}$-terms, the nonzero face set is the following eight-face set. With

$$E_\nu^M = -K_{\Theta_3}^M(\zeta_{M,\nu}), \qquad E_{\nu_3}^M = \mathsf{E}_{\theta_3,M},$$

the table is

ν	$\epsilon^{\mathrm{CE}}_{\Theta_3,\nu}$	$\zeta_{M,\nu}$	$\epsilon^{\mathrm{CE}}_{\Theta_3,\nu} E_\nu^M$
ν_{02}	$+2$	$u_{a,H_{0,1}} u_{b,H_{0,1}} c_{B_\theta,\rho_{0,2}}$	$2E^M_{\nu_{02}}$
ν_{20}	-2	$u_{a,H_{1,0}} u_{b,H_{1,0}} c_{B_\theta,\rho_{2,0}}$	$-2E^M_{\nu_{20}}$
ν_{03}	$+3$	$u_{a,H_{0,2}} u_{b,H_{0,1}} c_{B_\theta,\rho_{0,3}}$	$3E^M_{\nu_{03}}$
$\nu_{12}^{(1)}$	$+2$	$u_{a,H_{1,1}} u_{b,H_{0,1}} c_{B_\theta,\rho_{1,2}}$	$2E^M_{\nu_{12}^{(1)}}$
$\nu_{12}^{(2)}$	-1	$u_{a,H_{1,0}} u_{b,H_{0,2}} c_{B_\theta,\rho_{1,2}}$	$-E^M_{\nu_{12}^{(2)}}$
$\nu_3 = \nu_{21}^{(1)}$	$+1$	$u_{a,H_{2,0}} u_{b,H_{0,1}} c_{B_\theta,\rho_{2,1}}$	$\mathsf{E}_{\theta_3,M}$
$\nu_{21}^{(2)}$	-2	$u_{a,H_{1,0}} u_{b,H_{1,1}} c_{B_\theta,\rho_{2,1}}$	$-2E^M_{\nu_{21}^{(2)}}$
ν_{30}	-3	$u_{a,H_{1,0}} u_{b,H_{2,0}} c_{B_\theta,\rho_{3,0}}$	$-3E^M_{\nu_{30}}$

Hence the source-algebraic companion residual is

$$R^{\mathrm{cand}}_{\Theta_3,M} = 2E^M_{\nu_{02}} - 2E^M_{\nu_{20}} + 3E^M_{\nu_{03}} + 2E^M_{\nu_{12}^{(1)}} - E^M_{\nu_{12}^{(2)}} + \mathsf{E}_{\theta_3,M} - 2E^M_{\nu_{21}^{(2)}} - 3E^M_{\nu_{30}}.$$

This expression is the exact residual of the attempted companion route. Its vanishing is an additional identity.

The induced source-face transport is the diagonal cutoff

$$v^{M,N}_{\nu;\nu'} = \begin{cases} 1, & \nu = \nu' \text{ and } d(\nu) \le N, \\ 0, & \text{otherwise,} \end{cases}$$

where $d(\nu_{02}) = d(\nu_{20}) = 2$ and the other six faces have degree 3. It satisfies the cocycle identity $v^{M,P} = v^{M,N} v^{N,P}$. For $N = 2$, however, the transported residual is

$$\pi_{M,2} R^{\mathrm{cand}}_{\Theta_3,M} = 2E^2_{\nu_{02}} - 2E^2_{\nu_{20}}.$$

Thus a fixed-window cancellation at $M \geq 3$ gives a compatible tower only after the lower-window identity $E^2_{\nu_{02}} = E^2_{\nu_{20}}$ is proved in the chosen row basis, or after the lower-window row presentation is changed and its v-matrix is recomputed.

Consequently the companion route is a construction/obstruction theorem with the following exact missing marked Costello datum:

$$\left(d_{\mathrm{CE}}\zeta_M,\ \epsilon^{\mathrm{CE}}_{\Theta_3,\nu},\ K^M_{\Theta_3}(\zeta_{M,\nu}),\ \mu_\nu,\ v^{M,N}_{\nu;\nu'},\ C^{(2)}_{\nu,\nu'}\right)_{\nu,\nu'}.$$

Here μ_ν is the marked result and marker-transport entry, and $C^{(2)}_{\nu,\nu'}$ is the codimension-two source-face incidence/weight table: the two orders of each pair of boundary operations, the resulting graph, the orientation/Koszul coefficient, the transported marker, and the resulting renormalized row coordinate in the declared degree-one basis. Until this table is given and proves both $R^{\mathrm{cand}}_{\Theta_3,M} = 0$ and its transported lower-window identities, the proved conclusion remains the one-row obstruction of Proposition 8.33.

Proof. The Poisson rule $[H_{a,b}, H_{r,s}] = (as - br)H_{a+r-1,b+s-1}$ gives the displayed source-algebra faces from the differential of the two u-factors against the current label. In degree ≤ 3, constants are removed by the Hamiltonian quotient and the two $\rho_{1,1}$-faces have opposite coefficients, so the eight listed terms remain. Applying the row convention $R^{\mathrm{row}}_{\mathrm{CE}(\Theta_3,\nu),M} = \epsilon^{\mathrm{CE}}_{\Theta_3,\nu}E^M_\nu$ gives the signed residual formula, with the ν_3-summand equal to $\mathsf{E}_{\theta_3,M}$.

The diagonal transport is finite-window projection on the Hamiltonian degree of the c-label. Such diagonal cutoffs compose, proving the cocycle identity. Projecting to $N = 2$ deletes every degree-three face and leaves exactly the two degree-two summands. Therefore the lower-window residual is $2E^2_{\nu_{02}} - 2E^2_{\nu_{20}}$.

The passage from this source-algebra table to a Costello cancellation theorem is precisely the missing marked incidence/weight computation. Source coefficients alone leave the renormalized graph weights, marker transports, codimension-two closure signs, or lower-window row coordinates undetermined. Those entries are the displayed missing datum. They are the data by which Definition 8.24 gives a zero residual vector and Proposition 8.31 still sees the current one-row obstruction. □

Theorem 8.36 *(Finite-window non-scalar curvature criterion).* Fix a finite-window graph/QME system $(I, w, M, \mathcal{G}_M)$, and fix the labelled CE cycle ζ_M on which the curvature is evaluated. Write K^M_Γ for the component of $\kappa^M_{\mathcal{G},w,I}$ attached to a graph Γ; for the zero-internal-edge component this is $\kappa^{\mathrm{coef},M}_{\hbar,w,I}$, and for a positive graph it is the renormalized weight $W^{R,M}_{\Gamma,L,w}$. If $|\Gamma|_\hbar$ denotes the $\hbar$-order assigned by the finite graph system, then the order-n graph curvature is

$$\begin{aligned}
\mathrm{Ob}^{\mathrm{graph}}_{\mathcal{G},M}[n](\zeta_M) &= [\hbar^n]\Big(d_M\kappa^M_{\mathcal{G},w,I}(\zeta_M) - \kappa^M_{\mathcal{G},w,I}(d_{\mathrm{CE}}\zeta_M) \\
&\qquad\qquad + \frac{1}{2}\sum_{(\zeta_M)}\{\kappa^M_{\mathcal{G},w,I}(\zeta'_M), \kappa^M_{\mathcal{G},w,I}(\zeta''_M)\}_{\hbar,M}\Big) \\
&= \sum_{|\Gamma|_\hbar=n}\varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{|\Gamma'|_\hbar=n}\varepsilon^{\mathrm{CE}}_{\Gamma'}K^M_{\Gamma'} \\
&\qquad + \frac{1}{2}\sum_{|\Gamma_1|_\hbar+|\Gamma_2|_\hbar=n}\varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M}.
\end{aligned}$$

The signs are the orientation signs of the given finite graph differential and the CE coproduct. With lower counterterms $C_{<n,M}$, the cochain tested at order n is

$$R_{n,M} = \mathrm{Ob}^{\mathrm{graph}}_{\mathcal{G},M}[n] + [\hbar^n]\Big(d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big).$$

Its scalar condition is

$$\mathfrak{s}_{n,M} = \widehat{\sigma}_{\mathrm{sc},M}(R_{n,M}).$$

The first scalar-contact graph whose coefficient is computed in the present local model is the one-cross-contraction boundary graph with two Hamiltonian external legs and one open scalar loop. It has $\hbar$-order 1. For a full-equivariant marked boundary generator $X_{\Gamma,o,\mathfrak{m},M}$, the extracted order-one scalar condition is

$$\mathfrak{s}^{\mathrm{mk}}_{1,M}(X_{\Gamma,o,\mathfrak{m},M}) = \sum_{i:\, r(\beta_i)=1} (-1)^{i-1}\, \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m})\big)\, \omega(f_{\partial_{\beta_i}\Gamma}, g_{\partial_{\beta_i}\Gamma})\gamma_{\mathbf{1}}.$$

Before the coefficient extraction this is the $\hbar^1$-term $\hbar\, \mathfrak{s}^{\mathrm{mk}}_{1,M}$. If $V = \mathbb{C}^{N|N}$ and the markers are full $\mathfrak{gl}(V)$-equivariant, then

$$\mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m})\big) = 0$$

for every elementary boundary term by Lemma E.31; hence

$$\mathfrak{s}^{\mathrm{mk}}_{1,M} = 0.$$

This is a scalar-condition vanishing. A non-scalar primitive requires its graph-level construction.

After the scalar condition vanishes, the first non-scalar order is the least n_0 for which the cochain $R_{n_0,M}$ is defined for the chosen graph list and survives the lower-order choices. Its class is

$$[\mathfrak{o}^{\mathrm{fw}}_{n_0,M}] = [R_{n_0,M}] \in H^1\big(Q^{\bullet}_{\mathcal{G},M}(I), d_M\big).$$

It vanishes exactly when there exists $C_{n_0,M} \in Q^{0}_{\mathcal{G},M}(I)$ with $d_M C_{n_0,M} = -R_{n_0,M}$. In a compatible finite-window tower, after choosing any such windowwise primitives, the remaining compatibility obstruction is

$$\lambda^{\mathrm{fw}}_{n_0} = \big[\big([\pi_{M+1,M} C_{n_0,M+1} - C_{n_0,M}]\big)_M\big] \in \varprojlim\nolimits^1_M H^0\big(Q^{\bullet}_{\mathcal{G},M}(I)\big).$$

Thus compatible finite-window counterterms exist precisely when all fixed-window classes $[\mathfrak{o}^{\mathrm{fw}}_{n_0,M}]$ vanish and $\lambda^{\mathrm{fw}}_{n_0} = 0$.

The abstract existence of $\mathcal{G}_M$ gives the graph list but does not determine the coefficient array. The integer n_0, the class $[\mathfrak{o}^{\mathrm{fw}}_{n_0,M}]$, and the Roos class $\lambda^{\mathrm{fw}}_{n_0}$ are determined only after a valid finite-window coefficient array has been chosen in the sense of Definition 8.24:

$$\mathsf{A}^{\mathrm{fw}}_{n,M}$$

whose rows specify graph type, edge labels, vertex labels, marker tensor, incidence coefficient, $\hbar$-order, coefficient degree, scalar-contact filtration, scalar-condition term, row curvature value, and truncation coefficients in M. This array turns the formal obstruction symbol of Theorem 8.22 into a computation of n_0, a test of $[\mathfrak{o}^{\mathrm{fw}}_{n_0,M}] = 0$, and a defined $\lambda^{\mathrm{fw}}_{n_0}$.

Proof. The first displayed formula is the definition of the convolution curvature in Definition B.15, evaluated on ζ_M and then projected to the $\hbar^n$-coefficient. Expanding $-\kappa d_{\mathrm{CE}}$ and the CE coproduct bracket gives the three graph sums, with exactly the orientation signs given by the finite graph system. Adding the lower counterterm terms gives $R_{n,M}$, the residual of Theorem 8.22.

The order-one scalar-contact computation is Proposition E.23 applied to each first filtration-lowering marked boundary term of Definition E.28. The full-equivariant marker transport maps preserve the signed Brauer marker span. Lemma E.31 computes the cyclic supertrace of every such resulting marker as a power of sdim $\mathbb{C}^{N|N} = 0$, and the empty marker word gives the same zero coefficient. Therefore the scalar condition vanishes on the full-equivariant marked complex before any non-scalar cohomology class is formed.

Once the scalar condition is zero, $R_{n,M}$ lies in $Q^\bullet_{\mathcal{G},M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M}$ exactly as in the proof of Theorem 8.22. A degree-zero counterterm changes $R_{n,M}$ by a d_M-boundary, so the fixed-window solvability criterion is precisely the vanishing of its H^1-class. If primitives are chosen window by window, their truncation differences are closed degree-zero cochains; their classes form the standard Roos cocycle computing $\varprojlim^1_M H^0(Q^\bullet_{\mathcal{G},M}(I))$. Vanishing of this class is exactly the condition that the primitives can be adjusted to a compatible family.

The final assertion is a dependency statement. The formulas require a finite graph list with order assignment, CE cycle, signs, coproduct decomposition, graph weights, lower counterterms, scalar projection, and truncation maps. Omitting any of these removes a term from the displayed cochain $R_{n,M}$, so its first nonzero order and its cohomology class both become undefined. □

Remark 8.37 *(Scalar-condition boundary).* In the ordinary scalar-reduced $\mathfrak{gl}_N$ branch, the same two-Hamiltonian one-cross graph gives the order-one scalar class $N\,\omega(f,g)\gamma_1$, equivalently $N[\bar{c}]\gamma_1$ in cohomology after extracting the $\hbar^1$-coefficient. In the even block-diagonal marked complex a single marker $T = \lambda_0 P_{\bar{0}} + \lambda_1 P_{\bar{1}}$ gives $N(\lambda_0 - \lambda_1)\omega(f,g)\gamma_1$. These vanish only under the corresponding central-character, balanced, or full-equivariant hypotheses; they are scalar-contact rows.

Remark 8.38 *(Relation to regulator independence).* Theorem 5.82 proves that, after the finite-window Mittag-Leffler hypotheses of Definition 5.76 are imposed, the Hadamard criterion is independent of the admissible exponential weight. This is weight-independence among regulated coefficient products. The unweighted product/direct-sum pair is separated by the endpoint obstruction of Theorem 5.94: the required coefficient Casimir has an obstructed strict product/direct-sum representative.

Remark 8.39 *(Hadamard parametrix versus the Bochner-Martinelli-Koppelman kernel).* The Hadamard parametrix used in Theorem 8.17 controls Costello's heat-kernel propagator $P^{\mathrm{base}}_{\epsilon,L}$ on the bulk free elliptic complex $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$ by short-time asymptotics in Hadamard normal form, with Hörmander wave-front-set tracking [12, Ch. 5], [36, Ch. 7]. The Bochner-Martinelli-Koppelman kernel

$$K_{\mathrm{BM}}(\zeta,z) = \frac{1}{(2\pi i)^2}\frac{\overline{\eta_1}\,d\bar{\eta}_2 - \overline{\eta_2}\,d\bar{\eta}_1}{r^4} \wedge d\zeta_1 \wedge d\zeta_2, \qquad \eta_j = \zeta_j - z_j,\ r = |\zeta - z|,$$

is the several-complex-variables homotopy kernel of [38, §5] producing the classical Koppelman formula $\bar{\partial}H + H\bar{\partial} = \mathrm{id} - i_B p_B$ on bounded C^1 domains ([38, Def. 5.1, Thm. 5.2, Eq. (10)]; Laurent-Thiébaut [37, Ch. III, Def. 1.2, Lem. 1.3]). These are two distinct analytic technologies attached to two different sectors of the local theorem. The heat-kernel parametrix governs the Costello/QME graph expansion of Theorem 8.17; the BMK kernel governs the Dolbeault homotopy on the formal holomorphic polydisk $\widehat{\mathbb{C}^2}_0$ and the principal-part current sector recorded in Theorem B.9. The formulas remain separate: the

Hadamard parametrix uses the bulk Riemannian/Hermitian heat data on $\mathbb{R}^2 \times \mathbb{C}^2$, the BMK kernel uses the holomorphic homotopy data on $\mathbb{C}^2$ alone, and the local theorem combines their separate finite-window classes through the obstruction vectors stated below.

THEOREM 8.40 *(BMK finite-window obstruction for strict all-window Hamiltonian transfer).* Let $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ be the Hamiltonian Lie algebra of the formal symplectic polydisk $(\widehat{\mathbb{C}^2}_0, \omega)$, let

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 \wedge dz_2, \qquad \mathcal{P} = \bigoplus_{\substack{a,b\ge 0\\ a+b>0}} \mathbb{C}\rho_{a,b}, \qquad \mathcal{P}_{\le N} = \bigoplus_{0<a+b\le N} \mathbb{C}\rho_{a,b},$$

and write $\pi_N : \mathcal{P} \twoheadrightarrow \mathcal{P}_{\le N}$ for the finite-Matlis truncation. For every $N \ge 1$, the following $\mathbb{C}$-linear quotient problem has obstruction:

$$q_N : \mathcal{P} \longrightarrow \mathcal{P}_{\le N}$$

retains $\mathcal{P}_{\le N}$ pointwise, kills the high transverse currents $\rho_{a,b}$ with $a + b > N$, and intertwines the full Hamiltonian coadjoint action of $\mathfrak{h}$ on principal parts. In particular, the support-local quotient problem on the derivative-delta-current module $\Delta_{a,b} \in \mathcal{D}_0'^{0,2}(B)$ of Theorem B.9 has the same obstruction to carrying the Hamiltonian action onto a finite Matlis target. The obstruction is the Tate-cohomology class of the principal-part module recorded by

$$\mathrm{ob}_{\mathfrak{h},N} = \pi_N\big(\mathfrak{h} \cdot \ker \pi_N\big) \ne 0 \quad \text{in} \quad \mathcal{P}_{\le N}.$$

The class is exhibited by the explicit boundary-mode identity

$$z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\, \rho_{0,1}, \tag{29}$$

where the high transverse mode $\rho_{N+1,0}$ lies in the killed window $\ker \pi_N$ but the low brane-momentum mode $\rho_{0,1}$ is retained in $\mathcal{P}_{\le N}$. Equivalently, the projected finite-window L_∞-extension has nonzero unary block

$$\kappa_N(z_1^{N+2}, \rho_{N+1,0}) = -(N+2)\, \rho_{0,1},$$

and the projected Jacobiator on the triple $(z_1^{N+2}, z_2, \rho_{N,0})$ is

$$(N+1)(N+2)\, \rho_{0,1} \ne 0. \tag{30}$$

Therefore strict all-window finite-current Hamiltonian transfer

$$\mathbb{C}[z_1, z_2]_{\mathrm{red}} \longrightarrow \text{finite Matlis window}$$

has a nonzero obstruction for every N. The full obstruction vector $\mathrm{Ob}_N^{\text{BMK-curr}}$ of Theorem B.9, with the third entry the displayed $\mathrm{ob}_{\mathfrak{h},N}$, is the precise finite-window obstruction datum.

Proof. The Hamiltonian coadjoint action of Corollary B.8 gives, for $f = z_1^p z_2^q$ and $\rho_{i,j}$,

$$f \cdot \rho_{i,j} = -(pj - qi + p - q)\, \rho_{i-p+1,\, j-q+1}.$$

Specializing $(p,q) = (N+2, 0)$ and $(i,j) = (N+1, 0)$,

$$z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\, \rho_{(N+1)-(N+2)+1,\,0-0+1} = -(N+2)\, \rho_{0,1},$$

which is (29). The mode $\rho_{N+1,0}$ is killed by π_N since $N+1+0>N$; the mode $\rho_{0,1}$ is retained in $\mathcal{P}_{\le N}$ since $0+1\le N$ for every $N\ge 1$. Hence $\rho_{N+1,0}\in\ker\pi_N$ and $\pi_N(z_1^{N+2}\cdot\rho_{N+1,0})=-(N+2)\,\rho_{0,1}\neq 0$, so $\mathrm{ob}_{\mathfrak{h},N}\neq 0$ in $\mathcal{P}_{\le N}$.

A support-local current quotient q_N intertwining the Hamiltonian action would satisfy $q_N(\mathfrak{h}\cdot\xi)=\mathfrak{h}\cdot q_N(\xi)$ for every $\xi\in\mathcal{P}$; applied to $\xi=\rho_{N+1,0}$, with $q_N(\rho_{N+1,0})=0$ by the killing hypothesis, this would force

$$0=\mathfrak{h}\cdot q_N(\rho_{N+1,0})=q_N(\mathfrak{h}\cdot\rho_{N+1,0})\ni q_N(z_1^{N+2}\cdot\rho_{N+1,0})=-(N+2)\,q_N(\rho_{0,1})=-(N+2)\,\rho_{0,1},$$

contradicting $N+2\neq 0$ and $\rho_{0,1}\neq 0$ in $\mathcal{P}_{\le N}$. This is the obstruction.

The projected Jacobiator (30) is the nonzero Jacobiator of the truncated bracket: the unprojected coadjoint bracket on $\mathfrak{h}\ltimes\mathcal{P}[1]$ is a genuine semidirect Lie module and obeys Jacobi; its image after π_N has the displayed obstruction. Concretely, on the triple $(z_1^{N+2},z_2,\rho_{N,0})$ the three cyclic terms in the lifted Jacobi sum are computed by the displayed coadjoint rule together with the Hamiltonian Poisson rule $\{H_{a,b},H_{r,s}\}=(as-br)H_{a+r-1,b+s-1}$ of the polydisk:

$$\begin{aligned}
&\{z_1^{N+2},\{z_2,\rho_{N,0}\}\}=\{z_1^{N+2},(N+1)\rho_{N+1,0}\}=-(N+1)(N+2)\,\rho_{0,1},\\
&\{z_2,\{\rho_{N,0},z_1^{N+2}\}\}=\{z_2,0\}=0\quad\text{since }\{\rho_{N,0},z_1^{N+2}\}=-(z_1^{N+2}\cdot\rho_{N,0})=0\\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\text{by the negative-index rule of Proposition 8.7,}\\
&\{\rho_{N,0},\{z_1^{N+2},z_2\}\}=\{\rho_{N,0},(N+2)\,z_1^{N+1}\}=(N+1)(N+2)\,\rho_{0,1}.
\end{aligned}$$

The unprojected sum is zero, as required by Jacobi for the lifted bracket. Apply π_N entrywise: the first cycle's intermediate $\{z_2,\rho_{N,0}\}=(N+1)\rho_{N+1,0}$ lies in $\ker\pi_N$ since $N+1>N$, so the projected first cycle is $\pi_N\{z_1^{N+2},\pi_N(N+1)\rho_{N+1,0}\}=\pi_N\{z_1^{N+2},0\}=0$; the second cycle is already zero; the third cycle is $\pi_N\{\rho_{N,0},(N+2)z_1^{N+1}\}=(N+1)(N+2)\rho_{0,1}$ since $\rho_{0,1}\in\mathcal{P}_{\le N}$ is retained. The projected sum is therefore $(N+1)(N+2)\rho_{0,1}\neq 0$, which is (30). This is the projected escape-return Jacobiator recorded in Theorem B.9 on the same triple; any quotient q_N making this return exact as a unary boundary would also collapse $\rho_{0,1}$ and hence $\mathcal{P}_{\le N}$ itself.

The third entry $\mathrm{ob}_{\mathfrak{h},N}$ of $\mathrm{Ob}_N^{\mathrm{BMK\text{-}curr}}$ in Theorem B.9 is by construction the displayed class. The remaining entries – diagonal current sign, finite-versus-analytic moment kernel, collar factorization compatibility – are independent obstruction coordinates of the same finite-window obstruction statement and are recorded there. □

Corollary 8.41 *(Pro-Matlis retract as the compatible replacement target).* Theorem 8.40 forecloses every choice of finite-Matlis target for a Hamiltonian-equivariant principal-part transfer. Inside the category Curr^{BM} of compatible finite-window principal-part current systems with full $\mathfrak{h}$-coadjoint action, BMK homotopy data, and uniform $\mathfrak{m}$-adic moment bounds (Theorem 8.42), the compatible replacement target used here is the one-pair analytic pro-Matlis retract

$$\widehat{\mathcal{P}}_\Pi=\prod_{\substack{a,b\ge 0\\ a+b>0}}\mathbb{C}\rho_{a,b},\qquad \mathcal{P}_{\mathrm{an}}(B_1)=\tau\big(\mathcal{O}(B_1)^\vee_{\mathrm{red}}\big)\subset\widehat{\mathcal{P}}_\Pi,$$

with the normalized analytic contraction $H_\chi^\circ=(1-I_{\mathrm{an}}P_{\mathrm{an}})H_\chi(1-I_{\mathrm{an}}P_{\mathrm{an}})$ of Theorem B.9 satisfying

$$\bar\partial H_\chi^\circ+H_\chi^\circ\bar\partial=\mathrm{id}-I_{\mathrm{an}}P_{\mathrm{an}},$$

and the pro-analytic moment map $P_\Pi^{\mathrm{an}} = \tau P_{\mathrm{an}}$ with compatible finite projections

$$\pi_N P_\Pi^{\mathrm{an}} = p_N, \qquad \pi_{M,N} p_M = p_N \quad (M \geq N).$$

This gives the projection-compatible inverse-limit receiver for the fixed BMK moment system. Theorem 8.40 establishes the necessity of a pro-Matlis replacement in place of every finite-Matlis target.

THEOREM 8.42 *(Pro-Matlis inverse-limit receiver property).* Let Curr^{BM} be the category whose objects are $\mathbb{C}$-linear pairs (Q, Q^{an}) with $Q^{\mathrm{an}} \subset Q$, equipped with:

(a) compatible projections $\pi_N : Q \to \mathcal{P}_{\leq N}$ satisfying $\pi_N = \pi_{M,N} \circ \pi_M$ for $M \geq N$;
(b) a Hamiltonian $\mathfrak{h}$-action through full residue-coadjoint operators on the untruncated map $\mathfrak{h} \to \mathrm{End}(Q)$;
(c) a Dolbeault chain homotopy $H : Q \to Q[-1]$ with $\bar\partial H + H\bar\partial = \mathrm{id} - I \circ P$ for an analytic-submodule retraction $P : Q \to Q^{\mathrm{an}}$, $I : Q^{\mathrm{an}} \hookrightarrow Q$;
(d) a continuous analytic comparison $\pi^{\mathrm{an}} : Q^{\mathrm{an}} \to \mathcal{P}_{\mathrm{an}}(B_1)$, compatible with the restrictions of the finite-window projections to the analytic submodule;
(e) uniform $\mathfrak{m}$-adic moment bounds: for every $f \in \mathfrak{h}$ and every $\xi \in Q$ of $\mathfrak{m}$-adic order $\geq n$, $f \cdot \xi$ has $\mathfrak{m}$-adic order $\geq n - r(f)$ for an f-dependent constant $r(f)$.

Morphisms are $\mathbb{C}$-linear maps $(Q, Q^{\mathrm{an}}) \to (Q', Q'^{\mathrm{an}})$ commuting with all finite projections π_N, π'_N and with the analytic-submodule retractions and analytic comparisons. A morphism is $\mathfrak{h}$-equivariant only when the untruncated residue-coadjoint actions commute with it. Then the pro-Matlis pair $\big(\widehat{\mathcal{P}}_\Pi, \mathcal{P}_{\mathrm{an}}(B_1)\big)$ with the analytic moment map P_Π^{an} of Theorem B.9 receives, from every object (Q, Q^{an}), a unique morphism compatible with the fixed finite-window projections and analytic retractions:

$$(Q, Q^{\mathrm{an}}) \longrightarrow \big(\widehat{\mathcal{P}}_\Pi, \mathcal{P}_{\mathrm{an}}(B_1)\big).$$

The assertion is an inverse-limit receiver property for the fixed BMK system; it is not a terminality statement after the category is enlarged by spectator summands or by changing the analytic submodule.

Proof. Existence: condition (a) gives a compatible system $\pi_N : Q \to \mathcal{P}_{\leq N}$, defining a unique projection-compatible map $Q \to \varprojlim_N \mathcal{P}_{\leq N} = \widehat{\mathcal{P}}_\Pi$ by the universal property of the inverse limit. Condition (d) sends the analytic submodule to $\mathcal{P}_{\mathrm{an}}(B_1) = \tau(\mathcal{O}(B_1)^\vee_{\mathrm{red}})$ compatibly with the finite-window maps, and condition (e) ensures the lift is convergent in the $\mathfrak{m}$-adic topology of $\widehat{\mathcal{P}}_\Pi$. Uniqueness: any two such morphisms agree on every $\mathcal{P}_{\leq N}$-component by compatibility with π_N, hence on $\varprojlim_N \mathcal{P}_{\leq N}$. Chain-homotopy compatibility and the moment bounds are preserved by the inverse limit. Hamiltonian equivariance is a separate condition on the untruncated actions, not a consequence of the finite projections. This proves the projection-compatible receiver property. Theorem 8.40 excludes every finite-rank target (those have a strict finite-window restriction of $\mathfrak{h}$, violating (b)), and Corollary 8.41 verifies the pro-Matlis pair structure, the normalized Dolbeault contraction H_χ°, and the analytic moment map P_Π^{an} on $\big(\widehat{\mathcal{P}}_\Pi, \mathcal{P}_{\mathrm{an}}(B_1)\big)$. □

Proof of Corollary 8.41. The first sentence is Theorem 8.40: the finite-Matlis quotient has the Hamiltonian-action obstruction. The positive constructions of the pro-Matlis pair $(\widehat{\mathcal{P}}_\Pi, \mathcal{P}_{\mathrm{an}}(B_1))$, the normalized contraction H_χ°, and the moment map P_Π^{an} are Theorem B.9.

Fixed-projection characterization. An object in the stated Curr^{BM} slice with a $\mathbb{C}$-linear $\mathfrak{h}$-module structure and a Dolbeault chain homotopy compatible with p_N and i_N at every N carries the following data:

(a) accept the full Hamiltonian coadjoint action of $\mathfrak{h}$,
(b) admit the Dolbeault homotopy obtained from the BMK kernel after the diagonal orientation and collar quotient,
(c) project compatibly to the finite Matlis windows used by the arity-two coefficient comparison $\ell_2^{\mathrm{BM},N}(x,y) = p_N[ix,iy]$.

Condition (a) excludes every finite-rank quotient by Theorem 8.40; (b) selects the cohomology of the BMK contraction as the moment-restriction kernel, fixing the analytic submodule $\mathcal{P}_{\mathrm{an}}(B_1) = \tau(\mathcal{O}(B_1)^\vee_{\mathrm{red}})$ on which the kernel acts; (c) forces the fixed inverse-limit receiver for the finite projections to contain every $\rho_{a,b}$ and to map to $\mathcal{P}_{\le N}$, hence to be $\widehat{\mathcal{P}}_\Pi = \varprojlim_N \mathcal{P}_{\le N}$ in this slice. The pair $(\widehat{\mathcal{P}}_\Pi, \mathcal{P}_{\mathrm{an}}(B_1))$ is therefore the target used here with these fixed projections and BMK moment data. The strict all-window factorization-current transfer requires the additional vanishing of $\mathrm{Ob}^\Pi_{\mathrm{BM}}$ recorded in Theorem B.9, whose first three nonzero entries (direct-sum moment cokernel, missing compact-current section of the pro-product tower, divergent formal Hamiltonian action on the pro-product) are independent current-topology obstructions that remain alongside the analytic pro-Matlis retract. □

Remark 8.43 *(Physical interpretation: high transverse holomorphic modes leak back into low brane momentum modes).* The boundary-mode identity (29) of Theorem 8.40 carries a direct physical reading. In the Hamiltonian/Matlis dictionary the index $\rho_{a,b}$ labels a holomorphic transverse mode of bidegree (a,b) along $\mathbb{C}^2_{\mathrm{hol}}$; the brane line $L \subset \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ carries the low brane-momentum modes $\rho_{0,b}$ and their finite Matlis truncations. The class

$$z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\,\rho_{0,1}$$

asserts that under Hamiltonian flow a high transverse holomorphic mode at degree $(N+1,0)$ – formally above the cutoff window – transports back into a low brane-momentum mode at degree $(0,1)$ – inside the cutoff window. Physically: high transverse holomorphic modes leak back into low brane momentum modes.

This leak is the obstruction for every strict finite-Matlis cutoff: truncating high transverse modes severs the Hamiltonian return channel that would have lifted them to brane momenta, but the return channel is already a finite-window operation by (29), so the image remains. The pro-Matlis retract of Corollary 8.41 is the fixed-projection module used here and closed under this leak: the full Taylor tower retains every high transverse mode and lets the Hamiltonian flow act consistently, while the analytic submodule $\mathcal{P}_{\mathrm{an}}(B_1)$ gives the moment-restriction kernel required by the BMK contraction. The arity-two projection $\ell_2^{\mathrm{BM},N}(x,y) = p_N[ix,iy]$ of Theorem B.9 is then a finite-window readout of this consistent flow; the finite-rank operation appears after this readout.

The formal-Darboux trace-sector theorem at one Dirac brane vacuum, the six native obstruction classes, and the pro-Matlis coefficient target determine the local obstruction complex. The trace-sector datum $A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}(\mathfrak{gl}_N, \mathrm{Kosz}([\phi_1,\phi_2])), J(f) = \overline{\mathrm{Tr}\, f(\phi_1,\phi_2)}$, with formal completion understood, $c_f \mapsto \theta_f, u_f \mapsto J(f)$, is evaluated at the brane vacua of the central families. The boundary open–closed type theorem separates the six typed objects, names the open–closed comparison morphism, and assigns each obstruction class to its native source complex.

9 Boundary open–closed type theorem

The formal-Darboux stalk theorem of §5.3 gives one fibre of the mixed holomorphic–topological open–closed correspondence. The Hamiltonian BF bulk, evaluated at a Dirac brane vacuum, acts on the boundary Koszul algebra through $J(f) = \operatorname{Tr} f(\phi_1, \phi_2)$; the resulting Chevalley–Eilenberg/polyvector chart is the trace-sector coordinate of the local derived-centre comparison. The global comparison of Theorem 5.13 is the descent problem for these fibres over the Costello–Gwilliam brane-link boundary ∂X.

The descent problem separates six typed objects. The E_1-bar coalgebra of the brane algebra, its Koszul dual algebra, the derived mixed-HT centre, the physical bulk factorization algebra, the protected scalar trace with Capelli anomaly, and the filtered boundary operator algebra live in different categories and are related by definite morphisms. The functor Φ^{HT}, the open–closed comparison morphism OCA, the mixed-HT Deligne–Swiss-cheese action, scalar non-faithfulness, and the topologisation equations $T_i = [Q, G_i]$ are the structure maps between these objects. The type table records the category in which each object and map is defined.

9.1 *The six objects and their ambient categories*

Definition 9.1 *(Six boundary open–closed objects).* Fix the formal-Darboux local model $X = \mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{hol}}$, brane line $L = \mathbb{R}_t \times \{b\}$, reduced Hamiltonian Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$, residue dual

$$P = \mathfrak{h}^\vee_{\mathrm{cont}} = \mathrm{Ann}_{\mathrm{Res}}(\mathbb{C} \cdot 1) \subset H^2_{(z_1,z_2)}(\mathbb{C}[[z_1, z_2]])\, dz_1 dz_2 = \bigoplus_{a+b>0} \mathbb{C}\, \rho_{a,b},$$

shifted-cotangent BF Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes P[1]$, Capelli-projective extension $\widehat{\mathfrak{g}}_{\mathrm{cent}}$, boundary algebra $A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}(\mathfrak{gl}_N, \mathrm{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1]))$ with $Q\psi = [\phi_1, \phi_2]$, and admissible category $\mathcal{B}^{\mathrm{adm}}$ of Appendix G. The six boundary open–closed objects are:

- (C1) $\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N}) = T^c(s^{-1}\bar{A}^{\mathrm{cl}}_{\partial,N})$, the conilpotent E_1^{top}-bar coalgebra of the augmented brane algebra $\bar{A}^{\mathrm{cl}}_{\partial,N}$: a coassociative conilpotent dg coalgebra in $\mathcal{B}^{\mathrm{adm}}$ with deconcatenation coproduct and differential $d_{\bar{B}} = d_{\mathrm{int}} + d_m$ of (32);
- (C2) $(A^{\mathrm{cl}}_{\partial,N})^!$, the Verdier/Koszul dual algebra of $A^{\mathrm{cl}}_{\partial,N}$ on the Koszul locus, in the sense of Theorem 5.5;
- (C3) $Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C^{\mathrm{op}}_\partial)$, the derived mixed-HT E_2-centre of the primitive open boundary factorization category C^{op}_∂ on the Costello–Gwilliam brane-link boundary; the bulk action or OCA map is extra structure;
- (C4) $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$, the physical bulk mixed-HT Hamiltonian BF factorisation algebra of Definition 1.21;
- (C5) the protected scalar trace together with the projective anomaly class

  $$(\chi, \kappa_{\mathrm{Cap}}), \qquad \chi = \mathrm{sTr}\,(-) \colon \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \to \mathbb{Z}((q, \hbar)), \quad \kappa_{\mathrm{Cap}} = \hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})[[\hbar]].$$

  The Lie class controls the projective anomaly of the trace-sector action; the scalar trace is its protected character.
- (C6) the filtered boundary operator algebra $\mathrm{Op}^{\mathrm{HT}}_{\partial X}$, a filtered $\mathrm{SC}^{\mathrm{HT}}$-object on ∂X (§11.3).

Proposition 9.2 *(Boundary open–closed type table).* The six objects occupy six distinct cells of the type table below, with pairwise distinct roles; (C_1) is a coalgebra, (C_2) is a Koszul-dual algebra, (C_3) is a centre carrying an E_2-Gerstenhaber bracket distinct from any (C_i), (C_4) is a factorisation algebra realised through OCA, (C_5) is a graded scalar invariant valued in $\mathbb{Z}((q,\hbar))$, and (C_6) is a separately constructed filtered $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-object on ∂X:

Object	Type	Role	Comparison
$\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N})$	conilpotent E_1^{top}-coalgebra	classifies twisting morphisms	$\Omega^c\bar{B} \simeq A^{\mathrm{cl}}_{\partial,N}$
$(A^{\mathrm{cl}}_{\partial,N})^!$	Verdier/Koszul dual algebra	controls line/open dual modules	Koszul paired with $\bar{B}$
$Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C_\partial^{\mathrm{op}})$	derived E_2^{HT}-centre object	closed centre before physical realisation	realised by OCA
$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$	mixed-HT BF factorisation algebra	physical bulk if realised	$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \xrightarrow{\mathrm{OCA}} Z^{\mathrm{der}}$
$(\chi,\kappa_{\mathrm{Cap}})$	protected index $\times$ Lie 2-class	scalar shadow and projective anomaly	χ and κ_{Cap} lie in distinct target
$\mathrm{Op}^{\mathrm{HT}}_{\partial X}$	filtered $\mathrm{SC}^{\mathrm{HT}}$-object	boundary operator object	$\mathrm{sTr}(\mathrm{Op}^{\mathrm{HT}}_{\partial X}) = \chi$

Each equality between two objects factors through the comparison column. Bar coalgebras, derived centres, physical factorisation algebras, scalar traces, and filtered $\mathrm{SC}^{\mathrm{HT}}$ objects occupy different ambient categories.

Proof. The six cells are pairwise distinct. (C_1) is a coalgebra by (32) so cannot coincide with the algebra (C_2), the centre (C_3), or the bulk (C_4); (C_2) is the Verdier/Koszul dual algebra (different bracket from the Gerstenhaber bracket of (C_3) by Theorem 5.5); (C_3) is the derived E_2^{HT}-centre object whose underlying formal-stalk endomorphism object carries the E_2-Gerstenhaber bracket of the admissible Hochschild model (Theorem 5.6), and it is distinct from (C_4)'s factorisation product (Definition 1.21: (C_4) acts on the boundary through OCA, (C_3) is the abstract centre of the boundary category); (C_5) is a graded scalar invariant in $\mathbb{Z}((q,\hbar))$ and is non-faithful by Lemma 9.10; (C_6) is a chain-level filtered $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-object on ∂X determined by an independent filtered construction (Corollary 9.11), and whose scalar χ is a non-faithful image (Lemma 9.10). Distinct roles are read off the column entries, and the comparison column records the allowed maps between the six cells. □

9.2 *The boundary open–closed functor* Φ^{HT}

Definition 9.3 *(Boundary open–closed datum).* A *boundary open–closed datum* on the mixed-HT brane line $L \subset X$ is a tuple

$$(\bar{B},\ \Theta,\ r(z),\ Z^{\mathrm{der}},\ \nabla),$$

where $\bar{B}$ is a conilpotent dg coalgebra in $\mathcal{B}^{\mathrm{adm}}$; $\Theta \in \mathrm{Hom}^1_{\mathrm{prim}}(\bar{B},\bar{B})$ is a primitive degree-one element satisfying the Maurer–Cartan equation

$$d_{\bar{B}}\Theta + \tfrac{1}{2}[\Theta,\Theta] = 0;$$

$r(z) \in H^2_\Delta\big(\mathbb{C}[[z,w]]\big)\, d\zeta_1 d\zeta_2 \otimes \bar{B}^{\otimes 2}$ is the binary collision residue of Θ at the formal diagonal in $\mathbb{C}^2 \times \mathbb{C}^2$; Z^{der} is a derived E_2^{HT}-centre object whose formal-stalk endomorphism object carries a chiral-brace dg algebra structure; and ∇ is a flat Gauss–Manin connection on bar fibres along the formal Darboux moduli of b, compatible with Θ and inducing the Gauss–Manin connection on Z^{der}.

Theorem 9.4 *(Typed boundary open–closed realisation Φ^{HT} at N Dirac branes).* In $\mathcal{B}^{\mathrm{adm}}$, suppose the bar–cobar admissibility, the diagonal-kernel compatibility, the chiral brace operations, the centrality hierarchy, and the flat connection datum of Definition 9.3 are fixed. Then the assignment (31) extends from the E_1^{top}-brane algebra $A^{\mathrm{cl}}_{\partial,N}$ on the worldline $\mathbb{R}_t$ (with Koszul differential $Q\psi = [\phi_1,\phi_2]$) to the corresponding boundary open–closed datum.

There is a functor

$$\Phi^{\mathrm{HT}}\colon \mathbf{E_1}^{\mathrm{top}}\text{-}\mathbf{Brane}^{\mathrm{HT}}_{\mathbb{R}_t,\mathrm{Kosz}} \longrightarrow \mathbf{BdOC}^{\mathrm{HT}}_X$$

from the category of E_1^{top}-algebras on $\mathbb{R}_t$ with Koszul differential and HT decorations to the category of boundary open–closed data of Definition 9.3, sending $A^{\mathrm{cl}}_{\partial,N}$ to the boundary open–closed datum

$$\Phi^{\mathrm{HT}}\big(A^{\mathrm{cl}}_{\partial,N}\big) = \Big(\bar{B}^{E_1}\big(A^{\mathrm{cl}}_{\partial,N}\big),\ \Theta_{A^{\mathrm{cl}}_{\partial,N}},\ K^{(2)}_{\Delta},\ Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}\big(C^{\mathrm{op}}_{\partial}\big),\ \nabla^{\mathrm{HT}}\Big), \tag{31}$$

where

(i) $\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N}) = T^{\mathrm{c}}(s^{-1}\bar{A}^{\mathrm{cl}}_{\partial,N})$ is the conilpotent E_1-bar coalgebra with deconcatenation coproduct and differential

$$d_{\bar{B}} = d_{\mathrm{int}} + d_m \tag{32}$$

where d_{int} is the internal differential of $A^{\mathrm{cl}}_{\partial,N}$ (the Koszul differential $Q\psi = [\phi_1,\phi_2]$ plus the Chevalley–Eilenberg differential for $\mathfrak{gl}_N$) and d_m is the standard bar coderivation induced by the E_1-multiplication of $A^{\mathrm{cl}}_{\partial,N}$. The diagonal local-cohomology kernel $K^{(2)}_{\Delta} = d\zeta_1 d\zeta_2/(\zeta_1\zeta_2)$ of Proposition 5.7 is a chiral-kernel decoration of the bar object; it becomes a coderivation only after a coefficient action map $K^{(2)}_{\Delta} \otimes (A^{\mathrm{cl}}_{\partial,N})^{\otimes 2} \to A^{\mathrm{cl}}_{\partial,N}$ is fixed;

(ii) $\Theta_{A^{\mathrm{cl}}_{\partial,N}} \in \mathrm{Hom}^1_{\mathrm{prim}}\big(\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N}), \bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N})\big)$ is the primitive Maurer–Cartan element representing the E_1-product on $A^{\mathrm{cl}}_{\partial,N}$ inside its bar:

$$\Theta_{A^{\mathrm{cl}}_{\partial,N}} = \sum_{n\geq 1} m_n^{A^{\mathrm{cl}}_{\partial,N}}\colon \bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N})^{\otimes n} \to A^{\mathrm{cl}}_{\partial,N},$$

with m_2 the multiplication, m_3 the associator homotopy (zero in the strict case), and higher m_n the A_∞ corrections coming from boundary OPE contractions; Θ satisfies

$$d_{\bar{B}}\Theta_{A^{\mathrm{cl}}_{\partial,N}} + \tfrac{1}{2}[\Theta_{A^{\mathrm{cl}}_{\partial,N}}, \Theta_{A^{\mathrm{cl}}_{\partial,N}}] = 0; \tag{33}$$

(iii) $K^{(2)}_{\Delta} \in H^2_{\Delta}(\mathbb{C}[[z_1,z_2,w_1,w_2]])\, d\zeta_1 d\zeta_2$ is the scalar binary collision kernel. It is not a residue of the ordinary E_1-bar Maurer–Cartan element $\Theta_{A^{\mathrm{cl}}_{\partial,N}}$. After a coefficient action map $K^{(2)}_{\Delta} \otimes (A^{\mathrm{cl}}_{\partial,N})^{\otimes 2} \to A^{\mathrm{cl}}_{\partial,N}$ has been fixed, its line/Koszul-dual restriction gives the Arnold–Jacobi degree-three relation

$$\eta_{12}\wedge\eta_{23} + \eta_{23}\wedge\eta_{31} + \eta_{31}\wedge\eta_{12} = 0, \qquad \eta_{ij} = K^{(2)}_{\Delta}(z_i,z_j), \tag{34}$$

as the scalar kernel-associativity constraint. A classical Yang–Baxter equation is a structure on a Lie-coefficient r-matrix; the scalar diagonal-kernel datum contains no such coefficient;

(iv) $Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C^{\mathrm{op}}_{\partial})$ denotes the derived E_2^{HT}-centre object. On the formal-Darboux stalk it carries only the trace-sector generator action into the admissible Hochschild target

$$\begin{gathered}\Theta_{\mathrm{tr}}(c_f) = \theta_f, \qquad \Theta_{\mathrm{tr}}(u_f) = M_{J(f)},\\ \Theta_{\mathrm{tr}}\colon \langle c_f, u_f : f\in\bar{A}\rangle_{\mathrm{CE/PV}} \dashrightarrow \mathrm{RHom}_{A^{\mathrm{cl}}_{\partial,N}\otimes(A^{\mathrm{cl}}_{\partial,N})^{\mathrm{op}}}\big(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}\big) \ \simeq\ C^{\bullet}_{\mathrm{Hoch,adm}}\big(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}\big),\end{gathered} \tag{35}$$

A full centre identification requires the admissible centre-lift, equivariance homotopies, chiral E_2 operations, and descent data of Theorem 5.13; and

(v) ∇^{HT} is the flat Gauss–Manin connection on bar fibres along the formal Darboux moduli of b (parametrised by canonical transformations of $\widehat{\mathbb{C}^2}$), compatible with $\Theta_{A^{\mathrm{cl}}_{\partial,N}}$ and inducing the Gauss–Manin connection on $Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}_\partial^{\mathrm{op}})$ through (35).

The projection to $\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N})$ is the bar construction; the projection to $(\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N}), \Theta)$ is the Maurer–Cartan twisting datum; the $K_\Delta^{(2)}$-component is the external binary singular-product kernel; the projection to $Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}_\partial^{\mathrm{op}})$ is the centre model. None of these projections produces the physical bulk $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$; that identification is Theorem 9.6 below.

Proof. The construction of (31) uses the HT decorations and the CE/PV trace-sector map of Theorem 5.6.

(i) Cofree conilpotent bar coalgebra. $\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N})$ is the cofree conilpotent E_1-coalgebra on $s^{-1}\bar{A}^{\mathrm{cl}}_{\partial,N}$ by the standard bar/cobar adjunction in $\mathcal{B}^{\mathrm{adm}}$ (Lemma 5.166, Theorem 5.31). The internal differential d_{int} is the Koszul-CE differential of $A^{\mathrm{cl}}_{\partial,N}$. The term d_m is the standard bar coderivation determined by the E_1-multiplication. The diagonal kernel $K_\Delta^{(2)}$ enters as a chiral collision coefficient on the closed-colour action; an additional action map is required before it defines a coderivation of the bar coalgebra.

(ii) $d_{\bar{B}}^2 = 0$ and kernel associativity. The identity $d_{\bar{B}}^2 = 0$ is the usual bar identity: it is equivalent to $d_{\mathrm{int}}^2 = 0$, the Leibniz rule for d_{int} against m_2, and associativity of the E_1-product, with the higher A_∞ terms included in Θ when the boundary product is not strict. The scalar chiral-kernel decoration separately satisfies the iterated three-point relation on $\Delta_3 \subset (\mathbb{C}^2)^3$:

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0, \qquad \eta_{ij} = K_\Delta^{(2)}(z_i, z_j),$$

the two-complex-dimensional Arnold relation on three points, proved as the residue-form identity at Δ_3 by Proposition 5.7. The cross-term with the Koszul-CE differential is not a formal consequence of this scalar identity; it belongs to the chosen chiral-kernel action map. Under that action-map hypothesis, the Koszul homotopy for adjacent transpositions in traces (Lemma 3.30) gives the required compatibility. Hence $d_{\bar{B}}^2 = 0$ for the ordinary bar coalgebra, and the OPE residue data define a Maurer–Cartan element $\Theta_{A^{\mathrm{cl}}_{\partial,N}}$ of $\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N})$.

(iii) Binary collision residue equals $K_\Delta^{(2)}$. The binary collision residue of Θ at $\Delta \subset \mathbb{C}^2 \times \mathbb{C}^2$ is the arity-two OPE coefficient $K_\Delta^{(2)} = d\zeta_1 d\zeta_2/(\zeta_1\zeta_2)$ of (19). The degree-three component of (33), evaluated on three free generators x_1, x_2, x_3 of $\bar{A}^{\mathrm{cl}}_{\partial,N}$, produces the Arnold–Jacobi relation (34). This is the scalar operadic associativity constraint for the diagonal kernel. It becomes a classical Yang–Baxter equation only after an independent choice of a Lie-coefficient r-matrix.

(iv) Derived centre via bar resolution. The underlying bimodule endomorphism object of the derived E_2^{HT}-centre on the formal-Darboux stalk identifies with the chiral Hochschild cochain brace dg algebra by the bar resolution $\Omega^c\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N}) \to A^{\mathrm{cl}}_{\partial,N}$, the admissible Tate bar–cobar theorem (Theorem 5.31), the universal filtered cobar recognition criterion (Theorem 5.24), and the E_2-Gerstenhaber bracket match of (17). Equation (35) is the local form of the trace-sector Hochschild action $\mathrm{Obs}^{\mathrm{cl}}_b(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}) \to C^\bullet_{\mathrm{Hoch,adm}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ of Theorem 5.6; an equivalence with the whole derived centre requires the admissible centre-lift and descent data.

(v) Gauss–Manin compatibility. The flat Gauss–Manin connection ∇^{HT} is the Tate-continuous extension of the Kashiwara–Kawai connection on bar fibres along the formal-Darboux moduli of $b \in \widehat{\mathbb{C}^2}$. Flatness follows from the integrability of canonical transformations of $\widehat{\mathbb{C}^2}$ on $\mathrm{Sym}\,\mathfrak{h} \subset A^{\mathrm{cl}}_{\partial,N}$

(Proposition 3.22) and the GL_N-equivariance of the Koszul-CE differential. Compatibility with Θ is the statement that $\nabla^{\mathrm{HT}}\Theta \in \mathrm{Hom}^1_{\mathrm{prim}}(\bar{B}, \bar{B})$ satisfies the linearised Maurer–Cartan equation, which is the Bianchi identity for the bulk current $\mathsf{J}_x(z)$ of (19).

(Functoriality.) A morphism in $\mathbf{E_1}^{\mathrm{top}}\text{-}\mathbf{Brane}^{\mathrm{HT}}_{\mathbb{R}_t,\mathrm{Kosz}}$ is an E_1^{top}-morphism $f\colon A^{\mathrm{cl}}_{\partial,N} \to A^{\mathrm{cl}}_{\partial,N'}$ together with homotopies identifying the pulled-back diagonal residue, centre lift, centrality hierarchy, and Gauss–Manin connection. The bar functor sends f to a coalgebra morphism $\bar{B}(f)\colon \bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N}) \to \bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N'})$, and the chosen homotopies make it a morphism of boundary open–closed data. Thus Φ^{HT} is a functor on the decorated category. The decoration consists of the chiral E_2, centrality, descent, and connection data attached to the underlying E_1-morphism. □

9.3 The open–closed comparison morphism

Definition 9.5 *(Open–closed comparison datum).* At N Dirac branes a *pre-open–closed comparison datum* is the bulk-to-boundary morphism

$$\mathrm{preOCA}_N = \widetilde{\rho}_{\mathrm{bulk}\to\partial}\colon \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \longrightarrow \underline{\mathrm{End}}_{\mathrm{SC}^{\mathrm{HT}}_{(1,2)}}(\mathcal{C}^{\mathrm{op}}_\partial)$$

in $\mathcal{B}^{\mathrm{adm}}$, together with its formal-Darboux trace-sector fibre. If a chiral E_2 centrality morphism

$$\iota_{\mathrm{cent}}\colon \underline{\mathrm{End}}_{\mathrm{SC}^{\mathrm{HT}}_{(1,2)}}(\mathcal{C}^{\mathrm{op}}_\partial) \longrightarrow Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}^{\mathrm{op}}_\partial)$$

and the required $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-centrality homotopies are chosen, the datum is promoted to the *open–closed comparison morphism*

$$\mathrm{OCA}_N\colon \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \longrightarrow Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}^{\mathrm{op}}_\partial), \qquad \mathrm{OCA}_N = \iota_{\mathrm{cent}} \circ \mathrm{preOCA}_N.$$

Here

(i) $\widetilde{\rho}_{\mathrm{bulk}\to\partial}$ is the Costello–Gwilliam bulk–boundary factorisation pairing of [15, Vol. II §4–5] adjointed to the endomorphism algebra of $\mathcal{C}^{\mathrm{op}}_\partial$ as an $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-module: for nested $U \supset V \subset \partial X$, the pairing $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}(U) \otimes \mathcal{C}^{\mathrm{op}}_\partial(V) \to \mathcal{C}^{\mathrm{op}}_\partial(V)$ is the closed-acting-on-open operation of Theorem 9.9 item (iii), and $\widetilde{\rho}$ is its currying to the endomorphism algebra;

(ii) ι_{cent} is the centrality morphism into the derived centre object. Its universal property is part of the chiral E_2 centrality datum; the formal-Darboux trace theorem gives its trace-sector coordinate.

With these choices the composition is a morphism of $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-algebras on ∂X. When this morphism is a quasi-isomorphism in $\mathcal{B}^{\mathrm{adm}}$, it is the open–closed correspondence quasi-isomorphism.

THEOREM 9.6 *(*OCA *at the formal-Darboux stalk).* This is Theorem 5.6 viewed as the formal-Darboux trace-sector fibre of an open–closed comparison datum. In $\mathcal{B}^{\mathrm{adm}}$ the formal stalk assignment restricts on closed CE generators to

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \dashrightarrow C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}), \qquad c_f \mapsto \theta_f, \quad u_f \mapsto J(f).$$

After stable scalar reduction and admissible completion this restriction is the CE/PV isomorphism. At fixed finite N it is not the whole derived-centre equivalence. Its formal-Darboux coordinate is $J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2)$ for polynomial f, extended to formal f only after completion, with Capelli projective anomaly class $\hbar N[\bar{c}]$.

Proof. The pre-open–closed datum of Definition 9.5, evaluated at the formal-Darboux stalk b, contains the bulk-boundary factorisation pairing on the contractible polydisk neighbourhood of b. On the trace sector Theorem 5.6 computes this pairing by the CE/PV map $c_f \mapsto \theta_f$, $u_f \mapsto J(f)$. The binary diagonal kernel and the Capelli class are local data. The all-arity chiral E_2 extension, descent, and all-order QME are not produced by this local calculation. □

9.4 The mixed-HT chiral Deligne–Swiss-cheese action

The normal-defect geometry is encoded by a two-coloured mixed-HT Swiss-cheese operadic model. The open colour is E_1^{top} along the brane line $\mathbb{R}_t \subset L$. The normal closed colour is the mixed holomorphic-topological factorisation operad on $\mathbb{R}_s \times \mathbb{C}^2$. The full bulk colour on $\mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}^2$ is the tensor of this normal operad with the tangential E_1^{top} factor, equipped with the mixed coherence maps.

Definition 9.7 (Normal mixed-HT Swiss-cheese operad). The *normal mixed-HT Swiss-cheese operad of bidegree* $(1, 2)$, denoted $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$, is the two-coloured coloured dg operad with:

- closed colour c: the $\mathrm{HT}^{(1,2)}$ framed little disks operad on $\mathbb{R}^1_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$, equivalently $E_1^{\mathrm{top}} \widehat{\otimes} E_2^{\mathrm{hol}}$, modelled on the framed Fulton–MacPherson compactification of configurations of disks in $\mathbb{R}^1_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$;
- open colour o: the E_1^{top} little intervals operad on the brane line $\mathbb{R}_t$;
- mixed operations recording configurations of normal closed disks in $\mathbb{R}_{s\geq 0} \times \mathbb{C}^2_{\mathrm{hol}}$, parametrised along $\mathbb{R}_t$, approaching the boundary $\mathbb{R}_t \times \{s = 0\} \times \mathbb{C}^2_{\mathrm{hol}}$, with marked open intervals on the brane line $\mathbb{R}_t \times \{s = z_1 = z_2 = 0\}$, with composition by nesting closed disks and substitution of open intervals.

$\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$ is generated by binary closed operations carrying the diagonal local-cohomology kernel $K^{(2)}_{\Delta}$ on $\mathbb{C}^2$, binary open operations carrying the E_1-product on $\mathbb{R}_t$, and binary mixed operations $\mu^{\mathrm{mix}}_{\mathsf{c},\mathsf{o}}$ realising bulk-to-boundary insertion. An equivalence with E_3^{top} is precisely the topologisation datum of Theorem 9.13.

Remark 9.8 (Dunn additivity and mixed holomorphicity). Dunn additivity [41, Theorem 5.1.2.2] identifies $E_p^{\mathrm{top}} \otimes E_q^{\mathrm{top}} \simeq E_{p+q}^{\mathrm{top}}$ in topological algebras. The mixed-HT operad $E_1^{\mathrm{top}} \otimes E_2^{\mathrm{hol}}$ is the tensor of a topological E_1 factor with a holomorphic E_2 factor, and the passage from holomorphic E_2 to topological E_2 is given by the two topologisations $T_i = [Q, G_i]$. Hence $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$ is a two-coloured mixed-HT operadic model, and its E_3^{top} form is the conclusion of Theorem 9.13.

Theorem 9.9 *(Mixed-HT chiral Deligne–Swiss-cheese action).* Assume that the chiral brace operations and the equivariance homotopy hierarchy $\{H^{\mathrm{prod}}_{f,-}, H^{P_0}_{f,-}, H^3_{f,-,-}, \ldots\}$ exist in $\mathcal{B}^{\mathrm{adm}}$, compatibly with the binary trace-sector operation of Theorem 5.6. Then the assignment (36) below endows the open–closed pair $\big(Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C_\partial^{\mathrm{op}}),\ C_\partial^{\mathrm{op}}\big)$ of Definition 9.1 with the structure of an algebra over the mixed swiss-cheese operad $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$ at the formal-Darboux stalk; the global full-bulk form also requires the tangential E_1^{top} coherence. It is governed by the chiral E_2-Deligne obstruction $\mathrm{Ob}^{\mathrm{ch}\,E_2}_{\mathrm{Deligne}}$ and on the comparison-cone contraction. The route to the global statement is Theorem 5.13.

The pair

$$\big(Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C_\partial^{\mathrm{op}}),\, C_\partial^{\mathrm{op}}\big) \tag{36}$$

is, at the formal-Darboux stalk, an algebra over $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$ in $\mathcal{B}^{\mathrm{adm}}$, with:

(i) closed colour acting on $Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C_\partial^{\mathrm{op}})$ by the chiral brace operations on $C^\bullet_{\mathrm{Hoch,adm}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ of (35), with binary singular product (19);

(ii) open colour acting on C_∂^{op} through the E_1^{top}-product of $A^{\mathrm{cl}}_{\partial,N}$;

(iii) trace-sector mixed binary operation $\mu^{\mathrm{mix}}_{\mathrm{c,o}}(c \otimes a) = c \cdot a$ realising the action of the trace-sector closed generators on the open sector; the higher mixed operations, when a centre lift exists, are the equivariance homotopy hierarchy $\{H^{\mathrm{prod}}_{f,-}, H^{P_0}_{f,-}, H^3_{f,-,-}, \dots\}$ of (17).

The trace-sector action is bulk–boundary compatible: the Hochschild component sends c_f, u_f to $\theta_f, J(f) \cdot (-)$, with $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ for polynomial f, and by completed formal evaluation for $f \in \mathfrak{h}$. The scalar-contact component sends c_K to γ_1, while the projective central character sends K to $\mathrm{Tr}\,\mathbf{1} = N$. A full closed action of $Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}$ requires the centrality hierarchy.

If, in addition, an open–closed correspondence quasi-isomorphism $\mathrm{OCA}_N : \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \xrightarrow{\sim} Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C_\partial^{\mathrm{op}})$ exists (Theorem 5.13), with null-homotopies for $\mathrm{Ob}^{\mathrm{global},5}$ and a contracting homotopy for $\mathrm{Cone}(\mathrm{OCA}_N)$, then $(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, C_\partial^{\mathrm{op}})$ is the physical $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-bulk–boundary system of the mixed-HT theory on X at N Dirac branes.

Proof. Under the chiral brace datum, the closed-colour action is the E_2-Gerstenhaber bracket on $C^\bullet_{\mathrm{Hoch,adm}}$ of Theorem 5.6, refined to the chiral brace dg algebra structure through the bar resolution (35). The binary singular product is the diagonal local-cohomology kernel $K^{(2)}_\Delta$ of (19), established as a coloured dg operad action by Proposition 5.7. The open-colour action is the E_1^{top}-product on $A^{\mathrm{cl}}_{\partial,N}$ along the brane line.

The mixed binary operation is the standard closed-acting-on-open bulk–boundary factorisation pairing $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}(U) \otimes C_\partial^{\mathrm{op}}(V) \to C_\partial^{\mathrm{op}}(V)$ of Costello–Gwilliam [15, Vol. II §4–5]; restricted to the formal-Darboux stalk and post-composed with $\mathrm{OCA}_N|_b$ of Theorem 9.6, it agrees with the centrality arrow of the derived centre object acting on $A^{\mathrm{cl}}_{\partial,N}$ in the trace sector.

The higher mixed operations are the required equivariance homotopy hierarchy $\{H^{\mathrm{prod}}_{f,-}, H^{P_0}_{f,-}, H^3_{f,-,-}, \dots\}$ of (17). They are part of the admissible chiral Deligne–Tamarkin one-up $A^{\mathrm{cl}}_{\partial,N} \rightsquigarrow Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}$ in $\mathcal{B}^{\mathrm{adm}}$, not a consequence of the binary trace calculation alone. When this datum exists, each higher coherence in $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$ is represented by a chain-level homotopy in $C^\bullet_{\mathrm{Hoch,adm}}$, with the K-cocycle giving the projective correction. Operadic associativity is then checked on generators through the iterated three-point Arnold relation $\eta_{12}\eta_{23} + \eta_{23}\eta_{31} + \eta_{31}\eta_{12} = 0$ and the Jacobi identity for the Gerstenhaber bracket.

The compatibility with the trace pairing is (17): $c_f \mapsto \theta_f$ is the closed-on-open action, $u_f \mapsto J(f)$ is the open evaluation, $c_K \mapsto \gamma_1$ is the scalar-contact CE class, and the central character evaluates K by $K \mapsto \mathrm{Tr}\,\mathbf{1} = N$ only in the projective current algebra.

The global theorem under hypotheses follows from Theorem 5.13 and its typed refinement Theorem 9.16: once the components of $\mathrm{Ob}^{\mathrm{global},5}$ are represented and killed by compatible null-homotopies, OCA_N extends from the stalk to a quasi-isomorphism on ∂X; at that point $(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, C_\partial^{\mathrm{op}})$ becomes a global $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-algebra realising the physical bulk–boundary system. □

9.5 Scalar non-faithfulness

The graded Euler character is the protected scalar trace $\chi = \mathrm{sTr} \colon \mathrm{FiltCh}_{\mathrm{SC}^{\mathrm{HT}}_{(1,2)}} \to \mathbb{Z}((q,\hbar))$ defined on chain-level filtered $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-algebras with a compatible homological grading and a filtration by charge / weight / height. It is a shadow, not a chain-level invariant; scalar non-faithfulness is the elementary obstruction to recovering a chain-level $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-object from its scalar trace.

Lemma 9.10 *(Scalar non-faithfulness).* The graded Euler supertrace

$$\chi = \mathrm{sTr}(-)\colon \mathrm{FiltCh}_{\mathrm{SC}^{\mathrm{HT}}_{(1,2)}} \longrightarrow \mathbb{Z}((q,\hbar))$$

has nontrivial fibres: equality of scalar traces is compatible with inequivalent quasi-isomorphism types, with inequivalent $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-operations, and with distinct Hall-style products, Drinfeld-double pairings, current-envelope locality structures, derived-centre trace maps, and open–closed comparison morphisms. Consequently every scalar-to-chain lift of automorphic data includes a chain-level $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-object.

Proof. Let $C = \mathbb{C}$ be concentrated in degree 0 with the trivial filtered $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-structure. Let

$$V = \mathbb{C}e_0 \oplus \mathbb{C}e_1, \qquad |e_0| = 0, \quad |e_1| = 1, \quad d_V = 0,$$

and set $C' = \mathbb{C} \oplus V$ with $V^2 = 0$, unit $1 \in \mathbb{C}$, and all mixed operations on V square-zero. Then

$$\chi(C') = 1 + \chi(V) = 1 + 1 - 1 = \chi(C),$$

while $H^\bullet(C') = \mathbb{C} \oplus \mathbb{C}e_0 \oplus \mathbb{C}e_1$ is not isomorphic to $H^\bullet(C) = \mathbb{C}$. Thus the scalar trace takes the same value on two distinct quasi-isomorphism types. Keeping the same graded complex and changing square-zero products, Hall pairings, current envelopes, or open–closed operations on V gives further points of the same scalar fibre with different chain-level structures. □

Corollary 9.11 *(Mandatory chain-level supplements).* Every lift from automorphic scalar to physical operator — in particular of the Igusa cusp form Φ_{10} or its Borcherds square root $\Delta_5 = \Phi_{10}^{1/2}$ — consists of a chain-level filtered $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-object, for example the filtered boundary operator algebra $\mathrm{Op}^{\mathrm{HT}}_{\partial X}$ of Definition 9.1 (C6), together with an identification of its protected trace with the scalar invariant. The scalar identity is the trace value; the chain-level operator is the filtered object whose trace it is.

Proof. Direct from Lemma 9.10 applied to the chain-level objects whose protected traces are the scalar invariants in question. □

9.6 Topologisation criterion

The normal mixed-HT operad $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$ of Definition 9.7 is a two-coloured operad by construction; its passage to a topological E_d^{top}-operad on the closed sector is the *topologisation* of the holomorphic factor, under the two stress-tensor null-homotopies $T_i = [Q, G_i]$, $i = 1, 2$.

Definition 9.12 (Topologisation datum). A *topologisation datum* for the mixed-HT closed sector $(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, \mathrm{SC}^{\mathrm{HT}}_{(1,2)})$ is a tuple $(C_\rho, Q, G_1, G_2, T_1, T_2)$ where C_ρ is a complete filtered ambient symmetric-monoidal ∞-category refining $\mathcal{B}^{\mathrm{adm}}$ by a weight filtration ρ; Q is the BRST/BV differential of Statement 5.207; G_i are ρ-degree--1 operators on $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$; T_i are the two holomorphic translation stress-tensor components; and they satisfy

$$T_i = [Q, G_i] \quad (i = 1, 2) \quad \text{in } C_\rho. \tag{37}$$

Theorem 9.13 *(Topologisation by stress-tensor null-homotopy).* Inside the complete filtered refinement C_ρ of $\mathcal{B}^{\mathrm{adm}}$, the operadic restructuring (37) of Definition 9.7 turns the holomorphic-topological swiss-cheese closed colour $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$ on $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ to a topological closed colour $\mathrm{SC}^{\mathrm{top}}_{(1,2)}$ with E_3^{top} on the closed sector, under the six chain-level hypotheses (T1)–(T6) below.

Suppose a topologisation datum $(C_\rho, Q, G_1, G_2, T_1, T_2)$ of Definition 9.12 is given. Then the closed sector $(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, \mathrm{SC}^{\mathrm{HT}}_{(1,2)})$ is locally constant in the two holomorphic directions of $\mathbb{C}^2_{\mathrm{hol}}$ inside C_ρ after completion, and carries an E_3^{top}-algebra structure on the closed colour, intertwining the bulk currents (19) with the topological binary operation on E_3^{top}. The Capelli class $\hbar N[\bar{c}]$ remains the trace-sector projective Lie anomaly. The topologisation requires the chain-level hypotheses:

(T1) the operators G_1, G_2 are constructed and satisfy $[Q, G_i] = T_i$ as identities of chain-level operators in C_ρ;

(T2) the identities $T_i = [Q, G_i]$ are compatible with the weight filtration ρ and with the completion in C_ρ;

(T3) the two null-homotopies commute up to a specified secondary homotopy compatible with the holomorphic E_2-operad;

(T4) the topologisation has finite holomorphic propagation in C_ρ or, equivalently, the completed propagator respects the open–closed correspondence Definition 9.5;

(T5) the transferred E_3^{top}-operations are independent of the small-disks SDR transfer choices used to carry (37) across the bar resolution;

(T6) the anomaly class $\mathrm{Ob}^{\mathrm{anom}}$ is represented and killed by a null-homotopy in the operadic convolution complex of Definition 9.15, $\mathrm{Ob}^{\mathrm{anom}} \in H^2\big(\mathrm{Conv}_{\mathrm{SC}}\big(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C_\partial^{\mathrm{op}})\big)\big) = 0$, with secondary coherence class in $H^3\big(\mathrm{Conv}_{\mathrm{SC}}(\cdots)\big)$.

Proof. The local constancy in the two holomorphic directions follows from (37): the holomorphic translation generators $T_i = \partial_{z_i}$ are Q-exact in C_ρ under $[Q, G_i] = T_i$, so the bulk currents $\mathrm{J}_X(z_1, z_2)$ of (19) become locally constant in (z_1, z_2) in cohomology. Constant holomorphic dependence collapses the holomorphic factor E_2^{hol} to E_2^{top} on the level of operad actions [13, §3–4]. The normal closed colour $E_1^{\mathrm{top}} \otimes E_2^{\mathrm{top}}$ then identifies with E_3^{top} by Dunn additivity [41, Theorem 5.1.2.2].

The Capelli class $\hbar N[\bar{c}]$ is carried through the topologisation as a projective Lie central-extension class. Identifying it with determinant-line curvature or with a BCOV stress-tensor anomaly requires the determinant-line construction.

Conditions (T1)–(T6) are the standard homotopy-transfer equations for a stress-tensor null-homotopy: (T1) is the chain-level construction of G_1, G_2; (T2) is filtration compatibility; (T3) is the secondary commutation datum; (T4) is finite holomorphic propagation; (T5) is independence of SDR choices in the small-disks operad transfer; (T6) is the closed–open anomaly cancellation in the operadic convolution complex, sufficient for the higher coherences of the topologised $\mathrm{SC}^{\mathrm{top}}_{(1,2)}$-action. □

Remark 9.14 *(Class* L *versus class* M *topologisation).* For class L (affine Kac–Moody at non-critical level) boundary algebras the two topologisations $T_i = [Q, G_i]$ can be raw chain-level provided finite holomorphic propagation is proved. For class M (Virasoro / principal $\mathcal{W}$-families) the admissible topologisation is the weight-completed / pro / Banach ambient unless finite holomorphic propagation is separately verified. The finite-jet PVA models realize this distinction: the classical BV action is constructed, while the all-order QME, the boundary vertex algebra, and the E_3^{top}-lift remain governed by the filtered formal SDR, sectorial analytic regularity when analytic convergence is asserted, finite-window weight symmetry, and the local-functional lifts $T_i = d^L_{\mathrm{QME}} G_i$.

9.7 The typed global mixed-HT Deligne theorem

Theorem 5.13 states the OCA comparison on the Costello–Gwilliam brane-link boundary with a three-component obstruction. The typed refinement makes the five-component obstruction explicit, routes the comparison through OCA, and assigns each obstruction to the source complex in which its null-homotopy lives.

Definition 9.15 *(Open–closed centrality convolution complexes).* Let

$$Z_\partial = Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C_\partial^{\mathrm{op}}).$$

For a source object

$$S \in \left\{ \bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N}),\ \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \right\}$$

in $\mathcal{B}^{\mathrm{adm}}$, its centrality convolution complex with the derived centre has underlying Hom complex

$$\mathrm{Conv}(S, Z_\partial) = \underline{\mathrm{Hom}}^\bullet_{\mathcal{B}^{\mathrm{adm}}}(S, Z_\partial), \qquad d_{\mathrm{Conv}}\varphi = d_{Z_\partial}\varphi - (-1)^{|\varphi|}\varphi d_S.$$

For $S = \bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N})$ we write

$$\mathrm{Conv}_{\bar{B}}\big(\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N}), Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C_\partial^{\mathrm{op}})\big) \;=\; \underline{\mathrm{Hom}}^\bullet_{\mathcal{B}^{\mathrm{adm}}}\big(\bar{B}^{E_1}\big(A^{\mathrm{cl}}_{\partial,N}\big),\ Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}\big(C_\partial^{\mathrm{op}}\big)\big),$$

The obstruction theory for $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-algebra morphisms is the corresponding operadic convolution dg Lie, or L_∞, complex

$$\mathrm{Conv}_{\mathrm{SC}}(S, Z_\partial),$$

whose differential is the linearisation of the $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-Maurer–Cartan equation at the chosen source and target algebra structures, and whose bracket is induced by the cooperadic decomposition. The displayed Hom complex is its underlying graded mapping complex; obstruction classes for centrality, anomaly, and locality are taken in $\mathrm{Conv}_{\mathrm{SC}}$ after the operadic deformation problem has been fixed.

Theorem 9.16 *(Typed global mixed-HT Deligne criterion).* In $\mathcal{B}^{\mathrm{adm}}$ on the brane-link boundary ∂X (with possible refinement to C_ρ for topologisation), the pre-open–closed comparison datum preOCA_N of Definition 9.5 is promoted to an open–closed comparison morphism

$$\mathrm{OCA}_N : \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \to Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C_\partial^{\mathrm{op}})$$

of $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-algebras after the five native obstruction classes of (38) are represented in their governing complexes and killed by compatible null-homotopies. It is a quasi-isomorphism after $\mathrm{Cone}(\mathrm{OCA}_N)$ is equipped with a contracting homotopy in $\mathcal{B}^{\mathrm{adm}}$. A nonzero component, or a non-acyclic comparison cone, obstructs the typed OCA quasi-isomorphism. If topologisation of the holomorphic closed colour is included, the six hypotheses (T1)–(T6) and the secondary H^3-coherence class of Theorem 9.13 are part of the same obstruction datum.

The open–closed comparison $\mathrm{OCA}_N : \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \longrightarrow Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C_\partial^{\mathrm{op}})$ has native obstruction tuple

$$\mathrm{Ob}^{\mathrm{global},5} \;:=\; \left(\mathrm{Ob}^{\mathrm{ch}\,E_2}_{\mathrm{Deligne}}, \mathrm{Ob}^{\mathrm{desc}}_{\partial}, \mathrm{Ob}^{\mathrm{QME}}_{\text{all-order}}, \mathrm{Ob}^{\mathrm{anom}}, \mathrm{Ob}^{\mathrm{loc}}\right), \tag{38}$$

where the first three components are those of Theorem 5.13 and live in their own cohomology complexes, and the additional two are:

- $\mathrm{Ob}^{\mathrm{anom}} \in H^2\big(\mathrm{Conv}_{\mathrm{SC}}\big(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}^{\mathrm{op}}_{\partial})\big)\big)$, the anomaly-cancellation class in the operadic convolution complex of Definition 9.15, whose trace-sector projective anomaly is the Capelli class $\hbar N[\bar{c}]$;
- $\mathrm{Ob}^{\mathrm{loc}} = [d_{\mathrm{loc}}] \in H^1\big(\partial X, \underline{\mathrm{Hom}}^{\circ}_{\mathcal{B}^{\mathrm{adm}}}\big(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}^{\mathrm{op}}_{\partial})\big)\big)$, the locality class of the disjoint-union extension of OCA_N in the factorisation operad $\mathrm{Fact}_{\mathrm{SC}^{\mathrm{HT}}_{(1,2)}}$, as a Costello–Gwilliam Čech–Weiss–Ran cocycle on ∂X.

The vanishing of this tuple, together with chosen null-homotopies, is exactly the condition that OCA_N be an $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-morphism on ∂X. Contractibility of its cone gives the induced quasi-isomorphism. Before a common deformation complex is constructed, the five components are native cohomology classes in the following obstruction complexes. Here $d^L_{\mathrm{QME}} = Q + \{I[L], -\}_L + \hbar\Delta_L$ is the Costello scale-L quantum master differential on local functionals,

Class	Native source
$\mathrm{Ob}^{\mathrm{ch}\,E_2}_{\mathrm{Deligne}}$	linearised chiral E_2-Deligne deformation complex on $\mathbb{C}^2$
$\mathrm{Ob}^{\mathrm{desc}}_{\partial}$	$H^2(\partial X, \underline{\mathrm{Hom}}^{-1}(\mathcal{H}^{\bullet}, \mathcal{H}^{\bullet}))$
$\mathrm{Ob}^{\mathrm{QME}}_{\text{all-order}}$	$\big(\{o^{(r)}_{\sigma,w}\}_r,\ H^1(\ker\widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{QME}}),\ \varprojlim^1 H^0\big)$
$\mathrm{Ob}^{\mathrm{anom}}$	$H^2(\mathrm{Conv}_{\mathrm{SC}}(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}))$
$\mathrm{Ob}^{\mathrm{loc}}$	$H^1(\partial X, \underline{\mathrm{Hom}}^{\circ}_{\mathcal{B}^{\mathrm{adm}}}(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}))$

(39)

The common direct-sum language is valid only after comparison maps from these native sources to a master deformation complex have been fixed. Each displayed source carries the native null-homotopy for its component. The four-case obstruction table of Theorem 7.1 records the local curvature type only for components already represented in a fixed deformation complex: the Deligne and descent classes are target-construction obligations until the all-arity operad and descent object are built; their formal-Darboux trace-sector representatives are the arity-two local data, not an all-arity solution. The QME class lies in case (2) (counterterm coherence); the anomaly class lies in case (3) (projective coherence); and the locality class lies in case (2) (counterterm coherence).

Proof. Relation with the three source obstructions. The three source obstructions of Theorem 5.13 are the ordered triple

$$\big(\mathrm{Ob}^{\mathrm{ch}\,E_2}_{\mathrm{Deligne}}, \mathrm{Ob}^{\mathrm{desc}}_{\partial}, \mathrm{Ob}^{\mathrm{QME}}_{\text{all-order}}\big).$$

The QME entry is the scalar-contact reduction of the Costello local-functional obstruction tower in $H^1(\ker\widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{QME}})$. When a common filtered obstruction complex has been constructed, the ghost-number filtration separates the projective-anomaly representative

$$\mathrm{Ob}^{\mathrm{anom}} \in H^2(\mathrm{Conv}_{\mathrm{SC}}(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}))$$

whose trace-sector scalar is the Capelli class $\hbar N[\bar{c}]$, and the factorisation-degree filtration separates the locality representative

$$\mathrm{Ob}^{\mathrm{loc}} \in H^1(\partial X, \underline{\mathrm{Hom}}^{\circ}(\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}})).$$

The residual QME entry is a filtered tower: scalar-projection lift classes $o^{(r)}_{\sigma,w}$, the reduced all-order renormalisation-group class in $H^1(\ker\widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{QME}})$, and the finite-window $\varprojlim^1 H^0$ primitive-compatibility class. Thus (39) records five native sources before comparison, and records five graded summands only after the comparison maps to $\mathfrak{K}_{\mathcal{T}}$ have been fixed.

Independence. The five source complexes of (39) are native complexes. They become bidegrees of a master deformation complex $\mathfrak{K}_{\mathcal{T}}$ only after comparison maps into that complex have been fixed: the Deligne and descent classes are graded by their Čech–Weiss–Ran cover degree on ∂X, the QME class by the renormalisation-group filtration in $\ker \widehat{\sigma}_{\mathrm{sc}}$, the anomaly class by ghost number in the operadic convolution complex, and the locality class by factorisation degree in $\mathrm{Fact}_{\mathrm{SC}^{\mathrm{HT}}_{(1,2)}}$. After the comparison maps are fixed, no two complexes share both bidegree and graded piece, so the five-component obstruction $\mathrm{Ob}^{\mathrm{global},5}$ of (38) is represented by a direct sum of cohomology classes in visible bidegrees. A typed OCA datum is a compatible system of null-homotopies in these five summands together with a contracting homotopy of $\mathrm{Cone}(\mathrm{OCA}_N)$. The null-homotopies construct the typed comparison morphism; the cone contraction gives the quasi-isomorphism.

Costello–Li control. The Costello–Li construction (Statement 5.208, Statement 5.207) gives the renormalised local-functional complex in which the QME obstruction class is defined. The anomaly and locality rows require the operadic convolution complex, the locality cocycle, and the corresponding null-homotopies. If topologisation is imposed, the homotopy-transfer conditions are (T1)–(T6), including the secondary H^3-coherence class of Theorem 9.13. The Deligne and descent classes remain governed by the chiral E_2-Deligne problem and the Costello–Gwilliam descent spectral sequence, respectively.

Four obstruction mechanisms. $\mathrm{Ob}^{\mathrm{ch}\,E_2}_{\mathrm{Deligne}}$ and $\mathrm{Ob}^{\mathrm{desc}}_{\partial}$ lie in the target-replacement cell (Theorem 7.1 case (4)) until the all-arity chiral operad and the Costello–Gwilliam descent datum have been constructed. On a formal-Darboux polydisk their trace-sector representatives are the boundary values of the global null-homotopy problem; $\mathrm{Ob}^{\mathrm{QME}}_{\text{all-order}}$ and $\mathrm{Ob}^{\mathrm{loc}}$ are counterterm classes (case (2)) only after the Costello local counterterm complex and its locality homotopies have been fixed; $\mathrm{Ob}^{\mathrm{anom}}$ is the projective Lie-anomaly class (case (3)) represented by the Capelli central extension $\widehat{\mathfrak{g}}_{\mathrm{cent}}$ of (15). □

Remark 9.17 *(Local trivialisation of the five-component obstruction).* On the contractible formal-Darboux neighbourhood of a brane vacuum b, the trace-sector part of $\mathrm{Ob}^{\mathrm{global},5}$ is represented by explicit formal data: the binary diagonal kernel $K^{(2)}_{\Delta}$, the CE/PV map $c_f \mapsto \theta_f$, $u_f \mapsto J(f)$, and the Capelli central class. These are the arity-two and trace-sector boundary values of the five global components. The content of $\mathrm{Ob}^{\mathrm{global},5}$ is the extension of these formal representatives along ∂X, together with the all-arity chiral E_2-Deligne datum, the Costello–Gwilliam descent datum, and the all-order QME datum. The formal-Darboux stalk theorem (Theorem 9.6, = Theorem 5.6) is the trace-sector comparison fibre at b.

9.8 Local realization and descent under hypotheses

The formal-Darboux theorem supplies the local fibre for a global comparison datum whose formal restriction at one brane vacuum is the trace-sector map. A global open–closed quasi-isomorphism requires this fibre and five global data: an all-arity chiral E_2 action, a Costello–Gwilliam descent null-homotopy on the brane-link boundary, an all-scale QME solution, anomaly and locality null-homotopies, and a contracting homotopy of $\mathrm{Cone}(\mathrm{OCA}_N)$. Stein or affine hypotheses simplify ordinary sheaf cohomology after these complexes have been constructed.

9.8.1 Local trace-sector theorem at the formal-Darboux stalk

Theorem 5.6 states a trace-sector closed-to-open map inside the chosen admissible category $\mathcal{B}^{\mathrm{adm}}$. The admissibility conditions are structural hypotheses on the completion, not consequences of the finite-N brane algebra.

Lemma 9.18 *(Admissible formal-Darboux data).* In the formal-Darboux setting $X = \mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{hol}}$, the data used in Theorem 5.6 consist of the following admissible structures:

(i) the residue dual $P = \mathfrak{h}^\vee_{\mathrm{cont}}$ and the pairing $\mathfrak{h} \otimes P \to \mathbb{C}$, $\langle f, \omega\rangle = \mathbf{Res}_0 f\omega$, identified by Grothendieck–Matlis local duality on $H^2_{\mathfrak{m}}(R)$ [20][21];
(ii) a weighted or pro-Matlis topology in which the diagonal Casimir $C = \sum_{a+b>0} z_1^a z_2^b \otimes \rho_{a,b}$ and the Hamiltonian coadjoint action are continuous;
(iii) the diagonal local-cohomology kernel $K^{(2)}_\Delta = d\zeta_1 d\zeta_2/(\zeta_1\zeta_2)$ is bracket- and kernel-admissible by Proposition 5.7: the binary OPE of (19) factors through $K^{(2)}_\Delta$ with principal-part residue in the Grothendieck–Matlis dual;
(iv) a completed Hochschild-centrality convolution algebra in which $\{H^{\mathrm{prod}}_{f,-}, H^{P_0}_{f,-}, H^3_{f,-,-}, \ldots\}$ can be stated;
(v) Milnor/Roos compatibility for any counterterm tower used in a quantum or unreduced P_0-centrality extension.

Proof. The first item is Grothendieck–Matlis local duality. The remaining items are the explicit hypotheses under which the coordinate CE/PV identity is promoted to a bracketed and completed trace-sector map. The strict finite-N polynomial model is the algebraic quotient before these completions, kernels, and Milnor/Roos compatibility classes are chosen. □

COROLLARY 9.19 *(Local trace-sector closed-to-open theorem).* With the admissible data of Lemma 9.18, the formal-Darboux stalk has the trace-sector map of Theorem 5.6. After stable scalar reduction it is the continuous CE/PV isomorphism.

Proof. This is Theorem 5.6 with the admissible data spelled out in Lemma 9.18. □

9.8.2 GLOBAL CRITERION UNDER HYPOTHESES ON CONTRACTIBLE HOLOMORPHIC-SYMPLECTIC TARGETS

Contractibility and Stein vanishing act on the sheaf-cohomological rows of the obstruction complex. The chiral E_2 all-arity operation, the QME solution, and the comparison-cone contraction are additional chain-level data.

THEOREM 9.20 *(Contractible Stein descent criterion for the global Mixed HT Deligne map).* On a contractible Stein holomorphic-symplectic surface Y_{hol}, with brane-link boundary in $X = \mathbb{R}^2_{\mathrm{top}} \times Y_{\mathrm{hol}}$, the open–closed comparison

$$\mathrm{OCA}_N : \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \longrightarrow Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}^{\mathrm{op}}_\partial)$$

is a quasi-isomorphism in the strict admissible category under the following five hypotheses:

(i) an all-arity chiral E_2 diagonal-kernel action extending $K^{(2)}_\Delta$;
(ii) a Costello–Gwilliam descent datum on the brane-link boundary whose Čech–Weiss obstruction class is represented and killed by a descent null-homotopy;
(iii) a renormalised closed-to-open coupling satisfying the QME at all scales;
(iv) a null-homotopy of the anomaly class $\mathrm{Ob}^{\mathrm{anom}}$ and a locality homotopy killing $\mathrm{Ob}^{\mathrm{loc}}$;
(v) a contracting homotopy for $\mathrm{Cone}(\mathrm{OCA}_N)$ in $\mathcal{B}^{\mathrm{adm}}$ after the preceding data have been installed.

Under these hypotheses all components of $\mathrm{Ob}^{\mathrm{global},5}$ are null-homotopic in their governing complexes, and the comparison cone is acyclic. Replacing the anomaly null-homotopy in the fourth hypothesis by the Capelli central extension gives the projective comparison of Theorem 9.21.

Proof. The first three hypotheses give the three source components of Theorem 5.13. Contractibility removes monodromy for locally constant descent data, and Stein vanishing removes coherent sheaf cohomology obstructions once the relevant sheaves and counterterm complexes have been constructed. The Capelli class is not killed by contractibility; it is a Lie central extension. The fourth hypothesis is therefore the strict trivialisation of the anomaly and locality rows. The fifth hypothesis identifies the OCA map as a quasi-isomorphism. □

9.8.3 Projective descent criterion on Stein holomorphic-symplectic targets

On a general Stein holomorphic-symplectic surface Y, coherent cohomology vanishing controls the sheaf-theoretic rows after the chiral E_2, descent, QME, and locality complexes have been constructed. The Capelli central extension records the projective part of the trace anomaly.

Theorem 9.21 *(Projective Stein descent criterion).* On a Stein holomorphic-symplectic surface Y_{hol} (not necessarily contractible), let $X = \mathbb{R}^2_{\mathrm{top}} \times Y_{\mathrm{hol}}$, and let N Dirac branes sit along $L = \mathbb{R}_t \times \{b\}$. Suppose the all-arity chiral E_2 operation, the brane-link descent datum, the all-order QME solution, the locality homotopy for $\mathrm{Ob}^{\mathrm{loc}}$, and the comparison-cone contraction are fixed. Then the OCA map is a quasi-isomorphism in the projective category obtained from $\mathcal{B}^{\mathrm{adm}}$ by extension along the Capelli central extension $\widehat{\mathfrak{g}}_{\mathrm{cent}}$. The projective residual is the pair

$$\big(\mathrm{per}_X,\ \hbar N[\bar{c}]\big), \qquad \mathrm{per}_X \in H^1(X, \mathbb{C}_X), \quad [\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})[[\hbar]].$$

This pair is a class of the common obstruction complex after the comparison map into that complex has been constructed.

Proof. With these global data fixed, the chiral E_2, descent, QME, locality, and cone rows have the required null-homotopies or contractions. Stein vanishing applies inside the relevant sheaves and renormalised complexes. The residual scalar anomaly is the Lie central class $[\bar{c}]$ evaluated by the trace character, together with any period class of the chosen global Hamiltonian datum. Passing to the projective category records this class as the projective curvature of the trace-sector action. □

9.8.4 Concrete local realizations and global hypotheses

Corollary 9.22 *(Concrete local realizations and global hypotheses).* The formal-Darboux theorem applies at every smooth point of each target listed below. A global open–closed quasi-isomorphism is the conclusion of Theorem 9.20 or Theorem 9.21 under their stated hypotheses.

The following contractible Stein holomorphic-symplectic surfaces have no topological monodromy for the local trace-sector system:

(1) $Y = \mathbb{C}^2$ with $\omega_Y = dz_1 \wedge dz_2$ — the full local model of §1.4;
(2) $Y = D^2 \subset \mathbb{C}^2$ the open polydisk $\{|z_1| < 1, |z_2| < 1\}$ with the induced symplectic form;
(3) $Y = T^*\mathbb{C} = \mathbb{C}_p \times \mathbb{C}_q$ with $\omega_Y = dp \wedge dq$, the cotangent bundle of the affine line;
(4) $Y = T^*\mathbb{H}$ with $\mathbb{H}$ the upper half-plane, $\omega_Y = dp \wedge dq$ on the cotangent;
(5) $Y = B^4_r$ the open Stein ball of radius r in $\mathbb{C}^2$ with the standard symplectic form;
(6) $Y = \widehat{\mathbb{C}^2}_b$ the formal-Darboux completion at any smooth point b of any holomorphic-symplectic surface (the stalk case of Corollary 9.19).

The following Stein holomorphic-symplectic surfaces carry possible period or equivariant descent classes and therefore belong to the projective criterion:

(7) $Y = \mathbb{C} \times \mathbb{C}^*$ with $\omega_Y = dz \wedge d\log w$ — the $\mathbb{C}^*$-cylinder, with a period class along the circle;
(8) an affine smooth toric holomorphic-symplectic surface, with toric chart gluing recorded by the descent class.

The punctured surface $\mathbb{C}^2 \setminus \{0\}$ is excluded from the Stein list: by Hartogs extension its holomorphic functions extend to $\mathbb{C}^2$, so it is not Stein. The smooth locus of an affine ADE surface $\operatorname{Spec} \mathbb{C}[z_1, z_2]^\Gamma$ is likewise a punctured quotient surface, not a Stein target for the global criterion. The singular affine surface requires a derived or stacky target, and its minimal resolution is listed below as a chartwise local target with compact descent data.

The following holomorphic-symplectic surfaces have compact analytic subvarieties or compact descent data. The local theorem applies on Stein charts; the global statement requires compact descent:

(7′) $Y = T^*E$ for E an elliptic curve: holomorphic-symplectic but globally non-Stein (compact zero section $E \hookrightarrow T^*E$);
(8′) $Y = T^*\Sigma_{g\geq 1}$ for Σ_g a higher-genus Riemann surface: holomorphic-symplectic but globally non-Stein (compact zero section $\Sigma_g \hookrightarrow T^*\Sigma_g$);
(9′) $Y = \widetilde{\mathbb{C}^2/\Gamma}$ the minimal resolution of the Kleinian quotient: smooth holomorphic-symplectic but globally non-Stein (compact ADE exceptional divisor of $\mathbb{P}^1$ components);
(10′) $Y = \operatorname{Hilb}^N(\mathbb{C}^2)$ the Hilbert scheme of N points on $\mathbb{C}^2$: smooth quasi-projective but globally non-Stein for $N \geq 2$ (compact punctual locus $\operatorname{Hilb}^N(\mathbb{C}^2)_0 \hookrightarrow \operatorname{Hilb}^N(\mathbb{C}^2)$); the cyclic-framed stable branch of Lemma 3.3 carries the Nakajima ADHM presentation [46] as a GIT quotient on the stable framed locus where $\mathbb{C}[\phi_1, \phi_2] \cdot v = \mathbb{C}^N$; on each Stein chart of the Hilbert–Chow morphism $\operatorname{Hilb}^N(\mathbb{C}^2) \to S^N(\mathbb{C}^2)$ the local theorem gives the trace-sector chart; descent through the exceptional stratum is measured by a central obstruction class in $\widehat{\mathfrak{g}}_{\text{cent}}$. The trace map $J(f)$ descends to the $\operatorname{Hilb}^N(\mathbb{C}^2)$-coordinate ring through the cyclic framing vector [46].

Thus the catalogue gives local trace-sector realizations and the strict, projective, and compact-descent hypotheses for global comparisons.

Proof. Items (1)–(6) are formal or Stein contractible models. Items (7)–(8) are Stein or affine only after the smooth-target and descent data are specified. Hartogs extension excludes $\mathbb{C}^2 \setminus \{0\}$ and the smooth locus of the affine ADE surface from the Stein list. Items (7')–(10') contain compact analytic subvarieties and therefore cannot be Stein; the formal theorem applies chartwise, and the descent classes carry the global obstruction. □

COROLLARY 9.23 *(Stein-cover descent: explicit obstruction classes for the compact-containing realisations).* For each compact-containing realisation (7′)–(10′) of Corollary 9.22, choose the following Stein atlas or universal Stein cover and let Γ_Y denote its descent groupoid:

Item	**Target Y**	**Stein atlas or cover**	**Descent groupoid Γ_Y**
(7′)	T^*E	$\mathbb{C}^2 = T^*\mathbb{C}$ (universal cover of T^*E)	$\Lambda \cong \mathbb{Z}^2$, translation by the lattice $E = \mathbb{C}/\Lambda$
(8′)	$T^*\Sigma_{g\geq 1}$	$T^*\widetilde{\Sigma}_g$ ($\mathbb{C}^2$ for $g = 1$; $\mathbb{C} \times \mathbb{H}$ for $g \geq 2$)	$\pi_1(\Sigma_g)$
(9′)	$\widetilde{\mathbb{C}^2/\Gamma}$	affine Stein chart cover of the resolution	gluing groupoid on the ADE exceptional divisor
(10′)	$\operatorname{Hilb}^N(\mathbb{C}^2)$	Białynicki-Birula chart cover indexed by partitions $\lambda \vdash N$	combinatorial gluing on the punctual stratum

Assume first that the chart-level chiral E_2 operation and QME data of Theorem 9.21 have been fixed on the Stein atlas. The remaining obstruction to the OCA quasi-isomorphism on Y is the descent class

$$\mathrm{Ob}^{\mathrm{desc}}_{\Gamma_Y} \in \check{H}^2(\Gamma_Y; \widehat{\mathfrak{g}}_{\mathrm{cent}})$$

and the OCA descends after this class is represented and killed by a compatible descent null-homotopy; a nonzero class obstructs descent. The local theorem (Theorem 9.21) applies on each Stein chart of the atlas after the stated data are fixed; descent is the problem of gluing the chart-level OCA along the descent groupoid. After the coefficient groupoid and comparison maps are fixed, the representatives are the period class on $H^1(E)$ for (7′), the period class on $H^1(\Sigma_g)$ for (8′), the ADE root-lattice gluing class of the exceptional divisor for (9′), and the tautological gluing class along the Hilbert–Chow exceptional stratum for (10′). Each is recorded as a component of the compact comparison class $\mathrm{Ob}_{\mathrm{comparison}}$ of Theorem 11.4.

Proof. On each Stein chart $U \subset \widetilde{Y}$, the projective descent criterion (Theorem 9.21) gives the local OCA quasi-isomorphism once the chiral E_2, descent, and QME data are fixed. The descent groupoid Γ_Y preserves the holomorphic-symplectic form on overlaps by construction; the class $\mathrm{Ob}^{\mathrm{desc}}_{\Gamma_Y}$ is the obstruction to lifting the chart-level OCA quasi-isomorphisms to a Γ_Y-descent datum, equivalently the class in $\check{H}^2(\Gamma_Y; \widehat{\mathfrak{g}}_{\mathrm{cent}})$ classifying central extensions of the descent groupoid by the Capelli centre. Standard descent for the chosen Stein atlas gives the OCA quasi-isomorphism on Y after a null-homotopy of $\mathrm{Ob}^{\mathrm{desc}}_{\Gamma_Y}$; a nonzero class obstructs equivariant gluing. The named period classes are proposed cocycle representatives until the coefficient groupoid is fixed. □

Remark 9.24 *(Nakajima quiver-variety reading on* $\mathrm{Hilb}^N(\mathbb{C}^2)$*).* On the cyclic-framed stable branch of Corollary item (10′), the boundary algebra $A^{\mathrm{cl}}_{\partial,N}$ restricts to the coordinate ring of $\mathrm{Hilb}^N(\mathbb{C}^2)$ through the Nakajima ADHM presentation: the framed ADHM quiver Γ_N (one node carrying the gauge $\mathfrak{gl}_N$, one framed boundary node, two arrows for (ϕ_1, ϕ_2), one moment-map arrow for ψ) is the representation-variety presentation of the underlying derived commuting stack, and the trace map $J(f)$ is the trace of the arrow-monomial on the framed representation. The OCA trace-sector map on this branch is the restriction of Theorem 5.6 to the cyclic-framed stable locus, where $N = 1$ gives $\widehat{\mathcal{O}}_{\mathrm{Hilb}^1(\mathbb{C}^2),[b]} \cong \mathbb{C}[[z_1, z_2]]$. For $N > 1$, the Hilbert scheme has dimension $2N$, and its completed local ring is a $2N$-dimensional Darboux chart with tautological rank-N algebra; $A^{\mathrm{coord}}_b = \mathbb{C}[[z_1, z_2]]$ remains the target formal disk acting through that tautological algebra. The completed local ring of $\mathrm{Hilb}^N(\mathbb{C}^2)$ is the corresponding tautological $2N$-dimensional deformation algebra. This is a concrete chart-level realization of the trace map; global descent across the Hilbert–Chow exceptional locus is a compact comparison problem.

Corollary 9.25 *(Concrete local comparison loci).* At every smooth holomorphic-symplectic point of the following spaces, Theorem 5.6 gives the formal-Darboux trace-sector map:

(14) T^*S, for a contractible Stein open Riemann surface S;
(15) a contractible pseudoconvex Reinhardt domain in $\mathbb{C}^2$;
(16) $K3 \setminus C$, for an ample curve $C \subset K3$;
(17) a smooth two-dimensional character variety;
(18) a smooth two-dimensional Coulomb or Higgs branch of a $3d$ $\mathcal{N} = 4$ gauge theory;
(19) a smooth two-dimensional multiplicative quiver variety;
(20) a smooth two-dimensional hypertoric variety;

(21) a smooth affine wild character variety of complex dimension 2.

Items (14)–(15) fall under the contractible Stein criterion when the chiral E_2, descent, QME, and comparison-cone data are fixed. Items (16)–(21) fall under the projective Stein criterion when the same data are fixed and the Capelli central character is retained. Singular quotients require a derived or stacky replacement of the smooth-target hypothesis.

Proof. The listed spaces are Stein or affine on the stated smooth loci under the standard hypotheses in their construction. The local theorem is formal and depends only on the completed holomorphic-symplectic neighbourhood of the brane point. A global theorem requires the five hypotheses of Theorem 9.20; Steinness alone gives no chiral E_2 operad, no all-order QME solution, and no comparison-cone contraction. □

Remark 9.26 *(Compact comparison loci).* The comparison loci of Corollary 9.25 carry only the formal local conclusion unless the global comparison data exist:

- (16) $K3 \setminus C$: the holomorphic-symplectic chart gives only the local formal trace-sector map;
- (17)–(21): the character, Coulomb, Higgs, multiplicative quiver, hypertoric, and wild-character targets require their own Stein atlas, OCA comparison, QME solution, and descent null-homotopy.

In each case the formal-Darboux theorem is the local trace-sector closed-to-open map at a smooth brane point. The full derived E_2^{HT}-centre identification is the global problem under hypotheses.

Remark 9.27 *(Slodowy slices and deformations of Kleinian singularities).* Slodowy slices $\mathcal{S}_e = e + \ker \mathrm{ad}(f) \subset \mathfrak{g}$ at subregular nilpotent elements e in semisimple Lie algebras $\mathfrak{g}$ intersected with the nilpotent cone produce the Kleinian surface singularities:

$$\mathcal{S}_e \cap \mathcal{N} \simeq \mathbb{C}^2/\Gamma_e,$$

where $\Gamma_e \subset SL_2(\mathbb{C})$ is the finite ADE subgroup McKay-corresponding to the Dynkin type of $\mathfrak{g}$. The singular surface is affine and Poisson; its smooth locus is holomorphic-symplectic, and the minimal resolution is the smooth holomorphic-symplectic target of item (9′) of Corollary 9.22. The formal theorem applies on smooth formal Darboux charts; the singular point requires a derived or stacky replacement.

Remark 9.28 *(Compact and non-Stein targets).* For compact holomorphic-symplectic surfaces or compact Calabi–Yau 2-folds ($K3$ being the canonical example), Serre duality replaces Stein vanishing: by simple-connectedness $H^1(K3, \mathcal{O}) = 0$, but Serre duality and the holomorphic symplectic class $\sigma \in H^0(K3, \Omega^2)$ give

$$H^2(K3, \mathcal{O}) \;\cong\; H^0(K3, \Omega^2)^\vee \;\cong\; \mathbb{C},$$

the nonvanishing coherent class ($h^{2,0}(K3) = h^{0,2}(K3) = 1$; separately $h^{1,1}(K3) = 20$ sits in $H^1(K3, \Omega^1)$, not $H^1(\mathcal{O})$). The descent obstruction complex on ∂X has a possible Čech class sourced by this $H^2(\mathcal{O})$-class. The descent class $\mathrm{Ob}_\partial^{\mathrm{desc}}$ and the all-order QME class $\mathrm{Ob}_{\text{all-order}}^{\mathrm{QME}}$ therefore live in global cohomology groups not killed by Stein vanishing; representing them and choosing null-homotopies is the content of the Hall–Drinfeld–Borcherds filtered comparison complex (Conjecture 11.2). A spectral sequence appears only after its filtration, bidegrees, differentials, and abutment have been fixed. Conjecture 11.3 asks for the chain-level filtered boundary operator algebra of Definition 9.1 (C6) whose protected trace has scalar shadow the Igusa cusp form Φ_{10} and its Borcherds square root $\Delta_5 = \Phi_{10}^{1/2}$. By Lemma 9.10, a scalar partition identity such as $Z_{\mathrm{BPS,un}}^{K3\times E} = \Phi_{10}^{-1} = \Delta_5^{-2}$ or

$Z_{\mathrm{OP}}^{K3\times E} = -4096\,\Phi_{10}^{-1}$ acquires operator meaning only after its normalisation and such a chain-level $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-object has been constructed; the required object is the operator-valued lift of the modular line (§11.3).

The six typed objects impose the following chain-level constraints. Each constraint is either proved at the formal-Darboux stalk or represented by an obstruction class.

(V1) *Object typing.* Every equality names its source object, its target object, its ambient category, and its comparison map.

(V2) *Scalar non-faithfulness.* For any proposed chain-level $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-object, adjoin the acyclic filtered summand K of the proof of Lemma 9.10; the resulting object has the same χ but inequivalent chain-level structure. This is the obstruction to any scalar-determines-chain assertion.

(V3) *Maurer–Cartan identity.* The Maurer–Cartan identity $d_{\bar{B}}\Theta + \frac{1}{2}[\Theta,\Theta] = 0$ of (33) is the bar A_∞ identity on $\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N})$. The diagonal-kernel decoration also satisfies the Arnold relation $\eta_{12}\eta_{23} + \eta_{23}\eta_{31} + \eta_{31}\eta_{12} = 0$ after the chiral coefficient action map has been fixed.

(V4) *Bar–cobar equivalence.* The bar–cobar Quillen equivalence $\Omega^{c}\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N}) \xrightarrow{\sim} A^{\mathrm{cl}}_{\partial,N}$ is verified on the Koszul locus by Theorem 5.31 and Theorem 5.24. The Milnor $\varprojlim^1$ defect of Theorem 8.17 is the obstruction off the criterion's domain.

(V5) *Derived centre.* The bar-resolution computation $C^{\bullet}_{\mathrm{ch}}(A,A) = \mathrm{RHom}_{A\otimes A^{\mathrm{op}}}(A,A)$ on the chiral side is realised on the formal-Darboux stalk by (35); comparison with any proposed physical bulk $O^{\mathrm{phys}}_{\mathrm{bulk}}$ is mediated by the open–closed comparison OCA of Definition 9.5, with obstruction the non-vanishing of any component of $\mathrm{Ob}^{\mathrm{global},5}$.

(V6) *Topologisation.* Exhibit $T_i = [Q, G_i]$ of (37) chain-level in C_ρ, then verify the six conditions (T1)–(T6) of Theorem 9.13. The obstruction is a non-zero anomaly class $\mathrm{Ob}^{\mathrm{anom}} \neq 0$ in the operadic convolution complex.

(V7) *Object disjointness.* For each forbidden pair of objects (C_i, C_j) in the type table of Proposition 9.2, exhibit the identification that would collapse them and the corresponding replacement (§9.1).

9.9 Summary diagram

The six boundary open–closed objects fit into the following diagram; the upper square is the formal-Darboux operadic square, and the lower square is defined after the operator lift and topologisation data are fixed:

$$\begin{array}{ccc}
\bar{B}^{E_1}(A^{\mathrm{cl}}_{\partial,N}) & \xrightarrow{\text{Verdier/Koszul}} & (A^{\mathrm{cl}}_{\partial,N})^! \\
\downarrow \text{Hochschild} & & \downarrow \text{modules/lines} \\
Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(C^{\mathrm{op}}_{\partial}) & \xleftarrow{\text{OCA}} & \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \\
\downarrow \text{protected trace after operator lift} & & \downarrow \text{topologisation } T_i=[Q,G_i] \\
\chi(\mathrm{Op}^{\mathrm{HT}}_{\partial X}) & \xrightarrow[\text{under (T1–T6) of}]{\text{Theorem 9.13}} & (\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}, E_3^{\mathrm{top}})
\end{array}$$

The upper square is operadic, proved at the formal-Darboux stalk (Theorem 9.6) and global under null-homotopies for $\mathrm{Ob}^{\mathrm{global},5}$ of Theorem 9.16 and on the comparison-cone contraction. The lower square is the topologisation passage, under the operator lift and the topologisation datum of Definition 9.12. The scalar character χ is non-faithful by Lemma 9.10; the filtered boundary operator object $\mathrm{Op}^{\mathrm{HT}}_{\partial X}$ is a separate filtered $\mathrm{SC}^{\mathrm{HT}}$-object (§11.3) whose chain-level construction is controlled by the Hall–Borcherds residual class.

The diagram is the typed expression of the relative global mixed-HT Deligne problem: every assertion factors through the named morphisms Φ^{HT}, OCA, and the topologisation arrow; every scalar-to-operator lift factors through Lemma 9.10; every operator-level lift to the filtered boundary operator object requires its chain-level construction.

Remark 9.29 *(Formal-Darboux fibre and global K3×E Hall–Borcherds comparison).* The formal-Darboux stalk theorem (Theorem 5.6, equivalently Theorem 9.6 at the formal-Darboux fibre) is a *local* trace-sector E_2-map in the admissible filtered nuclear Tate / pro-Matlis / Capelli-projective category $\mathcal{B}^{\mathrm{adm}}$. Its correct global role is twofold:

(i) to give the *local model* for the open–closed comparison OCA at every brane vacuum $b \in \widehat{\mathbb{C}^2}$, entering the global Costello–Gwilliam bulk–boundary correspondence as the formal-Darboux fibre;

(ii) to give the *trace map* $J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2)$ of Theorem 6.8 as the Capelli-projective scalar of the local stalk.

The theorem contributes only the formal-Darboux fibre to the following global $K3 \times E$ constructions, each governed by its own chain-level theorem in the $d = 1$ algebraic-HT sector or the $d \geq 2$ Calabi–Yau-to-chiral comparison:

- the compact Hall descent $\widehat{\mathrm{CoHA}}^{\mathrm{red}}_{\Lambda^{2,1}_{\mathrm{II}}}(K3 \times E)$ on the polarized Borcherds source datum;
- the completed Drinfeld-double extension $\mathbf{D}\big(\mathcal{Y}^+_{\mathrm{Hall}}(K3 \times E)\big)$ with Hall pairing non-degeneracy on every recognised window;
- the current-envelope morphism $\mathrm{Cur}_E(\mathbf{D}(\mathcal{Y}^+_{\mathrm{Hall}})) \to \mathrm{Op}^{\mathrm{HT}}_{K3\times E}$ along the reference elliptic curve E;
- the $\mathrm{SC}^{\mathrm{HT}}_{(1,2)}$-trace compatibility whose curvature-section trace is $\Phi_{10} \in H^0(\overline{\mathcal{A}}_2, \lambda^{10})$; the unnormalised $K3 \times E$ BPS partition scalar is $Z^{K3\times E}_{\mathrm{BPS,un}} = \Phi_{10}^{-1} = \Delta_5^{-2}$, and the OP convention inserts its separate normalising factor.

The four items $(\beta, \delta, \varepsilon, \tau)$ above are the four-part chain comparison datum of the boundary-operator Hall–Borcherds conjecture in the $d = 1$ algebraic holomorphic-topological comparison; each carries its own finite-window hypotheses $(W1)$–$(W8)$, distinct from the seven admissibility conditions (i)–(vii) of Theorem 5.6. Confusing the formal-Darboux local OCA at a single brane vacuum with the global Hall–Borcherds chain comparison at K3×E is the precise collapse forbidden by the typed separation of source complexes: (C4) at one stalk does not imply (C6) on the whole target geometry.

10 Examples

At each closed point $p \in \mathbb{C}^2$ the formal-Darboux theorem gives the trace-sector map

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \dashrightarrow_{\Phi_N} C^\bullet_{\mathrm{Hoch}}\big(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}\big), \qquad c_f \mapsto \theta_f, \quad u_f \mapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

After stable scalar reduction and admissible completion this is the CE/PV coordinate isomorphism. The two examples below compute J at the two simplest brane multiplicities. Global realisations on Stein or compact geometries are governed by the descent and QME obstruction classes of Theorem 5.13. Vertex-algebra realisations (Heisenberg, $\widehat{\mathfrak{g}}_k$, $\beta\gamma$, Vir_c, W_N) arise only after a controlled curve restriction with pushed-forward bracket, BV pairing, brane image, and anomaly matching.

10.0.1 The single brane: $N = 1$

At $N = 1$ the matrices commute. The underived formal-Darboux chart is $A_b^{\mathrm{coord}} = \widehat{\mathcal{O}}_{\mathbb{C}^2,p} \cong \mathbb{C}[[z_1, z_2]]$ with $(\phi_1, \phi_2) = (z_1, z_2)$, and the trace is evaluation:

$$J(z_1^a z_2^b) = z_1^a z_2^b.$$

The full derived zero fibre also contains the closed scalar Koszul class $[\psi]$; the displayed chart is the zero-ψ trace-sector chart. The closed-to-open dictionary is $c_f \mapsto \theta_f$, $u_f \mapsto J(f)$. The formal Weyl algebra of one pair gives $[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_\star = \hbar$, the one-pair Heisenberg relation; the Capelli scalar $\hbar N[\bar{c}]\big|_{N=1} = \hbar[\bar{c}]$ is retained by the projective central extension; the ordinary scalar commutator is lost only in the unprojectivized scalar-reduced shadow.

10.0.2 The brane stack: $N \geq 2$

At a stack of $N \geq 2$ Dirac branes the trace reads

$$J(z_1^a z_2^b) = \overline{\mathrm{Tr}\ \phi_1^a \phi_2^b} \in \mathcal{O}\big(\mu_{\mathrm{der}}^{-1}(0)\big)^{GL_N}$$

on the derived commuting variety, whose coordinate cdga is

$$\big(\mathbb{C}[\mathfrak{gl}_N^{\oplus 2}] \otimes \Lambda(\mathfrak{gl}_N[1]),\ Q\psi = [\phi_1, \phi_2]\big);$$

its underived support is the commuting-pair locus. The Procesi–Razmyslov forcing makes J the unique GL_N-equivariant primitive single-trace Darboux-natural map. The Weyl-quantised trace representation carries the projective commutator

$$[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_{\mathrm{proj}} = \hbar N \cdot \mathbf{1},$$

and the Capelli scalar $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})[[\hbar]]$ (Theorem 6.8) is the projective Lie anomaly at the trace generator.

11 Compact comparison obstructions

Theorem 5.6 supplies the trace-sector formal-Darboux fibre for a relative global Mixed Holomorphic–Topological Deligne comparison datum whose closed-to-centre map has target

$$Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}_\partial^{\mathrm{op}}).$$

The existence of the global comparison is the obstruction problem of Theorem 5.13. By the formal-Darboux restriction theorem (Theorem 11.4), the chiral E_d Deligne conjecture at $d \geq 2$, the Hall–Drinfeld–Borcherds filtered comparison complex on compact non-toric CY_3, the operator-level lift of the modular line on $K3 \times E$, and the $W_\infty[\lambda]/E_\infty$ admissible endpoint identification are distinct theorems. The three remaining conjectures have the following mathematical form: the chiral E_2-Deligne conjecture contains $\mathrm{Ob}^{\mathrm{ch}\,E_2}_{\mathrm{Deligne}}$; the Hall–Drinfeld–Borcherds comparison requires a compact-target filtered comparison complex with descent differentials; the operator lift of Φ_{10} is the modular-line class $\mathrm{Ob}^V_{\mathrm{mod}}$.

11.1 Chiral E_d Deligne conjecture at $d \geq 2$

Conjecture 11.1 *(Chiral E_d Deligne).* Let $\mathcal{A}$ be a chiral E_d^{ch}-algebra on $\mathbb{C}^d$. There is a natural equivalence

$$Z^{\mathrm{der}}_{E_d^{\mathrm{ch}}}(\mathcal{A}) \;\simeq\; C^\bullet_{E_d^{\mathrm{ch}}}(\mathcal{A}, \mathcal{A})$$

in chiral E_{d+1}^{ch}-algebras, lifting the topological Deligne conjecture of Lurie [41, Thm. 5.3.1.14] along the chiral refinement.

The $d = 1$ case is the chiral Deligne theorem of Beilinson–Drinfeld [28] in the factorization-algebra formulation of Costello–Gwilliam [15, Vol. I]. At $d \geq 2$ the topological E_d-centre is constructed by Lurie [41, §5.3], but a chiral E_d-operad on $\mathbb{C}^d$ realising the holomorphic factorization structure at all arities is the datum measured by $\mathrm{Ob}^{\mathrm{ch}\,E_d}_{\mathrm{Deligne}}$; the corresponding chiral E_d-Hochschild cochain complex with its E_{d+1} structure is part of the same obstruction. The formal-Darboux stalk theorem (Theorem 5.6), together with Theorem 1.17, gives the arity-two $d = 2$ formal-coordinate operation as the diagonal kernel $K^{(2)}_\Delta\ \mathfrak{g}^{\otimes 2} \to \mathfrak{g}$; the all-arity extension is the content of Conjecture 11.1.

11.2 Hall–Drinfeld–Borcherds filtered comparison complex on compact non-toric CY_3

Conjecture 11.2 *(Hall–Drinfeld–Borcherds filtered comparison complex on compact non-toric CY_3).* On a compact non-toric Calabi–Yau threefold X with chosen Hall–Drinfeld–Borcherds datum of compatible conventions (Appendix 11.4, Theorem 11.13), construct a filtered complex

$$\mathcal{K}_{\mathrm{HDB}}(X, F^\bullet)$$

whose associated graded contains the scalar shadows defined by the chosen compact-target datum. When X carries a specified $K3$-Mukai component, as $K3 \times E$ does, these shadows are

$$\big(\kappa_{\mathrm{cat}},\ \kappa_{\mathrm{ch}}^{\mathrm{Hodge}},\ \kappa_{\mathrm{ch}}^{\mathrm{Heis}},\ \kappa_{\mathrm{BKM}},\ \kappa_{\mathrm{Mukai}}\big)(X).$$

The differential must realise the ordered Hall–Borcherds comparison residuals

$$\big(\mathrm{ob}_{\mathrm{Igusa/BKM}},\ \mathrm{ob}_{\mathrm{CY\text{-}ch}}\big)$$

of Theorem 11.4 after the comparison maps to $\mathcal{K}_{\mathrm{HDB}}(X, F^\bullet)$ have been fixed. These shadows become entries of a spectral-sequence page only after the filtered complex, bidegrees, differential, and abutment have been constructed.

On $X = K3 \times E$ the scalar shadows predicted by the matched Hall–Borcherds conventions are $(0, 0, 3, 5, 24)$. The entry 5 is the weight of the Borcherds square-root branch $\Delta_5 = \Phi_{10}^{1/2}$; the Igusa cusp form Φ_{10} has weight 10. The entry 24 is the rank of the extended Mukai lattice $\widetilde{H}^*(K3, \mathbb{Z})$; the fibre Euler datum $\chi(\mathcal{O}_{K3}) = 2$ is a distinct compact comparison scalar. Compact-target data on the S^3 framing, quintic DT/PT theory, and Gopakumar–Vafa closure belong to the compact Calabi–Yau threefold comparison.

11.3 Operator-level lift of the modular line on $K3 \times E$

Conjecture 11.3 *(Operator-level lift of Φ_{10}).* Let $L \subset K3 \times E$ be a chosen Dirac brane defect, let $\partial_{\mathrm{link}} N_L$ be the boundary of a tubular neighbourhood of L, and let b be a Dirac brane vacuum on this brane-link boundary. Choose a formal-normal chart at b and comparison maps $\chi_{\mathrm{cl}}, \chi_{\mathrm{op}}$ identifying its two-complex-dimensional formal restriction with the local Hamiltonian stalk. Let $\mathcal{B}_{K3\times E}$ be a compact-target holomorphic-topological bulk theory with boundary trace on C_∂^{op}. The level-2 projective modular class $\Omega_{\mathrm{central}}|_{\overline{\mathcal{A}}_2}$ lifts to an operator-valued cocycle $\widehat{\Omega}^{(2)}_{\mathrm{central}}$ on $\mathcal{B}_{K3\times E}$, whose protected scalar trace $\mathrm{Tr}_{C_\partial^{\mathrm{op}}}(\widehat{\Omega}^{(2)}_{\mathrm{central}})$ recovers the normalized modular section $\Phi_{10} = \chi_{10} \in H^0(\overline{\mathcal{A}}_2, \lambda^{10})$. The Hall–Borcherds comparison residual of Conjecture 11.2 is an obstruction class in the $K3 \times E$ operator-valued cocycle complex after that complex and its filtration have been defined.

The curvature-section convention and the partition-function convention are inverse scalar shadows. The protected trace of the modular curvature cocycle gives Φ_{10}; the unnormalised protected scalar BPS partition function, when the compact operator object exists, gives Φ_{10}^{-1}, while the OP-normalised scalar is $-4096\,\Phi_{10}^{-1}$. Thus an identity for $Z^{K3\times E}_{\mathrm{BPS}}$ has operator meaning only after its normalisation convention and operator-valued cocycle have been fixed. The branch $\Delta_5 = \Phi_{10}^{1/2}$ is the Borcherds square-root branch governing the paramodular product [32]. A construction satisfying the conjecture restricts near b to the formal trace-sector fibre controlled by Theorem 5.6; the formal-Darboux theorem does not construct the compact-target operator complex. The formal-Darboux restriction theorem (Theorem 11.4) records the matched-conventions class $\mathrm{ob}_{\mathrm{Igusa/BKM}}$ which must vanish for the lift to descend.

11.4 Formal-Darboux restriction and compact comparison

The local theorem of §5.3 is a formal-Darboux statement on

$$\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}} = \mathbb{R}_{\mathrm{brane}} \times \mathbb{R}_{\mathrm{top,normal}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}},$$

in holomorphic dimension $d = 2$, with brane line $\mathbb{R}_{\mathrm{brane}} \times \{0\} \times \{0\}$ as the one-dimensional factorization direction. A compact or arithmetic comparison changes the field content, topology, trace, anomaly complex, and period normalization. The compact objects considered below are: the BCOV/Kodaira–Spencer theory of a compact Calabi–Yau threefold [1]; the Calabi–Yau-to-chiral specialization $\Phi_d = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}}$; the Borcherds–Kac–Moody normalization of the Igusa cusp form Φ_{10} and of $\Delta_5 = \Phi_{10}^{1/2}$; the Gromov–Witten/Donaldson–Thomas/Pandharipande–Thomas/Maulik–Nekrasov–Okounkov–Pandharipande virtual trace; the Ooguri–Strominger–Vafa attractor comparison; the mixed-Dunn six-relation operadic structure; and global Weiss–Ran/Hamiltonian descent on a compact holomorphic-symplectic surface. The local theorem enters each comparison through the transported formal-Darboux stalk

$$\chi_{\mathrm{op}}^{-1}\,\Phi^{\mathrm{Ham}}_{\mathrm{fact}}\,\chi_{\mathrm{cl}},$$

and the comparison is obstructed by the class

$$\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}}) \in H^1(\mathcal{K}_{\mathrm{obs}}) \oplus H^1(\mathfrak{K}_{\mathrm{KD}})$$

of Theorem 11.13. A nonzero coordinate is the obstruction to the corresponding compact comparison.

In the three-functor chain

$$C_{\mathrm{CY}} \xrightarrow{\Phi_d^{\mathrm{FA}}} E_d\text{-HolFA}(X) \xrightarrow{\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}} A \xrightarrow{Z^{\mathrm{der}}_{E_d^{\mathrm{ch}}}} Z^{\mathrm{der}}_{E_d^{\mathrm{ch}}}(A) \xrightarrow{\text{HT realization}} \mathrm{Obs}_{\mathrm{bulk}}(T_A),$$

where the chiral derived E_d-centre functor on the right of the specialisation is the $d = 1$ chiral-algebra theorem ($Z^{\mathrm{der}}_{\mathrm{ch}} = C^\bullet_{\mathrm{ch}}$ on chiral algebras over curves; [28]) and at $d \geq 2$ the chiral E_d-Deligne conjecture (Conjecture 11.1, open). A matched target datum may identify the formal restriction of this global centre at a brane stack with the local CE model. The local theorem computes the subsequent trace-sector closed-to-open map

$$\Phi_N : C^\bullet_{\mathrm{CE}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]) \dashrightarrow C^\bullet_{\mathrm{Hoch,tr,adm}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}), \qquad u_f \mapsto J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)} \quad (f \in \bar{A}),$$

with formal f interpreted only in the completed brane algebra. After such a target comparison, the formal-stalk curvature has coordinates

$$(\hbar N[\bar{c}],\ \Omega^{\mathrm{rad}}_{a,b},\ [\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}],\ \mathrm{Ob}_{\mathrm{cent/QME/desc}},\ \mathrm{Ob}^{\Pi}_{\mathrm{BM}},\ \mathrm{Ob}_{\Phi_d\text{-descent}})$$

in a common deformation complex, when the comparison maps to $\mathfrak{K}_{\mathrm{KD}}$ have been constructed. In that complex, agreement means that the formal comparison and Φ_N admit a compatible null-homotopy; a nonzero coordinate is the named obstruction.

The derived chiral centre lives before the specialisation functor $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}$. On $K3 \times E$ the Heisenberg chiral scalar, the Borcherds weight of the square-root branch $\Delta_5 = \Phi_{10}^{1/2}$, and the Mukai-lattice rank are distinct scalar invariants until an operator-level lift is constructed. When that lift and its trace maps are part of $\mathfrak{C}_{\mathrm{BKM}}$, and $\mathrm{ob}_{\mathrm{Igusa/BKM}}$ has a chosen null-homotopy, the chosen traces of the operator object have these scalar values. The local theorem contributes the formal trace-sector fibre.

THEOREM 11.4 *(Formal-Darboux restriction for compact comparisons).* Let $\mathfrak{T}$ be one of the compact comparison objects listed above. The local theorem on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ contributes exactly the transported formal-Darboux stalk

$$\chi_{\mathrm{op}}^{-1}\, \Phi^{\mathrm{Ham}}_{\mathrm{fact}}\, \chi_{\mathrm{cl}}$$

to a morphism into $\mathfrak{T}$. Such a morphism exists only when a convention object $\mathfrak{C}_{\mathrm{tar}}$ (Definition 11.7) is fixed and the Koszul-descent obstruction class $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}})$ of Theorem 11.13 is null-homotopic. The compact geometry, modular normalization, automorphic operator, specialization functor, framing, descent datum, virtual cycle, attractor variable, or operadic structure is part of $\mathfrak{T}$, not of the formal-Darboux stalk. A nonzero coordinate of $\mathrm{Ob}_{\mathrm{comparison}}$ is the obstruction to the corresponding comparison.

Proof. The formal-stalk content of any cross-target morphism produced by the local theorem is identified by Lemma 11.6: only the displayed transported stalk passes from $\mathfrak{C}^{(2)}_{\mathrm{loc}}$ through the comparison morphisms. The matched-conventions criterion is Theorem 11.13: under vanishing of $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}})$ and a chosen null-homotopy, the target morphism exists and has the displayed formal-Darboux stalk; otherwise the nonzero coordinate is the exact obstruction. The divergence theorem for each compact object is given respectively by Theorems 11.21, 11.22, 11.23, 11.24, 11.25, 11.26, and 11.27. Each theorem names the local conventions, the compact conventions, and the obstruction class. Lemma 11.6 restricts the local contribution to the formal-Darboux stalk in every case. □

For a convention object $\mathfrak{C}_{\mathrm{tar}}$ with filtered deformation complexes, filtrations, curvature maps, and comparison morphisms, the obstruction class $\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}}$ has six coordinates common to the formal-Darboux fibre and seven coordinates belonging to the compact comparison:

$$\mathrm{Ob}^{\mathrm{univ}}_{\mathfrak{C}_{\mathrm{tar}}} = (\mathrm{per}_{\mathrm{tar}},\ \mathrm{ob}_{\mathrm{WR}},\ \mathrm{ob}_{\mathrm{anom}},\ \mathrm{ob}_{\mathrm{QME}},\ \mathrm{ob}_{\mathrm{conv}},\ \mathrm{ob}_{\mathrm{hyp}}),$$
$$\mathrm{Ob}^{\mathrm{tar}}_{\mathfrak{C}_{\mathrm{tar}}} = (\mathrm{ob}_{\mathrm{BCOV}},\ \mathrm{ob}_{\mathrm{CY\text{-}ch}},\ \mathrm{ob}_{\mathrm{Igusa/BKM}},\ \mathrm{ob}_{\mathrm{GW/DT}},\ \mathrm{ob}_{\mathrm{OSV}},\ \mathrm{ob}_{6r},\ \mathrm{ob}_{\mathrm{MD}}).$$

The shared coordinates are: period of locally Hamiltonian fields, Weiss/Ran descent of the local maps, scalar anomaly compatibility, brane-defect quantum master equation, inverse-limit convergence of the obstruction tower, and hyperdescent control of hidden $\varprojlim^1$ defects. The compact coordinates are the compact CY_3 BCOV holomorphic-anomaly equation; the compact CY-to-chiral target two-step specialization; Igusa / BKM Borcherds-product datum; Gromov–Witten / Donaldson–Thomas / Pandharipande–Thomas / MNOP virtual-cycle counts; OSV black-hole attractor identification; mixed topological–holomorphic six-relation factorization; mixed-Dunn operadic rectification. When $\mathfrak{C}_{\mathrm{tar}}$ contains a dg

Lie deformation complex, the obstruction class also carries the transported Maurer–Cartan curvature $\left[d\Theta_{\mathrm{KD}} + \frac{1}{2}[\Theta_{\mathrm{KD}}, \Theta_{\mathrm{KD}}]\right]$. A coordinate appears exactly for a structure present in $\mathfrak{C}_{\mathrm{tar}}$. It becomes a cohomology class exactly when the corresponding compact deformation complex and comparison map are part of $\mathfrak{C}_{\mathrm{tar}}$. The bare symbol is a named entry of the convention datum.

The seven compact comparison objects are bound to the seven compact coordinates of $\mathrm{Ob}^{\mathrm{tar}}_{\mathfrak{C}_{\mathrm{tar}}}$ by the following correspondence; the shared formal-stalk coordinates are included uniformly across all cases.

Compact object	**Coordinate**	**Divergence theorem**
Compact CY_3 BCOV / Kodaira–Spencer [1]	$\mathrm{ob}_{\mathrm{BCOV}}$	Theorem 11.21
the compact CY-to-chiral target specialization $\Phi_d = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi^{\mathrm{FA}}_d$	$\mathrm{ob}_{\text{CY-ch}}$	Theorem 11.22
Borcherds / BKM bridge to Igusa Φ_{10} and the Δ_5 branch	$\mathrm{ob}_{\mathrm{Igusa/BKM}}$	Theorem 11.23
GW / DT / PT / MNOP virtual cycles on a compact CY_3	$\mathrm{ob}_{\mathrm{GW/DT}}$	Theorem 11.24
OSV attractor identification $Z^{\mathrm{BH}} \sim \lvert Z^{\mathrm{top}}\rvert^2$	$\mathrm{ob}_{\mathrm{OSV}}$	Theorem 11.25
Mixed-Dunn / six-relation factorization on $\mathbb{R} \times \mathbb{C}^2$	ob_{6r}, $\mathrm{ob}_{\mathrm{MD}}$	Theorem 11.26
Global Hamiltonian descent on (X, ω, L)	$\mathrm{Ob}_{\mathrm{glob}}$	Theorem 11.27

Each compact object carries a divergence theorem comparing the local convention with the compact convention and an obstruction class given by the corresponding coordinate. These coordinates live in syntactically distinct cohomology groups ($H^1(\mathcal{K}_{\mathrm{obs}})$, $H^1(\mathfrak{K}_{\mathrm{KD}})$, and the appropriate comparison complexes); a strict cohomology-level coordinate-independence theorem is an additional compact comparison theorem.

Definition 11.5 *(Local Hamiltonian convention base).* The local theorem fixes the convention base

$$\mathfrak{C}^{(2)}_{\mathrm{loc}} = \left(d = 2,\ \mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}},\ \mathbb{R}_{\mathrm{brane}},\ \widehat{\mathfrak{h}},\ \widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[1],\ \mathbf{Ext}^*(\mathcal{O}_0, \mathcal{O}_0),\ [\bar{c}],\ \hbar_{\mathrm{Weyl}},\ \Phi^{\mathrm{Ham}}_{\mathrm{fact}}\right).$$

The superscript (2) is the holomorphic dimension $d = \dim_{\mathbb{C}} \widehat{\mathbb{C}^2}_0$. The completed Hamiltonian algebra is $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ (formal Hamiltonians modulo constants), $\widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[1]$ is its continuous cotangent shift, $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ is the scalar anomaly class carried by the brane line on the reduced Hamiltonian algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$, $\hbar_{\mathrm{Weyl}}$ is the Weyl/Moyal deformation parameter, and $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ is the brane-line interval factorization map produced by the local theorem. The datum records the brane-line factorization convention, the scalar quotient, and the reduced degree-zero Capelli/Moyal normalization.

Lemma 11.6 *(Formal-polydisk descent criterion).* The formal-Darboux theorem contributes the formal stalk used by a global or compact descent object $\mathfrak{T}$ exactly when $\mathfrak{T}$ carries a matched-conventions datum $\mathfrak{C}_{\mathrm{tar}}$ and a null-homotopy of the obstruction class

$$\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}}) = \left(\mathrm{Ob}^{\mathrm{univ}}_{\mathfrak{C}_{\mathrm{tar}}},\ \mathrm{Ob}^{\mathrm{tar}}_{\mathfrak{C}_{\mathrm{tar}}},\ [\, d\Theta_{\mathrm{KD}} + \tfrac{1}{2}[\Theta_{\mathrm{KD}}, \Theta_{\mathrm{KD}}]\,]\right)$$

of Theorem 11.13; a coordinate is present precisely when the corresponding structure is part of $\mathfrak{T}$. The admissible morphism has formal stalk

$$\chi_{\mathrm{op}}^{-1} \Phi^{\mathrm{Ham}}_{\mathrm{fact}} \chi_{\mathrm{cl}}.$$

Proof. The local datum consists of the formal Hamiltonian algebra $\mathbb{C}[[z_1, z_2]]/\mathbb{C}$, its continuous cotangent extension, the point brane Ext algebra, the brane-line factorization-current convention, the scalar quotient, and the Capelli/Weyl normalization. These data are invariant under the formal Darboux germ. Compact topology, a negative-cyclic Calabi–Yau class, a worldsheet or specialization curve, an S^3 framing, a global BV propagator, the holomorphic anomaly equation, Borcherds data, a Weyl chamber, a compact Hall orientation, virtual cycles, attractor counts, and compact chiral-homology objects therefore enter as components of $\mathfrak{C}_{\mathrm{tar}}$ accompanying the formal germ. The displayed coordinates record exactly the shared admissibility tests ($\mathrm{Ob}^{\mathrm{univ}}_{\mathfrak{C}_{\mathrm{tar}}}$: period, Weiss/Ran descent, scalar anomaly, brane-defect QME, inverse-limit convergence, hyperdescent), the target-specific admissibility tests ($\mathrm{Ob}^{\mathrm{tar}}_{\mathfrak{C}_{\mathrm{tar}}}$: BCOV, the compact CY-to-chiral target, Igusa/BKM, GW/DT/PT/MNOP, OSV, six-relation, mixed-Dunn), and the transported Maurer–Cartan curvature in $H^1(\mathfrak{K}_{\mathrm{KD}})$. Vanishing with null-homotopies is the matched-conventions admissibility datum; a nonzero coordinate is the corresponding obstruction class. □

Definition 11.7 *(Matched-conventions theorem datum).* A matched-conventions theorem datum is a tuple

$$\mathfrak{C}_{\mathrm{tar}} = \big(d, Y, \eta, (\Sigma_{d-1}, C), L, \tau, \lambda, \mathfrak{f}, \Lambda, \hbar_{\mathrm{tar}}, O_{\mathrm{tar}}, \chi_{\mathrm{cl}}, \chi_{\mathrm{op}}, \chi_{\mathrm{anom}}, \mathcal{K}_{\mathrm{obs}}, \mathfrak{K}_{\mathrm{KD}}, \Theta_{\mathrm{KD}}\big)$$

with the following meaning. Here $d = \dim_{\mathbb{C}} Y$; η is the negative-cyclic Calabi–Yau class, holomorphic volume form, or holomorphic-symplectic form used by the target theory; (Σ_{d-1}, C) is the worldsheet or factorization-homology specialization when a Φ_d-type theorem is asserted; L is the brane or boundary condition; τ is the trace pairing and scalar quotient; λ is the anomaly normalization; $\mathfrak{f}$ is the framing datum, including an S^3-framing for compact CY_3 statements; Λ is the coefficient topology; $\hbar_{\mathrm{tar}}$ is the deformation parameter; O_{tar} is the QME or obstruction hypothesis; χ_{cl}, χ_{op}, and χ_{anom} are the closed-field, open-field, and anomaly comparison morphisms from the target formal restriction to $\mathfrak{C}^{(2)}_{\mathrm{loc}}$; and $\mathcal{K}_{\mathrm{obs}}$ is the obstruction complex in which descent, QME, and operadic coherence are solved; $\mathfrak{K}_{\mathrm{KD}}$ is the filtered dg Lie algebra governing the target global or cross-target descent Maurer–Cartan problem; and Θ_{KD} is the transported local reduced central-operation Maurer–Cartan element whose curvature is the target obstruction class.

Definition 11.8 *(Compact Hamiltonian period complex).* Let (X, ω) be a compact connected holomorphic-symplectic surface. For a holomorphic symplectic vector field v, set

$$\alpha_v = -\iota_v \omega \in H^0(X, \Omega^1_{X,\mathrm{cl}}).$$

Choose a Stein cover $\mathfrak{U} = \{U_i\}$ and local holomorphic primitives $h_i \in O_X(U_i)$ with $dh_i = \alpha_v|_{U_i}$. The differences $h_i - h_j \in \mathbb{C}$ define a Čech cocycle, and its class

$$\mathrm{per}_X(v) = [\, h_i - h_j \,] \in H^1(X, \mathbb{C}_X)$$

is the Hamiltonian period obstruction. Equivalently, it is the connecting image of α_v for the holomorphic de Rham exact sequence

$$0 \longrightarrow \mathbb{C}_X \longrightarrow O_X \xrightarrow{d} \Omega^1_{X,\mathrm{cl}} \longrightarrow 0.$$

Definition 11.9 *(Global Hamiltonian descent complex).* Let L be the brane locus in a proposed global target and suppose that formal Darboux charts have produced local Hamiltonian maps Φ_U on a Weiss/Ran cover of L. Write $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}$ for the Hamiltonian closed-sector classical observables and $\mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}})$

for the P_0-factorization centre of the open observables, both taken from the local theorem's reduced Hamiltonian model. The descent target for these maps is the total complex

$$\mathfrak{G}_{\mathrm{WR}}(L) = \mathrm{Tot}\, C^{\bullet}_{\mathrm{WR}}\Big(\mathrm{Ran}(L); \underline{\mathrm{RHom}}\Big(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}, \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}})\Big)\Big).$$

The degree-zero Čech-Weiss cochain $\Phi_{\mathrm{loc}} = \{\Phi_U\}$ has first descent obstruction

$$\mathrm{ob}^{(1)}_{\mathrm{WR}}(\Phi_{\mathrm{loc}}) = [d_{\mathrm{WR}}\Phi_{\mathrm{loc}}] \in H^1(\mathfrak{G}_{\mathrm{WR}}(L)).$$

After choosing a filtration by number of factorization collisions and cotangent insertions, the higher coherence obstructions live in

$$\mathrm{ob}^{(r)}_{\mathrm{WR}}(\Phi_{\mathrm{loc}}) \in H^{r+1}\big(\mathrm{gr}^r\, \mathfrak{G}_{\mathrm{WR}}(L)\big), \qquad r \geq 1.$$

Vanishing of all these classes, together with chosen null-homotopies, is the exact assertion that the local Darboux maps descend to a global factorization-centre morphism.

Theorem 11.10 *(Sharp compact-surface period obstruction).* Let (X, ω) be a compact connected holomorphic-symplectic surface and v a global holomorphic symplectic vector field on X. Then

$$\mathrm{per}_X(v) = 0 \iff \alpha_v = 0 \iff v = 0.$$

In particular, every nonzero such v has nonzero period $\mathrm{per}_X(v) \in H^1(X, \mathbb{C}_X)$. This nonzero period is the global Hamiltonian obstruction before any descent or matched-conventions datum is fixed.

Proof. If $\mathrm{per}_X(v) = 0$, local Stein primitives h_i of $\alpha_v = -\iota_v\omega$ on a Stein cover differ on overlaps by constants exact in the de Rham connecting morphism, hence can be modified by constants to glue to a global holomorphic primitive h. Since X is compact and connected, $H^0(X, \mathcal{O}_X) = \mathbb{C}$, so $dh = 0$, hence $\alpha_v = 0$, and non-degeneracy of ω gives $v = 0$. The converses are immediate. □

Example 11.11 *(Abelian-surface translation).* A nonzero translation vector field v on an abelian surface (X, ω) has $\alpha_v \neq 0$, hence $\mathrm{per}_X(v) \neq 0$: local Darboux primitives exist on every chart and carry a nonzero gluing obstruction for a global holomorphic Hamiltonian.

Theorem 11.12 *(Formal local-to-global Hamiltonian descent theorem data).* Let (X, ω) be a compact connected holomorphic-symplectic surface and $L \subset X$ a brane locus for which the formal local Hamiltonian theorem is applied along Darboux charts. Fix the finite collection of locally Hamiltonian symplectic vector fields and local Darboux Hamiltonians used by the construction. Assume the target volume carries the complete pronilpotent descent complex, hyperdescent for the relevant Weiss/Ran cosheaf, and the exact inverse-limit conditions excluding $\varprojlim^1$ defects in the obstruction tower. Under those convergence and descent hypotheses, the following data are equivalent.

(i) A global Hamiltonian matched-conventions descent datum: a closed-open morphism

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}})$$

whose Darboux restrictions are the local maps Φ_{loc}, together with its interval, Weiss/Ran, scalar, and connected-stable coherence homotopies.

(ii) Vanishing of the obstruction vector

$$\mathrm{Ob}_{\mathrm{glob}}(X, L, \Phi_{\mathrm{loc}}) = \Big(\{\mathrm{per}_X(v)\}_v,\ \mathrm{ob}_{\mathrm{scal}},\ \mathrm{ob}^{(1)}_{\mathrm{WR}}(\Phi_{\mathrm{loc}}),\ \{\mathrm{ob}^{(r)}_{\mathrm{WR}}(\Phi_{\mathrm{loc}})\}_{r\geq 1}\Big)$$

with null-homotopies compatible with disjoint unions, interval inclusions, cotangent boundary realization, P_0-centrality, and the stable connected quotient.

The morphism in (i) is the global admissible Koszul-duality map

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}})$$

and the entries in (ii) are:

(a) every locally Hamiltonian symplectic vector field used by the construction has period class

$$\mathrm{per}_X(v) = \delta(-\iota_v\omega) \in H^1(X, \mathbb{C}_X)$$

equal to zero;

(b) the first Weiss/Ran class $\mathrm{ob}^{(1)}_{\mathrm{WR}}$ vanishes;

(c) all higher coherence classes $\mathrm{ob}^{(r)}_{\mathrm{WR}}$, $r \geq 1$, vanish after the cotangent boundary realization and P_0-centrality homotopies have been fixed;

(d) the scalar anomaly class agrees with the local class $[\bar{c}]$.

When these conditions hold, the local Darboux maps glue, uniquely up to the corresponding contractible space of choices, to the displayed global Hamiltonian central-operation morphism. When one included component is nonzero, that component is the displayed descent obstruction in the comparison complex; every component required by the asserted compact structure is part of the comparison datum.

On a compact connected X the period condition is sharp by Theorem 11.10: every nonzero global holomorphic symplectic vector field has nonzero $\mathrm{per}_X(v)$ and is rejected.

Proof. The period calculation is Definition 11.8, and the sharpness on compact connected X is Theorem 11.10: vanishing of $\mathrm{per}_X(v) = 0$ is the precise condition for gluing local holomorphic Hamiltonian primitives to a global one.

Once local primitives and Darboux charts are fixed, the remaining condition is descent for morphisms of factorization objects. On double overlaps the local-map defect is $d_{\mathrm{WR}}\Phi_{\mathrm{loc}}$, hence the first obstruction is the displayed class in $H^1(\mathfrak{G}_{\mathrm{WR}}(L))$. If it vanishes, a first homotopy can be chosen. The obstruction to making those homotopies coherent on higher overlaps and higher collision strata is computed by the associated graded pieces of the same filtered total complex, giving the classes $\mathrm{ob}^{(r)}_{\mathrm{WR}}$. This is the standard obstruction theory for a Maurer–Cartan element in a filtered dg Lie algebra of descent data. Scalar compatibility is separate because the local theorem has already quotiented by the distinguished scalar line. If all classes vanish and null-homotopies have been chosen, the Maurer–Cartan element is gauge equivalent to a global descent datum, which is exactly the displayed morphism. □

THEOREM 11.13 *(Matched-conventions transfer criterion).* Let $\mathfrak{C}_{\mathrm{tar}}$ be a matched-conventions datum whose comparison morphisms identify the formal restriction of the target closed and open theories with $\mathfrak{C}^{(2)}_{\mathrm{loc}}$:

$$\chi_{\mathrm{cl}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed,tar}}\big|_{\widehat{p}} \xrightarrow{\simeq} \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}, \qquad \chi_{\mathrm{op}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open,tar}}\big|_{\mathbb{R}_{\mathrm{brane}}} \xrightarrow{\simeq} \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},H}.$$

Suppose

$$\mathrm{Ob}^{\mathrm{univ}}_{\mathfrak{C}_{\mathrm{tar}}} = \big(\mathrm{per}_{\mathrm{tar}}, \mathrm{ob}_{\mathrm{WR}}, \mathrm{ob}_{\mathrm{anom}}, \mathrm{ob}_{\mathrm{QME}}, \mathrm{ob}_{\mathrm{conv}}, \mathrm{ob}_{\mathrm{hyp}}\big) = 0,$$
$$\mathrm{Ob}^{\mathrm{tar}}_{\mathfrak{C}_{\mathrm{tar}}} = \big(\mathrm{ob}_{\mathrm{BCOV}}, \mathrm{ob}_{\mathrm{CY\text{-}ch}}, \mathrm{ob}_{\mathrm{Igusa/BKM}}, \mathrm{ob}_{\mathrm{GW/DT}}, \mathrm{ob}_{\mathrm{OSV}}, \mathrm{ob}_{6r}, \mathrm{ob}_{\mathrm{MD}}\big) = 0,$$

with chosen null-homotopies and with target-specific coordinates present precisely for the corresponding BCOV, compact Calabi–Yau-to-chiral, Igusa/BKM, GW/DT/PT/MNOP, OSV, six-relation, or mixed-Dunn structure in $\mathfrak{C}_{\mathrm{tar}}$; the shared coordinates record the target period vanishing, the Weiss/Ran descent of $\mathfrak{G}_{\mathrm{WR}}(L)$, the scalar-anomaly identification $\chi_{\mathrm{anom}}([\mathrm{anom}_{\mathrm{tar}}]) = [\bar{c}]$, the QME class of $\mathcal{O}_{\mathrm{tar}}$, the convergence of the Maurer–Cartan obstruction tower, and the vanishing of $\varprojlim^1$ defects. When a mixed topological-holomorphic operadic comparison is asserted, $\delta_{m,n}$ satisfies Theorem 11.17. Let

$$\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}}) = \Big(\mathrm{Ob}_{\mathfrak{C}_{\mathrm{tar}}}, [d\Theta_{\mathrm{KD}} + \frac{1}{2}[\Theta_{\mathrm{KD}}, \Theta_{\mathrm{KD}}]]\Big) \in H^1(\mathcal{K}_{\mathrm{obs}}) \oplus H^1(\mathfrak{K}_{\mathrm{KD}})$$

be the matched-conventions obstruction class of the transported local reduced Hamiltonian central-operation Maurer–Cartan element.

If $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}}) = 0$ and a null-homotopy of this class has been chosen, then the target descent datum, using the local formal-stalk component, determines an admissible target factorization-centre morphism

$$\Phi_{\mathfrak{C}_{\mathrm{tar}}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed,tar}}\big|_L \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open,tar}})$$

whose formal restriction at p is

$$\chi_{\mathrm{op}}^{-1}\,\Phi^{\mathrm{Ham}}_{\mathrm{fact}}\,\chi_{\mathrm{cl}}.$$

If a specialization functor $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}$ is part of $\mathfrak{C}_{\mathrm{tar}}$, the specialized morphism is $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}(\Phi_{\mathfrak{C}_{\mathrm{tar}}})$. If the obstruction class is nonzero, that class is the obstruction to the compact comparison theorem in the global descent problem.

Proof. The comparison morphisms transport the local reduced central-operation map to each brane formal-normal chart where χ_{cl} and χ_{op} identify the target restriction with $\mathfrak{C}^{(2)}_{\mathrm{loc}}$. Anomaly compatibility identifies the scalar quotient with the local class $[\bar{c}]$, so the transported bracket and connected-trace normalization agree with the Hamiltonian theorem. The vanishing of the target period classes gives Hamiltonian primitives for the local symplectic vector fields in every chart where the Hamiltonian model is used, and the vanishing of the Weiss/Ran classes in $\mathfrak{G}_{\mathrm{WR}}(L)$ glues the local maps with coherent factorization homotopies by Theorem 11.12. The QME obstruction vanishing lifts the classical comparison to the target quantum normalization specified by $\mathcal{O}_{\mathrm{tar}}$. When a mixed topological-holomorphic operad is present, the mixed-Dunn criterion gives the required equivalence of algebra categories before applying specialization. The displayed formula is the definition of the glued map on the formal stalk. The remaining Maurer–Cartan curvature is exactly $d\Theta_{\mathrm{KD}} + \frac{1}{2}[\Theta_{\mathrm{KD}}, \Theta_{\mathrm{KD}}]$; a chosen null-homotopy makes the transported local datum a global matched-conventions global descent datum. Descent gives the global morphism. □

Definition 11.14 *(Functor of points of mixed six-relation factorization).* Let M be a real m-manifold and Y a complex n-fold. For an Artinian commutative dg algebra R, an R-point of $\mathrm{Fact}^{6r}_{\mathrm{mix}}(M, Y)$ is a datum

$$\mathcal{A}_R = (A_R, \mu_R, \ell_R, \bar{\partial}_R, \mathsf{W}_R, \mathsf{R}_R, \iota_R)$$

on product disks $D \times B \subset M \times Y$, where D is a real disk and B is a holomorphic polydisk: this is the product-disk site $\mathrm{Ran}_{\prod}(M, Y)$, generated by product disks and finite disjoint unions of such products. Cofinality of product-disk covers inside arbitrary Weiss covers of $M \times Y$, and descent on the ordinary $\mathrm{Ran}(M \times Y)$, belong to the unrestricted Weiss/Ran target. A target using arbitrary Weiss/Ran covers requires $\mathrm{ob}_{\mathrm{WR}}$, $\mathrm{ob}_{\mathrm{hyp}}$, and compatible null-homotopies in the matched-conventions obstruction complex.

(i) $A_R(D \times B)$ is a nilcomplete R-linear cochain complex, functorial for inclusions of product disks;
(ii) μ_R gives factorization products for pairwise disjoint finite families of product disks;
(iii) ℓ_R gives local-constancy homotopies for inclusions of real disks in the M-direction;
(iv) $\bar{\partial}_R$ gives Dolbeault null-homotopies for anti-holomorphic translations in the Y-direction;
(v) W_R gives Weiss descent in the real direction;
(vi) R_R gives Ran descent in the holomorphic direction;
(vii) ι_R gives the interchange homotopies between real collision products and holomorphic factorization products.

These data are required to satisfy six relations. Put

$$\mathsf{Rel}^{6r}(\mathcal{A}_R) = \big(\mathrm{Assoc}(\mu_R),\ \mathrm{LC}(\ell_R),\ \bar{\partial}_R^2,\ d_{\mathrm{WR}}\mathsf{W}_R,\ d_{\mathrm{Ran}}\mathsf{R}_R,\ \mathrm{Int}(\iota_R)\big).$$

An R-point is precisely a seven-tuple with $\mathsf{Rel}^{6r}(\mathcal{A}_R) = 0$: associativity of μ_R, homotopy invariance for ℓ_R, Dolbeault closedness for $\bar{\partial}_R$, Weiss cocycle descent for W_R, Ran cocycle descent for R_R, and the mixed interchange identity for ι_R. A k-simplex of the derived functor of points is the same datum over $R \otimes \Omega^\bullet(\Delta^k)$. Base change in R defines the functor

$$\mathrm{Fact}^{6r}_{\mathrm{mix}}(M, Y)\colon \mathrm{dgArt}_{\mathbb{C}} \longrightarrow \mathrm{sSet}.$$

This functor is the mixed factorization object used by global or compact descent. Its admissibility condition is the equation $\mathsf{Rel}^{6r} = 0$ together with coherent null-homotopies in square-zero families. For a partial lift $\widetilde{\mathcal{A}}_{R'}$ over a square-zero extension $R' \twoheadrightarrow R$ with kernel I, the six-relation obstruction is

$$\mathrm{ob}_{6r}(\widetilde{\mathcal{A}}_{R'}) = [\ \mathsf{Rel}^{6r}(\widetilde{\mathcal{A}}_{R'})\ ] \in H^2\Big(\mathrm{Der}^{\mathrm{fact,mix}}(\mathcal{A}_R) \otimes_R I\Big).$$

A mixed-Dunn comparison is a morphism of cofibrant operadic models

$$\eth_{m,n}\colon \mathbb{E}^{\mathrm{top}}_m \otimes \mathbb{E}^{\mathrm{hol}}_n \longrightarrow \mathbb{E}^{\mathrm{mix}}_{m,n}$$

whose induced algebra functor realizes this functor of points.

PROPOSITION 11.15 *(Recognition and tangent complex for mixed six-relation factorization).* Let $\mathcal{A}_R$ be an R-point of $\mathrm{Fact}^{6r}_{\mathrm{mix}}(M, Y)$. Then $\mathcal{A}_R$ determines a mixed factorization object on $\mathrm{Ran}_{\prod}(M, Y)$, locally constant in the M-directions and holomorphic in the Y-directions. Conversely, restriction to product disks sends every such mixed factorization object on this product-disk site to an R-point of $\mathrm{Fact}^{6r}_{\mathrm{mix}}(M, Y)$.

If $R' \twoheadrightarrow R$ is a square-zero extension with kernel I, the obstruction to lifting $\mathcal{A}_R$ to an R'-point is the six-relation class ob_{6r}, lying in

$$H^2\begin{pmatrix}\mathrm{Der}^{\mathrm{fact,mix}}(\mathcal{A}_R) \\ \otimes_R I\end{pmatrix},$$

and the space of lifts, when nonempty, is a torsor under

$$H^1\begin{pmatrix}\mathrm{Der}^{\mathrm{fact,mix}}(\mathcal{A}_R)\\ \otimes_R I\end{pmatrix}.$$

Proof. Product disks form the objects of the product-disk mixed Weiss/Ran site $\mathrm{Ran}_{\prod}(M, Y)$. The factorization products μ_R define the multiplication on disjoint configurations. The homotopies ℓ_R and $\bar{\partial}_R$ impose the two directional invariance conditions. The descent systems W_R and R_R glue the basis values to the mixed Ran object, and ι_R is exactly the compatibility between the two collision operations. Nilcompleteness over R identifies this glued object with the derived functor of points. Restricting a mixed factorization object to product disks recovers the seven pieces of data and the six relations.

For a square-zero extension $R' \to R$, deforming the seven structure maps while preserving the six relations is the Maurer–Cartan lifting problem in the dg Lie algebra of mixed factorization derivations $\mathrm{Der}^{\mathrm{fact,mix}}(\mathcal{A}_R) \otimes_R I$. The curvature of a partial lift is the degree-two component of Rel^{6r} for the lifted seven-tuple; its cohomology class is precisely ob_{6r} and is independent of choices. If this class vanishes, the remaining choices form the usual H^1-torsor. This proves the recognition criterion: a global or compact descent factorization object is exactly an R-point whose square-zero thickenings kill the six-relation obstruction tower. $\square$

Definition 11.16 *(Mixed-Dunn obstruction complex).* The deformation complex of the operad map $\delta_{m,n}$ is

$$\mathfrak{D}_{\mathrm{MD}}(m,n) = \mathrm{Cone}\Big(C^\bullet_{\mathrm{Operad}}(\mathbb{E}^{\mathrm{mix}}_{m,n}, \mathbb{E}^{\mathrm{mix}}_{m,n}) \longrightarrow C^\bullet_{\mathrm{Operad}}(\mathbb{E}^{\mathrm{top}}_m \otimes \mathbb{E}^{\mathrm{hol}}_n, \mathbb{E}^{\mathrm{mix}}_{m,n})\Big)[-1].$$

After a cofibrant filtration of the source operad has been chosen, the mixed-Dunn curvature is

$$F_\delta = d\delta + \frac{1}{2}[\delta,\delta] \in Z^1(\mathfrak{D}_{\mathrm{MD}}(m,n)).$$

Its class

$$\mathrm{ob}_{\mathrm{MD}}(\delta_{m,n}) = [F_\delta] \in H^1(\mathfrak{D}_{\mathrm{MD}}(m,n))$$

measures the simultaneous obstruction for the topological E_m operations, the holomorphic E_n operations, and the mixed interchange law to assemble into a strict operadic comparison. Higher rectification obstructions lie in $H^{r+1}(\mathrm{gr}^r\, \mathfrak{D}_{\mathrm{MD}}(m,n))$.

Theorem 11.17 *(Mixed-Dunn matched-conventions criterion).* Let $\delta_{m,n}$ be a morphism between cofibrant operadic models as in Definition 11.14, equipped with a complete exhaustive cofibrant filtration for which the obstruction tower below converges and has vanishing hidden $\varprojlim^1$ defect. The global or compact descent problem uses the mixed-Dunn object exactly under the following equivalent conditions.

(i) $\delta_{m,n}$ rectifies to a weak equivalence of operads and induces an equivalence

$$\mathrm{Alg}_{\mathbb{E}^{\mathrm{mix}}_{m,n}} \simeq \mathrm{Alg}_{\mathbb{E}^{\mathrm{top}}_m \otimes \mathbb{E}^{\mathrm{hol}}_n}$$

on the homotopy categories of algebras, with $\mathrm{Fact}^{6r}_{\mathrm{mix}}(M, Y)$ represented by the tensor product of the topological and holomorphic structures.

(ii) The aritywise homology map of $\delta_{m,n}$ is an isomorphism, the primary class $\mathrm{ob}_{\mathrm{MD}}(\delta_{m,n})$ vanishes, and every higher class in $H^{r+1}(\mathrm{gr}^r\, \mathfrak{D}_{\mathrm{MD}}(m,n))$ vanishes with chosen null-homotopies compatible with operadic composition.

A nonzero ordered tuple of these classes is the exact mixed-Dunn obstruction.

Proof. A weak equivalence between cofibrant operads induces a Quillen equivalence on their algebra categories [41, Thm. A.3.7.5]. This proves that (i) implies the aritywise homology isomorphism and that the curvature classes in $\mathfrak{D}_{\mathrm{MD}}(m,n)$ vanish.

Conversely, suppose (ii) holds. The complete exhaustive cofibrant filtration reduces rectification of $\eth_{m,n}$ to a convergent tower of lifting problems. The primary curvature $F_{\eth}$ is the first obstruction class; the associated-graded obstruction groups record the successive obstruction classes for operadic composition, holomorphic-side coherence in the $\mathbb{E}_n^{\mathrm{hol}}$ factor, and the topological-holomorphic interchange law. By hypothesis each obstruction class is zero and a compatible null-homotopy has been chosen, so the tower has a rectified operad map. The aritywise homology isomorphism makes the rectified map a weak equivalence. Applying the cofibrant-operad Quillen-equivalence theorem gives (i), and Proposition 11.15 identifies the resulting algebra objects with the displayed functor of points. □

COROLLARY 11.18 *(Matched-conventions factorization).* Every compact comparison theorem that uses the local Hamiltonian BF/Moyal theorem through a matched-conventions datum $\mathfrak{C}_{\mathrm{tar}}$ with $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}}) = 0$ and chosen null-homotopies has, as its formal-stalk component, the transported morphism

$$\Phi_{\mathfrak{C}_{\mathrm{tar}}} = \chi_{\mathrm{op}}^{-1}\,\Phi_{\mathrm{fact}}^{\mathrm{Ham}}\,\chi_{\mathrm{cl}}$$

constructed in Theorem 11.13. The remaining specialization, compactification, automorphic construction, and global descent maps are compact comparison data.

Proof. Theorem 11.13 constructs the formal-stalk component obtained from the local theorem and the comparison morphisms χ_{cl}, χ_{op}. Any subsequent operation belongs to the comparison datum: specialization along (Σ_{d-1}, C), compact BV integration, or the Borcherds automorphic construction. Thus the local theorem enters the compact proof only through the displayed formal-stalk component. □

COROLLARY 11.19 *(Compact matched-conventions criterion).* A compact BCOV, MNOP/Abel–Jacobi, Calabi–Yau-to-chiral, Igusa/Borcherds–Kac–Moody, or related comparison theorem may use the local Hamiltonian BF/Moyal formal stalk only through a global or compact descent datum $\mathfrak{C}_{\mathrm{tar}}$ with

$$\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}}) = 0$$

and chosen null-homotopies for all components of the obstruction tower. The compact comparison data include the formal stalk

$$\chi_{\mathrm{op}}^{-1}\,\Phi_{\mathrm{fact}}^{\mathrm{Ham}}\,\chi_{\mathrm{cl}},$$

the global descent datum, the QME/anomaly null-homotopies, and any mixed-Dunn or six-relation factorization null-homotopies asserted by the compact theorem. The exact residual of an asserted compact structure is the corresponding component of $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}})$. A null-homotopy of that class permits the compact theorem to use the formal-stalk component. The compact CY_3, global BCOV, MNOP/Abel–Jacobi growth, compact CY-to-chiral, Igusa, and BKM conclusions remain compact theorems with their own data.

Proof. Theorem 11.13 constructs the transported comparison morphism after the comparison datum has been fixed and the matched-conventions obstruction class has been null-homotoped. The class contains the period, descent, anomaly, QME, six-relation, and mixed-Dunn components. Lemma 11.6 identifies these components as compact theorem data attached to $\mathfrak{C}_{\mathrm{tar}}$. Therefore a nonzero component is the corresponding obstruction class, and a zero component with a null-homotopy is exactly the compact theorem datum. □

Definition 11.20 *(Compact comparison data).* The compact comparisons require the following matched-conventions data.

(i) *the compact CY-to-chiral comparison datum.*

$$\mathfrak{C}_{\mathrm{CY}\to\mathrm{ch}} = \big(d, Y, \eta_{\mathrm{CY}} \in HC_d^-(Y), (\Sigma_{d-1}, C), \Phi_d^{\mathrm{FA}}, \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}, \kappa_{\mathrm{cat}}, \tau, \mathfrak{f}, \mathcal{O}_{\mathrm{QME}}, \mathfrak{K}_{\mathrm{KD}}, \mathrm{Ob}_{\mathrm{comparison}}\big).$$

The comparison map from $\mathfrak{C}^{(2)}_{\mathrm{loc}}$ is the formal $d = 2$ Darboux normalization at the chosen brane point. The compact CY-to-chiral target convention is the two-step factorization

$$\Phi_d = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}}.$$

For a compact CY$_3$ target the total-space κ_{cat} is the Hodge supertrace $\sum_q (-1)^q h^{0,q}(Y)$; in particular $\kappa_{\mathrm{cat}}(K3 \times E) = 0$. This component is distinct from any Heisenberg, BKM, or fibre normalization in the compact CY-to-chiral target. Any BKM weight comparison is therefore a distinct component of $\mathrm{Ob}_{\mathrm{comparison}}$, with the unqualified total-space supertrace kept as its own component.

(ii) *Compact BCOV datum.*

$$\mathfrak{C}_{\mathrm{BCOV}} = \big(Y^3, \Omega_Y, \mathrm{PV}^{*,*}(Y)[[t]], Q_{\mathrm{BCOV}}, P_{\mathrm{prop}}, I_{\mathrm{ct}}, \mathcal{O}_{\mathrm{HAE}}, \mathfrak{K}_{\mathrm{KD}}, \mathrm{Ob}_{\mathrm{comparison}}\big).$$

Here Y^3 is the compact Calabi–Yau threefold, Ω_Y the holomorphic volume form, $\mathrm{PV}^{*,*}(Y)[[t]]$ the polyvector-field BCOV field complex, Q_{BCOV} the BCOV differential, P_{prop} the global gauge-fixed propagator, I_{ct} the counterterm scheme, and $\mathcal{O}_{\mathrm{HAE}}$ the holomorphic-anomaly/QME obstruction complex. The BCOV holomorphic anomaly equation $\bar\partial_{\bar i} F_g = \frac{1}{2}\bar C_{\bar i}^{jk}(D_j D_k F_{g-1} + \sum_{h+h'=g} D_j F_h\, D_k F_{h'})$ is a target equation in this datum [1]; it belongs to the comparison datum for global BCOV theory rather than to the local $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_0$ theorem. Compact Calabi–Yau threefolds and genus expansions enter in this datum. Open data such as $(L, \mathfrak{f}_{S^3}, h_{\mathfrak{f}}, \mathrm{Ob}_{\mathrm{frame}})$ form a separate brane datum.

(iii) *Igusa/BKM datum.*

$$\mathfrak{C}_{\mathrm{BKM}} = \big(\varphi_{\mathrm{Jac}}, \Lambda_{\mathrm{Borch}}, W, \rho, \mathrm{ch}_{\mathrm{Hall}}, \ell_{\mathrm{or}}, \pi_{\mathrm{prim}}, \kappa_{\mathrm{BKM}}, \tau_{\mathrm{DT/PT/OP}}, \mathrm{Ob}_{\mathrm{comparison}}\big).$$

The entries are the Jacobi or modular form, Borcherds lattice, Weyl chamber, character, Hall or chiral source, orientation line, primitive recognition map, and scalar DT/PT/OP normalization. For the Δ_5 branch the matched constants are

$$D_5 = 64^{-1}\Delta_5, \qquad \mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z),$$

$$\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 10/2 = 5, \qquad Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}.$$

These constants live in the BKM/Borcherds comparison for the Δ_5 construction; they fix the convention datum at which the matched-conventions theorem must match. The local Hamiltonian theorem on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ contributes the formal-Darboux stalk. The orientation comparison $\epsilon_o = \nu_{\Delta_5}$, compact Hall constants, and primitive BKM recognition are target obstructions outside the local $d = 2$ Hamiltonian theorem.

(iv) *Global Hamiltonian descent datum.*

$$\mathfrak{C}_{\mathrm{glob}} = \big(L, \mathcal{H}_X = \mathcal{O}_X/\mathbb{C}_X, \mathrm{per}(v) = \delta(-\iota_v\omega), \mathfrak{G}_{\mathrm{WR}}(L), \{\mathrm{ob}^{(r)}_{\mathrm{WR}}\}_{r\geq 1}, h_{\mathrm{cent}}, \mathrm{Ob}_{\mathrm{glob}}, \Theta_{\mathrm{KD}}\big).$$

The vanishing condition is the one in Theorem 11.12; its conclusion is the global matched-conventions descent morphism.

(v) *GW/DT/PT/MNOP datum.*

$$\mathfrak{C}_{\mathrm{GW/DT}} = \big(Y^3, \beta \in H_2(Y,\mathbb{Z}),\ F^{\mathrm{GW}}_{g,\beta},\ Z^{\mathrm{DT}}_{n,\beta},\ Z^{\mathrm{PT}}_{n,\beta},\ \mathfrak{f}_{\mathrm{vir}},\ \nu_{\mathrm{Behr}},\ \tau_{\mathrm{vir}},\ \Phi_{\mathrm{MNOP}},\ \mathrm{Ob}_{\mathrm{comparison}}\big).$$

The entries are the compact CY_3 target, the curve class, the generating series for the four enumerative theories, the virtual framing, Behrend's microlocal weight, the virtual trace pairing, and the MNOP correspondence map. When the required comparison theorem is available on Y, these data identify the GW, DT, PT, and MNOP series. Otherwise their failure to glue is a compact-target obstruction. These data belong to the compact target theory. The local $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$ theorem has a formal polydisk as holomorphic factor and carries only formal holomorphic coordinates.

(vi) *OSV attractor datum.*

$$\mathfrak{C}_{\mathrm{OSV}} = \big(Y^3, \Lambda_{\mathrm{att}}, Z^{\mathrm{BH}}_{Q,P},\ e^{-S_{\mathrm{att}}},\ \Phi_{\mathrm{OSV}},\ \mathrm{Ob}_{\mathrm{comparison}}\big).$$

The OSV conjecture $Z^{\mathrm{BH}} \sim |Z^{\mathrm{top}}|^2$ and any attractor identification of charges with topological-string genera is a target equation in this datum. Its derivation belongs to the OSV target theory, with the local theorem on $\widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$ contributing only the formal Hamiltonian stalk.

(vii) *Mixed-Dunn datum.*

$$\mathfrak{C}_{\mathrm{MD}} = \big(\mathbb{E}^{\mathrm{top}}_m, \mathbb{E}^{\mathrm{hol}}_n, \mathbb{E}^{\mathrm{mix}}_{m,n}, \mathrm{Fact}^{6r}_{\mathrm{mix}}, \mathsf{Rel}^{6r}, \delta_{m,n}, \mathrm{ob}_{6r}, \mathfrak{D}_{\mathrm{MD}}(m,n), \mathrm{ob}_{\mathrm{MD}}\big).$$

The vanishing condition is the simultaneous six-relation and rectification condition in Proposition 11.15 and Theorem 11.17.

Divergence theorems for compact comparisons

The seven pieces of matched-conventions data in Definition 11.20 carry seven divergence theorems. Each names the convention used in the local theorem, the convention used in the compact comparison, the obstruction class to identification, and the matched-conventions morphism obtained once that obstruction is killed by a chosen null-homotopy.

THEOREM 11.21 *(BCOV divergence and matched-conventions).* Let $\mathfrak{C}_{\mathrm{BCOV}}$ be the compact BCOV datum of Definition 11.20 (ii).

(i) *Divergence.* The local theorem on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$ asserts the formal-local Darboux stalk theorem. A compact Calabi–Yau threefold Y^3, a global gauge-fixed BCOV propagator P_{prop}, a genus expansion in F_g, and the BCOV holomorphic-anomaly equation are closed compact data. An S^3-framing $\mathfrak{f}_{S^3}$ belongs to a separate open-brane datum. The Bershadsky–Cecotti–Ooguri–Vafa equation

$$\bar\partial_{\bar i} F_g = \tfrac{1}{2}\, \bar C_{\bar i}^{\,jk}\Big(D_j D_k F_{g-1} + \sum_{h+h'=g} D_j F_h\, D_k F_{h'}\Big)$$

is a target equation in $\mathfrak{C}_{\mathrm{BCOV}}$. The BCOV paper gives the physical holomorphic-anomaly equation; a theorem in the mathematical BV sense requires the chosen compact BCOV field theory, gauge fixing, boundary data, and QME solution. The local Hamiltonian theorem contributes only the formal-stalk component.

(ii) *Obstruction class.* The obstruction class is the holomorphic-anomaly / counterterm class

$$\mathrm{ob}_{\mathrm{BCOV}} \in H^1(\mathcal{O}_{\mathrm{HAE}}),$$

computed in the BCOV obstruction complex $\mathcal{O}_{\mathrm{HAE}}$ of Definition 11.20 (ii).

(iii) *Matched-conventions criterion.* If $\mathrm{ob}_{\mathrm{BCOV}} = 0$ with a chosen null-homotopy and the shared coordinates $\mathrm{Ob}^{\mathrm{univ}}_{\mathfrak{C}_{\mathrm{tar}}}$ vanish, then by Theorem 11.13 the BCOV target receives the formal-Darboux component $\chi_{\mathrm{op}}^{-1}\Phi^{\mathrm{Ham}}_{\mathrm{fact}}\chi_{\mathrm{cl}}$ at every brane formal-normal chart where χ_{cl} and χ_{op} identify the target restriction with $\mathfrak{C}^{(2)}_{\mathrm{loc}}$. Compactification, propagator, framing, and genus-expansion data remain compact comparison data.

Proof. Clauses (i) and (ii) are Definition 11.20 (ii) read as a divergence and an obstruction class. Clause (iii) is the BCOV case of Theorem 11.13. □

Theorem 11.22 *(The compact CY-to-chiral divergence and matched conventions).* Let $\mathfrak{C}_{\mathrm{CY}\to\mathrm{ch}}$ be the compact CY-to-chiral comparison datum of Definition 11.20 (i).

(i) *Divergence.* The compact CY-to-chiral target convention is the two-step factorization $\Phi_d = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi^{\mathrm{FA}}_d$ with worldsheet / specialization curve (Σ_{d-1}, C), negative-cyclic CY class $\eta_{\mathrm{CY}} \in HC^-_d(Y)$, and total-space coefficient κ_{cat}. These data are target entries of $\mathfrak{C}_{\mathrm{CY}\to\mathrm{ch}}$. The local theorem on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ has a formal polydisk as holomorphic factor, and its Calabi–Yau datum is the trivial holomorphic symplectic form on $\mathbb{C}^2$, used only as a unimodularity datum.

(ii) *Obstruction class.* The obstruction class is the compact CY-to-chiral target specialization class

$$\mathrm{ob}_{\text{CY-ch}} \in H^1(\mathfrak{K}_{\mathrm{KD}})$$

attached to the comparison of the local $\mathfrak{C}^{(2)}_{\mathrm{loc}}$-stalk with the compact CY-to-chiral target formal restriction at the chosen brane point. In the compact CY_3 case the Hodge supertrace $\kappa_{\mathrm{cat}} = \sum_q (-1)^q h^{0,q}(Y)$ is target datum; the convention check $\kappa_{\mathrm{cat}}(K3 \times E) = 0$ is distinct from any Heisenberg, BKM, or fibre normalization in the compact CY-to-chiral target.

(iii) *Matched-conventions criterion.* If $\mathrm{ob}_{\text{CY-ch}} = 0$ with a chosen null-homotopy and the shared coordinates vanish, then by Theorem 11.13 the specialized morphism $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}(\Phi_{\mathfrak{C}_{\mathrm{tar}}})$ is defined and has formal stalk $\chi_{\mathrm{op}}^{-1}\Phi^{\mathrm{Ham}}_{\mathrm{fact}}\chi_{\mathrm{cl}}$. The remaining specialization, worldsheet, BKM-weight, and Hall-orientation maps are target data.

Proof. Clauses (i)–(ii) read off Definition 11.20 (i). Clause (iii) is the compact CY-to-chiral target case of Theorem 11.13, applied with the specialization functor $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}$ contained in $\mathfrak{C}_{\mathrm{tar}}$. □

Theorem 11.23 *(Igusa / BKM divergence and matched-conventions).* Let $\mathfrak{C}_{\mathrm{BKM}}$ be the Igusa/BKM datum of Definition 11.20 (iii).

(i) *Divergence.* The Igusa/BKM convention datum consists of the Jacobi form φ_{Jac}, Borcherds lattice Λ_{Borch}, Weyl chamber W, character ρ, Hall character $\mathrm{ch}_{\mathrm{Hall}}$, orientation line ℓ_{or}, primitive recognition map π_{prim}, and DT/PT/OP normalization $\tau_{\mathrm{DT/PT/OP}}$. These are entries of $\mathfrak{C}_{\mathrm{BKM}}$. For the Δ_5 branch, the constants

$$D_5 = 64^{-1}\Delta_5, \qquad \mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z),$$

$$\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 10/2 = 5, \qquad Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2},$$

live in the BKM/Borcherds comparison for the Igusa cusp-form construction; they fix the convention datum at which the admissibility theorem must match. The local Hamiltonian theorem on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ contributes the formal-Darboux stalk.

(ii) *Obstruction class.* The obstruction class is the Igusa/BKM Borcherds match class

$$\mathrm{ob}_{\mathrm{Igusa/BKM}} \in H^1(\mathfrak{K}_{\mathrm{KD}})$$

computed in the BKM matched-conventions complex of $\mathfrak{C}_{\mathrm{BKM}}$; it includes the orientation comparison $\epsilon_0 = \nu_{\Delta_5}$, compact Hall constants, and primitive BKM recognition. These are target obstructions in the BKM comparison datum.

(iii) *Matched-conventions criterion.* If $\mathrm{ob}_{\mathrm{Igusa/BKM}} = 0$ with a chosen null-homotopy and the shared coordinates vanish, then the target inherits the formal-Darboux component as in Theorem 11.13. The Δ_5 constants, Borcherds-product factorization, BKM weights, and modular invariance remain compact data.

Proof. Clauses (i)–(ii) read off Definition 11.20 (iii). Clause (iii) is the Igusa/BKM case of Theorem 11.13. □

THEOREM 11.24 *(GW / DT / PT / MNOP divergence and matched-conventions).* Let $\mathfrak{C}_{\mathrm{GW/DT}}$ be the GW/DT/PT/MNOP datum of Definition 11.20 (v).

(i) *Divergence.* The GW/DT/PT/MNOP convention datum consists of the compact CY_3 Y, a curve class $\beta \in H_2(Y, \mathbb{Z})$, the four generating series $F^{\mathrm{GW}}_{g,\beta}$, $Z^{\mathrm{DT}}_{n,\beta}$, $Z^{\mathrm{PT}}_{n,\beta}$, the virtual framing $\mathfrak{f}_{\mathrm{vir}}$, Behrend's microlocal weight ν_{Behr}, the virtual trace pairing τ_{vir}, and the MNOP correspondence map Φ_{MNOP}. These are entries of $\mathfrak{C}_{\mathrm{GW/DT}}$. The local theorem has the formal polydisk $\widehat{\mathbb{C}^2}_0$ and carries only formal holomorphic coordinates.

(ii) *Obstruction class.* The obstruction class is

$$\mathrm{ob}_{\mathrm{GW/DT}} \in H^1(\mathfrak{K}_{\mathrm{KD}})$$

attached to compatibility of the formal-Darboux stalk with the virtual GW/DT/PT/MNOP normalization at every brane formal-normal chart where $\mathfrak{C}_{\mathrm{GW/DT}}$ restricts to $\mathfrak{C}^{(2)}_{\mathrm{loc}}$.

(iii) *Matched-conventions criterion.* If $\mathrm{ob}_{\mathrm{GW/DT}} = 0$ with a chosen null-homotopy and the shared coordinates vanish, then the target inherits the formal-Darboux component as in Theorem 11.13. The virtual cycle, MNOP correspondence, and curve-class generating series remain compact data.

Proof. Clauses (i)–(ii) read off Definition 11.20 (v). Clause (iii) is the GW/DT/PT/MNOP case of Theorem 11.13. □

THEOREM 11.25 *(OSV attractor divergence and matched-conventions).* Let $\mathfrak{C}_{\mathrm{OSV}}$ be the OSV datum of Definition 11.20 (vi).

(i) *Divergence.* The OSV convention datum consists of the compact CY_3 Y, the attractor lattice Λ_{att}, the black-hole partition $Z^{\mathrm{BH}}_{Q,P}$, the attractor entropy $e^{-S_{\mathrm{att}}}$, and the OSV identification $\Phi_{\mathrm{OSV}}\colon Z^{\mathrm{BH}} \sim |Z^{\mathrm{top}}|^2$. The OSV identification is a target conjecture on a compact CY_3 and is an entry of $\mathfrak{C}_{\mathrm{OSV}}$. The local theorem has formal holomorphic factor $\widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$.

(ii) *Obstruction class.* The obstruction class is

$$\mathrm{ob}_{\mathrm{OSV}} \in H^1(\mathfrak{K}_{\mathrm{KD}})$$

attached to compatibility of the formal-Darboux stalk with the attractor identification, when an OSV target invokes the formal-Darboux component.

(iii) *Matched-conventions criterion.* If $\mathrm{ob}_{\mathrm{OSV}} = 0$ with a chosen null-homotopy and the shared coordinates vanish, then the target inherits the formal-Darboux component as in Theorem 11.13. Black-hole attractor counts and the OSV identification remain target data.

Proof. Clauses (i)–(ii) read off Definition 11.20 (vi). Clause (iii) is the OSV case of Theorem 11.13. □

Theorem 11.26 *(Mixed-Dunn divergence and matched-conventions).* Let $\mathfrak{C}_{\mathrm{MD}}$ be the mixed-Dunn datum of Definition 11.20 (vii). The local theorem carries the holomorphic E_2-type factorization algebra on $\mathbb{C}^2$ (the formal symplectic surface, with bidegree two in the holomorphic direction), the topological factorization algebra on $\mathbb{R}_{\mathrm{brane}}$ (one topological direction along the brane line), and a chosen brane-line interval factorization map $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ on product disks $D \times B \subset \mathbb{R} \times \mathbb{C}^2$.

(i) *Divergence.* The local product-disk site $\mathrm{Ran}_{\prod}(\mathbb{R}, \mathbb{C}^2)$ is generated by product disks $D \times B$; product-disk covers form a proper subsite of arbitrary Weiss covers of $\mathbb{R} \times \mathbb{C}^2$, and the local theorem leaves descent on the unrestricted Weiss site $\mathrm{Ran}(\mathbb{R} \times \mathbb{C}^2)$ to the unrestricted Weiss/Ran target. A target using arbitrary Weiss/Ran covers, the operadic interchange $\delta_{m,n} : \mathbb{E}^{\mathrm{top}}_m \otimes \mathbb{E}^{\mathrm{hol}}_n \to \mathbb{E}^{\mathrm{mix}}_{m,n}$ with (m, n) different from the local $(1, 2)$, or a strict mixed-operadic rectification, has these structures as components of $\mathfrak{C}_{\mathrm{MD}}$. The local theorem contributes only the product-disk $(1, 2)$ formal-stalk component.

(ii) *Obstruction class.* The obstruction class is the ordered pair

$$\big(\mathrm{ob}_{6r},\ \mathrm{ob}_{\mathrm{MD}}\big) \in H^2\big(\mathrm{Der}^{\mathrm{fact,mix}}(\mathcal{A}_R) \otimes I\big) \oplus H^1(\mathfrak{D}_{\mathrm{MD}}(m, n)).$$

The first coordinate is the six-relation deformation obstruction of Proposition 11.15; the second is the operadic deformation curvature of Definition 11.16. Higher operadic obstructions live in $H^{r+1}(\mathrm{gr}^r\, \mathfrak{D}_{\mathrm{MD}}(m, n))$.

(iii) *Matched-conventions criterion.* If both classes vanish with chosen null-homotopies and the shared coordinates vanish, then by Theorem 11.13 the target inherits the formal-Darboux component as a strict mixed-operadic factorization map. Operadic rectification at arities other than the local $(1, 2)$ and Weiss/Ran descent outside the product-disk site remain compact data.

Proof. Clauses (i)–(ii) read off Definitions 11.20 (vii), 11.14, and 11.16. Clause (iii) is the mixed-Dunn case of Theorem 11.13, combined with Theorem 11.17. □

Theorem 11.27 *(Global Hamiltonian descent divergence and matched-conventions).* Let $\mathfrak{C}_{\mathrm{glob}}$ be the global descent datum of Definition 11.20 (iv).

(i) *Divergence.* The local theorem is stated on the formal Darboux chart $\widehat{\mathbb{C}^2}_0$; the global descent datum $\mathfrak{C}_{\mathrm{glob}}$ lives on a compact connected holomorphic-symplectic surface (X, ω) with brane locus L, a Stein cover $\mathfrak{U}$, the de Rham extension $0 \to \mathbb{C}_X \to \mathcal{O}_X \to \Omega^1_{X,\mathrm{cl}} \to 0$, and the total complex $\mathfrak{G}_{\mathrm{WR}}(L)$ of Definition 11.9. These structures are the global entries of $\mathfrak{C}_{\mathrm{glob}}$; the formal polydisk supplies only the Darboux restriction.

(ii) *Obstruction class.* The obstruction class is the global descent vector

$$\mathrm{Ob}_{\mathrm{glob}}(X, L, \Phi_{\mathrm{loc}}) \in H^1(X, \mathbb{C}_X) \oplus H^1(\mathfrak{G}_{\mathrm{WR}}(L)) \oplus \bigoplus_{r \geq 1} H^{r+1}\big(\mathrm{gr}^r\, \mathfrak{G}_{\mathrm{WR}}(L)\big) \oplus H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}),$$

with first coordinate the period $\mathrm{per}_X(v)$ – sharp by Theorem 11.10 on a compact connected surface – the second coordinate the first Weiss/Ran descent class, the third the higher coherence classes, and the fourth the scalar anomaly compatibility class.

(iii) *Matched-conventions criterion.* Vanishing of $\mathrm{Ob}_{\mathrm{glob}}$ with chosen null-homotopies is equivalent to a global Hamiltonian central-operation morphism on L whose Darboux restrictions are the local maps Φ_{loc}, by Theorem 11.12.

Proof. Clauses (i)–(iii) are Definition 11.20 (iv) and Theorem 11.12 read as a divergence and a matched-conventions criterion. □

The three Tate-P residuals form a sequential matched-conventions criterion. The Hadamard / Mittag-Leffler class $\mathrm{Ob}_w^{\mathrm{HML}}$ of §7.5 controls weighted summation on the weighted Tate envelope; the ambient-restriction class $\mathrm{Ob}_{p,X}^{\mathrm{univ}}$ of §5.6 records whether an ambient twisted-M sector restricts to the same formal local model; the matched-conventions class $\mathrm{Ob}_{\mathrm{comparison}}(\mathfrak{C}_{\mathrm{tar}})$ records the compact comparison class for use of that formal local model by a compact, modular, automorphic, virtual-cycle, or operadic theorem. After the compact deformation complex and comparison maps are constructed, a compact theorem uses the local Hamiltonian formal-stalk component exactly when three classes vanish in sequence: $\mathrm{Ob}_w^{\mathrm{HML}}$ at the analytic regulator, $\mathrm{ob}_{p,X}^{\mathrm{univ}}$ at the ambient-to-formal restriction, and $\mathrm{Ob}_{\mathrm{comparison}}$ at the formal-to-target matched-conventions comparison. A nonvanishing class in any component is the exact obstruction.

The closing identification. The trace map $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is the formal Darboux coordinate of the trace-sector closed-to-open map at one Dirac brane vacuum; for $f \in \mathfrak{h}$ it is read in the completed brane algebra. On the stable scalar-reduced trace algebra the CE/PV dictionary identifies this trace sector with the corresponding Hochschild polyvectors. The mixed holomorphic–topological Deligne conjecture is the full derived-centre identification with the bulk $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_{b_p}$, whose arity-two local singular product is

$$\mathsf{J}_x(z)\mathsf{J}_y(w) \sim K_\Delta^{(2)}(z, w)\mathsf{J}_{[x,y]_{\mathfrak{g}}}(w),$$

generalising the curve-level chiral operation only after controlled restriction. The global identification at $d = 2$, the operator-level lift of the modular line on $K3 \times E$, and the chiral E_d Deligne theorem at $d \geq 2$ are controlled by the three residual obstruction classes.

ACKNOWLEDGEMENTS

This work belongs to the lineage of Bershadsky–Cecotti–Ooguri–Vafa 1993, Witten 1988, Costello 2011, and Beilinson–Drinfeld 2004. The Capelli–Howe scalar identities, Procesi–Razmyslov trace relations, and Harish-Chandra radial-parts theorem give the formal-stalk-chart side; Matlis 1958 and Hartshorne local cohomology give the principal-part dual; Kontsevich's deformation quantisation gives the all-order Moyal target.

A Higher factorization categories and chiral derived centres at $d \geq 2$

The local theorem (Theorem 5.6) is the formal-Darboux fibre of a higher open–closed problem. At the chosen Dirac brane vacuum $b_p \in C_\partial^{\mathrm{op}}$, the Hamiltonian BF bulk gives Chevalley–Eilenberg cochains, the boundary brane gives the E_1-Hochschild complex of $A^{\mathrm{cl}}_{\partial,N}$, and the trace map $J(f) = \operatorname{Tr} f(\phi_1, \phi_2)$ gives the coordinate closed-to-open morphism. The Mixed Holomorphic–Topological Deligne conjecture asserts that, before choosing the vacuum, this local Hochschild chart is the restriction of the chiral derived centre of the primitive open boundary category at $d = 2$.

The higher factorization-categorical setting consists of holomorphic factorization algebras, mixed holomorphic–topological factorization algebras, chiral derived centre objects for $d \geq 2$, and the formal-Darboux restriction functor. At $d = 1$ this language reduces to Beilinson–Drinfeld chiral algebras on curves [28]. At $d = 2$, the formal-Darboux trace theorem gives the trace-sector stalk and the arity-two holomorphic collision given by the diagonal local-cohomology singular product of Theorem 1.17; the all-arity chiral E_2-Deligne structure is the global extension problem.

A.1 Factorization E_d-algebras on smooth complex d-folds

Let X be a smooth complex variety of complex dimension d. Write $\operatorname{Ran}(X)$ for the Ran space of finite subsets of X. Let $\mathrm{Mfd}^{\mathrm{lc}}(X)$ be the symmetric monoidal ∞-category whose objects are open subsets $U \subset X$ and whose monoidal product is disjoint union of subsets with disjoint closures.

Definition A.1 *(Holomorphic factorization algebra).* A *holomorphic factorization algebra* on X is a Costello–Gwilliam prefactorization algebra valued in cochain complexes, with holomorphic locality data and Weiss descent. It assigns a cochain complex $U \mapsto \mathcal{F}(U)$ to each open $U \subset X$, structure maps

$$\mathcal{F}(U_1) \otimes \cdots \otimes \mathcal{F}(U_n) \longrightarrow \mathcal{F}(V)$$

for pairwise disjoint opens $U_i \subset V$, and extension maps for inclusions, satisfying the factorization axioms of Costello–Gwilliam [15, Vol. I, Defn. 3.1.4]. The holomorphic condition means that the local cochain models are Dolbeault resolutions and the structure maps commute with the holomorphic differential. An E_d-algebra object in the category of such factorization algebras is extra structure.

Remark A.2 *(Comparison presentations).* A holomorphic factorization algebra on X has three comparison languages [28][62][41, §5.5]:

(i) *Costello–Gwilliam.* A factorization algebra in the sense of [15, Vol. I, §3], with the holomorphic locality differential.

(ii) *Factorisable $\mathcal{D}$-module on* $\operatorname{Ran}(X)$. A factorisable $\mathcal{D}$-module $\mathcal{A}$ on $\operatorname{Ran}(X)$. At $d = 1$ this is the algebro-geometric construction of chiral algebras on curves [28, §3]. At $d \geq 2$ the corresponding Ran presentation is an additional comparison datum.

(iii) *Lurie–Ayala–Francis.* An E_d-algebra object of the ∞-category of factorisation algebras on X in the sense of [63]*lurie-ha*§5.5.4.

At $d = 1$ these presentations reduce to chiral algebras on curves in the algebro-geometric sense of [28, §3]. For $d \geq 2$ the Costello–Gwilliam holomorphic factorization-algebra presentation is the definition of the closed-string sector classical observables. The Ran $\mathcal{D}$-module comparison and the Lurie–Ayala–Francis comparison are hypotheses in the global chiral E_d-Deligne problem isolated in §11.1.

Definition A.3 *(Category of holomorphic factorization algebras).* Let $\mathrm{HolFA}_d(X)$ denote the symmetric monoidal ∞-category whose objects are holomorphic factorization algebras on X in the sense of

Definition A.1, and whose morphisms are natural transformations of prefactorization diagrams that intertwine the factorization structure maps. Its monoidal product is the pointwise tensor product of cochain complexes. An E_d-algebra object of $\mathrm{HolFA}_d(X)$ is extra structure.

A.2 Mixed holomorphic-topological refinement

The local geometry is $X = \mathbb{R}^k_{\mathrm{top}} \times M_{\mathrm{hol}}$ with M a smooth complex variety of complex dimension d ($k = 2$, $d = 2$ in the body). The factorization algebra carries three independent locality differentials: support locality on $\mathbb{R}^k$, holomorphic locality on M, and topological / locally constant in $\mathbb{R}^k$.

Definition A.4 *(Mixed-HT factorization E_d^{HT}-algebra).* A *mixed-HT factorization E_d^{HT}-algebra* on $X = \mathbb{R}^k_{\mathrm{top}} \times M_{\mathrm{hol}}$ is a Costello–Gwilliam prefactorization algebra on X, valued in cochain complexes, satisfying Weiss descent and carrying the three locality data:

(i) *Support locality on* $\mathbb{R}^k$. For any compact embedding $K \subset U \subset \mathbb{R}^k$, the section $\mathcal{F}$ is supported on K when restricted to $U \setminus K$ extends to zero.

(ii) *Topological locality on* $\mathbb{R}^k$. For any open $U \subset \mathbb{R}^k$ and any inclusion $U_0 \subset U$ with $\pi_0(U_0) \to \pi_0(U)$ a bijection, the structure map $\mathcal{F}(U_0 \times B) \to \mathcal{F}(U \times B)$ is a quasi-isomorphism for every $B \subset M$.

(iii) *Holomorphic locality on* M. Restriction to the holomorphic factor is a holomorphic factorization algebra in the sense of Definition A.1.

An E_d^{HT}-operadic action on such a factorization algebra is additional structure.

Definition A.5 *(Category* $\mathrm{ChirAlg}^{\mathrm{HT}}_{k,d}(X)$*).* Let $\mathrm{ChirAlg}^{\mathrm{HT}}_{k,d}(X)$ denote the symmetric monoidal ∞-category whose objects are mixed-HT factorization E_d^{HT}-algebras on X in the sense of Definition A.4, and whose morphisms intertwine all three locality structures. At $k = 0$ this reduces to $\mathrm{HolFA}_d(M)$; at $d = 0$ it reduces to topological factorization algebras on $\mathbb{R}^k$. The formal-Darboux local geometry is $k = 2$, $d = 2$.

A.3 The chiral derived centre functor

Definition A.6 *(Chiral derived centre at $d = 1$).* For a chiral algebra $\mathcal{A} \in \mathrm{HolFA}_1(C)$ on a smooth curve C, the *chiral derived centre* $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ is the chiral Hochschild cochain complex

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}) := \mathrm{RHom}_{\mathcal{A} \otimes \mathcal{A}^{\mathrm{op}}}(\mathcal{A}, \mathcal{A})$$

in the chiral monoidal structure of [28, §3.4]. It carries a natural E_2-algebra structure (chiral Deligne theorem at $d = 1$; [28, §4], Lurie HA [41, Thm. 5.3.1.14], Costello–Gwilliam [15, Vol. I, §6]). This is the proved $d = 1$ chiral Deligne theorem of Vol I.

Definition A.7 *(Chiral derived-centre model at $d \geq 2$).* For a holomorphic factorization algebra $\mathcal{A} \in \mathrm{HolFA}_d(X)$ on a smooth complex variety X of complex dimension $d \geq 2$, the *chiral derived-centre model at d* is the endomorphism object

$$Z^{\mathrm{der}}_{E_d^{\mathrm{ch}}}(\mathcal{A}) := \mathrm{RHom}_{\mathcal{A} \otimes \mathcal{A}^{\mathrm{op}}}(\mathcal{A}, \mathcal{A})$$

in a chosen ∞-category of factorization E_d-algebra bimodules on X. The notation records the chiral centre; the higher chiral monoidal structure and its E_{d+1}^{ch} operations are extra data, not part of this definition. At $d = 1$ this construction reduces to Definition A.6.

The obstruction at $d \geq 2$. The existence of an E^{ch}_{d+1}-structure on $Z^{\mathrm{der}}_{E^{\mathrm{ch}}_d}(\mathcal{A})$ extending the E_2-structure of the $d = 1$ case is precisely the datum asserted by the chiral E_d-Deligne conjecture (§11.1; Lurie [41, §5.3.1]); partial constructions are due to Hennion (chiral Lie algebroids on smooth schemes) and Francis (factorisable co-operads). The Mixed Holomorphic–Topological Deligne conjecture at $d = 2$ is the corresponding statement for $\mathrm{ChirAlg}^{\mathrm{HT}}_{2,2}(\mathbb{R}^2 \times \mathbb{C}^2)$.

Definition A.8 *(Mixed-HT chiral derived-centre model).* For $\mathcal{F} \in \mathrm{ChirAlg}^{\mathrm{HT}}_{k,d}(X)$, the *mixed-HT chiral derived-centre model* is

$$Z^{\mathrm{der}}_{E^{\mathrm{HT}}_d}(\mathcal{F}) = C^{\bullet}_{\mathrm{ch,HT}}(\mathcal{F}, \mathcal{F}),$$

the RHom of $\mathcal{F}$ with itself in the bimodule ∞-category over $\mathcal{F} \otimes \mathcal{F}^{\mathrm{op}}$ inside $\mathrm{ChirAlg}^{\mathrm{HT}}_{k,d}(X)$. For $d \geq 2$ the mixed-HT E_{d+1}-structure on this complex is an additional chiral E_d-Deligne datum.

A.4 The formal-Darboux stalk functor

For $X = \mathbb{R}^k_{\mathrm{top}} \times \mathbb{C}^d_{\mathrm{hol}}$ and a closed point $p \in \mathbb{C}^d$, the formal-Darboux stalk operation is restriction to the formal completion at p.

Definition A.9 *(Formal-Darboux stalk functor).* The *formal-Darboux stalk functor* at p is the symmetric monoidal ∞-functor

$$(-)|_p \colon \mathrm{ChirAlg}^{\mathrm{HT}}_{k,d}(\mathbb{R}^k \times \mathbb{C}^d) \longrightarrow \mathrm{ChirAlg}^{\mathrm{HT}}_{k,d}(\mathbb{R}^k \times \widehat{\mathbb{C}^d}_p),$$

sending $\mathcal{F} \mapsto \mathcal{F}|_{\mathbb{R}^k \times \widehat{\mathbb{C}^d}_p}$, the restriction of the factorisable structure to the formal neighbourhood $\widehat{\mathbb{C}^d}_p = \mathrm{Spf}\, \widehat{\mathcal{O}}_{\mathbb{C}^d,p}$. The holomorphic Darboux theorem trivialises the germ of $\mathbb{C}^d$ at p to the formal symplectic polydisk in coordinates $(z_1, \dots, z_d)$, so the value at p depends on p only through its formal isomorphism class — equivalently, only through the formal Darboux coordinates.

Proposition A.10 *(Equivariance under formal symplectomorphism).* The formal-Darboux stalk functor of Definition A.9 is equivariant under the formal symplectomorphism group $\mathrm{Symp}^{\mathrm{form}}(\mathbb{C}^d)$ acting on the source category by formal coordinate change at p and on the target by the same change of formal Darboux coordinates. At $d = 2$ on the holomorphic-symplectic factor, the stalk is canonical up to formal Darboux change.

Proof. The Darboux theorem in formal coordinates trivialises any holomorphic symplectic form to $\omega_p = \sum_i dz_{2i-1} \wedge dz_{2i}$, with the trivialisation defined uniquely up to the formal symplectomorphism group. The factorisation structure pulls back along this trivialisation by symmetric monoidal naturality. □

A.5 The bulk closed-string sector as an E^{HT}_2-factorization algebra

Construction A.11 *(Bulk classical observables on* $\mathbb{R}^2 \times \mathbb{C}^2$*).* Let $X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ and $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1]$ as in §2. The closed-string sector classical observables factorization algebra $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \in \mathrm{ChirAlg}^{\mathrm{HT}}_{2,2}(X)$ is the assignment

$$U \times B \longmapsto C^{\bullet}_{\mathrm{CE,cont}}\big(\Omega^{\bullet}_c(U) \widehat{\otimes} \Omega^{0,\bullet}_c(B) \widehat{\otimes} \mathfrak{g}\big),$$

for $U \subset \mathbb{R}^2$ and $B \subset \mathbb{C}^2$ open, with E_2-algebra structure from the symmetric algebra structure of the continuous CE cochains and factorisation isomorphisms inherited from extension-by-zero on disjoint opens. The three locality differentials of Definition A.4 are the de Rham differential on U, the Dolbeault differential $\bar{\partial}$ on B, and the internal CE differential dual to the $\mathfrak{g}$-bracket.

PROPOSITION A.12 *(Formal-Darboux stalk of the bulk).* At a closed point $p \in \mathbb{C}^2$, the formal-Darboux stalk of $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ is the continuous CE cochain algebra of the shifted-cotangent BF Lie algebra at the formal polydisk:

$$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_p = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}), \qquad \mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1], \quad \mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1.$$

The formal coordinate algebra $\widehat{\mathcal{O}}_{\mathbb{C}^2,p}$ gives the Hamiltonian labels $\mathfrak{h} = \widehat{\mathcal{O}}_{\mathbb{C}^2,p}/\mathbb{C} \cdot 1$; it is not the closed CE cochain algebra and it is not the brane endomorphism algebra. The arity-two singular operation on the closed stalk is the native diagonal singular product

$$\mathsf{J}_x(z)\mathsf{J}_y(w) \sim K^{(2)}_\Delta(z,w)\mathsf{J}_{[x,y]_\mathfrak{g}}(w), \qquad x, y \in \mathfrak{g},$$

with

$$K^{(2)}_\Delta = \frac{d\zeta_1 d\zeta_2}{\zeta_1\zeta_2}, \qquad \zeta_i = z_i - w_i.$$

Proof. Apply Definition A.9 to Construction A.11: the de Rham and Dolbeault Poincaré contractions on the formal polydisk reduce the compactly supported sheaf data to the continuous CE cochains of $\mathfrak{g}$ at p. Extension by zero gives the regular product on disjoint formal polydisks. When two formal insertions collide, the Dolbeault resolution of the diagonal leaves the local-cohomology quotient

$$H^2_\Delta(\mathbb{C}[[z_1, z_2, w_1, w_2]])\, d\zeta_1 d\zeta_2,$$

and the CE bracket of $\mathfrak{g}$ gives exactly the singular product of Theorem 1.17. Thus the formal completion retains the diagonal collision operation even though its disjoint-open product is the extension-by-zero product. □

A.6 The brane open category as factorization E_1^{HT}-module

Definition A.13 *(Boundary open category).* The *boundary open category* of the local theory is $\mathcal{C}^{\mathrm{op}}_\partial$, the category of boundary conditions on the brane-link boundary ∂X of the Dirac brane defect $L = \mathbb{R}_t \times \{p\} \subset X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ — the link sphere bundle over the worldline $\mathbb{R}_t$, equivalently the boundary of an excised tubular neighbourhood $X \setminus \nu(L)$. The bulk factorization algebra acts on $\mathcal{C}^{\mathrm{op}}_\partial$ by the bulk–boundary OPE. A module-category model for this action is extra structure, not the definition of the open category.

Construction A.14 *(Formal-Darboux vacuum and brane E_1-algebra).* The *formal-Darboux vacuum* at the closed point $p \in \mathbb{C}^2$ is an object $b_p \in \mathcal{C}^{\mathrm{op}}_\partial$ equipped with the formal coordinate chart $\widehat{\mathcal{O}}_{\mathbb{C}^2,p} \cong \mathbb{C}[[z_1, z_2]]$. The chart gives the substitution $z_i \mapsto \phi_i$. The derived endomorphism algebra of a skyscraper brane is the corresponding Ext–Koszul algebra, not the formal function ring itself. At N Dirac branes over b_p the brane E_1-algebra of classical observables on the worldline $\mathbb{R}_t$ is

$$A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}\big(\mathfrak{gl}_N,\ \mathrm{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])\big), \qquad Q\psi = [\phi_1, \phi_2],$$

with derived Chan–Paton substitution $z_i \mapsto \phi_i$ and Koszul differential $Q\psi = [\phi_1, \phi_2]$. On H^0 this substitution factors through the commuting-pair locus (Definition 1.1). Curve chiral algebras such as affine Kac–Moody, $\beta\gamma$, Virasoro, and $\mathcal{W}$-algebras enter through a curve restriction and anomaly matching before comparison with this formal disk.

Proposition A.15 *(Brane E_1-Hochschild as E_2-algebra).* The derived E_1-centre $Z^{\mathrm{der}}_{E_1}(A^{\mathrm{cl}}_{\partial,N})$ is naturally an E_2-algebra, equivalent to the Hochschild cochain complex $C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$; its cohomology is $HH^\bullet(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$ with Gerstenhaber bracket and cup product. This is Deligne's E_1-Hochschild theorem (Tamarkin [59], McClure–Smith [60], Kontsevich–Soibelman [61], Lurie [41, Thm. 5.3.1.14]).

A.7 The Mixed Holomorphic–Topological Deligne conjecture as a functorial statement

Conjecture A.16 *(Mixed Holomorphic–Topological Deligne, functorial form).* For the bulk closed-string sector $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \in \mathrm{ChirAlg}^{\mathrm{HT}}_{2,2}(\mathbb{R}^2 \times \mathbb{C}^2)$ of Construction A.11 and the boundary open category $\mathcal{C}^{\mathrm{op}}_\partial$ of Definition A.13, there is a canonical equivalence

$$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \;\simeq\; Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}(\mathcal{C}^{\mathrm{op}}_\partial)$$

of mixed-HT chiral algebras at $d = 2$, where the right-hand side is the Morita derived chiral centre of the open category. After choosing $b \in \mathcal{C}^{\mathrm{op}}_\partial$ this centre has the Hochschild chart $C^\bullet_{\mathrm{ch}}(A_b, A_b)$; the chart is not a definition of the category-level centre.

Known cases.

- At $d = 1$ (replacing $\mathbb{C}^2$ by a smooth curve C): proved (chiral Deligne theorem at $d = 1$, Vol I [64]; [28]).
- At $d \geq 2$ the datum is the chiral E_d-Deligne conjecture in the sense of Lurie [41, §5.3.1]. Its mixed-HT form is the holomorphic-topological enhancement. Partial constructions are recorded in Hennion's chiral Lie algebroids on smooth schemes and Francis's factorisable co-operads.
- *Formal-Darboux stalk:* the trace-sector map

  $$\Phi_N : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \dashrightarrow C^\bullet_{\mathrm{Hoch}}(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}), \qquad c_f \mapsto \theta_f, \quad u_f \mapsto J(f),$$

  at N Dirac branes over the formal-Darboux vacuum b_p. On the stable scalar-reduced trace algebra this is the Chevalley–Eilenberg/polyvector coordinate isomorphism after the admissible Tate/pro-Matlis completion. The whole finite-N derived-centre equivalence is governed by Hochschild centrality homotopies, finite-trace relations, and descent/QME hypotheses recorded below. The formal-coordinate arity-two singular product is computed by Theorem 1.17.

Theorem A.17 *(Formal-Darboux trace-sector stalk of Conjecture A.16).* At N Dirac branes over the formal-Darboux vacuum $b_p \in \mathcal{C}^{\mathrm{op}}_\partial$, the formal-Darboux stalk of Conjecture A.16 contains the trace-sector morphism

$$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_p \simeq C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \dashrightarrow C^\bullet_{\mathrm{Hoch}}\big(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N}\big)$$

with formal Darboux coordinate expression $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ on polynomials, extended to $\mathfrak{h} = \widehat{\mathcal{O}}_{\mathbb{C}^2,p}/\mathbb{C} \cdot 1$ only after formal completion, and reduced closed-to-open dictionary $c_f \mapsto \theta_f \in C^1$, $u_f \mapsto J(f) \cdot (-) \in C^0$ on Chevalley–Eilenberg generators of $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$. After scalar reduction and passage to the stable trace algebra this map is the admissible CE/PV coordinate isomorphism. For fixed finite N it is not an equivalence onto the whole Hochschild complex. The arity-two formal diagonal operation at the stalk is the singular product of Theorem 1.17. The chain-level lift Φ_N to a strict P_0-factorization-centre morphism exists under compatible null-homotopies of $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$ (Theorem E.53).

Proof. Combine Proposition A.12 (formal-Darboux stalk of the bulk), Proposition A.15 (Deligne's E_1-Hochschild theorem for the brane E_1-algebra), and the trace-sector closed-to-open construction of Theorem 5.6. The trace-map computation is Theorem 5.147 (Procesi–Razmyslov + Loday–Quillen–Tsygan); the dictionary is Theorem 4.20; the arity-two diagonal operation is Theorem 1.17; the chain-level lift with the stated cotangent hypotheses is Theorem E.53. □

Remark A.18 *(Formal-Darboux stalk and global conjecture).* Theorem A.17 computes the trace-sector formal-Darboux fibre of Conjecture A.16 at the formal-Darboux vacuum b_p. The global equivalence on $\mathbb{R}^2 \times \mathbb{C}^2$ between the bulk $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}$ and the mixed-HT chiral derived centre of the boundary open category at $d = 2$ is Conjecture A.16. The $d = 2$ chiral E_d-Deligne obstruction is $\mathrm{Ob}^{\mathrm{ch}\,E_2}_{\mathrm{Deligne}}$ (§11.1). Identification with any curve chiral algebra requires a curve-restriction theorem and matched-conventions transfer (Theorem 11.13).

A.8 The chiral E_d-operad: holomorphic little disks with principal-parts data

The functorial Mixed-HT Deligne conjecture (Conjecture A.16) requires an operadic action: the chiral derived centre $Z^{\mathrm{der}}_{E^{\mathrm{ch}}_d}(\mathcal{A})$ is the endomorphism object in a chiral E_d-module category. At $d = 1$ the chiral operad on curves gives this structure [28, §3.4]. At $d \geq 2$ the operad is controlled by higher-codimension diagonal local cohomology. The coefficient system controlling the formal-Darboux diagonal kernel has, at $d = 2$, the arity-two coefficient computed in Theorem 1.17. All-arity operadic composition is the all-arity hypothesis.

Definition A.19 *(Configuration $\mathcal{D}$-module).* Let X be a smooth complex variety of complex dimension d and I a non-empty finite set. Write X^I for the $|I|$-fold product and $j_I : \mathrm{Conf}_I(X) \hookrightarrow X^I$ for the open embedding of the configuration space of $|I|$ distinct points. The *configuration $\mathcal{D}$-module*

$$\omega^{\mathrm{ch}}_{X^I} = j_{I,*} j_I^! \omega_{X^I}$$

is the !-extension to X^I of the dualising sheaf of $\mathrm{Conf}_I(X)$.

Definition A.20 *(Chiral E_d coefficient system).* The *chiral E_d coefficient system* on $\mathbb{C}^d$ is the collection $\mathsf{hE}^{\mathrm{ch}}_d$ with single colour and coefficient object at arity $|I|$

$$\mathsf{hE}^{\mathrm{ch}}_d(I) = \omega^{\mathrm{ch}}_{(\mathbb{C}^d)^I},$$

and with composition given by the chiral product on !-pullback along the diagonal embedding $\Delta_A \colon (\mathbb{C}^d)^A \hookrightarrow (\mathbb{C}^d)^I$ for a partition $I = \bigsqcup_{a \in A} I_a$, following [28, §3.4] as the formal higher-codimension Ran-space analogue. At $d \geq 2$ the all-arity associativity and action on mixed-HT factorization algebras are additional hypotheses. The theorem proved here is the $d = 2$, arity-two diagonal local-cohomology coefficient.

Remark A.21 *(Comparison with curves at $d = 1$).* For $d = 1$ on a curve C, the configuration $\mathcal{D}$-module $\omega^{\mathrm{ch}}_{C^I}$ is the chiral operad on curves of [28, §3.4]. At $d \geq 2$ the diagonals $\Delta_{ij} \subset (\mathbb{C}^d)^I$ have complex codimension d, so the principal parts on $\omega^{\mathrm{ch}}_{(\mathbb{C}^d)^I}$ involve distributions of d-fold polar order; this is the higher-codimension chiral structure specific to $d \geq 2$.

Remark A.22 *(Arity two at $d = 2$).* For $X = \mathbb{C}^2$ and $I = \{1, 2\}$, the singular quotient of $\omega^{\mathrm{ch}}_{X^I}$ along the pairwise diagonal is

$$H^2_\Delta(\mathbb{C}[[z_1, z_2, w_1, w_2]])\, d\zeta_1 d\zeta_2 = \bigoplus_{a,b \geq 0} \mathbb{C}[[w_1, w_2]]\, \rho^\Delta_{a,b}.$$

The generator $\rho^{\Delta}_{0,0} = K^{(2)}_{\Delta}$ is the two-dimensional Cauchy kernel. For the Hamiltonian BF algebra, the associated binary operation sends $x \otimes y$ to $[x, y]_{\mathfrak{g}}$, giving the formal singular product of Theorem 1.17.

Construction A.23 *($\mathsf{hE}^{\mathrm{ch}}_d$ operation maps under the operadic action).* Assume the chiral E_d-Deligne operadic datum of §11.1 is supplied on $\mathrm{HolFA}_d(\mathbb{C}^d)$. Then a holomorphic factorization E_d-algebra $\mathcal{A} \in \mathrm{HolFA}_d(\mathbb{C}^d)$ (Definition A.1) carries structure maps

$$\mu^{\mathrm{ch}}_I : \mathsf{hE}^{\mathrm{ch}}_d(I) \otimes^! \mathcal{A}^{\boxtimes I}_{\mathrm{Ran}} \longrightarrow \Delta^*_I \mathcal{A}_{\mathrm{Ran}}$$

intertwining the chiral product of Definition A.20 with Δ^*_I, where $\mathcal{A}_{\mathrm{Ran}}$ denotes the corresponding Ran-space factorization object. At $d = 1$ these maps are the chiral operations on curves. At $d = 2$ and arity two, the formal-Darboux stalk coefficient is computed directly by Theorem 1.17.

Remark A.24 *(Operation maps).* The Costello–Gwilliam and Lurie–Ayala–Francis factorization formalisms give the symmetric monoidal factorization category and its factorization-homology presentation. The chiral E_d-Deligne equivalence at $d \geq 2$ is the all-arity theorem. Construction A.23 is only the global model; the proved local theorem is the arity-two diagonal local-cohomology operation of Theorem 1.17.

Construction A.25 *(Mixed-HT enhancement under the operadic action $\mathsf{hE}^{\mathrm{ch,HT}}_d$).* Adjoin to the system $\mathsf{hE}^{\mathrm{ch}}_d$ the topological E_k-coefficients on $\mathbb{R}^k_{\mathrm{top}}$. The resulting mixed-HT coefficient system $\mathsf{hE}^{\mathrm{ch,HT}}_d$ has arity-I coefficient

$$\mathsf{hE}^{\mathrm{ch,HT}}_d(I) = \mathrm{Conf}_I(\mathbb{R}^k) \times \omega^{\mathrm{ch}}_{(\mathbb{C}^d)^I},$$

with composition the product of topological little-disks composition on the $\mathbb{R}^k$ factor and the chiral composition of Definition A.20 on the $\mathbb{C}^d$ factor. An action of this system on a mixed-HT factorization algebra is an all-arity operadic structure.

A.9 Three hypotheses for chiral E_d Deligne at $d \geq 2$

The chiral E_d Deligne problem at $d \geq 2$ decomposes here into three obstruction classes for the coefficient system $\mathsf{hE}^{\mathrm{ch}}_d$ of Definition A.20. The arity-two diagonal kernel gives one operation; the all-arity operad, its Ran descent, and the formality morphism are distinct classes.

Definition A.26 *(Hypothesis F: formality of $\mathsf{hE}^{\mathrm{ch}}_d$).* The system $\mathsf{hE}^{\mathrm{ch}}_d$ is first promoted to an all-arity ∞-operad, and that operad is formal: there is a quasi-isomorphism

$$\mathsf{hE}^{\mathrm{ch}}_d \;\simeq\; H_*(\mathsf{hE}^{\mathrm{ch}}_d)$$

to its homology operad in chain complexes of $\mathcal{D}_{\mathrm{Ran}(\mathbb{C}^d)}$-modules. At $d = 1$ this is Tamarkin's formality theorem [59][61]; at $d \geq 2$ the obstruction is the deformation class $\mathrm{Ob}_F = [\mathfrak{m}_d]$.

Definition A.27 *(Hypothesis C: chiral centrality).* For $\mathcal{A} \in \mathrm{HolFA}_d(\mathbb{C}^d)$, there is a chiral bimodule ∞-category $\mathcal{A} \otimes \mathcal{A}^{\mathrm{op}}\text{-}\mathrm{Mod}^{\mathrm{ch}}$ and an endomorphism object

$$Z^{\mathrm{der}}_{\mathsf{hE}^{\mathrm{ch}}_d}(\mathcal{A}) = \mathrm{RHom}_{\mathcal{A} \otimes \mathcal{A}^{\mathrm{op}}}(\mathcal{A}, \mathcal{A}).$$

This object acts on $\mathcal{A}$ by homotopy-central chiral operations compatible with factorization products. At $d = 1$ this is the chiral-centre construction for chiral algebras on curves. For $d \geq 2$ its topological analogue is Lurie's E_d-centre construction [41, §5.3.1]; the holomorphic chiral enhancement is governed by the corresponding operadic obstruction problem.

Definition A.28 *(Hypothesis R: Ran !-descent).* There is a chosen Ran $\mathcal{D}$-module presentation $\mathcal{A}_{\mathrm{Ran}}$ of the Costello–Gwilliam factorization algebra $\mathcal{A}$, and $\mathcal{A}_{\mathrm{Ran}}$ satisfies !-descent on the Ran site. Equivalently, the comparison from the Costello–Gwilliam Weiss descent model to the Ran !-descent model is a quasi-isomorphism in the selected coefficient category. At $d = 1$ this is the Beilinson–Drinfeld factorization theorem for chiral algebras on curves [28, §3]. At $d \geq 2$ it is a hypothesis of the global chiral E_d-Deligne theorem.

THEOREM A.29 *(Chiral E_d Deligne under formality and Ran descent).* Under Hypotheses A.26, A.27, A.28, the chiral derived centre satisfies

$$Z^{\mathrm{der}}_{\mathsf{hE}^{\mathrm{ch}}_d}(\mathcal{A}) \;\simeq\; C^{\bullet}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$$

as E^{ch}_{d+1}-algebras. Hypotheses A.27 and A.28 are the chiral enhancements of known topological and curve-level theorems; at $d \geq 2$ they remain hypotheses of the chiral E_d-Deligne statement together with Hypothesis A.26.

Proof sketch. Hypothesis A.27 gives the bimodule category and identifies the centre with the stated RHom. The bar resolution computes this RHom only after that bimodule category is present. Hypothesis A.28 identifies the Costello–Gwilliam and Ran models in which the chiral Hochschild complex is computed. Hypothesis A.26 gives the all-arity E^{ch}_{d+1}-operations. The conclusion is the composition of these three structures. □

Remark A.30 *(Obstruction class for Hypothesis A.26).* Hypothesis A.26 is the assertion that the Maurer–Cartan element $\mathfrak{m}_d \in \mathrm{Def}\big(\mathsf{hE}^{\mathrm{ch}}_d \to H_*(\mathsf{hE}^{\mathrm{ch}}_d)\big)$ classifying the chiral E_d-operad relative to its homology operad is gauge-equivalent to zero. At $d = 1$ Tamarkin [59] proves the corresponding formality statement by associator methods. At $d \geq 2$ the corresponding chiral formality theorem is an independent datum. The class

$$\mathrm{Ob}_F = [\mathfrak{m}_d] \in H^*\big(\mathrm{Def}(\mathsf{hE}^{\mathrm{ch}}_d)\big)$$

is the formality obstruction at $d \geq 2$.

Remark A.31 *(Chiral E_d obstruction).* The global chiral E_d-Deligne hypothesis at $d \geq 2$ consists of the operadic action of Definition A.20 together with Hypotheses A.26, A.27, and A.28. Its obstruction is the class

$$\mathrm{Ob}_{\mathrm{ch}E_d} = \big(\mathrm{Ob}_F, \mathrm{Ob}_C, \mathrm{Ob}_R\big),$$

where Ob_C and Ob_R measure the obstruction to the chiral centrality and Ran-descent hypotheses. The formal-Darboux stalk theorem computes the arity-two $d = 2$ local operation on which any global theorem restricts; vanishing of $\mathrm{Ob}_{\mathrm{ch}E_d}$ is the additional global condition.

A.10 Descent over the Costello–Gwilliam brane-link boundary

The boundary open category $C^{\mathrm{op}}_{\partial}$ (Definition A.13) is defined globally on the Costello–Gwilliam brane-link boundary ∂X. This subsection records the Weiss descent datum for local boundary categories. A global mixed-HT Deligne equivalence requires this descent datum together with the chiral E_d, quantum master equation, and comparison-cone data.

Definition A.32 *(Costello–Gwilliam brane-link bundle).* For $X = \mathbb{R}^2_{\mathrm{top}} \times Y_{\mathrm{hol}}$ a mixed-HT spacetime and $L \subset X$ a Dirac brane defect, the *Costello–Gwilliam brane-link boundary* ∂X is the boundary of the excised tubular neighbourhood $\nu(L)$ of L in X [15, Vol. II],

$$\partial X \;=\; \partial(X \setminus \nu(L)) \;\cong\; L \times S^{c-1}_{\epsilon},$$

the unit sphere bundle of codimension $c = \operatorname{codim}_X L$ over L. At $L = \mathbb{R}_t \times \{p\} \subset \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ the codimension is $c = 5$ and $\partial X \cong \mathbb{R}_t \times S^4_{\epsilon}$.

Definition A.33 *(Boundary-category Weiss diagram on ∂X).* The boundary open category is locally encoded by a Weiss descent diagram $\underline{\mathcal{C}^{\mathrm{op}}_{\partial}}$ of mixed-HT factorization categories on ∂X: for $U \subset \partial X$ open,

$$\underline{\mathcal{C}^{\mathrm{op}}_{\partial}}(U) = \mathcal{C}^{\mathrm{op}}_{\partial}(U),$$

the category of boundary conditions supported on U, together with the restricted bulk–boundary action. A module-category model over $\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_U$ is an additional presentation. At a closed point $p \in \partial X$ the formal fibre is the formal-Darboux stalk of Definition A.9.

THEOREM A.34 *(Weiss descent criterion for the boundary open category).* Assume the boundary category is presented by a Costello–Gwilliam boundary factorization category and satisfies the Weiss cosheaf condition. Then, for any Weiss cover $\{U_a \subset \partial X\}$, the homotopy-colimit descent map

$$\operatorname{hocolim}\bigl(\textstyle\coprod_a \underline{\mathcal{C}^{\mathrm{op}}_{\partial}}(U_a) \rightrightarrows \coprod_{a,b} \underline{\mathcal{C}^{\mathrm{op}}_{\partial}}(U_a \cap U_b) \rightrightarrows \cdots\bigr) \xrightarrow{\sim} \underline{\mathcal{C}^{\mathrm{op}}_{\partial}}(\partial X)$$

is an equivalence.

Proof. This is the Weiss cosheaf axiom for Costello–Gwilliam factorization categories on a real manifold, applied to the brane-link boundary. The mixed-HT labels and the bulk–boundary action restrict to opens; compatibility with the displayed homotopy-colimit map is part of the descent datum. □

Definition A.35 *(Descent obstruction complex).* Choose a Weiss cover $\{U_a \subset \partial X\}_{a \in A}$ trivialising the local formal-Darboux structure. Let $\mathcal{E}^{\bullet}$ be the sheaf of cochain complexes governing homotopies of the local trace-sector closed-to-open maps on overlaps. The first descent obstruction is the Čech class

$$\mathrm{Ob}_{\mathrm{desc}} \in \check{H}^2(\{U_a\}; \mathcal{E}^{-1}),$$

defined by the failure of the overlap homotopies to satisfy the triple-overlap cocycle equation. Higher obstructions live in the successive Čech cohomology groups of the same obstruction complex.

THEOREM A.36 *(Descent obstruction at the formal-Darboux strata).* If the class $\mathrm{Ob}_{\mathrm{desc}}$ of Definition A.35 is represented and killed by a compatible Čech null-homotopy, then the trace-sector formal stalks glue across the first descent morphism of the Costello–Gwilliam brane-link boundary. The global equivalence

$$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \simeq Z^{\mathrm{der}}_{E_2^{\mathrm{HT}}}\bigl(\mathcal{C}^{\mathrm{op}}_{\partial}\bigr)$$

still requires the remaining chiral E_d, quantum master equation, and global comparison obstructions to vanish.

Proof. Theorem A.17 gives the trace-sector map on each formal-Darboux chart. On double overlaps the chosen comparison homotopies form a Čech 1-cochain in $\mathcal{E}^{\bullet}$. The failure of the triple-overlap equation is the Čech 2-cocycle $\mathrm{Ob}_{\mathrm{desc}}$. A null-homotopy of this class gives the first gluing datum. Identification of the global bulk with the global chiral derived centre requires the remaining obstruction classes and all higher descent obstructions to vanish. □

A.11 All-order QME obstruction criterion for finite-jet PVA

The classical PVA Jacobi identity holds at the formal-Darboux stalk (Theorem 4.20) and is the Jacobi identity for the native two-dimensional singular product of Theorem 1.17. Its quantum lift is a recursive quantum master equation in the Costello local-functional complex. At order $\hbar^n$ the obstruction lies in the cohomology of the quantum BV differential with scalar-contact projection, curved bulk-to-defect kernel, and compatible counterterms; it is not killed by the classical Jacobi identity. The hypotheses below name the exact additional data needed to transport this recursion to the trace-sector centre.

Convention. Under the binding convention of Conventions and notation for the local theorem, the symbol $C^\bullet_{\mathrm{ch}}(A_b, A_b)$ in this subsection denotes the brane-side Hochschild target receiving the trace-sector map Φ_N of Theorem 5.6. A full identification with the bulk stalk is the admissible centre-lift problem. Q_{ch} is the Hochschild differential on this target, transported to the trace summand only where Φ_N is an admissible quasi-isomorphism. The hypotheses below are formulated on that target.

Definition A.37 *(Hypothesis K: filtered formal SDR).* There is a pronilpotent filtered strong deformation retract

$$\left(C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}), Q_{\mathrm{tot}}\right) \underset{\text{retract}}{\xrightarrow{\text{deform}}} \left(C^\bullet_{\mathrm{ch}}(A_b, A_b), Q_{\mathrm{ch}}\right)$$

over $\mathbb{C}[[\hbar]]$, compatible with the scalar-contact filtration, finite PVA weight windows, and the admissible Hochschild centrality homotopies. This is additional homological datum, not a consequence of the formal-Darboux stalk theorem.

Remark A.38 *(Formal and analytic transfer).* The classical PVA Jacobi holds in the $\hbar$-adic topology of the formal disk. Costello's renormalization theorem gives, after a renormalization scheme has been fixed, a finite-order obstruction calculus. A formal SDR transfers the obstruction classes. It does not prove analytic convergence. Sectorial analytic convergence is the extra content of Hypothesis A.39 together with explicit graph estimates.

Definition A.39 *(Hypothesis S: sectorial Stokes regularity).* If an analytic refinement is sought, the renormalized graph sums and the transferred homological perturbation data form a sheaf over sectors $0 < |\hbar| < r$. On the sectors meeting the real-positive $\hbar$-axis the Stokes automorphisms act trivially on the chosen trace-sector image, and the sectorial lifts have the same asymptotic $\hbar$-adic series. No KZ connection is assumed unless it is constructed as part of this analytic sheaf.

Definition A.40 *(Hypothesis W: finite-window weight symmetry).* Under crossing $z \leftrightarrow w$ on the formal E_2 singular product of Theorem 1.17, the PVA weight grading on $C^\bullet_{\mathrm{ch}}(A_b, A_b)$ reflects: the operator

$$R\colon \mathrm{wt}_z \otimes \mathrm{wt}_w \longrightarrow \mathrm{wt}_w \otimes \mathrm{wt}_z$$

intertwines the arity-two singular product with itself in each finite weight window. This is S_2-equivariance of the binary kernel; it is not reflection positivity and it does not construct the all-arity chiral E_2 action.

Definition A.41 *(Hypothesis G: $T = [Q_{\mathrm{tot}}, G]$).* In the Costello local-functional complex the holomorphic stress tensor is quantum-BV exact modulo contact terms:

$$T = d^L_{\mathrm{QME}} G.$$

The primitive G is a degree-(-1) local functional compatible with the same filtration and scalar-contact projection used in the QME complex. Transport to Hochschild cochains is allowed only after the admissible quantum closed-to-open map has been constructed.

Theorem A.42 *(All-order QME obstruction criterion for the trace-sector assignment).* Assume Hypotheses A.37, A.40, and A.41. Assume also the brane-defect Costello local-functional complex with d^L_{QME}-stable scalar-contact projection, finite-window residual complexes, curved bulk-to-defect kernel, compatible local counterterms, zero Milnor/Roos obstruction, and quantum centrality homotopies for the centre lift. If the Costello obstruction classes vanish in every loop order with compatible primitives, the formal-Darboux stalk of Theorem A.17 admits a formal all-order quantization: there exists a quantum BV master action $S^{\mathrm{quant}} = S^{\mathrm{cl}} + \sum_{n\geq 1} \hbar^n S^{(n)}$ satisfying

$$QS^{\mathrm{quant}} + \frac{1}{2}\{S^{\mathrm{quant}}, S^{\mathrm{quant}}\}_L + \hbar\Delta_L S^{\mathrm{quant}} = 0$$

to all orders in $\hbar$, and a quantum trace-sector closed-to-open factorization-centre assignment

$$\Phi^\hbar_N : \mathcal{A}^{\mathrm{q,tr}}_{\mathrm{bulk}}|_p \longrightarrow Z^{P_0,\hbar}_{\mathrm{fact}}(\mathrm{Obs}^\hbar_{\mathrm{open}})|_b$$

whose classical limit is the chosen lift of Φ_N. After a quantum boundary A_∞-algebra $A^\hbar_{\partial,N}$ has been chosen, this target may be modelled by

$$C^\bullet_{\mathrm{Hoch,adm}}(A^\hbar_{\partial,N}, A^\hbar_{\partial,N}).$$

The trace map $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is defined on polynomials and extended to formal f after completion; its quantum correction has the form $J^{\mathrm{quant}}(f) = J(f) + \sum_n \hbar^n J^{(n)}(f)$. The assignment is a morphism of Hochschild complexes only in the chosen Hochschild model and precisely after the quantum equivariance homotopies are chosen. Descent to Hochschild cohomology and an all-order derived-centre equivalence require the additional Hochschild equivariance, centre-lift, and descent data of Definition 5.8.

Proof. Set $S^{(0)} = S^{\mathrm{cl}}$. At order n, after $S^{(1)}, \ldots, S^{(n-1)}$ have been chosen, the order-n part of the quantum master equation has the form

$$d^L_{\mathrm{cl}} S^{(n)} = -\frac{1}{2} \sum_{\substack{i+j=n \\ i,j<n}} \{S^{(i)}, S^{(j)}\}_L - \Delta_L S^{(n-1)}, \qquad d^L_{\mathrm{cl}} = Q + \{S^{(0)}[L], -\}_L.$$

The right-hand side is d^L_{cl}-closed by the Jacobi identity for the BV bracket and by $\Delta_L^2 = 0$. Its cohomology class is the Costello obstruction at loop order n. By hypothesis this class vanishes and a compatible primitive is chosen; induction constructs $S^{(n)}$ for every n.

The scalar-contact projection, finite-window residual complexes, curved bulk-to-defect kernel, compatible counterterms, and zero Milnor/Roos class supply the primitives in the brane-defect tower. Hypothesis A.37 transfers the formal tower to the trace-sector centre. Hypothesis A.40 is the finite-window S_2-compatibility of the binary kernel. Hypothesis A.41 supplies the local-functional homotopy for the stress tensor. Hypothesis A.39, when imposed, gives sectorial analytic compatibility; it is not used to create a formal primitive. □

Remark A.43 *(QME obstruction class).* Before comparison maps to a common local-functional complex are fixed, the QME obstruction classes form the tuple

$$\mathrm{Ob}_{\mathrm{QME}} = \big([\mathfrak{ob}_K], [\mathfrak{ob}_S], [\mathfrak{ob}_W], [\mathfrak{ob}_G]\big),$$

with entries the formal-SDR obstruction $[\mathfrak{ob}_K]$, the sectorial Stokes class $[\mathfrak{ob}_S]$, the binary weight-symmetry class $[\mathfrak{ob}_W]$, and the stress-tensor-exactness class $[\mathfrak{ob}_G]$. It becomes a single class in $H^*(\mathrm{Loc}_{\mathbb{R}^2\times\mathbb{C}^2})$ only after these entries have been represented in the same Costello local-functional deformation complex. At fixed loop order, only the algebraic finite-window entries are checked in finite PVA weight windows. Analytic convergence requires the sectorial estimates of Hypothesis A.39.

A.12 Operator-level lift of the Igusa section and its square-root branch

The Igusa cusp form Φ_{10} is the level-2 section of the modular line Ω_{central} on the Siegel period domain $\mathfrak{H}_2$ [31][32]. The square root $\Delta_5 = \Phi_{10}^{1/2}$ is a scalar modular form in the compact $K3 \times E$ comparison (§11.3). The formal-Darboux theorem constructs the local Capelli scalar. An operator-level lift is an operator-valued cocycle whose protected trace is Φ_{10} on the Siegel period domain, or Δ_5 after the square-root cover has been chosen, in the compact-target theory.

Conjecture A.44 *(Operator lift of the Igusa modular section).* There exists a graded vertex-algebraic or factorization-algebraic object V, an involution or real form σ, and a protected trace such that

$$\mathrm{Tr}_V\big(\sigma\, q^{L_0 - c/24} \prod_{i=1}^{2} z_i^{H_i}\big) \;=\; S(\tau, \rho, \nu), \qquad S \in \{\Phi_{10},\, \Delta_5 = \Phi_{10}^{1/2}\}$$

with $S = \Delta_5$ only on the square-root cover of the Siegel period domain $(\tau, \rho, \nu) \in \mathfrak{H}_2$, with the trace compatible with the modular line Ω_{central}. The required object is the operator-valued cocycle lifting this scalar modular section.

Scalar automorphy does not determine an operator-valued cocycle. The missing datum is a vertex algebra or a factorization algebra with a protected trace.

Construction A.45 *(Borcherds–Kac–Moody lift obstruction).* The Borcherds–Kac–Moody denominator formulas give automorphic products, including the fake-Monster denominator and the Gritsenko–Nikulin products [66][32]. These are scalar identities. Passing from a scalar denominator formula to a vertex-algebraic object whose graded trace is Δ_5 requires an additional construction.

Obstruction. $\mathrm{Ob}_{\mathrm{BKM}}$ is the obstruction to lifting the scalar Borcherds product to an operator trace with the required integral form, real structure, and modular covariance.

Construction A.46 *(Free-field realisation on $K3 \times E$).* The elliptic genus of $K3$ [30], its second-quantised product, and the Donaldson–Thomas / Pandharipande–Thomas comparison on $K3 \times E$ [67] give scalar modular data. The obstruction measures the failure to construct a free-field or factorization algebra on $K3 \times E$ whose protected trace is Δ_5.

Obstruction. Ob_{K3E} is the obstruction to constructing the compact target factorization object, its protected trace, and the descent of that trace to the level-2 modular line.

Construction A.47 *(Dabholkar–Murthy–Zagier lift obstruction).* The Dabholkar–Murthy–Zagier decomposition of meromorphic Jacobi forms separates polar and finite parts in the $1/\Phi_{10}$ dyonic counting problem [68]. This is a statement about mock modular and wall-crossing structures. An operator construction of Δ_5 is the lift to the modular-line cocycle.

Obstruction. $\mathrm{Ob}_{\mathrm{DMZ}}$ is the obstruction to lifting the mock-modular wall-crossing structure to an honest operator trace with the required holomorphic modular transformation law.

Remark A.48 *(Composition of operator-lift obstructions).* The scalar theories determine modular forms in the same comparison class, and the operator-valued cocycle is the common lifting datum. The symbol

Ob_V denotes the direct sum of the named obstruction complexes above. A cohomological product formula for these scalar obstruction classes requires a product on the direct-sum obstruction complex.

A.13 Obstruction taxonomy

The local theorem and the obstruction classes assemble into the global Mixed Holomorphic–Topological Deligne criterion.

Theorem A.49 *(Global Mixed-HT Deligne criterion).* Under the joint vanishing

$$\mathrm{Ob}_{\mathrm{global}} := \left(\mathrm{Ob}_{\mathrm{ch}E_d}, \mathrm{Ob}_{\mathrm{desc}}, \mathrm{Ob}_{\mathrm{QME}}\right) = 0,$$

with the three summands as in Remark A.31, Theorem A.36, and Remark A.43, the Mixed-HT Deligne conjecture A.16 holds globally on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ at $d = 2$, N Dirac branes, provided the construction hypotheses named in those theorems hold. The operator-level lift of Φ_{10} or $\Phi_{10}^{1/2}$ belongs to the $K3 \times E$ comparison problem.

Proof. Compose Theorem A.29 (chiral E_d Deligne via the operadic action and Hypotheses A.26–A.28), Theorem A.36 (Costello–Gwilliam brane-link descent via Definition A.35), Theorem A.42 (all-order QME obstruction criterion via Hypotheses A.37, A.40, and A.41, with Hypothesis A.39 only for analytic refinement). The three obstructions are independent cohomology classes living in distinct cohomology groups; their tuple becomes a single curvature class only after a common deformation complex and comparison maps have been constructed. □

Remark A.50 *(Reduction to obstruction classes).* The obstruction $\mathrm{Ob}_{\mathrm{QME}}$ is the all-order quantum master-equation obstruction for lifting the finite-jet PVA across all $\hbar$-orders. The chiral E_d Deligne obstruction $\mathrm{Ob}_{\mathrm{ch}E_d}$ at $d \geq 2$ is the obstruction class of Conjecture 11.1; partial results (chiral Lie algebroids on smooth schemes via Hennion, factorisable co-operads via Francis) give adjacent structures; the global theorem is Conjecture A.16. The brane-link descent obstruction $\mathrm{Ob}_{\mathrm{desc}}$ is the Čech class of Definition A.35. The formal-Darboux trace theorem (Theorem 5.6) is the formal-Darboux stalk that anchors the local side of these obstructions, and Theorem 1.17 gives its arity-two formal diagonal operation. Each relative theorem has the stalk theorem, including this diagonal singular product, as its base case.

Construction A.51 *(Global obstruction class).* The global criterion (Theorem A.49) translates the compact-comparison obstructions of §11 into cohomology classes in distinct source complexes:

- $\mathrm{Ob}_{\mathrm{ch}E_d}$: existence, centrality, Ran-descent, and formality for the chiral E_d-operadic structure at $d \geq 2$; its formality summand is the associator-type operadic homotopy class on $\mathrm{Conf}_I(\mathbb{C}^d)$, $|I| \geq 2$.
- $\mathrm{Ob}_{\mathrm{desc}}$: null-homotopy of the Čech class in the descent obstruction complex (Definition A.35) on the formal-Darboux stratum.
- $\mathrm{Ob}_{\mathrm{QME}}$: each of four checkable cohomology classes (Hypotheses A.37–A.41).

The formal-Darboux stalk theorem (Theorem 5.6) is the local trace-sector fibre. The operator lift of the modular line on $K3 \times E$ is governed by the compact-target comparison complex, rather than by a summand of $\mathrm{Ob}_{\mathrm{global}}$.

B Factorization current conventions for the CE/PV interface

The brane-line factorization-current sector of the local mixed HT string field theory on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ is the interval-wise CE/PV interface between Hamiltonian BF currents and boundary Hochschild cochains. Its obstruction surfaces are the classes that block the lift from the algebraic current model to a Costello local-functional graph theorem [12][15][16]. The BV degree, sign, completion, and obstruction conventions used by the body are fixed in this interface.

The interface is one universal triangle of brackets. On a brane interval $I \subset \mathbb{R}_{\mathrm{brane}}$, the Hamiltonian boundary observable $B_f(a)$ (degree 0) and the residual cotangent observable $\Theta_\rho(B)$ (degree 1) carry the open P_0-brackets

$$\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab), \qquad \{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0,$$

where $\{f, g\}$ is the Poisson bracket of the formal holomorphic symplectic polydisk $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$, $f \cdot \rho$ is its Matlis residue coadjoint action of Proposition 8.7, and aB is the multiplication of a smooth compactly supported test function by a compactly supported distributional current. The CE source coordinates are c_λ (degree 1) and $u_{a(t)dt\otimes f}$ (degree 0), with the universal generator dictionary

$$c_\lambda \longmapsto \theta_\lambda = \Theta_\rho(B), \qquad u_{a(t)dt\otimes f} \longmapsto B_f(a), \qquad \lambda = B \otimes \rho.$$

This real-current triangle is the interval contraction of the native two-dimensional holomorphic singular product computed in Theorem 1.17. Before smearing along the brane line, the holomorphic singular coefficient is

$$K_\Delta^{(2)}(z, w) = \frac{d\zeta_1 d\zeta_2}{\zeta_1\zeta_2}, \qquad \zeta_i = z_i - w_i,$$

and the displayed three brackets are the $(0, 0)$-mode products after pairing with the interval currents. The reduced algebraic current target is the freely adjoined completed graded-commutative algebra $A_\partial^{\mathrm{cur}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1])$ on these generators. All four bracket signs and the displayed generator dictionary are forced by the symplectic form $\Omega_{\mathrm{loc}} = dz_1 \wedge dz_2$, the residue pairing $\langle \rho_{a,b}, z_1^m z_2^n\rangle = \delta_{a,m}\delta_{b,n}$, and the equal-time bracket of Remark 4.10; the displayed signs match Lemmas C.2–C.9 of Appendix C verbatim.

Three notions of locality, formal apparatus. The body proves three independent locality statements in Proposition 1.36: support locality ($\delta(t - t')$ on the brane line, with (t, t') two brane-time points and s reserved for the transverse $\mathbb{R}_s$ coordinate of X), topological locality ($L_v = [Q, \iota_v]$ in the $\mathbb{R}^2_{\mathrm{top}}$ factor), and formal-holomorphic locality ($L_{\partial/\partial\bar{z}_i} = [Q, \iota_{\partial/\partial\bar{z}_i}]$ in the $\widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ factor). The formal apparatus is:

(i) *Support locality* is encoded by the cosheaf Lie-algebra $\mathfrak{h}_I = \Omega_c^\bullet(I)\widehat{\otimes}\mathfrak{h}$ and its current dual $\mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}$ of Definitions B.1, B.2. Multiplication of a current by a smooth compactly supported test function (Lemma B.4) is the only distributional operation used in the reduced current model; for disjoint supports it kills the bracket and produces the strict factorization product. The smeared form ab of the equal-time contact $\delta(t - t')$ of Remark 4.10 is the same operation read on the source side.

(ii) *Topological locality* is encoded by the de Rham differential $d_I = d_{\mathrm{dR}} \otimes 1$ on $\Omega_c^\bullet(I)$ and by the compact-support density shift of Lemma 4.8: the surviving cohomology along the brane line is the density line, so the factorization algebra is locally constant in the topological direction. The interval factor of every current operator below is a compactly supported de Rham object; topological motion changes it by a d_I-boundary.

(iii) *Formal-holomorphic locality* is encoded by the residue pairing $\langle \rho, f\rangle = \mathrm{Res}_0(\rho f)$ on the formal holomorphic polydisk $(\widehat{\mathbb{C}^2}_0, \Omega_{\mathrm{loc}})$ (Proposition B.3), realized through the Matlis principal-part module $\mathcal{P}$ of § 8. This produces the Taylor Hamiltonian source side and its residue-dual principal-part current target side. The polynomial cotangent realization is a different label system

(Corollary 5.173, Corollary C.8). For two holomorphic points the same residue formalism is the diagonal local-cohomology module $H^2_\Delta(\mathbb{C}[[z,w]])\, d\zeta_1 d\zeta_2$ of Theorem 1.17.

Each of the three localities is independent: the three operators $\delta(t-t')$, L_v, $L_{\partial/\partial\bar{z}_i}$ live in different directions and none implies the others. Convention agreement is therefore part of the factorization-current datum.

Prefactorization versus factorization. The structure is prefactorization in the sense of Costello–Gwilliam [15, Ch. 3]: cosheaf source on intervals, completed graded-commutative algebra, extension-by-zero factorization product, and strict vanishing of mixed brackets between disjoint-support generators. Weiss-cover descent and the passage to a factorization algebra in the Weiss-cover sense is proved by Theorems 4.11 and 5.186, which interpret the prefactorization datum as a factorization algebra on the brane line under the locality hypotheses of Proposition 1.36. The prefactorization datum is convention-compatible; the Weiss-cover extension is the content of the cited theorems. The native holomorphic singular product on $\mathbb{C}^2$ is already the formal diagonal calculation of Theorem 1.17; the Weiss-cover issue is the real-current and unreduced graph extension of that structure.

Convention table. Every factorization-current identity uses the following degree/sign/topology conventions. They match Appendix C verbatim and are repeated here because they fix every factorization-current statement.

(BV degrees) $|c_\lambda| = 1$, $|u_{a(t)dt\otimes f}| = 0$, $|\theta_\lambda| = |\Theta_\rho(B)| = 1$, $|B_f(a)| = 0$, $|\gamma_1| = 1$. The source CE differential d_{CE}, the target Schouten differential d_π, and the classical local-functional differential $d_{\mathrm{cl}} = Q + \{I_0, -\}$ have degree $+1$. In the finite-scale Costello complex the corresponding quantum BV differential is $d^L_{\mathrm{qBV}} = Q + \{I[L], -\}_L + \hbar\Delta_L$. The open P_0-bracket on the reduced current target has degree 0; the bulk-to-defect convolution curvature $\mathrm{Curv}_{\mathbf{K}}$ has degree $+1$; the BV Laplacian Δ has degree $+1$ wherever it is invoked.

(Hamiltonian) $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$, $\Omega_{\mathrm{loc}} = dz_1 \wedge dz_2$, $\iota_{X_f}\Omega_{\mathrm{loc}} = -df$, $\{f,g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$, $\{z_1, z_2\} = 1$, $\mathrm{div}_{\Omega_{\mathrm{loc}}} X_f = 0$. The Moyal first quantum correction has coefficient $1/24$ (Lemma C.6).

(Residue pairing) $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$, $a, b \geq 0$, $(a,b) \neq (0,0)$; residue normalization $\langle \rho_{a,b}, z_1^m z_2^n\rangle = \mathrm{Res}_0(\rho_{a,b} z_1^m z_2^n) = \delta_{a,m}\delta_{b,n}$; coadjoint sign $(f\cdot\rho)(g) = -\rho(\{f,g\})$; resulting monomial action $z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1, b-q+1}$. This is the universal cotangent normalization of Corollary C.8.

(Brane-line currents) $I \subset \mathbb{R}_{\mathrm{brane}} \subset \mathbb{R}^2_{\mathrm{top}}$ is an oriented open interval; $\Omega^\bullet_c(I)$ is the de Rham complex of compactly supported smooth forms (degree ≤ 1 on a 1-manifold); $\mathcal{D}'_c(I)$ is the space of compactly supported distributions in the duality $\mathcal{D}'(I) = C^\infty(I)^\vee$ of de Rham 0-currents ([36, Ch. 6]); the de Rham differential is $d_I = d_{\mathrm{dR}} \otimes 1$. The only distributional operation on the reduced current model is multiplication $B \mapsto aB$ of a current by a smooth compactly supported test function (Lemma B.4); products of two distributions occur only inside the explicitly flagged Costello graph complexes of Propositions B.17–B.20.

V generator dictionary) $c_\lambda \longmapsto \theta_\lambda$, $u_{a(t)dt\otimes f} \longmapsto B_f(a)$, with $\lambda = B\otimes\rho \in \mathcal{P}_I$; the symbol θ_λ is interchangeable with $\Theta_\rho(B)$, where the upper-case form is the boundary-observable physics-side notation and the lower-case form is the PV/CE-image side coordinate. Both are degree 1 and abelian on the cotangent current ideal.

sional singular product) The interval brackets are the support-local contraction of the two-dimensional holomorphic singular product of Theorem 1.17. In unsmeared holomorphic variables $z, w \in \mathbb{C}^2$,

$$\mathsf{B}_f(z)\mathsf{B}_g(w) \sim K^{(2)}_{\Delta}(z,w)\mathsf{B}_{\{f,g\}}(w), \qquad \mathsf{B}_f(z)\Theta_\rho(w) \sim K^{(2)}_{\Delta}(z,w)\Theta_{f\cdot\rho}(w),$$

and $\Theta_\rho(z)\Theta_\sigma(w) \sim 0$. The brane-line formulas above are obtained by pairing this diagonal coefficient with compactly supported interval currents and keeping the $(0,0)$ singular product.

ology and completions) All completed symmetric algebras are interpreted degreewise in the chosen current category: finite symmetric powers are formed before completion, and brackets extend continuously from the displayed length-one current formulas; tensor products are completed projective tensor products of nuclear Fréchet spaces in the de Rham/distribution direction and $\mathfrak{m}$-adic in the formal Hamiltonian direction (§ 8).

The factorization-current theorem uses the following algebraic source/target interface. The interface is used before the admissible bar-cobar/enveloping criterion: interval Hamiltonian currents form the CE source, distributional Matlis principal parts form the cotangent current target, and the CE/PV generator dictionary is the one fixed by the convention table above. The proved content is the reduced algebraic current model after de Rham-current contraction. The unreduced BV/QME current lift and the Costello brane-defect graph theorem are separate obstruction problems, isolated in Remark 5.190, Appendix E, and Problem 5.213. The geometry throughout is the formal-local mixed HT model on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$; the interval I is an oriented topological current interval in the brane line, and the Hamiltonian labels live on the formal holomorphic symplectic polydisk $(\widehat{\mathbb{C}^2}_0, \Omega_{\mathrm{loc}})$.

Definition B.1 *(Interval Hamiltonian current algebra).* For an open interval $I \subset \mathbb{R}$, set

$$\mathfrak{h}_I = \Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h},$$

with differential $d_I = d_{\mathrm{dR}} \otimes 1$. Here

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$$

is the Hamiltonian Lie algebra of the formal holomorphic symplectic disk $(\widehat{\mathbb{C}^2}_0, \Omega_{\mathrm{loc}})$, concentrated in degree 0. The graded bracket is

$$[\alpha \otimes f, \beta \otimes g] = \alpha \wedge \beta \otimes \{f, g\}, \qquad \alpha, \beta \in \Omega^\bullet_c(I),\ f, g \in \mathfrak{h}.$$

For degree-zero test functions $a, b \in \Omega^0_c(I)$,

$$[a \otimes f, b \otimes g] = ab \otimes \{f, g\}.$$

The cosheaf structure for inclusions of intervals is extension by zero on $\Omega^\bullet_c$, tensored with the identity on $\mathfrak{h}$. Polynomial formulas use the dense scalar-reduced subalgebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1 \subset \mathfrak{h}$. For $j : I \hookrightarrow J$, extension by zero is smooth because every element of $\Omega^\bullet_c(I)$ has support compactly contained in I. On CE cochains the induced map is the dual coproduct/pullback; the covariant factorization structure is the compact-support current target.

The notation $a(t)dt \otimes f$ is the oriented density label paired with distributional cotangent currents. The Hamiltonian generator in $\mathfrak{h}_I$ remains $a \otimes f$.

Definition B.2 *(Distributional Matlis current dual).* Let

$$\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C}\cdot 1) \cong \mathfrak{h}^{\vee}_{\mathrm{cont}}$$

be the reduced Matlis principal-part module of Section 8. Its interval current form is

$$\mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}.$$

This is the selected support-local current dual used after contraction. The full strong dual of $\Omega^{\bullet}_c(I)\widehat{\otimes}\mathfrak{h}$ is a separate functional-analytic object. The pairing with degree-zero Hamiltonian currents is

$$\langle B\otimes\rho,\ a\otimes f\rangle = B(a)\,\mathrm{Res}(\rho f), \qquad B\in\mathcal{D}'_c(I),\ \rho\in\mathcal{P}.$$

The residue is the formal residue determined by $dz_1\wedge dz_2$ on $\widehat{\mathbb{C}^2}_0$. If $B = b(t)dt$ is a regular compactly supported density, then

$$B(a) = \int_I a(t)b(t)\,dt.$$

The shifted cotangent interval Lie algebra is

$$\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1],$$

where $\mathfrak{h}_I$ acts on $\mathcal{P}_I$ by the product of the $C^\infty_c(I)$-module operation $B\mapsto aB$ on compactly supported currents and the Matlis coadjoint action on $\mathcal{P}$. On degree-zero labels this action is

$$(a\otimes f)\cdot(B\otimes\rho) = (aB)\otimes(f\cdot\rho).$$

PROPOSITION B.3 *(Local unimodular datum for the current kernels).* The current and kernel constructions rest on the following formal-local unimodular datum:

$$\left(\widehat{\mathbb{C}^2}_0, \Omega_{\mathrm{loc}} = dz_1\wedge dz_2\right).$$

This datum gives:

(i) the Poisson bracket on $\mathfrak{h} = \mathbb{C}[[z_1,z_2]]/\mathbb{C}\cdot 1$;

(ii) the Hamiltonian vector-field convention

$$\iota_{X_f}\Omega_{\mathrm{loc}} = -df, \qquad X_f = (\partial_{z_1}f)\partial_{z_2} - (\partial_{z_2}f)\partial_{z_1};$$

(iii) the divergence-free identity $\mathrm{div}_{\Omega_{\mathrm{loc}}} X_f = 0$, hence the residue integration-by-parts formula defining the Matlis coadjoint action;

(iv) the principal-part residue pairing

$$\mathrm{Res}_0\left(z_1^{-i-1}z_2^{-j-1}dz_1dz_2\ z_1^m z_2^n\right) = \delta_{i,m}\delta_{j,n}.$$

(v) the diagonal two-point principal-part module

$$H^2_\Delta(\mathbb{C}[[z_1,z_2,w_1,w_2]])\,d\zeta_1 d\zeta_2$$

and its Cauchy generator $K^{(2)}_\Delta$, which carry the native holomorphic singular product of Theorem 1.17.

At the reduced current-interface level the topological factor contributes the compactly supported de Rham current direction $I \subset \mathbb{R}_{\text{brane}}$, the operation $B \mapsto aB$, and extension by zero. In the graph complex it also contributes the local heat-kernel variables on $\mathbb{R}^2_{\text{top}}$. Finite-window and weighted kernels are completions of the formal Hamiltonian coefficient system.

Proof. The bracket in Definition B.1 is the inverse bivector of $dz_1 \wedge dz_2$, and constants are quotiented because they give the zero Hamiltonian vector field. The displayed X_f satisfies $X_f(g) = \{f, g\}$, and

$$\operatorname{div}_{\Omega_{\text{loc}}} X_f = -\partial_{z_1}\partial_{z_2} f + \partial_{z_2}\partial_{z_1} f = 0.$$

Thus Hamiltonian fields preserve the residue density. The transpose of this action under $\operatorname{Res}(\rho f)$ is exactly the Matlis coadjoint action used in Definition B.2 and Corollary B.8.

The interval variable is independent of the holomorphic polydisk. In the reduced current interface its operations are the de Rham differential, multiplication of a current by a smooth compactly supported test function, and extension by zero. In the graph complex the additional topological variables enter only through the local heat-kernel calculus on $\mathbb{R}^2_{\text{top}}$. The finite-window constructions truncate only the Hamiltonian degree in the formal polydisk and then pass through the weighted Mittag–Leffler inverse limit. None of these operations leaves the displayed local current category. □

LEMMA B.4 *(Multiplication of a current by a test function).* For $a \in C_c^\infty(I)$ and $B \in \mathcal{D}'_c(I)$, define

$$(aB)(\varphi) = B(a\varphi), \qquad \varphi \in C_c^\infty(I).$$

Then $aB \in \mathcal{D}'_c(I)$,

$$\operatorname{supp}(aB) \subset \operatorname{supp}(a) \cap \operatorname{supp}(B),$$

and $a(bB) = (ab)B$. If $B = b(t)dt$ is regular, then $aB = (ab)(t)dt$. The reduced current model uses this smooth-function action on distributions.

Proof. The function $a\varphi$ is a compactly supported smooth test function, so precomposition by $\varphi \mapsto a\varphi$ defines a compactly supported distribution. If φ is supported away from $\operatorname{supp}(a) \cap \operatorname{supp}(B)$, then $a\varphi$ is supported away from $\operatorname{supp}(B)$, hence $B(a\varphi) = 0$. Associativity follows from $a(b\varphi) = (ab)\varphi$. The regular-density formula is the displayed integral identity. □

Definition B.5 *(Current CE/PV coordinates).* Let $\lambda = B \otimes \rho \in \mathcal{P}_I$. The Hamiltonian CE coordinate dual to the Hamiltonian current through λ is written

$$c_\lambda, \qquad |c_\lambda| = 1.$$

Its PV/current target coordinate is

$$\theta_\lambda = \theta_{B\otimes\rho} := \Theta_\rho(B).$$

The label $a(t)dt \otimes f$ denotes the functional on $\mathcal{P}_I$

$$B \otimes \rho \longmapsto B(a)\operatorname{Res}(\rho f).$$

The CE coordinate dual to the shifted cotangent generator through this functional is

$$u_{a(t)dt\otimes f}, \qquad |u_{a(t)dt\otimes f}| = 0.$$

Its PV/current target coordinate is the smeared Hamiltonian boundary observable

$$O_{a,f} = B_f(a).$$

In the brane realization,

$$B_f(a) = \int_I a(t)\,\overline{\operatorname{Tr} f(\phi_1(t), \phi_2(t))}\,dt.$$

Definition B.6 *(Enlarged reduced defect-current target).* The reduced defect-current target attached to I is

$$A_\partial^{\mathrm{cur}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1]).$$

This is the reduced principal-part target denoted $A_\partial^{\mathrm{pp}}(I)$ in the formal-Darboux trace theorem; the superscript cur records its current-interface role. It enlarges the Hamiltonian target $A_\partial^{\mathrm{Ham}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$ by adjoining the odd principal-part current generators $\theta_\lambda = \Theta_\rho(B)$. The structure maps for inclusions of intervals are extension by zero on $\Omega_c^\bullet(I)$ and on $\mathcal{D}'_c(I)$. The factorization product on disjoint intervals is the completed graded-commutative product after extension by zero.

Theorem B.7 *(Universal algebraic current interface).* For each interval I, the generator assignment

$$\Phi_I^{\mathrm{cur}}(c_\lambda) = \theta_\lambda, \qquad \Phi_I^{\mathrm{cur}}(u_{a(t)dt\otimes f}) = B_f(a)$$

extends uniquely to a continuous morphism of completed graded-commutative algebras from the interval CE cofactorization source $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_I)$ to the reduced defect-current central-operation target $A_\partial^{\mathrm{cur}}(I)$. On length-one current generators the target open-current P_0 brackets are

$$\begin{aligned}
\{B_f(a), B_g(b)\} &= B_{\{f,g\}}(ab),\\
\{B_f(a), \theta_{B\otimes\rho}\} &= \theta_{aB\otimes(f\cdot\rho)} = \Theta_{f\cdot\rho}(aB),\\
\{\theta_{B\otimes\rho}, \theta_{C\otimes\sigma}\} &= 0.
\end{aligned}$$

The factorization product and the reduced current source-target P_0-equivariance defects in this reduced current target are strict:

$$H_{\mathrm{fact}} = 0, \qquad H_{P_0} = 0.$$

For overlapping supports, the nonzero Hamiltonian current bracket is the moment-map bracket displayed above. Thus $u_{a(t)dt\otimes f} \mapsto B_f(a)$ and $c_\lambda \mapsto \theta_\lambda$ form the current-level source/target interface used before the admissible bar-cobar/enveloping criterion.

Proof. The CE algebra of $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$ is the completed symmetric algebra on the continuous dual shifted by $[-1]$. The coordinate c_λ has degree 1, and the coordinate $u_{a(t)dt\otimes f}$ has degree 0. The target $A_\partial^{\mathrm{cur}}(I)$ is the completed symmetric algebra on the corresponding current coordinates θ_λ and $B_f(a)$. Hence the displayed assignment extends uniquely and continuously to the completed graded-commutative algebras.

The CE differential and the target current brackets use the same semidirect Lie structure constants. The Hamiltonian-current bracket in Definition B.1 gives

$$[a\otimes f, b\otimes g] = ab\otimes\{f,g\},$$

and the boundary realization gives the first displayed bracket. The action of $a \otimes f$ on $B \otimes \rho$ is $(aB) \otimes (f \cdot \rho)$, giving the second displayed bracket by negative transpose under the product pairing

$$\langle B \otimes \rho,\ a \otimes f \rangle = B(a)\,\mathrm{Res}(\rho f).$$

The shifted cotangent current ideal is abelian, giving the third bracket. These three identities are exactly the current form of the CE/PV generator identities $c_\lambda \mapsto \theta_\lambda$ and $u \mapsto O$. Since the CE differential is a derivation and the target bracket is a biderivation, the generator identities determine the full completed current interface.

Factorization compatibility is strict in the cofactorization-source convention. Extension by zero on current Lie algebras induces the CE coproduct, and the target product is compatible with pointwise products ab and with the $C_c^\infty(I)$-module operation $B \mapsto aB$. If two intervals are disjoint, the products have empty support and the mixed brackets vanish. The target product is graded commutative, so multiplication by an image generator has zero Hochschild product-centrality defect.

For reduced current P_0-equivariance, let x be a length-one source generator and let Y be a homogeneous monomial in the CE source. The defect

$$D_x(Y) = \{\Phi_I^{\mathrm{cur}}(x), \Phi_I^{\mathrm{cur}}(Y)\} - \Phi_I^{\mathrm{cur}}(\mathrm{ad}_x Y)$$

is a derivation in Y. The three generator bracket identities give $D_x(Y) = 0$ for every length-one Y. The Leibniz rule gives $D_x(Y) = 0$ for every monomial, and continuity extends this to the completed algebra. Thus H_{fact} and H_{P_0} are zero equivariance-defect homotopies in the reduced current model. □

COROLLARY B.8 *(Principal-part current brackets in monomial coordinates).* For $f = z_1^p z_2^q$, $a \in C_c^\infty(I)$, $B \in \mathcal{D}_c'(I)$, and $\rho_{i,j} = z_1^{-i-1} z_2^{-j-1} dz_1 dz_2$, one has

$$\{B_{z_1^p z_2^q}(a), \Theta_{\rho_{i,j}}(B)\} = -(pj - qi + p - q)\Theta_{\rho_{i-p+1, j-q+1}}(aB),$$

with negative-index targets equal to zero. If $B = b(t)dt$ is regular, then the interval factor is $(ab)(t)dt$.

Proof. Proposition 8.7 gives

$$z_1^p z_2^q \cdot \rho_{i,j} = -(pj - qi + p - q)\rho_{i-p+1, j-q+1},$$

with negative-index terms projected to zero in the principal-part module. Substituting this action into the second bracket of Theorem B.7 gives the displayed formula. The regular-current statement is Lemma B.4. □

The BMK obstruction calculation starts from the classical Bochner–Martinelli–Koppelman kernel gives a Dolbeault homotopy on $\mathbb{C}^2 \setminus \{0\}$ with explicit Hadamard estimates. One asks whether truncating to the finite Matlis window $\mathcal{P}_{\leq N}$ and quotienting by a section i_N, p_N yields a strict finite-window homotopy on the full compact-support/current complex. The obstruction is the precise four-component vector $\mathrm{Ob}_N^{\mathrm{BMK\text{-}curr}}$. The analytic Taylor tower gives a one-pair pro-Matlis retract, and support-local factorization-current transfer is governed by this obstruction vector. The proved content of the theorem is the BMK current data (H_χ, p_N, i_N), the finite-window obstruction vector, the positive one-pair analytic pro-Matlis retract, and the arity-two Hamiltonian/Matlis comparison.

Theorem B.9 *(BMK current data and finite-window obstruction vector).* Let $B_0 \Subset B_1 \Subset B_2 \Subset B \subset \mathbb{C}^2$ be nested polydisks around the collision point, and choose $\chi \in C_c^\infty(B_2)$ with $\chi = 1$ on B_1. Put

$$\mathcal{P}_{\le N} = \bigoplus_{\substack{a,b\ge 0\\ 0<a+b\le N}} \mathbb{C}\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2.$$

On $(B_\zeta \times B_z) \setminus \Delta$, set

$$\eta = \zeta - z, \qquad r^2 = |\eta_1|^2 + |\eta_2|^2, \qquad d\bar\eta_j = d\bar\zeta_j - d\bar z_j.$$

The Dolbeault homotopy kernel is the Bochner–Martinelli–Koppelman current

$$K_{\mathrm{BM}}(\zeta, z) = \frac{1}{(2\pi i)^2} \frac{\overline{\eta_1}\, d\bar\eta_2 - \overline{\eta_2}\, d\bar\eta_1}{r^4} \wedge d\zeta_1 \wedge d\zeta_2.$$

The $d\bar z_j$-terms are part of the Koppelman current. The Dolbeault homotopy uses the full Koppelman current.

Choose a radial cutoff $\vartheta_\epsilon(r)$, equal to 0 for $r \le \epsilon$ and to 1 for $r \ge 2\epsilon$, and put

$$K_{\chi,\epsilon} = \chi(\zeta)\chi(z)\vartheta_\epsilon(|\zeta - z|) K_{\mathrm{BM}}(\zeta, z).$$

The current limit

$$H_\chi \alpha(z) = \lim_{\epsilon\to 0} \int_{\zeta\in B} K_{\chi,\epsilon}(\zeta, z) \wedge \alpha(\zeta)$$

exists on every fixed finite configuration stratum. For separated supports it satisfies

$$|\nabla^k K_{\mathrm{BM}}(\zeta, z)| \le C_k |\zeta - z|^{-3-k}.$$

Let $i_N : \mathcal{P}_{\le N} \to \mathcal{D}'^{\,0,2}_0(B)$ be

$$i_N(\rho_{a,b}) = \Delta_{a,b} = \frac{(-1)^{a+b}}{a!\, b!} \partial_{z_1}^a \partial_{z_2}^b \delta_0\, d\bar z_1 \wedge d\bar z_2,$$

normalized by

$$\langle \Delta_{a,b}, f\rangle = \mathrm{Res}_0(\rho_{a,b} f).$$

Define the finite principal-part moment projection on top Dolbeault degree by

$$p_N(\alpha) = \sum_{0<a+b\le N} \left(\int_B z_1^a z_2^b\, \alpha^{0,2} \wedge dz_1 \wedge dz_2 \right) \rho_{a,b}.$$

The data above construct H_χ, p_N, and i_N. The finite-window homotopy on the full compact-support/current complex is the identity

$$\bar\partial H_{\chi,N} + H_{\chi,N}\bar\partial = \mathrm{id} - i_N p_N + E_{\chi,N} \tag{$*_N^{\mathrm{BMK}}$}$$

is blocked by

$$\mathrm{Ob}_N^{\text{BMK-curr}} = \Big(\mathrm{ob}_{\mathrm{diag}}, \mathrm{ob}_{\mathrm{an/fin},N}, \mathrm{ob}_{\mathfrak{h},N}, \mathrm{ob}_{\text{collar-fact},N} \Big).$$

The first entry is the annular diagonal sign and scalar of $\bar\partial\vartheta_\epsilon \wedge K_{\mathrm{BM}}$ relative to the derivative-delta Matlis convention. With the displayed kernel and real outward boundary orientation the angular mass is -1.

The second entry is the gap between the full analytic moment kernel $\ker(\mathcal{O}(B_1)^\vee \to \mathcal{P}_{\leq N})$ and the smaller support-at-zero span generated by the scalar and high derivative-delta currents. The third entry is

$$\mathrm{ob}_{\mathfrak{h},N} = \pi_N(\mathfrak{h} \cdot \ker \pi_N),$$

which is nonzero because

$$z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}.$$

Thus the full Hamiltonian action sends high modes back into the retained window, obstructing a strict finite support-local quotient that retains $\mathcal{P}_{\leq N}$. The fourth entry is the cosheaf/factorization compatibility of the collar relation with extension by zero, disjoint products, and the BMK homotopy.

A normalized finite-window identity is available only after changing the underlying complex. If the full analytic BMK contraction is fixed and the relative complex is quotiented by representatives of $\ker(\mathcal{O}(B_1)^\vee \to \mathcal{P}_{\leq N})$, then the normalized homotopy

$$(1 - I_{\mathrm{an}}P_{\mathrm{an}})H_\chi(1 - I_{\mathrm{an}}P_{\mathrm{an}})$$

descends and gives

$$\bar{\partial} H_{\mathrm{BM},N} + H_{\mathrm{BM},N}\bar{\partial} = \mathrm{id} - i_N p_N.$$

This is a one-pair analytic quotient. Support-locality on the factorization-current complex is measured by $\mathrm{Ob}_N^{\text{BMK-curr}}$.

Keeping the full Taylor tower gives the positive one-pair pro-Matlis retract. Put

$$\widehat{\mathcal{P}}_\Pi = \prod_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\rho_{a,b}, \qquad \mathcal{P}_{\mathrm{an}}(B_1) = \tau\big(\mathcal{O}(B_1)^\vee_{\mathrm{red}}\big) \subset \widehat{\mathcal{P}}_\Pi,$$

where

$$\tau(\lambda) = \big(\lambda(z_1^a z_2^b)\big)_{a+b>0}.$$

After the diagonal orientation is fixed and the collar quotient kills the cutoff error, the normalized analytic contraction $H_\chi^\circ = (1 - I_{\mathrm{an}}P_{\mathrm{an}})H_\chi(1 - I_{\mathrm{an}}P_{\mathrm{an}})$ satisfies

$$\bar{\partial} H_\chi^\circ + H_\chi^\circ \bar{\partial} = \mathrm{id} - I_{\mathrm{an}}P_{\mathrm{an}}.$$

The pro-analytic moment map $P_\Pi^{\mathrm{an}} = \tau P_{\mathrm{an}}$ has compatible finite projections

$$\pi_N P_\Pi^{\mathrm{an}} = p_N, \qquad \pi_{M,N} p_M = p_N.$$

This is a one-pair analytic pro-Matlis statement. A support-local factorization-current transfer additionally requires vanishing of $\mathrm{Ob}_N^{\text{BMK-curr}}$.

The binary coefficient operation

$$\ell_2^{\mathrm{BM},N}(x,y) = p_N[ix, iy]$$

is retained as the Hamiltonian/Matlis arity-two comparison, read in the codomain window $\mathcal{P}_{\leq N}$. For monomials,

$$\ell_2^{\mathrm{BM},N}(z_1^p z_2^q, z_1^r z_2^s) = (ps - qr)\ \mathrm{pr}_{\leq N}(z_1^{p+r-1} z_2^{q+s-1}) \pmod{\mathbb{C}},$$

and
$$\ell_2^{\mathrm{BM},N}(z_1^p z_2^q, \rho_{i,j}) = -(pj - qi + p - q)\ \mathrm{pr}_{\le N}\, \rho_{i-p+1,j-q+1},$$
with negative-index targets equal to zero, while
$$\ell_2^{\mathrm{BM},N}(\rho, \sigma) = 0.$$
Thus the arity-two cotangent coefficient comparison is the Matlis coadjoint action of Proposition 8.7.

With the finite-current condition $\mathrm{Ob}_N^{\text{BMK-curr}}$ killed, or replaced by a support-local factorization-current complex, the strict all-window and arity-three extension has the following obstruction vector. Set
$$Q_\oplus = \widehat{\mathcal{P}}_\Pi / \mathcal{P}.$$
For $\alpha \in \Omega_c^{0,2}(B)$, put
$$\mathrm{mom}_\infty(\alpha) = \left(\int_B z_1^a z_2^b\, \alpha^{0,2} \wedge dz_1 \wedge dz_2 \right)_{a+b>0} \in \widehat{\mathcal{P}}_\Pi,$$
and
$$\mathrm{ob}_\oplus(\alpha) = [\mathrm{mom}_\infty(\alpha)] \in Q_\oplus.$$
This class is zero exactly when the all-window moment sequence lands in the direct-sum Matlis module $\mathcal{P}$.

The pro-product tower leaves the current problem governed by the section obstruction. A support-local compact-current section
$$i_\Pi : \widehat{\mathcal{P}}_\Pi \to \mathcal{D}_0'^{\,0,2}(B)$$
extending the finite derivative-delta sections i_N would assign infinite-order point-supported distributions, while point-supported distributions have finite order. This is ob_{I_Π}. The action of the full formal Hamiltonian algebra on $\widehat{\mathcal{P}}_\Pi$ is obstructed at $\mathrm{ob}_{\mathfrak{h},\Pi}$. For $f = \sum_{p\ge 2} z_1^p$ and $\lambda = \sum_{p \ge 2} \rho_{p-1,0}$, the $\rho_{0,1}$-coefficient of $f \cdot \lambda$ collects $-\sum_{p\ge 2} p$, the divergent witness $\mathrm{ob}_{\mathfrak{h},\Pi}$.

The collar part is the finite-window family
$$\mathrm{ob}_{\mathrm{collar},\Pi,N} = E_{\chi,N} \in \mathrm{Hom}\big(\Omega_c^{0,\bullet}(B), \mathcal{D}'^{\,0,\bullet}(B)\big)_{\mathrm{supp}\, d\chi}.$$
The first arity-three obstruction is
$$\mathrm{ob}^\chi_{3,\Pi,N}(x,y,z) = p_N[H_{\mathrm{BM},N}([ix,iy] - i\ell_2^{\mathrm{BM},N}(x,y)), iz] + \mathrm{cyc}_{x,y,z}\,.$$
It vanishes on separated-support strata by the displayed estimates. On the coefficient collision stratum its non-collar part is the Jacobiator of $\mathfrak{h} \ltimes \mathcal{P}[1]$, hence zero. The remaining part is supported on the small diagonal and on the cutoff collar.

Finally, let $C_{s,N}$ be the optimal constants for the chosen finite-window seminorm estimates of $H_{\chi,N}$, p_N, $E_{\chi,N}$, and $\mathrm{ob}^\chi_{3,\Pi,N}$. The uniformity obstruction is
$$\mathrm{ob}_{\mathrm{unif},\Pi} = \big(\big[(C_{s,N})_{N\ge 1}\big]\big)_s \in \prod_s \left(\prod_{N \ge 1} \mathbb{R}_{\ge 0} / \ell_N^\infty \right).$$
Thus the all-window obstruction vector is
$$\mathrm{Ob}^\Pi_{\mathrm{BM}} = \Big(\mathrm{ob}_\oplus, \mathrm{ob}_{I_\Pi}, \mathrm{ob}_{\mathfrak{h},\Pi}, (\mathrm{ob}_{\mathrm{collar},\Pi,N})_N, (\mathrm{ob}^\chi_{3,\Pi,N})_N, \mathrm{ob}_{\mathrm{unif},\Pi}\Big).$$

Any strict all-window current E_2 transfer extending the displayed binary coefficient comparison forces this vector to vanish. The first, second, third, and sixth entries are nonzero in the current topology; the collar and ternary entries are unconstructed exact primitive classes. The proved theorem here is the BMK current data, the finite-window obstruction vector, the positive one-pair analytic pro-Matlis retract, and the arity-two Hamiltonian/Matlis comparison.

The principal-part face remains two-variable:

$$\frac{1}{(2\pi i)^2}\frac{d\zeta_1 \wedge d\zeta_2}{(\zeta_1 - z_1)(\zeta_2 - z_2)} = \sum_{a,b\geq 0} z_1^a z_2^b \rho_{a,b}(\zeta).$$

Setting $z_2 = 0$ kills the Hamiltonian bracket, and the $b = 0$ principal-part line leaves the subspace under $z_1 \cdot \rho_{a,0} = -\rho_{a,1}$; the principal-part face stays two-variable.

Proof. The Bochner–Martinelli–Koppelman identity gives the Dolbeault homotopy on the punctured product $B_\zeta \times B_z \setminus \Delta$. Multiplication by the cutoff gives the exact derivative split

$$\bar\partial(\chi(\zeta)\chi(z)\vartheta_\epsilon K_{\mathrm{BM}}) = \chi\chi\,\vartheta_\epsilon\,\bar\partial K_{\mathrm{BM}} + \bar\partial(\chi\chi)\,\vartheta_\epsilon K_{\mathrm{BM}} + \chi\chi\,\bar\partial\vartheta_\epsilon \wedge K_{\mathrm{BM}}.$$

The middle term is collar-supported. The last term is the annular diagonal current; its sign and scalar are exactly $\mathrm{ob}_{\mathrm{diag}}$. Since K_{BM} has size r^{-3} in real dimension four, the cutoff current has a limit on every fixed finite configuration stratum. This proves the current data. The finite-window identity on the full current complex is the additional obstruction equation displayed in the theorem.

The formula for i_N is fixed by integration by parts:

$$\left\langle \frac{(-1)^{a+b}}{a!\,b!}\partial_{z_1}^a\partial_{z_2}^b\delta_0, f\right\rangle = [z_1^a z_2^b]f = \mathrm{Res}_0(\rho_{a,b}f).$$

Hence $i_N p_N$ is the finite support-at-collision projector with the Matlis residue normalization, after the diagonal-current orientation is tied to that convention.

The finite-window identity on the full current complex is obstructed already on closed top currents in the kernel of the finite moment map. If $T = \Delta_{N+1,0}$, then $p_N(T) = 0$ but $\langle T, z_1^{N+1}\rangle = 1$. A putative identity $T = \bar\partial S$ would give $\langle T, z_1^{N+1}\rangle = 0$ by Stokes. The scalar $\Delta_{0,0}$ is the same obstruction for the removed reduced principal-part line. More strongly, a strict support-local quotient must be an $\mathfrak{h}$-submodule quotient; the calculation $z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}$ contradicts killing $\rho_{N+1,0}$ while retaining $\rho_{0,1}$.

The one-pair analytic quotient works for the reason stated in the theorem: after choosing the full analytic projector $I_{\mathrm{an}}P_{\mathrm{an}}$, the normalized homotopy kills representatives of $\ker \pi_N$, descends to the quotient, and identifies the analytic projector with $i_N p_N$. Keeping the Taylor tower instead gives the pro-analytic map $P_\Pi^{\mathrm{an}} = \tau P_{\mathrm{an}}$, and the equality $\pi_N P_\Pi^{\mathrm{an}} = p_N$ is functoriality of Taylor moments under finite truncation. Its dependence on analytic representatives and on the chosen pair makes it a one-pair analytic quotient.

The arity-two coefficient comparison is the residue transpose. For Hamiltonian labels it is the Poisson bracket modulo constants, truncated only after the resulting coefficient is computed. For cotangent principal parts, Proposition 8.7 gives

$$\mathrm{Res}_0((f\cdot\rho)g) = -\,\mathrm{Res}_0(\rho\,\{f,g\}),$$

which is the displayed coadjoint formula. The shifted cotangent ideal is abelian, so the (ρ,σ)-operation is zero.

The displayed ternary expression is only the next obstruction after the finite-current condition has been crossed. The separated-support estimate makes it vanish away from collisions. On the coefficient collision stratum the semidirect Lie algebra $\mathfrak{h} \ltimes \mathcal{P}[1]$ satisfies Jacobi. Therefore any surviving term is a collar, small-diagonal, or completed-window transfer residue, recorded as the finite-window representative $\mathrm{ob}^{\chi}_{3,\Pi,N}$ of the Π-tower class only after the finite-current obstruction has been removed.

The obstruction to a strict all-window current transfer has six components: replacement of the compatible family p_N by a single moment map landing in $\mathcal{P}$, construction of a compact-current section of the pro-product tower, definition of the full formal Hamiltonian action on that tower, removal of $E_{\chi,N}$ compatibly with extension by zero, annihilation of the ternary collar residue by compatible primitives, and uniform estimates in the completed window topology. These six requirements are precisely the six entries of $\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$. The all-window theorem concerns a refined factorization-current complex where these six witnesses exist, together with the displayed obstruction theorem. The finite-window calculation gives the data measured by $\mathrm{Ob}^{\text{BMK-curr}}_{N}$.

The Cauchy expansion is the Laurent expansion in the two holomorphic variables and records the same residue basis. The calculation $z_1 \cdot \rho_{a,0} = -\rho_{a,1}$, from Proposition 8.7, rules out the $b = 0$ principal-part truncation as an $\mathfrak{h}$-submodule. □

The brane-preserving normal Ω-background is the equivariant deformation relevant to the current sector. In the local dictionary the brane-preserving equivariant deformation is normal scaling of $N_L X = \mathbb{R}_s \oplus \mathbb{C}_{z_1} \oplus \mathbb{C}_{z_2}$, with t fixed. The fixed brane line selects this normal action. The current sector responds by an explicit grading on the generators $B_{p,q}(a)$ and $\Theta_{r,s}(B)$; the Hamiltonian/Matlis brackets become weighted by the inverse symplectic-character line $\mathbb{L}_{\omega}^{-1}$, or equivalently by adjoining a weight-χ_ω parameter $\hbar_\omega$. The theorem records the resulting triangle of length-one current brackets and the self-dual specialization $\epsilon_1 + \epsilon_2 = 0$. This is an algebraic identity in the freely adjoined current target. A normal-Ω graph/QME lift additionally requires the Q_Ω-Cartan data, normal-weight contraction, residue/Euler normalization choice, equivariant counterterms, and the scalar/non-scalar QME obstruction classes named below.

Theorem B.10 *(Normal Ω-weighted brane-current brackets).* Use the normal brane-preserving equivariant parameters of the local dictionary: t has weight 0, z_1, z_2 have weights ϵ_1, ϵ_2, and

$$\chi_\omega = \epsilon_1 + \epsilon_2, \qquad \mathrm{wt}_\Omega(\hbar_\omega) = \chi_\omega.$$

Put

$$B_{p,q}(a) := B_{z_1^p z_2^q}(a), \qquad \Theta_{r,s}(B) := \Theta_{\rho_{r,s}}(B),$$

with $p, q, r, s \geq 0$, $p + q > 0$, $r + s > 0$, and $\rho_{r,s} = z_1^{-r-1} z_2^{-s-1} dz_1 dz_2$. The compactly supported test function a and compactly supported current B live on the fixed brane line and have weight 0. Hence

$$\mathrm{wt}_\Omega\, B_{p,q}(a) = p\epsilon_1 + q\epsilon_2, \qquad \mathrm{wt}_\Omega\, \Theta_{r,s}(B) = -r\epsilon_1 - s\epsilon_2.$$

Off the self-dual locus the Hamiltonian current bracket is valued in the inverse symplectic-character line. After adjoining $\hbar_\omega$, the scalarized convention has length-one current brackets are

$$\begin{aligned}
\{B_{p,q}(a), B_{r,s}(b)\}_\Omega &= \hbar_\omega(ps - qr)\, B_{p+r-1,q+s-1}(ab),\\
\{B_{p,q}(a), \Theta_{r,s}(B)\}_\Omega &= -\hbar_\omega(ps - qr + p - q)\, \Theta_{r-p+1,s-q+1}(aB),\\
\{\Theta_{r,s}(B), \Theta_{u,v}(C)\}_\Omega &= 0.
\end{aligned}$$

Here $B_{\alpha,\beta} = 0$ if $\alpha < 0$, $\beta < 0$, or $(\alpha, \beta) = (0, 0)$, and $\Theta_{\alpha,\beta} = 0$ if $\alpha < 0$, $\beta < 0$, or $(\alpha, \beta) = (0, 0)$. The last convention only records the reduced scalar-annihilator target; when the coadjoint formula lands at $\rho_{0,0}$ its coefficient is already zero.

Equivalently, before adjoining $\hbar_\omega$, remove the displayed factor $\hbar_\omega$ and read the first two brackets as valued in the inverse symplectic-character line $\mathbb{L}_\omega^{-1}$, where $\mathrm{wt}_\Omega(\mathbb{L}_\omega) = \chi_\omega$. On the self-dual specialization $\epsilon_1 + \epsilon_2 = 0$, the line is trivial. After choosing the trivialization $\hbar_\omega = 1$, the formulas reduce to the non-equivariant brackets of Theorem B.7 and Corollary B.8. This specialization is a separate branch: first restrict to $\epsilon_1 + \epsilon_2 = 0$, then choose the trivialization.

For an inclusion of intervals $j : I \hookrightarrow J$, extension by zero has weight 0 and satisfies

$$j_! B_{p,q,I}(a) = B_{p,q,J}(j_! a), \qquad j_! \Theta_{r,s,I}(B) = \Theta_{r,s,J}(j_! B),$$

together with

$$j_! \{x, y\}_{\Omega,I} = \{j_! x, j_! y\}_{\Omega,J}$$

for the displayed length-one generators. For disjoint intervals the mixed brackets vanish after extension to a common interval.

These statements are algebraic current identities. They hold for all $B, C \in \mathcal{D}'_c(I)$ because the only distributional operation is multiplication of a current by a smooth compactly supported function. A normal-Ω graph/QME lift for arbitrary distributional labels is a further analytic theorem. A graph-level lift must still restrict cotangent labels to regular densities or to graphwise wavefront-admissible tuples as in Propositions B.17 and B.19, and must add the Q_Ω-Cartan data, normal-weight contraction, residue/Euler normalization choice, equivariant counterterms, and the scalar/non-scalar QME obstruction classes.

Proof. The weight assignments are forced by the normal torus action: $z_1^p z_2^q$ has character $p\epsilon_1 + q\epsilon_2$, while

$$z_1^{-r-1} z_2^{-s-1} dz_1 dz_2$$

has character $-(r+1)\epsilon_1 - (s+1)\epsilon_2 + \chi_\omega = -r\epsilon_1 - s\epsilon_2$. The brane-line variables and currents are in the fixed t-direction, so their equivariant character is 0.

The Poisson bivector $\partial_{z_1} \wedge \partial_{z_2}$ has character $-\chi_\omega$. Thus the ordinary Hamiltonian bracket has weight $\mathrm{wt}(f) + \mathrm{wt}(g) - \chi_\omega$. Multiplying by $\hbar_\omega$ makes the bracket scalar and weight-preserving. On monomials this gives

$$[z_1^p z_2^q, z_1^r z_2^s]_\Omega = \hbar_\omega (ps - qr) z_1^{p+r-1} z_2^{q+s-1},$$

followed by the quotient by constants. Substitution in the first bracket of Theorem B.7 gives the displayed B-B formula.

For the cotangent current, the coadjoint action of Proposition 8.7 gives

$$z_1^p z_2^q \cdot \rho_{r,s} = -(ps - qr + p - q)\rho_{r-p+1,s-q+1}.$$

Scalarizing the Hamiltonian action by the same $\hbar_\omega$ gives the B-Θ formula, and the shifted cotangent current ideal is abelian. Each right-hand side has total weight equal to the sum of the weights of the two arguments, so the formulas are T_Ω-homogeneous.

Extension by zero commutes with multiplication of compactly supported test functions and currents: $j_!(ab) = (j_! a)(j_! b)$ and $j_!(aB) = (j_! a)(j_! B)$. Hence it commutes with all three displayed brackets. If two intervals are disjoint inside a common interval, these products are zero, so the mixed brackets vanish.

Finally, the algebraic current formulas use multiplication of currents by smooth compactly supported functions. Costello graph weights are different: after pushing current labels through graph incidence maps one must multiply them by singular propagator boundary values and extend across collision diagonals. The wavefront and scaling-degree hypotheses in Propositions B.19 and B.20 are exactly the required analytic conditions. The Ω-weights and the factor $\hbar_\omega$ preserve those microlocal conditions. □

Corollary B.11 *(Smeared Hamiltonian source convention).* The smeared boundary Hamiltonian observable is sourced by the shifted-cotangent CE coordinate:

$$u_{a(t)dt\otimes f} \longmapsto B_f(a) = \int_I a(t)\, \overline{\operatorname{Tr} f(\phi_1(t), \phi_2(t))}\, dt.$$

The Hamiltonian CE coordinate c_λ maps to the PV/current vector θ_λ. Consequently

$$\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$$

is the degree-zero moment-map shadow of the CE/PV identity $c_\lambda \mapsto \theta_\lambda$, $u \mapsto O$.

Proof. The coordinate $u_{a(t)dt\otimes f}$ is dual to the shifted cotangent generator through the pairing with $\mathcal{P}_I$. Under the CE/PV isomorphism it maps to the function coordinate $O_{a,f}$, realized on the brane by the displayed trace integral. The coordinate c_λ is dual to a Hamiltonian current and maps to the vector/current coordinate θ_λ. The bracket identity is the first bracket of Theorem B.7; equivalently, it is the equal-time Poisson bracket smeared against $a(t)b(t')\delta(t-t')$ on two brane-time points, which produces the pointwise product ab. □

Remark B.12 *(Orientation reversal).* The orientation convention is the ordinary convention for oriented one-forms and current one-forms. The labels $a(t)dt$ and regular currents $B = b(t)dt$ transform by pullback, $a(\psi(s))\psi'(s)\, ds$ and $b(\psi(s))\psi'(s)\, ds$. Additional signs are governed only by the written one-form: $u_{a(t)dt\otimes f}$, c_λ, θ_λ, or the principal-part label ρ. If the written one-form is held fixed while only the chosen orientation is reversed, the sign is exactly the sign of dt.

Remark B.13 *(Boundary of the algebraic current theorem).* Theorem B.7 proves the algebraic distributional-current interface. The finite polynomial BV observable, primitive/indecomposable projection before reduction, full de Rham-current BV differential, mixed brane-defect QME, and Costello renormalized graph weights require the unreduced current complex, scalar cancellation data, Tate/weighted coefficient kernel, and brane-defect propagator specified in Appendix E and Problem 5.213.

The unreduced lift requires a weighted Costello local-functional complex. The algebraic interface above is degree-zero current data on the freely adjoined principal-part current sector; the associated graded target has the classical differential $d^w_{\mathrm{cl}} = Q + \{I^w_{\mathrm{o}}, -\}_\hbar$, while the genuine Costello target is the same filtered local-functional space with finite-scale BV Laplacian, finite-scale propagator, and quantum BV differential

$$d^L_{\mathrm{qBV}} = Q + \{I[L], -\}_L + \hbar\Delta_L.$$

Four complexes enter: the unreduced source and target (Definition B.14), the bulk-to-defect convolution Hom complex (Definition B.15) in which Maurer–Cartan / curvature lives, the coefficient-current kernel morphism (Proposition B.16) that solves the algebraic problem strictly, and the Costello graph complexes (Propositions B.17–B.20) that give the analytic extension under wavefront/scaling-degree hypotheses.

The obstruction map list and centrality homotopy criterion (Proposition B.21, Theorem B.22) then record exactly which classes block strengthening from the algebraic kernel to a Costello graph lift.

Definition B.14 *(Unreduced non-scalar current lift datum).* Fix an oriented interval $I \subset \mathbb{R}_{\mathrm{brane}} \subset \mathbb{R}^2_{\mathrm{top}}$, an admissible weight w on the Hamiltonian degree of $\widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$, and the Moyal parameter $\hbar$. A non-scalar Costello current lift of the reduced interface for the local mixed HT model consists of the following data.

(i) A source current complex

$$\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I} = \big(\Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}_{w,\hbar}\big) \ltimes \big(\mathcal{D}'_c(I)\widehat{\otimes}\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}\big)[1],$$

or, before scalar reduction, the corresponding central-character replacement of Definition E.8.

(ii) A target defect local-functional complex

$$\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I) = \Big(O^{\mathrm{bd}}_{\mathrm{loc}}\big(\mathcal{E}_w|_I;\ A^{\mathrm{pp},w}_{\partial,\hbar}(I)\big), d^w_{\mathrm{cl}} = Q + \{I^w_{\mathrm{o}}, -\}_\hbar\Big),$$

together with the finite-scale BV Laplacian and propagator in the formulation where the Costello scale-L differential d^L_{qBV} remains explicit. Below $d_{\mathfrak{Q}}$ denotes the chosen codomain differential: d^w_{cl} on the algebraic coefficient-current complex and d^L_{qBV} on the Costello scale-L complex. Here $\mathcal{E}_w$ is the weighted local mixed HT field complex on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{\mathrm{o,hol}}$, restricted to the brane-defect interval I; the coefficient algebra is the weighted Hamiltonian/Moyal algebra of the same formal holomorphic symplectic polydisk.

(iii) A continuous bulk-to-defect kernel, equivalently a Maurer–Cartan element in the completed convolution Lie algebra,

$$\kappa_{\hbar,w,I} : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}) \longrightarrow \mathfrak{Q}^\bullet_{w,\partial,\hbar}(I),$$

whose generator values are representatives

$$u_{a(t)dt\otimes f} \longmapsto B^{\mathrm{BV}}_f(a), \qquad c_{B,\rho} \longmapsto \Theta^{\mathrm{BV}}_\rho(B).$$

(iv) A $d_{\mathfrak{Q}}$-stable scalar-contact filtration and a filtered chain projection

$$\widehat{\sigma}_{\mathrm{sc}} : \mathfrak{Q}^\bullet_{w,\partial,\hbar}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]$$

whose associated graded is the scalar-symbol map of Definition E.3.

(v) Product centrality homotopies and P_0-bracket homotopies against arbitrary unreduced open observables, extending the checks on the freely adjoined current generators of the reduced target.

The reduced algebraic theorem is the image of this datum after scalar reduction, de Rham-current contraction, and passage to the added principal-part current target.

Definition B.15 *(Bulk-to-defect convolution Hom complex).* With $I, w, \hbar$ as in Definition B.14, and with all Hamiltonian coefficients taken from the local formal holomorphic polydisk of Proposition B.3, put

$$\mathbf{K}^\bullet_{w,\partial,\hbar}(I) = \mathrm{Hom}_{\mathrm{cont}}\Big(C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}), \mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)\Big).$$

The topology is the complete filtration topology generated by the $\hbar$-adic filtration, the scalar-contact filtration, and the finite-window Mittag–Leffler filtration. Thus a sequence of kernels converges exactly when all of its finite-window, scalar-contact, and $\hbar$-adic projections converge. The differential and bracket are the completed convolution operations

$$d_{\mathbf{K}}T = d_{\mathfrak{Q}}T - (-1)^{|T|}T\, d_{\mathrm{CE}}, \qquad [T_1, T_2]_{\mathbf{K}} = [-,-]_\hbar \circ (T_1 \widehat{\otimes} T_2) \circ \Delta_{\mathrm{CE}},$$

where Δ_{CE} is the completed CE coproduct and $[-,-]_\hbar$ is the codomain $P_{0,\hbar}$/BV bracket. For a degree-zero kernel κ, its curvature is

$$\mathrm{Curv}_{\mathbf{K}}(\kappa) = d_{\mathbf{K}}\kappa + \frac{1}{2}[\kappa,\kappa]_{\mathbf{K}} \in \mathbf{K}^1_{w,\partial,\hbar}(I).$$

On generators, the term $-\kappa d_{\mathrm{CE}}$ is exactly the source-bracket subtraction appearing in the bracket-defect formulas below.

Proposition B.16 *(Bulk-to-defect coefficient-current kernel morphism).* In the local mixed HT model on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$, there is a canonical degree-zero continuous coefficient-current kernel

$$\kappa^{\mathrm{coef}}_{\hbar,w,I} \in \mathbf{K}^0_{w,\partial,\hbar}(I)$$

whose generator values are

$$\kappa^{\mathrm{coef}}_{\hbar,w,I}(u_{a(t)dt\otimes f}) = \iota_I\big(B^w_f(a)\big), \qquad \kappa^{\mathrm{coef}}_{\hbar,w,I}(c_{B,\rho}) = \iota_I\big(\Theta^w_\rho(B)\big).$$

Here $a \in C^\infty_c(I)$, $f \in \mathfrak{h}_{w,\hbar}$, $B \in \mathcal{D}'_c(I)$, and $\rho \in \mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}$. The map ι_I is the coefficient inclusion of $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$ as field-independent defect-local functionals in $\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$. The degrees are

$$|u_{a(t)dt\otimes f}| = |B^w_f(a)| = 0, \qquad |c_{B,\rho}| = |\Theta^w_\rho(B)| = 1.$$

The kernel has the following proved properties.

(i) For each finite Hamiltonian window M in the formal holomorphic Hamiltonian degree, its restriction $\kappa^{\mathrm{coef},M}_{\hbar,w,I}$ lands in

$$\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}\big(\mathcal{E}^M_w|_I;\, A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)\big)$$

and the finite-window maps satisfy

$$\pi_{M,N}\kappa^{\mathrm{coef},M}_{\hbar,w,I} = \kappa^{\mathrm{coef},N}_{\hbar,w,I}\,\pi^{\mathrm{CE}}_{M,N}, \qquad M \geq N.$$

For admissible weights w, w', the diagonal transport maps satisfy

$$R^M_{w,w'}\kappa^{\mathrm{coef},M}_{\hbar,w,I} = \kappa^{\mathrm{coef},M}_{\hbar,w',I}\,R^{M,\mathrm{CE}}_{w,w'}.$$

Hence the displayed finite-window kernels determine the unique strict Mittag–Leffler inverse-limit kernel $\kappa^{\mathrm{coef}}_{\hbar,w,I}$.

(ii) After setting $\hbar = 0$, applying scalar reduction and the de Rham-current contraction, the kernel is exactly Φ^{cur}_I of Theorem B.7.

(iii) Inside the freely adjoined coefficient-current subalgebra, the generator $P_{0,\hbar}$-defects vanish:

$$\{B^w_f(a), B^w_g(b)\}_\hbar = B^w_{\{f,g\}_\hbar}(ab),$$

$$\{B^w_f(a), \Theta^w_\rho(B)\}_\hbar = \Theta^w_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB), \qquad \{\Theta^w_\rho(B), \Theta^w_\sigma(C)\}_\hbar = 0.$$

(iv) Let $\kappa_{\hbar,w,I}$ be any Costello-local-functional lift of this coefficient kernel. If its curved Maurer–Cartan equation has been solved through order $\hbar^{n-1}$, then the order-n component of $\mathrm{Curv}_{\mathbf{K}}(\kappa_{\hbar,w,I})$ is d^L_{qBV}-closed modulo the already solved lower-order terms. If the filtered scalar projection exists and its scalar image is zero, each fixed source-arity or graph component gives, by evaluation, a residual class in

$$H^1\Big(\ker \widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{qBV}}\Big).$$

Equivalently, the assembled curvature coefficient is a class in the corresponding Hom-valued obstruction complex; this is the class denoted $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]$ in Problem 5.220. Its compatible finite-window counterterm obstruction is precisely the inverse-limit H^1 class together with the $\varprojlim^1 H^0$ primitive-compatibility class of Theorem 5.84.

This proposition constructs the local coefficient-current kernel morphism. A Costello graph kernel for arbitrary $B \in \mathcal{D}'_c(I)$ requires the following analytic theorem: for every finite window and every defect-current-labelled graph, the products of the finite-scale propagators with the pushed-forward compactly supported currents must satisfy Hörmander wavefront transversality, admit local extensions across the relevant small diagonals, and produce counterterms compatible with interval extension, finite-window truncation, and admissible weight transport. The graph-level lift proved below applies after restricting the cotangent labels to regular compactly supported densities or to a wavefront-admissible class; the arbitrary distributional case is the obstruction problem.

Proof. By Proposition B.3, the source Lie brackets and principal-part pairings are the formal-disk Hamiltonian brackets and residues tensored with interval currents. The continuous CE cochain algebra is the completed graded-symmetric algebra on the continuous dual of $\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}[1]$. The target coefficient algebra $A^{\mathrm{pp},w}_{\partial,\hbar}(I)$ is the completed graded-symmetric algebra on the matching Hamiltonian-current generators $B^w_f(a)$ and cotangent-current generators $\Theta^w_\rho(B)$. The displayed assignment therefore extends uniquely to a continuous completed graded-commutative algebra map. Since the source and target generators have the displayed equal degrees, the resulting element of the convolution Hom complex has degree 0.

Windowwise, the same construction is finite-dimensional in the coefficient direction, with $D_{w,\leq M}$ and $A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)$ as in Definition 5.83. Truncation discards coefficient degrees $d > N$, while $R^M_{w,w'}$ rescales each surviving degree-d basis vector and its dual by inverse factors. The formulas for $B^w_f(a)$ and $\Theta^w_\rho(B)$ are degreewise identities, so they commute with truncation and with weight transport. The strict Mittag–Leffler universal property gives the inverse-limit kernel.

The reduction statement follows by forgetting the Moyal deformation, applying the scalar quotient, and contracting the de Rham current direction. The two generator formulas become $u_{a(t)dt\otimes f} \mapsto B_f(a)$ and $c_{B,\rho} \mapsto \Theta_\rho(B)$, which are precisely the generators of Theorem B.7.

The three displayed bracket identities are the weighted Moyal version of the semidirect-product calculation in Theorem B.7: the Hamiltonian-current bracket gives the pointwise product ab, the cotangent action is multiplication $B \mapsto aB$ together with the Moyal coadjoint action, and the shifted cotangent current ideal is abelian. By biderivation, the identities extend from generators to the completed coefficient-current subalgebra.

Finally, the convolution Hom complex is a filtered dg Lie algebra. The Bianchi identity

$$d_{\mathbf{K}}\,\mathrm{Curv}_{\mathbf{K}}(\kappa) + [\kappa, \mathrm{Curv}_{\mathbf{K}}(\kappa)]_{\mathbf{K}} = 0$$

implies, order by order in $\hbar$, that the first unsolved curvature coefficient is closed for the linearized differential. Under the chosen scalar projection, zero scalar image places every evaluated component in ker $\widehat{\sigma}_{\mathrm{sc}}$, equivalently places the assembled coefficient in the Hom-valued kernel. Adding a degree-zero local counterterm changes the first unsolved coefficient by the corresponding d^{L}_{qBV}-boundary, and the finite-window compatibility of such primitives is exactly the Milnor obstruction measured in Theorem 5.84. This proves the stated obstruction class and leaves only the analytic distributional graph-extension theorem named in the statement. □

The Costello graph complex is split by the analytic admissibility of the external cotangent labels. Three classes are constructed, in strict order of increasing label generality:

- *Regular densities* $\mathcal{D}^{\mathrm{reg}}_{c}(I)$ (Proposition B.17), where the distributional operation is multiplication of smooth heat-kernel propagators by smooth densities and the wavefront obstruction vanishes;
- *Wavefront-admissible distributional tuples* $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M)$ (Definition B.18, Proposition B.19), where the Hörmander product, pushforward, and finite-scaling extension calculus [36, Ch. 8], [12, Chs. 5–7] are defined, including the one-sided sign-conic finite-scaling classes $\mathcal{D}'_{c,\Lambda_{\pm},\mathrm{fs}}(I)$;
- *Arbitrary compactly supported distributions* $\mathcal{D}'_{c}(I)$ (Proposition B.20), where the algebraic kernel is the coefficient-current kernel of Proposition B.16, and the coincident-current boundary graph $B_1 = B_2 = \delta_{t_0}$ gives an explicit Hörmander product obstruction for general $\mathcal{D}'_{c}(I)$.

The first two classes give renormalized graph kernels compatible with interval extension, finite-window truncation, and admissible weight transport; the third names the residual obstruction problem.

Proposition B.17 *(Regular-density Costello graph kernel theorem).* Let

$$\mathcal{D}^{\mathrm{reg}}_{c}(I) = \{\, b(t)dt \mid b \in C^{\infty}_{c}(I) \,\} \subset \mathcal{D}'_{c}(I)$$

and let

$$\mathfrak{g}^{w,\mathrm{reg}}_{\hbar,I} = \big(\Omega^{\bullet}_{c}(I)\widehat{\otimes}\mathfrak{h}_{w,\hbar}\big) \ltimes \big(\mathcal{D}^{\mathrm{reg}}_{c}(I)\widehat{\otimes}\mathfrak{h}^{\vee,w}_{\hbar,\mathrm{cont}}\big)[1] \subset \mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}.$$

Put $\mathbf{K}^{\bullet,\mathrm{reg}}_{w,\partial,\hbar}(I)$ for the corresponding completed convolution Hom obtained from Definition B.15 by restricting the source to $C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}^{w,\mathrm{reg}}_{\hbar,I})$. Assume that, on every finite Hamiltonian window M of the local formal holomorphic Hamiltonian sector, the target $\mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$ is equipped with the regulator-admissible finite-scale Costello brane-defect graph calculus for $\mathbb{R}^{2}_{\mathrm{top}} \times \widehat{\mathbb{C}^{2}}_{0,\mathrm{hol}}$: the weighted propagator $P^{w,M}_{\epsilon,L}$, regularized BV Laplacian, local interaction $I^{w}_{0,M}$, interval extension by zero, truncation maps $\pi_{M,N}$, and admissible weight transports $R^{M}_{w,w'}$, satisfying the weighted RG criterion of Statement 5.79 and the graphwise Hadamard/Mittag–Leffler control of Theorem 8.17.

Then the restriction of $\kappa^{\mathrm{coef}}_{\hbar,w,I}$ to regular density labels has a graphwise Costello-local lift

$$\kappa^{\mathrm{graph,reg}}_{\hbar,w,I} \in \mathbf{K}^{0,\mathrm{reg}}_{w,\partial,\hbar}(I)$$

in the following precise sense. For every fixed connected Costello graph Γ, finite CE argument, finite window M, and external cotangent labels $B_j = b_j(t)dt \in \mathcal{D}^{\mathrm{reg}}_{c}(I)$, the finite-scale graph contribution

$$W^{\mathrm{reg},M}_{\Gamma,\epsilon,L}(a_{\bullet}, b_{\bullet}; f_{\bullet}, \rho_{\bullet}) \in \mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}\big(\mathcal{E}^{M}_{w}|_{I};\ A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)\big)$$

is a well-defined smooth local-functional kernel for $0 < \epsilon \le L < \infty$. Its small-$\epsilon$ expansion admits Costello local counterterms $C^{\mathrm{reg},M}_{\Gamma,w}(\epsilon; b_{\bullet})$, and the renormalized graph

$$W^{R,\mathrm{reg},M}_{\Gamma,L,w} = \lim_{\epsilon\to 0}\big(W^{\mathrm{reg},M}_{\Gamma,\epsilon,L} - C^{\mathrm{reg},M}_{\Gamma,w}(\epsilon; b_{\bullet})\big)$$

is again a brane-defect local functional with regular density coefficient labels. These graph kernels are compatible with interval extension by zero, finite-window truncation, and admissible weight transport:

$$\pi_{M,N} W^{R,\mathrm{reg},M}_{\Gamma,L,w} = W^{R,\mathrm{reg},N}_{\Gamma,L,w} \pi^{\mathrm{CE}}_{M,N}, \qquad R^{M}_{w,w'} W^{R,\mathrm{reg},M}_{\Gamma,L,w} = W^{R,\mathrm{reg},M}_{\Gamma,L,w'} R^{M,\mathrm{CE}}_{w,w'}.$$

On the zero-internal-edge part, and after scalar reduction and de Rham-current contraction, the lifted kernel is Φ^{cur}_{I}.

Consequently, wherever a filtered scalar projection $\widehat{\sigma}_{\mathrm{sc}}$ has been constructed on the chosen regular-density graph/counterterm subcomplex, the curvature

$$\mathrm{Curv}_{\mathbf{K}}(\kappa^{\mathrm{graph,reg}}_{\hbar,w,I})$$

defines the same finite-window non-scalar obstruction system as in Theorem 5.84. Thus the regular-density theorem removes the analytic wavefront/counterterm gap attached to the cotangent labels. The filtered scalar projection and vanishing of the residual class

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d^{L}_{\mathrm{qBV}}).$$

are the non-scalar QME data not contained in the regular-density graph theorem.

Extension from $\mathcal{D}^{\mathrm{reg}}_{c}(I)$ to $\mathcal{D}'_{c}(I)$ is the remaining analytic theorem: for every finite window M, every Costello graph Γ, and every compactly supported distributional label $B_j \in \mathcal{D}'_c(I)$, the product of the finite-scale brane-defect propagator kernels with the pushed-forward current labels must satisfy Hörmander wavefront transversality, admit local extensions across all graph small diagonals, and admit counterterms compatible with interval extension, $\pi_{M,N}$, and $R^{M}_{w,w'}$. This is a local brane-defect microlocal condition on the displayed finite-scale kernels.

Proof. The regular subspace is stable under the current operations used in the source. Indeed, if $B = b(t)dt$ and $a \in C^{\infty}_{c}(I)$, then Lemma B.4 gives $aB = (ab)(t)dt$, again a compactly supported smooth density. Extension by zero along $I \hookrightarrow J$ is smooth because the support is compactly contained in I. Thus the regular labels define the stated source subcomplex and the restriction of the convolution Hom complex.

Fix a finite window M, a graph Γ, and $0 < \epsilon \leq L < \infty$. In this window the coefficient part of the propagator is finite rank, and the spacetime part of $P^{w,M}_{\epsilon,L}$ is the finite-scale heat-kernel propagator of Costello's elliptic BV calculus [12, Chs. 2–4]. It is smooth at positive scale. After restriction to the brane line, multiplication by the smooth compactly supported functions a_i and densities $b_j(t)dt$ produces a smooth compactly supported graph integrand on the relevant configuration space and edge-parameter domain. Thus every product is a product of smooth densities at finite scale, and the graph integral is a well-defined element of the displayed brane-defect local-functional target.

As $\epsilon \to 0$, the singularities are the usual heat-kernel singularities along graph diagonals. Multiplication by smooth compact densities preserves wavefront sets and the Hadamard singular order. Therefore the local asymptotic expansion and local counterterm construction are exactly the windowwise Costello construction, tensored with the finite coefficient kernel and multiplied by smooth density coefficients. Theorem 8.17 gives the graphwise Hadamard estimates and strict Mittag–Leffler passage in the weighted coefficient direction. The counterterms remain local on the brane line; their coefficients are differential expressions in smooth compactly supported labels, hence regular density labels.

The compatibility identities are degreewise. The maps $\pi_{M,N}$ discard coefficient degrees above N, while $R^{M}_{w,w'}$ rescales the surviving weighted Hamiltonian and cotangent bases. The regular density factors live in the interval variable and are untouched by these coefficient maps. The strict Mittag–Leffler compatibility of the finite-window propagators and counterterms in Statement 5.79 therefore gives the two displayed identities. Interval extension compatibility is the same locality statement: all labels and counterterms are extended by zero, and local graph weights are supported on the same brane diagonals after extension.

The zero-internal-edge graph is the generator assignment of Proposition B.16. After forgetting the Moyal deformation, applying scalar reduction, and contracting the de Rham-current direction, it is Theorem B.7.

The curvature assertion is formal in the filtered convolution dg Lie algebra. The Bianchi identity used in the proof of Proposition B.16 applies to the regular restricted complex. Once a filtered scalar projection is available, the first unsolved curvature coefficient lands in $\ker \widehat{\sigma}_{\mathrm{sc}}$, and compatible counterterms change it by d^L_{qBV}-boundaries. The finite-window obstruction is exactly the inverse-limit H^1 class and the $\varprojlim^1 H^0$ primitive-compatibility class of Theorem 5.84.

If B is an arbitrary compactly supported distribution, the graph-level construction requires Hörmander transversality for the product of the pushed-forward brane current with propagator boundary values and a compatible extension across graph diagonals. This is exactly the residual theorem stated above. □

Definition B.18 *(Wavefront-admissible defect-current labels).* Fix a finite Hamiltonian window M, a connected Costello graph Γ, and a finite list of external cotangent-current labels $B_\bullet = (B_1, \dots, B_r)$, $B_j \in \mathcal{D}'_c(I)$. Let $X = \mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ and $L_I = I \times \{0\} \times \{p\} \subset X$. If $V_{\mathrm{bulk}}(\Gamma)$ and $V_\partial(\Gamma)$ are the bulk and defect vertices, set

$$X^\circ_\Gamma(I) = X^{V_{\mathrm{bulk}}(\Gamma)} \times L_I^{V_\partial(\Gamma)} \times E_{\Gamma,M},$$

where $E_{\Gamma,M}$ is the finite-dimensional coefficient and edge parameter space in the window M. The final graph space $X_\Gamma(I)$ is the open complement of the graph collision diagonals in $X^\circ_\Gamma(I)$ before local extension across those diagonals. Choose an attachment map $\ell : \{1, \dots, r\} \to V_\partial(\Gamma)$ for the current labels and define the incidence map

$$e_\Gamma : X^\circ_\Gamma(I) \longrightarrow I^r, \qquad e_{\Gamma,j}(x) = t_{\ell(j)}.$$

Repeated labels at one defect vertex are therefore diagonal pullbacks by e_Γ. For $B^\boxtimes = B_1 \boxtimes \cdots \boxtimes B_r$, write $B_\Gamma = e_\Gamma^* B^\boxtimes$ only when this pullback is defined. Let $\mathrm{Diag}(\Gamma)$ be the finite set of graph collision diagonals on which the small-ϵ Hadamard coefficients may be singular.

The tuple $B_\bullet$ is called (Γ, M)-wavefront-admissible if the following four operations are defined by the usual Hörmander calculus [36, Ch. 8] and by Costello's finite-window graph calculus [12, Chs. 5–7].

(i) The incidence pullback is non-characteristic:

$$N_{e_\Gamma} \cap \mathrm{WF}(B^\boxtimes) = \emptyset, \qquad N_{e_\Gamma} = \{(e_\Gamma(x), \xi) \mid de^t_{\Gamma,x}\xi = 0\}.$$

Then

$$\mathrm{WF}(B_\Gamma) \subset e_\Gamma^\# \mathrm{WF}(B^\boxtimes) = \{(x, de^t_{\Gamma,x}\xi) \mid (e_\Gamma(x), \xi) \in \mathrm{WF}(B^\boxtimes)\}.$$

The resulting support is proper over the graph integration variables.

(ii) For every Hadamard coefficient $K^M_{\Gamma,\alpha}$ appearing in the small-ϵ expansion of the finite-window graph kernel,

$$\big(\mathrm{WF}(K^M_{\Gamma,\alpha}) + \mathrm{WF}(B_\Gamma)\big) \cap T^*_{X_\Gamma(I)} X_\Gamma(I) = \emptyset.$$

Equivalently, the wavefront sets are transverse to the zero section after summation. Thus the product $K^{M}_{\Gamma,\alpha} B_\Gamma$ is a canonical distribution.

(iii) For each collision diagonal $\Delta \in \mathrm{Diag}(\Gamma)$, the products $K^{M}_{\Gamma,\alpha} B_\Gamma$ have finite scaling degree along Δ. Their allowed extensions across Δ differ by distributions supported on $\Delta \cap \mathrm{supp}(B_\Gamma)$, hence by brane-defect local counterterms with distributional current coefficients.

(iv) The final graph pushforward has wavefront contained in the brane-defect local-functional wavefront class already used in Theorem 8.17, after tensoring with the finite coefficient kernel.

Denote the resulting multilinear relation by

$$\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M) \subset \big(\mathcal{D}'_c(I)\big)^r.$$

The universal wavefront-admissible relation is the intersection of these relations over the finite windows, graphs, and finite CE arguments under consideration. This is a relation on finite tuples. A linear subspace $\mathcal{V} \subset \mathcal{D}'_c(I)$ gives an honest restricted CE source only when every finite tuple from $\mathcal{V}$ lies in the corresponding relation.

A useful source of such subspaces is a diagonal-admissible conic set $\Lambda \subset T^*I \setminus 0$: for every $n \geq 1$,

$$\xi_1, \ldots, \xi_n \in \Lambda \quad \Longrightarrow \quad \xi_1 + \cdots + \xi_n \neq 0.$$

Then

$$\mathcal{D}'_{c,\Lambda,\mathrm{fs}}(I) = \{\, B \in \mathcal{D}'_c(I) \mid \mathrm{WF}(B) \subset \Lambda,\ B \text{ has finite scaling degree along its singular support} \,\}$$

is the finite-scaling one-sided wavefront class. On an oriented interval, either open half-cone $\Lambda_+ = \{(t,\xi) \mid \xi > 0\}$ or $\Lambda_- = \{(t,\xi) \mid \xi < 0\}$ is diagonal-admissible. These two half-cones are maximal among sign-conic diagonal-admissible linear subspaces for ordinary collision-conormal kernels: any conic subset of $T^*I \setminus 0$ containing both signs admits, after positive rescaling, a finite nonempty covector sum equal to zero, so it can cancel a collision conormal. Compactly supported one-sided conormal boundary values, for example cutoffs of $(t - t_0 - i0)^{-m}$ after choosing the Fourier sign convention, lie in one of these distributional classes.

PROPOSITION B.19 *(Wavefront-admissible distributional graph kernel theorem).* Assume the same regulator-admissible finite-window Costello brane-defect graph calculus as in Proposition B.17. Let Γ be a fixed connected graph, M a finite Hamiltonian window, and $B_\bullet \in \mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M)$. Then the graph contribution with external cotangent labels $B_\bullet$ has the following distributional extension.

(i) For every $0 < \epsilon \leq L < \infty$, the finite-scale graph weight

$$W^{\mathrm{WF},M}_{\Gamma,\epsilon,L}(a_\bullet, B_\bullet; f_\bullet, \rho_\bullet)$$

is a well-defined brane-defect local-functional kernel with compactly supported distributional current coefficients. At this positive scale the propagator is smooth, so the only distributional operation is multiplication of smooth kernels by the pushed-forward currents.

(ii) The small-ϵ expansion admits local counterterms

$$C^{\mathrm{WF},M}_{\Gamma,w}(\epsilon; B_\bullet)$$

supported on

$$\bigcup_{\Delta \in \mathrm{Diag}(\Gamma)} \Delta \cap \mathrm{supp}(B_\Gamma).$$

The renormalized graph

$$W^{R,\mathrm{WF},M}_{\Gamma,L,w} = \lim_{\epsilon\to 0}\big(W^{\mathrm{WF},M}_{\Gamma,\epsilon,L} - C^{\mathrm{WF},M}_{\Gamma,w}(\epsilon;B_\bullet)\big)$$

is a brane-defect local functional with coefficients in the distributional current target $A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)$.

(iii) The construction is compatible with interval extension by zero, finite-window truncation, and admissible weight transport whenever the chosen tuple remains wavefront-admissible after applying the corresponding incidence maps:

$$\pi_{M,N}W^{R,\mathrm{WF},M}_{\Gamma,L,w} = W^{R,\mathrm{WF},N}_{\Gamma,L,w}\pi^{\mathrm{CE}}_{M,N}, \qquad R^M_{w,w'}W^{R,\mathrm{WF},M}_{\Gamma,L,w} = W^{R,\mathrm{WF},M}_{\Gamma,L,w'}R^{M,\mathrm{CE}}_{w,w'}.$$

In particular, every finite tuple of labels in $\mathcal{D}'_{c,\Lambda,\mathrm{fs}}(I)$ for a diagonal-admissible cone Λ is wavefront-admissible for graph kernels whose singular wavefronts are conormal to ordinary graph collision diagonals. This gives a non-regular distributional extension of Proposition B.17.

This is the exact extension supported by Costello graph products: the relation $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M)$ is the maximal subspace on which the multiplication, extension, and graph-pushforward operations all admit the cited wavefront calculus. The theorem extends to $\mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M)$; the obstruction and the admitted extension are entirely local on the brane interval and on the finite-window formal polydisk coefficient sector.

Proof. At positive scale the heat-kernel propagator $P^{w,M}_{\epsilon,L}$ is smooth in the spacetime and defect variables. Multiplication of a compactly supported distribution by a smooth function is always defined, and the incidence maps are proper on the supports by Definition B.18. This proves the finite-scale graph weight.

As $\epsilon \to 0$, Theorem 8.17 gives graphwise Hadamard coefficients with wavefront contained in the union of conormals to graph collision diagonals. The second condition in Definition B.18 is exactly Hörmander's product criterion [36, Ch. 8], so $K^M_{\Gamma,\alpha}B_\Gamma$ is a canonical distribution for every coefficient. The finite-scaling condition then gives the standard local extension across each collision diagonal. Its ambiguity is supported on the collision locus meeting the support of B_Γ; this is precisely a brane-defect local counterterm with distributional current coefficient. Subtracting these local counterterms gives the displayed renormalized graph.

The two compatibility identities are inherited degreewise from the finite-window Costello graph calculus. The maps $\pi_{M,N}$ and $R^M_{w,w'}$ act on the weighted Hamiltonian coefficient factors and preserve the interval wavefront cone. Interval extension by zero is allowed exactly when the extended incidence tuple satisfies the same non-characteristic and proper-support conditions. Hence compatibility holds on the stated wavefront-admissible relation.

For the one-sided conic classes, let Λ be diagonal-admissible and let $B_j \in \mathcal{D}'_{c,\Lambda,\mathrm{fs}}(I)$. A conormal covector to an n-fold collision diagonal has components $(\eta_1,\dots,\eta_n)$ whose sum is zero on each collision block. Cancellation by covectors from the external labels gives a finite nonempty sum of elements of Λ equal to zero, contradicting diagonal-admissibility. Thus the product criterion holds. The finite-scaling hypothesis gives the extension condition, and the remaining pushforward condition is the same proper-support condition as above.

Finally, the full compactly supported distribution space exceeds the graph-product domain. In the one-edge boundary graph take two external labels $B_1 = B_2 = \delta_{t_0}$. The singular boundary propagator has wavefront containing the conormal directions $(t_0,t_0;\tau,-\tau)$ to the diagonal $t = s$, while $\mathrm{WF}(\delta_{t_0}\boxtimes\delta_{t_0})$ contains the opposite directions $(t_0,t_0;-\tau,\tau)$. Hörmander's product criterion detects the obstruction. Equivalently, the obstructed graph is the diagonal value of the singular propagator against $\delta_{t_0}\boxtimes\delta_{t_0}$, which requires an additional renormalization rule. Thus current Costello graph products require a wavefront-admissible subspace of $\mathcal{D}'_c(I)$. □

PROPOSITION B.20 *(Current-compatible D'_c graph criterion).* Assume the regulator-admissible finite-window Costello brane-defect graph system of Proposition B.17. The construction compatible with compactly supported distributional cotangent labels is the following.

(i) For arbitrary $B \in \mathcal{D}'_c(I)$, the constructed object is the coefficient-current kernel

$$\kappa^{\mathrm{coef}}_{\hbar,w,I}(u_{a(t)dt\otimes f}) = \iota_I(B^w_f(a)), \qquad \kappa^{\mathrm{coef}}_{\hbar,w,I}(c_{B,\rho}) = \iota_I(\Theta^w_\rho(B)).$$

This is an algebraic current kernel. A Costello graph-kernel theorem for all $B \in \mathcal{D}'_c(I)$ is the additional analytic target.

(ii) For a fixed graph Γ, finite window M, and admissible tuple $B_\bullet \in \mathcal{D}'_{c,\mathrm{WF}}(I;\Gamma,M)$, put

$$T^M_{\Gamma,\alpha}(B_\bullet) = K^M_{\Gamma,\alpha}B_\Gamma.$$

Its wavefront is bounded by the Hörmander product formula

$$\begin{aligned}\mathrm{WF}\Big(T^M_{\Gamma,\alpha}(B_\bullet)\Big) \subset &\mathrm{WF}(K^M_{\Gamma,\alpha}) \cup \mathrm{WF}(B_\Gamma)\\ &\cup \big(\mathrm{WF}(K^M_{\Gamma,\alpha}) + \mathrm{WF}(B_\Gamma)\big),\end{aligned}$$

with the fibrewise sum taken over the same base point in $X_\Gamma(I)$. For the final graph projection $p_\Gamma : X_\Gamma(I) \to Y_\Gamma(I)$, proper on support, the pushed-forward graph has

$$\mathrm{WF}\big(p_{\Gamma,*}T^M_{\Gamma,\alpha}\big) \subset \{(y,\eta) \mid \exists x \in p_\Gamma^{-1}(y),\ (x, dp^t_{\Gamma,x}\eta) \in \mathrm{WF}(T^M_{\Gamma,\alpha})\}.$$

These are the only wavefront bounds used by the distributional graph complex.

(iii) Let $\Delta \in \mathrm{Diag}(\Gamma)$, let $c_\Delta = \operatorname{codim}\Delta$, and suppose

$$s_{\Gamma,\alpha,\Delta} = \mathrm{sd}_\Delta\Big(T^M_{\Gamma,\alpha}(B_\bullet)\Big) < \infty.$$

Set

$$N_{\Gamma,\alpha,\Delta} = \max\{-1,\ \lfloor s_{\Gamma,\alpha,\Delta} - c_\Delta\rfloor\}.$$

If $\widetilde{T}$ and $\widetilde{T}'$ are two local extensions of $T^M_{\Gamma,\alpha}(B_\bullet)$ across Δ, then locally in normal coordinates to Δ

$$\widetilde{T}' - \widetilde{T} = \sum_{|\beta|\le N_{\Gamma,\alpha,\Delta}} \partial^\beta_\nu \delta_\Delta \otimes S_{\Delta,\beta},$$

with the sum empty when $N_{\Gamma,\alpha,\Delta} = -1$. Here $S_{\Delta,\beta} \in \mathcal{D}'(\Delta)$ is supported in $\Delta \cap \operatorname{supp}(B_\Gamma)$ and carries the same finite-window Hamiltonian coefficient label. Thus the allowed counterterm ambiguity belongs to

$$\sum_{\Delta,\alpha}\ \sum_{|\beta|\le N_{\Gamma,\alpha,\Delta}} \partial^\beta_\nu \delta_\Delta\, \mathcal{D}'_{\Delta\cap\mathrm{supp}(B_\Gamma)}(\Delta)\widehat{\otimes}\mathcal{A}^{\mathrm{pp},w,M}_{\partial,\hbar}(I).$$

(iv) The finite-window counterterms are compatible precisely when the chosen extensions satisfy

$$j_* C^{\mathrm{WF},M}_{\Gamma,I}(\epsilon; B_\bullet) = C^{\mathrm{WF},M}_{\Gamma,J}(\epsilon; j_* B_\bullet),$$

for interval inclusions $j : I \hookrightarrow J$ for which the extended tuple remains admissible, and

$$\pi_{M,N} C^{\mathrm{WF},M}_{\Gamma,w}(\epsilon; B_\bullet) = C^{\mathrm{WF},N}_{\Gamma,w}(\epsilon; B_\bullet)\pi^{\mathrm{CE}}_{M,N},$$

$$R^M_{w,w'} C^{\mathrm{WF},M}_{\Gamma,w}(\epsilon; B_\bullet) = C^{\mathrm{WF},M}_{\Gamma,w'}(\epsilon; B_\bullet) R^{M,\mathrm{CE}}_{w,w'}.$$

Defect of the interval identity is a factorization-compatibility residual. Defect of the last two identities is exactly the finite-window primitive compatibility obstruction measured by Theorem 5.84 after passing to the relevant graph/counterterm complex.

(v) The theorem for all of $\mathcal{D}'_c(I)$ is an additional renormalization problem. In particular, the coincident-current boundary graph with $B_1 = B_2 = \delta_{t_0}$ violates the product condition because conormal covectors $(\tau, -\tau)$ to $t = s$ can be cancelled by $\mathrm{WF}(\delta_{t_0} \boxtimes \delta_{t_0})$. Thus the arbitrary-$D'_c$ graph theorem requires extra renormalization data beyond Costello's general finite-scale construction and the Costello–Gwilliam factorization vocabulary.

Hence the graph construction is proved on $\mathcal{D}^{\mathrm{reg}}_c(I)$, on finite graphwise wavefront-admissible tuples, and on the sign-conic finite-scaling classes $\mathcal{D}'_{c,\Lambda_\pm,\mathrm{fs}}(I)$ for ordinary collision-conormal graph kernels. The full distributional space $\mathcal{D}'_c(I)$ remains an open obstruction problem.

Proof. Item (1) is Proposition B.16: the assignment is degreewise algebraic in the compactly supported current B and uses only multiplication of a current by a smooth test function.

Items (2) and (3) are the standard Hörmander product, pushforward, and finite-scaling extension calculus applied to Definition B.18. The product transversality condition makes $K^M_{\Gamma,\alpha} B_\Gamma$ canonical. Properness of the graph projection gives the displayed pushforward wavefront bound. Finite scaling degree along Δ gives extensions across Δ; the structure theorem for distributions supported on a submanifold gives the finite normal-derivative formula for the difference of two extensions. The support statement follows because all extensions agree off $\Delta \cap \mathrm{supp}(B_\Gamma)$.

Item (4) is the compatibility condition needed for the counterterms to define one interval-local element of the completed finite-window and weight-transport tower. When the identities hold, the Bianchi argument of Proposition B.16 produces the Hom-valued obstruction class and the Milnor $\varprojlim^1 H^0$ primitive obstruction. Interval-extension defects are factorization-current residuals; finite-window or weight defects are primitive-compatibility obstructions.

Item (5) is the counterexample in the proof of Proposition B.19. It also matches the Costello graph formalism: the local sources support finite-scale BV/RG/QME graph calculus, local functionals, and local counterterms under their hypotheses. The current-valued arbitrary-$\mathcal{D}'_c$ brane-defect theorem requires the extra renormalization rule above. □

Four obstruction classes control the passage from the algebraic coefficient-current kernel (Proposition B.16) to a Costello local-functional graph theorem: the scalar projection of the curvature; the recovery condition under scalar reduction; the three unreduced bracket defects $D_{f,g}, D_{f,\rho}, D_{\rho,\sigma}$ of degree zero with their factorization / finite-window-compatible homotopies; and the residual non-scalar curvature class in $H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{qBV}})$. The binary centrality criterion of Theorem B.22 records the precise homotopy equation (40) that a non-scalar Costello local lift must satisfy on every pair of source generators, with the inverse-system and Milnor primitive classes that obstruct compatible window-by-window assembly.

PROPOSITION B.21 *(Obstruction maps for the non-scalar current lift).* The datum of Definition B.14 can realize the non-scalar Costello graph/QME theorem for the local mixed HT brane-defect model only if the following obstruction maps vanish, with the stated homotopies.

(i) The scalar projection of the curvature must vanish in the chosen scalar branch:

$$\widehat{\sigma}_{\mathrm{sc}}\big(\mathrm{Curv}_{\mathbf{K}}(\kappa_{\hbar,w,I})\big) = 0.$$

In the ordinary scalar-reduced $\mathfrak{gl}_N$ branch this condition is exactly the requirement that the Capelli contact class $\hbar N[\bar{c}]$ be cancelled by a closed-exchange term and a local counterterm. In the balanced supertrace branch it is the condition $\widehat{\sigma}_{\mathrm{sc}}(\mathrm{Ob}_{w,\partial,\hbar}) = 0$.

(ii) After scalar reduction and de Rham-current contraction, the kernel must recover the reduced current interface:

$$\overline{\kappa}_{\hbar,w,I}(u_{a(t)dt\otimes f}) = B_f(a), \qquad \overline{\kappa}_{\hbar,w,I}(c_{B,\rho}) = \Theta_\rho(B),$$

with brackets

$$\{B_f(a), B_g(b)\}_\hbar = B_{\{f,g\}_\hbar}(ab), \qquad \{B_f(a), \Theta_\rho(B)\}_\hbar = \Theta_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

and $\{\Theta_\rho(B), \Theta_\sigma(C)\}_\hbar = 0$. At $\hbar = 0$ this is Theorem B.7.

(iii) The unreduced bracket defects

$$D_{f,g}(a,b) = \{B_f^{\mathrm{BV}}(a), B_g^{\mathrm{BV}}(b)\}_\hbar - B^{\mathrm{BV}}_{\{f,g\}_\hbar}(ab),$$

$$D_{f,\rho}(a,B) = \{B_f^{\mathrm{BV}}(a), \Theta_\rho^{\mathrm{BV}}(B)\}_\hbar - \Theta^{\mathrm{BV}}_{\mathrm{ad}^\vee_{f,\hbar}\rho}(aB),$$

and

$$D_{\rho,\sigma}(B,C) = \{\Theta_\rho^{\mathrm{BV}}(B), \Theta_\sigma^{\mathrm{BV}}(C)\}_\hbar$$

must be d^L_{qBV}-exact by homotopies compatible with inclusions of intervals, disjoint factorization products, and the finite-window regulator maps.

(iv) After evaluation on each fixed finite CE argument ξ, equivalently on each fixed graph/source-arity component, the residual non-scalar curvature class

$$\big[\mathrm{Curv}_{\mathbf{K}}(\kappa_{\hbar,w,I})(\xi)\big] \in H^1\Big(\ker \widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{qBV}}\Big)$$

must vanish after the compatible finite-window counterterms $C^{w_0}_{\Gamma,w}(\epsilon)$ have been inserted.

These conditions are necessary. They are also sufficient for the stated local graph/QME realization once the Costello finite-scale BV propagator, BV Laplacian, locality estimates, and counterterm system in the chosen coefficient category have been constructed.

Proof. The source and target in Definition B.14 are the current versions of the weighted cotangent Hamiltonian source and the mixed brane-defect obstruction complex of Problem 5.220. A Costello graph realization is a solution of the QME in this local-functional complex, hence its curvature has zero scalar projection and zero residual class in $\ker \widehat{\sigma}_{\mathrm{sc}}$. This gives the first and fourth obstruction maps.

The second condition follows from compatibility with the already proved reduced current theorem. After contraction and scalar reduction the only allowed target brackets are the Moyal principal-part current brackets of Theorem 5.218; at $\hbar = 0$ these reduce to the three brackets of Theorem B.7. Any different reduction would change the theorem already proved.

The three defects in the third condition are the generator-level differences between the source semidirect Lie brackets and the target P_0-brackets. Since the CE differential and the target bracket are derivations, exactness of these generator defects, with homotopies compatible with factorization products and finite-window transition maps, is precisely the condition that the generator assignment extend to the homotopy P_0-central operation. The reduced algebraic target has zero defects by Theorem B.7; the unreduced polynomial BV target has the obstruction of Theorem E.6. Therefore the required theorem is the construction of these homotopies in a larger current-compatible Costello local-functional category. □

Theorem B.22 *(Curved kernel centrality homotopy criterion).* Fix the local mixed HT brane-defect model on $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$, a weighted current source $\mathfrak{g}^{w,\mathrm{cur}}_{\hbar,I}$, and a Costello local-functional target $\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$ as in Definition B.14. Let κ_M be a finite-window component of a degree-zero continuous kernel in the convolution Hom complex $\mathbf{K}^\bullet_{w,\partial,\hbar}(I)$, and assume that the chosen scalar-contact projection is a chain map on the complex under discussion.

For source generators

$$x \in \{u_{a(t)dt\otimes f},\ c_{B,\rho}\}, \qquad y \in \{u_{b(t)dt\otimes g},\ c_{C,\sigma}\},$$

define the binary closed-open curvature component by

$$\mathfrak{D}^M_\kappa(x, y) = \mathrm{Curv}_{\mathbf{K},M}(\kappa_M)(x \wedge y).$$

Before inserting a binary Taylor component, these are the three generator defects

$$D^M_{f,g}(a, b), \qquad D^M_{f,\rho}(a, B), \qquad D^M_{\rho,\sigma}(B, C)$$

displayed in Proposition B.21.

A compatible unreduced binary centrality homotopy for this kernel is exactly a family of cochains

$$H^M_{x,y} \in \mathcal{K}^{|x|+|y|-1}_{\mathrm{ns},M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M} \cap \mathfrak{Q}^{|x|+|y|-1}_{w,M,\partial,\hbar}(I)$$

satisfying

$$\mathfrak{D}^M_\kappa(x, y) + d_M H^M_{x,y} = 0 \tag{40}$$

for every such pair, together with the compatibility identities

$$j_* H^M_{I;x,y} = H^M_{J;j_*x,j_*y},$$

for interval inclusions $j : I \hookrightarrow J$ for which the labels remain admissible, and

$$\pi_{M,N} H^M_{x,y} = H^N_{\pi x,\pi y}, \qquad R^M_{w,w'} H^M_{w;x,y} = H^M_{w';Rx,Ry}.$$

Equivalently, for every pair (x, y), the inverse-system cohomology class

$$\big([\mathfrak{D}^M_\kappa(x, y)]\big)_M \in \varprojlim_M H^{|x|+|y|}\Big(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_M\Big)$$

and the Milnor primitive-compatibility class represented by

$$\Big[\pi_{M,N} h^M_{x,y} - h^N_{\pi x,\pi y}\Big] \in H^{|x|+|y|-1}\Big(\mathcal{K}^\bullet_{\mathrm{ns},N}(I), d_N\Big)$$

must both vanish, where $h^M_{x,y}$ is any windowwise primitive solving $\mathfrak{D}^M_\kappa(x,y) + d_M h^M_{x,y} = 0$.

For the algebraic coefficient-current kernel $\kappa^{\mathrm{coef}}_{\hbar,w,I}$ of Proposition B.16, all $\mathfrak{D}^M_\kappa(x,y)$ vanish in the freely adjoined current target, and $H^M_{x,y} = 0$. This is the strict homotopy of Theorem E.10. For a genuine unreduced Costello local-functional target, the same conclusion requires the finite-window graph row representing $\mathfrak{D}^M_\kappa(x,y)$, a scalar-zero condition for that row, and a local counterterm or binary Taylor cochain $H^M_{x,y}$ satisfying equation (40) with the displayed interval, window, and weight compatibilities.

Proof. The arity-two part of the curved Maurer–Cartan equation in the convolution dg Lie algebra is precisely

$$\mathrm{Curv}_{\mathbf{K},M}(\kappa_M)(x \wedge y) = 0.$$

Expanding this identity before choosing a binary component gives the target bracket of the two generator images minus the image of the source semidirect bracket, with the source term given by $-\kappa d_{\mathrm{CE}}$. These are exactly the three defects $D_{f,g}$, $D_{f,\rho}$, and $D_{\rho,\sigma}$ of Proposition B.21.

A binary homotopy is, by definition, a degree $|x| + |y| - 1$ primitive for this curvature component in the scalar-zero kernel complex. This gives equation (40). Compatibility with extension by zero, finite-window truncation, and admissible weight transport is exactly the statement that these primitives assemble to a homotopy in the completed local factorization target. The inverse-limit obstruction and the Milnor class are the usual Roos obstruction for choosing compatible primitives, in the same form as Theorem 5.84.

In the algebraic coefficient-current target the three brackets are identities, hence the curvature components vanish and the zero primitive is compatible with every structural map. In a Costello local-functional target the terms are graph weights, source faces, collision/contact terms, and counterterms. Supplying those rows and their primitive in the chosen finite-window system turns the displayed obstruction equation into a constructed homotopy. □

C Sign conventions for the Hamiltonian comparison

The two $\mathfrak{h}$-modules attached to the labels (a,b) go in opposite directions. The polynomial PBW special-fibre descendant $\widetilde{\Psi}_{a,b}$, symmetrised trace shadow of the open Hamiltonian action on the matrix Koszul algebra A_ψ, admits the lowering action $z_1 \cdot \widetilde{\Psi}_{a,b} = b\,\widetilde{\Psi}_{a,b-1}$, $z_2 \cdot \widetilde{\Psi}_{a,b} = -a\,\widetilde{\Psi}_{a-1,b}$. The Matlis principal-part vector $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$, residue dual of the formal Hamiltonian polydisk against its continuous Taylor dual, admits the raising action $z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}$, $z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}$. Diagonal rescaling $\rho_{a,b} \mapsto \lambda_{a,b}\rho_{a,b}$, $\widetilde{\Psi}_{a,b} \mapsto \mu_{a,b}\widetilde{\Psi}_{a,b}$ preserves the locally nilpotent lowering direction on descendants and the raising direction on principal parts; the directions of the index shifts already disagree.

The cohomological shift convention is $(V[1])^n = V^{n+1}$; the reduced matrix Koszul letter ψ is odd, and cyclic graded signs use its parity. Absolute BV grading enters only through the global shift of the cotangent coordinate (Definition E.1). BV degree, ghost number, and form degree are tracked separately: BV degree is cohomological, ghost number is the antifield/ghost grading of the classical fields (ϕ_1, ϕ_2, ψ) and $(\mathcal{A}, \mathcal{B})$, and form degree is the de Rham/Dolbeault exterior grading on $\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)$; none of the three is identified with another. Hochschild homology $HH_\bullet$, cyclic homology, and negative-cyclic homology $HC^-_\bullet$ are distinguished at every reference; the cyclic trace symbol Tr below is the graded cyclic quotient of Definition C.1. Its Hochschild shadow is obtained by forgetting the S^1-equivariance. Connes' B is the degree-+1 operator in the mixed complex and the connecting operator in the SBI sequence. The negative-cyclic class is the constellation-level Calabi–Yau pairing of the compact CY-to-chiral target.

The BCOV holomorphic-anomaly equation enters only as a target equation on a compact Calabi–Yau threefold, with the BCOV-strict sign of [1] preserved by the matched-conventions divergence row of Theorem 11.21; the local Hamiltonian theorem is the formal-Darboux stalk.

Definition C.1 *(Local symplectic and cyclic conventions).* Fix the formal-local holomorphic symplectic datum $(\widehat{\mathbb{C}^2}_0, \Omega_{\mathrm{loc}} = dz_1 \wedge dz_2)$ of Section 8. The Hamiltonian vector field of $f \in \mathbb{C}[[z_1, z_2]]$ is determined by

$$\iota_{X_f}\Omega_{\mathrm{loc}} = -df,$$

giving

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad X_{z_1} = \partial_{z_2}, \quad X_{z_2} = -\partial_{z_1}.$$

The Poisson bracket is

$$\{f, g\} = X_f(g) = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g,$$

in particular $\{z_1, z_2\} = 1$. Hamiltonian fields are divergence-free for Ω_{loc}: the volume-form scalar contribution is zero (Proposition E.2).

The reduced matrix Koszul algebra of Definition D.1 is

$$A_\psi = \mathbb{C}\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| \equiv 1 \pmod 2, \qquad Q\psi = xy - yx, \; Qx = Qy = 0.$$

The graded cyclic quotient uses the Koszul sign rule

$$[uv]_{\mathrm{cyc}} = (-1)^{|u||v|}[vu]_{\mathrm{cyc}},$$

and the trace symbol Tr is graded cyclic with the same sign. The only odd letter is ψ.

Lemma C.2 *(CE/PV cochain signs).* Let $\mathfrak{h}$ have basis e_I and structure constants $[e_I, e_J] = C^K_{IJ} e_K$. Let c^I be the CE ghost dual to e_I, and let u_K be the shifted cotangent coordinate paired with e_K under the right-adjoint action $e_J \cdot u_K = [u_K, e_J] = C^L_{KJ} u_L$. The Chevalley–Eilenberg differential is

$$d_{\mathrm{CE}} c^K = -\tfrac{1}{2} C^K_{IJ} c^I c^J, \qquad d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J.$$

On the polyvector model with even coordinates O_K and odd coordinates θ^I, Schouten bracket $[\theta^I, O_J] = \delta^I_J$, and BV interaction $\pi = \tfrac{1}{2} C^K_{IJ} O_K \theta^I \theta^J$, the differential $d_\pi = [\pi, -]$ acts by

$$d_\pi \theta^K = -\tfrac{1}{2} C^K_{IJ} \theta^I \theta^J, \qquad d_\pi O_K = C^L_{KJ} O_L \theta^J.$$

Therefore the assignment

$$c^I \longmapsto \theta^I, \qquad u_K \longmapsto O_K$$

is a chain map intertwining d_{CE} and d_π.

Proof. The CE differential on c^K is the standard Koszul dual of the bracket, with the sign convention $d_{\mathrm{CE}} c^K = -\tfrac{1}{2} C^K_{IJ} c^I c^J$ ensuring $d^2_{\mathrm{CE}} = 0$ by the Jacobi identity in $\mathfrak{h}$. The CE differential on u_K is the dual of the right-adjoint action $e_J \cdot u_K = C^L_{KJ} u_L$; applying d_{CE} once more to $C^L_{KJ} u_L c^J$, with the Leibniz rule and $|u_L| = 0$, yields

$$C^L_{KJ}(C^M_{LA} u_M c^A) c^J - \tfrac{1}{2} C^L_{KJ} u_L C^J_{AB} c^A c^B,$$

which vanishes by the Jacobi identity for the right-adjoint representation.

The polyvector identity $d_\pi^2 = 0$ is the master equation $[\pi, \pi] = 0$, which in the present notation is the Jacobi identity for $\mathfrak{h}$ repackaged as a quadratic relation among the structure constants. The match of d_π on generators with the CE formulas above is standard (Costello [12], Chapter 5): one verifies on a single non-abelian example that $\pi = \frac{1}{2} C_{IJ}^K O_K \theta^I \theta^J$ reproduces the right-adjoint CE differential on the cotangent coordinate, and the general case follows by linearity in π. Both d_{CE} and d_π are derivations, so the assignment $c^I \mapsto \theta^I$, $u_K \mapsto O_K$ extends from generators to a chain map on the full Chevalley–Eilenberg and polyvector algebras. $\square$

LEMMA C.3 *(Primitive one-ψ chain boundary).* In A_ψ of Definition C.1, for ordinary words u, v in x, y,

$$Q\ \mathrm{Tr}(\psi u \psi v) = \mathrm{Tr}\big(\psi v x y u - \psi v y x u - \psi u x y v + \psi u y x v\big).$$

Removing the common front ψ gives the boundary in the one-ψ chain model

$$\partial_2(u, v) = v x y u - v y x u - u x y v + u y x v.$$

Proof. The first displayed ψ contributes $+(xy - yx)$ under Q; the second contributes $-(xy - yx)$ because exactly one odd letter precedes it, so the Koszul sign of the derivation flips. The even word produced by the first replacement may be moved cyclically past the remaining ψ and the word v with sign $+1$: the Koszul sign $(-1)^{|u||v|}$ for moving an even string past a graded cyclic block is trivial. Removing the common front ψ identifies the result with the displayed boundary. Cross-checked against Lemma D.5. $\square$

LEMMA C.4 *(Residue coadjoint signs).* Let

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \qquad a, b \geq 0,$$

with residue pairing $\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}$ and coadjoint sign $(f \cdot \rho)(g) = -\rho(\{f, g\})$. Then

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1, b-q+1},$$

with negative-index targets and the scalar target $\rho_{0,0}$ set to zero. In particular,

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Proof. Divergence-free Hamiltonian fields (Definition C.1) make the residue pairing $\mathfrak{h}$-equivariant via integration by parts with zero volume-form correction, fixing the coadjoint convention $(f \cdot \rho)(g) = -\rho(\{f, g\})$. Pair against a test monomial $g = z_1^m z_2^n$:

$$(z_1^p z_2^q \cdot \rho_{a,b})(z_1^m z_2^n) = -\rho_{a,b}(\{z_1^p z_2^q, z_1^m z_2^n\}).$$

Direct computation gives $\{z_1^p z_2^q, z_1^m z_2^n\} = (pn - qm) z_1^{p+m-1} z_2^{q+n-1}$, and the residue pairing keeps the (a, b) Taylor coefficient. The unique (m, n) producing $z_1^a z_2^b$ is $m = a - p + 1$, $n = b - q + 1$; substitution yields coefficient $p(b - q + 1) - q(a - p + 1) = pb - qa + p - q$, and the leading minus from the coadjoint convention gives the displayed formula. Targets with $a - p + 1 < 0$ or $b - q + 1 < 0$ require $m < 0$ or $n < 0$ and are unreachable by polynomial test monomials, hence the cutoff at negative indices. The scalar target $\rho_{0,0}$ is the case $p = a + 1$, $q = b + 1$, at which the coefficient $(a+1)b - (b+1)a + (a+1) - (b+1) = 0$ vanishes automatically; the formula is therefore consistent with the exclusion of $\rho_{0,0}$ from $\mathcal{P} = \mathrm{Ann}_{\mathrm{res}}(\mathbb{C} \cdot 1)$. $\square$

Lemma C.5 *(PBW descendant signs).* The polynomial primitive one-ψ descendants $\widetilde{\Psi}_{a,b}$ of Corollary 5.173 carry the symmetric PBW special-fibre action

$$z_1^p z_2^q : \widetilde{\Psi}_{a,b} \longmapsto (pb - qa)\widetilde{\Psi}_{a+p-1,b+q-1}.$$

In particular, $z_1 \cdot \widetilde{\Psi}_{a,b} = b\widetilde{\Psi}_{a,b-1}$ and $z_2 \cdot \widetilde{\Psi}_{a,b} = -a\widetilde{\Psi}_{a-1,b}$: the action lowers the polynomial bidegree.

Proof. The Hamiltonian vector field of $z_1^p z_2^q$ is

$$X_{z_1^p z_2^q} = p z_1^{p-1} z_2^q \partial_{z_2} - q z_1^p z_2^{q-1} \partial_{z_1}.$$

Acting on the symmetrised trace with a factors of x and b factors of y, the first summand differentiates one of the b copies of y and contributes pb copies of the symmetrised target $\widetilde{\Psi}_{a+p-1,b+q-1}$; the second differentiates one of the a copies of x and contributes $-qa$ copies of the same target. The sum is the displayed coefficient. □

Lemma C.6 *(Moyal expansion coefficients).* With

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}, \qquad f * g = m \circ \exp\left(\frac{\hbar}{2} P\right)(f \otimes g),$$

the normalized Moyal bracket $\{f, g\}_\hbar = \hbar^{-1}(f * g - g * f)$ expands as

$$\{f, g\}_\hbar = \sum_{s \geq 0} \frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g),$$

with first quantum correction $\hbar^2 P^3(f, g)/24$:

$$\{f, g\}_\hbar = \{f, g\} + \frac{\hbar^2}{24} P^3(f, g) + O(\hbar^4).$$

Proof. Even powers of P are symmetric and odd powers antisymmetric under $f \leftrightarrow g$; the commutator therefore retains only the odd powers and doubles them:

$$f * g - g * f = 2 \sum_{s \geq 0} \frac{(\hbar/2)^{2s+1}}{(2s+1)!} P^{2s+1}(f, g).$$

Division by $\hbar$ gives the displayed series. The $s = 1$ coefficient is $1/(2^2 \cdot 3!) = 1/24$; the all-order identity is Theorem F.2, and exact rational coefficient checks through order eleven give the same sequence. □

Lemma C.7 *(Moyal coadjoint signs).* Let $(n)_r = n(n-1)\cdots(n-r+1)$, with $(n)_0 = 1$, denote the falling factorial. For $f = z_1^p z_2^q$, the normalized Moyal coadjoint action on principal parts is

$$f \cdot_\hbar \rho_{a,b} = -\sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{\hbar^{r-1}}{2^{r-1} r!} C_r(p, q; a-p+r, b-q+r)\, \rho_{a-p+r,b-q+r},$$

with negative-index targets and the scalar target $\rho_{0,0}$ set to zero. Here

$$C_r(p, q; m, n) = \sum_{\alpha=0}^{r} (-1)^\alpha \binom{r}{\alpha} (p)_{r-\alpha} (q)_\alpha (m)_\alpha (n)_{r-\alpha}$$

is the bidifferential coefficient of Theorem F.2. At $r = 1$ the formula recovers the residue coadjoint of Lemma C.4,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}.$$

Proof. Apply the bidifferential P^r to $z_1^p z_2^q \otimes z_1^m z_2^n$: Theorem F.2 gives $P^r(f,g) = C_r(p,q;m,n)\, z_1^{p+m-r} z_2^{q+n-r}$. By Lemma C.6 the normalized Moyal bracket keeps only odd r, with coefficient $\hbar^{r-1}/(2^{r-1}r!)$. The coadjoint convention $(f \cdot_\hbar \rho)(g) = -\rho(\{f,g\}_\hbar)$ introduces the leading minus sign and determines the target principal part: pairing with $\rho_{a,b}$ selects (m,n) such that $p+m-r=a$ and $q+n-r=b$, i.e. $m=a-p+r$, $n=b-q+r$. At $r=1$ the coefficient is $C_1(p,q;a-p+1,b-q+1) = p(b-q+1) - q(a-p+1) = pb - qa + p - q$, reproducing Lemma C.4. □

COROLLARY C.8 *(Universal residue-dual normalization).* With the residue pairing $\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}$ and the coadjoint convention

$$(f \cdot \rho)(g) = -\rho(\{f,g\}), \qquad (f \cdot_\hbar \rho)(g) = -\rho(\{f,g\}_\hbar),$$

the linear Hamiltonians act on $\mathcal{P}$ by

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b},$$

while the polynomial descendants of Lemma C.5 satisfy

$$z_1 \cdot \widetilde{\Psi}_{a,b} = b\,\widetilde{\Psi}_{a,b-1}, \qquad z_2 \cdot \widetilde{\Psi}_{a,b} = -a\,\widetilde{\Psi}_{a-1,b}.$$

Diagonal rescaling $\rho_{a,b} \mapsto \lambda_{a,b}\rho_{a,b}$, $\widetilde{\Psi}_{a,b} \mapsto \mu_{a,b}\widetilde{\Psi}_{a,b}$ with $\lambda_{a,b}, \mu_{a,b} \in \mathbb{C}^\times$ preserves the opposite index directions of the two actions. Adjustments of factorial normalizations or of the Moyal $1/24$ coefficient act diagonally and leave the index shifts intact.

Proof. The two displayed actions and the residue pairing are restated from Lemma C.4 and Lemma C.5 at $(p,q) \in \{(1,0),(0,1)\}$. A diagonal rescaling sends $z_1 \cdot \rho_{a,b}$ to $-(b+1)(\lambda_{a,b+1}/\lambda_{a,b})\rho_{a,b+1}$, preserving the raising target index, and sends $z_1 \cdot \widetilde{\Psi}_{a,b}$ to $b(\mu_{a,b-1}/\mu_{a,b})\widetilde{\Psi}_{a,b-1}$, preserving the lowering target index. Equality of the two actions forces the same target index for both, contradicting the directions $b \mapsto b+1$ versus $b \mapsto b-1$. Higher Moyal corrections $r \geq 3$ shift indices by $b \mapsto b-q+r \geq b+1$ for $(p,q)=(1,0)$ and by $a \mapsto a-p+r \geq a+1$ for $(p,q)=(0,1)$; they raise as well, so the obstruction persists at every order in $\hbar$. The finite-window homology calculation records this obstruction explicitly: writing $\delta_{a,b}$ for the residue functional $\rho_{a,b}$ under the Matlis identification of Lemma 8.4, the third-order Moyal coadjoint gives $\mathrm{ad}^*_{z_1^4}(\delta_{1,1}) = -24\hbar^2\delta_{0,4}$, while the polynomial one-ψ action sends $\Psi_{1,1}$ to $4\Psi_{4,0}$; even matched at the Capelli/Weyl shift $\hbar N$, the higher Moyal coadjoint terms have zero PBW descendant counterpart. □

LEMMA C.9 *(Brane-line current source convention).* For an interval $I \subset \mathbb{R}$, the Hamiltonian source complex is $\Omega^\bullet_c(I)\widehat{\otimes}\mathfrak{h}$ of Definition B.1. A degree-zero source density is written $a(t)dt \otimes f$, and the cotangent CE source coordinate maps to the boundary observable by

$$u_{a(t)dt\otimes f} \longmapsto B_f(a) = \int_I a(t) B_f(t)\, dt.$$

The companion ghost coordinate $c_{a\otimes f}$ maps to the PV coordinate $\theta_{a\otimes f}$. For $B \in \mathcal{D}'_c(I)$, the multiplication of the distribution B by a smooth test function a is the standard $C_c^\infty(I)$-module operation aB, defined by $(aB)(\varphi) = B(a\varphi)$. The interval current brackets are

$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0.$$

Proof. The de Rham factor gives the compactly supported density on the brane line; the Hamiltonian current factor gives f. Pairing this source with the boundary field gives the integral defining $B_f(a)$. The first bracket is the semidirect-product bracket of Definition B.2 for $\mathfrak{h}_I \ltimes \mathcal{P}_I[1]$: the action factors as the $C_c^\infty(I)$-module operation $B \mapsto aB$ tensored with the Matlis coadjoint action of Lemma C.4. The second bracket vanishes because the shifted principal-part summand is an abelian odd ideal. □

D Full Primitive Higher-ψ Koszul Homology

The polynomial PBW one-ψ descendant labels $\Upsilon_{a,b}$ of the open BV brane sector and the residual Matlis principal parts $\rho_{a,b}$ of § 8 share the bidegree index set (a, b) and carry different $\mathfrak{h}$-actions. The defect position is polynomial; its cotangent momentum is residual. The separation is the homology of a single universal calculation: the full primitive cyclic Koszul homology of $(A_\psi,\ d\psi = xy - yx)$ graded by ψ-count and by $(R, S) = (\#x + \#\psi, \#y + \#\psi)$. The surviving H_1-line is the special-fibre image of the PBW Rees lattice; the associated-graded morphism $\operatorname{gr}\Phi : \bar{A}_{\mathrm{desc}}[1] \to \mathcal{P}$ is the bidegree-only special-fibre relabelling.

The universal primitive single-trace Koszul complex is attached to the moment-map equation $[x, y] = 0$. Its universality is independent of any choice of Matlis target, PBW model, finite N trace identity, or quantum deformation: after stable trace invariant theory has identified primitive traces with graded cyclic words, the answer is the cellular homology of a cyclic-word complex. Every finite-bidegree stable primitive descendant calculation over $\mathbb{C}$, with $d\psi = xy - yx$ and graded cyclicity as the trace relation, is obtained from this one. Passage to completed coefficient systems requires an exact coefficient functor, or an explicit condition excluding new $\varprojlim^1$ classes.

Definition D.1 *(Graded cyclic word complex).* Let

$$A_\psi = \mathbb{C}\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| \equiv 1 \pmod 2,$$

and let d be the odd derivation determined by

$$d\psi = xy - yx, \qquad dx = dy = 0.$$

Put

$$\operatorname{Cyc}(A_\psi) = A_\psi/[A_\psi, A_\psi]_{\mathrm{gr}}, \qquad [uv] = (-1)^{|u||v|}[vu].$$

Give the letters weights

$$\mathrm{wt}(x) = (1, 0), \qquad \mathrm{wt}(y) = (0, 1), \qquad \mathrm{wt}(\psi) = (1, 1).$$

For $R, S \geq 0$ define

$$K^p_{R,S} = \mathbb{C}\{[w]_{\mathrm{cyc}} : \#\psi(w) = p,\ \#x(w) = R - p,\ \#y(w) = S - p\},$$

with $K^p_{R,S} = 0$ unless $0 \leq p \leq \min(R, S)$. If $w = a_1 \cdots a_n$, the boundary is

$$\partial[w] = \sum_{i : a_i = \psi} (-1)^{\#\{j<i : a_j = \psi\}} [a_1 \cdots a_{i-1} xy\, a_{i+1} \cdots a_n - a_1 \cdots a_{i-1} yx\, a_{i+1} \cdots a_n]_{\mathrm{cyc}}.$$

Thus $\partial : K^p_{R,S} \to K^{p-1}_{R,S}$.

LEMMA D.2 *(Stable super-cyclic trace presentation).* Fix a total word length d. In the stable range $N \geq d$, the primitive GL_N-invariant single-trace functions in the variables x, y, ψ, homogeneous of total word length d, are exactly the graded cyclic words in $\mathrm{Cyc}(A_\psi)$. Equivalently, after connected single-trace extraction, the only stable relation among words in x, y, ψ is

$$\mathrm{Tr}(uv) = (-1)^{|u||v|}\,\mathrm{Tr}(vu).$$

Proof. First replace the odd letter ψ by p distinct even marked matrices $\eta_1, \ldots, \eta_p$, and take the summand multilinear in the η_i's and homogeneous of the prescribed x, y-degree. The Procesi–Razmyslov stable invariant theorem [17][18] identifies the primitive single-trace invariants of total word length d, for $N \geq d$, with ordinary cyclic words in the letters $x, y, \eta_1, \ldots, \eta_p$, modulo only cyclicity of the trace.

The symmetric group $\mathbb{S}_p$ permutes the marked letters. Passing from p even marked letters to one odd letter is the sign-isotypic projection for this action, followed by the identification $\eta_i \mapsto \psi$. Since the ground field has characteristic zero, this projection is exact. Interchanging two marked occurrences contributes the sign of the transposition, and rotating a block with a marked letters past a block with b marked letters contributes $(-1)^{ab}$. This is precisely the graded cyclic rule for a single odd letter ψ. This is the complete stable trace relation in word length d. □

LEMMA D.3 *(Boundary signs and $\partial^2 = 0$).* The formula of Definition D.1 is well-defined on the graded cyclic quotient and satisfies $\partial^2 = 0$.

Proof. On the free algebra A_ψ, the displayed formula is the derivation rule for the odd derivation d: replacing the i-th occurrence of ψ crosses exactly the preceding odd letters and hence contributes $(-1)^{\#\{j<i : a_j = \psi\}}$. Since $d(x) = d(y) = 0$ and $d(\psi) = xy - yx$ is even and d-closed, one has $d^2 = 0$ on the generators, hence on all of A_ψ.

If u, v are homogeneous, then

$$d(uv - (-1)^{|u||v|}vu)$$

is a sum of graded commutators, using the odd Leibniz rule and the fact that d lowers the number of odd letters by one. Thus d preserves $[A_\psi, A_\psi]_{\mathrm{gr}}$, so it descends to $\mathrm{Cyc}(A_\psi)$. The descended operator is exactly ∂, and $\partial^2 = 0$ follows from $d^2 = 0$. □

LEMMA D.4 *(Exact p-face formula).* Let $u_1, \ldots, u_p$ be ordinary words in x, y. In $K^{p-1}_{R,S}$,

$$\partial[\psi u_1 \psi u_2 \cdots \psi u_p] = \sum_{j=1}^{p} (-1)^{j-1} [\psi u_1 \cdots \psi u_{j-1} xy\, u_j \psi u_{j+1} \cdots \psi u_p$$

$$- \psi u_1 \cdots \psi u_{j-1} yx\, u_j \psi u_{j+1} \cdots \psi u_p]_{\mathrm{cyc}},$$

where, for $j = 1$, the initial displayed $\psi u_1 \cdots \psi u_{j-1}$ is the empty word.

Proof. This is the boundary of Definition D.1 applied to the representative $\psi u_1 \psi u_2 \cdots \psi u_p$. The j-th displayed ψ has $j - 1$ preceding odd letters, and replacing it by the even word $xy - yx$ contributes sign $+1$ from the replacement. □

LEMMA D.5 *(Two-ψ boundary sign).* For ordinary words u, v,

$$\partial[\psi u \psi v] = [\psi v xy u] - [\psi v yx u] - [\psi u xy v] + [\psi u yx v].$$

After removing the common front ψ from each term, this is the boundary

$$vxyu - vyxu - uxyv + uyxv.$$

Proof. Lemma D.4 with $p = 2$ gives

$$[(xy - yx)u\psi v] - [\psi u(xy - yx)v].$$

The first two terms are cyclically rotated to put the remaining ψ in front:

$$[(xy)u\psi v] = [\psi vxyu], \qquad [(yx)u\psi v] = [\psi vyxu].$$

The moved block is even, so the Koszul sign is $+1$. The second replacement already has the surviving ψ in front, and its overall sign is the sign -1 from the preceding odd letter. Expanding gives the displayed formula.

Interchanging u and v gives $uxyv - uyxv - vxyu + vyxu$, the negative of the displayed expression. Thus the formula is compatible with the alternating two-ψ quotient used below. □

Construction D.6 *(One-ψ three-term skeleton).* For $k, l \geq 0$, let $\mathsf{W}_{k,l}$ be the ordinary words with k letters x and l letters y, and let $\mathsf{N}_{a,b}$ be the necklaces with a letters x and b letters y. The one-ψ skeleton in bidegree (k, l) is the finite complex

$$C_2(k,l) \xrightarrow{\partial_2} C_1(k,l) \xrightarrow{\partial_1} C_0(k,l),$$

with

$$C_1(k,l) = \mathbb{C}\{\mathsf{W}_{k,l}\}, \qquad C_0(k,l) = \mathbb{C}\{\mathsf{N}_{k+1,l+1}\},$$

and

$$C_2(k,l) = \big(\mathbb{C}\{(u,v) : \deg u + \deg v = (k-1, l-1)\}\big)_{\mathrm{alt}},$$

where $(u,v) = -(v,u)$ and $(u,u) = 0$. If $k = 0$ or $l = 0$, then $C_2(k,l) = 0$. The boundaries are

$$\partial_1(w) = [xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}},$$

and

$$\partial_2(u,v) = vxyu - vyxu - uxyv + uyxv.$$

This is the truncation $K^2_{k+1,l+1} \to K^1_{k+1,l+1} \to K^0_{k+1,l+1}$ after putting the remaining ψ at the front of a one-ψ cyclic word. Equivalently, it is the cellular chain complex in dimensions 2, 1, 0 of the state complex constructed below: C_0 records point configurations, C_1 records one edge-interior point, and C_2 records two independent edge-interior points with the orientation fixed by the ordered edges.

Proof. The identifications of C_1 and C_0 are exactly the graded cyclic normal forms in ψw and the ordinary necklaces. The source C_2 is the two-ψ summand of $K^2_{k+1,l+1}$: after cutting at one displayed ψ, a class is written $\psi u\psi v$, and moving the second ψ-block past the first contributes the sign -1. Lemma D.5 gives ∂_2, while the definition of ∂ gives ∂_1. □

Construction D.7 *(Labelled cyclic-word cell complex).* Assume $R, S > 0$. Let $\mathscr{C}_S$ be the oriented circle subdivided into S labelled oriented edges and S labelled vertices. Define $\widetilde{Y}_{R,S}$ as follows. A p-cell is a state with $R - p$ stationary indistinguishable points on vertices and p points in the interiors of p distinct edges. Several stationary points may occupy the same vertex. Each edge contains at most one interior point.

Encode such a state by traversing $\mathscr{C}_S$. At a vertex write one x for each stationary point sitting there. Then write y if the following edge is empty and ψ if the following edge contains a interior point. This gives a word with $R-p$ letters x, $S-p$ letters y, and p letters ψ, with a distinguished starting edge.

The p-cell is oriented by the increasing order of the labelled edges carrying interior points. Its j-th oriented coordinate has boundary

$$\text{terminal face } xy - \text{ initial face } yx.$$

Hence the cellular boundary in the labelled complex is

$$\sum_j (-1)^{j-1}(\text{replace the } j\text{-th moving edge by } xy - \text{replace it by } yx).$$

The cyclic group $\Gamma_S = \mathbb{Z}/S\mathbb{Z}$ rotates the edge labels and acts cellularly on $\widetilde{Y}_{R,S}$.

LEMMA D.8 *(Cellular chain identification with cyclic relabelling).* For $R, S > 0$, the cellular chain complex of $\widetilde{Y}_{R,S}$, after taking Γ_S-coinvariants, is naturally isomorphic to $K^\bullet_{R,S}$:

$$C_p(\widetilde{Y}_{R,S};\mathbb{C})_{\Gamma_S} \cong K^p_{R,S}.$$

Under this isomorphism the cellular boundary is the boundary ∂ of Definition D.1.

Proof. A labelled p-cell is equivalently a labelled cyclic sequence

$$x^{a_1}m_1x^{a_2}m_2\cdots x^{a_S}m_S, \qquad m_i \in \{y,\psi\},$$

with exactly p of the m_i's equal to ψ and $\sum_i a_i = R-p$. Every graded cyclic word in $K^p_{R,S}$ has a representative of this form: move the cut, if necessary, past even x-letters until it lies immediately before an edge marker y or ψ. This representative is unique after a starting edge has been chosen. The remaining ambiguity is rotation of the S edge markers, hence quotienting by Γ_S is exactly passage to graded cyclic words. The labelled complex orders the S edge markers, and in the coinvariant complex the only identifications are their rotations with the orientation character below.

The sign is determined as follows. Suppose a rotation moves a block containing a occurrences of ψ past the complementary block containing b occurrences of ψ. On cellular orientations this rotation moves a oriented interval coordinates past b oriented interval coordinates, hence changes the orientation by $(-1)^{ab}$. This is exactly the graded cyclic sign $[uv] = (-1)^{|u||v|}[vu]$, since x and y are even and every ψ is odd.

With this orientation convention, the cellular boundary in Construction D.7 is

$$\sum_j (-1)^{j-1}([xy]-[yx])$$

in the j-th moving edge. The terminal face is the state in which the point has crossed the edge and is read as xy; the initial face is read as yx. This is exactly Lemma D.4, term by term. □

LEMMA D.9 *(Abel map for labelled states).* For $R, S > 0$, the labelled state complex $\widetilde{Y}_{R,S}$ has the homology of a circle:

$$H_q(\widetilde{Y}_{R,S};\mathbb{C}) \cong \begin{cases} \mathbb{C}, & q=0,\\ \mathbb{C}, & q=1,\\ 0, & q\geq 2.\end{cases}$$

The cyclic rotation action of Γ_S is trivial on these two homology lines.

Proof. Work in the universal cover of $\mathscr{C}_S$, whose vertices are the integers and whose edges are the unit intervals. A lifted labelled cell is described by the positions of the R points: stationary points have integer positions, and an interior point on the edge $[m, m+1]$ has coordinate $m+t$ with $0 < t < 1$. The closure of the cell permits $t = 0, 1$, which are exactly the two vertex states yx and xy in the boundary convention of Construction D.7. Define the Abel map

$$A : \widetilde{Y}_{R,S} \longrightarrow \mathbb{R}/S\mathbb{Z}$$

by summing the point positions on the oriented circle, counted with multiplicity, and reducing modulo S. On every cell this map is affine.

Lift the target to $\mathbb{R}$ and fix a cyclic order of the R lifted points. A lifted fibre is described by weakly ordered coordinates

$$t_1 \le t_2 \le \cdots \le t_R \le t_1 + S$$

with fixed sum. This is a convex polytope. Its faces are cut out by the equalities $t_i \in \mathbb{Z}$ and by the choices of those unit intervals which contain an interior point; these faces are precisely the cells of the cellular subdivision meeting the fibre. Hence every lifted fibre is a contractible finite polyhedral complex. The map is a proper PL map of finite polyhedra with acyclic fibres. The Vietoris–Begle theorem, applied to this affine cellular map, gives a homology equivalence

$$H_\bullet(\widetilde{Y}_{R,S}; \mathbb{C}) \cong H_\bullet(S^1; \mathbb{C}).$$

A cyclic relabelling of the S edges adds a constant to every point coordinate, so it rotates the target circle by an orientation-preserving rotation. Therefore Γ_S acts trivially on $H_0(S^1; \mathbb{C})$ and $H_1(S^1; \mathbb{C})$, and hence on the two homology lines of $\widetilde{Y}_{R,S}$. □

Lemma D.10 *(Cyclic Abel quotient and stabilizer fibres).* Let $Y_{R,S}$ denote the cyclic relabelling quotient of $\widetilde{Y}_{R,S}$, with the orientation character encoded by the graded cyclic rule. Then the quotient chain complex $C_\bullet(\widetilde{Y}_{R,S}; \mathbb{C})_{\Gamma_S}$ has the homology of S^1. Moreover, every stabilizer quotient of an Abel fibre is contractible.

Proof. By Lemma D.9, the labelled Abel map is a homology equivalence from $\widetilde{Y}_{R,S}$ to the circle. A cyclic relabelling by $r \in \Gamma_S$ adds r to each of the R point positions, hence translates the Abel coordinate by Rr in $\mathbb{R}/S\mathbb{Z}$. The action on the target circle is by orientation-preserving rotations, so it is trivial on the two homology lines. Since $\mathbb{C}[\Gamma_S]$ is semisimple,

$$H_\bullet(C_\bullet(\widetilde{Y}_{R,S}; \mathbb{C})_{\Gamma_S}) \cong H_\bullet(\widetilde{Y}_{R,S}; \mathbb{C})_{\Gamma_S} \cong H_\bullet(S^1; \mathbb{C}).$$

Now fix an Abel value and a stabilizer subgroup $G \subset \Gamma_S$ of that value. In the universal cover, choose the lifted fibre before decomposing it into cells. It is a convex polytope cut out by weak ordering, the fixed-sum equation, and the cellular inequalities, and G acts affinely on it. Averaging a point over G gives a G-fixed point. Straight-line contraction to this fixed point is G-equivariant and respects the cellular subdivision. Hence the stabilizer quotient of the Abel fibre is contractible. □

Lemma D.11 *(S^1 generator).* For $R, S > 0$, the cycle

$$\Psi_{R-1,S-1} = \sum_{w \in \mathsf{W}_{R-1,S-1}} [\psi w]_{\mathrm{cyc}}$$

represents the fundamental H_1-line of the cyclic-word complex.

Proof. Its boundary is

$$\partial\Psi_{R-1,S-1} = \sum_{w \in \mathsf{W}_{R-1,S-1}} ([xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}}).$$

For a fixed necklace with R letters x and S letters y, the coefficient is the number of cyclic xy-transitions minus the number of cyclic yx-transitions. These two numbers are equal on a circle; hence $\Psi_{R-1,S-1}$ is closed.

Let $\epsilon : K^1_{R,S} \to \mathbb{C}$ send every one-ψ cyclic word to 1. Lemma D.5 shows that every two-ψ boundary has two positive and two negative terms, so it lies in $\ker\epsilon$. But

$$\epsilon(\Psi_{R-1,S-1}) = \binom{R+S-2}{R-1} \neq 0.$$

By Lemma D.10, the H_1-line is one-dimensional. Moreover, every one-cell in the displayed sum is oriented in the positive Abel direction: its terminal face has the moved point one edge farther along the oriented circle. Thus the displayed cycle maps to $\binom{R+S-2}{R-1}$ times the oriented circle class and spans the fundamental H_1-line. □

THEOREM D.12 *(Full primitive higher-ψ Koszul homology).* For every $R, S \geq 0$,

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C}\cdot[x^R y^S], & p = 0, \\ \mathbb{C}\cdot[\Psi_{R-1,S-1}], & p = 1,\ R, S \geq 1, \\ 0, & p \geq 2, \end{cases}$$

where

$$\Psi_{R-1,S-1} = \sum_{w \in \mathsf{W}_{R-1,S-1}} [\psi w]_{\mathrm{cyc}}.$$

Consequently the primitive stable Koszul homology is

$$\mathbb{C}[x,y] \oplus \mathbb{C}[x,y]\theta, \qquad |\theta| = 1, \quad \mathrm{wt}(\theta) = (1,1),$$

with the scalar Hamiltonian removed later if one passes to the reduced Hamiltonian sector.

Proof. If $R = 0$ or $S = 0$, then $K^p_{R,S} = 0$ for $p > 0$, and $K^0_{R,S}$ is spanned by the unique cyclic word x^R or y^S (including the empty word when $R = S = 0$). This gives the stated homology.

Assume $R, S > 0$. By Lemma D.8, $K^\bullet_{R,S}$ is the Γ_S-coinvariant cellular chain complex of $\widetilde{Y}_{R,S}$. Since $\mathbb{C}[\Gamma_S]$ is semisimple, taking coinvariants is an exact functor on finite-dimensional $\mathbb{C}[\Gamma_S]$-modules. Therefore

$$H_p(K_{R,S}) \cong H_p(\widetilde{Y}_{R,S}; \mathbb{C})_{\Gamma_S}.$$

Lemma D.10 identifies this homology with $H_\bullet(S^1; \mathbb{C})$. Thus $H_p(K_{R,S})$ is one-dimensional for $p = 0, 1$ and vanishes for $p \geq 2$. In particular, the three-term skeleton $C_2 \to C_1 \to C_0$ already computes H_1: cells of dimension $p \geq 3$ have boundaries in C_{p-1} and preserve $\ker(C_1 \to C_0)/\mathrm{im}(C_2 \to C_1)$.

The degree-zero line is generated by any necklace of multidegree (R, S). The functional sending every zero-ψ necklace to 1 kills every one-ψ boundary, so this line is nonzero; it is denoted $\mathbb{C}\cdot[x^R y^S]$.

Lemma D.11 identifies the degree-one generator as $\Psi_{R-1,S-1}$. □

Remark D.13 *(Finite-window homology check).* The finite-window calculation tabulates the same three-term skeleton over $\mathbb{Q}$, confirms $\partial_1\partial_2 = 0$, checks that $\Upsilon_{R-1,S-1}$ is closed and independent modulo boundaries, and verifies

$$\dim\ H_1(K_{R,S}) = 1$$

in each of the 36 bidegrees $0 \le R-1, S-1 \le 5$. The same calculation verifies the open Hamiltonian descendant action of Proposition 5.172 in 240 cases with exponents at most 3, and the closed coadjoint Taylor-dual / residue formula of Proposition 8.7 in 1225 cases with exponents at most 5. The PBW-versus-Moyal sweep records 1200 explicit mismatches and 958 cases where the coadjoint Moyal action carries genuine $\hbar^{\ge 2}$ higher-order terms; the 70 linear-generator leading-order tests with labels at most 5 identify the exact obstruction to any $\hbar$-adic deformation of the classical $\Upsilon \to \rho$ projection. The displayed homology lines and the separation in Theorem 5.181 are therefore proved at the bidegree level by an independent finite computation.

Construction D.14 *(Special-fibre PBW one-ψ label basis).* Let

$$\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\,\rho_{a,b},$$

with the Hamiltonian Lie algebra of Definition 8.3 and the Matlis principal-part module of Definition 8.5. Let

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$$

be the semidirect Hamiltonian shifted-cotangent Lie algebra of Proposition 2.7, and let

$$U_\hbar(\mathfrak{g}) \supset \mathbf{S}_\hbar(\mathfrak{h} \oplus \mathfrak{h}^\vee[1])$$

denote its Rees PBW lattice over $\mathbb{C}[[\hbar]]$, namely the Rees algebra of the canonical PBW filtration. The special fibre is

$$\mathbf{S}_\hbar(\mathfrak{h} \oplus \mathfrak{h}^\vee[1])\big|_{\hbar=0} \cong \mathbf{S}(\mathfrak{h} \oplus \mathfrak{h}^\vee[1]),$$

the symmetric algebra carrying the trivial $\mathfrak{h}$-action on generators. The *primitive one-ψ PBW label sector* is the one-tensor-factor piece in $\mathfrak{h}^\vee[1]$, tensored with the symmetric algebra of $\mathfrak{h}$, in bidegree (a, b):

$$\bar{A}_{\mathrm{desc}}[1] := \bigoplus_{a+b>0} \mathbb{C}\cdot e^{\mathrm{PBW}}_{a,b}[1], \qquad e^{\mathrm{PBW}}_{a,b}[1] \in \mathbf{S}^1(\mathfrak{h}^\vee[1]) \otimes \mathbf{S}(\mathfrak{h})\big|_{\hbar=0}.$$

The boundary pairing of Corollary 3.29 identifies this label sector with the surviving primitive H_1-lines of Theorem D.12: under

$$e^{\mathrm{PBW}}_{a,b}[1] \longmapsto [\Upsilon_{a,b}], \qquad a+b>0,$$

the $\Upsilon_{a,b}$ of Theorem D.12 are exactly the special-fibre representatives of the PBW label basis, with the binomial weight $\epsilon(\Upsilon_{a,b}) = \binom{a+b}{a}$ recording the totally symmetric averaging of the cyclic words.

Proposition D.15 *(Associated-graded label morphism).* Let

$$\widetilde{\Upsilon}_{a,b} := \binom{a+b}{a}^{-1} \Upsilon_{a,b} \in \bar{A}_{\mathrm{desc}}[1]$$

be the unit-normalised special-fibre PBW label, so that $\epsilon(\widetilde{\Upsilon}_{a,b}) = 1$, and let $\rho_{a,b} \in \mathcal{P}$ be the Matlis principal-part basis of Definition 8.5. The bidegree-only relabelling

$$\operatorname{gr}\Phi : \bar{A}_{\mathrm{desc}}[1] \xrightarrow{\ \sim\ } \mathcal{P}, \qquad \widetilde{\Upsilon}_{a,b} \longmapsto \rho_{a,b}, \quad a+b>0,$$

is an isomorphism of $\mathbb{Z}^2_{\geq 0}$-graded $\mathbb{C}$-vector spaces. It is the special-fibre associated-graded label match of Theorem 8.12 in the Ψ-presentation of Construction D.14.

This morphism is a bigraded vector-space isomorphism with an $\mathfrak{h}$-module obstruction. More precisely, on the linear Hamiltonians,

$$z_1 \cdot \widetilde{\Psi}_{a,b} = b\,\widetilde{\Psi}_{a,b-1}, \qquad z_2 \cdot \widetilde{\Psi}_{a,b} = -a\,\widetilde{\Psi}_{a-1,b},$$

while

$$z_1 \cdot \rho_{a,b} = -(b+1)\,\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\,\rho_{a+1,b}.$$

The two actions have opposite shift directions and unequal scalars; in particular the polynomial PBW action is locally nilpotent on every $\widetilde{\Psi}_{a,b}$, whereas every nonzero $\rho_{a,b}$ has an infinite principal-part coadjoint orbit. The obstruction to an $\hbar$-adic same-action lift of $\operatorname{gr}\Phi$ is the local-nilpotence dichotomy of Theorem 8.11 and Corollary 8.13.

Proof. The bidegree component of $\bar{A}_{\mathrm{desc}}[1]$ is the line $\mathbb{C}\cdot e^{\mathrm{PBW}}_{a,b}[1]$ for $a+b>0$; the bidegree component of $\mathcal{P}$ is $\mathbb{C}\cdot\rho_{a,b}$, $a+b>0$, by Definition 8.5. The displayed map is the bidegree identity, hence an isomorphism of bigraded vector spaces.

The polynomial Hamiltonian action on $\widetilde{\Psi}_{a,b}$ is Proposition 5.172: for $F_{p,q}=\operatorname{Tr}(\phi_1^p\phi_2^q)$, $F_{p,q}\cdot\widetilde{\Psi}_{a,b} = (pb-qa)\widetilde{\Psi}_{a+p-1,b+q-1}$. Setting $(p,q)=(1,0),(0,1)$ gives the displayed two formulas. The principal-part coadjoint action is Proposition 8.7: for $(p,q)=(1,0),(0,1)$ the formula $z_1^p z_2^q\cdot\rho_{a,b} = -(pb-qa+p-q)\rho_{a-p+1,b-q+1}$ gives the two displayed formulas with the principal-part shifts $b+1$ and $a+1$.

For local nilpotence: z_1 lowers the second index of $\widetilde{\Psi}_{a,b}$ by one and is killed by $\widetilde{\Psi}_{a,0}$ for $a\geq 0$; a finite linear combination is exhausted after at most $\max b+1$ iterations. By contrast, z_1 raises the second index of $\rho_{a,b}$ and produces a nonzero rising-factorial coefficient $(-1)^N(b+1)(b+2)\cdots(b+N)\rho_{a,b+N}$, which is never zero. The same dichotomy holds for z_2. Local nilpotence is an isomorphism invariant; therefore the displayed $\mathfrak{h}$-actions have a module-isomorphism obstruction.

The obstruction to extending $\operatorname{gr}\Phi$ to an $\hbar$-adic same-action map is exactly this local-nilpotence dichotomy, and is recorded as Theorem 8.11 together with the deformation-level Corollary 8.13. Theorem 8.12 gives the positive Fourier–Rees bridge that does identify the two label sets, but only after the Fourier twist of the Hamiltonian action. □

Remark D.16 *(Smallest explicit mismatch: $(p,q,a,b)=(4,0,1,1)$).* The polynomial PBW action on the descendant labels and the Moyal coadjoint action on the principal parts disagree already at one exponent pair. Take the quartic Hamiltonian $F_{4,0}=\operatorname{Tr}(\phi_1^4)$ and the bidegree-$(1,1)$ descendant target. On the polynomial PBW side Proposition 5.172 gives

$$F_{4,0} : \widetilde{\Psi}_{1,1} \longmapsto (4\cdot 1 - 0\cdot 1)\,\widetilde{\Psi}_{1+4-1,1-1} = 4\,\widetilde{\Psi}_{4,0}.$$

Direct calculation in the Moyal coadjoint module gives

$$\mathrm{ad}^*_{z_1^4}(\partial_{1,1}) = -24\,\hbar^2\,\partial_{0,4},$$

the only nonzero term of order ≤ 3 in the formal Moyal expansion. Fourier-untwisted relabelling has the bidegree obstruction $(1,1)\to(0,4)$ versus $(1,1)\to(4,0)$, so the polynomial PBW action and the Moyal coadjoint action remain distinct. Replacing the Moyal coadjoint by its classical truncation ($\hbar^0$) gives $\mathrm{ad}^*_{z_1^4}(\rho_{1,1})=0$ by the principal-part formula at this exponent, again disagreeing with the polynomial PBW term $4\widetilde{\Psi}_{4,0}$. Exact finite-window calculation confirms this mismatch and embeds it inside a sweep of 1200 such mismatches.

Theorem D.17 *(Universal primitive descendant calculation).* Let $\mathsf{K}_{\mathrm{univ}}$ be the stable primitive cyclic Koszul complex generated by two even letters x, y and one odd letter ψ, with $d\psi = xy - yx$, decomposed by the weights of Definition D.1. Then

$$H_\bullet(\mathsf{K}_{\mathrm{univ}}) \cong \mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta, \qquad |\theta| = 1, \quad \mathrm{wt}(\theta) = (1, 1),$$

with $x^a y^b \theta$ represented by the symmetrised primitive cycle $\Upsilon_{a,b}$. This is the universal primitive calculation in the following precise sense. If $E_\bullet$ is any stable primitive single-trace model whose generators are the images of x, y, ψ, whose differential sends ψ to $xy - yx$, and whose only stable primitive trace relation is graded cyclicity, then the induced map

$$\mathsf{K}_{\mathrm{univ}} \longrightarrow E_\bullet$$

factors on homology through the displayed two-line module. Hence the universal primitive higher-ψ classes are exactly the displayed degree-zero polynomial line and degree-one Υ-line.

Proof. The first assertion is Theorem D.12 summed over all bidegrees (R, S), with $x^a y^b \theta = \Upsilon_{a,b}$ and $R = a + 1$, $S = b + 1$ in degree one. For the universal property, Lemma D.2 says that the stable primitive trace span is exactly the graded cyclic quotient of the free algebra on x, y, ψ. Therefore any stable primitive model satisfying the stated hypotheses receives a unique chain map from $\mathsf{K}_{\mathrm{univ}}$ after choosing the images of x, y, ψ. The homology of the source is the two-line module computed above. Since $H_p(K_{R,S}) = 0$ for $p \geq 2$ in every bidegree, every universal primitive class in the target comes from the degree-zero polynomial line or the degree-one Υ-line. The statement is independent of the later choice of principal-part, PBW, or Moyal coefficient module. □

Theorem D.18 *(Hamiltonian-sector extraction).* The full primitive homology decomposes as

$$\mathbb{C} \cdot [1] \oplus \bar{A} \oplus \mathbb{C} \cdot [\Upsilon_{0,0}] \oplus \bar{A}_{\mathrm{desc}}[1],$$

where

$$\bar{A} = \bigoplus_{R+S>0} \mathbb{C} \cdot [x^R y^S], \qquad \bar{A}_{\mathrm{desc}}[1] = \bigoplus_{a+b>0} \mathbb{C} \cdot [\Upsilon_{a,b}].$$

The Hamiltonian comparison uses only $\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]$, and recognises this latter pair as the special-fibre, $\hbar = 0$, shadow of the Rees PBW lattice of Construction D.14: the closed Hamiltonian summand is the polynomial part $\mathbf{S}(\mathfrak{h})|_{\hbar=0}$, and the descendant copy is the primitive one-ψ PBW label sector $\mathbf{S}^1(\mathfrak{h}^\vee[1]) \otimes \mathbf{S}(\mathfrak{h})|_{\hbar=0}$. The complementary sectors are exactly the scalar unit $[1]$, the scalar odd primitive $\Upsilon_{0,0}$, and the vanishing higher-ψ primitive groups $H_p(K_{R,S}) = 0$ for $p \geq 2$.

Proof. The constant Hamiltonian has zero Hamiltonian vector field, so the reduced Hamiltonian algebra is $\bar{A} = \mathbb{C}[x, y]/\mathbb{C} \cdot 1$ degree by degree. This removes the degree-zero scalar class $[1]$. The odd scalar $\Upsilon_{0,0}$ is the one-ψ descendant of the same removed constant direction; it is a stable primitive trace class in the removed scalar sector. Theorem D.12 proves that primitive homology is concentrated in ψ-degrees 0 and 1. What remains is exactly the positive-degree Hamiltonian and one-ψ PBW label sector displayed above. The identification with the special-fibre Rees lattice is Construction D.14: the closed Hamiltonian summand is the polynomial special fibre of the Rees algebra of $\mathfrak{h}$, and the one-ψ descendant copy is the bidegree-graded $\widetilde{\Upsilon}_{a,b}$ sector of the cotangent-shifted $\mathfrak{h}^\vee[1]$ tensor. This is a vector-space homology statement. The Matlis principal-part module comparison is the separate $\mathfrak{h}$-module obstruction. □

COROLLARY D.19 *(Special-fibre defect polarization).* The polynomial one-ψ PBW labels $\Psi_{a,b}$ and the Matlis principal parts $\rho_{a,b}$ are joined by the bidegree-only associated-graded morphism gr Φ of Proposition D.15: this morphism intertwines the two index sets at the special fibre $\hbar = 0$ and only there. In particular, full primitive ψ-homology computes the special-fibre shadow of the cotangent module on the open brane sector, while the Matlis principal-parts module of § 8 is the deformation-level cotangent module itself. The formal-stalk chart measures positions polynomially through $\bar{A}$, and its cotangent momenta live as residues through $\mathcal{P}$; the shared bidegree label is associated-graded data.

Proof. Construction D.14 identifies $\widetilde{\Psi}_{a,b}$ with the unit-normalised $\hbar = 0$ class in $\mathbf{S}^1(\mathfrak{h}^\vee[1]) \otimes \mathbf{S}(\mathfrak{h})|_{\hbar=0}$ in bidegree (a, b). The bidegree-only relabelling gr Φ of Proposition D.15 is an isomorphism of bigraded vector spaces with the $\mathfrak{h}$-module obstruction of Theorem 8.11. The Matlis side is the continuous coadjoint module identified by Theorem 8.10 as the unique perfect continuous coadjoint pairing. Therefore $\bar{A}_{\mathrm{desc}}[1]$ is the special-fibre shadow and $\mathcal{P}$ is the deformation-level cotangent module, with the shared index set carrying associated-graded label data only. This is the separation displayed by Theorem 5.181. □

E UNREDUCED BV COTANGENT, SCALAR QME OBSTRUCTIONS, AND CANCELLATIONS

Fix the formal-local mixed geometry $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_0$ with $\Omega_{\mathrm{loc}} = dz_1 \wedge dz_2$. On a brane interval $I \subset L$ the open BV theory contributes scalar contact counterterms each order in $\hbar$; the residual non-scalar Costello quantum master equation lives in their orthogonal complement. The complex $\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$ is the local-functional cochain complex on I at weighted-Tate coefficient weight w, with the brane-tangential differential d_∂ and the $\hbar$-formal grading; $\widehat{\sigma}_{\mathrm{sc}}$ is the filtered scalar-symbol chain projection killing the scalar-contact filtration; $d_{\mathrm{cl}} = Q + \{I_0, -\}$ is the associated-graded classical BV differential; and $d^L_{\mathrm{QME}} = Q + \{I[L], -\}_L + \hbar\Delta_L$ is the scale-L quantum master differential of Costello's renormalization. When a fixed Costello scale is understood, d_{QME} denotes this quantum differential d^L_{QME}. The associated-graded scalar-symbol complexes below use d_{cl}. Before passage to the residual non-scalar complex, the unreduced curvature $\mathrm{Ob}_{w,\partial,\hbar}$ has scalar-contact symbol

$$\widehat{\sigma}_{\mathrm{sc}}(\mathrm{Ob}_{w,\partial,\hbar}) = \hbar N[\bar{c}]\,\gamma_1 \in H^2_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]].$$

Forgetting the central ghost coefficient gives the scalar class $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}; \mathbb{C})[[\hbar]]$. After this scalar component is killed, the residual non-scalar QME complex contains the cohomology class

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}] \in H^1(\ker \widehat{\sigma}_{\mathrm{sc}}, d^L_{\mathrm{QME}}),$$

together with its finite-window $\varprojlim^1 H^0$-primitive-compatibility class (the secondary Roos cohomology obstruction tracking compatibility of window-by-window primitives across the truncation tower). The Capelli class of Chapter 6 (Theorem 6.8) is the scalar projection of the unreduced obstruction before reduction to $\ker \widehat{\sigma}_{\mathrm{sc}}$, with sign and convention fixed by Appendix B and the canonical scheme $z_1 z_2 \mapsto \frac{1}{2}(\phi_1\phi_2 + \phi_2\phi_1)$ of the Moyal star product. Exact rational expansion gives the Moyal coefficient $\hbar N$.

The scalar anomaly is one cohomology class realized in four cohomologies. The algebraic central extension of $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$, the Lie central extension $\widehat{\mathfrak{h}}_{\mathrm{cent}}$, the trace-algebraic boundary central extension, and the quantum Weyl commutator $[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_\star = \hbar N$ carry the same class in $H^2(\bar{A}; \mathbb{C})[[\hbar]]$ under the canonical scalar projection; the cocycle $\omega(f, g) = [\{f, g\}]_0$ is the same in each (Theorem E.13). Hamiltonian vector fields annihilate the constant line, while the open trace pairs that line with $\mathrm{Tr}(\mathbf{1}) = N$ and produces the class $\hbar N[\bar{c}]$. Each cancellation branch acts on this single class

— ordinary $\mathfrak{gl}_N$ is obstructed by $\hbar N[\bar{c}]$, the unreduced central character contributes the CE coboundary $d_{\mathrm{CE}}(\hbar\chi_N) = -\hbar N\omega$, the balanced $\mathbb{C}^{N|N}$ supertrace rescales the open trace by sdim $\mathbb{C}^{N|N} = 0$, and full-equivariant marker tensors satisfy $\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}) = 0$. Modifying one branch displaces the same class.

At each finite window M (a Hamiltonian-degree truncation of the weighted Tate coefficient pair) the obstruction is the linear-algebra problem

$$A^M c^M = -r^M$$

of Theorem E.35, where A^M is the finite matrix whose columns enumerate the counterterm primitives given by the row span, c^M is the unknown counterterm vector, and r^M is the residual vector collecting the QME source faces. The row entries are graph-by-graph Costello data: K_Γ^M the windowed kernel of the Feynman graph Γ, ε_Γ its sign, $\zeta_{M,\nu}$ a CE test pairing labelled by the cycle ν, and the bracket $\{-,-\}_{\hbar,M}$ the windowed Schouten/BV bracket. In these terms,

$$R^{\mathrm{row}}_{d\Gamma,M} = \varepsilon_\Gamma d_M K_\Gamma^M, \qquad R^{\mathrm{row}}_{\mathrm{CE}(\Gamma,\nu),M} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K_\Gamma^M(\zeta_{M,\nu}),$$

$$R^{\mathrm{row}}_{b(\Gamma_1,\Gamma_2),M} = \tfrac{1}{2}\varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M},$$

and the order-n homogeneous lower-counterterm rows. A degree-zero counterterm c^M exists in the given row span exactly when the system is consistent; otherwise a separating covector $\ell A^M = 0$, $\ell(r^M) \neq 0$ detects the obstruction. Tower compatibility passes through truncation matrices $T^{M,N}$ on rows (restriction-of-graphs from window M to a smaller window N) and $P^{M,N}$ on primitive columns (the matched counterterm projector) and the Roos class $[P^{M,N}c^M - c^N] \in H^0(\mathcal{K}^\bullet_{\mathrm{ns},N}(I))$ (the cohomology class measuring compatibility of windowwise primitives across truncation).

The order-three row $\theta_3 = \mathrm{CE}(\Theta_3, \nu_3)$ is the first scalar-zero source-face row beyond the order-one one-cross scalar contact. It has

$$R^{\mathrm{row}}_{\theta_3,M} = -K^M_{\Theta_3}(\zeta_{M,\nu_3}) =: \mathsf{E}_{\theta_3,M}, \qquad S_{\theta_3,M} = 0,$$

and in the one-row complex $\mathcal{K}^0_{\theta_3,M} = 0, \mathcal{K}^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, d_{\theta_3,M} = 0$, the covector $\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$ is an exact obstruction (Proposition E.36). A named coefficient C_{θ_3} requires one of three data: a CE ancestor with controlled boundary; a scalar-zero Costello local counterterm $C_{\theta_3,M}$ with $d_{\mathrm{ns},M}C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$ and truncation-compatible primitives; or a complete companion-face table of Hom-valued companion rows whose signed sum moves the residual vector into the image of A^M. Tower compatibility passes through the truncation defect $\Delta^1_{M,N} = -\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N$ ($\mathfrak{b}^M$ the order-one Bianchi residual at window M, $\pi_{M,N}$ the truncation projector) and the secondary $\varprojlim^1 H^0$ primitive class (Proposition E.37). The unreduced factorization-center cotangent lift has a four-component obstruction vector. Its vanishing is necessary; it is sufficient only after the finite-window companion tables, centrality homotopies, counterterms, and descent comparison homotopies are included in the datum (Theorem E.53).

Definition E.1 *(BV degrees and signs).* The open polynomial BV/Koszul model uses

$$|\phi_1| = |\phi_2| = 0, \qquad |\psi = A^*| = -1, \qquad Q\psi = [\phi_1, \phi_2], \qquad |Q| = 1.$$

A Hamiltonian generator H_f has degree 0. A shifted cotangent generator $\eta_\rho = \rho[1]$ is odd of BV degree -1. The CE coordinates satisfy

$$|c_f| = 1, \qquad |u_\rho| = 0.$$

This is the cotangent-coordinate grading of the CE/PV block. The ordinary graded dual of the BV degree of $\rho[1]$ is a separate convention. The CE/PV cotangent–Schouten bracket has degree -1. The reduced Hamiltonian boundary P_0 bracket has degree 0. The QME obstruction differential $Q + \{I_0, -\}$ has degree 1. The Poisson sign convention is the one determined by $\Omega_{\mathrm{loc}} = dz_1 \wedge dz_2$, so $\{z_1, z_2\} = 1$. The principal-part coadjoint action is

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}.$$

PROPOSITION E.2 *(Local geometry of scalar shadows).* All scalar-contact, finite-window, marked-graph, and supertrace calculations below use only the formal-local mixed holomorphic-topological geometry

$$I \subset \mathbb{R}_{\mathrm{brane}} \subset \mathbb{R}^2_{\mathrm{top}}, \qquad \widehat{\mathbb{C}^2}_0 = \operatorname{Spf} \mathbb{C}[[z_1, z_2]], \qquad \Omega_{\mathrm{loc}} = dz_1 \wedge dz_2.$$

The inverse Poisson bracket is

$$\{f, g\}_{\Omega_{\mathrm{loc}}} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

With the convention $\iota_{X_f}\Omega_{\mathrm{loc}} = -df$, one has

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \operatorname{div}_{\Omega_{\mathrm{loc}}} X_f = 0.$$

Thus the scalar shadow has zero divergence and zero volume-form anomaly. The Hamiltonian scalar cocycle entering every calculation below is

$$\omega(f, g) = [\{f, g\}_{\Omega_{\mathrm{loc}}}]_0, \qquad \omega(z_1, z_2) = 1.$$

The open index contraction then multiplies this local Hamiltonian scalar by the Chan–Paton trace coefficient:

$$\hbar N\, \omega(f, g)\gamma_1 \quad \text{in the ordinary } \mathfrak{gl}_N \text{ branch},$$

$$\hbar\, \operatorname{sdim}(\mathbb{C}^{N|N})\, \omega(f, g)\gamma_1 = 0 \quad \text{in the balanced supertrace branch},$$

and, for a marked scalar loop,

$$\hbar\, \operatorname{Str}_{\mathrm{cyc}}(\mathfrak{m})\, \omega(f, g)\gamma_1.$$

A finite window M restricts only the polynomial Hamiltonian labels, graph labels, and counterterm table to a finite local coefficient system; it preserves Ω_{loc}, the Hamiltonian bracket, and the divergence-free condition.

Proof. The formula for X_f is obtained by solving $\iota_{X_f}(dz_1 \wedge dz_2) = -df$. Its divergence is

$$\partial_{z_1}(-\partial_{z_2} f) + \partial_{z_2}(\partial_{z_1} f) = 0$$

after writing $X_f = -(\partial_{z_2} f)\partial_{z_1} + (\partial_{z_1} f)\partial_{z_2}$. Therefore the local volume form is preserved and contributes zero additional scalar contact term. Taking the constant Taylor coefficient of the displayed Poisson bracket gives the cocycle ω, with $\omega(z_1, z_2) = 1$.

The remaining scalar factor is purely the open Chan–Paton index loop. The ordinary trace of the identity on $\mathbb{C}^N$ is N. The supertrace of the identity on $\mathbb{C}^{N|N}$ is $N - N = 0$. A marked loop replaces the identity by the cyclic product of its marker tensors and gives the cyclic supertrace shown above. Finite-window truncation is a restriction of labels and graph tables inside this same formal polydisk, so it preserves the local cocycle and the divergence computation. □

Definition E.3 *(Scalar-reduced brane-defect obstruction complex).* Let $\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}$ denote local functionals on the ordinary $\mathfrak{gl}_N$ brane-defect fields in the formal-local model $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_0$, with compact support on the brane line and with coefficients in the scalar-reduced Hamiltonian source $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$. The associated-graded differential controlling a mixed brane-defect scalar-symbol lift is

$$d_{\mathrm{cl}} = Q + \{I_0, -\}.$$

A scalar contact term is tested by adjoining the degree-one central ghost $\gamma_{\mathbf{1}}$ for the omitted constant Hamiltonian line. Its associated graded is the Chevalley–Eilenberg complex of $\bar{A}$ with trivial coefficients on this central line. Writing

$$\mathfrak{Q}^\bullet_{\mathrm{bd}} = (\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}, d_{\mathrm{cl}}),$$

the associated-graded scalar-symbol projection is

$$\sigma_{\mathrm{sc}}\colon\ \mathrm{gr}_{\mathrm{sc}}\,\mathfrak{Q}^\bullet_{\mathrm{bd}} \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{\mathbf{1}})[1].$$

A cone or long exact sequence is formed only after one has chosen a d_{cl}-stable scalar filtration and a filtered chain projection

$$\widehat{\sigma}_{\mathrm{sc}}\colon \mathfrak{Q}^\bullet_{\mathrm{bd}} \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{\mathbf{1}})[1]$$

whose associated graded is the displayed σ_{sc}. The assertion below is an associated-graded scalar obstruction statement until this extra projection is fixed. Thus a degree-one QME representative with two Hamiltonian arguments maps to a Lie two-cocycle. Explicitly,

$$\sigma_{\mathrm{sc}}\!\left(\kappa(f, g)\int_{\mathbb{R}} a(t)b(t)\gamma_{\mathbf{1}}(t)\,dt\right) = \kappa(f, g)\gamma_{\mathbf{1}}.$$

Lemma E.4 *(Filtered scalar projection obstruction).* Fix a complete decreasing scalar-contact filtration $F^\bullet\mathfrak{Q}^\bullet_{\mathrm{bd}}$ preserved by d_{cl}, and put the target $C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{\mathbf{1}})[1]$ in filtration degree 0. Suppose the associated graded carries the scalar-symbol projection

$$\sigma_{\mathrm{sc}}\colon\ \mathrm{gr}_F\,\mathfrak{Q}^\bullet_{\mathrm{bd}} \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{\mathbf{1}})[1].$$

Choose any filtered linear lift $s\colon \mathfrak{Q}^\bullet_{\mathrm{bd}} \to C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{\mathbf{1}})[1]$ of σ_{sc}. Its chain-defect

$$\delta_s = d_{\mathrm{CE}}s - s\,d_{\mathrm{cl}}$$

lowers the scalar-contact filtration. The first nonzero homogeneous piece defines an obstruction class

$$o^{(r)}_\sigma \in H^1\!\big(\mathrm{Hom}\big(\mathrm{gr}_F\,\mathfrak{Q}^\bullet_{\mathrm{bd}}, C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_{\mathbf{1}})[1]\big)_{-r}\big),$$

with the usual differential $D(\alpha) = d_{\mathrm{CE}}\alpha - (-1)^{|\alpha|}\alpha d_{\mathrm{gr}}$. A filtered chain projection $\widehat{\sigma}_{\mathrm{sc}}$ with associated graded σ_{sc} exists exactly when the obstruction classes $o^{(r)}_\sigma$ vanish successively for all $r > 0$, with compatible choices in the completed inverse limit.

Proof. Since the associated graded of s is already a chain map, the commutator $d_{\mathrm{CE}}s - sd_{\mathrm{cl}}$ strictly lowers the filtration. Its lowest nonzero component is closed for the Hom differential because $d^2_{\mathrm{CE}} = d^2_{\mathrm{cl}} = 0$. Replacing s by $s - \alpha$, where α lowers filtration by the same amount, changes this component by $D\alpha$. Thus the first defect can be removed exactly when its cohomology class vanishes. Iterating the same argument through the complete filtration gives a compatible sequence of corrections if and only if all successive classes vanish. □

PROPOSITION E.5 *(First scalar-lift obstruction in the ordinary branch).* Work in the ordinary scalar-reduced $\mathfrak{gl}_N$ weighted brane-defect local-functional complex $\mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$, with $N > 0$. Let J_{sc} be the closed ideal of functionals whose associated-graded scalar-contact symbol vanishes, and put $F^r = J^r_{\mathrm{sc}}$, completed $\hbar$-adically and in the finite-window inverse limit. For any filtered linear lift s of the scalar-symbol projection, the first chain-defect class has Hamiltonian two-argument evaluation

$$\mathrm{ev}_{f,g}\big(o^{(1)}_{\sigma,w}\big) = \hbar N\,\omega(f,g)\gamma_1 \in H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[[\hbar]].$$

In particular

$$\mathrm{ev}_{z_1,z_2}\big(o^{(1)}_{\sigma,w}\big) = \hbar N\,\gamma_1 \neq 0.$$

Hence a filtered chain lift of the scalar-symbol projection in the ordinary scalar-reduced branch requires additional data whose leading scalar class is $-\hbar N\,[\bar{c}]$, or a replacement of the source or open trace model before scalar reduction.

Proof. The filtration $F^\bullet$ is the scalar-contact filtration: its associated graded keeps the leading scalar contact term and discards higher scalar-contact corrections. The first test is the bracket of the two linear Hamiltonian boundary insertions. By Theorem E.12,

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\mathrm{open},\hbar} - I_\partial(ab,\{f,g\}_{\bar{A}}) = \hbar N\,\omega(f,g)\int_I a(t)b(t)\,dt$$

after insertion of γ_1. Applying the scalar-symbol lift gives the displayed representative of $d_{\mathrm{CE}}s - sd_{\mathrm{cl}}$. For $f = z_1$ and $g = z_2$, $\omega(z_1,z_2) = 1$.

If this first defect were a Hom coboundary, evaluation on the Hamiltonian two-argument subcomplex would give a linear primitive $\bar{\eta} : \bar{A} \to \mathbb{C}\gamma_1$ with $d_{\mathrm{CE}}\bar{\eta} = \hbar N\,\omega\gamma_1$. Pulling $\bar{\eta}$ back to $A = \mathbb{C}[z_1,z_2]$ and forcing it to vanish on the constant line contradicts Lemma E.11, equivalently $\omega(z_1,z_2) = 1$ while $\{z_1,z_2\} = 1$ is zero in $\bar{A}$. Thus $o^{(1)}_{\sigma,w}$ is nonzero.

The obstruction is detected in the finite window containing z_1 and z_2. The admissible weight transports of Theorem 5.82 identify this finite-window class, so changing w preserves its vanishing criterion. The primitive exists on the unreduced source A, where $\eta(f) = -[f]_0$, and the central-character replacement uses exactly this primitive. In the balanced supertrace branch the displayed ordinary-trace evaluation is multiplied by $\mathrm{sdim}\,\mathbb{C}^{N|N} = 0$. The scalar-contact subcomplex and chain projection constructed in Proposition E.18 kill this obstruction on the balanced scalar-contact branch. Extension to all local functionals remains the separate non-scalar QME obstruction problem. □

THEOREM E.6 *(Polynomial unreduced centrality obstruction).* Let M_{pol} be any Q-stable subcomplex of the unreduced polynomial open BV observables in which every element has finite polynomial degree in ϕ_1, ϕ_2, ψ. Any assignment

$$\rho_{a,b}[1] \longmapsto \Theta_{a,b} \in H^\bullet(M_{\mathrm{pol}}, Q)$$

together with centrality homotopies in M_{pol} has locally nilpotent linear Hamiltonian action on Q-cohomology, whereas the principal-part module $\mathcal{P}$ has non-locally-nilpotent linear Hamiltonian action. Equivalently, an unreduced polynomial lift of the reduced map $\rho_{a,b}[1] \mapsto \Theta_{a,b}$ is the datum of polynomial boundary representatives with homotopies satisfying

$$\{O_f, \Theta_\rho\} \simeq \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\} \simeq 0$$

for all polynomial Hamiltonians f.

Proof. The linear Hamiltonians act on polynomial open observables by the canonical scalar-translation derivations

$$O_{1,0} \mapsto \partial_{\phi_2}, \qquad O_{0,1} \mapsto -\partial_{\phi_1}.$$

These derivations commute with Q, since translating ϕ_1 or ϕ_2 by a scalar matrix leaves the moment-map equation $Q\psi = [\phi_1, \phi_2]$ unchanged. On every finite polynomial observable, repeated application of either derivation eventually vanishes. Hence their induced actions on $H^\bullet(\mathcal{M}_{\mathrm{pol}}, Q)$ are locally nilpotent.

On the principal-part module the same two Hamiltonians act by

$$z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Iterating gives

$$z_1^r \cdot \rho_{a,b} = (-1)^r (b+1)\cdots(b+r)\rho_{a,b+r} \neq 0,$$

and similarly for z_2, for every $r \geq 1$. Thus every nonzero principal part has non-locally-nilpotent linear Hamiltonian orbit.

Existence of the stated homotopies in $\mathcal{M}_{\mathrm{pol}}$ identifies, after passage to Q-cohomology, the action of $O_{1,0}$ and $O_{0,1}$ on the classes $[\Theta_{a,b}]$ with the displayed principal-part action. Local nilpotence is invariant under module isomorphism and under passage to a submodule. This is incompatible with the non-locally-nilpotent principal-part action above. Therefore the unreduced lift requires a category beyond polynomial open BV/Koszul observables. □

Lemma E.7 *(Distributional locality of principal-part currents).* Let $I \subset \mathbb{R}$, $a \in C_c^\infty(I)$, and $B \in \mathcal{D}'_c(I)$. The product

$$(aB)(\varphi) = B(a\varphi), \qquad \varphi \in C_c^\infty(I),$$

is a compactly supported distribution with $\mathrm{supp}(aB) \subset \mathrm{supp}(a) \cap \mathrm{supp}(B)$. Consequently the reduced principal-part current brackets

$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0$$

are distributionally local on the brane line. If the supports are disjoint, the brackets vanish.

Proof. Multiplication of a compactly supported distribution by a smooth function is the standard operation $B \mapsto aB$ defined by precomposition with multiplication by a. If φ is supported away from $\mathrm{supp}(a) \cap \mathrm{supp}(B)$, then $a\varphi$ is supported away from $\mathrm{supp}(B)$, hence $B(a\varphi) = 0$. This proves the support assertion. The first bracket is the negative transpose of the Hamiltonian action under the residue-current pairing, so its interval factor is exactly aB. The second bracket is zero because the principal-part current ideal is abelian. Disjoint-support vanishing follows from $aB = 0$. □

Definition E.8 *(Central-character principal-part current algebra).* Let $A = \mathbb{C}[z_1, z_2]$, $\bar{A} = A/\mathbb{C}\cdot 1$, and let $\chi\colon A \to \mathbb{C}$ be a linear central character on the constant Hamiltonian line. For an interval $I \subset \mathbb{R}$, write

$$\widetilde{\mathfrak{h}}_I = \Omega_c^\bullet(I)\widehat{\otimes}A, \qquad \mathcal{P}_I = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}.$$

The algebraic current target with central character is the completed graded-commutative algebra generated by symbols

$$B_f^\chi(a), \qquad \Theta_\rho(B), \qquad a \in \Omega_c^0(I),\ f \in A,\ B \in \mathcal{D}'_c(I),\ \rho \in \mathcal{P},$$

modulo

$$B^{\chi}_{f+c}(a) = B^{\chi}_{f}(a) + \chi(c \cdot 1) \int_I a(t)\, dt, \qquad \Theta_{\rho+\sigma}(B) = \Theta_\rho(B) + \Theta_\sigma(B),$$

and with P_0 brackets

$$\{B^{\chi}_{f}(a), B^{\chi}_{g}(b)\} = B^{\chi}_{\{f,g\}}(ab),$$
$$\{B^{\chi}_{f}(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB),$$
$$\{\Theta_\rho(B), \Theta_\sigma(C)\} = 0.$$

Constants act trivially on $\mathcal{P}$. The product is the ordinary graded-commutative factorization product after extension by zero on intervals. This is an algebraic current model. The finite polynomial open BV/Koszul complex is governed by Theorem E.6.

Construction E.9 *(Algebraic current presentation map).* Let

$$\widetilde{\mathfrak{g}}_I = \widetilde{\mathfrak{h}}_I \ltimes \mathcal{P}_I[1]$$

with bracket induced by the Hamiltonian bracket on A, the coadjoint action on $\mathcal{P}$, and the abelian bracket on $\mathcal{P}_I[1]$. In the CE algebra $C^{\bullet}_{\mathrm{CE,cont}}(\widetilde{\mathfrak{g}}_I)$, write

$$c_{B,\rho}$$

for the degree-one CE coordinate dual to Hamiltonian currents through the pairing

$$\langle B \otimes \rho, a(t)dt \otimes f \rangle = B(a)\rho(f),$$

and write

$$u_{a(t)dt\otimes f}$$

for the degree-zero CE coordinate dual to the shifted cotangent generator under the same pairing with $a(t)dt \otimes f \in \widetilde{\mathfrak{h}}_I$. Define

$$\Phi^{\chi}_I(c_{B,\rho}) = \Theta_\rho(B), \qquad \Phi^{\chi}_I(u_{a(t)dt\otimes f}) = B^{\chi}_f(a),$$

and extend continuously and multiplicatively.

THEOREM E.10 *(Algebraic current presentation with zero homotopies).* The maps Φ^{χ}_I of Construction E.9 define a morphism of P_0 factorization algebras from the interval CE source of $\widetilde{\mathfrak{g}}_I$ to the algebraic current target of Definition E.8. The factorization centrality homotopies and the P_0-bracket compatibility homotopies in this algebraic model are strict:

$$H_{\mathrm{fact}} = 0, \qquad H_{P_0} = 0.$$

Thus the unreduced source with central character constructs the P_0 current presentation as far as the distributional algebraic current model permits. A morphism into $Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}_{\mathrm{open}})$ for the genuine unreduced open BV factorization algebra requires the additional unreduced graph/QME data. This is compatible with the polynomial obstruction theorem: the image of $\Theta_\rho(B)$ lives in the added principal-part current sector.

Proof. The CE differential of $C^\bullet_{\mathrm{CE,cont}}(\widetilde{\mathfrak{g}}_I)$ is dual to exactly three brackets:

$$[\widetilde{\mathfrak{h}}_I, \widetilde{\mathfrak{h}}_I], \qquad \widetilde{\mathfrak{h}}_I \curvearrowright \mathcal{P}_I, \qquad [\mathcal{P}_I[1], \mathcal{P}_I[1]] = 0.$$

Under Φ_I^χ, these three structure maps become the three displayed P_0 brackets of Definition E.8. Hence the CE differential and the target Schouten differential agree on generators, while the degree-zero current P_0 brackets agree by the same source brackets. The biderivation property extends the equality to the completed symmetric algebra.

Factorization compatibility is also strict. Extension by zero sends a, B to their extensions as test functions and currents; the products ab and aB are computed after extension. If the supports are disjoint these products vanish, so mixed brackets vanish before any homotopy is introduced. Multiplication in the target is graded-commutative, hence multiplication by an image generator is a Hochschild central operation with zero product-centrality homotopy. The P_0-centrality defect on generators is the difference between the target bracket and the image of the corresponding source bracket; it is zero by the previous paragraph. Therefore H_{fact} and H_{P_0} may both be chosen to be zero.

The same statement holds against arbitrary monomials in the added algebraic current target, including monomials beyond length-one generators. Let $Y = Y_1 \cdots Y_r$ be a homogeneous source monomial and put $W = \Phi_I^\chi(Y)$, a monomial in the target generators $B_g^\chi(b)$ and $\Theta_\sigma(C)$. For a source generator x, the product-centrality defect is zero because the target product is graded-commutative:

$$\Phi_I^\chi(x)W - (-1)^{|\Phi_I^\chi(x)||W|} W \Phi_I^\chi(x) = 0.$$

Let ∂_x denote the derivation of the source symmetric algebra induced by the corresponding bracket in $\widetilde{\mathfrak{g}}_I$. The bracket-compatibility defect

$$D_x(Y) = \{\Phi_I^\chi(x), \Phi_I^\chi(Y)\} - \Phi_I^\chi(\partial_x Y)$$

is a graded derivation in Y. It vanishes when $\Phi_I^\chi(Y)$ is $B_g^\chi(b)$ or $\Theta_\sigma(C)$ by the three displayed brackets. Hence the Leibniz rule gives $D_x(Y) = 0$ for every monomial and, by continuity, for every element of the completed algebra. Thus the centrality homotopies against arbitrary algebraic current-target monomials are strict zero homotopies in this model. Here "arbitrary" means arbitrary monomial in the freely adjoined algebraic current target. The genuine unreduced open BV observable factorization algebra and the unreduced $Z_{\mathrm{fact}}^{P_0}$-morphism require the additional graph/QME data. □

Lemma E.11 *(Unreduced scalar primitive).* On the local unreduced Hamiltonian algebra $A = \mathbb{C}[z_1, z_2]$, the constant-coefficient cocycle

$$\omega(f, g) = [\{f, g\}_{\Omega_{\mathrm{loc}}}]_0$$

is the CE coboundary of the linear functional

$$\eta(f) = -[f]_0.$$

With the convention $(d_{\mathrm{CE}}\eta)(f, g) = -\eta(\{f, g\})$, one has $d_{\mathrm{CE}}\eta = \omega$. The primitive lives on the unreduced algebra A; scalar reduction to $\bar{A} = A/\mathbb{C} \cdot 1$ removes it.

Proof. The identity is immediate:

$$(d_{\mathrm{CE}}\eta)(f, g) = -\eta(\{f, g\}_{\Omega_{\mathrm{loc}}}) = [\{f, g\}_{\Omega_{\mathrm{loc}}}]_0 = \omega(f, g).$$

If η descended to $\bar{A}$, then it would vanish on constants. But $\eta(1) = -1$. Equivalently $\omega(z_1, z_2) = 1$ while $\{z_1, z_2\} = 1$ is zero in the scalar-reduced Hamiltonian quotient. □

THEOREM E.12 *(Local scalar contact QME obstruction).* Let

$$\omega(f,g) = [\{f,g\}_{\Omega_{\mathrm{loc}}}]_0$$

be the formal-local Hamiltonian scalar cocycle of Proposition E.2. On monomials,

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = \begin{cases} kn - lm, & (k+m, l+n) = (1,1), \\ 0, & \text{otherwise.} \end{cases}$$

For compactly supported $a, b \in \Omega_c^0(\mathbb{R}_{\mathrm{brane}})$, the unreduced scalar boundary contact in the local mixed HT brane-defect model is

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\mathrm{open}} = I_\partial(ab, \{f,g\}_{\bar{A}}) + N\,\omega(f,g) \int_{\mathbb{R}} a(t)b(t)\,dt.$$

In the quantum Weyl normalization it is equivalently represented by

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\ \mathrm{Tr}(A) \quad \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N).$$

Equivalently,

$$\mathrm{Tr}(A[X,Y]) = \hbar N\ \mathrm{Tr}(A) \quad \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N),$$

for the matrix Weyl commutator $[X,Y] = XY - YX$. After insertion of the central boundary ghost γ_1, $|\gamma_1| = 1$, the degree-one obstruction representative is

$$\mathrm{Ob}_{\mathrm{sc}}(\gamma_1; a, f; b, g) = \hbar N\,\omega(f,g) \int_{\mathbb{R}} a(t)b(t)\gamma_1(t)\,dt.$$

Its associated graded CE class is

$$\hbar N[\bar{c}]\,\gamma_1 \in H^2_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]].$$

Under the coefficient-forgetting map it becomes the formal scalar multiple $\hbar N[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}; \mathbb{C})[[\hbar]]$. This class is a nonzero scalar-reduced Hamiltonian cohomology class. Hence an internal scalar counterterm in the ordinary scalar-reduced $\mathfrak{gl}_N$ brane-defect model has leading-symbol obstruction $\hbar N[\bar{c}]$.

Proof. The monomial formula for ω follows by applying

$$\{z_1^k z_2^l, z_1^m z_2^n\} = (kn - lm) z_1^{k+m-1} z_2^{l+n-1}$$

and then keeping only the constant Taylor coefficient. The first displayed formula is Corollary 5.192. The second is the quantum version of the same central term: the Weyl convention gives $[X^i{}_j, Y^k{}_l] = \hbar\delta^i_l\delta^k_j$, and the normal ordered moment relation is $Y_\hbar X_\hbar - X_\hbar Y_\hbar + \hbar N I \equiv 0$. Hence $XY - YX = \hbar N I$ in the quantum Hamiltonian reduction, and tracing against A gives the stated sign. The matrix-commutator form is the same identity written as $\mathrm{Tr}(A(XY - YX))$.

The scalar contact itself has degree 0. Multiplication by the central ghost γ_1 makes the QME obstruction representative degree 1, the degree of the obstruction complex. The CE associated graded of this representative is the scalar extension cocycle $\hbar N\omega\,\gamma_1$, hence the class

$$\hbar N[\bar{c}]\,\gamma_1 \in H^2_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]].$$

If its coefficient-forgetting image were exact in the scalar-reduced Hamiltonian source, the leading scalar term would give a linear primitive $\bar{\eta} : \bar{A} \to \mathbb{C}$ with $\delta\bar{\eta} = \omega$. Pulling $\bar{\eta}$ back to $\mathbb{C}[z_1, z_2]$ and setting the value on 1 to zero would contradict the computation

$$\omega(z_1, z_2) = 1, \qquad \{z_1, z_2\} = 1.$$

Equivalently, Lemma E.11 gives the unreduced scalar primitive used here. It lives on the unreduced polynomial Hamiltonian algebra; scalar reduction to $\bar{A}$ removes that primitive. Thus a scalar-reduced counterterm is equivalent to an associated-graded primitive for the class. □

THEOREM E.13 *(Scalar anomaly transgression: four equivalent representatives).* In the formal-local mixed geometry $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_0$ with $\Omega_{\mathrm{loc}} = dz_1 \wedge dz_2$ of Proposition E.2, the scalar Hamiltonian cocycle

$$\omega(f, g) = [\{f, g\}_{\Omega_{\mathrm{loc}}}]_0, \qquad \omega(z_1, z_2) = 1,$$

is detected in four equivalent forms. All four witness one and the same nonzero class in $H^2(\bar{A}; \mathbb{C})[[\hbar]]$. Up to the quantum scalar $\hbar N$ given by the open Chan–Paton index loop, the cocycle is the same in each.

(i) *Algebraic central extension of the Hamiltonian Lie algebra.* The short exact sequence of Lie algebras

$$0 \longrightarrow \mathbb{C} \cdot 1 \longrightarrow A = \mathbb{C}[z_1, z_2] \longrightarrow \bar{A} = A/\mathbb{C} \cdot 1 \longrightarrow 0, \qquad A \text{ with bracket } \{f, g\}_{\Omega_{\mathrm{loc}}},$$

is the central extension whose Lie cocycle is exactly ω. Its class is $[\bar{c}] \in H^2(\bar{A}; \mathbb{C})$ with Lemma E.11.

(ii) *Lie-algebraic central extension.* Setting $\widehat{\mathfrak{h}}_{\mathrm{cent}} = \bar{A} \oplus \mathbb{C}\bar{c}$ with bracket

$$[\![\bar{f}, \bar{g}]\!]_{\mathrm{cent}} = \overline{\{f, g\}} + \omega(f, g)\bar{c}, \qquad [\![\bar{c}, -]\!]_{\mathrm{cent}} = 0,$$

produces a Lie central extension whose class is the same $[\bar{c}]$. This is the source-side $\widehat{\mathfrak{h}}_{\mathrm{cent}}$.

(iii) *Trace-algebraic central extension on the boundary.* With the unreduced boundary Hamiltonian generator $I_\partial(a, f)$ of Theorem E.12, the open Chan–Paton ordinary-trace open coupling produces the central extension

$$0 \longrightarrow \mathbb{C} \cdot \mathrm{Tr}(\mathbf{1}) \longrightarrow \mathrm{Tr}\,\mathbb{C}[\phi_1, \phi_2]_{\mathrm{red}} \longrightarrow A_{\partial,\mathrm{red}} \longrightarrow 0$$

of trace-reduced boundary algebras, whose Lie cocycle is $\hbar N\omega$ and whose class in $H^2(A_{\partial,\mathrm{red}}; \mathbb{C})[[\hbar]]$ is $\hbar N[\bar{c}]$.

(iv) *Quantum Weyl commutator.* In the Weyl quantization,

$$[\Phi_\hbar(z_1), \Phi_\hbar(z_2)]_\star = \hbar N \quad \mathrm{mod}\ \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N), \qquad W_1^{\mathrm{sc,ord}} = \hbar N[\bar{c}] \quad \text{in cohomology},$$

i.e. the Capelli shift makes the scalar commutator equal $\hbar N$ times the identity, modulo the Hamiltonian moment-map ideal of $\mathfrak{sl}_N$. The associated CE class is again $\hbar N[\bar{c}]$.

These four are equivalent in the strong sense that the same class $[\bar{c}] \in H^2(\bar{A}; \mathbb{C})$ is the image, under the canonical scalar projections, of all four cocycles:

$$\begin{array}{ccccc}
[\omega]_{(\mathrm{i})} & = & [\omega]_{(\mathrm{ii})} & \in & H^2(\bar{A}; \mathbb{C}), \\
\big\downarrow \hbar N \cdot & & \big\downarrow \hbar N \cdot & & \\
[\hbar N\omega]_{(\mathrm{iii})} & = & [\hbar N\omega]_{(\mathrm{iv})} & \in & H^2(\bar{A}; \mathbb{C})[[\hbar]],
\end{array}$$

with horizontal equalities of cocycles given by the Lie/algebraic identification of (i) with (ii) and the trace/Weyl identification of (iii) with (iv), and vertical multiplications by $\hbar N$ given by the open Chan–Paton trace pairing.

The radial Poisson field X_f annihilates the constant line in A (Proposition E.2), but the open trace pairs the same constant line against $\mathrm{Tr}(\mathbf{1}) = N$ and produces the Capelli shift $\hbar N[\bar{c}]$. Equivalently, the same class is the Lie cohomology class $\hbar N[\bar{c}]$. If a determinant line with connection exists and its Atiyah class pulls back to $[\bar{c}]$, its curvature restricts to $\hbar N[\bar{c}]$ at the trace generator $J(f) = \mathrm{Tr}\, f(\phi_1, \phi_2)$ by Theorem 6.8: the four representatives are the four forms in which the local-stalk Lie cohomology class enters the formal-Darboux computation, with the determinant-line interpretation of the central extension given by Remark 6.9.

Proof. Representative (i) is the content of Lemma E.11: ω is the CE coboundary of $\eta(f) = -[f]_0$ on the unreduced A, and scalar reduction to $\bar{A}$ removes η. Hence $[\omega]$ is nonzero on $\bar{A}$; the central extension presented in (i) is this nontrivial class.

Representative (ii) is the standard correspondence between Lie central extensions of $\bar{A}$ by $\mathbb{C}$ and elements of $H^2(\bar{A}; \mathbb{C})$. Setting $[[\bar{f}, \bar{g}]]_{\text{cent}} = \overline{\{f, g\}} + \omega(f, g)\bar{c}$ defines a Lie bracket because ω is a 2-cocycle, and the associated extension class is $[\omega] = [\bar{c}]$ by construction. This is the same class as (i) up to canonical identification.

Representative (iii) is the trace-algebraic content of Theorem E.12: the open Chan–Paton ordinary trace pairs $I_\partial(a, f) I_\partial(b, g)$ and produces the residual $\hbar N\, \omega(f, g) \int_{\mathbb{R}} a(t) b(t)\, dt$, which is the trace-algebraic cocycle. The boundary source map $A_{\partial,\text{red}} \to \bar{A}$ (induced by $I_\partial(a, f) \mapsto \bar{f}$ on the Hamiltonian factor and by evaluation against the test function) factors the trace cocycle as $\hbar N$ times the canonical $\bar{A}$-cocycle ω. Hence the class in $H^2(A_{\partial,\text{red}}; \mathbb{C})[[\hbar]]$ is $\hbar N$ times the same scalar $[\bar{c}]$ under this canonical pullback: the trace coefficient is the only scalar dependence beyond the source cocycle.

Representative (iv) is the quantum Weyl reformulation in the same theorem: the matrix Weyl commutator $[X, Y] = \hbar N I$ modulo $\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)$ is the Capelli normal-ordered scalar shift. Tracing against A and applying the associated-graded scalar-symbol projection produces the same scalar CE class $\hbar N[\bar{c}]$.

The three vertical/horizontal equalities of the diagram are induced by the canonical scalar-symbol projection (Proposition E.2) and the open Chan–Paton trace. The algebraic content: $X_f(1) = 0$ for every f (the constant line is annihilated by every Hamiltonian vector field), while the trace pairing $\mathrm{Tr}(\mathbf{1}) = N$ records the constant line as the open index loop charge $\hbar N$. The four representatives are the four forms of the same class $[\bar{c}]$ after pairing with the open trace, the source CE differential, the Lie bracket, or the boundary Weyl commutator. □

Remark E.14 *(Action of the four cancellation branches on the scalar anomaly class).* Each branch acts on the class $\hbar N[\bar{c}] \in H^2(\bar{A}; \mathbb{C})[[\hbar]]$ of Theorem E.13. The unreduced central character produces the CE coboundary $d_{\text{CE}}(\hbar N \eta) = \hbar N \omega$ on the unreduced source, with $\eta(f) = -[f]_0$ of Lemma E.11 admitted as a degree-zero element (Theorem E.15). The balanced $\mathbb{C}^{N|N}$ supertrace replaces the open trace coefficient by $\mathrm{sdim}\, \mathbb{C}^{N|N} = 0$, rescaling the cocycle to zero before its CE class is formed (Theorem E.16). Supertraceless monodromy extends the balanced computation to marked open loops with cyclic supertrace zero on every monodromy word (Theorem E.25). Full $\mathfrak{gl}(V)$-equivariant signed Brauer marker tensors satisfy $\mathrm{Str}_{\text{cyc}}(\mathfrak{m}) = 0$ on $\mathbb{C}^{N|N}$ (Lemma E.31, Theorem E.32). By Theorem E.52, any internal local counterterm in the ordinary scalar-reduced $\mathfrak{gl}_N$ branch must contribute leading scalar class $-\hbar N[\bar{c}]$. The class $[\bar{c}] \neq 0$, so the four branch replacements above are exactly the scalar cancellations available here.

Theorem E.15 *(Central-character source replacement for the scalar QME symbol).* Let $A = \mathbb{C}[z_1, z_2]$ and let $\bar{f}$ denote the image of f in $\bar{A} = A/\mathbb{C}\cdot 1$ for the local Hamiltonian algebra of Proposition E.2. Define the ordinary trace central character

$$\chi_N(f) = N[f]_0$$

and the unreduced boundary generator

$$I_\partial^\chi(a, f) = I_\partial(a, \bar{f}) + \chi_N(f) \int_{\mathbb{R}} a(t)\, dt.$$

Then, for all $a, b \in C_c^\infty(\mathbb{R})$ and all $f, g \in A$,

$$\{I_\partial^\chi(a, f), I_\partial^\chi(b, g)\}_{\text{open}} = I_\partial^\chi(ab, \{f, g\}).$$

In the quantum Weyl normalization the scalar part is likewise represented by the same central character:

$$J_N^\chi(1) = N, \qquad \mathrm{Tr}(A[X, Y]) = J_N^\chi([z_1, z_2]_*)\ \mathrm{Tr}(A) = \hbar J_N^\chi(1)\,\mathrm{Tr}(A).$$

Thus the local scalar QME symbol becomes exact after replacing the scalar-reduced source by the unreduced source with central character χ_N. This is a source replacement before scalar reduction. The original scalar-reduced QME complex has its own null-homotopy class. Equivalently, if $\eta(f) = -[f]_0$, then

$$d_{\mathrm{CE}}\eta = \omega, \qquad \chi_N(f) = N[f]_0 = -N\eta(f),$$

so

$$d_{\mathrm{CE}}\chi_N = -N\omega, \qquad \hbar N\omega + d_{\mathrm{CE}}(\hbar\chi_N) = 0$$

in the unreduced source.

Proof. Decompose the Poisson bracket into its scalar and reduced parts:

$$\{f, g\} = \overline{\{f, g\}} + \omega(f, g)\cdot 1.$$

By Theorem E.12,

$$\{I_\partial(a, \bar{f}), I_\partial(b, \bar{g})\}_{\text{open}} = I_\partial(ab, \overline{\{f, g\}}) + N\omega(f, g) \int_{\mathbb{R}} a(t)b(t)\, dt.$$

Since the central-character summand is scalar, it has zero open bracket with every boundary field. Therefore the right hand side is exactly

$$I_\partial(ab, \overline{\{f, g\}}) + \chi_N(\{f, g\}) \int_{\mathbb{R}} a(t)b(t)\, dt = I_\partial^\chi(ab, \{f, g\}).$$

The quantum formula is the same computation in the Weyl algebra: $[z_1, z_2]_* = \hbar$, $J_N^\chi(1) = N$, and the matrix moment relation gives $\mathrm{Tr}(A[X, Y]) = \hbar N\,\mathrm{Tr}(A)$. Hence the scalar Weyl commutator is the image of the unreduced Moyal scalar instead of an obstruction class.

This is the normal-ordering mechanism. The relation $YX - XY + \hbar N I \equiv 0$ says that an XY-to-YX interchange contributes the scalar $\hbar N I$. In the scalar-reduced source this scalar is invisible and therefore appears as $\mathrm{Ob}_{\mathrm{sc}}$. In the unreduced source with $J_N^\chi(1) = N$, the same scalar is $J_N^\chi([z_1, z_2]_*) = \hbar N$, so it is exact before forming the QME obstruction class. $\square$

THEOREM E.16 *(Local $\mathfrak{gl}(N|N)$ supertrace scalar QME cancellation).* Let $V = \mathbb{C}^{N|N}$, replace the ordinary trace by the supertrace, and use the canonical even Weyl relations on $\mathrm{End}(V)$ with the supertrace pairing in the same formal-local model of Proposition E.2. Put

$$\operatorname{sdim} V = \operatorname{Str}(\mathrm{id}_V) = 0.$$

The classical scalar contact term is

$$\{I_\partial^{\mathrm{str}}(a,f), I_\partial^{\mathrm{str}}(b,g)\}_{\mathrm{open}} = I_\partial^{\mathrm{str}}(ab, \{f,g\}_{\bar{A}}) + \operatorname{sdim}(V)\, \omega(f,g) \int_{\mathbb{R}} a(t)b(t)\, dt,$$

and hence has zero scalar component for $V = \mathbb{C}^{N|N}$. The quantum scalar commutator is

$$\operatorname{Str}(A[X,Y]) = \hbar \operatorname{sdim}(V) \operatorname{Str}(A) = 0 \quad \mod \mathcal{W}_{\mathrm{End}(V)}\widehat{\mu}(\mathfrak{gl}(V)).$$

Consequently the scalar part of the local mixed brane-defect QME vanishes in the balanced $\mathfrak{gl}(N|N)$ supertrace model by the supertrace calculation.

Proof. The scalar term in the classical boundary bracket is the contraction of the identity endomorphism in the trace pairing. For an ordinary N-dimensional vector space this contraction is N. For a super vector space it is $\operatorname{Str}(\mathrm{id}_V) = \operatorname{sdim} V$. The same local calculation as in Theorem E.12 therefore gives the displayed classical formula. For $V = \mathbb{C}^{N|N}$, this coefficient is zero.

In coordinates, the super-Weyl contraction carries the Koszul sign of the supertrace pairing. Summing the diagonal contraction gives

$$YX - XY + \hbar \operatorname{sdim}(V) I \equiv 0 \pmod{\mathcal{W}_{\mathrm{End}(V)}\widehat{\mu}(\mathfrak{gl}(V))}.$$

Thus

$$\operatorname{Str}(A[X,Y]) = \hbar \operatorname{sdim}(V) \operatorname{Str}(A),$$

which is zero for $V = \mathbb{C}^{N|N}$. These are precisely the scalar terms entering the degree-one representative $\mathrm{Ob}_{\mathrm{sc}}$; hence that representative is zero in the balanced supertrace theory. □

Definition E.17 *(Balanced scalar-contact subcomplex).* Let $V = \mathbb{C}^{N|N}$. A balanced scalar-contact subcomplex over an interval $I \subset \mathbb{R}_{\mathrm{brane}}$ is a complete filtered subcomplex

$$\mathfrak{H}^\bullet_{\mathrm{bal,sc}}(I) \subset \mathfrak{Q}^\bullet_{w,\partial,\hbar,\mathrm{str}}(I)$$

of the local supertrace brane-defect local-functional complex, with differential d_{QME}, satisfying the following conditions. First, the filtration $F^\bullet$ is the closed scalar-contact filtration generated by identity-endomorphism contractions in $\mathrm{End}(V)$. Second, the associated-graded scalar-contact extractor factors through the superdimension:

$$\sigma_{\mathrm{bal,sc}} = \operatorname{sdim}(V)\, \tau_{\mathrm{sc}} \colon\ \mathrm{gr}_F\, \mathfrak{H}^\bullet_{\mathrm{bal,sc}}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]$$

for a continuous $\mathbb{C}[[\hbar]]$-linear extractor τ_{sc}. Third, the same factorization holds after applying d_{QME}: every scalar-contact symbol in this subcomplex is obtained by contracting a single identity endomorphism in the supertrace pairing.

Proposition E.18 *(Balanced scalar-contact projection).* On every balanced scalar-contact subcomplex of Definition E.17 with $V = \mathbb{C}^{N|N}$, the scalar-symbol map is the zero associated-graded map, and it admits the filtered chain projection

$$\widehat{\sigma}_{\mathrm{bal,sc}} = 0 \colon \mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I) \longrightarrow C^{\bullet}_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]].$$

Its chain-map defect is

$$\partial_{\mathrm{bal}} = d_{\mathrm{CE}}\widehat{\sigma}_{\mathrm{bal,sc}} - \widehat{\sigma}_{\mathrm{bal,sc}} d_{\mathrm{QME}} = 0.$$

Hence all scalar-projection obstruction classes $o^{(r)}_{\sigma,w,\mathrm{bal}}(I)$ vanish on this subcomplex. In particular the first two-Hamiltonian-argument defect is

$$\mathrm{ev}_{f,g}\big(o^{(1)}_{\sigma,w,\mathrm{bal}}(I)\big) = \hbar\ \mathrm{sdim}(V)\ \omega(f,g)\gamma_1 = 0.$$

Proof. By definition of the subcomplex, every scalar-contact symbol factors as the ordinary scalar contact multiplied by the contraction of the identity endomorphism in the supertrace pairing. That contraction is

$$\mathrm{Str}(\mathrm{id}_V) = \mathrm{sdim}(V) = N - N = 0.$$

Therefore $\sigma_{\mathrm{bal,sc}} = \mathrm{sdim}(V)\tau_{\mathrm{sc}} = 0$ on the associated graded. Take the zero filtered lift to the stated scalar CE target. Since both composites $d_{\mathrm{CE}}\widehat{\sigma}_{\mathrm{bal,sc}}$ and $\widehat{\sigma}_{\mathrm{bal,sc}} d_{\mathrm{QME}}$ are zero, the chain-map defect is zero. Lemma E.4 then gives $o^{(r)}_{\sigma,w,\mathrm{bal}}(I) = 0$ for every $r > 0$.

Evaluating on two Hamiltonian boundary insertions gives the same local contraction as Theorem E.16: the ordinary coefficient N in $\hbar N\ \omega(f,g)\gamma_1$ is replaced by $\mathrm{sdim}(V)$. Thus

$$\hbar\ \mathrm{sdim}(V)\ \omega(f,g)\gamma_1 = 0,$$

including the test $(f,g) = (z_1, z_2)$. □

Remark E.19 *(Extension obstruction outside the balanced subcomplex).* Proposition E.18 kills the scalar-contact projection tower. The full non-scalar graph/QME residual is governed by the non-scalar graph/QME obstruction complex. Enlarging $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$ to a complete filtered subcomplex $\mathfrak{H}'$ carrying scalar symbols with unspecified $\mathrm{sdim}(V)$-factorization, the obstruction to extending the scalar projection is the first nonzero homogeneous component of

$$d_{\mathrm{CE}}s - s d_{\mathrm{QME}} \in \mathrm{Hom}\big(\mathrm{gr}_F\, \mathfrak{H}', C^{\bullet}_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]\big)$$

for a filtered scalar-symbol lift s. Its cohomology class is the $o^{(r)}_{\sigma}$-class of Lemma E.4. Thus an enlargement is admitted exactly when these classes vanish successively, with compatible finite-window choices.

Definition E.20 *(Balanced Costello graph/counterterm subcomplex).* Fix a Hamiltonian brane-defect specialization of the finite-scale Costello graph calculus and a finite-window tower for the formal-local mixed HT theory $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_0$. A Costello graph/counterterm enlargement

$$\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I) \subset \mathfrak{G}^{\bullet}_{\mathrm{bal,Cost}}(I) \subset \mathfrak{Q}^{\bullet}_{w,\partial,\hbar,\mathrm{str}}(I)$$

is a balanced graph subcomplex if it is a complete d_{QME}-stable filtered subcomplex generated, over $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$, by finite-window Costello graph contributions $G_{\Gamma,L,M}$ and local counterterm representatives

$C_{\Gamma,L,M}$ such that every scalar-contact associated-graded term is an identity-loop supertrace contraction. Equivalently, for each homogeneous scalar-contact piece,

$$\sigma_{\mathrm{sc}}\big(\mathrm{gr}_F\, G_{\Gamma,L,M}\big) = \mathrm{Str}(\mathrm{id}_V)\, \tau_{\Gamma,L,M}, \qquad \sigma_{\mathrm{sc}}\big(\mathrm{gr}_F\, C_{\Gamma,L,M}\big) = \mathrm{Str}(\mathrm{id}_V)\, \tau^C_{\Gamma,L,M},$$

for continuous scalar extractors $\tau_{\Gamma,L,M}$ and $\tau^C_{\Gamma,L,M}$, and the same factorization holds after applying d_{QME} to these graph and counterterm generators. The factorization is required to commute with finite-window truncation and admissible weight transport.

PROPOSITION E.21 *(Balanced graph/counterterm preservation).* Let $V = \mathbb{C}^{N|N}$. Every balanced graph subcomplex $\mathfrak{G}^\bullet_{\mathrm{bal,Cost}}(I)$ of Definition E.20 is a balanced scalar-contact subcomplex in the sense of Definition E.17. Hence the zero projection

$$\widehat{\sigma}_{\mathrm{bal,Cost}} = 0\colon \mathfrak{G}^\bullet_{\mathrm{bal,Cost}}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]$$

is a filtered chain map, and all scalar-projection obstruction classes vanish on this specified graph/counterterm subcomplex.

Proof. By Definition E.20, each new graph or counterterm scalar symbol is obtained by closing an identity endomorphism loop in the supertrace pairing. Its coefficient is therefore

$$\mathrm{Str}(\mathrm{id}_V) = \mathrm{sdim}(V) = N - N = 0.$$

Thus the associated-graded scalar extractor is zero on the graph and counterterm generators, and it is already zero on $\mathfrak{H}^\bullet_{\mathrm{bal,sc}}(I)$ by Proposition E.18. The same factorization after applying d_{QME} says that the differential creates scalar-contact symbols only in the identity-loop class. Therefore

$$d_{\mathrm{CE}}\widehat{\sigma}_{\mathrm{bal,Cost}} - \widehat{\sigma}_{\mathrm{bal,Cost}} d_{\mathrm{QME}} = 0.$$

The finite-window and weight-transport compatibility in the definition gives the same zero class in the completed inverse limit. Hence the scalar-contact projection obstruction tower of Lemma E.4 vanishes on $\mathfrak{G}^\bullet_{\mathrm{bal,Cost}}(I)$. □

THEOREM E.22 *(Graph-extension scalar obstruction).* Let

$$\mathfrak{H}^\bullet_{\mathrm{bal,sc}}(I) \subset \mathfrak{G}^\bullet_{\mathrm{Cost}}(I) \subset \mathfrak{Q}^\bullet_{w,\partial,\hbar,\mathrm{str}}(I)$$

be any complete d_{QME}-stable filtered Costello graph/counterterm enlargement. Put

$$\overline{\mathfrak{G}}^\bullet_{\mathrm{Cost}}(I) = \mathfrak{G}^\bullet_{\mathrm{Cost}}(I)/\mathfrak{H}^\bullet_{\mathrm{bal,sc}}(I), \qquad S^\bullet = C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]].$$

Let

$$\bar{\sigma}_{\mathrm{Cost}}\colon\ \mathrm{gr}_F\, \overline{\mathfrak{G}}^\bullet_{\mathrm{Cost}}(I) \longrightarrow S^\bullet$$

be the associated-graded scalar-symbol extractor of the added graph/counterterm terms. The zeroth obstruction to extending the balanced scalar-contact projection to $\mathfrak{G}^\bullet_{\mathrm{Cost}}(I)$ is the associated-graded chain-map defect

$$\partial^{(0)}_{\mathrm{Cost},\sigma}(I) = d_{\mathrm{CE}}\bar{\sigma}_{\mathrm{Cost}} - \bar{\sigma}_{\mathrm{Cost}}\, d_{\mathrm{gr}} \in \mathrm{Hom}^1_0\big(\mathrm{gr}_F\, \overline{\mathfrak{G}}^\bullet_{\mathrm{Cost}}(I), S^\bullet\big).$$

If this defect vanishes, choose a filtered lift s of $\bar{\sigma}_{\mathrm{Cost}}$, extended by zero on $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$. If the first nonzero homogeneous component of

$$\partial_s = d_{\mathrm{CE}}s - s d_{\mathrm{QME}}$$

lowers the scalar-contact filtration by $r > 0$, then it defines

$$\mathfrak{o}^{(r)}_{\mathrm{Cost},\sigma}(I) = \left[\partial_s^{(-r)}\right] \in H^1\!\Big(\mathrm{Hom}_{-r}\big(\mathrm{gr}_F\, \overline{\mathfrak{G}}^{\bullet}_{\mathrm{Cost}}(I), S^{\bullet}\big)\Big).$$

These classes are independent of the chosen lift up to the usual Hom coboundary. A filtered chain projection extending the balanced projection exists exactly when

$$\partial^{(0)}_{\mathrm{Cost},\sigma}(I) = 0, \qquad \mathfrak{o}^{(r)}_{\mathrm{Cost},\sigma}(I) = 0 \quad (r > 0)$$

successively, with compatible finite-window representatives.

Explicitly, if $X_{\Gamma,M}$ is an added graph contribution or counterterm representative and, modulo F^{p-r+1},

$$d_{\mathrm{QME}} X_{\Gamma,M} = \sum_{\Gamma'} \varepsilon_{\Gamma,\Gamma'} X_{\Gamma',M}, \qquad X_{\Gamma,M} \in F^p,$$

with the signs $\varepsilon_{\Gamma,\Gamma'}$ determined by the graph orientation, BV bracket, and counterterm convention, then the obstruction cocycle is computed by

$$\mathfrak{o}^{(r)}_{\mathrm{Cost},\sigma}(X_{\Gamma,M}) = \left[d_{\mathrm{CE}} s(X_{\Gamma,M}) - \sum_{\Gamma'} \varepsilon_{\Gamma,\Gamma'} s(X_{\Gamma',M}) \right]_{F^{p-r}/F^{p-r+1}}.$$

Thus a graph or counterterm term beyond the identity-loop superdimension class is admitted by the balanced theorem exactly after this displayed relative Hom class is killed by an explicit compatible counterterm correction.

Proof. The quotient by $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$ makes the statement relative to the already proved zero projection on the balanced scalar-contact subcomplex. If $d_{\mathrm{CE}}\bar{\sigma}_{\mathrm{Cost}} - \bar{\sigma}_{\mathrm{Cost}} d_{\mathrm{gr}}$ is nonzero, then the associated-graded scalar extractor has a chain-map defect, and every filtered chain lift carries that defect in its associated graded. This is the zeroth defect $\partial^{(0)}_{\mathrm{Cost},\sigma}(I)$.

Once the associated-graded defect is killed, the proof is the relative version of Lemma E.4. The commutator $d_{\mathrm{CE}}s - s d_{\mathrm{QME}}$ strictly lowers filtration. Its first nonzero homogeneous piece is closed for the Hom differential because both differentials square to zero and because the lower pieces have already been removed. Replacing s by $s - \alpha$, with α lowering filtration by the same amount and vanishing on $\mathfrak{H}^{\bullet}_{\mathrm{bal,sc}}(I)$, changes this component by the Hom coboundary $D\alpha$. Therefore the class is independent of choices and vanishes exactly when the next correction to s exists. Completeness of the filtration and compatibility of the finite-window tower give the stated successive criterion.

The displayed graph formula is this same chain-defect evaluated on a homogeneous graph or counterterm generator, with the d_{QME}-boundary written as the signed sum of its graph and counterterm boundary terms. Hence it is computable from the given Costello graph differential, orientation signs, and scalar symbols. If all added terms satisfy Definition E.20, then $s = 0$ is the required lift for $V = \mathbb{C}^{N|N}$, and all the displayed classes vanish, recovering Proposition E.21. □

Proposition E.23 *(First non-identity-loop scalar symbol).* Let $V = \mathbb{C}^{N|N}$. Fix a finite local Hamiltonian window M containing the polynomial Hamiltonian arguments under discussion, and let $T_\bullet = (T_1, \ldots, T_m)$ be an ordered word of even $\mathrm{End}(V)$-labels carried by one open index loop. Write $M(T_\bullet) = T_1 \cdots T_m$, with the empty product equal to id_V. Let $\Gamma^{T_\bullet}_{1,M}$ be the connected local one-cross-contraction boundary graph with two Hamiltonian external legs (a, f) and (b, g), one Weyl edge, and this marked scalar loop. Denote the degree-one scalar-contact representative obtained after inserting γ_1 by $X_{\Gamma^{T_\bullet}_{1,M}}(a, f; b, g)$.

Then $X_{\Gamma^{T_\bullet}_{1,M}}(a, f; b, g) \in F^1$, its leading term has F^1/F^2 coefficient equal to the displayed scalar, and

$$X_{\Gamma^{T_\bullet}_{1,M}}(a, f; b, g) = \hbar \, \mathrm{Str}\big(M(T_\bullet)\big) \, \omega(f, g) \int_I a(t) b(t) \gamma_1(t) \, dt \quad \mod F^2.$$

Equivalently,

$$\sigma_{\mathrm{sc}}\Big(\mathrm{gr}_F \, X_{\Gamma^{T_\bullet}_{1,M}}(a, f; b, g)\Big) = \hbar \, \mathrm{Str}\big(M(T_\bullet)\big) \, \omega(f, g)\gamma_1.$$

For monomials,

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = \begin{cases} kn - lm, & (k+m, l+n) = (1, 1), \\ 0, & \text{otherwise,} \end{cases}$$

so the first non-identity test is obtained by a single marker T:

$$\sigma_{\mathrm{sc}}\Big(\mathrm{gr}_F \, X_{\Gamma^{T}_{1,M}}(1, z_1; 1, z_2)\Big) = \hbar \, \mathrm{Str}(T)\gamma_1.$$

In particular, if P_+ is projection onto one even basis line, then

$$\sigma_{\mathrm{sc}}\Big(\mathrm{gr}_F \, X_{\Gamma^{P_+}_{1,M}}(1, z_1; 1, z_2)\Big) = \hbar \, \gamma_1 \neq 0.$$

Thus this is the first scalar symbol beyond the identity-loop computation. Its associated CE class is

$$\hbar \, \mathrm{Str}\big(M(T_\bullet)\big) \, [\bar{c}] \, \gamma_1.$$

It is nonzero exactly when $\mathrm{Str}(M(T_\bullet)) \neq 0$. If every scalar-contact loop monodromy word in a complete d_{QME}-stable graph/counterterm subcomplex, and every such word appearing after d_{QME}, has supertrace zero, then the zero scalar projection is again a filtered chain map on that subcomplex.

Proof. The graph has one local scalar contact on the brane line, so it lies in F^1. The one-edge graph has exactly one scalar contact, hence its associated graded is read in F^1/F^2. The boundary Wick normalization fixed in Theorem 5.224 gives the one-edge antisymmetrized coefficient $\hbar P(f, g)$. Taking the scalar Taylor coefficient gives

$$[P(f, g)]_0 = \omega(f, g),$$

with positive sign for the ordered pair (z_1, z_2), since $\{z_1, z_2\} = 1$. Reversing the two external legs changes the sign through $\omega(g, f) = -\omega(f, g)$.

The open index loop is computed in a homogeneous basis e_α of V, the closed marked loop contributes

$$\sum_\alpha (-1)^{|e_\alpha|} \big(M(T_\bullet)\big)^\alpha{}_\alpha = \mathrm{Str}\big(M(T_\bullet)\big).$$

For the empty word this is $\mathrm{Str}(\mathrm{id}_V) = \mathrm{sdim}\, V = 0$, which is exactly the identity-loop cancellation of Proposition E.21. For a non-identity marker T the same contraction gives $\mathrm{Str}(T)$. Projection onto one even basis line has supertrace 1, giving the displayed nonzero symbol.

The scalar contact has BV degree 0; multiplication by γ_1, of degree 1, puts the representative in the degree-one QME obstruction complex. The cocycle ω is the same scalar Hamiltonian cocycle as in Theorem E.12. That theorem proves $[\bar{c}] \neq 0$ on the scalar-reduced Hamiltonian source, so multiplying by the scalar $\hbar\ \mathrm{Str}(M(T_\bullet))$ gives a nonzero class exactly when this supertrace is nonzero.

Suppose the stated marked-loop subcomplex has only supertraceless monodromy words, before and after applying d_{QME}. The formula above then gives zero associated scalar symbol on every graph and counterterm generator and on every associated-graded boundary term. Hence $d_{\mathrm{CE}} 0 - 0\, d_{\mathrm{QME}} = 0$, so the zero scalar projection is a filtered chain map on that subcomplex. □

Definition E.24 *(Supertraceless-monodromy filtered subspace).* Let $V = \mathbb{C}^{N|N}$. For an ordered word $T_\bullet = (T_1, \ldots, T_m)$ of even $\mathrm{End}(V)$-labels on a scalar open index loop, set

$$\mathrm{st}(T_\bullet) = \mathrm{Str}(T_1 \cdots T_m), \qquad \mathrm{st}(\varnothing) = \mathrm{Str}(\mathrm{id}_V) = 0.$$

The word is supertraceless if $\mathrm{st}(T_\bullet) = 0$. This condition is imposed on the ordered product $T_1 \cdots T_m$.

Fix a complete d_{QME}-stable finite-window graph/counterterm enlargement of the local balanced supertrace brane-defect complex. The supertraceless-monodromy filtered subspace

$$\mathfrak{G}^{\bullet,\mathrm{pre}}_{\mathrm{st}=0}(I)$$

is the closed filtered span, over $\mathfrak{H}^\bullet_{\mathrm{bal,sc}}(I)$, of homogeneous graph/counterterm representatives whose every scalar-contact associated-graded loop word is supertraceless. Let

$$\Pi_{\mathrm{st}\neq 0}$$

denote the quotient of the associated-graded scalar-contact monodromy module by the closed span of supertraceless loop words. The d_{QME}-preservation obstruction is the family of homogeneous quotient classes

$$\mathcal{B}_{\mathrm{st}=0}(X;q) = \Pi_{\mathrm{st}\neq 0}\big([d_{\mathrm{QME}}X]_{F^q/F^{q+1}}\big), \qquad X \in \mathfrak{G}^{\bullet,\mathrm{pre}}_{\mathrm{st}=0}(I),$$

over all filtration degrees q. Write $\mathcal{B}_{\mathrm{st}=0} = 0$ when every class in this family vanishes. If $\mathcal{B}_{\mathrm{st}=0} = 0$, this filtered subspace is a complete d_{QME}-stable filtered subcomplex; in that case it is the supertraceless-monodromy graph/counterterm subcomplex and is denoted $\mathfrak{G}^\bullet_{\mathrm{st}=0}(I)$.

Theorem E.25 *(Supertraceless-monodromy scalar projection).* Assume the notation of Definition E.24. The filtered subspace is preserved by d_{QME} if and only if

$$\mathcal{B}_{\mathrm{st}=0} = 0.$$

Under this condition the zero scalar projection

$$\widehat{\sigma}_{\mathrm{st}=0} = 0\colon \mathfrak{G}^\bullet_{\mathrm{st}=0}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]$$

is a filtered chain map. Hence all scalar-projection obstruction classes of Lemma E.4 vanish on the supertraceless-monodromy subcomplex.

More explicitly, let $X_{\Gamma,M} \in F^p$ be a homogeneous generator of the filtered subspace and suppose that, modulo F^{p-r+1},

$$d_{\mathrm{QME}}X_{\Gamma,M} = \sum_{\Gamma',U_\bullet} \varepsilon_{\Gamma,\Gamma',U_\bullet} X_{\Gamma' U_\bullet, M},$$

where $U_\bullet$ runs over scalar-loop monodromy words appearing in the boundary terms. The first possible preservation obstruction is the quotient class

$$\mathcal{B}^{(r)}_{\mathrm{st}=0}(X_{\Gamma,M}) = \left[\sum_{\Gamma',U_\bullet} \varepsilon_{\Gamma,\Gamma',U_\bullet} X_{\Gamma'^{U_\bullet},M}\right]_{\mathrm{st}\neq 0,\ F^{p-r}/F^{p-r+1}}.$$

Its scalar CE image on a two-Hamiltonian-argument scalar loop is

$$\hbar \sum_{\Gamma',U_\bullet} \varepsilon_{\Gamma,\Gamma',U_\bullet} \operatorname{Str}\!\big(M(U_\bullet)\big)\, \omega(f_{\Gamma'}, g_{\Gamma'})\gamma_1.$$

In cohomology this becomes

$$\hbar \sum_{\Gamma',U_\bullet} \varepsilon_{\Gamma,\Gamma',U_\bullet} \operatorname{Str}\!\big(M(U_\bullet)\big)\ [\bar{c}]\ \gamma_1.$$

Thus a non-supertraceless boundary word is the exact obstruction to d_{QME}-preservation, and its scalar shadow is precisely the marked-loop class computed in Proposition E.23.

For the unmarked Hamiltonian boundary generators $I^{\mathrm{str}}_{\partial}(a, f)$ used in Theorem E.16, and for the identity-loop graph generators of Definition E.20, the obstruction $\mathcal{B}_{\mathrm{st}=0}$ vanishes. These generators carry only the empty monodromy word, and the displayed differential $Q + \{I_0, -\}$ contains only the identity-loop marker. Therefore applying $d_{\mathrm{cl}} = Q + \{I_0, -\}$ to this displayed Hamiltonian generator set preserves supertraceless monodromy. Preservation by the full scale-L differential d^L_{QME} further requires that $\hbar\Delta_L$ and $I[L] - I_0$ create only empty identity-loop scalar contacts with zero supertrace; equivalently, the Costello specialization has $\mathcal{B}_{\mathrm{st}=0} = 0$. A larger specialization with marked idempotents, holonomies, or counterterm tensors is governed by $\mathcal{B}_{\mathrm{st}=0}$ on those added generators.

Proof. By definition, $\Pi_{\mathrm{st}\neq 0}$ kills exactly the associated-graded scalar-contact monodromy terms whose loop words are supertraceless. Hence $\mathcal{B}_{\mathrm{st}=0} = 0$ says precisely that the d_{QME}-boundary of every generator of the filtered subspace has only supertraceless loop words in every associated-graded filtration degree. Since the filtration is complete, this homogeneous condition is equivalent to preservation of the closed filtered span. This proves the preservation criterion.

On the preserved subcomplex every scalar-contact loop word $T_\bullet$ satisfies $\operatorname{Str}(M(T_\bullet)) = 0$. Proposition E.23 computes the associated-graded scalar symbol of such a loop as

$$\hbar \operatorname{Str}(M(T_\bullet))\, \omega(f, g)\gamma_1,$$

and this is zero. The same is true after applying d_{QME}, by the preservation criterion. Therefore both composites $d_{\mathrm{CE}}\widehat{\sigma}_{\mathrm{st}=0}$ and $\widehat{\sigma}_{\mathrm{st}=0}d_{\mathrm{QME}}$ are zero, so the zero projection is a filtered chain map. Lemma E.4 then kills the whole scalar-projection obstruction tower on this subcomplex.

The formula for $\mathcal{B}^{(r)}_{\mathrm{st}=0}(X_{\Gamma,M})$ is the preceding quotient criterion evaluated on a homogeneous generator and then reduced modulo F^{p-r+1}. Applying the one-loop scalar computation of Proposition E.23 to each boundary term gives the displayed scalar CE representative. Passing to cohomology replaces ω by the nonzero class $[\bar{c}]$ from Theorem E.12.

For the unmarked Hamiltonian boundary generators, the open index loop carries the empty word $\varnothing$, so its coefficient is $\operatorname{Str}(\mathrm{id}_V) = 0$. The differential $Q + \{I_0, -\}$ acts on BV fields and takes brackets with the unmarked classical interaction; it preserves the empty marker word on an open scalar loop. Thus the image of the unmarked Hamiltonian generator set again has only empty loop words. The identity-loop graph subcomplex has the same property by Definition E.20, which requires the same factorization after applying d^L_{QME}. This proves the d_{cl}-preservation result on the displayed generator set. The full quantum statement follows precisely when the Δ_L and $I[L] - I_0$ contributions satisfy the same identity-loop condition; arbitrary marked graph/counterterm tensors remain governed by $\mathcal{B}_{\mathrm{st}=0}$. □

Proposition E.26 *(Marked $\mathcal{B}_{\mathrm{st}=0}$ and gauge-equivariant markers).* Let $V = \mathbb{C}^{N|N}$, and impose full $\mathfrak{gl}(V)$-gauge equivariance on open Chan-Paton labels. A complete computation of $\mathcal{B}_{\mathrm{st}=0}$ for a marked Costello graph/counterterm enlargement requires, for every homogeneous generator $X_{\Gamma,M} \in F^p$, a marked boundary expansion

$$d_{\mathrm{QME}} X_{\Gamma,M} = \sum_{\Gamma',\lambda} \varepsilon_{\Gamma,\Gamma',\lambda}\, X_{\Gamma' \mathfrak{m}_{\Gamma,\Gamma',\lambda},M} \mod F^{p-r+1},$$

where each

$$\mathfrak{m}_{\Gamma,\Gamma',\lambda} \in \left(\mathrm{End}(V)^{\otimes m_\lambda}\right)^{\mathfrak{gl}(V)}$$

is an even invariant marker tensor decorating the scalar open index loop. For $\mathfrak{m} = \sum_\alpha T_{\alpha,1} \otimes \cdots \otimes T_{\alpha,m}$, put

$$\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}) = \sum_\alpha \mathrm{Str}(T_{\alpha,1} \cdots T_{\alpha,m}).$$

Then the first preservation obstruction is

$$\mathcal{B}^{(r)}_{\mathrm{st}=0}(X_{\Gamma,M}) = \left[\sum_{\Gamma',\lambda} \varepsilon_{\Gamma,\Gamma',\lambda}\, X_{\Gamma' \mathfrak{m}_{\Gamma,\Gamma',\lambda},M}\right]_{\mathrm{st}\neq 0,\ F^{p-r}/F^{p-r+1}},$$

and its two-Hamiltonian-argument scalar shadow is

$$\hbar \sum_{\Gamma',\lambda} \varepsilon_{\Gamma,\Gamma',\lambda}\ \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}_{\Gamma,\Gamma',\lambda})\, \omega(f_{\Gamma'}, g_{\Gamma'})\gamma_1.$$

In cohomology this becomes

$$\hbar \sum_{\Gamma',\lambda} \varepsilon_{\Gamma,\Gamma',\lambda}\ \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}_{\Gamma,\Gamma',\lambda})\, [\bar{c}]\, \gamma_1.$$

On the displayed unmarked Hamiltonian generator set the differential is $Q + \{I_0, -\}$, with unmarked classical interaction; the balanced graph/counterterm subcomplex only allows identity-loop scalar contractions before and after d_{QME}. Hence

$$\mathcal{B}_{\mathrm{st}=0} = 0$$

on that generator set. For any larger marked Costello specialization equipped with an oriented finite-window marked boundary complex as in Definition E.28 has the scalar computation below. For a larger marked specialization, the missing datum is exactly equation (E.26), including its signs and invariant marker tensors.

In particular, a single fixed marker $T \in \mathrm{End}(V)_{\bar{0}}$ can occur in a full $\mathfrak{gl}(V)$-equivariant marked differential only when $T = \lambda\, \mathrm{id}_V$. Its scalar shadow is then

$$\hbar\, \lambda\ \mathrm{Str}(\mathrm{id}_V)\, \omega(f, g)\gamma_1 = 0.$$

Thus the nonzero P_+ test of Proposition E.23 is a genuine obstruction to a non-equivariant or externally marked enlargement. In the full $\mathfrak{gl}(N|N)$-equivariant Costello differential, an allowed single-marker term is a scalar multiple of the identity. If one imposes only the even block-diagonal group, or gives external marker tensors, the admissible datum is the invariant tensor family above, and the first nonzero scalar shadow is detected exactly by the displayed cyclic-supertrace coefficient.

Proof. The formula for $\mathcal{B}^{(r)}_{\mathrm{st}=0}$ is Definition E.24 applied to the marked boundary expansion equation (E.26). The scalar-shadow formula is the linear extension of Proposition E.23: a decomposable marker word contributes the cyclic supertrace of the ordered product around the open index loop, and an invariant tensor is a finite linear combination of such words. Passing to cohomology replaces ω by $[\bar{c}]$ as in Theorem E.25.

The equivariance constraint is the following. A marked local functional with one fixed marker T transforms under $g \in GL(V)$ by replacing the marker with gTg^{-1}. Equivariance of the differential therefore forces $gTg^{-1} = T$, or infinitesimally $[A, T] = 0$ for every $A \in \mathfrak{gl}(V)$. Since T is even, write it in the parity decomposition as $T = T_{\bar{0}} \oplus T_{\bar{1}}$. Commutation with $\mathfrak{gl}(N) \oplus \mathfrak{gl}(N)$ gives $T_{\bar{0}} = \lambda_0 \,\mathrm{id}$ and $T_{\bar{1}} = \lambda_1 \,\mathrm{id}$. Commutation with the odd elementary endomorphisms forces $\lambda_0 = \lambda_1$. Hence $T = \lambda \,\mathrm{id}_V$. The cyclic supertrace is then $\lambda \,\mathrm{sdim}(V) = 0$ for $V = \mathbb{C}^{N|N}$.

On the unmarked Hamiltonian generator set the associated-graded differential $d_{\mathrm{cl}} = Q + \{I_0, -\}$ carries the empty endomorphism label; together with Definition E.20, which requires identity-loop factorization after applying d_{QME}, these terms have cyclic supertrace $\mathrm{Str}(\mathrm{id}_V) = 0$, so their $\mathcal{B}_{\mathrm{st}=0}$-class vanishes. Any further marked graph/counterterm term is governed by its chosen oriented marked boundary complex; equation (E.26) records the residual data. □

Definition E.27 *(Finite-window marked Costello list).* Fix a finite-window graph/QME system for the local mixed HT brane-defect theory, $(I, w, M, \mathcal{G}_M)$ in the sense of Definition 8.21. A finite-window marked Costello list

$$\mathcal{G}^{\mathrm{mk}}_M$$

is a finite set of connected brane-defect graphs and local counterterm graphs with the following labels and boundary table.

The admissible vertex labels are exactly the finite local labels visible in the chosen system:

(L1) external CE labels $u_{a(t)dt\otimes f}$ and $c_{B,\rho}$, with f, ρ in the Hamiltonian window M, and with $B \in \mathcal{D}^{\mathrm{reg}}_c(I)$ or $B_\bullet \in \mathcal{D}'_{c,\mathrm{WF}}(I; \Gamma, M)$;

(L2) interaction vertices from the finite-window local interaction $I^w_{0,M}$;

(L3) renormalized graph-weight vertices $W^{R,M}_{\Gamma,L,w}(B^\Gamma_\bullet)$ and local counterterm vertices $C^M_{\Gamma,w}(\epsilon; B^\Gamma_\bullet)$ given by the system;

(L4) the central scalar-contact ghost γ_1, only on the scalar-contact boundary vertices already used in Theorem E.12.

The admissible edge labels are the finite-scale local propagator $P^{w,M}_{\epsilon,L}$, the corresponding regularized BV contraction $\Delta^{w,M}_{\epsilon,L}$, and the boundary Weyl edge whose scalar part on two Hamiltonians is the formal-local scalar cocycle of Proposition E.2,

$$(f, g) \longmapsto \omega(f, g)\gamma_1.$$

The open index loops may carry marker tensors, defined below in Definition E.28.

For each $\Gamma \in \mathcal{G}^{\mathrm{mk}}_M$ one specifies a finite ordered set

$$\mathsf{B}_M(\Gamma) = \{\beta_1, \ldots, \beta_s\}$$

of elementary codimension-one boundary operations. If a differential, bracket, collision, or counterterm expansion is a finite sum, each summand is a separate β_i. Each operation has a coefficient $\varepsilon_{\beta_i} \in \mathbb{C}[[\hbar]]$, a resulting graph $\partial_{\beta_i}\Gamma \in \mathcal{G}_M^{\mathrm{mk}}$, and a marker-transport map μ_{β_i}. The local-functional boundary operations are:

(B1) *field differential:* if a vertex has label ℓ_v and $Q\ell_v = \sum_\alpha q_\alpha \ell_{v,\alpha}$, then the summand (v, α) replaces ℓ_v by $\ell_{v,\alpha}$ and has $\varepsilon_{v,\alpha} = q_\alpha$, including the Koszul sign from moving Q to v;

(B2) *BV edge contraction:* if an oriented internal edge e joins vertices v_1, v_2 and

$$\{\ell_{v_1}, \ell_{v_2}\}_{\hbar,M} = \sum_\alpha q_{e,\alpha}\ell_{e,\alpha},$$

then the summand (e, α) contracts e, merges the two vertices with label $\ell_{e,\alpha}$, and has coefficient $q_{e,\alpha}$ multiplied by the ordinary BV orientation sign. The bracket with $I^w_{0,M}$ is this operation with one vertex labelled by $I^w_{0,M}$;

(B3) *boundary collision/contact:* if an ordered boundary collision stratum $S = (v_1, \ldots, v_k)$ has local expansion

$$\mathrm{Coll}_S(\ell_{v_1}, \ldots, \ell_{v_k}) = \sum_\alpha q_{S,\alpha}\ell_{S,\alpha},$$

then (S, α) collapses S to one boundary vertex labelled $\ell_{S,\alpha}$. For the two-Hamiltonian scalar contact this convention gives

$$\mathrm{Coll}^{\mathrm{sc}}((a, f), (b, g)) = \hbar\,\omega(f, g) \int_I a(t)b(t)\gamma_1(t)\,dt,$$

antisymmetric in (f, g);

(B4) *counterterm face:* if the small-diagonal extension of a graph subconfiguration has local counterterm expansion $C^M_{\Gamma,w}(\epsilon; B_\bullet) = \sum_\alpha q_{\Gamma,\alpha} C^M_{\Gamma,\alpha}$, then (Γ, α) replaces that subconfiguration by the local counterterm vertex $C^M_{\Gamma,\alpha}$;

For the Hom-valued graph kernel one adjoins one further operation: the source differential face $-\kappa d_{\mathrm{CE}}$, with the CE bracket coefficient included in ε_{β_i}. After evaluation on fixed CE cycles this extra face evaluates to zero.

The finite list is required to contain every result of (B1)–(B4) that appears in the chosen system, and also the source face when the Hom-valued kernel table is used. If a resulting graph, coefficient, marker transport, or counterterm vertex lies outside $\mathcal{G}_M^{\mathrm{mk}}$, that boundary entry is the omitted datum; the larger specialization contains an unconstructed primitive class.

The orientation line used for signs is

$$\mathrm{or}_M(\Gamma) = \det \mathsf{B}_M(\Gamma) \otimes \det E_{\mathrm{int}}(\Gamma) \otimes \bigotimes_v \mathrm{or}(\ell_v).$$

The first determinant gives the incidence sign in the marked differential below. The edge, vertex, BV, and CE Koszul signs are included in the coefficients ε_{β_i}.

The list is *codimension-two closed* if, for every $\beta_i \neq \beta_j$, the two composites are again entries of the finite list and, after the canonical identification of resulting orientation lines,

$$\partial_{\beta_j|\beta_i}\partial_{\beta_i}\Gamma = \partial_{\beta_i|\beta_j}\partial_{\beta_j}\Gamma,$$

$$\varepsilon_{\beta_i}\varepsilon_{\beta_j|\beta_i} = \varepsilon_{\beta_j}\varepsilon_{\beta_i|\beta_j}, \qquad \mu_{\beta_j|\beta_i}\mu_{\beta_i} = \mu_{\beta_i|\beta_j}\mu_{\beta_j}.$$

Here $\beta_j|\beta_i$ denotes the surviving operation induced by β_j after applying β_i. Thus, from a seed graph Γ, the finite window consists of the finitely many faces obtained by deleting subsets of the finite set $\mathsf{B}_M(\Gamma)$:

$$\mathcal{G}_M^{\mathrm{mk}} = \{\partial_J\Gamma \,|\, \Gamma \in \mathcal{S}_M,\ J \subset \mathsf{B}_M(\Gamma)\}, \qquad \partial_J = \partial_{\beta_{j_k}} \cdots \partial_{\beta_{j_1}}.$$

Here $\mathcal{S}_M$ is the finite seed set of graph and counterterm graphs, and the codimension-two closure equations make ∂_J independent of the chosen order of J. A defect in this displayed equality is precisely the finite graph-list obstruction.

Definition E.28 *(Oriented full-equivariant marked boundary complex).* Let $V = \mathbb{C}^{N|N}$. For $m \geq 1$, let

$$\mathcal{M}_m^{\mathrm{full}}(V) \subset \left(\mathrm{End}(V)^{\otimes m}\right)^{\mathfrak{gl}(V)}$$

be the finite span of the signed Brauer permutation tensors $\mathfrak{C}_\sigma$, $\sigma \in S_m$. Here $\mathfrak{C}_\sigma$ is the string-diagram tensor obtained by writing $\mathrm{End}(V) = V \otimes V^\vee$, pairing the $V^\vee$ factor in the i-th copy with the V factor in the $\sigma(i)$-th copy, and using the Koszul signs of the symmetric monoidal category of super vector spaces. For $m = 0$ the marker is the empty word.

A full-equivariant marked graph/counterterm generator in a finite window M is a tuple

$$X_{\Gamma,o,\mathfrak{m},M},$$

where $\Gamma \in \mathcal{G}_M^{\mathrm{mk}}$ for a finite-window marked Costello list of Definition E.27, o is an orientation of the finite set $\mathsf{B}_M(\Gamma)$, and $\mathfrak{m} = (\mathfrak{m}_L)_L$ assigns to each marked scalar open index loop L a tensor $\mathfrak{m}_L \in \mathcal{M}_{m_L}^{\mathrm{full}}(V)$.

If

$$o = \beta_1 \wedge \cdots \wedge \beta_s$$

is the ordered orientation of $\mathsf{B}_M(\Gamma)$, put $o/\beta_i = \beta_1 \wedge \cdots \widehat{\beta_i} \cdots \wedge \beta_s$. The marker-transport map

$$\mu_{\beta_i} \colon \bigotimes_L \mathcal{M}_{m_L}^{\mathrm{full}}(V) \longrightarrow \bigotimes_{L'} \mathcal{M}_{m'_{L'}}^{\mathrm{full}}(V)$$

is obtained by the signed evaluation, coevaluation, braiding, loop splitting, and loop concatenation maps that define the open index contractions. Concretely, loop merging sends two cyclic words to their boundary-ordered concatenation with the Koszul braiding sign; loop splitting inserts the coevaluation tensor at the split points; deletion of a closed marked loop applies the cyclic supertrace. The signed marked differential is

$$d_{\mathrm{mk}} X_{\Gamma,o,\mathfrak{m},M} = \sum_{i=1}^{s} (-1)^{i-1} \varepsilon_{\beta_i}\, X_{\partial_{\beta_i}\Gamma,\ o/\beta_i,\ \mu_{\beta_i}(\mathfrak{m}),\ M}.$$

Its scalar-contact filtration is

$$p(\Gamma,\mathfrak{m}) = \#\{\text{scalar-contact open loops in } \Gamma\}, \qquad F^p = \overline{\mathrm{Span}}\{X_{\Gamma,o,\mathfrak{m},M} : p(\Gamma,\mathfrak{m}) \geq p\}.$$

For an elementary boundary operation set

$$r(\beta_i) = p(\Gamma,\mathfrak{m}) - p(\partial_{\beta_i}\Gamma, \mu_{\beta_i}(\mathfrak{m})).$$

The first filtration-lowering marked boundary is therefore

$$\left[d_{\mathrm{mk}} X_{\Gamma,o,\mathfrak{m},M}\right]_{F^{p-r}/F^{p-r+1}} = \sum_{i\colon r(\beta_i)=r} (-1)^{i-1} \varepsilon_{\beta_i}\, X_{\partial_{\beta_i}\Gamma,\ o/\beta_i,\ \mu_{\beta_i}(\mathfrak{m}),\ M}.$$

Lemma E.29 *(Incidence signs).* If the finite marked Costello list is codimension-two closed in the sense of Definition E.27, then $d_{\mathrm{mk}}^2 = 0$.

Proof. This is the cellular boundary sign computation. The codimension-two result obtained from β_i and β_j, $i < j$, appears twice. The order β_i then β_j contributes

$$(-1)^{i-1}(-1)^{j-2}\varepsilon_{\beta_i}\varepsilon_{\beta_j|\beta_i},$$

because β_i has been removed before β_j is counted. The order β_j then β_i contributes

$$(-1)^{j-1}(-1)^{i-1}\varepsilon_{\beta_j}\varepsilon_{\beta_i|\beta_j}.$$

The incidence signs differ by a minus sign, while (E.27) identifies the coefficients, resulting graph, and marker tensor. Hence the two terms cancel. Summing over all unordered pairs gives $d_{\mathrm{mk}}^2 = 0$. □

Proposition E.30 *(Finite-window marked-complex compatibility).* Let $(I, w, M, \mathcal{G}_M)$ be a finite-window graph/QME system for the local mixed HT brane-defect theory and let $\mathcal{G}_M^{\mathrm{mk}}$ be a codimension-two closed finite-window marked Costello list whose labels are the admissible local labels of Definition E.27. Let

$$\mathfrak{G}^\bullet_{\mathrm{mk,full},M}(I)$$

be the completed span of the full-equivariant marked generators $X_{\Gamma,o,\mathfrak{m},M}$. Then

$$\mathfrak{G}^\bullet_{\mathrm{mk,full},M}(I) \subset \mathfrak{Q}^\bullet_{\mathcal{G},M}(I)$$

and d_{mk} is the restriction of the finite-window differential $d_M = Q + \{I^w_{\mathrm{o},M}, -\}_{\hbar,M}$ on the local-functional table. On the Hom-valued kernel table, adjoining the source face makes the same formula the convolution differential $d_{\mathbf{K}} = d_M - (-1)^{|\cdot|}(\cdot)d_{\mathrm{CE}}$ before evaluation. The graph kernel $\kappa^M_{\mathcal{G},w,I}$ and the local counterterms of the system land in this marked complex exactly when their graph components and counterterm faces occur in $\mathcal{G}_M^{\mathrm{mk}}$. Truncation $\pi_{M,N}$ and admissible weight transport $R^M_{w,w'}$ preserve the marked complex precisely when they send the finite boundary table, coefficients, and marker transports to the corresponding lower-window or transported table.

Proof. The vertex and edge labels are those of Definition 8.21: regular-density or wavefront-admissible graph labels, finite-window propagators, renormalized weights, and local counterterms. Hence the represented local functionals lie in $\mathfrak{Q}^\bullet_{\mathcal{G},M}(I)$. The boundary table (B1)–(B4) is exactly the expansion of Q, the BV bracket with $I^w_{\mathrm{o},M}$, collision/contact faces, and Costello counterterm faces on these generators. Thus the signed sum (E.28) is d_M on the local functional span. In the convolution kernel before CE evaluation, the additional source face is the missing source differential term in $d_{\mathbf{K}} = d_M - (-1)^{|\cdot|}(\cdot)d_{\mathrm{CE}}$; for the degree-zero kernel this is $-\kappa d_{\mathrm{CE}}$.

The inclusion of the graph kernel and counterterms is tautological from the same finite label list. Each omitted graph component or counterterm face is an explicit missing datum for the equality with d_M. The finite-window system requires compatibility of graph weights, counterterms, scalar projections, and kernels with $\pi_{M,N}$ and $R^M_{w,w'}$; applying this requirement to each entry of the finite boundary table gives the stated preservation criterion. □

Lemma E.31 *(Cyclic supertrace of full-equivariant marker tensors).* Let $\tau_m = (1\,2\,\cdots\,m)$. For the signed Brauer permutation tensor $\mathfrak{C}_\sigma \in \mathcal{M}^{\mathrm{full}}_m(V)$,

$$\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{C}_\sigma) = \big(\mathrm{sdim}\, V\big)^{c(\tau_m^{-1}\sigma)},$$

where $c(\pi)$ is the number of cycles of the permutation π. Hence, for $V = \mathbb{C}^{N|N}$,

$$\mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m}) = 0 \qquad \text{for every } \mathfrak{m} \in \mathcal{M}_m^{\mathrm{full}}(V)$$

and also for the empty marker word.

Proof. Compose the string diagram for $\mathfrak{C}_\sigma$ with the cyclic supertrace functional $(T_1, \dots, T_m) \mapsto \mathrm{Str}(T_1 \cdots T_m)$. The cyclic multiplication identifies the outgoing leg of the i-th endomorphism with the argument of the $\tau_m(i)$-th endomorphism, while $\mathfrak{C}_\sigma$ identifies it with the argument of the $\sigma(i)$-th endomorphism. The resulting closed diagram has $c(\tau_m^{-1}\sigma)$ closed supertraces. Each closed supertrace evaluates to $\mathrm{Str}(\mathrm{id}_V) = \mathrm{sdim}\, V$. The signs in $\mathfrak{C}_\sigma$ are exactly the Koszul signs needed for this diagrammatic evaluation, giving (E.31). Since $\mathrm{sdim}\, \mathbb{C}^{N|N} = 0$, every nonempty full-equivariant marker tensor has zero cyclic supertrace. The empty word gives the identity-loop coefficient $\mathrm{Str}(\mathrm{id}_V) = 0$. □

THEOREM E.32 *(Full-equivariant marked scalar shadow vanishing).* Let $V = \mathbb{C}^{N|N}$. Let $\mathfrak{G}^\bullet_{\mathrm{mk,full}}(I)$ be a complete finite-window inverse-limit graph/counterterm complex for the formal-local mixed HT brane-defect theory generated windowwise by codimension-two closed finite-window marked Costello lists of Definition E.27 and by the full-equivariant marked generators of Definition E.28, compatible with truncation and admissible weight transport as in Proposition E.30. This is a finite-list statement in every window. Then the marked preservation obstruction has zero cyclic-supertrace shadow:

$$\mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m})\big) = 0 \qquad \text{for every elementary boundary term } \beta_i.$$

Consequently, for every homogeneous $X_{\Gamma,o,\mathfrak{m},M} \in F^p$, the first scalar CE shadow is

$$\hbar \sum_{i:\, r(\beta_i)=r} (-1)^{i-1} \varepsilon_{\beta_i}\ \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m})\big)\omega(f_{\partial_{\beta_i}\Gamma}, g_{\partial_{\beta_i}\Gamma})\gamma_{\mathbf{1}} = 0,$$

and its cohomology class on the $[\bar{c}]$-line is

$$\hbar \sum_{i:\, r(\beta_i)=r} (-1)^{i-1} \varepsilon_{\beta_i}\ \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_i}(\mathfrak{m})\big)[\bar{c}]\ \gamma_{\mathbf{1}} = 0.$$

Thus the zero scalar projection is a filtered chain map on $\mathfrak{G}^\bullet_{\mathrm{mk,full}}(I)$, and all scalar projection obstruction classes vanish on this full-equivariant marked graph/counterterm complex.

Proof. The marker transport maps μ_{β_i} are built from the evaluation, coevaluation, braiding, loop splitting, and loop concatenation maps of the $\mathfrak{gl}(V)$-module V. Therefore they send full-equivariant signed Brauer marker tensors to full-equivariant signed Brauer marker tensors. Lemma E.31 gives zero cyclic supertrace for every such resulting tensor when $V = \mathbb{C}^{N|N}$.

Formula (E.28) gives the signed first filtration-lowering boundary. Applying the one-loop scalar computation of Proposition E.23 to each boundary term gives exactly the displayed scalar CE representative. Every coefficient in that sum is zero by the preceding paragraph. Passing to cohomology replaces ω by the nonzero class $[\bar{c}]$, but it is multiplied by the same zero coefficient. Thus $\widehat{\sigma} = 0$ satisfies $d_{\mathrm{CE}}\widehat{\sigma} - \widehat{\sigma} d_{\mathrm{mk}} = 0$. Proposition E.30 identifies d_{mk} with the restricted finite-window QME differential on the chosen finite table. Hence this is the required filtered chain projection, and Lemma E.4 kills the scalar projection tower on the complex. □

Definition E.33 *(Non-scalar kernel complex for a marked finite window).* Fix a finite-window graph/QME system $(I, w, M, \mathcal{G}_M)$, a codimension-two closed marked Costello list $\mathcal{G}_M^{\mathrm{mk}}$, and a scalar projection

$$\widehat{\sigma}_{\mathrm{sc},M} : \mathfrak{Q}^\bullet_{\mathcal{G},M}(I) \longrightarrow S^\bullet_M, \qquad S^\bullet_M = C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]].$$

The non-scalar obstruction complex in that window is

$$\mathcal{K}^\bullet_{\mathrm{ns},M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M}, \qquad d_{\mathrm{ns},M} = d_M|_{\mathcal{K}_{\mathrm{ns},M}}.$$

It is a complex precisely when $\widehat{\sigma}_{\mathrm{sc},M}$ is a filtered chain map. In the full-equivariant marked complex of Theorem E.32 one has $\widehat{\sigma}_{\mathrm{sc},M} = 0$, hence

$$\mathcal{K}^\bullet_{\mathrm{ns},M}(I) = \mathfrak{G}^\bullet_{\mathrm{mk,full},M}(I).$$

Let $\mathsf{A}^{\mathrm{fw}}_{n,M}$ be a valid row array in the sense of Definition 8.24. If the scalar condition vanishes,

$$\sum_{\alpha \in \mathsf{A}^{\mathrm{fw}}_{n,M}} S_{\alpha,M} = 0,$$

the order-n kernel residual is the scalar-zero row sum

$$R^{\mathrm{ns}}_{n,M} = \sum_{\alpha \in \mathsf{A}^{\mathrm{fw}}_{n,M}} R^{\mathrm{row}}_{\alpha,M} \in \mathcal{K}^1_{\mathrm{ns},M}(I).$$

Its obstruction class is

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} = [R^{\mathrm{ns}}_{n,M}] \in H^1\Big(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}\Big).$$

A degree-zero counterterm $C_{n,M} \in \mathcal{K}^0_{\mathrm{ns},M}(I)$ exists at this order if and only if

$$d_M C_{n,M} = -R^{\mathrm{ns}}_{n,M}, \qquad \mathfrak{o}^{\mathrm{ns}}_{n,M} = 0.$$

Definition E.34 *(Finite-window realization complex).* A Costello/QME realization complex for the mixed HT brane-defect model on

$$\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$$

is the following finite-window datum. First, one fixes a graph/QME system $(I, w, M, \mathcal{G}_M)$ in the sense of Definition 8.21, a codimension-two closed marked list $\mathcal{G}_M^{\mathrm{mk}}$ in the sense of Definition E.27, and the completed local-functional complex

$$\mathfrak{Q}^\bullet_{\mathcal{G},M}(I) = \Big(\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc},\mathcal{G},M}(\mathcal{E}^M_w|_I; A^{\mathrm{pp},w,M}_{\partial,\hbar}(I)), d_M\Big), \qquad d_M = Q + \{I^w_{0,M}, -\}_{\hbar,M}.$$

The topology is the complete topology generated by the $\hbar$-adic, scalar-contact, and finite-window filtrations. The complex is part of the datum: every d_M-boundary, collision face, BV bracket row, and local counterterm face needed for closure must occur in the finite table.

Second, the source is the labelled finite CE object $C^\bullet_{\mathrm{lab}}(\mathcal{G}_M)$. The Hom-valued kernel complex is

$$\mathbf{K}^\bullet_{\mathcal{G},M}(I) = \mathrm{Hom}_{\mathrm{cont}}\Big(C^\bullet_{\mathrm{lab}}(\mathcal{G}_M), \mathfrak{Q}^\bullet_{\mathcal{G},M}(I)\Big),$$

with

$$d_{\mathbf{K},M} T = d_M T - (-1)^{|T|} T\, d_{\mathrm{CE}}, \qquad \mathrm{Curv}_{\mathbf{K},M}(\kappa) = d_{\mathbf{K},M}\kappa + \frac{1}{2}[\kappa, \kappa]_{\mathbf{K},M}.$$

Thus the source-face term is part of the Hom differential. For a degree-zero kernel it contributes $-\kappa d_{\mathrm{CE}}$.

Third, the kernel has a fixed zero-edge part:

$$\kappa^{\mathrm{coef},M}_{\hbar,w,I}(u_{a(t)dt\otimes f}) = \iota_I(B^{w,M}_f(a)), \qquad \kappa^{\mathrm{coef},M}_{\hbar,w,I}(c_{B,\rho}) = \iota_I(\Theta^{w,M}_\rho(B)).$$

Here $a \in C^\infty_c(I)$, B is a regular compactly supported density or a graphwise wavefront-admissible current for the chosen finite row, and f, ρ lie in the window M. Positive-edge components are exactly the renormalized graph weights $K^M_\Gamma = W^{R,M}_{\Gamma,L,w}(B^\Gamma_\bullet)$ given by the system. A graph theorem for arbitrary $B \in \mathcal{D}'_c(I)$ is additional analytic data.

Fourth, one gives a filtered chain scalar projection

$$\widehat{\sigma}_{\mathrm{sc},M} : \mathfrak{Q}^\bullet_{\mathcal{G},M}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A};\mathbb{C}\gamma_1)[1][[\hbar]],$$

and puts

$$\mathcal{K}^\bullet_{\mathrm{ns},M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M}.$$

The ambient QME complex is always denoted $\mathfrak{Q}^\bullet_{\mathcal{G},M}(I)$. If the windowed scalar-zero QME notation $\boldsymbol{Q}^\bullet_{w,M}(I)$ is used for the same kernel, then

$$\boldsymbol{Q}^\bullet_{w,M}(I) = \mathcal{K}^\bullet_{\mathrm{ns},M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},M}.$$

The non-scalar kernel complex is formed after this projection is a chain map.

THEOREM E.35 *(Constructive non-scalar Costello/QME realization criterion).* Work in a finite-window realization complex of Definition E.34. Suppose that the QME has been solved through order $\hbar^{n-1}$ by counterterms $C_{<n,M}$. The order-n realization is decided by the following finite row construction.

For every valid row array $\mathsf{A}^{\mathrm{fw}}_{n,M}$ of Definition 8.24, the rows are precisely

$$R^{\mathrm{row}}_{d\Gamma,M} = \varepsilon_\Gamma\, d_M K^M_\Gamma,$$

$$R^{\mathrm{row}}_{\mathrm{CE}(\Gamma,\nu),M} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu}),$$

$$R^{\mathrm{row}}_{b(\Gamma_1,\Gamma_2),M} = \frac{1}{2}\varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M},$$

together with the homogeneous lower-counterterm rows in

$$[\hbar^n]\Big(d_M C_{<n,M} + \frac{1}{2}\{C_{<n,M}, C_{<n,M}\}_{\hbar,M}\Big).$$

Their scalar conditions are

$$S_{\alpha,M} = \widehat{\sigma}_{\mathrm{sc},M}(R^{\mathrm{row}}_{\alpha,M}),$$

and the scalar condition for the order is the single equation

$$\mathfrak{s}_{n,M} = \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} S_{\alpha,M} = 0.$$

The non-scalar problem at order n begins after (E.35) holds. Then

$$R^{\mathrm{ns}}_{n,M} = \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{n,M}} R^{\mathrm{row}}_{\alpha,M} \in \mathcal{K}^1_{\mathrm{ns},M}(I)$$

is a cocycle whenever the finite row table is codimension-two closed and the lower orders have been solved. Its obstruction class is

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} = [R^{\mathrm{ns}}_{n,M}] \in H^1(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}).$$

Choose degree-one row coordinates $\rho^M_1, \dots, \rho^M_a$ and degree-zero counterterm primitives $\eta^M_1, \dots, \eta^M_b$ inside the same finite system, and write

$$R^{\mathrm{ns}}_{n,M} = \sum_i r^M_i \rho^M_i, \qquad d_{\mathrm{ns},M}\eta^M_j = \sum_i A^M_{ij}\rho^M_i.$$

A fixed-window counterterm in the given span exists exactly when

$$A^M c = -r^M$$

is solvable. In the obstruction case, any covector

$$\ell A^M = 0, \qquad \ell(r^M) \neq 0$$

is a separating covector proving that $\mathfrak{o}^{\mathrm{ns}}_{n,M} \neq 0$ in the displayed finite row subcomplex. A compatible finite-window tower also requires truncation matrices $T^{M,N}$ on the rows and $P^{M,N}$ on primitive columns satisfying

$$T^{M,N} r^M = r^N, \qquad T^{M,N} A^M = A^N P^{M,N},$$

and the Roos class

$$[P^{M,N} c_M - c_N] \in H^0(\mathcal{K}^\bullet_{\mathrm{ns},N}(I))$$

must vanish.

Iterating these finite-window tests for all n, with graphwise regular-density or wavefront-admissible kernel data and with compatible counterterms, is equivalent to

$$\mathrm{Curv}_{\mathbf{K}}(\kappa_{\mathcal{G},w,I} + C_{\hbar,w}) = 0$$

in the completed brane-defect kernel complex. A larger Costello specialization requires its missing graph weight, source face, scalar condition, primitive column, or truncation matrix as explicit data for the criterion.

Proof. The first three displayed row formulas are the homogeneous terms in

$$d_M \kappa^M_{\mathcal{G},w,I} - \kappa^M_{\mathcal{G},w,I} d_{\mathrm{CE}} + \frac{1}{2}[\kappa^M_{\mathcal{G},w,I}, \kappa^M_{\mathcal{G},w,I}]_{\mathbf{K},M}.$$

The minus sign in the source-face row is the term $-\kappa d_{\mathrm{CE}}$ in Definition E.34. The last displayed family is the order-n contribution of lower counterterms to the same curved Maurer–Cartan equation.

Applying the chain scalar projection gives the scalar condition. A nonzero scalar condition places the residual outside $\ker \widehat{\sigma}_{\mathrm{sc},M}$. A zero scalar condition places the residual in the kernel complex. Codimension-two closure is exactly the finite-row form of the Bianchi identity: the two ways of taking two elementary faces have opposite incidence signs and the same resulting graph, coefficient, and marker transport. Since lower orders are solved, this makes $R^{\mathrm{ns}}_{n,M}$ a $d_{\mathrm{ns},M}$-cocycle.

A counterterm $C_{n,M}$ changes the residual by $d_{\mathrm{ns},M} C_{n,M}$. Expanding $C_{n,M}$ in the given degree-zero row basis gives the linear system (E.35). The solvability obstruction is the residual vector class in the cokernel of A^M; a left-null covector with nonzero value on r^M is precisely the finite-dimensional cokernel separation. The two truncation identities say that both residuals and boundaries commute with finite-window projection. After windowwise primitives are chosen, their compatibility with truncation is measured by the displayed Roos 1-cocycle in H^0. Thus the fixed-window H^1 class and the Roos primitive class are exactly the obstruction pair. Passing to all orders gives the completed curvature equation. □

PROPOSITION E.36 *(Closed rows and the first given open source-face row).* In the full-equivariant branch $V = \mathbb{C}^{N|N}$, the one-cross two-Hamiltonian scalar-contact row α_{sc} with arguments (z_1, z_2) has

$$R^{\mathrm{row}}_{\alpha_{\mathrm{sc}},M} = 0, \qquad S_{\alpha_{\mathrm{sc}},M} = 0, \qquad C_{\alpha_{\mathrm{sc}},M} = 0.$$

The labelled scalar-zero row α_{11} with arguments (z_1, z_1) has

$$R^{\mathrm{row}}_{\alpha_{11},M} = \mathrm{Str}_{\mathrm{cyc}}(\mu_{\beta_{11}}(\mathfrak{m}))\,\omega(z_1, z_1)\int_I a(t)b(t)\gamma_1(t)\,dt = 0,$$

$$S_{\alpha_{11},M} = 0, \qquad C_{\alpha_{11},M} = 0,$$

with truncation $t^{M,N}_{\alpha_{11}\alpha_{11}} = 1$ when the labelled zero row is retained and the lower window still contains z_1.

The first nonzero scalar-zero source-face row entering the row machinery is the order-three symbolic row $\theta_3 = \mathrm{CE}(\Theta_3, \nu_3)$, with

$$R^{\mathrm{row}}_{\theta_3,M} = -K^M_{\Theta_3}(\zeta_{M,\nu_3}) =: \mathsf{E}_{\theta_3,M}, \qquad S_{\theta_3,M} = 0.$$

If the row-closure datum $d_{\mathrm{ns},M}\mathsf{E}_{\theta_3,M} = 0$ holds, it defines

$$\mathfrak{o}^{\mathrm{ns}}_{\theta_3,M} = [\mathsf{E}_{\theta_3,M}] \in H^1(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}).$$

In the finite row subcomplex currently given for this row,

$$\mathcal{K}^0_{\theta_3,M} = 0, \qquad \mathcal{K}^1_{\theta_3,M} = \mathbb{C}\mathsf{E}_{\theta_3,M}, \qquad d_{\theta_3,M} = 0,$$

the covector

$$\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$$

detects a nonzero finite-row obstruction. To remove it one must construct either a degree-zero local counterterm $C_{\theta_3,M}$ with

$$d_{\mathrm{ns},M}C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M},$$

compatible under finite-window truncation, or additional Hom-valued companion rows whose sum changes the residual vector. These data make the order-three larger system a solved QME row.

Proof. The row α_{sc} is zero as a local functional by the cyclic supertrace computation of Lemma E.31; this is the closed row of Proposition 8.25. The row α_{11} is zero for the independent reason

$$\omega(z_1, z_1) = [\{z_1, z_1\}_{\Omega_{\mathrm{loc}}}]_0 = 0,$$

which is Proposition E.45. In both cases the zero counterterm is a primitive and the stated truncation identities send zero rows to zero rows.

The formula for θ_3 is the source-face row $-\kappa d_{\mathrm{CE}}$ from Theorem E.35, with the concrete row labels of Proposition 8.32. Its scalar condition is zero in the full-equivariant marked complex. In the one-row finite complex, the degree-zero primitive space is zero, so the boundary matrix is the 1×0 zero matrix and the residual vector is the basis vector $\mathsf{E}_{\theta_3,M}$. The displayed covector annihilates the image of the differential and pairs nontrivially with the residual. This proves nonvanishing only in the given finite row subcomplex. The displayed $C_{\theta_3,M}$ or companion-row datum is exactly what enlarges the complex enough to kill the row. □

Proposition E.37 *(Bianchi-exposed lower condition for θ_3).* In the lower window $N = 2$, let $E^2_{\nu_{02}}$ and $E^2_{\nu_{20}}$ be the two transported source-face rows attached to the θ_3 source differential. The canonical source-coordinate primitive is

$$\zeta_2^0 = u_{a,H_{1,0}} u_{b,H_{0,1}}, \qquad D^2_{02,20} = K^2_{\Theta_3}(\zeta_2^0),$$

with

$$d_{\mathrm{CE}}\zeta_2^0 = 2\zeta_{\nu_{02}} - 2\zeta_{\nu_{20}}.$$

The Hom-valued Bianchi defect is

$$\mathfrak{b}^2_{02,20} = d_{\mathrm{ns},2} K^2_{\Theta_3}(\zeta_2^0) - K^2_{\Theta_3}(d_{\mathrm{CE}}\zeta_2^0).$$

In the Bianchi-exposed basis

$$(E^2_{\nu_{02}}, E^2_{\nu_{20}}, \mathfrak{b}^2_{02,20}),$$

the current source-coordinate column and the lower transported residual are

$$A^2_D = (-2, 2, 1)^T, \qquad r_2 = (2, -2, 0)^T.$$

Hence

$$A^2_D c = -r_2$$

has a cokernel obstruction in the given lower local-functional row complex. The normalized cokernel

$$\widetilde{\lambda}_{02,\mathfrak{b}} = \frac{1}{2}(E^2_{\nu_{02}})^* + (\mathfrak{b}^2_{02,20})^*$$

satisfies

$$\widetilde{\lambda}_{02,\mathfrak{b}} A^2_D = 0, \qquad \widetilde{\lambda}_{02,\mathfrak{b}}(r_2) = 1.$$

The missing Bianchi-counterterm target is

$$-\mathfrak{b}^2_{02,20} = (0, 0, -1)^T,$$

and it has nonzero cokernel pairing against the current column A^2_D, since

$$\widetilde{\lambda}_{02,\mathfrak{b}}((0, 0, -1)^T) = -1.$$

One must therefore add a genuine scalar-zero local-functional column

$$A^2_B = (0, 0, -1)^T, \qquad d_{\mathrm{ns},2} B^2_{02,20} = -\mathfrak{b}^2_{02,20}, \qquad \widehat{\sigma}_{\mathrm{sc},2}(B^2_{02,20}) = 0,$$

with the required locality or wavefront admissibility. If such $B^2_{02,20}$ exists, then

$$d_{\mathrm{ns},2}(D^2_{02,20} + B^2_{02,20}) = -2E^2_{\nu_{02}} + 2E^2_{\nu_{20}} = -r_2.$$

For a compatible finite-window tower, write $\mathfrak{b}^M$ for the corresponding Bianchi defect in window M. The transport obstruction is

$$\Delta^1_{M,N} := -\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N.$$

A family B^M transports the fixed-window primitive only when

$$\Delta^1_{M,N} = d_{\mathrm{ns},N}(\pi_{M,N}B^M - B^N),$$

possibly after an explicitly given scalar-zero transport correction. Thus the H^1 class of $\Delta^1_{M,N}$ must vanish first, and the resulting primitive-compatibility class in

$$\varprojlim\nolimits^1_N H^0(\mathcal{K}^\bullet_{\mathrm{ns},N}(I))$$

must vanish next. These two conditions promote the lower θ_3 Bianchi primitive from a fixed-window matrix target to a solved Costello/QME counterterm tower.

Proof. The source differential gives the two lower source-face rows with coefficients 2 and -2. Applying $K^2_{\Theta_3}$ becomes a chain map after its Hom-valued Bianchi defect is killed, and

$$d_{\mathrm{ns},2}D^2_{02,20} = -2E^2_{\nu_{02}} + 2E^2_{\nu_{20}} + \mathfrak{b}^2_{02,20},$$

which is the column $A^2_D = (-2,2,1)^T$ in the displayed basis. The residual transported from the lower two-face row is $r_2 = (2,-2,0)^T$. If $A^2_D c = -r_2 = (-2,2,0)^T$, the first two coordinates force $c = 1$, while the Bianchi coordinate forces $c = 0$. Thus the equation is inconsistent. The same conclusion is the cokernel calculation: $\widetilde{\lambda}_{02,\mathfrak{b}}$ annihilates A^2_D and pairs nontrivially with r_2.

The target $(0,0,-1)^T$ is the boundary column required to remove only the Bianchi coordinate. Since the same cokernel evaluates to -1 on it, this boundary is a new scalar-zero local-functional column. Adding a scalar-zero local functional $B^2_{02,20}$ with the displayed boundary produces $A^2_D + A^2_B = (-2,2,0)^T = -r_2$, hence the lower residual is killed. This is only a formal finite-window primitive until the local functional and its admissibility are fixed.

For $M \geq N$, applying $d_{\mathrm{ns},N}$ to $\pi_{M,N}B^M - B^N$ gives the difference between the transported Bianchi boundary and the N-window Bianchi boundary, namely $-\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N$. Therefore the transported family exists after this degree-one class vanishes in $H^1(\mathcal{K}^\bullet_{\mathrm{ns},N}(I))$; after choosing primitives, their compatibility is measured by the usual Roos $\varprojlim^1 H^0$ obstruction. $\square$

THEOREM E.38 *(First non-scalar finite-window H^1 at the θ_3 row).* Work in the full-equivariant marked finite-window complex with $V = \mathbb{C}^{N|N}$ and the given minimal generator set at order ≤ 2 of Theorem E.47. Adjoin only the order-three CE-source row θ_3 of Proposition E.36. The non-scalar kernel subcomplex at θ_3 is the given finite row complex

$$\mathcal{K}^0_{\theta_3,M} = 0, \qquad \mathcal{K}^1_{\theta_3,M} = \mathbb{C}\,\mathsf{E}_{\theta_3,M}, \qquad d_{\theta_3,M} = 0.$$

The finite-window H^1 splits into two cases.

(a) *Counterterm omitted.* In the one-row complex above, the $d_{\mathrm{ns},M}$-cohomology in degree one is

$$H^1(\mathcal{K}^\bullet_{\theta_3,M}, d_{\theta_3,M}) = \mathbb{C}\cdot\mathsf{E}_{\theta_3,M} \neq 0,$$

and the obstruction class is the basis vector itself,

$$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]\big|_{\theta_3,M} = [\mathsf{E}_{\theta_3,M}] \neq 0,$$

detected by the unique covector $\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$ on the row basis. The given finite complex has zero degree-zero row space; the linear system $A^M c = -r^M$ of Theorem E.35 has empty column matrix and residual $r^M = \mathsf{E}_{\theta_3,M}$, so it is unsolvable.

(b) *Scalar-zero local counterterm $C_{\theta_3,M}$.* Suppose a degree-zero local functional $C_{\theta_3,M} \in \mathcal{K}^0_{\theta_3,M}$ with $\widehat{\sigma}_{\mathrm{sc},M}(C_{\theta_3,M}) = 0$ exists so that, in the enlarged finite complex

$$\widetilde{\mathcal{K}}^0_{\theta_3,M} = \mathbb{C}\, C_{\theta_3,M}, \qquad \widetilde{\mathcal{K}}^1_{\theta_3,M} = \mathbb{C}\, \mathsf{E}_{\theta_3,M}, \qquad \widetilde{d}\, C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M},$$

the differential matrix is the scalar -1. Then

$$H^1(\widetilde{\mathcal{K}}^\bullet_{\theta_3,M}, \widetilde{d}) = 0, \qquad [\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]\big|_{\theta_3,M} = 0,$$

and the unique counterterm $c = 1$ (i.e. adopting $C_{\theta_3,M}$ with coefficient one in the QME expansion) solves the row equation $\widetilde{A}^M c = -r^M = -\mathsf{E}_{\theta_3,M}$. Tower compatibility holds with the identity primitive transport $P^{M,N} = 1$, zero Roos class $[P^{M,N} c_M - c_N] = 0$, and zero Bianchi tower obstruction $\Delta^1_{M,N} = 0$ when the same $C_{\theta_3,M}$ is retained at all windows containing θ_3.

Each case is the result of the fixed-window matrix solver of Theorem E.35 on the corresponding row complex.

Proof. Case (a). The given finite complex has empty $\mathcal{K}^0_{\theta_3,M} = 0$, so the differential matrix $\big(d_{\mathrm{ns},M}\big|_{\mathcal{K}^0_{\theta_3,M}}\big)$ is the empty 1×0 matrix and the residual is $r^M = \mathsf{E}_{\theta_3,M}$. The scalar gate $S_{\theta_3,M} = 0$ holds by Proposition E.36, so the row lives in the kernel complex. Since $d_{\theta_3,M} = 0$ on the one-element basis, the row is closed. Its cohomology class equals the basis vector itself, and the unique covector $\ell_{\theta_3}(\mathsf{E}_{\theta_3,M}) = 1$ annihilates the (empty) image of $d_{\theta_3,M}$ but is nonzero on $\mathsf{E}_{\theta_3,M}$. This is the cokernel separation of Theorem E.35.

Case (b). Adjoining the degree-zero $C_{\theta_3,M}$ with $\widehat{\sigma}_{\mathrm{sc},M}(C_{\theta_3,M}) = 0$ keeps both rows in the non-scalar kernel. The differential matrix is now the scalar $\widetilde{A}^M = (-1)$, so $\widetilde{A}^M c = -1 \cdot c$ and $\widetilde{A}^M c = -r^M = -1$ has unique solution $c = 1$. The H^1 class is therefore zero. Tower compatibility: with identity transport $\pi_{M,N} C_{\theta_3,M} = C_{\theta_3,N}$ on every window containing θ_3, the Roos cocycle $P^{M,N} c_M - c_N = 1 - 1 = 0$, and the Bianchi tower obstruction $\Delta^1_{M,N} = -\pi_{M,N}\mathfrak{b}^M + \mathfrak{b}^N = 0$ by the same identity transport on E_{θ_3}.

Both cases are stable under finite-window truncation since the scalar gate $S_{\theta_3,N} = 0$ is preserved (the cyclic supertrace vanishes at every window), and the residual transports as $\pi_{M,N}\mathsf{E}_{\theta_3,M} = \mathsf{E}_{\theta_3,N}$ when both windows contain θ_3 and as 0 otherwise. □

Remark E.39 *(Finite-window computation at the θ_3 row).* The finite-window matrix calculation gives the dichotomy of Theorem E.38. If the Chevalley–Eilenberg ancestor and the scalar-zero counterterm are both omitted, $\mathsf{E}_{\theta_3,M}$ has no primitive and is separated by the cokernel covector $\ell(\mathsf{E}_{\theta_3,M}) = 1$. If one supplies a scalar-zero counterterm, a Chevalley–Eilenberg ancestor, or the complete companion-face table, the primitive exists with coefficient $[[C^{\mathrm{ct}}_{\theta_3,M}, 1]]$, $[[C^{\mathrm{CE}}_{\theta_3,M}, 1]]$, or the corresponding unit-coefficient signed companion sum.

Remark E.40 *(Dichotomy at θ_3).* The two cases of Theorem E.38 apply under disjoint algebraic conditions on the order-three finite-row complex at θ_3. Case (a) is the minimal-generator case: when the order-three scalar-zero counterterm $C_{\theta_3,M}$ is omitted, the finite-window $d_{\mathrm{ns},M}$-cohomology in degree one is $\mathbb{C}\,\mathsf{E}_{\theta_3,M}$, with explicit covector representative ℓ_{θ_3}. Case (b) is the counterterm-extended complex: when $C_{\theta_3,M}$ with $\widehat{\sigma}_{\mathrm{sc},M}(C_{\theta_3,M}) = 0$ and $d_{\mathrm{ns},M}C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$ is given, the finite-window H^1 vanishes. A named coefficient C_{θ_3} is admissible exactly in case (b), with one of the three primitive data — CE ancestor with controlled boundary, scalar-zero local counterterm $C_{\theta_3,M}$, or complete companion-face table whose signed sum moves the residual into the image of $\widetilde{A}^M$ (Proposition E.36).

Definition E.41 *(Normal Ω-equivariant Costello kernel datum).* Let

$$X = \mathbb{R}_t \times \mathbb{R}_s \times \mathbb{C}_{z_1} \times \mathbb{C}_{z_2}, \qquad L = \mathbb{R}_t \times \{s = z_1 = z_2 = 0\},$$

and let the normal torus be

$$T_\Omega = \mathbb{C}^*_{\epsilon_s} \times \mathbb{C}^*_{\epsilon_1} \times \mathbb{C}^*_{\epsilon_2}.$$

Its infinitesimal vector field is

$$V_\Omega = \epsilon_s s\partial_s + \epsilon_1 z_1\partial_{z_1} + \epsilon_2 z_2\partial_{z_2}, \qquad V_\Omega(t) = 0,$$

and the Cartan differential is

$$Q_\Omega = Q + \iota_{V_\Omega}, \qquad Q_\Omega^2 = L_{V_\Omega}.$$

All complexes below are taken in the T_Ω-invariant, or equivalently basic Cartan, subcomplex. Before this restriction the displayed operator is a curved Cartan operator with square L_{V_Ω}.

Put

$$\mathsf{X}_{N,M} \subset \{\, n\epsilon_s + a\epsilon_1 + b\epsilon_2,\ n\epsilon_s - a\epsilon_1 - b\epsilon_2 \,\}$$

for the finite set of nonzero moving normal characters occurring in the chosen N-brane, M-window system. The coefficient ring is

$$R_\Omega^{N,M} = \mathbb{C}[\epsilon_s, \epsilon_1, \epsilon_2, \hbar_\omega][\chi^{-1} \mid \chi \in \mathsf{X}_{N,M}], \qquad \mathrm{wt}(\hbar_\omega) = \epsilon_1 + \epsilon_2.$$

The self-dual branch is formed by base change to $\epsilon_1 + \epsilon_2 = 0$ before localization; only the surviving nonzero characters in the image of $\mathsf{X}_{N,M}$ are inverted. In particular $\epsilon_1 + \epsilon_2$ is never inverted and then specialized to zero. On that branch $\hbar_\omega$ has weight zero and may be retained as a scalar parameter or set to 1 after the specialization is explicit.

The three quantum parameters are distinct:

$$\hbar_{\mathrm{QME}}, \qquad \hbar_{\mathrm{W}}, \qquad \hbar_\omega.$$

The first counts Costello/BV graph order; the second deforms the boundary Weyl/Moyal algebra; the third scalarizes the equivariant Hamiltonian bracket. A Weyl branch may identify $\hbar_{\mathrm{W}} = \hbar_\omega$. Identification of $\hbar_{\mathrm{QME}}$ with either parameter is an explicit theorem specialization. The equivariant Hamiltonian bracket is the $R_\Omega^{N,M}$-linear bracket

$$\{f, g\}_\Omega = \hbar_\omega\big(\partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g\big).$$

A normal Ω-equivariant finite-window Costello kernel datum over $(I, w, M, \mathcal{G}_M)$ is a finite-window marked Costello system as in Definitions 8.21 and E.27, together with the following additional data.

(Ω1) A T_Ω-locally finite weighted field complex $\mathcal{E}^M_{w,\Omega}|_I$, source CE complex, local interaction $I^w_{0,M,\Omega}$, and invariant local-functional target

$$\mathfrak{Q}^\bullet_{\mathcal{G},\Omega,M}(I) \subset \mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}^M_{w,\Omega}|_I; A^{\mathrm{pp},w,M}_{\partial,\Omega,\hbar_{\mathrm{W}}}(I) \widehat{\otimes} R^{N,M}_\Omega[[\hbar_{\mathrm{QME}}]])$$

with differential

$$d_{M,\Omega} = Q_\Omega + \{I^w_{0,M,\Omega}, -\}_{\hbar_{\mathrm{QME}},M,\Omega}$$

on the invariant subcomplex. The source CE differential $d_{\mathrm{CE},\Omega}$ is formed from Q_Ω and the bracket (E.41).

(Ω2) For every localized normal character $\chi \in \mathsf{X}_{N,M}$, or for every surviving nonzero image of such a character on the self-dual branch, a normal contraction h_χ satisfying

$$Q_\Omega h_\chi + h_\chi Q_\Omega = \mathrm{id} - \mathrm{pr}_{\chi=0}$$

on the corresponding χ-weight summand, after inverting χ. This is the normal-mode acyclicity datum. Graph curvature classes and QME counterterms are independent finite-row data.

(Ω3) An equivariant finite-scale gauge fixing and propagator

$$P^{w,M}_{\epsilon,L,\Omega} = \int_\epsilon^L (Q^{\mathrm{GF}}_\Omega \otimes 1) K^{w,M}_{t,\Omega}\, dt$$

such that

$$[Q_\Omega, P^{w,M}_{\epsilon,L,\Omega}] = K^{w,M}_{\epsilon,\Omega} - K^{w,M}_{L,\Omega}, \qquad [L_{V_\Omega}, P^{w,M}_{\epsilon,L,\Omega}] = 0,$$

and such that the coefficient Casimir, wavefront conditions, and finite-window weighted bounds required by the nonequivariant system remain valid over $R^{N,M}_\Omega[[\hbar_{\mathrm{QME}}]]$.

(Ω4) A named normal-localization normalization ν_Ω. The residue branch has

$$\nu^{\mathrm{res}}_\Omega = 1.$$

The Euler branch has

$$\nu^{\mathrm{Eu}}_\Omega = \sigma_s(\epsilon_s \epsilon_1 \epsilon_2)^{-1},$$

after the corresponding Euler denominators have been admitted in the chosen localized ring. Scalar-contact rows, graph weights, and trace normalizations are always read with the selected branch. A comparison between the residue branch, Euler branch, and Euler-rescaled residue branch must state the real orientation sign σ_s and every denominator.

(Ω5) Equivariant local counterterm representatives

$$C^M_{\Gamma,w,\Omega}(\epsilon; B^\Gamma_\bullet) \in \mathfrak{Q}^0_{\mathcal{G},\Omega,M}(I)$$

which are T_Ω-invariant, local on the relevant bulk diagonal or brane-collision stratum, compatible with finite-window truncation and admissible weight transport, and whose scalar-contact associated graded is computed using the same ν_Ω.

(Ω6) An associated-graded scalar-symbol extractor

$$\sigma_{\mathrm{sc},\Omega,M}\colon\ \mathrm{gr}_F\, \mathfrak{Q}^\bullet_{\mathcal{G},\Omega,M}(I) \longrightarrow S^\bullet_{\Omega,M}, \qquad S^\bullet_{\Omega,M} = C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar_{\mathrm{QME}}]] \widehat{\otimes} R^{N,M}_\Omega,$$

which sends the two-Hamiltonian scalar contact to

$$\nu_\Omega\, \hbar_{\mathrm{QME}}\ \mathrm{Str}_{\mathrm{cyc}}(\mathfrak{m})\, [\{f, g\}_\Omega]_0\, \gamma_1.$$

A filtered chain lift $\widehat{\sigma}_{\mathrm{sc},\Omega,M}$ of $\sigma_{\mathrm{sc},\Omega,M}$ exists exactly when the Hom-obstruction test of Theorem E.42 vanishes.

(Ω7) A degree-zero Hom-valued graph kernel

$$\kappa^M_{\mathcal{G},\Omega,w,I} \in \mathrm{Hom}_{\mathrm{cont}}\Big(C^\bullet_{\mathrm{CE},\Omega,\mathrm{lab}}(\mathcal{G}_M), \mathfrak{Q}^\bullet_{\mathcal{G},\Omega,M}(I)\Big)$$

whose zero-edge part is the coefficient-current kernel and whose positive graph components are the renormalized Ω-equivariant graph weights.

This datum gives the equivariant structure: coefficient ring, Q_Ω, normal contractions, and equivariant propagator data. Equivariant QME and centrality obstructions are formed inside this structure by the criterion of Theorem E.42.

THEOREM E.42 *(Equivariant brane-defect QME obstruction criterion)*. Fix a normal Ω-equivariant finite-window system satisfying items (Ω1)–(Ω6) of Definition E.41. Let

$$\sigma_{\mathrm{sc},\Omega,M}:\ \mathrm{gr}_F\, \mathfrak{Q}^\bullet_{\mathcal{G},\Omega,M}(I) \longrightarrow S^\bullet_{\Omega,M}$$

be its associated-graded scalar extractor. The scalar projection is a chain projection exactly when the following relative Hom obstructions vanish. First,

$$\delta^{(0)}_{\Omega,\sigma,M} = d_{\mathrm{CE},\Omega}\sigma_{\mathrm{sc},\Omega,M} - \sigma_{\mathrm{sc},\Omega,M}\, d_{\mathrm{gr},\Omega} = 0.$$

If this holds and $s_{\Omega,M}$ is any filtered lift of $\sigma_{\mathrm{sc},\Omega,M}$, the first nonzero homogeneous piece of

$$d_{\mathrm{CE},\Omega} s_{\Omega,M} - s_{\Omega,M}\, d_{M,\Omega}$$

lowering the scalar-contact filtration by $r > 0$ gives

$$\mathfrak{o}^{(r)}_{\Omega,\sigma,M} \in H^1\Big(\mathrm{Hom}_{T_\Omega,-r}\big(\mathrm{gr}_F\, \mathfrak{Q}^\bullet_{\mathcal{G},\Omega,M}(I), S^\bullet_{\Omega,M}\big)\Big).$$

A filtered chain projection $\widehat{\sigma}_{\mathrm{sc},\Omega,M}$ exists if and only if $\delta^{(0)}_{\Omega,\sigma,M} = 0$ and the classes $\mathfrak{o}^{(r)}_{\Omega,\sigma,M}$ vanish successively, with compatible finite-window and weight-transport choices.

Once this projection is fixed, put

$$\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M}(I) = \ker \widehat{\sigma}_{\mathrm{sc},\Omega,M}, \qquad d_{\mathrm{ns},\Omega,M} = d_{M,\Omega}|_{\mathcal{K}_{\mathrm{ns},\Omega,M}}.$$

Suppose the Hom-valued kernel of item (Ω7) is fixed, and counterterms have been chosen through order $\hbar^{n-1}_{\mathrm{QME}}$. The order-$n$ equivariant residual is

$$R^{\mathrm{ns}}_{n,\Omega,M} = [\hbar^n_{\mathrm{QME}}]\bigg(\mathrm{Curv}_{\mathbf{K},\Omega,M}(\kappa^M_{\mathcal{G},\Omega,w,I}) + d_{M,\Omega} C_{<n,\Omega,M} + \frac{1}{2}\{C_{<n,\Omega,M}, C_{<n,\Omega,M}\}_{\hbar_{\mathrm{QME}},M,\Omega}\bigg).$$

The scalar condition at this order is

$$\mathfrak{s}_{n,\Omega,M} = \widehat{\sigma}_{\mathrm{sc},\Omega,M}(R^{\mathrm{ns}}_{n,\Omega,M}).$$

Only when $\mathfrak{s}_{n,\Omega,M} = 0$ and all lower orders have been solved does the residual define the non-scalar obstruction class

$$\mathfrak{o}^{\mathrm{ns}}_{n,\Omega,M} = [R^{\mathrm{ns}}_{n,\Omega,M}] \in H^1\Big(\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}(I), d_{\mathrm{ns},\Omega,M}\Big).$$

A compatible order-n counterterm exists precisely when

$$\big(\mathfrak{o}^{\mathrm{ns}}_{n,\Omega,M}\big)_M = 0 \quad \text{in} \quad \varprojlim_M H^1(\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}(I))$$

and the Milnor primitive-compatibility class

$$\lambda_{n,\Omega} \in {\varprojlim}^1_M H^0(\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}(I))$$

vanishes. Iterating this criterion for all n is equivalent to

$$\mathrm{Curv}_{\mathbf{K},\Omega}(\kappa_{\mathcal{G},\Omega,w,I} + C_{\Omega,w}) = 0$$

in the completed invariant kernel complex.

Therefore the normal Ω-background determines the equivariant structure. A QME proof requires, in addition, the chain scalar projection, the non-scalar row classes displayed above, explicit local counterterms killing those classes, compatible primitive transport in the ${\varprojlim}^1 H^0$ sense, and the Q_Ω-centrality homotopies of Proposition E.50 for the source rows whose centrality is used.

Proof. The first paragraph is the equivariant version of Lemma E.4. On the T_Ω-invariant subcomplex $Q^2_\Omega = L_{V_\Omega} = 0$, so the Hom differential is $D_\Omega(\alpha) = d_{\mathrm{CE},\Omega}\alpha - (-1)^{|\alpha|}\alpha d_{\mathrm{gr},\Omega}$. The associated-graded defect is the obstruction to even starting a filtered lift. After it vanishes, the first filtration-lowering chain defect is closed for D_Ω; changing the lift by a filtration-lowering cochain changes it by a Hom coboundary. Completeness gives the successive criterion and the compatibility condition in the inverse finite-window tower.

With a chain scalar projection fixed, its kernel is preserved by $d_{M,\Omega}$. The displayed residual is the order-n term of the curved Maurer–Cartan equation for the Hom-valued kernel plus lower counterterms. Applying the scalar projection gives the scalar condition. If the condition vanishes and the lower-order equations hold, the equivariant Bianchi identity in the invariant complex makes $R^{\mathrm{ns}}_{n,\Omega,M}$ a $d_{\mathrm{ns},\Omega,M}$-cocycle. Adding an order-n local counterterm changes this cocycle by a $d_{\mathrm{ns},\Omega,M}$-boundary, so the fixed-window obstruction is its H^1-class. Compatibility of primitives across the finite-window inverse system is exactly the usual ${\varprojlim}^1 H^0$ obstruction of Theorem 5.84. The final sentence follows because the normal contraction datum appears only in the construction of the equivariant propagator and localized coefficient complex. The curvature residual boundary, scalar-projection chain data, and source-row centrality homotopies are additional QME data. $\square$

Theorem E.43 *(Minimal full-equivariant kernel vanishing).* Work in the local $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ full-equivariant marked branch with $V = \mathbb{C}^{N|N}$. Let the finite marked list be the minimal codimension-two closed list generated by the strict current-interface rows and by the one-cross two-Hamiltonian scalar-contact row of Proposition 8.25. Assume that the list consists exactly of these rows through codimension two: it contains the $|\Gamma|_\hbar = 1$ scalar-contact row and its codimension-two closure, and it has zero additional $|\Gamma|_\hbar = 1$ graph rows, zero genuine $|\Gamma|_\hbar = 2$ seed graphs, zero order-2 CE-source faces, and zero order-2 graph counterterm faces.

Then the first kernel residuals are

$$R^{\mathrm{ns}}_{1,M} = 0, \qquad R^{\mathrm{ns}}_{2,M} = 0,$$

and the compatible counterterms are

$$C_{1,M} = 0, \qquad C_{2,M} = 0.$$

Thus the minimal full-equivariant finite-window array has zero genuinely non-scalar QME obstruction through the first codimension-two closure level. Its first possible non-scalar obstruction can only appear after a larger finite marked list gives an additional scalar-zero row.

Proof. The order-0 current-interface rows have zero curvature by Theorem E.10 and by the finite-window compatibility recorded in Proposition E.30. The only order-1 row in the minimal list is the one-cross scalar-contact row. Its scalar coefficient is the cyclic supertrace of the transported full-equivariant marker tensor. By Lemma E.31, this coefficient is zero for $V = \mathbb{C}^{N|N}$, including the empty word. Hence the row is zero as a local functional in the full-equivariant branch, so $R^{\mathrm{ns}}_{1,M} = 0$ and $C_{1,M} = 0$.

Codimension-two closure adds only the two ordered composites of two first scalar-contact boundary faces. Their incidence signs are opposite and their resulting graphs, coefficients, and marker transports are identified by Equation (E.27). This gives the pairwise cancellation of the marked differential. In the full-equivariant branch each scalar-loop factor is already zero by the same cyclic supertrace calculation. The minimal list has zero genuine order-2 seed graph, CE-source face, or counterterm face, so the kernel complex has zero remaining row. Thus $R^{\mathrm{ns}}_{2,M} = 0$, and the zero counterterm is compatible under every finite-window truncation. □

PROPOSITION E.44 *(First missing non-scalar row).* Let $\mathcal{G}^{\mathrm{mk}}_M$ be any larger finite marked list in the same full-equivariant local branch, with scalar projection killed as in Theorem E.32. The first genuinely non-scalar obstruction beyond scalar projection is the least order n for which the row sum of Definition E.33 is nonzero in cohomology:

$$\mathfrak{o}^{\mathrm{ns}}_{n,M} = \left[\sum_{\alpha \in \mathsf{A}^{\mathrm{fw}}_{n,M}} R^{\mathrm{row}}_{\alpha,M} \right] \in H^1\Big(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}\Big).$$

Since the minimal order-1 scalar-contact row vanishes, the first datum that can make $n = 1$ is an additional order-1 scalar-zero row

$$\alpha \in \mathsf{A}^{\mathrm{fw}}_{1,M} \setminus \{\alpha_{\mathrm{sc}}\}, \qquad S_{\alpha,M} = 0,$$

with its row value $R^{\mathrm{row}}_{\alpha,M}$. Equivalently, the missing row entry is one of the order-1 summands

$$\varepsilon_\Gamma d_M K^M_\Gamma, \qquad -\varepsilon^{\mathrm{CE}}_{\Gamma'} K^M_{\Gamma'}, \qquad \frac{1}{2}\varepsilon_{\Gamma_0,\Gamma_1}\{K^M_{\Gamma_0}, K^M_{\Gamma_1}\}_{\hbar,M},$$

together with its graph type, finite-window labels, full-equivariant marker tensor, incidence sign, scalar condition, and truncation coefficients $t^{M,N}_{\alpha\alpha'}$.

If the finite marked list explicitly has zero such order-1 rows, the next required datum is the order-2 scalar-zero row sum

$$\begin{aligned} R^{\mathrm{ns}}_{2,M} = &\sum_{|\Gamma|_\hbar=2} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{|\Gamma'|_\hbar=2} \varepsilon^{\mathrm{CE}}_{\Gamma'} K^M_{\Gamma'} \\ &+ \frac{1}{2} \sum_{|\Gamma_1|_\hbar=|\Gamma_2|_\hbar=1} \varepsilon_{\Gamma_1,\Gamma_2} \{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M} + [\hbar^2] d_M C_{<2,M} \\ &+ \frac{1}{2}[\hbar^2]\{C_{<2,M}, C_{<2,M}\}_{\hbar,M}. \end{aligned}$$

These row values, signs, scalar conditions, and truncation coefficients define the non-scalar obstruction class and counterterm problem beyond the minimal vanishing theorem above.

Proof. The scalar projection is a chain map on the full-equivariant marked complex by Theorem E.32, so its kernel is a d_M-stable complex. After the scalar condition vanishes, Theorem 8.22 puts the first unsolved residual in degree one of that kernel complex. Adding a degree-zero counterterm changes the residual by a d_M-boundary, and hence the displayed cohomology class is exactly the obstruction.

The minimal order-1 row is already zero by Theorem E.43. Thus an order-1 non-scalar class can only be given by an additional finite-window row outside α_{sc}. The three displayed forms are precisely the order-1 possibilities in the curvature expansion: target differential of an order-1 graph component, CE source face of such a component, or a convolution bracket whose orders add to 1. Each is a row only after the finite graph list gives its labels, signs, marker transport, scalar condition, and truncation coefficients.

With zero order-1 rows, the next possible residual is the order-2 curvature expansion, including genuine order-2 graph components, CE-source faces, brackets of order-1 components, and lower-counterterm terms. These data are exactly the row entries required by Definition 8.24. □

Proposition E.45 *(First scalar-zero order-one row outside the minimal array).* Work in the full-equivariant local branch of Theorem E.43. Let $M \geq 1$, and enlarge the minimal finite marked list by the labelled one-cross scalar-contact boundary operation α_{11} with ordered Hamiltonian arguments (a, z_1), (b, z_1), the boundary Weyl edge E_{Weyl}, the scalar-contact vertex $\gamma_{\mathbf{1}}$, and the same full-equivariant marker complex as before. Choose the orientation in which the displayed ordered pair has incidence coefficient +1.

Then α_{11} is a scalar-zero order-one row outside α_{sc}, with row data

$$|\alpha_{11}|_\hbar = 1, \qquad d_{\max}(\alpha_{11}) = 1, \qquad p_{\alpha_{11}} = 1, \qquad r_{\alpha_{11}} = 1, \qquad \iota_{\alpha_{11},M} = +1.$$

Its row value and scalar condition are

$$R^{\mathrm{row}}_{\alpha_{11},M} = \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_{11}}(\mathfrak{m})\big)\omega(z_1, z_1) \int_I a(t)b(t)\gamma_{\mathbf{1}}(t)\,dt = 0,$$

$$S_{\alpha_{11},M} = \mathrm{Str}_{\mathrm{cyc}}\big(\mu_{\beta_{11}}(\mathfrak{m})\big)\omega(z_1, z_1)\gamma_{\mathbf{1}} = 0.$$

Here the coefficient is the extracted $\hbar^1$-coefficient; the full scalar-contact representative has the additional factor $\hbar$. For $M \geq N \geq 1$, retain the same labelled row in window N and set

$$t^{M,N}_{\alpha_{11}\alpha_{11}} = 1, \qquad t^{M,N}_{\alpha_{11}\alpha'} = 0 \quad (\alpha' \neq \alpha_{11}).$$

These coefficients satisfy the truncation equations because $\pi_{M,N} R^{\mathrm{row}}_{\alpha_{11},M} = 0$ and $\pi_{M,N} S_{\alpha_{11},M} = 0$.

The associated kernel class is therefore

$$\mathfrak{o}^{\mathrm{ns}}_{1,M}(\alpha_{11}) = [R^{\mathrm{row}}_{\alpha_{11},M}] = 0 \in H^1\Big(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}\Big),$$

and the required counterterm contribution is $C_{1,M} = 0$.

Thus the finite closed graph-list axioms support explicit scalar-zero order-one rows beyond α_{sc}. The first explicit such row is harmless: it is zero before cohomology. If the row convention retains only nonzero summands, then α_{11} is omitted, and a nonzero order-one class still requires an additional given row $R^{\mathrm{row}}_{\alpha,M}$ of the form stated in Proposition E.44.

Proof. The scalar-contact boundary formula in Proposition 8.25 applies to every ordered pair of Hamiltonian labels in the finite window. For the pair (z_1, z_1),

$$\omega(z_1, z_1) = \big[\{z_1, z_1\}_{\Omega_{\mathrm{loc}}}\big]_0 = 0.$$

Multiplying by the transported full-equivariant cyclic supertrace preserves this zero coefficient. Hence both the local functional row and its scalar projection vanish.

Since the row value is zero, $d_{\mathrm{ns},M}$-closedness is automatic and the cohomology class is the zero class. The displayed truncation coefficients send the labelled row to the same labelled row in lower windows containing z_1, and both sides of the row and scalar truncation identities are zero. The zero counterterm solves the row equation. The final assertion is logical: the finite-list axioms allow this added labelled zero row; a nonzero order-one class is decided by the nonzero order-one row values. □

Remark E.46 *(Order two after zero nonzero order-one rows).* If zero rows such as α_{11} are suppressed and the finite marked list gives zero nonzero order-one rows outside α_{sc}, then $C_{1,M} = 0$ and every order-two bracket row containing α_{sc} is zero by Proposition 8.26. The order-two scalar-zero residual reduces to

$$\begin{aligned} R^{\mathrm{ns}}_{2,M} = &\sum_{|\Gamma|_\hbar=2} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{|\Gamma'|_\hbar=2} \varepsilon^{\mathrm{CE}}_{\Gamma'} K^M_{\Gamma'} \\ &+ \frac{1}{2} \sum_{\substack{|\Gamma_1|_\hbar=|\Gamma_2|_\hbar=1 \\ \Gamma_i\neq\alpha_{\mathrm{sc}}}} \varepsilon_{\Gamma_1,\Gamma_2} \big\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\big\}_{\hbar,M}. \end{aligned}$$

Under the stronger minimal hypotheses of Theorem E.43 the three sums are empty, and $R^{\mathrm{ns}}_{2,M} = 0$. In any larger finite system this displayed row sum is a cochain only after the genuine order-two graph weights, CE-source faces, bracket signs, scalar conditions, and truncation coefficients are fixed.

THEOREM E.47 *(Minimal full-equivariant all-order vanishing).* Work in the local $\mathbb{R}^2_{\mathrm{top}} \times \widehat{\mathbb{C}^2}_{0,\mathrm{hol}}$ full-equivariant marked branch with $V = \mathbb{C}^{N|N}$. Let $\mathcal{G}^{\mathrm{mk},0}_M$ be the finite marked list generated by the strict current-interface rows, the two-Hamiltonian scalar-contact boundary row α_{sc}, and its subset-deletion closure in the sense of Definition E.27. Assume that the finite-window system adds zero further positive-order seed graphs, CE-source faces, graph counterterm faces, or nonzero scalar-zero rows.

Then for every $n \geq 1$

$$R^{\mathrm{ns}}_{n,M} = 0, \qquad \mathfrak{o}^{\mathrm{ns}}_{n,M} = 0, \qquad C_{n,M} = 0$$

If the row convention suppresses zero rows, then $\mathsf{A}^{\mathrm{fw},0}_{n,M}$ is empty for every $n \geq 1$. If it retains labelled zero rows, every positive-order retained row has

$$R^{\mathrm{row}}_{\alpha,M} = 0, \qquad S_{\alpha,M} = 0,$$

and may be given the zero truncation row $t^{M,N}_{\alpha\alpha'} = 0$. In particular, the minimal full-equivariant order-three residual is zero.

Proof. Proposition 8.25 proves that the one-cross scalar-contact component satisfies $K^M_{\alpha_{\mathrm{sc}}} = 0$ as a local functional before scalar projection. The proof is the cyclic-supertrace calculation of Lemma E.31: every transported full-equivariant marker on a scalar open loop has cyclic supertrace zero, including the empty marker word.

Every positive-order boundary row in $\mathcal{G}^{\mathrm{mk},0}_M$ contains at least one such scalar-contact factor. Hence its local-functional value is zero. Bracket rows in the minimal system have at least one positive-order factor of this form, so bilinearity of the BV/convolution bracket gives zero. Lower counterterm rows vanish inductively because the chosen lower counterterms are zero. The minimality hypothesis sets the genuine dK, CE-source, and graph-counterterm row spaces to zero. Therefore the order-n scalar-zero row sum is zero for all $n \geq 1$, its class in $H^1(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M})$ is zero, and the zero counterterm is a compatible primitive. Truncation sends zero rows to zero rows, giving the displayed truncation choice. □

Proposition E.48 *(First order-three enlarged-system datum).* Keep the full-equivariant local branch, and suppose the minimal system has been closed through order two with $C_{1,M} = C_{2,M} = 0$ and zero nonzero order-one or order-two rows outside the scalar-contact closure. Then a nonzero order-three obstruction requires row data beyond the finite graph-list axioms. A nonzero order-three cochain exists only after the finite-window system gives at least one of the following row data:

$$\begin{aligned} R^{\mathrm{ns}}_{3,M} = \sum_{|\Gamma|_\hbar=3} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{|\Gamma'|_\hbar=3} \varepsilon^{\mathrm{CE}}_{\Gamma'} K^M_{\Gamma'} \\ + \frac{1}{2} \sum_{\substack{|\Gamma_1|_\hbar+|\Gamma_2|_\hbar=3 \\ K^M_{\Gamma_1}\neq 0,\ K^M_{\Gamma_2}\neq 0}} \varepsilon_{\Gamma_1,\Gamma_2} \{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M} \\ + [\hbar^3]\Big(d_M C_{<3,M} + \frac{1}{2}\{C_{<3,M}, C_{<3,M}\}_{\hbar,M}\Big). \end{aligned}$$

Under the preceding minimal hypotheses the bracket and lower-counterterm sums are empty. Thus the first missing order-three datum is a genuine $|\Gamma|_\hbar = 3$ graph-differential row

$$R^{\mathrm{row}}_{d\Gamma,M} = \varepsilon_\Gamma\, d_M K^M_\Gamma, \qquad S_{d\Gamma,M} = \varepsilon_\Gamma\, \widehat{\sigma}_{\mathrm{sc},M}(d_M K^M_\Gamma),$$

or an order-three CE-source row

$$R^{\mathrm{row}}_{\mathrm{CE}(\Gamma,\nu),M} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu} K^M_\Gamma(\zeta_{M,\nu}), \qquad S_{\mathrm{CE}(\Gamma,\nu),M} = -\varepsilon^{\mathrm{CE}}_{\Gamma,\nu}\widehat{\sigma}_{\mathrm{sc},M}\big(K^M_\Gamma(\zeta_{M,\nu})\big).$$

In a nonminimal system with given nonzero order-one and order-two components one must also give each bracket row

$$R^{\mathrm{row}}_{b(\Gamma_1,\Gamma_2),M} = \frac{1}{2}\varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M}, \qquad S_{b,M} = \widehat{\sigma}_{\mathrm{sc},M}(R^{\mathrm{row}}_{b(\Gamma_1,\Gamma_2),M}),$$

for $|\Gamma_1|_\hbar + |\Gamma_2|_\hbar = 3$, together with its truncation coefficients $t^{M,N}_{bb'}$.

The scalar condition for order three is the single condition

$$\mathfrak{s}_{3,M} = \sum_{\alpha\in\mathsf{A}^{\mathrm{fw}}_{3,M}} S_{\alpha,M} = 0.$$

Only after this condition is fixed and vanishes is the non-scalar obstruction class defined:

$$\mathfrak{o}^{\mathrm{ns}}_{3,M} = [R^{\mathrm{ns}}_{3,M}] \in H^1\Big(\mathcal{K}^{\bullet}_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}\Big).$$

A fixed-window counterterm $C_{3,M}$ exists exactly when this class vanishes; a compatible tower also requires the corresponding $\varprojlim^1 H^0$-compatibility class to vanish.

Proof. The order-three expansion is Theorem 8.36 with $n = 3$, written in the row notation of Definition 8.24. Rows containing α_{sc} vanish by Proposition 8.26, and rows containing any retained labelled zero scalar-contact row vanish by the same bilinearity argument. Since $C_{1,M} = C_{2,M} = 0$, the lower-counterterm contribution is zero in the minimal system. Every nonzero order-three cochain in this system therefore comes from a genuine order-three graph row, an order-three CE-source row, or a bracket row between already given nonzero lower-order components. The displayed formulas are precisely the row values and scalar conditions required by Definition 8.24. The final assertions are the kernel obstruction criterion of Definition E.33 and the finite-window compatibility criterion of Theorem 8.36. □

PROPOSITION E.49 *(Marked-row test for unreduced centrality homotopies).* Work in a finite-window marked Costello list for the local mixed HT brane-defect theory, with labels and source faces as in Definition E.27. Let $\kappa^M_{\mathcal{G},w,I}$ be the Hom-valued finite-window kernel carried by this list. For source generators $x, y \in \{u_{a(t)dt\otimes f}, c_{B,\rho}\}$, the marked row testing unreduced centrality is

$$R^{\mathrm{cent}}_{x,y,M} = \mathrm{Curv}_{\mathbf{K},M}(\kappa^M_{\mathcal{G},w,I})(x \wedge y).$$

In row notation this is the finite sum

$$\begin{aligned} R^{\mathrm{cent}}_{x,y,M} = &\sum_{\Gamma\in \mathsf{A}^d_{x,y,M}} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{\Gamma\in \mathsf{A}^{\mathrm{CE}}_{x,y,M}} \varepsilon^{\mathrm{CE}}_\Gamma K^M_\Gamma(\zeta_{\Gamma,x,y}) \\ &+ \frac{1}{2}\sum_{(\Gamma_1,\Gamma_2)\in \mathsf{A}^{\mathrm{br}}_{x,y,M}} \varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M} + R^{\mathrm{ct}}_{x,y,M}. \end{aligned}$$

Here the four summands are respectively the target differential rows, the CE-source rows, the convolution bracket rows, and the given counterterm rows of the marked table. The scalar condition is

$$S^{\mathrm{cent}}_{x,y,M} = \widehat{\sigma}_{\mathrm{sc},M}(R^{\mathrm{cent}}_{x,y,M}).$$

Only after this condition is zero does the centrality row define a class

$$\mathfrak{o}^{\mathrm{cent}}_{x,y,M} = [R^{\mathrm{cent}}_{x,y,M}] \in H^{|x|+|y|}(\mathcal{K}^{\bullet}_{\mathrm{ns},M}(I), d_{\mathrm{ns},M}).$$

A marked finite-window system gives an unreduced centrality homotopy for the pair (x, y) exactly when it also gives a degree $|x| + |y| - 1$ row $H^M_{x,y} \in \mathcal{K}^{|x|+|y|-1}_{\mathrm{ns},M}(I)$ with

$$R^{\mathrm{cent}}_{x,y,M} + d_M H^M_{x,y} = 0,$$

and these rows satisfy the truncation and weight identities of Theorem B.22. The centrality-homotopy row and its compatibility identities are additional data beyond scalar-shadow vanishing.

In the minimal full-equivariant branch of Theorem E.47, every positive-order row contains the zero scalar-contact factor $K^M_{\alpha_{\mathrm{sc}}} = 0$, and the marked table contains zero additional positive-order seed graph, CE-source face, graph counterterm face, or nonzero scalar-zero row. Hence

$$R^{\mathrm{cent}}_{x,y,M} = 0, \qquad H^M_{x,y} = 0$$

is a valid homotopy inside that minimal finite marked complex. This constructs the homotopy only for the minimal system. An enlarged unreduced Costello graph/counterterm target must contain the displayed row values, scalar condition, primitive $H^M_{x,y}$, and transition coefficients.

Proof. Proposition E.30 identifies the marked differential with the finite-window QME differential and, for Hom-valued kernels, adds the source face $-\kappa d_{\mathrm{CE}}$. Therefore the binary curvature on $x \wedge y$ is exactly the sum of the four row types displayed above: target differential, CE-source subtraction, convolution bracket, and counterterm contribution. Applying the scalar projection gives the scalar condition. If the condition vanishes, the row lies in the non-scalar kernel complex of Definition E.33; its cohomology class is the obstruction to finding a fixed-window primitive.

The equation $R^{\mathrm{cent}}_{x,y,M} + d_M H^M_{x,y} = 0$ is precisely the homotopy equation (40). The transition identities are necessary because the local kernel is the inverse limit over finite windows and admissible weights. Thus a row omitted from the marked table is additional graph, source-face, counterterm, or homotopy data beyond the scalar projection.

In the minimal full-equivariant system all positive-order scalar-contact rows are zero before cohomology by Theorem E.47. Since the minimal hypothesis allows zero other positive-order rows, the centrality row is zero and the zero primitive is compatible with all truncation maps. The final assertion is the contrapositive of the row formula: once an enlarged system adds a nonzero graph, source, bracket, or counterterm row, the corresponding scalar condition and primitive must be given explicitly. □

Proposition E.50 *(Q_Ω-centrality homotopy criterion).* Let $(I, w, M, \mathcal{G}_M)$ carry a normal Ω-equivariant Costello kernel datum in the sense of Definition E.41, and assume the scalar projection has passed the Hom-obstruction test of Theorem E.42. For T_Ω-invariant source generators

$$x, y \in \{u_{a(t)dt\otimes f}, c_{B,\rho}\},$$

define the equivariant centrality row

$$R^{\mathrm{cent}}_{x,y,\Omega,M} = \mathrm{Curv}_{\mathbf{K},\Omega,M}(\kappa^M_{\mathcal{G},\Omega,w,I})(x \wedge y).$$

Its finite-row expansion is

$$\begin{aligned} R^{\mathrm{cent}}_{x,y,\Omega,M} = & \sum_{\Gamma\in \mathsf{A}^d_{x,y,\Omega,M}} \varepsilon_\Gamma\, d_{M,\Omega} K^M_{\Gamma,\Omega} - \sum_{\Gamma\in \mathsf{A}^{\mathrm{CE}}_{x,y,\Omega,M}} \varepsilon^{\mathrm{CE}}_\Gamma K^M_{\Gamma,\Omega}(\zeta_{\Gamma,x,y,\Omega}) \\ & + \frac{1}{2} \sum_{(\Gamma_1,\Gamma_2)\in \mathsf{A}^{\mathrm{br}}_{x,y,\Omega,M}} \varepsilon_{\Gamma_1,\Gamma_2} \{K^M_{\Gamma_1,\Omega}, K^M_{\Gamma_2,\Omega}\}_{\hbar_{\mathrm{QME}},M,\Omega} + R^{\mathrm{ct}}_{x,y,\Omega,M}. \end{aligned}$$

The scalar condition is

$$S^{\mathrm{cent}}_{x,y,\Omega,M} = \widehat{\sigma}_{\mathrm{sc},\Omega,M}(R^{\mathrm{cent}}_{x,y,\Omega,M}).$$

Only after this condition vanishes does one obtain the obstruction class

$$\mathfrak{o}^{\mathrm{cent}}_{x,y,\Omega,M} = [R^{\mathrm{cent}}_{x,y,\Omega,M}] \in H^{|x|+|y|}(\mathcal{K}^\bullet_{\mathrm{ns},\Omega,M}(I), d_{\mathrm{ns},\Omega,M}).$$

A Q_Ω-centrality homotopy for (x, y) is exactly a T_Ω-invariant row

$$H^{\Omega}_{x,y,M} \in \mathcal{K}^{|x|+|y|-1}_{\mathrm{ns},\Omega,M}(I)$$

satisfying

$$R^{\mathrm{cent}}_{x,y,\Omega,M} + d_{M,\Omega} H^{\Omega}_{x,y,M} = 0, \qquad L_{V_\Omega} H^{\Omega}_{x,y,M} = 0,$$

together with the same interval, finite-window, and admissible-weight transition identities as in Theorem B.22. Equivalently, the inverse-system class

$$\left(\mathfrak{o}^{\mathrm{cent}}_{x,y,\Omega,M}\right)_M \in \varprojlim_M H^{|x|+|y|}(\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}(I))$$

and the corresponding $\varprojlim^1 H^{|x|+|y|-1}$-primitive compatibility class must both vanish.

If one works before passing to T_Ω-invariants, then $Q_\Omega^2 = L_{V_\Omega}$ gives a secondary obstruction

$$\ell^{\mathrm{cent}}_{x,y,\Omega,M} = [L_{V_\Omega} H_{x,y,\Omega,M}] \in H^{|x|+|y|}(\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}(I), d_{\mathrm{ns},\Omega,M})$$

for any chosen primitive. This class must vanish, or the primitive must be replaced by a basic one, before the homotopy is a Q_Ω-homotopy.

Proof. The row expansion is the Hom-valued curvature computation of Proposition E.49 with d_M, the CE differential, the bracket, and the counterterm table replaced by their equivariant counterparts from Definition E.41. The scalar condition is necessary because only its vanishing places the row in $\mathcal{K}^{\bullet}_{\mathrm{ns},\Omega,M}$. Once there, the equivariant Bianchi identity makes the row closed, so its cohomology class is the obstruction to a fixed-window primitive.

The displayed equation is precisely the centrality homotopy equation in the invariant Q_Ω-complex. Since $Q_\Omega^2 = L_{V_\Omega}$, T_Ω-invariance of the primitive is an independent condition beyond scalar-shadow vanishing. The inverse-limit assertion is the same primitive-compatibility argument as in Theorem B.22. If a primitive is chosen outside the basic subcomplex, applying $d_{M,\Omega}$ a second time records $L_{V_\Omega} H$; its cohomology class is exactly the secondary obstruction displayed above. □

PROPOSITION E.51 *(Even-block marker obstruction).* Replace full $\mathfrak{gl}(V)$-equivariance by the smaller even block-diagonal subalgebra

$$\mathfrak{gl}(V_{\bar 0}) \oplus \mathfrak{gl}(V_{\bar 1}) \subset \mathfrak{gl}(V), \qquad V_{\bar 0} \cong V_{\bar 1} \cong \mathbb{C}^N.$$

Then a single fixed even marker is admissible exactly when

$$T = \lambda_0 P_{\bar 0} + \lambda_1 P_{\bar 1},$$

where $P_{\bar 0}$ and $P_{\bar 1}$ are the parity block projections. Its scalar shadow is

$$\mathrm{Str}(T) = N(\lambda_0 - \lambda_1).$$

For a two-Hamiltonian scalar contact the obstruction class is

$$\hbar N(\lambda_0 - \lambda_1)\, \omega(f, g)\gamma_1, \qquad \hbar N(\lambda_0 - \lambda_1)[\bar c]\, \gamma_1 \text{ in cohomology.}$$

Since $[\bar c] \neq 0$ in the scalar-reduced source, this smaller subalgebra has zero $[\bar c]$-shadow for single markers if and only if $\lambda_0 = \lambda_1$, i.e. the marker is scalar and the subalgebra has collapsed back to the full-equivariant single-marker case. For higher marker tensors the strongest true positive statement is exactly the supertraceless-monodromy submodule of Theorem E.25; outside it the obstruction is the displayed $\mathrm{Str}_{\mathrm{cyc}}$-weighted $[\bar c]$-class.

Proof. The commutant of $\mathfrak{gl}(V_{\bar{0}}) \oplus \mathfrak{gl}(V_{\bar{1}})$ on $V_{\bar{0}} \oplus V_{\bar{1}}$ is $\mathbb{C}P_{\bar{0}} \oplus \mathbb{C}P_{\bar{1}}$, so the displayed form of T is necessary and sufficient. The supertrace is

$$\mathrm{Str}(\lambda_0 P_{\bar{0}} + \lambda_1 P_{\bar{1}}) = \lambda_0 \dim V_{\bar{0}} - \lambda_1 \dim V_{\bar{1}} = N(\lambda_0 - \lambda_1).$$

Proposition E.23 then gives the scalar CE representative and the cohomology class by multiplying ω, respectively $[\bar{c}]$, by this coefficient. The class $[\bar{c}]$ is nonzero by Theorem E.12; hence the class vanishes for all two-Hamiltonian tests exactly when $\lambda_0 = \lambda_1$. The higher tensor assertion is the same calculation with $\mathrm{Str}(T)$ replaced by $\mathrm{Str}_{\mathrm{cyc}}$ and with the preservation criterion given by Theorem E.25. □

THEOREM E.52 *(Ordinary scalar-reduced $\mathfrak{gl}_N$ obstruction criterion).* Work in the scalar-reduced source of Definition E.3 and keep the ordinary trace on the open $\mathfrak{gl}_N$ brane. Let W_{add} be any additional formal-local closed-exchange or brane-defect graph contribution whose leading scalar associated graded is $w_{\mathrm{add}} \in C^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})$. A cancellation

$$\mathrm{Ob}_{\mathrm{sc}} + W_{\mathrm{add}} + d_{\mathrm{QME}} C = 0$$

with $C \in \mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}$ can hold only if

$$[w_{\mathrm{add}}] = -\hbar N[\bar{c}] \qquad \text{in } H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]].$$

In particular, with $W_{\mathrm{add}} = 0$, or with w_{add} exact, the ordinary scalar-reduced $\mathfrak{gl}_N$ brane-defect model carries the scalar QME obstruction $\hbar N[\bar{c}]$.

Proof. Taking associated graded with respect to the scalar-contact filtration sends the equation

$$\mathrm{Ob}_{\mathrm{sc}} + W_{\mathrm{add}} + d_{\mathrm{QME}} C = 0$$

to

$$\hbar N\omega + w_{\mathrm{add}} + \partial\bar{\eta} = 0 \qquad \text{in } C^2_{\mathrm{Lie}}(\bar{A};\mathbb{C}),$$

where $\bar{\eta}$ is the linear functional represented by the leading scalar part of C. Passing to cohomology gives the stated necessary condition. If $W_{\mathrm{add}} = 0$, or if w_{add} is exact, this condition forces $\hbar N[\bar{c}] = 0$ in $H^2_{\mathrm{Lie}}(\bar{A};\mathbb{C})[[\hbar]]$. For $N > 0$ and formal $\hbar$, Theorem E.12 identifies $[\bar{c}] \neq 0$ on $\bar{A}$. Thus the ordinary scalar-reduced obstruction survives every internal local counterterm. □

THEOREM E.53 *(Unreduced cotangent lift obstruction criterion).* Fix a finite-window marked Costello system $(I, w, M, \mathcal{G}_M, \mathcal{G}^{\mathrm{mk}}_M)$ for the local mixed HT brane-defect theory and a filtered chain scalar projection $\widehat{\sigma}_{\mathrm{sc},M}$ (Definition E.34). Let $\Phi^{\mathrm{unred}}_{\mathrm{fact}}\colon C^\bullet_{\mathrm{CE}}(\widetilde{\mathfrak{g}}_I) \to Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}_{\mathrm{open}})$ be a factorization-center morphism extending the algebraic current presentation Φ^χ_I of Construction E.9 beyond the algebraic current target. Define the obstruction vector

$$\mathrm{Ob}_{\mathrm{cent/QME/descent}} = (\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}, \mathrm{ob}^{P_0}_{\mathrm{cent}}, \mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}, \partial^{\mathrm{Weiss}}_{\check{C}/R}),$$

with components defined as follows.

(i) *Product/tensor-completion centrality.* $\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}$ is the $H^{|x|+|y|}$-class of the product centrality defect $\Phi^{\mathrm{unred}}_{\mathrm{fact}}(x)\Phi^{\mathrm{unred}}_{\mathrm{fact}}(y) - (-1)^{|x||y|}\Phi^{\mathrm{unred}}_{\mathrm{fact}}(y)\Phi^{\mathrm{unred}}_{\mathrm{fact}}(x)$ in $\mathfrak{Q}^\bullet_{\mathcal{G},M}(I)/\mathrm{im}(d_M)$. In the algebraic current model (Theorem E.10) this defect is zero on generators by graded commutativity; the obstruction measures the corresponding product-centrality defect on the enlarged unreduced target.

(ii) P_0 / *shifted-cotangent refinement.* $\mathrm{ob}^{P_0}_{\mathrm{cent}}$ is the $H^{|x|+|y|}$-class of the P_0-bracket defect $\{\Phi^{\mathrm{unred}}_{\mathrm{fact}}(x), \Phi^{\mathrm{unred}}_{\mathrm{fact}}(y)\} - \Phi^{\mathrm{unred}}_{\mathrm{fact}}(\partial_x y)$ in the same cohomology, evaluated as a finite row sum

$$R^{\mathrm{cent}}_{x,y,M} = \sum_{\Gamma \in \mathsf{A}^d_{x,y,M}} \varepsilon_\Gamma\, d_M K^M_\Gamma - \sum_{\Gamma \in \mathsf{A}^{\mathrm{CE}}_{x,y,M}} \varepsilon^{\mathrm{CE}}_\Gamma K^M_\Gamma(\zeta_{\Gamma,x,y}) + \tfrac{1}{2} \sum_{(\Gamma_1,\Gamma_2)} \varepsilon_{\Gamma_1,\Gamma_2}\{K^M_{\Gamma_1}, K^M_{\Gamma_2}\}_{\hbar,M} + R^{\mathrm{ct}}_{x,y,M}$$

of the row machinery (Proposition E.49). A centrality homotopy $H^M_{x,y}$ with $R^{\mathrm{cent}}_{x,y,M} + d_M H^M_{x,y} = 0$ is the construction-level witness; the obstruction is the cohomology class of the row before such a witness exists.

(iii) *Boundary QME obstruction.* $\mathrm{ob}^{\mathrm{bd}}_{\mathrm{QME}}$ is the compatible inverse-system class

$$(\mathfrak{o}^{\mathrm{ns}}_{n,M})_M = ([R^{\mathrm{ns}}_{n,M}])_M \in \varprojlim_M H^1(\mathcal{K}^\bullet_{\mathrm{ns},M}(I), d_{\mathrm{ns},M})$$

of Theorem E.35, together with the secondary $\varprojlim^1 H^0$ primitive-compatibility class $\lambda_{n,M}$. The first nontrivial finite-window value occurs at the θ_3 row (Theorem E.38).

(iv) *Čěch–Roos / Weiss-versus-Ran descent gap.* $\delta^{\mathrm{Weiss}}_{\check{C}/R}$ is the codimension-one face class of the descent comparison between the Weiss cover $\mathcal{U}^{\mathrm{Weiss}}$ and the Ran-space cover $\mathcal{U}^R$: the first descent obstruction is the Čěch–Roos difference $\delta^{\mathrm{Weiss}}_{\check{C}/R}$ in the finite-window $\varprojlim^1 H^0$ of the corresponding boundary kernel. Sufficiency requires all higher Čěch/Ran compatibility classes and transition homotopies.

Then vanishing of $\mathrm{Ob}_{\mathrm{cent/QME/descent}}$ is necessary for an unreduced $Z^{P_0}_{\mathrm{fact}}$-morphism. If the finite-window Hom-valued companion tables, centrality homotopies, QME counterterms, and Čech–Roos descent homotopies specified below are part of the datum, then it is sufficient:

$$\boxed{\Phi^{\mathrm{unred}}_{\mathrm{fact}} \text{ lifts to an unreduced } Z^{P_0}_{\mathrm{fact}}\text{-morphism} \quad \Longleftarrow \quad \mathrm{Ob}_{\mathrm{cent/QME/descent}} = 0 \text{ with these choices.}}$$

At the associated-graded level, the primitive indecomposable P_0-shadow of Theorem 5.133 is the first associated-graded target for the centrality filtration. A spectral sequence converging to $\mathrm{ob}^{P_0}_{\mathrm{cent}}$ is obtained only after the filtration is complete, exhaustive, and compatible with the Hom-valued companion differential. In that case its first differential is the primitive Hamiltonian shadow acting on the source one-ψ label, and its first nonzero piece is the obstruction class to the strict centrality of the primitive-shadow lift.

The four components are independent in the following sense. Vanishing of (i) is a graded commutativity test on the unreduced target and is automatic in the algebraic current model. Vanishing of (ii) is the centrality homotopy test of Proposition E.49. Vanishing of (iii) is the boundary QME content of Theorem E.35, whose first finite-window obstruction is θ_3 (Theorem E.38). Vanishing of (iv) is the descent comparison; in the algebraic current model this gap is zero by the same finite-list axiomatization. For genuine unreduced $Z^{P_0}_{\mathrm{fact}}$-morphisms the Weiss/Ran comparison requires its own descent datum.

Proof. Necessity in each component.

Component (i). Any factorization-center morphism into $Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}_{\mathrm{open}})$ sends every source generator to a graded center element by definition, hence the product centrality defect is a d_M-boundary on every generator pair. The cohomology class $\mathrm{ob}^{\mathrm{prod}}_{\mathrm{cent}}$ of such a lift is the obstruction to this strict graded centrality; its vanishing is necessary. In the algebraic current model with graded-commutative target it is automatic (Theorem E.10).

Component (ii). A factorization-center morphism into $Z^{P_0}_{\mathrm{fact}}$ intertwines the source CE/Schouten bracket with the target P_0-bracket. The deviation from this intertwining is precisely the row sum $R^{\mathrm{cent}}_{x,y,M}$ of Proposition E.49; its cohomology class is the obstruction to choosing a centrality homotopy. Vanishing is necessary because the homotopy is exactly the datum making the source CE differential commute with the target P_0-bracket on the image of $\Phi^{\mathrm{unred}}_{\mathrm{fact}}$.

Component (iii). A factorization-center morphism produces a curved Maurer–Cartan element of the brane-defect kernel complex. Its curvature class is exactly the residual $R^{\mathrm{ns}}_{n,M}$ of Theorem E.35, in the inverse-limit version recorded as $(\mathfrak{o}^{\mathrm{ns}}_{n,M})_M$. Vanishing, together with the $\varprojlim^1 H^0$ compatibility class, gives the order-n obstruction condition after lower orders are solved and the finite row system is complete. The first finite-window obstruction occurs at the θ_3 row by Theorem E.38.

Component (iv). The factorization-center construction depends globally on the chosen Weiss/Ran cover; a different cover may give a different value of the same morphism on overlapping factorizations. The codimension-one Čech difference between Weiss and Ran covers is a class in the appropriate boundary-kernel H^0; its vanishing is the descent compatibility for the factorization-center morphism. In the algebraic current model with the displayed finite-list axiomatization this difference is zero, but the genuine unreduced $Z^{P_0}_{\mathrm{fact}}$-target carries the comparison nontrivially in general.

Sufficiency under given companion data. Conversely, suppose all four components vanish and the companion tables, centrality homotopies, QME counterterms, and descent homotopies are part of the datum. Construct the lift by induction on $\hbar_{\mathrm{QME}}$-order and finite window M. At order zero the morphism is the algebraic current presentation $\Phi^{\mathcal{X}}_I$ of Construction E.9, which already gives a strict (zero-homotopy) factorization-center morphism into the algebraic current target (Theorem E.10); hence (i) vanishes at order zero. At each order $n \geq 1$ and each window M:

- vanishing of (iii) gives a degree-zero counterterm $C_{n,M}$ with $d_{\mathrm{ns},M} C_{n,M} = -R^{\mathrm{ns}}_{n,M}$ (Theorem E.35); together with the $\varprojlim^1 H^0$-primitive-compatibility class, this assembles a compatible counterterm tower;
- vanishing of (ii) gives a centrality homotopy $H^M_{x,y}$ with $R^{\mathrm{cent}}_{x,y,M} + d_M H^M_{x,y} = 0$ for every source generator pair (Proposition E.49);
- vanishing of (iv) makes the resulting morphism agree on Weiss and Ran cover overlaps, by the codimension-one Čech–Roos comparison;
- vanishing of (i) at every window inductively proves strict graded centrality on the constructed lift.

Under these given finite-window data, the four vanishings are the rows of the inductive extension problem for the algebraic current presentation to a strict $Z^{P_0}_{\mathrm{fact}}$-morphism: each is a separate row in the curved Maurer–Cartan equation of Theorem E.35 and its Hom-valued kernel companion, and any one nonzero component blocks the inductive step at the corresponding row. Hence the obstructions are necessary, and they are sufficient for the $Z^{P_0}_{\mathrm{fact}}$-lift only with the stated companion and descent data.

The associated-graded statement. Take the primitive shadow filtration of Theorem 5.133. At the associated-graded level, the source differential is the CE differential and the target differential is the Hamiltonian/Schouten bracket on primitive trace classes. The If the primitive filtration is complete, exhaustive, and compatible with the Hom-valued companion differential, the E_1 page is this primitive-shadow target and its first differential is the Hamiltonian shadow on one-ψ primitive trace classes, as computed in Theorem 5.120. Without these filtration hypotheses the statement is only the associated-graded primitive obstruction. □

Remark E.54 *(Locus of each component of* $\mathrm{Ob}_{\mathrm{cent/QME/descent}}$*).* The component $\mathrm{ob}_{\mathrm{cent}}^{\mathrm{prod}}$ is nonzero outside the graded-commutative algebraic current target of Definition E.8, on a noncommutative $\hbar$-deformed target where the product centrality defect carries a Capelli-type contribution. The polynomial-obstruction boundary (Theorem E.6) exhibits the same obstruction in the polynomial open BV/Koszul category: polynomial open BV observables have locally nilpotent translation action, while the principal-part module requires non-locally-nilpotent action for strict product centrality.

The component $\mathrm{ob}_{\mathrm{cent}}^{P_0}$ is detected at the smallest window where the brane-defect bracket of two source generators carries a nonzero scalar-zero contribution whose primitive requires a compatible centrality homotopy $H_{x,y}^{M}$ (Proposition E.49). In the minimal full-equivariant branch the row vanishes and the zero homotopy suffices; the finite-window matrix calculation gives the same answer in the centrality-homotopy complex.

The component $\mathrm{ob}_{\mathrm{QME}}^{\mathrm{bd}}$ is nonzero at θ_3 in case (a) of Theorem E.38, where the counterterm is omitted; case (b) of the same theorem gives $\mathrm{ob}_{\mathrm{QME}}^{\mathrm{bd}} = 0$ at θ_3 under the given counterterm $C_{\theta_3,M}$.

The component $\partial_{\check{C}/R}^{\mathrm{Weiss}}$ is nonzero when the codimension-one face of the Weiss-versus-Ran cover comparison carries a non-vanishing boundary-kernel class; this is a separate descent comparison in the factorization-current convention complex.

F RADIAL PARTS AND THE MOYAL NORMALIZATION

Formal Weyl quantization of the constant symplectic plane and stable scalar-reduced trace map on the Chan–Paton brane stack disagree at each bidegree (a, b) by a single class. The disagreement lives in a finite-dimensional bidegree-local complex whose source records cyclic-word boundary letters and whose target combines cyclic-word incidence with the residual normal-diagram corrections. Concretely, $\mathsf{W}_{a,b}$ is the rational span of words containing the moment-map letter $[x, y]$ once; $\mathsf{V}_{a,b}$ is the cyclic-word target of the necklace boundary; $\mathsf{Diag}_{a,b}$ is the free indexed normal-diagram space carrying contraction-degree-≥ 1 (PBW) corrections; and $B_{a,b}$ is the combined coboundary $(\partial, C_{a,b}^{+})$ of Definition F.21. The discrepancy is then

$$\Omega_{a,b}^{\mathrm{rad}} \in \operatorname{coker} B_{a,b}, \qquad B_{a,b} \colon \mathsf{W}_{a,b} \longrightarrow \mathsf{V}_{a,b} \oplus \mathsf{Diag}_{a,b},$$

in the bidegree-local radial bicomplex $(\mathsf{W}_{a,b} \xrightarrow{\partial} \mathsf{V}_{a,b},\ \mathsf{W}_{a,b} \xrightarrow{C_{a,b}^{+}} \mathsf{Diag}_{a,b})$ of Definition F.21. This class vanishes in every bidegree: Theorem F.28 gives $\Omega_{a,b}^{\mathrm{rad}} = 0$ under the radial-parts Statement F.8, and Proposition F.31 gives verified vanishing on the total-degree-15-and-below window by stable Procesi–Razmyslov reduction.

The cyclic-word incidence complex $\partial : \mathsf{W}_{a,b} \to \mathsf{V}_{a,b}$ is the $\hbar^0$ skeleton of the radial bicomplex; the classical lift $\partial c = T_{a,b}$ is the ordinary necklace Stokes identity of Lemma F.24 ($T_{a,b}$ the necklace target of a moment-map insertion). The free indexed normal-diagram target $\mathsf{Diag}_{a,b}$ records the residual contraction-degree-≥ 1 (PBW) corrections and the Capelli $\hbar N$ term; the residual class $R_{a,b}^{\mathrm{free}} \in \mathsf{Diag}_{a,b}$ (the PBW residual of the Weyl/cyclic-trace comparison) lies entirely in this positive part. Its vanishing is proved by testing on the Cartan (Lemmas F.34 and F.11) against the Levasseur–Stafford radial-kernel hypothesis. A signed-row obstruction $(\phi, -\lambda) \in \ker B_{a,b}^{*}$ (a covector pairing a cyclic-word functional $\phi \in \mathsf{V}_{a,b}^{*}$ with a diagonal-target functional $\lambda \in \mathsf{Diag}_{a,b}^{*}$) of the $\lambda(E_{a,b}^{+}) - \phi(T_{a,b}) \neq 0$ type is excluded under that hypothesis.

The formal Weyl/Moyal coefficients on the constant symplectic plane and the Capelli triangular trace normalization on the matrix Weyl algebra reduce the comparison to a finite-dimensional linear-algebra problem in the free indexed normal-diagram algebra; finite-rank Procesi–Razmyslov stable injectivity extends free identities to all-N identities. The Vandermonde-conjugated Harish-Chandra radial-parts datum [8][9][10][11] is named in Statement F.8; the structural all-bidegree theorem uses clauses (1) and (2) (Levasseur–Stafford radial-kernel) and clause (3) only for the specific Moyal-bracket polynomial of bidegree (a, b). Pure-power clause (3) is tautological because Weyl-symmetric and normal-ordered traces of pure powers coincide. The reduced open-line Wick graph realization with midpoint propagator $\frac{1}{2}\operatorname{sgn}(t-s)$ realizes the same Moyal product to all orders and is the bridge to the open-line BV sector.

The edge strips $\mathfrak{o}_{2,b} = \mathfrak{o}_{a,2} = 0$ close in closed form (Proposition F.32). The Procesi–Razmyslov reduction (Proposition F.31) closes the listed interior bidegrees by finite rational linear algebra; the explicit list of bidegrees verified in this mode reaches through total degree 15 and is given in Remark F.42. An unlisted bidegree remains governed by the obstruction class $\Omega^{\mathrm{rad}}_{a,b} \in \operatorname{coker} B_{a,b}$, or is depends on Statement F.8. Reflection symmetry $(a,b) \leftrightarrow (b,a)$ is implemented by the symplectic involution $z_1 \leftrightarrow z_2$, $X \leftrightarrow Y$; both sides of the radial bicomplex are equivariant under it.

Definition F.1 *(Formal Weyl/Moyal convention).* Let

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

The formal Weyl product on $\mathbb{C}[[z_1, z_2]][[\hbar]]$ is

$$f * g = m \circ \exp\left(\frac{\hbar}{2} P\right)(f \otimes g),$$

and

$$[f, g]_{\mathrm{raw}} = f * g - g * f, \qquad \{f, g\}_\hbar = \hbar^{-1}[f, g]_{\mathrm{raw}}.$$

The reduced quantum Hamiltonian Lie algebra is the scalar quotient

$$\bar{A}_\hbar = \mathbb{C}[[z_1, z_2]][[\hbar]]/\mathbb{C}[[\hbar]]$$

as a Lie algebra. The construction uses this quotient only in the Lie-algebra sense.

THEOREM F.2 *(Exact formal Weyl/Moyal coefficients).* For $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$,

$$P^r(f, g) = \sum_{a=0}^{r} (-1)^a \binom{r}{a} (k)_{r-a} (l)_a (m)_a (n)_{r-a} z_1^{k+m-r} z_2^{l+n-r}.$$

Consequently

$$[f, g]_{\mathrm{raw}} = \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{2}{2^r r!} \hbar^r P^r(f, g),$$

equivalently, for $s \geq 0$, the raw coefficient of $\hbar^{2s+1}$ is

$$\frac{1}{2^{2s}(2s+1)!} P^{2s+1}(f, g),$$

and the normalized Hamiltonian bracket has coefficient

$$\frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g).$$

The first and third raw commutator coefficients are

$$\hbar P(f, g), \qquad \frac{\hbar^3}{24} P^3(f, g),$$

and all even raw commutator coefficients vanish. This is a theorem inside the formal Weyl algebra of the constant symplectic plane, independent of the finite-N radial-parts hypothesis and Costello graph specialization.

Proof. The two summands in $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$ commute as differential operators on the tensor product. The binomial theorem gives

$$P^r = \sum_{a=0}^{r} (-1)^a \binom{r}{a} \partial_{z_1}^{r-a} \partial_{z_2}^{a} \otimes \partial_{z_1}^{a} \partial_{z_2}^{r-a}.$$

Applying this to the two monomials gives the falling-factorial formula. Interchanging f and g changes $P^r(f, g)$ by $(-1)^r$, so the commutator retains exactly the odd powers and contributes the coefficient $2/(2^r r!)$ in order r. □

COROLLARY F.3 *(All odd Moyal coefficients).* The Weyl/Moyal commutator has only odd raw powers of $\hbar$. Its odd raw and normalized coefficients are

$$[f, g]^{(2s+1)}_{\mathrm{raw}} = \frac{\hbar^{2s+1}}{2^{2s}(2s+1)!} P^{2s+1}(f, g), \qquad \{f, g\}^{(2s)}_{\hbar} = \frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g).$$

In particular the next two raw coefficients after the third-order term are

$$\frac{\hbar^5}{1920} P^5(f, g), \qquad \frac{\hbar^7}{322560} P^7(f, g).$$

Proof. Put $r = 2s + 1$ in Theorem F.2. Then $2/(2^{2s+1}(2s + 1)!) = 1/(2^{2s}(2s + 1)!)$. Dividing by $\hbar$ gives the normalized coefficient. □

COROLLARY F.4 *(First and third formal graph weights).* In the ordered formal product $m \circ \exp(\hbar P/2)$, the one-cross-contraction term has weight $\hbar P(f, g)/2$, and the three-cross-contraction term has weight $\hbar^3 P^3(f, g)/48$. After antisymmetrising the ordered product, the commutator weights are

$$\hbar P(f, g), \qquad \frac{\hbar^3}{24} P^3(f, g).$$

Proof. The ordered exponential contributes

$$\frac{1}{r!} \left(\frac{\hbar}{2}\right)^r P^r(f, g)$$

in order r. For $r = 1$ this is $\hbar P(f, g)/2$; for $r = 3$ it is $\hbar^3 P^3(f, g)/(3!\, 2^3) = \hbar^3 P^3(f, g)/48$. Since $P(g, f) = -P(f, g)$ and $P^3(g, f) = -P^3(f, g)$, subtracting the reverse ordered product doubles these two ordered contributions. □

Example F.5 *(A higher-order coefficient check).* For

$$f = z_1^4 z_2^2, \qquad g = z_1^2 z_2^4,$$

the first nonzero odd contractions are

$$P(f, g) = 12 z_1^5 z_2^5, \qquad P^3(f, g) = -960 z_1^3 z_2^3, \qquad P^5(f, g) = 11520 z_1 z_2.$$

Hence

$$[f, g]_{\mathrm{raw}} = 12\hbar z_1^5 z_2^5 - 40\hbar^3 z_1^3 z_2^3 + 6\hbar^5 z_1 z_2.$$

The third coefficient is $-960/24 = -40$. The fifth coefficient is $11520/1920 = 6$. This example checks the numerical normalization beyond the first nontrivial term, independently of radial parts.

Proof. Apply the falling-factorial formula of Theorem F.2. For $r = 1$,

$$4 \cdot 4 - 2 \cdot 2 = 12.$$

For $r = 3$,

$$\begin{aligned} P^3(f, g) &= \big(24 \cdot 24 - 3 \cdot 12 \cdot 2 \cdot 2 \cdot 12 + 3 \cdot 4 \cdot 2 \cdot 2 \cdot 4\big) z_1^3 z_2^3 \\ &= -960 z_1^3 z_2^3. \end{aligned}$$

For $r = 5$, only the $a = 1$ and $a = 2$ summands survive, and they give

$$\big(-5 \cdot 24 \cdot 2 \cdot 2 \cdot 24 + 10 \cdot 24 \cdot 2 \cdot 2 \cdot 24\big) z_1 z_2 = 11520 z_1 z_2.$$

Substituting the odd Moyal coefficients 1, 1/24, and 1/1920 gives the displayed raw commutator. □

Definition F.6 *(Matrix Weyl algebra and Weyl traces).* Let $\mathcal{W}_N$ be the completed Weyl algebra generated by entries of two $N \times N$ matrices X, Y with

$$[X^i{}_j, Y^k{}_l] = \hbar\, \delta^i_l \delta^k_j, \qquad [X^i{}_j, X^k{}_l] = [Y^i{}_j, Y^k{}_l] = 0.$$

The normal-ordered quantum comoment for conjugation is

$$\widehat{\mu}_\epsilon = -\operatorname{Tr}([\epsilon, X]Y), \qquad \hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a.$$

For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, define the Weyl-ordered trace

$$J_N(f) = \operatorname{Tr}\operatorname{Op}_{\mathrm{W}}(f)(X, Y).$$

The degree-zero quantum Hamiltonian reduction uses the effective traceless comoment ideal

$$\mathrm{N}_{\mathcal{W}_N}(\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)) / \mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N).$$

The scalar Hamiltonian line survives this moment ideal; below it is removed by quotienting the empty trace before connected large-N extraction.

Lemma F.7 *(Capelli–Weyl triangular normalization).* Let $T_{a,b} = \operatorname{Tr}(X^a Y^b)$ be the normal-ordered trace. In the quantum Hamiltonian reduction, the locally proved Weyl-normal-ordering formula is

$$J_N(z_1^a z_2^b) \equiv \sum_{r=0}^{\min(a,b)} \left(-\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} T_{a-r,b-r}.$$

The inverse triangular system is

$$T_{a,b} \equiv \sum_{r=0}^{\min(a,b)} \left(\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} J_N(z_1^{a-r} z_2^{b-r}).$$

Proof. Total Weyl symmetrisation is obtained from normal ordering by commuting Y's past X's. Each contraction contributes the Capelli contact term coming from

$$YX - XY + \hbar N I \equiv 0 \pmod{\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)}.$$

Choosing r contracted X-slots and r contracted Y-slots gives $\binom{a}{r}\binom{b}{r}$, pairing them gives $r!$, and Weyl ordering contributes 2^{-r}. The sign is negative because the displayed moment relation rewrites a normal-ordering swap as the lower triangular term $-\hbar N$. The diagonal term is $r = 0$, hence the system is unipotent over $\mathbb{C}[[\hbar N]]$; inversion changes $-\hbar N/2$ to $+\hbar N/2$.

The cited Capelli literature gives the standard source for the Capelli contact phenomenon, and the displayed contraction count fixes the bivariate trace normalization used here. □

Statement F.8 *(Radial-parts hypotheses).* The finite-N radial-parts step uses the following Harish-Chandra radial-parts theorem for invariant differential operators, in the form refined by Wallach and by Levasseur–Stafford [8][9][10][11]. After identifying $\mathcal{W}_N \simeq \mathcal{D}_\hbar(\mathfrak{gl}_N)$, this theorem uses the Harish-Chandra map on invariants whose kernel is the quantum moment-map ideal and hence injects the quantum Hamiltonian reduction by conjugation into Weyl-invariant differential operators on the regular Cartan. With $\lambda_1, \dots, \lambda_N$ the Cartan coordinates and $\Delta = \prod_{i<j}(\lambda_i - \lambda_j)$, the radial-part map is conjugated by Δ.

The cited radial-parts literature sources the Harish-Chandra homomorphism, regular-Cartan landing, and kernel statement. The exact Weyl-trace image in item (3) is the matched normalization datum used in the formal-local comparison. The cited radial-parts sources give the Harish-Chandra kernel theorem; the exact Weyl/Capelli convention enters here as an additional normalization. The finite exact checks below test it in ranks 2 and 3; the all-N radial-parts theorem remains the structural hypothesis.

The theorem is used through the following three assertions.

(i) The Vandermonde-conjugated radial map is a homomorphism from the invariant quantum Hamiltonian reduction by conjugation to Weyl-invariant differential operators on the regular Cartan.
(ii) Its kernel is precisely the quantum moment-map ideal; hence it is injective on the reduced quotient.
(iii) The Weyl trace generators used here have image

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1}\left(\sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i})\right)\Delta.$$

The nonlinear-shear test in Lemma F.13 shows that pure z_1- and z_2-power images together with classical symbol-level shear compatibility determine only the pure-power datum. Item (3) requires a genuine quantum-corrected Egorov theorem in the Weyl algebra.

Definition F.9 *(Radially normalized Weyl trace class).* Assume only the homomorphism and injectivity clauses of Statement F.8. For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, a class $J_N^{\mathrm{rad}}(f)$ in the degree-zero quantum Hamiltonian reduction is called radially normalized if

$$\mathrm{Rad}_0(J_N^{\mathrm{rad}}(f)) = \Delta^{-1} S_N(f) \Delta.$$

Existence is a normalization hypothesis; uniqueness follows from radial injectivity. Statement F.8(3) is exactly the assertion that the ordinary Weyl trace $J_N(f)$ equals $J_N^{\mathrm{rad}}(f)$.

Definition F.10 *(Vandermonde gauge variables).* Set

$$\nabla_i^\Delta = \Delta^{-1}(-\hbar\partial_{\lambda_i})\Delta = -\hbar\partial_{\lambda_i} - \hbar\sum_{j\neq i}\frac{1}{\lambda_i-\lambda_j}.$$

The logarithmic summand is the type A Dunkl–Calogero correction in the Vandermonde gauge. For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, put

$$S_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i})$$

and

$$R_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, \nabla_i^\Delta) = \Delta^{-1}S_N(f)\Delta.$$

LEMMA F.11 *(Dunkl–Calogero correction cancellation).* The Vandermonde-gauge variables satisfy

$$[\lambda_i, \lambda_j] = 0, \qquad [\nabla_i^\Delta, \nabla_j^\Delta] = 0, \qquad [\lambda_i, \nabla_j^\Delta] = \hbar\,\delta_{ij}.$$

Consequently

$$\hbar^{-1}[R_N(f), R_N(g)] = R_N(\{f, g\}_\hbar)$$

for all f, g. In particular, all logarithmic Dunkl–Calogero terms introduced by expanding $\Delta^{-1}S_N(-)\Delta$ cancel in the commutator identity.

Proof. The operator ∇_i^Δ is conjugate to $-\hbar\partial_{\lambda_i}$ by multiplication by Δ. The three displayed commutation relations are therefore the ordinary Weyl relations for $(\lambda_i, -\hbar\partial_{\lambda_i})$, conjugated by Δ. The correction terms are a pure gauge connection: their curvature is zero because the unconjugated partial derivatives commute. Hence the i-th summand is again a one-particle Weyl representation of the formal Moyal algebra, and different summands commute. This gives the displayed commutator identity. The Vandermonde-conjugated form contains the Calogero potential. □

LEMMA F.12 *(Radial-map injectivity on the reduced quotient).* Under Statement F.8, if an invariant Weyl operator D in the normalizer of the quantum moment-map ideal satisfies

$$\mathrm{Rad}_0(D) = 0,$$

then D is zero in the degree-zero quantum Hamiltonian reduction.

Proof. Statement F.8 identifies the kernel of the Vandermonde-conjugated Harish-Chandra radial map with the quantum moment-map ideal. Passing to the normalizer quotient by that ideal gives an injective map into Weyl-invariant differential operators on the regular Cartan. Thus an element with zero radial image represents the zero class in the reduced quotient. □

LEMMA F.13 *(Nonlinear shear obstruction to weaker radial-parts hypotheses).* Pure-power radial images plus classical symbol-level compatibility with nonlinear shears determine only the pure-power Weyl-trace image normalization. In the one-particle Weyl algebra $[\lambda, y] = \hbar$, the coefficient of t in

$$(y + t\lambda^2)^4$$

is

$$4\lambda^2y^3 - 12\hbar\lambda y^2 + 8\hbar^2 y,$$

whereas

$$4\,\mathrm{Op}_{\mathrm{W}}(z_1^2 z_2^3) = 4\lambda^2 y^3 - 12\hbar\lambda y^2 + 6\hbar^2 y.$$

The difference $2\hbar^2 y$ is non-scalar. Thus Statement F.8(3) requires the quantum Egorov correction in addition to classical nonlinear shear covariance.

Proof. Expand to first order in t:

$$(y + t\lambda^2)^4 = y^4 + t\sum_{j=0}^{3} y^j \lambda^2 y^{3-j} + O(t^2).$$

Using $y\lambda = \lambda y - \hbar$, the sum normal-orders to $4\lambda^2 y^3 - 12\hbar\lambda y^2 + 8\hbar^2 y$. Weyl symmetrisation of $z_1^2 z_2^3$ gives $\lambda^2 y^3 - 3\hbar\lambda y^2 + \frac{3}{2}\hbar^2 y$, hence the displayed value after multiplication by 4. The discrepancy is a nonconstant one-particle operator, hence survives the scalar Hamiltonian quotient. □

PROPOSITION F.14 *(Quantum shear congruence suffices for the all-N image).* Assume the homomorphism and injectivity clauses of Statement F.8, and assume the pure power images

$$\mathrm{Rad}_0(J_N(z_1^r)) = \Delta^{-1} S_N(z_1^r)\Delta, \qquad \mathrm{Rad}_0(J_N(z_2^r)) = \Delta^{-1} S_N(z_2^r)\Delta$$

for all $r \geq 0$. Suppose moreover that, for all $a, b \geq 0$, the quantum shear congruence

$$\hbar^{-1}\big[J_N(z_1^{a+1}/(a+1)), J_N(z_2^{b+1})\big] \equiv J_N\Big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\Big) \pmod{I_N}$$

holds in the degree-zero quantum Hamiltonian reduction, where I_N is the quantum moment-map ideal. Then the exact Weyl-trace image in Statement F.8(3) holds for every monomial $z_1^a z_2^b$, hence for every polynomial f.

Proof. The normalized Moyal bracket gives

$$\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar = (b+1) z_1^a z_2^b + \sum_{s\geq 1} \frac{\hbar^{2s}(a+1)_{2s+1}(b+1)_{2s+1}}{(a+1)2^{2s}(2s+1)!} z_1^{a-2s} z_2^{b-2s}.$$

Every summand after the leading term has lower total degree. The pure power cases start the induction. Assume the asserted radial image is known in all total degrees $< a+b$. Applying the radial homomorphism to the quantum shear congruence identifies the radial image of the left-hand commutator with the Moyal commutator of the already known pure-power radial images. Lemma F.34 computes this as the Vandermonde-conjugated Weyl quantization of the displayed Moyal bracket. By the induction hypothesis all lower-degree terms have the required images, so the leading coefficient $(b+1)$ determines the image of $J_N(z_1^a z_2^b)$. Radial injectivity then identifies the class in the reduced quotient. □

LEMMA F.15 *(Moment-ideal primitive form of the quantum shear hypothesis).* In the notation of Proposition F.14, the quantum shear congruence is equivalent to the following explicit left-ideal membership problem. Put

$$E_{a,b,N} = \sum_{j=0}^{b} \mathrm{Tr}(Y^j X^a Y^{b-j}) - J_N\Big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\Big).$$

Then the congruence holds if and only if $E_{a,b,N} \in \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$. Equivalently, after choosing normal-ordered matrix generators $\widehat{\mu}^i{}_j$ for the traceless quantum moment-map ideal, one must construct coefficients $A^j{}_i(a,b,N) \in \mathcal{W}_N$ such that

$$E_{a,b,N} = \sum_{i,j} A^j{}_i(a,b,N)\widehat{\mu}^i{}_j.$$

Proof. Since the pure powers are already Weyl ordered, $J_N(z_1^{a+1}/(a+1)) = \mathrm{Tr}(X^{a+1})/(a+1)$ and $J_N(z_2^{b+1}) = \mathrm{Tr}(Y^{b+1})$. The Weyl relations give

$$\hbar^{-1}\big[\mathrm{Tr}(X^{a+1})/(a+1), Y\big] = X^a.$$

Applying the Leibniz rule to Y^{b+1} gives

$$\hbar^{-1}\big[J_N(z_1^{a+1}/(a+1)), J_N(z_2^{b+1})\big] = \sum_{j=0}^{b} \mathrm{Tr}(Y^jX^aY^{b-j}).$$

Subtracting the J_N-image of the normalized Moyal bracket gives exactly $E_{a,b,N}$. Reduction modulo I_N is reduction modulo the left ideal $\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$, so the displayed congruence is precisely the asserted membership. The coefficient formula is the same membership written after a choice of generators for that left ideal. □

Statement F.16 *(Free trace-diagram quantum-lift obstruction).* Put $M = YX - XY + \hbar NI$, and interpret

$$\mathrm{Tr}(WM) = \sum_{i,j} W^j{}_iM^i{}_j$$

as a left-ideal expression. Let $\mathsf{C}_{a,b}$ be the rational vector space spanned by words W with $a-1$ letters X and $b-1$ letters Y, and let $\mathsf{Cyc}_{a,b}$ be the cyclic word space with a letters X and b letters Y. The leading term of such a primitive is the cyclic boundary equation

$$\partial c = \sum_{j=0}^{b}[Y^jX^aY^{b-j}] - \frac{b+1}{\binom{a+b}{a}} \sum_{U\in\mathrm{Sh}(a,b)} [U], \qquad \partial W = [WYX] - [WXY].$$

An all-N primitive is stronger. It requires equality after free indexed normal ordering, including every contraction term which splits a trace into multi-traces:

$$\mathcal{N}_{\mathrm{free}}\left(\sum_W c_W \,\mathrm{Tr}(WM)\right) = \mathcal{N}_{\mathrm{free}}(E_{a,b,N}).$$

The cyclic equation is the associated-graded part of this quantum equation.

In bidegree $(a,b) = (4,4)$, the lexicographic pivot solution of the cyclic equation is

$$\begin{aligned} C^{\mathrm{cl}}_{4,4} = {} & \frac{31}{7}\,\mathrm{Tr}(XXXYYY\,M) - \frac{4}{7}\,\mathrm{Tr}(XXYXYY\,M) - \frac{4}{7}\,\mathrm{Tr}(XXYYXY\,M) \\ & + \frac{27}{7}\,\mathrm{Tr}(XXYYYX\,M) + \frac{23}{7}\,\mathrm{Tr}(XYXXYY\,M) + \frac{1}{7}\,\mathrm{Tr}(XYXYXY\,M) \\ & + \mathrm{Tr}(XYXYYX\,M) - \frac{2}{7}\,\mathrm{Tr}(XYYXXY\,M) + \frac{15}{7}\,\mathrm{Tr}(XYYXYX\,M). \end{aligned}$$

Its quantum residual is nonzero. In rank $N = 2$, exact normal ordering gives

$$\begin{aligned} R^{\mathrm{cl}}_{4,4,2} &:= E_{4,4,2} - C^{\mathrm{cl}}_{4,4} \\ &= \frac{43}{70}\hbar^2\big(4\,\mathrm{Tr}(XXYY) - 5\,\mathrm{Tr}(XYXY) + 2\,\mathrm{Tr}(XYYX) \\ &\qquad - 5\,\mathrm{Tr}(YXYX) + 4\,\mathrm{Tr}(YYXX)\big) \neq 0. \end{aligned}$$

After expanding the displayed traces into normal-ordered matrix entries, the coefficient of $\hbar^2 X^1{}_1 X^1{}_1 Y^1{}_2 Y^2{}_1$ is $43/7$.

The residual is killed by the classical-kernel correction

$$\begin{aligned} K_{4,4} = \frac{43}{14}\big(&\mathrm{Tr}(XXYXYY\,M) - \mathrm{Tr}(XYXXYY\,M) \\ &- \mathrm{Tr}(XYYXYX\,M) + \mathrm{Tr}(XYYYXX\,M)\big), \end{aligned}$$

whose cyclic boundary is zero. The free normal-form computation below proves that this correction cancels the $(4,4)$ residual before any finite-rank trace identity is imposed. The decorated PBW Stokes identity closes uniformly: by Theorem F.28, for every bidegree

$$\Omega^{\mathrm{rad}}_{a,b} = 0 \quad \Longleftrightarrow \quad D^{\square}_{a,b} H_{a,b} = R^{\mathrm{free}}_{a,b}, \qquad D^{\square}_{a,b} = C^{+}_{a,b}\partial_2,$$

under Statement F.8. Unconditionally, the bidegrees in Remark F.42 are closed by Procesi–Razmyslov verification (Proposition F.31); the normalized signed-dual-row obstruction set is empty there.

Proof. The leading $\hbar^0$ component of $\mathrm{Tr}(WM)$ is $[WYX] - [WXY]$ in the cyclic trace quotient; comparing it with the leading term of $E_{a,b,N}$ gives the displayed cyclic equation. Normal ordering in the indexed Weyl algebra contracts every occurrence of a Y-entry moved past an X-entry. These contractions identify matrix indices and may split one cyclic trace into a product of cyclic traces. Hence equality in the cyclic quotient is only the associated graded condition.

The $(4,4)$ coefficients displayed above solve the cyclic equation in the stated pivot gauge. Direct exact normal ordering in $\mathcal{W}_2$ gives the displayed residual. Its nonvanishing follows from the indicated normal monomial coefficient. Applying ∂ to the four words in $K_{4,4}$ gives two copies of each resulting cyclic word with opposite signs, so $K_{4,4} \in \ker \partial$. The finite computation proves the obstruction to the uncorrected cyclic lift. The all-N identity for $K_{4,4}$ is given by the free trace-diagram normal form below. □

Definition F.17 *(Free indexed normal diagrams).* Let $u = L_1 \cdots L_m$ be a word in X, Y. Write the trace $\mathrm{Tr}(u)$ as the directed m-gon with vertices $v_0, \ldots, v_{m-1}$ and edge $e_r : v_r \longrightarrow v_{r+1}$, labelled L_r, with indices read modulo m. A contraction matching Γ is a partial matching of pairs

$$p < q, \qquad L_p = Y, \qquad L_q = X.$$

For each pair $(p,q) \in \Gamma$, delete the two edges e_p, e_q and identify

$$v_q \sim v_{p+1}, \qquad v_p \sim v_{q+1}.$$

Let $e(u,\Gamma)$ be the number of isolated vertices left after these deletions and identifications. Let $\mathsf{D}(u,\Gamma)$ be the remaining directed labelled graph, with all X-edges placed before all Y-edges in the ambient commutative normal-ordered algebra of matrix entries. Disconnected components are retained; they are the split-trace terms. The empty graph is denoted 1.

Theorem F.18 *(Free indexed normal-ordering formula).* In the universal indexed matrix Weyl algebra with

$$Y^k{}_l X^i{}_j = X^i{}_j Y^k{}_l - \hbar\, \delta^i_l \delta^k_j,$$

the normal-ordered trace word is

$$\mathcal{N}_{\text{free}}\, \mathrm{Tr}(u) = \sum_{\Gamma} (-\hbar)^{|\Gamma|} N^{e(u,\Gamma)} \mathsf{D}(u,\Gamma),$$

where the sum runs over all contraction matchings Γ of Y-before-X inversions in u. Consequently

$$\begin{aligned}\mathcal{N}_{\text{free}}\, \mathrm{Tr}(WM) &= \mathcal{N}_{\text{free}}\, \mathrm{Tr}(WYX) - \mathcal{N}_{\text{free}}\, \mathrm{Tr}(WXY)\\ &\quad + \hbar N\, \mathcal{N}_{\text{free}}\, \mathrm{Tr}(W),\end{aligned}$$

with $M = YX - XY + \hbar N I$, interpreted as the left-ideal expression $\mathrm{Tr}(WM) = \sum_{i,j} W^j{}_i M^i{}_j$.

Proof. Move the X-entries leftward through the Y-entries by the PBW relation displayed above. Choosing the commutator summand at a swap selects one pair Y_p, X_q, contributes the factor $-\hbar$, and imposes exactly the two Kronecker identifications $v_q \sim v_{p+1}$ and $v_p \sim v_{q+1}$. Choosing the ordered summand leaves both edges and continues the PBW ordering. Since a generator can be contracted only once, the possible choices are precisely the partial matchings of original inversions. After all X-entries have been moved left, the uncontracted entries form the normal diagram $\mathsf{D}(u,\Gamma)$. Each isolated index loop contributes a free summation over $1, \dots, N$, hence N. This proves the formula for trace words. The formula for $\mathrm{Tr}(WM)$ is the defining expansion of M, with the scalar matrix term contributing $\hbar N\, \mathrm{Tr}(W)$. □

Lemma F.19 *(Stable injectivity for normal trace diagrams).* Fix d. Let $\mathsf{Diag}_{\le d}$ be the rational span of free indexed normal diagrams with at most d labelled nonempty edges, with coefficients in $\mathbb{Q}[\hbar, N]$. The joint specialization map to $N \times N$ matrices for all ranks $N \ge d$ is injective. Thus a free normal-form equality in $\mathsf{Diag}_{\le d}$ is an all-N matrix Weyl identity. A nonzero free residual remains visible until the stable Procesi–Razmyslov range is reached.

Proof. Polarize in the labelled edges. A linear relation among polarized closed directed trace diagrams of edge degree at most d is a trace identity for generic matrices of degree at most d. The Procesi–Razmyslov second fundamental theorem says that the trace identities are generated by the fundamental antisymmetrizer on $N+1$ index lines. Hence every identity of degree at most d is zero in ranks $N \ge d$. Since the coefficients are polynomials in the formal scalar N, vanishing in every rank $N \ge d$ forces each coefficient polynomial to vanish. Depolarizing gives the stated injectivity. □

Proposition F.20 *(First free residual and its kernel correction).* Use the normal-diagram notation of Definition F.17; for instance $\mathsf{D}(X_{0\to 1} Y_{1\to 0})$ denotes the directed two-edge normal diagram with one X-edge and one Y-edge. Then

$$\begin{aligned}\mathcal{N}_{\text{free}}(E_{4,4,N} - C^{\text{cl}}_{4,4}) = \frac{43}{7}\hbar^2 \Big(&\mathsf{D}(X_{0\to1}X_{1\to2}Y_{2\to3}Y_{3\to0})\\ &- \mathsf{D}(X_{0\to1}X_{2\to3}Y_{1\to2}Y_{3\to0})\\ &- \hbar\, \mathsf{D}(X_{0\to0} \mid Y_{0\to0}) + \hbar N\, \mathsf{D}(X_{0\to1}Y_{1\to0})\Big).\end{aligned}$$

Moreover

$$\mathcal{N}_{\text{free}}(K_{4,4}) = \mathcal{N}_{\text{free}}(E_{4,4,N} - C^{\text{cl}}_{4,4}),$$

and therefore

$$\mathcal{N}_{\mathrm{free}}(E_{4,4,N} - C^{\mathrm{cl}}_{4,4} - K_{4,4}) = 0.$$

This is an all-N identity in the free indexed normal-diagram algebra, proved before any rank specialization.

Proof. Apply Theorem F.18 to the nine trace words in $C^{\mathrm{cl}}_{4,4}$ and to the Weyl-trace expression defining $E_{4,4,N}$. The finite matching enumeration is rational and takes place in the free diagram algebra. It leaves exactly the four displayed normal diagrams. Applying the same formula to the four trace words in $K_{4,4}$ gives the same four-diagram vector. Exact enumeration in the free trace-diagram basis gives the four-term residual above and gives zero residual after adding $K_{4,4}$. □

Definition F.21 *(Kernel-correction residual and decorated necklace complex).* Let $\mathsf{W}_{a,b}$ be the rational span of words with $a-1$ letters X and $b-1$ letters Y, and let $\mathsf{V}_{a,b} = \mathsf{Cyc}_{a,b}$ be the rational span of cyclic words with a letters X and b letters Y. Define the two-colour necklace multigraph $G_{a,b}$ by taking $\mathsf{V}_{a,b}$ as its vertex span and by assigning to $W \in \mathsf{W}_{a,b}$ the oriented edge

$$e_W : [WXY] \longrightarrow [WYX].$$

Its incidence map is

$$\partial : \mathsf{W}_{a,b} \to \mathsf{V}_{a,b}, \qquad \partial W = [WYX] - [WXY],$$

and $Z_{a,b} := \ker \partial$ is the cycle space.

Let $\mathsf{Diag}_{a,b}$ denote the positive free indexed normal-diagram target in bidegree (a,b): the uncontracted $\hbar^0$ cyclic component is removed and all PBW contraction, split-trace, and Capelli $\hbar N$-terms are retained. Write π_+ for this projection. Define the decorated contraction cochain by

$$C^+_{a,b}(W) = \pi_+ \mathcal{N}_{\mathrm{free}} \operatorname{Tr}(WM), \qquad M = YX - XY + \hbar N I.$$

The correction map is

$$C_{a,b} = A_{a,b} := C^+_{a,b}|_{Z_{a,b}} : Z_{a,b} \longrightarrow \mathsf{Diag}_{a,b}.$$

Put

$$T_{a,b} = \sum_{j=0}^{b} [Y^j X^a Y^{b-j}] - \frac{b+1}{\binom{a+b}{a}} \sum_{U \in \mathrm{Sh}(a,b)} [U].$$

For any solution $c^{\mathrm{cl}}_{a,b}$ of $\partial c = T_{a,b}$, define

$$E^+_{a,b} = \pi_+ \mathcal{N}_{\mathrm{free}}(E_{a,b,N})$$

and put

$$R^{\mathrm{free}}_{a,b} = E^+_{a,b} - C^+_{a,b}(c^{\mathrm{cl}}_{a,b})$$

and define

$$\mathfrak{o}_{a,b} = [R^{\mathrm{free}}_{a,b}] \in \operatorname{coker} C_{a,b}.$$

This is the kernel-correction class after a choice of classical lift. The radial obstruction itself is the lift-independent class $\Omega^{\mathrm{rad}}_{a,b}$ and the signed-row criterion of Proposition F.25.

Statement F.22 *(Decorated kernel-correction Hodge criterion).* Give $Z_{a,b}$ and $\mathsf{Diag}_{a,b}$ the inner products for which the chosen word-cycle and free normal-diagram bases are orthonormal. Let $A = A_{a,b}$ and define

$$\Delta^{\mathrm{dec}}_{a,b} = AA^*, \qquad \mathsf{H}^{\mathrm{dec}}_{a,b} = \ker A^* = (\operatorname{im} A)^{\perp}.$$

Let

$$\Pi^{\mathrm{harm}}_{a,b} \colon \mathsf{Diag}_{a,b} \to \mathsf{H}^{\mathrm{dec}}_{a,b}$$

be the orthogonal projection. For every $a, b \geq 1$, the following conditions are equivalent.

(i) There exists $K_{a,b} \in Z_{a,b}$ such that

$$C^{+}_{a,b}(K_{a,b}) = R^{\mathrm{free}}_{a,b}.$$

(ii) The kernel-correction class $\mathfrak{o}_{a,b}$ vanishes.

(iii) The decorated harmonic projection vanishes:

$$\Pi^{\mathrm{harm}}_{a,b} R^{\mathrm{free}}_{a,b} = 0.$$

(iv) Every functional $\lambda \in \mathsf{Diag}^*_{a,b}$ with

$$\lambda C^{+}_{a,b}(K) = 0 \qquad (K \in Z_{a,b})$$

also satisfies $\lambda(R^{\mathrm{free}}_{a,b}) = 0$.

When these equivalent conditions hold, the canonical Hodge correction is

$$K_{a,b} = A^*_{a,b}(\Delta^{\mathrm{dec}}_{a,b})^{-1} R^{\mathrm{free}}_{a,b},$$

with the inverse taken on $\operatorname{im} A_{a,b}$. Thus the fixed-lift kernel-correction equation is equivalent to

$$\Pi^{\mathrm{harm}}_{a,b} R^{\mathrm{free}}_{a,b} = 0 \qquad \text{in the chosen bidegree.}$$

The lift-independent obstruction $\Omega^{\mathrm{rad}}_{a,b}$ and its normalized signed-row form are given by Proposition F.25.

Proof. If c and c' solve the cyclic equation, then $c - c' \in Z_{a,b}$. Hence $R(c') - R(c) = -A_{a,b}(c' - c)$, so the class in $\operatorname{coker} A_{a,b}$, and its harmonic projection, are independent of the chosen cyclic lift.

The finite-dimensional Hodge decomposition for A gives

$$\mathsf{Diag}_{a,b} = \operatorname{im} A \oplus \ker A^*.$$

Therefore $R^{\mathrm{free}}_{a,b} \in \operatorname{im} A$ if and only if its projection to $\ker A^*$ is zero. This proves the equivalence of the correction equation, the intermediate cokernel class, and the harmonic projection. The functional criterion is the annihilator form of the same statement: $R \in \operatorname{im} A$ if and only if every $\lambda \in (\operatorname{im} A)^{\perp} = \ker A^*$ vanishes on R. If $R \in \operatorname{im} A$, then $AA^*(AA^*)^{-1}R = R$ on $\operatorname{im} A$, so the displayed $K_{a,b}$ is a correction. Conversely, any correction places $R^{\mathrm{free}}_{a,b}$ in $\operatorname{im} A$. □

Definition F.23 *(Square-cell PBW Stokes complex).* Let $\mathsf{S}_{a,b}$ be the rational span of square cells obtained from two commuting adjacent transpositions in a cyclic word with a letters X and b letters Y. A based representative has the form

$$uXYvXYw,$$

and its oriented square is

$$uXYvXYw \to uYXvXYw \to uYXvYXw \to uXYvYXw \to uXYvXYw.$$

Each side is read as the corresponding edge $e_W : [WXY] \to [WYX]$ of the necklace graph. This defines

$$\partial_2 \colon \mathsf{S}_{a,b} \longrightarrow \mathsf{W}_{a,b}.$$

The decorated PBW Stokes map is

$$D^{\square}_{a,b} = C^{+}_{a,b}\partial_2 \colon \mathsf{S}_{a,b} \longrightarrow \mathsf{Diag}_{a,b}.$$

LEMMA F.24 *(Ordinary necklace Stokes theorem).* For $a, b \geq 1$,

$$\operatorname{im} \partial_2 = \ker \partial.$$

Proof. Work first with based words. A binary word with a letters X and b letters Y is the same as an order ideal in the rectangle poset $[a] \times [b]$, via its inversion set. An adjacent $XY \leftrightarrow YX$ move toggles one extremal box. The cubical complex of the distributive lattice of order ideals, with squares generated by commuting toggles, is contractible. Hence its rational one-cycles are generated by square boundaries. Passing to cyclic words is the quotient by a finite cellular rotation action; over $\mathbb{Q}$, transfer identifies the quotient rational homology with the invariant rational homology upstairs. Thus every necklace cycle is a rational sum of square boundaries. □

PROPOSITION F.25 *(Dual-potential form of the radial obstruction).* Define the finite-dimensional map

$$B_{a,b} \colon \mathsf{W}_{a,b} \longrightarrow \mathsf{V}_{a,b} \oplus \mathsf{Diag}_{a,b}, \qquad B_{a,b}(W) = (\partial W, C^{+}_{a,b}(W)),$$

and the target

$$\eta_{a,b} = (T_{a,b}, E^{+}_{a,b}).$$

Let

$$\Omega^{\mathrm{rad}}_{a,b} = [\eta_{a,b}] \in \operatorname{coker} B_{a,b}.$$

For fixed (a, b), the following are equivalent.

(i) $\Omega^{\mathrm{rad}}_{a,b} = 0$.

(ii) For one, equivalently every, classical lift $c^{\mathrm{cl}}_{a,b}$ with $\partial c^{\mathrm{cl}}_{a,b} = T_{a,b}$, there is an $H_{a,b} \in \mathsf{S}_{a,b}$ such that

$$D^{\square}_{a,b} H_{a,b} = R^{\mathrm{free}}_{a,b}.$$

(iii) For one, equivalently every, classical lift $c^{\mathrm{cl}}_{a,b}$, the residual $R^{\mathrm{free}}_{a,b}$ lies in $\operatorname{im}(C^{+}_{a,b}|_{\ker \partial})$.

(iv) $\Pi^{\mathrm{harm}}_{a,b} R^{\mathrm{free}}_{a,b} = 0$.

(v) For every pair

$$(\phi, \lambda) \in \mathsf{V}^{*}_{a,b} \oplus \mathsf{Diag}^{*}_{a,b}$$

satisfying

$$\lambda C^{+}_{a,b}(W) = \phi(\partial W) \qquad (W \in \mathsf{W}_{a,b}),$$

one has

$$\lambda(E^{+}_{a,b}) = \phi(T_{a,b}).$$

Thus

$$\Omega^{\mathrm{rad}}_{a,b} = 0 \quad \Longleftrightarrow \quad D^{\square}_{a,b} H_{a,b} = R^{\mathrm{free}}_{a,b}$$

for one, equivalently every, classical lift. The obstruction in bidegree (a, b) is exactly a signed dual row

$$(\phi, -\lambda) \in \ker B^{*}_{a,b}$$

with

$$\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) \neq 0.$$

Equivalently, after rescaling, it is a normalized row

$$\lambda \in \ker(D^{\square}_{a,b})^{*}, \qquad \lambda(R^{\mathrm{free}}_{a,b}) = 1.$$

Proof. The pair (ϕ, λ) in the statement is the signed dual row $(\phi, -\lambda)$ on $\mathsf{V}_{a,b} \oplus \mathsf{Diag}_{a,b}$. It annihilates $B_{a,b}(W)$ precisely when $\lambda C^{+}_{a,b}(W) = \phi(\partial W)$. Since the spaces are finite dimensional,

$$\eta_{a,b} \in \operatorname{im} B_{a,b} \quad \Longleftrightarrow \quad \ell(\eta_{a,b}) = 0 \text{ for every } \ell \in \ker B^{*}_{a,b}.$$

Evaluating $(\phi, -\lambda)$ on $\eta_{a,b}$ gives $\phi(T_{a,b}) - \lambda(E^{+}_{a,b})$, proving the equivalence of $\Omega^{\mathrm{rad}}_{a,b} = 0$ with the signed-row condition.

If $c^{\mathrm{cl}}_{a,b}$ is a classical lift, then

$$\eta_{a,b} - B_{a,b}(c^{\mathrm{cl}}_{a,b}) = (0, R^{\mathrm{free}}_{a,b}).$$

Adding $B_{a,b}(K)$ with $K \in \ker \partial$ changes only the decorated diagram component by $C^{+}_{a,b}(K)$. Hence $\eta_{a,b} \in \operatorname{im} B_{a,b}$ if and only if $R^{\mathrm{free}}_{a,b} \in \operatorname{im}(C^{+}_{a,b}|_{\ker \partial})$. By Lemma F.24, $\ker \partial = \operatorname{im} \partial_2$, so this image is exactly $\operatorname{im} D^{\square}_{a,b}$. The Hodge equivalence is the decomposition of Statement F.22.

Finally, finite-dimensional duality for $D^{\square}_{a,b}$ identifies the cokernel of $D^{\square}_{a,b}$ with $\ker(D^{\square}_{a,b})^{*}$. A nonzero residual class therefore has a row $\lambda \in \ker(D^{\square}_{a,b})^{*}$ with $\lambda(R^{\mathrm{free}}_{a,b}) \neq 0$. Rescale the row to make the value 1. The equality $\lambda C^{+}_{a,b}(W) = \phi(\partial W)$ identifies this row with the signed row $(\phi, -\lambda) \in \ker B^{*}_{a,b}$, and evaluating that signed row on $\eta_{a,b}$ gives $\phi(T_{a,b}) - \lambda(E^{+}_{a,b})$. $\square$

Remark F.26 *(Potential form and the bare graph Green operator).* A functional $\lambda \in \mathsf{Diag}^{*}_{a,b}$ satisfies $\lambda C^{+}_{a,b}|_{\ker \partial} = 0$ precisely when the edge cochain

$$W \longmapsto \lambda C^{+}_{a,b}(W)$$

has zero periods on the necklace graph $G_{a,b}$. The graph is connected for $a, b > 0$, since adjacent $XY \leftrightarrow YX$ moves connect all two-colour necklaces. Hence this zero-period condition is equivalent to a vertex potential $\phi_\lambda \in \mathsf{V}^{*}_{a,b}$, unique up to a constant, with

$$\lambda C^{+}_{a,b}(W) = \phi_\lambda([WYX]) - \phi_\lambda([WXY]) = \phi_\lambda(\partial W).$$

Because $T_{a,b}$ has total coefficient zero, $\phi_\lambda(T_{a,b})$ is independent of the additive constant. The radial obstruction vanishes, equivalently $\Omega^{\mathrm{rad}}_{a,b} = 0$, if and only if every such λ satisfies the potential identity

$$\lambda \pi_{+} \mathcal{N}_{\mathrm{free}}(E_{a,b,N}) = \phi_\lambda(T_{a,b}).$$

By Lemma F.24, the same row condition is $\lambda D^{\square}_{a,b} = 0$, and

$$\lambda(R^{\mathrm{free}}_{a,b}) = \lambda E^{+}_{a,b} - \lambda C^{+}_{a,b}(c^{\mathrm{cl}}_{a,b}) = \lambda E^{+}_{a,b} - \phi_{\lambda}(T_{a,b}).$$

The ordinary graph Green operator for the incidence map ∂ is therefore insufficient. It constructs a classical cyclic lift $c^{\mathrm{cl}}_{a,b}$ of $T_{a,b}$, i.e. it solves only the uncontracted $\hbar^{0}$ incidence equation. The quantum correction equation is instead $A_{a,b}K_{a,b} = R^{\mathrm{free}}_{a,b}$, equivalently $D^{\square}_{a,b}H_{a,b} = R^{\mathrm{free}}_{a,b}$, inside the positive PBW diagram target, where contraction diagrams, split traces, and the Capelli term are visible. A canonical inverse from an arbitrary positive diagram residual to a necklace edge cochain is extra data. The finite-dimensional Green operator used here is the decorated square-cell one attached to $D^{\square}_{a,b}(D^{\square}_{a,b})^{*}$.

Definition F.27 *(Radial bicomplex and PBW ħ-filtration).* Fix $a, b \geq 1$. The *radial bicomplex* in bidegree (a, b) is the augmented diagram

$$\mathcal{R}_{a,b}: \quad \mathsf{S}_{a,b} \xrightarrow{\partial_2} \mathsf{W}_{a,b} \xrightarrow{\partial} \mathsf{V}_{a,b}, \qquad \mathsf{W}_{a,b} \xrightarrow{C^{+}_{a,b}} \mathsf{Diag}_{a,b}$$

with ∂, ∂_2, $C^{+}_{a,b}$ as in Definitions F.21 and F.23. Equip $\mathsf{Diag}_{a,b}$ with the *contraction filtration*

$$F^{p}\mathsf{Diag}_{a,b} = \bigoplus_{p' \geq p} \mathsf{Diag}^{(p')}_{a,b},$$

where $\mathsf{Diag}^{(p)}_{a,b}$ is the rational span of free indexed normal diagrams with exactly p PBW contractions. The Capelli scalar contributes one unit of $\hbar$ and is recorded in $\mathsf{Diag}^{(0)}_{a,b}$ with an explicit factor $\hbar N$. Consequently $C^{+}_{a,b}$ and its restriction $D^{\square}_{a,b} = C^{+}_{a,b}\partial_2$ take values in $F^{1}\mathsf{Diag}_{a,b}$, and the residual $R^{\mathrm{free}}_{a,b}$ lies in $F^{1}\mathsf{Diag}_{a,b}$. The graded pieces $\mathrm{gr}^{p}\mathsf{Diag}_{a,b}$ carry the Cartan one-particle Moyal action of Lemma F.34.

THEOREM F.28 *(Radial bicomplex acyclicity).* Assume Statement F.8 in full. Then for every interior bidegree (a, b) with $a, b \geq 1$,

$$\Omega^{\mathrm{rad}}_{a,b} = 0, \qquad D^{\square}_{a,b}H_{a,b} = R^{\mathrm{free}}_{a,b} \text{ has a solution } H_{a,b} \in \mathsf{S}_{a,b}.$$

Equivalently the radial bicomplex $\mathcal{R}_{a,b}$ is acyclic at the residual class $R^{\mathrm{free}}_{a,b}$ on the decorated side: every element of $\mathsf{Diag}_{a,b}$ of the form $R^{\mathrm{free}}_{a,b}$ lies in $\operatorname{im} D^{\square}_{a,b}$. Equivalently, the normalized signed-dual-row obstruction set is empty.

Proof. The argument proceeds in three steps. Step 1 uses only clause (1) (radial homomorphism) and the Cartan-one-particle Moyal homomorphism of Lemma F.34; Step 2 uses the mixed-monomial image clause (3) of Statement F.8 for the Moyal-bracket polynomial of bidegree (a, b), which is the radial-parts hypothesis; Step 3 uses clause (2) (Levasseur–Stafford radial-kernel theorem) and stable Procesi–Razmyslov injectivity.

Step 1 (Cartan-side commutator). Compute the Vandermonde-conjugated Harish-Chandra radial image of the matrix Weyl commutator $\hbar^{-1}[J_N(z_1^{a+1}/(a+1)), J_N(z_2^{b+1})]$. Pure-power Weyl traces equal normal-ordered traces, so $J_N(z_1^{a+1}) = \mathrm{Tr}(X^{a+1})$ and $J_N(z_2^{b+1}) = \mathrm{Tr}(Y^{b+1})$ with zero contractions; clause (3) for pure powers is therefore tautological:

$$\mathrm{Rad}_0(J_N(z_1^{a+1})) = \Delta^{-1}S_N(z_1^{a+1})\Delta = \sum_{i=1}^{N} \lambda_i^{a+1}, \qquad \mathrm{Rad}_0(J_N(z_2^{b+1})) = \Delta^{-1}S_N(z_2^{b+1})\Delta = \sum_{i=1}^{N} (\nabla_i^{\Delta})^{b+1}.$$

Both pre-images are GL_N-invariant, so by clause (1) the radial map is a homomorphism on their commutator. Different Cartan summands commute by the third commutation relation of Lemma F.11, hence

$$\mathrm{Rad}_0\big(\hbar^{-1}[J_N(z_1^{a+1}/(a+1)), J_N(z_2^{b+1})]\big) = \sum_{i=1}^{N} \hbar^{-1}\big[\lambda_i^{a+1}/(a+1), (\nabla_i^{\Delta})^{b+1}\big].$$

Lemma F.34 identifies each summand with the Cartan one-particle Weyl quantization of the normalized Moyal bracket:

$$\hbar^{-1}\big[\lambda_i^{a+1}/(a+1), (\nabla_i^{\Delta})^{b+1}\big] = \mathrm{Op}_{\mathrm{W}}\big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_{\hbar}\big)(\lambda_i, \nabla_i^{\Delta}).$$

Summing over i and using the displayed formula in Definition F.10 gives

$$\mathrm{Rad}_0\big(\hbar^{-1}[J_N(z_1^{a+1}/(a+1)), J_N(z_2^{b+1})]\big) = \Delta^{-1}S_N\big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_{\hbar}\big)\Delta.$$

Up to this point only the pure-power part of clause (3) has entered.

Step 2 (Spherical-reduction equality). By Statement F.8 clause (3) applied to the Moyal-bracket polynomial $\{z_1^{a+1}/(a+1), z_2^{b+1}\}_{\hbar}$,

$$\mathrm{Rad}_0\big(J_N(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_{\hbar})\big) = \Delta^{-1}S_N\big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_{\hbar}\big)\Delta.$$

Subtracting the two displayed identities, $\mathrm{Rad}_0(E_{a,b,N}) = 0$.

Step 3 (Universal lift via Procesi–Razmyslov). By Statement F.8 clause (2) (Levasseur–Stafford radial-kernel theorem), the kernel of the Vandermonde-conjugated Harish-Chandra map on GL_N-invariant operators is exactly the moment-map ideal $\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$. Since $E_{a,b,N}$ is GL_N-invariant and has zero radial image,

$$E_{a,b,N} \in \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N) \qquad (\text{all } N \geq 1).$$

By Lemma F.15, the membership $E_{a,b,N} \in \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$ produces coefficients $A^j{}_i(N) \in \mathcal{W}_N$ with $E_{a,b,N} = \sum_{i,j} A^j{}_i(N)\widehat{\mu}^i{}_j$; the coefficients depend on N and on $\hbar$ a priori. Apply the GL_N-symmetrization average to the lift; $E_{a,b,N}$ is GL_N-invariant, so the symmetrization preserves the equality and produces GL_N-equivariant tensor coefficients. By the Procesi classical fundamental theorem on trace ring generators (which embeds in stable Procesi–Razmyslov, Lemma F.19), every GL_N-equivariant tensor coefficient with values in $\mathcal{W}_N$ of total edge degree $\leq a+b$ is a $\mathbb{Q}[\hbar]$-linear combination of word matrix-elements $W^j{}_i$. Polarizing in N reduces the resulting trace-form identity

$$E_{a,b,N} = \sum_{W \in \mathsf{W}_{a,b}} c_W(a,b)\,\mathrm{Tr}(WM)$$

to a finite-dimensional rational linear system in $c = (c_W)_{W\in\mathsf{W}_{a,b}} \in \mathbb{Q}^{|\mathsf{W}_{a,b}|}$, decomposed into independent homogeneous components by the contraction filtration of Definition F.27 and the Capelli factor. The system is consistent in every homogeneous component because, for every $N \geq a+b$, the matrix-Weyl-rank specialization has a solution; stable Procesi–Razmyslov injectivity in those degrees extends the rank-N consistency to free-side rational consistency. Choose any $\mathbb{Q}$-rational solution $c = \sum_W c_W W \in \mathsf{W}_{a,b}$.

Project the resulting free identity through the cyclic-word target. At contraction degree 0, $\sum_W c_W \pi_0 \mathcal{N}_{\mathrm{free}}\,\mathrm{Tr}(WM) = \sum_W c_W([WYX] - [WXY]) = \partial(\sum_W c_W W)$, so the underlying word $c \in \mathsf{W}_{a,b}$ is a classical lift, $\partial c = T_{a,b}$. Subtracting any preferred classical lift $c^{\mathrm{cl}}_{a,b}$, the difference $K_{a,b} = c - c^{\mathrm{cl}}_{a,b}$ is a cycle in $\ker\partial$, and on the decorated side

$\pi_+\mathcal{N}_{\mathrm{free}}(\sum_W c_W \operatorname{Tr}(WM)) = E^+_{a,b} = C^+_{a,b}(c^{\mathrm{cl}}_{a,b}) + C^+_{a,b}(K_{a,b})$, so $C^+_{a,b}(K_{a,b}) = R^{\mathrm{free}}_{a,b}$. By the ordinary necklace Stokes Lemma F.24, every cycle in $\ker\partial$ lies in $\operatorname{im}\partial_2$, so there exists $H_{a,b} \in \mathsf{S}_{a,b}$ with $\partial_2 H_{a,b} = K_{a,b}$; hence $D^{\square}_{a,b}H_{a,b} = C^+_{a,b}\partial_2 H_{a,b} = C^+_{a,b}(K_{a,b}) = R^{\mathrm{free}}_{a,b}$. By Proposition F.25, $\Omega^{\mathrm{rad}}_{a,b} = 0$ and the normalized signed-dual-row obstruction set is empty. □

Remark F.29 *(Cartan one-particle Moyal lift).* The proof of Theorem F.28 runs the same Cartan one-particle reduction in each bidegree. At contraction degree 0, the residual is the cyclic shear sum minus the Weyl average; the necklace Stokes Lemma F.24 is the assertion that it is a coboundary in ∂. At each higher contraction degree $p \geq 1$, the residual is the p-contraction component of the matrix-Weyl shear identity *after* subtracting the classical lift; the Vandermonde-conjugated radial map evaluates this on the Cartan as a sum of one-particle Moyal commutators $\hbar^{-1}[\lambda_i^{a+1}/(a+1), (\nabla_i^{\Delta})^{b+1}]$, and Lemma F.34 replaces each by its Weyl quantization of the Moyal-bracket polynomial. Statement F.8 clause (3) for that polynomial closes the equality with $\mathrm{Rad}_0(J_N(\{\dots\}_\hbar))$; clause (2) places the matrix-Weyl difference in the moment ideal. Stable Procesi–Razmyslov polarization extracts the universal cycle $K_{a,b}$, and the necklace Stokes lemma gives the square-cell primitive. The contraction-filtered dimension of $\mathrm{gr}^{\geq 1}\mathsf{Diag}_{a,b}$ grows polynomially in (a, b), matching the empirical $K_{a,b}$-correction support count in Remark F.42.

COROLLARY F.30 *(Bidegree obstruction equals the clause-(3) defect on the Moyal-bracket polynomial).* Assume Statement F.8 clauses (1) and (2) (Levasseur–Stafford radial homomorphism and radial-kernel theorem). For each bidegree (a, b) with $a, b \geq 1$, the following are equivalent.

(i) $\Omega^{\mathrm{rad}}_{a,b} = 0$.
(ii) Statement F.8 clause (3) holds modulo the moment ideal for the Moyal-bracket polynomial $\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar$:

$$\mathrm{Rad}_0\big(J_N(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar)\big) \equiv \Delta^{-1}S_N\big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\big)\Delta \pmod{I_N}.$$

Equivalently, the bidegree-local cokernel obstruction is exactly the spherical-reduction defect of the radial image on the Moyal-bracket polynomial.

Proof. Step 1 of the proof of Theorem F.28 shows that, under clauses (1) and (2),

$$\mathrm{Rad}_0\big(\hbar^{-1}[J_N(z_1^{a+1}/(a+1)), J_N(z_2^{b+1})]\big) = \Delta^{-1}S_N\big(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar\big)\Delta.$$

Subtracting the radial image of $J_N(\{\dots\}_\hbar)$,

$$\mathrm{Rad}_0(E_{a,b,N}) = \Delta^{-1}S_N\big(\{\dots\}_\hbar\big)\Delta - \mathrm{Rad}_0\big(J_N(\{\dots\}_\hbar)\big).$$

By clause (2), $E_{a,b,N} \in I_N$ iff $\mathrm{Rad}_0(E_{a,b,N}) = 0$, iff the displayed clause-(3) identity holds. Lemma F.15 identifies $E_{a,b,N} \in I_N$ with $\Omega^{\mathrm{rad}}_{a,b} = 0$ via Proposition F.25; the intermediate Procesi–Razmyslov reduction is in Step 3 of the bicomplex acyclicity proof. □

PROPOSITION F.31 *(Procesi–Razmyslov exact verification window).* For each bidegree (a, b) and each $N \geq a + b$, the finite-rank moment-ideal membership $E_{a,b,N} \in \mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)$ holds if and only if the free indexed normal-diagram identity $\mathcal{N}_{\mathrm{free}}(E_{a,b,N} - K_{a,b}) = 0$ holds for some cycle $K_{a,b} \in \ker\partial$, and the latter is independent of N. Hence the all-N closure for fixed (a, b) reduces to a finite rational linear-algebra problem on the $(\mathsf{W}_{a,b} \oplus \mathsf{S}_{a,b})$-side, decided by exact rational normal-form calculation. This verification closes $\Omega^{\mathrm{rad}}_{a,b} = 0$ by exact rational normal-form computation, independently of Statement F.8, for the bidegrees in Remark F.42.

Proof. Stable Procesi–Razmyslov injectivity (Lemma F.19) makes the free indexed normal-diagram algebra a faithful joint model of all matrix Weyl algebras of rank $N \geq a+b$. Hence a free identity holds if and only if it holds in $\mathcal{W}_N$ for every $N \geq a+b$. The kernel-correction equation $\partial K_{a,b} = 0$, $\mathcal{N}_{\text{free}}(E_{a,b,N} - C^{+}_{a,b}(K_{a,b})) = \mathcal{N}_{\text{free}}(C^{+}_{a,b}(c^{\text{cl}}_{a,b}))$ is finite-dimensional rational linear algebra on $(\mathsf{W}_{a,b} \oplus \mathsf{S}_{a,b})$ and gives the coefficient array $K_{a,b}$, then the Stokes primitive $H_{a,b} = \partial_2^{-1} K_{a,b}$ on the necklace square-cell complex by Lemma F.24. Existence of $K_{a,b}$ is therefore an exact finite computation in rational arithmetic. □

Proposition F.32 *(Edge PBW telescoping primitive).* For every $b \geq 2$, set

$$C^{\text{edge}}_{2,b} = \sum_{r=0}^{\lfloor (b-2)/2 \rfloor} (b-1-2r)\,\mathrm{Tr}(Y^r X Y^{b-1-r} M).$$

For every $a \geq 2$, set

$$C^{\text{edge}}_{a,2} = \frac{3}{a+1} \sum_{r=0}^{\lfloor (a-2)/2 \rfloor} (a-1-2r)\,\mathrm{Tr}(X^{a-1-r} Y X^r M).$$

Then the free indexed normal-diagram identities

$$\mathcal{N}_{\text{free}}(E_{2,b,N} - C^{\text{edge}}_{2,b}) = 0, \qquad \mathcal{N}_{\text{free}}(E_{a,2,N} - C^{\text{edge}}_{a,2}) = 0$$

hold for all displayed edge bidegrees. Hence

$$\mathfrak{d}_{2,b} = 0, \qquad \mathfrak{d}_{a,2} = 0$$

with zero separate kernel correction in this edge gauge.

Proof. Take the $(2,b)$ strip. Put

$$\Theta^{(b)}_s = \mathsf{D}(XY^s XY^{b-s}), \qquad 0 \leq s \leq b,$$

and

$$\Phi^{(b)}_s = \mathsf{D}(XY^s) \mid \mathsf{D}(Y^{b-1-s}), \qquad 0 \leq s \leq b-1,$$

where the vertical bar denotes disjoint union and $\mathsf{D}(Y^0)$ is the isolated index loop, hence contributes the scalar factor N. For

$$W_r = Y^r X Y^{b-1-r}, \qquad 0 \leq r \leq b-1,$$

Theorem F.18 gives the PBW column identity

$$\mathcal{N}_{\text{free}}\,\mathrm{Tr}(W_r M) = \Theta^{(b)}_r - \Theta^{(b)}_{r+1} - \hbar\Phi^{(b)}_r + \hbar\Phi^{(b)}_{b-1-r}.$$

Expand

$$\mathrm{Tr}(W_r M) = \mathrm{Tr}(W_r YX) - \mathrm{Tr}(W_r XY) + \hbar N\,\mathrm{Tr}(W_r).$$

PBW matchings in the first two trace words away from the adjacent crossing given by M cancel in pairs. The matchings using that adjacent crossing are cancelled, including their secondary contractions, by the scalar term $\hbar N\,\mathrm{Tr}(W_r)$. The uncontracted part leaves the boundary $\Theta^{(b)}_r - \Theta^{(b)}_{r+1}$. The two endpoint contraction families are the left and reflected right blocks, giving $-\hbar\Phi^{(b)}_r$ and $+\hbar\Phi^{(b)}_{b-1-r}$. These are exactly the surviving terms.

The target $E_{2,b,N}$ contains only the first and third Moyal terms:

$$J_N\Big(\{z_1^3/3, z_2^{b+1}\}_\hbar\Big) = (b+1)J_N(z_1^2 z_2^b) + \frac{(b+1)b(b-1)}{12}\hbar^2 \operatorname{Tr}(Y^{b-2}).$$

Counting PBW matchings in the shear sum and in the Weyl average gives

$$\mathcal{N}_{\text{free}}(E_{2,b,N}) = \sum_{r=0}^{\lfloor (b-2)/2 \rfloor} (b-1-2r)\Big(\Theta_r^{(b)} - \Theta_{r+1}^{(b)} - \hbar\Phi_r^{(b)} + \hbar\Phi_{b-1-r}^{(b)}\Big).$$

In contraction degree 0, this is the discrete boundary coefficient of the two-X cyclic separation. In contraction degree 1, the coefficient of $\Phi_s^{(b)}$ is $-(b+1)+2(s+1) = 2s-(b-1)$, the antisymmetric half-sum above. In contraction degree 2, the difference between the shear count and the first-Moyal Weyl count is exactly

$$\frac{(b+1)b(b-1)}{12}\,\mathcal{N}_{\text{free}} \operatorname{Tr}(Y^{b-2}),$$

which is cancelled by the third Moyal term in the displayed target. Combining this target expansion with the column identity proves $\mathcal{N}_{\text{free}}(E_{2,b,N} - C_{2,b}^{\text{edge}}) = 0$.

The $(a,2)$ strip is the same one-moving-letter calculation with Y moving through an X-string. For $V_r = X^{a-1-r}YX^r$,

$$\mathcal{N}_{\text{free}} \operatorname{Tr}(V_r M) = \widetilde{\Theta}_r^{(a)} - \widetilde{\Theta}_{r+1}^{(a)} - \hbar\widetilde{\Phi}_r^{(a)} + \hbar\widetilde{\Phi}_{a-1-r}^{(a)}.$$

Here

$$\widetilde{\Theta}_s^{(a)} = \mathsf{D}(X^{a-s}YX^sY), \qquad \widetilde{\Phi}_s^{(a)} = \mathsf{D}(X^{a-1-s}) \mid \mathsf{D}(X^sY),$$

with $\mathsf{D}(X^0)$ again interpreted as the isolated index loop. The first and third Moyal coefficients for $\{z_1^{a+1}/(a+1), z_2^3\}_\hbar$ are

$$3, \qquad \frac{a(a-1)}{4},$$

namely $3/(a+1)$ times the corresponding $(2,a)$ coefficients. The same count therefore gives the stated $C_{a,2}^{\text{edge}}$ and proves the second identity. □

PROPOSITION F.33 *(Free residual vanishings beyond* $(4,4)$*).* In the free indexed normal-diagram algebra, $\mathfrak{d}_{3,5} = \mathfrak{d}_{5,3} = 0$, and the lexicographic cyclic lifts already have zero quantum residual. Explicitly,

$$\begin{aligned} E_{3,5,N} = {} & \frac{36}{7}\operatorname{Tr}(XXYYYY\,M) + \frac{24}{7}\operatorname{Tr}(XYXYYY\,M) \\ & - \frac{6}{7}\operatorname{Tr}(XYYXYY\,M) - \frac{6}{7}\operatorname{Tr}(XYYYXY\,M) \\ & + \frac{30}{7}\operatorname{Tr}(XYYYYX\,M) + \frac{12}{7}\operatorname{Tr}(YXYXYY\,M), \end{aligned}$$

and

$$\begin{aligned} E_{5,3,N} = {} & \frac{24}{7}\operatorname{Tr}(XXXXYY\,M) - \frac{4}{7}\operatorname{Tr}(XXXYXY\,M) \\ & + \frac{20}{7}\operatorname{Tr}(XXXYYX\,M) + \frac{12}{7}\operatorname{Tr}(XXYXXY\,M) \\ & - \frac{8}{7}\operatorname{Tr}(XXYXYX\,M) + \frac{16}{7}\operatorname{Tr}(XXYYXX\,M). \end{aligned}$$

In total degree 9 the same free computation gives $\mathfrak{o}_{3,6} = \mathfrak{o}_{6,3} = 0$. The cyclic lifts already have zero free residual:

$$\begin{aligned}E_{3,6,N} = {} & \frac{25}{4}\operatorname{Tr}(XXYYYYYM) + \frac{19}{4}\operatorname{Tr}(XYXYYYYM) \\ & - \frac{3}{4}\operatorname{Tr}(XYYXYYYM) - \frac{3}{4}\operatorname{Tr}(XYYYXYYM) \\ & - \frac{3}{4}\operatorname{Tr}(XYYYYXYM) + \frac{11}{2}\operatorname{Tr}(XYYYYYXM) \\ & + \frac{13}{4}\operatorname{Tr}(YXYXYYYM) + \frac{1}{4}\operatorname{Tr}(YXYYXYYM) \\ & + \frac{7}{4}\operatorname{Tr}(YXYYYXYM),\end{aligned}$$

and

$$\begin{aligned}E_{6,3,N} = {} & \frac{25}{7}\operatorname{Tr}(XXXXXYYM) - \frac{3}{7}\operatorname{Tr}(XXXXYXYM) \\ & + \frac{22}{7}\operatorname{Tr}(XXXXYYXM) - \frac{3}{7}\operatorname{Tr}(XXXYXXYM) \\ & - \frac{6}{7}\operatorname{Tr}(XXXYXYXM) + \frac{19}{7}\operatorname{Tr}(XXXYYXXM) \\ & + \frac{16}{7}\operatorname{Tr}(XXYXXXYM) + \frac{1}{7}\operatorname{Tr}(XXYXXYXM) \\ & + \operatorname{Tr}(XXYXYXXM).\end{aligned}$$

Proof. Apply Theorem F.18. For $(3,5)$ and $(5,3)$, the displayed coefficients solve the cyclic equation and the free normal-form residual is zero. The total-degree-9 formulas for $(3,6)$ and $(6,3)$ are checked in the same way; their residual is also zero, so $K_{3,6} = K_{6,3} = 0$ in this pivot gauge. Exact normal-form enumeration gives the $(3,5)$, $(5,3)$, $(3,6)$, $(6,3)$, $(2,7)$, and $(7,2)$ cases and separately checks the edge theorem. The calculation takes place in the free diagram basis before finite-rank trace identities enter. □

Lemma F.34 *(Cartan one-particle Moyal representation).* Put

$$S_N(f) = \sum_{i=1}^{N} \operatorname{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

Then, for all $f, g \in \mathbb{C}[z_1, z_2][[\hbar]]$,

$$\hbar^{-1}[S_N(f), S_N(g)] = S_N(\{f,g\}_\hbar).$$

Proof. The i-th summand of $S_N(f)$ lies in the one-particle Weyl algebra generated by λ_i and $-\hbar\partial_{\lambda_i}$. Weyl quantization is an algebra isomorphism from the formal Moyal algebra of the constant Poisson plane to this one-particle Weyl algebra, so

$$\hbar^{-1}[\operatorname{Op}_{\mathrm{W}}(f)_i, \operatorname{Op}_{\mathrm{W}}(g)_i] = \operatorname{Op}_{\mathrm{W}}(\{f,g\}_\hbar)_i.$$

Summands with different i commute, because they use disjoint Cartan variables. Summing over i gives the displayed identity. □

THEOREM F.35 *(Finite-N radial-parts identity).* Under the radial-parts hypothesis of Statement F.8, put

$$S_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

Then

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1} S_N(f)\Delta,$$

and for all f, g

$$D_N(f,g) = \hbar^{-1}[J_N(f), J_N(g)] - J_N(\{f,g\}_\hbar)$$

lies in the quantum moment-map ideal. Equivalently, in the Vandermonde gauge,

$$\mathrm{Rad}_0(D_N(f,g)) = \hbar^{-1}[R_N(f), R_N(g)] - R_N(\{f,g\}_\hbar) = 0.$$

Thus $D_N(f,g)$ vanishes in degree-zero quantum Hamiltonian reduction.

Proof. The first displayed identity is the exact image normalization included in Statement F.8. The same radial-parts hypothesis makes the conjugated radial map injective on the reduced invariant quotient, so the commutator defect can be tested on the Cartan. Lemma F.34 gives $\hbar^{-1}[S_N(f), S_N(g)] = S_N(\{f,g\}_\hbar)$, and Lemma F.11 gives the same identity after Vandermonde conjugation:

$$\hbar^{-1}[R_N(f), R_N(g)] = R_N(\{f,g\}_\hbar).$$

Thus $\mathrm{Rad}_0(D_N(f,g)) = 0$. The radial-map injectivity of Lemma F.12 places $D_N(f,g)$ in the quantum moment-map ideal.

Lemma F.13 marks the exact theorem behind the full displayed formula: mixed Weyl images in this convention require the quantum correction beyond classical nonlinear shear covariance. The all-bidegree decorated PBW Stokes closure $\Omega^{\mathrm{rad}}_{a,b} = 0$ is given uniformly by Theorem F.28 under the same statement hypotheses and by exact computation on the bidegrees of Remark F.42 by Proposition F.31; a nonzero obstruction is a normalized signed row $(\phi, -\lambda) \in \ker B^*_{a,b}$, structurally excluded under Statement F.8 and excluded by exact computation on the verified bidegrees. □

Definition F.36 *(Stable primitive extraction).* Fix a total polynomial degree bound d and take $N \geq d$. Let

$$\kappa_{N,d}$$

denote the connected single-trace primitive/indecomposable projection on the stable trace algebra in degrees $\leq d$: it kills the empty trace and all decomposable multi-trace products, and it sends the Capelli-renormalized generator

$$J_N(z_1^a z_2^b), \qquad 0 < a+b \leq d,$$

to its single-trace indecomposable class. The stable map $\kappa_{\infty,d}$ is the stable-tail identification of these projections in the Procesi–Razmyslov stable range, after passing through the renormalized associated-graded J_N-basis transition system.

LEMMA F.37 *(Connected-trace projection after Capelli triangularization).* Fix a polynomial degree bound d. In the stable range $N \geq d$, the Capelli triangular system

$$J_N(z_1^a z_2^b) \equiv \sum_{r=0}^{\min(a,b)} \left(-\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} T_{a-r,b-r}$$

identifies the span of nonempty Weyl traces $J_N(z_1^a z_2^b)$, $a + b \le d$, with the span of nonempty normal-ordered traces $T_{a,b}$, modulo the empty trace and decomposable multi-traces. The primitive/indecomposable projection

$$\kappa_{N,d}\big(J_N(z_1^a z_2^b)\big) \longmapsto z_1^a z_2^b$$

is therefore degreewise injective on the scalar-reduced connected single-trace quotient in the stable range.

Proof. The displayed Capelli formula is triangular for total degree: its $r = 0$ term is $T_{a,b}$, and every $r \ge 1$ term has total degree $a + b - 2r$. The diagonal coefficient is 1, hence the triangular matrix is unipotent and invertible over $\mathbb{C}[[\hbar N]]$. The only term which can become the empty trace is the term with $a - r = b - r = 0$, and the scalar-reduced connected quotient removes it. Decomposable products are removed by the connected projection. Stable Procesi–Razmyslov trace independence in degree d gives linear independence of the remaining connected trace generators for $N \ge d$. Thus the stated projection is injective degreewise in the stable range. □

Theorem F.38 *(Connected-trace bracket compatibility and stable primitive projections).* Under the same radial-parts hypothesis as Theorem F.35, for each fixed polynomial degree the classes

$$\overline{J_N(f)}$$

have a stable connected large-N limit after removing the empty trace and disconnected products. This limit is the renormalized associated-graded connected transition system associated to the finite-rank normal-ordered trace algebras over $\mathbb{C}[[\hbar]]$. In that limit,

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f, g\}_\hbar).$$

More explicitly, for $N \ge d$ and all f, g with all monomials in $f, g, \{f, g\}_\hbar$ of total degree $\le d$,

$$\kappa_{N,d}\big(\hbar^{-1}[J_N(f), J_N(g)]\big) = \kappa_{N,d}\big(J_N(\{f, g\}_\hbar)\big),$$

and these identities are compatible under $N \mapsto N + 1$.

Proof. Lemma F.37 gives the degreewise stable connected single-trace projection and its injectivity after removing the empty trace and decomposable products. The finite-N identity of Theorem F.35 gives equality in the reduced quantum Hamiltonian quotient. Applying $\kappa_{N,d}$ gives the displayed connected-trace identity, since the primitive projection kills only the empty trace and decomposable products and is injective on the Capelli-renormalized single-trace span by Lemma F.37. The identity is compatible with adjoining extra Cartan variables, because $S_{N+k}(f)$ is obtained from $S_N(f)$ by adding commuting one-particle summands. Therefore the bracket identity stabilizes degreewise in N. □

Definition F.39 *(Reduced open-line Wick graph data).* In the reduced first-order open-line model with action

$$S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2),$$

use Weyl-ordered boundary vertices $f(\phi_1, \phi_2)$, and choose the zero-mode-removed midpoint propagator

$$G(t,s) = \frac{1}{2}\,\mathrm{sgn}(t-s), \qquad \langle \phi_1(t)\phi_2(s)\rangle = \hbar G(t,s), \qquad \langle \phi_2(t)\phi_1(s)\rangle = -\hbar G(t,s).$$

Self-contractions at a Weyl vertex are set to zero. The only connected two-vertex graphs are r-fold cross-contractions between the two boundary vertices.

THEOREM F.40 *(Reduced open-line Wick graph expansion).* With the graph data of Definition F.39, the ordered local product for two boundary Hamiltonians at $t > s$ is the formal Moyal product

$$F \star G = m \circ \exp\left(\frac{\hbar}{2} P\right)(F \otimes G).$$

More precisely, the graph with exactly r cross-edges contributes

$$W_r^{>}(f, g) = \frac{1}{r!}\left(\frac{\hbar}{2}\right)^r P^r(f, g),$$

while the reversed ordering $s > t$ contributes

$$W_r^{<}(f, g) = \frac{1}{r!}\left(-\frac{\hbar}{2}\right)^r P^r(f, g).$$

Hence the antisymmetrized factorization commutator is

$$[f, g]_{\text{graph}} = \sum_{r \geq 0} \frac{1 - (-1)^r}{2^r r!} \hbar^r P^r(f, g) = \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{2}{2^r r!} \hbar^r P^r(f, g).$$

The one-edge and three-edge commutator coefficients are exactly

$$\hbar P(f, g), \qquad \frac{\hbar^3}{24} P^3(f, g).$$

All even commutator orders vanish.

Proof. For ordered insertions with $t > s$, the midpoint propagator gives $G(t, s) = 1/2$. A single cross-contraction therefore contributes

$$\frac{\hbar}{2}\left(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}\right) = \frac{\hbar}{2} P.$$

Wick expansion of the free first-order model exponentiates the cross-contraction operator. Weyl vertices have zero self-contractions; hence a connected two-vertex graph is determined by its number r of cross-edges. The exponential gives the symmetry factor $1/r!$, and the ordered propagator gives $(\hbar/2)^r$. This proves the formula for $W_r^{>}$. Reversing the order changes the propagator from $1/2$ to $-1/2$, giving $W_r^{<}$. Subtracting the two orderings gives the displayed commutator series. The coefficients in orders 1 and 3 are $2/(2 \cdot 1!) = 1$ and $2/(2^3 \cdot 3!) = 1/24$. □

COROLLARY F.41 *(All-order reduced Wick realization of Moyal).* The reduced open-line Wick graph expansion of Theorem F.40 realizes the Weyl/Moyal Lie bracket of Definition F.1 to all orders:

$$[f, g]_{\text{graph}} = [f, g]_{\text{raw}}, \qquad \hbar^{-1}[f, g]_{\text{graph}} = \{f, g\}_\hbar.$$

Proof. Compare the graph commutator series in Theorem F.40 with the formal Weyl commutator series in Theorem F.2. The coefficient of P^r is $2\hbar^r/(2^r r!)$ for odd r and 0 for even r in both series. □

Remark F.42 *(Verified bidegree window).* Exact rational linear algebra over the indexed-diagram basis proves $\Omega^{\text{rad}}_{a,b} = 0$ on every edge strip, every non-edge interior bidegree of total degree ≤ 13, every total-degree-14 bidegree with one component at most 6, and a partial total-degree-15 sublist; the balanced $(7, 7)$ is the first undecided case. The reduction is the Procesi–Razmyslov kernel-correction route of Proposition F.31. The all-bidegree closure is Theorem F.28.

Exact rational reduction

Example F.43 *(The first nonzero balanced cyclic-pivot residual).* At $(a,b)=(2,2)$ and $(3,3)$ the classical lift already lies in im $D^{\square}_{a,b}$, and $\Omega^{\mathrm{rad}}_{a,b}=0$ with zero correction. The first bidegree where the cyclic pivot produces a residual is $(4,4)$, with residual

$$\tfrac{43}{14}\Big(\overline{\mathrm{Tr}(XXYXYY)}-\overline{\mathrm{Tr}(XYXXYY)}-\overline{\mathrm{Tr}(XYYXYX)}+\overline{\mathrm{Tr}(XYYYXX)}\Big)$$

killed by a four-term correction in $\ker\partial$.

Theorem F.44 *(Per-bidegree exact rational reduction).* Each bidegree datum is the triple

$$\big(\beta_{a,b},\ B_{a,b}\in M_{r_0\times r_1}(\mathbb{Q}),\ v_{a,b}\in\mathbb{Q}^{r_0}\big),$$

where $\beta_{a,b}$ is the indexed-diagram basis ordering, $r_1=\dim\mathsf{W}_{a,b}=\binom{a+b-2}{a-1}$, $r_0=\dim\big(\mathsf{W}_{a,b}\oplus\mathsf{S}_{a,b}\big)$, $B_{a,b}$ is the Stokes boundary in the basis $\beta_{a,b}$, and $v_{a,b}=(T_{a,b},E^{+}_{a,b})$ is the radial comparison vector. Bareiss elimination over $\mathbb{Q}$ decides whether $v_{a,b}\in\operatorname{im}B_{a,b}$ in exact rational arithmetic and yields the dimensions

$$\operatorname{rank}B_{a,b},\qquad\dim\ker B^{*}_{a,b},\qquad\dim\operatorname{coker}B_{a,b}.$$

When $v_{a,b}\in\operatorname{im}B_{a,b}$, the elimination returns the unique correction $C^{+}_{a,b}\in\mathbb{Q}^{r_1}$ with $B_{a,b}C^{+}_{a,b}=v_{a,b}$, giving $\Omega^{\mathrm{rad}}_{a,b}=0$ at that bidegree. In the obstruction case, the elimination returns the obstruction covector $(\phi,-\lambda)\in\ker B^{*}_{a,b}$ with the strict pairing $\lambda(E^{+}_{a,b})-\phi(T_{a,b})\neq 0$; the obstruction class $\Omega^{\mathrm{rad}}_{a,b}\in\operatorname{coker}B_{a,b}$ is its equivalence class. Every arithmetic step is exact rational; the same elimination on the same basis ordering returns the same matrix entries.

Corollary F.45 *(Vanishing on the listed range).* For every bidegree (a,b) listed in Remark F.42, the rational reduction of Theorem F.44 gives $v_{a,b}\in\operatorname{im}B_{a,b}$: the comparison vector lies in the image of the Stokes boundary, with explicit correction $C^{+}_{a,b}$ and dimensions $\dim\ker B^{*}_{a,b}$, $\dim\operatorname{coker}B_{a,b}$ read off from the elimination. In particular $\Omega^{\mathrm{rad}}_{a,b}=0$ on the listed bidegrees. For an unlisted bidegree, the per-bidegree class $\Omega^{\mathrm{rad}}_{a,b}\in\operatorname{coker}B_{a,b}$ is the exact obstruction whose vanishing is the all-bidegree statement.

Remark F.46 *(Scalar quotient and the $U(1)$ anomaly).* Constants are central in the Moyal Lie algebra and are killed in $\bar{A}_{\hbar}$. Before this scalar reduction the linear commutator carries the central term $[z_1,z_2]_*=\hbar$, and on the brane side the corresponding matrix trace is the Capelli shift $\hbar N$. The reduced theorem is the theorem after quotienting by this scalar line. The unreduced line is the $U(1)$ anomaly class of Theorem 6.3.

Proposition F.47 *(Moyal–Capelli normalization and realization boundary).* The following statements are used by the degree-zero Moyal sector.

(i) The formal Weyl algebra on the constant symplectic plane gives the raw commutator coefficients

$$[f,g]^{(2s+1)}_{\mathrm{raw}}=\frac{\hbar^{2s+1}}{2^{2s}(2s+1)!}P^{2s+1}(f,g),\qquad [f,g]^{(2s)}_{\mathrm{raw}}=0.$$

In particular the third raw coefficient is $\hbar^3P^3(f,g)/24$.

(ii) The finite-rank brane algebra is the degree-zero normalizer quotient of the matrix Weyl algebra by the traceless quantum moment ideal. With the normal-ordering convention of Definition F.6, the contact relation is

$$YX-XY+\hbar NI\equiv 0\pmod{\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)}.$$

This scalar Capelli shift is represented by the triangular change of basis $T_{a,b} \leftrightarrow J_N(z_1^a z_2^b)$, and the remaining scalar Hamiltonian line is removed only after passing to $\bar{A}_\hbar$ and to the empty-trace quotient.

(iii) The equality

$$\hbar^{-1}[J_N(f), J_N(g)] = J_N(\{f,g\}_\hbar)$$

in degree-zero quantum Hamiltonian reduction holds in every bidegree under Statement F.8 by Theorem F.28, and holds by exact computation on the bidegrees listed in Remark F.42 by Proposition F.31. The obstruction-theoretic equivalence

$$\Omega^{\mathrm{rad}}_{a,b} = 0 \quad \Longleftrightarrow \quad D^{\square}_{a,b} H_{a,b} = R^{\mathrm{free}}_{a,b}$$

holds in every bidegree by Proposition F.25. The moment-ideal primitive $E_{a,b,N} = \sum A^j{}_i(a,b,N)\widetilde{\mu}^i{}_j$ of Lemma F.15 and the class $\mathfrak{o}_{a,b}$ of Definition F.21 are equivalent intermediate forms after choosing a classical lift.

(iv) The quantum principal-part target is the continuous $\bar{A}_\hbar$-coadjoint module. It is defined by

$$(\mathrm{ad}^\vee_{f,\hbar}\,\rho)(g) = -\rho(\{f,g\}_\hbar).$$

For $f = z_1^p z_2^q$ and the principal-part basis $\rho_{a,b}$, this gives the finite formula

$$z_1^p z_2^q \cdot_\hbar \rho_{a,b} = -\sum_{\substack{r\geq 1\\ r\ \mathrm{odd}}} \frac{\hbar^{r-1}}{2^{r-1}r!}\, C_r(p,q;a-p+r,b-q+r)\,\rho_{a-p+r,b-q+r},$$

where

$$C_r(p,q;m,n) = \sum_{\alpha=0}^{r}(-1)^\alpha\binom{r}{\alpha}(p)_{r-\alpha}(q)_\alpha(m)_\alpha(n)_{r-\alpha},$$

and terms with negative indices or scalar target $(0,0)$ are zero. The $r=1$ part is the classical Matlis coadjoint action. The $r\geq 3$ terms are genuine Moyal corrections; the polynomial one-ψ PBW action records only the $r=1$ term.

(v) The reduced open-line Wick graph computation realizes the same formal Moyal product with midpoint propagator. An unreduced Costello graph/QME theorem additionally requires the brane-defect propagator, counterterms, anomaly cancellation, and d_{QME}-homotopies in the local BV complex; those data belong to the unreduced BV target.

Proof. The five items collect already-proven content. The raw Moyal commutator coefficients are Theorem F.2, with the 1/24 and 1/1920 check at Example F.5. The Capelli triangular contact relation is the normal-ordered comoment calculation of Definition F.6 and Lemma F.7, with the scalar quotient of Definition F.1 and Remark F.46. The Cartan one-particle commutator and the all-bidegree compatibility are Theorem F.35 and Theorem F.28; the obstruction- theoretic equivalence is Proposition F.25 with primitive form Lemma F.15 and fixed-lift Hodge presentation Statement F.22; the edge strips are Proposition F.32, the exact finite window Proposition F.31, with interior anchors Propositions F.20, F.33. The residue-dual coadjoint formula in (4) uses the same P^r-coefficients as (1); falling factorials vanish for $r > p+q$, and the scalar target is removed by the Hamiltonian quotient (special fibre Proposition 5.176; quantum target Construction 5.135). The open-line midpoint Moyal realisation is Theorem F.40 with Corollary F.41; the separation from the Costello graph/QME realization belongs to Theorem 5.217. □

Remark F.48 *(Frontier of the all-bidegree obstruction).* Theorem F.28 gives $\Omega^{\mathrm{rad}}_{a,b} = 0$ for every interior bidegree (a, b) under Statement F.8; clauses (1) and (2) are the Levasseur–Stafford radial-kernel theorem, and clause (3) enters only through the Moyal-bracket polynomial $\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar$ of the chosen bidegree. Corollary F.30 refines this to a logical equivalence: under clauses (1) and (2), $\Omega^{\mathrm{rad}}_{a,b} = 0$ is equivalent to clause (3) modulo the moment ideal for that Moyal-bracket polynomial. The bidegree-local obstruction is concentrated in clause (3) for the Moyal-bracket polynomial of the chosen bidegree.

Independently of clause (3), Proposition F.31 closes $\Omega^{\mathrm{rad}}_{a,b} = 0$ by exact rational linear algebra against the moment ideal, independently of the statement hypotheses. The bidegrees enumerated in Remark F.42 (edges and interior cases through total degree 15) are closed by this exact rational normal-form route.

In the bidegrees verified by the exact finite-window route, the signed-row obstruction set of Proposition F.25 is empty; in the remaining bidegrees, its vanishing is equivalent to Statement F.8 clause (3) for the corresponding Moyal-bracket polynomial. A defect in that clause produces a signed dual row $(\phi, -\lambda) \in \ker B^*_{a,b}$ with $\lambda(E^+_{a,b}) - \phi(T_{a,b}) \neq 0$ identical with the clause-(3) defect inside $\mathsf{Diag}^*_{a,b}$. An explicit normal-form formula realizing clause (3) for every mixed monomial of degree $\leq a + b - 2$ in the Capelli/Weyl convention is fixed by the Levasseur–Stafford spherical-Hecke methods.

Reflection symmetry $(a, b) \leftrightarrow (b, a)$ is implemented by the symplectic involution $z_1 \leftrightarrow z_2$, $X \leftrightarrow Y$, with sign $\omega(z_1, z_2) \to -\omega(z_2, z_1)$ recorded in Proposition E.2, and is preserved by the radial-bicomplex argument.

G First-principles completion of the formal-local theorem

The topology is part of the formal-Darboux theorem. The residue dual $\mathfrak{h}^\vee_{\mathrm{cont}}$, the diagonal Casimir, the Hamiltonian coadjoint action, the Hochschild equivariance homotopies, and the heat-kernel propagator live in one completed filtered tensor category; outside that category the same formulas are only polynomial shadows. The category C is the category in which the local Hamiltonian BF theory, the brane trace J, and the Chevalley–Eilenberg/polyvector dictionary are continuous objects.

The obstruction classes of the local theorem are then honest cohomology classes in filtered L_∞ deformation complexes. They measure separate lifts: radial Weyl compatibility, Bochner–Martinelli current transfer, Tate bar-cobar admissibility, Hochschild equivariance and centre-lift, the scalar-contact chain map, quantum master equation counterterms, the brane-defect propagator, and the large-N trace state. A theory $\mathcal{T}$ compares its anomalies with the local theorem by a morphism to this completed deformation complex. The resulting objects are $\mathfrak{U}_{\mathcal{T}}$ and the local trace-sector Maurer–Cartan element Θ_{loc}.

G.1 The host filtered nuclear Tate dg category

Tate algebra gives formal power series, residue duals, and infinite Matlis sums. Renormalisation gives heat kernels, wavefront conditions, and distributional products against test functions. The common category is the filtered nuclear Tate dg category.

Definition G.1 *(Filtered nuclear Tate dg category).* $\mathsf{C} = \mathrm{Ch}^{\otimes,\mathrm{fil}}_{\mathrm{Tate,nuc},\mathbb{C}}$ is the symmetric monoidal dg category of complete decreasingly filtered cochain complexes $V = (V, F^\bullet V, d_V)$ with

$$V \xrightarrow{\sim} \varprojlim_r V/F^r V, \qquad \bigcap_r F^r V = 0,$$

each associated graded $\mathrm{gr}^r_F V = F^r V/F^{r+1} V$ carried by a nuclear locally convex cochain complex tensored with an admissible Tate $\mathbb{C}$-coefficient module (linearly compact, projective system of finite-dimensional $\mathbb{C}$-vector spaces). The completed tensor product is the strict inverse limit of the term-wise nuclear projective tensor product on the analytic factor with the term-wise Tate tensor product on the coefficient factor:

$$V \widehat{\otimes} W = \varprojlim_r \big((V/F^r V) \otimes_\pi (W/F^r W)\big).$$

The continuous dual is $V^\vee_{\mathrm{cont}} = \mathrm{Hom}^{\mathrm{cts}}_{\mathbb{C}}(V, \mathbb{C})$, computed in the same projective topology.

LEMMA G.2 *(Axioms of* C*).* C satisfies the following four properties.

C1 (filtered Tate exactness) For every inverse system $\{V_M\}_{M \geq 0}$ used in the constructions of §G.5–§G.11, the transition maps are continuous and strict. When ordinary inverse limits are used, Mittag–Leffler holds; when derived inverse limits $\mathbf{R}\varprojlim$ are used, the corresponding $\varprojlim^1$-defects are recorded as obstruction classes in the master complex. In every case the basic exact sequence

$$0 \to \varprojlim_M{}^1 H^{k-1}(V_M) \to H^k(\mathbf{R}\varprojlim_M V_M) \to \varprojlim_M H^k(V_M) \to 0$$

holds.

C2 (continuous duality) For an admissible perfect pair $(V, V^\vee_{\mathrm{cont}})$ in C, the evaluation $V^\vee_{\mathrm{cont}} \widehat{\otimes} V \to \mathbb{C}$ is continuous. The coevaluation $\mathbb{C} \to V \widehat{\otimes} V^\vee_{\mathrm{cont}}$ exists in a weighted or kernel-admissible completion (Casimir convergence, §G.6); perfect pairing and convergent Casimir are distinguished; the second map belongs to the completed category.

C3 (local-functional subcategory) A subspace $\mathcal{O}_{\mathrm{loc}}(\mathcal{E}) \subset \mathcal{O}(\mathcal{E})$ of local functionals on a BV field complex is closed under Q, the BV bracket $\{-,-\}_{\mathrm{BV}}$, the regularized BV Laplacian Δ_L, Costello renormalization-group flow, and counterterm subtraction. For a brane-defect theory the brane-local subspace is $\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}_X; \mathcal{E}_L)$ consisting of functionals supported on X, on L, or on controlled collision strata involving L.

C4 (distributional current admissibility) The reduced principal-part current sector uses $\mathcal{D}'_c(I) \widehat{\otimes} \mathcal{P}$ with multiplication $C^\infty_c(I) \times \mathcal{D}'_c(I) \to \mathcal{D}'_c(I)$, $(a, B) \mapsto aB$, while the product $\mathcal{D}'_c(I) \times \mathcal{D}'_c(I) \to \mathcal{D}'_c(I)$ is excluded from the admissible operations. The reduced principal-part bracket of the brane current sector factors through this restricted multiplication; the unreduced brane-defect QME requires a wavefront condition (§G.12).

Proof. C1 is the Milnor sequence [54]. C2 is the standard distinction between a perfect pairing in linearly compact / Tate spaces and the completion of the diagonal tensor needed to write down the BV pairing inverse: a continuous evaluation map exists for any continuous dual, but a coevaluation map exists only after the convergent regulator of §G.6. C3 is the local functional subcategory of Costello [12, Ch. 5] adapted to brane-defect support strata; closure under Δ_L is the Costello effective-action calculus and closure under counterterm subtraction follows from the regulator-independence of local divergences. C4 is Hörmander's product theorem [36, Thm. 8.2.10]: the smooth-times-distribution product is everywhere defined, while two distributions multiply under the standard wavefront transversality condition. The reduced principal-part bracket of the brane current sector respects exactly this restriction. □

All subsequent constructions are performed in C.

G.2 The local Hamiltonian BF theory as a genuine BV theory

The classical master equation $\{S_H, S_H\}_{\mathrm{BV}} = 0$ of the formal-local Hamiltonian BF theory holds in the completed local-functional space as an honest local-functional identity. The proof uses the classical BV bracket and avoids the Costello renormalization complex and the brane-defect propagator; it is the foundation on which every quantum construction below depends.

Theorem G.3 *(Classical master equation).* Let $A = \mathbb{C}[[z_1, z_2]]$, $\mathfrak{m} = (z_1, z_2)$, $\mathfrak{h} = A/\mathbb{C} \cdot 1$, and let $\mathcal{P} = \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{a+b>0} \mathbb{C}\rho_{a,b}$ with $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2$. Define

$$\mathcal{E}_H(U \times B) = \Omega^\bullet_c(U) \widehat{\otimes} \Omega^{0,\bullet}_c(B) \widehat{\otimes} \big(\mathfrak{h}[1] \oplus \mathcal{P}[2]\big)$$

on $U \subset \mathbb{R}^2_{t,s}$, $B \subset \mathbb{C}^2_{z_1,z_2}$, with differential $Q_H = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2} + d_{\mathfrak{g}}$, (-1)-shifted symplectic pairing

$$\omega_H(\alpha \otimes f, \beta \otimes \rho) = \int_{\mathbb{R}^2 \times \mathbb{C}^2} \alpha \wedge \beta \wedge dz_1\, dz_2 \cdot \mathrm{Res}_0(f\rho),$$

and classical action

$$S_H(\mathcal{A}, \mathcal{B}) = \int_X \big\langle \mathcal{B}, (d + \bar{\partial})\mathcal{A} + \tfrac{1}{2}[\mathcal{A}, \mathcal{A}]_{\mathfrak{h}} \big\rangle.$$

Then $\{S_H, S_H\}_{\mathrm{BV}} = 0$ in $\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}(\mathcal{E}_H)$.

Proof. The bracket expansion of $\{S_H, S_H\}$ is the Maurer–Cartan Bianchi identity for the connection $(d + \bar{\partial})\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}]_{\mathfrak{h}}$:

$$D\big(D\mathcal{A} + \tfrac{1}{2}[\mathcal{A}, \mathcal{A}]\big) + [\mathcal{A}, D\mathcal{A} + \tfrac{1}{2}[\mathcal{A}, \mathcal{A}]] = 0.$$

The four ingredients are $D^2 = 0$ ($(d + \bar{\partial})^2 = 0$ on $\mathbb{R}^2 \times \mathbb{C}^2$), the graded Leibniz rule $D[\mathcal{A}, \mathcal{A}] = 2[D\mathcal{A}, \mathcal{A}]$, the Jacobi identity in $\mathfrak{h}$, and continuity of the residue pairing. Continuity is the C2 axiom: the Tate dual $\mathcal{P} \subset \mathfrak{h}^\vee_{\mathrm{cont}}$ is paired with $\mathfrak{h}$ by the continuous functional $\langle f, \rho \rangle = \mathrm{Res}_0(f\rho)$, which descends through the integration by parts identity $\int_X \langle D\mathcal{B}, \mathcal{A} \rangle + (-1)^{|\mathcal{B}|} \int_X \langle \mathcal{B}, D\mathcal{A} \rangle = 0$ for compactly supported test forms. Therefore the bracket $\{S_H, S_H\}$ reduces to a sum of terms each of which vanishes by one of the four ingredients. The resulting equality holds in the Tate-completed local functional space, since each ingredient is a continuous operation in C. □

Remark G.4 *(BV degree, Koszul sign, propagator weight).* The conventions are those of $d = \dim_{\mathbb{C}} \mathbb{C}^2 = 2$ holomorphic dimensions and $d_{\mathbb{R}^2} = 2$ topological dimensions; BV degree $\deg_{\mathrm{BV}} \mathcal{A} = 1$, $\deg_{\mathrm{BV}} \mathcal{B} = 2$ following Costello [12]; the propagator $P_{\varepsilon,L}$ has BV degree -1; the holomorphic volume form is $dz_1\, dz_2$ (purely holomorphic); the worldline $L = \mathbb{R}_t \subset X$ is unframed in the local model and acquires the S^3-framing only after a matched-conventions transfer (§11.4).

G.3 Hamiltonian shears and the trace uniqueness theorem

The trace uniqueness theorem (Theorem 5.42) states that $J(z_1^a z_2^b) = \mathrm{Tr}(\phi_1^a \phi_2^b)$ for every bidegree. The proof uses linear shears to relate same-degree coefficients and one nonlinear Hamiltonian shear to relate different total degrees. Linear shears alone leave one constant per total degree undetermined; the nonlinear shear closes the proof.

Theorem G.5 *(Hamiltonian-shear forcing of coefficients).* Let $J : \mathfrak{h} \to \mathcal{O}(\mathrm{Rep}^\wedge_N(A))^{GL_N}$ be a linear, stable GL_N-equivariant, Dirac-descended, scalar-reduced, primitive single-trace map satisfying $J(z_i) = \overline{\mathrm{Tr}\,\phi_i}$ and Darboux-naturality under the formal symplectomorphism group of $(\mathbb{C}^2, dz_1 \wedge dz_2)$. Then $J(z_1^a z_2^b) = \overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}$ for every $a, b \geq 0$ with $a + b > 0$.

Proof. By stable invariant theory [17][18], primitive block-additivity, and the Koszul homotopy for adjacent transpositions on the moment-map zero fibre (Lemma 3.1),

$$J(z_1^a z_2^b) = c_{a,b}\overline{\mathrm{Tr}(\phi_1^a \phi_2^b)}$$

for some scalar $c_{a,b} \in \mathbb{C}$. It suffices to prove $c_{a,b} = 1$ for every (a, b) with $a + b > 0$; the linear case $c_{1,0} = c_{0,1} = 1$ is the normalization hypothesis $J(z_i) = \overline{\mathrm{Tr}\,\phi_i}$.

Linear shears within fixed total degree. The Darboux flow generated by $H_{\mathrm{lin}} = z_1^2/2$, at time t, is $z_1 \mapsto z_1$, $z_2 \mapsto z_2 + tz_1$, so $(\Phi^* z_1^a z_2^b) = \sum_{j=0}^{b} \binom{b}{j} t^j z_1^{a+j} z_2^{b-j}$. Equivariance $J(\Phi^* f) = \Phi^* J(f)$ on the brane side gives $\Phi^*\overline{\mathrm{Tr}(\phi_1^a\phi_2^b)} = \sum_j \binom{b}{j} t^j \overline{\mathrm{Tr}(\phi_1^{a+j}\phi_2^{b-j})}$, and comparing coefficients of t^j yields $c_{a+j,b-j} = c_{a,b}$ within fixed total degree $a + b$.

Nonlinear shear across degrees. Consider the Hamiltonian $H_k = z_2^{k+1}/(k + 1)$, whose flow is $z_1 \mapsto z_1 - tz_2^k$, $z_2 \mapsto z_2$. Apply Darboux-naturality to the linear observable z_1, already normalized to $c_{1,0} = 1$:

$$J(z_1 - tz_2^k) = \overline{\mathrm{Tr}(\phi_1)} - t\, c_{0,k}\overline{\mathrm{Tr}(\phi_2^k)},$$

while the brane-side flow gives $\overline{\mathrm{Tr}(\phi_1)} - t\overline{\mathrm{Tr}(\phi_2^k)}$. Comparing coefficients of t yields $c_{0,k} = 1$, and linear shears propagate this to $c_{a,b} = 1$ for every (a, b) with $a + b = k$. Since $k \geq 1$ is arbitrary the assertion follows for every total degree. □

Remark G.6 *(Multiplicativity Is Forced).* The hypothesis "primitive single-trace" is the Loday–Quillen–Tsygan condition that the Lie algebra map J lands in the connected primitive sector of the cyclic invariant algebra [2][3]; this is the structural condition that produces the trace word for an arbitrary monomial. The natural-correction obstruction lemma (§G.3) is then the assertion that once the linear normalization $c_{1,0} = c_{0,1} = 1$ is fixed and nonlinear shears have propagated this to every total degree, admissible CE/PV-natural correction terms are exhausted. Multiplicativity in the sense "$J(z_1^a z_2^b)$ is the trace of a product of matrices" is the conclusion of the Procesi trace word, the Koszul resolution of $[\phi_1, \phi_2] = 0$, and the single-trace primitivity.

G.4 The decorated PBW–Stokes complex and $\Omega_{a,b}^{\mathrm{rad}} = 0$

The all-bidegree Weyl/radial compatibility theorem (Theorem 5.217) depends on the Statement F.8 clauses (1) and (3); clause (1) is the Harish-Chandra homomorphism with kernel the quantum moment-map ideal, and clause (3) is the exact Weyl-trace image $\mathrm{Rad}_0(J_N(f)) = \Delta^{-1} S_N(f)\Delta$ in the Weyl/Capelli normalization of the formal-Darboux theorem. The decorated PBW–Stokes complex below, in which the per-bidegree obstruction $\Omega_{a,b}^{\mathrm{rad}} = [(T_{a,b}, E_{a,b}^+)] \in \operatorname{coker} B_{a,b}$ becomes the boundary of an explicit contracting homotopy, is the construction.

Definition G.7 *(Decorated PBW–Stokes complex).* Fix $a, b \geq 0$ with $a + b > 0$. Let $\mathcal{W}_N$ be the matrix Weyl algebra on X_j^i, Y_ℓ^k of Definition F.6, with quantum moment-map ideal $I_N = \mathcal{W}_N\,\mu(\mathfrak{sl}_N)$. Let $C_{a,b}^0 = \big(\Omega_{\mathrm{PBW}}^{a+b+2}(X, Y; \hbar)\big) \oplus \mathbb{C}$, the rank-$b + 2$ free $\mathcal{W}_N$-module of PBW $(Y^j X^a Y^{b-j})$-words decorated by Vandermonde–Stokes generators $s_{a,b}^{(\alpha)} \in C_{a,b}^0$ carrying the boundary contributions of the type-A Dunkl–Calogero correction (Lemma F.11). Let $C_{a,b}^{-1} = \mathcal{W}_N^{\oplus(b+1)}$ be the corresponding free module of homotopy primitives, and define

$$B_{a,b} : C_{a,b}^{-1} \longrightarrow C_{a,b}^0, \qquad B_{a,b}(\xi^{(j)})_{j=0}^b = \Big(\sum_{j=0}^{b} A_{a,b,N}^{(j)} \xi^{(j)},\ \sum_{j,\alpha} \partial_{j,a,b}^{(\alpha)} \xi^{(j)}\Big),$$

where $A^{(j)}_{a,b,N}$ are the explicit primitives produced by the Vandermonde-conjugated radial homomorphism (§G.4.1) and $\partial^{(\alpha)}_{j,a,b}$ are the Stokes-decoration boundary operators recording the Vandermonde-pole-stripping data on the α-th coordinate chart of the regular Cartan. The transported shear vector is $(T_{a,b}, E^+_{a,b}) \in C^0_{a,b}$ with $T_{a,b} = \sum_j \operatorname{Tr}(Y^j X^a Y^{b-j})$ and $E^+_{a,b}$ the $J_N(\{z_1^{a+1}/(a+1), z_2^{b+1}\}_\hbar)$ shear-error term.

G.4.1 Explicit primitives $A^{(j)}_{a,b,N}$ from the Vandermonde-conjugated radial map

Under the homomorphism and injectivity hypotheses of Statement F.8(1)–(2), the Vandermonde-conjugated radial map sends X^i_j, Y^k_ℓ into formal differential operators on the regular Cartan in the Vandermonde gauge ∇^Δ_i of Definition F.10. In this gauge the Calogero–Dunkl correction terms cancel pairwise (Lemma F.11), so the radial image of $\sum_{j=0}^b \operatorname{Tr}(Y^j X^a Y^{b-j})$ coincides with $\Delta^{-1} \sum_{i=1}^N \operatorname{Op}_{\mathrm{W}}(z_1^a z_2^b)(\lambda_i, -\hbar\partial_{\lambda_i})\Delta$ up to a Stokes correction supported on the boundary of the regular Cartan.

Lemma G.8 *(Stokes correction in the Vandermonde gauge).* For every $a, b \geq 0$ with $a + b > 0$ there exists an explicit collection of differential operators $A^{(j)}_{a,b,N} \in \mathcal{W}_N$ and a Stokes correction $\tau_{a,b} \in C^{-1}_{a,b}$ such that

$$E_{a,b,N} = \sum_{i,j} A^{(j)}_{a,b,N} \mu^j_i + B_{a,b}(\tau_{a,b}),$$

where μ^j_i are generators of the quantum moment-map ideal. The data $(A^{(j)}_{a,b,N}, \tau_{a,b})$ are computed by exact rational arithmetic in the radial coordinates and are independent of any radial normalization beyond Statement F.8(1)–(2).

Proof. The Vandermonde gauge variables satisfy $[\lambda_i, \nabla^\Delta_j] = \hbar\delta_{ij}$, so the algebra they generate is the standard scalar Weyl algebra on N symbols; products of ∇^Δ_i factors expand by Leibniz into $(-\hbar\partial_{\lambda_i})$ factors plus the Vandermonde boundary terms $\hbar \sum_{j \neq i} 1/(\lambda_i - \lambda_j)$. In the symmetric sum $\sum_{i=1}^N (-\hbar\partial_{\lambda_i})^k$, every Vandermonde boundary term cancels pairwise across (i, j)-pairs in the two-body interactions with $j \neq i$ (Lemma F.11); the remaining higher-body interactions cancel against the corresponding Stokes-corrected boundary contributions on the chambers of the regular Cartan, producing $\sum_i \operatorname{Op}_{\mathrm{W}}(z_1^a z_2^b)(\lambda_i, -\hbar\partial_{\lambda_i}) = \sum_i \operatorname{Op}_{\mathrm{W}}(z_1^a z_2^b)(\lambda_i, \nabla^\Delta_i) + \partial(\tau_{a,b})$ for an explicit chamber-Stokes primitive $\tau_{a,b}$. The kernel of the radial map is the quantum moment-map ideal (Statement F.8(2)), so any element of $\mathcal{W}_N$ whose image is the radial expression $\Delta^{-1} \sum_i \operatorname{Op}_{\mathrm{W}}(z_1^a z_2^b)(\lambda_i, -\hbar\partial_{\lambda_i})\Delta$ differs from $E_{a,b,N} + \partial(\tau_{a,b})$ by an element of I_N. Choosing this element to be $\sum_{i,j} A^{(j)}_{a,b,N} \mu^j_i$ expresses the difference in the standard moment-map generators. The arithmetic is rational and finite, performed in the $\mathbb{C}[X, Y][[\hbar]]$-Weyl algebra of bidegree $\leq a + b + 1$. □

Theorem G.9 *(Decorated radial reformulation of the all-bidegree problem).* On the decorated PBW-Stokes complex $C^\bullet_{a,b}$ of Definition G.7, the per-bidegree obstruction $\Omega^{\mathrm{rad}}_{a,b} = [(T_{a,b}, E^+_{a,b})] \in \operatorname{coker} B_{a,b}$ vanishes if and only if there exist explicit primitives $(A^{(j)}_{a,b,N}, \tau_{a,b})$ of Lemma G.8, equivalently if and only if the radial image of the trace map satisfies clause (3) of Statement F.8. In either formulation the all-bidegree hypothesis is converted into a finite-rank exact-rational existence problem at each bidegree (a, b), independent of the radial normalization beyond clauses (1)–(2). The contracting-homotopy formulation (Route A of Remark G.11) and the dual kernel formulation (Route B) are equivalent, and the finite exact-rational calculations of Appendix F verify both routes through the recorded ranges of (a, b); the all-(a, b) statement is the uniform existence of these primitives on the entire decorated bicomplex.

Proof. By Lemma G.8, the existence of $(A^{(j)}_{a,b,N}, \tau_{a,b})$ gives $E_{a,b,N} - B_{a,b}(\tau_{a,b}) = \sum_{i,j} A^{(j)}_{a,b,N}\mu^j_i \in I_N$, so $[(T_{a,b}, E^+_{a,b})] = 0$ in coker $B_{a,b}$. Conversely, if the class vanishes, the cokernel computation produces such primitives by finite-rank linear algebra in the decorated bicomplex. The correspondence with clause (3) of Statement F.8 is the equality $\mathrm{Rad}_0(J_N(f)) = \Delta^{-1}S_N(f)\Delta$ read off from the Vandermonde-gauge expansion: the symmetric sum $\sum_i \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, \nabla^{\Delta}_i)$ equals $\Delta^{-1}S_N(f)\Delta$ by Lemma F.11, so the radial image of $J_N(f)$ is determined by clause (3) up to the quantum moment-map ideal, hence by injectivity (clause (2)). Equivalence of Routes A and B follows from the duality coker $B_{a,b} = (\ker B^*_{a,b})^\vee$ on the finite-rank decorated bicomplex. □

Remark G.10 *(All-bidegree obstruction).* Theorem G.9 converts the Statement F.8(3) into a constructible primitive-existence problem on a finite-rank decorated bicomplex. The remaining datum is the uniform construction of $(A^{(j)}_{a,b,N}, \tau_{a,b})$ for all (a, b); the finite exact-rational calculations of Appendix F give these primitives in the recorded ranges, and the all-(a, b) statement is the assertion that the bicomplex has a Lagrangian splitting compatible with the shears. This reformulation is mathematically stronger than the radial normalization because it identifies the obstruction with a coker-class on a concrete finite-rank Weyl target and identifies the all-N construction as a theorem; it makes that construction the existence of a contracting homotopy or a dual-kernel certification at each bidegree.

Remark G.11 *(Route B: dual kernel criterion).* The contracting homotopy of Theorem G.9 is Route A of the two acceptable proof routes; the equivalent Route B is the dual kernel criterion $(\varphi, -\lambda) \in \ker B^*_{a,b} \Rightarrow \lambda(E^+_{a,b}) - \varphi(T_{a,b}) = 0$, proved by pairing both sides with the same Vandermonde-conjugated functionals. Either route closes the radial obstruction in the bidegree under the homomorphism and injectivity hypotheses.

G.5 The pro-Matlis weighted Köthe envelope

The attempted strict all-window Bochner–Martinelli current transfer meets an obstruction in every finite window, given by the residue-coadjoint leak $z_1^{N+2} \cdot \rho_{N+1,0} = -(N+2)\rho_{0,1}$ (Theorem 8.40). The replacement target is the pro-Matlis tower; its analytic content is a continuous Hamiltonian action, which requires a topology stronger than the bare inverse limit of the finite Matlis windows.

Definition G.12 *(Weighted Köthe Hamiltonian space).* Fix $q > 1$. The weighted Köthe Hamiltonian space is

$$\mathfrak{h}_q = \Big\{ f = \sum_{a+b>0} f_{a,b} z_1^a z_2^b \in \mathbb{C}[[z_1, z_2]] \;:\; p_n(f) = \sup_{a,b} |f_{a,b}| q^{a+b}(1 + a + b)^n < \infty \;\forall\, n \Big\},$$

a Fréchet algebra under $\{p_n\}_{n\geq 0}$ and the Poisson bracket of $\mathfrak{h}$. The pro-Matlis dual is the strong continuous dual $\mathcal{P}^{\Pi}_q = \mathrm{Hom}^{\mathrm{cts}}_{\mathbb{C}}(\mathfrak{h}_q, \mathbb{C})$, with coordinate functionals dual to the $z_1^a z_2^b$ and topology dual to the $\{p_n\}$. It is the corresponding weighted strong-dual coefficient space. The finite Matlis windows project to it, but the bare product $\prod_{a+b>0} \mathbb{C}\rho_{a,b}$ is not identified with $\mathcal{P}^{\Pi}_q$.

PROPOSITION G.13 *(Continuity estimate).* There exist constants $C_n \geq 0$ and integers $r = r(n) \geq 0$ such that for all $f \in \mathfrak{h}_q$ and $\rho \in \mathcal{P}^{\Pi}_q$,

$$p^*_n(f \cdot \rho) \leq C_n\, p_{n+r}(f)\, p^*_{n+r}(\rho),$$

where $f \cdot \rho$ is the residue coadjoint action. Consequently the bracket $\mathfrak{h}_q \widehat{\otimes} \mathcal{P}_q^{\Pi} \to \mathcal{P}_q^{\Pi}$ is continuous, the action descends to a continuous $\mathfrak{h}_q$-module structure, and the modified coherence law $\pi_N(f \cdot \xi) = \pi_N(f \cdot \pi_{N+r(f)}\xi)$ holds for every $\xi \in \mathcal{P}_q^{\Pi}$.

Proof. The residue coadjoint action is $z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1}$ with negative-index terms zero (Theorem 5.179). For f, ρ of bidegree (p,q) and (a,b), the coefficient of $(f \cdot \rho)_{m,n}$ is $-(pb - qa + p - q) f_{p,q}\rho_{a,b}\delta_{m,a-p+1}\delta_{n,b-q+1}$, bounded by $C \cdot (p+q+a+b) \cdot |f_{p,q}| \cdot |\rho_{a,b}|$. Summing over (p,q) and (a,b) of fixed total $m+n+1$ gives

$$\left|(f \cdot \rho)_{m,n}\right| q^{m+n+1}(1+m+n)^n \le C(m+n+1) \sum_{p+a=m+1} |f_{p,q}|\,|\rho_{a,b}| q^{p+q+a+b} \cdots,$$

and standard estimates in Köthe spaces [55] bound this by $C_n\, p_{n+r}(f)\, p^*_{n+r}(\rho)$ for some integer shift $r \ge 1$. The shift r is the content of "current transfer is coherent only in the pro-tower": every action by f loses finitely many low modes, and these are recovered only after truncating one window deeper. The modified coherence law follows by applying π_N and using the continuity of the bracket. □

Theorem G.14 *(Native pro-Matlis replacement target).* The pair $(\mathfrak{h}_q, \mathcal{P}_q^{\Pi})$ is the chosen replacement target for formal Hamiltonian actions on residue-current systems with Bochner–Martinelli leakage. It carries a continuous coadjoint action and the modified finite-window coherence law of Proposition G.13. The theorem does not assert terminality in the category $\mathrm{Curr}_{\mathrm{BM}}$ of inverse systems $(\rho_N : \mathcal{P}_N \twoheadrightarrow \mathcal{P}_{N-1})$ of $\mathfrak{h}_q$-modules with $\mathfrak{h}_q$-equivariant truncations and continuous inverse limit.

Proof. The truncations $\pi_N : \mathcal{P}_q^{\Pi} \to \mathcal{P}_{\le N}$ satisfy the modified coherence law $\pi_N(f \cdot \xi) = \pi_N(f \cdot \pi_{N+r(f)}\xi)$ of Proposition G.13. Continuity of the bracket on $\mathcal{P}_q^{\Pi}$ gives a coherent action on the weighted pro-tower. Finite windows are not strict modules for the full coadjoint action; the theorem gives the corrected target in which the window loss is recorded by the shift $r(f)$. □

Remark G.15 *(The category determines the replacement datum).* The category $\mathrm{Curr}_{\mathrm{BM}}$ has not been shown to have $\mathcal{P}_q^{\Pi}$ as a terminal object. Direct sums $\mathcal{P}_q^{\Pi} \oplus W$ with spectator modules show why the replacement target must be specified together with its projections and coherence law.

G.6 Bracket-admissible B and kernel-admissible $\mathcal{K}_B$

Levels 2 and 3 of the CE/PV ladder hold under a bracket-admissible subspace $B \subset \mathrm{PV}_{\mathrm{coord}}(A^{\mathrm{cl}}_{\partial,\mathrm{coord}})$ and a kernel-admissible completion $\mathcal{K}_B$ in which the diagonal Casimir converges. The pro-Matlis weighted Köthe topology gives both.

Definition G.16 *(Bracket-admissible B_q).* Fix $q > 1$. Let $B_q \subset \mathrm{PV}_{\mathrm{coord}}(A^{\mathrm{cl}}_{\partial,\mathrm{coord}})$ be the closure of the polynomial subalgebra $\mathbb{C}[O_f, \theta_f, d_\pi O_f \; : \; f \in \mathfrak{h}_q]$ in the weighted Köthe topology of Definition G.12. Then B_q is complete Hausdorff, closed under multiplication, closed under the Schouten bracket $[B_q, B_q]_{\mathrm{Sch}} \subset B_q$ (because each generator is itself a continuous functional with bracket compatible to the Köthe semi-norms), closed under the de Rham differential $d_\pi B_q \subset B_q$, and contains $O_f, \theta_f, d_\pi O_f$ for every $f \in \mathfrak{h}_q$. The continuous inclusion $B_q \hookrightarrow \mathrm{PV}_{\mathrm{coord}}$ detects the coordinate generators.

Theorem G.17 *(P_0-bracket lift on B_q).* The reduced CE/PV cochain map Φ_{coord} of Theorem 4.20 restricts to a P_0-isomorphism $\Phi^{B_q}_{P_0} : \Phi^{-1}_{\mathrm{coord}}(B_q) \xrightarrow{\sim} B_q$ onto the bracket-admissible B_q.

Proof. By Definition G.16, B_q is complete Hausdorff, multiplicatively closed, Schouten-closed, d_π-closed, and contains the coordinate generators $O_f, \theta_f, d_\pi O_f$ for $f \in \mathfrak{h}_q$. Each of the five conditions in Definition 4.18 is satisfied; the P_0-bracket lift criterion of Theorem 5.5(2) applies and produces the isomorphism. □

Definition G.18 *(Kernel-admissible completion $\mathcal{K}_{B_q}$).* Let $\mathcal{K}_{B_q} = \mathrm{Conv}^{\mathrm{cts}}_{\mathsf{C}}\big(C^{\mathrm{CE}}_\bullet(\mathfrak{g}), B_q\big)$ be the continuous convolution dg Lie algebra of CE chains of $\mathfrak{g} = \mathfrak{h}_q \ltimes \mathcal{P}^{\Pi}_q[1]$ into B_q, with bracket and differential induced by the operad structure on B_q. The diagonal tensor $\Theta_{B_q} = \sum_I f_I \otimes \theta_{f_I} + \sum_I \eta_I \otimes O_I$ lies in $\mathfrak{h}_q \widehat{\otimes} B_q$ by the Köthe continuity estimate (Proposition G.13), and the sum converges in $\mathcal{K}_{B_q}$.

THEOREM G.19 *(Maurer–Cartan kernel).* The diagonal tensor Θ_{B_q} satisfies the Maurer–Cartan equation $d\Theta_{B_q} + \frac{1}{2}[\Theta_{B_q}, \Theta_{B_q}]_{\mathcal{K}_{B_q}} = 0$ in the kernel-admissible completion $\mathcal{K}_{B_q}$ of Definition G.18.

Proof. The bracket $[\Theta_{B_q}, \Theta_{B_q}]$ decomposes into the Schouten bracket of the bracket-admissible subspace $[B_q, B_q]_{\mathrm{Sch}} \subset B_q$ and the residue pairing of $\mathfrak{h}_q$ and $\mathcal{P}^{\Pi}_q$, each computed by the formula $\{O_f, O_g\}_{\mathrm{Sch}} = O_{\{f,g\}}$ in the bracket-admissible B_q. Continuity of the bracket (Proposition G.13) makes the infinite sum converge, and the Maurer–Cartan equation reduces to the Jacobi identity in $\mathfrak{h}_q$ plus the operadic P_0-compatibility on B_q. □

G.7 The Tate bar-cobar admissible envelope

Level 4 of the CE/PV ladder is the bar-cobar comparison $\Omega C^{\mathrm{CE}}_\bullet(\mathfrak{g}_{\mathrm{adm}}) \xrightarrow{\sim} U(\mathfrak{g}_{\mathrm{adm}})$ for an admissible Lie algebra replacement.

Definition G.20 *(Admissible Tate envelope).* The admissible Tate envelope of $\mathfrak{g} = \mathfrak{h}_q \ltimes \mathcal{P}^{\Pi}_q[1]$ is the inverse system $\mathfrak{g}_{\mathrm{adm}} = \varprojlim_M \mathfrak{g}_M$ of finite nilpotent quotients $\mathfrak{g}_M = \mathfrak{h}_q/\mathfrak{h}^M_q \ltimes \mathcal{P}_{\leq M}[1]$ with strict transition maps $\mathfrak{g}_{M+1} \twoheadrightarrow \mathfrak{g}_M$. The CE chain complex is $C^{\mathrm{CE}}_\bullet(\mathfrak{g}_{\mathrm{adm}}) = \varprojlim_M C^{\mathrm{CE}}_\bullet(\mathfrak{g}_M)$, conilpotent in the filtration by the kernel of the augmentation, and the universal enveloping algebra is $U(\mathfrak{g}_{\mathrm{adm}}) = \varprojlim_M U(\mathfrak{g}_M)$, PBW-complete in the same filtration.

LEMMA G.21 *(Milnor exact sequence for $C^{\mathrm{CE}}_\bullet$).* For the Tate envelope $\mathfrak{g}_{\mathrm{adm}}$ of Definition G.20,

$$0 \to \varprojlim_M{}^1 H^{\bullet-1}(C^{\mathrm{CE}}_\bullet(\mathfrak{g}_M)) \to H^\bullet(C^{\mathrm{CE}}_\bullet(\mathfrak{g}_{\mathrm{adm}})) \to \varprojlim_M H^\bullet(C^{\mathrm{CE}}_\bullet(\mathfrak{g}_M)) \to 0,$$

and $\varprojlim^1 H^{-1}(C^{\mathrm{CE}}_\bullet(\mathfrak{g}_M)) = 0$ because the transition maps are surjective on H^0 and the Mittag–Leffler condition holds in that degree. In other degrees the $\varprojlim^1$-term is part of the completed comparison datum unless degreewise Mittag–Leffler hypotheses hold.

Proof. The strict transition maps $\mathfrak{g}_{M+1} \twoheadrightarrow \mathfrak{g}_M$ induce surjections on the corresponding CE chain complexes, hence the Mittag–Leffler condition holds in degree 0; the basic Milnor exact sequence (axiom C1) gives the displayed sequence and the $\varprojlim^1$-vanishing in that degree. The displayed exact sequence retains the remaining derived-limit terms. □

THEOREM G.22 *(Tate bar-cobar comparison with derived-limit obstruction).* At each finite quotient the map

$$\Omega C^{\mathrm{CE}}_\bullet(\mathfrak{g}_M) \to U(\mathfrak{g}_M)$$

is a quasi-isomorphism. The completed map $\Omega C^{\mathrm{CE}}_\bullet(\mathfrak{g}_{\mathrm{adm}}) \to U(\mathfrak{g}_{\mathrm{adm}})$ is a quasi-isomorphism in C only after the derived-limit obstruction groups for the bar-cobar and enveloping systems vanish and the completed bar-cobar functor commutes with the chosen inverse limit.

Proof. Each $\mathfrak{g}_M$ is finite nilpotent, hence the bar-cobar comparison $\Omega C^{\mathrm{CE}}_\bullet(\mathfrak{g}_M) \to U(\mathfrak{g}_M)$ is a quasi-isomorphism of dg algebras [42][7]. The Milnor sequence of Lemma G.21 shows the exact derived-limit terms that must be controlled before the finite-quotient comparison passes to the inverse limit. Under degreewise Mittag–Leffler vanishing for both systems and compatibility of bar-cobar with the chosen completion, the inverse limit map is a quasi-isomorphism in the filtered topology of C. □

G.8 Open-trace A_∞-Koszul homotopy datum

Level 5 of the CE/PV ladder requires an $(A_{\mathrm{op}}, C_{\mathrm{op}}, B_{\mathrm{CE/PV}}, \tau_{\mathrm{op}})$ datum.

Definition G.23 *(Open-trace A_∞-Koszul datum).* The open-trace filtered cyclic A_∞ algebra is $A_{\mathrm{op}} = A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}(\mathfrak{gl}_N, R^{\mathrm{poly}}_N)$ of Definition 1.24, with cyclic structure $\mathrm{Tr}_{\mathrm{open}} : A_{\mathrm{op}} \to \mathbb{C}[-1]$ given by the ghost-zero matrix trace $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ extended cyclically. The Koszul-dual conilpotent filtered coalgebra is $C_{\mathrm{op}} = BA_{\mathrm{op}} = T(s\overline{A}_{\mathrm{op}})$ (the bar construction), with the filtration by tensor length. The bracket-admissible CE/PV target is $B_{\mathrm{CE/PV}} = B_q$ of Definition G.16. The twisting cochain is $\tau_{\mathrm{op}} : C_{\mathrm{op}} \to A_{\mathrm{op}}$, defined on C_{op} by composing the bar-cobar comparison $\tau : BA \to A$ with the cyclic structure (Loday–Quillen–Tsygan [2][3]).

Theorem G.24 *(Twisting cochain τ_{op} satisfies MC).* The twisting cochain τ_{op} of Definition G.23 satisfies the Maurer–Cartan equation

$$d\tau_{\mathrm{op}} + \mu_2(\tau_{\mathrm{op}} \otimes \tau_{\mathrm{op}})\Delta_2 + \sum_{k\geq 3} \mu_k(\tau_{\mathrm{op}}^{\otimes k})\Delta_k = 0,$$

and the bar-cobar comparison $\Omega C_{\mathrm{op}} \xrightarrow{\sim} A_{\mathrm{op}}$ is a filtered cyclic A_∞ quasi-isomorphism. The induced map $H^0(A_{\mathrm{op}})_{\mathrm{prim,conn}} \xrightarrow{\sim} \mathrm{span}\{J(f) : f \in \mathfrak{h}_q\}$ identifies the primitive connected open-trace cohomology with the linear span of the trace maps.

Proof. The twisting-cochain equation is the Maurer–Cartan condition for the bar-cobar pair (BA, A) in the convolution dg Lie algebra $\mathrm{Conv}(BA, A)$ [24]; for the cyclic-augmented case the cyclic compatibility adds the higher Koszul terms while preserving the Maurer–Cartan condition. The bar-cobar comparison is the standard Koszul duality quasi-isomorphism for the cyclic A_∞ operad [56], transported through the filtration by tensor length. The primitive connected identification is the Loday–Quillen–Tsygan theorem applied to the trace algebra of $\mathfrak{gl}_N$-modules, restricted to the connected primitive sector by the single-trace projection. □

G.9 Hochschild equivariance homotopies for Φ_N

The reduced CE/PV map $c_f \mapsto \theta_f, u_f \mapsto O_f = J(f)$ is the generator-level shadow of the chain map $\Phi_N : \mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk},0} \to Z^{\mathrm{der}}_{E_1}(A^{\mathrm{cl}}_{\partial,N})$. The derived E_1-centre is $Z^{\mathrm{der}}_{E_1}(A) \simeq CC^\bullet(A, A)$, so the chain map must produce actual Hochschild cochains and centrality homotopies, extending the derivation and multiplication data.

Definition G.25 *(Centre-map cochains and homotopies).* On generators of $\mathfrak{g} = \mathfrak{h}_q \ltimes \mathcal{P}^\Pi_q[1]$, define

$$\Phi_N(c_f) = \Theta^{(1)}_f \in CC^1(A_{\mathrm{op}}, A_{\mathrm{op}}), \qquad \Theta^{(1)}_f(a) = \theta_f(a),$$

the Hochschild 1-cochain given by the Schouten action of c_f on observables; and

$$\Phi_N(u_f) = M_{J(f)} \in CC^0(A_{\mathrm{op}}, A_{\mathrm{op}}), \qquad M_{J(f)}(1) = J(f),$$

multiplication by the boundary observable. For the cotangent sector $\rho \in \mathcal{P}_q^{\Pi}$ define $\Phi_N(\eta_\rho) = \Theta_\rho^{\mathrm{BV}}$ given by the brane-defect propagator extension (§G.12). Equivariance data $H_{x,a}^{\mathrm{prod}}, H_{x,a}^{P_0}, H_3, \ldots \in CC^\bullet(A_{\mathrm{op}}, A_{\mathrm{op}})$ are required to satisfy the Hochschild action equations

$$[\Phi_N(x), M_a]_G - M_{\rho_x(a)} = dH_{x,a}^{\mathrm{prod}}, \qquad \{\Phi_N(x), M_a\}_{P_0} - M_{\rho_x^{P_0}(a)} = dH_{x,a}^{P_0}$$

for every closed-side generator x and every open observable a, where ρ_x and $\rho_x^{P_0}$ are the induced closed-on-open actions. Actual centre classes are the special case in which these actions vanish. The secondary Jacobiator dH_3 = associator/Jacobiator of the chosen primary homotopies.

THEOREM G.26 *(Hochschild equivariance criterion on the bracket-admissible sector).* A Hochschild lift of the bracket-admissible PV sector B_q is exactly a continuous assignment of Hochschild cochains $\Theta_f^{(1)}, M_{J(f)}$ and homotopies $H_{f,g}, H_{f|g}, H_{f,g}^{\mathrm{mult}}$, compatible with the weighted Köthe filtration, satisfying

$$[\Theta_f^{(1)}, \Theta_g^{(1)}]_G = \Theta_{\{f,g\}}^{(1)} + bH_{f,g}, \qquad [\Theta_f^{(1)}, M_{J(g)}]_G = M_{J(\{f,g\})} + bH_{f|g}, \qquad [M_{J(f)}, M_{J(g)}]_G = 0 + bH_{f,g}^{\mathrm{mult}},$$

for all $f, g \in \mathfrak{h}_q$, together with the corresponding Jacobiator homotopy H_3. For such a lift the obstruction component $\mathrm{ob}_{\mathrm{cent}}^{\mathrm{prod}} \in H^2(\mathfrak{K}_{\mathrm{cent}})$ of the four-component $\mathrm{Ob}_{\mathrm{cent/QME/desc}}$ vanishes on the bracket-admissible sector. Conversely, vanishing of this component is the obstruction-theoretic statement that such Hochschild homotopies may be chosen in the selected Hochschild model.

Proof. The definition of $\mathfrak{K}_{\mathrm{cent}}$ is the mapping cone for the Gerstenhaber product, the P_0-bracket, and the first Jacobiator of the candidate centre map. The displayed equations are the degree-two components of its Maurer–Cartan equation after the PV identities on B_q have been transported to the chosen Hochschild model. Thus a choice of $H_{f,g}, H_{f|g}, H_{f,g}^{\mathrm{mult}}$ and H_3 is precisely a primitive for $\mathrm{ob}_{\mathrm{cent}}^{\mathrm{prod}}$. No assertion is made that the PV closure of B_q alone constructs these Hochschild cochains; the construction of the Hochschild model is part of the lift. □

Remark G.27 *(Cotangent centrality and P_0-bracket lifts).* The cotangent centrality $\{\Phi_N(\eta_\rho), a\}_{P_0} = dh_{\rho,a}^{P_0}$ and the unmixed multiplication centrality $\Phi_N(\eta_\rho) \cdot a - (-1)^{|\eta_\rho||a|} a \cdot \Phi_N(\eta_\rho) = dh_{\rho,a}^{\mathrm{prod}}$ require the brane-defect propagator extension and the given principal-part current cochains $T_\rho \in CC^\bullet(\mathrm{Obs}_{\partial,N}, \mathrm{Obs}_{\partial,N})$ of §G.12.

G.10 Protected-sector cyclic BV strong deformation retract

The local theorem on $\mathbb{R}^2 \times \mathbb{C}^2$ is presented as a formal Hamiltonian stalk; for the universal completed theorem the local fields must arise from a parent string-field-theory target by an actual deformation retract.

Definition G.28 *(Admissible protected target).* An admissible protected mixed holomorphic-topological string-field target is a tuple $\mathcal{T} = (Y, \Omega_Y, \mathcal{B}, p, \mathcal{E}_T, S_T, \omega_T, Q, C_{\mathcal{T}})$ where Y is the target geometry, Ω_Y is the holomorphic-symplectic volume datum, $\mathcal{B}$ is the brane category, $p \in Y$ is the brane point, $(\mathcal{E}_T, Q_T, \omega_T, S_T)$ is the parent BV theory, Q is the protected supercharge or BRST operator, and $C_{\mathcal{T}}$ is the matched-conventions datum (§11.4).

Definition G.29 *(Protected-sector cyclic BV strong deformation retract).* A protected-sector cyclic BV strong deformation retract for a target $\mathcal{T}$ is a triple of Q_T-equivariant continuous chain maps and homotopy (i_T, p_T, h_T): $i_T : \mathcal{E}_H \hookrightarrow \mathcal{E}_T$, $p_T : \mathcal{E}_T \twoheadrightarrow \mathcal{E}_H$, $h_T : \mathcal{E}_T \to \mathcal{E}_T[-1]$, satisfying

$$p_T i_T = \mathrm{id}_{\mathcal{E}_H}, \qquad Q_T h_T + h_T Q_T = \mathrm{id}_{\mathcal{E}_T} - i_T p_T, \qquad i_T^* \omega_T = \omega_H, \qquad h_T^\vee = \pm h_T$$

relative to the BV pairing. The transferred interaction is $S_T^{\mathrm{trans}} = S_H + \sum_{k\geq 3} S_k^{\mathrm{trans}}$, with the L_∞-transferred higher operations S_k^{trans} computed by the Homological Perturbation Lemma applied to (i_T, p_T, h_T).

Theorem G.30 *(Protected-sector obstruction).* For an admissible protected target $\mathcal{T}$ and a protected-sector strong deformation retract (SDR) (i_T, p_T, h_T), the protected-sector obstruction

$$\mathrm{Ob}_{\mathrm{prot}}(\mathcal{T}) = \Big[\sum_{k\geq 3} S_k^{\mathrm{trans}}\Big] \in H^1\big(\mathfrak{K}_{\mathrm{prot}}(\mathcal{T})\big), \qquad \mathfrak{K}_{\mathrm{prot}}(\mathcal{T}) = \mathrm{Def}_{\mathrm{cycBV}}(\mathcal{E}_H \rightleftarrows \mathcal{E}_T),$$

is the curvature of the transferred interaction in the cyclic BV deformation complex. Strict universality $\mathrm{Ob}_{\mathrm{prot}}(\mathcal{T}) = 0$ holds when the higher S_k^{trans} vanish; counterterm universality $\mathrm{Ob}_{\mathrm{prot}}(\mathcal{T}) = d_{\mathfrak{K}} \tau_{\mathrm{prot}}$ holds when they are exact in the local-functional complex; projective universality holds when they admit a central representative.

Proof. The Homological Perturbation Lemma [58][57] applied to the SDR (i_T, p_T, h_T) produces S_k^{trans} by the explicit formula $S_k^{\mathrm{trans}} = p_T \circ S_T \circ \cdots \circ S_T \circ h_T \circ S_T \circ i_T$ with $k-1$ propagators h_T interleaved by S_T. The cyclicity hypothesis $h_T^\vee = \pm h_T$ implies S_k^{trans} is cyclic. The deformation complex $\mathfrak{K}_{\mathrm{prot}}(\mathcal{T})$ records the obstruction for the transferred interaction to equal S_H; its first cohomology class is the obstruction. The three universality cases follow from the obstruction-twist dichotomy (Theorem 7.1) applied to $\mathfrak{K}_{\mathrm{prot}}$. □

Remark G.31 *(Existence of the protected retract).* Theorem G.30 gives the obstruction calculus once an SDR exists. For a specific compact CY_3 target an admissible protected SDR is part of the compact field theory and is constrained by the matched-conventions obstruction (§11.4). The local theorem is the formal Hamiltonian normal-slice test of the bulk produced by such an SDR.

G.11 Scalar-contact projection and the QME counterterm tower

The reduced non-scalar Costello QME class $[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]$ is defined on $\ker \sigma_{\mathrm{sc}}$ for a filtered chain map σ_{sc}, refining the associated graded scalar symbol. The construction below gives σ_{sc} and the counterterm tower.

Definition G.32 *(Scalar-contact filtration and projection).* The scalar-contact filtration $F^\bullet \mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$ is generated by the powers of the scalar Capelli line $\mathbb{C}\cdot \mathbf{1} \subset \mathfrak{h}_{\hbar,N}$: F^r consists of local functionals whose expansion in the Capelli generators contains at least r copies of the scalar. The associated graded is $\mathrm{gr}^\bullet_F \mathfrak{Q}^\bullet_{w,\partial,\hbar}(I)$, with associated-graded scalar symbol $\sigma^{\mathrm{gr}}_{\mathrm{sc}} : \mathrm{gr}^\bullet_F \mathfrak{Q}^\bullet_{w,\partial,\hbar}(I) \to C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]]$. The filtered chain map is

$$\sigma_{\mathrm{sc}} : \mathfrak{Q}^\bullet_{w,\partial,\hbar}(I) \longrightarrow C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]],$$

with $\mathrm{gr}_F(\sigma_{\mathrm{sc}}) = \sigma^{\mathrm{gr}}_{\mathrm{sc}}$.

Theorem G.33 *(Filtered chain map σ_{sc} and obstruction tower).* On an unreduced polynomial or balanced branch with a fixed chain scalar projection, work in a separated complete filtration for which the order-r correction lies in F^r. The associated-graded scalar symbol $\sigma_{\mathrm{sc}}^{\mathrm{gr}}$ lifts to a filtered chain map σ_{sc} inside this fixed filtered complex if and only if the scalar-lift obstruction tower $o_{\sigma,w}^{(r)}(I) \in H^1\big(\mathrm{Hom}(\mathrm{gr}_F^r \mathfrak{Q}_{w,\partial,\hbar}^\bullet(I), C_{\mathrm{Lie}}^\bullet(\bar{A}; \mathbb{C}\gamma_1)[1][[\hbar]])_{-r}\big)$ vanishes for every $r \geq 0$. At order $r = 1$, the obstruction $o_{\sigma,w}^{(1)}$ is the Capelli scalar cocycle $\omega(f, g) = [\{f, g\}]_0$. On the unreduced polynomial algebra $\mathfrak{h}_{\mathrm{poly}} = \mathbb{C}[z_1, z_2]$ it has primitive $\eta(f) = -[f]_0$ with $d_{\mathrm{CE}}\eta = \omega$. On the reduced Hamiltonian algebra $\bar{A}$ the class is $[\bar{c}] \in H_{\mathrm{Lie}}^2(\bar{A}, \mathbb{C})$ and is retained as a projective central class. At higher orders, the obstruction lies in the iterated extension of the Capelli scalar line and is the class defined by the preceding choices in the exact couple of the filtration. In the finite-window prototype θ_3 row, the displayed obstruction is killed after adjoining a scalar-zero windowwise primitive ($d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}, c = 1$), confirming the enlarged construction in that row; the all-order case is the separate assertion that the same windowwise data extend Milnor-compatibly across the inverse system of windows (Theorem G.36).

Proof. By Definition G.32, the obstruction to lifting $\sigma_{\mathrm{sc}}^{\mathrm{gr}}$ by one filtration order is a class in $H^1\big(\mathrm{Hom}(\mathrm{gr}_F^r, C_{\mathrm{Lie}}^\bullet)_{-r}\big)$. At order $r = 1$ the class is $[\omega]$. The primitive η exists only on the unreduced algebra (Lemma 6.2); after reduction the surviving class is $[\bar{c}]$ and the target must be the central extension. At higher orders the class lives on the r-fold symmetric product of the Capelli scalar line. The sequence of corrections converges in the filtered topology because the r-th correction lies in F^r and the filtration is separated and complete (Definition G.1); without this pronilpotence the statement is only finite order. The all-order existence reduces to the Milnor compatibility of the windowwise counterterms across the inverse system of finite Hamiltonian windows, controlled by Theorem G.36. □

Definition G.34 *(Residual non-scalar complex and obstruction).* The residual non-scalar complex is $\mathfrak{Q}_{\mathrm{red}}^\bullet(I) = \ker \sigma_{\mathrm{sc}}$ of Theorem G.33. The reduced non-scalar Costello QME class is $[\mathrm{Ob}_{w,\partial,\hbar}^{\mathrm{red}}] \in H^1(\mathfrak{Q}_{\mathrm{red}}^\bullet(I), d_{\mathrm{QME}})$, defined as the curvature of the reduced interaction in the kernel of the scalar-contact projection.

G.11.1 Finite-window QME counterterm recursion

Definition G.35 *(Finite-window residual complex).* For each Hamiltonian window M and each loop/order n, let $Q_{w,M}^\bullet(I) = \ker(\sigma_{\mathrm{sc},M}) \subset O_{\mathrm{loc}}^{\mathrm{bd}}(\mathcal{E}_w^M|_I; A_{\partial,\hbar}^{\mathrm{pp},w,M}(I))$. At order n the window obstruction is $\mathfrak{o}_{n,M}^{\mathrm{ns}} \in Q_{w,M}^1(I)$. A compatible counterterm system is $\{C_{n,M} \in Q_{w,M}^0(I)\}_{n,M}$ with $d_M C_{n,M} = -\mathfrak{o}_{n,M}^{\mathrm{ns}}$ and $\pi_{M,N} C_{n,M} = C_{n,N}$ for $M \geq N$.

THEOREM G.36 *(Milnor obstruction vector and counterterm criterion).* Assume the finite-window residual complexes form an inverse system closed under the boundary and bracket terms of the QME, the graph weights are summable in the chosen weighted topology, and the residuals $\mathfrak{o}_{n,M}^{\mathrm{ns}}$ are compatible under truncation. The obstruction to a compatible counterterm system at fixed order n is the Milnor pair

$$\mathfrak{D}_n^{\mathrm{ns}} = \big(([\mathfrak{o}_{n,M}^{\mathrm{ns}}])_M, \lambda_n \big) \in \varprojlim_M H^1(Q_{w,M}^\bullet(I)) \oplus \varprojlim_M{}^1 H^0(Q_{w,M}^\bullet(I)),$$

where the first component is the windowwise primitive class and the second is the Milnor compatibility class. Under the displayed finite-to-limit hypotheses, an all-order QME counterterm tower as a formal $\hbar$-adic inverse-limit family exists if and only if $\mathfrak{D}_n^{\mathrm{ns}} = 0$ for every $n \geq 1$; when this holds the formal counterterm $C_{\hbar,w}^{\mathrm{red}} = \sum_{n\geq 1} \hbar^n C_n$, with $C_n \in \mathbf{R}\varprojlim_M Q_{w,M}^0(I)$, solves the QME $d_{\mathrm{QME}} C_{\hbar,w}^{\mathrm{red}} +$

$\frac{1}{2}\{C^{\mathrm{red}}_{\hbar,w}, C^{\mathrm{red}}_{\hbar,w}\} + \cdots = -\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}$ at all orders as a formal power series. No analytic convergence in $\hbar$ is asserted.

Proof. The Milnor exact sequence (axiom C1) for the inverse system $\{Q^\bullet_{w,M}\}$ is

$$0 \to \varprojlim_M{}^1 H^0(Q^\bullet_{w,M}) \to H^1(\mathbf{R}\varprojlim_M Q^\bullet_{w,M}) \to \varprojlim_M H^1(Q^\bullet_{w,M}) \to 0.$$

The window obstructions $[\mathfrak{o}^{\mathrm{ns}}_{n,M}]$ form a class in $\varprojlim_M H^1(Q^\bullet_{w,M})$; they are killed by an inverse system of windowwise counterterms $\{C_{n,M}\}$ precisely when this class is zero. The Milnor compatibility class $\lambda_n \in \varprojlim^1 H^0(Q^\bullet_{w,M})$ measures the obstruction to compatibility of the windowwise counterterms across windows; it vanishes when the truncations $\pi_{M,N}$ are compatible with the chosen primitives. The combined vanishing $\mathfrak{D}^{\mathrm{ns}}_n = 0$ is the fixed-order inverse-limit criterion; the all-order statement requires the same condition for all n. For the prototype θ_3 row, after a scalar-zero local counterterm $C_{\theta_3,M}$ with $d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$ is adjoined to the finite-window complex, the coefficient $c = 1$ is windowwise primitive and Milnor-compatible, so $\mathfrak{D}^{\mathrm{ns}}_3 = 0$ for that given row (Theorem E.38). ☐

G.12 Costello brane-defect graph QME

The brane-defect Costello graph QME requires a heat-kernel propagator with controlled wavefront, removable divergences, the RG equation, and the QME after counterterms.

Definition G.37 *(Brane-defect heat-kernel propagator).* Fix a flat metric on $\mathbb{R}^2$, a Hermitian metric on $\mathbb{C}^2$, and the gauge-fixing operator $Q^{\mathrm{GF}} = d^*_{\mathbb{R}^2} + \bar\partial^*_{\mathbb{C}^2}$ on $\mathcal{E}_H$. The generalized Laplacian is $D = [Q_H, Q^{\mathrm{GF}}] = \Delta_{\mathbb{R}^2} + \Delta_{\mathbb{C}^2} + L_{\mathfrak{g}}$, with $L_{\mathfrak{g}}$ the CE Laplacian of $\mathfrak{g}$. The heat kernel is $K_t = e^{-tD}$, and the brane-defect propagator at scale (ε, L) is

$$P_{\varepsilon,L} = \int_\varepsilon^L \tfrac{1}{2}(Q^{\mathrm{GF}} \otimes 1 + 1 \otimes Q^{\mathrm{GF}})\, K_t\, dt,$$

restricted to the brane-defect support stratum $L \times L \subset X \times X$.

Theorem G.38 *(Wavefront control of $P_{\varepsilon,L}$ on $L \times L$).* The propagator restriction $P_{\varepsilon,L}|_{L\times L}$ is a distributional kernel with wavefront set $\mathrm{WF}(P_{\varepsilon,L}|_{L\times L}) \subset \{((t_1,t_2),(\xi,-\xi)) : t_1 = t_2\}$, the conormal of the diagonal in $L \times L$. In particular $\mathrm{WF}(P_{\varepsilon,L}|_{L\times L})$ is compatible with distributional current insertions in $\mathcal{D}'_c(L)\widehat{\otimes}\mathcal{P}^\Pi_q$ by axiom C4: products of two unregularized defect distributions are excluded.

Proof. The heat kernel K_t on $X = \mathbb{R}^2 \times \mathbb{C}^2$ factorizes as $K_t^{\mathbb{R}^2} \otimes K_t^{\mathbb{C}^2} \otimes \mathrm{id}_{\mathfrak{g}}$, each factor with explicit Gaussian and Mehler-type fundamental solutions. Restriction to $L \times L = \mathbb{R}_t \times \mathbb{R}_t$ integrates out the $\mathbb{R}_s$-direction and the holomorphic $\mathbb{C}^2$-directions, producing a kernel on $\mathbb{R}_t \times \mathbb{R}_t$ whose singularity is concentrated at the diagonal. The wavefront set is the conormal of the diagonal by general elliptic-operator theory [36]. Compatibility with distributional current insertions follows from C4. ☐

Theorem G.39 *(Graph weights and QME after counterterms).* For the brane-defect interaction $I = I_{\mathrm{Ham}}$ of the local Hamiltonian BF theory and the propagator $P_{\varepsilon,L}$ of Definition G.37, assume a brane-defect Costello locality theorem for this support stratum, a fixed chain scalar projection, finite row arrays, graph summability/wavefront admissibility, and $\mathfrak{D}^{\mathrm{ns}}_n = 0$ for every order n, with compatible primitives. Then the given graph weights $W_\Gamma(P_{\varepsilon,L}, I)$ admit local asymptotic expansions as $\varepsilon \to 0$; divergences are removable by local counterterms $C_{\hbar,w,I}$; the renormalized limit $\lim_{\varepsilon\to 0} \sum_\Gamma \hbar^{b_1(\Gamma)}/|\mathrm{Aut}\,\Gamma| W_\Gamma$ exists in the weighted Tate topology of $\mathfrak{K}_{\mathcal{T}}$; wavefront sets restrict cleanly by Theorem G.38; and the renormalized effective interaction $I[L]$ satisfies the renormalized quantum master equation

$$Q_H I[L] + \tfrac{1}{2}\{I[L], I[L]\}_L + \hbar\Delta_L I[L] = 0$$

in the corrected residual complex. Before the compatible primitives are inserted, the curvature is the residual non-scalar QME class $\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}$ of Definition G.34. The bulk-to-defect kernel $\kappa_{\hbar,w,I} : C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}^{\mathrm{cur}}_{w,\hbar,I}) \to \mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I)$ is defined on generators by $u_{a(t)dt\otimes f} \mapsto B^{\mathrm{BV}}_f(a)$, $c_{B,\rho} \mapsto \Theta^{\mathrm{BV}}_{\rho}(B)$, and the counterterm $C_{\hbar,w,I} = \sum_n \hbar^n C_{n,w,I}$ of Theorem G.36 produces the QME-compatible kernel $K_{\hbar,w,I} = \kappa_{\hbar,w,I} + C_{\hbar,w,I}$ satisfying $d_{\mathrm{QME}}K_{\hbar,w,I} + \frac{1}{2}[K_{\hbar,w,I}, K_{\hbar,w,I}]_{\hbar} = 0$.

Proof. Local asymptotic expansion as $\varepsilon \to 0$ is the given brane-defect version of Costello's renormalization theorem [12, Thm. 6.1.1]; the expansion has only finitely many non-renormalizable divergences in each loop order, removable by local counterterms in $\mathcal{O}^{\mathrm{bd}}_{\mathrm{loc}}$ by the locality hypothesis. The renormalized limit converges in the weighted Tate topology because the Köthe weights p_n of Definition G.12 dominate the heat kernel decay (Theorem 5.58, applied to the brane-defect restriction). Wavefront control is Theorem G.38. The QME after compatible counterterms is the standard Costello effective-action calculus [12, Ch. 5] applied to the brane-defect support stratum, with the residual non-scalar class $\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}$ measuring the curvature before the compatible primitive system is inserted. The kernel construction $K_{\hbar,w,I} = \kappa_{\hbar,w,I} + C_{\hbar,w,I}$ follows from the Milnor exact sequence (Theorem G.36) once the windowwise counterterms are compatible. □

Remark G.40 *(θ_3 prototype).* The order-three CE-source row θ_3 is the prototype: in each finite window $M \ni \theta_3$ the bare $H^1 = \mathbb{C}\cdot \mathsf{E}_{\theta_3,M}$ is killed in the enlarged complex after a scalar-zero local counterterm $C_{\theta_3,M}$ with $d_M C_{\theta_3,M} = -\mathsf{E}_{\theta_3,M}$ is adjoined and taken with coefficient $c = 1$; the counterterms are Milnor-compatible because the truncations preserve the scalar normalization, and the corresponding component of the all-order $C_{\hbar,w,I}$ is the windowwise sum of $C_{\theta_3,M}$ (Theorem E.38). This is the explicit construction of the QME-compatible kernel on the order-three row.

G.13 Stratified factorization algebra on (X, L)

The closed-open object is a stratified factorization algebra: $\mathrm{Obs}^{\mathrm{str}}(X, L)$; Weiss descent relates the prefactorization datum (Appendix B) to the full factorization theory.

THEOREM G.41 *(Stratified factorization on (X, L)).* The bulk observables Obs_X, the brane-line observables $\mathrm{Obs}_L = \mathrm{Obs}_{\partial,N}$, the bulk-to-brane collision module $\mathcal{M}_{X/L}$, the extension-by-zero functor $j_* : \mathrm{Obs}_{X\setminus L} \to \mathrm{Obs}_X$, the restriction $i^! : \mathrm{Obs}_X \to \mathcal{M}_{X/L}$, and the stratified factorization product $\mu^{\mathrm{str}}_{U_1,\ldots,U_k;I} : \mathrm{Obs}_X(U_1) \otimes \cdots \otimes \mathrm{Obs}_X(U_k) \otimes \mathrm{Obs}_L(I) \to \mathrm{Obs}_L(J)$ (whenever the U_i approach the brane interval $I \subset J$) together form a stratified prefactorization algebra on (X, L). Weiss descent is the additional assertion that the homotopy-colimit map

$$\operatorname*{hocolim}_{\mathcal{U}\in \mathrm{Weiss}(O)} \mathrm{Obs}^{\mathrm{str}}(\mathcal{U}) \xrightarrow{\sim} \mathrm{Obs}^{\mathrm{str}}(O)$$

is an equivalence for every admissible open $O \subset X$. After quantization this remains governed by the renormalized brane-defect propagator and compatible descent homotopies.

Proof. The bulk factorization algebra Obs_X is constructed by Costello–Gwilliam [15] for the Hamiltonian BF theory on $X = \mathbb{R}^2 \times \mathbb{C}^2$; the holomorphic factor satisfies Dolbeault descent on $\mathbb{C}^2$ and the topological factor satisfies de Rham descent on $\mathbb{R}^2$, so the product satisfies Weiss descent on the mixed open $O = U_{\mathrm{top}} \times B_{\mathrm{hol}} \subset X$. The brane-line theory is the E_1-factorization algebra $\mathrm{Obs}_{\partial,N}$ of Definition 1.24, satisfying E_1-Weiss descent on $L = \mathbb{R}_t$. The collision module $\mathcal{M}_{X/L}$ is constructed by extension by zero from $X \setminus L$ and restriction to L via the costalk $i^!$; the stratified factorization product is the bulk-to-brane operator-product expansion in the local Hamiltonian BF theory, extracted from the Wick contractions of the brane-defect propagator (Theorem G.39). These data give the stratified prefactorization maps. Weiss descent requires the homotopy-colimit descent equivalence on each stratum and compatibility of the structure maps with the $j_*/i^!$-adjunction; after quantization the same statement requires the renormalized propagator and descent homotopies. □

G.14 Large-N trace state and connected cumulants

The stable primitive trace algebra becomes a physical large-N string state only in the presence of an actual state and connected cumulants.

Definition G.42 *(Large-N trace state).* For each $N \geq 1$, a large-N trace state is a continuous $\mathbb{C}$-linear functional $\omega_{N,\Omega} : \mathcal{A}^q_{\partial,N}(I) \to \mathbb{C}[[\hbar_{\mathrm{QME}}, \hbar_W, \epsilon]]$ satisfying:

(i) $\omega_{N,\Omega}((Q + \hbar\Delta + \{I, -\})a) = 0$ (Ward identity);

(ii) $\omega_{N,\Omega}(ab) = (-1)^{|a||b|}\omega_{N,\Omega}(ba)$ (cyclicity, up to a declared anomaly line);

(iii) $\omega_{N,\Omega}(\mu_{I_1,\dots,I_k}(a_1 \otimes \cdots \otimes a_k)) = \omega_{N,\Omega}(a_1 \cdots a_k)$ (factorization compatibility).

The connected cumulants are $\kappa^N_k(a_1, \dots, a_k) = \sum_{\pi \in \mathrm{Part}(k)} (-1)^{|\pi|-1}(|\pi| - 1)! \prod_{B \in \pi} \omega_{N,\Omega}(\prod_{i \in B} a_i)$, and the large-$N$ limit is $\kappa^\infty_k(f_1, \dots, f_k) = \lim_{N\to\infty} N^{\beta(k)} \kappa^N_k(J_N(f_1), \dots, J_N(f_k))$ for some scaling exponent $\beta(k)$ and rescaling $J_N(f) = N^{-\alpha}\,\mathrm{Tr}\, f(\phi_1, \phi_2)$.

Theorem G.43 *(Large-N trace state).* The assembled datum $\mathfrak{U}_{\mathcal{T}}$ does not by itself construct a state $\omega_{N,\Omega}$ or the limits κ^∞_k. A large-N sector consists of the additional state of Definition G.42, Ward and cyclicity identities, a factorization condition, scaling exponents $\alpha, \beta(k)$, and convergence of the rescaled connected cumulants.

Proof. The cyclic trace, the QME counterterms, and the weighted Köthe estimate are necessary algebraic constraints on a possible state. They are not a measure, a Hamiltonian, a localization cycle, an asymptotic expansion, or a tightness theorem. These analytic structures belong to the large-N theory. □

G.15 The comparison datum and master Maurer–Cartan equation

Definition G.44 *(Comparison datum).* The comparison datum is

$$\mathfrak{U}_{\mathcal{T}} = (\mathsf{C}, \mathcal{E}_{\mathcal{T}}, \mathcal{E}_H, \mathrm{SDR}_{\mathcal{T}}, \mathcal{F}^q_{\mathrm{bulk}}, \mathcal{A}^q_{\partial,N}, \Phi^q_N, K_{\hbar,w,I}, \mathcal{R}_{\mathrm{rad}}, \mathcal{P}^\Pi_q, C_{\mathcal{T}}, \omega_\infty),$$

where C is the host filtered nuclear Tate dg category (§G.1); $(\mathcal{E}_{\mathcal{T}}, \mathcal{E}_H)$ are the parent and local Hamiltonian BV complexes (§G.2, §G.10); $\mathrm{SDR}_{\mathcal{T}}$ is the protected-sector cyclic BV strong deformation retract (Definition G.29); $\mathcal{F}^q_{\mathrm{bulk}}$ is the renormalized quantum bulk factorization algebra (Theorem G.41 and Theorem G.39); $\mathcal{A}^q_{\partial,N}$ is the quantum open Dirac brane factorization algebra (Theorem G.24); Φ^q_N is the quantum closed-to-open derived-centre morphism (§G.9); $K_{\hbar,w,I}$ is the all-order brane-defect QME kernel (Theorem G.39); $\mathcal{R}_{\mathrm{rad}}$ is the all-bidegree Weyl/radial normal-form theorem (Theorem G.9); $\mathcal{P}^\Pi_q$ is the native pro-Matlis current target (§G.5); $C_{\mathcal{T}}$ is the matched-conventions/global-descent datum (§11.4); ω_∞ is the large-N trace-state cumulant datum (§G.14).

Definition G.45 *(Assembled deformation complex).* The assembled target deformation complex is the product filtered L_∞ algebra

$$\mathfrak{K}_{\mathrm{univ},\mathcal{T}} = \mathfrak{K}_{\mathrm{prot},\mathcal{T}} \times \mathfrak{K}_{\mathrm{cent}} \times \mathfrak{K}_{\mathrm{bar/cobar}} \times \mathfrak{K}_{\mathrm{QME}} \times \mathfrak{K}_{\mathrm{rad}} \times \mathfrak{K}_{\mathrm{BM}} \times \mathfrak{K}_{\mathrm{glob},\mathcal{T}} \times \mathfrak{K}_{N\to\infty},$$

with

$$\begin{aligned}
\mathfrak{K}_{\mathrm{prot},\mathcal{T}} &= \mathrm{Def}_{\mathrm{cycBV}}(\mathcal{E}_H \rightleftarrows \mathcal{E}_{\mathcal{T}}), & \mathfrak{K}_{\mathrm{cent}} &= \mathrm{Conv}^{\mathrm{cts}}(C^{\mathrm{CE}}_{\bullet}(\mathfrak{g}_I), CC^{\bullet}(\mathrm{Obs}_{\partial,N}, \mathrm{Obs}_{\partial,N} \\
\mathfrak{K}_{\mathrm{bar/cobar}} &= \mathrm{Cone}\bigl(\Omega C^{\mathrm{CE}}_{\bullet}(\mathfrak{g}_{\mathrm{adm}}) \to U(\mathfrak{g}_{\mathrm{adm}})\bigr) \oplus \mathrm{Def}(A_{\mathrm{op}}, C_{\mathrm{op}}, \tau_{\mathrm{op}}), & \mathfrak{K}_{\mathrm{QME}} &= \mathfrak{Q}^{\bullet}_{w,\partial,\hbar}(I), \\
\mathfrak{K}_{\mathrm{rad}} &= \prod_{a,b} \mathrm{Cone}(B_{a,b})[-1], & \mathfrak{K}_{\mathrm{BM}} &= \mathbf{R}\varprojlim_{N} CC^{\bullet}(P_{\le N}, P_{\le N}), \\
\mathfrak{K}_{\mathrm{glob},\mathcal{T}} &= \mathbf{R}\Gamma(Y, \mathcal{K}_{\mathcal{T}}), & \mathfrak{K}_{N\to\infty} &= \mathrm{Def}\bigl(\{\omega_{N,\Omega}\}_{N\ge 1}, \omega_\infty\bigr).
\end{aligned}$$

THEOREM G.46 *(Closed-open formal Hamiltonian brane table).* Let $\mathcal{T}$ be an admissible protected mixed holomorphic-topological string-field-theory target with assembled datum $\mathfrak{U}_{\mathcal{T}}$ of Definition G.44. The local theorem of Theorem 5.147 produces a transported local trace-sector datum Θ_{loc}. The realization of this datum in $\mathcal{T}$ is governed by the following component obstruction table:

(i) Strict realization ($\tau = 0$): all named component obstructions vanish and the comparison-cone contraction exists. Then the formal Darboux brane stalk of $\mathcal{T}$ realizes the local Hamiltonian BF/Dirac-brane trace-sector model on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{0,\mathrm{hol}}$.

(ii) Counterterm realization: $\tau = \sum \tau_i$ with each $\tau_i \in F^1\mathfrak{K}^{\circ}_i$ cancelling the i-th component curvature; concrete constructions are τ_{rad} (decorated PBW-Stokes, by Theorem G.9), $\tau_{\mathrm{QME}} = C^{\mathrm{red}}_{\hbar,w}$ (Milnor counterterm, by Theorem G.36), τ_{cent} (Hochschild equivariance homotopies, by Theorem G.26).

(iii) Projective/descent-torsor realization: the obstruction lands in a central summand and the corrected element solves the master equation in a central extension; the scalar Capelli anomaly $\hbar N[\bar{c}]$ is of this type. The global Hamiltonian period $\mathrm{per}_X(v)$ is a descent torsor unless a degree-two central extension is constructed.

(iv) Target replacement: the curvature is non-absorptive and forces a substitution; the attempted strict all-window Bochner–Martinelli current transfer is of this type, with replacement $\mathcal{P} \rightsquigarrow \mathcal{P}^{\Pi}_q$ by Theorem G.14.

The formal trace map is $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$, selected by Theorem G.5; the cotangent module is the Matlis principal-part module $\mathcal{P}^{\Pi}_q$; and the quantum closed-to-open derived-centre map is defined under the centrality and QME data: $\Phi^q_N : \mathrm{Obs}^q_{\mathrm{closed},H}|_L \to Z^q_{E_1}(\mathrm{Obs}^q_{\mathrm{open},N})$ of §G.9.

Proof. Each component complex of Definition G.45 is the deformation complex constructed in the corresponding section, and the local datum Θ_{loc} is the assembled image of the local trace map J, the closed Hamiltonian sector $\mathfrak{g}$, the brane factorization algebra $\mathcal{A}^{\mathrm{cl}}_{\partial,N}$, and the protected SDR $i_{\mathcal{T}}$. The eight named obstruction classes are $(\mathrm{Ob}_{\mathrm{prot}}, \mathrm{Ob}_{\mathrm{cent/QME/desc}}, \Omega^{\mathrm{rad}}, \mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}, \mathrm{Ob}^{\Pi}_{\mathrm{BM}}, \mathrm{per}_X(v), \hbar N[\bar{c}], \mathrm{Ob}_{N\to\infty})$. A single curvature class exists only after a common deformation complex and comparison maps have been constructed. The realization modes are read component-by-component, with the explicit twists given by the constructions of §§G.4–§G.14. The formal trace map is Theorem G.5; the pro-Matlis cotangent module is Theorem G.14; the derived-centre map is defined under the assembly of the Hochschild equivariance homotopies (Theorem G.26) with the all-order QME kernel (Theorem G.39). □

G.16 Compact comparison reading

The local Hamiltonian BF/Dirac-brane theorem (Theorem G.46) is the formal-Darboux trace-sector map

$$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}}|_b \longrightarrow HH^\bullet(A^{\mathrm{cl}}_{\partial,N}, A^{\mathrm{cl}}_{\partial,N})$$

at the Dirac brane vacuum. It identifies the stable scalar-reduced trace sector after admissible completion. A full equivalence with the brane-side derived E_1-Hochschild object requires the Hochschild equivariance, QME, and descent data of Theorems G.26, G.39, and G.41; under the binding convention of Conventions and notation for the local theorem the notation $Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ is used only when that chiral centre structure has been constructed. Volume I constructs $Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ algebraically through ordered bar-cobar duality; Volume II identifies it with the bulk observables of a 3d holomorphic-topological boundary-bulk system; Volume III gives the chiral algebra $A = \mathrm{SpCh}_{\Sigma_{d-1},C} \circ \Phi^{\mathrm{FA}}_d(C)$ through the composite Calabi–Yau-to-chiral functor Φ_d; the formal-Darboux trace theorem computes the formal Hamiltonian normal-slice restriction of that bulk to a finite Chan–Paton brane stack via the trace map $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$. On that normal slice the holomorphic collision product is explicit: Theorem 1.17 computes the diagonal OPE with kernel $K^{(2)}_\Delta$. The cross-target comparison below must preserve this arity-two local cohomology operation before adding descent, QME, or compact-target period data.

Theorem G.47 *(Cross-target formal-slice principle).* Let C be a CY-d category satisfying the hypotheses of the Volume III functor $\Phi_d = \mathrm{SpCh}_{\Sigma_{d-1},C} \circ \Phi^{\mathrm{FA}}_d$, and let $A = \Phi_d(C)$. Assume the Volume II holomorphic-topological realization $\mathcal{T}_A$ admits an admissible protected Hamiltonian normal sector at a brane L, with formal holomorphic-symplectic normal slice $(\widehat{\mathbb{C}^2}_0, dz_1 \wedge dz_2)$ and a GL_N-Chan–Paton Dirac brane formal-stalk chart. Then the protected formal-normal trace sector of $Z^{\mathrm{der}}_{\mathrm{ch}}(A) \simeq \mathrm{Obs}_{\mathrm{bulk}}(\mathcal{T}_A)$ has the shifted-cotangent CE/PV coordinate model

$$C^\bullet_{\mathrm{CE}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]) \longrightarrow PV(S\bar{A}_\partial),$$

with Hochschild/PV generator component

$$c_f \longmapsto \theta_f, \qquad u_f \longmapsto M_{J(f)}, \qquad J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)},$$

and scalar-contact component

$$c_K \longmapsto \gamma_1 \quad \text{in} \quad C^\bullet_{\mathrm{Lie}}(\bar{A}; \mathbb{C}\gamma_1)[1].$$

The full derived-centre comparison beyond this formal trace sector is governed by the obstruction classes

$$\left(\hbar N[\bar{c}],\ \Omega^{\mathrm{rad}}_{a,b},\ [\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}],\ \mathrm{Ob}_{\mathrm{cent/QME/desc}},\ \mathrm{Ob}^{\Pi}_{\mathrm{BM}},\ \mathrm{Ob}_{\Phi_d\text{-descent}}\right)$$

of Definition G.45. These entries become a curvature vector only after a common filtered deformation complex and comparison maps from the native complexes have been constructed.

Proof. The Volume II identification $Z^{\mathrm{der}}_{\mathrm{ch}}(A) \simeq \mathrm{Obs}_{\mathrm{bulk}}(\mathcal{T}_A)$ gives the bulk factorization algebra; the protected-sector SDR (§G.10) restricts this bulk to the formal Hamiltonian normal slice $\mathcal{E}_H$; the shifted-cotangent CE/PV model (Theorem 4.20) computes the algebraic trace sector of $\mathcal{E}_H$; the trace map (Theorem G.5) identifies the closed-to-open coordinate $u_f \mapsto M_{J(f)}$; the Hamiltonian action gives $c_f \mapsto \theta_f$; and the scalar-contact quotient gives $c_K \mapsto \gamma_1$. The obstruction vector is the component table of Theorem G.46 augmented by the Φ_d-descent component recording the matched-conventions obstruction (§11.4). □

Remark G.48 *(Modular and Borcherds/BKM bridges).* The local Hamiltonian scalar cocycle $\hbar N[\bar{c}]$ and the Borcherds weight κ_{BKM} of the level-two modular section Δ_5 are distinct data. Their identification is an operator-level trace statement; scalar weights are protected trace shadows of such a cocycle. The Borcherds/BKM comparison in this formal neighbourhood is the matched-conventions comparison of scalar shadows. Likewise, the BCOV holomorphic anomaly, Gopakumar–Vafa, Donaldson–Thomas, and MNOP classes are compact-target comparison data transported through Theorem G.47 and §11.4; globalization is the corresponding descent theorem.

G.17 Obstruction classes

THEOREM G.49 *(Obstruction classes of the assembled datum).* The assembled datum of Definition G.45 has the following eight named obstruction components. They become one curvature vector only inside a common deformation complex with comparison maps.

Class	Construction	Mode	Reference
$\hbar N[\bar{c}]$	scalar Capelli cocycle on $\bar{A}$	projective	Theorem E.13, thi
$\Omega^{\mathrm{rad}}_{a,b}$	decorated PBW-Stokes contracting homotopy	counterterm (vanishing)	Thm. G.9
$[\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}]$	Milnor obstruction vector $\mathfrak{D}^{\mathrm{ns}}$	counterterm (Milnor-compatible)	Thm. G.36
$\mathrm{Ob}_{\mathrm{cent/QME/desc}}$	Hochschild equivariance homotopies	counterterm (homotopy)	Thm. G.26
$\mathrm{Ob}^{\Pi}_{\mathrm{BM}}$	pro-Matlis weighted Köthe $\mathcal{P}^{\Pi}_{q}$	target replacement	Thm. G.14
$\mathrm{per}_X(v)$	Hamiltonian descent torsor	descent	§G.16
$\mathrm{Ob}_{\mathrm{prot}}$	cyclic BV strong deformation retract	L_∞-transferred bracket / counterterm	Thm. G.30
$\mathrm{Ob}_{N\to\infty}$	large-N state $\omega_{N,\Omega}$ and limits	state extension	Thm. G.43

The hypotheses of the local theorem are expressed as explicit obstruction classes, relative constructions, or replacement objects in Theorem G.49. The remaining hypotheses are the matched-conventions data $C_{\mathcal{T}}$, the protected SDR $(i_{\mathcal{T}}, p_{\mathcal{T}}, h_{\mathcal{T}})$, the all-bidegree radial primitive outside the verified finite windows, the all-order QME compatibility, and the large-N state.

Proof. Each row is a corollary of the cited theorem, with the realization mode read off from the curvature mechanism: counterterm-vanishing via contracting homotopy when fixed (radial), Milnor counterterm-compatibility (QME), Hochschild equivariance homotopy (centre), pro-target replacement (BM), central extension (Capelli scalar), descent torsor (period), and SDR construction (protected sector). The matched-conventions datum is controlled by §11.4; the additional analytic and large-N data are those named in the theorem. □

Remark G.50 *(Remaining data).* After Theorem G.49, the remaining data are: (i) the parent BV theory $(\mathcal{E}_{\mathcal{T}}, Q_{\mathcal{T}}, \omega_{\mathcal{T}}, S_{\mathcal{T}})$ and the protected supercharge $\boldsymbol{Q}$ for a specific compact target $\mathcal{T}$; (ii) the matched-conventions datum $C_{\mathcal{T}}$ (residue versus equivariant Euler normalization, orientation, framing on S^3, genus expansion endpoints) for that target; (iii) the uniform decorated PBW-Stokes contracting homotopy $(A^{(j)}_{a,b,N}, \tau_{a,b})$ at every bidegree, of which the finite-window checks give the recorded ranges (Remark G.10); and (iv) the Statement F.8(1)–(2) homomorphism and injectivity theorems from the cited radial-parts literature; (v) the all-order QME compatibility across the inverse system; and (vi) the large-N state and cumulant convergence data. The remaining clauses are represented by constructed objects or by obstruction classes in the assembled datum $\mathfrak{U}_{\mathcal{T}}$.

G.18 Proofs and obstruction classes

Each component of the assembled datum is classified by the theorem that proves it, the hypotheses under which it is formulated, and the cohomology class that obstructs its extension.

Theorem G.3 (classical master equation). Proved from first principles, in the host category C, using $D^2 = 0$, graded Leibniz, Jacobi in $\mathfrak{h}$, and continuity of the residue pairing. The proof is internal to the host category.

Theorem G.5 (trace uniqueness). Proved from the linear normalization $c_{1,0} = c_{0,1} = 1$ and Darboux-naturality. The nonlinear Hamiltonian shear $H_k = z_2^{k+1}/(k+1)$ closes the proof; multiplicativity is forced.

Theorem G.9 (decorated radial reformulation). This is a reformulation toward first-principles closure. The all-bidegree statement is converted into a primitive-existence problem on a finite-rank decorated bicomplex; exact rational calculation gives the primitives at the recorded bidegrees, and the all-(a, b) statement is the assertion that this primitive construction is uniform. The remaining obstruction is the all-bidegree decorated primitive.

Theorem G.14 (pro-Matlis target). Proved as a replacement target with continuous action and modified window coherence. The continuity estimate (Proposition G.13) is by standard Köthe-space techniques. Terminality in $\mathrm{Curr}_{\mathrm{BM}}$ is not asserted.

Theorem G.17 and Theorem G.19 (CE/PV levels 2 and 3). Proved from the bracket-admissible B_q of Definition G.16 and the kernel-admissible $\mathcal{K}_{B_q}$ of Definition G.18, both constructed in the Köthe topology. Soundness depends on the continuity estimate, which is standard.

Theorem G.22 (CE/PV level 4, bar-cobar). Proved at each finite quotient by Quillen rational-homotopy bar-cobar. The completed comparison requires the derived-limit obstruction terms to vanish and the bar-cobar functor to commute with the chosen completion.

Theorem G.24 (CE/PV level 5, open-trace A_∞-Koszul). Proved by Koszul duality in the cyclic A_∞ operad (Loday–Vallette, Getzler–Jones). Soundness depends on the cyclicity hypothesis on A_{op}, which holds for the brane factorization algebra $A^{\mathrm{cl}}_{\partial,N}$.

Theorem G.26 (Hochschild equivariance, bracket-admissible sector). On the bracket-admissible sector B_q, the Koszul homotopy for $[\phi_1, \phi_2] = 0$ gives the explicit Hochschild homotopies. The unrestricted centre lift on arbitrary cotangent currents requires the brane-defect propagator extension (Theorem G.39), acknowledged in Remark G.27.

Theorem G.30 (protected-sector SDR obstruction). Proved as an obstruction theorem given an SDR (i_T, p_T, h_T); existence of the SDR for a specific compact target is a theorem in the compact field theory, constrained by the matched-conventions obstruction.

Theorem G.33 (σ_{sc} filtered chain map). The unreduced scalar cocycle has primitive η. On the reduced Hamiltonian algebra the class $[\bar{c}]$ remains projective. The filtered lift is governed by the obstruction tower of Theorem G.36.

Theorem G.36 (Milnor obstruction vector). The Milnor exact sequence is applied to the inverse system of finite Hamiltonian windows. Vanishing of $\mathfrak{D}_n^{\mathrm{ns}}$ for every n is the all-order QME criterion; the θ_3 row verifies this at $n = 3$.

Theorem G.39 (Costello brane-defect graph QME). Proved by Costello's renormalization theorem adapted to the brane-defect support stratum, plus Hörmander wavefront control of the heat-kernel propagator restriction. The QME is solved modulo the residual non-scalar class $\mathrm{Ob}^{\mathrm{red}}_{w,\partial,\hbar}$, which is killed by the Milnor counterterm tower; soundness depends on the explicit Costello finite-scale graph expansion in the weighted Tate topology, which holds for the Köthe weights of Definition G.12.

Theorem G.41 (stratified factorization on (X, L)). Proved at the prefactorization level by extension by zero plus brane-defect OPE. The holomorphic part of this OPE is the native diagonal local-cohomology operation of Theorem 1.17; the brane-defect term adds the interval support current and defect restriction. Weiss descent follows from descent on each stratum and $j_*/i^!$-adjunction. The post-quantization statement requires the renormalized brane-defect propagator, which is Theorem G.39.

Theorem G.43 (large-N trace state). This theorem assumes a state, Ward identity, cyclicity, factorization, scaling exponents, and convergence of connected cumulants. Köthe continuity and QME counterterms are necessary algebraic constraints, not an $N \to \infty$ theorem.

Theorem G.46 (closed-open formal Hamiltonian brane comparison). This assembles the preceding obstruction classes and relative constructions into the comparison datum $\mathfrak{U}_{\mathcal{T}}$. The remaining hypotheses are the matched-conventions datum, the SDR existence, the radial all-bidegree primitive outside verified windows, the all-order QME compatibility, and the large-N state.

Theorem G.47 (cross-target formal-slice principle). Proved as the formal restriction of the trace sector of the Volumes I/II derived chiral centre to the protected Hamiltonian normal sector, with the component obstruction classes recording the global, quantum, chain-level, all-window, and Calabi–Yau comparison obstructions. It is formulated under the Volume III categorical theorem and Volume II HT realization.

Remark G.51 *(Algebraic and analytic hypotheses).* The classical master equation and the trace uniqueness theorem are identities in C. The pro-Matlis replacement target, the finite-quotient bar-cobar comparison, the open-trace A_∞-Koszul comparison, the Milnor obstruction vector, and the universal obstruction vector carry their derived-limit obstruction classes explicitly. The CE/PV levels 2–5, the Hochschild equivariance homotopies, the σ_{sc} chain map, the brane-defect graph QME, and the stratified factorization algebra require bracket-admissibility, kernel-admissibility, SDR, and decorated-Stokes hypotheses. A compact target also requires a parent BV theory and protected supercharge, the matched-conventions datum, a uniform decorated PBW-Stokes contracting homotopy in every bidegree, clauses (1)–(2) of Statement F.8, all-order QME compatibility, and a large-N trace state.

H Coordinate constructions

Six coordinate constructions recover the local trace substitution J, the closed Hamiltonian BF theory, the CE/PV comparison, the Matlis coadjoint action, quantization at a fixed bidegree, and global transport along a Čech–Weiss–Ran cover.

Construction 1 (The local trace substitution J)

Data. A Chan–Paton rank $N \in \mathbb{Z}_{\geq 1}$; the formal symplectic disk $(D, \omega) = (\mathrm{Spf}\, \mathbb{C}[[z_1, z_2]], dz_1 \wedge dz_2)$.

Construction. (i) Set the brane phase space $\mathcal{P}_N = \mathfrak{gl}_N^{\oplus 2}$.

(ii) Define the brane symplectic form $\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2)$ and the Liouville one-form $\theta_N = \mathrm{Tr}(\phi_1\, d\phi_2)$.

(iii) Define the GL_N-action by simultaneous conjugation $g \cdot (\phi_1, \phi_2) = (g\phi_1 g^{-1}, g\phi_2 g^{-1})$ and the infinitesimal vector field $\partial_\epsilon \phi_i = [\epsilon, \phi_i]$.

(iv) Compute the moment map from $d\langle \mu, \epsilon\rangle = \iota_{\partial_\epsilon}\Omega_N$: $\langle \mu(\phi_1, \phi_2), \epsilon\rangle = -\operatorname{Tr}\epsilon[\phi_1, \phi_2]$.

(v) Construct the BV antifield $\psi \in \mathfrak{gl}_N[1]$ with $Q\psi = [\phi_1, \phi_2]$; the algebra of functions on the derived zero fibre is $\mathcal{O}(\mu_{\text{der}}^{-1}(0)) = \operatorname{Kosz}\big(\mathcal{O}(\mathfrak{gl}_N^{\oplus 2}), [\phi_1, \phi_2]\big)$.

(vi) Take ghost-zero GL_N-invariant cohomology to obtain the stable trace ring $R_N = H^0\big(\mathcal{O}(\mu_{\text{der}}^{-1}(0))^{GL_N}, Q\big)$.

(vii) Reduce stable invariants to commutative monomial traces by the Koszul Q-exactness of adjacent transpositions on the moment-map zero-fibre, obtaining the basis $\{\operatorname{Tr}(\phi_1^a \phi_2^b)\}_{a,b\geq 0}$.

(viii) Define the trace map on monomials: $J(z_1^a z_2^b) = \overline{\operatorname{Tr}(\phi_1^a \phi_2^b)}$, where the bar denotes scalar reduction.

(ix) Extend by linearity to polynomial Hamiltonians. Extend further by $\mathfrak{m}$-adic continuity only after completing the brane trace algebra at the formal Darboux point: $J(f) = \sum f_{a,b}\overline{\operatorname{Tr}(\phi_1^a \phi_2^b)}$.

Result. The trace substitution on polynomials, and its completed formal extension $J: \mathfrak{h} \to \widehat{\mathbf{S}}(\bar{A})$, $f \mapsto \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$, with $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$.

Construction 2 (The closed Hamiltonian BF theory)

Data. The reduced Hamiltonian Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1$ with Poisson bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$.

Construction. (i) Compute the continuous dual $\mathfrak{h}^\vee_{\text{cont}} = \mathcal{P} = \bigoplus_{a+b>0}\mathbb{C}\rho_{a,b}$ by the Matlis residue identification $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$ under the residue pairing $\langle \rho_{a,b}, z_1^c z_2^d\rangle = \delta_{a,c}\delta_{b,d}$ (Construction 4).

(ii) Form the shifted cotangent Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\text{cont}}[1]$ with bracket $[(f, \eta), (g, \xi)] = (\{f, g\}, \operatorname{ad}^\vee_f \xi - (-1)^{|\eta||g|}\operatorname{ad}^\vee_g \eta)$ and $\operatorname{ad}^\vee_f$ the Hamiltonian coadjoint action computed by Construction 4.

(iii) On a product open $U \times B \subset \mathbb{R}^2_{\text{top}} \times \mathbb{C}^2_{\text{hol}}$ define the closed BV fields $\mathcal{E}_{\text{cl}}(U \times B) = \Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}(\mathfrak{h}[1] \oplus \mathfrak{h}^\vee_{\text{cont}}[2])$, with components $\alpha \in \Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{h}[1]$ and $\beta \in \Omega^\bullet_c(U)\widehat{\otimes}\Omega^{0,\bullet}_c(B)\widehat{\otimes}\mathfrak{h}^\vee_{\text{cont}}[2]$.

(iv) Differential $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$; BV pairing $\langle \beta, \alpha\rangle_{\text{BV}} = \int_{U\times B}\langle \beta, \alpha\rangle_{\mathfrak{h}^\vee \otimes \mathfrak{h}}\, dz_1 dz_2$.

(v) Action $S_{\text{BF}} = \int_{U\times B}\langle \beta, D\alpha + \frac{1}{2}\{\alpha, \alpha\}\rangle\, dz_1 dz_2$.

(vi) Verify the classical master equation $\{S_{\text{BF}}, S_{\text{BF}}\}_{\text{BV}} = 0$ from $D^2 = 0$, D-derivation property of $\{\cdot,\cdot\}$, Jacobi for $\{\cdot,\cdot\}$, and coadjointness of $\mathfrak{h}^\vee_{\text{cont}}$.

Result. The mixed holomorphic-topological Hamiltonian BF factorization algebra $F^{\text{mix}}_{\text{Ham}}$ on $\mathbb{R}^2_{\text{top}} \times \mathbb{C}^2_{\text{hol}}$, satisfying CME.

Construction 3 (CE/PV comparison)

Data. The shifted cotangent Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\text{cont}}[1]$ of Construction 2; an admissibility class (Tate / nilpotent / weighted / relative) for the higher levels of the comparison.

Construction. (i) Choose ordered bases $\{e_i\}$ of $\mathfrak{h}$ (e.g. $e_{a,b} = z_1^a z_2^b$) and dual residues $\{e^i\} \subset \mathfrak{h}^\vee_{\text{cont}}$ (e.g. $e^{a,b} = \rho_{a,b}$).

(ii) Introduce CE generators c^I (degree-1 duals to e_I) and u_I (degree-1 duals to e^I) for the cochain complex $C^\bullet_{\text{CE,coord}}(\mathfrak{g})$.

(iii) Introduce PV generators θ^I (Schouten polyvectors on $\mathbf{S}(\bar{A})$) and O_I (boundary observables on the trace ring) for the polyvector complex $\mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$.
(iv) Define the cochain comparison map on generators: $c^I \mapsto \theta^I$, $u_I \mapsto O_I$.
(v) Verify the differential intertwiners: under CE differential d_{CE} and Schouten differential d_π^{coord}, $d_{\mathrm{CE}}c^I \leftrightarrow [\theta^I, \theta^I]$, $d_{\mathrm{CE}}u_I \leftrightarrow \theta^I(O_I)$, with the second identity yielding the decisive computation $\theta_f(O_g) = O_{\{f,g\}}$.
(vi) Complete in the chosen coordinate window or weighted-Tate topology. The cochain identity (level 1) uses no extra datum; the P_0-bracket clause (level 2) holds on a bracket-admissible subtarget B; the Maurer–Cartan kernel clause (level 3) holds on a kernel-admissible datum $(B, \mathcal{K}_B)$; the bar-cobar / enveloping clause (level 4) holds on a Tate / nilpotent / weighted admissibility datum with conilpotent endpoint and $\varprojlim^1$-vanishing; the open-trace A_∞-Koszul clause (level 5) holds on a homotopy datum.
(vii) Record which levels converge in the chosen completion; the remainder is the obstruction vector Ob_T of §7.5.

Result. The cochain CE/PV map $\Phi_{\mathrm{coord}}\colon C^\bullet_{\mathrm{CE,coord}}(\mathfrak{g}) \to \mathrm{PV}_{\mathrm{coord}}(A_{\partial,\mathrm{coord}})$, together with its admissible extensions and obstruction classes.

Construction 4 (The Matlis coadjoint action)

Data. A Hamiltonian $f = z_1^p z_2^q \in \mathfrak{h}$ and a residue functional $\rho_{a,b} \in \mathcal{P}$.

Construction. (i) Compute the Hamiltonian vector field $X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1} = p z_1^{p-1} z_2^q \partial_{z_2} - q z_1^p z_2^{q-1}\partial_{z_1}$.
(ii) For any $g \in \mathfrak{h}$, evaluate the residue pairing $\langle f \cdot \rho_{a,b}, g\rangle = -\langle \rho_{a,b}, \{f, g\}\rangle$.
(iii) Expand $\{f, g\} = X_f(g)$ in the basis $\{z_1^c z_2^d\}$ and read off the Taylor coefficients.
(iv) Apply the residue-coadjoint formula

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\, \rho_{a-p+1,b-q+1}$$

of Theorem 5.179.
(v) For an arbitrary $f = \sum_{p,q} f_{p,q} z_1^p z_2^q$, expand by linearity: $f \cdot \rho_{a,b} = \sum_{p,q} f_{p,q}(z_1^p z_2^q \cdot \rho_{a,b})$. The result is a finite linear combination of residues $\rho_{a-p+1,b-q+1}$; terms with $a - p + 1 < 0$ or $b - q + 1 < 0$ vanish by support.

Result. The Matlis coadjoint action $f \cdot \rho_{a,b} \in \mathcal{P}$ as a finite linear combination of basis residues. This is the calculation that detects the Bochner–Martinelli finite-window leak (Theorem 8.40) at the canonical row $(p, q, a, b) = (N + 2, 0, N + 1, 0)$: $z_1^{N+2} \cdot \rho_{N+1,0} = -(N + 2)\rho_{0,1}$.

Construction 5 (Quantization at bidegree (a, b))

Data. A bidegree (a, b); the formal Moyal star product on the symplectic disk to the order required by (a, b).

Construction. (i) Generate the radial PBW basis of $C^0_{a,b}$ (normal monomials on a letters X and b letters Y).
(ii) Generate the Weyl-normal basis of the same cohomological degree by Moyal-symmetrizing each PBW monomial.
(iii) Compute the Moyal terms of $z_1^a \star z_2^b$ up to the order required by (a, b) using $f \star g = fg + \frac{\hbar}{2}P(f, g) + \frac{\hbar^2}{8}P^2(f, g) + \cdots$.

(iv) Construct the matrix $B_{a,b}\colon C^{1}_{a,b} \to C^{0}_{a,b}$ of adjacent-transposition relations and the radial comparison vector $v_{a,b} = (T_{a,b}, E^{+}_{a,b})$ (Weyl-normal lift minus Moyal-normal lift).

(v) Compute $v_{a,b}$ over $\mathbb{Q}$ by exact rational arithmetic.

(vi) Decide whether $v_{a,b} \in \operatorname{im} B_{a,b}$ by Bareiss row reduction.

(vii) Obtain either the unique correction $C^{+}_{a,b}$ with $B_{a,b}C^{+}_{a,b} = v_{a,b}$, or a covector $(\phi, -\lambda) \in \ker B^{*}_{a,b}$ with $\lambda(E^{+}_{a,b}) - \phi(T_{a,b}) \neq 0$ representing $\Omega^{\mathrm{rad}}_{a,b} \in \operatorname{coker} B_{a,b}$. The same data are read by the exact rational reduction of Exact rational reduction.

Result. Either $\Omega^{\mathrm{rad}}_{a,b} = 0$, with explicit correction, or $\Omega^{\mathrm{rad}}_{a,b} \neq 0$, with explicit cokernel witness.

Construction 6 (Global transport)

Data. A holomorphic-symplectic ambient (X, ω); a brane $L \subset X$; a Chan–Paton rank N; a deformation functor $\mathcal{T}$ – global Hamiltonian descent, Costello QME, unreduced P_0-factorization-centre lift, decorated radial Stokes, BM current transfer, modular open-closed clutching, or CY-to-chiral specialisation.

Construction. (i) Choose a Darboux cover $\{U_i\}$ of the brane neighbourhood such that each completed normal patch is symplectomorphic to $\operatorname{Spf} \mathbb{C}[[z_1, z_2]]$.

(ii) On each U_i, build the local trace substitution $J_i(f_i) = \overline{\operatorname{Tr} f_i(\phi_{1,i}, \phi_{2,i})}$ by Construction 1.

(iii) On overlaps U_{ij}, compute transition symplectomorphisms $g_{ij} \in \mathrm{Symp}_{\mathrm{formal}}(D)$ and lift them to local Hamiltonians h_{ij} where possible: $g_{ij} = \exp(X_{h_{ij}})$.

(iv) Build the Čech–Weiss–Ran deformation complex $\mathfrak{K}_{\mathcal{T}} = \operatorname{Tot} C^{\bullet}_{\check{C}/\mathrm{WR}}(\{U_i\}, \mathfrak{K}_{\mathrm{loc}})$.

(v) Insert the local Maurer–Cartan element $\Theta_{\mathrm{loc}} = (\Theta_i, \alpha_{ij}, \beta_{ijk}, \dots)$, where Θ_i is the Construction 1 trace-substitution datum on U_i, α_{ij} is the lift to local Hamiltonians, and higher components encode descent homotopies.

(vi) Compute the curvature $F_{\mathcal{T}}(\Theta_{\mathrm{loc}}) = d\Theta_{\mathrm{loc}} + \frac{1}{2}\ell_2(\Theta_{\mathrm{loc}}, \Theta_{\mathrm{loc}}) + \cdots$ in $F^2\mathfrak{K}^2_{\mathcal{T}}$.

(vii) Decompose into named components: $\mathrm{Ob}_{\mathcal{T}} = \big(\mathrm{per}_X(v), \mathrm{ob}_{\mathrm{scal}}, \mathrm{ob}^{(1)}_{\mathrm{WR}}, \{\mathrm{ob}^{(r)}_{\mathrm{WR}}\}_{r\geq 1}\big)$ and similar component lists for the chosen target.

(viii) Solve recursively for filtered twists $\tau = \tau_1 + \tau_2 + \cdots$: $d_\Theta \tau_n = -F_n - \frac{1}{2}\sum_{i+j=n}[\tau_i, \tau_j] - \cdots$, applying Theorem 7.1 of § 7.

(ix) The curvature alternatives are: zero curvature $\Rightarrow$ clause (1), strict global theorem; exact curvature with chosen homotopy $\Rightarrow$ clause (2), counterterm coherence; central but inexact curvature $\Rightarrow$ clause (3), pass to gerbe-valued/projective deformation functor; non-absorptive curvature $\Rightarrow$ clause (4), force functor replacement and record the matched-conventions data.

Result. Either a strict or twisted global $\mathcal{T}$-coherence together with its given null-homotopy data, or an obstruction theorem together with the curvature class blocking the comparison.

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